

AMC 12

§1 AMC 12 12B 2020

1. (AMC 12 12B 2020) What is the value in simplest form of the following expression?

$$\sqrt{1} + \sqrt{1+3} + \sqrt{1+3+5} + \sqrt{1+3+5+7}$$

- (A) 5 (B) $4 + \sqrt{7} + \sqrt{10}$ (C) 10 (D) 15 (E) $4 + 3\sqrt{3} + 2\sqrt{5} + \sqrt{7}$

Answer: C

Solution

We have

$$\sqrt{1} + \sqrt{1+3} + \sqrt{1+3+5} + \sqrt{1+3+5+7} = \sqrt{1} + \sqrt{4} + \sqrt{9} + \sqrt{16} = 1 + 2 + 3 + 4 = \boxed{\text{(C)} 10}.$$

Note: This comes from the fact that the sum of the first n odds is n^2 .

2. (AMC 12 12B 2020) What is the value of the following expression?

$$\frac{100^2 - 7^2}{70^2 - 11^2} \cdot \frac{(70 - 11)(70 + 11)}{(100 - 7)(100 + 7)}$$

- (A) 1 (B) $\frac{9951}{9950}$ (C) $\frac{4780}{4779}$ (D) $\frac{108}{107}$ (E) $\frac{81}{80}$

Answer: A

Solution

Using difference of squares to factor the left term, we get

$$\frac{100^2 - 7^2}{70^2 - 11^2} \cdot \frac{(70 - 11)(70 + 11)}{(100 - 7)(100 + 7)} = \frac{(100 - 7)(100 + 7)}{(70 - 11)(70 + 11)} \cdot \frac{(70 - 11)(70 + 11)}{(100 - 7)(100 + 7)}.$$

Cancelling all the terms, we get $\boxed{\text{(A)} 1}$ as the answer.

3. (AMC 12 12B 2020) The ratio of w to x is $4 : 3$, the ratio of y to z is $3 : 2$, and the ratio of z to x is $1 : 6$. What is the ratio of w to y ?

- (A) $4 : 3$ (B) $3 : 2$ (C) $8 : 3$ (D) $4 : 1$ (E) $16 : 3$

Answer: E

Solution 1 (One Sentence)

We have

$$\frac{w}{y} = \frac{w}{x} \cdot \frac{x}{z} \cdot \frac{z}{y} = \frac{4}{3} \cdot \frac{6}{1} \cdot \frac{2}{3} = \frac{16}{3},$$

from which $w : y = \boxed{\text{(E)} 16 : 3}$.

Solution 2

We need to somehow link all three of the ratios together. We can start by connecting the last two ratios together by multiplying the last ratio by two.

$z : x = 1 : 6 = 2 : 12$, and since $y : z = 3 : 2$, we can link them together to get $y : z : x = 3 : 2 : 12$.

Finally, since $x : w = 3 : 4 = 12 : 16$, we can link this again to get: $y : z : x : w = 3 : 2 : 12 : 16$, so $w : y = \boxed{\text{(E)} 16 : 3}$.

Solution 3

We have the equations $\frac{w}{x} = \frac{4}{3}$, $\frac{y}{z} = \frac{3}{2}$, and $\frac{z}{x} = \frac{1}{6}$. Clearing denominators, we have $3w = 4x$, $2y = 3z$, and $6z = x$. Since we want $\frac{w}{y}$, we look to find y in terms of x since we know the relationship between x and w . We begin by multiplying both sides of $2y = 3z$ by two, obtaining $4y = 6z$. We then substitute that into $6z = x$ to get $4y = x$. Now, to be able to substitute this into our first equation, we need to have $4x$ on the RHS. Multiplying both sides by 4, we have $16y = 4x$. Substituting this into our first equation, we have $3w = 16y$, or $\frac{w}{y} = \frac{16}{3}$, so our answer is $\boxed{\text{(E)} 16 : 3}$.

Solution 4

WLOG, let $w = 4$ and $x = 3$.

Since the ratio of z to x is $1 : 6$, we can substitute in the value of x to get $\frac{z}{3} = \frac{1}{6} \implies z = \frac{1}{2}$.

The ratio of y to z is $3 : 2$, so $\frac{y}{\frac{1}{2}} = \frac{3}{2} \implies y = \frac{3}{4}$.

The ratio of w to y is then $\frac{4}{\frac{3}{4}} = \frac{16}{3}$ so our answer is $\boxed{\text{(E)} 16 : 3}$.

Solution 5

We have $\frac{w}{x} = \frac{4}{3}$, $\frac{y}{z} = \frac{3}{2}$, $\frac{z}{x} = \frac{1}{6}$. Find LCD: $\frac{w}{x} = \frac{8}{6}$, $\frac{z}{x} = \frac{1}{6}$, we have $\frac{w}{z} = \frac{8}{1}$. Similarly, $\frac{w}{z} = \frac{16}{2}$, $\frac{y}{z} = \frac{3}{2}$, so $\frac{w}{y} = \frac{16}{3}$, $\boxed{\text{(E)} 16 : 3}$.

4. (AMC 12 12B 2020) The acute angles of a right triangle are a° and b° , where $a > b$ and both a and b are prime numbers. What is the least possible value of b ?
- (A) 2 (B) 3 (C) 5 (D) 7 (E) 11

Answer: D

Solution 1

Since the three angles of a triangle add up to 180° and one of the angles is 90° because it's a right triangle, $a^\circ + b^\circ = 90^\circ$.

The greatest prime number less than 90 is 89. If $a = 89^\circ$, then $b = 90^\circ - 89^\circ = 1^\circ$, which is not prime.

The next greatest prime number less than 90 is 83. If $a = 83^\circ$, then $b = 7^\circ$, which IS prime, so we have our answer $\boxed{\text{(D)} 7}$ quacker88

Solution 2

Looking at the answer choices, only 7 and 11 are coprime to 90. Testing 7, the smaller angle, makes the other angle 83 which is prime, therefore our answer is $\boxed{\text{(D)} 7}$

Solution 3 (Euclidean Algorithm)

It is clear that $\gcd(a, b) = 1$. By the Euclidean Algorithm, we have

$$\gcd(a, b) = \gcd(a + b, b) = \gcd(90, b) = 1,$$

so 90 and b are relatively prime.

The least such prime number b is 7, from which $a = 90 - b = 83$ is also a prime number. Therefore, the answer is **(D) 7**.

5. (AMC 12 12B 2020) Teams A and B are playing in a basketball league where each game results in a win for one team and a loss for the other team. Team A has won $\frac{2}{3}$ of its games and team B has won $\frac{5}{8}$ of its games. Also, team B has won 7 more games and lost 7 more games than team A . How many games has team A played?
- (A) 21 (B) 27 (C) 42 (D) 48 (E) 63

Answer: C

Solution 1 (One Variable)

Suppose team A has played g games in total so that it has won $\frac{2}{3}g$ games. It follows that team B has played $g + 14$ games in total so that it has won $\frac{2}{3}g + 7$ games.

We set up and solve an equation for team B 's win ratio:

$$\begin{aligned}\frac{\frac{2}{3}g + 7}{g + 14} &= \frac{5}{8} \\ \frac{16}{3}g + 56 &= 5g + 70 \\ \frac{1}{3}g &= 14 \\ g &= \boxed{\text{(C) } 42}.\end{aligned}$$

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Solution 2 (Two Variables)

If we consider the number of games team B has played as x and the number of games that team A has played as y , then we can set up the following system of equations:

$$\begin{aligned}\frac{5}{8}x &= \frac{2}{3}y + 7, \\ \frac{3}{8}x &= \frac{1}{3}y + 7.\end{aligned}$$

The first system equated the number of wins of each team, while the second system equates the number of losses by each team. By multiplying the second equation by 2 and solving the system, we get $y = 42$ or answer choice **(C) 42**.

Solution 3 (Two Variables)

First, let us assign some variables. Let

$$A_w = 2x, \quad A_l = x, \quad A_g = 3x,$$

$$B_w = 5y, \quad B_l = 3y, \quad B_g = 8y,$$

where X_w denotes number of games won, X_l denotes number of games lost, and X_g denotes total games played for $X \in \{A, B\}$. Using the given information, we can set up the following two equations:

$$\begin{aligned} B_w &= A_w + 7 \implies 5y = 2x + 7, \\ B_l &= A_l + 7 \implies 3y = x + 7. \end{aligned}$$

We can solve through substitution, as the second equation can be written as $x = 3y - 7$, and plugging this into the first equation gives $5y = 6y - 7 \implies y = 7$, which means $x = 3(7) - 7 = 14$. Finally, we want the total number of games team A has played, which is $A_g = 3(14) = \boxed{\text{(C) } 42}$.

Solution 4 (Answer Choices: Substitutions)

Using the information from the problem, we can note that team A has lost $\frac{1}{3}$ of their matches. Using the answer choices, we can construct the following list of possible win-lose scenarios for A , represented in the form (w, l) for convenience:

- (A) $\implies (14, 7)$
- (B) $\implies (18, 9)$
- (C) $\implies (28, 14)$
- (D) $\implies (32, 16)$
- (E) $\implies (42, 21)$

Thus, we have 5 matching B scenarios, simply adding 7 to w and l . We can then test each of the five B scenarios for $\frac{w}{w+l} = \frac{5}{8}$ and find that $(35, 21)$ fits this description. Then working backwards and subtracting 7 from w and l gives us the point $(28, 14)$, making the answer $\boxed{\text{(C) } 42}$.

Solution 5 (Answer Choices: Observations)

Let's say that team A plays n games in total. Therefore, team B must play $n + 14$ games in total (7 wins, 7 losses) Since the ratio of A is

$$\frac{2}{3} \implies n \equiv 0 \pmod{3}$$

Similarly, since the ratio of B is

$$\frac{5}{8} \implies n + 14 \equiv 0 \pmod{8}$$

Now, we can go through the answer choices and see which ones work:

- (A) $21 \implies 21 + 14 = 35 \not\equiv 0 \pmod{8}$
- (B) $27 \implies 27 + 14 = 41 \not\equiv 0 \pmod{8}$
- (C) $42 \implies 42 + 14 = 56 \equiv 0 \pmod{8}$
- (D) $48 \implies 48 + 14 = 62 \not\equiv 0 \pmod{8}$
- (E) $63 \implies 63 + 14 = 77 \not\equiv 0 \pmod{8}$

So we can see $\boxed{\text{(C) } 42}$ is the only valid answer.

6. (AMC 12 12B 2020) For all integers $n \geq 9$, the value of

$$\frac{(n+2)! - (n+1)!}{n!}$$

is always which of the following?

- (A) a multiple of 4 (B) a multiple of 10 (C) a prime number (D) a perfect square

Answer: D

Solution 1

We first expand the expression:

$$\frac{(n+2)! - (n+1)!}{n!} = \frac{(n+2)(n+1)n! - (n+1)n!}{n!}.$$

We can now divide out a common factor of $n!$ from each term of the numerator:

$$(n+2)(n+1) - (n+1).$$

Factoring out $(n+1)$, we get

$$[(n+2) - 1](n+1) = (n+1)^2,$$

which proves that the answer is (D) a perfect square.

Solution 2

In the numerator, we factor out an $n!$ to get

$$\frac{(n+2)! - (n+1)!}{n!} = (n+2)(n+1) - (n+1).$$

Now, without loss of generality, test values of n until only one answer choice is left valid:

- $n = 1 \implies (3)(2) - (2) = 4$, knocking out (B), (C), and (E).
- $n = 2 \implies (4)(3) - (3) = 9$, knocking out (A).

This leaves (D) a perfect square as the only answer choice left.

This solution does not consider the condition $n \geq 9$. The reason is that, with further testing it becomes clear that for all n , we get

$$(n+2)(n+1) - (n+1) = (n+1)^2,$$

as proved in Solution 1. The condition $n \geq 9$ was added most likely to encourage picking (B) and discourage substituting smaller values into n .

7. (AMC 12 12B 2020) Two nonhorizontal, non vertical lines in the xy -coordinate plane intersect to form a 45° angle. One line has slope equal to 6 times the slope of the other line. What is the greatest possible value of the product of the slopes of the two lines?

- (A) $\frac{1}{6}$ (B) $\frac{2}{3}$ (C) $\frac{3}{2}$ (D) 3 (E) 6

Answer: C

Solution 1 (complex)

Let the intersection point be the origin. Let (a, b) be a point on the line of lesser slope. The multiplication of $a + bi$ by $\text{cis } 45^\circ$ gives $(a + bi)(\frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}}) = \frac{1}{\sqrt{2}}((a - b) + (a + b)i)$ and therefore $(a - b, a + b)$ lies on the line of greater slope.

Then, the rotation of (a, b) by 45 degrees gives a line of slope $\frac{a+b}{a-b}$.

We get the equation $\frac{6b}{a} = \frac{a+b}{a-b} \implies a^2 - 5ab + 6b^2 = (a - 3b)(a - 2b) = 0$ and this gives our answer to be $(\mathbf{C}) \frac{3}{2}$.

Solution 2 (polar coordinates)

Let the lines intersect at the origin. The problem is basically saying a line $y = ax$ is rotated 45° and becomes a new line $y = 6ax$. In polar form, $x = r \cos \theta$ and $y = r \sin \theta$. Rotating 45° counterclockwise gives us

$$x' = r \cos(\theta + 45^\circ) = r(\cos \theta \cos 45^\circ - \sin \theta \sin 45^\circ) = \frac{\sqrt{2}}{2}(x - y) = \frac{\sqrt{2}}{2}x(1 - a)$$

$$y' = r \sin(\theta + 45^\circ) = r(\sin \theta \cos 45^\circ + \cos \theta \sin 45^\circ) = \frac{\sqrt{2}}{2}(x + y) = \frac{\sqrt{2}}{2}x(1 + a)$$

So the slope of the new line is $\frac{y'}{x'} = \frac{1+a}{1-a}$.

Now we have $\frac{1+a}{1-a} = 6a$.

Solving gives us $a = \frac{1}{2}$ or $\frac{1}{3}$. The product is maximized when $a = \frac{1}{2}$, $6a = 3$. So the answer is $\boxed{\frac{3}{2}}$.

Solution 3 (vector products)

Intersect at the origin and select a point on each line to define vectors $\mathbf{v}_i = (x_i, y_i)$. Note that $\theta = 45^\circ$ gives equal magnitudes of the vector products

$$\mathbf{v}_1 \cdot \mathbf{v}_2 = v_1 v_2 \cos \theta \quad \text{and} \quad |\mathbf{v}_1 \times \mathbf{v}_2| = v_1 v_2 \sin \theta.$$

Substituting coordinate expressions for vector products, we find

$$\mathbf{v}_1 \cdot \mathbf{v}_2 = |\mathbf{v}_1 \times \mathbf{v}_2| \implies x_1 x_2 + y_1 y_2 = x_1 y_2 - x_2 y_1.$$

Divide this equation by $x_1 x_2$ to obtain

$$1 + m_1 m_2 = m_2 - m_1,$$

where $m_i = y_i/x_i$ is the slope of line i . Taking $m_2 = 6m_1$, we obtain

$$6m_1^2 - 5m_1 + 1 = 0 \implies m_1 \in \left\{\frac{1}{3}, \frac{1}{2}\right\}.$$

The latter solution gives the largest product of slopes $m_1 m_2 = 6m_1^2 = \frac{3}{2}$. $\boxed{(\mathbf{C})}$

Solution 4 (bash)

Place on coordinate plane. Lines are $y = mx$, $y = 6mx$. The intersection point at the origin. Goes through $(0, 0)$, $(1, m)$, $(1, 6m)$, $(1, 0)$. So by law of sines, $\frac{5m}{\sin 45^\circ} = \frac{\sqrt{1+m^2}}{1/(\sqrt{1+36m^2})}$, letting $a = m^2$ we want $6a$. Simplifying gives $50a = (1 + a)(1 + 36a)$,

so $36a^2 - 13a + 1 = 0 \implies 36(a - 1/4)(a - 1/9) = 0$, so $\max a = 1/4$, and $6a = 3/2$ (C).

Law of sines on the green triangle with the red angle (45 deg) and blue angle, where sine blue angle is $1/(\sqrt{1+36m^2})$ from right triangle w vertices $(0,0), (1,0), (1,6m)$.

Solution 5 (tan)

Let one of the lines have equation $y = ax$. Let θ be the angle that line makes with the x-axis, so $\tan(\theta) = a$. The other line will have a slope of $\tan(45^\circ + \theta) = \frac{\tan(45^\circ) + \tan(\theta)}{1 - \tan(45^\circ)\tan(\theta)} = \frac{1+a}{1-a}$. Since the slope of one line is 6 times the other, and a is the smaller slope, $6a = \frac{1+a}{1-a} \implies 6a - 6a^2 = 1 + a \implies 6a^2 - 5a + 1 = 0 \implies a = \frac{1}{2}, \frac{1}{3}$.

If $a = \frac{1}{2}$, the other line will have slope $\frac{1+\frac{1}{2}}{1-\frac{1}{2}} = 3$. If $a = \frac{1}{3}$, the other line will have slope $\frac{1+\frac{1}{3}}{1-\frac{1}{3}} = 2$. The first case gives the bigger product of $\frac{3}{2}$, so our answer is

$$\text{(C) } \frac{3}{2}.$$

Solution 6 (Cheating)

Let the smaller slope be m , then the larger slope is $6m$. Since we want the greatest product we begin checking each answer choice, starting with (E).

$$6m^2 = 6.$$

$$m^2 = 1.$$

This gives $m = 1$ and $6m = 6$. Checking with a protractor we see that this does not form a 45 degree angle.

Using this same method for the other answer choices, we eventually find that the answer is (C) $\frac{3}{2}$ since our slopes are $\frac{1}{2}$ and 3 which forms a perfect 45 degree angle.

Solution 7

If you have this formula memorized then it will be very easy to do: If θ is the angle between two lines with slopes m_1 and m_2 , then $\tan(\theta) = \frac{m_1 - m_2}{1 + m_1 m_2}$.

Now let the smaller slope be m , thus the other slope is $6m$. Using our formula above:

$$\tan(45^\circ) = \frac{6m - m}{1 + 6m^2} \implies 6m^2 - 5m + 1 = 0 \implies (2m - 1)(3m - 1) = 0.$$

Therefore the two possible values for m are $\frac{1}{2}$ and $\frac{1}{3}$. We choose the larger one and thus our answer is

$$6m^2 = 6 \cdot \frac{1}{4} = \frac{3}{2} \implies \text{(C)}.$$

8. (AMC 12 12B 2020) How many ordered pairs of integers (x, y) satisfy the equation

$$x^{2020} + y^2 = 2y?$$

- (A) 1 (B) 2 (C) 3 (D) 4 (E) infinitely many

Answer: D

Solution 1

Rearranging the terms and completing the square for y yields the result $x^{2020} + (y - 1)^2 = 1$. Then, notice that x can only be 0, 1, or -1 because any value of x^{2020} that is greater than 1 will cause the term $(y - 1)^2$ to be less than 0, which is impossible as y must be real. Therefore, plugging in the above values for x gives the ordered pairs $(0, 0)$, $(1, 1)$, $(-1, 1)$, and $(0, 2)$ gives a total of $\boxed{\text{(D)} 4}$ ordered pairs.

Solution 2

Bringing all of the terms to the LHS, we see a quadratic equation

$$y^2 - 2y + x^{2020} = 0$$

in terms of y . Applying the quadratic formula, we get

$$y = \frac{2 \pm \sqrt{4 - 4 \cdot 1 \cdot x^{2020}}}{2} = \frac{2 \pm \sqrt{4(1 - x^{2020})}}{2}.$$

In order for y to be real, which it must be given the stipulation that we are seeking integral answers, we know that the discriminant, $4(1 - x^{2020})$ must be nonnegative. Therefore,

$$4(1 - x^{2020}) \geq 0 \implies x^{2020} \leq 1.$$

Here, we see that we must split the inequality into a compound inequality, resulting in $-1 \leq x \leq 1$.

The only integers that satisfy this are $x \in \{-1, 0, 1\}$. Plugging these values back into the quadratic equation, we see that $x = \{-1, 1\}$ both produce a discriminant of 0, meaning that there is only 1 solution for y . If $x = \{0\}$, then the discriminant is nonzero, resulting in two solutions for y .

Thus, the answer is $2 \cdot 1 + 1 \cdot 2 = \boxed{\text{(D)} 4}$.

Solution 3: Solve for x first

Set it up as a quadratic in terms of y :

$$y^2 - 2y + x^{2020} = 0$$

Then the discriminant is

$$\Delta = 4 - 4x^{2020}$$

This will clearly only yield real solutions when $|x^{2020}| \leq 1$, because the discriminant must be positive. Then $x = -1, 0, 1$. Checking each one: -1 and 1 are the same when raised to the 2020th power:

$$y^2 - 2y + 1 = (y - 1)^2 = 0$$

This has only has solutions 1, so $(\pm 1, 1)$ are solutions. Next, if $x = 0$:

$$y^2 - 2y = 0 \implies y(y - 2) = 0$$

Which has 2 solutions, so $(0, 2)$ and $(0, 0)$.

These are the only 4 solutions, so our answer is $\boxed{\text{(D)} 4}$.

Solution 4: Solve for y first

Move the y^2 term to the other side to get $x^{2020} = 2y - y^2 = y(2 - y)$.

Because $x^{2020} \geq 0$ for all x , then $y(2 - y) \geq 0 \implies y = 0, 1, 2$.

If $y = 0$ or $y = 2$, the right side is 0 and therefore $x = 0$.

When $y = 1$, the right side become 1, therefore $x = 1, -1$.

Our solutions are $(0, 2), (0, 0), (1, 1), (-1, 1)$. These are the only 4 solutions, so the answer is (D) 4

- wwt7535

Solution 5: Similar to solution 4

Since x^{2020} and y^2 are perfect squares, they are both nonnegative. That means y^2 plus a nonnegative number equals $2y$, which means $y^2 \leq 2y$. The only possible integer values for y are 0, 1, 2.

For $y = 0$, x can only be 0.

For $y = 1$, $x^2 = 1$ so $x = 1, -1$.

For $y = 2$, x can only be 0 as well.

This gives us the solutions $(0, 0), (1, 1), (-1, 1)$, and $(0, 2)$. These are the only solutions, so there is a total of (D) 4 ordered pairs.

- kc1374

Solution 6: (Casework)

We see that x has to be 1, 0, or -1 , as any other integer would make this value too large. We also know that because 2020 is even, both -1 , and 1 for x will yield the same x^{2020} value of 1.

Case 1: $x = 0$. This gives us that $y^2 = 2y$. Dividing both sides by y gives us $y = 2$. Additionally, we know intuitively that $y = 0$ is also a case, which gives us 2 possibilities for this case.

Case 2: $x = 1$ or -1 . This gives us that $1 + y^2 = 2y$. Bringing the $2y$ to the other side, we have a simple quadratic. $y^2 - 2y + 1 = 0$. Factor to get $(y - 1)^2 = 0$ so $y = 1$. Because this works for x as -1 and 1, there are 2 possibilities for this case.

Adding the cases gets us our final answer of (D) 4 ordered pairs. iluvme

Solution 7: (Casework)

We can move the y^2 term to the other side to obtain $x^{2020} = 2y - y^2$, which can be easily factored into $x^{2020} = (2 - y)y$. If we raise each side to the power $1/2020$, we get $x = ((2 - y)(y))^{1/2020}$. Since we're dealing with integer values, the inside of the square root and thus $(2 - y)y$ must be greater than 0. For this to happen, y must be between 0 and 2 inclusive. Testing each integer value, we get the cases:

Case 1: When $y = 0$, then $y(2 - y)$ is also 0, and since the square root of 0 is 0, x must also be 0. So our first pair of solutions is $(0, 0)$.

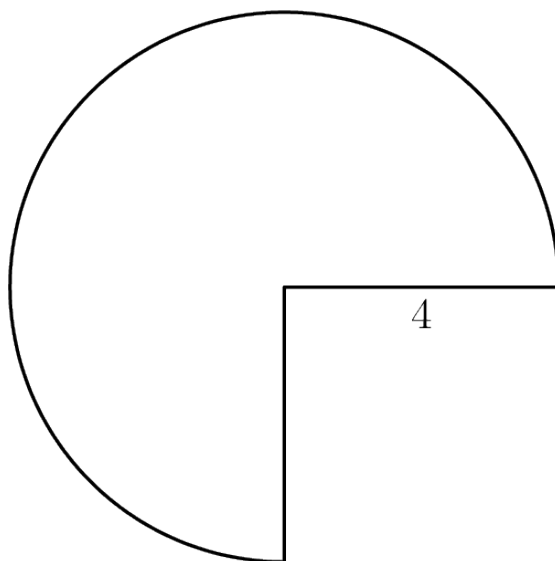
Case 2: When $y = 1$, then $y(2 - y)$ is also 1. The possible values for x are 1 and -1 , so we get 2 more pairs of solutions: $(1, -1)$ and $(1, 1)$.

Case 3: When $y = 2$, then $y(2 - y)$ is 0, and since the square root of 0 is 0, x must also be 0 - so we get one more pair of solutions, $(2, 0)$

If we count all these pairs $((0, 0), (1, -1), (1, 1), (2, 0))$, we have 4 pairs of solutions, so our answer is (D) 4 ordered pairs.

- youtube.com/indianmathguy

9. (AMC 12 12B 2020) A three-quarter sector of a circle of radius 4 inches together with its interior can be rolled up to form the lateral surface of a right circular cone by taping together along the two radii shown. What is the volume of the cone in cubic inches?



- (A) $3\pi\sqrt{5}$ (B) $4\pi\sqrt{3}$ (C) $3\pi\sqrt{7}$ (D) $6\pi\sqrt{3}$ (E) $6\pi\sqrt{7}$

Answer: C

Solution 1

Notice that when the cone is formed, the 2 shown radii, when joined, will form the slant height of the cone, and the intact circumference of the circle will become the circumference of the base of the cone.

We can calculate that the intact circumference of the circle is $8\pi \cdot \frac{3}{4} = 6\pi$. Since that is also equal to the cone's circumference, the radius of the cone is 3. We also have that the slant height of the cone is 4. Therefore, we use the Pythagorean Theorem to calculate that the height of the cone is $\sqrt{4^2 - 3^2} = \sqrt{7}$. The volume of the cone is $\frac{1}{3} \cdot \pi \cdot 3^2 \cdot \sqrt{7} = \boxed{\text{(C)} 3\pi\sqrt{7}}$

-PCChess minor edits by WrenMath

Solution 2 (Last Resort/Cheap)

Using a ruler, measure a circle of radius 4 and cut out the circle and then the quarter missing. Then, fold it into a cone and measure the diameter to be 6 cm $\implies r = 3$. You can form a right triangle with sides 3, 4, and then through the Pythagorean theorem the height h is found to be $h^2 = 4^2 - 3^2 \implies h = \sqrt{7}$. The volume of a cone is $\frac{1}{3}\pi r^2 h$. Plugging in we find $V = 3\pi\sqrt{7} \implies \boxed{\text{(C)}}$

- DBlack2021

Solution 3

The radius of the given $\frac{3}{4}$ - circle will end up being the slant height of the cone. Thus, the radius and height of the cone are legs of a right triangle with hypotenuse 4. The volume of a cone is $\frac{1}{3}\pi r^2 h$. Using this with the options, we can take out the π and multiply the coefficient of the radical by 3 to get $r^2 h$. We can then use the r and h values to see that the only option that satisfies $r^2 + h^2 = 4^2 = 16$ is $\boxed{\text{(C)}}$

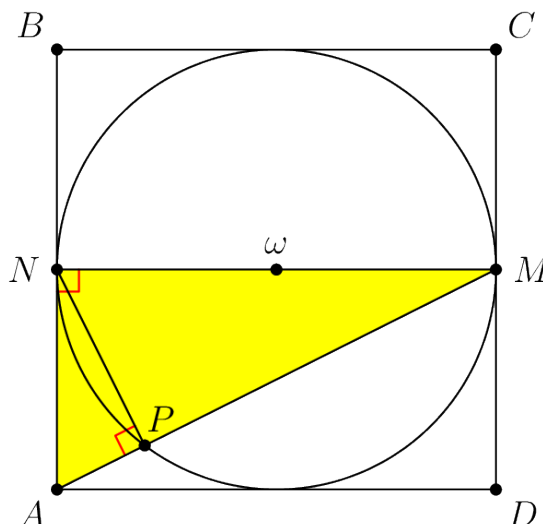
- ColtsFan10

10. (AMC 12 12B 2020) In unit square $ABCD$, the inscribed circle ω intersects $\overline{CD}$ at M , and $\overline{AM}$ intersects ω at a point P different from M . What is AP ?
 (A) $\frac{\sqrt{5}}{12}$ (B) $\frac{\sqrt{5}}{10}$ (C) $\frac{\sqrt{5}}{9}$ (D) $\frac{\sqrt{5}}{8}$ (E) $\frac{2\sqrt{5}}{15}$

Answer: B

Solution 1 (Similar Triangles)

Call the midpoint of $\overline{AB}$ point N . Draw in $\overline{NM}$ and $\overline{NP}$. Note that $\angle NPM = 90^\circ$ due to Thales's Theorem.



Using the Pythagorean theorem, $AM = \frac{\sqrt{5}}{2}$. Now we just need to find AP using similar triangles $\triangle APN \sim \triangle ANM$:

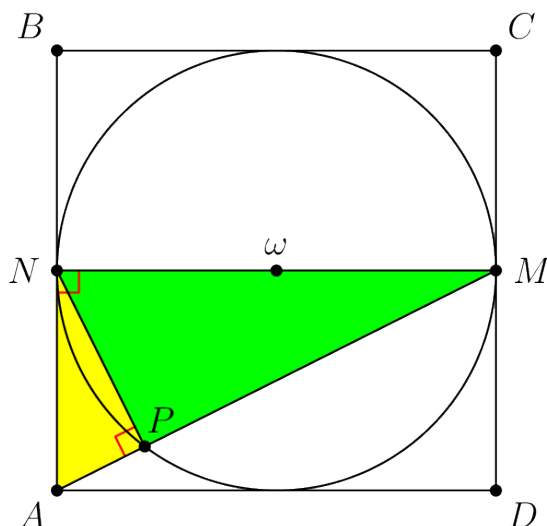
$$\begin{aligned} \frac{AP}{AN} &= \frac{AN}{AM} \\ \frac{AP}{1/2} &= \frac{1/2}{\sqrt{5}/2} \\ AP &= \boxed{\text{(B)} \frac{\sqrt{5}}{10}}. \end{aligned}$$

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Solution 2 (Inscribed Angle Theorem and Pythagorean Theorem)

Let N be the midpoint of $\overline{AB}$, so $\angle ANM = 90^\circ$. Note that $\overline{NM}$ is a diameter of ω , so $\angle NPM = 90^\circ$ by the inscribed angle theorem.

We have the following diagram:



Since $AN = \frac{1}{2}$ and $NM = 1$, we get $AM = \frac{\sqrt{5}}{2}$ by the Pythagorean theorem.

Let $AP = x$. It follows that $PM = \frac{\sqrt{5}}{2} - x$. Applying the Pythagorean theorem to right $\triangle ANP$ gives $NP^2 = \left(\frac{1}{2}\right)^2 - x^2$, and applying the Pythagorean theorem to right $\triangle MNP$ gives $NP^2 = 1^2 - \left(\frac{\sqrt{5}}{2} - x\right)^2$. Equating the expressions for NP^2 produces

$$\begin{aligned} \left(\frac{1}{2}\right)^2 - x^2 &= 1^2 - \left(\frac{\sqrt{5}}{2} - x\right)^2 \\ \frac{1}{4} - x^2 &= 1 - \frac{5}{4} + \sqrt{5}x - x^2 \\ \frac{1}{2} &= \sqrt{5}x. \end{aligned}$$

Finally, dividing both sides by $\sqrt{5}$ and then rationalizing the denominator, we obtain $x = \frac{1}{2\sqrt{5}} = \boxed{\text{(B)} \frac{\sqrt{5}}{10}}$.

Solution 3 (Power of a Point)

Let circle ω intersect $\overline{AB}$ at point N . By Power of a Point, we have $AN^2 = AP \cdot AM$. We know $AN = \frac{1}{2}$ because N is the midpoint of $\overline{AB}$, and we can easily find AM by the Pythagorean Theorem, which gives us $AM = \sqrt{1^2 + \left(\frac{1}{2}\right)^2} = \frac{\sqrt{5}}{2}$. Our equation

is now $\frac{1}{4} = AP \cdot \frac{\sqrt{5}}{2}$, or $AP = \frac{2}{4\sqrt{5}} = \frac{1}{2\sqrt{5}} = \frac{\sqrt{5}}{2 \cdot 5}$, thus our answer is $\boxed{\text{(B)} \frac{\sqrt{5}}{10}}$.

Solution 4 (Trigonometry)

Let O be the center of the circle and the point of tangency between ω and $\overline{AD}$ be represented by K . We know that $\overline{AK} = \overline{KD} = \overline{DM} = \frac{1}{2}$. Consider the right triangle $\triangle ADM$. Let $\angle AMD = \theta$.

Since ω is tangent to $\overline{DC}$ at M , $\angle PMO = 90 - \theta$. Now, consider $\triangle POM$. This triangle is isosceles because $\overline{PO}$ and $\overline{OM}$ are both radii of ω . Therefore, $\angle POM = 180 - 2(90 - \theta) = 2\theta$.

We can now use Law of Cosines on $\angle POM$ to find the length of PM and subtract it from the length of AM to find AP . Since $\cos \theta = \frac{1}{\sqrt{5}}$ and $\sin \theta = \frac{2}{\sqrt{5}}$, the double angle formula tells us that $\cos 2\theta = -\frac{3}{5}$. We have

$$PM^2 = \frac{1}{2} - \frac{1}{2} \cos 2\theta \implies PM = \frac{2\sqrt{5}}{5}.$$

By Pythagorean theorem, we find that $AM = \frac{\sqrt{5}}{2} \implies AP = \boxed{\text{(B)} \frac{\sqrt{5}}{10}}.$

Solution 5 (Trigonometry)

Take O as the center and draw segment ON perpendicular to AM , $ON \cap AM = N$, link OM . Then we have $OM \parallel AD$. So $\angle DAM = \angle OMA$. Since $AD = 2AM = 2OM = 1$, we have $\cos \angle DAM = \cos \angle OMP = \frac{2}{\sqrt{5}}$. As a result, $NM = OM \cos \angle OMP = \frac{1}{2} \cdot \frac{2}{\sqrt{5}} = \frac{1}{\sqrt{5}}$. Thus $PM = 2NM = \frac{2}{\sqrt{5}} = \frac{2\sqrt{5}}{5}$. Since $AM = \frac{\sqrt{5}}{2}$,

we have $AP = AM - PM = \boxed{\text{(B)} \frac{\sqrt{5}}{10}}.$

Solution 6 (Coordinate Geometry)

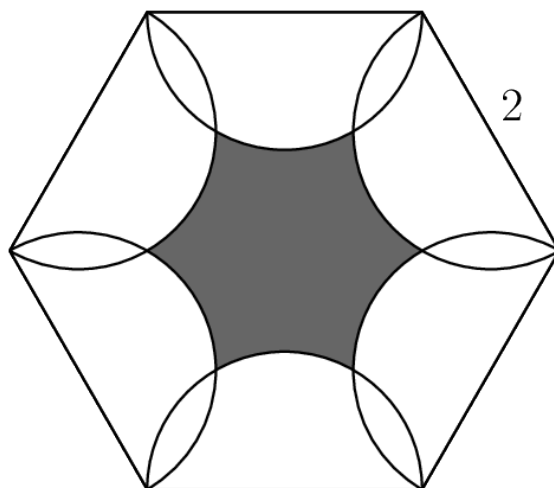
Place circle ω in the Cartesian plane such that the center lies on the origin. Then we can easily find the equation for ω as $x^2 + y^2 = \frac{1}{4}$, because it is not translated and the radius is $\frac{1}{2}$.

We have $A = \left(-\frac{1}{2}, \frac{1}{2}\right)$ and $M = \left(0, -\frac{1}{2}\right)$. The slope of the line passing through these two points is $\frac{\frac{1}{2} + \frac{1}{2}}{-\frac{1}{2} - 0} = \frac{1}{-\frac{1}{2}} = -2$, and the y -intercept is simply M . This gives us the line passing through both points as $y = -2x - \frac{1}{2}$.

We substitute this into the equation for the circle to get $x^2 + \left(-2x - \frac{1}{2}\right)^2 = \frac{1}{4}$, or $x^2 + 4x^2 + 2x + \frac{1}{4} = \frac{1}{4}$. Simplifying gives $x(5x + 2) = 0$. The roots of this quadratic are $x = 0$ and $x = -\frac{2}{5}$, but if $x = 0$ we get point M , so we only want $x = -\frac{2}{5}$.

We plug this back into the linear equation to find $y = \frac{3}{10}$, and so $P = \left(-\frac{2}{5}, \frac{3}{10}\right)$. Finally, we use distance formula on A and P to get $AP = \sqrt{\left(-\frac{5}{10} + \frac{4}{10}\right)^2 + \left(\frac{5}{10} - \frac{3}{10}\right)^2} = \sqrt{\frac{1}{100} + \frac{4}{100}} = \boxed{\text{(B)} \frac{\sqrt{5}}{10}}.$

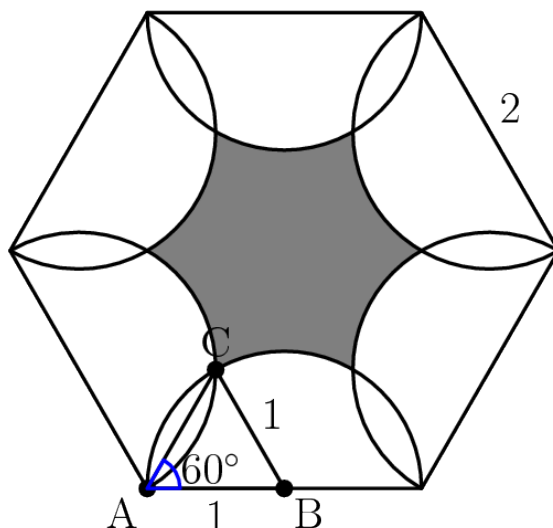
11. (AMC 12 12B 2020) As shown in the figure below, six semicircles lie in the interior of a regular hexagon with side length 2 so that the diameters of the semicircles coincide with the sides of the hexagon. What is the area of the shaded region—inside the hexagon but outside all of the semicircles?



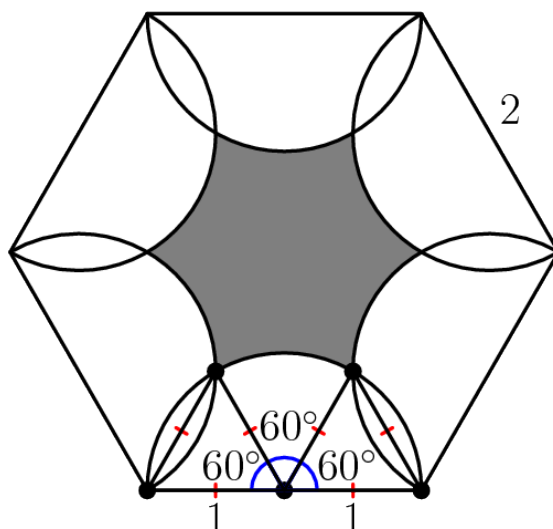
- (A) $6\sqrt{3}-3\pi$ (B) $\frac{9\sqrt{3}}{2}-2\pi$ (C) $\frac{3\sqrt{3}}{2}-\frac{\pi}{3}$ (D) $3\sqrt{3}-\pi$ (E) $\frac{9\sqrt{3}}{2}-\pi$

Answer: D

Solution 1



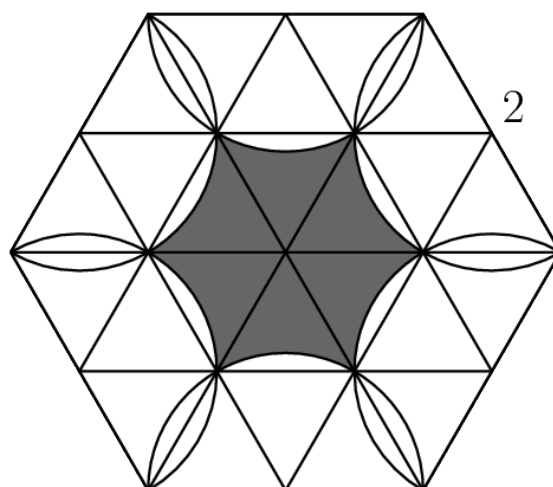
Let point A be a vertex of the regular hexagon, let point B be the midpoint of the line connecting point A and a neighboring vertex, and let point C be the second intersection of the two semicircles that pass through point A . Then, $BC = 1$, since B is the center of the semicircle with radius 1 that C lies on, $AB = 1$, since B is the center of the semicircle with radius 1 that A lies on, and $\angle BAC = 60^\circ$, as a regular hexagon has angles of 120° , and $\angle BAC$ is half of any angle in this hexagon. Now, since $AB = BC$, $\angle ACB = \angle BAC = 60^\circ$. Since the angles in a triangle sum to 180° , $\angle ABC$ is also 60° . Therefore, $\triangle ABC$ is an equilateral triangle with side lengths of 1.



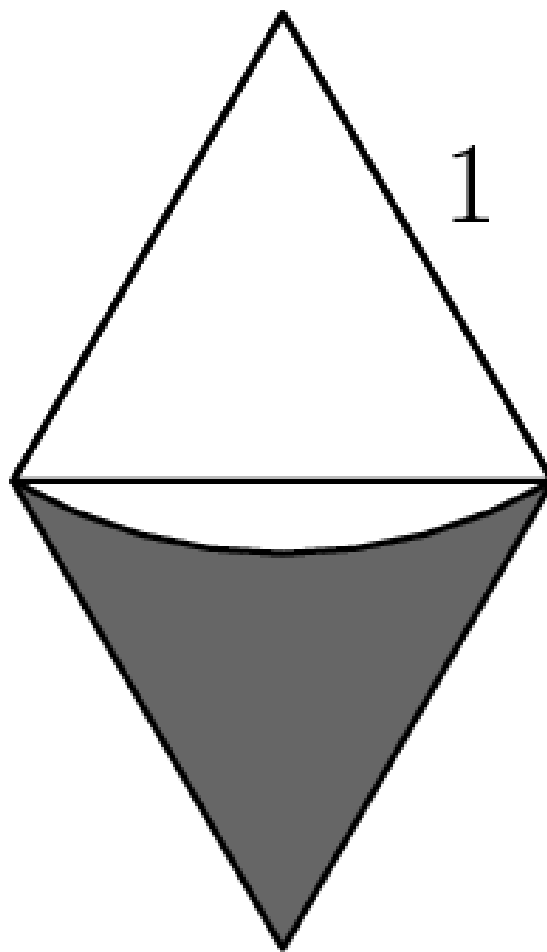
Since the area of a regular hexagon can be found with the formula $\frac{3\sqrt{3}s^2}{2}$, where s is the side length of the hexagon, the area of this hexagon is $\frac{3\sqrt{3}(2^2)}{2} = 6\sqrt{3}$. Since the area of an equilateral triangle can be found with the formula $\frac{\sqrt{3}}{4}s^2$, where s is the side length of the equilateral triangle, the area of an equilateral triangle with side lengths of 1 is $\frac{\sqrt{3}}{4}(1^2) = \frac{\sqrt{3}}{4}$. Since the area of a circle can be found with the formula πr^2 , the area of a sixth of a circle with radius 1 is $\frac{\pi(1^2)}{6} = \frac{\pi}{6}$. In each sixth of the hexagon, there are two equilateral triangles colored white, each with an area of $\frac{\sqrt{3}}{4}$, and one-sixth of a circle with radius 1 colored white, with an area of $\frac{\pi}{6}$. The rest of the sixth is colored gray. Therefore, the total area that is colored white in each sixth of the hexagon is $2\left(\frac{\sqrt{3}}{4}\right) + \frac{\pi}{6}$, which equals $\frac{\sqrt{3}}{2} + \frac{\pi}{6}$, and the total area colored white is $6\left(\frac{\sqrt{3}}{2} + \frac{\pi}{6}\right)$, which equals $3\sqrt{3} + \pi$. Since the area colored gray equals the total area of the hexagon minus the area colored white, the area colored gray is $6\sqrt{3} - (3\sqrt{3} + \pi)$, which equals **(D) $3\sqrt{3} - \pi$** .

Solution 2

First, subdivide the hexagon into 24 equilateral triangles with side length 1:



Now note that the entire shaded region is just 6 times this part:



The entire rhombus is just 2 equilateral triangles with side lengths of 1, so it has an area of:

$$2 \cdot \frac{\sqrt{3}}{4} = \frac{\sqrt{3}}{2}$$

The sector that is not included has an area of:

$$\frac{1}{6} \cdot \pi \cdot 1^2 = \frac{\pi}{6}$$

Hence, the area of the shaded region in that section is

$$\frac{\sqrt{3}}{2} - \frac{\pi}{6}$$

For a final area of:

$$6 \left(\frac{\sqrt{3}}{2} - \frac{\pi}{6} \right) = 3\sqrt{3} - \pi \Rightarrow \boxed{(D)}$$

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Solution 3

We can see there are six congruent "leaves" on the original figure, and let us say the total area of them is S . If we make a circle with radius one in the circle (the dashed one), we can get another six congruent "leaves", and they are the same as the old six ones. So the area of the six new leaves can also be represented by S . Then the area of the shaded region can be expressed in the following two ways:

[1]. **Area** of hexagon - (**Area** of six semicircles - **Area** of six **old** "leaves") = $\frac{\sqrt{3}}{4} \cdot 4 \cdot 6 - (3\pi - S)$; [2]. **Area** of dashed circle - **Area** of six **new** "leaves" = $\pi - S$;

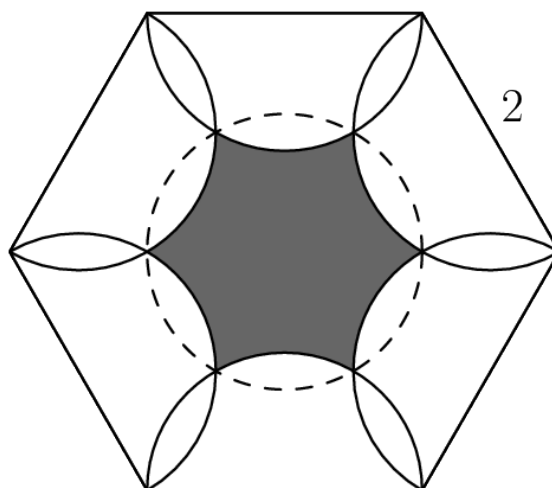
Then we have $\frac{\sqrt{3}}{4} \cdot 4 \cdot 6 - (3\pi - S) = \pi - S$ $S = 2\pi - 3\sqrt{3}$

Plugging S back in to either [1] or [2], we can solve for the shaded area = $3\sqrt{3} - \pi$

→ (D)

–RyanZ@BRS

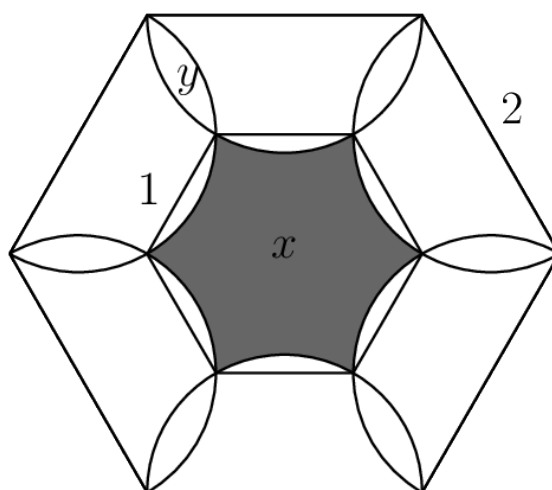
–Extremelysupercooldude (Minor latex and grammar edits)



Solution 4 (Variables)

We can set several variables.

Let x be the area of the shaded region, and let y be the area of one of the 'leaf-shaped' regions.



We then can connect the vertices of x , and we end up with another hexagon, with side length 1. Since each of the regions that are outside of x but inside of the smaller hexagon is half of y , their total area is $\frac{6y}{2} = 3y$. The smaller hexagon minus this region would be x . So,

$$x = \frac{3\sqrt{3}}{2} \cdot 1^2 - 3y$$

Also, the area of the larger hexagon minus the semicircles *then add* back $6y$ would also be x . (Note that by just subtracting the semicircles from the larger hexagon would result in an overlap of $6y$.) So,

$$x = \frac{3\sqrt{3}}{2} \cdot 2^2 - 3\pi + 6y$$

Equating the equations, we have

$$\begin{aligned} \frac{3\sqrt{3}}{2} \cdot 1^2 - 3y &= \frac{3\sqrt{3}}{2} \cdot 2^2 - 3\pi + 6y \\ \frac{3\sqrt{3}}{2} - 3y &= 6\sqrt{3} - 3\pi + 6y \end{aligned}$$

Solving for y , we have

$$\begin{aligned} 9y &= \frac{3\sqrt{3}}{2} - 6\sqrt{3} + 3\pi \\ &= -\frac{9\sqrt{3}}{2} + 3\pi \end{aligned}$$

Then, we can plug this back into

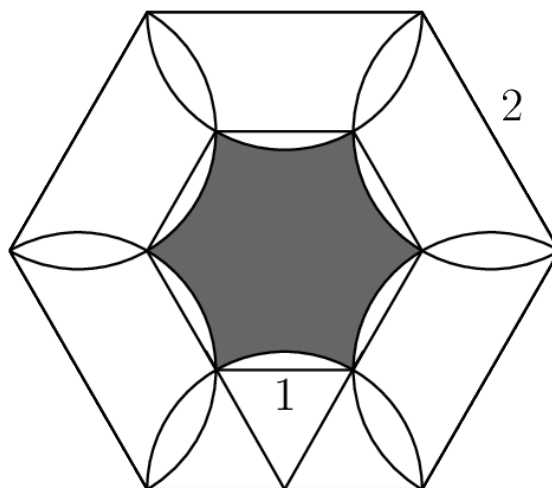
$$x = \frac{3\sqrt{3}}{2} \cdot 1^2 - 3y$$

Since $9y = -\frac{9\sqrt{3}}{2} + 3\pi$, dividing by 3 gives

$$\begin{aligned} 3y &= -\frac{9\sqrt{3}}{2} \times \frac{1}{3} + \pi \\ &= -\frac{3\sqrt{3}}{2} + \pi \end{aligned}$$

Therefore, $x = \frac{3\sqrt{3}}{2} - (-\frac{3\sqrt{3}}{2} + \pi) \implies x = \boxed{\text{(D)} 3\sqrt{3} - \pi}$.

Solution 5



First we connect every vertex of the shaded portion to another, forming an equilateral hexagon. Each interior angle of this hexagon is 120° , the interior angles of the new triangle are all 60° . As the radius of the semicircle is 1, we know that this hexagon has side length 1, so its area is given by the combined area of six equilateral triangles with side length 1, or $\frac{\sqrt{3}}{4} \cdot 6 = \frac{3\sqrt{3}}{2}$.

Now, we have to subtract off the six mini-sectors between the triangle and the shaded area. The area of one of these mini-sectors is the difference between a 60° sector of the semicircle and the equilateral we just drew. The area of a sector is just $\frac{\pi}{6}$, and the area of a triangle is $\frac{\sqrt{3}}{4}$. Therefore our combined area of the mini-sectors is

$$6 \left(\frac{\pi}{6} - \frac{\sqrt{3}}{4} \right) = \pi - \frac{3\sqrt{3}}{2}.$$

Subtracting this from the area of the (smaller) hexagon, the desired area is

$$\frac{3\sqrt{3}}{2} - \left(\pi - \frac{3\sqrt{3}}{2} \right) = \boxed{\text{(D)} \ 3\sqrt{3} - \pi}$$

-Benedict T (countmath1)

12. (AMC 12 12B 2020) Let $\overline{AB}$ be a diameter in a circle of radius $5\sqrt{2}$. Let $\overline{CD}$ be a chord in the circle that intersects $\overline{AB}$ at a point E such that $BE = 2\sqrt{5}$ and $\angle AEC = 45^\circ$. What is $CE^2 + DE^2$? (A) 96 (B) 98 (C) $44\sqrt{5}$ (D) $70\sqrt{2}$ (E) 100

Answer: E

Solution 1 (Pythagorean Theorem)

Let O be the center of the circle, and X be the midpoint of $\overline{CD}$. Let $CX = a$ and $EX = b$. This implies that $DE = a - b$. Since $CE = CX + EX = a + b$, we now want to find $(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$. Since $\angle CXO$ is a right angle, by Pythagorean theorem $a^2 + b^2 = CX^2 + OX^2 = (5\sqrt{2})^2 = 50$. Thus, our answer is $2 \times 50 = \boxed{\text{(E)} \ 100}$.

Solution 2 (Power of a Point)

Let O be the center of the circle, and X be the midpoint of CD . Draw triangle OCD , and median OX . Because $OC = OD$, OCD is isosceles, so OX is also an altitude of OCD . $OE = 5\sqrt{2} - 2\sqrt{5}$, and because angle OEC is 45 degrees and triangle

OXE is right, $OX = EX = \frac{5\sqrt{2}-2\sqrt{5}}{\sqrt{2}} = 5 - \sqrt{10}$. Because triangle OXC is right, $CX = \sqrt{(5\sqrt{2})^2 - (5 - \sqrt{10})^2} = \sqrt{15 + 10\sqrt{10}}$. Thus, $CD = 2\sqrt{15 + 10\sqrt{10}}$.

We are looking for $CE^2 + DE^2$ which is also $(CE + DE)^2 - 2 \cdot CE \cdot DE$.

Because $CE + DE = CD = 2\sqrt{15 + 10\sqrt{10}}$, $(CE + DE)^2 = CD^2 = 4(15 + 10\sqrt{10}) = 60 + 40\sqrt{10}$.

By Power of a Point, $CE \cdot DE = AE \cdot BE = 2\sqrt{5} \cdot (10\sqrt{2} - 2\sqrt{5}) = 20\sqrt{10} - 20$, so $2 \cdot CE \cdot DE = 40\sqrt{10} - 40$.

Finally, $CE^2 + DE^2 = (CE + DE)^2 - 2 \cdot CE \cdot DE = (60 + 40\sqrt{10}) - (40\sqrt{10} - 40) = \boxed{\text{(E)} 100}$.

Solution 3 (Law of Cosines)

Let O be the center of the circle. Notice how $OC = OD = r$, where r is the radius of the circle. By applying the law of cosines on triangle OCE ,

$$r^2 = CE^2 + OE^2 - 2(CE)(OE) \cos 45 = CE^2 + OE^2 - (CE)(OE)\sqrt{2}.$$

Similarly, by applying the law of cosines on triangle ODE ,

$$r^2 = DE^2 + OE^2 - 2(DE)(OE) \cos 135 = DE^2 + OE^2 + (DE)(OE)\sqrt{2}.$$

By subtracting these two equations, we get

$$CE^2 - DE^2 - (CE)(OE)\sqrt{2} - (DE)(OE)\sqrt{2} = 0.$$

We can rearrange it to get

$$CE^2 - DE^2 = (CE)(OE)\sqrt{2} + (DE)(OE)\sqrt{2} = (CE + DE)(OE\sqrt{2}).$$

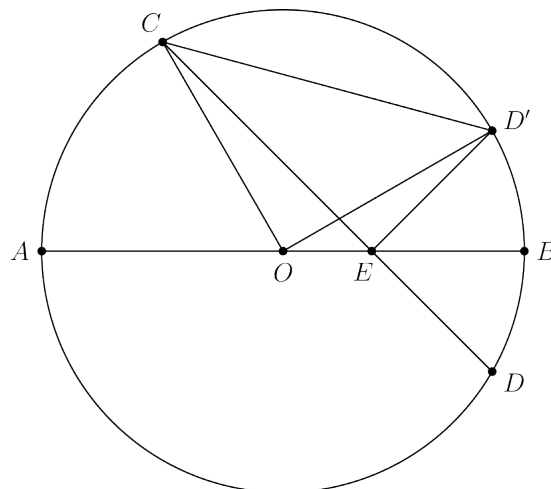
Because both CE and DE are both positive, we can safely divide both sides by $(CE + DE)$ to obtain $CE - DE = OE\sqrt{2}$. Because $OE = OB - BE = 5\sqrt{2} - 2\sqrt{5}$,

$$(CE - DE)^2 = CE^2 + DE^2 - 2(CE)(DE) = (OE\sqrt{2})^2 = 2(5\sqrt{2} - 2\sqrt{5})^2 = 140 - 40\sqrt{10}.$$

Through power of a point, we can find out that $(CE)(DE) = 20\sqrt{10} - 20$, so

$$CE^2 + DE^2 = (CE - DE)^2 + 2(CE)(DE) = (140 - 40\sqrt{10}) + 2(20\sqrt{10} - 20) = \boxed{\text{(E)} 100}.$$

Solution 4 (Reflections)



Let O be the center of the circle. By reflecting D across the line AB to produce D' , we have that $\angle BED' = 45$. Since $\angle AEC = 45$, $\angle CED' = 90$. Since $DE = ED'$, by the Pythagorean Theorem, our desired solution is just CD'^2 . Looking next to circle arcs, we know that $\angle AEC = \frac{6.0\text{ptAC} + 6.0\text{ptBD}}{2} = 45$, so $6.0\text{ptAC} + 6.0\text{ptBD} = 90$. Since $6.0\text{ptBD}' = 6.0\text{ptBD}$, and $6.0\text{ptAC} + 6.0\text{ptBD}' + 6.0\text{ptCD}' = 180$, $6.0\text{ptCD}' = 90$. Thus, $\angle COD' = 90$. Since $OC = OD' = 5\sqrt{2}$, by the Pythagorean Theorem, the desired $CD'^2 = \boxed{\text{(E) } 100}$.

Solution 5 (Basically Solution 2 With Motivation)

Basically, by PoP, you have that

$$CE \times DE = (10\sqrt{2} - 2\sqrt{5})(2\sqrt{5}) = 20\sqrt{10} - 20.$$

Therefore, as $CE^2 + DE^2 = (CD)^2 - 2(CE \times DE)$, basically, once you find CD^2 , the problem is done. Now, this is an IMPORTANT concept: If you have a circle which you know the radius of and you want to find the length of a chord of that circle, drop an altitude from the center of the circle to the chord to find distance between the center of the circle and the chord.

In this case, let M be the midpoint of chord CD . Notice that now we can use our 45° angle, since OME is a $45^\circ - 45^\circ - 90^\circ$ triangle so that $ME = x$ and $OE = x\sqrt{2}$. However, we have that $OE = 5\sqrt{2} - 2\sqrt{5}$, so that $x = 5 - \sqrt{10}$. Now, notice that $x^2 = 35 - 10\sqrt{10}$, so that

$$CM^2 = 50 - x^2 = 50 - (35 - 10\sqrt{10}) = 15 + 10\sqrt{10}$$

and

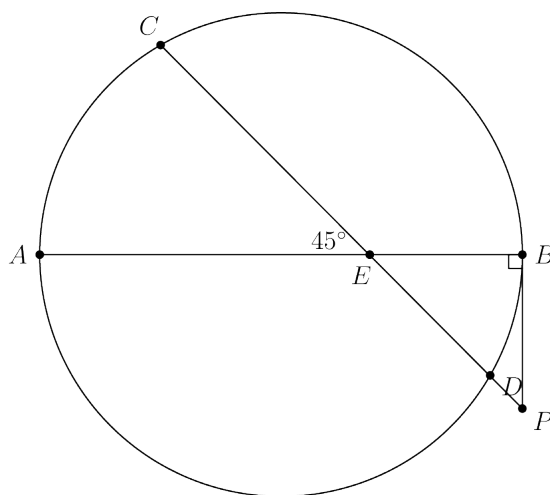
$$CD^2 = 60 + 40\sqrt{10}.$$

Therefore,

$$CE^2 + DE^2 = CD^2 - 2(CE \times DE) = (60 + 40\sqrt{10}) - 2(2\sqrt{10} - 20) = (60 + 40\sqrt{10}) + (40 - 40\sqrt{10}) = \boxed{\text{(E) } 100}$$

This may not be the “shortest solution”, but in my opinion is very well motivated and doesn’t require much creativity. [Not requiring much creativity, it also saves more time than you’d think. ;)]

Solution 6 (Double Power of a Point)



For ease of notation, let $x = DE$ and $y = EC$. Extend $\overline{CD}$ to point P until $\overline{BP}$ is perpendicular to AB . It's given that $\angle AEC = 45^\circ$, so, by vertical angles, we have $\angle BEP = \angle EPB = 45^\circ$.

Since PEB is a $45 - 45 - 90$ right triangle, we have $BE = BP = 2\sqrt{5}$ and $PE = 2\sqrt{10}$. Hence, $PD = 2\sqrt{10} - x$.

By Power of a Point, we have

$$\begin{aligned} PB^2 &= PD \cdot PC \\ 20 &= (2\sqrt{10} - x)(y + 2\sqrt{10}). \end{aligned}$$

Isolating the variables after expanding gives $x - y = 2\sqrt{10} - 10$.

Using Power of a Point again, we have

$$\begin{aligned} DE \cdot EC &= BE \cdot EA \\ xy &= 2\sqrt{5}(10\sqrt{2} - 2\sqrt{5}) \\ xy &= 20\sqrt{10} - 20. \end{aligned}$$

To get $x^2 + y^2$, we can perform the operation $(x - y)^2 + 2xy$. Plugging these values in,

$$\begin{aligned} (x - y)^2 + 2xy &= (2\sqrt{10} - 10)^2 + 2(20\sqrt{10} - 20) \\ &= 40 + 100 - 40\sqrt{10} + 40\sqrt{10} - 40 = \boxed{\text{(E)} 100} \end{aligned}$$

-Benedict T (countmath1)

Solution 7 (Cheating)

Perhaps not reliable in general, but very useful as a last resort. The choice of the radius $5\sqrt{2}$ is strange, and is probably motivated by a nice answer in the end, so we only consider integer options. Notice that a 5 also appears in the condition $BE = 2\sqrt{5}$, therefore it will likely be present in the answer as well; the only integer containing a factor of 5 amongst the choices is 100, thus the answer is **(E)** 100

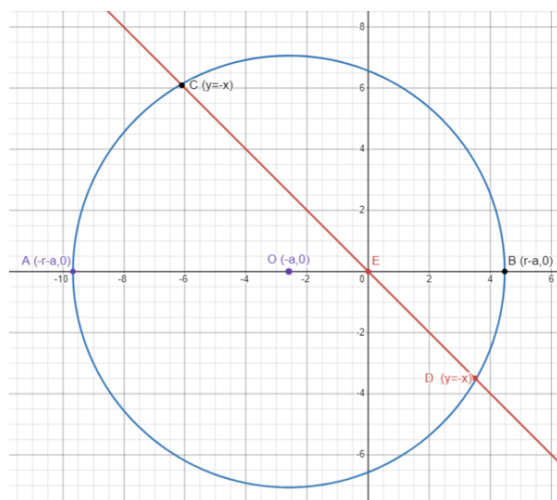
Solution 7 (Coordinate Geometry + Vieta)

set the origin at E (instead of circle center O) will reduce calculation. line CD is $y = -x$, let $OE = a$, circle radius $r = 5\sqrt{2}$, circle $(x + a)^2 + y^2 = r^2$ intersects line $y = -x$ at 2 points $C(X_c, Y_c)$ and $D(X_d, Y_d)$ $X_c = -Y_c$, $X_d = -Y_d$

substitute y with $-x$, circle becomes $(x + a)^2 + y^2 = (x + a)^2 + (-x)^2 = 2x^2 + 2ax + a^2 - r^2 = 0$ X_c, X_d will be 2 roots of above quadratic and we will apply vieta $X_c + X_d, X_c \cdot X_d$ below

$$CE^2 + DE^2 = ((-X_c) + Y_c)^2 + (X_d + (-Y_d))^2 = ((-X_c) + (-X_c))^2 + (X_d + (X_d))^2 = 4(X_c^2 + X_d^2) = 4((X_c + X_d)^2 - 2X_c X_d)$$

plug in $a = OE = OB - BE = 5\sqrt{2} - 2\sqrt{5}$ and $r = 5\sqrt{2}$ we get answer **(E)** 100.



13. (AMC 12 12B 2020) Which of the following is the value of $\sqrt{\log_2 6 + \log_3 6}$?
 (A) 1 (B) $\sqrt{\log_5 6}$ (C) 2 (D) $\sqrt{\log_2 3} + \sqrt{\log_3 2}$ (E) $\sqrt{\log_2 6} + \sqrt{\log_3 6}$

Answer: D

Solution 1 (Properties of Logarithms)

Recall that:

- a) $\log_b (uv) = \log_b u + \log_b v$.
- b) $\log_b u \cdot \log_u b = 1$.

We use these properties of logarithms to rewrite the original expression:

$$\begin{aligned} \sqrt{\log_2 6 + \log_3 6} &= \sqrt{(\log_2 2 + \log_2 3) + (\log_3 2 + \log_3 3)} \\ &= \sqrt{2 + \log_2 3 + \log_3 2} \\ &= \sqrt{\left(\sqrt{\log_2 3} + \sqrt{\log_3 2}\right)^2} \\ &= \boxed{\text{(D)} \sqrt{\log_2 3} + \sqrt{\log_3 2}}. \end{aligned}$$

MRENTHUSIASM (Solution)

Solution 2 (Change of Base Formula)

First,

$$\sqrt{\log_2 6 + \log_3 6} = \sqrt{\frac{\log 6}{\log 2} + \frac{\log 6}{\log 3}} = \sqrt{\frac{\log 6 \cdot \log 3 + \log 6 \cdot \log 2}{\log 3 \cdot \log 2}} = \sqrt{\frac{\log 6(\log 2 + \log 3)}{\log 2 \cdot \log 3}}.$$

From here,

$$\begin{aligned} \sqrt{\frac{\log 6(\log 2 + \log 3)}{\log 2 \cdot \log 3}} &= \sqrt{\frac{(\log 2 + \log 3)(\log 2 + \log 3)}{\log 2 \cdot \log 3}} = \sqrt{\frac{(\log 2 + \log 3)^2}{\log 2 \cdot \log 3}} = \frac{\log 2}{\sqrt{\log 2 \cdot \log 3}} + \frac{\log 3}{\sqrt{\log 2 \cdot \log 3}} \\ &= \sqrt{\frac{\log 2}{\log 3}} + \sqrt{\frac{\log 3}{\log 2}} \\ &= \sqrt{\log_3 2} + \sqrt{\log_2 3}. \end{aligned}$$

Answer: **(D)** $\sqrt{\log_2 3} + \sqrt{\log_3 2}$

Note that in this solution, even the most minor steps have been written out. On the actual test, this solution would be quite fast, and much of it could easily be done in your head.

Solution 3 (Observations)

Using the knowledge of the powers of 2 and 3, we know that $\log_2 6 > 2.5$ and $\log_3 6 > 1.5$. Therefore,

$$\sqrt{\log_2 6 + \log_3 6} > \sqrt{2.5 + 1.5} = 2.$$

Only choices **(D)** and **(E)** are greater than 2, but **(E)** is certainly incorrect: If we compare the squares of the original expression and **(E)**, then they are clearly not equal. So, the answer is **(D)** $\sqrt{\log_2 3} + \sqrt{\log_3 2}$.

Solution 4 (Solution 3 but More Detailed)

Note: Only use this method if all else fails and you cannot find a way to simplify the logarithms.

We can see that $\log_2 6$ is greater than 2 and less than 3. Additionally, since 6 is halfway between 2^2 and 2^3 , knowing how exponents increase more the larger x is, we can deduce that $\log_2 6$ is just above halfway between 2 and 3. We can guesstimate this as $\log_2 6 \approx 2.55$. (It's actually about 2.585.)

Next, we think of $\log_3 6$. This is greater than 1 and less than 2. As 6 is halfway between 3^1 and 3^2 , and similar to the logic for $\log_2 6$, we know that $\log_3 6$ is just above halfway between 1 and 2. We guesstimate this as $\log_3 6 \approx 1.55$. (It's actually about 1.631.)

So, $\log_2 6 + \log_3 6$ is approximately 4.1. The square root of that is just above 2, maybe 2.02. We cross out all choices below **(C)** since they are less than 2, and **(E)** can't possibly be true unless either $\log_2 6$ and/or $\log_3 6$ is 0 (You can prove this by squaring.). Thus, the only feasible answer is **(D)** $\sqrt{\log_2 3} + \sqrt{\log_3 2}$.

14. (AMC 12 12B 2020) Bela and Jenn play the following game on the closed interval $[0, n]$ of the real number line, where n is a fixed integer greater than 4. They take turns playing, with Bela going first. At his first turn, Bela chooses any real number in the interval $[0, n]$. Thereafter, the player whose turn it is chooses a real number that is more than one unit away from all numbers previously chosen by either player. A player unable to choose such a number loses. Using optimal strategy, which player will win the game?

(A) Bela will always win. **(B)** Jenn will always win. **(C)** Bela will win if and only if n is odd.
(D) Jenn will win if and only if n is odd. **(E)** Jenn will win if and only if $n > 8$.

Answer: A

Solution 1 (Game Theory)

We use game theory.

Notice how they say Optimal Strategy. This means that each player plays the best move in any scene.

We then solve through cases.

We set $n > 4$, n is even. Say $n = 10$. Then, we see that Bela could pick either 3 or 4, and then Jenn would be forced to choose either 1(2), 5, or 6. But notice how you can only make one move, and 1 and $[5, 6]$ are on opposite sides. So therefore, Bela will always win.

We set $n > 4$, n is odd. Say $n = 9$. This is even easier! Bela could just choose the middle, and then again, Jenn would be forced to pick a move that Bela could just replicate and win. So, Bela will also always win.

Through deduction, our answer is (A) Bela will always win. *But that's just a theory, a game theory!*

Solution 2 (Symmetry)

If Bela selects the middle number in the range $[0, n]$ and then mirror whatever number Jenn selects, then if Jenn can select a number within the range, so can Bela. Jenn will always be the first person to run out of a number to choose, so the answer is (A) Bela will always win.

Solution 3 (Educated Guess)

First of all, realize that the parity of n likely has no effect on the strategy, since Bela obviously wins when $n = 1$, and it's not hard to find a winning strategy for $n = 2$ or $n = 3$ (start with $n/2$). Similarly, there is no reason the strategy would change when $n > 8$.

So we are left with (A) and (B). From here it is best to try out random numbers and try to find the strategy that will let Bela win, but if you can't find it, realize that it is more likely the answer is (A) Bela will always win since Bela has the first move and thus has more control.

Solution 4 (Simulation)

We can just play the game. We can draw a number line to 9 and assume Bela and Jenn will only play the integer values. From now after playing one round of the game, Bela will win so answers (B), (D), and (E) are eliminated. Now we want to test if Bela will win if n is even so we can draw another number line this time up to 10. Now after playing the game, we can find that Bela won yet again so the answer is (A) Bela will always win.

-minor edits by jjbros

Solution 5 (Answer Choices)

Play out $n = 5$ and $n = 6$. In the first case, Bela can choose 0 and win, and in the second case, Bela can choose 1 and win (try it out). This rules out (B), (C), and (D).

Next, consider (E). Assume for the sake of contradiction that this is true; an alternate way of stating this is that for $n > 8$, the second player always wins (we ignore the "only if" statement). Thus, $n = 9$ is a win for the second player. Consider $n = 11$. If Bela chooses 11, then the game becomes an $n = 9$ case where the first player is instead Jenn. However, by the claim that the second player should win $n > 8$, this game should be a win for Bela, but we have already stated that any games with $n > 8$ should be a win for Jenn, which is a contradiction. Thus (E) is incorrect, and so the answer is (A) Bela will always win

Solution 6 (Similar to Solution 2)

Bela can choose the exact midpoint of the number line. Whatever Jenn does, Bela can copy her moves in the opposite side half of the number line. Jenn will always run out of space first.

15. (*AMC 12 12B 2020*) There are 10 people standing equally spaced around a circle. Each person knows exactly 3 of the other 9 people: the 2 people standing next to her or him, as well as the person directly across the circle. How many ways are there for the 10 people to split up into 5 pairs so that the members of each pair know each other?

(A) 11 (B) 12 (C) 13 (D) 14 (E) 15

Answer: C

Solution

Consider the 10 people to be standing in a circle, where two people opposite of each other form a diameter of the circle.

Let us use casework on the number of pairs that form a diameter of the circle.

Case 1: 0 diameters

There are 2 ways: either 1 pairs with 2, 3 pairs with 4, and so on or 10 pairs with 1, 2 pairs with 3, etc. So $\boxed{2}$ cases.

Case 2: 1 diameter

There are 5 possible diameters to draw (everyone else pairs with the person next to them). So $\boxed{5}$ cases.

Note that there cannot be 2 diameters since there would be one person on either side that will not have a pair adjacent to them. The only scenario forced is when the two people on either side would be paired up across a diameter. Thus, a contradiction will arise.

Case 3: 3 diameters

There are 5 possible sets of 3 diameters to draw. Notice we are technically choosing the number of ways to choose a pair of two diameters that are neighbors to each other. This means we can choose the first diameter in the pair, and have only two diameters to choose from for the second in the pair. This means we have $5 \cdot 2 = 10$ possibilities for choosing 5 neighboring diameters. However, notice that there are duplicates, so we divide the 10 possibilities by 2 to get $\boxed{5}$.

Note that there cannot be a case with 4 diameters because then there would have to be 5 diameters for the two remaining people as they have to be connected with a diameter. A contradiction arises.

Case 4: 5 diameters

There is only $\boxed{1}$ way to do this.

Thus, in total there are $2 + 5 + 5 + 1 = \boxed{(C) 13}$ possible ways.

16. (*AMC 12 12B 2020*) An urn contains one red ball and one blue ball. A box of extra red and blue balls lie nearby. George performs the following operation four times: he draws a ball from the urn at random and then takes a ball of the same color from the box and returns those two matching balls to the urn. After the four iterations the urn contains six balls. What is the probability that the urn contains three balls of each color?

- (A) $\frac{1}{6}$ (B) $\frac{1}{5}$ (C) $\frac{1}{4}$ (D) $\frac{1}{3}$ (E) $\frac{1}{2}$

Answer: B

Solution 1 (The Shortcut Passed Every Barrier)

Suppose that we have a deck, currently containing just one black card. We then insert n red cards one-by-one into the deck at random positions. It is easy to see using induction, that the black card is randomly situated in the deck.

Now, suppose that we have this deck again, with only one black card. Each time we pick a red ball, we place a card above the black card, and each time we pick a blue ball, we place a card below the black card. It is easy to see that the probability that the card is inserted into the top part of the deck is simply equal to the number of red balls divided by the total number of balls, and the probability that the card is inserted into the bottom part of the deck is equal to the number of blue balls divided by the total number of balls. Therefore, this is equivalent to inserting the card randomly into the deck.

Finally, four more red cards will be inserted into the deck, and so the black card can be in five possible positions. Only one corresponds to having three balls of each type. Our probability is thus $\frac{1}{5}$, and so the answer is **B**.

Solution 2

Let R denote the action where George selects a red ball and B denote the action where he selects a blue one. Now, in order to get 3 balls of each color, he needs 2 more of both R and B .

There are 6 cases: $RRBB, RRRB, RBBR, BBRR, BRBR, BRRB$ (we can confirm that there are only 6 since $\binom{4}{2} = 6$). However we can clump $RRBB + BBRR$, $RRRB + BRBR$, and $RBBR + BRRB$ together since they are equivalent by symmetry.

CASE 1: $RRBB$ and $BBRR$

Let's find the probability that he picks the balls in the order of $RRBB$.

The probability that the first ball he picks is red is $\frac{1}{2}$.

Now there are 2 reds and 1 blue in the urn. The probability that he picks red again is now $\frac{2}{3}$.

There are 3 reds and 1 blue now. The probability that he picks a blue is $\frac{1}{4}$.

Finally, there are 3 reds and 2 blues. The probability that he picks a blue is $\frac{2}{5}$.

So the probability that the $RRBB$ case happens is $\frac{1}{2} \cdot \frac{2}{3} \cdot \frac{1}{4} \cdot \frac{2}{5} = \frac{1}{30}$. However, since the $BBRR$ case is the exact same by symmetry, case 1 has a probability of $\frac{1}{30} \cdot 2 = \frac{1}{15}$ chance of happening.

CASE 2: $RRRB$ and $BRBR$

Let's find the probability that he picks the balls in the order of $RRRB$.

The probability that the first ball he picks is red is $\frac{1}{2}$.

Now there are 2 reds and 1 blue in the urn. The probability that he picks blue is $\frac{1}{3}$.

There are 2 reds and 2 blues now. The probability that he picks a red is $\frac{1}{2}$.

Finally, there are 3 reds and 2 blues. The probability that he picks a blue is $\frac{2}{5}$.

So the probability that the $RRRB$ case happens is $\frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{2} \cdot \frac{2}{5} = \frac{1}{30}$. However, since the $BRBR$ case is the exact same by symmetry, case 2 has a probability of $\frac{1}{30} \cdot 2 = \frac{1}{15}$ chance of happening.

CASE 3: $RBBR$ and $RRBB$

Let's find the probability that he picks the balls in the order of $RBBR$.

The probability that the first ball he picks is red is $\frac{1}{2}$.

Now there are 2 reds and 1 blue in the urn. The probability that he picks blue is $\frac{1}{3}$.

There are 2 reds and 2 blues now. The probability that he picks a blue is $\frac{1}{2}$.

Finally, there are 2 reds and 3 blues. The probability that he picks a red is $\frac{2}{5}$.

So the probability that the $RBBR$ case happens is $\frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{2} \cdot \frac{2}{5} = \frac{1}{30}$. However, since the $RRBB$ case is the exact same by symmetry, case 3 has a probability of $\frac{1}{30} \cdot 2 = \frac{1}{15}$ chance of happening.

Adding up the cases, we have $\frac{1}{15} + \frac{1}{15} + \frac{1}{15} = \boxed{\text{(B)} \frac{1}{5}}$ quacker88

Solution 3

We know that we need to find the probability of adding 2 red and 2 blue balls in some order. There are 6 ways to do this, since there are $\binom{4}{2} = 6$ ways to arrange $RRBB$ in some order. We will show that the probability for each of these 6 ways is the same.

We first note that the denominators should be counted by the same number. This number is $2 \cdot 3 \cdot 4 \cdot 5 = 120$. This is because 2, 3, 4, and 5 represent how many choices there are for the four steps. No matter what the k -th step involves $k+1$ numbers to choose from.

The numerators are the number of successful operations. No matter the order, the first time a red is added will come from 1 choice and the second time will come from 2 choices, since that is how many reds are in the urn originally. The same goes for the blue ones. The numerator must equal $(1 \cdot 2)^2$.

Therefore, the probability for each of the orderings of $RRBB$ is $\frac{4}{120} = \frac{1}{30}$. There are 6 of these, so the total probability is $\boxed{\text{(B)} \frac{1}{5}}$.

Solution 4

First, notice that when George chooses a ball he just adds another ball of the same color. On George's first move, he either chooses the red or the blue with a $\frac{1}{2}$ chance each. We can assume he chooses Red (chance $\frac{1}{2}$), and then multiply the final answer by two for symmetry. Now, there are two red balls and one blue ball in the urn. Then, he can either choose another Red (chance $\frac{2}{3}$), in which case he must choose two blues to get three of each, with probability $\frac{1}{4} \cdot \frac{2}{5} = \frac{1}{10}$ or a blue for two blue and two red in the urn, with chance $\frac{1}{3}$. If he chooses blue next, he can either choose a red then a blue, or a blue then a red. Each of these has a $\frac{1}{2} \cdot \frac{2}{5}$ for total of $2 \cdot \frac{1}{5} = \frac{2}{5}$. The total probability that he ends up with three red and three

blue is $2 \cdot \left(\frac{1}{2} \left(\frac{2}{3} \cdot \frac{1}{10} + \frac{1}{3} \cdot \frac{2}{5} \right) \right) = \frac{1}{15} + \frac{2}{15} = \boxed{\text{(B)} \frac{1}{5}}$.

Solution 4 Alternate

A solution pretty similar to the above is as follows.

Your only two cases of having 3R and 3B after 4 iterations are (without regard to order; if you do order, you multiply by 2 on both the top and the bottom and yield the same result):

1|1 -> 2|1 -> 3|1 -> 3|2 -> 3|3, and this occurs with probability $(1)(2/3)(1/4)(2/5)=1/15$.

1|1 -> 2|1 -> 2|2 -> 3|2 -> 3|3, and this occurs with probability $(1)(1/3)(1)(2/5)=2/15$.

Probability of 2|2->3|2 is 1 because it doesn't matter what you do. We don't count for order here so either ball you put from the box into of the urn is going to get you a 3|2 distribution. Same argument goes for the iteration from 1|1 to 2|1.

Summing these two up we get $\frac{1}{15} + \frac{2}{15} = \boxed{\text{(B)} \frac{1}{5}}$.

mathboy282

Solution 5

Let the probability that the urn ends up with more red balls be denoted $P(R)$. Since this is equal to the probability there are more blue balls, the probability there are equal amounts is $1 - 2P(R)$. $P(R)$ = the probability no more blues are chosen plus the probability only 1 more blue is chosen. The first case, $P(\text{no more blues}) = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \cdot \frac{4}{5} = \frac{1}{5}$.

The second case, $P(1 \text{ more blue}) = 4 \cdot \frac{1 \cdot 1 \cdot 2 \cdot 3}{2 \cdot 3 \cdot 4 \cdot 5} = \frac{1}{5}$. Thus, the answer is $1 -$

$$2\left(\frac{1}{5} + \frac{1}{5}\right) = 1 - \frac{4}{5} = \boxed{\text{(B)} \frac{1}{5}}.$$

Solution 6

Here X stands for R or B, and Y for the remaining color. After 3 rounds one can either have a 4 + 1 configuration ($XXXXY$), or 3 + 2 configuration ($XXXY$). The probability of getting to $XXXXYY$ from $XXXY$ is $\frac{2}{5}$. Observe that the probability of arriving to 4 + 1 configuration is

$$\frac{2}{3} \cdot \frac{3}{4} = \frac{1}{2}$$

($\frac{2}{3}$ to get from XXY to $XXXY$, $\frac{3}{4}$ to get from $XXXY$ to $XXXXY$). Thus the probability of arriving to 3 + 2 configuration is also $\frac{1}{2}$, and the answer is

$$\frac{1}{2} \cdot \frac{2}{5} = \boxed{\text{(B)} \frac{1}{5}}.$$

Solution 7

We can use dynamic programming to solve this problem.

We let $dp[i][j]$ be the probability that we end up with i red balls and j blue balls. Notice that there are only two ways that we can end up with i red balls and j blue balls: one is by fetching a red ball from the urn when we have $i - 1$ red balls and j blue balls and the other is by fetching a blue ball from the urn when we have i red balls and $j - 1$ blue balls.

Then we have $dp[i][j] = \frac{i-1}{i-1+j} dp[i-1][j] + \frac{j-1}{i-1+j} dp[i][j-1]$

Then we start can with $dp[1][1] = 1$ and try to compute $dp[3][3]$.

$i \setminus j$	1	2	3
1	1	1/2	1/3
2	1/2	1/3	1/4
3	1/3	1/4	1/5

The answer is (B) $\frac{1}{5}$.

(Solution by CircleOO)

Solution 8 (Easier Solution)

Basically, George is picking a ball and adding another ball of that color to the urn. Basically, he needs to pick a red ball twice and a blue ball twice to get 3 red and 3 blue balls in the urn. The number of ways to choose a red ball twice is $1 \cdot 2$ (at the start there's 1 red ball, then he needs to choose it again when there's 2, and then there is 3 red balls so he doesn't need to choose it again). Similarly the number of ways to choose blue balls twice is $2 \cdot 2 = 4$ ways to choose 2 red balls and 2 blue balls (not considering order). The total number of ways is $2 \cdot 3 \cdot 4 \cdot 5$. Adding order to the 4 ways to choose 2 red and 2 blue balls, it is $4C2$ because out of the 4 ball choosings, 2 of them must be red, and we automatically make the other ones blue. $\frac{\binom{4}{2} \cdot 2 \cdot 2}{2 \cdot 3 \cdot 4 \cdot 5} = \frac{1}{5}$.

Solution 9 (Desperate Guess)

Notice that the answer can either be (1R, 5B) (2R, 4B) (3R, 3B) (4R, 2B) or (5R, 1B). Also, the answer choices are all really basic. Just guess that each of these cases are equal by symmetry so the probability of (3R, 3B) is (B) $\frac{1}{5}$.

-unhappyfarmer

17. (AMC 12 12B 2020) How many polynomials of the form $x^5 + ax^4 + bx^3 + cx^2 + dx + 2020$, where a, b, c , and d are real numbers, have the property that whenever r is a root, so is $\frac{-1+i\sqrt{3}}{2} \cdot r$? (Note that $i = \sqrt{-1}$)
 (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Answer: C

Solution 1

Let $P(x) = x^5 + ax^4 + bx^3 + cx^2 + dx + 2020$. We first notice that $\frac{-1+i\sqrt{3}}{2} = e^{2\pi i/3}$. That is because of Euler's Formula: $e^{ix} = \cos(x) + i \cdot \sin(x)$. $\frac{-1+i\sqrt{3}}{2} = -\frac{1}{2} + i \cdot \frac{\sqrt{3}}{2} = \cos(120^\circ) + i \cdot \sin(120^\circ) = e^{2\pi i/3}$.

In order r to be a root of P , $re^{2\pi i/3}$ must also be a root of P , meaning that 3 of the roots of P must be $r, re^{i\frac{2\pi}{3}}, re^{i\frac{4\pi}{3}}$. However, since P is degree 5, there must be two additional roots. Let one of these roots be w , if w is a root, then $we^{2\pi i/3}$ and $we^{4\pi i/3}$ must also be roots. However, P is a fifth degree polynomial, and can therefore only have 5 roots. This implies that w is either $r, re^{2\pi i/3}$, or $re^{4\pi i/3}$. Thus we know that the polynomial P can be written in the form $(x-r)^m(x-re^{2\pi i/3})^n(x-re^{4\pi i/3})^p$. Moreover, by Vieta's, we know that there is only one possible value for the magnitude of r as $|r|^5 = 2020$, meaning that the amount of possible polynomials P is equivalent to the possible sets (m, n, p) . In order for the coefficients of the polynomial to all be real, $n = p$ due to $re^{2\pi i/3}$ and $re^{4\pi i/3}$ being conjugates and since $m + n + p = 5$, (as the polynomial is 5th degree) we have two possible solutions for (m, n, p) which are $(1, 2, 2)$ and $(3, 1, 1)$ yielding two possible polynomials. The answer is thus (C) 2.

Solution 2

Let $x_1 = r$, then

$$x_2 = \frac{-1 + i\sqrt{3}}{2}r,$$

$$x_3 = \left(\frac{-1 + i\sqrt{3}}{2} \right)^2 r = \left(\frac{-1 - i\sqrt{3}}{2} \right) r,$$

$$x_4 = \left(\frac{-1 + i\sqrt{3}}{2} \right)^3 r = r,$$

which means x_4 is the same as x_1 .

Now we have 3 different roots of the polynomial, x_1 , x_2 , and x_3 . Next, we will prove that all 5 roots of the polynomial must be chosen from those 3 roots. Let us assume that there is one root $x_4 = p$ which is different from the three roots we already know, then there must be two other roots,

$$x_5 = \left(\frac{-1 + i\sqrt{3}}{2} \right)^2 p = \left(\frac{-1 - i\sqrt{3}}{2} \right) p,$$

$$x_6 = \left(\frac{-1 + i\sqrt{3}}{2} \right)^3 p = p,$$

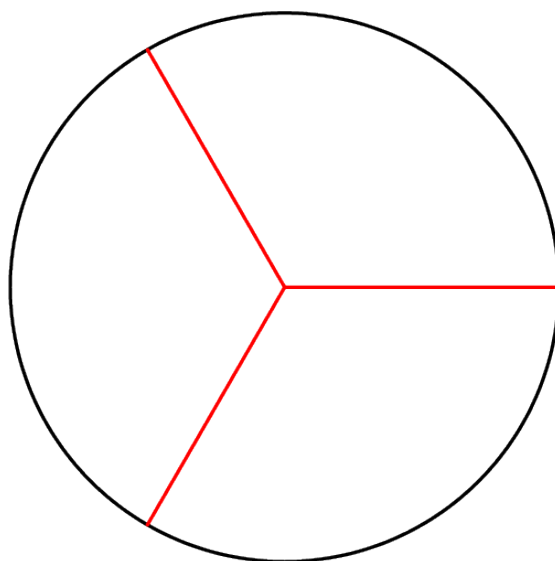
different from all known roots. We get 6 different roots for the polynomial, which contradicts the limit of 5 roots. Therefore the assumption of a different root is wrong, thus the roots must be chosen from x_1 , x_2 , and x_3 .

The polynomial then can be expressed as $f(x) = (x - x_1)^m(x - x_2)^n(x - x_3)^q$, where m , n , and q are non-negative integers and $m + n + q = 5$. Since a , b , c and d are real numbers, then n must be equal to q . By Vieta's we have $mr + n\left(\frac{-1+i\sqrt{3}}{2}\right) + q\left(\frac{-1-i\sqrt{3}}{2}\right) = a$ is a real number. Therefore (m, n, q) can only be $(1, 2, 2)$ or $(3, 1, 1)$, so the answer is **(C)2**.

Solution 3

Call $\frac{-1+i\sqrt{3}}{2} = q$. We have r , qr , and q^2r having to be roots, and since $q^3 = 1$, we must choose 2 more roots out of these three such that the condition that a , b , c , and d are all real. There are **2** ways to do this because of the fundamental theorem of algebra, these being (qr, qr^2) and (r, r) because to satisfy FTA if z is a root then so must its conjugate.

Solution 4 (Similar but using visualization)



Due to Euler's Formula, $\frac{-1+i\sqrt{3}}{2}$ is a 120 degree rotation CCW; therefore, the problem stipulates that a root must be followed by a root rotated 120 degrees CCW, and that the rotated root must also have yet another root 240 degrees CCW. Therefore, all roots must come similar groups of 3. We cannot have 6 roots from 2 groups, and we cannot have 0 roots, so the 3 distinct roots, when drawn (with 2 with multiplicities not shown), must look like the figure above.

As for the magnitudes of the roots, as all roots pictured above have the same magnitude, there is 1 possible magnitude for each root, which is $\sqrt[5]{2020}$, meaning we don't have to care about the scale of the diagram.

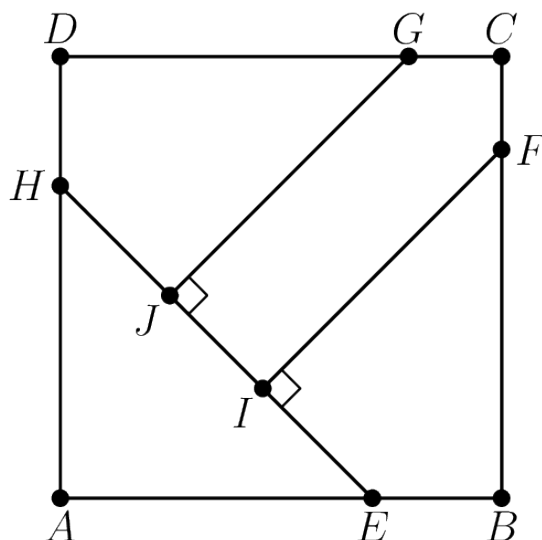
To satisfy the complex conjugate root theorem (that if the coefficients are real, any root of $a+bi$ must have a root of $a-bi$), the figure depicting all 5 roots must be symmetrical across the horizontal axis. Without consideration for roots with multiplicities, there are 3 axes of symmetry for this shape, and two unique ways to align one of the axes of symmetry such that it's parallel to the horizontal axis, the other such case created by reflecting across the vertical axis.

However, not only must the magnitudes multiply to 2020, but the product of the roots must be *positive* 2020. Only one of the above cases can satisfy this condition (If there exists a configuration that multiplies to -2020 , then reflecting across the vertical axis flips the sign creating the $+2020$ case, with the converse also being true. As each case guarantees the existence of the other, and while at least one case is guaranteed to exist in the first place, we know there is exactly 1 of each case that remain, with 1 prevailing $+2020$ case to use.

Now, when taking roots with multiplicities back into consideration, the singular case has 2 ways to assign multiplicities while preserving symmetry: either both collating to the horizontal root, or through one being distributed to each of the angled roots (This corresponds to the $(2, 2, 1)$ and $(3, 1, 1)$ cases discussed earlier).

Therefore, the final answer is $\boxed{(C)2}$.

18. (AMC 12 12B 2020) In square $ABCD$, points E and H lie on $\overline{AB}$ and $\overline{DA}$, respectively, so that $AE = AH$. Points F and G lie on $\overline{BC}$ and $\overline{CD}$, respectively, and points I and J lie on $\overline{EH}$ so that $\overline{FI} \perp \overline{EH}$ and $\overline{GJ} \perp \overline{EH}$. See the figure below. Triangle AEH , quadrilateral $BFIE$, quadrilateral $DHJG$, and pentagon $FCGJI$ each has area 1. What is FI^2 ?



- (A) $\frac{7}{3}$ (B) $8 - 4\sqrt{2}$ (C) $1 + \sqrt{2}$ (D) $\frac{7}{4}\sqrt{2}$ (E) $2\sqrt{2}$

Answer: B

Solution 1

Since the total area is 4, the side length of square $ABCD$ is 2. We see that since triangle HAE is a right isosceles triangle with area 1, we can determine sides HA and AE both to be $\sqrt{2}$. Now, consider extending FB and IE until they intersect. Let the point of intersection be K . We note that EBK is also a right isosceles triangle with side $2 - \sqrt{2}$ and find its area to be $3 - 2\sqrt{2}$. Now, we notice that FIK is also a right isosceles triangle (because $\angle EKB = 45^\circ$) and find its area to be $\frac{1}{2}FI^2$. This is also equal to $1 + 3 - 2\sqrt{2}$ or $4 - 2\sqrt{2}$. Since we are looking for FI^2 , we want two times this. That gives **(B)** $8 - 4\sqrt{2}$. TLiu

Solution 2

Draw the auxiliary line AC . Denote by M the point it intersects with HE , and by N the point it intersects with GF . Last, denote by x the segment FN , and by y the segment FI . We will find two equations for x and y , and then solve for y^2 .

Since the overall area of $ABCD$ is 4 $\implies AB = 2$, and $AC = 2\sqrt{2}$. In addition, the area of $\triangle AME = \frac{1}{2} \implies AM = 1$.

The two equations for x and y are then:

• Length of AC : $1 + y + x = 2\sqrt{2} \implies x = (2\sqrt{2} - 1) - y$ • Area of CMIF: $\frac{1}{2}x^2 + xy = \frac{1}{2} \implies x(x + 2y) = 1$.

Substituting the first into the second, yields $[(2\sqrt{2} - 1) - y] \cdot [(2\sqrt{2} - 1) + y] = 1$

Solving for y^2 gives **(B)** $8 - 4\sqrt{2}$ DrB

Solution 3

Plot a point F' such that $F'I$ and AB are parallel and extend line FB to point B' such that $FIB'F'$ forms a square. Extend line AE to meet line $F'B'$ and point E' is the intersection of the two. The area of this square is equivalent to FI^2 . We see that the area of square $ABCD$ is 4, meaning each side is of length 2. The area of the pentagon $EIFF'E'$ is 2. Length $AE = \sqrt{2}$, thus $EB = 2 - \sqrt{2}$. Triangle $EB'E'$ is isosceles, and the area of this triangle is $\frac{1}{2}(4 - 2\sqrt{2})(2 - \sqrt{2}) = 6 - 4\sqrt{2}$. Adding these two areas, we get

$$2 + 6 - 4\sqrt{2} = 8 - 4\sqrt{2} \rightarrow \textbf{(B)} \ 8 - 4\sqrt{2}$$

. -OGBooger

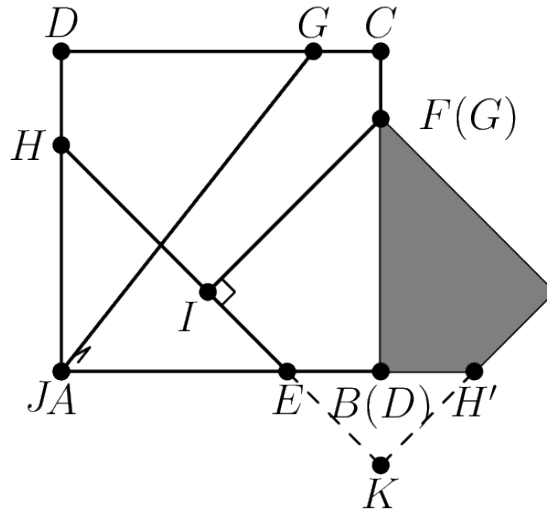
Solution 4 (HARD Calculation)

We can easily observe that the area of square $ABCD$ is 4 and its side length is 2 since all four regions that build up the square has area 1. Extend FI and let the intersection with AB be K . Connect AC , and let the intersection of AC and HE be L . Notice that since the area of triangle AEH is 1 and $AE = AH$, $AE = AH = \sqrt{2}$, therefore $BE = HD = 2 - \sqrt{2}$. Let $CG = CF = m$, then $BF = DG = 2 - m$. Also notice that $KB = 2 - m$, thus $KE = KB - BE = 2 - m - (2 - \sqrt{2}) = \sqrt{2} - m$. Now use the condition that the area of quadrilateral $BFIE$ is 1, we can set up the following equation: $\frac{1}{2}(2 - m)^2 - \frac{1}{4}(\sqrt{2} - m)^2 = 1$. We solve the equation and yield $m = \frac{8 - 2\sqrt{2} - \sqrt{64 - 32\sqrt{2}}}{2}$. Now notice that $FI = AC - AL - \frac{m}{\sqrt{2}} =$

$$2\sqrt{2} - 1 - \frac{\sqrt{2}}{2} \cdot \frac{8 - 2\sqrt{2} - \sqrt{64 - 32\sqrt{2}}}{2} = 2\sqrt{2} - 1 - \frac{8\sqrt{2} - 4 - \sqrt{128 - 64\sqrt{2}}}{4} = \frac{\sqrt{128 - 64\sqrt{2}}}{4}. \text{Hence}$$

$$FI^2 = \frac{128 - 64\sqrt{2}}{16} = 8 - 4\sqrt{2}. \text{ -HarryW}$$

Solution 5 (Basically Same as Solution 3)



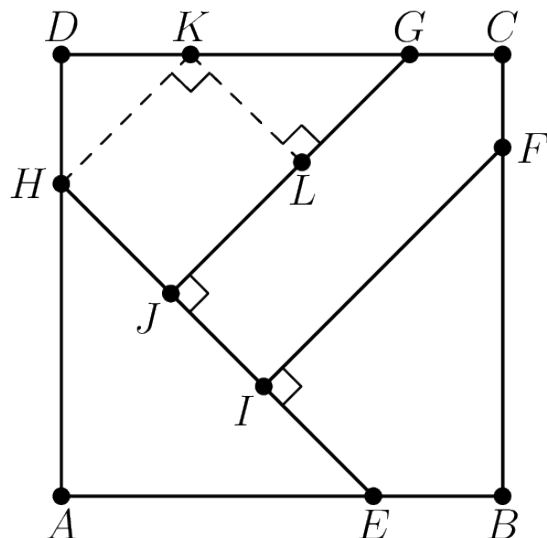
Easily, we can find that: quadrilateral $BFIE$ and $DHJG$ are congruent with each other, so we can move $DHJG$ to the shaded area (F and G , B and D overlapping) to form a square $FIKJ'$ ($DG = FB$, $CG = FC$, $\angle CGF = \angle CFG = 45^\circ$ so $\angle IFJ' = 90^\circ$). Then we can solve $AH = AE = \sqrt{2}$, $EB = 2 - \sqrt{2}$, $EK = 2\sqrt{2} - 2$.

$$FI^2 = \text{area of } BFIE + \text{area of } FJ'H'B + \text{area of } EH'K$$

$$= 1 + 1 + \frac{1}{2}(2\sqrt{2} - 2)^2 = 8 - 4\sqrt{2} \rightarrow \boxed{\text{(B)} 8 - 4\sqrt{2}}$$

-Ryan Zhang @BRS-Minor edit by WhySean38

Solution 6



$$[ABCD] = 4, AB = 2, [AHE] = 1, AH = AE = \sqrt{2}, DH = 2 - \sqrt{2}, JL = HK = \sqrt{2} \cdot DH = 2\sqrt{2} - 2$$

Because $ABCD$ is a square and $AH = AE$, AC is the line of symmetry of pentagon

$CDHEB$. Because $[DHJG] = [BFIE]$, $DHJG$ is the reflection of $BFIE$ about line AC

Let $FI = GJ = x$, $KL = LG = GJ - LJ = x - 2\sqrt{2} + 2$ $[DHK] = \frac{(2-\sqrt{2})^2}{2} = 3 - 2\sqrt{2}$
 $[GKL] = \frac{(x-2\sqrt{2}+2)^2}{2} = \frac{x^2}{2} + 2x - 2x\sqrt{2} - 4\sqrt{2} + 6$ $[HKJL] = (x - 2\sqrt{2} + 2) \cdot (2\sqrt{2} - 2) = 2x\sqrt{2} - 2x + 8\sqrt{2} - 12$

$$[DHK] + [GKL] + [HKJL] = [DHJG]$$

$$3 - 2\sqrt{2} + \frac{x^2}{2} + 2x - 2x\sqrt{2} - 4\sqrt{2} + 6 + 2x\sqrt{2} - 2x + 8\sqrt{2} - 12 = 1$$

$$\frac{x^2}{2} + 2\sqrt{2} - 4 = 0$$

$$x^2 = 8 - 4\sqrt{2}$$

$$FI^2 = \boxed{\text{(B)} \ 8 - 4\sqrt{2}}$$

Solution 7 (Easy to See)

Note that the side length of $ABCD$ is 2 and thus the diagonal is of length $2\sqrt{2}$. However, the height to side HE in triangle HAE is 1, implying that $CM = 2\sqrt{2} - 1$ where M is the midpoint of JI . From here suppose that N is the midpoint of $\overline{FG}$ and let $x = NC$, which means $FG = 2x$. The area of the pentagon is then

$$[FIJG] + [GCF] = GF \cdot FI + x^2 = (2x)(2\sqrt{2} - 1 - x) + x^2 = 1$$

Solving this quadratic for x yields $x = 2\sqrt{2} - 1 \pm \sqrt{8 - 4\sqrt{2}}$ (technically the smaller value is the correct one but it doesn't matter for our purposes). We can then calculate $FI^2 = (2\sqrt{2} - 1 - x)^2 = \boxed{\text{(B)} \ 8 - 4\sqrt{2}}$.

Solution 8

We extend $\overline{FB}$ and $\overline{IE}$ to meet at a point X . Since $\angle AEI = \angle BEX$ and FX is parallel to DA , we know that $\triangle BEX \sim \triangle AEH$, and because $\angle BXE = \angle IXF$ and $\angle XBE = \angle XIF$, we can conclude that $\triangle BEX \sim \triangle AEH \sim \triangle IFX$.

Now, because $\triangle AEH$ is isosceles, right, and has an area of 1, we can conclude that $AE = AH = \sqrt{2}$ and that $BE = 2 - \sqrt{2}$. Armed with this knowledge, and setting $IF = a$ and the area of $\triangle BEX = b$, we can use similarity to say that

$$\left(\frac{a}{2 - \sqrt{2}}\right)^2 = \frac{1 + b}{b}$$

Since we know the side lengths of $\triangle BEX$ due to the fact that it is also an isosceles right triangle, we know that the area is $\frac{(2-\sqrt{2})^2}{2}$. Simplifying further and plugging in values, we have

$$\frac{a^2}{(2 - \sqrt{2})^2} = 1 + \frac{2}{(2 - \sqrt{2})^2}$$

Multiplying by $(2 - \sqrt{2})^2$ on both sides, we get

$$a^2 = (2 - \sqrt{2})^2 + 2 = \boxed{\text{(B)} \ 8 - 4\sqrt{2}}$$

yingkai_0_

19. (AMC 12 12B 2020) Square $ABCD$ in the coordinate plane has vertices at the points $A(1, 1)$, $B(-1, 1)$, $C(-1, -1)$, and $D(1, -1)$. Consider the following four transformations:

- L , a rotation of 90° counterclockwise around the origin;
- R , a rotation of 90° clockwise around the origin;
- H , a reflection across the x -axis; and
- V , a reflection across the y -axis.

Each of these transformations maps the squares onto itself, but the positions of the labeled vertices will change. For example, applying R and then V would send the vertex A at $(1, 1)$ to $(-1, -1)$ and would send the vertex B at $(-1, 1)$ to itself. How many sequences of 20 transformations chosen from $\{L, R, H, V\}$ will send all of the labeled vertices back to their original positions? (For example, R, R, V, H is one sequence of 4 transformations that will send the vertices back to their original positions.)

- (A) 2^{37} (B) $3 \cdot 2^{36}$ (C) 2^{38} (D) $3 \cdot 2^{37}$ (E) 2^{39}

Answer: C

Solution 1

For each transformation:

- a) Each labeled vertex will move to an adjacent position.
- b) The labeled vertices will maintain the consecutive order $ABCD$ in either direction (clockwise or counterclockwise).
- c) L and R will retain the direction of the labeled vertices, but H and V will alter the direction of the labeled vertices.

After the 19th transformation, vertex A will be at either $(1, -1)$ or $(-1, 1)$. All possible configurations of the labeled vertices are shown below:

Configuration	$\begin{matrix} C & B \\ D & A \end{matrix}$	$\begin{matrix} A & D \\ B & C \end{matrix}$	$\begin{matrix} C & D \\ B & A \end{matrix}$	$\begin{matrix} A & B \\ D & C \end{matrix}$
The 20th Transformation	L	R	H	V

Each sequence of 19 transformations generates one valid sequence of 20 transformations. Therefore, the answer is $4^{19} = \boxed{(C) 2^{38}}$.

Solution 2

Let $(+)$ denote counterclockwise/starting orientation and $(-)$ denote clockwise orientation. Let 1, 2, 3, and 4 denote which quadrant A is in.

Realize that from any odd quadrant and any orientation, the 4 transformations result in some permutation of $(2+, 2-, 4+, 4-)$.

The same goes that from any even quadrant and any orientation, the 4 transformations result in some permutation of $(1+, 1-, 3+, 3-)$.

We start our first 19 moves by doing whatever we want, 4 choices each time. Since 19 is odd, we must end up on an even quadrant.

As said above, we know that exactly one of the four transformations will give us $(1+)$, and we must use that transformation.

Thus, the answer is $4^{19} = \boxed{(C) 2^{38}}$.

IGNORE THIS SOLUTION IT IS WRONG Notice that any **pair** of transformations either swaps the x and y -coordinates, negates the x and y -coordinates, swaps and negates the x and y -coordinates, or leaves the original unchanged. Furthermore, notice that for each of these results, if we apply another pair of transformations, one of these results will happen again, and with equal probability. Therefore, no matter what state we are in after we apply the first 9 pairs of transformations, there is a $\frac{1}{4}$ chance the last pair of transformations will return the figure to its original position. Therefore, the answer is $\frac{4^{20}}{8} = 2^{37} = \boxed{(A) 2^{37}}$.

note that this solution a few flaws and the answer is wrong- Happymaths

Solution 4

The total number of sequences is $4^{20} = 2^{40}$.

Note that there can only be an even number of reflections since they result in the same anti-clockwise orientation of the vertices A, B, C, D . Therefore, the probability of having the same anti-clockwise orientation with the original arrangement after the transformation is $\frac{1}{2}$.

Next, the even number of reflections means that there must be an even number of rotations since their sum is even. Even rotations result in only the original position or a 180° rotation of it.

Since rotation R and rotation L cancel each other out, the difference between the numbers of them define the final position. The probability of the transformation returning the vertices to the original position, given that there are even number of rotations, is equivalent to the probability that

$$|n(R) - n(L)| \equiv 0 \pmod{4} \text{ when } |n(H) - n(V)| \equiv 0 \pmod{4}$$

or

$$|n(R) - n(L)| \equiv 2 \pmod{4} \text{ when } |n(H) - n(V)| \equiv 2 \pmod{4}$$

which is again, $\frac{1}{2}$.

$$\text{Therefore, the answer is } 2^{40} \cdot \frac{1}{2} \cdot \frac{1}{2} = \boxed{(C) 2^{38}}.$$

Solution 5 (Group Theory)

This problem is a **Dihedral Group** problem, D_4 , in **Group Theory**.

The transformation has associativity, for $x, y, z \in \{L, R, H, V\}$, $(x \circ y) \circ z = x \circ (y \circ z)$.

Let I be the initial state of the square $A(1, 1), B(-1, 1), C(-1, -1)$, and $D(1, -1)$.

$$R \circ L = L \circ R = V \circ V = H \circ H = I \tag{1}$$

$$V \circ H = R \circ R = H \circ V = L \circ L \tag{2}$$

$$H \circ R = L \circ H = V \circ L = R \circ V \tag{3}$$

$$V \circ R = L \circ V = H \circ L = R \circ H \tag{4}$$

It's not hard to see that after a series of transformations from initial state I to initial state I , the number of transformations must be even. Denote $f(2n)$ be the number of sequences of $2n$ transformations from initial state I to initial state I . We are going to prove $f(2n) = 4^{2n-1}$.

For each transformation composite operator, there are 4 replacements.

For example, when $n = 2$:

$$R \circ R \circ R \circ R = (R \circ R) \circ R \circ R = I.$$

From (2), $R \circ R = V \circ H = H \circ V = L \circ L$, so $R \circ R$ can be replaced with $V \circ H$, $H \circ V$, $L \circ L$ without changing the result. Suppose we choose $V \circ H$, then

$$(R \circ R) \circ R \circ R = (V \circ H) \circ R \circ R = V \circ (H \circ R) \circ R.$$

From (3), $H \circ R = L \circ H = V \circ L = R \circ V$, so $H \circ R$ can be replaced with $L \circ H$, $V \circ L$, $R \circ V$ without changing the result. Suppose we choose $R \circ V$, then

$$V \circ (H \circ R) \circ R = V \circ (R \circ V) \circ R = V \circ R \circ (V \circ R).$$

From (4), $V \circ R = L \circ V = H \circ L = R \circ H$, so $V \circ R$ can be replaced with $L \circ V$, $H \circ L$, $R \circ H$ without changing the result. Suppose we choose $H \circ L$, then

$$V \circ R \circ (V \circ R) = V \circ R \circ (H \circ L) = V \circ R \circ H \circ L = I.$$

So, we have $f(4) = 4^{4-1}$:

$$\underbrace{R \circ R \circ R \circ R \circ \dots \circ R \circ R \circ R}_{2n}.$$

With $2n$ R transformations, it will go from initial state to initial state. There are $2n - 1$ transformation composite operators $\circ$ between the transformations, and each pair of transformations surrounding the transformations composite operator $\circ$ have 4 options. So, we have $f(2n) = 4^{2n-1}$, from which $f(20) = 4^{20-1} = \boxed{(C) 2^{38}}$.

<u>**Side Note**</u>

Equations (2), (3), (4) are equivalent. Here I will prove that (2) is equivalent to (3).

From (1), $R \circ L = V \circ V$. We have

$$H \circ (R \circ L) = H \circ (V \circ V) = (H \circ V) \circ V.$$

From (2), $H \circ V = V \circ H = L \circ L$. We have

$$(H \circ V) \circ V = (V \circ H) \circ V = V \circ (H \circ V) = V \circ (L \circ L).$$

So,

$$(H \circ R) \circ L = H \circ (R \circ L) = V \circ (L \circ L) = (V \circ L) \circ L.$$

It follows that $H \circ R = V \circ L$, which is (3).

Solution 6 (Quick)

The number of possible ways to orient a square using this method 20 times would be 4^{20} , as for each of the transformations we can only use the four transformations L , V , H , or R .

After 20 transformations, there are only 4 possible outcomes, as parity prohibits the other 4 outcomes. By symmetry, all 4 of these possibilities are equally likely. Thus, our answer is $\frac{4^{20}}{4} = 4^{19} = \boxed{(C) 2^{38}}$.

Solution 7 (Generating Functions)

Let x represent rotating $\overline{AC}$ 90° counterclockwise about the origin and let y represent rotating $\overline{BD}$ 90° counterclockwise about the origin. Then, note that the

transformations L, V, H , and R can be encoded by $xy, \frac{1}{xy}, \frac{y}{x}$, and $\frac{x}{y}$, respectively. Thus, we wish to compute the sum of the coefficients of the terms of the form $cx^{4a}y^{4b}$ for $a, b, c \geq 0$ in the polynomial

$$\left(xy + \frac{1}{xy} + \frac{y}{x} + \frac{x}{y}\right)^{20}.$$

We can factor the polynomial to get

$$\left(x + \frac{1}{x}\right)\left(y + \frac{1}{y}\right)^{20} = \left(x + \frac{1}{x}\right)^{20}\left(y + \frac{1}{y}\right)^{20}.$$

We can multiply the polynomial by $x^{20}y^{20}$ and it wouldn't affect the sum of the coefficients of the terms of the form $cx^{4a}y^{4b}$, so we can now work with

$$(x^2 + 1)^{20}(y^2 + 1)^{20}.$$

Now, it is very evident that we will get a term of the form $cx^{4a}y^{4b}$ exactly when the first part of the polynomial contributes an even number of x^2 's and the second part of the polynomial contributes an even number of y^2 's, so the answer is

$$\begin{aligned} \left(\binom{20}{0} + \binom{20}{2} + \dots + \binom{20}{20}\right) \left(\binom{20}{0} + \binom{20}{2} + \dots + \binom{20}{20}\right) &= 2^{19} \cdot 2^{19} \\ &= \boxed{(C) \ 2^{38}}. \end{aligned}$$

□

20. (AMC 12 12B 2020) Two different cubes of the same size are to be painted, with the color of each face being chosen independently and at random to be either black or white. What is the probability that after they are painted, the cubes can be rotated to be identical in appearance?

(A) $\frac{9}{64}$ (B) $\frac{289}{2048}$ (C) $\frac{73}{512}$ (D) $\frac{147}{1024}$ (E) $\frac{589}{4096}$

Answer: D

Solution

Define two ways of painting to be in the same *class* if one can be rotated to form the other.

We can count the number of ways of painting for each specific *class*.

Case 1: Black-white color distribution is 0-6 (out of 6 total faces)

Trivially $1^2 = 1$ way to paint the cubes.

Case 2: Black-white color distribution is 1-5

Trivially all $\binom{6}{5} = 6$ ways belong to the same *class*, so 6^2 ways to paint the cubes.

Case 3: Black-white color distribution is 2-4

There are two *classes* for this case: the *class* where the two black faces are touching and the other *class* where the two black faces are on opposite faces. There are 3 members of the latter *class* since there are 3 unordered pairs of 2 opposite faces of a cube. Thus, there are $\binom{6}{4} - 3 = 12$ members of the former *class*. Thus, $12^2 + 3^2$ ways to paint the cubes for this case.

Case 4: Black-white color distribution is 3-3

By simple intuition, there are also two *classes* for this case, the *class* where the three black faces meet at a single vertex, and the other class where the three black faces are in a "straight line" along the edges of the cube. Note that since there are 8 vertices in a cube, there are 8 members of the former class and $\binom{6}{3} - 8 = 12$ members of the latter class. Thus, $12^2 + 8^2$ ways to paint the cubes for this case. Note that by symmetry (since we are only switching the colors), the number of ways to paint the cubes for black-white color distributions 4-2, 5-1, and 6-0 is 2-4, 1-5, and 0-6 (respectively).

Thus, our total answer is

$$\frac{2(6^2 + 1^2 + 12^2 + 3^2) + 12^2 + 8^2}{2^{12}} = \frac{588}{4096} = \boxed{\text{(D)} \frac{147}{1024}}.$$

-fidgetboss_4000

21. (AMC 12 12B 2020) How many positive integers n satisfy

$$\frac{n + 1000}{70} = \lfloor \sqrt{n} \rfloor?$$

(Recall that $\lfloor x \rfloor$ is the greatest integer not exceeding x .)

(A) 2 (B) 4 (C) 6 (D) 30 (E) 32

Answer: C

Solution 1 (Fakesolve)

We can first consider the equation without a floor function:

$$\frac{n + 1000}{70} = \sqrt{n}$$

Multiplying both sides by 70 and then squaring:

$$n^2 + 2000n + 1000000 = 4900n$$

Moving all terms to the left:

$$n^2 - 2900n + 1000000 = 0$$

Now we can determine the factors:

$$(n - 400)(n - 2500) = 0$$

This means that for $n = 400$ and $n = 2500$, the equation will hold without the floor function.

Now we can simply check the multiples of 70 around 400 and 2500 in the original equation, which we abbreviate as $L = R$.

For $n = 330$, $L = 19$ but $18^2 < 330 < 19^2$ so $R = 18$

For $n = 400$, $L = 20$ and $R = 20$

For $n = 470$, $L = 21$, $R = 21$

For $n = 540$, $L = 22$ but $540 > 23^2$ so $R = 23$

Now we move to $n = 2500$

For $n = 2430$, $L = 49$ and $49^2 < 2430 < 50^2$ so $R = 49$

For $n = 2360$, $L = 48$ and $48^2 < 2360 < 49^2$ so $R = 48$

For $n = 2290$, $L = 47$ and $47^2 < 2360 < 48^2$ so $R = 47$

For $n = 2220$, $L = 46$ but $47^2 < 2220$ so $R = 47$

For $n = 2500$, $L = 50$ and $R = 50$

For $n = 2570$, $L = 51$ but $2570 < 51^2$ so $R = 50$

Therefore we have 6 total solutions, $n = 400, 470, 2290, 2360, 2430, 2500 = \boxed{\text{(C) } 6}$

Solution 2

This is my first solution here, so please forgive me for any errors.

We are given that

$$\frac{n + 1000}{70} = \lfloor \sqrt{n} \rfloor.$$

We know $\lfloor \sqrt{n} \rfloor$ must be an integer, which means that $n + 1000$ is divisible by 70. As $1000 \equiv 20 \pmod{70}$, this means that $n \equiv 50 \pmod{70}$, so we can write $n = 70k + 50$ for $k \in \mathbb{Z}$. Note that n has to be nonnegative for $\sqrt{n}$ to be defined. Thus, $n = 70k + 50 \geq 0$. Simplifying this, we have $k \geq -5/7$. In other words, k also has to be nonnegative.

Therefore,

$$\frac{n + 1000}{70} = \frac{70k + 1050}{70} = k + 15 = \lfloor \sqrt{70k + 50} \rfloor.$$

Also, we can say that $\sqrt{70k + 50} - 1 < k + 15$ and $k + 15 \leq \sqrt{70k + 50}$.

Since k is nonnegative, both sides of the second inequality are nonnegative, so we can square them to get $k^2 + 30k + 225 \leq 70k + 50 \implies k^2 - 40k + 175 \leq 0 \implies (k - 5)(k - 35) \leq 0 \implies 5 \leq k \leq 35$.

Similarly, solving the first inequality gives us $k < 19 - \sqrt{155}$ or $k > 19 + \sqrt{155}$.

We know that $\sqrt{155}$ is larger than 12 and smaller than 13, so instead, we can say $k \leq 6$ or $k \geq 32$.

Combining this with $5 \leq k \leq 35$, we get $k = 5, 6, 32, 33, 34, 35$ are all solutions for k that give a valid solution for n , meaning that our answer is $\boxed{\text{(C) } 6}$.-Solution By Qqqwerw

Solution 3

We start with the given equation

$$\frac{n + 1000}{70} = \lfloor \sqrt{n} \rfloor$$

From there, we can start with the general inequality that $\lfloor \sqrt{n} \rfloor \leq \sqrt{n} < \lfloor \sqrt{n} \rfloor + 1$. This means that

$$\frac{n + 1000}{70} \leq \sqrt{n} < \frac{n + 1070}{70}$$

Solving each inequality separately gives us two inequalities:

$$n - 70\sqrt{n} + 1000 \leq 0 \rightarrow (\sqrt{n} - 50)(\sqrt{n} - 20) \leq 0 \rightarrow 20 \leq \sqrt{n} \leq 50$$

$$n - 70\sqrt{n} + 1070 > 0 \rightarrow \sqrt{n} < 35 - \sqrt{155}, \sqrt{n} > 35 + \sqrt{155}$$

Simplifying and approximating decimals yields 2 solutions for one inequality and 4 for the other. Hence, the answer is $2 + 4 = \boxed{\text{(C)} 6}$.

Solution 4

Let n be uniquely of the form $n = k^2 + r$ where $0 \leq r \leq 2k$ ★. Then,

$$\frac{k^2 + r + 1000}{70} = k$$

Rearranging and completeing the square gives

$$\begin{aligned} (k - 35)^2 + r &= 225 \\ \Rightarrow r &= (k - 20)(50 - k) \end{aligned}$$

This gives us

$$(k - 35)^2 \leq (k - 35)^2 + r = 225 \leq (k - 35)^2 + 2k$$

Solving the left inequality shows that $20 \leq k \leq 50$. Combing this with the right inequality gives that

$$(k - 35)^2 + r = 225 \leq (k - 35)^2 + 2k \leq (k - 35)^2 + 100$$

which implies either $k \geq 47$ or $k \leq 23$. By directly computing the cases for $k = 20, 21, 22, 23, 47, 48, 49, 50$ using , it follows that only $k = 22, 23$ yield and invalid r from ★. Since each k corresponds to one r and thus to one n (from and the original form), there must be 6 such n .

Solution 5

Since the right-hand-side is an integer, so must be the left-hand-side. Therefore, we must have $n \equiv -20 \pmod{70}$; let $n = 70j - 20$. The given equation becomes

$$j + 14 = \lfloor \sqrt{70j - 20} \rfloor$$

Since $\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1$ for all real x , we can take $x = \sqrt{70j - 20}$ with $\lfloor x \rfloor = j + 14$ to get

$$j + 14 \leq \sqrt{70j - 20} < j + 15$$

We can square the inequality to get

$$196 + 28j + j^2 \leq 70j - 20 < 225 + 30j + j^2$$

The left inequality simplifies to $(j - 36)(j - 6) \leq 0$, which yields

$$6 \leq j \leq 36.$$

The right inequality simplifies to $(j - 20)^2 - 155 > 0$, which yields

$$j < 20 - \sqrt{155} < 8 \quad \text{or} \quad j > 20 + \sqrt{155} > 32$$

Solving $j < 8$, and $6 \leq j \leq 36$, we get $6 \leq j < 8$, for 2 values $j \in \{6, 7\}$.

Solving $j > 32$, and $6 \leq j \leq 36$, we get $32 < j \leq 36$, for 4 values $k \in \{33, \dots, 36\}$.

Thus, our answer is $2 + 4 = \boxed{\text{(C)} 6}$

Solution 6

Set $x = \sqrt{n}$ in the given equation and solve for x to get $x^2 = 70 \cdot \lfloor x \rfloor - 1000$. Set $k = \lfloor x \rfloor \geq 0$; since $\lfloor x \rfloor^2 \leq x^2 < (\lfloor x \rfloor + 1)^2$, we get

$$k^2 \leq 70k - 1000 < k^2 + 2k + 1.$$

The left inequality simplifies to $(k - 20)(k - 50) \leq 0$, which yields

$$20 \leq k \leq 50.$$

The right inequality simplifies to $(k - 34)^2 > 155$, which yields

$$k < 34 - \sqrt{155} < 22 \quad \text{or} \quad k > 34 + \sqrt{155} > 46$$

Solving $k < 22$, and $20 \leq k \leq 50$, we get $20 \leq k < 22$, for 2 values $k \in \{20, 21\}$.

Solving $k > 46$, and $20 \leq k \leq 50$, we get $46 < k \leq 50$, for 4 values $k \in \{47, \dots, 50\}$.

Thus, our answer is $2 + 4 = \boxed{\text{(C)} 6}$

Solution 7

If n is a perfect square, we can write $n = k^2$ for a positive integer k , so $\lfloor \sqrt{n} \rfloor = \sqrt{n} = k$. The given equation turns into

$$\begin{aligned} \frac{k^2 + 1000}{70} &= k \\ k^2 - 70k + 1000 &= 0 \\ (k - 20)(k - 50) &= 0, \end{aligned}$$

so $k = 20$ or $k = 50$, so $n = 400, 2500$.

If n is not square, then we can say that, for a positive integer k , we have

$$\begin{aligned} k^2 &< n < (k + 1)^2 \\ k^2 + 1000 &< n + 1000 = 70\lfloor \sqrt{n} \rfloor = 70k < (k + 1)^2 + 1000 \\ k^2 + 1000 &< 70k < (k + 1)^2 + 1000. \end{aligned}$$

To solve this inequality, we take the intersection of the two solution sets to each of the two inequalities $k^2 + 1000 < 70k$ and $70k < (k + 1)^2 + 1000$. To solve the first one, we have

$$\begin{aligned} k^2 - 70k + 1000 &< 0 \\ (k - 20)(k - 50) &< 0 \end{aligned}$$

$k \in (20, 50)$, because the portion of the parabola between its two roots will be negative.

The second inequality yields

$$\begin{aligned} 70k &< k^2 + 2k + 1 + 1000 \\ 0 &< k^2 - 68k + 1001. \end{aligned}$$

This time, the inequality will hold for all portions of the parabola that are not on or between the its two roots, which are $34 + \sqrt{155} > 46$ and $34 - \sqrt{155} < 22$ (they are roughly equal, but this is to ensure that we do not miss any solutions).

Notation wise, we need all integers k such that

$$k \in (20, 50) \cap (-\infty, 34 - \sqrt{155})$$

or

$$k \in (20, 50) \cap (34 + \sqrt{155}, \infty).$$

For the first one, since our upper bound is a little less than 22, the k that works is 21. For the second, our lower bound is a little more than 46, so the k that work are 47, 48, and 49.

(C) 6 total solutions for n , since each value of k corresponds to exactly one value of n .

-Benedict T (countmath1)

Solution 8

First, express $\lfloor \sqrt{n} \rfloor = \sqrt{n} - \{\sqrt{n}\}$ and simplify the equation out:

$$n + 1000 = 70\sqrt{n} - 70\{\sqrt{n}\}$$

For simplicity, define $a = \sqrt{n}$. Then, $a^2 + 1000 = 70a - 70\{a\}$. Rearranging and factoring: $(a - 20)(a - 50) = -70\{a\}$. We know that $0 \leq \{a\} < 1$, so we can set $\{a\} = 0$ to obtain $a = 20, 50 \implies n = 400, 2500$. For the next solutions, we can use the quadratic formula on the equation $a^2 - 70a + (1000 + 70\{a\}) = 0$:

$$a = 35 \pm \sqrt{1225 - (1000 + 70\{a\})} = 35 \pm \sqrt{225 - 70\{a\}}$$

Since $a = 35 \pm \sqrt{225 - 70\{a\}}$ must produce real values of a that satisfy $\lfloor a \rfloor = a - \{a\}$, the expression inside the square root must correspond to a value of a whose integer part stays consistent. Therefore, we only consider values of $\{a\}$ that make a land near an integer, which happens when $\sqrt{225 - 70\{a\}}$ is close to an integer.

From earlier, we can use the bounds of $\{a\}$ to find out that $225 - 70\{a\}$ will approach a minimum of $225 - 70 = 155$ but never reach it. Now, we can just list out the perfect squares between 155 and 225, as setting $225 - 70\{a\}$ equal to that perfect square will result in an integer value of a corresponding to an integer value of n .

The two perfect squares that satisfy these restrictions are $k = 13$ and $k = 14$. Then, $a = 35 \pm 14 \implies a = 49, 21$ and $a = 35 \pm 13 \implies a = 48, 22$.

We know that each value of $a = \sqrt{n}$ corresponds to exactly one value of n , so we have a total of $2 + 4 = \mathbf{(C) 6}$ total solutions for n .

22. (AMC 12 12B 2020) What is the maximum value of $\frac{(2^t-3t)t}{4^t}$ for real values of t ?
 (A) $\frac{1}{16}$ (B) $\frac{1}{15}$ (C) $\frac{1}{12}$ (D) $\frac{1}{10}$ (E) $\frac{1}{9}$

Answer: C

Solution 1

We proceed by using AM-GM. We get $\frac{(2^t-3t)+3t}{2} \geq \sqrt{(2^t-3t)(3t)}$. Thus, squaring gives us that $4^{t-1} \geq (2^t-3t)(3t)$. Remembering what we want to find, we divide both sides of the inequality by the positive amount of $\frac{1}{3 \cdot 4^t}$. We get the maximal values as $\boxed{(C) \frac{1}{12}}$, and we are done.

Solution 2

Set $u = t2^{-t}$. Then the expression in the problem can be written as

$$R = -3t^2 4^{-t} + t2^{-t} = -3u^2 + u = -3(u - 1/6)^2 + \frac{1}{12} \leq \frac{1}{12}.$$

It is easy to see that $u = \frac{1}{6}$ is attained for some value of t between $t = 0$ and $t = 1$, thus the maximal value of R is (C) $\frac{1}{12}$.

Solution 3 (Calculus Needed)

We want to maximize $f(t) = \frac{(2^t-3t)t}{4^t} = \frac{t \cdot 2^t - 3t^2}{4^t}$. We can use the first derivative test. Use quotient rule to get the following:

$$\begin{aligned} \frac{(2^t + t \cdot \ln 2 \cdot 2^t - 6t)4^t - (t \cdot 2^t - 3t^2)4^t \cdot 2 \ln 2}{(4^t)^2} &= 0 \implies 2^t + t \cdot \ln 2 \cdot 2^t - 6t = (t \cdot 2^t - 3t^2)2 \ln 2 \\ \implies 2^t + t \cdot \ln 2 \cdot 2^t - 6t &= 2t \ln 2 \cdot 2^t - 6t^2 \ln 2 \\ \implies 2^t(1 - t \cdot \ln 2) &= 6t(1 - t \cdot \ln 2) \implies 2^t = 6t \end{aligned}$$

Therefore, we plug this back into the original equation to get $\boxed{(C) \frac{1}{12}}$

Solution 4

First, substitute $2^t = x(\log_2 x = t)$ so that

$$\frac{(2^t - 3t)t}{4^t} = \frac{x \log_2 x - 3(\log_2 x)^2}{x^2}$$

Notice that

$$\frac{x \log_2 x - 3(\log_2 x)^2}{x^2} = \frac{\log_2 x}{x} - 3\left(\frac{\log_2 x}{x}\right)^2.$$

When seen as a function, $\frac{\log_2 x}{x} - 3\left(\frac{\log_2 x}{x}\right)^2$ is a synthesis function that has $\frac{\log_2 x}{x}$ as its inner function.

If we substitute $\frac{\log_2 x}{x} = p$, the given function becomes a quadratic function that has a maximum value of $\frac{1}{12}$ when $p = \frac{1}{6}$.

Now we need to check if $\frac{\log_2 x}{x}$ can have the value of $\frac{1}{6}$ in the range of real numbers.

In the range of (positive) real numbers, function $\frac{\log_2 x}{x}$ is a continuous function whose value gets infinitely smaller as x gets closer to 0 (as $\log_2 x$ also diverges toward negative infinity in the same condition). When $x = 2$, $\frac{\log_2 x}{x} = \frac{1}{2}$, which is larger than $\frac{1}{6}$.

Therefore, we can assume that $\frac{\log_2 x}{x}$ equals to $\frac{1}{6}$ when x is somewhere between 1 and 2 (at least), which means that the maximum value of $\frac{(2^t-3t)t}{4^t}$ is $\boxed{\text{(C)} \frac{1}{12}}$.

Solution 5

Let the maximum value of the function be m . Then we have

$$\frac{(2^t-3t)t}{4^t} = m \implies m2^{2t} - t2^t + 3t^2 = 0.$$

Solving for 2^t , we see

$$2^t = \frac{t}{2m} \pm \frac{\sqrt{t^2 - 12mt^2}}{2m} = \frac{t}{2m} \pm \frac{t\sqrt{1-12m}}{2m}.$$

We see that $1 - 12m \geq 0 \implies m \leq \frac{1}{12}$. Therefore, the answer is $\boxed{\text{(C)} \frac{1}{12}}$.

Solution 6

Let $z = 2^t$. Then,

$$\frac{(2^t-3t)t}{4^t} = \frac{(z-3t)t}{z^2} = \frac{-3t^2 + zt}{z^2}.$$

Upon inspection, the numerator of this expression grows at a relatively faster rate than the denominator, when t is close to 0.

As the numerator is a quadratic in t with a negative leading coefficient, its maximum value occurs at $t = \frac{-z}{2 \cdot -3} = \frac{z}{6}$, or when $6t = 2^t$. Therefore,

$$\frac{(2^t-3t)t}{4^t} = \frac{(6t-3t)t}{(6t)^2} = \frac{3t^2}{36t^2} = \boxed{\text{(C)} \frac{1}{12}}.$$

-Benedict T (countmath1)

Solution 7 (fast)

Note that

$$\frac{(2^t-3t)t}{4^t} = \frac{t \cdot 2^t}{4^t} - \frac{3t^2}{4^t} = \frac{t}{2^t} - 3\left(\frac{t}{2^t}\right)^2.$$

Let $x = \frac{t}{2^t}$. Then, the expression becomes $x - 3x^2$, which is minimized at $x = \frac{1}{6}$, giving a value of $\boxed{\text{(C)} \frac{1}{12}}$.

Note: $\frac{t}{2^t} = \frac{1}{6}$ is possible as $\frac{t}{2^t}$ is continuous and that $\frac{0}{2^0} = 0$ and $\frac{2}{2^2} = \frac{1}{2}$ so by the Intermediate Value Theorem (or just by intuition), there must be a t between 0 and 2 that satisfies $\frac{t}{2^t} = \frac{1}{6}$.

23. (AMC 12 12B 2020) How many integers $n \geq 2$ are there such that whenever $z_1, z_2, \dots, z_n$ are complex numbers such that

$$|z_1| = |z_2| = \dots = |z_n| = 1 \text{ and } z_1 + z_2 + \dots + z_n = 0,$$

then the numbers $z_1, z_2, \dots, z_n$ are equally spaced on the unit circle in the complex plane?

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Answer: B

Solution

For $n = 2$, we see that if $z_1 + z_2 = 0$, then $z_1 = -z_2$, so they are evenly spaced along the unit circle.

For $n = 3$, WLOG, we can set $z_1 = 1$. Notice that now $\operatorname{Re}(z_2 + z_3) = -1$ and $\operatorname{Im}\{z_2\} = -\operatorname{Im}\{z_3\}$. This forces z_2 and z_3 to be equal to $e^{i\frac{2\pi}{3}}$ and $e^{-i\frac{2\pi}{3}}$, meaning that all three are equally spaced along the unit circle.

We can now show that we can construct complex numbers when $n \geq 4$ that do not satisfy the conditions in the problem.

Suppose that the condition in the problem holds for some $n = k$. We can now add two points z_{k+1} and z_{k+2} anywhere on the unit circle such that $z_{k+1} = -z_{k+2}$, which will break the condition. Now that we have shown that $n = 2$ and $n = 3$ works, by this construction, any $n \geq 4$ does not work, making the answer **(B)2**.

-Solution by Qqqwerw-Minor edit made by HappySharks

24. (AMC 12 12B 2020) Let $D(n)$ denote the number of ways of writing the positive integer n as a product

$$n = f_1 \cdot f_2 \cdots f_k,$$

where $k \geq 1$, the f_i are integers strictly greater than 1, and the order in which the factors are listed matters (that is, two representations that differ only in the order of the factors are counted as distinct). For example, the number 6 can be written as 6, $2 \cdot 3$, and $3 \cdot 2$, so $D(6) = 3$. What is $D(96)$?

(A) 112 (B) 128 (C) 144 (D) 172 (E) 184

Answer: A

Solution 1

Note that $96 = 2^5 \cdot 3$. Since there are at most six not necessarily distinct factors > 1 multiplying to 96, we have six cases: $k = 1, 2, \dots, 6$. Now we look at each of the six cases.

$k = 1$: We see that there is 1 way, merely 96.

$k = 2$: This way, we have the 3 in one slot and 2 in another, and symmetry. The four other 2's leave us with 5 ways and symmetry doubles us so we have 10.

$k = 3$: We have 3, 2, 2 as our baseline. We need to multiply by 2 in 3 places, and see that we can split the remaining three powers of 2 in a manner that is 3 - 0 - 0, 2 - 1 - 0 or 1 - 1 - 1. A 3 - 0 - 0 split has $6 + 3 = 9$ ways of happening (24 - 2 - 2 and symmetry; 2 - 3 - 16 and symmetry), a 2 - 1 - 0 split has $6 \cdot 3 = 18$ ways of happening (due to all being distinct) and a 1 - 1 - 1 split has 3 ways of happening (6 - 4 - 4 and symmetry) so in this case we have $9 + 18 + 3 = 30$ ways.

$k = 4$: We have 3, 2, 2, 2 as our baseline, and for the two other 2's, we have a 2 - 0 - 0 - 0 or 1 - 1 - 0 - 0 split. The former grants us $4 + 12 = 16$ ways (12 - 2 - 2 - 2 and symmetry and 3 - 8 - 2 - 2 and symmetry) and the latter grants us also $12 + 12 = 24$ ways (6 - 4 - 2 - 2 and symmetry and 3 - 4 - 4 - 2 and symmetry) for a total of $16 + 24 = 40$ ways.

$k = 5$: We have 3, 2, 2, 2, 2 as our baseline and one place to put the last two: on another two or on the three. On the three gives us 5 ways due to symmetry and on

another two gives us $5 \cdot 4 = 20$ ways due to symmetry. Thus, we have $5 + 20 = 25$ ways.

$k = 6$: We have $3, 2, 2, 2, 2, 2$ and symmetry and no more twos to multiply, so by symmetry, we have 6 ways.

Thus, adding, we have $1 + 10 + 30 + 40 + 25 + 6 = \boxed{\text{(A)} 112}$.

Solution 2

As before, note that $96 = 2^5 \cdot 3$, and we need to consider 6 different cases, one for each possible value of k , the number of factors in our factorization. However, instead of looking at each individually, find a general form for the number of possible factorizations with k factors. First, the factorization needs to contain one factor that is itself a multiple of 3. There are k to choose from. That leaves $k - 1$ slots left to fill, each of which must contain at least one factor of 2. Once we have filled in a 2 to each of the remaining slots, we're left with $5 - (k - 1) = 6 - k$ twos.

Consider the remaining $6 - k$ factors of 2 left to assign to the k factors. Using stars and bars, the number of ways to do this is:

$$\binom{(6 - k) + k - 1}{6 - k} = \binom{5}{6 - k}$$

This makes $k \binom{5}{6 - k}$ possibilities for each k .

To obtain the total number of factorizations, add all possible values for k :

$$\sum_{k=1}^6 k \binom{5}{6 - k} = 1 + 10 + 30 + 40 + 25 + 6 = \boxed{\text{(A)} 112}.$$

Solution 3

Begin by examining f_1 . f_1 can take on any value that is a factor of 96 except 1. For each choice of f_1 , the resulting $f_2 \dots f_k$ must have a product of $96/f_1$. This means the number of ways the rest f_a , $1 < a \leq k$ can be written by the scheme stated in the problem for each f_1 is equal to $D(96/f_1)$, since the product of $f_2 \cdot f_3 \dots \cdot f_k = x$ is counted as one valid product if and only if $f_1 \cdot x = 96$, the product x has the properties that factors are greater than 1, and differently ordered products are counted separately.

For example, say the first factor is 2. Then, the remaining numbers must multiply to 48, so the number of ways the product can be written beginning with 2 is $D(48)$. To add up all the number of solutions for every possible starting factor, $D(96/f_1)$ must be calculated and summed for all possible f_1 , except 96 and 1, since a single 1 is not counted according to the problem statement. The 96 however, is counted, but only results in 1 possibility, the first and only factor being 96. This means

$$D(96) = D(48) + D(32) + D(24) + D(16) + D(12) + D(8) + D(6) + D(4) + D(3) + D(2) + 1.$$

Instead of calculating D for the larger factors first, reduce $D(48)$, $D(32)$, and $D(24)$ into sums of $D(m)$ where $m \leq 16$ to ease calculation. Following the recursive definition $D(n) = (\text{sums of } D(c)) + 1$ where c takes on every divisor of n except for 1 and itself, the sum simplifies to

$$D(96) = (D(24) + D(16) + D(12) + D(8) + D(6) + D(4) + D(3) + D(2) + 1) + (D(16) + D(8) + D(4) + D(2) + 1) + D(24) + D(16) + D(12) + D(8) + D(6) + D(4) + D(3) + D(2) + 1.$$

$D(24) = D(12) + D(8) + D(6) + D(4) + D(3) + D(2) + 1$, so the sum further simplifies to

$D(96) = 3D(16) + 4D(12) + 5D(8) + 4D(6) + 5D(4) + 4D(3) + 5D(2) + 5$, after combining terms. From quick casework,

$D(16) = 8, D(12) = 8, D(8) = 4, D(6) = 3, D(4) = 2, D(3) = 1$ and $D(2) = 1$. Substituting these values into the expression above,

$$D(96) = 3 \cdot 8 + 4 \cdot 8 + 5 \cdot 4 + 4 \cdot 3 + 5 \cdot 2 + 4 \cdot 1 + 5 \cdot 1 + 5 = \boxed{\text{(A)} 112}.$$

Solution 4

Note that $96 = 3 \cdot 2^5$, and that D of a perfect power of a prime is relatively easy to calculate. Also note that you can find $D(96)$ from $D(32)$ by simply totaling the number of ways there are to insert a 3 into a set of numbers that multiply to 32.

First, calculate $D(32)$. Since $32 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$, all you have to do was find the number of ways to divide the 2's into groups, such that each group has at least one 2. By stars and bars, this results in 1 way with five terms, 4 ways with four terms, 6 ways with three terms, 4 ways with two terms, and 1 way with one term. (The total, 16, is not needed for the remaining calculations.)

Then, to get $D(96)$, in each possible $D(32)$ sequence, insert a 3 somewhere in it, either by placing it somewhere next to the original numbers (in one of $n + 1$ ways, where n is the number of terms in the $D(32)$ sequence), or by multiplying one of the numbers by 3 (in one of n ways). There are $2 + 1 = 3$ ways to do this with one term, $3 + 2 = 5$ with two, 7 with three, 9 with four, and 11 with five.

The resulting number of possible sequences is $3 \cdot 1 + 5 \cdot 4 + 7 \cdot 6 + 9 \cdot 4 + 11 \cdot 1 = \boxed{\text{(A)} 112}$.

[emerald_block](#)

Solution 5 (Minimal Casework)

Consider the arrangement of the prime factors of 96 in a line $(2, 2, 2, 2, 2, 3)$. An arrangement of factors can be created by placing "dividers" to group primes. For example, $(2, 2, |, 2, 2, 2, |, 3)$ is equivalent to the arrangement $4 \cdot 8 \cdot 3$. Because there are 6 ways to order the prime factors, and 2^5 ways to place dividers, this gives us an initial $6 \cdot 2^5$ ways to arrange dividers.

However, through this method, we overcount cases where 3 is combined with another factor. For example, the arrangement $4 \cdot 6 \cdot 4$ can be written as $(2, 2, |, 2, 3, |, 2, 2)$ or $(2, 2, |, 3, 2, |, 2, 2)$. Precisely, we double count any case with 6 as a factor, triple count any case with 12, quadruple count any case with 24, etc.

Now, consider all cases where 3 must be grouped with at least one 2. This can be expressed in the same "line" format as $(2, 2, 2, 2, 6)$, where dividers can again be placed to group divisors. In this case, there are 5 ways to order divisors, and 2^4 ways to place dividers, so we have an $5 \cdot 2^4$ possible sequences for this case. Notice that in this format, we double count cases where 12 is a factor, we triple count cases where 24 is a factor, etc. Precisely, for any case counted n times in the first step, it is counted $n - 1$ times in this step. Thus, if we subtract, we count each case exactly once.

So, we get:

$$6 \cdot 2^5 - 5 \cdot 2^4 = \boxed{\text{(A)} 112}. \quad \text{hdai1122}$$

Solution 6 (Another Fast Way)

First we factor 32 into m numbers $g_1, \dots, g_m$ where $g_i > 1, i = 1, \dots, m$. By applying stars and bars there are $\binom{5-1}{m-1}$ ways. Then we can either insert 3 into each of the $m+1$ spaces between (or beyond) g_i 's, or multiply it to one of the g_i 's, a total of $2m+1$ ways. Hence the answer to the problem is

$$\sum_{m=1}^5 (2m+1) \binom{5-1}{m-1} = \sum_{n=0}^4 (2n+3) \binom{4}{n} = 8 \sum_{n=0}^4 \frac{n}{4} \binom{4}{n} + 3 \sum_{n=0}^4 \binom{4}{n} = 8 \sum_{n=0}^4 \binom{3}{n-1} + 3 \sum_{n=0}^4 \binom{4}{n} = 8 \cdot 2^3 + 3 \cdot 2^4 = 8 \cdot 8 + 3 \cdot 16 = 64 + 48 = 112$$

Solution 7 (Integer Partition)

Note that $96 = 2^5 \cdot 3$. $D(n)$ depends on dividing 2^5 into different terms, which is the **integer partition** of 5.

Divide 96 into 1 term:

There is only one way. 1

Divide 96 into 2 terms:

Case 1\$: 3\$ is alone $(2^5, 3)$ has 2! different arrangements.

Case 2\$: 3\$ is with 2^n $5 = 1 + 4 = 2 + 3$. For $(2, 2^4)$ and $(2^2, 2^3)$, 3\$

$$2 + 8 = \underline{10}$$

Divide 96 into 3 terms:

Case 1\$: 3\$ is alone $5 = 2 + 3 = 1 + 4$. For $(2^2, 2^3, 3)$ and $(2, 2^4, 3)$, th

Case 2\$: 3\$ is with 2^n $5 = 1 + 1 + 3 = 1 + 2 + 2$. For $(2, 2, 2^3)$ and $(2, 2$

$$12 + 18 = \underline{30}$$

Divide 96 into 4 terms:

Case 1\$: 3\$ is alone $5 = 1 + 1 + 3 = 1 + 2 + 2$. For $(2, 2, 2^3, 3)$ and $(2, 2^2$

Case 2\$: 3\$ is with 2^n $5 = 1 + 1 + 1 + 2$. For $(2, 2, 2, 2^2)$, 3\$ can be wit

$$24 + 16 = \underline{40}$$

Divide 96 into 5 terms:

When dividing 5 into 5 parts there are 2 cases.

Case 1\$: 3\$ is alone $5 = 1 + 1 + 1 + 2$. For $(2, 2, 2, 2^2, 3)$, there are $\frac{5!}{2!}$

Case 2\$: 3\$ is with 2^n $5 = 1 + 1 + 1 + 1 + 1$. For $(2, 2, 2, 2, 2)$, 3\$ can o

$$20 + 5 = \underline{25}$$

Divide 96 into 6 terms:

$5 = 1 + 1 + 1 + 1 + 1$. The number of arrangements of $(2, 2, 2, 2, 2, 3)$ is $\frac{6!}{5!} = \underline{6}$

$$1 + 10 + 30 + 40 + 25 + 6 = \boxed{\text{(A)} 112}$$

Solution 8

Ignore the 3 first and first count $2^{x_1+x_2+\dots+x_n} = 32$ which $x_1 + x_2 + \dots + x_n = 5$. This implies that n is less than or equal to 5. Now, we can see that 3 can lie between the two x_i, x_{i+1} , or contribute to one of them. This gives $2k+1$ if $x_1 + \dots + x_k = 5$. Now, just sum up gives $\binom{5-1}{5-1} \cdot 11 + \binom{5-1}{4-1} \cdot 9 + \binom{5-1}{3-1} \cdot 7 + \binom{5-1}{2-1} \cdot 5 + \binom{5-1}{1-1} \cdot 3 = \boxed{112}$.

Solution 9 (Cleanest Algebra)

Consider how we partition the factors of $96 = 2^5 \cdot 3$. For each k , there are two cases. Either we can put the 2s into k nonzero parts, so that the 3 shares a partition with some 2s, which can be done in $k \binom{5-(k-1)+(k-1)-1}{k-2} = k \binom{4}{k-2}$ ways, or we can put the 2s into $k-1$ nonzero parts and put the 3 in its own partition, which can be done in $k \binom{5-k+k-1}{k-1} = k \binom{4}{k-1}$ ways. Summing over all k , we have

$$\sum_{k=1}^6 k \binom{4}{k-2} + \sum_{k=1}^6 k \binom{4}{k-1}.$$

But $\sum_{k=1}^6 k \binom{4}{k-2} = 2 \binom{4}{0} + 3 \binom{4}{1} + \dots + 6 \binom{4}{4} = 4 \cdot \sum_{i=0}^4 \binom{4}{i} = 4 \cdot 2^4 = 64$. Similarly, $\sum_{k=1}^6 k \binom{4}{k-1} = 3 \cdot 2^4 = 48$. So our answer is $64 + 48 = \boxed{\text{(A)} 112}$.

Solution 10 (Word Arrangements and Brute Forcing)

After careful inspection we notice that for each respect value of k , there are a certain number of unique set of numbers that form total word arrangements. Using case work we can deduce that for each respective k value:

$k = 1: 96$

$k = 2: 2 \ 48, 3 \ 32, 4 \ 24, 6 \ 16, 8 \ 12$

$k = 3: 2 \ 6 \ 8, 2 \ 3 \ 16, 3 \ 4 \ 8, 2 \ 4 \ 12, 2 \ 2 \ 24, 4 \ 4 \ 6$

$k = 4: 2 \ 2 \ 3 \ 8, 2 \ 3 \ 4 \ 4, 2 \ 2 \ 2 \ 12, 2 \ 2 \ 4 \ 6$

$k = 5: 2 \ 2 \ 2 \ 3 \ 4, 2 \ 2 \ 2 \ 2 \ 6$

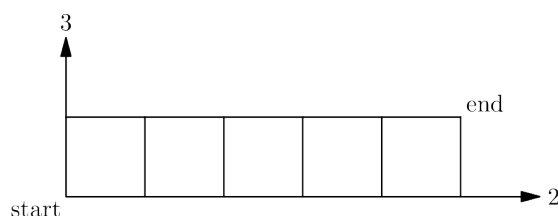
$k = 6: 2 \ 2 \ 2 \ 2 \ 2 \ 3$

Using the word arrangement formula $(\frac{n!}{f_1!f_2!\dots f_n!})$, where n is the total number of letters in the word, and f is the frequency of each respective letter for instance in "aabc" the letter "a" has a frequency of 2, b has 1, and c has 1) for each respective set of unique integers we obtain the sum $1 + 10 + 30 + 40 + 25 + 6 = 112 = \boxed{\text{(A)} 112}$

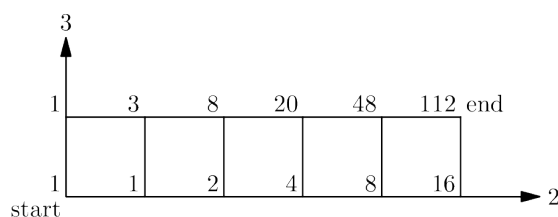
Solution 11 (Grid with Labels)

This question is like [2010 AMC 8 problem 25](#), if we treat each prime factor of $96 = 2^5 \cdot 3$ as a step, this is a recursive problem of climbing stairs.

The stair can be designed like



In this grid, we can only take any number of steps to the right or up, so the number of each point is the sum of all numbers to its left and below it. We label the points to obtain number of paths.



The result is **(A) 112**

Solution 12 (The simplest solution)

There are $2^4 = 16$ ways of placing (or not placing) $\times$'s between five 2's. These placements can be grouped into 8 pairs such that in each pair, the total number of $\times$'s is 4.

In each pair, there are $4 + 2 = 6$ powers of 2 and $6 + 2 = 8$ spaces between/beyond these powers. We can either multiply 3 with one of the powers, or place 3 in the spaces between/beyond the powers (so that we have a single 3 in the product), giving a total of $6 + 8 = 14$ ways of decomposing 96 into products greater than 1.

The answer is then $8 \cdot 14 = 112$.

25. (AMC 12 12B 2020) For each real number a with $0 \leq a \leq 1$, let numbers x and y be chosen independently at random from the intervals $[0, a]$ and $[0, 1]$, respectively, and let $P(a)$ be the probability that

$$\sin^2(\pi x) + \sin^2(\pi y) > 1$$

What is the maximum value of $P(a)$? **(A)** $\frac{7}{12}$ **(B)** $2 - \sqrt{2}$ **(C)** $\frac{1+\sqrt{2}}{4}$ **(D)** $\frac{\sqrt{5}-1}{2}$ **(E)** $\frac{5}{8}$

Answer: B

Solution 1

Let's start first by manipulating the given inequality.

$$\begin{aligned} \sin^2(\pi x) + \sin^2(\pi y) &> 1 \\ \sin^2(\pi x) &> 1 - \sin^2(\pi y) = \cos^2(\pi y) \end{aligned}$$

Let's consider the boundary cases: $\sin(\pi x) = \pm \cos(\pi y)$.

$$\sin(\pi x) = \pm \cos(\pi y) = \sin\left(\frac{1}{2}\pi \pm \pi y\right)$$

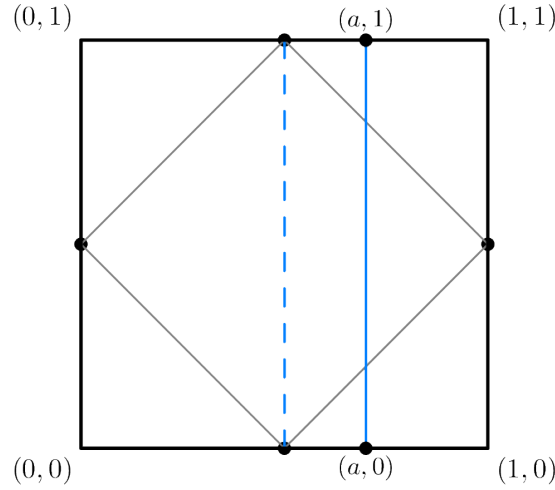
Solving the first case gives us

$$y = \frac{1}{2} - x \quad \text{and} \quad y = x - \frac{1}{2}.$$

Solving the second case gives us

$$y = x + \frac{1}{2} \quad \text{and} \quad y = \frac{3}{2} - x.$$

If we graph these equations in $[0, 1] \times [0, 1]$, we get a rhombus shape.



Testing points in each section tells us that the inside of the rhombus satisfies the inequality in the problem statement.

From the region graph, notice that in order to maximize $P(a)$, $a \geq \frac{1}{2}$. We can solve the rest with geometric probability.

Instead of maximizing $P(a)$, we minimize $Q(a) = 1 - P(a)$. $Q(a)$ consists of two squares (each broken into two triangles), one of area $\frac{1}{4}$ and another of area $(a - \frac{1}{2})^2$. To calculate $Q(a)$, we divide this area by a , so

$$Q(a) = \frac{1}{a} \left(\frac{1}{4} + (a - \frac{1}{2})^2 \right) = \frac{1}{a} \left(\frac{1}{2} + a^2 - a \right) = a + \frac{1}{2a} - 1.$$

By AM-GM, $a + \frac{1}{2a} \geq 2\sqrt{\frac{a}{2a}} = \sqrt{2}$, which we can achieve by setting $a = \frac{\sqrt{2}}{2}$.

Therefore, the maximum value of $P(a)$ is $1 - \min(Q(a))$, which is $1 - (\sqrt{2} - 1) =$

(B) $2 - \sqrt{2}$

Solution 2 (Trigonometry Identities)

$$\sin^2(\pi x) + \sin^2(\pi y) > 1, \quad \sin^2(\pi x) - (1 - \sin^2(\pi y)) > 0, \quad \sin^2(\pi x) - \cos^2(\pi y) > 0$$

$$(\sin(\pi x) + \cos(\pi y))(\sin(\pi x) - \cos(\pi y)) > 0, \quad \left(\sin(\pi x) + \sin\left(\frac{\pi}{2} - \pi y\right) \right) \left(\sin(\pi x) - \sin\left(\frac{\pi}{2} - \pi y\right) \right) >$$

$$2 \sin\left(\frac{\pi x + \frac{\pi}{2} - \pi y}{2}\right) \cos\left(\frac{\pi x - \frac{\pi}{2} + \pi y}{2}\right) 2 \sin\left(\frac{\pi x - \frac{\pi}{2} + \pi y}{2}\right) \cos\left(\frac{\pi x + \frac{\pi}{2} - \pi y}{2}\right) > 0$$

$$2 \sin\left(\frac{\pi x + \frac{\pi}{2} - \pi y}{2}\right) \cos\left(\frac{\pi x + \frac{\pi}{2} - \pi y}{2}\right) 2 \sin\left(\frac{\pi x - \frac{\pi}{2} + \pi y}{2}\right) \cos\left(\frac{\pi x - \frac{\pi}{2} + \pi y}{2}\right) > 0$$

$$\sin\left(\pi x + \frac{\pi}{2} - \pi y\right) \sin\left(\pi x - \frac{\pi}{2} + \pi y\right) > 0$$

If $\sin(\pi x + \frac{\pi}{2} - \pi y) > 0$ and $\sin(\pi x - \frac{\pi}{2} + \pi y) > 0$

$$0 < \pi x + \frac{\pi}{2} - \pi y < \pi \quad \text{and} \quad 0 < \pi x - \frac{\pi}{2} + \pi y < \pi$$

$$0 < x + \frac{1}{2} - y < 1 \quad \text{and} \quad 0 < x - \frac{1}{2} + y < 1$$

$$x - \frac{1}{2} < y < x + \frac{1}{2} \quad \text{and} \quad -x + \frac{1}{2} < y < -x + \frac{3}{2}$$

If $\sin(\pi x + \frac{\pi}{2} - \pi y) < 0$ and $\sin(\pi x - \frac{\pi}{2} + \pi y) < 0$

$$\pi < \pi x + \frac{\pi}{2} - \pi y < 2\pi \quad \text{and} \quad \pi < \pi x - \frac{\pi}{2} + \pi y < 2\pi$$

$$1 < x + \frac{1}{2} - y < 2 \quad \text{and} \quad 1 < x - \frac{1}{2} + y < 2$$

$$x - \frac{3}{2} < y < x - \frac{1}{2} \quad \text{and} \quad -x + \frac{3}{2} < y < -x + \frac{5}{2}$$

Notice that $\sin(\pi x + \frac{\pi}{2} - \pi y) < 0$ and $\sin(\pi x - \frac{\pi}{2} + \pi y) < 0$ cannot be true, because that means x is in the interval $[1, 2]$. Thus, y is in the boundaries $x - \frac{1}{2} < y < x + \frac{1}{2}$, and $-x + \frac{1}{2} < y < -x + \frac{3}{2}$.

Finishing like Solution 1, 3, or 4 gives $P(a) = \boxed{\text{(B)} 2 - \sqrt{2}}$

Solution 3 (Calculus finish)

We find the same region as in the first solution or the second solution, and again notice we must have $a \geq \frac{1}{2}$.

We now express P as a function of $b = (1 - a)$. The triangle on the right of the line $x = b$ is an isosceles right triangle with altitude b , so it has area b^2 . The total area of the region to the left of $x = b$ has area $1 - b$. So

$$P(b) = \frac{1/2 - b^2}{1 - b}$$

Differentiating using the quotient rule, we find P has local extrema at

$$P'(b) = \frac{(1 - b)(-2b) - (-1/2 - b^2)(-1)}{(1 - b)^2} = 0$$

Setting the numerator equal to 0 and solving the quadratic, we find P has extrema at $b = 1 \pm \frac{\sqrt{2}}{2}$. Only $b = 1 - \frac{\sqrt{2}}{2}$ is within the desired region. Plugging in, we get $P(1 - \frac{\sqrt{2}}{2}) = \boxed{\text{(B)} 2 - \sqrt{2}}$ as our solution. We also need to check $P(b = 0) = 1/2$. But $1/2 < 2 - \sqrt{2}$, and if this isn't immediately obvious, $1/2$ isn't an answer choice anyways.

Solution 4 (Calculus Finish)

We find the same region as in the first solution or the second solution, and again notice $a \geq \frac{1}{2}$.

The numerator of $P(a)$ is the area of the triangle with vertices $(0, \frac{1}{2})$, $(\frac{1}{2}, 0)$, $(\frac{1}{2}, 1)$ plus the area of the rectangle with vertices $(\frac{1}{2}, 0)$, $(a, 0)$, $(a, 1)$, $(\frac{1}{2}, 1)$ subtract the area of the isosceles right triangle between $x = a$, $y = 0$, $x - \frac{1}{2}$ and the isosceles right triangle between $x = a$, $y = 1$, $-x + \frac{3}{2}$. The denominator of $P(a)$ is the area of rectangle with vertices $(0, 0)$, $(a, 0)$, $(a, 1)$, $(0, 1)$.

$$P(a) = \frac{\frac{1}{4} + a - \frac{1}{2} - (a - \frac{1}{2})^2}{a} = \frac{a - \frac{1}{4} - a^2 + a - \frac{1}{4}}{a} = \frac{-a^2 + 2a - \frac{1}{2}}{a} = -a + 2 - \frac{1}{2a}$$

By the power rule, $P'(a) = -1 - \frac{1}{2}(-1)\frac{1}{a^2} = -1 + \frac{1}{2a^2}$. Solving $P'(a) = 0$, we find that $a = \frac{\sqrt{2}}{2} \therefore P'(a) = 0$ when $a = \frac{\sqrt{2}}{2}$, $P'(a)$ is positive when $a < \frac{\sqrt{2}}{2}$, and $P'(a)$ is negative when $a > \frac{\sqrt{2}}{2} \therefore P(a)$ has a local maximum when $a = \frac{\sqrt{2}}{2}$.

$$P\left(\frac{\sqrt{2}}{2}\right) = -\frac{\sqrt{2}}{2} + 2 - \frac{1}{2 \cdot \frac{\sqrt{2}}{2}} = \boxed{\text{(B)} 2 - \sqrt{2}}$$

Solution 5 (Calculus)

Comparing the diagram of $\sin^2(\pi x)$ and $\cos^2(\pi y)$, we can see if we let $x = k$, then $y \in (0 + |\frac{1}{2} - k|, 1 - |\frac{1}{2} - k|)$ will satisfy the above inequality.

Thus, we can write that

$$P(a) = \frac{1}{a} \left(\int_0^a 1 - 2\left|\frac{1}{2} - x\right| dx \right)$$

$$P(a) = \frac{1}{a} \left(\int_0^{\frac{1}{2}} 2x dx + \int_{\frac{1}{2}}^a 2 - 2x dx \right) = -a + 2 - \frac{1}{2a}$$

Then it is easy to find that $P(a)$ is maximized when $a = \frac{\sqrt{2}}{2}$, which give us

$$P\left(\frac{\sqrt{2}}{2}\right) = \boxed{\text{(B)} 2 - \sqrt{2}}$$

§2 AMC 12 12A 2020

1. (AMC 12 12A 2020) Carlos took 70% of a whole pie. Maria took one third of the remainder. What portion of the whole pie was left?

(A) 10% (B) 15% (C) 20% (D) 30% (E) 35%

Answer: C

Solution 1

If Carlos took 70% of the pie, there must be $(100 - 70)\% = 30\%$ left. After Maria takes $\frac{1}{3}$ of the remaining 30%, $1 - \frac{1}{3} = \frac{2}{3}$ of the remaining 30% is left.

Therefore, the answer is $30\% \cdot \frac{2}{3} = \boxed{\text{(C)} 20\%}$.

Solution 2

Like solution 1, it is clear that there is 30% of the pie remaining. Since Maria takes $\frac{1}{3}$ of the remainder, she takes $\frac{1}{3} \cdot 30\% = 10\%$, meaning that there is $30\% - 10\% =$

$\boxed{\text{(C)} 20\%}$ left.

Solution 3 (One Sentence)

We have

$$(100\% - 70\%) \cdot \left(1 - \frac{1}{3}\right) = 30\% \cdot \frac{2}{3} = \boxed{\text{(C)} 20\%}$$

of the whole pie left.

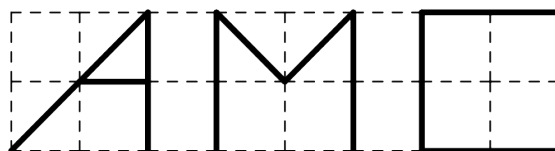
Solution 4 (Just Fractions One Sentence)

Conversion to fractions and simplification yields

$$\frac{3}{10} \cdot \frac{2}{3} = \frac{1}{5}$$

that remain for the pie. In percent, that ratio is just $\boxed{20\%}$

2. (AMC 12 12A 2020) The acronym AMC is shown in the rectangular grid below with grid lines spaced 1 unit apart. In units, what is the sum of the lengths of the line segments that form the acronym AMC?



- (A) 17 (B) $15 + 2\sqrt{2}$ (C) $13 + 4\sqrt{2}$ (D) $11 + 6\sqrt{2}$ (E) 21

Answer: C

Solution 1

Each of the straight line segments have length 1 and each of the slanted line segments have length $\sqrt{2}$ (this can be deduced using $45-45-90$, pythag, trig, or just sense)

There are a total of 13 straight line segments and 4 slanted line segments. The sum is $\boxed{\text{(C)} 13 + 4\sqrt{2}}$

Solution 2

Either count the straight or diagonals and deduce from the answers that the only answer possible is $\boxed{\text{(C)} 13 + 4\sqrt{2}}$.

3. (AMC 12 12A 2020) A driver travels for 2 hours at 60 miles per hour, during which her car gets 30 miles per gallon of gasoline. She is paid \$0.50 per mile, and her only expense is gasoline at \$2.00 per gallon. What is her net rate of pay, in dollars per hour, after this expense?

- (A) 20 (B) 22 (C) 24 (D) 25 (E) 26

Answer: E

Solution 1

Since the driver travels 60 miles per hour and each hour she uses 2 gallons of gasoline, she spends \$4 per hour on gas. If she gets \$0.50 per mile, then she gets \$30 per hour of driving. Subtracting the gas cost, her net rate of money earned per hour is $\boxed{\text{(E)} 26}$. mathsmiley

Solution 2 (longer)

The driver is driving for 2 hours at 60 miles per hour, she drives 120 miles. Therefore, she uses $\frac{120}{30} = 4$ gallons of gasoline. So, total she has $\$0.50 \cdot 120 - \$2.00 \cdot 4 = \$60 - \$8 = \$52$. So, her rate is $\frac{52}{2} = \boxed{\text{(E)} 26}$. sosiaops

4. (AMC 12 12A 2020) How many 4-digit positive integers (that is, integers between 1000 and 9999, inclusive) having only even digits are divisible by 5? (A) 80 (B) 100 (C) 125

Answer: B

Solution

The units digit, for all numbers divisible by 5, must be either 0 or 5. However, since all digits are even, the units digit must be 0. The middle two digits can be 0, 2, 4, 6, or 8, but the thousands digit can only be 2, 4, 6, or 8 since it cannot be zero. There is one choice for the units digit, 5 choices for each of the middle 2 digits, and 4 choices for the thousands digit, we have a total of $4 \cdot 5 \cdot 5 \cdot 1 = \boxed{\text{(B)} 100}$ numbers.

5. (AMC 12 12A 2020) The 25 integers from -10 to 14 , inclusive, can be arranged to form a 5-by-5 square in which the sum of the numbers in each row, the sum of the numbers in each column, and the sum of the numbers along each of the main diagonals are all the same. What is the value of this common sum?

(A) 2 (B) 5 (C) 10 (D) 25 (E) 50

Answer: C

Solution 1

Without loss of generality, consider the five rows in the square. Each row must have the same sum of numbers, meaning that the sum of all the numbers in the square divided by 5 is the total value per row. The sum of the 25 integers is $-10 + -9 + \dots + 14 = 11 + 12 + 13 + 14 = 50$, and the common sum is $\frac{50}{5} = \boxed{\text{(C)} 10}$.

Solution 2

Take the sum of the middle 5 values of the set (they will turn out to be the mean of each row). We get $0 + 1 + 2 + 3 + 4 = \boxed{\text{(C)} 10}$ as our answer. Baolan

Solution 3

Taking the average of the first and last terms, -10 and 14 , we have that the mean of the set is 2. There are 5 values in each row, column or diagonal, so the value of the common sum is $5 \cdot 2$, or $\boxed{\text{(C)} 10}$. Arctic_Bunny, edited by KINGLOGIC

Solution 4

Let us consider the horizontal rows. Since there are five of them, each with constant sum x , we can add up the 25 numbers in 5 rows for a sum of $5x$. Since the sum of the 25 numbers used is $-10 - 9 - 8 - \dots + 12 + 13 + 14 + 15 = 11 + 12 + 13 + 14 + 15 = 50$, $5x = 50$ and $x = \boxed{\text{(C)} 10}$. cw357

Solution 5

The mean of the set of numbers is $(14 - 10) \div 2 = 2$. The numbers around it must be equal (i.e. if the mean of 1, 2, 3, 4, and 5 is 3, then $2 + 4 = 1 + 5$.) One row of the square would be

$\square\square 2\square\square$

Adding the numbers would be

0, 1, 2, 3, 4

with a sum of $\boxed{(C) 10}$.

Solution 6

If the sum of each row, column, and diagonal is x , then we have a total of $12x$ for the sum. The sum of the rows and columns is the sum of all the numbers doubled, which is $50 \cdot 2 = 100$. Therefore $100 + 2x = 12x$, $100 = 10x$, and $x = \boxed{(C) 10}$. MC413551

Solution 7

A cool thing to find is that with magic squares, the smallest and largest number are either in the same row, column and diagonal to balance the equation out. To continue doing this, the other numbers in that same line form an arithmetic sequence. With our A term generally being 1, and our last term being n^2 . The $d = n + 1$, by working out the $a + (n - 1)d$ formula. For a 5 by 5, $d = 6$. To apply this we start with A being -10 , and our last term being 14. We increase by increments by 6 to get the 5 numbers in our line is $-10, -4, 2, 8, 14$. This adds up to $\boxed{(C) 10}$.

(Note the goal of this solution was just to point out something I found that I thought was cool, I don't advise using it in actual competitions)

-CubiksRube

Solution 8

We can solve this by counting how many times each square is included when summing all possible lines. The number of possible lines that can be drawn is 12 (5 rows + 5 columns + 2 main diagonals).

First, the sum of all 25 integers in the set $\{-10, -9, \dots, 14\}$ is 50. Let S be the common sum of a single line. The sum of all 12 lines combined is $12S$.

However, not all squares are used an equal amount of times in these 12 lines. Let's categorize them:

- The center square is used in 4 lines. Let its value be a .
- The remaining 8 squares on the main diagonals are used in 3 lines each. Let the sum of these 8 squares be b .
- The remaining 16 squares are used in 2 lines each. Let the sum of these 16 squares be c .

We can set up a system of equations based on this information:

1. The sum of all squares is 50:

$$a + b + c = 50$$

2. The sum of all 12 lines counts each square multiple times based on its position:

$$12S = 4a + 3b + 2c$$

This means $4a + 3b + 2c \equiv 0 \pmod{12}$.

We just need to find a combination of a , b , and c such that $a + b + c$ is a factor of 50, and then scale it up. A simple combination that works is $a = 0$, $b = 2$, and $c = 3$.

The sum here is $0 + 2 + 3 = 5$. Since we need the sum to be 50, we can multiply our combination by 10 to get $a = 0$, $b = 20$, and $c = 30$. (Note: $0 + 20 + 30 = 50$, satisfying our first equation).

Now, plug this combination back into our line sum equation:

$$12S = 4(0) + 3(20) + 2(30)$$

$$12S = 0 + 60 + 60$$

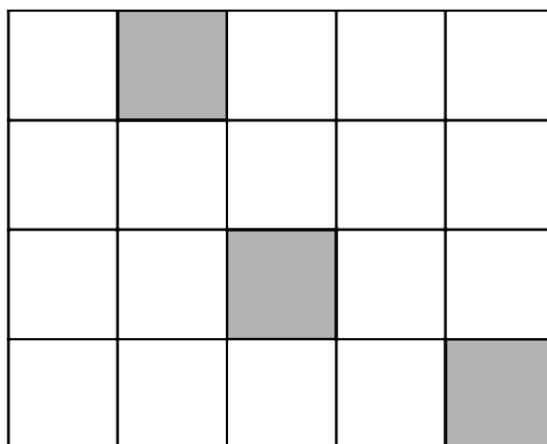
$$12S = 120$$

$$S = 10$$

The common sum is thus (C) 10.

-PandaGod7

6. (AMC 12 12A 2020) In the plane figure shown below, 3 of the unit squares have been shaded. What is the least number of additional unit squares that must be shaded so that the resulting figure has two lines of symmetry?

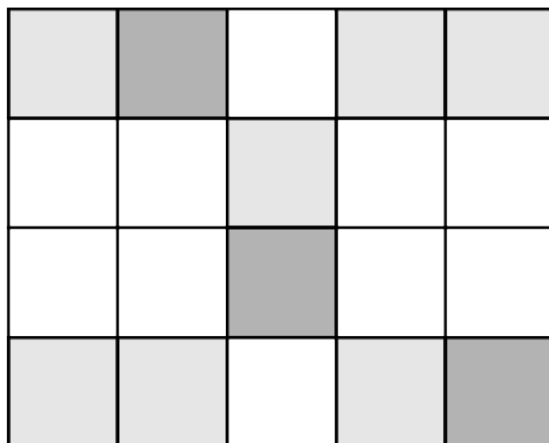


- (A) 4 (B) 5 (C) 6 (D) 7 (E) 8

Answer: D

Solution 1 (Graphical)

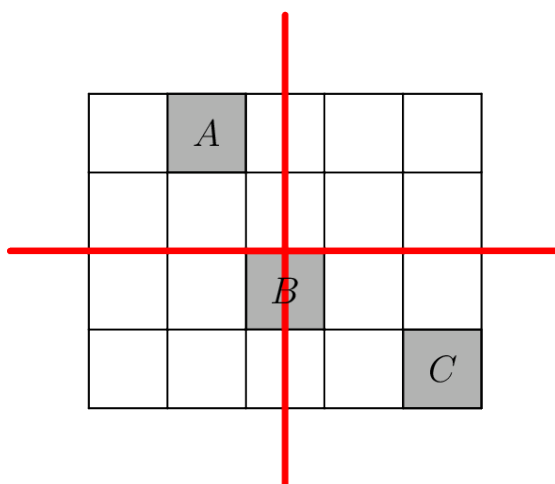
The two lines of symmetry must be horizontally and vertically through the middle. We can then fill the boxes in like so:



where the light gray boxes are the ones we have filled. Counting these, we get **(D) 7** total boxes.

Solution 2 (Analytical)

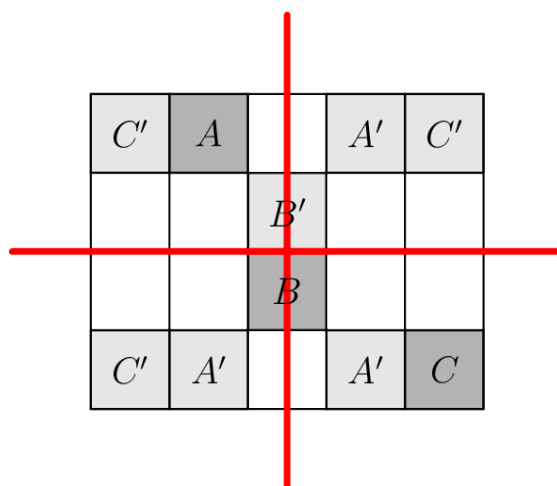
We label the three shaded unit squares A , B , and C , then construct the two lines of symmetry of the resulting figure, as shown below:



Note that:

- Since the centers of A and C are on neither line of symmetry, A and C each contribute 4 shaded unit squares to the resulting figure.
- Since the center of B is on one line of symmetry, B contributes 2 shaded unit squares to the resulting figure.

The shaded unit squares contributed by A , B , and C are all distinct, so we need to shade at least $4 + 4 + 2 - 3 = \mathbf{(D) 7}$ unit squares in addition, as shown below:



MRENTHUSIASM

7. (AMC 12 12A 2020) Seven cubes, whose volumes are 1, 8, 27, 64, 125, 216, and 343 cubic units, are stacked vertically to form a tower in which the volumes of the cubes decrease from bottom to top. Except for the bottom cube, the bottom face of each cube lies completely on top of the cube below it. What is the total surface area of the tower (including the bottom) in square units?
- (A) 644 (B) 658 (C) 664 (D) 720 (E) 749

Answer: B

Solution 1

The volume of each cube follows the pattern of n^3 , for n is between 1 and 7.

We see that the total surface area can be comprised of three parts: the sides of the cubes, the tops of the cubes, and the bottom of the $7 \times 7 \times 7$ cube (which is just $7 \times 7 = 49$). The sides areas can be measured as the sum $4 \sum_{n=1}^7 n^2$, giving us 560. Structurally, if we examine the tower from the top, we see that it really just forms a 7×7 square of area 49. Therefore, we can say that the total surface area is $560 + 49 + 49 = \boxed{\text{(B) } 658}$. Alternatively, for the area of the tops, we could have found the sum $\sum_{n=2}^7 ((n)^2 - (n-1)^2)$, giving us 49 as well.

Note: The area on top and bottom are 49 because the largest area is 49, and the other cubes are "inscribed" in it.

Solution 2

It can quickly be seen that the side lengths of the cubes are the integers from 1 to 7 inclusive.

First, we will calculate the total surface area of the cubes, ignoring overlap. This value is $6(1^2 + 2^2 + \dots + 7^2) = 6 \sum_{n=1}^7 n^2 = 6 \left(\frac{7(7+1)(2 \cdot 7+1)}{6} \right) = 7 \cdot 8 \cdot 15 = 840$. Then, we need to subtract out the overlapped parts of the cubes. Between each consecutive pair of cubes, one of the smaller cube's faces is completely covered, along with an equal area of one of the larger cube's faces. The total area of the overlapped parts of the cubes is thus equal to $2 \sum_{n=1}^6 n^2 = 182$. Subtracting the overlapped surface area from the total surface area, we get $840 - 182 = \boxed{\text{(B) } 658}$.

emerald_block

Solution 3 (a bit more tedious than other solutions)

It can be seen that the side lengths of the cubes using cube roots are all integers from 1 to 7, inclusive.

Only the cubes with side length 1 and 7 have 5 faces in the surface area and the rest have 4. Also, since the cubes are stacked, we have to find the difference between each n^2 and $(n-1)^2$ side length as n ranges from 7 to 2.

We then come up with this: $5(49) + 13 + 4(36) + 11 + 4(25) + 9 + 4(16) + 7 + 4(9) + 5 + 4(4) + 3 + 5(1)$.

We then add all of this and get **(B) 658**.

Solution 4

Notice that the surface area of the top cube is $6s^2$ and the others are $4s^2$. Then we can directly compute. The edge length for the first cube is 7 and has a surface area of 294. The surface area of the next cube is 144. The surface area of the next cube 100. The surface area of the next cube is 64. The surface area of the next cube is 36. The surface area of the next cube is 16. The surface area of the next cube is 4. We then sum up $294 + 144 + 100 + 64 + 36 + 16 + 4$ to get **(B) 658**. smartatmath

Solution 5

First of all, compute the area of the sides, excluding the top and bottoms, of the cubes. The side lengths (cube root the volumes) are 1, 2, 3, 4, 5, 6, 7. Each cube's area of the sides can be calculated with $4(\text{area of one side}) = 4(l^2)$ so in total that is $4(1 + 4 + 16 + \dots + 49)$ so $4(140) = 560$ the area of all the sides of the cubes is 560. Then, calculate the bottom face of the largest cube, $7 * 7 = 49$. Now, notice that if you stack the cubes up on top of each other, and look directly down on them, the tops of the cubes showing add up to the area of the bottom cube, the 7×7 . Therefore, the sum of the area of the tops of the cubes is $7 * 7 = 49$.

Now add them all up: $49 + 49 + 560 = 658$. Therefore, the answer is **(B) 658**.

Solution 6 (Similar to Solution 2, without notation)

Firstly, calculate the total surface area for all the cubes. Knowing that the area of each square of each cube are $2^2, 3^2, 4^2, 5^2, 6^2, 7^2$, we have $6(1 + 4 + 9 + 16 + 25 + 36 + 49) = 6(140) = 840$. When you stack these cubes, the top square of each are subtracted by the square area of the cube above it. Meanwhile, the bottom square of each (except the $7 \times 7 \times 7$ cube) won't be counted because of the stacking. So, we have to subtract $2(1 + 4 + 9 + 16 + 25 + 36) = 2(91) = 182$ from 840, getting **(B) 658**.

Solution 7 (Going from Bottom to Top)

The surface area of the bottom cube is the bottom face, the 4 side faces, and the top face minus the bottom face of the next cube. Therefore, the surface area of the bottom cube is 6 faces minus the bottom face of the next cube. That will be $6 \cdot 49 - 36$. For the next cube, we take the 4 side faces and the top face, and subtract the bottom face of the next cube. Therefore, the surface area of the next cube will be 5 of its faces minus the bottom face of the next cube. That will be $5 \cdot 36 - 25$. We can do the next cube the same way. It will end up as $5 \cdot 25 - 16$. From here we start noticing a pattern. The next will be $5 \cdot 16 - 9$, and then $5 \cdot 9 - 4$, next $5 \cdot 4 - 1$, and to include the top cube, we have to add $5 \cdot 1$, which is basically $5 \cdot 1 - 0$. Now, putting it all together, we get $6 \cdot 49 - 36 + 5 \cdot 36 - 25 + 5 \cdot 25 - 16 + 5 \cdot 16 - 9 + 5 \cdot 9 - 4 + 5 \cdot 4 - 1 + 5 \cdot 1 - 0 = 6 \cdot 49 + 5(36 + 25 + 16 + 9 + 4 + 1) - (36 + 25 + 16 + 9 + 4 + 1) = 6 \cdot 49 + 4(36 + 25 + 16 + 9 + 4 + 1) = 294 + 4(91) = 294 + 364 = \mathbf{(B) 658}$.

8. (AMC 12 12A 2020) What is the median of the following list of 4040 numbers?

$$1, 2, 3, \dots, 2020, 1^2, 2^2, 3^2, \dots, 2020^2$$

- (A) 1974.5 (B) 1975.5 (C) 1976.5 (D) 1977.5 (E) 1978.5

Answer: C

Solution 1

We can see that $44^2 = 1936$ which is less than 2020. Therefore, there are $2020 - 44 = 1976$ of the 4040 numbers greater than 2020. Also, there are $2020 + 44 = 2064$ numbers that are less than or equal to 2020.

Since there are 44 duplicates/extras, it will shift up our median's placement down 44. Had the list of numbers been $1, 2, 3, \dots, 4040$, the median of the whole set would be $\frac{1 + 4040}{2} = 2020.5$.

Thus, our answer is $2020.5 - 44 = \boxed{\text{(C) } 1976.5}$.

Solution 2

As we are trying to find the median of a 4040-term set, we must find the average of the 2020th and 2021st terms.

Since $45^2 = 2025$ is slightly greater than 2020, we know that the 44 perfect squares 1^2 through 44^2 are less than 2020, and the rest are greater. Thus, from the number 1 to the number 2020, there are $2020 + 44 = 2064$ terms. Since 44^2 is $44 + 45 = 89$ less than $45^2 = 2025$ and 84 less than 2020, we will only need to consider the perfect square terms going down from the 2064th term, 2020, after going down 84 terms. Since the 2020th and 2021st terms are only 44 and 43 terms away from the 2064th term, we can simply subtract 44 from 2020 and 43 from 2020 to get the two terms, which are 1976 and 1977. Averaging the two, we get $\boxed{\text{(C) } 1976.5}$.

Solution 3

We want to know the 2020th term and the 2021st term to get the median.

We know that $44^2 = 1936$. So, numbers $1^2, 2^2, \dots, 44^2$ are in between 1 and 1936. So, the sum of 44 and 1936 will result in 1980, which means that 1936 is the 1980th number.

Also, notice that $45^2 = 2025$, which is larger than 2021.

Then the 2020th term will be $1936 + 40 = 1976$, and similarly the 2021st term will be 1977.

Solving for the median of the two numbers, we get $\boxed{\text{(C) } 1976.5}$

Solution 4

We note that $44^2 = 1936$, which is the largest square less than 2020, which means that there are 44 additional terms before 2020. This makes 2020 the 2064th term. To find the median, we need the 2020th and 2021st term. We note that every term before 2020 is one less than the previous term (That is, we subtract 1 to get the previous term.). If 2020 is the 2064th term, then $2020 - 44$ is the $(2064 - 44)$ th term. So, the 2020th term is 1976, and the 2021st term is 1977, and the average of these two terms is the median, or $\boxed{\text{(C) } 1976.5}$.

Solution 5 (Decreasing Order)

To find the median, we sort the 4040 numbers in decreasing order, then average the 2020th and the 2021st numbers of the sorted list.

Since $45^2 = 2025$ and $44^2 = 1936$, the first 2021 numbers of the sorted list are

$$\underbrace{2020^2, 2019^2, 2018^2, \dots, 46^2, 45^2}_{1976 \text{ numbers}}, \underbrace{2020, 2019, 2018, \dots, 1977, 1976}_{45 \text{ numbers}},$$

from which the answer is $\frac{1977+1976}{2} = \boxed{\text{(C)} 1976.5}$.

9. (AMC 12 12A 2020) How many solutions does the equation $\tan(2x) = \cos\left(\frac{x}{2}\right)$ have on the interval $[0, 2\pi]$? (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Answer: E

Solution

We count the intersections of the graphs of $y = \tan(2x)$ and $y = \cos\left(\frac{x}{2}\right)$:

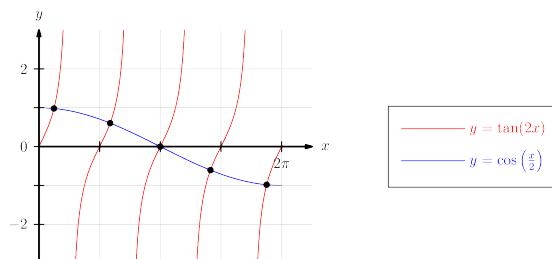
- a) The graph of $y = \tan(2x)$ has a period of $\frac{\pi}{2}$, asymptotes at $x = \frac{\pi}{4} + \frac{k\pi}{2}$, and zeros at $x = \frac{k\pi}{2}$ for some integer k . On the interval $[0, 2\pi]$, the graph has five branches:

$$\left[0, \frac{\pi}{4}\right), \left(\frac{\pi}{4}, \frac{3\pi}{4}\right), \left(\frac{3\pi}{4}, \frac{5\pi}{4}\right), \left(\frac{5\pi}{4}, \frac{7\pi}{4}\right), \left(\frac{7\pi}{4}, 2\pi\right].$$

Note that $\tan(2x) \in [0, \infty)$ for the first branch, $\tan(2x) \in (-\infty, \infty)$ for the three middle branches, and $\tan(2x) \in (-\infty, 0]$ for the last branch. Moreover, all branches are strictly increasing.

- b) The graph of $y = \cos\left(\frac{x}{2}\right)$ has a period of 4π and zeros at $x = \pi + 2k\pi$ for some integer k . On the interval $[0, 2\pi]$, note that $\cos\left(\frac{x}{2}\right) \in [-1, 1]$. Moreover, the graph is strictly decreasing.

The graphs of $y = \tan(2x)$ and $y = \cos\left(\frac{x}{2}\right)$ intersect once on each of the five branches of $y = \tan(2x)$, as shown below:



Therefore, the answer is $\boxed{\text{(E)} 5}$.

10. (AMC 12 12A 2020) There is a unique positive integer n such that

$$\log_2(\log_{16} n) = \log_4(\log_4 n).$$

What is the sum of the digits of n ? (A) 4 (B) 7 (C) 8 (D) 11 (E) 13

Answer: E

Solution 1 (Properties of Logarithms)

We can use the fact that $\log_{a^b} c = \frac{1}{b} \log_a c$. This can be proved by using **Change of Base Formula** to base a .

So, the original equation $\log_2(\log_{2^4} n) = \log_{2^2}(\log_{2^2} n)$ becomes

$$\log_2\left(\frac{1}{4}\log_2 n\right) = \frac{1}{2}\log_2\left(\frac{1}{2}\log_2 n\right).$$

Using log property of addition, we expand both sides and then simplify:

$$\begin{aligned}\log_2 \frac{1}{4} + \log_2(\log_2 n) &= \frac{1}{2} \left[\log_2 \frac{1}{2} + \log_2(\log_2 n) \right] \\ \log_2 \frac{1}{4} + \log_2(\log_2 n) &= \frac{1}{2} [-1 + \log_2(\log_2 n)] \\ -2 + \log_2(\log_2 n) &= -\frac{1}{2} + \frac{1}{2}(\log_2(\log_2 n)).\end{aligned}$$

Subtracting $\frac{1}{2}(\log_2(\log_2 n))$ from both sides and adding 2 to both sides gives us

$$\frac{1}{2}(\log_2(\log_2 n)) = \frac{3}{2}.$$

Multiplying by 2, exponentiating, and simplifying gives us

$$\begin{aligned}\log_2(\log_2 n) &= 3 \\ \log_2 n &= 8 \\ n &= 256.\end{aligned}$$

Adding the digits together, we have $2 + 5 + 6 = \boxed{\text{(E)} 13}$.

Solution 2 (Properties of Logarithms)

We will apply the following logarithmic identity:

$$\log_{p^k} q^k = \log_p q,$$

which can be proven by the **Change of Base Formula**:

$$\log_{p^k} q^k = \frac{\log_p q^k}{\log_p p^k} = \frac{k \log_p q}{k} = \log_p q.$$

Note that $\log_{16} n \neq 0$, so we rewrite the original equation as follows:

$$\begin{aligned}\log_4(\log_{16} n)^2 &= \log_4(\log_4 n) \\ (\log_{16} n)^2 &= \log_4 n \\ (\log_{16} n)^2 &= \log_{16} n^2 \\ (\log_{16} n)^2 &= 2 \log_{16} n \\ \log_{16} n &= 2,\end{aligned}$$

from which $n = 16^2 = 256$. The sum of its digits is $2 + 5 + 6 = \boxed{\text{(E)} 13}$.

Solution 3 (Properties of Logarithms)

Using the change of base formula on the RHS of the initial equation yields

$$\log_2(\log_{16} n) = \frac{\log_2(\log_4 n)}{\log_2 4}.$$

This means we can multiply each side by 2 for

$$\log_2(\log_{16} n)^2 = \log_2(\log_4 n).$$

Canceling out the logs gives

$$(\log_{16} n)^2 = \log_4 n.$$

We use change of base on the RHS to see that

$$\begin{aligned} (\log_{16} n)^2 &= \frac{\log_{16} n}{\log_{16} 4} \\ (\log_{16} n)^2 &= 2 \log_{16} n. \end{aligned}$$

Substituting in $m = \log_{16} n$ gives $m^2 = 2m$, so m is either 0 or 2. Since $m = 0$ yields no solution for n (since this would lead to use taking the log of 0), we get $2 = \log_{16} n$, or $n = 16^2 = 256$, for the digit-sum of $2 + 5 + 6 = \boxed{\text{(E)} 13}$.

Solution 4 (Exponential Form)

Suppose $\log_2(\log_{16} n) = k \implies \log_{16} n = 2^k \implies n = 16^{2^k}$. Similarly, we have $\log_4(\log_4 n) = k \implies \log_4 n = 4^k \implies n = 4^{4^k}$. Thus, we have

$$16^{2^k} = (4^2)^{2^k} = 4^{2^{k+1}}$$

and

$$4^{4^k} = 4^{2^{2k}},$$

so $k + 1 = 2k \implies k = 1$. Plugging this in to either one of the expressions for n gives 256, and the requested answer is $2 + 5 + 6 = \boxed{\text{(E)} 13}$.

Solution 5 (Guess and Check)

We know that, as the answer is an integer, n must be some power of 16. Testing 16 yields

$$\begin{aligned} \log_2(\log_{16} 16) &= \log_4(\log_4 16) \\ \log_2 1 &= \log_4 2 \\ 0 &= \frac{1}{2}, \end{aligned}$$

which does not work. We then try 256, giving us

$$\begin{aligned} \log_2(\log_{16} 256) &= \log_4(\log_4 256) \\ \log_2 2 &= \log_4 4 \\ 1 &= 1, \end{aligned}$$

which holds true. Thus, $n = 256$, so the answer is $2 + 5 + 6 = \boxed{\text{(E)} 13}$.

(Don't use this technique unless you absolutely need to! Guess and check methods aren't helpful for learning math.)

11. (AMC 12 12A 2020) A frog sitting at the point $(1, 2)$ begins a sequence of jumps, where each jump is parallel to one of the coordinate axes and has length 1, and the direction of each jump (up, down, right, or left) is chosen independently at random. The sequence ends when the frog reaches a side of the square with vertices $(0, 0)$, $(0, 4)$, $(4, 4)$, and $(4, 0)$. What is the probability that the sequence of jumps ends on a vertical side of the square? (A) $\frac{1}{2}$ (B) $\frac{5}{8}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$ (E) $\frac{7}{8}$

Answer: B

Solution 1

Drawing out the square, it's easy to see that if the frog goes to the left, it will immediately hit a vertical end of the square. Therefore, the probability of this happening is $\frac{1}{4} \cdot 1 = \frac{1}{4}$. If the frog goes to the right, it will be in the center of the square at $(2, 2)$, and by symmetry (since the frog is equidistant from all sides of the square), the chance it will hit a vertical side of a square is $\frac{1}{2}$. The probability of this happening is $\frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}$.

If the frog goes either up or down, it will hit a line of symmetry along the corner it is closest to and furthest to, and again, is equidistant relating to the two closer sides and also equidistant relating the two further sides. The probability for it to hit a vertical wall is $\frac{1}{2}$. Because there's a $\frac{1}{2}$ chance of the frog going up or down, the total probability for this case is $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$ and summing up all the cases,

$$\frac{1}{4} + \frac{1}{8} + \frac{1}{4} = \frac{5}{8} \implies \boxed{\text{(B)} \frac{5}{8}}.$$

Solution 2

Let's say we have our five by five grid and we work this out by casework. A is where the frog is, while B and C are possible locations for his second jump, while O is everything else. If we land on a C, we have reached the vertical side. However, if we land on a B, the probability of landing on a C is $\frac{1}{4}$, while B is $\frac{3}{4}$. Since C means that we have "succeeded", while B means that we have a half chance, we compute $1 \cdot C + \frac{1}{2} \cdot B$.

$$1 \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{3}{4} \\ \frac{1}{4} + \frac{3}{8}$$

We get $\frac{5}{8}$, or B

O	O	O	O	O
O	B	O	O	O
C	A	B	O	O
O	B	O	O	O
O	O	O	O	O

-yeskay

Solution 3

If the frog is on one of the 2 diagonals, the chance of landing on vertical or horizontal each becomes $\frac{1}{2}$. Since it starts on $(1, 2)$, there is a $\frac{3}{4}$ chance (up, down, or right) it will reach a diagonal on the first jump and $\frac{1}{4}$ chance (left) it will reach the vertical

side. The probability of landing on a vertical is $\frac{1}{4} + \frac{3}{4} \cdot \frac{1}{2} = \boxed{\text{(B)} \frac{5}{8}}$.- Lingjun

Solution 4 (Complete States)

Let $P_{(x,y)}$ denote the probability of the frog's sequence of jumps ends with it hitting a vertical edge when it is at (x, y) . Note that $P_{(1,2)} = P_{(3,2)}$ by reflective symmetry over the line $x = 2$. Similarly, $P_{(1,1)} = P_{(1,3)} = P_{(3,1)} = P_{(3,3)}$, and $P_{(2,1)} = P_{(2,3)}$.

Now we create equations for the probabilities at each of these points/states by considering the probability of going either up, down, left, or right from that point:

$$P_{(1,2)} = \frac{1}{4} + \frac{1}{2}P_{(1,1)} + \frac{1}{4}P_{(2,2)}$$

$$P_{(2,2)} = \frac{1}{2}P_{(1,2)} + \frac{1}{2}P_{(2,1)}$$

$$P_{(1,1)} = \frac{1}{4} + \frac{1}{4}P_{(1,2)} + \frac{1}{4}P_{(2,1)}$$

$$P_{(2,1)} = \frac{1}{2}P_{(1,1)} + \frac{1}{4}P_{(2,2)}$$

We have a system of 4 equations in 4 variables, so we can solve for each of these probabilities. Plugging the second equation into the fourth equation gives

$$P_{(2,1)} = \frac{1}{2}P_{(1,1)} + \frac{1}{4} \left(\frac{1}{2}P_{(1,2)} + \frac{1}{2}P_{(2,1)} \right)$$

$$P_{(2,1)} = \frac{8}{7} \left(\frac{1}{2}P_{(1,1)} + \frac{1}{8}P_{(1,2)} \right) = \frac{4}{7}P_{(1,1)} + \frac{1}{7}P_{(1,2)}$$

Plugging in the third equation into this gives

$$P_{(2,1)} = \frac{4}{7} \left(\frac{1}{4} + \frac{1}{4}P_{(1,2)} + \frac{1}{4}P_{(2,1)} \right) + \frac{1}{7}P_{(1,2)}$$

$$P_{(2,1)} = \frac{7}{6} \left(\frac{1}{7} + \frac{2}{7}P_{(1,2)} \right) = \frac{1}{6} + \frac{1}{3}P_{(1,2)} \quad (*)$$

Next, plugging in the second and third equation into the first equation yields

$$P_{(1,2)} = \frac{1}{4} + \frac{1}{2} \left(\frac{1}{4} + \frac{1}{4}P_{(1,2)} + \frac{1}{4}P_{(2,1)} \right) + \frac{1}{4} \left(\frac{1}{2}P_{(1,2)} + \frac{1}{2}P_{(2,1)} \right)$$

$$P_{(1,2)} = \frac{3}{8} + \frac{1}{4}P_{(1,2)} + \frac{1}{4}P_{(2,1)}$$

Now plugging in (*) into this, we get

$$P_{(1,2)} = \frac{3}{8} + \frac{1}{4}P_{(1,2)} + \frac{1}{4} \left(\frac{1}{6} + \frac{1}{3}P_{(1,2)} \right)$$

$$P_{(1,2)} = \frac{3}{2} \cdot \frac{5}{12} = \boxed{\text{(B)} \frac{5}{8}}$$

-mathisawesome2169

Note: We can also think of each state $P_{(x,y)}$ as the probability the frog will reach a vertical side given that it is x away from a vertical side and y away from a horizontal side. This is just another way to think about the reflective symmetry.
-GardenRock10

Solution 5 (States)

this is basically another version of solution 4; shoutout to mathisawesome2169 :D

First, we note the different places the frog can go at certain locations in the square:

If the frog is at a border vertical point $((1,2), (3,2))$, it moves with probability $\frac{1}{4}$ to a vertical side of the square, probability $\frac{1}{4}$ to the center of the square, and probability $\frac{1}{2}$ to a corner square.

If the frog is at a border horizontal point $((2, 1), (2, 3))$, it moves with probability $\frac{1}{4}$ to a horizontal side of the square, probability $\frac{1}{4}$ to the center of the square, and probability $\frac{1}{2}$ to a corner square.

If the frog is at a center square $((2, 2))$, it moves with probability $\frac{1}{2}$ to a border horizontal point and probability $\frac{1}{2}$ to a border vertical point.

If the frog is at a corner $((1, 1), (1, 3), (3, 3), (3, 1))$, it moves with probability $\frac{1}{4}$ to a vertical side of the square, probability $\frac{1}{4}$ to a horizontal side, probability $\frac{1}{4}$ to a border horizontal point, and probability $\frac{1}{4}$ to a border vertical point.

Now, let x denote the probability of the frog reaching a vertical side when it is at a border vertical point. Similarly, let y denote the probability of the frog reaching a vertical side when it is at a border horizontal point. Now, the probability of the frog reaching a vertical side of the square at any location inside the square can be expressed in terms of x and y .

First, the two easier ones: $P_{center} = \frac{1}{2}x + \frac{1}{2}y$, and $P_{corner} = \frac{1}{4} + \frac{1}{4}x + \frac{1}{4}y$. Now, we can write x and y in terms of x and y , allowing us to solve a system of two variables:

$$x = \frac{1}{4} + \frac{1}{4}P_{center} + \frac{1}{2}P_{corner} = \frac{1}{4} + \frac{1}{4}\left(\frac{1}{2}x + \frac{1}{2}y\right) + \frac{1}{2}\left(\frac{1}{4} + \frac{1}{4}x + \frac{1}{4}y\right)$$

and

$$y = \frac{1}{4}P_{center} + \frac{1}{2}P_{corner} = \frac{1}{4}\left(\frac{1}{2}x + \frac{1}{2}y\right) + \frac{1}{2}\left(\frac{1}{4} + \frac{1}{4}x + \frac{1}{4}y\right).$$

From these two equations, it is apparent that $y = x - \frac{1}{4}$. We can then substitute this value for y back into any of the two equations above to get

$$x = \frac{1}{4} + \frac{1}{4}\left(\frac{1}{2}x + \frac{1}{2}\left(x - \frac{1}{4}\right)\right) + \frac{1}{2}\left(\frac{1}{4} + \frac{1}{4}x + \frac{1}{4}\left(x - \frac{1}{4}\right)\right).$$

Although this certainly looks intimidating, we can expand the parentheses and multiply both sides by 16 to eliminate the fractions, which upon simplification yields the equation

$$16x = 5 + 8x,$$

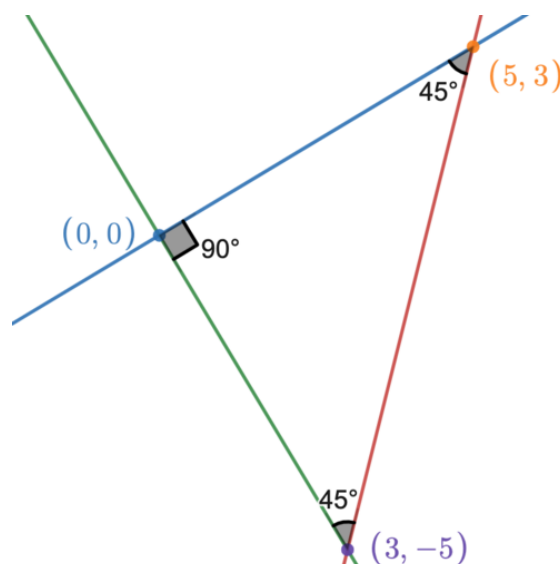
giving us the desired probability $x = \frac{5}{8}$. The answer is then **(B)** $\frac{5}{8}$.

- curiousmind888 TGSN

12. (AMC 12 12A 2020) Line ℓ in the coordinate plane has the equation $3x - 5y + 40 = 0$. This line is rotated 45° counterclockwise about the point $(20, 20)$ to obtain line k . What is the x -coordinate of the x -intercept of line k ? **(A)** 10 **(B)** 15 **(C)** 20 **(D)** 25 **(E)**

Answer: B

Solution 1



The slope of the line is $\frac{3}{5}$. We must transform it by 45° .

45° creates an isosceles right triangle, since the sum of the angles of the triangle must be 180° and one angle is 90° . This means the last leg angle must also be 45° .

In the isosceles right triangle, the two legs are congruent. We can therefore construct an isosceles right triangle with a line of $\frac{3}{5}$ slope on graph paper. That line with $\frac{3}{5}$ slope starts at $(0,0)$ and will go to $(5,3)$, the vector $\langle 5,3 \rangle$.

Construct another line from $(0,0)$ to $(3,-5)$, the vector $\langle 3,-5 \rangle$. This is $\perp$ and equal to the original line segment. The difference between the two vectors is $\langle 2,8 \rangle$, which is the slope 4, and that is the slope of line k .

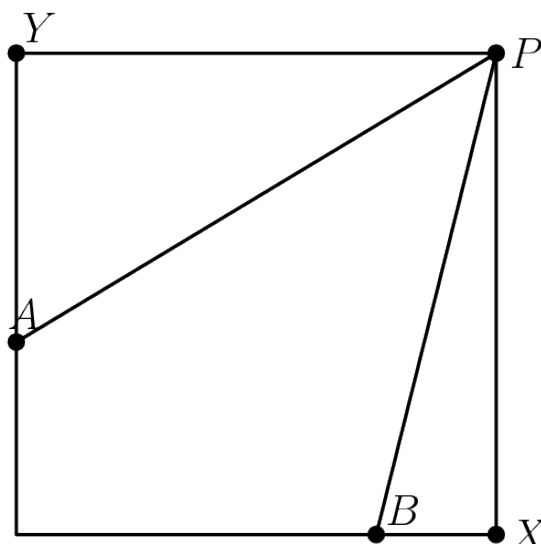
Furthermore, the equation $3x - 5y + 40 = 0$ passes straight through $(20,20)$ since $3(20) - 5(20) + 40 = 60 - 100 + 40 = 0$, which means that any rotations about $(20,20)$ would contain $(20,20)$. We can create a line of slope 4 through $(20,20)$. The x -intercept is therefore $20 - \frac{20}{4} = \boxed{\text{(B) } 15}$. lopkiloim ShawnX (diagram)

Solution 2

Since the slope of the line is $\frac{3}{5}$, and the angle we are rotating around is x , then $\tan x = \frac{\frac{3}{5}\tan(x + 45^\circ)}{1 - \frac{3}{5}\tan(x + 45^\circ)} = \frac{0.6+1}{1-0.6} = \frac{1.6}{0.4} = 4$

Hence, the slope of the rotated line is 4. Since we know the line intersects the point $(20,20)$, then we know the line is $y = 4x - 60$. Set $y = 0$ to find the x -intercept, and so $x = \boxed{15}$

Solution 3



Let P be $(20, 20)$ and X, Y be $(20, 0)$ and $(0, 20)$ respectively. Since the slope of the line is $3/5$ we know that $\tan \angle YPA = 3/5$. Segments $\overline{PA}$ and $\overline{PB}$ represent the before and after of rotating l by 45° counterclockwise. Thus, $\angle XPB = 45^\circ - \angle YPA$ and

$$BX = 20 \tan \angle XPB = 20 \cdot \frac{1 - 3/5}{1 + 3/5} = 5$$

by tangent addition formula. Since BX is 5 and the sidelength of the square is 20 the answer is $20 - 5 \Rightarrow \boxed{\text{B}}$.

Solution 4 (Cheap)

Using the protractor you brought, carefully graph the equation and rotate the given line 45° counter-clockwise about the point $(20, 20)$. Scaling everything down by a factor of 5 makes this process easier.

It should then become fairly obvious that the x intercept is $x = \boxed{15}$ (only use this as a last resort).

Solution 5 (Rotation Matrix)

First note that the given line goes through $(20, 20)$ with a slope of $\frac{3}{5}$. This means that $(25, 23)$ is on the line. Now consider translating the graph so that $(20, 20)$ goes to the origin, then $(25, 23)$ becomes $(5, 3)$. We now rotate the line 45° about the origin using a rotation matrix. This maps $(5, 3)$ to

$$\begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} \sqrt{2} \\ 4\sqrt{2} \end{bmatrix}$$

The line through the origin and $(\sqrt{2}, 4\sqrt{2})$ has slope 4. Translating this line so that the origin is mapped to $(20, 20)$, we find that the equation for the new line is $4x - 60$, meaning that the x -intercept is $x = \frac{60}{4} = \boxed{\text{(B)} 15}$.

Solution 6 (Angle Bisector)

Note $(20, 20)$ is on the line. Construct the perpendicular line $5x + 3y - 160 = 0$. This creates a right triangle that intersects the x -axis at $(-\frac{40}{3}, 0)$ and $(32, 0)$ a distance of $136/3$ apart. The 45° transformation will bisect the right angle. The angle bisector theorem tells us the 136 will split in ratio to the lengths of the sides. These are

$\sqrt{12^2 + 20^2}$ and $\sqrt{\frac{100^2}{3} + 20^2} = 4\sqrt{34}$ and $\frac{20}{3}\sqrt{34}$. Thus the x intercept will split the line from $\frac{-40}{3}$ to 32 into a ratio of 5 : 3 making the x-intercept 15.

Solution 7 (Complex Numbers)

Converting to the complex plane, we can see that two numbers on the line are $8i$ and $5 + 11i$. Translating $20 + 20i$ to the origin, we get $8i - 20 - 20i = -20 - 12i$ and $5 + 11i - 20 - 20i = -15 - 9i$. Multiplying each of them by $e^{\pi i/4}$, we get $-4\sqrt{2} - 16\sqrt{2}i$ and $-3\sqrt{2} - 12\sqrt{2}i$. This line has a slope of 4. Now, back to the cartesian plane. We have a line passing through $(20, 20)$ with slope 4 which gives the equation as $y = 4x - 60$ which implies the x coordinate of the x intercept is $60/4 = 15$.

Solution 8 (quick)

A quick check tells us that $(20, 20)$ falls on the given line. Common sense tells us that if the slope of the original line is 1, or 45 degrees from the horizontal, a 45 counter clockwise rotation will result in a vertical line, and the x-intercept will be $(20, 0)$. Thinking of a 45 degree counter clockwise rotation as a 90 degree counter clockwise rotation that is bisected will help in visualizing this line of reasoning. Therefore, it follows that if the original line is made steeper, then the x-intercept will move away from $(20, 0)$ to the right. If the original line is made lower, then the opposite will happen. Our given line has slope $3/5$, so the answer must be A or B. A can be eliminated because an x-intercept of $(10, 0)$ can only occur when the original line is horizontal. In conclusion, the answer must be **(B) 15**.

13. (AMC 12 12A 2020) There are integers a , b , and c , each greater than 1, such that

$$\sqrt[a]{N \sqrt[b]{N \sqrt[c]{N}}} = \sqrt[36]{N^{25}}$$

for all $N > 1$. What is b ?

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Answer: B

Solution 1

$\sqrt[a]{N \sqrt[b]{N \sqrt[c]{N}}}$ can be simplified to $N^{\frac{1}{a} + \frac{1}{ab} + \frac{1}{abc}}$.

The equation is then $N^{\frac{1}{a} + \frac{1}{ab} + \frac{1}{abc}} = N^{\frac{25}{36}}$ which implies that $\frac{1}{a} + \frac{1}{ab} + \frac{1}{abc} = \frac{25}{36}$. a has to be 2 since $\frac{25}{36} > \frac{7}{12}$. $\frac{7}{12}$ is the result when a, b , and c are 3, 2, and 2 b being 3 will make the fraction $\frac{2}{3}$ which is close to $\frac{25}{36}$.

Finally, with c being 6, the fraction becomes $\frac{25}{36}$. In this case a, b , and c work, which means that b must equal **(B) 3**.

Solution 2

As above, notice that you get $\frac{1}{a} + \frac{1}{ab} + \frac{1}{abc} = \frac{25}{36}$.

Now, combine the fractions to get $\frac{bc+c+1}{abc} = \frac{25}{36}$.

Let us assume $bc + c + 1 = 25$ and $abc = 36$ as this is the most convenient. (EDIT: This used to say WLOG but that is inaccurate)

From the first equation, we get $c(b + 1) = 24$. Note also that from the second equation, b and c must be factors of 36.

After listing out the factors of 36 and utilising trial and error, we find that $c = 6$ and $b = 3$ works, with $a = 2$. So our answer is **(B)** 3.

Edits by Snore, Swaggetgotcha

Solution 3

Collapsed, $\sqrt[a]{N} \sqrt[b]{N} \sqrt[c]{N} = \sqrt[abc]{N^{bc+c+1}}$. Comparing this to $\sqrt[36]{N^{25}}$, observe that $bc + c + 1 = 25$ and $abc = 36$. The first can be rewritten as $c(b + 1) = 24$. Then, $b + 1$ has to factor into 24 while 1 less than that also must factor into 36. The prime factorizations are as follows $36 = 2^2 3^2$ and $24 = 2^3 3$. Then, $b = \mathbf{(B)}3$, as only 4 and 3 factor into 36 and 24 while being 1 apart.

Note: 3 and 2 also factor into 24 and 36 while being 1 apart, but we can eliminate by quick guess and check. Edit by Soupnoodle

Solution 4

$\sqrt[a]{N} \sqrt[b]{N} \sqrt[c]{N} = \sqrt[36]{N^{25}}$ can be rewritten as $\frac{\frac{1}{c}+1}{a} = \frac{25}{36}$. The trick is to focus on the properties of the denominators and numerators. a must be either be a factor of 36, because in any other situation the denominator of the final expression is preserved, and b is forced to be 36, which is not an answer choice. Next, realize that the numerator of $\frac{1}{c}+1$ must be 25, and even more importantly, that the denominator must be large enough so that when 1 is subtracted from it, it will form a fraction less than 1. The only number large enough to do this will also being a fraction of 36 is 18. This means that a is 2. Moving onward, we are left with $\frac{1}{c} = \frac{7}{18}$. Since b must be a factor of 18, just plug in values from the answer choices to find that $c = 6$ and $b = 3$.

14. (AMC 12 12A 2020) Regular octagon $ABCDEFGH$ has area n . Let m be the area of quadrilateral $ACEG$. What is $\frac{m}{n}$? **(A)** $\frac{\sqrt{2}}{4}$ **(B)** $\frac{\sqrt{2}}{2}$ **(C)** $\frac{3}{4}$ **(D)** $\frac{3\sqrt{2}}{5}$ **(E)** $\frac{2\sqrt{2}}{3}$

Answer: B

Solution 1

$ACEG$ is a square. WLOG $AB = 1$, then using Law of Cosines, $AC^2 = [ACEG] = 1^2 + 1^2 - 2 \cos 135 = 2 + \sqrt{2}$. The area of the octagon is just $[ACEG]$ plus the area of the four congruent (by symmetry) isosceles triangles, all an angle of 135 in between two sides of length 1. Now,

$$\frac{m}{n} = \frac{2 + \sqrt{2}}{2 + \sqrt{2} + 4 \cdot \frac{1}{2} \sin 135} = \frac{2 + \sqrt{2}}{2 + 2\sqrt{2}} = \frac{\sqrt{2}}{2}.$$

The answer is **(B)**.

Solution 2

Refer to the diagram. Call one of the side lengths of the square s . Since quadrilateral $ACEG$ is a square, the area of the square would just be s^2 , which we can find by applying Law of Cosines on one of the four triangles. Assume each of the sides of the octagon has length 1. Since each angle measures 135° in an octagon, then $s^2 = 1^2 + 1^2 - 2 \cos(135^\circ) = 2 + \sqrt{2}$

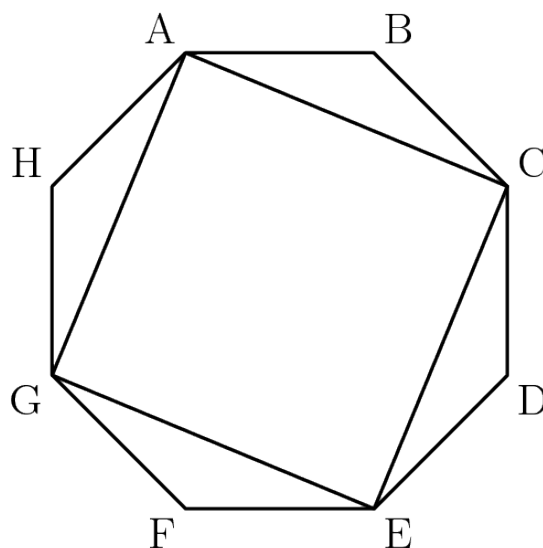
There are many ways to find the area of the octagon, but one way is to split the octagon into two trapezoids and one rectangle. We can easily compute AF to be

$1 + \sqrt{2}$ from splitting one of the sides into two $45 - 45 - 90$ triangles. So the area of the octagon is $2 \cdot \frac{1+\sqrt{2}}{2} + 1 + \sqrt{2} \Rightarrow 2 + 2\sqrt{2}$.

Hence,

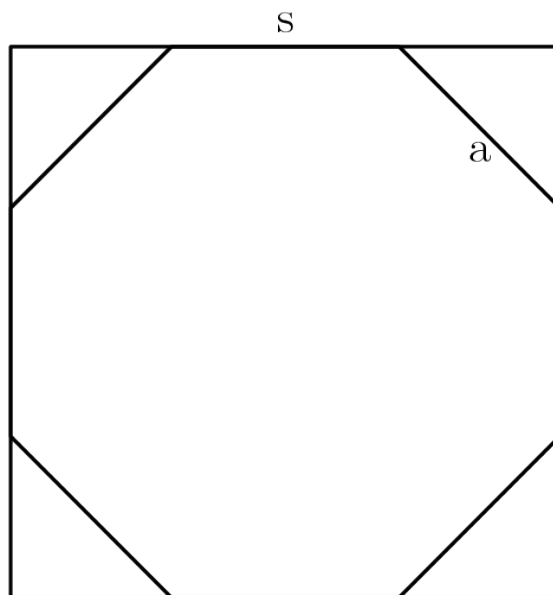
$$\begin{aligned} \frac{m}{n} &= \frac{2 + \sqrt{2}}{2 + 2\sqrt{2}} \\ \Rightarrow \frac{2 + \sqrt{2}}{2 + 2\sqrt{2}} \cdot \frac{2 - 2\sqrt{2}}{2 - 2\sqrt{2}} \\ &\Rightarrow \frac{-2\sqrt{2}}{-4} \\ &\Rightarrow \boxed{\text{(B)} \frac{\sqrt{2}}{2}} \end{aligned}$$

Solution 3 (Deriving Formulas)



The first thing to notice is that $ACEG$ is a square. This is because, as $\triangle ABC \cong \triangle CDE \cong \triangle EFG \cong \triangle GHE$, they all have the same base, meaning that $AC = DE = EG = GA$. Hence, we have that it is a square. To determine the area of this square, we can determine the length of its diagonals.

In order to do this, we first determine the area of the octagon. Letting the side length be a , we can create a square of length s around it (see figure).



Creating a small square of side length a from the corners of this figure gives us an area of a^2 . Thus, $s^2 - a^2 = n$ where n is the area of the octagon. We know from the Pythagorean Theorem that $s = a + \frac{a}{\sqrt{2}}$, meaning that $n = (a + \frac{a}{\sqrt{2}})^2 - a^2 = 2a^2(1 + \sqrt{2})$.

Dividing this by 8 gives us the area of each triangular segment which makes up the octagon. Further dividing by 2 gives us the area of a smaller segment consisting of the right triangle with legs of the apothem and $\frac{a}{2}$. Using the area of a triangle as $\frac{1}{2}bh$, we can determine the length of apothem r from

$$\begin{aligned}\frac{2a^2(1 + \sqrt{2})}{8 \times 2} &= \frac{\frac{a}{2} \times r}{2} \\ \frac{4a^2(1 + \sqrt{2})}{8} &= ar \\ \frac{a(1 + \sqrt{2})}{2} &= r\end{aligned}$$

From the apothem, we can once again use the Pythagorean Theorem, giving us the length of the circumradius R .

$$\begin{aligned}R^2 &= \left(\frac{a(1 + \sqrt{2})}{2}\right)^2 + \left(\frac{a}{2}\right)^2 \\ R^2 &= \frac{a^2(1 + \sqrt{2})^2}{4} + \frac{a^2}{4} \\ R^2 &= \frac{a^2(3 + 2\sqrt{2})}{4} + \frac{a^2}{4} \\ R^2 &= \frac{a^2(4 + 2\sqrt{2})}{4} \\ R &= \sqrt{\frac{a^2(4 + 2\sqrt{2})}{4}} = \frac{a\sqrt{4 + 2\sqrt{2}}}{2}\end{aligned}$$

Doubling this gives us the diagonal of both the square and the octagon. From here, we can use the formula $A = \frac{1}{2}d^2$ for the area of the square:

$$A = \frac{(a\sqrt{4+2\sqrt{2}})^2}{2}$$

$$A = \frac{a^2(4+2\sqrt{2})}{2} = m$$

Thus we now only need to find the ratio $\frac{m}{n}$. This can be easily done through some algebra:

$$\frac{m}{n} = \frac{a^2(4+2\sqrt{2})}{2(2a^2(1+\sqrt{2}))}$$

$$\frac{m}{n} = \frac{4+2\sqrt{2}}{4+4\sqrt{2}}$$

$$\frac{m}{n} = \frac{2+\sqrt{2}}{2+2\sqrt{2}}$$

Rationalizing the denominator by multiplying by the conjugate, we get $\frac{m}{n} =$

(B) $\frac{\sqrt{2}}{2}$

ciceronii

- **Note:** this can more easily be done if you know any of these formulas. This was an entire derivation of the area of an octagon, it's apothem, and it's circumradius, but it can be much simpler if you have any of these memorized.
- **Note No.2:** The formula $[ABCDEFGH] = (\sqrt{2})[ACEG]$ is one from the Intro to Geometry Textbook: Problem 16.46.
- **Note No.3:** after trying my self, setting the sides as 1 would be much easier to solve as numbers easily cancel out

Solution 3 (Elementary Geometry)

WLOG, let $AE = 1$. Let the intersection of AF and GD be point I . GIF is an isosceles right triangle ($m\angle HGF = 135^\circ$), so $IG = IF = \frac{1}{\sqrt{2}}$. The distance between each side and the center is then $IF + \frac{1}{2}GH = \frac{1}{2} + \frac{1}{\sqrt{2}}$. $ABCDEFGH$ is 8 triangles of base 1 and altitude $\frac{1}{2} + \frac{1}{\sqrt{2}}$, so its area is $8(\frac{1}{2})(\frac{1}{2} + \frac{1}{\sqrt{2}})$ or

$$2 + 2\sqrt{2}$$

Similarly, $ACEG$ is clearly a square of area GA^2 . By the Pythagorean Theorem, $GA^2 = GI^2 + IA^2 = (\frac{1}{\sqrt{2}})^2 + (1 + \frac{1}{\sqrt{2}})^2$ or

$$2 + \sqrt{2}$$

$$\frac{m}{n} = \frac{2 + \sqrt{2}}{2 + 2\sqrt{2}}$$

From some fast manipulations, see that it is **B)** $\frac{\sqrt{2}}{2}$

Solution 4 (Quick n' Easy)

This solution is quite efficient and unconvoluted. Though the explanation appears long, this is only because I have tried to make it very thorough.

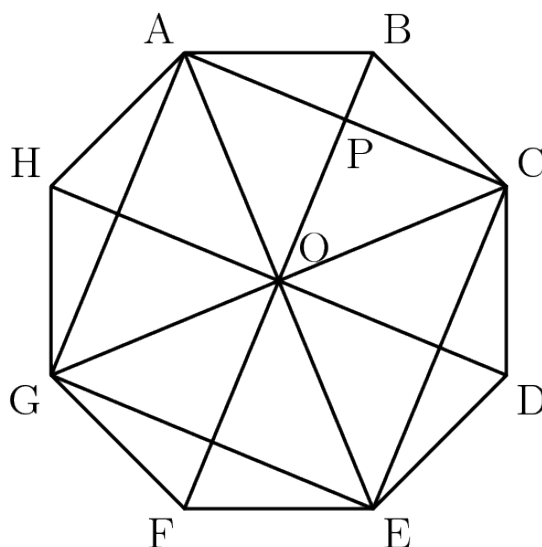
Because we are finding the ratios of the areas of the quadrilateral, and the octagon, let us assume, for simplicity's sake, that the octagon has side length 1. Call O the center of the octagon. Because it is a regular octagon, $\angle AOB$ is $\frac{360}{8} = 45$. Draw the apothem, and call where it intersects AB , I . As the apothem bisects $\angle AOB$, $\angle AIB$ is $\frac{45}{2} = 22.5$. Then, as the apothem is perpendicular to AB , $\angle IBO$ is $90 - 22.5 = 67.5$. As the apothem bisects AB , segment IB is $\frac{1}{2}$. Thus, the apothem has length $\frac{1}{2} \cdot \tan(67.5)$. By the formula $\text{area} = \text{apothem} \cdot \frac{\text{perimeter}}{2}$, the area of the octagon is then $\frac{1}{2} \cdot \tan(67.5) \cdot \frac{8}{2}$.

Now, we move on to the quadrilateral. Let us take triangle AGH . By the interior angle formula, $m\angle AGH = \frac{180(8-2)}{8} = 135$. Because the octagon is regular, AGH is isosceles, with assumed sidelength 1. Drop an altitude from H to AG , and call the intersection point, P . As AGH is isosceles, this altitude bisects both segment AG and angle AHG . Thus, the $m\angle AHP = \frac{135}{2} = 67.5$. Let us call the length of AG , which is also a side of the quadrilateral, s . Then, AP is $\frac{s}{2}$. Thus, $\sin(67.5) = \frac{\frac{s}{2}}{1}$, so $s = 2 \cdot \sin(67.5)$. We can do the same for each of the other sides of the quadrilateral and find that they are of the same length. One can easily see that triangle ABC is congruent to triangle AGH . Because triangle AGH is isosceles, $m\angle HAG = \frac{180-135}{2} = 22.5$. The same can be said of angles BAC . Thus, angle CAG , an interior angle of the quadrilateral is 90. We can do the same for the other interior angles of $CEGA$ and find that they are also 90. Thus, $CEGA$ is a square, so its area is $s^2 = 4 \cdot \sin^2(67.5)$.

Now, we must find the ratios of the two areas. $\frac{\text{Area}(\text{quadrilateral})}{\text{Area}(\text{hexagon})} = 4 \cdot \frac{\sin^2(67.5)}{2 \cdot \tan(67.5)} = \frac{2 \cdot \sin^2(67.5)}{\tan(67.5)} = 2 \cdot \sin(67.5) \cdot \cos(67.5) = \sin(135) = \frac{\sqrt{2}}{2}$, the answer.

-jyatvitskiy :)

Solution 5 (Simplest)



The ratio $\frac{m}{n} = \frac{[OPC]}{[OBC]} = \frac{OP}{OB} = \frac{OP}{OC} = \frac{\sqrt{2}}{2}$.

«ERMS Coach»

Solution 6 (Law of Sines, similar to Solution 5)

Let O be the center of the octagon. Let AO with HB intersect at X . By symmetry, the problem reduces to finding the ratio between the area of the triangle ABO and

OXB , which is equal to $\frac{\frac{1}{2}XO \cdot BO \sin \theta}{\frac{1}{2}AO \cdot BO \sin \theta} = \frac{XO}{AO} = \frac{XO}{BO} = \boxed{\frac{\sqrt{2}}{2}}$. Here, θ represents the angle AOB .

15. (AMC 12 12A 2020) In the complex plane, let A be the set of solutions to $z^3 - 8 = 0$ and let B be the set of solutions to $z^3 - 8z^2 - 8z + 64 = 0$. What is the greatest distance between a point of A and a point of B ? (A) $2\sqrt{3}$ (B) 6 (C) 9 (D) $2\sqrt{21}$ (E) $9 + \sqrt{3}$

Answer: D

Solution 1

We solve each equation separately:

- a) We solve $z^3 - 8 = 0$ by De Moivre's theorem. Let $z = r(\cos \theta + i \sin \theta) = r \operatorname{cis} \theta$, where r is the magnitude of z such that $r \geq 0$, and θ is the argument of z such that $0 \leq \theta < 2\pi$.

We have

$$z^3 = r^3 \operatorname{cis}(3\theta) = 8(1),$$

from which

$$\begin{aligned} r^3 = 8, \text{ so } r = 2. \quad & \begin{cases} \cos(3\theta) = 1 \\ \sin(3\theta) = 0 \end{cases} \quad \text{so } 3\theta = 0, 2\pi, 4\pi, \text{ or} \\ \theta = 0, \frac{2\pi}{3}, \frac{4\pi}{3}. \end{aligned}$$

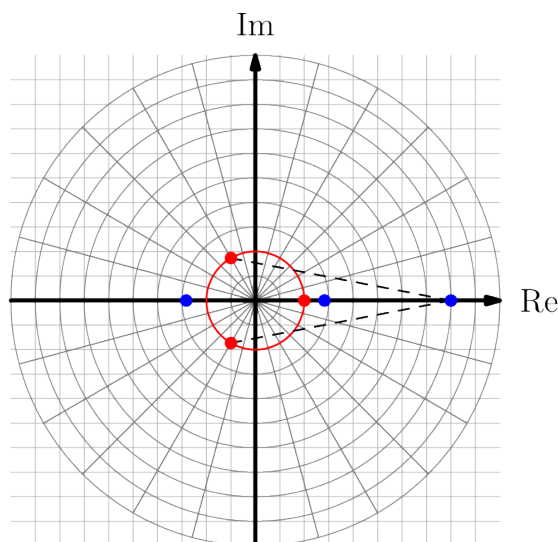
The set of solutions to $z^3 - 8 = 0$ is $A = \{2, -1 + \sqrt{3}i, -1 - \sqrt{3}i\}$. In the complex plane, the solutions form the vertices of an equilateral triangle whose circumcircle has center 0 and radius 2.

- b) We solve $z^3 - 8z^2 - 8z + 64 = 0$ by factoring by grouping. We have

$$\begin{aligned} z^2(z - 8) - 8(z - 8) &= 0 \\ (z^2 - 8)(z - 8) &= 0. \end{aligned}$$

The set of solutions to $z^3 - 8z^2 - 8z + 64 = 0$ is $B = \{2\sqrt{2}, -2\sqrt{2}, 8\}$.

In the graph below, the points in set A are shown in red, and the points in set B are shown in blue. The greatest distance between a point of A and a point of B is the distance between $-1 \pm \sqrt{3}i$ to 8, as shown in the dashed line segments.



By the distance formula, the answer is

$$\sqrt{(-1-8)^2 + (\pm\sqrt{3}-0)^2} = \sqrt{84} = \boxed{\text{(D)} 2\sqrt{21}}.$$

lopkiloinm MRENTHUSIASM

Solution 2

Alternatively, we can solve $z^3 - 8 = 0$ by the difference of cubes:

$$(z-2)(z^2+2z+4) = 0.$$

- If $z-2=0$, then $z=2$.
- If $z^2+2z+4=0$, then $z=-1 \pm \sqrt{3}i$ by either completing the square or the quadratic formula.

The set of solutions to $z^3 - 8 = 0$ is $A = \{2, -1 + \sqrt{3}i, -1 - \sqrt{3}i\}$.

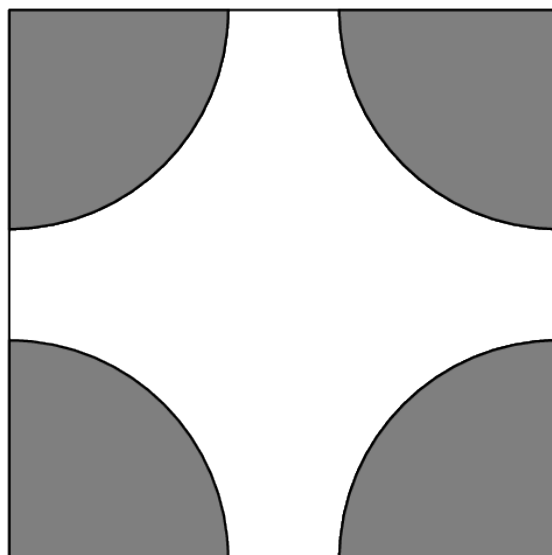
Following the rest of Solution 1 gives the answer $\boxed{\text{(D)} 2\sqrt{21}}$.

16. (AMC 12 12A 2020) A point is chosen at random within the square in the coordinate plane whose vertices are $(0, 0)$, $(2020, 0)$, $(2020, 2020)$, and $(0, 2020)$. The probability that the point is within d units of a lattice point is $\frac{1}{2}$. (A point (x, y) is a lattice point if x and y are both integers.) What is d to the nearest tenth?
 (A) 0.3 (B) 0.4 (C) 0.5 (D) 0.6 (E) 0.7

Answer: B

Solutions

Diagram



The diagram represents each unit square of the given 2020×2020 square.

Solution 1

We consider an individual one-by-one block.

If we draw a quarter of a circle from each corner (where the lattice points are located), each with radius d , the area covered by the circles should be 0.5. Because of this, and the fact that there are four circles, we write

$$4 \cdot \frac{1}{4} \cdot \pi d^2 = \frac{1}{2}$$

Solving for d , we obtain $d = \frac{1}{\sqrt{2\pi}}$, where with $\pi \approx 3$, we get $d \approx \frac{1}{\sqrt{6}} \approx \frac{1}{2.5} = \frac{10}{25} = \frac{2}{5}$, and from here, we see that $d \approx 0.4 \implies \boxed{\text{(B)} 0.4}$. **Note:** To be more rigorous, note that $d < 0.5$ since if $d \geq 0.5$ then clearly the probability is greater than $\frac{1}{2}$. This would make sure the above solution works, as if $d \geq 0.5$ there is overlap with the quartercircles. - **Emathmaster**

Solution 2

As in the previous solution, we obtain the equation $4 \cdot \frac{1}{4} \cdot \pi d^2 = \frac{1}{2}$, which simplifies to $\pi d^2 = \frac{1}{2} = 0.5$. Since π is slightly more than 3, d^2 is slightly less than $\frac{0.5}{3} = 0.1\bar{6}$. We notice that $0.1\bar{6}$ is slightly more than $0.4^2 = 0.16$, so d is roughly $\boxed{\text{(B)} 0.4}$.
emerald_block

Solution 3 (Estimating)

As above, we find that we need to estimate $d = \frac{1}{\sqrt{2\pi}}$.

Note that we can approximate $2\pi \approx 6.28318 \approx 6.25$ and so $\frac{1}{\sqrt{2\pi}} \approx \frac{1}{\sqrt{6.25}} = \frac{1}{2.5} = 0.4$.

And so our answer is $\boxed{\text{(B)} 0.4}$.

Solution 4 (Estimating but a bit different)

We only need to figure out the probability for a unit square, as it will scale up to the 2020×2020 square. Since we want to find the probability that a point inside a unit square that is d units away from a lattice point (a corner of the square) is $\frac{1}{2}$, we can find which answer will come the closest to covering $\frac{1}{2}$ of the area.

Since the closest is 0.4 which turns out to be $(0.4)^2 \times \pi = 0.16 \times \pi$ which is about 0.502, we find that the answer rounded to the nearest tenth is 0.4 or **(B)**.

Solution 5 (Estimating but differently again)

As per the above diagram, realize that $\pi d^2 = \frac{1}{2}$, so $d = \frac{1}{(\sqrt{2})(\sqrt{\pi})}$.

$$\sqrt{2} \approx 1.4 = \frac{7}{5}.$$

$\sqrt{\pi}$ is between 1.7 and 1.8 ($(1.7)^2 = 2.89$ and $(1.8)^2 = 3.24$), so we can say $\sqrt{\pi} \approx 1.75 = \frac{7}{4}$.

So $d \approx \frac{1}{(\frac{7}{5})(\frac{7}{4})} = \frac{1}{\frac{49}{20}} = \frac{20}{49}$. This is slightly above **(B) 0.4**, since $\frac{20}{49} \approx \frac{2}{5}$.

-Solution by Joeya

Solution 6 (Estimating but differently again, again)

As above, we have the equation $\pi d^2 = \frac{1}{2}$, and we want to find the most accurate value of d . We resort to the answer choices and can plug those values of d in and see which value of d will lead to the most accurate value of π .

Starting off in the middle, we try option C with $d = 0.5$. Plugging this in, we get $\pi \left(\frac{1}{2}\right)^2 = \frac{1}{2}$, and after simplifying we get $\pi = \frac{1}{2} \cdot 4 = 2$. That's not very good. We know $\pi \approx 3.14$.

Let's see if we can do better. Trying option A with $d = 0.3$, we get $\pi = \frac{1}{2} \cdot \frac{100}{9} = \frac{50}{9} = 5\frac{5}{9}$.

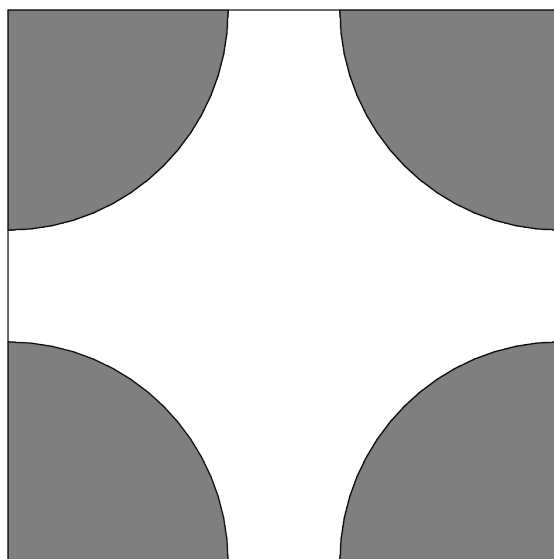
Hm, let's try option B with $d = 0.4$. We get $\pi = \frac{1}{2} \cdot \frac{25}{4} = \frac{25}{8} = 3\frac{1}{8}$. This is very close to π and is the best estimate for π of the 5 options.

Therefore, the answer is **(B) 0.4**. epiconan

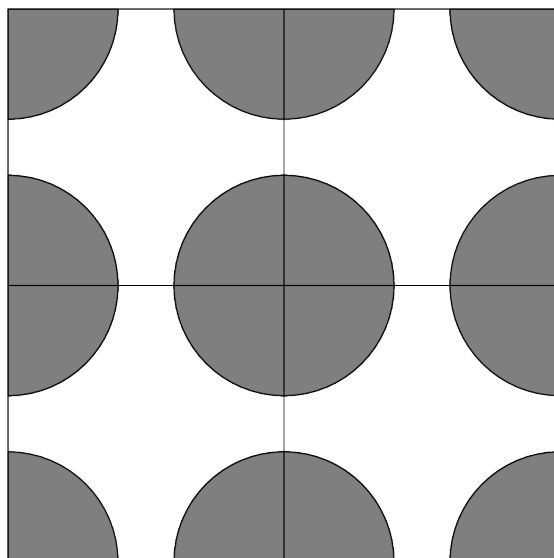
Solution 7 (Sol. 1, but rigorous (and excessive))

PLEASE NOTE: Solution 1 IS rigorous. Say there are k unit squares (it doesn't matter how many). There is a $\frac{1}{k}$ probability the point is in some unit square. There is a πd^2 probability the point is in the shaded region. So, there is a $\frac{1}{k} \pi d^2 \cdot k = \pi d^2$ probability the point is in any shaded region (since there are k unit squares).

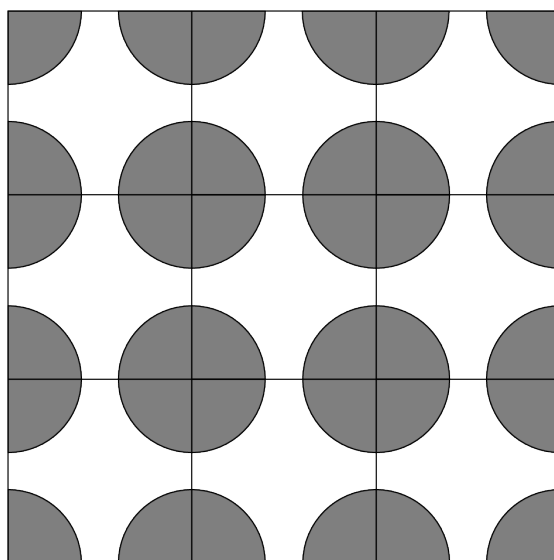
Let n be the side length of a square. When $n = 1$, the shaded areas represent half of the total area:



When $n = 2$:



For $n = 3$:



We can calculate the total number of shaded circles given some n . There are $(n-1)^2$ full circles on the inside, $4(n-1)$ semicircles on the sides, and 4 quarter circles for the corners.

Full circles are, of course, worth one circle. Semicircles are worth half a circle each, and quarter circles are worth $\frac{1}{4}$ of a circle. Thus, weighing our sum gives $(n-1)^2 + \frac{4(n-1)}{2} + \frac{4}{4} = n^2 - 2n + 1 + 2(n-1) + 1 = n^2 - 2n + 2n + 2 - 2 = n^2$. Thus, there is $n^2 \cdot \pi r^2$ worth of the shaded area for any n , and since the area of each circle is πr^2 if r is the radius of each.

We want the ratio of this shaded area to the entire to be $\frac{1}{2}$. The area of the entire square is n^2 , so dividing, we see that $\frac{n^2 \cdot \pi r^2}{n^2} = \pi r^2 = \frac{1}{2}$.

The rest is the same as solution 1.

Solution 8 (most simple, easy ,estimation)

First, let's test what happens if $d = 0.5$. If $d = 0.5$, the area of the circle around each lattice point is $\pi(0.5)^2 = 0.25\pi$. Since each lattice point corresponds to a 1×1 square of area, the probability of the point falling within 0.5 units is exactly $\frac{0.25\pi}{1} \approx 0.785$. Because this strictly exceeds our target probability of $\frac{1}{2}$, we know that $d < 0.5$. This conveniently eliminates 0.5, 0.6, and 0.7, and proves that the circles centered at the lattice points do not overlap.

Since the circles do not overlap, we can simply find the combined favorable area of the circles (and partial circles) inside the 2020×2020 square.

We can break this down into three types of lattice points: There are 2019^2 interior lattice points, each contributing a full circle of area πd^2 . There are $4 \cdot 2019$ edge lattice points (excluding corners), each contributing a semicircle of area $\frac{1}{2}\pi d^2$. There are 4 corner lattice points, each contributing a quarter-circle of area $\frac{1}{4}\pi d^2$.

Summing these gives the total favorable area: $2019^2\pi d^2 + 4(2019)\left(\frac{1}{2}\pi d^2\right) + 4\left(\frac{1}{4}\pi d^2\right)$

Factoring out the πd^2 , we get a perfect square: $= (2019^2 + 2(2019) + 1)\pi d^2 = (2019 + 1)^2\pi d^2 = 2020^2\pi d^2$.

The total area of the bounding 2020×2020 square is 2020^2 . Therefore, the probability is the ratio of the favorable area to the total area: $\frac{2020^2\pi d^2}{2020^2} = \pi d^2$.

We are given that the probability is $\frac{1}{2}$, so we equate and solve: $\pi d^2 = \frac{1}{2} \implies d^2 = \frac{1}{2\pi}$.

Solving for d gives $d = \sqrt{\frac{1}{2\pi}} \approx \sqrt{0.159}$. Since $0.4^2 = 0.16$, we can clearly see that $d \approx 0.4$.

by chundoowan

17. (AMC 12 12A 2020) The vertices of a quadrilateral lie on the graph of $y = \ln x$, and the x -coordinates of these vertices are consecutive positive integers. The area of the quadrilateral is $\ln \frac{91}{90}$. What is the x -coordinate of the leftmost vertex?
- (A) 6 (B) 7 (C) 10 (D) 12 (E) 13

Answer: D

Solution 1

Let the coordinates of the quadrilateral be $(n, \ln(n))$, $(n+1, \ln(n+1))$, $(n+2, \ln(n+2))$, $(n+3, \ln(n+3))$. We have by shoelace's theorem, that the area is

$$\begin{aligned} & \frac{\ln(n)(n+1) + \ln(n+1)(n+2) + \ln(n+2)(n+3) + n \ln(n+3)}{2} - \frac{\ln(n+1)(n) + \ln(n+2)(n+1) + \ln(n+3)(n+2) + (n+3)\ln(n)}{2} \\ &= \frac{\ln \left(\frac{n^{n+1}(n+1)^{n+2}(n+2)^{n+3}(n+3)^n}{(n+1)^n(n+2)^{n+1}(n+3)^{n+2}n^{n+3}} \right)}{2} \\ &= \ln \left(\sqrt{\frac{(n+1)^2(n+2)^2}{n^2(n+3)^2}} \right) \end{aligned}$$

$$\begin{aligned}
 &= \ln \left(\frac{(n+1)(n+2)}{n(n+3)} \right) \\
 &= \ln \left(\frac{91}{90} \right).
 \end{aligned}$$

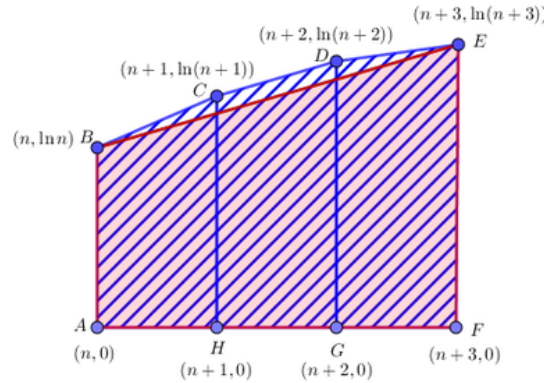
We know that the numerator must have a factor of 13, so given the answer choices, n is either 12 or 11. If $n = 11$, the expression $\frac{(n+1)(n+2)}{n(n+3)}$ does not evaluate to $\frac{91}{90}$, but if $n = 12$, the expression evaluates to $\frac{91}{90}$. Hence, our answer is $\boxed{12}$.

Solution 2

Like above, use the shoelace formula to find that the area of the quadrilateral is equal to $\ln \frac{(n+1)(n+2)}{n(n+3)}$. Because the final area we are looking for is $\ln \frac{91}{90}$, the numerator factors into 13 and 7, which one of $n+1$ and $n+2$ has to be a multiple of 13 and the other has to be a multiple of 7. Clearly, the only choice for that is $\boxed{12}$

Solution 3

How $f(x) = \ln(x)$ is a concave function, then:



Therefore $[BCDE] = [ABCH] + [HCDG] + [GDEF] - [ABEF]$, all quadrilaterals of side right are trapezium

$$[BCDE] = \frac{\ln(n+1) + \ln n}{2} + \frac{\ln(n+2) + \ln(n+1)}{2} + \frac{\ln(n+3) + \ln(n+2)}{2} - \frac{3(\ln(n+3) + \ln n)}{2}$$

$$[BCDE] = \frac{2\ln(n+1) + 2\ln(n+2) - 2\ln(n+3) - 2\ln n}{2} = \ln(n+1) + \ln(n+2) - \ln(n+3) - \ln n = \ln \frac{(n+1)(n+2)}{n(n+3)}$$

$$\implies \ln \frac{(n+1)(n+2)}{n(n+3)} = \ln \frac{91}{90} \implies \frac{(n+1)(n+2)}{n(n+3)} = \frac{91}{90}$$

$$\implies n^2 + 3n - 180 = 0 \implies (n+15)(n-12) = 0 \implies n = 12$$

comment by observer: brilliant way! I find this method much easier than the shoe lace one as it is pretty geometry.

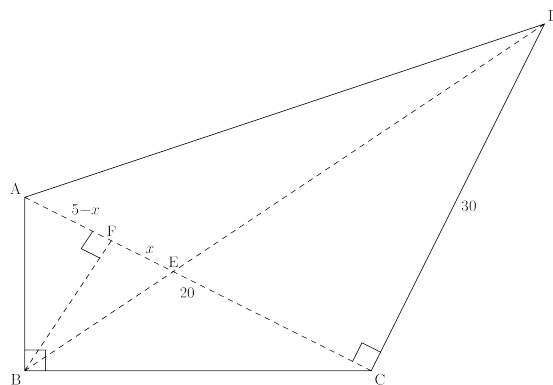
comment by lamont: i love this solution! It doesn't require as much algebra as the shoelace theorem (which i have no idea what it is lol)

18. (AMC 12 12A 2020) Quadrilateral $ABCD$ satisfies $\angle ABC = \angle ACD = 90^\circ$, $AC = 20$, and $CD = 30$. Diagonals $\overline{AC}$ and $\overline{BD}$ intersect at point E , and $AE = 5$. What is the area of quadrilateral $ABCD$?

(A) 330 (B) 340 (C) 350 (D) 360 (E) 370

Answer: D

Solution 1



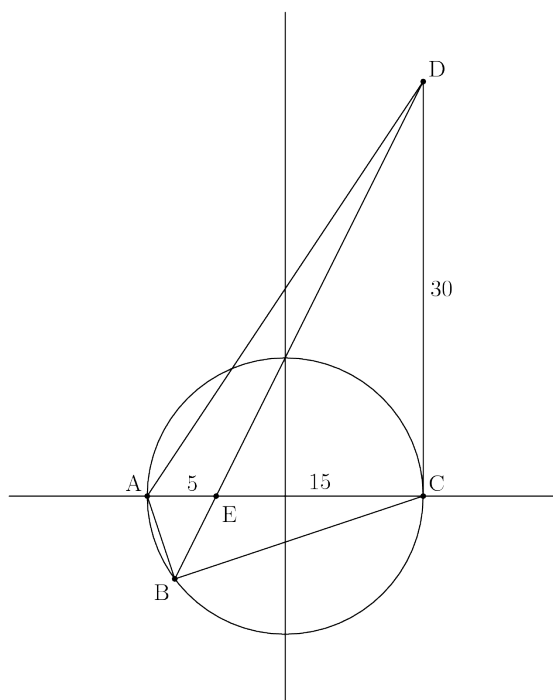
It's crucial to draw a good diagram for this one. Since $AC = 20$ and $CD = 30$, we get $[ACD] = 300$. Now we need to find $[ABC]$ to get the area of the whole quadrilateral. Drop an altitude from B to AC and call the point of intersection F . Let $FE = x$. Since $AE = 5$, then $AF = 5 - x$.

By dropping this altitude, we can also see two similar triangles, $\triangle BFE \sim \triangle DCE$. Since EC is $20 - 5 = 15$, and $DC = 30$, we get that $BF = 2x$.

Now, if we redraw another diagram just of ABC , we get that $(2x)^2 = (5 - x)(15 + x)$ because of the altitude geometric mean theorem which states that in any right triangle, the altitude squared is equal to the product of the two lengths that it divides the base into.

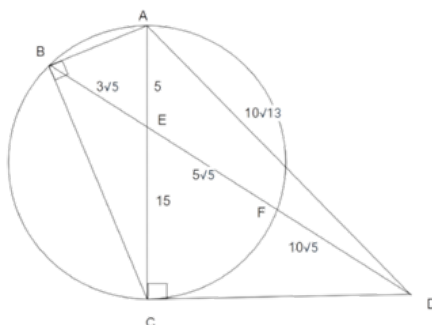
Expanding, simplifying, and dividing by the GCF, we get $x^2 + 2x - 15 = 0$. This factors to $(x + 5)(x - 3)$, which has roots of $x = -5, 3$. Since lengths cannot be negative, $x = 3$. Since $x = 3$, that means the altitude $BF = 2 \cdot 3 = 6$, or $[ABC] = 60$. Thus $[ABCD] = [ACD] + [ABC] = 300 + 60 = \boxed{\text{(D) } 360}$

Solution 2 (Coordinates)



Let the points be $A(-10, 0)$, $B(x, y)$, $C(10, 0)$, $D(10, 30)$, and $E(-5, 0)$, respectively. Since B lies on line DE , we know that $y = 2x + 10$. Furthermore, since $\angle ABC = 90^\circ$, B lies on the circle with diameter AC , so $x^2 + y^2 = 100$. Solving for x and y with these equations, we get the solutions $(0, 10)$ and $(-8, -6)$. We immediately discard the $(0, 10)$ solution as y should be negative. Thus, we conclude that $[ABCD] = [ACD] + [ABC] = \frac{20 \cdot 30}{2} + \frac{20 \cdot 6}{2} = \boxed{\text{(D) } 360}$.

Solution 3 (Power of A Point)



Draw the circumcircle of $\triangle ABC$. Let $BE = x$, and let F be the point where the circumcircle of $\triangle ABC$ meets BD . By the Pythagorean Theorem, $ED = \sqrt{1125}$, so $\frac{75}{x} + DF = \sqrt{1125} \implies DF = \sqrt{1125} - \frac{75}{x}$. ($EF = \frac{75}{x}$ by the Intersecting Chords Theorem.) From Power of A Point we have $CD^2 = 30^2 = FD \cdot BD = FD^2 + FD \cdot BF = (\sqrt{1125} - \frac{75}{x})^2 + (\sqrt{1125} - \frac{75}{x})(x + \frac{75}{x}) = (\sqrt{1125} - \frac{75}{x})(\sqrt{1125} + x)$. Solving we get $x = 3\sqrt{5}$. $\sin \angle CED = \frac{30}{15\sqrt{5}}$. The area of a quadrilateral is half the product of its diagonals times the sine of the angle between them, so the answer is $18\sqrt{5} \cdot 20 \cdot \frac{30}{15\sqrt{5}} \cdot \frac{1}{2} = 180 \cdot 2 = \boxed{360}$.

Solution 4 (Trigonometry)

Let $\angle C = \angle ACB$ and $\angle B = \angle CBE$. Using Law of Sines on $\triangle BCE$ we get

$$\frac{BE}{\sin C} = \frac{CE}{\sin B} = \frac{15}{\sin B}$$

and LoS on $\triangle ABE$ yields

$$\frac{BE}{\sin(90 - C)} = \frac{5}{\sin(90 - B)} = \frac{BE}{\cos C} = \frac{5}{\cos B}.$$

Divide the two to get $\tan B = 3 \tan C$. Now,

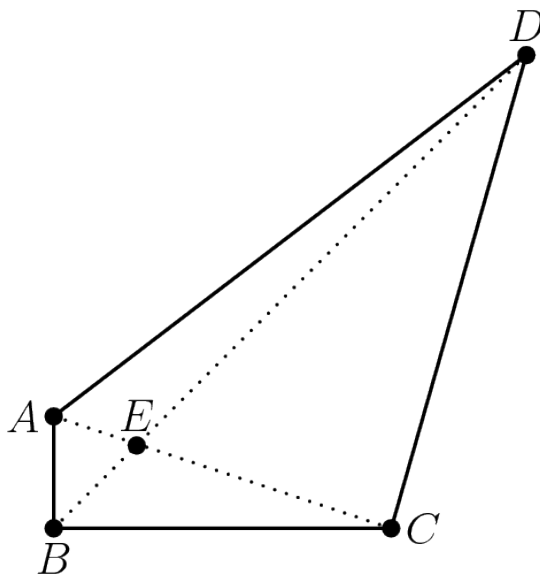
$$\tan \angle CED = 2 = \tan \angle B + \angle C = \frac{4 \tan C}{1 - 3 \tan^2 C}$$

and solve the quadratic, taking the positive solution (C is acute) to get $\tan C = \frac{1}{3}$. So if $AB = a$, then $BC = 3a$ and $[ABC] = \frac{3a^2}{2}$. By Pythagorean Theorem, $10a^2 = 400 \iff \frac{3a^2}{2} = 60$ and the answer is $300 + 60 \iff \boxed{\text{(D)}}$.

(This solution is incomplete, can someone complete it please)Edit by kc5170

We could use the famous m-n rule in trigonometry in $\triangle ABC$ with Point E [Unable to write it here. Could anybody write the expression]. We will find that $\overrightarrow{BD}$ is an angle bisector of $\triangle ABC$ (because we will get $\tan(x) = 1$). Therefore by converse of angle bisector theorem $AB : BC = 1 : 3$. By using Pythagorean theorem, we have values of AB and AC . Computing $AB \cdot AC = 120$. Adding the areas of ABC and ACD , hence the answer is $\boxed{\text{(D) } 360}$.

Solution 5 (Law of Cosines)



Denote EB as x . By the Law of Cosines:

$$AB^2 = 25 + x^2 - 10x \cos(\angle DEC)$$

$$BC^2 = 225 + x^2 + 30x \cos(\angle DEC)$$

Adding these up yields:

$$400 = 250 + 2x^2 + 20x \cos(\angle DEC) \implies x^2 + \frac{10x}{\sqrt{5}} - 75 = 0$$

Observe:

•

Solution 6 (Vectors / Coordinates)

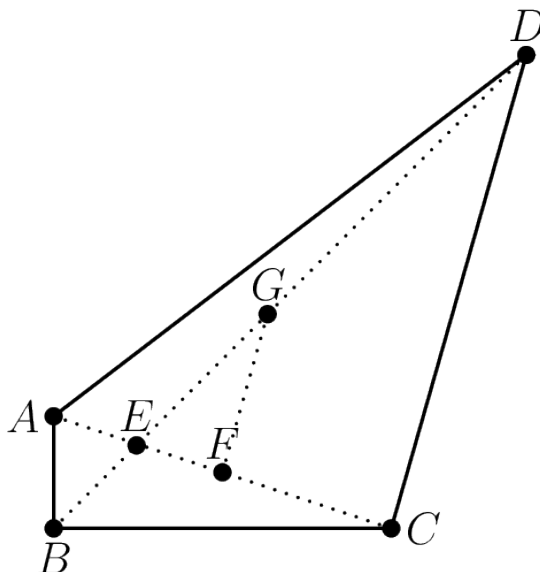
$$(x, 2x + 30).$$

We determine $\overrightarrow{\mathbf{BC}}$ and $\overrightarrow{\mathbf{BA}}$ to be $\begin{pmatrix} -x \\ -2x - 30 \end{pmatrix}$ and $\begin{pmatrix} -20 - x \\ -2x - 30 \end{pmatrix}$, respectively. (To get this, we use the fact that $\overrightarrow{\mathbf{BC}} = \overrightarrow{\mathbf{C}} - \overrightarrow{\mathbf{B}}$ and similarly, $\overrightarrow{\mathbf{BA}} = \overrightarrow{\mathbf{A}} - \overrightarrow{\mathbf{B}}$.)

From here, there are multiple ways to get the area of $\triangle ABC$ to be 60, and since the area of $\triangle ACD$ is 300, we get our final answer to be

$$60 + 300 = \boxed{(D) 360}.$$

Solution 7 (Power of a Point)



Let F be the midpoint of AC , and draw $FG \parallel CD$ where G is on BD . We have $EF = 5, FC = 10$.

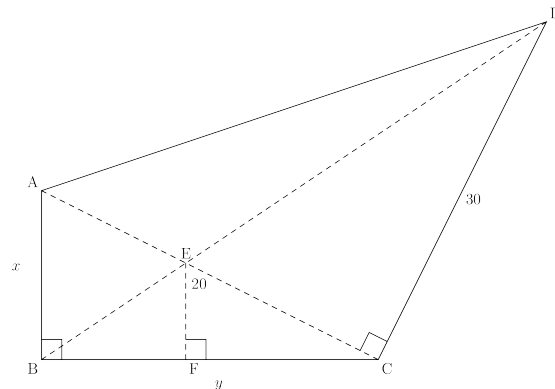
$\triangle EFG \sim \triangle ECD \implies FG = 10 = FA = FC$. Therefore $ABCG$ is a cyclic quadrilateral.

Notice that $\angle EFG = 90^\circ, EG = \sqrt{5^2 + 10^2} = 5\sqrt{5} \implies BE = \frac{AE \cdot EC}{EG} = \frac{5 \cdot 15}{5\sqrt{5}} = 3\sqrt{5}$ via Power of a Point.

The altitude from B to AC is then equal to $GF \cdot \frac{BE}{GE} = 10 \cdot \frac{3\sqrt{5}}{5\sqrt{5}} = 6$.

Finally, the total area of $ABCD$ is equal to $\frac{1}{2} \cdot 20(30 + 6) = \boxed{(D) 360}$.

Solution 8 (Solving Equations)



Let $AB = x, BC = y$

Looking at the diagram we have $x^2 + y^2 = 20^2, DE = \sqrt{30^2 + 15^2} = 15\sqrt{5}, [ACD] = \frac{1}{2} \cdot 20 \cdot 30 = 300$

Because $\triangle CEF \sim \triangle CAB, EF = AB \cdot \frac{CE}{CA} = x \cdot \frac{15}{20} = \frac{3x}{4} BF = BC - CF = BC - BC \cdot \frac{CE}{CA} = \frac{1}{4} \cdot BC = \frac{y}{4} BE = \sqrt{\left(\frac{3x}{4}\right)^2 + \left(\frac{y}{4}\right)^2} = \frac{\sqrt{9x^2 + y^2}}{4}$, substituting $x^2 + y^2 = 400$, we get $BE = \frac{\sqrt{8x^2 + 400}}{4} = \frac{\sqrt{2x^2 + 100}}{2} [ABC] = \frac{1}{2} \cdot x \cdot y$

Because $\triangle ABC$ and $\triangle ACD$ share the same base, $\frac{[ABC]}{[ACD]} = \frac{BE}{DE} [ABC] = [ACD] \cdot$

$$\frac{BE}{DE} = 300 \cdot \frac{\frac{\sqrt{2x^2 + 100}}{2}}{15\sqrt{5}} \cdot \frac{1}{2} \cdot x \cdot y = 20 \cdot \frac{\sqrt{2x^2 + 100}}{\sqrt{5}} xy = 4\sqrt{10x^2 + 500}$$

By $x^2 + y^2 = 400, y = \sqrt{400 - x^2}$. So, $x \cdot \sqrt{400 - x^2} = 4\sqrt{10x^2 + 500} x^2(400 - x^2) = 16(10x^2 + 500)$

Let $x^2 = a, a(400 - a) = 16(10a + 500), 400a - a^2 = 160a + 8000, a^2 - 240a + 8000 = 0, (a - 200)(a - 40) = 0$

Because $x < 20, a$ can only equal 40. $a = 40, x = 2\sqrt{10}, y = 6\sqrt{10} [ABC] = \frac{1}{2} \cdot 2\sqrt{10} \cdot 6\sqrt{10} = 60 [ABCD] = [ABC] + [ACD] = 60 + 300 = \boxed{(D) 360}$

Solution 9

Drop perpendiculars $\overline{AF}$ and $\overline{CG}$ to $\overline{BD}$. Notice that since $\angle AEF = \angle CEG$ (since they are vertical angles) and $\angle AFE = \angle CGE = 90^\circ$, triangles AEF and CEG are similar. Therefore, we have

$$x/EF = CE/AE = 15/5 = 3,$$

where $EG = x$. Therefore, $EF = x/3$.

Additionally, angle chasing shows that triangles CEG and DCG are also similar. This gives $CG/x = DC/CE = 30/15 = 2$, so $CG = 2x$. Thus, applying the Pythagorean Theorem to triangle CEG gives

$$x^2 + (2x)^2 = 15^2,$$

so $EG = x = 3\sqrt{5}$. Our pairs of similar triangles then allow us to fill in the following lengths (in this order):

$$EF = x/3 = \sqrt{5}, CG = 2x = 6\sqrt{5}, AF = CG/3 = 2\sqrt{5}, DG = 2 \cdot CG = 12\sqrt{5}.$$

Now, let $BF = y$. Angle chasing shows that triangle ABF and BCG are similar, so $BG/AF = CG/BF$. Plugging in known lengths gives

$$\frac{y + 4\sqrt{5}}{2\sqrt{5}} = \frac{6\sqrt{5}}{y}.$$

This gives $y = 2\sqrt{5}$. Now we know all the lengths that make up BD , which allows us to find

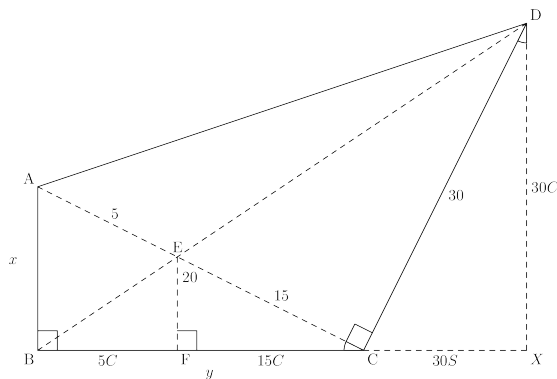
$$BD = 2\sqrt{5} + \sqrt{5} + 3\sqrt{5} + 12\sqrt{5} = 18\sqrt{5}.$$

Therefore,

$$\begin{aligned} [ABCD] &= [ABD] + [CBD] \\ &= (BD)(AF)/2 + (BD)(CG)/2 \\ &= (18\sqrt{5})(2\sqrt{5})/2 + (18\sqrt{5})(6\sqrt{5})/2 \\ &= \boxed{(D) \ 360}. \end{aligned}$$

–vaporwave

Solution 10 (Trigonometry)



set

$$\angle ACB = \theta, C = \cos(\theta), S = \sin(\theta)$$

$$\frac{E_y}{E_x} = \frac{30C}{20C + 30S} = \frac{15S}{20C - 15C}$$

$$2SC = \frac{3}{5}$$

$$\begin{aligned} [ABCD] &= [ABD] + [CBD] \\ &= \frac{1}{2} \cdot 20C \cdot 30C + \frac{1}{2} \cdot 20S(20C + 30S) \\ &= 100 \cdot 2SC + 300 \\ &= \boxed{\text{(D) } 360}. \end{aligned}$$

19. (AMC 12 12A 2020) There exists a unique strictly increasing sequence of nonnegative integers $a_1 < a_2 < \dots < a_k$ such that

$$\frac{2^{289} + 1}{2^{17} + 1} = 2^{a_1} + 2^{a_2} + \dots + 2^{a_k}.$$

What is k ? (A) 117 (B) 136 (C) 137 (D) 273 (E) 306

Answer: C

Solution 1 (Simplest)

If we begin with a common identity or just try polynomial long division on $\frac{2^{289}+1}{2^{17}+1}$, we quickly find it is equal to $(2^{272} - 2^{255} \pm \dots + 1)$. We recognize that we want all of these differences to turn into sums, and so we express this as $(2^{255} + 2^{221} + \dots + 2^{17})(2^{17} - 1) + 1$ (we grouped every two terms and factored out $2^{17} - 1$). If we keep this product in its current form, the -1 term on the second factor will create subtraction which we don't want, so we express this as the sum of a geometric sequence, which is $2^{16} + 2^{15} + \dots + 2^0$, so we have $(2^{255} + 2^{221} + \dots + 2^{17})(2^{16} + 2^{15} + \dots + 2^0) + 1$, and when we expand there are $8(17) + 1 = 137$ terms in this sequence, so our answer is **(C) 137**

- SumDumNerd

Solution 2

First, substitute 2^{17} with x . Then, the given equation becomes $\frac{x^{17}+1}{x+1} = x^{16} - x^{15} + x^{14} \dots - x^1 + x^0$ by sum of powers factorization. Now consider only $x^{16} - x^{15}$. This equals $x^{15}(x - 1) = x^{15} \cdot (2^{17} - 1)$. Note that $2^{17} - 1$ equals $2^{16} + 2^{15} + \dots + 1$, by difference of powers factorization (or by considering the expansion of $2^{17} = 2^{16} + 2^{15} + \dots + 2^2 + 2 + 2$). Thus, we can see that $x^{16} - x^{15}$ forms the sum of 17 different powers of 2. Applying the same method to each of $x^{14} - x^{13}$, $x^{12} - x^{11}$, ..., $x^2 - x^1$, we can see that each of the pairs forms the sum of 17 different powers of 2. This gives us $17 \cdot 8 = 136$. But we must count also the x^0 term. Thus, Our answer is $136 + 1 = \boxed{\text{(C) } 137}$.

Solution 3 (Intuitive)

(There is actually an error...) Multiply both sides by $2^{17} + 1$ to get

$$2^{289} + 1 = 2^{a_1} + 2^{a_2} + \dots + 2^{a_k} + 2^{a_1+17} + 2^{a_2+17} + \dots + 2^{a_k+17}.$$

20. (AMC 12 12A 2020) Let T be the triangle in the coordinate plane with vertices $(0, 0)$, $(4, 0)$, and $(0, 3)$. Consider the following five isometries (rigid transformations) of the plane: rotations of 90° , 180° , and 270° counterclockwise around the origin, reflection across the x -axis, and reflection across the y -axis. How many of the 125 sequences of three of these transformations (not necessarily distinct) will return T to its original position? (For example, a 180° rotation, followed by a reflection across the x -axis, followed by a reflection across the y -axis will return T to its original position, but a 90° rotation, followed by a reflection across the x -axis, followed by another reflection across the x -axis will not return T to its original position.)
- (A) 12 (B) 15 (C) 17 (D) 20 (E) 25

Answer: A

Solution 1 (Simplest)

Label each rotation A, B, C, D , and E respectively.

One can notice that any rotation on any figure is formed by the transformation on its vertices. Therefore, instead of thinking about the entire triangle, just think of the point (x, y) .

We proceed with casework on the 180° rotation (B)

Case A: We don't consider B (180° degrees)

We have A, C, D , and E . Notice that we cannot have two reflections with one another as that will finish the sequence in two moves, not three.

We also may not have two rotations, as it will cause the sequence to end in two moves as well.

So we have one rotation and one reflection, but now we have a dilemma. We must choose either another rotation or another reflection.

However, we deduced this is impossible, as it will not take the coordinate back to its original position. Therefore, there are 0 cases that work here.

Case B: We consider the 180° degree

We first look at reflections. If we have two reflections of one kind D or E, we see that the sequence will finish in 2 rounds. Therefore, we must have one reflection D, one reflection E, and one 180° rotation. This means we have a transformation set $\{B, D, E\}$. The elements of this set can be swapped in $3! = 6$ ways.

We now look at rotations. If we have two rotations of A, that is essentially $90^\circ + 90^\circ = 180^\circ$ of rotation, and therefore mixed in with a 180° rotation guarantees the same coordinate (x, y) . So, we have a transformation set $\{A, A, B\}$, which can be arranged in $\frac{3!}{2!} = 3$ ways. What about C ? We can just see that C is just a rotation 90° clockwise, so if we apply 2 of these, we again get $90^\circ + 90^\circ = 180^\circ$ of rotation, giving us enough room for transformation B. Then, our set is $\{C, C, B\}$, which can be again arranged in $\frac{3!}{2!} = 3$ ways.

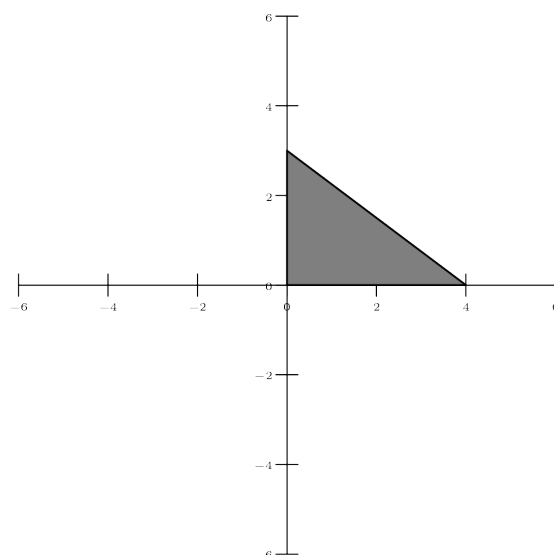
Do any other cases exist? Well, if we have two rotations, then it is impossible, as the rotations negate one another into a 360° rotation in two moves. More importantly, we also see that if we have a mix between a rotation and a reflection, we can see that one will swap the places of x and y $(x, y) \rightarrow (-y, x)$ for instance, and the only other way to change that is another rotation that is not B , which we already deemed was impossible. We already concluded in case A that we cannot have A, C , and then

either D or E , therefore, we can only have B , but either through inspection or seeing that B negates the values of x and y while a rotation of A and C brings them back to normal, therefore, this can also not work.

Thus the total number of cases is $3 + 3 = 6$

We have examined all the cases, and therefore our answer is $6 + 6 = \boxed{\text{(A)}12}$

Solution 2



First, any combination of motions we can make must reflect T an even number of times. This is because every time we reflect T , it changes orientation. Once T has been flipped once, no combination of rotations will put it back in place because it is the mirror image; however, flipping it again changes it back to the original orientation. Since we are only allowed 3 transformations and an even number of them must be reflections, we either reflect T 0 times or 2 times.

Case 1: 0 reflections on T .

In this case, we must use 3 rotations to return T to its original position. Notice that our set of rotations, $\{90^\circ, 180^\circ, 270^\circ\}$, contains every multiple of 90° except for 0° . We can start with any two rotations a, b in $\{90^\circ, 180^\circ, 270^\circ\}$ and there must be exactly one $c \equiv -a - b \pmod{360^\circ}$ such that we can use the three rotations (a, b, c) which ensures that $a + b + c \equiv 0^\circ \pmod{360^\circ}$. That way, the composition of rotations a, b, c yields a full rotation. For example, if $a = b = 90^\circ$, then $c \equiv -90^\circ - 90^\circ = -180^\circ \pmod{360^\circ}$, so $c = 180^\circ$ and the rotations $(90^\circ, 90^\circ, 180^\circ)$ yields a full rotation.

The only case in which this fails is when c would have to equal 0° . This happens when (a, b) is already a full rotation, namely, $(a, b) = (90^\circ, 270^\circ), (180^\circ, 180^\circ)$, or $(270^\circ, 90^\circ)$. However, we can simply subtract these three cases from the total. Selecting (a, b) from $\{90^\circ, 180^\circ, 270^\circ\}$ yields $3 \cdot 3 = 9$ choices, and with 3 that fail, we are left with 6 combinations for case 1.

Case 2: 2 reflections on T .

In this case, we first eliminate the possibility of having two of the same reflection. Since two reflections across the x -axis maps T back to itself, inserting a rotation before, between, or after these two reflections would change T 's final location,

meaning that any combination involving two reflections across the x-axis would not map T back to itself. The same applies to two reflections across the y-axis.

Therefore, we must use one reflection about the x-axis, one reflection about the y-axis, and one rotation. Since a reflection about the x-axis changes the sign of the y component, a reflection about the y-axis changes the sign of the x component, and a 180° rotation changes both signs, these three transformation composed (in any order) will suffice. It is therefore only a question of arranging the three, giving us $3! = 6$ combinations for case 2.

Combining both cases we get $6 + 6 = \boxed{(A)12}$.

Solution 3 (Rewording of Solution 2)

As in the previous solution, note that we must have either 0 or 2 reflections because of orientation since reflection changes orientation that is impossible to fix by rotation. We also know we can't have the same reflection twice, since that would give a net of no change and would require an identity rotation.

Suppose there are no reflections. Denote 90° as 1, 180° as 2, and 270° as 3, just for simplification purposes. We want a combination of 3 of these that will sum to either 4 or 8 (0 and 12 are impossible since the minimum is 3 and the max is 9). 4 can be achieved with any permutation of $(1 - 1 - 2)$ and 8 can be achieved with any permutation of $(2 - 3 - 3)$. This case can be done in $3 + 3 = 6$ ways.

Suppose there are two reflections. As noted already, they must be different, and as a result will take the triangle to the opposite side of the origin if we don't do any rotation. We have 1 rotation left that we can do though, and the only one that will return to the original position is 2, which is 180° AKA reflection across origin. Therefore, since all 3 transformations are distinct. The three transformations can be applied anywhere since they are commutative (think quadrants). This gives 6 ways.

$6 + 6 = \boxed{(A)12}$.

Solution 5 (Direct Casework)

This is a straightforward approach that relies on counting possibilities based on the starting move.

Let the five transformations be R_{90} , R_{180} , R_{270} , R_x , and R_y . We count the valid sequences based on the starting transformation.

Case 1: Starts with R_{90}

- $R_{90} \rightarrow R_{90} \rightarrow R_{180}$
- $R_{90} \rightarrow R_{180} \rightarrow R_{90}$

This gives 2 ways.

Case 2: Starts with R_{180}

- $R_{180} \rightarrow R_{90} \rightarrow R_{90}$
- $R_{180} \rightarrow R_{270} \rightarrow R_{270}$
- $R_{180} \rightarrow R_x \rightarrow R_y$
- $R_{180} \rightarrow R_y \rightarrow R_x$

This gives 4 ways.

Case 3: Starts with R_{270}

- $R_{270} \rightarrow R_{180} \rightarrow R_{270}$
- $R_{270} \rightarrow R_{270} \rightarrow R_{180}$

This gives 2 ways.

Case 4: Starts with R_x

- $R_x \rightarrow R_{180} \rightarrow R_y$
- $R_x \rightarrow R_y \rightarrow R_{180}$

This gives 2 ways.

Case 5: Starts with R_y

- $R_y \rightarrow R_{180} \rightarrow R_x$
- $R_y \rightarrow R_x \rightarrow R_{180}$

This gives 2 ways.

Adding up the possibilities across all five cases:

$$2 + 4 + 2 + 2 + 2 = \mathbf{12}$$

Thus the answer is **(A) 12**.

21. (AMC 12 12A 2020) How many positive integers n are there such that n is a multiple of 5, and the least common multiple of $5!$ and n equals 5 times the greatest common divisor of $10!$ and n ? **(A)** 12 **(B)** 24 **(C)** 36 **(D)** 48 **(E)** 72

Answer: D

Solution 1

We set up the following equation as the problem states:

$$\text{lcm}(5!, n) = 5\text{gcd}(10!, n).$$

Breaking each number into its prime factorization, we see that the equation becomes

$$\text{lcm}(2^3 \cdot 3 \cdot 5, n) = 5\text{gcd}(2^8 \cdot 3^4 \cdot 5^2 \cdot 7, n).$$

We can now determine the prime factorization of n . We know that its prime factors belong to the set $\{2, 3, 5, 7\}$, as no factor of $10!$ has 11 in its prime factorization, nor anything greater. Next, we must find exactly how many different possibilities exist for each.

There can be anywhere between 3 and 8 2's and 1 to 4 3's. However, since n is a multiple of 5, and we multiply the gcd by 5, there can only be 3 5's in n 's prime factorization. Finally, there can either 0 or 1 7's.

Thus, we can multiply the total possibilities of n 's factorization to determine the number of integers n which satisfy the equation, giving us $6 \times 4 \times 1 \times 2 = \mathbf{(D) 48}$.

Solution 2

We do casework based on the values of n to uniquely determine the values of $\text{lcm}(5!, n)$.

Case 1: $n \leq 5!$

If $n \leq 5!$, then $\text{lcm}(n, 5!) = 120$. Then, we have

$$120 = 5 \gcd(10!, n)$$

Or, $24 = \gcd(10!, n)$. However, this cannot be possible, as $10!$ and n share a common 5. Then, the gcd must also be a multiple of 5, but 24 is not a multiple. Thus, the second case must be true.

Case 2: $n > 5!$.

If $n > 5!$, we have $\text{lcm}(n, 5!) = n$. Then, we have

$$n = 5 \gcd(10!, n)$$

The prime factorization of $10!$ is $2^8 \cdot 3^4 \cdot 5^2 \cdot 7$. Then, the gcd is the highest power of the number of factors between $10!$ and n . The prime factorization of n must have 5^j . Then, we see that $j = 2$, because if $j = 1$ then we have that when we divide 5 on both sides, one side will contain no power of 5 and the other will have a power of 5. Thus, n has a prime factorization $2^a 3^b 5^2 7^c$. Then, we have $n > 120$. To account for this, we divide by 120 on both sides to obtain a number with the prime factorization $2^5 \cdot 3^3 \cdot 5 \cdot 7$. Thus, n is determined by the factors of this number (excluding 5, since n must have 5). Then, the number of unique n is $(5 + 1)(3 + 1)(1 + 1) = 6 \times 4 \times 2 = \boxed{048}$.

This solution incorrectly assumes $\text{lcm}(n, 5!) = 120$ and $\text{lcm}(n, 5!) = n$. In case 1, $\text{lcm}(n, 5!) = 120$ but this assumes n must be a factor of $5!$, for example $n = 25, 35, 45, 50, 55$ (and more) are all excluded. Case 2 claims that if $n > 5!$, then $\text{lcm}(n, 5!) = n$. $\text{lcm}(n, 5!) = n$ is only true if $5! \mid n$. Merely being larger than 120 ($n > 120$) does not force 120 to divide n . For example, $n = 125$ satisfies $n > 5!$ and $5 \mid n$, but $\text{lcm}(125, 120) = 3000$ (which is not 125). This solution is not rigorous but it works by coincidence. $\square$

-Akshaj S

Solution 3

Like the Solution 1, we start from the equation:

$$\text{lcm}(5!, n) = 5 \gcd(10!, n).$$

Assume $\text{lcm}(5!, n) = k \cdot 5!$, with some integer k . It follows that $k \cdot 4! = \gcd(10!, n)$. It means that n has a divisor $4!$. Since n is a multiple of 5, n has a divisor $5!$. Thus, $\text{lcm}(5!, n) = n = k \cdot 5!$. The equation can be changed as

$$k \cdot 5! = 5 \gcd(10!, k \cdot 5!)$$

$$k = 5 \gcd(6 \cdot 7 \cdot 8 \cdot 9 \cdot 10, k)$$

We can see that k is also a multiple of 5, with a form of $5 \cdot m$. Substituting it in the above equation, we have

$$m = 5 \gcd(6 \cdot 7 \cdot 8 \cdot 9 \cdot 2, m)$$

Similarly, m is a multiple of 5, with a form of $5 \cdot s$. We have

$$s = \gcd(6 \cdot 7 \cdot 8 \cdot 9 \cdot 2, 5 \cdot s) = \gcd(2^5 \cdot 3^3 \cdot 7, s)$$

The equation holds, if s is a divisor of $2^5 \cdot 3^3 \cdot 7$, which has $(5+1)(3+1)(1+1) = \boxed{\text{(D)} 48}$ divisors. Linty Huang.shen k.kai

Solution 4

As in the previous solutions, we start with

$$\text{lcm}(5!, n) = 5\text{gcd}(10!, n)$$

From this we have that $\text{lcm}(5!, n) \mid 5\text{gcd}(10!, n)$, and in particular, $n \mid 5\text{gcd}(10!, n)$. However, $\text{gcd}(10!, n) \mid n$, so we must have $\text{gcd}(10!, n) = n$ or $\text{gcd}(10!, n) = n/5$. If $\text{gcd}(10!, n) = n$, then we have $\text{lcm}(5!, n) = 5n$; because $5 \mid 5!$, this implies that 5 does not divide n , so we must have $\text{gcd}(10!, n) = n/5$.

Now we have $\text{lcm}(5!, n) = n$, implying that $5! \mid n$, and $n/5 \mid 10!$. Writing out prime factorizations, this gives us

$$\begin{aligned} 2^3 \cdot 3 \cdot 5 &\mid n \\ n &\mid 2^8 \cdot 3^4 \cdot 5^2 \cdot 7 \end{aligned}$$

So n can have 3, 4, 5, 6, 7, or 8 factors of 2; 1, 2, 3, or 4 factors of three; and 0 or 1 factors of 7. Note that $\text{gcd}(2^8 \cdot 3^4 \cdot 5^2 \cdot 7, n) = n/5$ implies that n has 2 factors of 5. Thus, there are $6 \cdot 4 \cdot 2 = 48$ possible choices for n , and our answer is **(D) 48**.

-gumbymoo

22. (AMC 12 12A 2020) Let (a_n) and (b_n) be the sequences of real numbers such that

$$(2+i)^n = a_n + b_n i$$

for all integers $n \geq 0$, where $i = \sqrt{-1}$. What is

$$\sum_{n=0}^{\infty} \frac{a_n b_n}{7^n} ?$$

- (A) $\frac{3}{8}$ (B) $\frac{7}{16}$ (C) $\frac{1}{2}$ (D) $\frac{9}{16}$ (E) $\frac{4}{7}$

Answer: B

Solution 1

Square the given equality to yield

$$(3+4i)^n = (a_n + b_n i)^2 = (a_n^2 - b_n^2) + 2a_n b_n i,$$

so $a_n b_n = \frac{1}{2} \text{Im}((3+4i)^n)$ and

$$\sum_{n \geq 0} \frac{a_n b_n}{7^n} = \frac{1}{2} \text{Im} \left(\sum_{n \geq 0} \frac{(3+4i)^n}{7^n} \right) = \frac{1}{2} \text{Im} \left(\frac{1}{1 - \frac{3+4i}{7}} \right) = \boxed{\frac{7}{16}}.$$

comment: lol, idk who write this but it is brilliant

Solution 2 (DeMoivre's Formula)

Note that $(2+i) = \sqrt{5} \cdot \left(\frac{2}{\sqrt{5}} + \frac{1}{\sqrt{5}} i \right)$. Let $\theta = \arctan(1/2)$, then, we know that

$$(2+i) = \sqrt{5} \cdot (\cos \theta + i \sin \theta),$$

so

$$(2+i)^n = (\cos(n\theta) + i\sin(n\theta))(\sqrt{5})^n.$$

Therefore,

$$\begin{aligned} \sum_{n=0}^{\infty} \frac{a_n b_n}{7^n} &= \sum_{n=0}^{\infty} \frac{\cos(n\theta) \sin(n\theta) (5)^n}{7^n} \\ &= \frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{5}{7}\right)^n \sin(2n\theta) \\ &= \frac{1}{2} \operatorname{Im} \left(\sum_{n=0}^{\infty} \left(\frac{5}{7}\right)^n e^{2i\theta n} \right). \end{aligned}$$

Aha! $\sum_{n=0}^{\infty} \left(\frac{5}{7}\right)^n e^{2i\theta n}$ is a geometric sequence that evaluates to $\frac{1}{1 - \frac{5}{7}e^{2\theta i}}$. Now we can quickly see that

$$\sin(2\theta) = 2 \cdot \sin \theta \cdot \cos \theta = 2 \cdot \frac{2}{\sqrt{5}} \cdot \frac{1}{\sqrt{5}} = \frac{4}{5},$$

$$\cos(2\theta) = \cos^2 \theta - \sin^2 \theta = \frac{4}{5} - \frac{1}{5} = \frac{3}{5}.$$

Therefore,

$$\frac{1}{1 - \frac{5}{7}e^{2\theta i}} = \frac{1}{1 - \frac{5}{7} \left(\frac{3}{5} + \frac{4}{5}i\right)} = \frac{7}{8} + \frac{7}{8}i.$$

The imaginary part is $\frac{7}{8}$, so our answer is $\frac{1}{2} \cdot \frac{7}{8} = \boxed{\frac{7}{16}} \Rightarrow \text{(B)}.$

Solution 3

Clearly $a_n = \frac{(2+i)^n + (2-i)^n}{2}$, $b_n = \frac{(2+i)^n - (2-i)^n}{2i}$. So we have $\sum_{n \geq 0} \frac{a_n b_n}{7^n} = \sum_{n \geq 0} \frac{((2+i)^n + (2-i)^n)((2+i)^n - (2-i)^n)}{4i(7^n)}$.

By linearity, we have the latter is equivalent to $\frac{1}{4i} \sum_{n \geq 0} \frac{[(2+i)^n + (2-i)^n][(2+i)^n - (2-i)^n]}{7^n}$. Expanding the summand yields

$$\begin{aligned} \frac{1}{4i} \sum_{n \geq 0} \frac{(3+4i)^n - (3-4i)^n}{7^n} &= \frac{1}{4} \left[\frac{1}{1 - \left(\frac{3+4i}{7}\right)} - \frac{1}{1 - \left(\frac{3-4i}{7}\right)} \right] \\ &= \frac{1}{4i} \left[\frac{7}{7 - (3+4i)} - \frac{7}{7 - (3-4i)} \right] \\ &= \frac{1}{4} \left[\frac{7}{4-4i} - \frac{7}{4+4i} \right] \\ &= \frac{1}{4i} \left[\frac{7(4+4i)}{32} - \frac{7(4-4i)}{32} \right] = \frac{1}{4} \cdot \frac{56}{32} \\ &= \boxed{\frac{7}{16}} \text{ (B)} \end{aligned}$$

-vsamc

23. (AMC 12 12A 2020) Jason rolls three fair standard six-sided dice. Then he looks at the rolls and chooses a subset of the dice (possibly empty, possibly all three dice) to reroll. After rerolling, he wins if and only if the sum of the numbers face up on the three dice is exactly 7. Jason always plays to optimize his chances of winning. What is the probability that he chooses to reroll exactly two of the dice?

- (A) $\frac{7}{36}$ (B) $\frac{5}{24}$ (C) $\frac{2}{9}$ (D) $\frac{17}{72}$ (E) $\frac{1}{4}$

Answer: A

Solution 1

Consider the probability that rolling two dice gives a sum of s , where $s \leq 7$. There are $s - 1$ pairs that satisfy this, namely $(1, s - 1), (2, s - 2), \dots, (s - 1, 1)$, out of $6^2 = 36$ possible pairs. The probability is $\frac{s-1}{36}$.

Therefore, if one die has a value of a and Jason rerolls the other two dice, then the probability of winning is $\frac{7-a-1}{36} = \frac{6-a}{36}$.

In order to maximize the probability of winning, a must be minimized. This means that if Jason rerolls two dice, he must choose the two dice with the maximum values.

Thus, we can let $a \leq b \leq c$ be the values of the three dice. Consider the case when $a + b < 7$. If $a + b + c = 7$, then we do not need to reroll any dice. Otherwise, if we reroll one die, we can reroll any one die in the hope that we get the value that makes the sum of the three dice 7. This happens with probability $\frac{1}{6}$. If we reroll two dice, we will roll the two maximum dice, and the probability of winning is $\frac{6-a}{36}$, as stated above. However, $\frac{1}{6} > \frac{6-a}{36}$, so rolling one die is always better than rolling two dice if $a + b < 7$.

Now consider the case where $a + b \geq 7$. Rerolling one die will not help us win since the sum of the three dice will always be greater than 7. If we reroll two dice, the probability of winning is, once again, $\frac{6-a}{36}$. To find the probability of winning if we reroll all three dice, we can let each die have 1 dot and find the number of ways to distribute the remaining 4 dots. By stars and bars, there are $\binom{6}{2} = 15$ ways to do this, making the probability of winning $\frac{15}{6^3} = \frac{5}{72}$.

In order for rolling two dice to be more favorable than rolling three dice, $\frac{6-a}{36} > \frac{5}{72} \rightarrow a \leq 3$.

Thus, rerolling two dice is optimal if and only if $a \leq 3$ and $a + b \geq 7$. The possible triplets (a, b, c) that satisfy these conditions, and the number of ways they can be permuted, are

$(3, 4, 4) \rightarrow 3$ ways.

$(3, 4, 5) \rightarrow 6$ ways.

$(3, 4, 6) \rightarrow 6$ ways.

$(3, 5, 5) \rightarrow 3$ ways.

$(3, 5, 6) \rightarrow 6$ ways.

$(3, 6, 6) \rightarrow 3$ ways.

$(2, 5, 5) \rightarrow 3$ ways.

$(2, 5, 6) \rightarrow 6$ ways.

$(2, 6, 6) \rightarrow 3$ ways.

$(1, 6, 6) \rightarrow 3$ ways.

There are $3 + 6 + 6 + 3 + 6 + 3 + 3 + 6 + 3 + 3 = 42$ ways in which rerolling two dice is optimal, out of $6^3 = 216$ possibilities. Therefore, the probability that Jason

will reroll two dice is $\frac{42}{216} = \boxed{\text{(A)} \frac{7}{36}}$.

Solution 2 (Illustration)

We conclude all of the following after the initial roll:

- a) Jason rerolls exactly zero dice if and only if the sum of the three dice is 7, in which the probability of winning is always 1.
- b) If Jason rerolls exactly one die, then the sum of the two other dice must be 2, 3, 4, 5, or 6. The probability of winning is always $\frac{1}{6}$, as exactly 1 of the 6 possible outcomes of the die rerolled results in a win.
- c) If Jason rerolls exactly two dice, then the outcome of the remaining die must be 1, 2, 3, 4, or 5. Applying casework to the remaining die produces the following table:

Remaining Die	Sum Needed for the Two Dice Rerolled	Probability of Winning
1	6	$5/36$
2	5	$4/36$
3	4	$3/36$
4	3	$2/36$
5	2	$1/36$

The probability of winning is at most $\frac{5}{36}$.

- d) If Jason (re)rolls all three dice, then the probability of winning is always $\frac{\binom{6}{2}}{6^3} = \frac{15}{216} = \frac{5}{72}$. For the denominator, rolling three dice gives a total of $6^3 = 216$ possible outcomes.

For the numerator, this is the same as counting the ordered triples of positive integers (a, b, c) for which $a + b + c = 7$. Suppose that 7 balls are lined up in a row. There are 6 gaps between the balls, and placing dividers in 2 of the gaps separates the balls into 3 piles. From left to right, the numbers of balls in the piles correspond to a, b , and c , respectively. There are $\binom{6}{2} = 15$ ways to place the dividers. Note that the dividers' positions and the ordered triples have one-to-one correspondence, and $1 \leq a, b, c \leq 6$ holds for all such ordered triples.

The optimal strategy is that:

- If Jason needs to reroll at least zero dice to win, then he rerolls exactly zero dice.
- If Jason needs to reroll at least one die to win, then he rerolls exactly one die.
- If Jason needs to reroll at least two dice to win, then he rerolls exactly two dice if and only if the probability of winning is greater than $\frac{5}{72}$, the probability of winning for rerolling all three dice. The first three cases in the table above satisfy this requirement. We will analyze these cases by considering the initial outcomes of the two dice rerolled. Note that any of the three dice can be the remaining die, so we need a factor of 3 for all counts in the third column:

Remaining Die	Initial Outcomes of the Two Dice Rerolled	# of Ways
1	Both are in $\{6\}$, not necessarily distinct.	$3 \cdot 1^2 = 3$
2	Both are in $\{5, 6\}$, not necessarily distinct.	$3 \cdot 2^2 = 12$
3	Both are in $\{4, 5, 6\}$, not necessarily distinct.	$3 \cdot 3^2 = 27$

Finally, the requested probability is

$$\frac{3 + 12 + 27}{6^3} = \frac{42}{216} = \boxed{\text{(A)} \frac{7}{36}}.$$

Solution 3 (Similar to Solution 2)

We count the numerator. Jason will pick up no dice if he already has a 7 as a sum. We need to assume he does not have a 7 to begin with. If Jason decides to pick up all the dice to re-roll, by the stars and bars rule ways to distribute, $\binom{n+k-1}{k-1}$, there will be 2 bars and 4 stars (3 of them need to be guaranteed because a roll is at least 1) for a probability of $\frac{15}{216} = \frac{2.5}{36}$. If Jason picks up 2 dice and leaves a die showing k , he will need the other two to sum to $7 - k$. This happens with probability

$$\frac{6 - k}{36}$$

for integers $1 \leq k \leq 6$. If the roll is not 7, Jason will pick up exactly one die to re-roll if there can remain two other dice with sum less than 7, since this will give him a $\frac{1}{6}$ chance which is a larger probability than all the cases unless he has a 7 to begin with. We have

$$\frac{1}{6} > \frac{5, 4, 3}{36} > \frac{2.5}{36} > \frac{2, 1, 0}{36}.$$

We count the underlined part's frequency for the numerator without upsetting the probability greater than it. Let a be the roll we keep. We know $a \leq 3$ since $a = 4$ would cause Jason to pick up all the dice. When $a = 1$, there are 3 choices for whether it is rolled 1st, 2nd, or 3rd, and in this case the other two rolls have to be at least 6 (or he would have only picked up 1). This gives $3 \cdot 1^2 = 3$ ways. Similarly, $a = 2$ gives $3 \cdot 2^2 = 12$ because the 2 can be rolled in 3 places and the other two rolls are at least 5. $a = 3$ gives $3 \cdot 3^2 = 27$. Summing together gives the numerator of 42. The denominator is $6^3 = 216$, so we have $\frac{42}{216} = \boxed{\text{(A)} \frac{7}{36}}$.

Solution 4

We can quickly write out all possible rolls that result in a total of 7, to get that there is a $\frac{5}{72}$ chance that rolling three dice will result in a total of 7.

Claim 1: If all of Jason's initial outcomes are 4, 5 or 6, Jason is best off rerolling everything.

Proof of if part: If Jason rerolls two dice, he would like for the sum of the two outcomes to be 7 - (The initial outcome of the untouched dice). Since in this case, every dice's initial outcome is 4, 5 or 6, this value is 1, 2 or 3. By basic dice probability, the probability that they add to one is 0; the probability that they add to two is $\frac{1}{36}$; the probability that they add to three is $\frac{1}{18}$. Each of these probabilities is less than $\frac{5}{72}$, (the probability of winning by rerolling everything) so Jason should not reroll one die in this scenario. Rolling one die is also nonsense in this situation, since the other two dice will already add up to something greater than 7.

Proof of only-if part: If Jason rolls a 1, 2 or 3, even just once, then, the probability that rerolling the other two dice will result in win is $\frac{5}{36}$, $\frac{1}{9}$ and $\frac{1}{12}$, by basic die probability. Each of these is greater than $\frac{5}{72}$, so if Jason rolls a 1, 2 or 3, he should not reroll everything.

Claim 2: If the sum of any two of the initial dice is less than 7, Jason is best off rerolling one or zero die.

Proof of if part: If the sum of any two of the initial dice is less than 7, when the other die is rolled, there is always a fixed $\frac{1}{6}$ chance that this results in a win for Jason (or, he has already won, so he doesn't have to roll anything). Rerolling that one die is better than rerolling all three dice, since, as we have seen there is a $\frac{5}{72}$ chance that Jason wins from that. Rerolling that one die is also better than rerolling two dice: by basic die probability, if Jason wishes for the sum of two rerolled dice to equal one value, the maximum probability this happens is $\frac{1}{6}$, and this is when he wishes for the sum to be 7; however, since he wishes for the sum of all three dice to be 7, this will never happen, so the probability that rerolling two dice results in a win is always less than $\frac{1}{6}$.

Proof of only-if part: If the sum of each pair of the three dice is at least 7, it would make no sense to reroll one die, since the final sum of all three dice will always be greater than 7. Also, if the sum of each pair of the three dice is at least 7, the total initial sum must also be greater than 7, so Jason should not reroll zero dice.

From having shown these two claims, we know that Jason should reroll two dice if at least one of them is a 1, 2 or 3, and if the sum of any two of the initial dice is at least 7. We can write out all the possibilities of initial rolls, account for their permutations, and add everything together to see that there are 42 total outcomes, where Jason should reroll two dice. Therefore, our final answer is $\frac{42}{216} = \boxed{\text{(A)} \frac{7}{36}}$.

Solution 5

If the sum of the dice is 7, we reroll no dice. If the sum of any two dice is ≤ 6 , we reroll 1 die because rolling 1 die will give Jason a $\frac{1}{6}$ chance of winning. Thus, we consider the scenario where the sum of any two dice is 7 or greater.

If the smallest die is 1, the other two dice must be 6, which has 3 possible permutations.

If the smallest die is 2, the other two dice can be 5 or 6. There are 3 ways to choose the position of the smallest die, 2 ways each to choose the value of the other dice, so there are $2 \cdot 2 \cdot 3 = 12$ possible permutations.

If the smallest die is 3, the other two dice can be 4, 5, or 6. Similarly, there are 3 ways to choose the position of the smallest die, and 3 ways each to choose the value of the other dice, so there are $3 \cdot 3 \cdot 3 = 27$ possible permutations.

If all dice are 4 or above, then we reroll 3 dice. When we reroll all 3 dice, the winning rolls are 1,1,5; 1,2,4; 1,3,3; 2,2,3 in some arrangement, for a total of 15 permutations. Thus, the probability of winning after rerolling 3 dice is $\frac{15}{6^3} = \frac{5}{72}$. If we reroll only 2 dice with the smallest die being 4, we would need to roll 1,2 or 2,1 to win, which gives a $\frac{2}{36} = \frac{1}{18}$ chance of winning, so this is worse.

If the smallest die is 5 or 6, rerolling 2 dice gives us even worse odds.

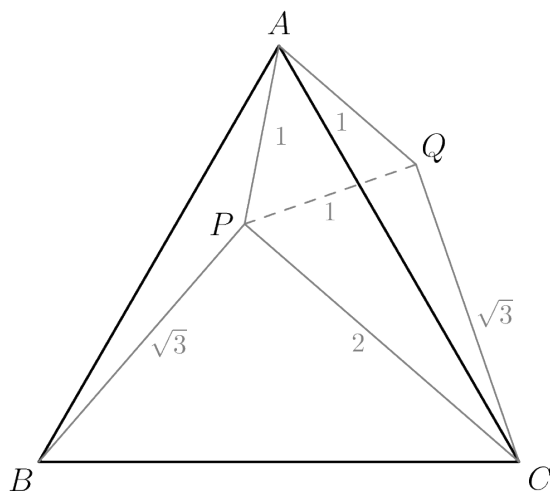
Therefore, the cases where 1, 2, or 3 are the smallest die covers all cases where 2 rerolls is optimal. There are $3 + 12 + 27 = 42$ permutations out of 216 total permutations, giving us a probability of $\frac{42}{216} = \boxed{\frac{7}{36}}$.

24. (AMC 12 12A 2020) Suppose that $\triangle ABC$ is an equilateral triangle of side length s , with the property that there is a unique point P inside the triangle such that $AP = 1$, $BP = \sqrt{3}$, and $CP = 2$. What is s ? (A) $1 + \sqrt{2}$ (B) $\sqrt{7}$ (C) $\frac{8}{3}$ (D) $\sqrt{5 + \sqrt{5}}$ (E) $2\sqrt{2}$

Answer: B

Solution 1(a)

We begin by rotating $\triangle APB$ counterclockwise by 60° about A , such that $P \mapsto Q$ and $B \mapsto C$. We see that $\triangle APQ$ is equilateral with side length 1, meaning that $\angle APQ = 60^\circ$. We also see that $\triangle CPQ$ is a 30-60-90 right triangle, meaning that $\angle CPQ = 60^\circ$. Thus, by adding the two together, we see that $\angle APC = 120^\circ$.



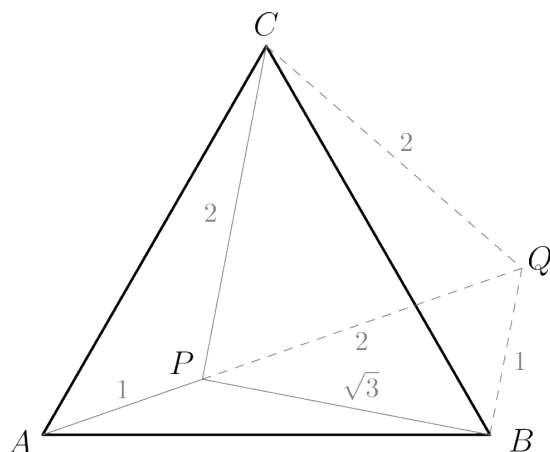
We can now use the law of cosines as following:

$$\begin{aligned} s^2 &= (AP)^2 + (CP)^2 - 2 \cdot AP \cdot CP \cdot \cos \angle APC \\ &= 1 + 4 - 2 \cdot 1 \cdot 2 \cdot \cos 120^\circ \\ &= 5 - 4 \left(-\frac{1}{2} \right) \\ &= 7, \end{aligned}$$

giving us that $s = \boxed{\sqrt{7} \text{ (B)}}$.

Solution 1(b)

Rotate $\triangle CPA$ counterclockwise 60° around point C to $\triangle CQB$. Then $CP = CQ$, $\angle PCQ = 60^\circ$, so $\triangle CPQ$ is an equilateral triangle.



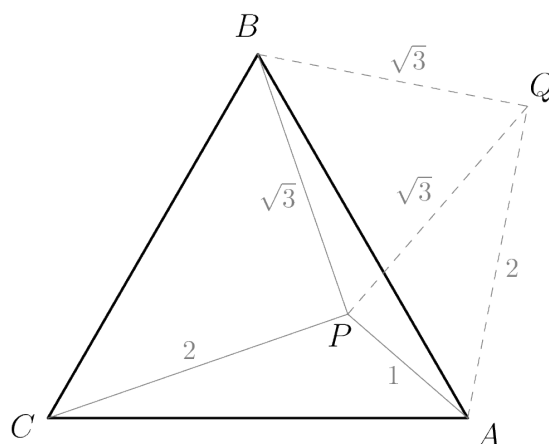
Note that $\triangle PQB$ is a 30° - 60° - 90° triangle, hence $\angle BPQ = 30^\circ$, and $\angle BPC = 90^\circ$, so

$$BC^2 = PC^2 + PB^2 = 2^2 + 3 = 7,$$

and the answer is **(B)** $\sqrt{7}$.

Solution 1(c)

Rotate $\triangle BPC$ counterclockwise by 60° around point B to $\triangle BQA$. Then $BP = BQ$, and $\angle PBQ = 60^\circ$, so $\triangle BPQ$ is an equilateral triangle.

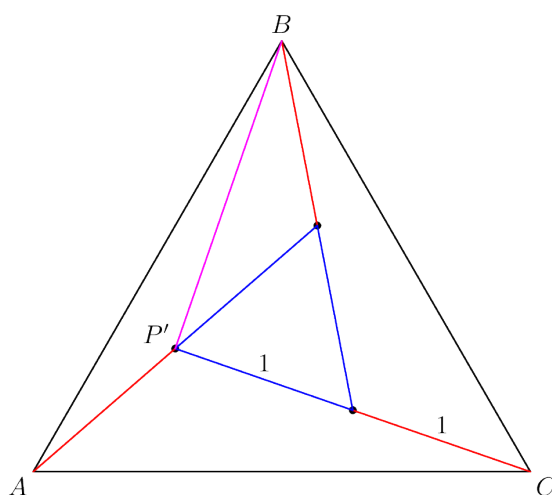


Note that $\triangle QAP$ is a 30° - 60° - 90° triangle, hence $\angle PQA = 30^\circ$, and $\angle BQA = 90^\circ$, so

$$AB^2 = PQ^2 + AQ^2 = 2^2 + 3 = 7,$$

and the answer is **(B)** $\sqrt{7}$.

Solution 2 (Intuition)



Suppose that triangle ABC had three segments of length 2, emanating from each of its vertices, making equal angles with each of its sides, and going into its interior. Suppose each of these segments intersected the segment clockwise to it precisely at its other endpoint and inside ABC (as pictured in the diagram above). Clearly $s > 2$ and the triangle defined by these intersection points will be equilateral (pictured by the blue segments).

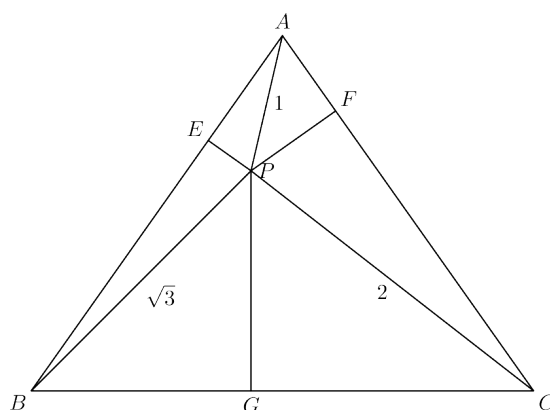
Take this equilateral triangle to have side length 1. The portions of each segment outside this triangle (in red) have length 1. Take P' to be the intersection of the segments emanating from A and C . By Law of Cosines,

$$BP' = \sqrt{1 + 1 - 2 \cos 120^\circ} = \sqrt{3}.$$

So, P' actually satisfies the conditions of the problem, and we can obtain again by Law of Cosines

$$s = \sqrt{4 + 1 - 4 \cos 120^\circ} = \boxed{\text{(B)}\sqrt{7}}.$$

Solution 3 (Answer Choices)



We begin by dropping altitudes from point P down to all three sides of the triangle as shown above. We can therefore make equations regarding the areas of triangles $\triangle APC$, $\triangle APB$, and $\triangle BPC$. Let s be the side of the equilateral triangle, we use the Heron's formula:

$$\begin{aligned} \triangle APC &= \frac{s \cdot PF}{2} = \sqrt{\frac{s+3}{2} \left(\frac{s+3}{2} - s \right) \left(\frac{s+3}{2} - 1 \right) \left(\frac{s+3}{2} - 2 \right)} \\ \implies PF &= \frac{\sqrt{10s^2 - s^4 - 9}}{2s} \end{aligned}$$

Similarly, we obtain:

$$\begin{aligned} PE &= \frac{\sqrt{8s^2 - s^4 - 4}}{2s} \\ PG &= \frac{\sqrt{14s^2 - s^4 - 1}}{2s} \end{aligned}$$

By Viviani's theorem,

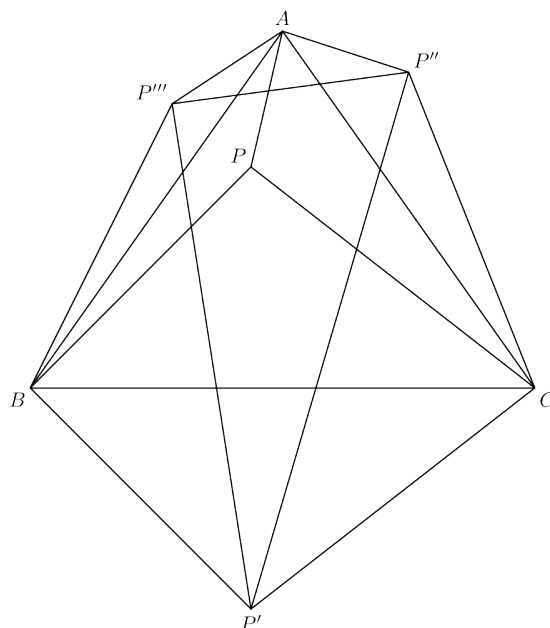
$$\begin{aligned} \frac{\sqrt{10s^2 - s^4 - 9}}{2s} + \frac{\sqrt{8s^2 - s^4 - 4}}{2s} + \frac{\sqrt{14s^2 - s^4 - 1}}{2s} &= \frac{\sqrt{3}}{2}s \\ \sqrt{10s^2 - s^4 - 9} + \sqrt{8s^2 - s^4 - 4} + \sqrt{14s^2 - s^4 - 1} &= \sqrt{3}s^2 \end{aligned}$$

Note that from now on, the algebra will get extremely ugly and almost impossible to do by hand within the time frame. However, we do see that it's extremely easy

to check the answer choices with the equation in this form. Testing $s = \sqrt{7}$, We obtain $7\sqrt{3}$ on both sides, revealing that our answer is in fact **(B)** $\sqrt{7}$

Solution 4 (Area)

Instead of directly finding the side length of the equilateral triangle, we instead find the area and use it to find the side length. Begin by reflecting P over each of the sides. Label these reflected points P', P'', P''' . Connect these points to the vertices of the equilateral triangle, as well as to each other.



Observe that the area of the equilateral triangle ABC is half that of the hexagon $AP''CP'BP'''$.

Note that $AP = AP'' = AP'''$. The same goes for the other vertices. This means that $AP''P'''$ is isosceles. Using either the Law of Cosines or simply observing that $AP''P'''$ is comprised of two 30-60-90 triangles, we find that $P''P''' = \sqrt{3}$. Similarly (pun intended), $P'P''' = 3$ and $P'P'' = 2\sqrt{3}$. Using the previous observation that $AP''P'''$ is two 30-60-90 triangles (as are the others) we find the areas of $AP''P'''$ to be $\frac{\sqrt{3}}{4}$. Again, using similarity we find the area of $BP'P'''$ to be $\frac{3\sqrt{3}}{4}$ and the area of $CP'P''$ to be $\sqrt{3}$.

Next, observe that $P'P''P'''$ is a 30-60-90 right triangle. This right triangle therefore has an area of $\frac{3\sqrt{3}}{2}$.

Adding these areas together, we get the area of the hexagon as $\frac{7\sqrt{3}}{2}$. This means that the area of ABC is $\frac{7\sqrt{3}}{4}$.

The formula for the area of an equilateral triangle with side length s is $\frac{s^2\sqrt{3}}{4}$ (if you don't have this memorized it's not hard to derive). Comparing this formula to the area of ABC , we can easily find that $s^2 = 7$, which means that the side length of ABC is **(B)** $\sqrt{7}$.

While this approach feels rather convoluted in comparison to Solution 1 (which only works for isosceles triangles), it is more flexible and can actually be generalized

for any point in a general triangle (although that requires use of Heron's, and potentially Law of Sines and Cosines).

Solution 5 (Coordinate Bashing)

Suppose $A(0, \sqrt{3}a)$, $B(-a, 0)$, $C(a, 0)$ and $P(x, y)$. So $s = 2a$. Since $BP = \sqrt{3}$ and $CP = 2$, we have

$$(x + a)^2 + y^2 = 3$$

$$(x - a)^2 + y^2 = 4$$

Solving the equations, we have

$$x = -\frac{1}{4a}, \quad y = \sqrt{\frac{7}{2} - a^2 - \frac{1}{16a^2}}$$

From $AP = 1$ (and a fair amount of algebra), we can have $a = \sqrt{7}/2$. The answer is **(B)** $\sqrt{7}$.

Solution 6 (Diagram Nuke)

Drawing out a rough sketch, it appears that $\angle BPC = 90^\circ$. By Pythagorean, our answer is $\sqrt{\sqrt{3}^2 + 2^2} = \mathbf{(B) \sqrt{7}}$.

Proof of this fact can be found in the Video Solution by Richard Rusczyk below.

Solution 7 (Theorem Nuke)

We can use the following theorem:

$\text{\textit{For a point } } P \text{ \textit{ inside of an equilateral triangle } } \triangle ABC \text{ \textit{ then } } \\ \left[3(PA^4 + PB^4 + PC^4 + s^4) = (PA^2 + PB^2 + PC^2 + s^2)^2 \right]$

We know that $PA = 1$, $PB = \sqrt{3}$, and $CP = 2$. Plugging these into our formula, we get $3(1+9+16+s^4) = (1+3+4+s^2)^2 \Rightarrow 78+3s^4 = 64+16s^2+s^4 \Rightarrow 2s^4-16s^2+14=0$. Let $s^2 = u$. Then, we have $2u^2-16u+14=0$. Solving for u , we get $\frac{16 \pm \sqrt{144}}{4} = 4 \pm 3$. If u is equal to 1, then we have $s = 1$, but this is not possible since P is inside of the triangle. This means that $u = 7$, and therefore $s = \mathbf{(B) \sqrt{7}}$.

Solution 8 (More Coordinate Bashing)

Set the points $A(0, 0)$, $B\left(\frac{s}{2}, \frac{s\sqrt{3}}{2}\right)$, $C(s, 0)$. Then, we want to find the intersection of the three circles

$$x^2 + y^2 = 1, (x - s)^2 + y^2 = 4, \left(x - \frac{s}{2}\right)^2 + \left(y - \frac{s\sqrt{3}}{2}\right)^2 = 3.$$

Subtracting circle 1 from circle 2 yields $s^2 - 2xs = 3$, which can be rewritten as $x = \frac{s^2-3}{2s}$. Subtracting circle 1 from circle 3 yields $s^2 - sx - ys\sqrt{3} = 2$, or $2s^2 - 2sx - 2ys\sqrt{3} = 4$. Subtracting $s^2 - 2xs = 3$ from this yields $s^2 - 2ys\sqrt{3} = 1$, or $y = \frac{s^2-1}{2s\sqrt{3}}$.

We then substitute our values for x and y back into $x^2 + y^2 = 1$, which gives us

$$\left(\frac{s^2-3}{2s}\right)^2 + \left(\frac{s^2-1}{2s\sqrt{3}}\right)^2 = 1.$$

Solving this equation (the algebra is surprisingly not bad!) gives us $s = \pm 1, \pm\sqrt{7}$.

Since s must be greater than 1, the answer is **(B)** $\sqrt{7}$.

- curiousmind888 TGSN

Solution 9 (Trig Bash)

Let $\angle APC = \alpha$, $\angle CPB = \beta$, by the Law of Cosine

$$1^2 + 2^2 - 2 \cdot 1 \cdot 2 \cdot \cos \alpha = 2^2 + (\sqrt{3})^2 - 2 \cdot 2 \cdot \sqrt{3} \cdot \cos \beta$$

$$2^2 + (\sqrt{3})^2 - 2 \cdot 2 \cdot \sqrt{3} = (\sqrt{3})^2 + 1^2 - 2 \cdot \sqrt{3} \cdot 1 \cdot \cos(360^\circ - \alpha - \beta)$$

Simplifying the two equations we get

$$\sqrt{3} \cos \beta - \cos \alpha = \frac{1}{2}$$

$$3 - 4\sqrt{3} \cos \beta = -2\sqrt{3}(\cos \alpha \cos \beta - \sin \alpha \sin \beta)$$

Let $\cos \alpha = x$, $\cos \beta = y$, by simplifying we get:

$$y\sqrt{3} - x = \frac{1}{2}$$

$$3 - 4y\sqrt{3} = -2\sqrt{3} \left(xy - \sqrt{(1-x^2)(1-y^2)} \right)$$

Solving the first equation for y we get $y = \frac{2x\sqrt{3} + \sqrt{3}}{6}$

Substituting this into the second equation gives

$$3 - 4\sqrt{3} \cdot \frac{2x\sqrt{3} + \sqrt{3}}{6} = -2\sqrt{3} \left(x \cdot \frac{2x\sqrt{3} + \sqrt{3}}{6} - \sqrt{(1-x^2)(1 - (\frac{2x\sqrt{3} + \sqrt{3}}{6})^2)} \right)$$

$$3 - 2(2x + 1) = -2\sqrt{3} \cdot x \cdot \frac{2x\sqrt{3} + \sqrt{3}}{6} + 2\sqrt{3} \cdot \sqrt{(1-x^2)(1 - (\frac{2x\sqrt{3} + \sqrt{3}}{6})^2)}$$

$$3 - 2x - 2 = -2x^2 - x + 2\sqrt{3} \cdot \sqrt{(1-x^2)(1 - (\sqrt{3})^2(\frac{2x+1}{6})^2)}$$

$$3 - 2x - 2 + 2x^2 + x = \sqrt{12(1-x^2)(1 - \frac{4x^2 + 4x + 1}{12})}$$

$$1 - 3x + 2x^2 = \sqrt{(1-x^2)(11 - 4x^2 - 4x)}$$

$$(1 - 3x + 2x^2)^2 = (1 - x^2)(11 - 4x^2 - 4x)$$

$$1 - 3x + 2x^2 - 3x + 9x^2 - 6x^3 + 2x^2 - 6x^3 + 4x^4 = 11 - 4x - 4x^2 - 11x^2 + 4x^3 + 4x^4$$

$$1 - 6x + 13x^2 - 12x^3 = 11 - 4x - 15x^2 + 4x^3$$

$$8x^3 - 14x^2 + x + 5 = 0$$

$$(x-1)(4x-5)(2x+1) = 0$$

$$\cos \alpha < 1, \quad \cos \alpha = x = -\frac{1}{2}, \quad s = \sqrt{5 - 4 \cos \alpha} = \boxed{\text{(B)} \sqrt{7}}$$

25. (AMC 12 12A 2020) The number $a = \frac{p}{q}$, where p and q are relatively prime positive integers, has the property that the sum of all real numbers x satisfying

$$\lfloor x \rfloor \cdot \{x\} = a \cdot x^2$$

is 420, where $\lfloor x \rfloor$ denotes the greatest integer less than or equal to x and $\{x\} = x - \lfloor x \rfloor$ denotes the fractional part of x . What is $p+q$? (A) 245 (B) 593 (C) 929 (D) 1331

Answer: C

Solution 1 (Solves for Floor(x))

Let $w = \lfloor x \rfloor$ and $f = \{x\}$ denote the whole part and the fractional part of x , respectively, for which $0 \leq f < 1$ and $x = w + f$.

We rewrite the given equation as

$$w \cdot f = a \cdot (w + f)^2. \quad (1)$$

Since $a \cdot (w + f)^2 \geq 0$, it follows that $w \cdot f \geq 0$, from which $w \geq 0$.

We expand and rearrange (1) as

$$af^2 + (2a - 1)wf + aw^2 = 0, \quad (2)$$

which is a quadratic with either f or w .

For simplicity purposes, we will treat w as some fixed nonnegative integer so that (2) is a quadratic with f . By the Quadratic Formula, we have

$$f = w \left(\frac{1 - 2a \pm \sqrt{1 - 4a}}{2a} \right). \quad (3)$$

If $w = 0$, then $f = 0$. We get $x = w + f = 0$, which does not affect the sum of the solutions. Therefore, we consider the case for $w \geq 1$:

Recall that $0 \leq f < 1$, so $\frac{1 - 2a \pm \sqrt{1 - 4a}}{2a} \geq 0$. From the discriminant, we require that $0 \leq 1 - 4a < 1$, or

$$0 < a \leq \frac{1}{4}. \quad (4)$$

We consider each part of $0 \leq f < 1$ separately:

- a) $f \geq 0$ From (2), note that $a > 0$, $(2a - 1)w < 0$, and $aw^2 > 0$. By Descartes' Rule of Signs, we deduce that (2) must have two positive roots, so $f \geq 0$ is always valid.

Alternatively, from (3) and (4), note that all values of a for which $0 < a \leq \frac{1}{4}$ satisfy $1 - 2a > \sqrt{1 - 4a}$. We deduce that both roots in (3) must be positive, so $f \geq 0$ is always valid.

- b) $f < 1$ We rewrite (3) as

$$f = w \left(\frac{1}{2a} - 1 \pm \frac{\sqrt{1 - 4a}}{2a} \right).$$

From (4), it follows that $\frac{1}{2a} \geq \frac{1}{1/2} = 2$. The larger root is

$$f \geq w \left(2 - 1 + 2\sqrt{1 - 4a} \right) \geq 1 \left(2 - 1 + 2\sqrt{1 - 4 \cdot \frac{1}{4}} \right) = 1,$$

which contradicts $f < 1$. So, we take the smaller root, from which

$$f = w \left(\frac{1}{2a} - 1 - \frac{\sqrt{1-4a}}{2a} \right)$$

for some constant $k = \frac{1}{2a} - 1 - \frac{\sqrt{1-4a}}{2a} > 0$. We rewrite f as

$$f = wk,$$

in which $f < 1$ is valid as long as $k < \frac{1}{w}$. Note that the solutions of x are generated at

$$w = 1, 2, 3, \dots, W,$$

up to some value $w = W$ such that $\frac{1}{W+1} \leq k < \frac{1}{W}$.

Now, we express x in terms of w and k :

$$x = w + f = w + wk = w(1 + k).$$

The sum of all solutions to the original equation is

$$\sum_{w=1}^W w(1 + k) = (1 + k) \cdot \sum_{w=1}^W w = (1 + k) \cdot \frac{W(W+1)}{2} = 420. \quad (\star)$$

As $1 + k < 1 + \frac{1}{W}$, we conclude that $1 + k$ is slightly above 1 so that $\frac{W(W+1)}{2}$ is slightly below 420, or $W(W+1)$ is slightly below 840. By observations, we get $W = 28$. Substituting this into $(\star)$ produces $k = \frac{1}{29}$, which satisfies $\frac{1}{W+1} \leq k < \frac{1}{W}$, as required.

Finally, we solve for a in $k = \frac{1}{2a} - 1 - \frac{\sqrt{1-4a}}{2a}$:

$$\begin{aligned} \frac{1}{29} &= \frac{1}{2a} - 1 - \frac{\sqrt{1-4a}}{2a} \\ \frac{2}{29}a &= 1 - 2a - \sqrt{1-4a} \\ \frac{60}{29}a - 1 &= -\sqrt{1-4a} \\ \frac{60^2}{29^2}a^2 - \frac{120}{29}a + 1 &= 1 - 4a \\ \frac{60^2}{29^2}a^2 - \frac{4}{29}a &= 0 \\ a \left(\frac{60^2}{29^2}a - \frac{4}{29} \right) &= 0. \end{aligned}$$

Since $a > 0$, we obtain $\frac{60^2}{29^2}a - \frac{4}{29} = 0$, from which

$$a = \frac{4}{29} \cdot \frac{29^2}{60^2} = \frac{29}{900}.$$

The answer is $29 + 900 = \boxed{(C) 929}$.

Solution 2 (Solves for x)

Let x_n be a root in the interval $(n, n+1)$. In this interval, $\lfloor x_n \rfloor = n$ and $\{x_n\} = x_n - n$, so we must have $ax_n^2 = nx_n - n^2$, i.e., $ax_n^2 - nx_n + n^2 = 0$. We can

homogenize this equation by setting $x_n = n\zeta$; then $x_1 = \zeta$, and ζ is a root of $a\zeta^2 - \zeta + 1 = 0$.

Suppose N is the largest integer for which there is such a root; we have, for $n = 1, 2, \dots, N$,

$$n < x_n = n\zeta < n + 1$$

Summing over $n \in \{1, 2, \dots, N\}$ we get

$$\frac{1}{2}N(N+1) < 420 = \frac{1}{2}N(N+1)\zeta < \frac{1}{2}N(N+3)$$

From the right inequality we get $27 < N$ and from the left one we get $N < 29$. Thus $N = 28$. Using this in the middle equality we get $\zeta = \frac{30}{29}$. Since ζ satisfies $a\zeta^2 - \zeta + 1 = 0$, we get

$$a = \zeta^{-2}(\zeta - 1) = \frac{29^2}{30^2} \cdot \frac{1}{29} = \frac{29}{900}.$$

The answer is $29 + 900 = \boxed{(C) 929}$.

Solution 3 (Solves for x)

First note that $\lfloor x \rfloor \cdot \{x\} < 0$ when $x < 0$ while $ax^2 \geq 0 \forall x \in \mathbb{R}$. Thus we only need to look at positive solutions ($x = 0$ doesn't affect the sum of the solutions). Next, we break $\lfloor x \rfloor \cdot \{x\}$ down for each interval $[n, n+1)$, where n is a positive integer. Assume $\lfloor x \rfloor = n$, then $\{x\} = x - n$. This means that when $x \in [n, n+1)$, $\lfloor x \rfloor \cdot \{x\} = n(x - n) = nx - n^2$. Setting this equal to ax^2 gives

$$nx - n^2 = ax^2 \implies ax^2 - nx + n^2 = 0 \implies x = \frac{n \pm \sqrt{n^2 - 4an^2}}{2a}$$

We're looking at the solution with the positive x , which is $x = \frac{n - n\sqrt{1-4a}}{2a} = \frac{n}{2a}(1 - \sqrt{1-4a})$. Note that if $\lfloor x \rfloor = n$ is the greatest n such that $\lfloor x \rfloor \cdot \{x\} = ax^2$ has a solution, the sum of all these solutions is slightly over $\sum_{k=1}^n k = \frac{n(n+1)}{2}$, which is 406 when $n = 28$, just under 420. Checking this gives

$$\begin{aligned} \sum_{k=1}^{28} \frac{k}{2a} (1 - \sqrt{1-4a}) &= \frac{1 - \sqrt{1-4a}}{2a} \cdot 406 = 420 \\ \frac{1 - \sqrt{1-4a}}{2a} &= \frac{420}{406} = \frac{30}{29} \\ 29 - 29\sqrt{1-4a} &= 60a \\ 29\sqrt{1-4a} &= 29 - 60a \\ 29^2 - 4 \cdot 29^2 a &= 29^2 + 3600a^2 - 120 \cdot 29a \\ 3600a^2 &= 116a \\ a &= \frac{116}{3600} = \frac{29}{900} \implies \boxed{(C) 929}. \end{aligned}$$

ktong

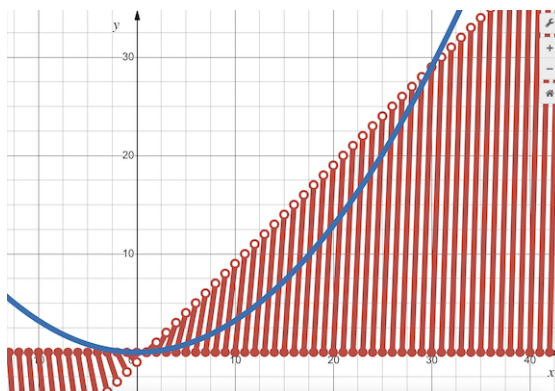
- Note: Using the Binomial expansion one can see that $\frac{1 - \sqrt{1-4a}}{2a}$ is close to 1 when a is close to 0.

Remark Let $f(x) = \lfloor x \rfloor \cdot \{x\}$ and $g(x) = a \cdot x^2$.

We make the following table of values:

x	$\lfloor x \rfloor$	$f(x)$	Equation
$[0, 1)$	0	0	$y = 0$
$[1, 2)$	1	$[0, 1)$	$y = x - 1$
$[2, 3)$	2	$[0, 2)$	$y = 2x - 4$
$[3, 4)$	3	$[0, 3)$	$y = 3x - 9$
$[4, 5)$	4	$[0, 4)$	$y = 4x - 16$
$\dots$	$\dots$	$\dots$	$\dots$
$[m, m + 1)$	m	$[0, m)$	$y = mx - m^2$

We graph $f(x)$ (in red, by branches) and $g(x)$ (in blue, for $a = \frac{29}{900}$) as shown below.



Graph in Desmos: <https://www.desmos.com/calculator/ouvaiqjdzj>

§3 AMC 12 12B 2021

1. (AMC 12 12B 2021) How many integer values of x satisfy $|x| < 3\pi$? (A) 9 (B) 10 (C) 18 (D) 19

Answer: D

Solution 1

Since $3\pi \approx 9.42$, we multiply 9 by 2 for the integers from 1 to 9 and the integers from -1 to -9 and add 1 to account for 0 to get **(D) 19**.

Solution 2

$|x| < 3\pi \iff -3\pi < x < 3\pi$. Since π is approximately 3.14, 3π is approximately 9.42. We are trying to solve for $-9.42 < x < 9.42$, where $x \in \mathbb{Z}$. Hence, $-9.42 < x < 9.42 \implies -9 \leq x \leq 9$, for $x \in \mathbb{Z}$. The number of integer values of x is $9 - (-9) + 1 = 19$. Therefore, the answer is **(D) 19**.

Solution 3

$3\pi \approx 9.4$. There are two cases here.

When $x > 0$, $|x| > 0$, and $x = |x|$. So then $x < 9.4$

When $x < 0$, $|x| > 0$, and $x = -|x|$. So then $-x < 9.4$. Dividing by -1 and flipping the sign, we get $x > -9.4$.

From case 1 and 2, we know that $-9.4 < x < 9.4$. Since x is an integer, we must have x between -9 and 9 . There are a total of

$$9 - (-9) + 1 = \boxed{\text{(D)} 19} \text{ integers.}$$

PureSwag

Solution 4

Looking at the problem, we see that instead of directly saying x , we see that it is $|x|$. That means all the possible values of x in this case are positive and negative. Rounding π to 3 we get $3(3) = 9$. There are 9 positive solutions and 9 negative solutions: $9 + 9 = 18$. But what about zero? Even though zero is neither negative nor positive, but we still need to add it into the solution. Hence, the answer is $9 + 9 + 1 = 18 + 1 = \boxed{\text{(D)} 19}$.

Solution 5

There are an odd number of integer solutions x to this inequality since if any non-zero integer x satisfies this inequality, then so does $-x$, and we must also account for 0, which gives us the desired. Then, the answer is either **(A)** or **(D)**, and since $3\pi > 3 \cdot 3 > 9$, the answer is at least $9 \cdot 2 + 1 = 19$, yielding **(D) 19**.

2. (AMC 12 12B 2021) At a math contest, 57 students are wearing blue shirts, and another 75 students are wearing yellow shirts. The 132 students are assigned into 66 pairs. In exactly 23 of these pairs, both students are wearing blue shirts. In how many pairs are both students wearing yellow shirts?
- (A) 23 (B) 32 (C) 37 (D) 41 (E) 64

Answer: B

Solution

There are 46 blue students paired with a blue partner. The other 11 students wearing blue shirts must each be paired with a partner wearing a shirt of the opposite color. There are 64 yellow students remaining. Therefore the requested number of pairs is $\frac{64}{2} = \boxed{\text{(B)} 32}$

3. (AMC 12 12B 2021) Suppose

$$2 + \frac{1}{1 + \frac{1}{2 + \frac{2}{3+x}}} = \frac{144}{53}.$$

What is the value of x ? (A) $\frac{3}{4}$ (B) $\frac{7}{8}$ (C) $\frac{14}{15}$ (D) $\frac{37}{38}$ (E) $\frac{52}{53}$

Answer: A

Solution

Subtracting 2 from both sides and taking reciprocals gives $1 + \frac{1}{2 + \frac{2}{3+x}} = \frac{53}{38}$. Subtracting 1 from both sides and taking reciprocals again gives $2 + \frac{2}{3+x} = \frac{38}{15}$. Subtracting 2 from both sides and taking reciprocals for the final time gives $\frac{x+3}{2} = \frac{15}{8}$ or $x = \frac{3}{4} \implies \boxed{\text{(A)} 3/4}$.

4. (AMC 12 12B 2021) Ms. Blackwell gives an exam to two classes. The mean of the scores of the students in the morning class is 84, and the afternoon class's mean score is 70. The ratio of the number of students in the morning class to the number of students in the afternoon class is $\frac{3}{4}$. What is the mean of the score of all the students?

(A) 74 (B) 75 (C) 76 (D) 77 (E) 78

Answer: C

Solution 1 (One Variable)

Let there be $3x$ students in the morning class and $4x$ students in the afternoon class. The total number of students is $3x + 4x = 7x$. The average is $\frac{3x \cdot 84 + 4x \cdot 70}{7x} = 76$. Therefore, the answer is **(C) 76**.

Solution 2 (Two Variables)

Suppose the morning class has m students and the afternoon class has a students. We have the following table:

	# of Students	Mean	Total
Morning	m	84	$84m$
Afternoon	a	70	$70a$

We are also given that $\frac{m}{a} = \frac{3}{4}$, which rearranges as $m = \frac{3}{4}a$.

The mean of the scores of all the students is

$$\frac{84m + 70a}{m + a} = \frac{84\left(\frac{3}{4}a\right) + 70a}{\frac{3}{4}a + a} = \frac{133a}{\frac{7}{4}a} = 133 \cdot \frac{4}{7} = \boxed{\text{(C) } 76}.$$

MRENTHUSIASM

Solution 3 (Ratio)

Of the average, $\frac{3}{3+4} = \frac{3}{7}$ of the scores came from the morning class and $\frac{4}{7}$ came from the afternoon class. The average is $\frac{3}{7} \cdot 84 + \frac{4}{7} \cdot 70 = \boxed{\text{(C) } 76}$.

Solution 4 (Convenient Values)

WLOG, assume there are 3 students in the morning class and 4 in the afternoon class. Then the average is $\frac{3 \cdot 84 + 4 \cdot 70}{7} = \boxed{\text{(C) } 76}$.

Solution 5 (Basic manipulation)

Let S_1 be the sum of the morning class's scores, and S_2 be the sum of the afternoon class's scores. Let $3x$ be the number of students in the morning class, and $4x$ be the number of students in the afternoon class. We can write $\frac{S_1}{3x} = 84$ and $\frac{S_2}{4x} = 70$, so $S_2 = 70 \cdot 4x$ and $S_1 = 84 \cdot 3x$. Adding these, we get $S_1 + S_2 = x(70 \cdot 4 + 84 \cdot 3)$. We want to find the mean, which is $\frac{S_1 + S_2}{7x}$, so dividing by $7x$ on both sides of the equation, we get the mean to be $10 \cdot 4 + 12 \cdot 3 = \boxed{\text{(C) } 76}$.

5. (AMC 12 12B 2021) The point $P(a, b)$ in the xy -plane is first rotated counter-clockwise by 90° around the point $(1, 5)$ and then reflected about the line $y = -x$. The image of P after these two transformations is at $(-6, 3)$. What is $b - a$?
- (A) 1 (B) 3 (C) 5 (D) 7 (E) 9

Answer: D

Solution 1 (Transformation Rules)

The final image of P is $(-6, 3)$. We know the reflection rule for reflecting over $y = -x$ is $(x, y) \rightarrow (-y, -x)$. So before the reflection and after rotation the point is $(-3, 6)$.

By definition of rotation, the slope between $(-3, 6)$ and $(1, 5)$ must be perpendicular to the slope between (a, b) and $(1, 5)$. The first slope is $\frac{5-6}{1-(-3)} = \frac{-1}{4}$. This means the slope of P and $(1, 5)$ is 4.

Rotations also preserve distance to the center of rotation, and since we only "travelled" up and down by the slope once to get from $(-3, 6)$ to $(1, 5)$ it follows we shall only use the slope once to travel from $(1, 5)$ to P .

Therefore point P is located at $(1 + 1, 5 + 4) = (2, 9)$. The answer is $9 - 2 = 7 =$

(D) 7.

-abhinavg0627

Solution 2 (Complex Numbers)

Let us reconstruct that coordinate plane as the complex plane. Then, the point $P(a, b)$ becomes $a + b \cdot i$. A 90° rotation around the point $(1, 5)$ can be done by translating the point $(1, 5)$ to the origin, rotating around the origin by 90° , and then translating the origin back to the point $(1, 5)$.

$$a + b \cdot i \implies (a-1) + (b-5) \cdot i \implies ((a-1) + (b-5) \cdot i) \cdot i = 5 - b + (a-1)i \implies 5 + 1 - b + (a-1+5)i = 6 - b + (a+4)i$$

By basis reflection rules, the reflection of $(-6, 3)$ about the line $y = -x$ is $(-3, 6)$. Hence, we have

$$6 - b + (a + 4)i = -3 + 6i \implies b = 9, a = 2,$$

from which $b - a = 9 - 2 =$ **(D) 7**.

Solution 3 (Reverso)

The problem gives a series of transformations and proceeds to give the resultant point, being $(-6, 3) = P$. Therefore, all we must do is reverse it. First, we reverse the last transformation by computing the distance from the point $(-6, 3)$ to $y = -x$ by using the formula $d = \frac{|Ax + By + C|}{\sqrt{A^2 + B^2}}$. Where $Ax + By + C$ is the standard form of a line. Computing, we obtain that $d = \frac{3\sqrt{2}}{2}$. We also know that this is magnitude is at an angle of $\frac{\pi}{4}$. Therefore, to do the transformation, we double that vector and add it to the point. We get:

$$P' = P + 2\vec{v} \implies P' = (-6, 3) + \left\langle 3\sqrt{2} \cdot \frac{\sqrt{2}}{2}, 3\sqrt{2} \cdot \frac{\sqrt{2}}{2} \right\rangle \implies P' = (-3, 6).$$

Now, we must reverse the second transformation. To do so, realize that $P' - (1, 5) \Leftrightarrow \vec{v}_2 = \langle -4, 1 \rangle$. Simply make this vector perpendicular by switching the x and y components and switching the sign of the initial y component. Therefore, we get $\vec{v}_{2\perp} = \langle 1, 4 \rangle$. Therefore, adding this vector to $(1, 5)$ yields $(2, 9)$, which leads us to $9 - 2 =$ **(D) 7**.

Solution 4 (Vector Dot Product)

Using the same method as in Solution 1, we can obtain that the point before the reflection is $(-3, 6)$. If we let the original point be (x, y) , then we can use that the starting point is $(1, 5)$ to obtain two vectors $\langle -4, 1 \rangle$ and $\langle x - 1, y - 5 \rangle$. We know

that two vectors are perpendicular if their dot product is equal to 0, and that both points are the same distance ($\sqrt{17}$) from $(1, 5)$.

Therefore, we can write two equations using these vectors: $(x - 1)^2 + (y - 5)^2 = 17$ (from distance and pythagorean theorem) and $-4x + y - 1 = 0$ (from dot product)

Solving, we simplify the second equation to $y = 4x + 1$, and plug it into the first equation. We obtain $(x - 1)^2 + (4x - 4)^2 = 17$. We can simplify this to the quadratic $17x^2 - 34x = 0$. When we factor out $17x$, we find that $x = 2$ or $x = 0$. However, x cannot equal 0. Therefore, $x = 2$, and plugging this into the second equation gives us that $y = 9$. Since the point is $(9, 2)$, we compute $9 - 2 = \boxed{\text{(D)} 7}$.

Solution 5 (Vector Dot Product scuffed version)

Using the same method as in Solution 1 reflecting $(-6, 3)$ about the line $y = -x$ gives us $(-3, 6)$.

Let the original point be $\langle x, y \rangle$. From point $(1, 5)$, we form the vectors $\langle -4, 1 \rangle$ and $\langle x - 1, y - 5 \rangle$ that extend out from the initial point. If they are perpendicular, we know that their dot product has to equal zero. Therefore,

$$\langle -4, 1 \rangle \cdot \langle x - 1, y - 5 \rangle = 0 \implies -4x + y - 1 = 0.$$

Now, we have to do some guess and check from the multiple choices. Let $y - x = A$ where A is one of the answer choices. Then, $A - 3x = 1$. By intuition and logical reasoning we deduce that A must be 1 (mod 3) so that brings our potential answers down to **(A)** and **(C)**. If $A = 1$ from **(A)**, then $x = 0$, which we can quickly rule out since we know that P rotated counterclockwise not clockwise. Hence, **(D) 7** is the answer.

Solution 6 (Working Backwards Approach, Super Easy!)

The last thing that happened was the reflection over the line $y = -x$, so we reverse that by reflecting $(-6, 3)$ over $y = -x$ to get $(-3, 6)$. Now, we have to rotate it **clockwise**, to reverse the first step. To rotate $(-3, 6)$ over the point $(1, 5)$ clockwise 90 degrees, we can subtract 1 from each x-coordinate and subtract 5 from each y-coordinate. That way, we only have to rotate $(-4, 1)$ clockwise 90 degrees over the origin $(0, 0)$. This is pretty easy to do in your head, so you get $(1, 4)$. But we are not done, we still have to add 1 to each x-coordinate and add 5 to each y-coordinate, to get $(2, 9)$, so our final answer is $9 - 2 = 7$, which is **(D) 7**.

6. (AMC 12 12B 2021) An inverted cone with base radius 12cm and height 18cm is full of water. The water is poured into a tall cylinder whose horizontal base has a radius of 24cm. What is the height in centimeters of the water in the cylinder?

(A) 1.5 **(B)** 3 **(C)** 4 **(D)** 4.5 **(E)** 6

Answer: A

Solution 1

The volume of a cone is $\frac{1}{3} \cdot \pi \cdot r^2 \cdot h$ where r is the base radius and h is the height. The water completely fills up the cone so the volume of the water is $\frac{1}{3} \cdot 18 \cdot 144\pi = 6 \cdot 144\pi$. The volume of a cylinder is $\pi \cdot r^2 \cdot h$ so the volume of the water in the cylinder would be $24 \cdot 24 \cdot \pi \cdot h$.

We can equate these two expressions because the water volume stays the same like this $24 \cdot 24 \cdot \pi \cdot h = 6 \cdot 144\pi$. We get $4h = 6$ and $h = \frac{6}{4}$.

So the answer is (A) 1.5.

Solution 2

The water completely fills up the cone. For now, assume the radius of both cone and cylinder are the same. Then the cone has $\frac{1}{3}$ of the volume of the cylinder, and so the height is divided by 3. Then, from the problem statement, the radius is doubled, meaning the area of the base is quadrupled (since $2^2 = 4$).

Therefore, the height is divided by 3 and divided by 4, which is $18 \div 3 \div 4 = \span style="border: 1px solid black; padding: 2px;">(A) 1.5.$

7. (AMC 12 12B 2021) Let $N = 34 \cdot 34 \cdot 63 \cdot 270$. What is the ratio of the sum of the odd divisors of N to the sum of the even divisors of N ? (A) 1 : 16 (B) 1 : 15 (C) 1 : 14 (D) 1 : 8 (E) 1 : 3

Answer: C

Solution 1

Prime factorize N to get $N = 2^3 \cdot 3^5 \cdot 5 \cdot 7 \cdot 17^2$. For each odd divisor n of N , there exist even divisors $2n, 4n, 8n$ of N , therefore the ratio is $1 : (2 + 4 + 8) = \span style="border: 1px solid black; padding: 2px;">(C) 1 : 14$

Solution 2

Prime factorizing N , we see $N = 2^3 \cdot 3^5 \cdot 5 \cdot 7 \cdot 17^2$. The sum of N 's odd divisors are the sum of the factors of N without 2, and the sum of the even divisors is the sum of the odds subtracted by the total sum of divisors.

BY SUM OF FACTORS FORMULA (search it up if you don't know): The formula is actually for all factors, but we can just take out the 2^3 , so we have:

The sum of odd divisors is given by

$$a = (1 + 3 + 3^2 + 3^3 + 3^4 + 3^5)(1 + 5)(1 + 7)(1 + 17 + 17^2)$$

and the total sum of divisors is

$$(1 + 2 + 4 + 8)(1 + 3 + 3^2 + 3^3 + 3^4 + 3^5)(1 + 5)(1 + 7)(1 + 17 + 17^2) = 15a.$$

Thus, our ratio is

$$\frac{a}{15a - a} = \frac{a}{14a} = \span style="border: 1px solid black; padding: 2px;">(C) 1 : 14.$$

Solution 3

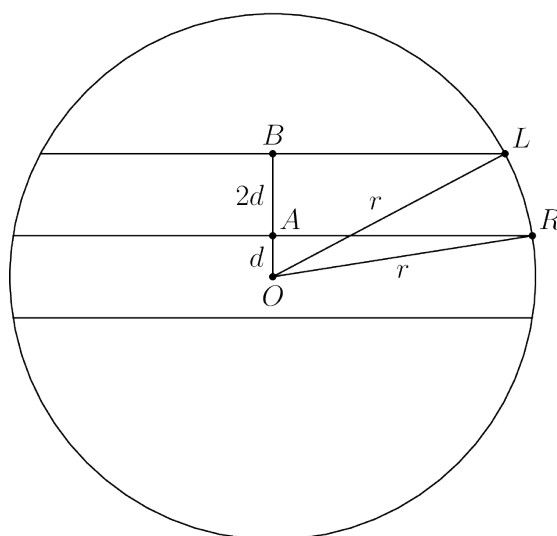
Prime factorizing N , we have that there is 2^3 in our factorization. Now, call the sum of the odd divisors k . We know that if we multiply k by 2, we will have even divisors. So, we can multiply k by 2, $2^2, 2^3$ respectively to get $14k$ as the sum of the even divisors. Therefore, the answer is

$$\frac{k}{14k} = \span style="border: 1px solid black; padding: 2px;">(C) 1 : 14.$$

MC

8. (AMC 12 12B 2021) Three equally spaced parallel lines intersect a circle, creating three chords of lengths 38, 38, and 34. What is the distance between two adjacent parallel lines?
(A) $5\frac{1}{2}$ (B) 6 (C) $6\frac{1}{2}$ (D) 7 (E) $7\frac{1}{2}$

Answer: B

Solution 1 (Pythagorean Theorem)

Since two parallel chords have the same length (38), they must be equidistant from the center of the circle. Let the perpendicular distance of each chord from the center of the circle be d . Thus, the distance from the center of the circle to the chord of length 34 is

$$2d + d = 3d$$

and the distance between each of the chords is just $2d$. Let the radius of the circle be r . Drawing radii to the points where the lines intersect the circle, we create two different right triangles:

- One with base $\frac{38}{2} = 19$, height d , and hypotenuse r ($\triangle RAO$ on the diagram)
- Another with base $\frac{34}{2} = 17$, height $3d$, and hypotenuse r ($\triangle LBO$ on the diagram)

By the Pythagorean theorem, we can create the following system of equations:

$$19^2 + d^2 = r^2$$

$$17^2 + (2d + d)^2 = r^2$$

Solving, we find $d = 3$, so $2d = \boxed{\text{(B)} 6}$.

Solution 2 (Coordinates)

Because we know that the equation of a circle is $(x - a)^2 + (y - b)^2 = r^2$ where the center of the circle is (a, b) and the radius is r , we can find the equation of this circle by centering it on the origin. Doing this, we get that the equation is $x^2 + y^2 = r^2$. Now, we can set the distance between the chords as $2d$ so the distance from the chord with length 38 to the diameter is d .

Therefore, the following points are on the circle as the y -axis splits the chord in half, that is where we get our x value:

$$(19, d) \quad (19, -d) \quad (17, -3d)$$

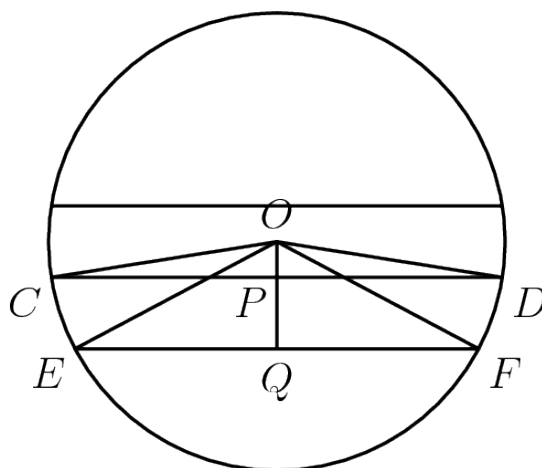
Now, we can plug one of the first two value in as well as the last one to get the following equations:

$$19^2 + d^2 = r^2$$

$$17^2 + (3d)^2 = r^2$$

Subtracting these two equations, we get $19^2 - 17^2 = 8d^2$ - therefore, we get $72 = 8d^2 \rightarrow d^2 = 9 \rightarrow d = 3$. We want to find $2d = 6$ because that's the distance between two chords. So, our answer is $\boxed{B}$.

Solution 3 (Stewart's Theorem)



If d is the requested distance, and r is the radius of the circle, Stewart's Theorem applied to $\triangle OCD$ with cevian $\overleftrightarrow{OP}$ gives

$$19 \cdot 38 \cdot 19 + \frac{1}{2}d \cdot 38 \cdot \frac{1}{2}d = 19r^2 + 19r^2.$$

This simplifies to $13718 + \frac{19}{2}d^2 = 38r^2$. Similarly, another round of Stewart's Theorem applied to $\triangle OEF$ with cevian $\overleftrightarrow{OQ}$ gives

$$17 \cdot 34 \cdot 17 + \frac{3}{2}d \cdot 34 \cdot \frac{3}{2}d = 17r^2 + 17r^2.$$

This simplifies to $9826 + \frac{153}{2}d^2 = 34r^2$. Dividing the top equation by 38 and the bottom equation by 34 results in the system of equations

$$\begin{aligned} 361 + \frac{1}{4}d^2 &= r^2 \\ 289 + \frac{9}{4}d^2 &= r^2 \end{aligned}$$

By transitive, $361 + \frac{1}{4}d^2 = 289 + \frac{9}{4}d^2$. Therefore $(\frac{9}{4} - \frac{1}{4})d^2 = 361 - 289 \rightarrow 2d^2 = 72 \rightarrow d^2 = 36 \rightarrow d = \boxed{\text{(B) } 6}$.

9. (AMC 12 12B 2021) What is the value of

$$\frac{\log_2 80}{\log_{40} 2} - \frac{\log_2 160}{\log_{20} 2}?$$

- (A) 0 (B) 1 (C) $\frac{5}{4}$ (D) 2 (E) $\log_2 5$

Answer: D

Solution

$$\frac{\log_2 80}{\log_{40} 2} - \frac{\log_2 160}{\log_{20} 2}$$

Note that $\log_{40} 2 = \frac{1}{\log_2 40}$, and similarly $\log_{20} 2 = \frac{1}{\log_2 20}$

$$= \log_2 80 \cdot \log_2 40 - \log_2 160 \cdot \log_2 20$$

$$= (\log_2 4 + \log_2 20)(\log_2 2 + \log_2 20) - (\log_2 8 + \log_2 20) \log_2 20$$

$$= (2 + \log_2 20)(1 + \log_2 20) - (3 + \log_2 20) \log_2 20$$

Expanding,

$$2 + 2 \log_2 20 + \log_2 20 + (\log_2 20)^2 - 3 \log_2 20 - (\log_2 20)^2$$

All the log terms cancel, so the answer is $2 \implies \boxed{\text{(D)}}$.

10. (AMC 12 12B 2021) Two distinct numbers are selected from the set $\{1, 2, 3, 4, \dots, 36, 37\}$ so that the sum of the remaining 35 numbers is the product of these two numbers. What is the difference of these two numbers?

- (A) 5 (B) 7 (C) 8 (D) 9 (E) 10

Answer: E

Solution

The sum of the first n integers is given by $\frac{n(n+1)}{2}$, so $\frac{37(37+1)}{2} = 703$.

Therefore, $703 - x - y = xy$

Rearranging, $xy + x + y = 703$. We can factor this equation by **SFFT** to get

$$(x+1)(y+1) = 704$$

Looking at the possible divisors of $704 = 2^6 \cdot 11$, 22 and 32 are within the constraints of $0 < x \leq y \leq 37$ so we try those:

$$(x+1)(y+1) = 22 \cdot 32 \quad x+1 = 22, y+1 = 32 \quad x = 21, y = 31$$

Therefore, the difference $y - x = 31 - 21 = \boxed{\text{(E) } 10}$.

11. (AMC 12 12B 2021) Triangle ABC has $AB = 13$, $BC = 14$ and $AC = 15$. Let P be the point on $\overline{AC}$ such that $PC = 10$. There are exactly two points D and E on line BP such that quadrilaterals $ABCD$ and $ABCE$ are trapezoids. What is the distance DE ? (A) $\frac{42}{5}$ (B) $6\sqrt{2}$ (C) $\frac{84}{5}$ (D) $12\sqrt{2}$ (E) 18

Answer: D

Solution 1 (Analytic Geometry)

Toss on the Cartesian plane with $A = (5, 12)$, $B = (0, 0)$, and $C = (14, 0)$. Then $\overline{AD} \parallel \overline{BC}$, $\overline{AB} \parallel \overline{CE}$ by the trapezoid condition, where $D, E \in \overline{BP}$. Since $PC = 10$, point P is $\frac{10}{15} = \frac{2}{3}$ of the way from C to A and is located at $(8, 8)$. Thus line BP has equation $y = x$. Since $\overline{AD} \parallel \overline{BC}$ and $\overline{BC}$ is parallel to the ground, we know D has the same y -coordinate as A , except it'll also lie on the line $y = x$. Therefore, $D = (12, 12)$. ■

To find the location of point E , we need to find the intersection of $y = x$ with a line parallel to $\overline{AB}$ passing through C . The slope of this line is the same as the slope of $\overline{AB}$, or $\frac{12}{5}$, and has equation $y = \frac{12}{5}x - \frac{168}{5}$. The intersection of this line with $y = x$ is $(24, 24)$. Therefore point E is located at $(24, 24)$. ■

The distance DE is equal to the distance between $(12, 12)$ and $(24, 24)$, which is $\boxed{\text{(D)} 12\sqrt{2}}$.

Solution 2 (Stewart's Theorem)

Using Stewart's Theorem we find $BP = 8\sqrt{2}$. From the similar triangles $BPA \sim EPC$ and $BPC \sim DPA$, we have

$$DP = BP \cdot \frac{PA}{PC} = \frac{1}{2}BP$$

$$EP = BP \cdot \frac{PC}{PA} = 2BP$$

So,

$$DE = \frac{3}{2}BP = \boxed{\text{(D)} 12\sqrt{2}}.$$

Solution 3

Let x be the length PE . From the similar triangles $BPA \sim EPC$ and $BPC \sim DPA$ we have

$$BP = \frac{PA}{PC}x = \frac{1}{2}x$$

$$PD = \frac{PA}{PC}BP = \frac{1}{4}x$$

Therefore $BD = DE = \frac{3}{4}x$. Now extend line CD to the point Z on AE , forming parallelogram $ZABC$. As $BD = DE$ we also have $EZ = ZC = 13$ so $EC = 26$.

We now use the Law of Cosines to find x (the length of PE):

$$x^2 = EC^2 + PC^2 - 2(EC)(PC)\cos(\angle PCE) = 26^2 + 10^2 - 2 \cdot 26 \cdot 10 \cos(\angle PCE)$$

As $\angle PCE = \angle BAC$, we have (by Law of Cosines on triangle BAC)

$$\cos(\angle PCE) = \frac{13^2 + 15^2 - 14^2}{2 \cdot 13 \cdot 15}.$$

Therefore

$$\begin{aligned} x^2 &= 26^2 + 10^2 - 2 \cdot 26 \cdot 10 \cdot \frac{198}{2 \cdot 13 \cdot 15} \\ &= 776 - 264 \\ &= 512 \end{aligned}$$

And $x = 16\sqrt{2}$. The answer is then $\frac{3}{4}x = \boxed{\text{(D)} 12\sqrt{2}}$.

Solution 4 (Heron's Formula, Pythagorean Theorem, Similar Triangles)

Let the brackets denote areas. By Heron's Formula, we have

$$\begin{aligned} [ABC] &= \sqrt{\frac{13+14+15}{2} \left(\frac{13+14+15}{2} - 13 \right) \left(\frac{13+14+15}{2} - 14 \right) \left(\frac{13+14+15}{2} - 15 \right)} \\ &= \sqrt{21(21-13)(21-14)(21-15)} \\ &= \sqrt{21(8)(7)(6)} \\ &= \sqrt{(3 \cdot 7)(2^3)(7)(2 \cdot 3)} \\ &= 2^2 \cdot 3 \cdot 7 \\ &= 84. \end{aligned}$$

It follows that the height of $ABCD$ is $\frac{2[ABC]}{14} = 12$.

Next, we drop the altitudes $\overline{AF}$ and $\overline{DG}$ of $ABCD$. By the Pythagorean Theorem on right $\triangle AFB$, we get $BF = 5$. By the AA Similarity, we have $\triangle ADP \sim \triangle CBP$, with the ratio of similitude $1 : 2$. It follows that $AD = 7$. Since $ADGF$ is a rectangle, we get $FG = AD = 7$. By the Pythagorean Theorem on right $\triangle DGB$, we get $BD = 12\sqrt{2}$.

By $\triangle ADP \sim \triangle CBP$ again, we get $BP = 8\sqrt{2}$ and $DP = 4\sqrt{2}$. Also, by the AA Similarity, we have $\triangle ABP \sim \triangle CEP$, with the ratio of similitude $1 : 2$. It follows that $EP = 16\sqrt{2}$.

Finally, we obtain $DE = EP - DP = \boxed{\text{(D)} 12\sqrt{2}}$. Remark

It is well known that the area of a 13-14-15 triangle is 84.

Solution 5 (Barycentric Coordinates)

(For those unfamiliar with barycentric coordinates, consider reading the barycentric coordinates article written by Evan Chen and Max Schindler here: <https://artofproblemsolving.com/wiki>

We can find P in barycentric coordinates as $\left(\frac{2}{3}, 0, \frac{1}{3}\right)$. We can then write $\overline{BP}$ as $x - 2z = 0$, where (x, y, z) defines a point in barycentric coordinates. We have $\overline{AD} \parallel \overline{BC}$ as $y + z = 0$ and $\overline{CE} \parallel \overline{AB}$ as $x + y = 0$. We can then compute D and E by intersecting lines:

$$\begin{cases} x - 2z = 0 \\ y + z = 0 \\ x + y + z = 1 \end{cases}$$

Which gives us $D = \left(1, -\frac{1}{2}, \frac{1}{2}\right)$. We can get E with:

$$\begin{cases} x - 2z = 0 \\ x + y = 0 \\ x + y + z = 1 \end{cases}$$

Which gives us $E = (2, -2, 1)$. Then, finding the displacement vector, we have $\overrightarrow{ED} = \left(1, -\frac{3}{2}, \frac{1}{2}\right)$. Using the barycentric distance formula:

$$\text{dist}(D, E) = \sqrt{-a^2yx - b^2zx - c^2xy}$$

$$= \sqrt{-14^2 \left(-\frac{3}{2}\right) \left(\frac{1}{2}\right) - 15^2 \left(1\right) \left(\frac{1}{2}\right) - 13^2 \left(-\frac{3}{2}\right) \left(\frac{1}{2}\right)}$$

$$= \boxed{\text{(D)} 12\sqrt{2}}.$$

12. (AMC 12 12B 2021) Suppose that S is a finite set of positive integers. If the greatest integer in S is removed from S , then the average value (arithmetic mean) of the integers remaining is 32. If the least integer in S is also removed, then the average value of the integers remaining is 35. If the greatest integer is then returned to the set, the average value of the integers rises to 40. The greatest integer in the original set S is 72 greater than the least integer in S . What is the average value of all the integers in the set S ?

(A) 36.2 (B) 36.4 (C) 36.6 (D) 36.8 (E) 37

Answer: D

Solution 1

We can then say that $\frac{A+S(n)}{n+1} = 32$, $\frac{S(n)}{n} = 35$, and $\frac{B+S(n)}{n+1} = 40$.

Expanding gives us $A + S(n) = 32n + 32$, $S(n) = 35n$, and $B + S(n) = 40n + 40$.

Substituting $S(n) = 35n$ to all gives us $A + 35n = 32n + 32$ and $B + 35n = 40n + 40$.

Solving for A and B gives $A = -3n + 32$ and $B = 5n + 40$.

We now need to find $\frac{S(n)+A+B}{n+2}$. We substitute everything to get $\frac{35n+(-3n+32)+(5n+40)}{n+2}$, or $\frac{37n+72}{n+2}$.

Say that the answer to this is Z . Then, Z needs to be a number that makes n a positive integer. The only options that work is $\boxed{\text{(C)} 36.6}$ and $\boxed{\text{(D)} 36.8}$.

However, if 36.6 is an option, we get $n = 3$. So that means that A is 23 and B is 55, and $S(n) = 105$. But if there is 3 terms, then the middle number is 105, but we said that B is the largest number in the set, so therefore our answer cannot be $\boxed{\text{(C)} 36.6}$ and is instead $\boxed{\text{(D)} 36.8}$ and now, we're finished!

Solution 1 Alternative Ending

If A is smaller than B by 72 therefore from the equation on the top you can find out that $N = 8$ using substitution the plug it in to the equation $\frac{37n+72}{n+2}$ then you will get that $Z = \boxed{\text{(D)} 36.8}$.

This was more efficient then the last part.

Solution 2

Let the lowest value be L and the highest G , and let the sum be Z and the amount of numbers n . We have $\frac{Z-G}{n-1} = 32$, $\frac{Z-L-G}{n-2} = 35$, $\frac{Z-L}{n-1} = 40$, and $G = L + 72$. Clearing denominators gives $Z - G = 32n - 32$, $Z - L - G = 35n - 70$, and $Z - L = 40n - 40$. We use $G = L + 72$ to turn the first equation into $Z - L = 32n + 40$. Since $Z - L = 40(n - 1)$ we substitute it into the equation which gives $n = 10$. Turning the second into $Z - 2L = 35n + 2$ using $G = L + 72$ we see $L = 8$ and $Z = 368$ so the average is $\frac{Z}{n} = \boxed{\text{(D)} 36.8}$ aop2014

Solution 3

Let x be the greatest integer, y be the smallest, z be the sum of the numbers in S excluding x and y , and k be the number of elements in S .

Then, $S = x + y + z$

First, when the greatest integer is removed, $\frac{S-x}{k-1} = 32$

When the smallest integer is also removed, $\frac{S-x-y}{k-2} = 35$

When the greatest integer is added back, $\frac{S-y}{k-1} = 40$

We are given that $x = y + 72$

After you substitute $x = y + 72$, you have 3 equations with 3 unknowns S , y and k .

$$S - y - 72 = 32k - 32 \quad S - 2y - 72 = 35k - 70 \quad S - y = 40k - 40$$

This can be easily solved to yield $k = 10$, $y = 8$, $S = 368$.

$$\therefore \text{average value of all integers in the set} = S/k = 368/10 = \boxed{\text{(D)} 36.8}$$

Solution 4

We should plug in 36.2 and assume everything is true except the 35 part. We then calculate that part and end up with 35.75. We also see with the formulas we used with the plug in that when you increase by 0.2 the 35.75 part decreases by 0.25. The answer is then $\boxed{\text{(D)} 36.8}$. You can work backwards because it is multiple choice and you don't have to do critical thinking. Lopkiloim

Solution 5

Let $S = \{a_1, a_2, a_3, \dots, a_n\}$ with $a_1 < a_2 < a_3 < \dots < a_n$. We are given the following:

$$\begin{cases} \sum_{i=1}^{n-1} a_i = 32(n-1) = 32n - 32, \\ \sum_{i=2}^n a_i = 40(n-1) = 40n - 40, \\ \sum_{i=2}^{n-1} a_i = 35(n-2) = 35n - 70, \\ a_n - a_1 = 72 \implies a_1 + 72 = a_n. \end{cases}$$

Subtracting the third equation from the sum of the first two, we find that

$$\sum_{i=1}^n a_i = (32n - 32) + (40n - 40) - (35n - 70) = 37n - 2.$$

Furthermore, from the fourth equation, we have

$$\sum_{i=2}^n a_i - \sum_{i=1}^{n-1} a_i = \left[(a_1 + 72) + \sum_{i=2}^{n-1} a_i \right] - \left[(a_1) + \sum_{i=2}^{n-1} a_i \right] = (40n - 40) - (32n - 32).$$

Combining like terms and simplifying, we have

$$72 = 8n - 8 \implies 8n = 80 \implies n = 10.$$

Thus, the sum of the elements in S is $37 \cdot 10 - 2 = 368$, and since there are 10 elements in S , the average of the elements in S is $\frac{368}{10} = \boxed{\text{(D)} 36.8}$

Solution 6

Let n be the number of elements in S , $m_l = 32$, $\Sigma_l = m_l \cdot (n-1)$, $m_g = 40$, $\Sigma_g = m_g \cdot (n-1)$, $m_{lg} = 35$, $\Sigma_{lg} = m_{lg} \cdot (n-2)$.

$$\Sigma_g - \Sigma_l = 72 = (n-1) \cdot (m_g - m_l) \implies n = 1 + \frac{72}{m_g - m_l} = 1 + \frac{72}{40 - 32} = 10.$$

$$m = \frac{\Sigma_g + \Sigma_l - \Sigma_{lg}}{n} = \frac{40 \cdot 9 + 32 \cdot 9 - 35 \cdot 8}{10} = 36.8.$$

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13. (AMC 12 12B 2021) How many values of θ in the interval $0 < \theta \leq 2\pi$ satisfy

$$1 - 3 \sin \theta + 5 \cos 3\theta = 0?$$

- (A) 2 (B) 4 (C) 5 (D) 6 (E) 8

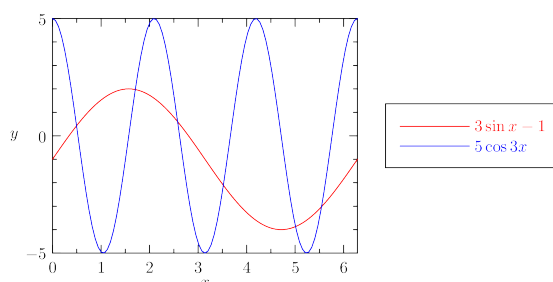
Answer: D

Solution

We rearrange to get

$$5 \cos 3\theta = 3 \sin \theta - 1.$$

We can graph two functions in this case: $y = 5 \cos 3x$ and $y = 3 \sin x - 1$. Using transformation of functions, we know that $5 \cos 3x$ is just a cosine function with amplitude 5 and period $\frac{2\pi}{3}$. Similarly, $3 \sin x - 1$ is just a sine function with amplitude 3 and shifted 1 unit downward:



So, we have **(D) 6** solutions.

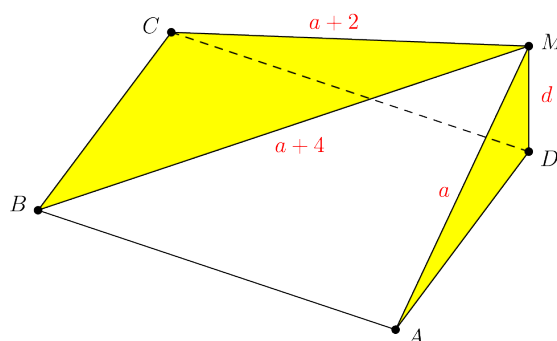
14. (AMC 12 12B 2021) Let $ABCD$ be a rectangle and let $\overline{DM}$ be a segment perpendicular to the plane of $ABCD$. Suppose that $\overline{DM}$ has integer length, and the lengths of $\overline{MA}$, $\overline{MC}$, and $\overline{MB}$ are consecutive odd positive integers (in this order). What is the volume of pyramid $MABCD$? (A) $24\sqrt{5}$ (B) 60 (C) $28\sqrt{5}$ (D) 66 (E) $8\sqrt{70}$

Answer: A

Solution 1

Let $MA = a$ and $MD = d$. It follows that $MC = a + 2$ and $MB = a + 4$.

As shown below, note that $\triangle MAD$ and $\triangle MBC$ are both right triangles.



By the Pythagorean Theorem, we have

$$AD^2 = MA^2 - MD^2 = a^2 - d^2,$$

$$BC^2 = MB^2 - MC^2 = (a+4)^2 - (a+2)^2.$$

Since $AD = BC$ in rectangle $ABCD$, we equate the expressions for AD^2 and BC^2 , then rearrange and factor:

$$\begin{aligned} a^2 - d^2 &= (a + 4)^2 - (a + 2)^2 \\ a^2 - d^2 &= 4a + 12 \\ a^2 - 4a - d^2 &= 12 \\ (a - 2)^2 - d^2 &= 16 \\ (a + d - 2)(a - d - 2) &= 16. \end{aligned}$$

As $a + d - 2$ and $a - d - 2$ have the same parity, we get $a + d - 2 = 8$ and $a - d - 2 = 2$, from which $(a, d) = (7, 3)$.

Applying the Pythagorean Theorem to right $\triangle MAD$ and right $\triangle MCD$, we obtain $AD = 2\sqrt{10}$ and $CD = 6\sqrt{2}$, respectively.

Let the brackets denote areas. Together, the volume of pyramid $MABCD$ is

$$\frac{1}{3} \cdot [ABCD] \cdot MD = \frac{1}{3} \cdot (AD \cdot CD) \cdot MD = \boxed{\text{(A)} \ 24\sqrt{5}}.$$

Lopkiloim MRENTHUSIASM

Solution 2

Let $AD = b$, $CD = a$, $MD = x$, $MC = t$. It follows that $MA = t - 2$ and $MB = t + 2$.

We have three equations:

$$\begin{aligned} a^2 + x^2 &= t^2, \\ a^2 + b^2 + x^2 &= t^2 + 4t + 4, \\ b^2 + x^2 &= t^2 - 4t + 4. \end{aligned}$$

Substituting the first and third equations into the second equation, we get:

$$\begin{aligned} t^2 - 8t - x^2 &= 0 \\ (t - 4)^2 - x^2 &= 16 \\ (t - 4 - x)(t - 4 + x) &= 16. \end{aligned}$$

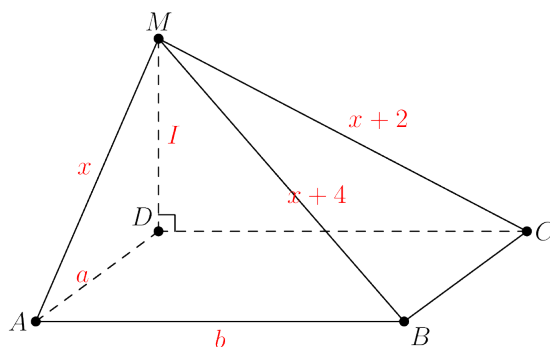
Therefore, we have $t = 9$ and $x = 3$.

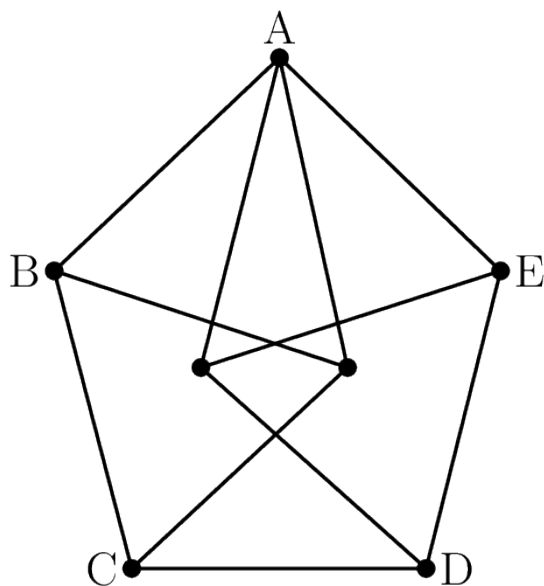
Solving for other values, we get $b = 2\sqrt{10}$, $a = 6\sqrt{2}$. The volume is then

$$\frac{1}{3}abx = \boxed{\text{(A)} \ 24\sqrt{5}}.$$

Solution 3

First and foremost, the diagram must be drawn.

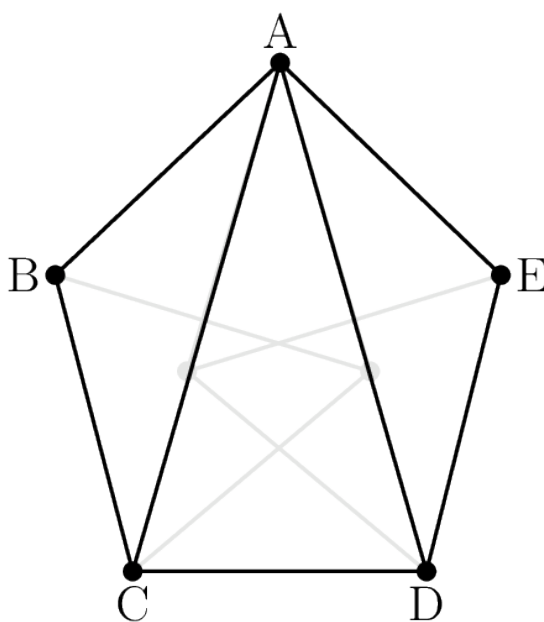




- (A) 20 (B) 21 (C) 22 (D) 23 (E) 24

Answer: D

Solution 1



Draw diagonals AC and AD to split the pentagon into three parts. We can compute the area for each triangle and sum them up at the end. For triangles ABC and ADE , they each have area $2 \cdot \frac{1}{2} \cdot \frac{4\sqrt{3}}{4} = \sqrt{3}$.* For triangle ACD , we can see that $AC = AD = 2\sqrt{3}$ and $CD = 2$. Using Pythagorean Theorem, the altitude for this triangle is $\sqrt{11}$, so the area is $\sqrt{11}$. Adding each part up, we get $2\sqrt{3} + \sqrt{11} = \sqrt{12} + \sqrt{11} \Rightarrow \boxed{\text{(D) } 23}$.

Note: Another way to find the areas would be to use the formula $A = \frac{1}{2}ab \sin C$ -
erringbubble

* A more intuitive way to find this: We know $\angle B = \angle E = 120^\circ$ because of equilateral triangles, so the triangles can each be partitioned into 2 30-60-90 right triangles with side lengths 1, $\sqrt{3}$, and 2. Hence, the area of the each triangle is $2\sqrt{3} \cdot 1 \cdot \frac{1}{2} = \sqrt{3}$. belindazhu13

16. (AMC 12 12B 2021) Let $g(x)$ be a polynomial with leading coefficient 1, whose three roots are the reciprocals of the three roots of $f(x) = x^3 + ax^2 + bx + c$, where $1 < a < b < c$. What is $g(1)$ in terms of a, b , and c ? (A) $\frac{1+a+b+c}{c}$ (B) $1 + a + b + c$ (C) $\frac{1+a+b+c}{c^2}$ (D) $\frac{a+b+c}{c^2}$ (E) $\frac{1+a+b+c}{a+b+c}$

Answer: A

Solution 1

Note that $f(1/x)$ has the same roots as $g(x)$, if it is multiplied by some monomial so that the leading term is x^3 they will be equal. We have

$$f(1/x) = \frac{1}{x^3} + \frac{a}{x^2} + \frac{b}{x} + c$$

so we can see that

$$g(x) = \frac{x^3}{c} f(1/x)$$

Therefore

$$g(1) = \frac{1}{c} f(1) = \boxed{\text{(A)} \frac{1 + a + b + c}{c}}$$

Solution 2 (Vieta's bash)

Let the three roots of $f(x)$ be d, e , and f . (Here e does NOT mean 2.7182818...) We know that $a = -(d + e + f)$, $b = de + ef + df$, and $c = -def$, and that $g(1) = 1 - \frac{1}{d} - \frac{1}{e} - \frac{1}{f} + \frac{1}{de} + \frac{1}{ef} + \frac{1}{df} - \frac{1}{def}$ (Vieta's). This is equal to $\frac{def - de - df - ef + d + e + f - 1}{def}$,

which equals $\boxed{\text{(A)} \frac{1 + a + b + c}{c}}$. -dstanz5

Solution 3 (Fakesolve)

Because the problem doesn't specify what the coefficients of the polynomial actually are, we can just plug in any arbitrary polynomial that satisfies the constraints. Let's take $f(x) = (x + 5)^3 = x^3 + 15x^2 + 75x + 125$. Then $f(x)$ has a triple root of $x = -5$. Then $g(x)$ has a triple root of $-\frac{1}{5}$, and it's monic, so $g(x) = \left(x + \frac{1}{5}\right)^3 = \frac{125x^3 + 75x^2 + 15x + 1}{125}$. We can see that this is $\frac{1+a+b+c}{c}$, which is answer choice $\boxed{\text{(A)}}$.

-Darren Yao

Solution 4

If we let p, q , and r be the roots of $f(x)$, $f(x) = (x - p)(x - q)(x - r)$ and $g(x) = \left(x - \frac{1}{p}\right)\left(x - \frac{1}{q}\right)\left(x - \frac{1}{r}\right)$. The requested value, $g(1)$, is then

$$\left(1 - \frac{1}{p}\right)\left(1 - \frac{1}{q}\right)\left(1 - \frac{1}{r}\right) = \frac{(p-1)(q-1)(r-1)}{pqr}$$

The numerator is $-f(1)$ (using the product form of $f(x)$) and the denominator is $-c$, so the answer is

$$\frac{f(1)}{c} = \boxed{\text{(A)} \frac{1 + a + b + c}{c}}$$

- gting

Solution 5 (Good at Guessing)

The function $g(1)$ = sum of coefficients. If it's $(x-r)(x-s)(x-t)$, then it becomes $(x-\frac{1}{r})(x-\frac{1}{s})(x-\frac{1}{t})$. So, $-rst$ becomes $-\frac{1}{rst}$, so c becomes $\frac{1}{c}$. Also, there is a x^3 so the answer must include 1. The only answer having both of these is A.

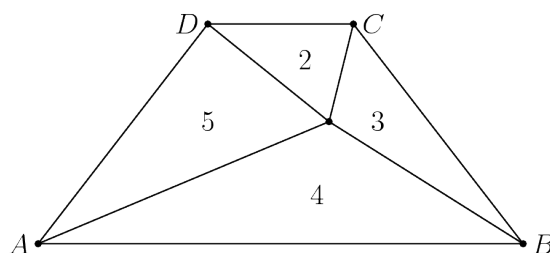
-Extremelysupercooldude (Minor Latex Edits and Grammar)

Solution 6

It is well known that reversing the order of the coefficients of a polynomial turns each root into its corresponding reciprocal. Thus, a polynomial with the desired roots may be written as $cx^3 + bx^2 + a + 1$. As the problem statement asks for a monic polynomial, our answer is

$$\boxed{\text{(A)} \frac{1+a+b+c}{c}}$$

17. (AMC 12 12B 2021) Let $ABCD$ be an isosceles trapezoid having parallel bases $\overline{AB}$ and $\overline{CD}$ with $AB > CD$. Line segments from a point inside $ABCD$ to the vertices divide the trapezoid into four triangles whose areas are 2, 3, 4, and 5 starting with the triangle with base $\overline{CD}$ and moving clockwise as shown in the diagram below. What is the ratio $\frac{AB}{CD}$?



- (A) 3 (B) $2 + \sqrt{2}$ (C) $1 + \sqrt{6}$ (D) $2\sqrt{3}$ (E) $3\sqrt{2}$

Answer: B

Solution

Without the loss of generality, let $\mathcal{T}$ have vertices A , B , C , and D , with $AB = r$ and $CD = s$. Also denote by P the point in the interior of $\mathcal{T}$.

Let X and Y be the feet of the perpendiculars from P to AB and CD , respectively. Observe that $PX = \frac{8}{r}$ and $PY = \frac{4}{s}$. Now using the formula for the area of a trapezoid yields

$$14 = \frac{1}{2} \cdot XY \cdot (AB + CD) = \frac{1}{2} \left(\frac{8}{r} + \frac{4}{s} \right) (r + s) = 6 + 2 \cdot \frac{r}{s} + 4 \cdot \frac{s}{r}.$$

Thus, the ratio $\rho := \frac{r}{s}$ satisfies $\rho + 2\rho^{-1} = 4$; solving yields $\rho = \boxed{\text{(B)} 2 + \sqrt{2}}$.

(Observe that the given areas of 3 and 5 are irrelevant to the ratio $\frac{AB}{CD}$.)

Solution 2

Let b_1 be the bottom base, b_2 be the top base, h_1 be the height of the bottom triangle, h_2 be the height of the top triangle. Thus, $b_1 h_1 = 8$, $b_2 h_2 = 4$, $(b_1 + b_2)(h_1 + h_2) = 28$,

so $b_1h_2 + b_2h_1 = 16$. Let $b_2 = 1, h_2 = 4$, so we get $b_1h_1 = 8, 4b_1 + h_1 = 16$. This gives us a quadratic in b_1 , ie. $4b_1^2 + 8 = 16b_1$, so $b_1 = \boxed{\text{(B)} 2 + \sqrt{2}}$.

- Solution by MathAwesome123, added by ccx09

18. (AMC 12 12B 2021) Let z be a complex number satisfying $12|z|^2 = 2|z+2|^2 + |z^2+1|^2+31$. What is the value of $z+\frac{6}{z}$? (A) -2 (B) -1 (C) $\frac{1}{2}$ (D) 1 (E) 4

Answer: A

Solution 1

Using the fact $z\bar{z} = |z|^2$, the equation rewrites itself as

$$\begin{aligned} 12z\bar{z} &= 2(z+2)(\bar{z}+2) + (z^2+1)(\bar{z}^2+1) + 31 \\ -12z\bar{z} + 2z\bar{z} + 4(z+\bar{z}) + 8 + z^2\bar{z}^2 + (z^2+\bar{z}^2) + 32 &= 0 \\ \left((z^2+2z\bar{z}+\bar{z}^2) + 4(z+\bar{z}) + 4\right) + \left(z^2\bar{z}^2 - 12z\bar{z} + 36\right) &= 0 \\ (z+\bar{z}+2)^2 + (z\bar{z}-6)^2 &= 0. \end{aligned}$$

As the two quantities in the parentheses are real, both quantities must equal 0 so

$$z + \frac{6}{z} = z + \bar{z} = \boxed{\text{(A)} -2}.$$

Solution 2

Let $z = a + bi$, $z^2 = a^2 - b^2 + 2abi$

By the equation given in the problem

$$12(a^2 + b^2) = 2((a+2)^2 + b^2) + ((a^2 - b^2 + 1)^2 + (2ab)^2) + 31$$

$$12a^2 + 12b^2 = 2a^2 + 8a + 8 + 2b^2 + a^4 + b^4 + 1 + 2a^2 - 2b^2 - 2a^2b^2 + 4a^2b^2 + 31$$

$$a^4 + b^4 - 8a^2 - 12b^2 + 2a^2b^2 + 8a + 40 = 0$$

$$(a^2 + b^2)^2 - 12(a^2 + b^2) + 4(a^2 + 2a + 1) + 36 = 0$$

$$(a^2 + b^2 - 6)^2 + 4(a+1)^2 = 0$$

Therefore, $a^2 + b^2 - 6 = 0$ and $a + 1 = 0$ $a = -1$, $b^2 = 6 - 1 = 5$, $b = \sqrt{5}$

$$z + \frac{6}{z} = \frac{a^2 - b^2 + 6 + 2abi}{a + bi} = \frac{1 - 5 + 6 + 2(-1)\sqrt{5}i}{-1 + i\sqrt{5}} = \frac{2 - 2i\sqrt{5}}{-1 + i\sqrt{5}} = \boxed{\text{(A)} -2}$$

should be $b = \sqrt{5}$ or $b = -\sqrt{5}$ but it won't affect the result. eddybai

Solution 3

Let $x = z + \frac{6}{z}$. Then $z = \frac{x \pm \sqrt{x^2 - 24}}{2}$. From the answer choices we know that x is real and $x^2 < 24$, so $z = \frac{x \pm i\sqrt{24 - x^2}}{2}$. Then we have

$$|z|^2 = 6$$

$$|z + 2|^2 = \left(\frac{x}{2} + 2\right)^2 + \frac{24 - x^2}{4} = 2x + 10$$

$$|z^2 + 1|^2 = |xz - 6 + 1|^2 = \left(\frac{x^2}{2} - 5\right)^2 + \frac{x^2(24 - x^2)}{4} = x^2 + 25$$

Plugging the above back to the original equation, we have

$$12 \cdot 6 = 2(2x + 10) + x^2 + 25 + 31$$

$$(x + 2)^2 = 0$$

So $x = \boxed{(\mathbf{A}) - 2}$.

Solution 4 (Funny Observations)

There are actually several ways to see that $|z|^2 = 6$. I present two troll ways of seeing it, and a legitimate way of checking.

Rewrite using $w\bar{w} = |w|^2$ $12z\bar{z} + 2(z + 2)(\bar{z} + 2) + (z^2 + 1)(\bar{z}^2 + 1) + 3112z\bar{z} = 2z\bar{z} + 4z + 4\bar{z} + 8 + z^2\bar{z}^2 + z^2 + \bar{z}^2 + 1 + 31.12z\bar{z} = 4(z + \bar{z}) + (z\bar{z})^2 + (z + \bar{z})^2 + 40$.

Symmetric in z and $\bar{z}$, so if w is a sol, then so is $\bar{w}$

TROLL OBSERVATION #1: ALL THE ANSWERS ARE REAL. THUS, $z + \frac{6}{z} \in \mathbb{R}$, which means they must be conjugates and so $|z|^2 = 6$.

TROLL OBSERVATION #2: Note that $z + \frac{6}{z} = \bar{z} + \frac{6}{\bar{z}}$ because either solution must give the same answer! which means that $|z|^2 = 6$.

Alternatively, you can check: Let $a = w + \bar{w} \in \mathbb{R}$, and $r = |w|^2 \in \mathbb{R}$. Thus, we have $a^2 + 4a + 40 + r^2 - 12r = 0$, and the discriminant of this must be nonnegative as a is real. Thus, $16 - 4(40 + r^2 - 12r) \geq 0$ or $(r - 6)^2 \leq 0$, which forces $r = 6$, as claimed.

Thus, we plug in $z\bar{z} = 6$, and get: $72 = 4(z + \bar{z}) + 76 + (z + \bar{z})^2$, ie. $(z + \bar{z})^2 + 4(z + \bar{z}) + 4 = 0$, or $(z + \bar{z} + 2)^2 = 0$, which means $z + \bar{z} = \boxed{(\mathbf{A}) - 2}$ and that's our answer since we know $\bar{z} = 6/z$

- ccx09

Solution 5

Observe that all the answer choices are real. Therefore, z and $\frac{6}{z}$ must be complex conjugates as this is the only way for both their sum (one of the answer choices) and their product (6) to be real. Thus $|z| = \left|\frac{6}{z}\right| = \sqrt{6}$. We will test all the answer choices, starting with **(A)**. Suppose the answer is **(A)**. If $z + \frac{6}{z} = -2$ then $z^2 + 2z + 6 = 0$ and $z = \frac{-2 \pm \sqrt{4 - 24}}{2} = -1 \pm \sqrt{5}i$. Note that if $z = -1 + \sqrt{5}i$ works, then so does $-1 - \sqrt{5}i$. It is relatively easy to see that if $z = -1 + \sqrt{5}i$, then $12|z|^2 = 12 \cdot 6 = 72$, $2|z + 2|^2 = 2|1 + \sqrt{5}i|^2 = 2 \cdot 6 = 12$, $|z^2 + 1|^2 = |-3 - 2\sqrt{5}i|^2 = 29$, and $72 = 12 + 29 + 31$. Thus the condition

$$12|z|^2 = 2|z + 2|^2 + |z^2 + 1|^2 + 31$$

is satisfied for $z + \frac{6}{z} = -2$, and the answer is $\boxed{(\mathbf{A}) - 2}$.

Solution 6

Using $z\bar{z} = |z|^2$, we have

$$\begin{aligned} 12z\bar{z} &= 2(z+2)(\bar{z}+2) + (z^2+1)(\bar{z}^2+1) + 31 \implies z^2\bar{z}^2 + z^2 + \bar{z}^2 + 1 + 2z\bar{z} + 4z + 4\bar{z} + 8 + 31 - 12z\bar{z} \\ &\implies z^2\bar{z}^2 - 10z\bar{z} + (z^2 + \bar{z}^2) + 4(z + \bar{z}) + 40 = 0 \end{aligned}$$

Let $p = z\bar{z}$ and $s = z + \bar{z}$. Then we get

$$p^2 - 10p + s^2 - 2p + 4s + 40 = 0 \implies p^2 - 12p + s^2 + 4s + 40 = 0$$

Completing the square, we get

$$(p-6)^2 - 36 + (s+2)^2 - 4 + 40 = (p-6)^2 + (s+2)^2 = 0$$

Therefore, $p = 6$ and $s = -2$. So, $z + \bar{z} = -2$ and $z\bar{z} = 6$. Plugging into $z + \frac{6}{z}$, we get

$$z + \frac{6}{z} = z + \frac{6\bar{z}}{z\bar{z}} = z + \frac{6\bar{z}}{6} = z + \bar{z} = \boxed{\text{(A)} - 2}$$

19. (AMC 12 12B 2021) Two fair dice, each with at least 6 faces are rolled. On each face of each die is printed a distinct integer from 1 to the number of faces on that die, inclusive. The probability of rolling a sum of 7 is $\frac{3}{4}$ of the probability of rolling a sum of 10, and the probability of rolling a sum of 12 is $\frac{1}{12}$. What is the least possible number of faces on the two dice combined?

(A) 16 (B) 17 (C) 18 (D) 19 (E) 20

Answer: B

Solution 1

Suppose the dice have a and b faces, and WLOG $a \geq b$. Since each die has at least 6 faces, there will always be 6 ways to sum to 7. As a result, there must be $\frac{4}{3} \cdot 6 = 8$ ways to sum to 10. There are at most nine distinct ways to get a sum of 10, which are possible whenever $a, b \geq 9$. To achieve exactly eight ways, b must have 8 faces, and $a \geq 9$. Let n be the number of ways to obtain a sum of 12, then $\frac{n}{8a} = \frac{1}{12} \implies n = \frac{2}{3}a$. Since $b = 8$, $n \leq 8 \implies a \leq 12$. In addition to $3 \mid a$, we only have to test $a = 9, 12$, of which both work. Taking the smaller one, our answer becomes $a + b = 9 + 8 = \boxed{\text{(B)} 17}$.

Solution 2

Suppose the dice have a and b faces, and WLOG $a \geq b$. Note that if $a + b = 12$ since they are both 6, there is one way to make 12, and incrementing a or b by one will add another way. This gives us the probability of making a 12 as

$$\frac{a+b-11}{ab} = \frac{1}{12}$$

Cross-multiplying, we get

$$12a + 12b - 132 = ab$$

Simon's Favorite Factoring Trick now gives

$$(a-12)(b-12) = 12$$

This narrows the possibilities down to 3 ordered pairs of (a, b) , which are $(13, 24)$, $(6, 10)$, and $(8, 9)$. We can obviously ignore the first pair and test the next two straightforwardly. The last pair yields the answer:

$$\frac{6}{72} = \frac{3}{4} \left(\frac{9 + 8 - 9}{72} \right)$$

The answer is then $a + b = 8 + 9 = \boxed{\text{(B)} 17}$.

Solution 2.1

notice that if a or $b \geq 12$ then this relationship doesn't stand. the value of a and b should be discussed. Hence the final solution should be $(6, 10)$, and $(8, 9)$ and $(12, b)$ ($b \leq 11$)

$$\frac{\min(a, 11) + \min(b, 11) - 11}{ab} = \frac{1}{12}$$

and to check $P(7)$ and $P(10)$, simply use

$$\frac{\min(a, 6) + \min(b, 6) - 6}{ab}$$

and

$$\frac{\min(a, 9) + \min(b, 9) - 9}{ab}$$

eddybai

Solution 3 (Logic)

Notice that

$$\begin{aligned} 7 &= 1 + 6 \\ &\vdots \\ &= 6 + 1 \end{aligned}$$

are the only cases that could possibly form 7. In another words, regardless of the number of faces for each dice, 6 is the number of cases that could form a rolling sum of 7.

Because the value of the total combination for both are the same, we could infer that the number of cases of having a rolling sum of 10 is $\frac{4}{3} \cdot 6 = 8$. With the number 8, we could also deduce that one dice has 8 sides and the other has at least 9 sides. Thence trial and error could be utilized.

Since 9 is the next smallest number, the case could be tested.

$$\begin{aligned} 12 &= 3 + 9 \\ &= 4 + 8 \\ &\vdots \\ &= 8 + 4 \end{aligned}$$

$\frac{6}{8 \cdot 9} = \frac{1}{12}$ is also true. Therefore, $8 + 9 = \boxed{\text{(B)} 17}$

Solution 4

Let the two dice have at least six faces each, and let their face counts sum to N . We seek the minimum possible N such that $p(\text{sum} = 12) = \frac{1}{12}$ and $p(\text{sum} = 7) = \frac{3}{4}p(\text{sum} = 10)$.

First try $N = 16$. Then the possible face-count pairs (m, n) with $m \leq n$ are $(6, 10)$, $(7, 9)$, $(8, 8)$. For $p(\text{sum} = 12) = \frac{1}{12}$ to be possible, the total outcomes mn must be a multiple of 12. But $7 \cdot 9 = 63$ and $8 \cdot 8 = 64$ are not multiples of 12, so $(7, 9)$ and $(8, 8)$ are impossible.

For $(6, 10)$, there are 60 outcomes. The combinations that sum to 12 are $(2, 10)$, $(3, 9)$, $(4, 8)$, $(5, 7)$, $(6, 6)$, five in total, giving $p(12) = \frac{5}{60} = \frac{1}{12}$. The combinations that sum to 10 are $(1, 9)$, $(2, 8)$, $(3, 7)$, $(4, 6)$, $(5, 5)$, $(6, 4)$, six in total, giving $p(10) = \frac{6}{60} = \frac{1}{10}$. The combinations that sum to 7 are $(1, 6)$, $(2, 5)$, $(3, 4)$, $(4, 3)$, $(5, 2)$, $(6, 1)$, also six in total, giving $p(7) = \frac{6}{60} = \frac{1}{10}$. So $p(7) = p(10)$, which does not satisfy the condition. Hence $N = 16$ is impossible.

Next try $N = 17$. The possible pairs are $(6, 11)$, $(7, 10)$, $(8, 9)$. Again, mn must be a multiple of 12, but $6 \cdot 11 = 66$ and $7 \cdot 10 = 70$ are not; discard these.

For $(8, 9)$, there are 72 outcomes. The combinations that sum to 12 are $(3, 9)$, $(4, 8)$, $(5, 7)$, $(6, 6)$, $(7, 5)$, $(8, 4)$, six in total, so $p(12) = \frac{6}{72} = \frac{1}{12}$. The combinations that sum to 10 are $(1, 9)$, $(2, 8)$, $(3, 7)$, $(4, 6)$, $(5, 5)$, $(6, 4)$, $(7, 3)$, $(8, 2)$, eight in total, so $p(10) = \frac{8}{72} = \frac{1}{9}$. The combinations that sum to 7 are $(1, 6)$, $(2, 5)$, $(3, 4)$, $(4, 3)$, $(5, 2)$, $(6, 1)$, six in total, so $p(7) = \frac{6}{72} = \frac{1}{12}$.

Thus $p(12) = \frac{1}{12}$ and $p(7) = \frac{1}{12} = \frac{3}{4} \cdot \frac{1}{9} = \frac{3}{4}p(10)$.

Therefore $N = 17$ works, and since $N = 16$ is impossible, the minimum is (B) 17.

20. (AMC 12 12B 2021) Let $Q(z)$ and $R(z)$ be the unique polynomials such that

$$z^{2021} + 1 = (z^2 + z + 1)Q(z) + R(z)$$

and the degree of R is less than 2. What is $R(z)$? (A) $-z$ (B) -1 (C) 2021 (D) $z + 1$ (E) $2z + 1$

Answer: A

Solution 1 (Difference of Cubes)

Let $z = s$ be a root of $z^2 + z + 1$ so that $s^2 + s + 1 = 0$. It follows that

$$(s - 1)(s^2 + s + 1) = s^3 - 1 = 0,$$

from which $s^3 = 1$, but $s \neq 1$.

Note that

$$\begin{aligned} s^{2021} + 1 &= s^{3 \cdot 673 + 2} + 1 \\ &= (s^3)^{673} \cdot s^2 + 1 \\ &= s^2 + 1 \\ &= (s^2 + s + 1) - s \\ &= -s. \end{aligned}$$

Since $z^{2021} + 1 = -z$ for each root $z = s$ of $z^2 + z + 1$, the remainder when $z^{2021} + 1$ is divided by $z^2 + z + 1$ is $R(z) = \boxed{\text{(A)} -z}$.

Solution 2 (Finds $Q(z)$ Using Patterns)

Note that the equation above is in the form of polynomial division, with $z^{2021} + 1$ being the dividend, $z^2 + z + 1$ being the divisor, and $Q(x)$ and $R(x)$ being the

quotient and remainder respectively. Since the degree of the dividend is 2021 and the degree of the divisor is 2, that means the degree of the quotient is $2021 - 2 = 2019$. Note that $R(x)$ can't influence the degree of the right hand side of this equation since its degree is either 1 or 0. Since the coefficients of the leading term in the dividend and the divisor are both 1, that means the coefficient of the leading term of the quotient is also 1. Thus, the leading term of the quotient is z^{2019} . Multiplying z^{2019} by the divisor gives $z^{2021} + z^{2020} + z^{2019}$. We have our z^{2021} term but we have these unnecessary terms like z^{2020} . We can get rid of these terms by adding $-z^{2018}$ to the quotient to cancel out these terms, but this then gives us $z^{2021} - z^{2018}$. Our first instinct will probably be to add z^{2017} , but we can't do this as although this will eliminate the $-z^{2018}$ term, it will produce a z^{2019} term. Since no other term of the form z^n where n is an integer less than 2017 will produce a z^{2019} term when multiplied by the divisor, we can't add z^{2017} to the quotient. Instead, we can add z^{2016} to the coefficient to get rid of the $-z^{2018}$ term. Continuing this pattern, we get the quotient as

$$z^{2019} - z^{2018} + z^{2016} - z^{2015} + \dots - z^2 + 1.$$

The last term when multiplied with the divisor gives $z^2 + z + 1$. This will get rid of the $-z^2$ term but will produce the expression $z + 1$, giving us the dividend as $z^{2021} + z + 1$. Note that the dividend we want is of the form $z^{2021} + 1$. Therefore, our remainder will have to be $-z$ in order to get rid of the z term in the expression and give us $z^{2021} + 1$, which is what we want. Therefore, the remainder is $\boxed{(\mathbf{A}) - z}$.

Solution 3 (Modular Arithmetic in Polynomials)

Note that

$$z^3 - 1 \equiv 0 \pmod{z^2 + z + 1}$$

so if $F(z)$ is the remainder when dividing by $z^3 - 1$,

$$F(z) \equiv R(z) \pmod{z^2 + z + 1}.$$

Now,

$$z^{2021} + 1 = (z^3 - 1)(z^{2018} + z^{2015} + \dots + z^2) + z^2 + 1$$

So $F(z) = z^2 + 1$, and

$$R(z) \equiv F(z) \equiv -z \pmod{z^2 + z + 1}$$

The answer is $\boxed{(\mathbf{A}) - z}$.

Solution 4 (Complex Numbers)

One thing to note is that $R(z)$ takes the form of $Az + B$ for some constants A and B . Note that the roots of $z^2 + z + 1$ are part of the solutions of $z^3 - 1 = 0$. They can be easily solved with roots of unity:

$$\begin{aligned} z^3 &= 1 \\ z^3 &= e^{i0} \\ z &= e^{i0}, e^{i\frac{2\pi}{3}}, e^{i(-\frac{2\pi}{3})}. \end{aligned}$$

Obviously the right two solutions are the roots of $z^2 + z + 1 = 0$. We substitute $e^{i\frac{2\pi}{3}}$ into the original equation, and $z^2 + z + 1$ becomes 0. Using De Moivre's Theorem, we get

$$e^{i\frac{4042\pi}{3}} + 1 = A \cdot e^{i\frac{2\pi}{3}} + B$$

$$e^{i\frac{4\pi}{3}} + 1 = A \cdot e^{i\frac{2\pi}{3}} + B.$$

Expanding into rectangular complex number form:

$$\frac{1}{2} - \frac{\sqrt{3}}{2}i = \left(-\frac{1}{2}A + B\right) + \frac{\sqrt{3}}{2}Ai.$$

Comparing the real and imaginary parts, we get $(A, B) = (-1, 0)$. So, the answer is **(A)** $-z$.

Solution 5 (Complex Numbers but Easier)

We have motivation to get rid of the $Q(z)$ term by substituting in either $e^{\frac{2\pi i}{3}}$ or $e^{\frac{4\pi i}{3}}$ for z , as this sets $z^2 + z + 1$ equal to 0. Doing so,

$$\begin{aligned} \left(e^{\frac{2\pi i}{3}}\right)^{2021} + 1 &= R\left(e^{\frac{2\pi i}{3}}\right) \\ e^{\frac{4042\pi i}{3}} + 1 &= R\left(e^{\frac{2\pi i}{3}}\right) \\ e^{\frac{4\pi i}{3}} + 1 &= R\left(e^{\frac{2\pi i}{3}}\right) \\ \frac{1}{2} - \frac{\sqrt{3}}{2}i &= R\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right). \end{aligned}$$

This immediately eliminates choices **(B)** and **(C)**, as they are constants. Upon a quick check, $R(z) = 2z + 1$ and $z + 1$ both don't work, as the sign of $\frac{\sqrt{3}}{2}$ has to be negative. Checking $R(z)$ (or realizing that it's the only one that actually works), we see that **(A)** $-z$ is the right answer.

-Benedict T (countmath1)

Solution 6 (More Complex Numbers)

Since $z^{2021} + 1 = (z^2 + z + 1)Q(z) + R(z)$, we can plug in either $\text{cis}(240^\circ)$ or $\text{cis}(120^\circ)$. Assuming $R(z) = ax + b$, we have $a(\text{cis}(240^\circ)) + b = \frac{1}{2} + \frac{i\sqrt{3}}{2}$ and $a(\text{cis}(120^\circ)) + b = \frac{1}{2} - \frac{i\sqrt{3}}{2}$. Solving gives us $a = -1$ and $b = 0$, thus $R(z) = \mathbf{(A)} -z$.

Solution 7 (Factorization)

Notice that $(z^{2018} + z^{2015} + z^{2012} + \dots + z^5 + z^2)(z^2 + z + 1) = z^{2020} + z^{2019} + \dots + z^4 + z^3 + z^2$, so

$$\begin{aligned} z^{2021} + 1 &= z^{2021} - 1 + 2 = (z - 1)(z^{2020} + z^{2019} + \dots + z^2 + z + 1) + 2 \\ &= (z - 1)[(z^{2018} + z^{2015} + z^{2012} + \dots + z^5 + z^2)(z^2 + z + 1) + z + 1] + 2 \\ &= (z - 1)(z^{2018} + z^{2015} + z^{2012} + \dots + z^5 + z^2)(z^2 + z + 1) + (z - 1)(z + 1) + 2 \\ &= (z - 1)(z^{2018} + z^{2015} + z^{2012} + \dots + z^5 + z^2)(z^2 + z + 1) + z^2 + 1 \\ &= (z - 1)(z^{2018} + z^{2015} + z^{2012} + \dots + z^5 + z^2)(z^2 + z + 1) + z^2 + z + 1 - z \\ &= (z^2 + z + 1)[(z^{2018} + z^{2015} + z^{2012} + \dots + z^5 + z^2)(z - 1) + 1] - z. \end{aligned}$$

Therefore, we have

$$Q(z) = (z^{2018} + z^{2015} + z^{2012} + \dots + z^5 + z^2)(z - 1) + 1,$$

from which $R(z) = \mathbf{(A)} -z$.

Solution 8

Let $P(z) = z^{2021} + 1$ and r be a root of $z^2 + z + 1$, then note that

$$\begin{aligned} r^{3k} &= 1 \\ r^{3k+1} &= r \\ r^{3k+2} &= -r - 1 \end{aligned}$$

for all non-negative integers k . Since

$$P(r) = (r^2 + r + 1)Q(r) + R(r) = R(r)$$

and $P(r) = r^{2021} + 1 = (-r - 1) + 1 = -r$, $R(r) = -r$. Therefore, $\boxed{\text{(A)} - z}$ is the only feasible option.

Solution 9 (Modular Arithmetic in less than a minute, basically a slightly modified solution 1)

Notice that $z^3 - 1 \equiv 0 \pmod{z^2 + z + 1}$ because $z^3 - 1 = (z - 1)(z^2 + z + 1)$. So we can add this to $z^{2021} + 1$:

$$\begin{aligned} z^{2021} + z^3 &\equiv z^{2021} + 1 \pmod{z^2 + z + 1} \\ \implies z^3(z^{2018} + 1) &\equiv 1 * (z^{2018} + 1) \pmod{z^2 + z + 1} \end{aligned}$$

We can generalize this by saying $z^{2021-3n} + 1 \equiv z^{2021} + 1 \pmod{z^2 + z + 1}$ as long as $2021 - 3n$ is positive. So we can minimize $2021 - 3n$ by setting $n = 673$. Then we get:

$$z^2 + 1 \equiv z^{2021} + 1 \pmod{z^2 + z + 1}$$

And this is clearly $\boxed{\text{(A)} - z}$.

Solution 10 (Easy Modular arithmetic, Similar to solution 9)

Similarly $z^3 - 1 \equiv 0 \pmod{z^2 + z + 1}$ because $z^3 - 1 = (z - 1)(z^2 + z + 1)$.

So,

$$\begin{aligned} z^3 &\equiv 1 \pmod{z^2 + z + 1} \\ \implies R(z) &\equiv z^{2021} + 1 \equiv (z^3)^{673} \cdot z^2 + 1 \equiv z^2 + 1 \pmod{z^2 + z + 1} \end{aligned}$$

Meaning,

$$R(z) \equiv z^2 + 1 \pmod{z^2 + z + 1}$$

Notice $z^2 \equiv -(z + 1) \pmod{z^2 + z + 1} \implies z^2 + 1 \equiv -(z + 1) + 1 \equiv -z \pmod{z^2 + z + 1}$ so

$$R(z) \equiv -z \pmod{z^2 + z + 1}$$

Gives our answer $\boxed{\text{(A)} - z}$.

21. (AMC 12 12B 2021) Let S be the sum of all positive real numbers x for which

$$x^{2^{\sqrt{2}}} = \sqrt{2}^{2^x}.$$

Which of the following statements is true?

- (A) $S < \sqrt{2}$ (B) $S = \sqrt{2}$ (C) $\sqrt{2} < S < 2$ (D) $2 \leq S < 6$ (E) $S \geq 6$

Answer: D

Solution 1

Note that

$$\begin{aligned} x^{2^{\sqrt{2}}} &= \sqrt{2}^{2^x} \\ 2^{\sqrt{2}} \log_2 x &= 2^x \log_2 \sqrt{2}. \end{aligned}$$

(At this point we see by inspection that $x = \sqrt{2}$ is a solution.)

We simplify the RHS, then take the base-2 logarithm for both sides:

$$\begin{aligned} 2^{\sqrt{2}} \log_2 x &= 2^{x-1} \\ \log_2 (2^{\sqrt{2}} \log_2 x) &= x - 1 \\ \sqrt{2} + \log_2 \log_2 x &= x - 1 \\ \log_2 \log_2 x &= x - 1 - \sqrt{2}. \end{aligned}$$

The RHS is a line; the LHS is a concave curve that looks like a logarithm and has x intercept at $(2, 0)$.

There are at most two solutions, one of which is $\sqrt{2}$. But note that at $x = 2$, we have $\log_2 \log_2 2 = 0 > 2 - 1 - \sqrt{2}$, meaning that the log log curve is above the line, so it must intersect the line again at a point $x > 2$. Now we check $x = 4$ and see that $\log_2 \log_2 4 = 1 < 4 - 1 - \sqrt{2}$, which means at $x = 4$ the line is already above the log log curve. Thus, the second solution lies in the interval $(2, 4)$. The answer is

(D) $2 \leq S < 6$.

Solution 2

We rewrite the right side without using square roots, then take the base-2 logarithm for both sides:

$$\begin{aligned} x^{2^{\sqrt{2}}} &= \left(2^{\frac{1}{2}}\right)^{2^x} \\ x^{2^{\sqrt{2}}} &= 2^{\frac{1}{2} \cdot 2^x} \\ x^{2^{\sqrt{2}}} &= 2^{2^{x-1}} \\ \log_2 (x^{2^{\sqrt{2}}}) &= \log_2 (2^{2^{x-1}}) \\ 2^{\sqrt{2}} \log_2 x &= 2^{x-1}. \end{aligned} \tag{*}$$

By observations, $x = \sqrt{2}$ is one solution. Graphing $f(x) = 2^{\sqrt{2}} \log_2 x$ and $g(x) = 2^{x-1}$, we conclude that (*) has two solutions, with the smaller solution $x = \sqrt{2}$.

We construct the following table of values:

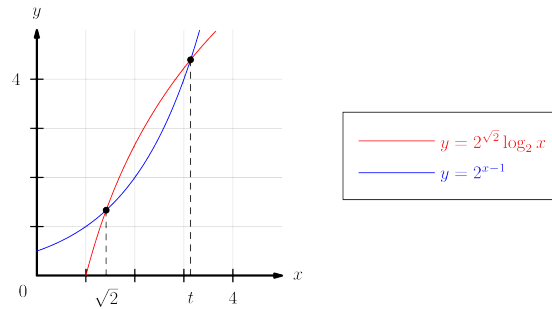
x	$f(x) = 2^{\sqrt{2}} \log_2 x$	$g(x) = 2^{x-1}$	Comparison
1	0	1	
$\sqrt{2}$	$2^{\sqrt{2}-1}$	$2^{\sqrt{2}-1}$	$f(\sqrt{2}) = g(\sqrt{2})$
2	$2^{\sqrt{2}}$	2	$f(2) > g(2)$
3	$2^{\sqrt{2}} \log_2 3$	2^2	
4	$2^{\sqrt{2}+1}$	2^3	$f(4) < g(4)$

Let $x = t$ be the larger solution. Since exponential functions outgrow logarithmic functions, we have $f(x) < g(x)$ for all $x > t$. By the **Intermediate Value Theorem**, we get $t \in (2, 4)$, from which

$$S = \sqrt{2} + t \in (\sqrt{2} + 2, \sqrt{2} + 4).$$

Finally, approximating with $\sqrt{2} \approx 1.414$ results in **(D)** $2 \leq S < 6$.

The graphs of $y = f(x)$ and $y = g(x)$ are shown below:



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Solution 3 (No Logarithm)

While utilizing log may seem conventional, graphing may also be used. Notice that $x^{2^{\sqrt{2}}}$ is a U-shaped, differentiable curve. Moreover, $\sqrt{2}^{2^x}$ is an exponentially increasing function. Furthermore, we could notice that the two graphs meet at $x = \sqrt{2}$.

x	$x^{2^{\sqrt{2}}}$	$\sqrt{2}^{2^x}$	Comparison
0	0	$\sqrt{2}$	$x^{2^{\sqrt{2}}} < \sqrt{2}^{2^x}$
1	$2 < 2^{\sqrt{2}} < 4$	2	$x^{2^{\sqrt{2}}} < \sqrt{2}^{2^x}$
$\sqrt{2}$	$\sqrt{2}^{2^{\sqrt{2}}}$	$\sqrt{2}^{2^{\sqrt{2}}}$	$x^{2^{\sqrt{2}}} = \sqrt{2}^{2^x}$
2	$4 < 2^{2^{\sqrt{2}}} < 16$	4	$x^{2^{\sqrt{2}}} > \sqrt{2}^{2^x}$
3	$9 < 3^{2^{\sqrt{2}}} < 81$	16	NA
4	$16 < 4^{2^{\sqrt{2}}} < 256$	256	$x^{2^{\sqrt{2}}} < \sqrt{2}^{2^x}$

Because the $x^2 < x^{2^{\sqrt{2}}} < x^4$ and $\sqrt{2}^{2^x}$ is an exponentially increasing function, $x^{2^{\sqrt{2}}}$ can never meet or catch up $\sqrt{2}^{2^x}$ from $x \geq 4$. In another words, only two intersections, or the roots of the equation, exists. The roots are $\sqrt{2}$ and a constant

in the interval $2 < r_2 < 4$. Let $S = \sqrt{2} + r_2$. It is evident that the possible range for S is **(D)** $2 \leq S < 6$.

22. (AMC 12 12B 2021) Arjun and Beth play a game in which they take turns removing one brick or two adjacent bricks from one "wall" among a set of several walls of bricks, with gaps possibly creating new walls. The walls are one brick tall. For example, a set of walls of sizes 4 and 2 can be changed into any of the following by one move: (3, 2), (2, 1, 2), (4), (4, 1), (2, 2), or (1, 1, 2).



Arjun plays first, and the player who removes the last brick wins. For which starting configuration is there a strategy that guarantees a win for Beth?

- (A) (6, 1, 1) (B) (6, 2, 1) (C) (6, 2, 2) (D) (6, 3, 1) (E) (6, 3, 2)

Answer: B

Solution 1

We say that a game state is an N-position if it is winning for the next player (the player to move), and a P-position if it is winning for the other player. We are trying to find which of the given states is a P-position.

First we note that symmetrical positions are P-positions, as the second player can win by mirroring the first player's moves. It follows that (6, 1, 1) is an N-position, since we can win by moving to (2, 2, 1, 1); this rules out **(A)**. We next look at (6, 2, 1). The possible next states are

(6, 2), (6, 1, 1), (6, 1), (5, 2, 1), (4, 2, 1, 1), (4, 2, 1), (3, 2, 2, 1), (3, 2, 1, 1), (2, 2, 2, 1).

None of these are symmetrical, so we might reasonably suspect that they are all N-positions. Indeed, it just so happens that for all of these states except (6, 2) and (6, 1), we can win by moving to (2, 2, 1, 1); it remains to check that (6, 2) and (6, 1) are N-positions.

To save ourselves work, it would be nice if we could find a single P-position directly reachable from both (6, 2) and (6, 1). We notice that (3, 2, 1) is directly reachable from both states, so it would suffice to show that (3, 2, 1) is a P-position. Indeed, the possible next states are

(3, 2), (3, 1, 1), (3, 1), (2, 2, 1), (2, 1, 1, 1), (2, 1, 1),

which allow for the following refutations:

(3, 2) $\rightarrow$ (2, 2), (3, 1, 1) $\rightarrow$ (1, 1, 1, 1), (3, 1) $\rightarrow$ (1, 1),
 (2, 2, 1) $\rightarrow$ (2, 2), (2, 1, 1, 1) $\rightarrow$ (1, 1, 1, 1), (2, 1, 1) $\rightarrow$ (1, 1).

Hence, (3, 2, 1) is a P-position, so (6, 2) and (6, 1) are both N-positions, along with all other possible next states from (6, 2, 1) as noted before. Thus, (6, 2, 1) is a P-position, so our answer is **(B)**. (For completeness, we could also rule out **(C)**, **(D)** and **(E)** as in Solution 2.)

-Note: In general, this game is very complicated. For example $(8, 7, 5, 3, 2)$ is winning for the first player but good luck showing that.

Solution 2 (Elimination)

$(6, 1, 1)$ can be turned into $(2, 2, 1, 1)$ by Arjun, which is symmetric, so Beth will lose.

$(6, 3, 1)$ can be turned into $(3, 1, 3, 1)$ by Arjun, which is symmetric, so Beth will lose.

$(6, 2, 2)$ can be turned into $(2, 2, 2, 2)$ by Arjun, which is symmetric, so Beth will lose.

$(6, 3, 2)$ can be turned into $(3, 2, 3, 2)$ by Arjun, which is symmetric, so Beth will lose.

That leaves $(6, 2, 1)$ or **(B)**.

Solution 3 (Nim-values)

Let the nim-value of the ending game state, where someone has just removed the final brick, be 0. Then, any game state with a nim-value of 0 is losing. It is well-known that the nim-value of a supergame (a combination of two or more individual games) is the binary xor function on the nim-values of the individual games that compose the supergame. Therefore, we calculate the nim-values of the states with a single wall up to 6 bricks long (since the answer choices only go up to 6).

First, the game with 1 brick has a nim-value of 1.

Similarly, the game with 2 bricks has a nim-value of 2.

Next, we consider a 3 brick wall. After the next move, the possible resulting game states are 1 brick, a 2 brick wall, or 2 separate bricks. The first two options have nim-values of 1 and 2. The final option has a nim-value of $1 \oplus 1 = 0$, so the nim-value of this game state is 3.

Next, the 4 brick wall. The possible states are a 2 brick wall, a 3 brick wall, a 2 brick wall and a 1 brick wall, or a 1 brick wall and a 1 brick wall. The nim-values of these states are 2, 3, 3, and 0, respectively, and hence the nim-value of this game state is 1.

[Why is the nim-value of it 1? - awesomediabrine

[https://en.wikipedia.org/wiki/Mex_\(mathematics\)](https://en.wikipedia.org/wiki/Mex_(mathematics))]

The possible game states after the 5 brick wall are the following: a 3 brick wall, a 4 brick wall, a 3 brick wall and a 1 brick wall, a two 2 brick walls, and a 2 brick wall plus a 1 brick wall. The nim-values of these are 3, 1, 2, 0, and 3, respectively, meaning the nim-value of a 5 brick wall is 4.

Finally, we find the nim-value of a 6 brick wall. The possible states are a 5 brick wall, a 4 brick wall and a 1 brick wall, a 3 brick wall and a 2 brick wall, a 4 brick wall, a 3 brick wall and a 1 brick wall, and finally two 2 brick walls. The nim-values of these game states are 4, 0, 1, 1, 2, and 0, respectively. This means the 6 brick wall has a nim-value of 3.

The problem is asking which of the answer choices is losing, or has a nim-value of 0. We see that option **(A)** has a nim-value of $3 \oplus 1 \oplus 1 = 3$, option **(B)** has a nim-value of $3 \oplus 2 \oplus 1 = 0$, option **(C)** has a nim-value of $3 \oplus 2 \oplus 2 = 3$, option **(D)**

has a nim-value of $3 \oplus 3 \oplus 1 = 1$, and option **(E)** has a nim-value of $3 \oplus 3 \oplus 2 = 2$, so the answer is **(B)** $(6, 2, 1)$.

This method can also be extended to solve the note after the first solution. The nim-values of the 7 brick wall and the 8 brick wall are 2 and 1, using the same method as above. The nim-value of $(8, 7, 5, 3, 2)$ is therefore $1 \oplus 2 \oplus 4 \oplus 3 \oplus 2 = 6$, which is winning.

Solution 4

Consider the following, much simpler game.

Arjun and Beth can each either take 1 or 2 bricks from the right-hand-side of a continuous row of initially n bricks. It is easy to see that for n a multiple of 3, Beth can "mirror" whatever Arjun plays: if he takes 1, she takes 2, and if he takes 2, she takes 1. With this strategy, Beth always takes the last brick. If n is not a multiple of 3, then Arjun takes whichever amount puts Beth in the losing position.

The total number of bricks in the initial states given by the answer choices is 8, 9, 10, 10, 11. Thus, answer choice **(B)** appears promising as a winning position for Beth. The difference between this game and the simplified game is that in certain positions, namely those consisting of fragments of size only 1, taking 2 bricks is not allowed. We can assume that for the starting position $(6, 2, 1)$, Beth always has a move to ensure that she can continue to mirror Arjun throughout the game. (This could be proven rigorously with lots of casework. In particular, she must avoid providing Arjun with a position of 3 continuous bricks, because then he could take the middle block and force a win.) The assumption seems reasonable, so the answer is **(B)**.

Solution 5

We can start by guessing and checking our solutions. Let's start with **A**, $(6, 1, 1)$. Arjun goes first, and he has a winning strategy. His strategy is to cut the 6 block in half, so whatever Beth does, Arjun will copy. If we see this in practice, Arjun has made the block in to $(2, 2, 1, 1)$. If Beth takes 1, he will take one. If Beth takes 2, he will take 2 as well.

Then, we can guess solution **B**, $(6, 2, 1)$. If Arjun starts by taking 1 of the blocks, he will either have $(6, 1, 1)$, $(6, 2)$, $(x, y, 2, 1)$ (Where $x + y = 5$). If he has $(6, 1, 1)$, Beth can take 2 out of the middle of $(6, 1, 1)$ to get $(2, 2, 1, 1)$ which we know Arjun will lose because we know by symmetry from answer choice **A**. If he has $(6, 2)$, Beth will take out one from the edge, and we get $(5, 2)$. Beth can now copy whatever Arjun does, so she will win.

Next, if he takes one out to get $(x, y, 2, 1)$, we can either have $(4, 1, 2, 1)$ or $(3, 2, 2, 1)$. If it is the first case, Beth can take the edge 2 out of the 4, which gives us $(2, 1, 2, 1)$ and whatever Arjun does, Beth can do, so she has established a win. If it is the 2nd case, Beth can take out 2 from the 3 to get $(1, 2, 2, 1)$, in which, Beth can copy whatever Arjun does, so Beth will win. So in all 4 cases, Beth wins. So our answer is **(B)**.

Solution 6 (Mirroring)

In many game problems, the winning strategy is to simply copy all of your opponent's moves. Since you will always have a valid move if you mirror a valid move, you must move last and win.

Then, if Arjun can make the game "mirrorable" in his first move, then he can just copy all of Beth's moves after that.

$(6, 1, 1)$ can be turned into $(2, 2, 1, 1)$ by Arjun. Any move Beth makes, Arjun can just copy on the other block with same value. For example, if Beth took one block off the first 2, Arjun takes one block off the second 2.

$(6, 2, 2)$ turns into $(2, 2, 2, 2)$.

$(6, 3, 1)$ turns into $(3, 1, 3, 1)$.

$(6, 3, 2)$ turns into $(3, 2, 3, 2)$.

There are definitely other ways to split, but Arjun must win these games. This leaves answer $\boxed{B(6, 2, 1)}$.

Solution 7 (Side Games)

We will call the 6-wall the main game and the remaining walls the side game. It is clear that if there is no side game, Arjun can win by taking the middle two walls out. However, if Beth is ever able to force Arjun to take two turns in a row on the main game, their roles will swap and she will win. To do so, she must force Arjun to take two turns in a row, which means she needs to take the last wall in the side game. In other words, we can reduce the problem into Beth winning a game with starting configuration ignoring the 6-wall.

From here, it's clear that Beth loses $(1, 1)$, $(2, 2)$, and $(3, 2)$. However, Beth does win both $(3, 1)$ and $(2, 1)$. However, with the $(3, 1)$ case, Arjun can instead start the game by splitting the $(3, 1)$ into $(1, 1, 1)$, which would swap their roles. Since Arjun now wins the side game, Beth will lose the $(3, 1)$ case, making the answer

$\boxed{B(6, 2, 1)}$

23. (AMC 12 12B 2021) Three balls are randomly and independently tossed into bins numbered with the positive integers so that for each ball, the probability that it is tossed into bin i is 2^{-i} for $i = 1, 2, 3, \dots$. More than one ball is allowed in each bin. The probability that the balls end up evenly spaced in distinct bins is $\frac{p}{q}$, where p and q are relatively prime positive integers. (For example, the balls are evenly spaced if they are tossed into bins 3, 17, and 10.) What is $p + q$?
 (A) 55 (B) 56 (C) 57 (D) 58 (E) 59

Answer: A

Solution 1

"Evenly spaced" just means the bins form an arithmetic sequence.

Suppose the middle bin in the sequence is x . There are $x - 1$ different possibilities for the first bin, and these two bins uniquely determine the final bin. Now, the probability that these 3 bins are chosen is $6 \cdot 2^{-3x} = 6 \cdot \frac{1}{8^x}$, so the probability x is the middle bin is $6 \cdot \frac{x-1}{8^x}$. Then, we want the sum

$$\begin{aligned} 6 \sum_{x=2}^{\infty} \frac{x-1}{8^x} &= \frac{6}{8} \left[\frac{1}{8} + \frac{2}{8^2} + \frac{3}{8^3} \cdots \right] \\ &= \frac{3}{4} \left[\left(\frac{1}{8} + \frac{1}{8^2} + \frac{1}{8^3} + \cdots \right) + \left(\frac{1}{8^2} + \frac{1}{8^3} + \frac{1}{8^4} + \cdots \right) + \cdots \right] \\ &= \frac{3}{4} \left[\frac{1}{7} \cdot \left(1 + \frac{1}{8} + \frac{1}{8^2} + \frac{1}{8^3} + \cdots \right) \right] \\ &= \frac{3}{4} \cdot \frac{8}{49} \end{aligned}$$

$$= \frac{6}{49}$$

The answer is $6 + 49 = \boxed{\text{(A)} 55}$.

Solution 2

As in solution 1, note that "evenly spaced" means the bins are in arithmetic sequence. We let the first bin be a and the common difference be d . Further note that each (a, d) pair uniquely determines a set of 3 bins.

We have $a \geq 1$ because the leftmost bin in the sequence can be any bin, and $d \geq 1$, because the bins must be distinct.

This gives us the following sum for the probability:

$$\begin{aligned} 6 \sum_{a=1}^{\infty} \sum_{d=1}^{\infty} 2^{-3a-3d} &= 6 \sum_{a=1}^{\infty} \sum_{d=1}^{\infty} 2^{-3a} \cdot 2^{-3d} \\ &= 6 \left(\sum_{a=1}^{\infty} 2^{-3a} \right) \left(\sum_{d=1}^{\infty} 2^{-3d} \right) \\ &= 6 \left(\sum_{a=1}^{\infty} 8^{-a} \right) \left(\sum_{d=1}^{\infty} 8^{-d} \right) \\ &= 6 \left(\frac{1}{7} \right) \left(\frac{1}{7} \right) \\ &= \frac{6}{49}. \end{aligned}$$

Therefore the answer is $6 + 49 = \boxed{\text{(A)} 55}$.

-Darren Yao

Solution 3

This is a slightly messier variant of solution 2. If the first ball is in bin i and the second ball is in bin $j > i$, then the third ball is in bin $2j - i$. Thus the probability is

$$\begin{aligned} 6 \sum_{i=1}^{\infty} \sum_{j=i+1}^{\infty} 2^{-i} 2^{-j} 2^{-2j+i} &= 6 \sum_{i=1}^{\infty} \sum_{j=i+1}^{\infty} 2^{-3j} \\ &= 6 \sum_{i=1}^{\infty} \left(\frac{2^{-3(i+1)}}{1 - \frac{1}{8}} \right) \\ &= 6 \sum_{i=1}^{\infty} \frac{8}{7} \cdot 2^{-3} \cdot 2^{-3i} \\ &= \frac{6}{7} \sum_{i=1}^{\infty} 2^{-3i} \\ &= \frac{6}{7} \cdot \frac{2^{-3}}{1 - \frac{1}{8}} \\ &= \frac{6}{49}. \end{aligned}$$

Therefore the answer is $6 + 49 = \boxed{\text{(A)} 55}$.

Solution 4 (Table)

Based on the value of n , we construct the following table:

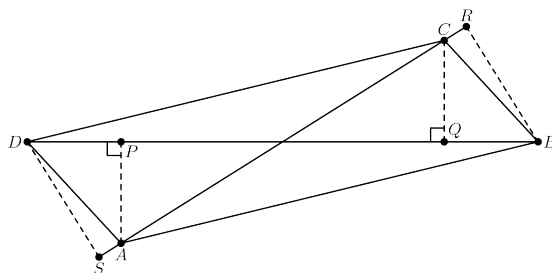
Exactly n Spaces Apart	Bin #s	Expression	Prob. of One Such Perm.
$n = 1$	1, 2, 3	$2^{-1} \cdot 2^{-2} \cdot 2^{-3}$	2^{-6}
	2, 3, 4	$2^{-2} \cdot 2^{-3} \cdot 2^{-4}$	2^{-9}
	3, 4, 5	$2^{-3} \cdot 2^{-4} \cdot 2^{-5}$	2^{-12}
	4, 5, 6	$2^{-4} \cdot 2^{-5} \cdot 2^{-6}$	2^{-15}
	...	...	...
$n = 2$	1, 3, 5	$2^{-1} \cdot 2^{-3} \cdot 2^{-5}$	2^{-9}
	2, 4, 6	$2^{-2} \cdot 2^{-4} \cdot 2^{-6}$	2^{-12}
	3, 5, 7	$2^{-3} \cdot 2^{-5} \cdot 2^{-7}$	2^{-15}
	4, 6, 8	$2^{-4} \cdot 2^{-6} \cdot 2^{-8}$	2^{-18}
	...	...	...
$n = 3$	1, 4, 7	$2^{-1} \cdot 2^{-4} \cdot 2^{-7}$	2^{-12}
	2, 5, 8	$2^{-2} \cdot 2^{-5} \cdot 2^{-8}$	2^{-15}
	3, 6, 9	$2^{-3} \cdot 2^{-6} \cdot 2^{-9}$	2^{-18}
	4, 7, 10	$2^{-4} \cdot 2^{-7} \cdot 2^{-10}$	2^{-21}
	...	...	...
...	...	...	...

Since three balls have $3! = 6$ permutations, the requested probability is

$$\begin{aligned}
 6 \left(\sum_{k=0}^{\infty} 2^{-6-3k} + \sum_{k=0}^{\infty} 2^{-9-3k} + \sum_{k=0}^{\infty} 2^{-12-3k} + \dots \right) &= 6 \left(2^{-6} \sum_{k=0}^{\infty} 2^{-3k} + 2^{-9} \sum_{k=0}^{\infty} 2^{-3k} + 2^{-12} \sum_{k=0}^{\infty} 2^{-3k} + \dots \right) \\
 &= \left(6 \sum_{k=0}^{\infty} 2^{-3k} \right) \cdot (2^{-6} + 2^{-9} + 2^{-12} + \dots) \\
 &= \left(6 \sum_{k=0}^{\infty} 2^{-3k} \right) \cdot \left(\sum_{k=0}^{\infty} 2^{-6-3k} \right) \\
 &= \left(6 \sum_{k=0}^{\infty} 2^{-3k} \right) \cdot \left(2^{-6} \sum_{k=0}^{\infty} 2^{-3k} \right) \\
 &= \frac{6}{1 - 2^{-3}} \cdot \frac{2^{-6}}{1 - 2^{-3}} \\
 &= \frac{6}{49}
 \end{aligned}$$

by infinite geometric series, from which the answer is $6 + 49 = \boxed{\text{(A)} 55}$.

24. (AMC 12 12B 2021) Let $ABCD$ be a parallelogram with area 15. Points P and Q are the projections of A and C , respectively, onto the line BD ; and points R and S are the projections of B and D , respectively, onto the line AC . See the figure, which also shows the relative locations of these points.



Suppose $PQ = 6$ and $RS = 8$, and let d denote the length of $\overline{BD}$, the longer diagonal of $ABCD$. Then d^2 can be written in the form $m + n\sqrt{p}$, where m, n , and p are positive integers and p is not divisible by the square of any prime. What is $m + n + p$? (A) 81 (B) 89 (C) 97 (D) 105 (E) 113

Answer: A

Solution 1

Let X denote the intersection point of the diagonals AC and BD . Remark that by symmetry X is the midpoint of both $\overline{PQ}$ and $\overline{RS}$, so $XP = XQ = 3$ and $XR = XS = 4$. Now note that since $\angle APB = \angle ARB = 90^\circ$, quadrilateral $ARPB$ is cyclic, and so

$$XR \cdot XA = XP \cdot XB,$$

which implies $\frac{XA}{XB} = \frac{XP}{XR} = \frac{3}{4}$.

Thus let $x > 0$ be such that $XA = 3x$ and $XB = 4x$. Then Pythagorean Theorem on $\triangle APX$ yields $AP = \sqrt{AX^2 - XP^2} = 3\sqrt{x^2 - 1}$, and so

$$[ABCD] = 2[ABD] = AP \cdot BD = 3\sqrt{x^2 - 1} \cdot 8x = 24x\sqrt{x^2 - 1} = 15$$

Solving this for x^2 yields $x^2 = \frac{1}{2} + \frac{\sqrt{41}}{8}$, and so

$$(8x)^2 = 64x^2 = 64 \left(\frac{1}{2} + \frac{\sqrt{41}}{8} \right) = 32 + 8\sqrt{41}.$$

The requested answer is $32 + 8 + 41 = \boxed{\text{(A) } 81}$.

Solution 2 (Trig)

Let X denote the intersection point of the diagonals AC and BD , and let $\theta = \angle COB$. Then, by the given conditions, $XR = 4$, $XQ = 3$, $[XCB] = \frac{15}{4}$. So,

$$\begin{aligned} XC &= \frac{3}{\cos \theta} \\ XB \cos \theta &= 4 \\ \frac{1}{2} XBXC \sin \theta &= \frac{15}{4} \end{aligned}$$

Combining the above 3 equations, we get

$$\frac{\sin \theta}{\cos^2 \theta} = \frac{5}{8}.$$

Since we want to find $d^2 = 4XB^2 = \frac{64}{\cos^2 \theta}$, we let $x = \frac{1}{\cos^2 \theta}$. Then

$$\frac{\sin^2 \theta}{\cos^4 \theta} = \frac{1 - \cos^2 \theta}{\cos^4 \theta} = x^2 - x = \frac{25}{64}.$$

Solving this, we get $x = \frac{4+\sqrt{41}}{8}$, so $d^2 = 64x = 32 + 8\sqrt{41} \rightarrow 32 + 8 + 41 = \boxed{\text{(A)} 81}$

Solution 3 (Similar Triangles and Algebra)

Let X be the intersection of diagonals AC and BD . By symmetry $[\triangle XCB] = \frac{15}{4}$, $XQ = 3$ and $XR = 4$, so now we have reduced all of the conditions one quadrant. Let $CQ = x$. $XC = \sqrt{x^2 + 9}$, $RB = \frac{4x}{3}$ by similar triangles and using the area condition we get $\frac{4}{3} \cdot x \cdot \sqrt{x^2 + 9} = \frac{15}{2}$. Note that it suffices to find $XB = \frac{4}{3}\sqrt{x^2 + 9}$ because we can double and square it to get d^2 . Solving for $a = x^2$ in the above equation, and then using $d^2 = \frac{64}{9}(x^2 + 9) = 8\sqrt{41} + 32 \Rightarrow 8 + 41 + 32 = \boxed{\text{(A)} 81}$

Solution 4 (Similar Triangles)

Again, Let X be the intersection of diagonals AC and BD . Note that triangles $\triangle QXC$ and $\triangle BXR$ are similar because they are right triangles and share $\angle CXQ$. First, call the length of $XB = \frac{d}{2}$. By the definition of an area of a parallelogram, $CQ \cdot 2XB = 15$, so $CQ = \frac{15}{d}$. Using similar triangles on $\triangle QXC$ and $\triangle BXR$, $\frac{CQ}{XQ} = \frac{BR}{XR}$. Therefore, finding BR , $BR = \frac{XR}{XQ} \cdot CQ = \frac{4}{3} \cdot \frac{15}{d} = \frac{20}{d}$. Now, applying the Pythagorean theorem once, we find $(\frac{20}{d})^2 + (4)^2 = (\frac{d}{2})^2$. Solving this equation for d^2 , we find $d^2 = \frac{64 + \sqrt{4096 + 6400}}{2} = 32 + 8\sqrt{41} \rightarrow 32 + 8 + 41 = \boxed{\text{(A)} 81}$

Solution 5

Let $BQ = PD = x$. We know that the area of the parallelogram is 15, so it follows that $[\triangle BCD] = [\triangle BAD] = \frac{15}{2}$ and the height of each triangle, which are also the lengths of QC and AP , is $\frac{15}{2(x+3)}$. Suppose that $E = RS \cap BD$. Because $\angle BRE = \angle CQE$ and $\angle BER = \angle CQD$, we have $\triangle BRE \sim \triangle CQE$. The length of CE , by the Pythagorean Theorem is $\sqrt{3^2 + (\frac{15}{2(x+3)})^2}$ and the length of BR , by the Pythagorean Theorem on $\triangle BRE$, is $\sqrt{(x+3)^2 - 4}$. Note that

$$\sin \angle QEC = \frac{CQ}{CE} = \frac{BR}{BE}$$

Substituting in our values,

$$\frac{\frac{15}{2(x+3)}}{\sqrt{9 + (\frac{15}{2(x+3)})^2}} = \frac{\sqrt{(x+3)^2 - 4^2}}{x+3}$$

To rid unnecessary computation, we let $(x+3)^2 = a$. The equation simplifies, after cross multiplying, to

$$\begin{aligned} \sqrt{9 + \frac{15^2}{4a}} \sqrt{a - 16} &= \frac{15}{2} \\ 36a^2 - 576a - 15^2 \cdot 16 &= 0 \\ a^2 - 16a - 100 &= 0 \end{aligned}$$

By the quadratic formula, $a \in \{\frac{16-\sqrt{656}}{2}, \frac{16+\sqrt{656}}{2}\}$, so we discard the negative solution. The value of BD^2 is

$$BD^2 = (2x+6)^2 = 4(x+3)^2 = 4a = 4 \cdot \frac{16+\sqrt{656}}{2} = 32 + 8\sqrt{41}$$

and the desired answer is $32 + 8 + 41 = \boxed{\text{(A)} 81}$ skyscraper

Solution 6 (Similar Triangles Pythagorean Theorem)

Let the intersection of RS and BD be X .

$\because \angle APX = \angle DSX$ and $\angle AXP = \angle DXS$, $\triangle APX \sim \triangle DSX$ by AA $\therefore \frac{PA}{DS} = \frac{6}{8} = \frac{3}{4}$, $DS = \frac{4}{3} \cdot PA$

By the Pythagorean theorem and the property of projection, $BD^2 = (DS + BR)^2 + RS^2 = 4DS^2 + 64 = 4(\frac{4}{3} \cdot PA)^2 + 64 = \frac{64}{9} \cdot PA^2 + 64$, $\frac{64}{9} \cdot PA^2 = BD^2 - 64$
 $\therefore [ABCD] = PA \cdot BD = 15$, $\therefore PA = \frac{15}{BD}$

$$\frac{64}{9} \left(\frac{15}{BD} \right)^2 = BD^2 - 64$$

$$\frac{1600}{BD^2} = BD^2 - 64$$

$$BD^4 - 64BD^2 - 1600 = 0$$

$$BD^2 = \frac{64 + \sqrt{64^2 - 4(-1600)}}{2} = 32 + 8\sqrt{41}$$

Therefore, the answer is $32 + 8 + 41 = \boxed{\text{(A)} 81}$.

Solution 7

Let O be the intersection of the diagonals in quadrilateral $ABCD$ and let the origin be at O .

Then, let BD be on the x-axis, making $\vec{Q} = \begin{pmatrix} 3 \\ 0 \end{pmatrix}$, $\vec{C} = \begin{pmatrix} 3 \\ y \end{pmatrix}$, $\vec{B} = \begin{pmatrix} x \\ 0 \end{pmatrix}$, and $\vec{D} = \begin{pmatrix} -x \\ 0 \end{pmatrix}$.

The area of $\triangle BDC$ is half the area of $ABCD$, so,

$$\frac{1}{2}(\overline{BD})(\overline{QC}) = \frac{1}{2}(ABCD)$$

$$\frac{1}{2}(2x)(y) = \frac{1}{2}(15)$$

$$xy = \frac{15}{2}$$

$$y = \frac{15}{2x}$$

.

Also, $\vec{R}$ is the projection of $\vec{B}$ onto $\vec{AC}$, which is the same as projecting it onto $\vec{C}$. We get:

$$\vec{R} = \text{proj}_{\vec{C}}(\vec{B}) = \frac{\vec{B} \cdot \vec{C}}{\vec{C} \cdot \vec{C}}(\vec{C}) = \frac{\begin{pmatrix} x \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ y \end{pmatrix}}{\begin{pmatrix} 3 \\ y \end{pmatrix} \cdot \begin{pmatrix} 3 \\ y \end{pmatrix}} \begin{pmatrix} 3 \\ y \end{pmatrix} = \frac{3x}{y^2 + 9} \begin{pmatrix} 3 \\ y \end{pmatrix}$$

We are given that $\overline{RS} = 8$, so $||\vec{RS}|| = 8$. Because diagonals bisect each other in a parallelogram, $||\vec{R}|| = \frac{8}{2} = 4$. Substituting $||\vec{R}||$ with the previous equation gives:

$$||\frac{3x}{y^2+9} \begin{pmatrix} 3 \\ y \end{pmatrix}|| = 4$$

$$\frac{3x}{y^2+9} \sqrt{y^2+9} = 4$$

$$\frac{3x}{\sqrt{y^2+9}} = 4$$

Squaring both sides gives:

$$\frac{9x^2}{y^2+9} = 16$$

$$9x^2 = 16y^2 + 144$$

Substituting $y = \frac{15}{2x}$:

$$9x^2 = 16 \left(\frac{15}{2x} \right)^2 + 144$$

$$9x^2 = 16 \left(\frac{225}{4x^2} \right) + 144$$

$$9x^4 = 144x^2 + 900$$

$$x^4 - 16x^2 - 100 = 0$$

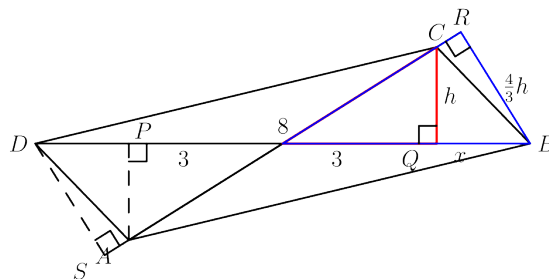
Solving the quadratic for x^2 gives:

$$x^2 = \frac{16 \pm \sqrt{16^2 - 4(1)(-100)}}{2} = 8 \pm 2\sqrt{41}$$

But x^2 is nonnegative, so x^2 must be $8 + 2\sqrt{41}$. We are looking for $\overline{BD}^2$, which is $4x^2$. $4(8 + 2\sqrt{41}) = 32 + 8\sqrt{41}$, so $a + b + c = 32 + 8 + 41 = \boxed{\text{(A)} 81}$ askeww (My first solution ever!)

Solution 8

First and foremost, the condition given in the problem could be represented. Moreover, let $QB = x$ and $CQ = h$ for convenience.



Using the property of a parallelogram, it could be inferred that $\frac{h(3+x)}{2} = \frac{15}{4}$. Moreover, Pythagorean Theorem manifests that $\left(\frac{4}{3}h\right)^2 + 4^2 = (3+x)^2$.

$$3+x = \frac{15}{2h}$$

$$\begin{aligned}\left(\frac{4}{3}h\right)^2 + 4^2 &= \left(\frac{15}{2h}\right)^2 \\ \frac{16h^2}{9} + 16 &= \frac{225}{4h^2} \\ 64h^4 + 576h^2 - 2025 &= 0 \\ h^2 &\Rightarrow \frac{9\sqrt{41} - 36}{8}\end{aligned}$$

$$\begin{aligned}\left(\frac{4}{3}h\right)^2 + 4^2 &= (3+x)^2 \\ \frac{16}{9} \cdot \frac{9\sqrt{41} - 36}{8} + 4^2 &= (3+x)^2 \\ 2\sqrt{41} - 8 + 16 &= (3+x)^2 \\ (3+x)^2 &= 8 + 2\sqrt{41} \\ d^2 = 4(3+x)^2 &= 32 + 8\sqrt{41}\end{aligned}$$

Thus, $32 + 8 + 41 = \boxed{\text{(A)} \ 81}$.

Solution 9(Similar to other solutions)

We first note that DS and BR are perpendicular to AC . Then, let $DS = BR = h$. Furthermore, let $AS = x$ and the intersections of diagonals AC and BD be T . We note that triangle APT is similar to triangle STD . We will also show that the areas of triangles ADT, BCT, CDT, ABT are the same namely $\frac{15}{4}$. Let $\angle DTA = \theta$. Then, $\angle DTC = \pi - \angle DTA = \pi - \theta$. Let $DP = QB = y$. Obviously, $PT = QT = \frac{6}{2} = 3$. Also, $AT = CT = 4 - x$. Then the area of triangle ADT is just $\frac{1}{2} \cdot (y+3)(4-x) \cdot \sin(\theta)$. It is easy to see that CBT is congruent to ADT and ABT is congruent to CDT so it suffices to show triangle CDT has the same area as triangle ADT . In fact, the area equals $\frac{1}{2} \cdot (y+3)(4-x) \cdot \sin(\pi - \theta) = \frac{1}{2} \cdot (y+3)(4-x) \cdot \sin(\theta)$ which indeed is equal to the area of triangle ADT . Hence, each of these triangles breaks the area of the entire parallelogram into 4 pieces so each area is $\frac{15}{4}$. Look at triangle ADT . The base is $y+3$. Let the height equal m . Hence:

$$\frac{1}{2} \cdot m \cdot (y+3) = \frac{15}{4} \implies m = \frac{15}{2(y+3)}$$

Remember that we established the fact that triangle APT is similar to triangle STD and so:

$$\frac{4}{3} = \frac{\frac{15}{2(y+3)}}{h} \implies h(y+3) = 10$$

$$\begin{aligned}\text{Also by the Pythagorean Theorem on triangle } STD \quad h^2 + 16 &= (y+3)^2 \implies \\ h^2 + 16 &= \frac{100}{h^2} \implies h^4 + 16h^2 - 100 = 0 \implies h^2 = \frac{-16 \pm \sqrt{16 \cdot 16 - 4(-100)}}{2} = \\ &= \frac{-16 \pm \sqrt{16 \cdot 16 + 16 \cdot 25}}{2} = \frac{-16 \pm 4\sqrt{41}}{2} = -8 \pm 2\sqrt{41}.\end{aligned}$$

Since h^2 is positive, $h^2 = -8 + 2\sqrt{41}$. We need to find $BD^2 = (y+3+3+y)^2 = 4(y+3)^2$ so remember $(y+3)^2 = \frac{100}{h^2} = \frac{100}{-8+2\sqrt{41}} = 8 + 2\sqrt{41}$. But BD^2 is 4 times this so the answer is $4(8 + 2\sqrt{41}) = 32 + 8\sqrt{41}$. Hence $32 + 8 + 41 = 81 \implies \boxed{\text{(A)}}$

25. (AMC 12 12B 2021) Let S be the set of lattice points in the coordinate plane, both of whose coordinates are integers between 1 and 30, inclusive. Exactly 300 points

in S lie on or below a line with equation $y = mx$. The possible values of m lie in an interval of length $\frac{a}{b}$, where a and b are relatively prime positive integers. What is $a + b$? (A) 31 (B) 47 (C) 62 (D) 72 (E) 85

Answer: E

Solution 1

First, we find a numerical representation for the number of lattice points in S that are under the line $y = mx$. For any value of x , the highest lattice point under $y = mx$ is $\lfloor mx \rfloor$. Because every lattice point from $(x, 1)$ to $(x, \lfloor mx \rfloor)$ is under the line, the total number of lattice points under the line is $\sum_{x=1}^{30} (\lfloor mx \rfloor)$.

Now, we proceed by finding lower and upper bounds for m . To find the lower bound, we start with an approximation. If 300 lattice points are below the line, then around $\frac{1}{3}$ of the area formed by S is under the line. By using the formula for a triangle's area, we find that when $x = 30, y \approx 20$. Solving for m assuming that $(30, 20)$ is a point on the line, we get $m = \frac{2}{3}$. Plugging in m to $\sum_{x=1}^{30} (\lfloor mx \rfloor)$, we get:

$$\sum_{x=1}^{30} (\lfloor \frac{2}{3}x \rfloor) = 0 + 1 + 2 + 2 + 3 + \cdots + 18 + 18 + 19 + 20$$

We have a repeat every 3 values (every time $y = \frac{2}{3}x$ goes through a lattice point). Thus, we can use arithmetic sequences to calculate the value above:

$$\begin{aligned} \sum_{x=1}^{30} (\lfloor \frac{2}{3}x \rfloor) &= 0 + 1 + 2 + 2 + 3 + \cdots + 18 + 18 + 19 + 20 \\ &= \frac{20(21)}{2} + 2 + 4 + 6 + \cdots + 18 \\ &= 210 + \frac{20}{2} \cdot 9 \\ &= 300 \end{aligned}$$

This means that $\frac{2}{3}$ is a possible value of m . Furthermore, it is the lower bound for m . This is because $y = \frac{2}{3}x$ goes through many points (such as $(21, 14)$). If m was lower, $y = mx$ would no longer go through some of these points, and there would be less than 300 lattice points under it.

Now, we find an upper bound for m . Imagine increasing m slowly and rotating the line $y = mx$, starting from the lower bound of $m = \frac{2}{3}$. The upper bound for m occurs when $y = mx$ intersects a lattice point again (look at this problem to get a better idea of what's happening: https://artofproblemsolving.com/wiki/index.php/2011_AMC_10B_Problems/P

In other words, we are looking for the first $m > \frac{2}{3}$ that is expressible as a ratio of positive integers $\frac{p}{q}$ with $q \leq 30$. For each $q = 1, \dots, 30$, the smallest multiple of $\frac{1}{q}$ which exceeds $\frac{2}{3}$ is $1, \frac{2}{2}, \frac{3}{3}, \frac{3}{4}, \frac{4}{5}, \dots, \frac{19}{27}, \frac{19}{28}, \frac{20}{29}, \frac{21}{30}$ respectively, and the smallest of these is $\frac{19}{28}$.

Alternatively, see the method of finding upper bounds in solution 2.

The lower bound is $\frac{2}{3}$ and the upper bound is $\frac{19}{28}$. Their difference is $\frac{1}{84}$, so the answer is $1 + 84 = \boxed{85}$.

An alternative would be using Farey fractions and the mediant theorem to find the upper bound. $\frac{2}{3}$ and $\frac{7}{10}$ gives $\frac{9}{13}$, and so on using Farey addition.

The above alternative can be made quicker by using a different Farey sequence rule. The Farey sequence rule states that if $\frac{a}{b}$ and $\frac{c}{d}$ are adjacent in a Farey sequence, we have $bc - ad = 1$ (see https://en.wikipedia.org/wiki/Farey_sequence#Farey_neighbours). We use the rule with $a = 2$, $b = 3$, so we have $3c - 2d = 1$. Starting with $d = 30$ (since $d \leq 30$), we see that c would be noninteger and the same can be shown for $d = 29$. When $d = 28$, $c = 19$ so the range is $[\frac{2}{3}, \frac{19}{28}] \Rightarrow \frac{1}{84} \Rightarrow \boxed{85}$

An alternative approach with the same methodology can be done using Pick's Theorem. Wikipedia page: https://en.wikipedia.org/wiki/Pick%27s_theorem. It's a formula to find the number of lattice points strictly inside a polygon. Approximation of the lower bound is still necessary.

Solution 2

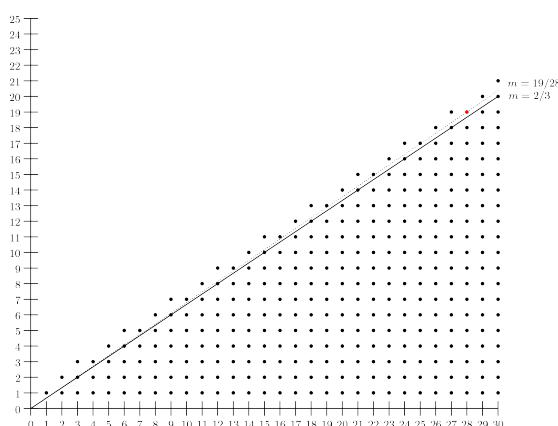
Based on area ratios between a square of side length 30 and a triangle with base 30, we estimate that the slope m of the line we want is approximately $\frac{2}{3}$. Following this estimate, we see if there are approximately $30 \cdot 30 - 300 = 600$ lattice points above the line $y = \frac{2}{3}x$.

Counting the number of lattice points with $x = 1$ above the line, the number of lattice points with $x = 2$ above the line, and so on, we find that the total number of lattice points above the line is $30 + 29 + 28 + 28 + 27 + 26 + 26 \dots + 10$, with the even integers repeating every third term. We see that the average of the 30 terms is 20, which means that exactly $20 \cdot 30 = 600$ lattice points above the line as desired. This gives a lower bound because any decrease in the slope of the line would cause points that were already on the line to shift to being above it.

To find the upper bound, notice that each lattice point less than 1 unit above the line is either $\frac{1}{3}$ or $\frac{2}{3}$ above. Since the slope through a point is the y-coordinate divided by the x-coordinate, a shift in the slope will increase the y-value of the higher x-coordinates more than the y-value of the lower x-coordinates. So, we turn our attention to $x = 28, 29, 30$ for which the line $y = \frac{2}{3}x$ intersects at $y = \frac{56}{3}, \frac{58}{3}, 20$. The point $(30, 20)$ is already counted, and we can clearly see that if we slowly increase the slope of the line, we will first hit the point $(28, 19)$ since $(28, \frac{56}{3})$ is the closest to it. The equation of the line which goes through both the origin and $(28, 19)$ is $y = \frac{19}{28}x$. This gives an upper bound of $m = \frac{19}{28}$.

Taking the upper bound of m and subtracting the lower bound yields $\frac{19}{28} - \frac{2}{3} = \frac{1}{84}$. The answer is therefore $1 + 84 = \boxed{\text{(E)} 85}$.

Diagram



Solution 3

As the procedure shown in the Solution 1, the lower bound of m is $\frac{2}{3}$. Here I give a more logic way to show how to find the upper bound of m . Denote $N = \sum_{x=1}^{30} \lfloor mx \rfloor$ as the number of lattice points in S .

$$N = \lfloor m \rfloor + \lfloor 2m \rfloor + \lfloor 3m \rfloor + \cdots + \lfloor 30m \rfloor = 300.$$

Let $m = \frac{2}{3} + k$. for $\forall x_i \leq 30, x \in N^*, \lfloor mx_i \rfloor = \lfloor \frac{2}{3}x + xk \rfloor$.

Our target is finding the minimum value of k which can increase one unit of $\lfloor mx_i \rfloor$.

Notice that:

When $x_i = 3n, \lfloor mx_i \rfloor = \lfloor 2n + (3n)k \rfloor$ We don't have to discuss this case.

When $x_i = 3n + 1, \lfloor mx_i \rfloor = \lfloor 2n + \frac{2}{3} + (3n + 1)k \rfloor, k_{min1} = \frac{1}{3(3n+1)}.$

When $x_i = 3n + 2, \lfloor mx_i \rfloor = \lfloor 2n + 1 + \frac{1}{3} + (3n + 2)k \rfloor, k_{min2} = \frac{2}{3(3n+2)}.$

Here $n \in N^*, n \leq 9$.

Denote $k_{min} = \min \{k_{min1}, k_{min2}\}.$

Obviously k_{min1} and k_{min2} are decreasing. Let's considering the situation when $n = 9$. $k_{min} = \min \left\{ \frac{1}{84}, \frac{2}{87} \right\} = \frac{1}{84}.$

This means that the answer is just $\frac{1}{84}$, so $a + b = 85$. Choose $\boxed{E}$.

Solution 4

It's easier to calculate the number of lattice points inside a rectangle with vertices $(0, 0), (p, 0), (p, q), (0, q)$. Those lattice points are divided by the diagonal $y = \frac{p}{q} \cdot x$ into 2 halves. In this problem the number of lattice points on or below the diagonal and $x \geq 1$ is

$$\frac{1}{2} [(p+1)(q+1) - d] + d - (p+1), \quad d \text{ is the number of lattice points on the diagonal}$$

$(p+1)(q+1)$ is the total number of lattice points inside the rectangle. Subtract the number of lattice points on the diagonal, divided by 2 is the number of lattice points below the diagonal, add the number of lattice points on the diagonal, and subtract the lattice points on the x axis, then we get the total number of lattice points on or below the diagonal and $x \geq 1$.

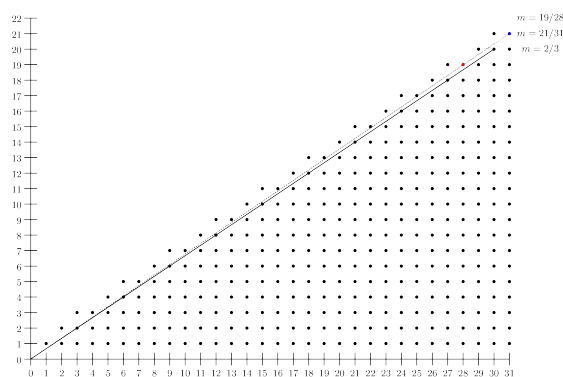
There are 900 lattice points in total. 300 is $\frac{1}{3}$ of 900. The x coordinate of the top-right vertex of the rectangle is 30, $\frac{1}{2} \cdot 30 \cdot 20 = 300$. I guess the y coordinate of the top-right vertex of the rectangle is 20. Now I am going to verify that. The slope of the diagonal is $\frac{20}{30} = \frac{2}{3}$, there are 11 lattice points on the diagonal. Substitute $(p, q) = (30, 20), d = 11$ to the above formula:

$$\frac{1}{2} [(30+1)(20+1) - 11] + 11 - (30+1) = 300$$

Because there are 11 lattice points on line $y = \frac{2}{3}x$, if $m < \frac{2}{3}$, then the number of lattice points on or below the line is less than 300. So $m = \frac{2}{3}$ is the lower bound.

Now I am going to calculate the upper bound. From $\frac{b}{a} < \frac{b+1}{a+1}$,

$$\frac{2}{3} = \frac{18}{27} < \frac{19}{28} \quad \frac{2}{3} = \frac{20}{30} < \frac{21}{31} < \frac{19}{28}$$



If $m = \frac{21}{31}$, I will calculate by using the rectangle with blue vertex $(p, q) = (31, 21)$, then subtract lattice points on line $x = 31$, which is 21. There are 2 lattice points on the diagonal, $d = 2$.

$$\frac{1}{2} [(31+1)(21+1) - 2] + 2 - (31+1) - 21 = 300, \text{ same as that of } m = \frac{19}{28}$$

If $m = \frac{19}{28}$, I will calculate by using the rectangle with red vertex $(p, q) = (28, 19)$, then add lattice points on line $x = 29$ and $x = 30$, which is $19 + 20 = 39$. There are 2 lattice points on the diagonal, $d = 2$.

$$\frac{1}{2} [(28+1)(19+1) - 2] + 2 - (28+1) + 39 = 301, \$1 \text{ more than that of } m = \frac{21}{31}$$

When m increases, more lattice points falls below the line $y = mx$. Any value larger than $\frac{19}{28}$ has more than 301 lattice points on or below $y = \frac{19}{28}x$. So the upper bound is $\frac{19}{28}$.

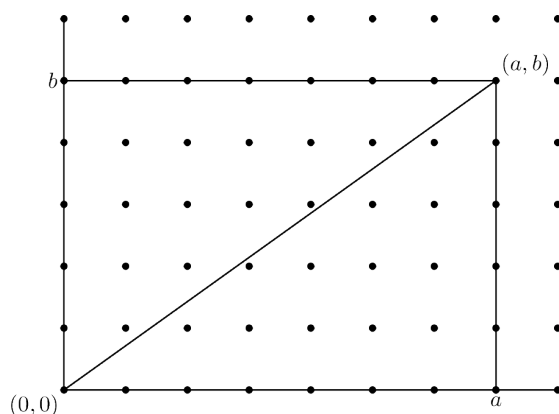
$$\frac{19}{28} - \frac{2}{3} = \frac{1}{84}, \quad \boxed{\text{(E) } 85}.$$

Solution 5

The lower bound of m is $\frac{2}{3} = \frac{20}{30}$. Inside the rectangle with vertices $(0, 0)$, $(30, 0)$, $(30, 20)$, $(0, 20)$ and diagonal $y = \frac{2}{3}x$, there are $(30 - 1)(20 - 1) = 551$ lattice points inside, not including the edges. There are 9 lattice points on diagonal $y = \frac{2}{3}x$ inside the rectangle, $551 + 9 = 560$. Half of the 560 lattice points are below diagonal $y = \frac{2}{3}x$, $560 \cdot \frac{1}{2} = 280$. There are 20 lattice points on edge $x = 30$, $280 + 20 = 300$. Once $m < \frac{2}{3}$, the 9 lattice points on diagonal $y = \frac{2}{3}x$ will be above the new diagonal, making the number of lattice points on and below the diagonal less than 300.

Now we are going to calculate the upper bound by the following formula:

The number of lattice points inside rectangle $(0, 0)$, $(a, 0)$, (a, b) , $(0, b)$ is



$$\frac{2}{3} = \frac{20}{30} < \frac{21}{31} < \frac{19}{28}$$

When $a = 31$, $b = 21$, $\frac{(31-1)(21-1)}{2} = 300$.

When $a = 28$, $b = 19$, $\frac{(28-1)(19-1)}{2} = 243$. Below the line $y = \frac{19}{28}x$, there are 19 lattice points on line $x = 28$, 19 lattice points on line $x = 29$, 20 lattice points on line $x = 30$, $243 + 19 + 19 + 20 = 301$.

More lattice points fall below the line $y = mx$ as m increases. There are more than 301 lattice points on and below the line for any m greater than $\frac{19}{28}$. Therefore, the upper bound is $\frac{19}{28}$.

$$\frac{19}{28} - \frac{2}{3} = \frac{1}{84}, \text{ so } 1 + 84 = \boxed{\text{(E) } 85}.$$

Remark $\lfloor \frac{b}{a}k \rfloor$ is the number of lattice points on line $x = k$, below line $y = \frac{b}{a}x$ and above the x axis, where k is an integer and $0 < k < a$. Therefore, $\sum_{k=1}^{a-1} \lfloor \frac{b}{a}k \rfloor$ is the number of lattice points inside the rectangle $(0, 0)$, $(a, 0)$, (a, b) , $(0, b)$, below diagonal $y = \frac{b}{a}x$. If a and b are relatively prime, $\sum_{k=1}^{a-1} \lfloor \frac{b}{a}k \rfloor = \frac{(a-1)(b-1)}{2}$, as explained in solution 5. This problem is about finding the upper and lower bound of $\frac{b}{a}$, given $\sum_{k=1}^{30} \lfloor \frac{b}{a}k \rfloor = 300$. The same problem can have geometric representation as stated in the original problem, or algebraic representation as stated here.

Solution 6

As the procedure shown in the Solution 1, the lower bound of m is $\frac{2}{3}$, whose equation is $2x - 3y = 0$.

Now, we are going to find the upper bound of m .

The signed distance between a point (x_0, y_0) and the line $2x - 3y = 0$ is

$$d = \frac{2x_0 - 3y_0}{\sqrt{2^2 + 3^2}}$$

If (x_0, y_0) is the lowest lattice point above the line, then it should have the largest values of coordinates among all the solutions of $2x_0 - 3y_0 = 1$, because among the points with the shortest distance from the line, the slope of the farthest point to the origin is the smallest. Solve the inequality:

$$2x - 3y = 1, \quad 1 \leq x \leq 30$$

We obtain $(x_0, y_0) = (28, 19)$, so the upper bound of $m = \frac{19}{28}$.

$$\frac{19}{28} - \frac{2}{3} = \frac{1}{84}, \quad \boxed{\text{(E) } 85}.$$

§4 AMC 12 12A 2021

1. (AMC 12 12A 2021) What is the value of

$$2^{1+2+3} - (2^1 + 2^2 + 2^3)?$$

- (A) 0 (B) 50 (C) 52 (D) 54 (E) 57

Answer: B

Solution

We evaluate the given expression to get that

$$2^{1+2+3} - (2^1 + 2^2 + 2^3) = 2^6 - (2^1 + 2^2 + 2^3) = 64 - 2 - 4 - 8 = 50 \implies \boxed{\text{(B)}}$$

2. (AMC 12 12A 2021) Under what conditions is $\sqrt{a^2 + b^2} = a + b$ true, where a and b are real numbers?

- (A) It is never true.
 (B) It is true if and only if $ab = 0$.
 (C) It is true if and only if $a + b \geq 0$.
 (D) It is true if and only if $ab = 0$ and $a + b \geq 0$.
 (E) It is always true.

Answer: D

Solution 1 (Algebra)

One can square both sides to get $a^2 + b^2 = a^2 + 2ab + b^2$. Then, $0 = 2ab \implies ab = 0$. Also, it is clear that both sides of the equation must be nonnegative. The answer is **(D)**.

Solution 2 (Algebra)

Complete the square of the left side by rewriting the radical to be

$$\sqrt{(a + b)^2 - 2ab}.$$

From there it is evident for the square root of the left to be equal to the right, ab must be equal to zero. Also, we know that the equivalency of square root values only holds true for nonnegative values of $a + b$, making the correct answer **(D)**.

Solution 3 (Process of Elimination)

The left side of the original equation is the **arithmetic square root**, which is always nonnegative. So, we need $a + b \geq 0$, which refutes **(B)** and **(E)**. Next, picking $(a, b) = (0, 0)$ refutes **(A)**, and picking $(a, b) = (1, 2)$ refutes **(C)**. By POE (Process of Elimination), the answer is **(D)**.

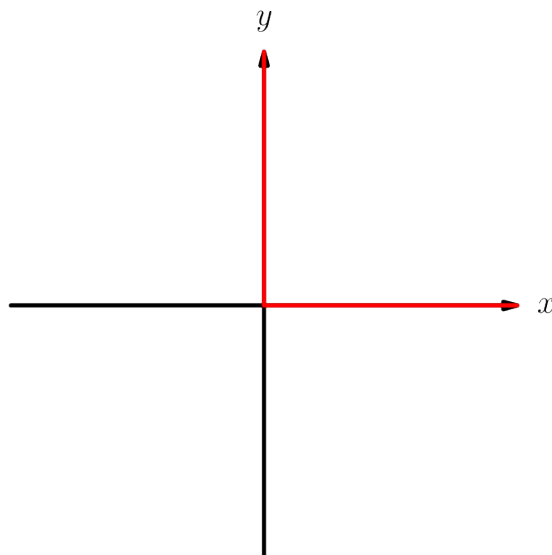
Solution 4 (Graphing)

If we graph $\sqrt{x^2 + y^2} = x + y$, then we get the union of:

- positive x -axis
- positive y -axis
- origin

Therefore, the answer is **(D)**.

The graph of $\sqrt{x^2 + y^2} = x + y$ is shown below.



MRENTHUSIASM (credit given to TheAMCHub)

3. (*AMC 12 12A 2021*) The sum of two natural numbers is 17,402. One of the two numbers is divisible by 10. If the units digit of that number is erased, the other number is obtained. What is the difference of these two numbers?

(A) 10,272 (B) 11,700 (C) 13,362 (D) 14,238 (E) 15,426

Answer: D

Solution 1 (Algebra)

The units digit of a multiple of 10 will always be 0. We add a 0 whenever we multiply by 10. So, removing the units digit is equal to dividing by 10.

Let the smaller number (the one we get after removing the units digit) be a . This means the bigger number would be $10a$.

We know the sum is $10a + a = 11a$ so $11a = 17402$. So $a = 1582$. The difference is $10a - a = 9a$. So, the answer is $9(1582) = \boxed{\text{(D) } 14,238}$.

Solution 2 (Arithmetic)

Since the unit's place of a multiple of 10 is 0, the other integer must end with a 2, for both integers sum up to a number ending in a 2. Thus, the unit's place of the difference must be $10 - 2 = 8$, and the only answer choice that ends with an 8 is **(D) 14,238**.

Another quick solution is to realize that the sum represents a number n added to $10n$. The difference is $9n$, which is $\frac{9}{11}$ of the given sum.

Solution 3 (Vertical Addition and Logic)

Let the larger number be $\overline{AB,CD0}$. It follows that the smaller number is $\overline{A,BCD}$. Adding vertically, we have

$$\begin{array}{r} \\ \\ + \\ \hline 1 7 4 0 2 \end{array}$$

Working from right to left, we get

$$D = 2 \implies C = 8 \implies B = 5 \implies A = 1.$$

The larger number is 15,820 and the smaller number is 1,582. Their difference is $15,820 - 1,582 = \boxed{\text{(D)} 14,238}$.

Solution 4 (Logic)

We know that the larger number has a units digit of 0 since it is divisible by 10. If D is the ten's digit of the larger number, then D is the units digit of the smaller number. Since the sum of the natural numbers has a unit's digit of 2, $D = 2$.

The units digit of the larger number is 0 and the units digit of the smaller number is 2, so the positive difference between the numbers is 8. There is only one answer choice that has this units digit, and that is $\boxed{\text{(D)} 14,238}$.

(Similar to MRENTHUSIASM's solution)

4. (AMC 12 12A 2021) Tom has a collection of 13 snakes, 4 of which are purple and 5 of which are happy. He observes that

- all of his happy snakes can add,
- none of his purple snakes can subtract, and
- all of his snakes that can't subtract also can't add.

Which of these conclusions can be drawn about Tom's snakes?

- (A) Purple snakes can add.
 (B) Purple snakes are happy.
 (C) Snakes that can add are purple.
 (D) Happy snakes are not purple.
 (E) Happy snakes can't subtract.

Answer: D

Solution 1 (Comprehensive Explanation of Logic)

We are given that

$$\text{happy} \implies \text{can add}, \quad (1)$$

$$\text{purple} \implies \text{cannot subtract}, \quad (2)$$

$$\text{cannot subtract} \implies \text{cannot add}. \quad (3)$$

Two solutions follow from here:

Solution 1.1 (Intuitive)

Combining (2) and (3) gives

$$\text{happy} \implies \text{can add}, \quad (1)$$

$$\underbrace{\text{purple} \implies \text{cannot subtract}}_{(2)} \overset{(3)}{\implies} \text{cannot add}. \quad (*)$$

Clearly, the answer is $\boxed{\text{(D)}}$.

Solution 1.2 (Rigorous)

Recall that every conditional statement $p \implies q$ is always logically equivalent to its contrapositive $\neg q \implies \neg p$.

Combining (1), (2) and (3) gives

$$\underbrace{\text{purple} \implies \text{cannot subtract}}_{(2)} \overset{(3)}{\implies} \underbrace{\text{cannot add} \implies \text{not happy.}}_{\text{Contrapositive of (1)}} \quad (**)$$

Applying the hypothetical syllogism to (**), we conclude that

$$\text{purple} \implies \text{not happy},$$

whose contrapositive is

$$\text{happy} \implies \text{not purple}.$$

Therefore, the answer is (D). Remark

The conclusions in the other choices do not follow from (**): **(A)** $\text{purple} \implies \text{can add}$ **(B)** $\text{purple} \implies \text{happy}$ **(C)** $\text{can add} \implies \text{purple}$ **(E)** $\text{happy} \implies \text{cannot subtract}$

Solution 2 (Process of Elimination)

From Solution 1.1, we can also see this through the process of elimination. Statement *A* is false because purple snakes cannot add. *B* is false as well because since happy snakes can add and purple snakes can not add, purple snakes are not happy snakes. *E* is false using the same reasoning, purple snakes are not happy snakes so happy snakes can subtract since purple snakes cannot subtract. *C* is false since snakes that can add are happy, not purple. That leaves statement D. (D) is the only correct statement.

Solution 3 (Rigorous)

We first convert each statement to "If X, then Y" form:

- If a snake is happy, then it can add.
- If a snake is purple, then it can't subtract.
- If a snake can't subtract, then it can't add.

Now, we simply check the truth value for each statement:

- a) Combining the last two propositions, we have

If a snake is purple, then it can't add.

Thus, **(A)** is never true.

- b) From the last part, we found that

If a snake is purple, then it can't add.

Also, since the contrapositive of a proposition has the same truth value as the proposition itself, we know, from the first statement, that

If a snake can't add, then it isn't happy.

Combining these two propositions, we find that

If a snake is purple, then it isn't happy. Purple snakes are not happy.

Thus, **(B)** is never true.

- c) From part (A), we found that "If a snake is purple, then it can't add." This implies its contrapositive, "If a snake can add, then it is not purple." is true, meaning (C) is NEVER true. [Thanks again to MRENTHUSIASM for pointing this out!]

- d) From the first statement, we have

If a snake is happy, then it can add.

From the contrapositive of the third statement, we have

If a snake can add, then it can subtract.

Then, from the contrapositive of the second statement, we have

If a snake can subtract, then it is not purple.

Combining all of these yields

If a snake is happy, then it is not purple.

Thus, (D) is always true.

- e) From the first proposition, we have

If a snake is happy, then it can add.

From the contrapositive of the third proposition, we have

If a snake can add, then it can subtract.

Combining these two propositions gives

If a snake is happy, then it can subtract.

Thus, (E) is never true.

Therefore, (D) is our answer.

5. (AMC 12 12A 2021) When a student multiplied the number 66 by the repeating decimal,

$$1.\underline{a}\underline{b}\underline{a}\underline{b}\dots = 1.\underline{a}\overline{b},$$

where a and b are digits, he did not notice the notation and just multiplied 66 times $1.\underline{a}\underline{b}$. Later he found that his answer is 0.5 less than the correct answer. What is the 2-digit number $\underline{a}\underline{b}$? (A) 15 (B) 30 (C) 45 (D) 60 (E) 75

Answer: E

Solution 1

We are given that $66\left(1.\overline{\underline{a}\underline{b}}\right) - 0.5 = 66\left(1.\underline{a}\underline{b}\right)$, from which

$$66\left(1.\overline{\underline{a}\underline{b}}\right) - 66\left(1.\underline{a}\underline{b}\right) = 0.5$$

$$66\left(\overline{\underline{a}\underline{b}} - \underline{a}\underline{b}\right) = 0.5$$

$$66\left(\overline{0.0\underline{a}\underline{b}}\right) = 0.5$$

$$66\left(\frac{1}{100} \cdot \overline{0.\underline{a}\underline{b}}\right) = \frac{1}{2}$$

$$\overline{0.\underline{a}\underline{b}} = \frac{25}{33}$$

$$\overline{0.\underline{a}\underline{b}} = 0.\overline{75}$$

$$\underline{a} \, \underline{b} = \boxed{\text{(E)} \, 75}.$$

MRENTHUSIASM

Solution 2

It is known that $0.\overline{\underline{a} \, \underline{b}} = \frac{a \, b}{99}$ and $0.\underline{a} \, \underline{b} = \frac{a \, b}{100}$.

Let $x = \underline{a} \, \underline{b}$. We have

$$66 \left(1 + \frac{x}{99} \right) - 66 \left(1 + \frac{x}{100} \right) = 0.5.$$

Expanding and simplifying give $\frac{x}{150} = 0.5$, so $x = \boxed{\text{(E)} \, 75}$.

Solution 3 (Similar to Solution 2)

We have

$$66 \cdot \left(1 + \frac{10a + b}{100} \right) + \frac{1}{2} = 66 \cdot \left(1 + \frac{10a + b}{99} \right).$$

Expanding both sides, we have

$$66 + \frac{33(10a + b)}{50} + \frac{1}{2} = 66 + \frac{2(10a + b)}{3}.$$

Subtracting 66 from both sides, we have

$$\frac{33(10a + b)}{50} + \frac{1}{2} = \frac{2(10a + b)}{3}.$$

Multiplying both sides by $50 \cdot 3 = 150$, we have

$$99(10a + b) + 75 = 100(10a + b).$$

Thus, the answer is $10a + b = \boxed{\text{(E)} \, 75}$.

By letting $x = \underline{a} \, \underline{b} = 10a + b$, this solution is similar to Solution 2. In this solution, we solve for $10a + b$ as a whole.

-mathboy282 (Solution)

Solution 4 (Answer Choices & Modular)

Let $\underline{a} \, \underline{b}$ represent the two-digit number ab , not the product of the digits a and b . We can construct fractions for the values $1.\underline{a} \, \underline{b}$, and $1.\overline{\underline{a} \, \underline{b}}$, which are $\frac{1a \, b}{100}$ and $\frac{1a \, b}{99}$ respectively. Multiplying by 66 on both sides and adding $1/2$ to $\frac{1a \, b}{100}$ and simplifying, we get this:

$$\frac{33 * \underline{ab} + 25}{50} = \frac{2ab}{3}.$$

Looking at the answer choices, we notice that all of them are divisible by 3. This means that since the right-hand side will result in an integer, the left-hand side needs to as well. This means that the numerator of the left-hand side fraction has to be divisible by 50. So, we get this expression:

$$33 * \underline{ab} + 25 \equiv 0 \pmod{50}. \quad 33 * \underline{ab} \equiv -25 \pmod{50}. \quad 33 * \underline{ab} \equiv 25 \pmod{50}.$$

This means that the product of 33 and $\underline{ab}$ must have a remainder of 25 when divided by 50. Since it must have a remainder of 25, the product should have a units digit of 5, which eliminates **(B)** and **(D)**. Multiplying 33 to the rest of the answer choices, the only one which fills this requirement is 75, which is $\boxed{\text{(E)} \, 75}$.

Solution 5 (A different way to calculate)

$1.\overline{ab} = 1 + \frac{\overline{ab}_{10}}{10^2 - 1} = 1 + \frac{10a + b}{99}$ and $1.ab = 1 + \frac{a}{10} + \frac{b}{100}$, so our equation is

$$66 \times \left(1 + \frac{10a + b}{99}\right) = 66 \times \left(1 + \frac{a}{10} + \frac{b}{100}\right) + 0.5,$$

so now let's simplify!

$$\begin{aligned} 66 \times \left(1 + \frac{10a + b}{99}\right) &= 66 \times \left(1 + \frac{a}{10} + \frac{b}{100}\right) + 0.5 \\ 1 + \frac{10a + b}{99} &= 1 + \frac{a}{10} + \frac{b}{100} + \frac{1}{132} \\ \frac{10a + b}{99} &= \frac{10a + b}{100} + \frac{1}{132}. \end{aligned}$$

Notice that the 2-digit number $\overline{ab}$ is just $10a + b$. Therefore, let's just find the value of $10a + b$.

$$\begin{aligned}\left(\frac{1}{99} - \frac{1}{100}\right)(10a + b) &= \frac{1}{132} \\ 10a + b &= \frac{\frac{1}{132}}{\frac{1}{99} - \frac{1}{100}} \\ 10a + b &= \frac{\frac{1}{132}}{\frac{1}{100 \times 99}} \\ 10a + b &= \frac{9900}{132} \\ 10a + b &= 75.\end{aligned}$$

We just got $10a + b = 75$, and 75 is indeed a 2-digit number, so the two-digit number $\overline{ab}$ is $\boxed{75}$.

6. (AMC 12 12A 2021) A deck of cards has only red cards and black cards. The probability of a randomly chosen card being red is $\frac{1}{3}$. When 4 black cards are added to the deck, the probability of choosing red becomes $\frac{1}{4}$. How many cards were in the deck originally?
- (A) 6 (B) 9 (C) 12 (D) 15 (E) 18

Answer: C

Solution 1 (Algebra)

If the probability of choosing a red card is $\frac{1}{3}$, the red and black cards are in ratio 1 : 2. This means at the beginning there are x red cards and $2x$ black cards.

After 4 black cards are added, there are $2x+4$ black cards. This time, the probability of choosing a red card is $\frac{1}{4}$ so the ratio of red to black cards is 1 : 3. This means in the new deck the number of black cards is also $3x$ for the same x red cards.

So, $3x = 2x + 4$ and $x = 4$ meaning there are 4 red cards in the deck at the start and $2(4) = 8$ black cards.

So, the answer is $8 + 4 = 12 = \boxed{(C) 12}$.

–abhinavg0627

Solution 2 (Arithmetic)

In terms of the number of cards, the original deck is 3 times the red cards, and the final deck is 4 times the red cards. So, the final deck is $\frac{4}{3}$ times the original deck. We are given that adding 4 cards to the original deck is the same as increasing the original deck by $\frac{1}{3}$ of itself. Since 4 cards are equal to $\frac{1}{3}$ of the original deck, the original deck has $4 \cdot 3 = \boxed{(C) 12}$ cards.

Solution 3 (Observations)

Suppose there were x cards in the deck originally. Now, the deck has $x + 4$ cards, which must be a multiple of 4.

Only $12 + 4 = 16$ is a multiple of 4, so the answer is $x = \boxed{(C) 12}$.

Solution 4 (Ratios and Proportions)

By looking at the ratio of black cards to the total number of cards, we can say that there are $2x$ black cards and $3x$ cards in total. We can write the probability of getting a black card as $\frac{2x}{3x}$. But when we increase the number of black cards by 4 the probability becomes $\frac{3}{4}$. So the equation is: $\frac{2x+4}{3x+4} = \frac{3}{4}$, $8x + 16 = 9x + 12$, $x = 4$. The answer is $3x = \boxed{\text{(C) } 12}$.

Solution 5 (Simpler Ratios)

This could in a way be considered a more efficient/alternate version of Solution 4. Since the problem mentions that there is initially a $\frac{1}{3}$ chance of getting a red card, we can represent the number of red cards as $\frac{x}{3}$, where x is the number of cards in the deck originally. Since the modification to the deck makes no changes to the number of red cards, you can see that if there is a $\frac{1}{4}$ probability of having a red card in a deck of $x + 4$ cards, $\frac{x}{3} = \frac{x+4}{4}$. Solving this for x gives $x = \boxed{\text{(C) } 12}$.

Solution 6 (Observation)

Notice that in a deck with 1 red card and 2 black cards, the probability of a red ball is $\frac{1}{3}$

Adding a single black card would make the chance $\frac{1}{4}$

If we scale this deck by 4, making so that there are 4 red cards and 8 black cards, the original chance is still $\frac{1}{3}$

Adding 4 black cards will make the chance $\frac{1}{4}$

The answer is $\boxed{\text{(C) } 12}$.

7. (AMC 12 12A 2021) What is the least possible value of $(xy - 1)^2 + (x + y)^2$ for all real numbers x and y ? (A) 0 (B) $\frac{1}{4}$ (C) $\frac{1}{2}$ (D) 1 (E) 2

Answer: D

Solution 1 (Expand)

Expanding, we get that the expression is $x^2 + 2xy + y^2 + x^2y^2 - 2xy + 1$ or $x^2 + y^2 + x^2y^2 + 1$. By the Trivial Inequality (all squares are nonnegative) the minimum value for this is $\boxed{\text{(D) } 1}$, which can be achieved at $x = y = 0$.

Solution 2 (Expand and then Factor)

We expand the original expression, then factor the result by grouping:

$$\begin{aligned}(xy - 1)^2 + (x + y)^2 &= (x^2y^2 - 2xy + 1) + (x^2 + 2xy + y^2) \\ &= x^2y^2 + x^2 + y^2 + 1 \\ &= x^2(y^2 + 1) + (y^2 + 1) \\ &= (x^2 + 1)(y^2 + 1).\end{aligned}$$

Clearly, both factors are positive. By the Trivial Inequality, we have

$$(x^2 + 1)(y^2 + 1) \geq (0 + 1)(0 + 1) = \boxed{\text{(D) } 1}.$$

Note that the least possible value of $(xy - 1)^2 + (x + y)^2$ occurs at $x = y = 0$.

Solution 3 (Beyond Overkill)

Like solution 1, expand and simplify the original equation to $x^2 + y^2 + x^2y^2 + 1$ and let $f(x, y) = x^2 + y^2 + x^2y^2 + 1$. To find local extrema, find where $\nabla f(x, y) = \mathbf{0}$.

First, find the first partial derivative with respect to x and y and find where they are 0:

$$\frac{\partial f}{\partial x} = 2x + 2xy^2 = 2x(1 + y^2) = 0 \implies x = 0$$

$$\frac{\partial f}{\partial y} = 2y + 2yx^2 = 2y(1 + x^2) = 0 \implies y = 0$$

Thus, there is a local extremum at $(0, 0)$. Because this is the only extremum, we can assume that this is a minimum because the problem asks for the minimum (though this can also be proven using the partial second derivative test) and the global minimum since it's the only minimum, meaning $f(0, 0)$ is the minimum of $f(x, y)$. Plugging $(0, 0)$ into $f(x, y)$, we find $1 \implies \boxed{\text{(D)} 1}$.

Remark: Honestly, this isn't even that overkill. I did it this way, and it was not very bashy. vaishnav

8. (AMC 12 12A 2021) A sequence of numbers is defined by $D_0 = 0, D_1 = 0, D_2 = 1$ and $D_n = D_{n-1} + D_{n-3}$ for $n \geq 3$. What are the parities (evenness or oddness) of the triple of numbers $(D_{2021}, D_{2022}, D_{2023})$, where E denotes even and O denotes odd?

(A) (O, E, O) (B) (E, E, O) (C) (E, O, E) (D) (O, O, E) (E) (O, O, O)

Answer: C

Solution

We construct the following table:

Term	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	$\dots$
Value	0	0	1	1	1	2	3	4	6	9	$\dots$
Parity	E	E	O	O	O	E	O	E	E	O	$\dots$

Note that (D_7, D_8, D_9) and (D_0, D_1, D_2) have the same parities, so the parity is periodic with period 7. Since the remainders of $(2021 \div 7, 2022 \div 7, 2023 \div 7)$ are $(5, 6, 7)$, we conclude that $(D_{2021}, D_{2022}, D_{2023})$ and (D_5, D_6, D_7) have the same parities, namely $\boxed{\text{(C)} (E, O, E)}$.

9. (AMC 12 12A 2021) Which of the following is equivalent to

$$(2 + 3)(2^2 + 3^2)(2^4 + 3^4)(2^8 + 3^8)(2^{16} + 3^{16})(2^{32} + 3^{32})(2^{64} + 3^{64})?$$

(A) $3^{127} + 2^{127}$ (B) $3^{127} + 2^{127} + 2 \cdot 3^{63} + 3 \cdot 2^{63}$ (C) $3^{128} - 2^{128}$ (D) $3^{128} + 2^{128}$
 (E) 5^{127}

Answer: C

Solution 1

By multiplying the entire equation by $3 - 2 = 1$, all the terms will simplify by difference of squares, and the final answer is $\boxed{\text{(C)} 3^{128} - 2^{128}}$.

Additionally, we could also multiply the entire equation (we can let it be equal to x) by $2 - 3 = -1$. The terms again simplify by difference of squares. This time, we get $-x = 2^{128} - 3^{128} \Rightarrow x = 3^{128} - 2^{128}$. Both solutions yield the same answer.

Note: Also notice when you multiply it by the first pair $(2 + 3)$, it immediately factors. Notice how it "domino" effects to the others, ultimately collapsing into

$$\boxed{\text{(C)} 3^{128} - 2^{128}}.$$

Solution 2

If you weren't able to come up with the $(3 - 2)$ insight, then you could just notice that the answer is divisible by $2 + 3 = 5$, and $2^2 + 3^2 = 13$. We can then use Fermat's Little Theorem for $p = 5, 13$ on the answer choices to determine which of the answer choices are divisible by both 5 and 13. This is $\boxed{(C) \ 3^{128} - 2^{128}}$.

-mewto

Solution 3 (Units Digit and Multiples)

Notice that simplifying the first term gives 5, which means that the units digit of the entire expression must be 5. Also, note how the numbers appear to form multiples of 2, so therefore our answer should contain a power of 2. The only one that satisfies the above properties is $\boxed{(C) \ 3^{128} - 2^{128}}$.

Solution 4

After expanding the first few terms, the result after each term appears to be $2^{2^n-1} + 2^{2^n-2} \cdot 3^1 + 2^{2^n-3} \cdot 3^2 + \dots + 2^1 \cdot 3^{2^n-2} + 3^{2^n-1}$ where n is the number of terms expanded. We can prove this using mathematical induction. The base step is trivial. When expanding another term, all of the previous terms multiplied by $2^{2^{n-1}}$ would give $2^{2^n-1} + 2^{2^n-2} \cdot 3^1 + 2^{2^n-3} \cdot 3^2 + \dots + 2^{2^{n-1}+1} \cdot 3^{2^{n-1}-1} + 2^{2^{n-1}} \cdot 3^{2^{n-1}}$, and all the previous terms multiplied by $3^{2^{n-1}}$ would give $3^{2^n-1} + 3^{2^n-2} \cdot 2^1 + 3^{2^n-3} \cdot 2^2 + \dots + 3^{2^{n-1}+1} \cdot 2^{2^{n-1}-1} + 3^{2^{n-1}} \cdot 2^{2^{n-1}}$. Their sum is equal to $2^{2^n-1} + 2^{2^n-2} \cdot 3^1 + 2^{2^n-3} \cdot 3^2 + \dots + 2^1 \cdot 3^{2^n-2} + 3^{2^n-1}$, so the proof is complete. Since $\frac{3^{2^n}-2^{2^n}}{3-2}$ is equal to $2^{2^n-1} + 2^{2^n-2} \cdot 3^1 + 2^{2^n-3} \cdot 3^2 + \dots + 2^1 \cdot 3^{2^n-2} + 3^{2^n-1}$, the answer is $\frac{3^{2^7}-2^{2^7}}{3-2} = \boxed{(C) \ 3^{128} - 2^{128}}$.

Solution 5 (Engineer's Induction)

We can compute some of the first few partial products, and notice that $\prod_{k=0}^{2^n-1} (2^{2^k} + 3^{2^k}) = 3^{2^{n+1}} - 2^{2^{n+1}}$. As we don't have to prove this, we get the product is $3^{2^7} - 2^{2^7} = 3^{128} - 2^{128}$, and click $\boxed{(C) \ 3^{128} - 2^{128}}$.

Solution 6 (Difference of Squares)

We notice that the first term is equal to $3^2 - 2^2$. If we multiply this by the second term, then we will get $(3^2 - 2^2)(3^2 + 2^2)$, and we can simplify by using difference of squares to obtain $3^4 - 2^4$. If we multiply this by the third term and simplify using difference of squares again, we get $3^8 - 2^8$. We can continue down the line until we multiply by the last term, $3^{64} + 2^{64}$, and get $\boxed{(C) \ 3^{128} - 2^{128}}$.

Solution 7 (Generalization)

Let's generalize to $x = 3$ and $y = 2$. Then we get:

$$(y+x)(y^2+x^2)(y^4+x^4)(y^8+x^8)(y^{16}+x^{16})(y^{32}+x^{32})(y^{64}+x^{64}).$$

We see that the first term is $y+x$, and the next is y^2+x^2 . We realize that if we multiply the first term by $y-x$, the result by difference of squares will be y^2-x^2 ; we can proceed to use difference of squares on that. In other words, the equation has the domino effect, and all we need to get started is to multiply the whole equation by $y-x$ (keeping in mind to divide $y-x$ at the end.)

Now we get:

$$(y-x)(y+x)(y^2+x^2)(y^4+x^4)(y^8+x^8)(y^{16}+x^{16})(y^{32}+x^{32})(y^{64}+x^{64})$$

$$\begin{aligned}
&= (y^2 - x^2)(y^2 + x^2)(y^4 + x^4)(y^8 + x^8)(y^{16} + x^{16})(y^{32} + x^{32})(y^{64} + x^{64}) \\
&= (y^4 - x^4)(y^4 + x^4)(y^8 + x^8)(y^{16} + x^{16})(y^{32} + x^{32})(y^{64} + x^{64}) \\
&\vdots \\
&= (y^{64} - x^{64})(y^{64} + x^{64}) \\
&= y^{128} - x^{128}.
\end{aligned}$$

Now we can plug in $y = 2$ and $x = 3$:

$$(2 - 3)(2 + 3)(2^2 + 3^2)(2^4 + 3^4)(2^8 + 3^8)(2^{16} + 3^{16})(2^{32} + 3^{32})(2^{64} + 3^{64}) = 2^{128} - 3^{128}.$$

However, we must not forget to divide by $2 - 3 = -1$ at the end! Dividing, we get the answer of

$$3^{128} - 2^{128} = \boxed{(C) \ 3^{128} - 2^{128}}.$$

KingRavi

Solution 8 (Generalization)

More generally, we have

$$(a^n - b^n)(a^n + b^n) = a^{2n} - b^{2n} \quad (\star)$$

by the difference of squares.

Multiplying the original expression by $1 = 3 - 2$ and then applying $(\star)$ repeatedly, we get

$$\begin{aligned}
&(2 + 3)(2^2 + 3^2)(2^4 + 3^4)(2^8 + 3^8)(2^{16} + 3^{16})(2^{32} + 3^{32})(2^{64} + 3^{64}) \\
&= \underbrace{\left(\begin{array}{c} 1 \\ 3-2 \end{array} \right)}_{3-2} \underbrace{(2 + 3)(2^2 + 3^2)(2^4 + 3^4)(2^8 + 3^8)(2^{16} + 3^{16})(2^{32} + 3^{32})(2^{64} + 3^{64})}_{\text{Apply the Commutative Property of Addition to all factors.}} \\
&= \underbrace{(3 - 2)(3 + 2)}_{\substack{\text{Apply } (\star) \\ \text{to get } 3^2 - 2^2.}} (3^2 + 2^2)(3^4 + 2^4)(3^8 + 2^8)(3^{16} + 2^{16})(3^{32} + 2^{32})(3^{64} + 2^{64}) \\
&= \underbrace{(3^2 - 2^2)(3^2 + 2^2)}_{\substack{\text{Apply } (\star) \\ \text{to get } 3^4 - 2^4.}} (3^4 + 2^4)(3^8 + 2^8)(3^{16} + 2^{16})(3^{32} + 2^{32})(3^{64} + 2^{64}) \\
&= \underbrace{(3^4 - 2^4)(3^4 + 2^4)}_{\substack{\text{Apply } (\star) \\ \text{to get } 3^8 - 2^8.}} (3^8 + 2^8)(3^{16} + 2^{16})(3^{32} + 2^{32})(3^{64} + 2^{64}) \\
&\vdots \\
&= \boxed{(C) \ 3^{128} - 2^{128}}.
\end{aligned}$$

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Solution 9 (Desperate Cheese, Similar to Solution 5)

One would probably assume that the factorization that this is equal to can be generalized for the pattern

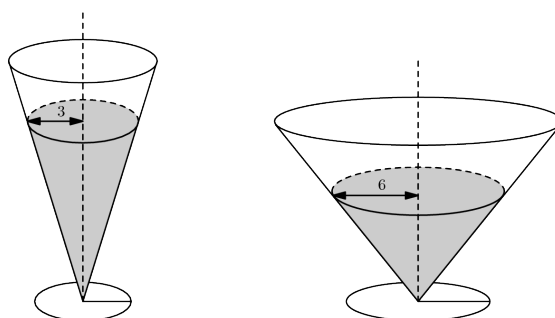
$$(2 + 3)(2^2 + 3^2)(2^4 + 3^4) \cdots (2^n + 3^n).$$

So we can take a smaller portion and apply the answer choices there. We get

$$(2 + 3)(2^2 + 3^2) = (5)(13) = 65.$$

We notice that the patterns have a relationship where its either the sum of the powers or the next power not included that is incorporated. So we change the numbers from 127 to 3, 128 to 4, and 63 to 1 (since 63 is the sum of the powers, divided by 2, rounded down). Checking the values of the other patterns, the only one that gives 65 is $3^4 - 2^4$, which means the answer is **(C)** $3^{128} - 2^{128}$.

10. (AMC 12 12A 2021) Two right circular cones with vertices facing down as shown in the figure below contain the same amount of liquid. The radii of the tops of the liquid surfaces are 3 cm and 6 cm. Into each cone is dropped a spherical marble of radius 1 cm, which sinks to the bottom and is completely submerged without spilling any liquid. What is the ratio of the rise of the liquid level in the narrow cone to the rise of the liquid level in the wide cone?



- (A) 1 : 1 (B) 47 : 43 (C) 2 : 1 (D) 40 : 13 (E) 4 : 1

Answer: E

Solution 1 (Algebra)

Initial Scenario

Let the heights of the narrow cone and the wide cone be h_1 and h_2 , respectively. We have the following table:

	Base Radius	Height	Volume
Narrow Cone	3	h_1	$\frac{1}{3}\pi(3)^2h_1 = 3\pi h_1$
Wide Cone	6	h_2	$\frac{1}{3}\pi(6)^2h_2 = 12\pi h_2$

Equating the volumes gives $3\pi h_1 = 12\pi h_2$, which simplifies to $\frac{h_1}{h_2} = 4$.

Furthermore, by similar triangles:

- For the narrow cone, the ratio of the base radius to the height is $\frac{3}{h_1}$, which always remains constant.
- For the wide cone, the ratio of the base radius to the height is $\frac{6}{h_2}$, which always remains constant.

Two solutions follow from here:

Solution 1.1 (Properties of Fractions)

Final Scenario

For the narrow cone and the wide cone, let their base radii be $3x$ and $6y$ (for some $x, y > 1$), respectively. By the similar triangles discussed above, their heights must

be h_1x and h_2y , respectively. We have the following table:

	Base Radius	Height	Volume
Narrow Cone	$3x$	h_1x	$\frac{1}{3}\pi(3x)^2(h_1x) = 3\pi h_1x^3$
Wide Cone	$6y$	h_2y	$\frac{1}{3}\pi(6y)^2(h_2y) = 12\pi h_2y^3$

Recall that $\frac{h_1}{h_2} = 4$. Equating the volumes gives $3\pi h_1x^3 = 12\pi h_2y^3$, which simplifies to $x^3 = y^3$, or $x = y$.

Finally, the requested ratio is

$$\frac{h_1x - h_1}{h_2y - h_2} = \frac{h_1(x - 1)}{h_2(y - 1)} = \frac{h_1}{h_2} = \boxed{\text{(E) } 4 : 1}.$$

<u>**Remarks**</u>

- This solution uses the following property of fractions: For unequal positive numbers a, b, c and d , if $\frac{a}{b} = \frac{c}{d} = k$, then $\frac{a \pm c}{b \pm d} = \frac{bk \pm dk}{b \pm d} = \frac{(b \pm d)k}{b \pm d} = k$.
- This solution shows that, regardless of the shape or the volume of the solid dropped into each cone, the requested ratio stays the same as long as the solid sinks to the bottom and is completely submerged without spilling any liquid.

Solution 1.2 (Bash)

<u>**Final Scenario**</u>

For the narrow cone and the wide cone, let their base radii be r_1 and r_2 , respectively; let their rises of the liquid levels be Δh_1 and Δh_2 , respectively. We have the following table:

	Base Radius	Height	Volume
Narrow Cone	r_1	$h_1 + \Delta h_1$	$\frac{1}{3}\pi r_1^2(h_1 + \Delta h_1)$
Wide Cone	r_2	$h_2 + \Delta h_2$	$\frac{1}{3}\pi r_2^2(h_2 + \Delta h_2)$

By the similar triangles discussed above, we get

$$\frac{3}{h_1} = \frac{r_1}{h_1 + \Delta h_1} \implies r_1 = \frac{3}{h_1}(h_1 + \Delta h_1), \quad (1)$$

$$\frac{6}{h_2} = \frac{r_2}{h_2 + \Delta h_2} \implies r_2 = \frac{6}{h_2}(h_2 + \Delta h_2). \quad (2)$$

The volume of the marble dropped into each cone is $\frac{4}{3}\pi(1)^3 = \frac{4}{3}\pi$.

Now, we set up an equation for the volume of the narrow cone, then express Δh_1 in terms of h_1 :

$$\begin{aligned} \frac{1}{3}\pi r_1^2(h_1 + \Delta h_1) &= 3\pi h_1 + \frac{4}{3}\pi \\ \frac{1}{3}r_1^2(h_1 + \Delta h_1) &= 3h_1 + \frac{4}{3} \\ \frac{1}{3}\left(\frac{3}{h_1}(h_1 + \Delta h_1)\right)^2(h_1 + \Delta h_1) &= 3h_1 + \frac{4}{3} && \text{by (1)} \\ \frac{3}{h_1^2}(h_1 + \Delta h_1)^3 &= 3h_1 + \frac{4}{3} \end{aligned}$$

$$(h_1 + \Delta h_1)^3 = h_1^3 + \frac{4h_1^2}{9}$$

$$\Delta h_1 = \sqrt[3]{h_1^3 + \frac{4h_1^2}{9}} - h_1.$$

Next, we set up an equation for the volume of the wide cone, then express Δh_2 in terms of h_2 :

$$\frac{1}{3}\pi r_2^2(h_2 + \Delta h_2) = 12\pi h_2 + \frac{4}{3}\pi.$$

Using a similar process from above, we get

$$\Delta h_2 = \sqrt[3]{h_2^3 + \frac{h_2^2}{9}} - h_2.$$

Recall that $\frac{h_1}{h_2} = 4$. Therefore, the requested ratio is

$$\begin{aligned} \frac{\Delta h_1}{\Delta h_2} &= \frac{\sqrt[3]{h_1^3 + \frac{4h_1^2}{9}} - h_1}{\sqrt[3]{h_2^3 + \frac{h_2^2}{9}} - h_2} \\ &= \frac{\sqrt[3]{(4h_2)^3 + \frac{4(4h_2)^2}{9}} - 4h_2}{\sqrt[3]{h_2^3 + \frac{h_2^2}{9}} - h_2} \\ &= \frac{\sqrt[3]{4^3 \left(h_2^3 + \frac{h_2^2}{9}\right)} - 4h_2}{\sqrt[3]{h_2^3 + \frac{h_2^2}{9}} - h_2} \\ &= \frac{4\sqrt[3]{h_2^3 + \frac{h_2^2}{9}} - 4h_2}{\sqrt[3]{h_2^3 + \frac{h_2^2}{9}} - h_2} \\ &= \boxed{\text{(E)} \ 4 : 1}. \end{aligned}$$

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Solution 2 (Approximate Cones with Cylinders)

The heights of the cones are not given, so suppose the heights are very large (i.e. tending towards infinity) in order to approximate the cones as cylinders with base radii 3 and 6 and infinitely large height. Then the base area of the wide cylinder is 4 times that of the narrow cylinder. Since we are dropping a ball of the same volume into each cylinder, the water level in the narrow cone/cylinder should rise (E) 4 times as much.

Solution 3 (Calculus)

The volume of the shorter cone can be expressed as $V_1 = 3\pi h_1$, and the volume of the larger cone can be expressed as $V_2 = 12\pi h_2$. We also know that the volume changes by $\frac{4\pi}{3}$, because the volume of the 1cm sphere is $\frac{4\pi}{3}$.

Taking the derivative of the equations, we get: $dV_1 = 3\pi(dh_1) = \frac{4\pi}{3}$ and $dV_2 = 12\pi(dh_2) = \frac{4\pi}{3}$.

Therefore, $dh_1 = \frac{4\pi}{3} \cdot \frac{1}{3\pi} = \frac{4}{9}$ and $dh_2 = \frac{4\pi}{3} \cdot \frac{1}{12\pi} = \frac{1}{9}$. The ratio is $\frac{4}{9} : \frac{1}{9}$ giving us the answer of (E) 4 : 1.

Solution 4 (Extremely Quick Observation)

Note that as long as the volumes are equal, the ratio of the heights of Cone 2 to Cone 1 is always 1 : 4. We can prove this as follows:

If we fix Cone 2 to have a certain height, then the volume of the cone is $\frac{1}{3} \cdot 36\pi h_1 = 12\pi h_1$. Now if Cone 1 has the same volume, its height would be $\frac{12\pi h_1}{\frac{1}{3} \cdot 9\pi} = \frac{12\pi h_1}{3\pi} = 4h_1$.

So the answer is (E) 4 : 1

Minor edits by: T3CHN0B14D3

Remark You can cheese this by letting r_1 and r_2 be some small integers. I recommend 4 and 1.

11. (AMC 12 12A 2021) A laser is placed at the point $(3, 5)$. The laser beam travels in a straight line. Larry wants the beam to hit and bounce off the y -axis, then hit and bounce off the x -axis, then hit the point $(7, 5)$. What is the total distance the beam will travel along this path?

(A) $2\sqrt{10}$ (B) $5\sqrt{2}$ (C) $10\sqrt{2}$ (D) $15\sqrt{2}$ (E) $10\sqrt{5}$

Answer: C

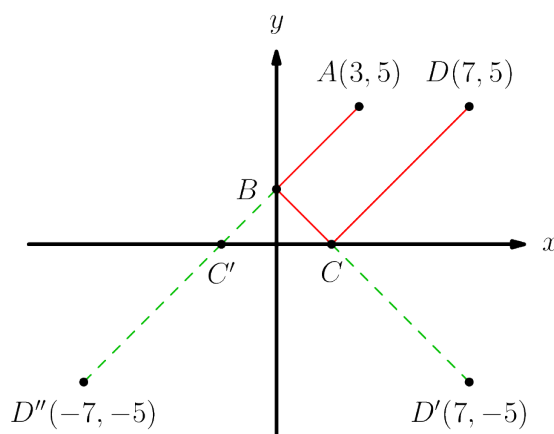
Solution 1 (Reflections)

Let $A = (3, 5)$ and $D = (7, 5)$. Suppose that the beam hits and bounces off the y -axis at B , then hits and bounces off the x -axis at C .

When the beam hits and bounces off a coordinate axis, the angle of incidence and the angle of reflection are congruent. Therefore, we straighten up the path of the beam by reflections:

- a) We reflect $\overline{BC}$ about the y -axis to get $\overline{BC'}$.
- b) We reflect $\overline{CD}$ about the x -axis to get $\overline{CD'}$ with $D' = (7, -5)$, then reflect $\overline{CD'}$ about the y -axis to get $\overline{C'D''}$ with $D'' = (-7, -5)$.

We obtain the following diagram:



The total distance that the beam will travel is

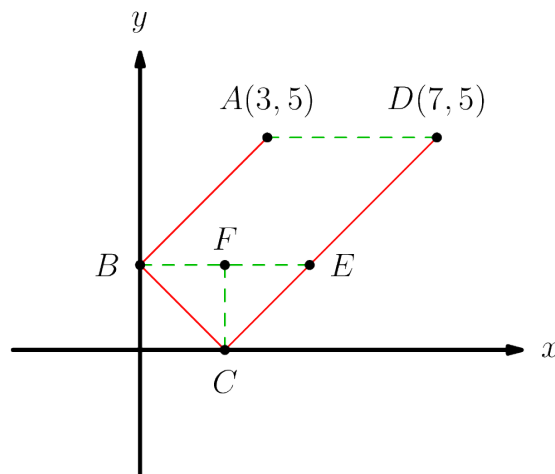
$$\begin{aligned}
 AB + BC + CD &= AB + BC + CD' \\
 &= AB + BC' + C'D'' \\
 &= AD'' \\
 &= \sqrt{((3 - (-7))^2 + (5 - (-5))^2)}
 \end{aligned}$$

$$= \boxed{(C) \ 10\sqrt{2}}.$$

MRENTHUSIASM (Solution)

Solution 2 (Parallelogram)

Define points A, B, C , and D as Solution 1 does. Moreover, let E be a point on $\overline{CD}$ such that $\overline{BE}$ is perpendicular to the y -axis, and F be a point on $\overline{BE}$ such that $\overline{CF}$ is perpendicular to the x -axis, as shown below.



When the beam hits and bounces off a coordinate axis, the angle of incidence and the angle of reflection are congruent, from which $\angle ABF = \angle CBF$ and $\angle BCF = \angle ECF$. We conclude that $\triangle BCF \cong \triangle ECF$ by ASA, so $\angle CBF = \angle CEF$. It follows that $\angle ABF = \angle CEF$ by transitive, so $\overline{AB} \parallel \overline{CD}$ by the Converse of the Alternate Interior Angles Theorem.

Note that $\overline{AD} \parallel \overline{BE}$. Since the opposite sides are parallel, quadrilateral $ABED$ is a parallelogram. From $\triangle BCF \cong \triangle ECF$, we get $BF = EF = 2$, so $C = (2, 0)$.

Let $B = (0, b)$. We equate the slopes of $\overline{AB}$ and $\overline{DC}$:

$$\frac{5 - b}{3 - 0} = \frac{5 - 0}{7 - 2},$$

from which $b = 2$, or $B = (0, 2)$.

By the Distance Formula, we have $AB = 3\sqrt{2}$, $BC = 2\sqrt{2}$, and $CD = 5\sqrt{2}$. The total distance that the beam will travel is

$$AB + BC + CD = \boxed{(C) \ 10\sqrt{2}}.$$

<u>Remark</u>

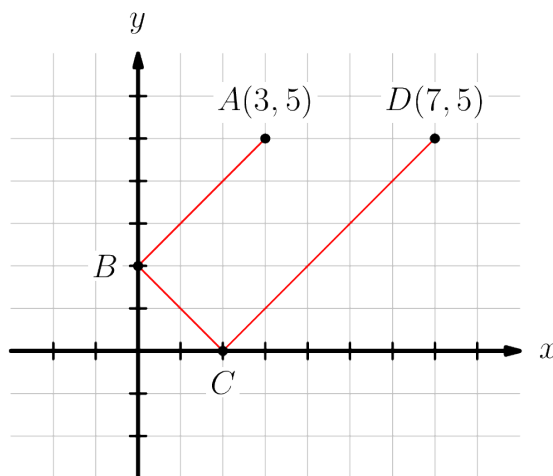
When a straight line hits and bounces off a coordinate axis at point P , the ray entering P and the ray leaving P always have negative slopes. In this problem, $\overline{AB}$ and $\overline{BC}$ have negative slopes; $\overline{BC}$ and $\overline{CD}$ have negative slopes. So, $\overline{AB}$ and $\overline{CD}$ have the same slope, or $\overline{AB} \parallel \overline{CD}$.

Solution 3 (Educated Guess)

Define points A, B, C , and D as Solution 1 does.

Since choices (B), (C), and (D) all involve $\sqrt{2}$, we suspect that one of them is the correct answer. We take a guess in faith that $\overline{AB}$, $\overline{BC}$, and $\overline{CD}$ all form 45°

angles with the coordinate axes, from which $B = (0, 2)$ and $C = (2, 0)$. The given condition $D = (7, 5)$ verifies our guess, as shown below.



Following the last paragraph of Solution 2 gives the answer (C) $10\sqrt{2}$.

Solution 4 (System of Linear Equations)

Denote $(3, 5)$ as point P and $(7, 5)$ as point S . Define the point Q on the y -axis that the laser hits as $(0, y_1)$ and the point R on the x -axis that the laser hits as $(x_1, 0)$. The laser will bounce off of the two axes at right angles, so we have that line $PQ \parallel RS$, meaning they have the same slope which we will denote as k . Line QR will be perpendicular to both PQ and RS , meaning it will have slope $-\frac{1}{k}$.

We can write the equations for lines PQ and RS in point-slope form, and then plug in points Q and R into those respective equations to get equations in terms of our three variables. We can also write the equation for line QR in slope-intercept form and plug in point R . Doing this yields

$$\begin{aligned} y_1 - 5 &= k(0 - 3) \\ 0 - 5 &= k(x_1 - 7) \\ 0 &= -\frac{1}{k}x_1 + y_1. \end{aligned}$$

We can simplify this into the following cubic equation in terms of k :

$$-3k^3 + 5k^2 - 7k + 5 = 0.$$

Luckily, we notice that the coefficients sum up to zero, so $k = 1$. Now we can solve for the other two, yielding $x_1 = y_1 = 2$. Now we simply have three 45° - 45° - 90° triangles with leg lengths of 3, 2, and 5, so our final distance is

$$3\sqrt{2} + 2\sqrt{2} + 5\sqrt{2} = \text{(C) } 10\sqrt{2}.$$

Mooshiros

12. (AMC 12 12A 2021) All the roots of the polynomial $z^6 - 10z^5 + Az^4 + Bz^3 + Cz^2 + Dz + 16$ are positive integers, possibly repeated. What is the value of B ?

(A) -88 (B) -80 (C) -64 (D) -41 (E) -40

Answer: A

Solution 1

By Vieta's formulas, the sum of the six roots is 10 and the product of the six roots is 16. By inspection, we see the roots are 1, 1, 2, 2, 2, and 2, so the function is $(z-1)^2(z-2)^4 = (z^2-2z+1)(z^4-8z^3+24z^2-32z+16)$. Therefore, calculating just the z^3 terms, we get $B = -32 - 48 - 8 = \boxed{\text{(A)} -88}$.

Solution 2

Using the same method as Solution 1, we find that the roots are 2, 2, 2, 2, 1, and 1. Note that B is the negation of the 3rd symmetric sum of the roots. Using casework on the number of 1's in each of the $\binom{6}{3} = 20$ products $r_a \cdot r_b \cdot r_c$, we obtain

$$B = - \left(\binom{4}{3} \binom{2}{0} \cdot 2^3 + \binom{4}{2} \binom{2}{1} \cdot 2^2 \cdot 1 + \binom{4}{1} \binom{2}{2} \cdot 2 \right) = -(32 + 48 + 8) = \boxed{\text{(A)} -88}.$$

ike.chen

Solution 3 (brute force)

Same as solution 1, we find the roots are 1, 1, 2, 2, 2, 2 by Vieta's formula. This next part is brute force expansion. $-B =$

$$\begin{aligned} & r_1 r_2 r_3 + r_1 r_2 r_4 + r_1 r_2 r_5 + r_1 r_2 r_6 + \\ & r_1 r_3 r_4 + r_1 r_3 r_5 + r_1 r_3 r_6 + \\ & r_1 r_4 r_5 + r_1 r_4 r_6 + \\ & r_1 r_5 r_6 + \\ & r_2 r_3 r_4 + r_2 r_3 r_5 + r_2 r_3 r_6 + \\ & r_2 r_4 r_5 + r_2 r_4 r_6 + \\ & r_2 r_5 r_6 + \\ & r_3 r_4 r_5 + r_3 r_4 r_6 + \\ & r_3 r_5 r_6 + \\ & r_4 r_5 r_6 \\ & = 2 \times 4 + 4 \times 12 + 8 \times 4 \\ & = 8 + 48 + 32 \\ & B = \boxed{\text{(A)} -88} \end{aligned}$$

13. (AMC 12 12A 2021) Of the following complex numbers z , which one has the property that z^5 has the greatest real part?

(A) -2 (B) $-\sqrt{3} + i$ (C) $-\sqrt{2} + \sqrt{2}i$ (D) $-1 + \sqrt{3}i$ (E) $2i$

Answer: B

Solution 1 (De Moivre's Theorem: Degrees)

First, (B) is $2\text{cis}(150)$, (C) is $2\text{cis}(135)$, (D) is $2\text{cis}(120)$.

Taking the real part of the 5th power of each we have:

(A): $(-2)^5 = -32$ (B): $32 \cos(750) = 32 \cos(30) = 16\sqrt{3}$ (C): $32 \cos(675) = 32 \cos(-45) = 16\sqrt{2}$ (D): $32 \cos(600) = 32 \cos(240) < 0$ (E): $(2i)^5 = 32i$, whose real part is 0

Thus, the answer is $\boxed{\text{(B)} -\sqrt{3} + i}$.

Solution 2 (De Moivre's Theorem: Radians)

We rewrite each answer choice to the polar form

$$z = r \text{cis } \theta = r(\cos \theta + i \sin \theta),$$

where r is the magnitude of z such that $r \geq 0$, and θ is the argument of z such that $0 \leq \theta < 2\pi$.

By **De Moivre's Theorem**, the real part of z^5 is

$$\operatorname{Re}(z^5) = r^5 \cos(5\theta).$$

We construct a table as follows:

Choice	r	θ	$\operatorname{Re}(z^5)$
(A)	2	π	$32 \cos(5\pi) = 32 \cos \pi = 32(-1)$
(B)	2	$\frac{5\pi}{6}$	$32 \cos \frac{25\pi}{6} = 32 \cos \frac{\pi}{6} = 32\left(\frac{\sqrt{3}}{2}\right)$
(C)	2	$\frac{3\pi}{4}$	$32 \cos \frac{15\pi}{4} = 32 \cos \frac{7\pi}{4} = 32\left(\frac{\sqrt{2}}{2}\right)$
(D)	2	$\frac{2\pi}{3}$	$32 \cos \frac{10\pi}{3} = 32 \cos \frac{4\pi}{3} = 32\left(-\frac{1}{2}\right)$
(E)	2	$\frac{\pi}{2}$	$32 \cos \frac{5\pi}{2} = 32 \cos \frac{\pi}{2} = 32(0)$

Clearly, the answer is (B) $-\sqrt{3} + i$.

Solution 3 (Binomial Theorem)

We evaluate the fifth power of each answer choice:

- For (A), we have $(-2)^5 = -32$, from which $\operatorname{Re}((-2)^5) = -32$.
- For (E), we have $(2i)^5 = 32i$, from which $\operatorname{Re}((2i)^5) = 0$.

We will apply the Binomial Theorem to each of (B), (C), and (D).

More generally, let

$$z = a + bi$$

for some real numbers a and b .

Two solutions follow from here:

Solution 3.1 (Real Parts Only)

To find the real part of z^5 , we only need the terms with even powers of i :

$$\begin{aligned} \operatorname{Re}(z^5) &= \operatorname{Re}\left((a + bi)^5\right) \\ &= \sum_{k=0}^2 \binom{5}{2k} a^{5-2k} (bi)^{2k} \\ &= \binom{5}{0} a^5 (bi)^0 + \binom{5}{2} a^3 (bi)^2 + \binom{5}{4} a^1 (bi)^4 \\ &= a^5 - 10a^3b^2 + 5ab^4. \end{aligned}$$

We find the real parts of (B), (C), and (D) directly:

- For (B), we have $\operatorname{Re}\left(\left(-\sqrt{3} + i\right)^5\right) = 16\sqrt{3}$.
- For (C), we have $\operatorname{Re}\left(\left(-\sqrt{2} + \sqrt{2}i\right)^5\right) = 16\sqrt{2}$.

- For **(D)**, we have $\operatorname{Re} \left((-1 + \sqrt{3}i)^5 \right) = -16$.

Therefore, the answer is $\boxed{\text{(B)} -\sqrt{3} + i}$.

Solution 3.2 (Full Expansions)

The full expansion of z^5 is

$$\begin{aligned} z^5 &= (a + bi)^5 \\ &= \sum_{k=0}^5 \binom{5}{k} a^{5-k} (bi)^k \\ &= \binom{5}{0} a^5 (bi)^0 + \binom{5}{1} a^4 (bi)^1 + \binom{5}{2} a^3 (bi)^2 + \binom{5}{3} a^2 (bi)^3 + \binom{5}{4} a^1 (bi)^4 + \binom{5}{5} a^0 (bi)^5 \\ &= a^5 + 5a^4 bi - 10a^3 b^2 - 10a^2 b^3 i + 5ab^4 + b^5 i \\ &= (a^5 - 10a^3 b^2 + 5ab^4) + (5a^4 b - 10a^2 b^3 + b^5) i. \end{aligned}$$

We find the full expansions of **(B)**, **(C)**, and **(D)**, then extract their real parts:

- For **(B)**, we have $(-\sqrt{3} + i)^5 = 16\sqrt{3} + 16i$, from which $\operatorname{Re} \left((-\sqrt{3} + i)^5 \right) = 16\sqrt{3}$.
- For **(C)**, we have $(-\sqrt{2} + \sqrt{2}i)^5 = 16\sqrt{2} - 16\sqrt{2}i$, from which $\operatorname{Re} \left((-\sqrt{2} + \sqrt{2}i)^5 \right) = 16\sqrt{2}$.
- For **(D)**, we have $(-1 + \sqrt{3}i)^5 = -16 - 16\sqrt{3}i$, from which $\operatorname{Re} \left((-1 + \sqrt{3}i)^5 \right) = -16$.

Therefore, the answer is $\boxed{\text{(B)} -\sqrt{3} + i}$.

14. (AMC 12 12A 2021) What is the value of

$$\left(\sum_{k=1}^{20} \log_{5^k} 3^{k^2} \right) \cdot \left(\sum_{k=1}^{100} \log_{9^k} 25^k \right)?$$

(A) 21 (B) $100 \log_5 3$ (C) $200 \log_3 5$ (D) 2,200 (E) 21,000

Answer: E

Solution 1 (Properties of Logarithms)

We will apply the following logarithmic identity:

$$\log_{p^n} q^n = \log_p q,$$

which can be proven by the Change of Base Formula:

$$\log_{p^n} q^n = \frac{\log_p q^n}{\log_p p^n} = \frac{n \log_p q}{n} = \log_p q.$$

Now, we simplify the expressions inside the summations:

$$\log_{5^k} 3^{k^2} = \log_{5^k} (3^k)^k$$

$$\begin{aligned}
 &= k \log_{5^k} 3^k \\
 &= k \log_5 3,
 \end{aligned}$$

and

$$\begin{aligned}
 \log_{9^k} 25^k &= \log_{3^{2k}} 5^{2k} \\
 &= \log_3 5.
 \end{aligned}$$

Using these results, we evaluate the original expression:

$$\begin{aligned}
 \left(\sum_{k=1}^{20} \log_{5^k} 3^{k^2} \right) \cdot \left(\sum_{k=1}^{100} \log_{9^k} 25^k \right) &= \left(\sum_{k=1}^{20} k \log_5 3 \right) \cdot \left(\sum_{k=1}^{100} \log_3 5 \right) \\
 &= \left(\log_5 3 \cdot \sum_{k=1}^{20} k \right) \cdot \left(\log_3 5 \cdot \sum_{k=1}^{100} 1 \right) \\
 &= \left(\sum_{k=1}^{20} k \right) \cdot \left(\sum_{k=1}^{100} 1 \right) \\
 &= \frac{21 \cdot 20}{2} \cdot 100 \\
 &= \boxed{\text{(E) } 21,000}.
 \end{aligned}$$

MRENTHUSIASM (Solution)

Solution 2 (Properties of Logarithms)

First, we can get rid of the k exponents using properties of logarithms:

$$\log_{5^k} 3^{k^2} = k^2 \cdot \frac{1}{k} \cdot \log_5 3 = k \log_5 3 = \log_5 3^k.$$

(Leaving the single k in the exponent will come in handy later). Similarly,

$$\log_{9^k} 25^k = k \cdot \frac{1}{k} \cdot \log_9 25 = \log_9 5^2.$$

Then, evaluating the first few terms in each parentheses, we can find the simplified expanded forms of each sum using the additive property of logarithms:

$$\begin{aligned}
 \sum_{k=1}^{20} \log_5 3^k &= \log_5 3^1 + \log_5 3^2 + \cdots + \log_5 3^{20} \\
 &= \log_5 3^{(1+2+\cdots+20)} \\
 &= \log_5 3^{\frac{20(20+1)}{2}} \\
 &= \log_5 3^{210}, \\
 \sum_{k=1}^{100} \log_9 5^2 &= \log_9 5^2 + \log_9 5^2 + \cdots + \log_9 5^2 \\
 &= \log_9 5^{2(100)} \\
 &= \log_9 5^{200}.
 \end{aligned}
 \tag{*}$$

In (*), we use the triangular numbers equation:

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2} = \frac{20(20+1)}{2} = 210.$$

Finally, multiplying the two logarithms together, we can use the chain rule property of logarithms to simplify:

$$\log_a b \log_x y = \log_a y \log_x b.$$

Thus,

$$\begin{aligned} (\log_5 3^{210}) (\log_{3^2} 5^{200}) &= (\log_5 5^{200}) (\log_{3^2} 3^{210}) \\ &= (\log_5 5^{200}) (\log_3 3^{105}) \\ &= (200)(105) \\ &= \boxed{\text{(E)} 21,000}. \end{aligned}$$

Joeya (Solution)

Solution 3

Focus the inner term $\log_{5^k}(3^{k^2})$. That is,

$$\begin{aligned} \frac{1}{k} \cdot \log_5(3^{k^2}) \\ &= k \log_5(3) \end{aligned}$$

The sum becomes

$$\log_5(3) \cdot \sum_{k=1}^{20} k = 210 \log_5(3)$$

Focus the next inner term, particularly

$$\begin{aligned} \log_{9^k}(25^k) \\ &= \frac{1}{k} \cdot \log_9(25^k) \\ &= \log_9(25) \end{aligned}$$

So the sum becomes

$$\sum_{k=1}^{100} \log_9(25) = 100 \log_9(25) = 200 \log_9(5) = 100 \log_3(5)$$

.

So we desire

$$\begin{aligned} 210 \log_5(3) \cdot 100 \log_3(5) \\ &= \frac{210}{\log_3(5)} \cdot 100 \log_3(5) \\ &= 210 \cdot 100 \\ &= \boxed{21000} \end{aligned}$$

.

Solution 4 (Estimations and Answer Choices)

In $\sum_{k=1}^{20} \log_{5^k} 3^{k^2}$, note that the addends are greater than 1 for all $k \geq 2$.

In $\sum_{k=1}^{100} \log_{9^k} 25^k$, note that the addends are greater than 1 for all $k \geq 1$.

We have the inequality

$$\left(\sum_{k=1}^{20} \log_{5^k} 3^{k^2} \right) \cdot \left(\sum_{k=1}^{100} \log_{9^k} 25^k \right) > \left(\sum_{k=2}^{20} 1 \right) \cdot \left(\sum_{k=1}^{100} 1 \right) = 19 \cdot 100 = 1,900,$$

which eliminates choices **(A)**, **(B)**, and **(C)**. We get the answer **(E)** 21,000 by either an educated guess or a continued approximation:

Observe that $\sum_{k=1}^{20} \log_{5^k} 3^{k^2} >> \sum_{k=2}^{20} 1 = 19$ and $\sum_{k=1}^{100} \log_{9^k} 25^k \approx \sum_{k=1}^{100} \log_{9^k} 27^k = \sum_{k=1}^{100} \frac{3}{2} = 150$. Therefore, we obtain the following rough underestimation:

$$\left(\sum_{k=1}^{20} \log_{5^k} 3^{k^2} \right) \cdot \left(\sum_{k=1}^{100} \log_{9^k} 25^k \right) > \left(\sum_{k=2}^{20} 1 \right) \cdot \left(\sum_{k=1}^{100} \frac{3}{2} \right) = 19 \cdot 150 = 2,850.$$

From here, it should be safe to guess that the answer is **(E)**.

Solution 5 (Properties of Logarithms)

Using the identity

$$\log_{p^n} q^n = \log_p q,$$

simplify

$$\begin{aligned} \log_{5^k} 3^{k^2} &= \log_{5^k} (3^k)^k \\ &= \log_5 3^k \end{aligned}$$

and

$$\begin{aligned} \log_{9^k} 25^k &= \log_{3^{2k}} 5^{2k} \\ &= \log_3 5. \end{aligned}$$

. Now we have the product:

$$\left(\sum_{k=1}^{20} \log_5 3^k \right) \cdot \left(\sum_{k=1}^{100} \log_3 5 \right)$$

$$\begin{aligned} \sum_{k=1}^{20} \log_5 3^k &= \log_5 3^1 + \log_5 3^2 + \cdots + \log_5 3^{20} \\ &= \log_5 3^{(1+2+\cdots+20)} \\ &= \log_5 3^{\frac{20(20+1)}{2}} \\ &= \log_5 3^{210} \\ &= 210 \cdot \log_5 3, \end{aligned}$$

$$\sum_{k=1}^{100} \log_3 5 = 100 \cdot \log_3 5.$$

With the reciprocal rule the logarithms cancel out leaving: **(E)** 21,000.

15. (AMC 12 12A 2021) A choir director must select a group of singers from among his 6 tenors and 8 basses. The only requirements are that the difference between the numbers of tenors and basses must be a multiple of 4, and the group must have at least one singer. Let N be the number of different groups that could be selected. What is the remainder when N is divided by 100?

(A) 47 (B) 48 (C) 83 (D) 95 (E) 96

Answer: D

Solution 1 (Bijection)

Suppose that t tenors and b basses are selected. The requirements are $t \equiv b \pmod{4}$ and $(t, b) \neq (0, 0)$.

It follows that $b' = 8 - b$ basses are not selected. **Replacing the set of selected basses by its complement gives a bijection between the original groups and choices consisting of the selected tenors together with the unselected basses.** The requirements become $t \equiv 8 - b' \pmod{4}$ and $(t, 8 - b') \neq (0, 0)$, which simplify to $t + b' \equiv 0 \pmod{4}$ and $(t, b') \neq (0, 8)$, respectively.

As $t + b' \in \{0, 4, 8, 12\}$, the total number of such groups is

$$\begin{aligned} N &= \binom{14}{0} + \binom{14}{4} + \left[\binom{14}{8} - 1 \right] + \binom{14}{12} \\ &= \binom{14}{0} + \binom{14}{4} + \left[\binom{14}{6} - 1 \right] + \binom{14}{2} \\ &= 1 + 1001 + [3003 - 1] + 91 \\ &= 4095, \end{aligned}$$

from which $N \equiv \boxed{\text{(D) } 95} \pmod{100}$.

Solution 2 (Vandermonde's Identity)

Suppose that t tenors and b basses are selected. The requirements are $t \equiv b \pmod{4}$ and $(t, b) \neq (0, 0)$.

Note that $\binom{6}{t}\binom{8}{b}$ different groups can be formed by selecting t tenors and b basses. Since $t - b \in \{-8, -4, 0, 4\}$, we apply casework:

- If $t - b = -8$, then $\binom{6}{0}\binom{8}{8}$ different group can be formed.
- If $t - b = -4$, then $\sum_{k=0}^4 \binom{6}{k}\binom{8}{k+4}$ different groups can be formed.
- If $t - b = 0$, then $\left[\sum_{k=0}^6 \binom{6}{k}\binom{8}{k} \right] - 1$ different groups can be formed, recalling that $(t, b) \neq (0, 0)$.
- If $t - b = 4$, then $\sum_{k=0}^2 \binom{6}{k+4}\binom{8}{k}$ different groups can be formed.

By the combinatorial identity $\binom{n}{k} = \binom{n}{n-k}$ and Vandermonde's Identity $\sum_{k=0}^r \binom{m}{k}\binom{n}{r-k} = \binom{m+n}{r}$, we find the total number of such groups:

$$\begin{aligned} N &= \binom{6}{0}\binom{8}{8} + \left[\sum_{k=0}^4 \binom{6}{k}\binom{8}{k+4} \right] + \left[\left[\sum_{k=0}^6 \binom{6}{k}\binom{8}{k} \right] - 1 \right] + \left[\sum_{k=0}^2 \binom{6}{k+4}\binom{8}{k} \right] \\ &= \binom{6}{0}\binom{8}{0} + \left[\sum_{k=0}^4 \binom{6}{k}\binom{8}{4-k} \right] + \left[\left[\sum_{k=0}^6 \binom{6}{6-k}\binom{8}{k} \right] - 1 \right] + \left[\sum_{k=0}^2 \binom{6}{2-k}\binom{8}{k} \right] \end{aligned}$$

$$\begin{aligned}
 &= \binom{14}{0} + \binom{14}{4} + \left[\binom{14}{6} - 1 \right] + \binom{14}{2} \\
 &= 1 + 1001 + [3003 - 1] + 91 \\
 &= 4095,
 \end{aligned}$$

from which $N \equiv \boxed{\text{(D) } 95} \pmod{100}$.

Solution 3 (Generating Functions)

The problem can be done using a roots of unity filter. Let $f(x, y) = (1+x)^8(1+y)^6$. By expanding the binomials and distributing, $f(x, y)$ is the generating function for different groups of basses and tenors. That is,

$$f(x, y) = \sum_{m=0}^8 \sum_{n=0}^6 a_{mn} x^m y^n,$$

where a_{mn} is the number of groups of m basses and n tenors. What we want to do is sum up all values of a_{mn} for which $4 \mid m - n$ except for $a_{00} = 1$. To do this, define a new function

$$g(x) = f(x, x^{-1}) = \sum_{m=0}^8 \sum_{n=0}^6 a_{mn} x^{m-n} = (1+x)^8(1+x^{-1})^6.$$

Now we just need to sum all coefficients of $g(x)$ for which $4 \mid m - n$. Consider a monomial $h(x) = x^k$. If $4 \mid k$, then

$$h(i) + h(-1) + h(-i) + h(1) = 1 + 1 + 1 + 1 = 4.$$

Otherwise,

$$h(i) + h(-1) + h(-i) + h(1) = 0.$$

$g(x)$ is a sum of these monomials so this gives us a method to determine the sum we're looking for:

$$\frac{g(i) + g(-1) + g(-i) + g(1)}{4} = 2^{12} = 4096.$$

(since $g(-1) = 0$ and it can be checked that $g(i) = -g(-i)$). Hence, the answer is $4096 - 1 = 4095 \equiv \boxed{\text{(D) } 95} \pmod{100}$.

Solution 4 (Enumeration)

Note that $\binom{6}{t}\binom{8}{b}$ different groups can be formed by selecting t tenors and b basses.

By casework, we construct the following table:

# of Tenors	# of Bases	# of Groups	Evaluate # of Groups
0	4	$\binom{6}{0}\binom{8}{4}$	70
0	8	$\binom{6}{0}\binom{8}{8}$	1
1	1	$\binom{6}{1}\binom{8}{1}$	48
1	5	$\binom{6}{1}\binom{8}{5}$	336
2	2	$\binom{6}{2}\binom{8}{2}$	420
2	6	$\binom{6}{2}\binom{8}{6}$	420
3	3	$\binom{6}{3}\binom{8}{3}$	1120
3	7	$\binom{6}{3}\binom{8}{7}$	160
4	0	$\binom{6}{4}\binom{8}{0}$	15
4	4	$\binom{6}{4}\binom{8}{4}$	1050
4	8	$\binom{6}{4}\binom{8}{8}$	15
5	1	$\binom{6}{5}\binom{8}{1}$	48
5	5	$\binom{6}{5}\binom{8}{5}$	336
6	2	$\binom{6}{6}\binom{8}{2}$	28
6	6	$\binom{6}{6}\binom{8}{6}$	28

We find the total number of such groups:

$$N = 70 + 1 + 48 + 336 + 420 + 420 + 1120 + 160 + 15 + 1050 + 15 + 48 + 336 + 28 + 28 = 4095,$$

from which $N \equiv \boxed{\text{(D)} 95} \pmod{100}$.

Alternatively, since the answer choices have different units digits, it suffices to find the units digit of N only.

Solution 5 (Symmetry Applied Twice)

Consider the set of all $2^{8+6} = 2^{14}$ possible choirs that can be formed. For a given choir let D be the difference in the number of tenors and bases modulo 4, so $D = T - B \pmod{4}$. Exactly half of all choirs have either $D = 0$ or $D = 2$. To see this, pick one of the tenors and note that including or removing him from a choir changes D by ± 1 . Of those 2^{13} choirs with $D = 0$ or $D = 2$, we claim exactly half have $D = 0$. To see this, for any choir having $D = 0$ or $D = 2$, we can replace the T tenors with the $6 - T$ tenors who were not in the choir, thereby sending $D \mapsto D + 2 \pmod{4}$. Excluding the empty choir, there are $2^{12} - 1 = 4095$ choirs that meet the conditions of the problem, and the answer is $\boxed{\text{(D)} 95}$.

16. (AMC 12 12A 2021) In the following list of numbers, the integer n appears n times in the list for $1 \leq n \leq 200$.

$$1, 2, 2, 3, 3, 3, 4, 4, 4, \dots, 200, 200, \dots, 200$$

What is the median of the numbers in this list?

(A) 100.5 (B) 134 (C) 142 (D) 150.5 (E) 167

Answer: C

Solution 1

There are $1 + 2 + \dots + 199 + 200 = \frac{(200)(201)}{2} = 20100$ numbers in total. Let the median be k . We want to find the median k such that

$$\frac{k(k+1)}{2} = 20100/2,$$

or

$$k(k+1) = 20100.$$

Note that $\sqrt{20100} \approx 142$. Plugging this value in as k gives

$$\frac{1}{2}(142)(143) = 10153.$$

$10153 - 142 < 10050$, so 142 is the 152nd and 153rd numbers, and hence, our desired answer. (C) 142.

Note that we can derive $\sqrt{20100} \approx 142$ through the formula

$$\sqrt{n} = \sqrt{a+b} \approx \sqrt{a} + \frac{b}{2\sqrt{a}+1},$$

where a is a perfect square less than or equal to n . We set a to 19600, so $\sqrt{a} = 140$, and $b = 500$. We then have $n \approx 140 + \frac{500}{2(140)+1} \approx 142$. approximation by ciceronii

Note by Fasolinka (use answer choices): Once you know that the answer is in the 140s range (200,000 is around 14^2 times 10^2) by the approximation, it is highly improbable for the answer to be anything but C.

Solution 2

The x th number of this sequence is $\left\lceil \frac{-1 \pm \sqrt{1+8x}}{2} \right\rceil$ via the quadratic formula. We can see that if we halve x we end up getting $\left\lceil \frac{-1 \pm \sqrt{1+4x}}{2} \right\rceil$. This is approximately the number divided by $\sqrt{2}$. $\frac{200}{\sqrt{2}} = 141.4$ and since 142 looks like the only number close to it, it is answer (C)142 Lopkiloim

Solution 3 (Answer Choices)

We can look at answer choice (C), which is 142 first. That means that the number of numbers from 1 to 142 is roughly the number of numbers from 143 to 200.

The number of numbers from 1 to 142 is $\frac{142(142+1)}{2}$ which is approximately 10000. The number of numbers from 143 to 200 is $\frac{200(200+1)}{2} - \frac{142(142+1)}{2}$ which is approximately 10000 as well. Therefore, we can be relatively sure the answer choice is (C) 142.

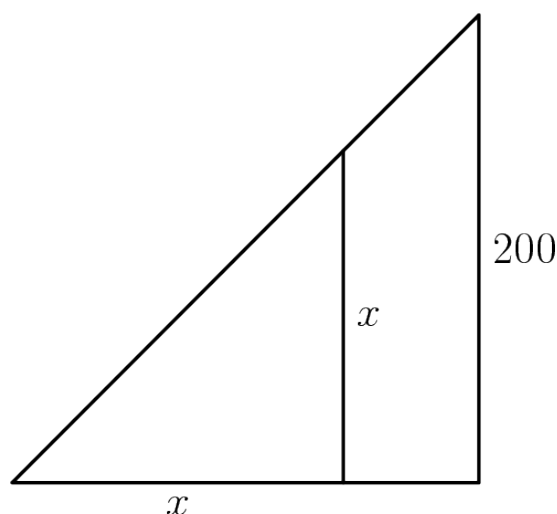
-PureSwag

Solution 4 (Geometry)

We can arrange the numbers in the following pattern:

$$\begin{array}{ccccccc} & & & & & & 200 \\ & & & & & 199 & 200 \\ & & & & \vdots & & \vdots \\ & & \ddots & & \vdots & & \vdots \\ & 2 & \cdots & 199 & 200 & & \\ 1 & 2 & \cdots & 199 & 200 & & \end{array}$$

We can see this as a isosceles right triangle, with legs of length 200.



Let x be the side length such that both sides of the triangle have the same area. The desired answer is then around x because about half of the numbers in the list fall on each side.

Solving for x yields:

$$\begin{aligned}\frac{x^2}{2} &= \frac{1}{2} \cdot \frac{200^2}{2} \\ x^2 &= \frac{1}{2} \cdot 200^2 \\ x &= \frac{200}{\sqrt{2}} = 100\sqrt{2} \approx 141.\end{aligned}$$

We see that $\boxed{(C) 142}$ is the closest to x by far, and thus, can be relatively certain this is the answer.

17. (AMC 12 12A 2021) Trapezoid $ABCD$ has $\overline{AB} \parallel \overline{CD}$, $BC = CD = 43$, and $\overline{AD} \perp \overline{BD}$. Let O be the intersection of the diagonals $\overline{AC}$ and $\overline{BD}$, and let P be the midpoint of $\overline{BD}$. Given that $OP = 11$, the length of AD can be written in the form $m\sqrt{n}$, where m and n are positive integers and n is not divisible by the square of any prime. What is $m + n$?

(A) 65 (B) 132 (C) 157 (D) 194 (E) 215

Answer: D

Solution 1 (Similar Triangles and Pythagorean Theorem)

Angle chasing* reveals that $\triangle BPC \sim \triangle BDA$, therefore

$$2 = \frac{BD}{BP} = \frac{AB}{BC} = \frac{AB}{43},$$

or $AB = 86$.

Additional angle chasing shows that $\triangle ABO \sim \triangle CDO$, therefore

$$2 = \frac{AB}{CD} = \frac{BO}{OD} = \frac{BP + 11}{BP - 11},$$

or $BP = 33$ and $BD = 66$.

Since $\triangle ADB$ is right, the Pythagorean theorem implies that

$$AD = \sqrt{86^2 - 66^2} = 4\sqrt{190}.$$

The answer is $4 + 190 = \boxed{\text{(D)} 194}$.

- Angle Chasing: If we set $\angle DBC = \alpha$, then we know that $\angle DCB = 180^\circ - 2\alpha$ because $\triangle DBC$ is isosceles. Then, $\angle BCP = 90^\circ - \alpha$ (since P is the midpoint of $\overline{BD}$, and therefore line $\overline{CP}$ is a median, which then means $\angle BCP$ bisects $\angle DBC$), so $\angle BPC$ is a right angle. Because $\angle BDC = \alpha$ and $\overline{AB} \parallel \overline{DC}$, we conclude that $\angle ABD = \alpha$ too. Lastly, because $\triangle BPC$ and $\triangle BDA$ are both right triangles, they are similar by AA.

=Another way to angle chase It is easy to show that triangles CDO and ODA are similar through AA. From there, we already know that through isosceles triangles CPD and CPB are congruent. Hence we can simply see that angle PCO and angle OAD are congruent through similar triangles. Also, angle OAB and angle OCD are congruent through parallelism. Thus, angle BAD and angle DCP are congruent. Then, through AA, $\triangle BCP \cong \triangle DCP \sim \triangle BDA$.

This skips the unnecessary step of angle chasing via similar angles of an isosceles triangle and skips the step of manually counting each angle's measure in terms of α . However, no discredit to the original angle chasing comment, it has its merits as well (using the angle measures in the above angle chasing technique proves useful for showing other statements [and cooler facts] about the problem).

Solution 2 (Similar Triangles, Areas, Pythagorean Theorem)

Since $\triangle BCD$ is isosceles with base $\overline{BD}$, it follows that median $\overline{CP}$ is also an altitude. Let $OD = x$ and $CP = h$, so $PB = x + 11$.

Since $\angle AOD = \angle COP$ by vertical angles, we conclude that $\triangle AOD \sim \triangle COP$ by AA, from which $\frac{AD}{CP} = \frac{OD}{OP}$, or

$$AD = CP \cdot \frac{OD}{OP} = h \cdot \frac{x}{11}.$$

Let the brackets denote areas. Notice that $[AOD] = [BOC]$ (By the same base and height, we deduce that $[ACD] = [BDC]$. Subtracting $[OCD]$ from both sides gives $[AOD] = [BOC]$). Doubling both sides produces

$$\begin{aligned} 2[AOD] &= 2[BOC] \\ OD \cdot AD &= OB \cdot CP \\ x \left(\frac{hx}{11} \right) &= (x + 22)h \\ x^2 &= 11(x + 22). \end{aligned}$$

Rearranging and factoring result in $(x - 22)(x + 11) = 0$, from which $x = 22$.

Applying the Pythagorean Theorem to right $\triangle CPB$, we have

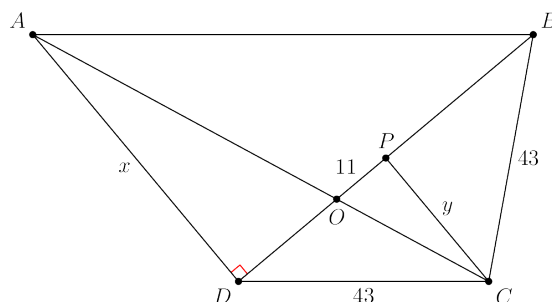
$$h = \sqrt{43^2 - 33^2} = \sqrt{(43 + 33)(43 - 33)} = \sqrt{760} = 2\sqrt{190}.$$

Finally, we get

$$AD = h \cdot \frac{x}{11} = 4\sqrt{190},$$

so the answer is $4 + 190 = \boxed{\text{(D)} 194}$.

Solution 3 (Short)



Let $CP = y$. CP is a perpendicular bisector of DB . Then, let $DO = x$, thus $DP = PB = 11 + x$.

(1) $\triangle CPO \sim \triangle ADO$, so we get $\frac{AD}{x} = \frac{y}{11}$, or $AD = \frac{xy}{11}$.

(2) Applying Pythagorean Theorem on $\triangle CDP$ gives $(11 + x)^2 + y^2 = 43^2$.

(3) $\triangle BPC \sim \triangle BDA$ with ratio $1 : 2$, so $AD = 2y$ using the fact that P is the midpoint of BD .

Thus, $\frac{xy}{11} = 2y$, or $x = 22$. And $y = \sqrt{43^2 - 33^2} = 2\sqrt{190}$, so $AD = 4\sqrt{190}$ and the answer is $4 + 190 = \boxed{\text{(D)} 194}$.

Solution 4 (Law of Sines and Similar Triangles)

This solution will refer to the diagram section at the top of this page.

We know $\angle ABD = \angle BDC = \angle DBC$. Let $PB = 11 + x$. Notice that $\triangle ABD$ is similar to $\triangle DOC$, so $\frac{AB}{43} = \frac{22+x}{x} \implies AB = \frac{43(22+x)}{x}$. Now we apply the Law of Sines to isosceles triangle $\triangle DCB$:

$$\frac{\sin(\angle ABD)}{43} = \frac{\sin(180^\circ - 2\angle ABD)}{22 + 2x} = \frac{2\sin(\angle ABD)\cos(\angle ABD)}{22 + 2x}$$

$$\frac{1}{43} = \frac{\cos(\angle ABD)}{11 + x} \implies \cos(\angle ABD) = \frac{11 + x}{43}$$

From right triangle $\triangle BAD$ we already know that $\cos(\angle ABD) = \frac{BD}{AB} = \frac{(22+2x)x}{43(22+x)}$.

Setting these equations equal to each other gives us:

$$\frac{(22 + 2x)x}{43(22 + x)} = \frac{11 + x}{43}$$

$$x = 22 \implies BD = 2(11 + x) = 66, AB = \frac{43 \cdot 44}{22} = 86.$$

So $AD = \sqrt{86^2 - 66^2} = 4\sqrt{190}$. Ans = $\boxed{194}$.

Solution 5 (Extending the Line)

Observe that $\triangle BPC$ is congruent to $\triangle DPC$; both are similar to $\triangle BDA$. Let's extend $\overline{AD}$ and $\overline{BC}$ past points D and C respectively, such that they intersect at a point E . Observe that $\angle BDE$ is 90 degrees, and that $\angle DBE \cong \angle PBC \cong$

$\angle DBA \implies \angle DBE \cong \angle DBA$. Thus, by ASA, we know that $\triangle ABD \cong \triangle EBD$, thus, $AD = ED$, meaning D is the midpoint of AE . Let M be the midpoint of $\overline{DE}$. Note that $\triangle CME$ is congruent to $\triangle BPC$, thus $BC = CE$, meaning C is the midpoint of $\overline{BE}$.

Therefore, $\overline{AC}$ and $\overline{BD}$ are both medians of $\triangle ABE$. This means that O is the centroid of $\triangle ABE$; therefore, because the centroid divides the median in a 2:1 ratio, $\frac{BO}{OD} = \frac{AO}{OE} = \frac{BD}{DO}$. Recall that P is the midpoint of BD ; $DP = \frac{BD}{2}$. The question tells us that $OP = 11$; $DP - DO = 11$; we can write this in terms of DB ; $\frac{DB}{2} - \frac{DB}{3} = \frac{DB}{6} = 11 \implies DB = 66$.

We are almost finished. Each side length of $\triangle ABD$ is twice as long as the corresponding side length $\triangle CBP$ or $\triangle CPD$, since those triangles are similar; this means that $AB = 2 \cdot 43 = 86$. Now, by Pythagorean theorem on $\triangle ABD$, $AB^2 - BD^2 = AD^2 \implies 86^2 - 66^2 = AD^2 \implies AD = \sqrt{3040} \implies AD = 4\sqrt{190}$.

The answer is $4 + 190 = \boxed{\text{(D)} 194}$.

Solution 6

Since P is the midpoint of isosceles triangle BCD , it would be pretty easy to see that $CP \perp BD$. Since $AD \perp BD$ as well, $AD \parallel CP$. Connecting AP , it's obvious that $[ADC] = [ADP]$. Since $DP = BP$, $[APB] = [ADC]$.

Since P is the midpoint of BD , the height of $\triangle APB$ on side AB is half that of $\triangle ADC$ on CD . Since $[APB] = [ADC]$, $AB = 2CD$.

As a basic property of a trapezoid, $\triangle AOB \sim \triangle COD$, so $\frac{OB}{OD} = \frac{AB}{CD} = 2$, or $OB = 2OD$. Letting $OD = x$, then $PB = DP = 11 + x$, and $OB = 22 + x$. Hence $22 + x = 2x$ and $x = 22$.

Since $\triangle AOD \sim \triangle COP$, $\frac{AD}{PC} = \frac{OD}{OP} = 2$. Since $PD = 11 + 22 = 33$, $PC = \sqrt{43^2 - 33^2} = \sqrt{760}$.

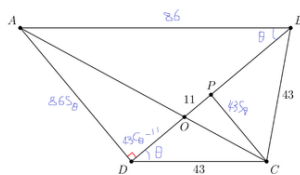
So, $AD = 2\sqrt{760} = 4\sqrt{190}$. The correct answer is $\boxed{\text{(D)} 194}$.

Solution 7 (Coordinate Geometry)

Let D be the origin of the cartesian coordinate plane, B lie on the positive x -axis, and A lie on the negative y -axis. Then let the coordinates of $B = (2a, 0)$, $A = (0, -2b)$. Then the slope of AB is $\frac{b}{a}$. Since $AB \parallel CD$ the slope of CD is the same. Note that as $\triangle DCB$ is isosceles C lies on $x = a$. Thus since CD has equation $y = \frac{b}{a}x$ (D is the origin), $C = (a, b)$. Therefore AC has equation $y = \frac{3b}{a}x - 2b$ and intersects BD (x -axis) at $O = (\frac{2}{3}a, 0)$. The midpoint of BD is $P = (a, 0)$, so $OP = \frac{a}{3} = 11$, from which $a = 33$. Then by Pythagorean theorem on $\triangle DPC$ ($\triangle DBC$ is isosceles), we have $b = \sqrt{43^2 - 33^2} = 2\sqrt{190}$, so $2b = 4\sqrt{190}$.

Finally, the answer is $4 + 190 = \boxed{\text{(D)} 194}$.

Solution 8 (Trigonometry)



set $\angle BDC = \theta$ $BD = 2 \cdot DP = 2 \cdot 43 \cdot \cos \theta$ $AB = \frac{BD}{\cos \angle DBA} = \frac{BD}{\cos \theta} = 2 \cdot 43 \cdot \frac{\cos \theta}{\cos \theta} = 86$
 $\frac{OP}{DO} = \frac{CP}{AD} \cdot \frac{11}{43 \cdot \cos \theta - 11} = \frac{43 \cdot \sin \theta}{86 \cdot \sin \theta} \cos \theta = \frac{33}{43}$ $AD = 86 \cdot \sin \theta = 2\sqrt{760} = 4\sqrt{190} =$
(D) 194.

18. (AMC 12 12A 2021) Let f be a function defined on the set of positive rational numbers with the property that $f(a \cdot b) = f(a) + f(b)$ for all positive rational numbers a and b . Suppose that f also has the property that $f(p) = p$ for every prime number p . For which of the following numbers x is $f(x) < 0$?

(A) $\frac{17}{32}$ (B) $\frac{11}{16}$ (C) $\frac{7}{9}$ (D) $\frac{7}{6}$ (E) $\frac{25}{11}$

Answer: E

Solution 1 (Intuitive)

From the answer choices, note that

$$\begin{aligned} f(25) &= f\left(\frac{25}{11} \cdot 11\right) \\ &= f\left(\frac{25}{11}\right) + f(11) \\ &= f\left(\frac{25}{11}\right) + 11. \end{aligned}$$

On the other hand, we have

$$\begin{aligned} f(25) &= f(5 \cdot 5) \\ &= f(5) + f(5) \\ &= 5 + 5 \\ &= 10. \end{aligned}$$

Equating the expressions for $f(25)$ produces

$$f\left(\frac{25}{11}\right) + 11 = 10,$$

from which $f\left(\frac{25}{11}\right) = -1$. Therefore, the answer is **(E)** $\frac{25}{11}$. **Remark**

Similarly, we can find the outputs of f at the inputs of the other answer choices:

$$\begin{aligned} \text{(A)} \quad f\left(\frac{17}{32}\right) &= 7 \\ \text{(B)} \quad f\left(\frac{11}{16}\right) &= 3 \\ \text{(C)} \quad f\left(\frac{7}{9}\right) &= 1 \\ \text{(D)} \quad f\left(\frac{7}{6}\right) &= 2 \end{aligned}$$

Alternatively, refer to Solutions 2 and 4 for the full processes.

Solution 2 (Specific)

We know that $f(p) = f(p \cdot 1) = f(p) + f(1)$. By transitive, we have

$$f(p) = f(p) + f(1).$$

Subtracting $f(p)$ from both sides gives $0 = f(1)$. Also

$$f(2) + f\left(\frac{1}{2}\right) = f(1) = 0 \implies 2 + f\left(\frac{1}{2}\right) = 0 \implies f\left(\frac{1}{2}\right) = -2$$

$$f(3) + f\left(\frac{1}{3}\right) = f(1) = 0 \implies 3 + f\left(\frac{1}{3}\right) = 0 \implies f\left(\frac{1}{3}\right) = -3$$

$$f(11) + f\left(\frac{1}{11}\right) = f(1) = 0 \implies 11 + f\left(\frac{1}{11}\right) = 0 \implies f\left(\frac{1}{11}\right) = -11$$

In **(A)** we have $f\left(\frac{17}{32}\right) = 17 + 5f\left(\frac{1}{2}\right) = 17 - 5(2) = 7$.

In **(B)** we have $f\left(\frac{11}{16}\right) = 11 + 4f\left(\frac{1}{2}\right) = 11 - 4(2) = 3$.

In **(C)** we have $f\left(\frac{7}{9}\right) = 7 + 2f\left(\frac{1}{3}\right) = 7 - 2(3) = 1$.

In **(D)** we have $f\left(\frac{7}{6}\right) = 7 + f\left(\frac{1}{2}\right) + f\left(\frac{1}{3}\right) = 7 - 2 - 3 = 2$.

In **(E)** we have $f\left(\frac{25}{11}\right) = 10 + f\left(\frac{1}{11}\right) = 10 - 11 = -1$.

Thus, our answer is **(E)** $\frac{25}{11}$.

Solution 3 (Generalized)

Consider the rational $\frac{a}{b}$, for a, b integers. We have $f(a) = f\left(\frac{a}{b} \cdot b\right) = f\left(\frac{a}{b}\right) + f(b)$. So $f\left(\frac{a}{b}\right) = f(a) - f(b)$. Let p be a prime. Notice that $f(p^k) = kf(p)$. And $f(p) = p$. So if $a = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$, $f(a) = a_1 p_1 + a_2 p_2 + \cdots + a_k p_k$. We simply need this to be greater than what we have for $f(b)$. Notice that for answer choices **(A)**, **(B)**, **(C)**, and **(D)**, the numerator has fewer prime factors than the denominator, and so they are less likely to work. We check **(E)** first, and it works, therefore the answer is

(E) $\frac{25}{11}$.

Solution 4 (Generalized)

We derive the following properties of f :

a) By induction, we have

$$f\left(\prod_{k=1}^n a_k\right) = \sum_{k=1}^n f(a_k)$$

for all positive rational numbers a_k and positive integers n . Since positive powers are just repeated multiplication of the base, it follows that

$$f(a^n) = f\left(\prod_{k=1}^n a\right) = \sum_{k=1}^n f(a) = nf(a)$$

for all positive rational numbers a and positive integers n .

b) For all positive rational numbers a , we have

$$f(a) = f(a \cdot 1) = f(a) + f(1),$$

from which $f(1) = 0$.

c) For all positive rational numbers a , we have

$$f(a) + f\left(\frac{1}{a}\right) = f\left(a \cdot \frac{1}{a}\right) = f(1) = 0,$$

from which $f\left(\frac{1}{a}\right) = -f(a)$.

For all positive integers x and y , suppose $\prod_{k=1}^m p_k^{d_k}$ and $\prod_{k=1}^n q_k^{e_k}$ are their respective prime factorizations. We get

$$\begin{aligned} f\left(\frac{x}{y}\right) &= f(x) + f\left(\frac{1}{y}\right) \\ &= f(x) - f(y) && \text{by Property 3} \\ &= f\left(\prod_{k=1}^m p_k^{d_k}\right) - f\left(\prod_{k=1}^n q_k^{e_k}\right) \\ &= \left[\sum_{k=1}^m f(p_k^{d_k})\right] - \left[\sum_{k=1}^n f(q_k^{e_k})\right] && \text{by Property 1} \\ &= \left[\sum_{k=1}^m d_k f(p_k)\right] - \left[\sum_{k=1}^n e_k f(q_k)\right] && \text{by Property 1} \\ &= \left[\sum_{k=1}^m d_k p_k\right] - \left[\sum_{k=1}^n e_k q_k\right]. \end{aligned}$$

We apply f to each fraction in the answer choices:

$$\begin{aligned} \text{(A)} \quad f\left(\frac{17}{32}\right) &= f\left(\frac{17^1}{2^5}\right) &= [1(17)] - [5(2)] &= 7 \\ \text{(B)} \quad f\left(\frac{11}{16}\right) &= f\left(\frac{11^1}{2^4}\right) &= [1(11)] - [4(2)] &= 3 \\ \text{(C)} \quad f\left(\frac{7}{9}\right) &= f\left(\frac{7^1}{3^2}\right) &= [1(7)] - [2(3)] &= 1 \\ \text{(D)} \quad f\left(\frac{7}{6}\right) &= f\left(\frac{7^1}{2^1 \cdot 3^1}\right) &= [1(7)] - [1(2) + 1(3)] &= 2 \\ \text{(E)} \quad f\left(\frac{25}{11}\right) &= f\left(\frac{5^2}{11^1}\right) &= [2(5)] - [1(11)] &= -1 \end{aligned}$$

Therefore, the answer is **(E)** $\frac{25}{11}$.

Solution 5 (Quick, Dirty, and Frantic Last Hope)

Note that answer choices **(A)** through **(D)** are $\frac{\text{prime}}{\text{composite}}$, whereas **(E)** is $\frac{\text{composite}}{\text{prime}}$. Because the functional equation is related to primes, we hope that the uniqueness of answer choice **(E)** $\frac{25}{11}$ is enough.

Solution 6 (Rushed Generalization)

If $f(a \cdot b) = f(a) + f(b)$, and if $f(p) = p$, then $f(p \cdot p) = 2p$. You can do this multiple times (Ex: $f(p^3) = 3p$). You can quickly assume then, that $f(p^n) = np$. Thus the answer choices can then be rewritten as the product of a prime and another prime to the negative power. Answer choices A-C are straightforward. For

D, you can rewrite $\frac{1}{6}$ as $\frac{1}{2} \cdot \frac{1}{3}$. When you get to E, you get $f(25) + f(\frac{1}{11})$, which is $10 - 11$, which is -1 . So the answer is **(E)** $\frac{25}{11}$

Solution 7 (Generalized)

Note that for each of the answer choices we can multiply the fractions by their denominators to be left with only the numerator and also have the prime factorization of the denominators, then set the function equal to the numerator. This will be shown throughout the solution.

For answer choice (A) $\frac{17}{32}$, we have $f\left(\frac{17}{32}\right)$. Now, we can set up an equation by multiplying $\frac{17}{32}$ by 32 and setting it equal to 17. This will give $f\left(32 \cdot \frac{17}{32}\right) = f(17) \rightarrow f(32) + f\left(\frac{17}{32}\right) = f(17)$. Our goal is to find $f\left(\frac{17}{32}\right)$, therefore we must find $f(32)$ and $f(17)$. Since $f(p) = p$ for any prime p , $f(17) = 17$. Taking the prime factorization of 32 gives 2^5 , so $f(32) = f(2 \cdot 2 \cdot 2 \cdot 2 \cdot 2) = f(2) + f(2) + f(2) + f(2) + f(2) = 5f(2) = 5 \cdot 2 = 10$. Therefore, $10 + f\left(\frac{17}{32}\right) = 17$ and $f\left(\frac{17}{32}\right) = 7$ (B) $\frac{11}{16}$: $f\left(\frac{11}{16}\right) \rightarrow f\left(16 \cdot \frac{11}{16}\right) = f(11) \rightarrow f(16) + f\left(\frac{11}{16}\right) = 11 \rightarrow 4f(2) + f\left(\frac{11}{16}\right) = 11 \rightarrow f\left(\frac{11}{16}\right) = 3$ (C) $\frac{7}{9}$: $f\left(\frac{7}{9}\right) \rightarrow f\left(9 \cdot \frac{7}{9}\right) = f(7) \rightarrow f(9) + f\left(\frac{7}{9}\right) = 7 \rightarrow 2f(3) + f\left(\frac{7}{9}\right) = 7 \rightarrow f\left(\frac{7}{9}\right) = 1$ (D) $\frac{7}{6}$: $f\left(\frac{7}{6}\right) \rightarrow f\left(6 \cdot \frac{7}{6}\right) = f(7) \rightarrow f(6) + f\left(\frac{7}{6}\right) = 7 \rightarrow f(2) + f(3) + f\left(\frac{7}{6}\right) = 7 \rightarrow f\left(\frac{7}{6}\right) = 2$ (E) $\frac{25}{11}$: $f\left(\frac{25}{11}\right) \rightarrow f\left(11 \cdot \frac{25}{11}\right) = f(25) \rightarrow f(11) + f\left(\frac{25}{11}\right) = 2f(5) \rightarrow 11 + f\left(\frac{25}{11}\right) = 10 \rightarrow f\left(\frac{25}{11}\right) = -1$

Therefore, the answer is **(E)** $\frac{25}{11}$

Note: The general strategy here was the setting up of equations to find $f(x)$. By setting it equal to $f(a)$ where a was an integer, we could take the prime factorization to find the value of $f(a)$ and also set up an equation involving $f(\text{denominator})$ because the denominator was also an integer, therefore we could take the prime factorization and find its value

Solution 8 (Generalized But Fast)

Let p be a prime number. We have that $f(p) = p$. Now, consider $f\left(\frac{1}{p^a}\right)$ where a is a positive integer. Notice that

$$f\left(p^{a+1} \cdot \frac{1}{p^a}\right) = f(p) = p \quad \text{and} \quad f\left(p^{a+1} \cdot \frac{1}{p^a}\right) = f(p^{a+1}) + f\left(\frac{1}{p^a}\right) = (a+1)p + f\left(\frac{1}{p^a}\right)$$

and hence

$$(a+1)p + f\left(\frac{1}{p^a}\right) = p \implies f\left(\frac{1}{p^a}\right) = -ap.$$

Thus, we see that

$$f\left(\frac{25}{11}\right) = 2f(5) + f\left(\frac{1}{11}\right) = 10 - 11 = -1 < 0 \implies \boxed{\textbf{(E)} \frac{25}{11}}$$

shockfront99

19. (AMC 12 12A 2021) How many solutions does the equation $\sin\left(\frac{\pi}{2}\cos x\right) = \cos\left(\frac{\pi}{2}\sin x\right)$ have in the closed interval $[0, \pi]$?

(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Answer: C

Solution 1 (Inverse Trigonometric Functions)

The ranges of $\frac{\pi}{2}\sin x$ and $\frac{\pi}{2}\cos x$ are both $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, which is included in the range of $\arcsin$, so we can use it with no issues.

$$\begin{aligned}\frac{\pi}{2}\cos x &= \arcsin\left(\cos\left(\frac{\pi}{2}\sin x\right)\right) \\ \frac{\pi}{2}\cos x &= \arcsin\left(\sin\left(\frac{\pi}{2} - \frac{\pi}{2}\sin x\right)\right) \\ \frac{\pi}{2}\cos x &= \frac{\pi}{2} - \frac{\pi}{2}\sin x \\ \cos x &= 1 - \sin x \\ \cos x + \sin x &= 1.\end{aligned}$$

This only happens at $x = 0, \frac{\pi}{2}$ on the interval $[0, \pi]$, because one of $\sin$ and $\cos$ must be 1 and the other 0. Therefore, the answer is (C) 2.

Solution 2 (Equality Conditions)

Hmmm, those values $\frac{\pi}{2}\cos(x)$ and $\frac{\pi}{2}\sin(x)$ look scary. If only there were a way to simplify it...

Well, thanks to trigonometry, we can! Notice the equality condition $\sin\left(\frac{\pi}{2}\cos(x)\right) = \cos\left(\frac{\pi}{2}\sin(x)\right)$. Now, say $\frac{\pi}{2}\cos(x) = \mathcal{A}$ and $\frac{\pi}{2}\sin(x) = \mathcal{B}$. Then, we get $\sin(\mathcal{A}) = \cos(\mathcal{B})$.

Aha! We know that $\sin\left(\frac{\pi}{4}\right) = \cos\left(\frac{\pi}{4}\right)$! So, the solution set to $\mathcal{A}$ or $\mathcal{B}$ is $(\mathcal{A}, \mathcal{B}) = \frac{\pi}{4} + \pi k : k \in \mathbb{Z}$.

Now, we substitute the values of $\mathcal{A}$ and $\mathcal{B}$, primarily, $\frac{\pi}{2}\cos(x)$ and $\frac{\pi}{2}\sin(x)$ respectively. We then have the two equalities:

$$\frac{\pi}{2}\cos(x) = \frac{\pi}{4} + \pi k : k \in \mathbb{Z}$$

$$\frac{\pi}{2}\sin(x) = \frac{\pi}{4} + \pi k : k \in \mathbb{Z}$$

In which multiplying by $\frac{2}{\pi}$ on both sides rewards us with

$$\cos(x) = \frac{1}{2} + 2k : k \in \mathbb{Z}$$

$$\sin(x) = \frac{1}{2} + 2k : k \in \mathbb{Z}$$

Aha! The values between the interval $[0, \pi]$ lie when $k = 0$ (because we originally had πk , so setting $k = 1$ results in π , in which because we have the value $\frac{1}{2}$ at the right, we obtain $\frac{1}{2} + \pi$, which is beyond the interval end bound π , so the only values that lie within that interval are when $k = 0$). Simplification results in

$$\cos(x) = \frac{1}{2}$$

$$\sin(x) = \frac{1}{2}$$

The solutions to $\sin(x) = \frac{1}{2}$ is $x = \frac{\pi}{6} + \pi k$ and $x = \frac{5\pi}{6} + \pi k$. We negate $\frac{5\pi}{6} + \pi k$ because that is not in the interval $[0, \pi]$

Looking at $\cos(x) = \frac{1}{2}$, the solutions are $\frac{\pi}{3} + \pi k$ and $\frac{5\pi}{3} + \pi k$. Again, we negate $\frac{5\pi}{3} + \pi k$ as it is not in the interval $[0, \pi]$

Now, we have one more thing to consider. The equality condition. Hmm, the two values of x seem to be different... Does this affect the equation to have no solutions?

No! Notice how we have $\sin(\cos(x)) = \cos(\sin(y))$! Additionally, notice how $\frac{\pi}{2} - \frac{\pi}{6} = \frac{\pi}{3}$! Now, assume $x = \frac{\pi}{3}$ and $y = \frac{\pi}{6}$. We already know $\sin(x) = \cos(90 - x)$! So, we just have $\cos(\cos(x)) = \cos(\cos(\frac{\pi}{2} - y))$, but $\frac{\pi}{2} - y = x$! So, we achieve $\cos(\cos(x)) = \cos(\cos(x))$, which is always true!

Thus, any solution to this equation is in the form $x = \frac{\pi}{6} + \pi k, \frac{\pi}{3} + \pi k$.

To obtain the degree measure over the interval $[0, \pi]$, we must have $k = 0$. So, we achieve the values $x = \frac{\pi}{6}, \frac{\pi}{3}$, or $x = 30^\circ, 60^\circ$

There are (C) 2 values that satisfy the equation.

Solution 3 (Cofunction Identity)

By the Cofunction Identity $\cos \theta = \sin(\frac{\pi}{2} - \theta)$, we rewrite the given equation:

$$\sin\left(\frac{\pi}{2} \cos x\right) = \sin\left(\frac{\pi}{2} - \frac{\pi}{2} \sin x\right).$$

Recall that if $\sin \theta = \sin \phi$, then $\theta = \phi + 2n\pi$ or $\theta = \pi - \phi + 2n\pi$ for some integer n . Therefore, we have two cases:

- a) $\frac{\pi}{2} \cos x = \left(\frac{\pi}{2} - \frac{\pi}{2} \sin x\right) + 2n\pi$ for some integer n We rearrange and simplify:

$$\sin x + \cos x = 1 + 4n.$$

By rough constraints, we know that $-2 < \sin x + \cos x < 2$, from which $-2 < 1 + 4n < 2$. The only possibility is $n = 0$, so

$$\begin{aligned} \sin x + \cos x &= 1 & (*) \\ \sin^2 x + \cos^2 x + 2 \sin x \cos x &= 1 \\ 2 \sin x \cos x &= 0 \\ \sin(2x) &= 0 \\ 2x &= k\pi \\ x &= \frac{k\pi}{2} \end{aligned}$$

for some integer k .

We get $x = 0, \frac{\pi}{2}$ for this case. Note that $x = \pi$ is an extraneous solution by squaring (*).

- b) $\frac{\pi}{2} \cos x = \pi - \left(\frac{\pi}{2} - \frac{\pi}{2} \sin x\right) + 2n\pi$ for some integer n Similar to Case 1, we conclude that $n = 0$, so

$$\cos x - \sin x = 1.$$

We get $x = 0$ for this case.

Together, we obtain (C) 2 solutions: $x = 0, \frac{\pi}{2}$.

Solution 4 (Graphs and Analyses)

This problem is equivalent to counting the intersections of the graphs of $y = \sin\left(\frac{\pi}{2} \cos x\right)$ and $y = \cos\left(\frac{\pi}{2} \sin x\right)$ in the closed interval $[0, \pi]$. We construct a table of values, as shown below:

	$x = 0$	$x = \frac{\pi}{2}$	$x = \pi$
$\cos x$	1	0	-1
$\frac{\pi}{2} \cos x$	$\frac{\pi}{2}$	0	$-\frac{\pi}{2}$
$\sin\left(\frac{\pi}{2} \cos x\right)$	1	0	-1
$\sin x$	0	1	0
$\frac{\pi}{2} \sin x$	0	$\frac{\pi}{2}$	0
$\cos\left(\frac{\pi}{2} \sin x\right)$	1	0	1

For $x \in [0, \pi]$, note that:

- $\frac{\pi}{2} \cos x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, so $\sin\left(\frac{\pi}{2} \cos x\right) \in [-1, 1]$.
- $\frac{\pi}{2} \sin x \in \left[0, \frac{\pi}{2}\right]$, so $\cos\left(\frac{\pi}{2} \sin x\right) \in [0, 1]$.

For the graphs to intersect, we need $\sin\left(\frac{\pi}{2} \cos x\right) \in [0, 1]$. This occurs when $\frac{\pi}{2} \cos x \in \left[0, \frac{\pi}{2}\right]$.

By the Cofunction Identity $\cos \theta = \sin\left(\frac{\pi}{2} - \theta\right)$, we rewrite the given equation:

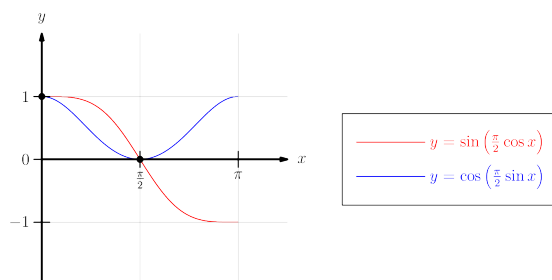
$$\sin\left(\frac{\pi}{2} \cos x\right) = \sin\left(\frac{\pi}{2} - \frac{\pi}{2} \sin x\right).$$

Since $\frac{\pi}{2} \cos x \in \left[0, \frac{\pi}{2}\right]$ and $\frac{\pi}{2} \sin x \in \left[0, \frac{\pi}{2}\right]$, it follows that $x \in \left[0, \frac{\pi}{2}\right]$ and $\frac{\pi}{2} - \frac{\pi}{2} \sin x \in \left[0, \frac{\pi}{2}\right]$.

We can apply the arcsine function to both sides, then rearrange and simplify:

$$\begin{aligned} \frac{\pi}{2} \cos x &= \frac{\pi}{2} - \frac{\pi}{2} \sin x \\ \sin x + \cos x &= 1. \end{aligned}$$

From Case 1 in Solution 2, we conclude that $(0, 1)$ and $\left(\frac{\pi}{2}, 0\right)$ are the only points of intersection, as shown below:



Therefore, the answer is **(C) 2**.

Solution 5 (No Graphing)

For $\sin(a) = \cos(b)$, a and b must be complementary angles, or $a + b = \frac{\pi}{2}$. So we have

$$\frac{\pi}{2}(\sin(x) + \cos(x)) = \frac{\pi}{2} \implies \sin(x) + \cos(x) = 1.$$

We know $\sin^2(x) + \cos^2(x) = 1$, so we can express $\cos(x)$ in terms of $\sin(x)$ and set those expressions equal to each other:

$$\sqrt{1 - \sin^2(x)} = 1 - \sin(x) \implies 1 - \sin^2(x) = (1 - \sin(x))^2 \implies \sin(x) = 1.$$

So now the problem becomes the number of values of x on the interval $[0, \pi]$ such that $\sin(x) = 1$. It's pretty easy to see that x can equal 0 or $\frac{\pi}{2}$, so the answer is

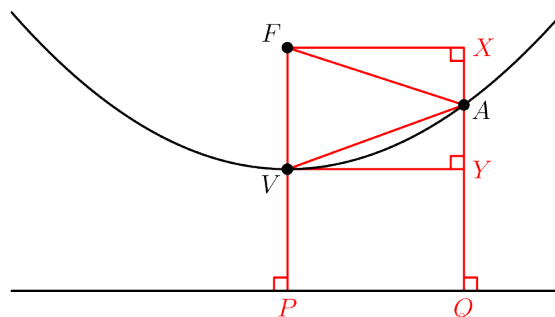
(C) 2.

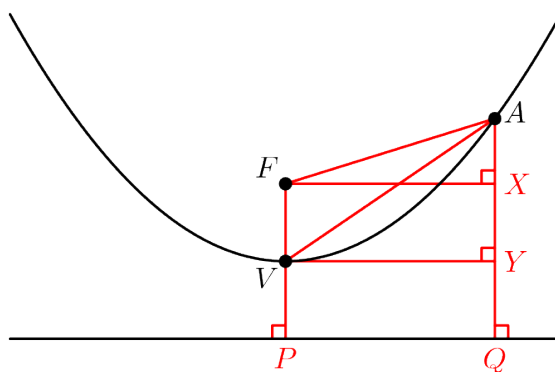
20. (AMC 12 12A 2021) Suppose that on a parabola with vertex V and a focus F there exists a point A such that $AF = 20$ and $AV = 21$. What is the sum of all possible values of the length FV ? **(A)** 13 **(B)** $\frac{40}{3}$ **(C)** $\frac{41}{3}$ **(D)** 14 **(E)** $\frac{43}{3}$

Answer: B

Solution 1

Let ℓ be the directrix of $\mathcal{P}$; recall that $\mathcal{P}$ is the set of points T such that the distance from T to ℓ is equal to TF . Let P and Q be the orthogonal projections of F and A onto ℓ , and further let X and Y be the orthogonal projections of F and V onto AQ . Because $AF < AV$, there are two possible configurations which may arise, and they are shown below.





Set $d = FV$, which by the definition of a parabola also equals VP . Then as $AQ = AF = 20$, we have $AY = 20 - d$ and $AX = |20 - 2d|$. Since $FXYV$ is a rectangle, $FX = VY$, so by Pythagorean Theorem on triangles AFX and AVY ,

$$\begin{aligned} 21^2 - (20 - d)^2 &= AV^2 - AY^2 = VY^2 \\ &= FX^2 = AF^2 - AX^2 = 20^2 - (20 - 2d)^2 \end{aligned}$$

This equation simplifies to $3d^2 - 40d + 41 = 0$, which has solutions $d = \frac{20 \pm \sqrt{277}}{3}$. Both values of d work - the smaller solution with the bottom configuration and the larger solution with the top configuration - and so the requested answer is

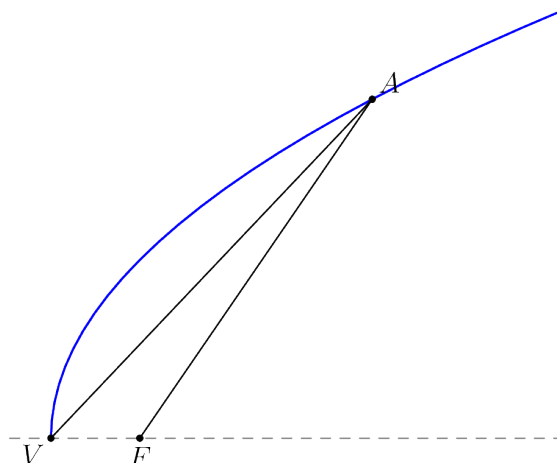
(B) $\frac{40}{3}$.

Solution 2

Let $\mathcal{P}$ be the parabola, let V be the origin, F lie on the positive y axis, and $d = FV$. The equation of the parabola is then $x^2 = 4dy$. If the coordinates of A are (p, q) , then $p^2 + q^2 = 441$ since the distance from the origin to A is 21. Note also that the parabola is the set of all points equidistant from F and a line known as its directrix, which in this case is a horizontal line d units below the origin. Since the distance from A to its directrix is equal to AF , then A is 20 units above this line and therefore $q = 20 - d$. Substituting for p and q yields $4d(20 - d) + (20 - d)^2 = 441$ or $(20 - d)(20 + 3d) = 441$, which simplifies to $3d^2 - 40d + 41 = 0$. Therefore the sum of all possible values of d is $\frac{-b}{a} =$ (B) $\frac{40}{3}$ by Viète's Formulas.

Solution 3

Two possible cases for A exist, either, it lies on the latus rectum endpoints, in which $VF = 10$, or it doesn't, in which $VF = 20 - x$ where $A(x, y)$. The first case is not allowed as $AF = 21$ and $VF = 10$ forces $FA \neq 20$. Therefore, the second case must be true.



Suppose $A(x, y)$. Then, if V is $(0, 0)$ for instance it follows $F(0, p)$ for some p . Also, $x^2 + y^2 = 441$, and $x^2 + (y - p)^2 = 400$. Finally, $x^2 = 4py$, so $\frac{x^2}{4p} = y$. Solving for p , one could obtain the quartic $9p^4 - 1354p^2 + 1681 = 0$, which solves nicely as

$$p = \frac{\pm 20 \pm \sqrt{277}}{3}$$

The only two positive solutions (as distance is never negative) is

$$p = \frac{20 \pm \sqrt{277}}{3}$$

, whose sum is $\boxed{\frac{40}{3}}$.

21. (AMC 12 12A 2021) The five solutions to the equation

$$(z - 1)(z^2 + 2z + 4)(z^2 + 4z + 6) = 0$$

may be written in the form $x_k + y_k i$ for $1 \leq k \leq 5$, where x_k and y_k are real. Let $\mathcal{E}$ be the unique ellipse that passes through the points $(x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4)$, and (x_5, y_5) . The eccentricity of $\mathcal{E}$ can be written in the form $\sqrt{\frac{m}{n}}$, where m and n are relatively prime positive integers. What is $m + n$? (Recall that the eccentricity of an ellipse $\mathcal{E}$ is the ratio $\frac{c}{a}$, where $2a$ is the length of the major axis of $\mathcal{E}$ and $2c$ is the distance between its two foci.)

- (A) 7 (B) 9 (C) 11 (D) 13 (E) 15

Answer: A

Solution 1 (Perpendicular Bisectors)

The solutions to this equation are $z = 1$, $z = -1 \pm i\sqrt{3}$, and $z = -2 \pm i\sqrt{2}$. Consider the five points $(1, 0)$, $(-1, \pm\sqrt{3})$, and $(-2, \pm\sqrt{2})$; these are the five points which lie on $\mathcal{E}$. Note that since these five points are symmetric about the x -axis, so must $\mathcal{E}$.

Now let $r = b/a$ denote the ratio of the length of the minor axis of $\mathcal{E}$ to the length of its major axis. Remark that if we perform a transformation of the plane which scales every x -coordinate by a factor of r , $\mathcal{E}$ is sent to a circle $\mathcal{E}'$. Thus,

the problem is equivalent to finding the value of r such that $(r, 0)$, $(-r, \pm\sqrt{3})$, and $(-2r, \pm\sqrt{2})$ all lie on a common circle; equivalently, it suffices to determine the value of r such that the circumcenter of the triangle formed by the points $P_1 = (r, 0)$, $P_2 = (-r, \sqrt{3})$, and $P_3 = (-2r, \sqrt{2})$ lies on the x -axis.

Recall that the circumcenter of a triangle ABC is the intersection point of the perpendicular bisectors of its three sides. The equations of the perpendicular bisectors of the segments $\overline{P_1P_2}$ and $\overline{P_1P_3}$ are

$$y = \frac{\sqrt{3}}{2} + \frac{2r}{\sqrt{3}}x \quad \text{and} \quad y = \frac{\sqrt{2}}{2} + \frac{3r}{\sqrt{2}}\left(x + \frac{r}{2}\right)$$

respectively. These two lines have different slopes for $r \neq 0$, so indeed they will intersect at some point (x_0, y_0) ; we want $y_0 = 0$. Plugging $y = 0$ into the first equation yields $x = -\frac{3}{4r}$, and so plugging $y = 0$ into the second equation and simplifying yields

$$-\frac{1}{3r} = x + \frac{r}{2} = -\frac{3}{4r} + \frac{r}{2}.$$

Solving yields $r = \sqrt{\frac{5}{6}}$.

Finally, recall that the lengths a , b , and c (where c is the distance between the foci of $\mathcal{E}$) satisfy $c = \sqrt{a^2 - b^2}$. Thus the eccentricity of $\mathcal{E}$ is $\frac{c}{a} = \sqrt{1 - \left(\frac{b}{a}\right)^2} = \sqrt{\frac{1}{6}}$ and the requested answer is **(A) 7**.

Solution 2 (Three Variables, Three Equations)

Completing the square in the original equation, we have

$$(z - 1)\left((z + 1)^2 + 3\right)\left((z + 2)^2 + 2\right) = 0,$$

from which $z = 1, -1 \pm \sqrt{3}i, -2 \pm \sqrt{2}i$.

Now, we will find the equation of an ellipse $\mathcal{E}$ that passes through $(1, 0)$, $(-1, \pm\sqrt{3})$, and $(-2, \pm\sqrt{2})$ in the xy -plane. By symmetry, the center of $\mathcal{E}$ must be on the x -axis.

The formula of $\mathcal{E}$ is

$$\frac{(x - h)^2}{a^2} + \frac{y^2}{b^2} = 1, \tag{★}$$

with the center $(h, 0)$ and the axes' lengths $2a$ and $2b$.

Plugging the points $(1, 0)$, $(-1, \sqrt{3})$, and $(-2, \sqrt{2})$ into (★), respectively, we have the following system of equations:

$$\begin{aligned} \frac{(1 - h)^2}{a^2} &= 1, \\ \frac{(-1 - h)^2}{a^2} + \frac{\sqrt{3}^2}{b^2} &= 1, \\ \frac{(-2 - h)^2}{a^2} + \frac{\sqrt{2}^2}{b^2} &= 1. \end{aligned}$$

Since $t^2 = (-t)^2$ holds for all real numbers t , we clear fractions and simplify:

$$(1 - h)^2 = a^2, \tag{1}$$

$$b^2(1+h)^2 + 3a^2 = a^2b^2, \quad (2)$$

$$b^2(2+h)^2 + 2a^2 = a^2b^2. \quad (3)$$

Applying the Transitive Property to (2) and (3), we isolate a^2 :

$$\begin{aligned} b^2(1+h)^2 + 3a^2 &= b^2(2+h)^2 + 2a^2 \\ a^2 &= b^2 \left((2+h)^2 - (1+h)^2 \right) \\ a^2 &= b^2(2h+3). \end{aligned} \quad (*)$$

Substituting (1) and (*) into (2), we solve for h :

$$\begin{aligned} b^2(1+h)^2 + 3 \underbrace{b^2(2h+3)}_{\text{by } (*)} &= \underbrace{(1-h)^2}_{\text{by (1)}} b^2 \\ (1+h)^2 + 3(2h+3) &= (1-h)^2 \\ 1 + 2h + h^2 + 6h + 9 &= 1 - 2h + h^2 \\ 10h &= -9 \\ h &= -\frac{9}{10}. \end{aligned}$$

Substituting this into (1), we get $a^2 = \frac{361}{100}$.

Substituting the current results into (*), we get $b^2 = \frac{361}{120}$.

Finally, we obtain

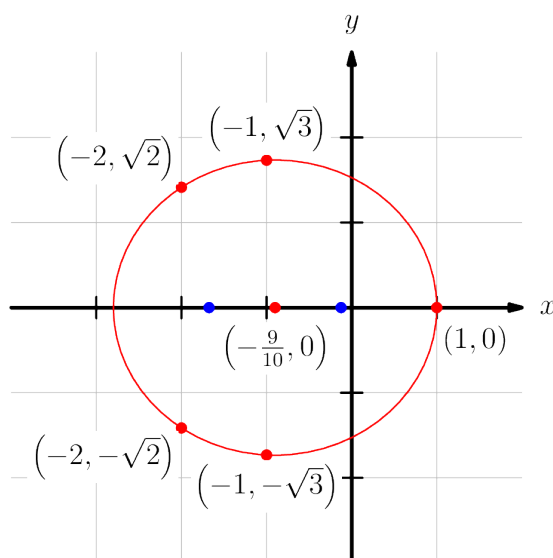
$$c^2 = a^2 - b^2 = 361 \left(\frac{1}{100} - \frac{1}{120} \right) = \frac{361}{600},$$

from which

$$\frac{c}{a} = \sqrt{\frac{c^2}{a^2}} = \sqrt{\frac{361/600}{361/100}} = \sqrt{\frac{1}{6}}.$$

The answer is $1 + 6 = \boxed{\text{(A)} 7}$.

The graph of $\mathcal{E}$ is shown below. Note that the foci are at $(h \pm c, 0) = \left(-\frac{9}{10} \pm \frac{19\sqrt{6}}{60}, 0 \right)$, as shown in the blue points.



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Solution 3 (Alternate Version of Solution 2)

Starting from this system of equations from Solution 2:

$$\begin{aligned}\frac{(1-h)^2}{a^2} &= 1, \\ \frac{(-1-h)^2}{a^2} + \frac{\sqrt{3}^2}{b^2} &= 1, \\ \frac{(-2-h)^2}{a^2} + \frac{\sqrt{2}^2}{b^2} &= 1.\end{aligned}$$

Let $A = a^{-2}$ and $B = b^{-2}$. Therefore, the system can be rewritten as:

$$\begin{aligned}(h^2 - 2h + 1)A &= 1, & (1) \\ (h^2 + 2h + 1)A + 3B &= 1, & (2) \\ (h^2 + 4h + 4)A + 2B &= 1. & (3)\end{aligned}$$

Subtracting (1) from (2) and (3), we get

$$4hA + 3B = 0 \quad \text{and} \quad 3A - 6hA + 3B = 0.$$

Plugging the former into the latter and simplifying yields $6A = 5B$. Hence $a^2 : b^2 = 6 : 5$. Since $c^2 = a^2 - b^2$, we get $a^2 = 6c^2$, so the eccentricity is $\frac{c}{a} = \sqrt{\frac{1}{6}}$.

Therefore, the answer is $1 + 6 = \boxed{\text{(A) } 7}$.

Solution 4 (Four Variables, Three Equations)

The five roots are $1, -1 + i\sqrt{3}, -1 - i\sqrt{3}, -2 + i\sqrt{2}, -2 - i\sqrt{2}$.

So, we express this conic in the form $ax^2 + by^2 + cx + z = 0$. Note that this conic cannot have the ky term since the roots are symmetric about the x -axis.

Now we have equations

$$\begin{aligned}a + c + z &= 0, \\ a + 3b - c + z &= 0, \\ 4a + 2b - 2c + z &= 0,\end{aligned}$$

from which $a : b : c = 5 : 6 : 9$.

So, the conic can be written in the form $5x^2 + 6y^2 + 9x = 14$. If it is written in the form of $\frac{(x-m)^2}{r^2} + \frac{y^2}{s^2} = 1$, then $r^2 : s^2 = 6 : 5$.

Therefore, the desired eccentricity is $\sqrt{\frac{\sqrt{6-5}}{6}} = \sqrt{\frac{1}{6}}$, and the answer is $1 + 6 = \boxed{\text{(A) } 7}$.

Solution 5 (Transformations)

After calculating the 5 points that lie on $\mathcal{E}$, we try to find a transformation that sends $\mathcal{E}$ to the unit circle. Scaling about $(1, 0)$ works, since $(1, 0)$ is already on the unit circle and such a transformation will preserve the ellipse's symmetry about the x -axis. If $2a$ and $2b$ are the lengths of the major and minor axes, respectively, then the ellipse will be scaled by a factor of $r := \frac{1}{a}$ in the x -dimension and $s := \frac{1}{b}$ in the y -dimension.

The transformation then sends the points $(-1, \pm\sqrt{3})$ and $(-2, \pm\sqrt{2})$ to the points $(1 - 2r, \pm s\sqrt{3})$ and $(1 - 3r, \pm s\sqrt{2})$, respectively. These points are on the unit circle, so

$$(1 - 2r)^2 + 3s^2 = 1 \quad \text{and} \quad (1 - 3r)^2 + 2s^2 = 1.$$

This yields

$$4r^2 + 3s^2 = 4r \quad \text{and} \quad 9r^2 + 2s^2 = 6r,$$

from which

$$\begin{aligned} 12r^2 + 9s^2 &= 18r^2 + 4s^2 \\ \frac{r^2}{s^2} &= \frac{5}{6}. \end{aligned}$$

Recalling that $r = \frac{1}{a}$ and $s = \frac{1}{b}$, this implies $\frac{b^2}{a^2} = \frac{5}{6}$. From this, we get

$$\frac{c^2}{a^2} = \frac{a^2 - b^2}{a^2} = 1 - \frac{b^2}{a^2} = \frac{1}{6},$$

so $\frac{c}{a} = \sqrt{\frac{1}{6}}$, giving an answer of $1 + 6 = \boxed{\text{(A)} 7}$.

Solution 6 (Similar to Others)

Zero product property gives three cases.

$z - 1 = 0$ from which $z = 1 + 0i$ is valid, $(x_1, y_1) = (1, 0)$.

$z^2 + 2z + 4 = 0$. Through the quadratic formula, $z = -1, +i\sqrt{3}, -1 - i\sqrt{3}$, yielding $(x_2, y_2) = (-1, \sqrt{3})$; $(x_3, y_3) = (-1, -\sqrt{3})$ $z^2 + 4z + 6 = 0$. Through the quadratic formula, $z = -2 + i\sqrt{2}, -2 - i\sqrt{2}$, yielding $(x_4, y_4) = (-2, \sqrt{2})$; $(x_5, y_5) = (-2, -\sqrt{2})$.

Assume the ellipse has general form

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

From substituting the given points,

$$A + D + F = 0$$

$$A + 3C - D \pm (-B + E)\sqrt{3} + F = 0$$

Subtraction yields $-B + E = 0 \implies E = B$. Adding yields $A + 3C - D + F = 0$.

Substituting the last remaining values of z ,

$$4A + 2C - 2D + F \pm (-2B + E)\sqrt{2} = 0$$

.

Subtracting gives $-2B + E = 0$. Since $E = B$, $B = E = 0$. From the first equation, $F = -A - D$. Substituting into the third equation gives $A + 3C - D - A - D = 0$ or $3C = 2D$. Using $z = (x_4, y_4)$, $4A + 2C - 2D + F = 0$, or $3A + 2C - 3D = 0$. Recall $D = \frac{3}{2}C$ so $3A + 2C - \frac{9}{2}C = 0 \implies A = \frac{5}{6}C$. From here, $F = -A - D = -\frac{5}{6}C - \frac{3}{2}C = -\frac{7}{3}C$. Therefore, the ratios of all integral values are $A : C : D : F = 5 : 6 : 9 : -14$, and because all z are relatively prime, $5x^2 + 6y^2 + 9x - 14 = 0$ is the conic. In completed form,

$$\frac{\left(x + \frac{9}{10}\right)^2}{\frac{361}{100}} + \frac{y^2}{\frac{361}{120}} = 1.$$

This implies

$$c = \frac{19\sqrt{6}}{60}$$

and

$$a = \frac{19}{10}$$

From which

$$\mathcal{E} = \frac{\sqrt{6}}{6} = \sqrt{\frac{6}{36}} \implies \sqrt{\frac{1}{6}} \implies \boxed{7}$$

22. (AMC 12 12A 2021) Suppose that the roots of the polynomial $P(x) = x^3 + ax^2 + bx + c$ are $\cos \frac{2\pi}{7}$, $\cos \frac{4\pi}{7}$, and $\cos \frac{6\pi}{7}$, where angles are in radians. What is abc ?

- (A) $-\frac{3}{49}$ (B) $-\frac{1}{28}$ (C) $\frac{\sqrt[3]{7}}{64}$ (D) $\frac{1}{32}$ (E) $\frac{1}{28}$

Answer: D

Solution 1 (Complex Numbers and Vieta's Formulas)

Let $z = e^{\frac{2\pi i}{7}}$. Since z is a 7th root of unity, we have $z^7 = 1$. For all integers k , note that

$$\cos \frac{2k\pi}{7} = \operatorname{Re} \left(z^k \right) = \operatorname{Re} \left(z^{-k} \right).$$

It follows that

$$\begin{aligned} \cos \frac{2\pi}{7} &= \frac{z + z^{-1}}{2} = \frac{z + z^6}{2}, \\ \cos \frac{4\pi}{7} &= \frac{z^2 + z^{-2}}{2} = \frac{z^2 + z^5}{2}, \\ \cos \frac{6\pi}{7} &= \frac{z^3 + z^{-3}}{2} = \frac{z^3 + z^4}{2}. \end{aligned}$$

By geometric series, we conclude that

$$\sum_{k=0}^6 z^k = \frac{1 - 1}{1 - z} = 0.$$

Alternatively, recall that the 7th roots of unity satisfy the equation $z^7 - 1 = 0$. By Vieta's formulas, the sum of these seven roots is 0.

As a result, we get

$$\sum_{k=1}^6 z^k = -1.$$

Let $(r, s, t) = \left(\cos \frac{2\pi}{7}, \cos \frac{4\pi}{7}, \cos \frac{6\pi}{7} \right)$. By Vieta's formulas, the answer is

$$\begin{aligned} abc &= [-(r + s + t)](rs + st + tr)(-rst) \\ &= (r + s + t)(rs + st + tr)(rst) \end{aligned}$$

$$\begin{aligned}
 &= \left(\frac{\sum_{k=1}^6 z^k}{2} \right) \left(\frac{2 \sum_{k=1}^6 z^k}{4} \right) \left(\frac{1 + \sum_{k=0}^6 z^k}{8} \right) \\
 &= \frac{1}{32} \left(\sum_{k=1}^6 z^k \right) \left(\sum_{k=1}^6 z^k \right) \left(1 + \sum_{k=0}^6 z^k \right) \\
 &= \frac{1}{32} (-1)(-1)(1) \\
 &= \boxed{\text{(D)} \frac{1}{32}}.
 \end{aligned}$$

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Solution 2 (Complex Numbers and Trigonometric Identities)

Let $z = e^{\frac{2\pi i}{7}}$. In Solution 1, we conclude that $\sum_{k=1}^6 z^k = -1$, so

$$\sum_{k=1}^6 \operatorname{Re}(z^k) = \sum_{k=1}^6 \cos \frac{2k\pi}{7} = -1.$$

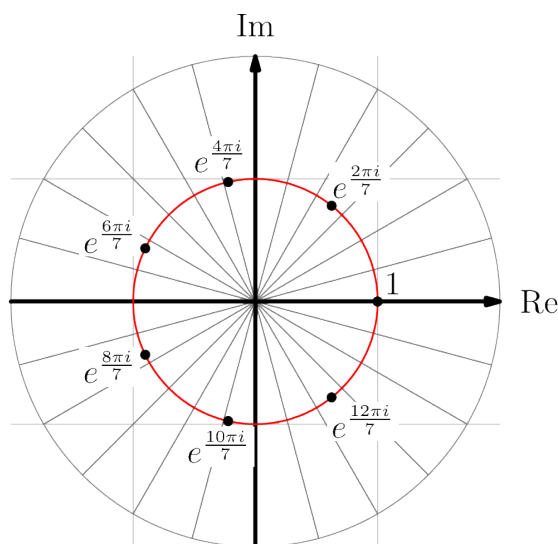
Since $\cos \theta = \cos(2\pi - \theta)$ holds for all θ , this sum becomes

$$\begin{aligned}
 2 \left(\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} \right) &= -1 \\
 \cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} &= -\frac{1}{2}.
 \end{aligned}$$

Note that $\theta = \frac{2\pi}{7}, \frac{4\pi}{7}, \frac{6\pi}{7}$ are roots of

$$\cos \theta + \cos(2\theta) + \cos(3\theta) = -\frac{1}{2}, \quad (\star)$$

as they can be verified either algebraically (by the identity $\cos \theta = \cos(-\theta) = \cos(2\pi - \theta)$) or geometrically (by the graph below).



Let $x = \cos \theta$. We apply the Double-Angle and Triple-Angle Identities to rewrite $(\star)$ in terms of x , and then rearrange:

$$x + (2x^2 - 1) + (4x^3 - 3x) = -\frac{1}{2}$$

$$4x^3 + 2x^2 - 2x - \frac{1}{2} = 0$$

$$x^3 + \frac{1}{2}x^2 - \frac{1}{2}x - \frac{1}{8} = 0.$$

Recall that the roots are $x = \cos \frac{2\pi}{7}, \cos \frac{4\pi}{7}, \cos \frac{6\pi}{7}$.

Therefore, we obtain $(a, b, c) = \left(\frac{1}{2}, -\frac{1}{2}, -\frac{1}{8}\right)$, from which $abc = \boxed{\text{(D)} \frac{1}{32}}$.

Solution 3 (Trigonometric Identities)

We solve for a, b , and c separately:

- a) Solve for a : By Vieta's Formulas, we have $a = -\left(\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7}\right)$. The real parts of the 7th roots of unity are $1, \cos \frac{2\pi}{7}, \cos \frac{4\pi}{7}, \cos \frac{6\pi}{7}, \cos \frac{8\pi}{7}, \cos \frac{10\pi}{7}, \cos \frac{12\pi}{7}$ and they sum to 0.

Note that $\cos \theta = \cos(2\pi - \theta)$ for all θ . Excluding 1, the other six roots add to

$$2\left(\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7}\right) = -1,$$

from which

$$\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} = -\frac{1}{2}.$$

Therefore, we get $a = -\left(-\frac{1}{2}\right) = \frac{1}{2}$.

- b) Solve for b : By Vieta's Formulas, we have $b = \cos \frac{2\pi}{7} \cos \frac{4\pi}{7} + \cos \frac{2\pi}{7} \cos \frac{6\pi}{7} + \cos \frac{4\pi}{7} \cos \frac{6\pi}{7}$. Note that $\cos \alpha \cos \beta = \frac{\cos(\alpha+\beta) + \cos(\alpha-\beta)}{2}$ for all α and β . Therefore, we get

$$b = \frac{\cos \frac{6\pi}{7} + \cos \frac{2\pi}{7}}{2} + \frac{\cos \frac{6\pi}{7} + \cos \frac{4\pi}{7}}{2} + \frac{\cos \frac{4\pi}{7} + \cos \frac{2\pi}{7}}{2} = \cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} = -\frac{1}{2}.$$

- c) Solve for c : By Vieta's Formulas, we have $c = -\cos \frac{2\pi}{7} \cos \frac{4\pi}{7} \cos \frac{6\pi}{7} = -\cos \frac{2\pi}{7} \cos \frac{4\pi}{7} \cos \frac{8\pi}{7}$. We multiply both sides by $8 \sin \frac{2\pi}{7}$, then repeatedly apply the angle addition formula for sine:

$$\begin{aligned} c \cdot 8 \sin \frac{2\pi}{7} &= -8 \sin \frac{2\pi}{7} \cos \frac{2\pi}{7} \cos \frac{4\pi}{7} \cos \frac{8\pi}{7} \\ &= -4 \sin \frac{4\pi}{7} \cos \frac{4\pi}{7} \cos \frac{8\pi}{7} \\ &= -2 \sin \frac{8\pi}{7} \cos \frac{8\pi}{7} \\ &= -\sin \frac{16\pi}{7} \\ &= -\sin \frac{2\pi}{7}. \end{aligned}$$

Therefore, we get $c = -\frac{1}{8}$.

Finally, the answer is $abc = \frac{1}{2} \cdot \left(-\frac{1}{2}\right) \cdot \left(-\frac{1}{8}\right) = \boxed{\text{(D)} \frac{1}{32}}$.

Solution 4 (Product-to-Sum Identity)

Note that the sum of the roots of unity equal zero, so the sum of their real parts equal zero, and $\operatorname{Re}(\omega^m) = \operatorname{Re}(\omega^{-m})$. We have

$$\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} = \frac{1}{2}(0 - \cos 0) = -\frac{1}{2},$$

so $a = \frac{1}{2}$.

By the Product-to-Sum Identity, we have

$$\begin{aligned} \cos \frac{2\pi}{7} \cos \frac{4\pi}{7} + \cos \frac{2\pi}{7} \cos \frac{6\pi}{7} + \cos \frac{4\pi}{7} \cos \frac{6\pi}{7} &= \frac{1}{2} \left(2 \cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} + \cos \frac{8\pi}{7} + \cos \frac{10\pi}{7} \right) \\ &= \frac{1}{2} \left(2 \cos \frac{2\pi}{7} + 2 \cos \frac{4\pi}{7} + 2 \cos \frac{6\pi}{7} \right) \\ &= \cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} \\ &= -\frac{1}{2}, \end{aligned}$$

so $b = -\frac{1}{2}$.

By the Product-to-Sum Identity, we have

$$\begin{aligned} \cos \frac{2\pi}{7} \cos \frac{4\pi}{7} \cos \frac{6\pi}{7} &= \frac{1}{2} \cos \frac{6\pi}{7} \left(\cos \frac{2\pi}{7} + \cos \frac{6\pi}{7} \right) \\ &= \frac{1}{4} \left(\cos \frac{4\pi}{7} + \cos \frac{8\pi}{7} \right) + \frac{1}{4} \left(1 + \cos \frac{12\pi}{7} \right) \\ &= \frac{1}{4} \left(1 + \cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} \right) \\ &= \frac{1}{8}, \end{aligned}$$

so $c = -\frac{1}{8}$.

Finally, we get $abc = \boxed{\text{(D)} \frac{1}{32}}$.

23. (AMC 12 12A 2021) Frieda the frog begins a sequence of hops on a 3×3 grid of squares, moving one square on each hop and choosing at random the direction of each hop up, down, left, or right. She does not hop diagonally. When the direction of a hop would take Frieda off the grid, she "wraps around" and jumps to the opposite edge. For example if Frieda begins in the center square and makes two hops "up", the first hop would place her in the top row middle square, and the second hop would cause Frieda to jump to the opposite edge, landing in the bottom row middle square. Suppose Frieda starts from the center square, makes at most four hops at random, and stops hopping if she lands on a corner square. What is the probability that she reaches a corner square on one of the four hops?

(A) $\frac{9}{16}$ (B) $\frac{5}{8}$ (C) $\frac{3}{4}$ (D) $\frac{25}{32}$ (E) $\frac{13}{16}$

Answer: D

Solution 1 (Complementary Counting)

We will use complementary counting. First, the frog can go left with probability $\frac{1}{4}$. We observe symmetry, so our final answer will be multiplied by 4 for the 4 directions, and since $4 \cdot \frac{1}{4} = 1$, we will ignore the leading probability.

From the left, she either goes left to another edge ($\frac{1}{4}$) or back to the center ($\frac{1}{4}$). Time for some casework.

Case 1: She goes back to the center.

Now, she can go in any 4 directions, and then has 2 options from that edge. This gives $\frac{1}{2}$. –End case 1

Case 2: She goes to another edge (rightmost).

Subcase 1: She goes back to the left edge. She now has 2 places to go, giving $\frac{1}{2}$

Subcase 2: She goes to the center. Now any move works.

$\frac{1}{4} \cdot \frac{1}{2} + \frac{1}{4} \cdot 1 = \frac{1}{8} + \frac{1}{4} = \frac{3}{8}$ for this case. –End case 2

She goes back to the center in Case 1 with probability $\frac{1}{4}$, and to the right edge with probability $\frac{1}{4}$

So, our answer is $\frac{1}{4} \cdot \frac{1}{2} + \frac{1}{4} \cdot \frac{3}{8} = \frac{1}{4}(\frac{1}{2} + \frac{3}{8}) = \frac{1}{4} \cdot \frac{7}{8} = \frac{7}{32}$

But, don't forget complementary counting. So, we get $1 - \frac{7}{32} = \frac{25}{32} \implies \boxed{D}$.
firebolt360

Solution 2 (Direct Counting and Probability States)

We can draw a state diagram with three states: center, edge, and corner. Denote center by M, edge by E, and corner by C. There are a few ways Frieda can reach a corner in four or less moves: EC, EEC, EEEEC, EMEC. Then, calculating the probabilities of each of these cases happening, we have $1 \cdot \frac{1}{2} + 1 \cdot \frac{1}{4} \cdot \frac{1}{2} + 1 \cdot \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{2} + 1 \cdot \frac{1}{4} \cdot 1 \cdot \frac{1}{2} = \frac{25}{32}$, so the answer is $\boxed{D}$. IceWolf10

Solution 3 (Finds Numerator and Denominator Separately)

Suppose Frieda makes four independent hops without stopping so that each outcome is equally likely.

Denominator

There are $4^4 = 256$ ways for Frieda to make four hops without restrictions.

Numerator

We perform casework on which hop Frieda reaches a corner for the first time:

- a) The second hop (The third and the fourth hops have no restrictions.) The four hops have 4, 2, 4, 4 options, respectively. So, this case has $4 \cdot 2 \cdot 4 \cdot 4 = 128$ ways.
 - b) The third hop (The fourth hop has no restrictions.) No matter which direction the first hop takes, the second hop must "wrap around".
The four hops have 4, 1, 2, 4 options, respectively. So, this case has $4 \cdot 1 \cdot 2 \cdot 4 = 32$ ways.
 - c) The fourth hop (There are two subcases based on the second hop.)
 - The second hop "wraps around". It follows that the third hop must also "wrap around". The four hops have 4, 1, 1, 2 options, respectively. So, this subcase has $4 \cdot 1 \cdot 1 \cdot 2 = 8$ ways.
 - The second hop returns to the center. The four hops have 4, 1, 4, 2 options, respectively. So, this subcase has $4 \cdot 1 \cdot 4 \cdot 2 = 32$ ways.
- Together, this case has $8 + 32 = 40$ ways.

The numerator is $128 + 32 + 40 = 200$. **Probability**

The requested probability is $\frac{200}{256} = \boxed{\text{(D)} \frac{25}{32}}$. **Remark**

This problem is quite similar to [1995 AIME Problem 3](#).

Solution 4

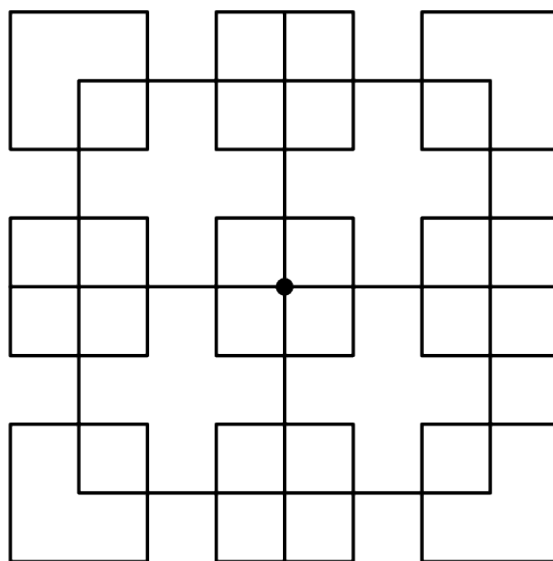
Let C_n be the probability that Frieda is on the central square after n moves, E_n be the probability that Frieda is on one of the four squares on the middle of the edges after n moves, and V_n (V for vertex) be the probability that Frieda is on a corner after n moves. The only way to reach the center is by moving in 1 specific direction out of 4 total directions from the middle of an edge, so $C_{n+1} = \frac{E_n}{4}$. The ways to reach the middle of an edge are by moving in any direction from the center or by moving in 1 specific direction from the middle of an edge, so $E_{n+1} = C_n + \frac{E_n}{4}$. The ways to reach a corner are by simply staying there after reaching there in a previous move or by moving in 2 specific directions from the middle of an edge, so $V_{n+1} = V_n + \frac{E_n}{2}$. Since Frieda always start from the center, $C_0 = 1$, $E_0 = 0$, and

$V_0 = 0$. We use the previous formulas to work out V_4 and find it to be $\boxed{\text{(D)} \frac{25}{32}}$.

-SmileKat32

Solution 5

Imagine an infinite grid of 2 by 2 squares such that there is a 2 by 2 square centered at $(3x, 3y)$ for all ordered pairs of integers (x, y) .



It is easy to see that the problem is equivalent to Frieda moving left, right, up, or down on this infinite grid starting at $(0, 0)$ (minus the teleportations). Since counting the complement set is easier, we'll count the number of 4-step paths such that Frieda never reaches a corner point.

In other words, since the reachable corner points are $(\pm 1, \pm 1)$, $(\pm 1, \pm 2)$, $(\pm 2, \pm 1)$, and $(\pm 2, \pm 2)$, Frieda can only travel along the collection of points included in S , where S is all points on $x = 0$ and $y = 0$ such that $|y| < 4$ and $|x| < 4$, respectively, plus all points on the big square with side length 6 centered at $(0, 0)$. We then can proceed with casework:

Case 1: Frieda never reaches $(0, \pm 3)$ nor $(\pm 3, 0)$.

When Frieda only moves horizontally or vertically for her four moves, she can do so in $2^4 - 4 = 12$ ways for each case. Thus, $12 \cdot 2$ total paths for the subcase of staying in one direction. (For instance, all length 4 combinations of F and B except $FFFF$, $BBBB$, $FFFB$, and $BBBF$ for the horizontal direction.)

There is another subcase where she changes directions during her path. There are four symmetric cases for this subcase depending on which of the four quadrants Frieda hugs. For the first quadrant, the possible paths are $FBUD$, $FBUU$, $UDFB$, and $UDFF$. Thus, a total of $4 \cdot 4 = 16$ ways for this subcase.

Total for Case 1: $24 + 16 = 40$

Case 2: Frieda reaches $(0, \pm 3)$ or $(\pm 3, 0)$.

Once Frieda reaches one of the points listed above (by using three moves), she has four choices for her last move. Thus, a total of $4 \cdot 4 = 16$ paths for this case.

Our total number of paths never reaching coroners is thus $16 + 40 = 56$, making for an answer of

$$\frac{4^4 - 56}{4^4} = \boxed{\text{(D)} \frac{25}{32}}.$$

-fidgetboss_4000

Solution 6 (Casework)

We take cases on the number of hops needed to reach a corner. For simplicity, denote E as a move that takes Frieda to an edge, W as wrap-around move and C as a corner move. Also, denote O as a move that takes us to the center.

2 Hops

Then, Frieda will have to (E, C) as her set of moves. There are 4 ways to move to an edge, and 2 corners to move to, for a total of $4 \cdot 2 = 8$ cases here. Then, there are 4 choices for each move, for a probability of $\frac{8}{4 \cdot 4} = \frac{1}{2}$.

3 Hops

In this case, Frieda must wrap-around. There's only one possible combination, just (E, W, C) . There are 4 ways to move to an edge, 1 way to wrap-around (you must continue in the same direction) and 2 corners, for a total of $4 \cdot 1 \cdot 2 = 8$ cases here. Then, there are 4 choices for each move, for a probability of $\frac{8}{4 \cdot 4 \cdot 4} = \frac{1}{8}$.

4 Hops

Lastly, there are two cases we must consider here. The first case is (E, O, E, C) , and the second is (E, W, W, C) . For the first case, there are 4 ways to move to an edge, 1 way to return to the center, 4 ways to move to an edge once again, and 2 ways to move to a corner. Hence, there is a total of $4 \cdot 1 \cdot 4 \cdot 2 = 32$ cases here. Then, for the second case, there are 4 ways to move to a corner, 1 way to wrap-around, 1 way to wrap-around the way Frieda just came from, and 2 ways to move to a corner. This implies there are $4 \cdot 1 \cdot 1 \cdot 2 = 8$ cases here. Then, there is a total of $8 + 32 = 40$ cases, out of a total of $4^4 = 256$ cases, for a probability of $\frac{40}{256} = \frac{5}{32}$.

Then, the total probability that Frieda ends up on a corner is $\frac{1}{2} + \frac{1}{8} + \frac{5}{32} = \frac{25}{32}$,

corresponding to choice $\boxed{\text{(D)} \frac{25}{32}}$.

Solution 7

I denote 3x3 grid by

- HOME square (x1)
- CORN squares (x4)
- SIDE squares (x4)

Transitions:

- HOME always move to SIDE
- CORN is DONE
- SIDE move to HOME with $p = 1/4$, move to SIDE with $p = 1/4$, and move to CORN with $p = 1/2$.

After one move, will be on SIDE square

After two moves, will be $1/2 + 1/4(\text{HOME}) + 1/4(\text{SIDE})$

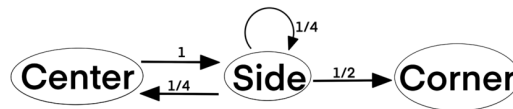
After three moves, will be $1/2 + 1/4(\text{SIDE}) + 1/4(1/2 + 1/4(\text{SIDE}) + 1/4(\text{HOME}))$

After four moves, probability on CORN will be $1/2 + 1/4(1/2) + 1/4(1/2 + 1/4(1/2)) = 1/2 + 1/8 + 5/32 = 25/32$.

Solution 8 (Markov Chain)

(i) <i>Main article: [Markov Chains](#)</i>

The grid is symmetrical. Denote the center cell as state Center, the 4 middle side cells as state Side, the 4 corner cells as state Corner. We can draw the following State Transition Diagram with [Markov Chain](#). The numbers on the transition arc are the transition probability.



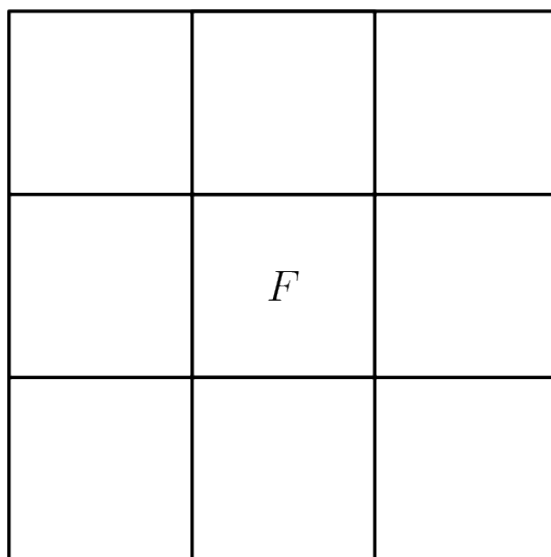
We calculate the probability of different states in each round in the following table by iterating a few more times:

Round	Center	Side	Corner
1	0	1	0
2	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{2}$
3	$\frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16}$	$\frac{1}{4} + \frac{1}{4} \cdot \frac{1}{4} = \frac{5}{16}$	$\frac{1}{4} \cdot \frac{1}{2} + \frac{1}{2} = \frac{5}{8}$
4	$\frac{5}{16} \cdot \frac{1}{4} = \frac{5}{64}$	$\frac{1}{16} + \frac{5}{16} \cdot \frac{1}{4} = \frac{9}{64}$	$\frac{5}{16} \cdot \frac{1}{2} + \frac{5}{8} = \frac{25}{32}$

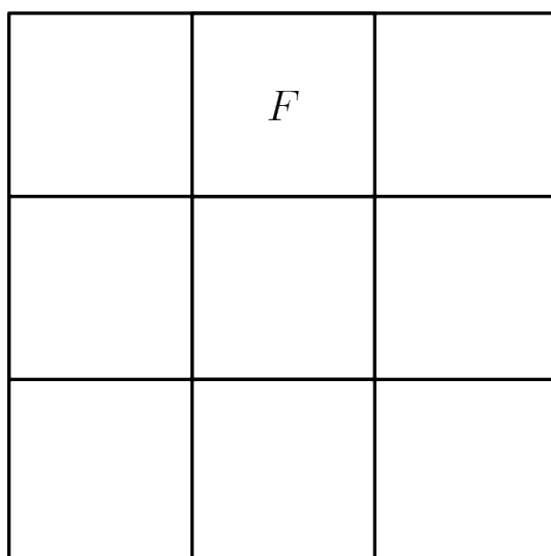
Therefore, the answer is (D) $\frac{25}{32}$.

Solution 9

Let us go through the frog's jumps step by step to find the probability that it will land in a corner. Denote the frog's position as F :

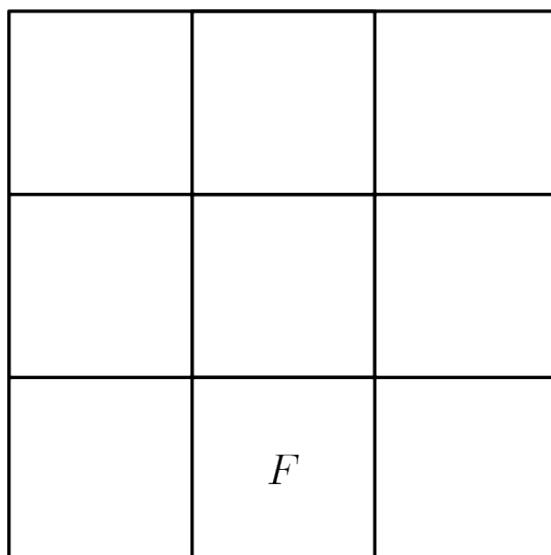


First notice that since the frog only moves up, down, left or right, it's original hop will not matter by symmetry, because all first hops will bring the frog at the edge next to two corners. WLOG we let the frog hop up:



Notice that in its second hop, there are two ways for it to move to a corner (left and right) out of four, and thus the probability of Frieda reaching a corner on the second hop is $\frac{1}{2}$.

There are two other cases for Frieda's second hop: it could either wrap around to another edge square or move back to the middle. We analyze each of these cases individually, noting that each has probability $\frac{1}{4}$:



In this case, there are again 2 choices to go to a corner on the third hop, go back to the previous square, or go to the middle. Here there is a $\frac{1}{2}$ probability that Frieda will reach a corner on the third hop. If Frieda hops back with a $\frac{1}{4}$ probability, she will again have a $\frac{1}{2}$ chance of reaching a corner on her fourth hop, but after four hops she stops hopping, so there are no more chances for her to reach a corner if she hops back. If she hops to the middle, then her fourth hop cannot reach a corner and that case yields 0 probability of reaching a corner.

To sum up this case, the total probability of her reaching a corner is $\frac{1}{4} \left(\frac{1}{2} + \frac{1}{4} \cdot \frac{1}{2} \right)$.

Finally suppose Frieda goes back to her initial position after 2 steps. Then her third step again doesn't matter because whatever she does, she will be next to 2 corners, so the probability of her reaching a corner on her fourth hop in this case is $\frac{1}{4} \cdot \frac{1}{2}$.

Summing up all the cases, we get

$$\frac{1}{2} + \frac{1}{4} \left(\frac{1}{2} + \frac{1}{4} \cdot \frac{1}{2} \right) + \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{2} + \frac{1}{4} \left(\frac{1}{2} + \frac{1}{8} \right) + \frac{1}{8} = \frac{5}{4} \cdot \left(\frac{1}{2} + \frac{1}{8} \right) = \frac{5}{4} \cdot \frac{5}{8} = \boxed{\text{(D)} \frac{25}{32}}.$$

KingRavi

Solution 10 (Similar to the States Solution)

Denote the center square as C , the edge squares as E , and the corner squares as Co . On the first move, the probability of Frieda on the C is 0 because she's moving away from the center square. The probability she's on E is 1 because no matter which direction she goes, she's always going to end up on an edge square. Probability she's on Co equals $1 - (0 + 1) = 0$ by complementary counting. Notice that the probability Frieda will be on the C can be expressed as: $P(C) = \frac{1}{4} \cdot P(E \text{ from last move})$ because one-fourth of the moves Frieda takes from an edge square sends her back to C . Also, she can't reach C from a Co square. Now $P(E) = P(C \text{ from last move}) + \frac{1}{4} \cdot P(E \text{ from last move})$ simply from the same logic. We can find $P(Co)$ using complementary counting. On the second move, $P(C) = \frac{1}{4} \cdot 1 = \frac{1}{4}$ and $P(E) = 0 + \frac{1}{4} \cdot 1 = \frac{1}{4}$. Now $P(Co) = 1 - \frac{1}{4} - \frac{1}{4} = \frac{1}{2}$. On the third move, $P(C) = \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16}$ and $P(E) = \frac{1}{4} + \frac{1}{4} \cdot \frac{1}{4} = \frac{5}{16}$. $P(Co) = 1 - \frac{1}{16} - \frac{5}{16} = \frac{5}{8}$. Now on the fourth and last move, $P(C) = \frac{1}{4} \cdot \frac{5}{16} = \frac{5}{64}$ and $P(E) = \frac{1}{16} + \frac{1}{4} \cdot \frac{5}{16} = \frac{9}{64}$. Finally, $P(Co) = 1 - \frac{5}{64} - \frac{9}{64} = 1 - \frac{14}{64} = \frac{50}{64} = \frac{25}{32}$.

24. (AMC 12 12A 2021) Semicircle Γ has diameter $\overline{AB}$ of length 14. Circle Ω lies tangent to $\overline{AB}$ at a point P and intersects Γ at points Q and R . If $QR = 3\sqrt{3}$ and $\angle QPR = 60^\circ$, then the area of $\triangle PQR$ equals $\frac{a\sqrt{b}}{c}$, where a and c are relatively prime positive integers, and b is a positive integer not divisible by the square of any prime. What is $a + b + c$?

(A) 110 (B) 114 (C) 118 (D) 122 (E) 126

Answer: D

Solution 1 (Synthetic Geometry)

Let $O = \Gamma$ be the center of the semicircle and $X = \Omega$ be the center of the circle.

Applying the extended law of sines to $\triangle PQR$, we find the radius of $\odot X$:

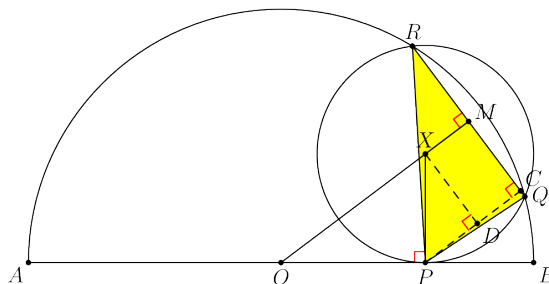
$$XP = \frac{QR}{2 \cdot \sin \angle QPR} = \frac{3\sqrt{3}}{2 \cdot \frac{\sqrt{3}}{2}} = 3.$$

Alternatively, by the inscribed angle theorem, $\triangle QRX$ is a 30° - 30° - 120° triangle with base $QR = 3\sqrt{3}$. Dividing $\triangle QRX$ into two congruent 30° - 60° - 90° triangles, we get that the radius of $\odot X$ is $XQ = XR = 3$ by the side-length ratios.

Let M be the midpoint of $\overline{QR}$. By the converse of the perpendicular chord theorem, we have $\overline{OM} \perp \overline{QR}$ and $\overline{XM} \perp \overline{QR}$. Together, points O, X , and M must be collinear.

By the SAS congruence, we have $\triangle QXM \cong \triangle RXM$, both of which are 30° - 60° - 90° triangles. By the side-length ratios, we obtain $RM = \frac{3\sqrt{3}}{2}$, $RX = 3$, and $XM = \frac{3}{2}$. By the Pythagorean theorem on right $\triangle ORM$, we get $OM = \frac{13}{2}$ and $OX = OM - XM = 5$. By the Pythagorean theorem on right $\triangle OXP$, we get $OP = 4$.

Let C be the foot of the perpendicular from P to $\overline{QR}$, and D be the foot of the perpendicular from X to $\overline{PC}$, as shown below:



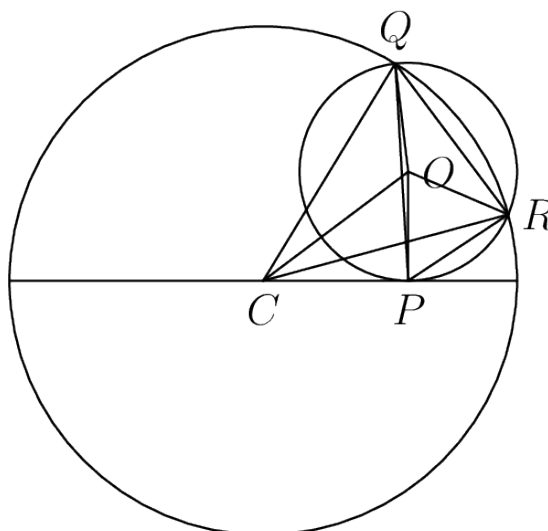
Clearly, quadrilateral $XDCM$ is a rectangle. Since $\angle XPD = \angle OXP$ by alternate interior angles, we have $\triangle XPD \sim \triangle OXP$ by the AA similarity, with the ratio of similitude $\frac{XP}{OX} = \frac{3}{5}$. Therefore, we get $PD = \frac{9}{5}$ and $PC = PD + DC = PD + XM = \frac{9}{5} + \frac{3}{2} = \frac{33}{10}$.

The area of $\triangle PQR$ is

$$\frac{1}{2} \cdot QR \cdot PC = \frac{1}{2} \cdot 3\sqrt{3} \cdot \frac{33}{10} = \frac{99\sqrt{3}}{20},$$

from which the answer is $99 + 3 + 20 = \boxed{\text{(D) } 122}$.

Solution 2 (Trigonometry)



Suppose we label the points as shown in the diagram above, where C is the center of the semicircle and O is the center of the circle tangent to $\overline{AB}$. Since $\angle QPR = 60^\circ$, we have $\angle QOR = 2 \cdot 60^\circ = 120^\circ$ and $\triangle QOR$ is a $30-30-120$ triangle, which can be split into two $30-60-90$ triangles by the altitude from O . Since $QR = 3\sqrt{3}$, we know $OQ = OR = \frac{3\sqrt{3}}{\sqrt{3}} = 3$ by $30-60-90$ triangles. The area of this part of $\triangle PQR$ is $\frac{1}{2}bh = \frac{3\sqrt{3}}{2} \cdot \frac{3}{2} = \frac{9\sqrt{3}}{4}$. We would like to add this value to the sum of the areas of the other two parts of $\triangle PQR$.

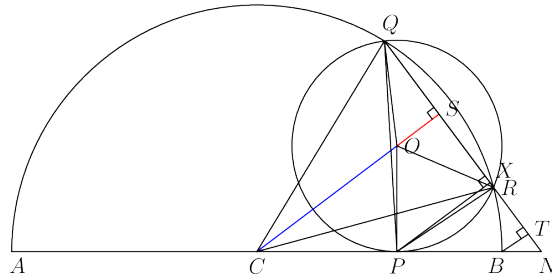
To find the areas of the other two parts of $\triangle PQR$ using the sin area formula, we need the sides and included angles. Here we know the sides but what we don't know are the angles. So it seems like we will have to use an angle from another triangle and combine them with the angles we already know to find these angles easily. We know that $\angle QOR = 120^\circ$ and triangles $\triangle COQ$ and $\triangle COR$ are congruent as they share a side, $CQ = CR$, and $OQ = OR$. Therefore $\angle COQ = \angle COR = 120^\circ$. Suppose $CO = x$. Then $3^2 + x^2 - 6x \cos 120^\circ = 7^2$, and since $\cos 120^\circ = -\frac{1}{2}$, this simplifies to $x^2 + 3x = 7^2 - 3^2 \rightarrow x^2 + 3x - 40 = 0$. This factors nicely as $(x - 5)(x + 8) = 0$, so $x = 5$ as x can't be -8 . Since $CO = 5$, $OP = 3$ and $\angle OPC = 90^\circ$, we now know that $\triangle OPC$ is a $3-4-5$ right triangle. This may be useful info for later as we might use an angle in this triangle to find the areas of the other two parts of $\triangle PQR$.

Let $\angle POC = \alpha$. Then $\sin \alpha = \frac{4}{5}$, $\cos \alpha = \frac{3}{5}$, $\angle QOP = 120 + \alpha$, and $\angle POR = 120 - \alpha$. The sum of the areas of $\triangle QOP$ and $\triangle POR$ is $3 \cdot 3 \cdot \frac{1}{2} \cdot [\sin(120 - \alpha) + \sin(120 + \alpha)] = \frac{9}{2} [\sin(120 - \alpha) + \sin(120 + \alpha)]$, which we will add to $\frac{9\sqrt{3}}{4}$ to get the area of $\triangle PQR$. Observe that

$$\sin(120 - \alpha) = \sin 120 \cos \alpha - \sin \alpha \cos 120 = \frac{\sqrt{3}}{2} \cdot \frac{3}{5} - \frac{4}{5} \cdot \frac{-1}{2} = \frac{3\sqrt{3}}{10} + \frac{4}{10} = \frac{3\sqrt{3} + 4}{10}$$

and similarly $\sin(120 + \alpha) = \frac{3\sqrt{3} - 4}{10}$. Adding these two gives $\frac{3\sqrt{3}}{5}$ and multiplying that by $\frac{9}{2}$ gets us $\frac{27\sqrt{3}}{10}$, which we add to $\frac{9\sqrt{3}}{4}$ to get $\frac{54\sqrt{3} + 45\sqrt{3}}{20} = \frac{99\sqrt{3}}{20}$. The answer is $99 + 3 + 20 = 102 + 20 = \boxed{\text{(D)} 122}$.

Solution 3 (Weighted Averages and Similar Triangles)



Define points as shown above, where $N = \overleftrightarrow{PA} \cap \overleftrightarrow{QR}$. The area of $\triangle PQR$ is simply

$$\frac{1}{2}PX \cdot QR = \frac{3\sqrt{3}}{2}PX;$$

it remains to compute the value of PX . Note that PX is simply a weighted average of BT and CS ; it is $\frac{CP}{BP}$ times closer to BT than it is to CS . Observe that

$$CS = \sqrt{CQ^2 - \left(\frac{1}{2}QR\right)^2} = \sqrt{7^2 - \left(\frac{3\sqrt{3}}{2}\right)^2} = 6.5$$

since the radius of Γ is 7 as its diameter is 14. Note also by the Extended Law of Sines the radius of Ω is $\frac{3\sqrt{3}}{2\sin 60^\circ} = 3$, so $OS = 3\cos 60^\circ = 1.5$. Since C , O , and S are collinear by symmetry we have $CO = CS - OS = 5$, so $CP = \sqrt{5^2 - 3^2} = 4$ and $BP = 7 - 4 = 3$. Therefore, $\triangle OPC$ is a 3-4-5 right triangle; $\triangle OPC \sim \triangle NSC$ since $\angle OPC = \angle CSN = 90^\circ$ and $\angle OCP = \angle NCS = \sin^{-1}\left(\frac{3}{5}\right)$. Therefore $\frac{CN}{CS} = \frac{CO}{CP} = \frac{5}{4}$ so $CN = \frac{5}{4}CS = \frac{65}{8}$. Since $\triangle BTN \sim \triangle CSN$, we have $\frac{BT}{BN} = \frac{CS}{CN} = \frac{4}{5}$. Therefore

$$BT = \frac{4}{5}BN = \frac{4}{5}(CN - 7) = \frac{4}{5} \cdot \frac{9}{8} = \frac{36}{40} = 0.9;$$

so PX is $\frac{4}{3}$ times as close to 0.9 as to 6.5; we can compute $PX = \frac{4}{7}BT + \frac{3}{7}CS = \frac{4}{7} \cdot 0.9 + \frac{3}{7} \cdot 6.5 = 3.3$. The area of $\triangle PQR$ is

$$\frac{3\sqrt{3}}{2} \cdot 3.3 = \frac{99\sqrt{3}}{20}$$

and $99 + 3 + 20 = \boxed{\text{(D)} 122}$.

Solution 4 (Similar Triangles)

Let O_1 be the center of $\odot\Gamma$, O_2 be the center of $\odot\Omega$, and M be the midpoint of $\overline{QR}$. We have $O_1M = \sqrt{7^2 - \left(\frac{3\sqrt{3}}{2}\right)^2} = \frac{13}{2}$ and by Extended Law of Sines, the radius of $\odot\Omega$ is $\frac{3\sqrt{3}}{2\sin 60^\circ} = 3$ so $O_2M = 3\cos 60^\circ = \frac{3}{2}$. Therefore $O_1O_2 = O_1M - O_2M = 5$ and $O_1P = \sqrt{5^2 - 3^2} = 4$.

Let $X = \overline{AB} \cap \overline{QR}$. Obviously

$$\angle O_1PO_2 = \angle O_1MX = 90^\circ \text{ and } \angle PO_1O_2 = \angle MO_1X = \arcsin\left(\frac{3}{5}\right)$$

so $\triangle PO_1O_2 \sim \triangle MO_1X$ with ratio $\frac{PO_1}{MO_1} = \frac{4}{\frac{13}{2}} = \frac{8}{13}$. Therefore $O_1X = \frac{13}{8} \cdot O_1O_2 = \frac{13}{8} \cdot 5 = \frac{65}{8}$ and $MX = \frac{13}{8} \cdot PO_2 = \frac{13}{8} \cdot 3 = \frac{39}{8}$.

Let H denote the foot of the altitude from P to $\overline{QR}$. Because $\overline{PH} \parallel \overline{O_1M}$, it follows that $\triangle PHX \sim \triangle O_1MX$. This similarity has ratio

$$\frac{PX}{O_1X} = 1 - \frac{O_1P}{O_1X} = 1 - \frac{4}{\frac{65}{8}} = 1 - \frac{32}{65} = \frac{33}{65}.$$

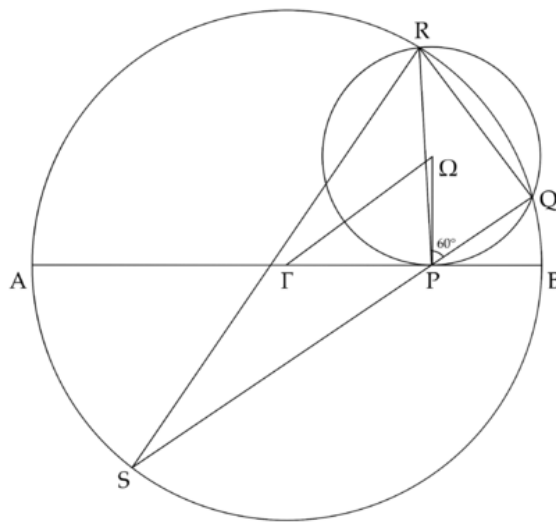
We therefore have $PH = \frac{33}{65} \cdot O_1M = \frac{33}{65} \cdot \frac{13}{2} = \frac{33}{10}$.

Finally, the area of $\triangle PQR$ is

$$\frac{1}{2} \cdot QR \cdot PH = \frac{1}{2} \cdot 3\sqrt{3} \cdot \frac{33}{10} = \frac{99\sqrt{3}}{20},$$

so the answer is $99 + 3 + 20 = \boxed{\text{(D) } 122}$.

Solution 5 (Law of Sine and Power of a Point)



By the Law of Sine in $\triangle PQR$ and its circumcircle $\odot\Omega$, $2r_\Omega = \frac{QR}{\sin 60^\circ} = \frac{3\sqrt{3}}{\frac{\sqrt{3}}{2}} = 6$,
 $r_\Omega = 3$

$$\Gamma\Omega = \sqrt{r_\Gamma^2 - \left(\frac{QR}{2}\right)^2} - \sqrt{r_\Omega^2 - \left(\frac{QR}{2}\right)^2} = \sqrt{7^2 - \left(\frac{3\sqrt{3}}{2}\right)^2} - \sqrt{3^2 - \left(\frac{3\sqrt{3}}{2}\right)^2} = \frac{13}{2} - \frac{3}{2} = 5, \quad \Gamma P = \sqrt{r_\Gamma^2 - \left(\frac{QR}{2}\right)^2} = \sqrt{7^2 - \left(\frac{3\sqrt{3}}{2}\right)^2} = \frac{13}{2}.$$

By Power of a Point in $\odot \Gamma$, $PQ \cdot PS = PA \cdot PB = (7 + 4)(7 - 4) = 33$.

By the Law of Sine in $\triangle PRS$, $\frac{PR}{PS} = \frac{\sin \angle PSR}{\sin \angle PRS}$

By the Law of Sine in $\triangle QRS$ and its circumcircle $\odot\Gamma$, $\frac{QR}{\sin \angle PSR} = 14$, $\frac{3\sqrt{3}}{\sin \angle PSR} = 14$, $\sin \angle PSR = \frac{3\sqrt{3}}{14}$, $\cos \angle PSR = \frac{13}{14}$

$$\sin \angle PRS = \sin(60^\circ - \angle PSR) = \sin 60^\circ \cos \angle PSR - \sin \angle PSR \cos 60^\circ = \frac{\sqrt{3}}{2} \cdot \frac{13}{14} - \frac{3\sqrt{3}}{14} \cdot \frac{1}{2} = \frac{5\sqrt{3}}{14}$$

$$\frac{PR}{PS} = \frac{\frac{3\sqrt{3}}{14}}{\frac{5\sqrt{3}}{14}} = \frac{3}{5}, \quad PQ \cdot PR = PQ \cdot PS \cdot \frac{PR}{PS} = 33 \cdot \frac{3}{5} = \frac{99}{5}$$

$$[PQR] = \frac{1}{2} \cdot \sin 60^\circ \cdot PQ \cdot PR = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{99}{5} = \frac{99\sqrt{3}}{20}, \quad 99 + 3 + 20 = \boxed{\text{(D)} 122}$$

Solution 6 (Analytic Geometry)

Following Solution 4, We have $O_1 (0,0)$, $O_2 (4,3)$. We can write the equation of the two circles as:

$$\odot\Gamma : x^2 + y^2 = 7^2 \dots (1)$$

$$\odot\Omega : (x - 4)^2 + (y - 3)^2 = 3^2 \dots (2)$$

By substituting (1) into (2), we get

$$8x + 6y - 65 = 0 \dots (3)$$

Notice (3) is the relationship between x value and y value, in other words, (3) is the linear equation that go through R and Q . Let the height drops from P to QR at H . Therefore, we have

$$\text{Area}\triangle PQR = \frac{1}{2} \cdot QR \cdot PH$$

So

$$QR = 3\sqrt{3}$$

And by distance formula, PH is the distance from $P (4,0)$ to $\overline{QR}$.

$$PH = \frac{|8 \cdot 4 + 6 \cdot 0 - 65|}{\sqrt{8^2 + 6^2}} = \frac{33}{10}$$

Thus, We get

$$\text{Area}\triangle PQR = \frac{1}{2} \cdot 3\sqrt{3} \cdot \frac{33}{10} = \frac{99\sqrt{3}}{20}$$

So the answer is $99 + 3 + 20 = \boxed{\text{(D)} 122}$.

25. (AMC 12 12A 2021) Let $d(n)$ denote the number of positive integers that divide n , including 1 and n . For example, $d(1) = 1$, $d(2) = 2$, and $d(12) = 6$. (This function is known as the divisor function.) Let

$$f(n) = \frac{d(n)}{\sqrt[3]{n}}.$$

There is a unique positive integer N such that $f(N) > f(n)$ for all positive integers $n \neq N$. What is the sum of the digits of N ? (A) 5 (B) 6 (C) 7 (D) 8 (E) 9

Answer: E

Solution 1 (Prime-by-Prime Optimization)

We consider the prime factorization of n :

$$n = \prod_{i=1}^k p_i^{e_i}.$$

By the Multiplication Principle, we have

$$d(n) = \prod_{i=1}^k (e_i + 1).$$

Now, we rewrite $f(n)$ as

$$f(n) = \frac{d(n)}{\sqrt[3]{n}} = \frac{\prod_{i=1}^k (e_i + 1)}{\prod_{i=1}^k p_i^{e_i/3}} = \prod_{i=1}^k \frac{e_i + 1}{p_i^{e_i/3}}.$$

As $f(n) > 0$ for all positive integers n , note that $f(a) > f(b)$ if and only if $f(a)^3 > f(b)^3$ for all positive integers a and b . So, $f(n)$ is maximized if and only if

$$f(n)^3 = \prod_{i=1}^k \frac{(e_i + 1)^3}{p_i^{e_i}}$$

is maximized.

We also allow $e = 0$, corresponding to the prime p not appearing in the prime factorization.

For each prime p , define

$$g_p(e) = \frac{(e + 1)^3}{p^e},$$

so maximizing $f(n)^3$ becomes choosing e to maximize $g_p(e)$ for each prime independently. For consecutive integers e , we have

$$\frac{g_p(e + 1)}{g_p(e)} = \frac{(e + 2)^3/p^{e+1}}{(e + 1)^3/p^e} = \frac{1}{p} \cdot \frac{(e + 2)^3}{(e + 1)^3}.$$

Now, $\frac{e+2}{e+1}$ strictly decreases, so does $\frac{g_p(e+1)}{g_p(e)}$. For each prime p , we conclude that $g_p(e)$ increases while $\frac{g_p(e+1)}{g_p(e)} > 1$ and, because these ratios strictly decrease, it decreases forever once the ratio drops below 1. Thus the first e for which $\frac{g_p(e+1)}{g_p(e)} < 1$ is the maximizing exponent.

We construct the following table:

i	p_i	e_i	$\frac{(e_i + 1)^3}{p_i^{e_i}}$	Max?
1	2	0	1	✓
		1	4	
		2	27/4	
		3	8	
		4	125/16	
2	3	0	1	✓
		1	8/3	
		2	3	
		3	64/27	
3	5	0	1	✓
		1	8/5	
		2	27/25	
4	7	0	1	✓
		1	8/7	
		2	27/49	
≥ 5	≥ 11	0	1	✓
		≥ 1	$\leq 8/11$	

Finally, the positive integer we seek is $N = 2^3 \cdot 3^2 \cdot 5^1 \cdot 7^1 = 2520$. The sum of its digits is $2 + 5 + 2 + 0 = \boxed{\text{(E)} 9}$.

Alternatively, once we notice that 3^2 is a factor of N , we can conclude that the sum of the digits of N must be a multiple of 9. Only choice **(E)** is possible.

Solution 2 (Solution 1 but Fewer Notations)

The question statement asks for the value of N that maximizes $f(N)$. Let N start out at 1; we will find what factors to multiply N by, in order for N to maximize the function.

First, we will find what power of 2 to multiply N by. If we multiply N by 2^a , the numerator of f , $d(N)$, will multiply by a factor of $a + 1$; this is because the number 2^a has $a + 1$ divisors. The denominator, $\sqrt[3]{N}$, will simply multiply by $\sqrt[3]{2^a}$. Therefore, the entire function multiplies by a factor of $\frac{a+1}{\sqrt[3]{2^a}}$. We want to find the integer value of a that maximizes this value. By inspection, this is 3. Therefore, we multiply N by 8; right now, N is 8.

Next, we will find what power of 3 to multiply N by. Similar to the previous step, we wish to find the integer value of a that maximizes $\frac{a+1}{\sqrt[3]{3^a}}$. This value, also by inspection, is 2.

We can repeat this step on the rest of the primes to get

$$N = 2^3 \cdot 3^2 \cdot 5 \cdot 7$$

but from 11 on, $a = 0$ will maximize the value of the function, so the prime is not a factor in N . We evaluate N to be 2520, so the answer is $\boxed{\text{(E)} 9}$.

Solution 3 (Fast)

Using the answer choices to our advantage, we can show that N must be divisible by 9 without explicitly computing N , by exploiting the following fact:

Claim: If n is not divisible by 3, then $f(9n) > f(3n) > f(n)$.

Proof: Since $d(\cdot)$ is a **multiplicative function**, we have $d(3n) = d(3)d(n) = 2d(n)$ and $d(9n) = 3d(n)$. Then

$$f(3n) = \frac{2d(n)}{\sqrt[3]{3n}} \approx 1.38f(n)$$

$$f(9n) = \frac{3d(n)}{\sqrt[3]{9n}} \approx 1.44f(n)$$

Note that the values $\frac{2}{\sqrt[3]{3}}$ and $\frac{3}{\sqrt[3]{9}}$ do not have to be explicitly computed; we only need the fact that $\frac{3}{\sqrt[3]{9}} > \frac{2}{\sqrt[3]{3}} > 1$ which is easy to show by hand.

The above claim automatically implies N is a multiple of 9: if N was not divisible by 9, then $f(9N) > f(N)$ which is a contradiction, and if N was divisible by 3 and not 9, then $f(3N) > f(N) > f\left(\frac{N}{3}\right)$, also a contradiction. Then the sum of digits of N must be a multiple of 9, so only choice **(E) 9** works.

-scrabbler94

Solution 4 (Guesswork)

The problem mentions the sum of digits - recall that if a number is divisible by 9, then so is the sum of its digits. Guess that the answer must therefore be **(E) 9**.

- youtube.com/indianmathguy

Solution 5 (Simple Explanation)

The aim of this problem is to maximize $f(N)$.

Let's start with when N has one prime factor. If this is the case, let a be its exponent. Clearly, 2 must be the prime factor because regardless of what prime we use, the numerator will only depend on a , so we must minimize the denominator. Therefore, in this case, we want to maximize $\frac{a+1}{\sqrt[3]{2^a}}$ (since the N will have $a+1$ factors). By inspection, we find that $a=3$ does the job.

Now notice that we can add another prime factor to N . Since we can split up prime factors and multiplication under prime factorization and cube roots, we just need to maximize the same process for the second prime factor. Again, we go with the next least prime factor to minimize the denominator which is 3. To clarify, we want to maximize $\frac{b+1}{\sqrt[3]{3^b}}$ (letting the exponent be b). Inspection again gives that $b=2$ is maximal.

We repeat this process two more times, adding a prime factor for 5 and 7. We find by inspection that the maximizing power of 5 is 1 and the maximizing power of 7 is also 1.

Now, notice that if $n > 8$, then regardless of what exponent we put in the numerator, $f(n) = \frac{d(n)}{\sqrt[3]{n}}$ will be less than 1. This is bad, because then our maximized value of $f(N)$ will decrease. Noting that our next least prime factor is 11, we know that we are done.

We evaluate $N = 2^3 * 3^2 * 5^1 * 7^1$ to be 2520, so the answer is $2+5+2+0 = \mathbf{(E) 9}$.

Calculus Remark We can find the maximum of the function $f(x) = \frac{x+1}{a^{x/3}}$ for some prime a by taking the derivative and setting it equal to 0. We get $f'(x) = e^{-\frac{x \ln a}{3}} - \frac{\ln a}{3} x e^{-\frac{x \ln a}{3}} - \frac{\ln a}{3} e^{-\frac{x \ln a}{3}} = 0 \rightarrow 1 - \frac{\ln a}{3} x - \frac{\ln a}{3} = 0 \rightarrow \frac{\ln a}{3} x = 1 - \frac{\ln a}{3} \rightarrow x = \frac{3}{\ln a} \left(1 - \frac{\ln a}{3}\right) = \frac{3}{\ln a} - 1$. This gives the exponents you need for each prime in the prime factorization of N ; however, it outputs real numbers, not positive integers.

§5 AMC 12 12B Fall 2021

1. (AMC 12 12B Fall 2021) What is the value of $1234 + 2341 + 3412 + 4123$?
 (A) 10,000 (B) 10,010 (C) 10,110 (D) 11,000 (E) 11,110

Answer: E

Solution 1

We see that 1, 2, 3, and 4 each appear in the ones, tens, hundreds, and thousands digit exactly once. Since $1 + 2 + 3 + 4 = 10$, we find that the sum is equal to

$$10 \cdot (1 + 10 + 100 + 1000) = \boxed{\text{(E)} 11,110}.$$

Note that it is equally valid to manually add all four numbers together to get the answer.

Solution 2

We have

$$1234 + 2341 + 3412 + 4123 = 1111(1 + 2 + 3 + 4) = \boxed{\text{(E)} 11,110}.$$

Steven Chen (www.professorchen.edu.com)

Solution 3

We see that the units digit must be 0, since $4 + 3 + 2 + 1$ is 0. But every digit from there, will be a 1 since we have that each time afterwards, we must carry the 1 from the previous sum. The answer choice that satisfies these conditions is $\boxed{\text{(E)} 11,110}$.

Solution 4 (Brute Force)

We can simply add the numbers. $1234 + 2341 + 3412 + 4123 = 11110 \Rightarrow \boxed{\text{(E)}}.$

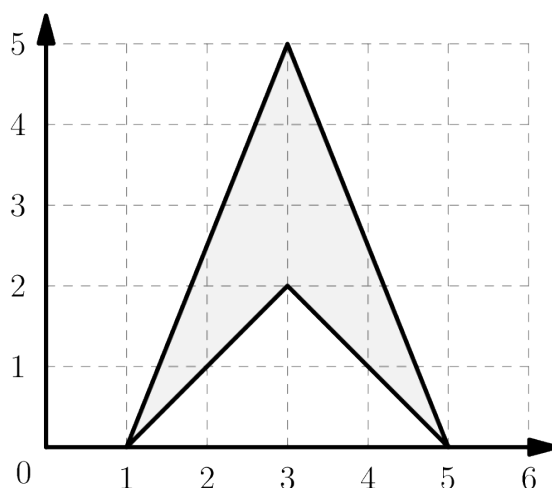
Note: Although this would not take terribly long, it is not recommended to do this in a real contest.

Solution 5 (Primary School But Also Similar to Solution 4)

We can simply just add each place value.

$$1000 + 2000 + 3000 + 4000 = 10,000 + 200 + 300 + 400 + 100 = 1,000 + 30 + 40 + 10 + 20 = 100 + 4 + 1 + 2 + 3 = 10 \quad 10,000 + 1,000 + 100 + 10 = \boxed{\text{(E)} 11,110}.$$

2. (AMC 12 12B Fall 2021) What is the area of the shaded figure shown below?



- (A) 4 (B) 6 (C) 8 (D) 10 (E) 12

Answer: B

Solution 1 (Area Addition)

The line of symmetry divides the shaded figure into two congruent triangles, each with base 3 and height 2.

Therefore, the area of the shaded figure is

$$2 \cdot \left(\frac{1}{2} \cdot 3 \cdot 2 \right) = 2 \cdot 3 = \boxed{\text{(B)} 6}.$$

MRENTHUSIASM Wilhelm Z

Solution 2 (Area Subtraction)

To find the area of the shaded figure, we subtract the area of the smaller triangle (base 4 and height 2) from the area of the larger triangle (base 4 and height 5):

$$\frac{1}{2} \cdot 4 \cdot 5 - \frac{1}{2} \cdot 4 \cdot 2 = 10 - 4 = \boxed{\text{(B)} 6}.$$

MRENTHUSIASM Steven Chen (www.professorchen.edu.com)

Solution 3 (Shoelace Theorem)

The consecutive vertices of the shaded figure are $(1, 0)$, $(3, 2)$, $(5, 0)$, and $(3, 5)$. By the **Shoelace Theorem**, the area is

$$\frac{1}{2} \cdot |(1 \cdot 2 + 3 \cdot 0 + 5 \cdot 5 + 3 \cdot 0) - (0 \cdot 3 + 2 \cdot 5 + 0 \cdot 3 + 5 \cdot 1)| = \frac{1}{2} \cdot 12 = \boxed{\text{(B)} 6}.$$

Taco12 I-AM-DA-KING

Solution 4 (Pick's Theorem)

We have 4 lattice points in the interior and 6 lattice points on the boundary. By **Pick's Theorem**, the area of the shaded figure is

$$4 + \frac{6}{2} - 1 = 4 + 3 - 1 = \boxed{\text{(B)} 6}.$$

danprathab

3. (*AMC 12 12B Fall 2021*) At noon on a certain day, Minneapolis is N degrees warmer than St. Louis. At 4:00 the temperature in Minneapolis has fallen by 5 degrees while the temperature in St. Louis has risen by 3 degrees, at which time the temperatures in the two cities differ by 2 degrees. What is the product of all possible values of N ? (A) 10 (B) 30 (C) 60 (D) 100 (E) 120

Answer: C

Solution 1 (Two Variables)

At noon on a certain day, let M and L be the temperatures (in degrees) in Minneapolis and St. Louis, respectively. It follows that $M = L + N$.

At 4:00, we get

$$\begin{aligned} |(M - 5) - (L + 3)| &= 2 \\ |M - L - 8| &= 2 \\ |N - 8| &= 2. \end{aligned}$$

We have two cases:

- a) If $N - 8 = 2$, then $N = 10$.
- b) If $N - 8 = -2$, then $N = 6$.

Together, the product of all possible values of N is $10 \cdot 6 = \boxed{\text{(C) } 60}$.

Solution 2 (One Variable)

At noon on a certain day, the difference of temperatures in Minneapolis and St. Louis is N degrees.

At 4:00, the difference of temperatures in Minneapolis and St. Louis is $N - 8$ degrees.

It follows that

$$|N - 8| = 2.$$

We continue with the casework in Solution 1 to get the answer $\boxed{\text{(C) } 60}$.

Solution 3

Let the temperature in Minneapolis be N . Then St. Louis is 0 degrees. $N - 5 = 3 \pm 2$, $N = -10, -6$, $-10 \cdot -6 = \boxed{\text{(C) } 60}$.

4. (*AMC 12 12B Fall 2021*) Let $n = 8^{2022}$. Which of the following is equal to $\frac{n}{4}$? (A) 4^{1010} (B) 2^{2022} (C) 8^{2018} (D) 4^{3031} (E) 4^{3032}

Answer: E

Solution 1

We have

$$n = 8^{2022} = \left(8^{\frac{2}{3}}\right)^{3033} = 4^{3033}.$$

Therefore,

$$\frac{n}{4} = \boxed{\text{(E) } 4^{3032}}.$$

kingofpineapplz

Solution 2

The requested value is

$$\frac{8^{2022}}{4} = \frac{2^{6066}}{4} = \frac{2^{6066}}{2^2} = 2^{6064} = \boxed{\text{(E)} 4^{3032}}.$$

NH14

Solution 3

If we rewrite everything in powers of 2, we get: $n = 8^{2022} = (2^3)^{2022} = 2^{6064} = (4^{\frac{1}{2}})^{6064} = 4^{\frac{6064}{2}} = \boxed{\text{(E)} 4^{3032}}.$

- abed_nadir (youtube.com/@indianmathguy)

Solution 4

$$n = 8^{2022} \cdot 8^{2022}/4 = 2 * 8^{2021} = 2 * 2^{6063} = 2^{6064} = \boxed{\text{(E)} 4^{3032}}.$$

-sphericalfox

5. (AMC 12 12B Fall 2021) Call a fraction $\frac{a}{b}$, not necessarily in the simplest form, *special* if a and b are positive integers whose sum is 15. How many distinct integers can be written as the sum of two, not necessarily different, special fractions?
 (A) 9 (B) 10 (C) 11 (D) 12 (E) 13

Answer: C

Solution 1

The special fractions are

$$\frac{1}{14}, \frac{2}{13}, \frac{3}{12}, \frac{4}{11}, \frac{5}{10}, \frac{6}{9}, \frac{7}{8}, \frac{8}{7}, \frac{9}{6}, \frac{10}{5}, \frac{11}{4}, \frac{12}{3}, \frac{13}{2}, \frac{14}{1}.$$

We rewrite them in the simplest form:

$$\frac{1}{14}, \frac{2}{13}, \frac{1}{4}, \frac{4}{11}, \frac{1}{2}, \frac{2}{3}, \frac{7}{8}, 1\frac{1}{7}, 1\frac{1}{2}, 2, 2\frac{3}{4}, 4, 6\frac{1}{2}, 14.$$

Note that two unlike fractions in the simplest form cannot sum to an integer. So, we only consider the fractions whose denominators appear more than once:

$$\frac{1}{4}, \frac{1}{2}, 1\frac{1}{2}, 2, 2\frac{3}{4}, 4, 6\frac{1}{2}, 14.$$

For the set $\{2, 4, 14\}$, two elements (not necessarily different) can sum to 4, 6, 8, 16, 18, 28.

For the set $\{\frac{1}{2}, 1\frac{1}{2}, 6\frac{1}{2}\}$, two elements (not necessarily different) can sum to 1, 2, 3, 7, 8, 13.

For the set $\{\frac{1}{4}, 2\frac{3}{4}\}$, two elements (not necessarily different) can sum to 3.

Together, there are $\boxed{\text{(C)} 11}$ distinct integers that can be written as the sum of two, not necessarily different, special fractions:

$$1, 2, 3, 4, 6, 7, 8, 13, 16, 18, 28.$$

KingRavi samrocksnature Wilhelm Z MRENTHUSIASM Steven Chen (www.professorchen.edu.com)

Solution 2

Let $a = 15 - b$, so the special fraction is

$$\frac{a}{b} = \frac{15 - b}{b} = \frac{15}{b} - 1.$$

We can ignore the -1 part and only focus on $\frac{15}{b}$.

The integers are $\frac{15}{1}, \frac{15}{3}, \frac{15}{5}$, which are 15, 5, 3, respectively. We get 30, 20, 18, 10, 8, 6 from this group of numbers.

The halves are $\frac{15}{2}, \frac{15}{6}, \frac{15}{10}$, which are $7\frac{1}{2}, 2\frac{1}{2}, 1\frac{1}{2}$, respectively. We get 15, 10, 9, 5, 4, 3 from this group of numbers.

The quarters are $\frac{15}{4}, \frac{15}{12}$, which are $3\frac{3}{4}, 1\frac{1}{4}$, respectively. We get 5 from this group of numbers.

Note that 10 and 5 each appear twice. Therefore, the answer is **(C) 11**.

Solution 3 (Casework)

We split this up into two cases:

Case 1: integer + integer

The whole numbers we have are $\frac{10}{5}$ (or 2), $\frac{12}{3}$ (or 4), and $\frac{14}{1}$ (or 14). There are $\binom{3}{2} = 3$ ways to choose different-numbered pairs and 3 ways to choose the same-numbered pairs. So, $3 + 3 = 6$.

Case 2: fraction + fraction

The fractions we have are $\frac{5}{10}$ (or $\frac{1}{2}$), $\frac{9}{6}$ (or $\frac{3}{2}$), and $\frac{13}{2}$. Similarly, there are $\binom{3}{2} = 3$ ways to choose different-numbered pairs and 3 ways to choose the same-numbered pairs. So, $3 + 3 = 6$.

Thus, $6 + 6 = 12$.

So now you would just go ahead and innocently choose **(D) 12**, right? No! We overcounted 8, as $\frac{13}{2} + \frac{9}{6} = \frac{12}{3} + \frac{12}{3} = 8$. Therefore, the correct answer is actually $12 - 1 = \mathbf{(C) 11}$.

6. (AMC 12 12B Fall 2021) The greatest prime number that is a divisor of 16,384 is 2 because $16,384 = 2^{14}$. What is the sum of the digits of the greatest prime number that is a divisor of 16,383?

(A) 3 (B) 7 (C) 10 (D) 16 (E) 22

Answer: C

Solution

We have

$$\begin{aligned} 16383 &= 2^{14} - 1 \\ &= (2^7 + 1)(2^7 - 1) \\ &= 129 \cdot 127 \end{aligned}$$

Since 129 is composite, 127 is the largest prime which can divide 16383. The sum of 127's digits is $1 + 2 + 7 = \mathbf{(C) 10}$.

Note Note that you can quickly tell that $2^7 - 1$ is prime because it is a **Mersenne prime**.

7. (AMC 12 12B Fall 2021) Which of the following conditions is sufficient to guarantee that integers x , y , and z satisfy the equation

$$x(x - y) + y(y - z) + z(z - x) = 1?$$

- (A) $x > y$ and $y = z$ (B) $x = y - 1$ and $y = z - 1$ (C) $x = z + 1$ and $y = x + 1$
 (D) $x = z$ and $y - 1 = x$ (E) $x + y + z = 1$

Answer: D

Solution 1 (Completing the Square)

It is obvious x , y , and z are symmetrical. We are going to solve the problem by Completing the Square.

$$x^2 + y^2 + z^2 - xy - yz - zx = 1 \quad 2x^2 + 2y^2 + 2z^2 - 2xy - 2yz - 2zx = 2 \quad (x - y)^2 + (y - z)^2 + (z - x)^2 = 2$$

Because

8. (AMC 12 12B Fall 2021) The product of the lengths of the two congruent sides of an obtuse isosceles triangle is equal to the product of the base and twice the triangle's height to the base. What is the measure, in degrees, of the vertex angle of this triangle?

- (A) 105 (B) 120 (C) 135 (D) 150 (E) 165

Answer: D

Solution 1 (Area)

Let the lengths of the two congruent sides of the triangle be x , then the product desired is x^2 .

Notice that the product of the base and twice the height is 4 times the area of the triangle.

Set the vertex angle to be a , we derive the equation:

$$x^2 = 4 \left(\frac{1}{2} x^2 \sin(a) \right) \sin(a) = \frac{1}{2}$$

As the triangle is obtuse, $a = 150^\circ$ only. We get (D) 150.

Solution 2

Denote by a the length of each congruent side. Denote by θ the degree measure of each acute angle. Denote by ϕ the degree measure of the obtuse angle.

Hence, this problem tells us the following relationship:

$$a^2 = 2a \cos \theta \cdot 2a \sin \theta.$$

Hence,

$$\begin{aligned} 1 &= 2 \cdot 2 \sin \theta \cos \theta \\ &= 2 \sin 2\theta \\ &= 2 \sin (180^\circ - 2\theta) \\ &= 2 \sin \phi. \end{aligned}$$

Hence, $\phi = 150^\circ$.

Therefore, the answer is (D) 150.

9. (AMC 12 12B Fall 2021) Triangle ABC is equilateral with side length 6. Suppose that O is the center of the inscribed circle of this triangle. What is the area of the circle passing through A , O , and C ?

(A) 9π (B) 12π (C) 18π (D) 24π (E) 27π

Answer: B

Solution 1 (Cosine Rule)

Construct the circle that passes through A , O , and C , centered at X .

Also notice that $\overline{OA}$ and $\overline{OC}$ are the angle bisectors of angle $\angle BAC$ and $\angle BCA$ respectively. We then deduce $\angle AOC = 120^\circ$.

Consider another point M on Circle X opposite to point O .

As $AOCM$ is an inscribed quadrilateral of Circle X , $\angle AMC = 180^\circ - 120^\circ = 60^\circ$.

Afterward, deduce that $\angle AXC = 2\angle AMC = 120^\circ$.

By the Cosine Rule, we have the equation: (where r is the radius of circle X)

$$2r^2(1 - \cos(120^\circ)) = 6^2 \quad r^2 = 12$$

The area is therefore $\pi r^2 = \boxed{\text{(B)} 12\pi}$.

Solution 2

We have $\angle AOC = 120^\circ$.

Denote by R the circumradius of $\triangle AOC$. In $\triangle AOC$, the law of sines implies

$$2R = \frac{AC}{\sin \angle AOC} = 4\sqrt{3}.$$

Hence, the area of the circumcircle of $\triangle AOC$ is

$$\pi R^2 = 12\pi.$$

Therefore, the answer is $\boxed{\text{(B)} 12\pi}$.

Solution 3

As in the previous solution, construct the circle that passes through A , O , and C , centered at X . Let Y be the intersection of $\overline{OX}$ and $\overline{AB}$.

Note that since $\overline{OA}$ is the angle bisector of $\angle BAC$ that $\angle OAC = 30^\circ$. Also by symmetry, $\overline{OX} \perp \overline{AB}$ and $AY = 3$. Thus $\tan(30^\circ) = \frac{OY}{3}$ so $OY = \sqrt{3}$.

Let r be the radius of circle X , and note that $AX = OX = r$. So $\triangle AYX$ is a right triangle with legs of length 3 and $r - \sqrt{3}$ and hypotenuse r . By Pythagoras, $3^2 + (r - \sqrt{3})^2 = r^2$. So $r = 2\sqrt{3}$.

Thus the area is $\pi r^2 = \boxed{\text{(B)} 12\pi}$.

-SharpeMind

Solution 5 (SIMPLE)

The semiperimeter is $\frac{6+6+6}{2} = 9$ units. The area of the triangle is $9\sqrt{3}$ units squared. By the formula that says that the area of the triangle is its semiperimeter times its inradius, the inradius $r = \sqrt{3}$. As $\angle AOC = 120^\circ$, we can form an altitude from point O to side AC at point M , forming two 30-60-90 triangles. As $CM = MA = 3$, we can solve for $OC = 2\sqrt{3}$. Now, call the center of the circle we are looking for X .

From here we can now utilize the fact that since the inscribed angle $\angle AOC = 120^\circ$ that the arclength that the angle describes is 240° . This leaves us 120° ($360 - 240$) left to be formed by the central angle, which by the Central Angle Theorem must be 120° as well. Note that now we have a parallelogram with the side OC (which is $2\sqrt{3}$) being opposite and therefor equivalent to the radius XA . Now, the area of the circle is just $\pi * (2 * \sqrt{3})^2 = 12\pi$. Select **[B]**.

Note For Solution 5 One must be careful as one can easily accidentally get this problem CORRECT by prematurely saying the radius is $2\sqrt{3}$ BEFORE proving that OC is a part of a parallelogram. Otherwise, your statement is simply saying that the Circumcircle's area is 12π . Meaning that although your answer was correct the reasoning was not.

Solution 6 (Ptolemy)

Call the diameter of the circle d . If we extend points A and C to meet at a point on the circle and call it E , then $\triangle OAE = \triangle OCE$. Note that both triangles are right, since their hypotenuse is the diameter of the circle. Therefore, $CE = AE = \sqrt{d^2 - 12}$. We know this since $OC = OA = OB$ and OC is the hypotenuse of a $30 - 60 - 90$ right triangle, with the longer leg being $\frac{6}{2} = 3$ so $OC = 2\sqrt{3}$. Applying Ptolemy's Theorem on cyclic quadrilateral $OCEA$, we get $2(\sqrt{d^2 - 12}) \cdot 2\sqrt{3} = 6d$. Squaring and solving we get $d^2 = 48 \implies (2r)^2 = 48$ so $r^2 = 12$. Therefore, the area of the circle is **[12π]**

10. (AMC 12 12B Fall 2021) What is the sum of all possible values of t between 0 and 360 such that the triangle in the coordinate plane whose vertices are

$$(\cos 40^\circ, \sin 40^\circ), (\cos 60^\circ, \sin 60^\circ), \text{ and } (\cos t^\circ, \sin t^\circ)$$

is isosceles?

- (A) 100 (B) 150 (C) 330 (D) 360 (E) 380

Answer: E

Solution

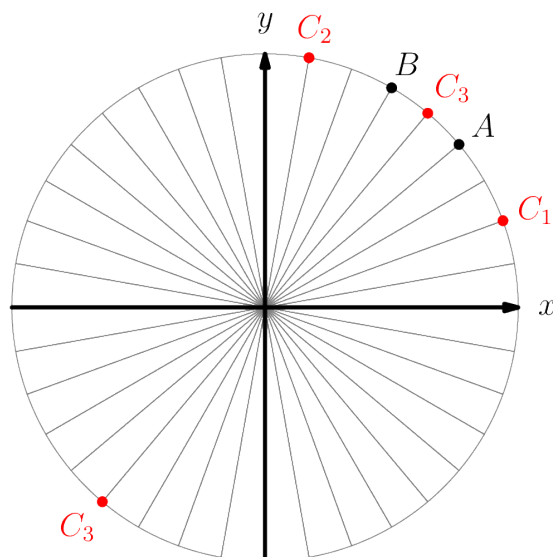
Let $A = (\cos 40^\circ, \sin 40^\circ)$, $B = (\cos 60^\circ, \sin 60^\circ)$, and $C = (\cos t^\circ, \sin t^\circ)$. We apply casework to the legs of isosceles $\triangle ABC$:

- $AB = AC$ Note that A must be the midpoint of $\widehat{BC}$. It follows that $C = (\cos 20^\circ, \sin 20^\circ)$, so $t = 20$.
- $BA = BC$ Note that B must be the midpoint of $\widehat{AC}$. It follows that $C = (\cos 80^\circ, \sin 80^\circ)$, so $t = 80$.
- $CA = CB$ Note that C must be the midpoint of $\widehat{AB}$. It follows that $C = (\cos 50^\circ, \sin 50^\circ)$ or $C = (\cos 230^\circ, \sin 230^\circ)$, so $t = 50$ or $t = 230$.

Together, the sum of all such possible values of t is $20 + 80 + 50 + 230 = \mathbf{(E) 380}$.

Remark

The following diagram shows all possible locations of C :



11. (AMC 12 12B Fall 2021) Una rolls 6 standard 6-sided dice simultaneously and calculates the product of the 6 numbers obtained. What is the probability that the product is divisible by 4? (A) $\frac{3}{4}$ (B) $\frac{57}{64}$ (C) $\frac{59}{64}$ (D) $\frac{187}{192}$ (E) $\frac{63}{64}$

Answer: C

Solution

We will use complementary counting to find the probability that the product is not divisible by 4. Then, we can find the probability that we want by subtracting this from 1. We split this into 2 cases.

Case 1: The product is not divisible by 2.

We need every number to be odd, and since the chance we roll an odd number is $\frac{1}{2}$, our probability is $\left(\frac{1}{2}\right)^6 = \frac{1}{64}$.

Case 2: The product is divisible by 2, but not by 4.

We need 5 numbers to be odd, and one to be divisible by 2, but not by 4. There is a $\frac{1}{2}$ chance that an odd number is rolled, a $\frac{1}{3}$ chance that we roll a number satisfying the second condition (only 2 and 6 work), and 6 ways to choose the order in which the even number appears.

Our probability is $\left(\frac{1}{2}\right)^5 \left(\frac{1}{3}\right) \cdot 6 = \frac{1}{16}$.

Therefore, the probability the product is not divisible by 4 is $\frac{1}{64} + \frac{1}{16} = \frac{5}{64}$.

Our answer is $1 - \frac{5}{64} = \boxed{\text{(C)} \frac{59}{64}}$.

Solution 2(Pretty much same as above)

Complementary counting implies that we can have either 0 factors of 2 or just 1 factor of 2.

Case 1: 0 factors of 2

The only odd numbers are 1, 3, 5 on a die so we have these 3 odd numbers with 6 positions so 3^6 ways here.

Case 2: 1 factor of 2

The only even numbers that have only one factor of 2 on a die are 2, 6. On the 6 positions, there are 6 ways to place one of these even numbers and the rest of the 5 positions must be one of the three odd numbers. Since we could have either 2 or 6, we have

$$2 \cdot 6 \cdot 3^5 = 4 \cdot 3^6.$$

Thus, we have $3^6 + 4 \cdot 3^6 = 5 \cdot 3^6$ cases that don't work.

By complementary counting we get

$$1 - \frac{5 \cdot 3^6}{6^6} = 1 - \frac{5}{2^6} = \boxed{\frac{59}{64}}$$

12. (AMC 12 12B Fall 2021) For n a positive integer, let $f(n)$ be the quotient obtained when the sum of all positive divisors of n is divided by n . For example,

$$f(14) = (1 + 2 + 7 + 14) \div 14 = \frac{12}{7}$$

What is $f(768) - f(384)$? (A) $\frac{1}{768}$ (B) $\frac{1}{192}$ (C) 1 (D) $\frac{4}{3}$ (E) $\frac{8}{3}$

Answer: B

Solution 1

The prime factorizations of 768 and 384 are $2^8 \cdot 3$ and $2^7 \cdot 3$, respectively. Note that $f(n)$ is the sum of all fractions of the form $\frac{1}{d}$, where d is a positive divisor of n . By geometric series, it follows that

$$\begin{aligned} f(768) &= \left(\sum_{k=0}^8 \frac{1}{2^k} \right) + \left(\sum_{k=0}^7 \frac{1}{2^k \cdot 3} \right) = \frac{511}{256} + \frac{511}{768} = \frac{2044}{768}, \\ f(384) &= \left(\sum_{k=0}^7 \frac{1}{2^k} \right) + \left(\sum_{k=0}^6 \frac{1}{2^k \cdot 3} \right) = \frac{255}{128} + \frac{255}{384} = \frac{1020}{384}. \end{aligned}$$

Therefore, the answer is $f(768) - f(384) = \boxed{\text{(B)} \frac{1}{192}}$.

Solution 2

The prime factorization of 384 is $2^7 \cdot 3$, so each of its positive divisors is of the form 2^m or $2^m \cdot 3$ for some integer m such that $0 \leq m \leq 7$. We will use this fact to calculate the sum of all its positive divisors. Note that

$$2^m + 2^m \cdot 3 = 2^m \cdot (1 + 3) = 2^m \cdot 4 = 2^m \cdot 2^2 = 2^{m+2}$$

is the sum of the two forms of positive divisors for all such m . By geometric series, the sum of all positive divisors of 384 is

$$\sum_{k=2}^9 2^k = \frac{2^{10} - 2^2}{2 - 1} = 1020,$$

from which $f(384) = \frac{1020}{384}$. Similarly, since the prime factorization of 768 is $2^8 \cdot 3$, we have $f(768) = \frac{2044}{768}$.

Therefore, the answer is $f(768) - f(384) = \boxed{\text{(B)} \frac{1}{192}}$.

Solution 3

Let $\sigma(n)$ denotes the sum of all positive divisors of n , so $f(n) = \sigma(n) \div n$.

Suppose that $n = \prod_{i=1}^k p_i^{e_i}$ is the prime factorization of n . Since $\sigma(n)$ is multiplicative, we have

$$\sigma(n) = \sigma\left(\prod_{i=1}^k p_i^{e_i}\right) = \prod_{i=1}^k \sigma(p_i^{e_i}) = \prod_{i=1}^k \left(\sum_{j=0}^{e_i} p_i^j\right) = \prod_{i=1}^k \frac{p_i^{e_i+1} - 1}{p_i - 1}$$

by geometric series.

The prime factorizations of 768 and 384 are $2^8 \cdot 3$ and $2^7 \cdot 3$, respectively. Note that

$$\begin{aligned} f(768) &= \left(\frac{2^9 - 1}{2 - 1} \cdot \frac{3^2 - 1}{3 - 1}\right) \div 768 = \frac{2044}{768}, \\ f(384) &= \left(\frac{2^8 - 1}{2 - 1} \cdot \frac{3^2 - 1}{3 - 1}\right) \div 384 = \frac{1020}{384}. \end{aligned}$$

Therefore, the answer is $f(768) - f(384) = \boxed{\text{(B)} \frac{1}{192}}$.

Solution 4

Note that

$$\begin{aligned} 384 &= 2^7 \cdot 3, \\ 768 &= 2^8 \cdot 3. \end{aligned}$$

Both numbers take the form $2^n \cdot 3$. Define a function $S(n)$ as being the sum of the factors for any number $2^n \cdot 3$, where n is a nonnegative integer.

If you list out the factors of a number $2^n \cdot 3$, all the factors are even except for 1 and 3. So to produce a list of factors for the number $2^{n+1} \cdot 3$, you can multiply all the factors on the first list by 2 and then add on 1 and 3. Thus, $S(n) = 2S(n-1) + 4$.

We need to find $\frac{S(8)}{768} - \frac{S(7)}{384}$. This can be written as $\frac{2S(7)+4-2S(7)}{768}$. Our recursive formula states that $S(n) = 2S(n-1) + 4$ so $S(n) - 2S(n-1) = 4$. Thus, the fraction simplifies to $\frac{4}{768} = \boxed{\text{(B)} \frac{1}{192}}$. Note that we never needed to figure out what $S(8)$ or $S(7)$ were, sparing us a lot of calculation.

Solution 5

We know that

$$\begin{aligned} f(768) &= \frac{1 + 2 + 3 + 4 + \cdots + 768}{768}, \\ f(384) &= \frac{1 + 2 + 3 + 4 + \cdots + 384}{384}. \end{aligned}$$

We want to find the difference of these so

$$\frac{1 + 2 + 3 + 4 + \cdots + 768}{768} - \frac{2(1 + 2 + 3 + 4 + \cdots + 384)}{2 \cdot 384} = \frac{1 + 2 + 3 + 4 + \cdots + 768}{768} - \frac{2 + 4 + 6 + 8 + \cdots + 768}{768}$$

We are only left with the even divisors of 768 on the fraction on the right so the answer would be the sum of all odd divisors of 768 (1 and 3) over 768:

$$\frac{1 + 3}{768} = \boxed{\text{(B)} \frac{1}{192}}.$$

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Solution 6

As in Solution 1, the prime factorization of 768 is $2^8 \cdot 3$, and the prime factorization of 384 is $2^7 \cdot 3$. Recalling the sum of factors formula and sum of the powers of 2 formula, we find that the sum of the factors of 768 is $(1 + 2^1 + 2^2 + 2^3 + 2^4 + 2^5 + 2^6 + 2^7 + 2^8)(1 + 3^1) = (2^9 - 1)(4) = (511)(4)$, and the sum of factors of 384 is $1 + 2^1 + 2^2 + 2^3 + 2^4 + 2^5 + 2^6 + 2^7)(1 + 3^1) = (2^8 - 1)(4) = (255)(4)$. Then $f(768) = \frac{(511)(4)}{768} = \frac{511}{192}$, and $f(384) = \frac{(255)(4)}{384} = \frac{510}{192}$. So, $f(768) - f(384) = \frac{511}{192} - \frac{510}{192} = \boxed{\text{(B)} \frac{1}{192}}$.

Note Note that you can cut away some of the arithmetic by realizing that $(2^8 - 1) \cdot 2$ is 1 less than $2^9 - 1$. bird987

Solution 7

Observe that $f(n)$ equals the sum of the reciprocals of the positive divisors of n . Since 768 is a multiple of 384, every divisor of 384 will also be a divisor of 768. Then $f(768) - f(384)$ equals the sum of the reciprocals of the positive divisors of 768 which are not divisors of 384. Since $768 = 2^8 \cdot 3$ and $384 = 2^7 \cdot 3$, the divisors of 768 which are not divisors of 384 are $2^8 = 256$ and $2^8 \cdot 3 = 768$. Therefore

13. (AMC 12 12B Fall 2021) Let $c = \frac{2\pi}{11}$. What is the value of

$$\frac{\sin 3c \cdot \sin 6c \cdot \sin 9c \cdot \sin 12c \cdot \sin 15c}{\sin c \cdot \sin 2c \cdot \sin 3c \cdot \sin 4c \cdot \sin 5c}?$$

- (A) -1 (B) $-\frac{\sqrt{11}}{5}$ (C) $\frac{\sqrt{11}}{5}$ (D) $\frac{10}{11}$ (E) 1

Answer: E

Solution

Plugging in c , we get

$$\frac{\sin 3c \cdot \sin 6c \cdot \sin 9c \cdot \sin 12c \cdot \sin 15c}{\sin c \cdot \sin 2c \cdot \sin 3c \cdot \sin 4c \cdot \sin 5c} = \frac{\sin \frac{6\pi}{11} \cdot \sin \frac{12\pi}{11} \cdot \sin \frac{18\pi}{11} \cdot \sin \frac{24\pi}{11} \cdot \sin \frac{30\pi}{11}}{\sin \frac{2\pi}{11} \cdot \sin \frac{4\pi}{11} \cdot \sin \frac{6\pi}{11} \cdot \sin \frac{8\pi}{11} \cdot \sin \frac{10\pi}{11}}.$$

Since $\sin(x + 2\pi) = \sin(x)$, $\sin(2\pi - x) = \sin(-x)$, and $\sin(-x) = -\sin(x)$, we get

$$\frac{\sin \frac{6\pi}{11} \cdot \sin \frac{12\pi}{11} \cdot \sin \frac{18\pi}{11} \cdot \sin \frac{24\pi}{11} \cdot \sin \frac{30\pi}{11}}{\sin \frac{2\pi}{11} \cdot \sin \frac{4\pi}{11} \cdot \sin \frac{6\pi}{11} \cdot \sin \frac{8\pi}{11} \cdot \sin \frac{10\pi}{11}} = \frac{\sin \frac{6\pi}{11} \cdot \sin \frac{-10\pi}{11} \cdot \sin \frac{-4\pi}{11} \cdot \sin \frac{2\pi}{11} \cdot \sin \frac{8\pi}{11}}{\sin \frac{2\pi}{11} \cdot \sin \frac{4\pi}{11} \cdot \sin \frac{6\pi}{11} \cdot \sin \frac{8\pi}{11} \cdot \sin \frac{10\pi}{11}} = \boxed{\text{(E)} 1}.$$

Solution 2

Eisenstein used such a quotient in his proof of **quadratic reciprocity**. Let $c = \frac{2\pi}{p}$ where p is an odd prime number and q is any integer.

Then $\frac{\sin(qc) \sin(2qc) \cdots \sin(\frac{p-1}{2}qc)}{\sin(c) \sin(2c) \cdots \sin(\frac{p-1}{2}c)}$ is the Legendre symbol $\left(\frac{q}{p}\right)$. Legendre symbol

is calculated using quadratic reciprocity which is $\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2} \frac{q-1}{2}}$. The Legendre symbol $\left(\frac{3}{11}\right) = (-1) \left(\frac{11}{3}\right) = (-1) \left(\frac{-1}{3}\right) = (-1)(-1) = \boxed{\text{(E)} 1}$

Solution 3

We have that $5^2 \equiv 3 \pmod{11}$, so 3 is a quadratic residue mod 11. For quadratic residues, their Legendre symbol which we know is the answer from Solution 2 is

(E) 1

Solution 4

We have ζ be an 11th primitive root of unity. Then the quotient becomes

$$\frac{(\zeta^3 - \zeta^{-3})(\zeta^6 - \zeta^{-6})(\zeta^9 - \zeta^{-9})(\zeta^{12} - \zeta^{-12})(\zeta^{15} - \zeta^{-15})}{(\zeta^1 - \zeta^{-1})(\zeta^2 - \zeta^{-2})(\zeta^3 - \zeta^{-3})(\zeta^4 - \zeta^{-4})(\zeta^5 - \zeta^{-5})}$$

which we can use modular arithmetic to become

$$\frac{(\zeta^3 - \zeta^{-3})(\zeta^{-5} - \zeta^5)(\zeta^{-2} - \zeta^2)(\zeta^1 - \zeta^{-1})(\zeta^4 - \zeta^{-4})}{(\zeta^1 - \zeta^{-1})(\zeta^2 - \zeta^{-2})(\zeta^3 - \zeta^{-3})(\zeta^4 - \zeta^{-4})(\zeta^5 - \zeta^{-5})}$$

and we see that is (E) 1

Solution 5 (guess)

Since we can't calculate the exact value of $\sin \frac{2\pi}{11}$, it's unlikely the answer will have radicals in it. This leaves us with $-1, 1$, and $\frac{10}{11}$. The fraction is too precise for it to be a likely answer, so that leaves us with 1 or -1 . Note that if the angle is greater than π , the sine will be negative, and if it's less than π it'll be positive. The numerator and denominator are therefore both positive, so we eliminate -1 to get (E) 1.

14. (AMC 12 12B Fall 2021) Suppose that $P(z)$, $Q(z)$, and $R(z)$ are polynomials with real coefficients, having degrees 2, 3, and 6, respectively, and constant terms 1, 2, and 3, respectively. Let N be the number of distinct complex numbers z that satisfy the equation $P(z) \cdot Q(z) = R(z)$. What is the minimum possible value of N ?
- (A) 0 (B) 1 (C) 2 (D) 3 (E) 5

Answer: B

Solution

The answer cannot be 0, as every nonconstant polynomial has at least 1 distinct complex root (Fundamental Theorem of Algebra). Since $P(z) \cdot Q(z)$ has degree $2 + 3 = 5$, we conclude that $R(z) - P(z) \cdot Q(z)$ has degree 6 and is thus nonconstant.

It now suffices to illustrate an example for which $N = 1$: Take

$$\begin{aligned} P(z) &= z^2 + 1, \\ Q(z) &= z^3 + 2, \\ R(z) &= (z + 1)^6 + P(z) \cdot Q(z). \end{aligned}$$

Note that $R(z)$ has degree 6 and constant term 3, so it satisfies the conditions.

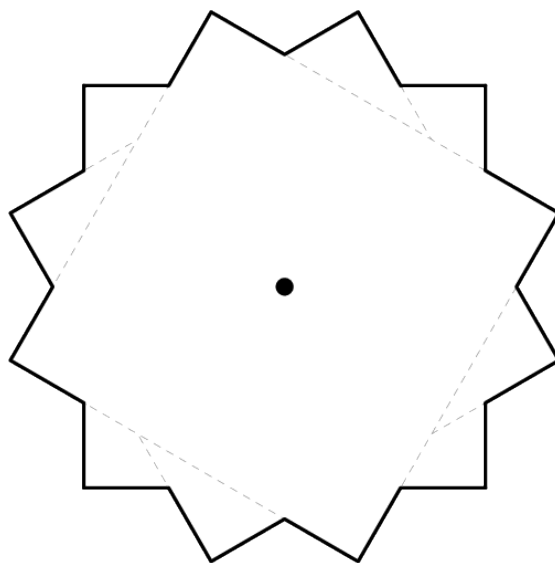
We need to find the solutions to

$$\begin{aligned} P(z) \cdot Q(z) &= (z + 1)^6 + P(z) \cdot Q(z) \\ 0 &= (z + 1)^6. \end{aligned}$$

Clearly, the only distinct complex root is -1 , so our answer is $N =$ (B) 1.

15. (AMC 12 12B Fall 2021) Three identical square sheets of paper each with side length 6 are stacked on top of each other. The middle sheet is rotated clockwise 30° about its center and the top sheet is rotated clockwise 60° about its center, resulting in the 24-sided polygon shown in the figure below. The area of this polygon can be

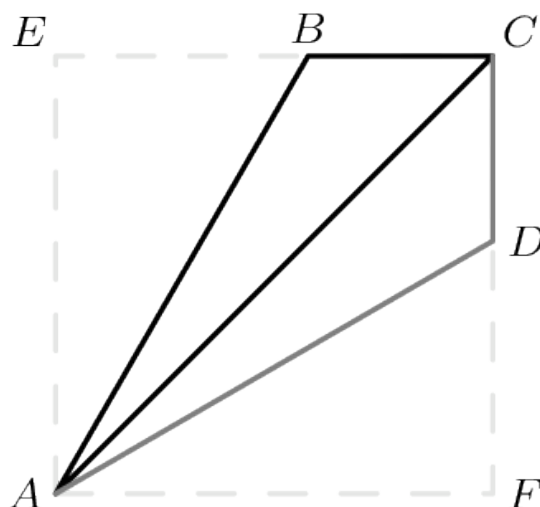
expressed in the form $a - b\sqrt{c}$, where a , b , and c are positive integers, and c is not divisible by the square of any prime. What is $a + b + c$?



- (A) 75 (B) 93 (C) 96 (D) 129 (E) 147

Answer: E

Solution 1



The 24-sided polygon is made out of 24 shapes like $\triangle ABC$. Then $\angle BAC = 360^\circ/24 = 15^\circ$, and $\angle EAC = 45^\circ$, so $\angle EAB = 30^\circ$. Then $EB = AE \tan 30^\circ = \sqrt{3}$; therefore $BC = EC - EB = 3 - \sqrt{3}$. Thus

$$[ABC] = \frac{BC}{EC} \cdot [ACE] = \frac{3 - \sqrt{3}}{3} \cdot \frac{9}{2}$$

and the required area is $24 \cdot [ABC] = 108 - 36\sqrt{3}$. Finally $108 + 36 + 3 = \boxed{\text{(E)} 147}$.

Note: Drop an altitude from A to $\overline{BC}$ to construct point E . This creates right triangles.

Solution 2

As shown in [Image:2021_AMC_12B_\(Nov\)_Problem_15,_sol.png](#), all 12 vertices of three squares form a regular dodecagon (12-gon). Denote by O the center of this dodecagon.

Hence, $\angle AOB = \frac{360^\circ}{12} = 30^\circ$.

Because the length of a side of a square is 6, $AO = 3\sqrt{2}$.

Hence, $AB = 2AO \sin \frac{\angle AOB}{2} = 3(\sqrt{3} - 1)$.

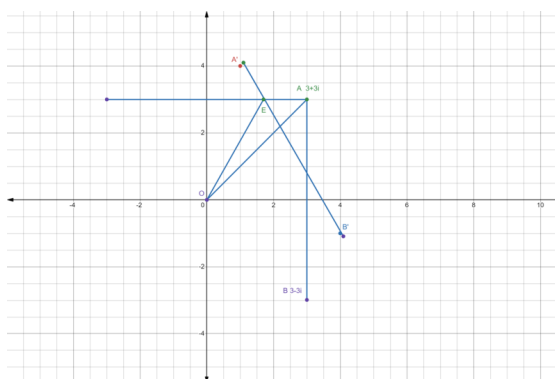
We notice that $\angle MAB = \angle MBA = 30^\circ$. Hence, $AM = \frac{AB}{2 \cos \angle MAB} = 3 - \sqrt{3}$.

Therefore, the area of the region that three squares cover is

$$\begin{aligned} & \text{Area } ABCDEFGHIJKL - 12 \text{Area } \triangle MAB \\ &= 12 \text{Area } \triangle OAB - 12 \text{Area } \triangle MAB \\ &= 12 \cdot \frac{1}{2} OA \cdot OB \sin \angle AOB - 12 \cdot \frac{1}{2} MA \cdot MB \sin \angle AMB \\ &= 6OA^2 \sin \angle AOB - 6MA^2 \sin \angle AMB \\ &= 108 - 36\sqrt{3}. \end{aligned}$$

Therefore, the answer is **(E) 147**.

Solution 3 (complex number coordinate geometry)



set $A = 3+3i$, A' , B' rotate 30 degree from A , B

$$A' = A \cdot e^{i30^\circ} = (3+3i) \cdot (\sqrt{3}/2 + i/2) = (\sqrt{3}/2 - 1/2) + (1/2 + \sqrt{3}/2)i$$

$$\text{line } A'B', \frac{y - (1/2 + \sqrt{3}/2)}{x - (\sqrt{3}/2 - 1/2)} = \tan(90^\circ + 30^\circ) = -\sqrt{3}$$

intersect with line $y=3$ at point $E_x = \sqrt{3}$, then length $AE = A_x - E_x = 3 - \sqrt{3}$,

use shoelace or $\triangle OAE = 1/2 * AE * AB/2 = 1/2 * (3 - \sqrt{3}) * 3$

total area = $24 * \triangle OAE = 108 - 36\sqrt{3}$ the answer is **(E) 147**.

16. (AMC 12 12B Fall 2021) Suppose a, b, c are positive integers such that

$$a + b + c = 23$$

and

$$\gcd(a, b) + \gcd(b, c) + \gcd(c, a) = 9.$$

What is the sum of all possible distinct values of $a^2 + b^2 + c^2$?

- (A) 259 (B) 438 (C) 516 (D) 625 (E) 687

Answer: B

Solution 1 (Observation)

Because $a + b + c$ is odd, a, b, c are either one odd and two evens or three odds.

Case 1: a, b, c have one odd and two evens.

Without loss of generality, we assume a is odd and b and c are even.

Hence, $\gcd(a, b)$ and $\gcd(a, c)$ are odd, and $\gcd(b, c)$ is even. Hence, $\gcd(a, b) + \gcd(b, c) + \gcd(c, a)$ is even. This violates the condition given in the problem.

Therefore, there is no solution in this case.

Case 2: a, b, c are all odd.

In this case, $\gcd(a, b), \gcd(a, c), \gcd(b, c)$ are all odd.

Without loss of generality, we assume

$$\gcd(a, b) \leq \gcd(b, c) \leq \gcd(c, a).$$

Case 2.1: $\gcd(a, b) = 1, \gcd(b, c) = 1, \gcd(c, a) = 7$.

The only solution is $(a, b, c) = (7, 9, 7)$.

Hence, $a^2 + b^2 + c^2 = 179$.

Case 2.2: $\gcd(a, b) = 1, \gcd(b, c) = 3, \gcd(c, a) = 5$.

The only solution is $(a, b, c) = (5, 3, 15)$.

Hence, $a^2 + b^2 + c^2 = 259$.

Case 2.3: $\gcd(a, b) = 3, \gcd(b, c) = 3, \gcd(c, a) = 3$.

There is no solution in this case.

Therefore, putting all cases together, the answer is $179 + 259 = \boxed{\text{(B)} 438}$.

Solution 2 (Enumeration)

Let $\gcd(a, b) = x, \gcd(b, c) = y, \gcd(c, a) = z$. Without the loss of generality, let $x \leq y \leq z$. We can split this off into cases:

$x = 1, y = 1, z = 7$: let $a = 7A, c = 7C$, we can try all possibilities of A and C to find that $a = 7, b = 9, c = 7$ is the only solution.

$x = 1, y = 2, z = 6$: No solutions. By y and z , we know that a, b , and c have to all be divisible by 2. Therefore, x cannot be equal to 1.

$x = 1, y = 3, z = 5$: Note that c has to be both a multiple of 3 and 5. Therefore, c has to be a multiple of 15. The only solution for this is $a = 5, b = 3, c = 15$.

$x = 1, y = 4, z = 4$: No solutions. By y and z , we know that a, b , and c have to all be divisible by 4. Therefore, x cannot be equal to 1.

$x = 2, y = 2, z = 5$: No solutions. By x and y , we know that a, b , and c have to all be divisible by 2. Therefore, z cannot be equal to 5.

$x = 2, y = 3, z = 4$: No solutions. By x and z , we know that a, b , and c have to all be divisible by 2. Therefore, y cannot be equal to 3.

$x = 3, y = 3, z = 3$: No solutions. As a, b , and c have to all be divisible by 3, $a + b + c$ has to be divisible by 3. This contradicts the sum $a + b + c = 23$.

Putting these solutions together, we have $(7^2 + 9^2 + 7^2) + (5^2 + 3^2 + 15^2) = 179 + 259 =$
(B) 438.

Solution 3

Since $a + b + c = 23$, $\gcd(a, b, c) = 23$ or $\gcd(a, b, c) = 1$.

As $\gcd(a, b) + \gcd(b, c) + \gcd(c, a) = 9$, it is impossible for $\gcd(a, b, c) = 23$, so $\gcd(a, b, c) = 1$.

This means that $\gcd(a, b)$, $\gcd(b, c)$, and $\gcd(c, a)$ must all be coprime. The only possible ways for this to be true are $1 + 1 + 7 = 9$ and $1 + 3 + 5 = 9$.

Without loss of generality, let $a \leq b \leq c$. Since $a + b + c = 23$, then $a = 7, b = 7, c = 9$ or $a = 3, b = 5, c = 15$.

$$(7^2 + 7^2 + 9^2) + (3^2 + 5^2 + 15^2) = 179 + 259 = \textbf{(B) 438}.$$

17. (AMC 12 12B Fall 2021) A bug starts at a vertex of a grid made of equilateral triangles of side length 1. At each step the bug moves in one of the 6 possible directions along the grid lines randomly and independently with equal probability. What is the probability that after 5 moves the bug never will have been more than 1 unit away from the starting position?

(A) $\frac{13}{108}$ (B) $\frac{7}{54}$ (C) $\frac{29}{216}$ (D) $\frac{4}{27}$ (E) $\frac{1}{16}$

Answer: A

Solution 1 (Recursion)

Let $S(n)$ be the number of paths of n moves such that the bug never will have been more than 1 unit away from the starting position. Clearly, by symmetry, there are two possible states here, the bug being on the center and the bug being on one of the vertices of the unit hexagon around the center. Let $C(n)$ be the number of paths with the aforementioned restriction that end on the center. Let $V(n)$ be the number of paths with the aforementioned restriction that end on a vertex of the surrounding unit hexagon. We have $S(n) = 6C(n - 1) + 3V(n - 1)$, since from the center, there are 6 possible points to land to and from a vertex there are 3 possible points to land to (the two adjacent vertices and the center). We also have $C(n) = V(n - 1)$, since to get to the center the bug must have come from a vertex, and $V(n) = 2V(n - 1) + 6C(n - 1)$, since from a vertex there are two vertices to move to, and from the center there are 6 vertices to move to. We can construct a recursion table using the base cases $V(1) = 6$ and $C(1) = 0$ and our recursive rules for $C(n)$ and $V(n)$ as follows:

n	$V(n)$	$C(n)$
1	6	0
2	12	6
3	60	12
4	192	60

Then, $S(5) = 6C(4) + 3V(4) = 6 \cdot 60 + 3 \cdot 192 = 936$, and the desired probability is

thus $\frac{936}{6^5} = \textbf{(A) } \frac{13}{108}.$

Solution 2 (Recursion)

Let $p(i)$ be such probability after i moves. $p(1) = 1$, $p(2) = \frac{1}{2}$. Then, $p(3) = \frac{1}{3} \cdot \frac{1}{2} + \frac{1}{6} \cdot 1 = \frac{1}{3}$. Then, we can prove the recursive formula

$$p(x) = \frac{1}{3}p(x-1) + \frac{1}{6}p(x-2).$$

Now, we evaluate $p(5) = \boxed{\text{(A)} \frac{13}{108}}$.

Solution 3 (Recursion)

We use (n, t) to denote the bug's current state. We wish to find $P(0, 5)$.

The first argument n denotes the bug's current position. We use $n = 0$ to denote the bug's starting point. We use $n = 1$ to denote any point whose distance to the bug's starting point is 1.

The second argument t denotes the remaining number of moves the bug has.

For $n = 0$ and $t \geq 1$, we have

$$P(0, t) = P(1, t-1).$$

For $n = 1$ and $t \geq 1$, we have

$$P(1, t) = \frac{1}{6}P(0, t-1) + \frac{1}{3}P(1, t-1).$$

For $n \in \{0, 1\}$ and $t = 0$, we have

$$P(n, 0) = 1.$$

We solve this recursive equation by using backward induction:

$$\begin{aligned} P(0, 1) &= 1, & P(1, 1) &= \frac{1}{2}, \\ P(0, 2) &= \frac{1}{2}, & P(1, 2) &= \frac{1}{3}, \\ P(0, 3) &= \frac{1}{3}, & P(1, 3) &= \frac{7}{36}, \\ P(0, 4) &= \frac{7}{36}, & P(1, 4) &= \frac{13}{108}. \end{aligned}$$

Therefore, the answer is $P(0, 5) = P(1, 4) = \boxed{\text{(A)} \frac{13}{108}}$.

Solution 4 (Generating Function)

Use a generating function, define $c_n \cdot x^n$ be c_n ways for the destination be n units away from the origin.

We conclude that:

- If the current point is origin, then we need to multiply by $6x$.
- If the current point on vertex of the unit hexagon, then we need to multiply by $x^{-1} + 2$, where there is 1 way to return to the origin and there are two ways to keep distance 1.

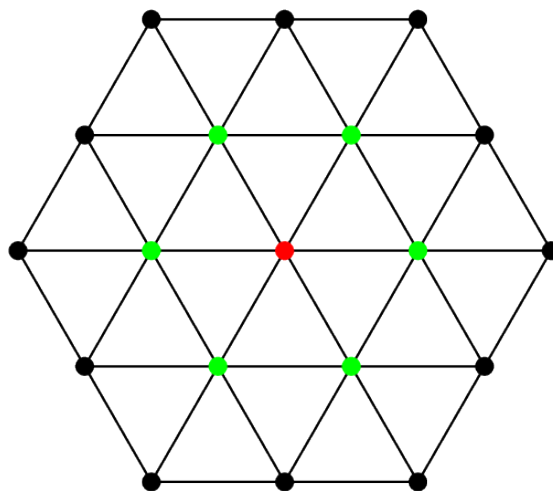
Now let's start with $p(x) = 1$.

1st step: $p(x) = 6x$ 2nd step: $p(x) = 6x \cdot (x^{-1} + 2) = 6 + 12x$ 3rd step: $p(x) = 6 \cdot 6x + 12x \cdot (x^{-1} + 2) = 12 + 60x$ 4th step: $p(x) = 12 \cdot 6x + 60x \cdot (x^{-1} + 2) = 60 + 192x$ 5th step: $p(x) = 60 \cdot 6x + 192x \cdot (x^{-1} + 2) = 192 + 744x$

So, there are $192 + 744 = 936$ ways for the bug never moves more than 1 unit away from origin. The answer is $\frac{936}{6^5} = \boxed{\text{(A)} \frac{13}{108}}$.

Solution 5 (Casework)

In the following diagram, let A denote the vertex where the bug starts (shown in red) and B denote one of the 6 adjacent vertices (shown in green).



Note that:

- If the bug is at A , then the probability that it moves to B next is 1.
- If the bug is at B , then the probability that it moves to A next is $\frac{1}{6}$.
- If the bug is at B , then the probability that it moves to B next is $\frac{1}{3}$.

We apply casework to the possible paths of the bug:

- $A \rightarrow B \rightarrow A \rightarrow B \rightarrow A \rightarrow B$ The probability for this case is $1 \cdot \frac{1}{6} \cdot 1 \cdot \frac{1}{6} \cdot 1 = \frac{1}{36}$.
- $A \rightarrow B \rightarrow A \rightarrow B \rightarrow B \rightarrow A$ The probability for this case is $1 \cdot \frac{1}{6} \cdot 1 \cdot \frac{1}{3} \cdot \frac{1}{6} = \frac{1}{108}$.
- $A \rightarrow B \rightarrow B \rightarrow A \rightarrow B \rightarrow A$ The probability for this case is $1 \cdot \frac{1}{3} \cdot \frac{1}{6} \cdot 1 \cdot \frac{1}{6} = \frac{1}{108}$.
- $A \rightarrow B \rightarrow A \rightarrow B \rightarrow B \rightarrow B$ The probability for this case is $1 \cdot \frac{1}{6} \cdot 1 \cdot \frac{1}{3} \cdot \frac{1}{3} = \frac{1}{54}$.
- $A \rightarrow B \rightarrow B \rightarrow A \rightarrow B \rightarrow B$ The probability for this case is $1 \cdot \frac{1}{3} \cdot \frac{1}{6} \cdot 1 \cdot \frac{1}{3} = \frac{1}{54}$.
- $A \rightarrow B \rightarrow B \rightarrow B \rightarrow A \rightarrow B$ The probability for this case is $1 \cdot \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{1}{6} \cdot 1 = \frac{1}{54}$.
- $A \rightarrow B \rightarrow B \rightarrow B \rightarrow B \rightarrow A$ The probability for this case is $1 \cdot \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{1}{6} = \frac{1}{162}$.
- $A \rightarrow B \rightarrow B \rightarrow B \rightarrow B \rightarrow B$ The probability for this case is $1 \cdot \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{1}{3} = \frac{1}{81}$.

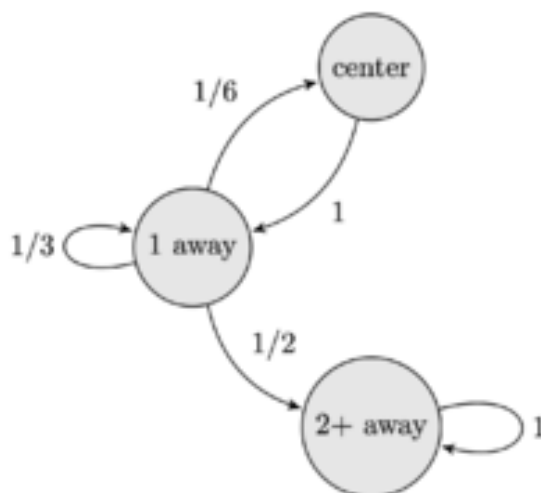
Together, the answer is

$$\frac{1}{36} + \frac{1}{108} + \frac{1}{108} + \frac{1}{54} + \frac{1}{54} + \frac{1}{54} + \frac{1}{162} + \frac{1}{81} = \boxed{\text{(A)} \frac{13}{108}}.$$

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Solution 6 (Markov Chain)

We can use a **Markov chain** to represent the different states of the bug.



Let "center" (C) represent the starting point, "1 away" (O) represent a point on the grid 1 move and 1 unit away from the starting point, and "2+ away" (T) represent the case where the bug has ventured more than 1 unit away from its starting point.

Notice that the bug must move to the 1 away state from the center. From there, it moves back to the center with a $\frac{1}{6}$ probability, to a point 2 units away from the center with $\frac{1}{2}$ probability, and to another point 1 unit away with $\frac{1}{3}$ probability.

We are trying to find the probability that the bug does not reach the 2+ away state in 5 moves. To accomplish this, we can find the complement by listing out the cases that the bug does reach the 2+ away state in a specified number of moves.

Clearly, the bug cannot reach the 2+ away (T) state in 0 moves.

In 1 move, the bug can only reach the 1 away (O) state, so it cannot reach the T state in 1 move either.

In 2 moves, the bug can move to the O state followed by the T state with probability $1 * \frac{1}{2} = \frac{1}{2}$.

In 3 moves, the bug can follow the pattern OOT with probability $1 * \frac{1}{3} * \frac{1}{2} = \frac{1}{6}$.

In 4 moves, the bug can follow the paths of OOOT or OCOT, with probabilities $\frac{1}{18}$ and $\frac{1}{12}$ respectively.

In 5 moves, the bug can follow the paths of OOOOT, OCOOT, or OOCOT, with probabilities $\frac{1}{54}$, $\frac{1}{36}$, and $\frac{1}{36}$.

Adding up these probabilities, we obtain $\frac{95}{108}$. Finding the complement of this, we subtract from 1 to get the answer of $(A) \frac{13}{108}$.

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18. (AMC 12 12B Fall 2021) Set $u_0 = \frac{1}{4}$, and for $k \geq 0$ let u_{k+1} be determined by the recurrence

$$u_{k+1} = 2u_k - 2u_k^2.$$

This sequence tends to a limit; call it L . What is the least value of k such that

$$|u_k - L| \leq \frac{1}{2^{1000}}?$$

- (A) 10 (B) 87 (C) 123 (D) 329 (E) 401

Answer: A

Solution 1

Note that terms of the sequence (u_k) lie in the interval $(0, \frac{1}{2})$ and are strictly increasing.

Since the sequence (u_k) tends to the limit L , we set $u_{k+1} = u_k = L > 0$.

The given equation becomes

$$L = 2L - 2L^2,$$

from which $L = \frac{1}{2}$.

The given inequality becomes

$$\frac{1}{2} - \frac{1}{2^{1000}} \leq u_k \leq \frac{1}{2} + \frac{1}{2^{1000}},$$

and we only need to consider $\frac{1}{2} - \frac{1}{2^{1000}} \leq u_k$.

We have

$$\begin{aligned} u_0 &= \frac{1}{4} = \frac{2^1 - 1}{2^2}, \\ u_1 &= \frac{3}{8} = \frac{2^2 - 1}{2^3}, \\ u_2 &= \frac{15}{32} = \frac{2^4 - 1}{2^5}, \\ u_3 &= \frac{255}{512} = \frac{2^8 - 1}{2^9}, \\ &\vdots \end{aligned}$$

By induction, it can be proven that

$$u_k = \frac{2^{2^k} - 1}{2^{2^{k+1}}} = \frac{1}{2} - \frac{1}{2^{2^{k+1}}}.$$

We substitute this into the inequality, then solve for k :

$$\begin{aligned} \frac{1}{2} - \frac{1}{2^{1000}} &\leq \frac{1}{2} - \frac{1}{2^{2^{k+1}}} \\ -\frac{1}{2^{1000}} &\leq -\frac{1}{2^{2^{k+1}}} \\ 2^{1000} &\leq 2^{2^{k+1}} \\ 1000 &\leq 2^k + 1. \end{aligned}$$

Since $2^9 + 1 \leq 1000 \leq 2^{10} + 1$, the least such value of k is **(A)** 10.

Solution 2

If we list out the first few values of k , we get the series $\frac{1}{4}, \frac{3}{8}, \frac{15}{32}, \frac{255}{512}$, which always seems to be a negative power of 2 away from $\frac{1}{2}$. We can test this out by setting $u_k = \frac{1}{2} - \frac{1}{2^{n_k}}$, where $n_0 = 2$.

Now, we get

$$u_{k+1} = 2 \cdot \left(\frac{1}{2} - \frac{1}{2^{n_k}} \right) - 2 \cdot \left(\frac{1}{2} - \frac{1}{2^{n_k}} \right)^2$$

$$\begin{aligned}
 &= 1 - \frac{1}{2^{n_k-1}} - 2 \cdot \left(\frac{1}{4} - \frac{1}{2^{n_k}} + \frac{1}{2^{2 \cdot n_k}} \right) \\
 &= 1 - \frac{1}{2^{n_k-1}} - \frac{1}{2} + \frac{1}{2^{n_k-1}} - \frac{1}{2^{2 \cdot n_k-1}} \\
 &= \frac{1}{2} - \frac{1}{2^{2 \cdot n_k-1}}.
 \end{aligned}$$

This means that this series approaches $\frac{1}{2}$, as the second term is decreasing. In addition, we find that $n_{k+1} = 2 \cdot n_k - 1$.

We claim that $n_k = 2^k + 1$, which can be proven by induction:

Base Case

We have $n_0 = 2 = 2^0 + 1$.

Induction Step

Assuming that the claim is true, we have $n_{k+1} = 2 \cdot (2^k + 1) - 1 = 2^{k+1} + 1$.

It follows that $n_{10} = 2^{10} + 1 > 1000$ and $n_9 = 2^9 + 1 < 1000$. Therefore, the least value of k would be **(A) 10**.

Solution 3

We are given $u_{k+1} = 2u_k - 2u_k^2$. Multiply this equation by 2 and subtract 1 from both sides. The equations can then be written nicely as $2u_{k+1} - 1 = -(2u_k - 1)^2$. Let $v_k = 2u_k - 1$ so that $v_{k+1} = -(v_k)^2$.

Clearly, $v_0 = 2u_0 - 1 = -\frac{1}{2}$. Since the magnitude of v_0 is less than 1 and because our recursive relation for v_k squares the previous term (and negates it), we see that as $k \rightarrow \infty$, $2u_k - 1 = v_k \rightarrow 0$. This means $u_k \rightarrow \frac{1}{2}$, so $L = \frac{1}{2}$.

Isolating u_k in our relation $2u_k - 1 = v_k$ gives us $u_k = \frac{v_k+1}{2}$. Substituting into the inequality, we have $\left| \frac{v_k+1}{2} - \frac{1}{2} \right| \leq \frac{1}{2^{1000}}$. Rewriting this, we get $|v_k| \leq \frac{1}{2^{999}}$.

The sequence $\{v_k\}$ is much easier to handle because of its simple recursive relation. Writing out a few terms shows that $|v_k| = \frac{1}{2^{2^k}}$. Now it just comes down to having $2^k > 999$, so $k = \mathbf{(A) 10}$.

19. (AMC 12 12B Fall 2021) Regular polygons with 5, 6, 7, and 8 sides are inscribed in the same circle. No two of the polygons share a vertex, and no three of their sides intersect at a common point. At how many points inside the circle do two of their sides intersect?
- (A) 52 (B) 56 (C) 60 (D) 64 (E) 68

Answer: E

Solution

Imagine we have 2 regular polygons with m and n sides and $m > n$ inscribed in a circle without sharing a vertex. We see that each side of the polygon with n sides (the polygon with fewer sides) will be intersected twice. (We can see this because to have a vertex of the m -gon on an arc subtended by a side of the n -gon, there will be one intersection to “enter” the arc and one to “exit” the arc. KingRavi)

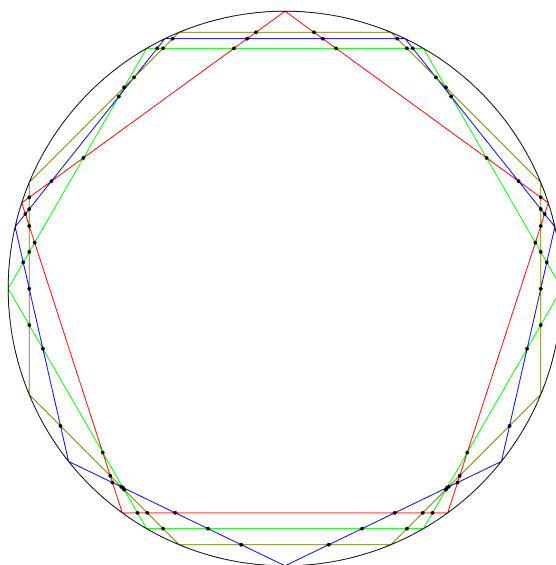
This means that we will end up with 2 times the number of sides in the polygon with fewer sides.

If we have polygons with 5, 6, 7, and 8 sides, we need to consider each possible pair of polygons and count their intersections.

Throughout 6 of these pairs, the 5-sided polygon has the least number of sides 3 times, the 6-sided polygon has the least number of sides 2 times, and the 7-sided polygon has the least number of sides 1 time.

Therefore the number of intersections is $2 \cdot (3 \cdot 5 + 2 \cdot 6 + 1 \cdot 7) = \boxed{\text{(E)} 68}$.

Remark For regular polygons with 5, 6, 7, and 8 sides, the 68 points of intersection inside the circle are shown below:



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20. (AMC 12 12B Fall 2021) A cube is constructed from 4 white unit cubes and 4 blue unit cubes. How many different ways are there to construct the $2 \times 2 \times 2$ cube using these smaller cubes? (Two constructions are considered the same if one can be rotated to match the other.)

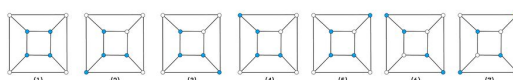
(A) 7 (B) 8 (C) 9 (D) 10 (E) 11

Answer: A

Solution 1 (Graph Theory)

This problem is about the relationships between the white unit cubes and the blue unit cubes, which can be solved by **Graph Theory**. We use a **Planar Graph** to represent the larger cube. Each vertex of the planar graph represents a unit cube. Each edge of the planar graph represents a shared face between 2 neighboring unit cubes. Each face of the planar graph represents a face of the larger cube.

Now the problem becomes a **Graph Coloring** problem of how many ways to assign 4 vertices blue and 4 vertices white with **Topological Equivalence**. For example, in Figure (1), as long as the 4 blue vertices belong to the same planar graph face, the different planar graphs are considered to be topological equivalent by rotating the larger cube.



Here is how the 4 blue unit cubes are arranged:

In Figure (1): 4 blue unit cubes are on the same layer (horizontal or vertical).

In Figure (2): 4 blue unit cubes are in T shape.

In Figure (3) and (4): 4 blue unit cubes are in S shape.

In Figure (5): 3 blue unit cubes are in L shape, and the other is isolated without a shared face.

In Figure (6): 2 pairs of neighboring blue unit cubes are isolated from each other without a shared face.

In Figure (7): 4 blue unit cubes are isolated from each other without a shared face.

So the answer is (A) 7

Solution 2 (Casework, Counting Down)

Let's split the cube into two layers; a bottom and top. Note that there must be four of each color, so however many number of one color are in the bottom, there will be four minus that number of the color on the top. We do casework on the color distribution of the bottom layer.

Case 1: 4, 0

In this case, there is only one possibility for the top layer - all of the other color - $\binom{4}{4}$. Therefore there is 1 construction from this case.

Case 2: 3, 1

In this case, the top layer has four possibilities, because there are four different ways to arrange it so that it also has a 3, 1 color distribution - $\binom{4}{3}$. Therefore there are 4 constructions from this case.

Case 3: 2, 2

In this case, the top layer has six possibilities of arrangement - $\binom{4}{2}$. However, having adjacent colors one way can be rotated to having adjacent colors any other way, so there is only one construction for the adjacent colors subcase and similarly, only one for the diagonal color subcase. Therefore the total number of constructions for this case is 2.

The total number of constructions for the cube is thus $1 + 4 + 2 = 7 = \span style="border: 1px solid black; padding: 0 5px;">A$

Solution 3 (Casework, Counting Up)

Divide the $2 \times 2 \times 2$ cube into two layers, say, front and back. Any possible construction can be rotated such that the front layer has the same or greater number of white cubes than blue cubes, so we only need to count the number of cases given that is true.

- a) **Case 1: Each layer contains 2 cubes of each color.** Note that we only need to consider the configuration of the white cubes because all the other cubes will be blue cubes. There are 2 ways that the 2 white cubes in each layer can be arranged: adjacent or diagonal to each other.
 - Case 1.1: Both layers have 2 white cubes adjacent to each other.** Rotate the cube such that there are white cubes along the top edge of the front layer. Now, the white cubes in the back layer can be along the top, bottom, right, or left edges. So, case 1.1 results in 4 constructions.
 - Case 1.2: One layer has 2 white cubes adjacent to each other, and the other has 2 white cubes diagonal from each other.** Rotate the cube such that there are white cubes along the top of the front layer. The white cubes in the back layer can be at the

top-left and bottom-right or at the top-right and bottom-left. If we rotate the former case by 90 degrees clockwise into the page, it becomes the same as the latter case. So, case 1.2 results in 1 additional construction.

Case 1.3: Both layers have white cubes diagonal from each other. Rotate the cube such that there is a white cube at the top-left and bottom-right of the front layer. The back layer could also have white cubes at the top-left and bottom-right, but this is the same as case 1.1 with the white cubes in the back layer along the bottom edge. Alternatively, the back layer could have white cubes at the top-right and the bottom-left. This is a distinct case. So, case 1.3 results in 1 additional construction.

So, case 1 results in $4 + 1 + 1 = 6$ distinct constructions.

- b) **Case 2: The front layer contains 3 white cubes, and the back layer contains one white cube.** In this case, unless the sole white and sole blue cubes in the front and back layers are on opposite corners of the $2 \times 2 \times 2$ cube, then that cube can be split into left and right layers with 2 cubes of each color in each layer. Those constructions were counted in case 1, so case 2 results in 1 additional construction.
- c) **Case 3: The front layer contains 4 white cubes, and the back layer contains no white cubes.** If we split this construction into its left and right layers, then each layer will have 2 cubes of each color. So, this construction was counted in case 1, and case 3 results in 0 additional constructions.

Therefore, our answer is $6 + 1 + 0 = \boxed{(A) 7}$.

Solution 4 (Burnside Lemma)

Burnside lemma is used to counting number of orbit where the element on the same orbit can be achieved by the defined operator, naming rotation, reflection and etc.

The facts for Burnside lemma are

1. the sum of stabilizer on the same orbit equals to the # of operators;
2. the sum of stabilizer can be counted as $fix(g)$
3. the sum of the $fix(g)/|G|$ equals the # of orbit.

Let's start with defining the operator for a cube,

1. e (identity)

For identity, there are $\frac{8!}{4!4!} = 70$

2. $\mathbf{r}^1, \mathbf{r}^2, \mathbf{r}^3$ to be the rotation axis along three pair of opposite face,

each contains $r_{90}^i, r_{180}^i, r_{270}^i$ where $i = 1, 2, 3$ $fix(r_{90}^i) = fix(r_{270}^i) = 2 \cdot 1 = 2$
 $fix(r_{180}^i) = \frac{4!}{2! \cdot 2!} = 6$

therefore $fix(\mathbf{r}^i) = 2 + 2 + 6 = 10$, and $fix(\mathbf{r}^1) + fix(\mathbf{r}^2) + fix(\mathbf{r}^3) = 30$

3. $\mathbf{r}^4, \mathbf{r}^5, \mathbf{r}^6, \mathbf{r}^7$ to be the rotation axis along four cube diagonals.

each contains r_{120}^i, r_{240}^i where $i = 4, 5, 6, 7$ $fix(r_{120}^i) = fix(r_{240}^i) = 2 \cdot 1 \cdot 2 \cdot 1 = 4$

therefore $fix(\mathbf{r}^i) = 4 + 4 = 8$, and $fix(\mathbf{r}^4) + fix(\mathbf{r}^5) + fix(\mathbf{r}^6) + fix(\mathbf{r}^7) = 32$

4. $\mathbf{r}^8, \mathbf{r}^9, \mathbf{r}^{10}, \mathbf{r}^{11}, \mathbf{r}^{12}, \mathbf{r}^{13}$ to be the rotation axis along 6 pairs of diagonally opposite sides

each contains r_{180}^i where $i = 8, 9, 10, 11, 12, 13$ $fix(r_{180}^i) = \frac{4!}{2! \cdot 2!} = 6$

therefore $fix(\mathbf{r}^8) + fix(\mathbf{r}^9) + fix(\mathbf{r}^{10}) + fix(\mathbf{r}^{11}) + fix(\mathbf{r}^{12}) + fix(\mathbf{r}^{13}) = 36$

5. The total number of operators are

$$|G| = 1 + 3 \cdot 3 + 4 \cdot 2 + 6 \cdot 1 = 24$$

Based on 1, 2, 3, 4 the total number of stabilizers is $70 + 30 + 32 + 36 = 168$

$$\text{therefore the number of orbits} = \frac{168}{|G|=24} = \boxed{7}$$

Solution 5 (Quick 'n Easy but not rigorous)

Since rotations of a single pattern are considered indistinguishable, we can assume that the forward upper right corner of the 2-by-2-by-2 cube is a blue cube (since we can always rotate the big cube to place a blue cube in that spot).

Once we've assigned this cube to be blue, we note that 3 1-by-1-by-1 cubes share a side with it, 3 1-by-1-by-1 cubes share a corner with it, and 1 1-by-1-by-1 cube does not touch the assigned cube at all, from the perspective of someone who can only see the cube's faces. We'll call the first 3 "adjacent", the second 3 "cornering", and the last one "opposite."

We can use a little bit of intuition to confirm that due to the rotation condition, we should treat all adjacents as indistinguishable, all cornerings as indistinguishable, and of course the opposite one is unique from all the others. Thus, we can list out like so (keeping in mind that there are 3 adjacents, 3 cornerings, and 1 opposite, and that we're choosing the positions of the remaining 3 blue cubes):

OAA, OAC, OCC, CCC, CAA, CCA, AAA.

This gives the answer to be $\boxed{\text{(A)} 7}$.

21. (AMC 12 12B Fall 2021) For real numbers x , let

$$P(x) = 1 + \cos(x) + i \sin(x) - \cos(2x) - i \sin(2x) + \cos(3x) + i \sin(3x)$$

where $i = \sqrt{-1}$. For how many values of x with $0 \leq x < 2\pi$ does

$$P(x) = 0?$$

(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Answer: A

Solution 1

Let $a = \cos(x) + i \sin(x)$. Now $P(a) = 1 + a - a^2 + a^3$. $P(-1) = -2$ and $P(0) = 1$ so there is a real root a_1 between -1 and 0 . The other a 's must be complex conjugates since all coefficients of the polynomial are real. The magnitude of those complex a 's squared is $-\frac{1}{a_1}$, which is greater than 1. If x is real number then a must have magnitude of 1, but none of the solutions for a have magnitude of 1, so the answer is $\boxed{\text{(A)} 0}$ lopekiloinm

Solution 2

For $\text{Im}(P(x)) = 0$, we get

$$\sin(2x) = \sin(x) + \sin(3x) = 2 \sin(2x) \cos(x)$$

So either $\sin(2x) = 0$, i.e. $x \in \{0, \pi\}$ or $\cos(x) = \frac{1}{2}$, i.e. $x \in \{\pi/3, 5\pi/3\}$.

For none of these values do we get $\text{Re}(P(x)) = 0$.

Therefore, the answer is $\boxed{\text{(A)} 0}$.

Solution 3

We have

$$P(x) = 1 + e^{ix} - e^{i2x} + e^{i3x}.$$

Denote $y = e^{ix}$. Hence, this problem asks us to find the number of y with $|y| = 1$ that satisfy

$$1 + y - y^2 + y^3 = 0. \quad (1)$$

Taking imaginary part of both sides, we have

$$\begin{aligned} 0 &= \operatorname{Im} (1 + y - y^2 + y^3) \\ &= \frac{1}{2i} (y - \bar{y} - y^2 + \bar{y}^2 + y^3 - \bar{y}^3) \\ &= \frac{y - \bar{y}}{2i} (1 - y - \bar{y} + y^2 + y\bar{y} + \bar{y}^2) \\ &= \operatorname{Im} y (1 - (y + \bar{y}) + (y + \bar{y})^2 - y\bar{y}) \\ &= \operatorname{Im} y (1 - 2\operatorname{Re} y + 4(\operatorname{Re} y)^2 - |y|^2) \\ &= \operatorname{Im} y (1 - 2\operatorname{Re} y + 4(\operatorname{Re} y)^2 - 1) \\ &= 2\operatorname{Im} y \cdot \operatorname{Re} y (2\operatorname{Re} y - 1) \end{aligned}$$

The sixth equality follows from the property that $|y| = 1$.

Therefore, we have either $\operatorname{Re} y = 0$ or $\operatorname{Im} y = 0$ or $2\operatorname{Re} y - 1 = 0$.

Case 1: $\operatorname{Re} y = 0$.

Because $|y| = 1$, $y = \pm i$.

However, these solutions fail to satisfy Equation (1).

Therefore, there is no solution in this case.

Case 2: $\operatorname{Im} y = 0$.

Because $|y| = 1$, $y = \pm 1$.

However, these solutions fail to satisfy Equation (1).

Therefore, there is no solution in this case.

Case 3: $2\operatorname{Re} y - 1 = 0$.

Because $|y| = 1$, $y = \frac{1}{2} \pm \frac{\sqrt{3}}{2}i = e^{i\pm\frac{\pi}{3}}$.

However, these solutions fail to satisfy Equation (1).

Therefore, there is no solution in this case.

All cases above imply that there is no solution in this problem.

Therefore, the answer is (A) 0.

Solution 4

Let $a = \cos(x) + i\sin(x)$, so by De Moivre $P(x) = a^3 - a^2 + a + 1$. The problem essentially asks for the number of real roots of P which lie on the complex unit circle. Let $|r| = 1$ be a root of P , and note that we can't have $r^3 - r^2 + r = 0$, else $P(r) = 0$. Thus, suppose henceforth that $r^3 - r^2 + r \neq 0$. We then have

$r^3 - r^2 + r = r^2(r + \frac{1}{r} - 1) = a^2(2\operatorname{Re}(r) - 1)$, hence the argument of $r^3 - r^2 + r$ is either the argument of a^2 , or the argument of $-a^2$. Since $r^3 - r^2 + r = -1$ is real, it follows that $a^2 = \pm 1 \implies a \in \{1, i, -1, -i\}$. Now, we can check all of these values and find that none of them work, yielding an answer of **(A) 0**.

-IAmTheHazard

Solution 5 (Geometry)

$P(x)$ can be written equivalently as $P(x) = 1 + \operatorname{cis}(x) - \operatorname{cis}(2x) + \operatorname{cis}(3x)$. Thus, we aim to find x such that the sum of the vectors $\operatorname{cis}(x)$, $-\operatorname{cis}(2x)$, and $\operatorname{cis}(3x)$ is -1 . Notice that $\operatorname{cis}(x)$, $-\operatorname{cis}(2x)$, $\operatorname{cis}(3x)$ all lie on the unit circle in the complex plane, and the vector $\operatorname{cis}(x) + \operatorname{cis}(3x)$ is collinear with $-\operatorname{cis}(2x)$. Since $|-\operatorname{cis}(2x)| = 1$ and we want the three vectors to sum to -1 , we either have $-\operatorname{cis}(2x) = 1$ and $\operatorname{cis}(x) + \operatorname{cis}(3x) = -2$, or $-\operatorname{cis}(2x) = -1$ and $\operatorname{cis}(x) + \operatorname{cis}(3x) = 0$. If the first condition is true, $\operatorname{cis}(x) = \operatorname{cis}(3x) = -1$. This will imply that $x = \pi$. But then $-\operatorname{cis}(2x) = -1$, which violates the condition. Similarly, we can show that the second condition cannot be met either. Thus $P(x)$ does not have any solutions on the interval $[0, 2\pi]$. Therefore, the answer is **(A) 0**.

-mathy_mathema

Solution 6 (Trig)

$$P(x) = 1 + \cos x - \cos 2x + \cos 3x + i(\sin x - \sin 2x + \sin 3x)$$

In other words, $1 + \cos x - \cos 2x + \cos 3x = 0$ and $\sin x - \sin 2x + \sin 3x = 0$. Using trigonometric identities, we know that

$$\begin{aligned} 1 + \cos x - \cos 2x + \cos 3x &= 0 \\ &= 1 + \cos x - (2\cos^2 x - 1) + (4\cos^3 x - 3\cos x) \\ &= 4\cos^3 x - 2\cos^2 x - 2\cos x + 2 \\ &= 2\cos^3 x - \cos^2 x - \cos x + 1 \\ \sin x - \sin 2x + \sin 3x &= 0 \\ &= \sin x - 2\sin x \cos x + 3\sin x - 4\sin^3 x \\ &= 4\sin^3 x - 4\sin x + 2\sin x \cos x \\ &= \sin x(2\sin^2 x - 2 + \cos x) \\ &= \sin x(-2\cos^2 x + \cos x). \end{aligned}$$

$1 + \cos x - \cos 2x + \cos 3x = 0$ must satisfy when $\sin x = 0$, $\cos x = 0$ or $\cos x = \frac{1}{2}$. The cases when $\cos x = 0$ and $\cos x = \frac{1}{2}$ does not work. Moreover, for the case $\sin x = 0$ to work, $x = 0, \pi, 2\pi$. However, none of the cases work. Therefore, **(A) 0** is the answer.

22. (AMC 12 12B Fall 2021) Right triangle ABC has side lengths $BC = 6$, $AC = 8$, and $AB = 10$. A circle centered at O is tangent to line BC at B and passes through A . A circle centered at P is tangent to line AC at A and passes through B . What is OP ?

(A) $\frac{23}{8}$ (B) $\frac{29}{10}$ (C) $\frac{35}{12}$ (D) $\frac{73}{25}$ (E) 3

Answer: C

Solution 1

Let M be the midpoint of AB ; so $BM = AM = 5$. Let D be the point such that $ABCD$ is a rectangle. Then $MO \perp AB$ and $MP \perp AB$. Let $\theta = \angle BAC$; so $\tan \theta = \frac{6}{8} = \frac{3}{4}$. Then

$$OP = MP - MO = AM \cot \theta - BM \tan \theta = 5\left(\frac{4}{3} - \frac{3}{4}\right) = \boxed{\text{(C)} \frac{35}{12}}.$$

Solution 2

This one uses the same diagram as Solution 1, except we draw BP . After doing angle chasing we find $\triangle BPM \sim \triangle BAC$ and $\frac{BM}{BC} = \frac{PM}{AC}$, resulting in $PM = \frac{20}{3}$.

We also find that $\triangle BOM \sim \triangle ABC$ and $\frac{BM}{AC} = \frac{OM}{BC}$, resulting in $OM = \frac{15}{4}$.
 $OP = PM - OM = \frac{35}{12}$.

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Solution 3 (Analytic Geometry)

In a Cartesian plane, let C, B , and A be $(0, 0), (0, 6), (8, 0)$ respectively.

By analyzing the behaviors of the two circles, we set O to be $(a, 6)$ and P be $(b, 8)$.

Hence derive the two equations:

$$(x - a)^2 + (y - 6)^2 = a^2 \quad (x - 8)^2 + (y - b)^2 = b^2$$

Considering the coordinates of A and B for the two equations respectively, we get:

$$(8 - a)^2 + (0 - 6)^2 = a^2 \quad (0 - 8)^2 + (6 - b)^2 = b^2$$

Solve to get $a = \frac{25}{4}$ and $b = \frac{25}{3}$

Through using the distance formula,

$$OP = \sqrt{\left(8 - \frac{25}{4}\right)^2 + \left(\frac{25}{3} - 6\right)^2} = \boxed{\text{(C)} \frac{35}{12}}.$$

Solution 4

Because the circle with center O passes through points A and B and is tangent to line BC at point B , O is on the perpendicular bisector of segment AB and $OB \perp BC$.

Because the circle with center P passes through points A and B and is tangent to line AC at point A , P is on the perpendicular bisector of segment AB and $PA \perp AC$.

Let lines OB and AP intersect at point D . Hence, $ACBD$ is a rectangle.

Denote by M the midpoint of segment AB . Hence, $BM = \frac{AB}{2} = 5$. Because O and P are on the perpendicular bisector of segment AB , points M, O, P are collinear with $OM \perp AB$.

We have $\triangle MOB \sim \triangle CBA$. Hence, $\frac{BO}{AB} = \frac{BM}{AC}$. Hence, $BO = \frac{25}{4}$. Hence, $OD = BD - BO = \frac{7}{4}$.

We have $\triangle DOP \sim \triangle CBA$. Hence, $\frac{OP}{BA} = \frac{DO}{CB}$. Therefore, $OP = \frac{35}{12}$.

Therefore, the answer is $\boxed{\text{(C)} \frac{35}{12}}$.

Solution 5

Let C be the origin, making $B = (0, 6)$ and $A = (8, 0)$. Let D be the midpoint of AB ; $D = (4, 3)$.

Notice that both O and P must be on the perpendicular bisector l of AB . The slope of AB is $-\frac{3}{4}$, making the l 's slope be $\frac{4}{3}$. Since l passes through D , the equation for l becomes

$$y - 3 = \frac{4}{3}(x - 4),$$

using the slope intercept form. Since OB is perpendicular to AC and AP is perpendicular to AC (cause of tangencies), the y -coordinate for O is 6 and the x -coordinate for P is 8. Plugging these numbers in the equation for l gives $O = \left(\frac{25}{4}, 6\right)$ and $P = \left(8, \frac{25}{3}\right)$. Thus,

$$OP = \sqrt{\left(\frac{7}{4}\right)^2 + \left(\frac{7}{3}\right)^2} = 7\sqrt{\frac{3^2 + 4^2}{(3^2)(4^2)}} = 7 \cdot \frac{5}{12} = \boxed{\frac{35}{12}} \text{ (C)}$$

23. (AMC 12 12B Fall 2021) What is the average number of pairs of consecutive integers in a randomly selected subset of 5 distinct integers chosen from the set $\{1, 2, 3, \dots, 30\}$? (For example the set $\{1, 17, 18, 19, 30\}$ has 2 pairs of consecutive integers.)
- (A) $\frac{2}{3}$ (B) $\frac{29}{36}$ (C) $\frac{5}{6}$ (D) $\frac{29}{30}$ (E) 1

Answer: A

Solution 1

There are 29 possible pairs of consecutive integers, namely $p_1 = \{1, 2\}, \dots, p_{29} = \{29, 30\}$. Define a random variable X_i , with $X_i = 1$, if p_i is part of the 5-element subset, and 0 otherwise. Then the number of pairs of consecutive integers in a 5-element selection is given by the sum $X_1 + \dots + X_{29}$. By linearity of expectation, the expected value is equal to the sum of the $\mathbb{E}[X_i]$:

$$\mathbb{E}[X_1 + \dots + X_{29}] = \mathbb{E}[X_1] + \dots + \mathbb{E}[X_{29}]$$

To compute $\mathbb{E}[X_i]$, note that $X_i = 1$ for a total of $\binom{28}{3}$ out of $\binom{30}{5}$ possible selections. Thus

$$\mathbb{E}[X_i] = \frac{\binom{28}{3}}{\binom{30}{5}} = \frac{1}{29} \cdot \frac{2}{3}, \quad \text{which implies} \quad \mathbb{E}[X_1 + \dots + X_{29}] = \boxed{\text{(A)} \frac{2}{3}}.$$

kingofpineapplz minor edit by sujoybec51

Solution 2

Alternatively, using linearity of expectation on the chosen subset: there are $\binom{5}{2}$ pairs of integers. There are 29 pairs of consecutive integers out of $\binom{30}{2}$ possible pairs, thus the probability that any given pair is consecutive is $\frac{29}{\binom{30}{2}}$. By linearity of

expectation, there are $\frac{\binom{5}{2} 29}{\binom{30}{2}} = \frac{10 \cdot 29}{\frac{30 \cdot 29}{2}} = \boxed{\text{(A)} \frac{2}{3}}$ such pairs on average.

Solution 3 (casework bash)

We will proceed with some casework. Let n be the number of sets of consecutive numbers in the subset. Note that the maximum possible value of n is 4. Case 1: $n = 4$ There is only one way to arrange the distribution of the number of elements

in each block here. There are $\binom{26}{1} = 26$ ways to arrange where that block of 5 numbers will go, so a total of $1 \cdot 26 = 26$ ways for this case. Case 2: $n = 3$ There are 4 ways to arrange the distribution of the number of elements in each block here. (4-1, 3-2, 2-3, 1-4). There are $\binom{26}{2}$ ways to arrange where those two blocks of numbers go among the other numbers such that no two of those blocks are adjacent to each other. Total for this case: $4 \cdot 325 = 1300$. Case 3: $n = 2$ By Stars and Bars, there are $\binom{4}{2} = 6$ ways to arrange the distribution of the number of elements in each block here. There are $\binom{26}{3}$ ways to arrange where those three blocks of numbers go among the other numbers such that no two of those blocks are adjacent to each other. Total for this case: $6 \cdot 2600 = 15600$. Case 4: $n = 1$ By Stars and Bars, there are $\binom{4}{3} = 4$ ways to arrange the distribution of the number of elements in each block here. There are $\binom{26}{4}$ ways to arrange where those four blocks of numbers go among the other numbers such that no two of those blocks are adjacent to each other. Total for this case: $4 \cdot 14950 = 59800$. Since the last case $n = 0$ doesn't contribute to the expected value sum, we can ignore it. Using the expected value formula, our desired value is

$$\frac{4 \cdot 26 + 3 \cdot 1300 + 2 \cdot 15600 + 1 \cdot 59800}{\binom{30}{5}} = \frac{95004}{142506} = \boxed{\frac{2}{3}}.$$

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Solution 4

We define an outcome as $(a_1, \dots, a_5)$ with $1 \leq a_1 < a_2 < a_3 < a_4 < a_5 \leq 30$.

We denote by Ω the sample space. Hence. $|\Omega| = \binom{30}{5}$.

Case 1: There is only 1 pair of consecutive integers.

Case 1.1: (a_1, a_2) is the single pair of consecutive integers.

We denote by E_{11} the collection of outcomes satisfying this condition. Hence, $|E_{11}|$ is the number of outcomes satisfying

$$\begin{cases} a_1 \geq 1 \\ a_3 \geq a_1 + 3 \\ a_4 \geq a_3 + 2 \\ a_5 \geq a_4 + 2 \\ a_5 \leq 30 \\ a_1, a_3, a_4, a_5 \in \mathbb{N} \end{cases}.$$

Denote $b_1 = a_1 - 1$, $b_2 = a_3 - a_1 - 3$, $b_3 = a_4 - a_3 - 2$, $b_4 = a_5 - a_4 - 2$, $b_5 = 30 - a_5$. Hence, $|E_{11}|$ is the number of outcomes satisfying

$$\begin{cases} b_1 + b_2 + b_3 + b_4 + b_5 = 22 \\ b_1, b_2, b_3, b_4, b_5 \text{ are non-negative integers} \end{cases}.$$

Therefore, $|E_{11}| = \binom{22+5-1}{5-1} = \binom{26}{4}$.

Case 1.2: (a_2, a_3) is the single pair of consecutive integers.

We denote by E_{12} the collection of outcomes satisfying this condition. Hence, $|E_{12}|$ is the number of outcomes satisfying

$$\begin{cases} a_1 \geq 1 \\ a_2 \geq a_1 + 2 \\ a_4 \geq a_2 + 3 \\ a_5 \geq a_4 + 2 \\ a_5 \leq 30 \\ a_1, a_2, a_4, a_5 \in \mathbb{N} \end{cases}.$$

Similar to our analysis for Case 1.1, $|E_{12}| = \binom{22+5-1}{5-1} = \binom{26}{4}$.

Case 1.3: (a_3, a_4) is the single pair of consecutive integers.

We denote by E_{13} the collection of outcomes satisfying this condition. Hence, $|E_{13}|$ is the number of outcomes satisfying

$$\begin{cases} a_1 \geq 1 \\ a_2 \geq a_1 + 2 \\ a_3 \geq a_2 + 2 \\ a_5 \geq a_3 + 3 \\ a_5 \leq 30 \\ a_1, a_2, a_3, a_5 \in \mathbb{N} \end{cases}.$$

Similar to our analysis for Case 1.1, $|E_{13}| = \binom{22+5-1}{5-1} = \binom{26}{4}$.

Case 1.4: (a_4, a_5) is the single pair of consecutive integers.

We denote by E_{14} the collection of outcomes satisfying this condition. Hence, $|E_{14}|$ is the number of outcomes satisfying

$$\begin{cases} a_1 \geq 1 \\ a_2 \geq a_1 + 2 \\ a_3 \geq a_2 + 2 \\ a_4 \geq a_3 + 2 \\ a_4 \leq 29 \\ a_1, a_2, a_3, a_4 \in \mathbb{N} \end{cases}.$$

Similar to our analysis for Case 1.1, $|E_{14}| = \binom{22+5-1}{5-1} = \binom{26}{4}$.

Case 2: There are 2 pairs of consecutive integers.

Case 2.1: (a_1, a_2) and (a_2, a_3) are two pairs of consecutive integers.

We denote by E_{21} the collection of outcomes satisfying this condition. Hence, $|E_{21}|$ is the number of outcomes satisfying

$$\begin{cases} a_1 \geq 1 \\ a_4 \geq a_1 + 4 \\ a_5 \geq a_4 + 2 \\ a_5 \leq 30 \\ a_1, a_4, a_5 \in \mathbb{N} \end{cases}.$$

Similar to our analysis for Case 1.1, $|E_{21}| = \binom{23+4-1}{4-1} = \binom{26}{3}$.

Case 2.2: (a_1, a_2) and (a_3, a_4) are two pairs of consecutive integers.

We denote by E_{22} the collection of outcomes satisfying this condition. Hence, $|E_{22}|$ is the number of outcomes satisfying

$$\begin{cases} a_1 \geq 1 \\ a_3 \geq a_1 + 3 \\ a_5 \geq a_3 + 3 \\ a_5 \leq 30 \\ a_1, a_3, a_5 \in \mathbb{N} \end{cases}.$$

Similar to our analysis for Case 1.1, $|E_{22}| = \binom{23+4-1}{4-1} = \binom{26}{3}$.

Case 2.3: (a_1, a_2) and (a_4, a_5) are two pairs of consecutive integers.

We denote by E_{23} the collection of outcomes satisfying this condition. Hence, $|E_{23}|$ is the number of outcomes satisfying

$$\begin{cases} a_1 \geq 1 \\ a_3 \geq a_1 + 3 \\ a_4 \geq a_3 + 2 \\ a_4 \leq 29 \\ a_1, a_3, a_4 \in \mathbb{N} \end{cases}.$$

Similar to our analysis for Case 1.1, $|E_{23}| = \binom{23+4-1}{4-1} = \binom{26}{3}$.

Case 2.4: (a_2, a_3) and (a_3, a_4) are two pairs of consecutive integers.

We denote by E_{24} the collection of outcomes satisfying this condition. Hence, $|E_{24}|$ is the number of outcomes satisfying

$$\begin{cases} a_1 \geq 1 \\ a_2 \geq a_1 + 2 \\ a_5 \geq a_2 + 4 \\ a_5 \leq 30 \\ a_1, a_2, a_5 \in \mathbb{N} \end{cases}.$$

Similar to our analysis for Case 1.1, $|E_{24}| = \binom{23+4-1}{4-1} = \binom{26}{3}$.

Case 2.5: (a_2, a_3) and (a_4, a_5) are two pairs of consecutive integers.

We denote by E_{25} the collection of outcomes satisfying this condition. Hence, $|E_{25}|$ is the number of outcomes satisfying

$$\begin{cases} a_1 \geq 1 \\ a_2 \geq a_1 + 2 \\ a_4 \geq a_2 + 3 \\ a_4 \leq 29 \\ a_1, a_2, a_4 \in \mathbb{N} \end{cases}.$$

Similar to our analysis for Case 1.1, $|E_{25}| = \binom{23+4-1}{4-1} = \binom{26}{3}$.

Case 2.6: (a_3, a_4) and (a_4, a_5) are two pairs of consecutive integers.

We denote by E_{26} the collection of outcomes satisfying this condition. Hence, $|E_{26}|$ is the number of outcomes satisfying

$$\begin{cases} a_1 \geq 1 \\ a_2 \geq a_1 + 2 \\ a_3 \geq a_2 + 2 \\ a_3 \leq 28 \\ a_1, a_2, a_3 \in \mathbb{N} \end{cases}.$$

Similar to our analysis for Case 1.1, $|E_{26}| = \binom{23+4-1}{4-1} = \binom{26}{3}$.

Case 3: There are 3 pairs of consecutive integers.

Case 3.1: (a_1, a_2) , (a_2, a_3) and (a_3, a_4) are three pairs of consecutive integers.

We denote by E_{31} the collection of outcomes satisfying this condition. Hence, $|E_{31}|$ is the number of outcomes satisfying

$$\begin{cases} a_1 \geq 1 \\ a_5 \geq a_1 + 5 \\ a_5 \leq 30 \\ a_1, a_5 \in \mathbb{N} \end{cases}.$$

Similar to our analysis for Case 1.1, $|E_{31}| = \binom{24+3-1}{3-1} = \binom{26}{2}$.

Case 3.2: (a_1, a_2) , (a_2, a_3) and (a_4, a_5) are three pairs of consecutive integers.

We denote by E_{32} the collection of outcomes satisfying this condition. Hence, $|E_{32}|$ is the number of outcomes satisfying

$$\begin{cases} a_1 \geq 1 \\ a_4 \geq a_1 + 4 \\ a_4 \leq 29 \\ a_1, a_4 \in \mathbb{N} \end{cases}.$$

Similar to our analysis for Case 1.1, $|E_{32}| = \binom{24+3-1}{3-1} = \binom{26}{2}$.

Case 3.3: (a_1, a_2) , (a_3, a_4) and (a_4, a_5) are three pairs of consecutive integers.

We denote by E_{33} the collection of outcomes satisfying this condition. Hence, $|E_{33}|$ is the number of outcomes satisfying

$$\begin{cases} a_1 \geq 1 \\ a_3 \geq a_1 + 3 \\ a_3 \leq 28 \\ a_1, a_3 \in \mathbb{N} \end{cases}.$$

Similar to our analysis for Case 1.1, $|E_{33}| = \binom{24+3-1}{3-1} = \binom{26}{2}$.

Case 3.4: (a_2, a_3) , (a_3, a_4) and (a_4, a_5) are three pairs of consecutive integers.

We denote by E_{34} the collection of outcomes satisfying this condition. Hence, $|E_{34}|$ is the number of outcomes satisfying

$$\begin{cases} a_1 \geq 1 \\ a_2 \geq a_1 + 2 \\ a_2 \leq 27 \\ a_1, a_2 \in \mathbb{N} \end{cases}.$$

Similar to our analysis for Case 1.1, $|E_{34}| = \binom{24+3-1}{3-1} = \binom{26}{2}$.

Case 4: There are 4 pairs of consecutive integers.

In this case, $(a_1, a_2, a_3, a_4, a_5)$ are consecutive integers.

We denote by E_4 the collection of outcomes satisfying this condition. Hence, $|E_4|$ is the number of outcomes satisfying

$$\begin{cases} a_1 \geq 1 \\ a_1 \leq 27 \\ a_1 \in \mathbb{N} \end{cases}.$$

Hence, $|E_4| = 26$.

Therefore, the average number of pairs of consecutive integers is

$$\begin{aligned} & \frac{1}{|\Omega|} \left(1 \cdot \sum_{i=1}^4 |E_{1i}| + 2 \cdot \sum_{i=1}^6 |E_{2i}| + 3 \cdot \sum_{i=1}^4 |E_{3i}| + 4 \cdot |E_4| \right) \\ &= \frac{1}{\binom{30}{5}} \left(4 \binom{26}{4} + 12 \binom{26}{3} + 12 \binom{26}{2} + 4 \cdot 26 \right) \\ &= \frac{2}{3}. \end{aligned}$$

Therefore, the answer is $\boxed{\text{(A)} \frac{2}{3}}$.

Solution 5

Let a_1, a_2, a_3, a_4, a_5 be the five numbers chosen. Then, we can consider the first number and the four differences, $a_1, a_2 - a_1, a_3 - a_2, a_4 - a_3, a_5 - a_4$. Now, each must be positive integers and the sum is less than or equal to 30, which there are $\binom{30}{5}$ by Hockey-Stick Identity. Now, we can use the Linearity of Expectation on $a_2 - a_1, a_3 - a_2, a_4 - a_3, a_5 - a_4$ since each of them being 1 represents a pair of consecutive integer. For each of them, we have $\binom{28}{3} + \binom{27}{3} + \dots + 1 = \binom{29}{4}$. Now, we have four such numbers to consider, and by Linearity of Expectation, its $4 \cdot \binom{29}{4}$.

Now, we divide by $\binom{30}{5}$ to get the answer of $\boxed{\text{(A)} \frac{2}{3}}$.

Solution 6

We will first list out the possible pairs:

$$(1, 2), (2, 3), (3, 4), (4, 5), \dots, (29, 30)$$

. Now, we can find the probability of each one and multiply it by 29 because of Linearity of Expectation. So, we have:

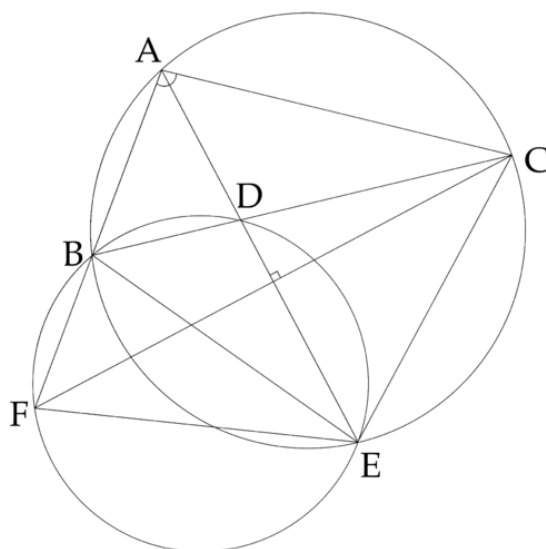
$$29 \cdot \frac{\binom{28}{3}}{\binom{30}{5}} = \boxed{\text{(A)} \frac{2}{3}}$$

24. (AMC 12 12B Fall 2021) Triangle ABC has side lengths $AB = 11$, $BC = 24$, and $CA = 20$. The bisector of $\angle BAC$ intersects $\overline{BC}$ in point D , and intersects the circumcircle of $\triangle ABC$ in point $E \neq A$. The circumcircle of $\triangle BED$ intersects the line AB in points B and $F \neq B$. What is CF ?

- (A) 28 (B) $20\sqrt{2}$ (C) 30 (D) 32 (E) $20\sqrt{3}$

Answer: C

Solution 1 (Reflection)



By the Law of Cosine $\cos A = \frac{AC^2 + AB^2 - BC^2}{2 \cdot AC \cdot AB} = \frac{20^2 + 11^2 - 24^2}{2 \cdot 20 \cdot 11} = -\frac{1}{8}$

As $ABCE$ is a cyclic quadrilateral, $\angle CEA = \angle CBA$. As $BDEF$ is a cyclic quadrilateral, $\angle CBA = \angle FEA$.

$\therefore \angle CEA = \angle FEA$ and $\angle CAE = \angle FAE \therefore \triangle AFE \cong \triangle ACE$ by ASA

Hence, $AF = AC = 20$

By the Law of Cosine $CF = \sqrt{20^2 + 20^2 - 2 \cdot 20 \cdot 20(-\frac{1}{8})} = \sqrt{900} = \boxed{\text{C } 30}$

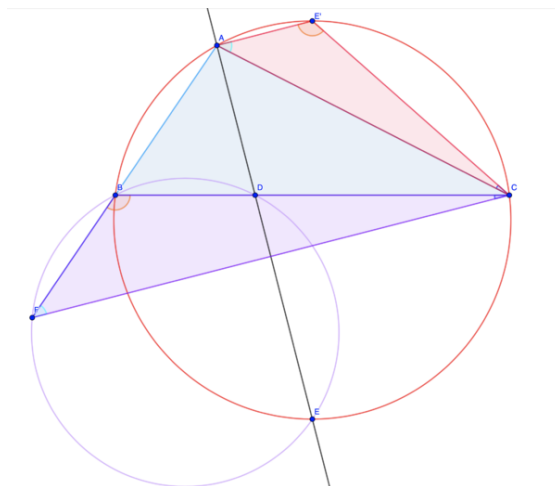
Note that F is C 's reflection over line AE , quadrilateral $ACEF$ is a kite symmetrical by line AE , $AE \perp CF$.

Solution 2 (Olympiad Solution using Spiral Similarity)

Construct the E -antipode, $E' \in (ABC)$. Notice $\triangle CE'A \stackrel{\pm}{\sim} \triangle CBF$ by spiral similarity at C , thus $CF = \frac{CB \cdot CA}{CE'} = \frac{480}{CE'}$. Let $CE' = x$; by symmetry $BE' = x$ as well and $\cos \angle BE'C = \cos \angle A = \frac{11^2 + 20^2 - 24^2}{2 \cdot 11 \cdot 20} = -\frac{1}{8}$ from Law of Cosines in $\triangle ABC$, so by Law of Cosines in $\triangle BE'C$ we have

$$x^2 + x^2 + (2x^2) \left(-\frac{1}{8}\right) = 24^2$$

from which $x = 16$. Now, $CF = \frac{480}{16} = \boxed{\text{C } 30}$.



Solution 3

Claim: $\triangle ADC \sim \triangle ABE$.

Proof: Note that $\angle CAD = \angle CAE = \angle EAB$ and $\angle DCA = \angle BCA = \angle BEA$ meaning that our claim is true by AA similarity.

Because of this similarity, we have that

$$\frac{AC}{AD} = \frac{AE}{AB} \implies AB \cdot AC = AD \cdot AE = AB \cdot AF$$

by Power of a Point. Thus, $AC = AF = 20$.

Two solution methods follow from here.

Solution 3.1 (Stewart's theorem)

Applying **Stewart's theorem** on $\triangle ABC$ with cevian $\overline{CF}$ using the **directed lengths** $AF = AC = 20$ and $FB = 11 - 20 = -9$, we obtain

$$(20)(-9)(11) + (CF)(11)(CF) = (24)(20)(24) + (20)(-9)(20)$$

$$11CF^2 - 1980 = 11520 - 3600$$

$$\text{so } CF = \sqrt{\frac{11520 - 3600 + 1980}{11}} = \sqrt{\frac{9900}{11}} = \sqrt{900} = \boxed{\text{(C) } 30}.$$

Solution 3.2 (Double Cosine Law)

Note that $\angle CAF = \angle CAB$ so we may plug into Law of Cosines to find the angle's cosine:

$$AB^2 + AC^2 - 2 \cdot AB \cdot AC \cdot \cos(\angle CAB) = BC^2 \rightarrow \cos(\angle CAB) = -\frac{1}{8}.$$

So, we observe that we can use Law of Cosines again to find CF :

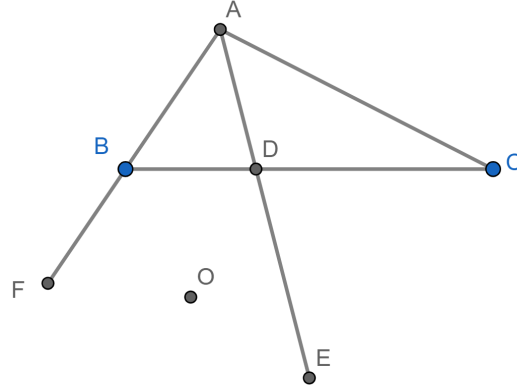
$$CF^2 = AF^2 + AC^2 - 2 \cdot AF \cdot AC \cdot \cos(\angle CAF) = 900 \rightarrow CF = \boxed{\text{(C) } 30}$$

both ways.

- Kevinmathz

Solution 4

This solution is based on this figure:



Denote by O the circumcenter of $\triangle BED$. Denote by R the circumradius of $\triangle BED$. In $\triangle BCF$, following from the law of cosines, we have

$$\begin{aligned} CF^2 &= BC^2 + BF^2 - 2BC \cdot BF \cos \angle CBF \\ &= BC^2 + BF^2 + 2BC \cdot BF \cos \angle ABC. \end{aligned} \quad (1)$$

For BF , we have

$$\begin{aligned} BF &= 2R \cos \angle FBO \\ &= 2R \cos (180^\circ - \angle ABC - \angle CBO) \\ &= 2R \cos \left(180^\circ - \angle ABC - \frac{180^\circ - \angle BOD}{2} \right) \\ &= 2R \cos \left(180^\circ - \angle ABC - \frac{180^\circ - 2\angle BED}{2} \right) \\ &= 2R \cos \left(180^\circ - \angle ABC - \frac{180^\circ - 2\angle BCA}{2} \right) \\ &= 2R \cos (90^\circ - \angle ABC + \angle BCA) \\ &= 2R \sin (\angle ABC - \angle BCA) \\ &= \frac{BD}{\sin \angle BED} \sin (\angle ABC - \angle BCA) \\ &= \frac{BD}{\sin \angle BCA} \sin (\angle ABC - \angle BCA) \\ &= BD (\sin \angle ABC \cot \angle BCA - \cos \angle ABC). \end{aligned} \quad (2)$$

The fourth equality follows from the property that B, D, E are concyclic. The fifth and the ninth equalities follow from the property that A, B, C, E are concyclic.

Because AD bisects $\angle BAC$, following from the angle bisector theorem, we have

$$\frac{BD}{CD} = \frac{AB}{AC}.$$

Hence, $BD = \frac{24 \cdot 11}{31}$.

In $\triangle ABC$, following from the law of cosines, we have

$$\cos \angle ABC = \frac{AB^2 + BC^2 - AC^2}{2AB \cdot BC}$$

$$= \frac{9}{16}$$

and

$$\begin{aligned}\cos \angle BCA &= \frac{AC^2 + BC^2 - AB^2}{2AC \cdot BC} \\ &= \frac{57}{64}.\end{aligned}$$

Hence, $\sin \angle ABC = \frac{5\sqrt{7}}{16}$ and $\sin \angle BCA = \frac{11\sqrt{7}}{64}$. Hence, $\cot \angle BCA = \frac{57}{11\sqrt{7}}$.

Now, we are ready to compute BF whose expression is given in Equation (2). We get $BF = 9$.

Now, we can compute CF whose expression is given in Equation (1). We have $CF = 30$.

Therefore, the answer is (C) 30.

Solution 5

Denote $B = (0, 0)$ and $C = (24, 0)$. Note that by Heron's formula the area of $\triangle ABC$ is $\frac{165\sqrt{7}}{4}$ so the y -coordinate of A (height of A above the x -axis) is easily computed by the base-height formula as $\frac{55\sqrt{7}}{16}$.

Now, since $AB = 11$, the x -coordinate of A satisfies $x^2 + (\frac{55\sqrt{7}}{16})^2 = 11^2$ and solving gives $x = \frac{99}{16}$.

The circumcircle of $\triangle ABC$ has radius $\frac{abc}{4A} = \frac{11 \cdot 24 \cdot 20}{165\sqrt{7}} = \frac{32}{\sqrt{7}}$. We know by the perpendicular bisector rule that the circumcenter O is located directly below the midpoint of $\overline{BC}$ (x -coordinate 12).

So, the negative y coordinate of O satisfies $12^2 + y^2 = (\frac{32}{\sqrt{7}})^2$ and solving gives $y = -\frac{4}{\sqrt{7}}$.

It's also clear that point E is going to be located directly below O on the circle, because the angle bisector intersects the circumcircle at the midpoint of the arc (Fact 5). Since the radius of the circle is $\frac{32}{\sqrt{7}}$, we have the coordinates of $E = (12, -\frac{36}{\sqrt{7}})$.

Solving for point D (the point on the x -axis between A and E), we get that $D = (\frac{264}{31}, 0)$.

So now we know six of the critical points: $A = (\frac{99}{16}, \frac{55\sqrt{7}}{16})$; $B = (0, 0)$; $C = (24, 0)$; $D = (\frac{264}{31}, 0)$; $E = (12, -\frac{36}{\sqrt{7}})$; $O = (12, -\frac{4}{\sqrt{7}})$.

We are now ready to add in the circumcircle of $\triangle BDE$, which has radius $\frac{BD \cdot DE \cdot BE}{4[BDE]}$.

From the above information, $BD = \frac{264}{31}$, $DE = \sqrt{(\frac{108}{31})^2 + (\frac{36}{\sqrt{7}})^2}$, and $BE = \sqrt{12^2 + (\frac{36}{\sqrt{7}})^2}$.

After a bit of simplification we end up with $DE = \frac{1152}{31\sqrt{7}}$ and $BE = \frac{48}{\sqrt{7}}$.

For the area of $\triangle BDE$, the altitude dropped from vertex E has height $\frac{36}{\sqrt{7}}$, and the base $\overline{BD}$ has length $\frac{264}{31}$, so its area is $\frac{1}{2} \cdot \frac{36}{\sqrt{7}} \cdot \frac{264}{31} = \frac{4752}{31\sqrt{7}}$.

Thus, $\frac{BD \cdot DE \cdot BE}{4[BDE]} = \frac{\frac{264}{31} \cdot \frac{1152}{31\sqrt{7}} \cdot \frac{48}{\sqrt{7}}}{4 \cdot \frac{4752}{31\sqrt{7}}}$ which after tons of cancellations becomes $\frac{768}{31\sqrt{7}}$.

We know from the perpendicular bisector rule that the circumcenter P of $\triangle BDE$ is located directly below the midpoint of $\overline{BD}$ (x -coordinate $\frac{132}{31}$).

So, the negative y -coordinate of P satisfies $(\frac{132}{31})^2 + y^2 = (\frac{768}{31\sqrt{7}})^2$, and solving gives $y = -\frac{684}{31\sqrt{7}}$. Thus, the equation of the circumcircle of $\triangle BDE$ is $(x - \frac{132}{31})^2 + (y + \frac{684}{31\sqrt{7}})^2 = (\frac{768}{31\sqrt{7}})^2$.

Point F is the intersection of this circle and the line $\overline{AB}$, which has equation $y = \frac{5\sqrt{7}}{9}x$. So, we substitute $y = \frac{5\sqrt{7}}{9}x$ into the equation of the circle to get $(x - \frac{132}{31})^2 + (\frac{5\sqrt{7}}{9}x + \frac{684}{31\sqrt{7}})^2 = (\frac{768}{31\sqrt{7}})^2$.

After simplifying, we have $\frac{256}{81}x^2 + 16x = 0$ (the $\frac{768}{31\sqrt{7}}$'s cancel out), whose solutions are $x = 0$ and $x = -\frac{81}{16}$. The first corresponds to the origin, and the second corresponds to point F . Thus the coordinates of F are $(-\frac{81}{16}, \frac{5\sqrt{7}}{9} \cdot -\frac{81}{16}) = (-\frac{81}{16}, -\frac{45\sqrt{7}}{16})$.

The coordinates of C are $(24, 0)$, so

$$CF = \sqrt{(24 + \frac{81}{16})^2 + (\frac{45\sqrt{7}}{16})^2} = \sqrt{(\frac{465}{16})^2 + (\frac{45\sqrt{7}}{16})^2} = \frac{\sqrt{465^2 + (45\sqrt{7})^2}}{16} = \frac{\sqrt{(15 \cdot 31)^2 + (15 \cdot 3\sqrt{7})^2}}{16}$$

25. (AMC 12 12B Fall 2021) For n a positive integer, let $R(n)$ be the sum of the remainders when n is divided by 2, 3, 4, 5, 6, 7, 8, 9, and 10. For example, $R(15) = 1 + 0 + 3 + 0 + 3 + 1 + 7 + 6 + 5 = 26$. How many two-digit positive integers n satisfy $R(n) = R(n+1)$? (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Answer: C

Solution 1

Note that we can add 9 to $R(n)$ to get $R(n+1)$, but must subtract k for all $k|n+1$. Hence, we see that there are four ways to do that because $9 = 7 + 2 = 6 + 3 = 5 + 4 = 4 + 3 + 2$. Note that only $7 + 2$ is a plausible option, since $4 + 3 + 2$ indicates $n + 1$ is

§6 AMC 12 12A Fall 2021

1. (AMC 12 12A Fall 2021) What is the value of $\frac{(2112-2021)^2}{169}$? (A) 7 (B) 21 (C) 49 (D) 64 (E) 91

Answer: C

Solution 1 (Laws of Exponents)

We have

$$\frac{(2112 - 2021)^2}{169} = \frac{91^2}{169} = \frac{91^2}{13^2} = \left(\frac{91}{13}\right)^2 = 7^2 = \boxed{\text{(C) } 49}.$$

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Solution 2 (Difference of Squares)

We have

$$\frac{(2112 - 2021)^2}{169} = \frac{91^2}{169} = \frac{(10^2 - 3^2)^2}{13^2} = \frac{((10 + 3)(10 - 3))^2}{13^2} = \frac{(13 \cdot 7)^2}{13^2} = \frac{13^2 \cdot 7^2}{13^2} = 7^2 = \boxed{\text{(C) } 49}.$$

Solution 3 (Estimate)

We know that $2112 - 2021 = 91$. Approximate this as 100 as it is pretty close to it. Also, approximate 169 to 170. We then have

$$\frac{(2112 - 2021)^2}{169} \approx \frac{100^2}{170} \approx \frac{1000}{17} \approx 58.$$

Now check the answer choices. The two closest answers are 49 and 64. As the numerator is actually bigger than it should be, it should be the smaller answer, or **(C) 49**.

2. (AMC 12 12A Fall 2021) Menkara has a 4×6 index card. If she shortens the length of one side of this card by 1 inch, the card would have area 18 square inches. What would the area of the card be in square inches if instead she shortens the length of the other side by 1 inch?
- (A) 16 (B) 17 (C) 18 (D) 19 (E) 20

Answer: E

Solution

We construct the following table:

Scenario	Length	Width	Area
Initial	4	6	24
Menkara shortens one side.	3	6	18
Menkara shortens other side instead.	4	5	20

Therefore, the answer is **(E) 20**.

3. (AMC 12 12A Fall 2021) Mr. Lopez has a choice of two routes to get to work. Route A is 6 miles long, and his average speed along this route is 30 miles per hour. Route B is 5 miles long, and his average speed along this route is 40 miles per hour, except for a $\frac{1}{2}$ -mile stretch in a school zone where his average speed is 20 miles per hour. By how many minutes is Route B quicker than Route A?
- (A) $2\frac{3}{4}$ (B) $3\frac{3}{4}$ (C) $4\frac{1}{2}$ (D) $5\frac{1}{2}$ (E) $6\frac{3}{4}$

Answer: B

Solution 1

If Mr. Lopez chooses Route A, then he will spend $\frac{6}{30} = \frac{1}{5}$ hour, or $\frac{1}{5} \cdot 60 = 12$ minutes.

If Mr. Lopez chooses Route B, then he will spend $\frac{9/2}{40} + \frac{1/2}{20} = \frac{11}{80}$ hour, or $\frac{11}{80} \cdot 60 = 8\frac{1}{4}$ minutes.

Therefore, Route B is quicker than Route A by $12 - 8\frac{1}{4} = \mathbf{(B) 3\frac{3}{4}}$ minutes.

Solution 2

We use the equation $d = st$ to solve this problem. Recall that 1 mile per hour is equal to $\frac{1}{60}$ mile per minute.

For Route A, the distance is 6 miles and the speed to travel this distance is $\frac{1}{2}$ mile per minute. Thus, the time it takes on Route A is 12 minutes.

For Route B, we have to use the equation twice: once for the distance of $5 - \frac{1}{2} = \frac{9}{2}$ miles with a speed of $\frac{2}{3}$ mile per minute and a distance of $\frac{1}{2}$ miles at a speed of $\frac{1}{3}$

mile per minute. Thus, the time it takes to go on Route B is $\frac{9}{2} \cdot \frac{3}{2} + \frac{1}{2} \cdot 3 = \frac{27}{4} + \frac{3}{2} = \frac{33}{4}$ minutes. Thus, Route B is $12 - \frac{33}{4} = \frac{15}{4} = 3\frac{3}{4}$ faster than Route A. Thus, the answer is (B) $3\frac{3}{4}$.

4. (AMC 12 12A Fall 2021) The six-digit number $\underline{2}\underline{0}\underline{2}\underline{1}\underline{0}\underline{A}$ is prime for only one digit A . What is A ? (A) 1 (B) 3 (C) 5 (D) 7 (E) 9

Answer: E

Solution 1

First, modulo 2 or 5, $\underline{20210A} \equiv A$. Hence, $A \neq 0, 2, 4, 5, 6, 8$.

Second modulo 3, $\underline{20210A} \equiv 2 + 0 + 2 + 1 + 0 + A \equiv 5 + A$. Hence, $A \neq 1, 4, 7$.

Third, modulo 11, $\underline{20210A} \equiv A + 1 + 0 - 0 - 2 - 2 \equiv A - 3$. Hence, $A \neq 3$.

Therefore, the answer is (E) 9.

Solution 2 (Elimination)

Any number ending in 5 is divisible by 5. So we can eliminate option (C).

If the sum of the digits of a number is divisible by 3, the number is divisible by 3. The sum of the digits of this number is $2 + 0 + 2 + 1 + 0 + A = 5 + A$. If $5 + A$ is divisible by 3, the number is divisible by 3. Thus we can eliminate options (A) and (D).

So the correct option is either (B) or (E). Let's try dividing the number with some integers.

$20210A/7 = 2887x$, where x is $1A/7$. Since 13 and 19 are both indivisible by 7, this does not help us narrow the choices down.

$20210A/11 = 1837x$, where x is $3A/11$. Since $33/11 = 3$, option (B) would make $20210A$ divisible by 11. Thus, by elimination, the correct choice must be option (E) 9.

Solution 3

$202100 \implies$ divisible by 2.

$202101 \implies$ divisible by 3.

$202102 \implies$ divisible by 2.

$202103 \implies$ divisible by 11.

$202104 \implies$ divisible by 2.

$202105 \implies$ divisible by 5.

$202106 \implies$ divisible by 2.

$202107 \implies$ divisible by 3.

$202108 \implies$ divisible by 2.

This leaves only $A =$ (E) 9.

Sidenote The divisibility test for 11 is if the difference between the sum of the alternating digits is 0 or 11, then that number is divisible by 11. For 202103, we have $2+2-1-3=0$.

For 7, the divisibility test can be slow. Take the last digit, double it, and subtract it from the rest of the number. If it is divisible by 7, then the original number

is divisible by 7. If the number is too large to determine whether or not it is a multiple of 7, then we repeat the process until we can determine whether or not the number is divisible by 7.

5. (*AMC 12 12A Fall 2021*) Elmer the emu takes 44 equal strides to walk between consecutive telephone poles on a rural road. Oscar the ostrich can cover the same distance in 12 equal leaps. The telephone poles are evenly spaced, and the 41st pole along this road is exactly one mile (5280 feet) from the first pole. How much longer, in feet, is Oscar's leap than Elmer's stride?
- (A) 6 (B) 8 (C) 10 (D) 11 (E) 15

Answer: B

Solution 1

There are $41 - 1 = 40$ gaps between the 41 telephone poles, so the distance of each gap is $5280 \div 40 = 132$ feet.

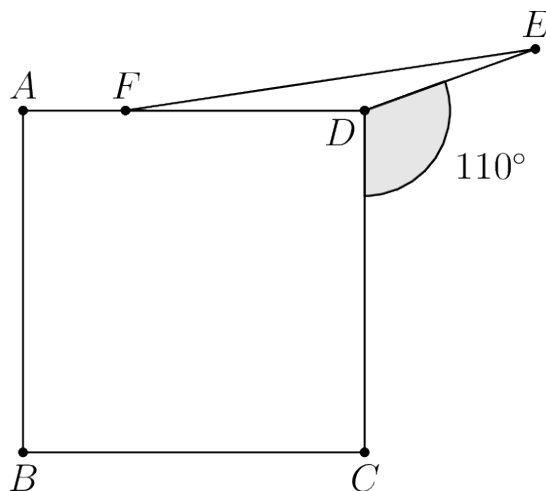
Each of Oscar's leaps covers $132 \div 12 = 11$ feet, and each of Elmer's strides covers $132 \div 44 = 3$ feet.

Therefore, Oscar's leap is $11 - 3 = \boxed{\text{(B) } 8}$ feet longer than Elmer's stride.

Solution 2

There are $41 - 1 = 40$ gaps between the 41 telephone poles, so Elmer takes $44 \cdot 40 = 1760$ strides in total, and Oscar takes $12 \cdot 40 = 480$ leaps in total. Therefore, the answer is $(5280 \div 480) - (5280 \div 1760) = 11 - 3 = \boxed{\text{(B) } 8}$.

6. (*AMC 12 12A Fall 2021*) As shown in the figure below, point E lies on the opposite half-plane determined by line CD from point A so that $\angle CDE = 110^\circ$. Point F lies on $\overline{AD}$ so that $DE = DF$, and $ABCD$ is a square. What is the degree measure of $\angle AFE$?



- (A) 160 (B) 164 (C) 166 (D) 170 (E) 174

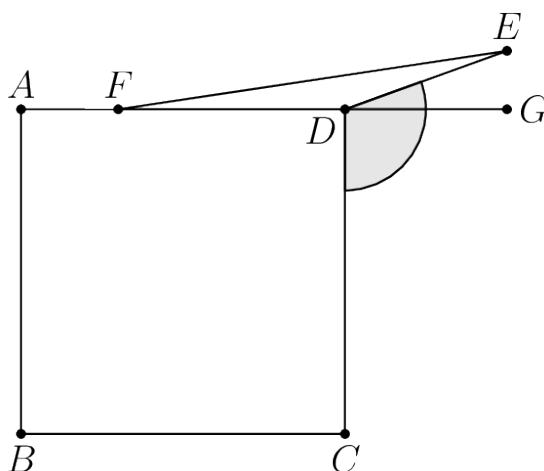
Answer: D

Solution 1

By angle subtraction, we have $\angle ADE = 360^\circ - \angle ADC - \angle CDE = 160^\circ$. Note that $\triangle DEF$ is isosceles, so $\angle DFE = \frac{180^\circ - \angle ADE}{2} = 10^\circ$. Finally, we get $\angle AFE = 180^\circ - \angle DFE = \boxed{\text{(D) } 170}$ degrees.

Solution 2 (Extension)

We can extend $\overline{AD}$ to G , making $\angle CDG$ a right angle. It follows that $\angle GDE$ is $110^\circ - 90^\circ = 20^\circ$, as shown below.



Since $\angle DFE = \angle DEF$, we see that $\angle DFE = \angle DEF = \frac{20}{2} = 10^\circ$. Thus, $\angle AFE = 180^\circ - 10^\circ = \boxed{\text{(D)} 170}$ degrees.

7. (AMC 12 12A Fall 2021) A school has 100 students and 5 teachers. In the first period, each student is taking one class, and each teacher is teaching one class. The enrollments in the classes are 50, 20, 20, 5, and 5. Let t be the average value obtained if a teacher is picked at random and the number of students in their class is noted. Let s be the average value obtained if a student was picked at random and the number of students in their class, including the student, is noted. What is $t - s$?

(A) -18.5 (B) -13.5 (C) 0 (D) 13.5 (E) 18.5

Answer: B

Solution

The formula for expected values is

$$\text{Expected Value} = \sum (\text{Outcome} \cdot \text{Probability}).$$

We have

$$\begin{aligned} t &= 50 \cdot \frac{1}{5} + 20 \cdot \frac{1}{5} + 20 \cdot \frac{1}{5} + 5 \cdot \frac{1}{5} + 5 \cdot \frac{1}{5} \\ &= (50 + 20 + 20 + 5 + 5) \cdot \frac{1}{5} \\ &= 100 \cdot \frac{1}{5} \\ &= 20, \\ s &= 50 \cdot \frac{50}{100} + 20 \cdot \frac{20}{100} + 20 \cdot \frac{20}{100} + 5 \cdot \frac{5}{100} + 5 \cdot \frac{5}{100} \\ &= 25 + 4 + 4 + 0.25 + 0.25 \\ &= 33.5. \end{aligned}$$

Therefore, the answer is $t - s = \boxed{\text{(B)} -13.5}$.

8. (AMC 12 12A Fall 2021) Let M be the least common multiple of all the integers 10 through 30, inclusive. Let N be the least common multiple of M , 32, 33, 34, 35, 36, 37, 38, 39, and 40. What is the value of $\frac{N}{M}$? (A) 1 (B) 2 (C) 37 (D) 74 (E) 2886

Answer: D

Solution

By the definition of least common mutiple, we take the greatest powers of the prime numbers of the prime factorization of all the numbers, that we are taking the lcm of. In this case,

$$M = 2^4 \cdot 3^3 \cdot 5^2 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23 \cdot 29.$$

Now, using the same logic, we find that

$$N = M \cdot 2 \cdot 37,$$

because we have an extra power of 2 and an extra power of 37. Thus, $\frac{N}{M} = 2 \cdot 37 =$

(D) 74.

Solution 2

We know that between 32 and 40, there is one prime number, namely 37. This was not included in the previous M , so we know that the value of $\frac{N}{M}$ will include a new 37. Then, let's consider each individual number.

$$32 = 2^5$$

has a new power of 2 now included in M , so we add that to our list.

$$33 = 3 \cdot 11$$

, which was included in the previous M . Similarly, we can go through the numbers 32 to 40 and find that the only new powers of numbers added are

$$2 \cdot 37 = 74$$

, which gives $\frac{N}{M} = 2 \cdot 37 =$ **(D) 74**.

9. (AMC 12 12A Fall 2021) A right rectangular prism whose surface area and volume are numerically equal has edge lengths $\log_2 x$, $\log_3 x$, and $\log_4 x$. What is x ? (A) $2\sqrt{6}$ (B) $6\sqrt{6}$ (C) 24 (D) 48 (E) 576

Answer: E

Solution 1

The surface area of this right rectangular prism is $2(\log_2 x \log_3 x + \log_2 x \log_4 x + \log_3 x \log_4 x)$.

The volume of this right rectangular prism is $\log_2 x \log_3 x \log_4 x$.

Equating the numerical values of the surface area and the volume, we have

$$2(\log_2 x \log_3 x + \log_2 x \log_4 x + \log_3 x \log_4 x) = \log_2 x \log_3 x \log_4 x.$$

Dividing both sides by $\log_2 x \log_3 x \log_4 x$, we get

$$2 \left(\frac{1}{\log_4 x} + \frac{1}{\log_3 x} + \frac{1}{\log_2 x} \right) = 1. \quad (\star)$$

Recall that $\log_b a = \frac{1}{\log_a b}$ and $\log_b (a^n) = n \log_b a$, so we rewrite (★) as

$$\begin{aligned} 2(\log_x 4 + \log_x 3 + \log_x 2) &= 1 \\ 2 \log_x 24 &= 1 \\ \log_x 576 &= 1 \\ x &= \boxed{\text{(E) } 576}. \end{aligned}$$

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Solution 2

We know that the formulae for the surface area and volume of a right rectangular prism are $S = 2lw + 2wh + 2lh$ and $V = lwh$, respectively. It is given that these values are numerically equal, so we can set $2lw + 2wh + 2lh = lwh$. Dividing both sides by lwh gives $\frac{2}{h} + \frac{2}{l} + \frac{2}{w} = 1$. We then substitute values for l , w , and h , which yields $\frac{2}{\log_2 x} + \frac{2}{\log_3 x} + \frac{2}{\log_4 x} = 1$. Then, we can recall the logarithm base-change formula $\log_b a = \frac{\log_x a}{\log_x b}$. We can use this formula to change all the sidelengths to log base 2 as follows:

$$\begin{aligned} \log_3 x &= \frac{\log_2 x}{\log_2 3} \\ \log_4 x &= \frac{\log_2 x}{\log_2 4} \end{aligned}$$

Then substitute back into our equation to get

$$1 = \frac{2}{\log_2 x} + \frac{2}{\frac{\log_2 x}{\log_2 3}} + \frac{2}{\frac{\log_2 x}{\log_2 4}}$$

.Simplifying this equation, as well as the fact that $\log_2 4 = 2$, we get:

$$1 = \frac{2}{\log_2 x} + \frac{2 \log_2 3}{\log_2 x} + \frac{4}{\log_2 x}$$

Further simplification yields:

$$\begin{aligned} 1 &= \frac{2 + 2 \log_2 3 + 4}{\log_2 x} \\ 1 &= \frac{6 + \log_2 9}{\log_2 x} \\ \log_2 x &= 6 + \log_2 9 \\ \log_2 x - \log_2 9 &= 6 \\ \log_2 \frac{x}{9} &= 6 \\ \frac{x}{9} &= 2^6 \\ x &= 9 \cdot 64 = \boxed{\text{(E) } 576} \end{aligned}$$

10. (AMC 12 12A Fall 2021) The base-nine representation of the number N is $27,006,000,052_9$. What is the remainder when N is divided by 5? (A) 0 (B) 1 (C) 2 (D) 3

Answer: D

Solution 1 (Modular Arithmetic)

Recall that $9 \equiv -1 \pmod{5}$. We expand N by the definition of bases:

$$\begin{aligned}
 N &= 27,006,000,052_9 \\
 &= 2 \cdot 9^{10} + 7 \cdot 9^9 + 6 \cdot 9^6 + 5 \cdot 9 + 2 \\
 &\equiv 2 \cdot (-1)^{10} + 7 \cdot (-1)^9 + 6 \cdot (-1)^6 + 2 \pmod{5} \\
 &\equiv 2 - 7 + 6 + 2 \pmod{5} \\
 &\equiv \boxed{\text{(D)} 3} \pmod{5}.
 \end{aligned}$$

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Solution 2 (Powers of 9)

We need to first convert N into a regular base-10 number:

$$N = 27,006,000,052_9 = 2 \cdot 9^{10} + 7 \cdot 9^9 + 6 \cdot 9^6 + 5 \cdot 9 + 2.$$

Now, consider how the last digit of 9 changes with changes of the power of 9 :

$$\begin{aligned}
 9^0 &= 1, \\
 9^1 &= 9, \\
 9^2 &= \dots 1, \\
 9^3 &= \dots 9, \\
 9^4 &= \dots 1, \\
 &\vdots
 \end{aligned}$$

Note that if x is odd, then $9^x \equiv 4 \pmod{5}$. On the other hand, if x is even, then $9^x \equiv 1 \pmod{5}$.

Therefore, we have

$$\begin{aligned}
 N &\equiv 2 \cdot (1) + 7 \cdot (4) + 6 \cdot (1) + 5 \cdot (4) + 2 \cdot (1) \pmod{5} \\
 &\equiv 2 + 28 + 6 + 20 + 2 \pmod{5} \\
 &\equiv 58 \pmod{5} \\
 &\equiv \boxed{\text{(D)} 3} \pmod{5}.
 \end{aligned}$$

Note that for the odd case, $9^x \equiv -1 \pmod{5}$ may simplify the process further, as given by Solution 1.

Remark By the long division algorithm, you can work from left to right accumulating the answer, and don't need to count the digits to get the power. You only need to take the current leading digit modulo 5, negate it, and add it to the next digit, and repeat the process until you reach the units digit.

11. (AMC 12 12A Fall 2021) Consider two concentric circles of radius 17 and 19. The larger circle has a chord, half of which lies inside the smaller circle. What is the length of the chord in the larger circle?

(A) $12\sqrt{2}$ (B) $10\sqrt{3}$ (C) $\sqrt{17 \cdot 19}$ (D) 18 (E) $8\sqrt{6}$

Answer: E

Solution 1 (Pythagorean Theorem)

Label the center of both circles O . Label the chord in the larger circle as $\overline{ABCD}$, where A and D are on the larger circle and B and C are on the smaller circle. Construct the radius perpendicular to the chord and label their intersection as M . Because a radius that is perpendicular to a chord bisects the chord, M is the midpoint of the chord.

Construct segments $\overline{AO}$ and $\overline{BO}$. These are radii with lengths 17 and 19 respectively. Then, use the **Pythagorean Theorem**. In $\triangle OMA$, we have

$$\begin{aligned} OM^2 &= OA^2 - AM^2 \\ &= OA^2 - \left(\frac{AD}{2}\right)^2 \\ &= 19^2 - \frac{AD^2}{4}. \end{aligned}$$

In $\triangle OMB$, we have

$$\begin{aligned} OM^2 &= OB^2 - BM^2 \\ &= OB^2 - \left(\frac{BC}{2}\right)^2 \\ &= OB^2 - \left(\frac{AD}{4}\right)^2 \\ &= 17^2 - \frac{AD^2}{16}. \end{aligned}$$

Equating these two expressions, we get

$$19^2 - \frac{AD^2}{4} = 17^2 - \frac{AD^2}{16}$$

and $AD = \boxed{\text{(E)} 8\sqrt{6}}$.

Solution 2 (Power of a Point)

Draw the diameter perpendicular to the chord. Notice that by symmetry this diameter bisects the chord.

Call the intersection between that diameter and the chord A . In the smaller circle, let the shorter piece of the diameter cut by the chord be x , making the longer piece $34 - x$. In that same circle, let the y be the length of the portion of the chord that is cut by the diameter.

The radius of the larger circle is 2 more than the radius of the small circle. So, in the larger circle, the shorter piece of the diameter cut by the chord is of length $x + 2$ and the longer piece is $36 - x$. As given in the problem, the bisected length of the chord in the larger circle is twice as much, so it is of length $2y$. By **Power of a Point**, we can construct a system of equations

$$\begin{aligned} x(34 - x) &= y^2 \\ (x + 2)(36 - x) &= (2y)^2. \end{aligned}$$

Expanding both equations, we get

$$\begin{aligned} 34x - x^2 &= y^2 \\ 36x - x^2 + 72 - 2x &= 4y^2, \end{aligned}$$

in which the $34x$ and $-x^2$ terms magically cancel when we subtract the first equation from the second equation. Thus, now we have

$$72 = 3y^2 \implies y = 2\sqrt{6} \implies 4y = \boxed{\text{(E)} 8\sqrt{6}}.$$

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Solution 3 (Coordinate Geometry)

Represent the circles as $x^2 + y^2 = 17^2$, and $x^2 + y^2 = 19^2$. Solving for x in these equations we obtain $x = \sqrt{17^2 - y^2}$ and $x = \sqrt{19^2 - y^2}$. Because half of the chord is in the smaller circle, the larger circle should have an x value that is twice as big as the smaller circle's x value. We set up the equation:

$$\begin{aligned} 2\sqrt{289 - y^2} &= \sqrt{361 - y^2} \\ 4(289 - y^2) &= 361 - y^2 \\ y^2 &= 265 \end{aligned}$$

Substituting y^2 into $x^2 + y^2 = 19^2$, we obtain $x = \sqrt{96} = 4\sqrt{6}$. However, this is only half of the chord length, so we must double it to obtain $\boxed{\text{(E)} 8\sqrt{6}}$.

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12. (AMC 12 12A Fall 2021) What is the number of terms with rational coefficients among the 1001 terms in the expansion of $(x\sqrt[3]{2} + y\sqrt{3})^{1000}$? (A) 0 (B) 166 (C) 167 (D) 500

Answer: C

Solution 1

By the Binomial Theorem, each term in the expansion is of the form

$$\binom{1000}{k} (x\sqrt[3]{2})^k (y\sqrt{3})^{1000-k} = \binom{1000}{k} 2^{\frac{k}{3}} 3^{\frac{1000-k}{2}} x^k y^{1000-k},$$

where $k \in \{0, 1, 2, \dots, 1000\}$.

This problem is equivalent to counting the values of k such that both $\frac{k}{3}$ and $\frac{1000-k}{2}$ are integers. Note that k must be a multiple of 3 and a multiple of 2, so k must be a multiple of 6. There are $\boxed{\text{(C)} 167}$ such values of k :

$$6(0), 6(1), 6(2), \dots, 6(166).$$

Solution 2 (Rough Conceptual Approach)

Similar to Solution 1, however, instead of applying the Binomial Theorem, we can notice that it "takes" two $\sqrt{3}$ terms multiplied by each other to get a rational coefficient for y and it "takes" three $\sqrt[3]{2}$ terms multiplied by each other to get a rational coefficient for x . Since the only way to get a total rational coefficient (that results when you multiply the final coefficient of y by that of for x) is for both x 's and y 's coefficients to be rational, we can notice that the solution simply comes down to the number of multiples of 6 between 0 and 1000, which turns out to be

$\boxed{\text{(C)} 167}$.

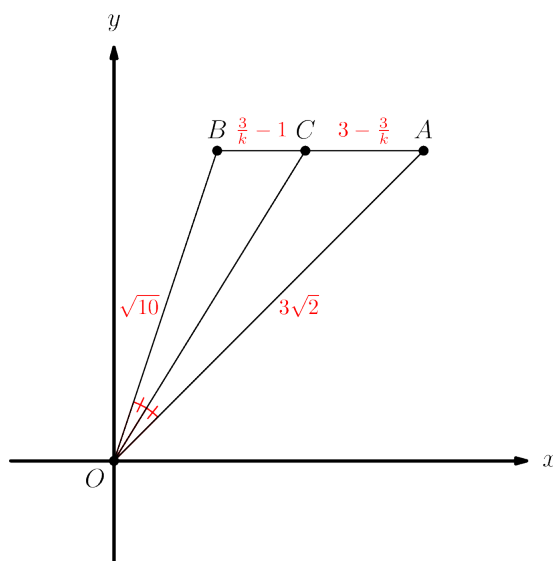
13. (AMC 12 12A Fall 2021) The angle bisector of the acute angle formed at the origin by the graphs of the lines $y = x$ and $y = 3x$ has equation $y = kx$. What is k ?
 (A) $\frac{1+\sqrt{5}}{2}$ (B) $\frac{1+\sqrt{7}}{2}$ (C) $\frac{2+\sqrt{3}}{2}$ (D) 2 (E) $\frac{2+\sqrt{5}}{2}$

Answer: A

Solution 1 (Angle Bisector Theorem)

This solution refers to the **Diagram** section.

Let $O = (0, 0)$, $A = (3, 3)$, $B = (1, 3)$, and $C = \left(\frac{3}{k}, 3\right)$. As shown below, note that $\overline{OA}$, $\overline{OB}$, and $\overline{OC}$ are on the lines $y = x$, $y = 3x$, and $y = kx$, respectively. By the Distance Formula, we have $OA = 3\sqrt{2}$, $OB = \sqrt{10}$, $AC = 3 - \frac{3}{k}$, and $BC = \frac{3}{k} - 1$.



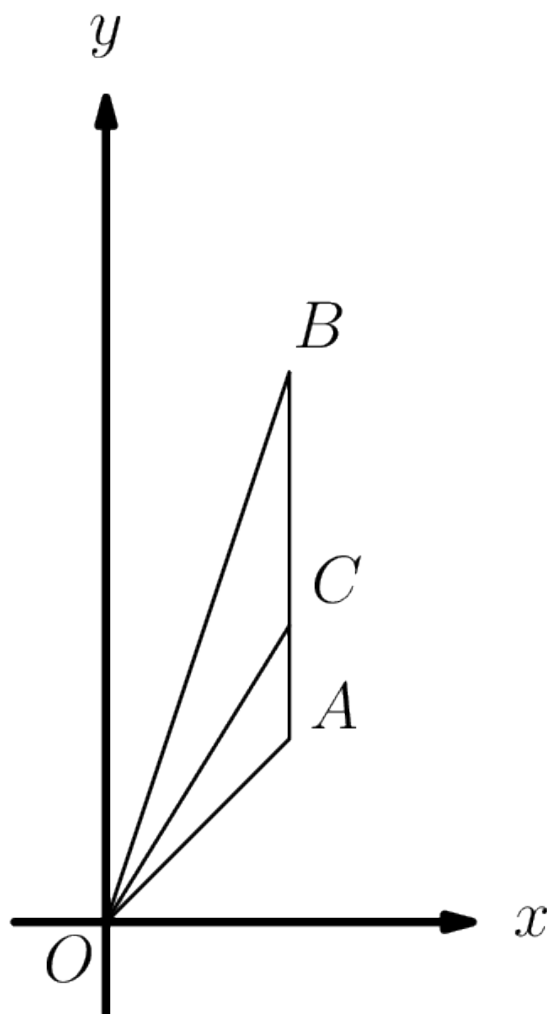
By the Angle Bisector Theorem, we get $\frac{OA}{OB} = \frac{AC}{BC}$, or

$$\begin{aligned}\frac{3\sqrt{2}}{\sqrt{10}} &= \frac{3 - \frac{3}{k}}{\frac{3}{k} - 1} \\ \frac{3\sqrt{2}}{\sqrt{10}} &= \frac{3k - 3}{3 - k} \\ \frac{\sqrt{2}}{\sqrt{10}} &= \frac{k - 1}{3 - k} \\ \frac{1}{5} &= \frac{(k - 1)^2}{(3 - k)^2} \\ 5(k - 1)^2 &= (3 - k)^2 \\ 4k^2 - 4k - 4 &= 0 \\ k^2 - k - 1 &= 0 \\ k &= \frac{1 \pm \sqrt{5}}{2}.\end{aligned}$$

Since $k > 0$, the answer is $k = \boxed{\text{(A)} \frac{1 + \sqrt{5}}{2}}$. **Remark**

The value of k is known as the **Golden Ratio**: $\phi = \frac{1+\sqrt{5}}{2} \approx 1.61803398875$.

Solution 2 (Analytic and Plane Geometry)



Consider the graphs of $f(x) = x$ and $g(x) = 3x$. Since it will be easier to consider at unity, let $x = 1$, then we have $f(1) = 1$ and $g(1) = 3$.

Now, let O be $(0, 0)$, A be $(1, 1)$, and B be $(1, 3)$. Cutting through side AB of triangle OAB is the angle bisector OC where C is on side AB .

Hence, by the Angle Bisector Theorem, we get $\frac{OB}{OA} = \frac{BC}{AC}$.

By the Pythagorean Theorem, $OA = \sqrt{2}$ and $OB = \sqrt{10}$. Therefore, $\frac{BC}{AC} = \sqrt{5} \implies BC = \sqrt{5}AC$.

Since $AB = AC + BC = 2$, it is easy derive $AC + \sqrt{5}AC = 2 \implies AC = \frac{2}{1+\sqrt{5}} = \frac{-1+\sqrt{5}}{2}$.

The vertical distance between the x -axis and C is $\frac{-1+\sqrt{5}}{2} + 1 = \frac{1+\sqrt{5}}{2}$. Because the x -coordinate of point C is 1, the slope we need to find is just the y -coordinate

(A) $\frac{1 + \sqrt{5}}{2}$.

Solution 3 (Analytic and Plane Geometry)

Let's begin by drawing a triangle that starts at the origin. Assume that the base of the triangle goes to the point $x = 1$. The line $x = y$ is the hypotenuse of a right

triangle with side length 1. The hypotenuse' length is $\sqrt{2}$. Then, let's draw the line $x = 3y$. We extend it to when $x = 1$. The length of the hypotenuse of the larger triangle is $\sqrt{10}$ with legs 1, 3. We then draw the angle bisector. We should label the triangle, so here we go. AC is 1. BC is 3. AB is $\sqrt{10}$. When the line with angle 45° intersects the line $x = 1$, call the point D . When the angle bisector intersects the line $x = 1$, call the point E . By Angle Bisector Theorem, $\frac{DE}{DB} = \frac{\sqrt{2}}{\sqrt{10}}$. Since BC is 3 and DC is 1, we have that BD is 2. Solving for DE , we get that DE is $\frac{\sqrt{5}-1}{2}$.

Since DE is $\frac{\sqrt{5}-1}{2}$, we have that CE is just one more than that. Therefore, CE is $\frac{1+\sqrt{5}}{2}$. Since AC is 1, we get that k is $\boxed{\text{(A)} \frac{1+\sqrt{5}}{2}}$.

<u>**Remark**</u>

The answer turns out to be the golden ratio or phi (ϕ). Phi has many properties and is related to the [Fibonacci sequence](#). See [Phi](#).

Solution 4 (Distance Between a Point and a Line)

Note that the distance between the point (m, n) to line $Ax + By + C = 0$, is $\frac{|Am+Bn+C|}{\sqrt{A^2+B^2}}$. Because line $y = kx$ is a perpendicular bisector, a point on the line $y = kx$ must be equidistant from the two lines ($y = x$ and $y = 3x$), call this point $P(z, w)$. Because, the line $y = kx$ passes through the origin, our requested value of k , which is the slope of the angle bisector line, can be found when evaluating the value of $\frac{w}{z}$. By the Distance from Point to Line formula we get the equation,

$$\frac{|3z - w|}{\sqrt{10}} = \frac{|z - w|}{\sqrt{2}}.$$

Note that $|3z - w| \geq 0$, because $y = 3x$ is higher than P and $|z - w| \leq 0$, because $y = x$ is lower to P . Thus, we solve the equation,

$$(3z - w)\sqrt{2} = (w - z)\sqrt{10} \Rightarrow 3z - w = \sqrt{5} \cdot (w - z) \Rightarrow (\sqrt{5} + 1)w = (3 + \sqrt{5})z.$$

Thus, the value of $\frac{w}{z} = \frac{3+\sqrt{5}}{1+\sqrt{5}} = \frac{1+\sqrt{5}}{2}$. Thus, the answer is $\boxed{\text{(A)} \frac{1+\sqrt{5}}{2}}$.

(Fun Fact: The value $\frac{1+\sqrt{5}}{2}$ is the golden ratio ϕ .)

Solution 5 (Trigonometry)

This problem can be trivialized using basic trig identities. Let the angle made by $y = x$ and the x -axis be θ_1 and the angle made by $y = 3x$ and the x -axis be θ_3 . Note that $\tan(\theta_1) = 1$ and $\tan(\theta_3) = 3$, and this is why we named them as such. Let the angle made by $y = kx$ be denoted as θ_k . Since $y = kx$ bisects the two lines, notice that

$$\theta_k - \theta_1 = \theta_3 - \theta_k.$$

Now, we can take the tangent and apply the tangent subtraction formula:

$$\begin{aligned} \tan(\theta_k - \theta_1) &= \tan(\theta_3 - \theta_k) \\ \frac{\tan(\theta_k) - \tan(\theta_1)}{1 + \tan(\theta_k) \tan(\theta_1)} &= \frac{\tan(\theta_3) - \tan(\theta_k)}{1 + \tan(\theta_3) \tan(\theta_k)} \\ \frac{k - 1}{1 + k} &= \frac{3 - k}{1 + 3k} \end{aligned}$$

$$(k-1)(1+3k) = (1+k)(3-k)$$

$$3k^2 - 2k - 1 = -k^2 + 2k + 3$$

$$4k^2 - 4k - 4 = 0$$

$$k^2 - k - 1 = 0$$

$$\implies k = \frac{1 \pm \sqrt{5}}{2}$$

Since $y = kx$ is increasing, $k > 0$; thus, $k = \boxed{\text{(A)} \frac{1 + \sqrt{5}}{2}}$.

Solution 6 (Trigonometry)

Denote by $\alpha_1, \alpha_2, \alpha_3$ the acute angles formed between the x -axis and lines $y = x$, $y = 3x$, $y = kx$, respectively. Hence, $\tan \alpha_1 = 1$, $\tan \alpha_2 = 3$, $\tan \alpha_3 = k$.

Denote by θ the acute angle formed by lines $y = x$ and $y = 3x$.

Hence,

$$\begin{aligned} \tan \theta &= \tan(\alpha_2 - \alpha_1) \\ &= \frac{\tan \alpha_2 - \tan \alpha_1}{1 + \tan \alpha_1 \tan \alpha_2} \\ &= \frac{3 - 1}{1 + 1 \cdot 3} \\ &= \frac{1}{2}. \end{aligned}$$

Following from the double-angle identity, we have

$$\tan \theta = \frac{2 \tan \frac{\theta}{2}}{1 - \tan^2 \frac{\theta}{2}}.$$

Hence, $\tan \frac{\theta}{2} = -2 \pm \sqrt{5}$.

Because θ is acute, $\frac{\theta}{2}$ is acute. Hence, $\tan \frac{\theta}{2} > 0$. Hence, $\tan \frac{\theta}{2} = -2 + \sqrt{5}$.

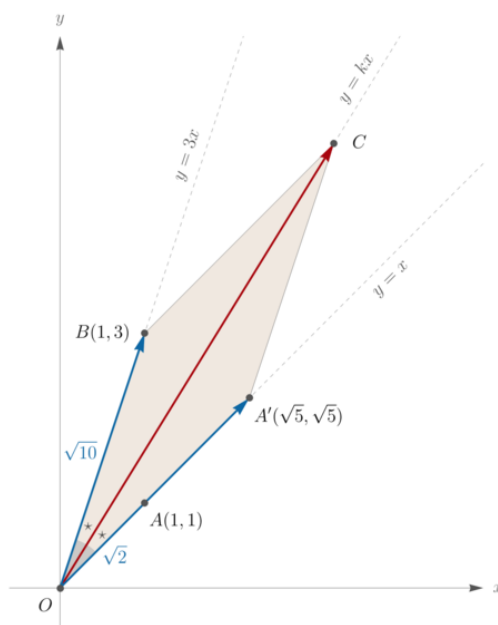
Because line $y = kx$ is the angle bisector of θ , the angle between lines $y = x$ and $y = kx$ is $\frac{\theta}{2}$.

Hence,

$$\begin{aligned} \tan \alpha_3 &= \tan \left(\alpha_1 + \frac{\theta}{2} \right) \\ &= \frac{\tan \alpha_1 + \tan \frac{\theta}{2}}{1 - \tan \alpha_1 \tan \frac{\theta}{2}} \\ &= \frac{1 + (-2 + \sqrt{5})}{1 - 1 \cdot (-2 + \sqrt{5})} \\ &= \frac{\sqrt{5} - 1}{3 - \sqrt{5}} \\ &= \frac{1 + \sqrt{5}}{2}. \end{aligned}$$

Therefore, the answer is $\boxed{\text{(A)} \frac{1 + \sqrt{5}}{2}}$.

Solution 7 (Vectors)



When drawing the lines $y = x$ and $y = 3x$, it is natural to choose points $A(1, 1)$ and $B(1, 3)$ together with origin O . See the figure attached. We utilize the fact that if $\mathbf{u}$ and $\mathbf{v}$ are vectors of same length, then $\mathbf{u} + \mathbf{v}$ bisects the angle between $\mathbf{u}$ and $\mathbf{v}$. In particular, we scale the vector $\overrightarrow{OA} = (1, 1)$ by the factor of $\frac{OB}{OA} = \frac{\sqrt{10}}{\sqrt{2}} = \sqrt{5}$ to get $\overrightarrow{OA'} := \sqrt{5} \overrightarrow{OA} = (\sqrt{5}, \sqrt{5})$. So by adding vectors $\overrightarrow{OA'}$ and $\overrightarrow{OB} = (1, 3)$ we get

$$\overrightarrow{OC} := \overrightarrow{OA'} + \overrightarrow{OB} = (1 + \sqrt{5}, 3 + \sqrt{5})$$

which bisects the acute angle formed by lines $OA : y = x$ and $OB : y = 3x$. (In other words, quadrilateral $OBCA'$ is a rhombus.) Finally, observe that $C(1 + \sqrt{5}, 3 + \sqrt{5})$ lies on the line $y = kx$ whose slope is

$$k = \frac{3 + \sqrt{5}}{1 + \sqrt{5}} = \frac{1 + \sqrt{5}}{2}.$$

Thus, the answer is $\boxed{\text{(A)} \frac{1 + \sqrt{5}}{2}}$. ■

Solution 8 (Dot Product)

We notice that the line $y = x$ can be represented as the vector $\vec{A} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $y = 3x$ as $\vec{B} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ as the "slope" of both vectors represent the coefficient of x .

Then, we can represent $y = kx$ as $\vec{C} = \begin{bmatrix} 1 \\ k \end{bmatrix}$ and notice that since C is in essence an angle bisector,

$$\theta = \angle \vec{AC} = \angle \vec{BC}$$

$$A \cdot C = |A| \cdot |C| \cos(\theta_1)$$

where $\theta_1 = \angle \vec{AC} = \theta$

$$B \cdot C = |B| \cdot |C| \cos(\theta_2)$$

where $\theta_2 = \angle \vec{BC} = \theta$

Since both θ_i 's are equivalent, we may simply represent them with θ .

Simplifying both equations by performing the necessary operations, we get

$$1 + k = \sqrt{2} \cdot \sqrt{k^2 + 1} \cdot \cos(\theta) \implies \frac{1 + k}{\sqrt{2}} = \sqrt{k^2 + 1} \cos(\theta)$$

$$1 + 3k = \sqrt{10} \cdot \sqrt{k^2 + 1} \cdot \cos(\theta)$$

Substituting the first into the second, we get

$$1 + 3k = \sqrt{10} \cdot \frac{1 + k}{\sqrt{2}} \implies k = \boxed{\text{(A)} \frac{1 + \sqrt{5}}{2}}$$

Solution 9 (Trig by Averaging the Angles)

Let θ_1 and θ_2 be the angles that the lines $y = x$ and $y = 3x$ make with the positive x -axis, so $\tan \theta_1 = 1$ and $\tan \theta_2 = 3$.

The slope k of the angle bisector is given by $k = \tan \theta$, where $\theta = \frac{\theta_1 + \theta_2}{2}$. Using the tangent addition formula, we find $\tan(2\theta)$:

$$\tan(2\theta) = \tan(\theta_1 + \theta_2) = \frac{\tan \theta_1 + \tan \theta_2}{1 - \tan \theta_1 \tan \theta_2} = \frac{1 + 3}{1 - (1 \cdot 3)} = -2$$

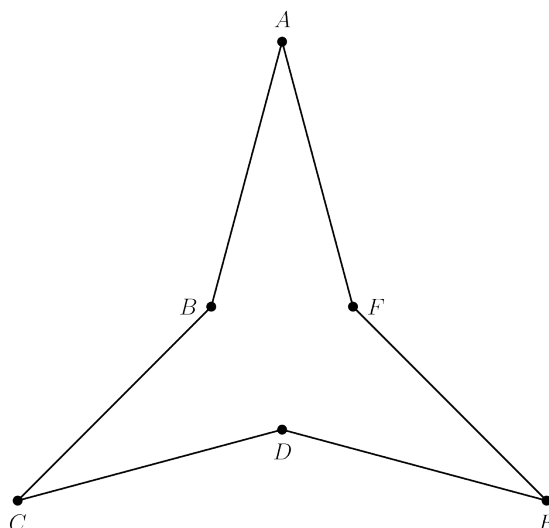
Applying the double-angle identity $\tan(2\theta) = \frac{2 \tan \theta}{1 - \tan^2 \theta}$ and substituting $k = \tan \theta$:

$$\frac{2k}{1 - k^2} = -2 \implies -k = 1 - k^2 \implies k^2 - k - 1 = 0$$

Solving this quadratic equation yields $k = \frac{1 \pm \sqrt{5}}{2}$. Since both lines lie in the first quadrant, their acute angle bisector must have a positive slope. Thus, we select the positive root:

$$k = \frac{1 + \sqrt{5}}{2}$$

14. (AMC 12 12A Fall 2021) In the figure, equilateral hexagon $ABCDEF$ has three nonadjacent acute interior angles that each measure 30° . The enclosed area of the hexagon is $6\sqrt{3}$. What is the perimeter of the hexagon?



- (A) 4 (B) $4\sqrt{3}$ (C) 12 (D) 18 (E) $12\sqrt{3}$

Answer: E

Solution 1

Divide the equilateral hexagon $ABCDEF$ into isosceles triangles ABF , CBD , and EDF and triangle BDF . The three isosceles triangles are congruent by SAS congruence. By **CPCTC**, $BF = BD = DF$, so triangle BDF is equilateral.

Let the side length of the hexagon be s . The area of each isosceles triangle is

$$\frac{1}{2}ab \sin \angle C = \frac{1}{2} \cdot s \cdot s \cdot \sin 30^\circ = \frac{1}{4}s^2.$$

By the **Law of Cosines** on triangle ABF ,

$$BF^2 = s^2 + s^2 - 2s^2 \cos 30^\circ = 2s^2 - \sqrt{3}s^2.$$

Hence, the

15. (AMC 12 12A Fall 2021) Recall that the conjugate of the complex number $w = a + bi$, where a and b are real numbers and $i = \sqrt{-1}$, is the complex number $\bar{w} = a - bi$. For any complex number z , let $f(z) = 4i\bar{z}$. The polynomial

$$P(z) = z^4 + 4z^3 + 3z^2 + 2z + 1$$

has four complex roots: z_1 , z_2 , z_3 , and z_4 . Let

$$Q(z) = z^4 + Az^3 + Bz^2 + Cz + D$$

be the polynomial whose roots are $f(z_1)$, $f(z_2)$, $f(z_3)$, and $f(z_4)$, where the coefficients A , B , C , and D are complex numbers. What is $B + D$? (A) -304 (B) -208 (C) $12i$ (D)

Answer: D

Solution 1

By Vieta's formulas, $z_1 z_2 z_3 z_4 = 1$, and $D = (4i)^4 \bar{z}_1 \bar{z}_2 \bar{z}_3 \bar{z}_4$.

Since $\bar{a} \cdot \bar{b} = \overline{ab}$,

$$D = (4i)^4 \overline{z_1 z_2 z_3 z_4} = 256(\bar{1}) = 256$$

By Vieta's formulas, $z_1 z_2 + z_1 z_3 + \cdots + z_3 z_4 = 3$, and $B = (4i)^2 (\bar{z}_1 \bar{z}_2 + \bar{z}_1 \bar{z}_3 + \cdots + \bar{z}_3 \bar{z}_4)$. Since $\bar{a} \cdot \bar{b} = \overline{ab}$,

$$B = (4i)^2 (\overline{z_1 z_2} + \overline{z_1 z_3} + \overline{z_1 z_4} + \overline{z_2 z_3} + \overline{z_2 z_4} + \overline{z_3 z_4}).$$

Since $\bar{a} + \bar{b} = \overline{a + b}$,

$$B = (4i)^2 (\overline{z_1 z_2 + z_1 z_3 + \cdots + z_3 z_4}) = -16(\bar{3}) = -48$$

Our answer is $B + D = 256 - 48 = \boxed{\text{(D)} 208}$.

Solution 2

Since the coefficients of P are real, the roots of P can also be written as $\bar{z}_1, \bar{z}_2, \bar{z}_3, \bar{z}_4$. With this observation, it's easy to see that the polynomials $P(z)$ and $Q(4iz)$ have the same roots. Hence, there exists some constant K such that

$$P(z) = K * Q(4iz)$$

By comparing coefficients, it's easy to see that $K = \frac{1}{(4i)^4}$. Hence $\frac{B * (4i)^2}{(4i)^4} = 3$ and $\frac{D}{(4i)^4} = 1$. Hence $B = -48$, $D = 256$, so $B + D = 208$ and our answer is $\boxed{\text{(D)} 208}$.

16. (AMC 12 12A Fall 2021) An organization has 30 employees, 20 of whom have a brand A computer while the other 10 have a brand B computer. For security, the computers can only be connected to each other and only by cables. The cables can only connect a brand A computer to a brand B computer. Employees can communicate with each other if their computers are directly connected by a cable or by relaying messages through a series of connected computers. Initially, no computer is connected to any other. A technician arbitrarily selects one computer of each brand and installs a cable between them, provided there is not already a cable between that pair. The technician stops once every employee can communicate with each other. What is the maximum possible number of cables used?
- (A) 190 (B) 191 (C) 192 (D) 195 (E) 196

Answer: B

Solution

We claim that to maximize the number of cables used, we isolate one computer and connect all cables for the remaining 29 computers, then connect one more cable for the isolated computer.

If a brand A computer is isolated, then the technician can use at most $19 \cdot 10 + 1 = 191$ cables. If a brand B computer is isolated instead, then the technician can use at most $20 \cdot 9 + 1 = 181$ cables. Therefore, the answer is $\boxed{\text{(B)} 191}$. **Remark**

Suppose that the computers are labeled $A_1, A_2, \dots, A_{20}$ and $B_1, B_2, \dots, B_{10}$. We will prove the claim in the first paragraph of this solution:

- a) **With exactly 190 cables, it is possible that some computers cannot communicate with each other.** From this solution, it is clear that this claim is true: We isolate A_{20} and connect all cables for $A_1, A_2, \dots, A_{19}$ and $B_1, B_2, \dots, B_{10}$ to get $19 \cdot 10 = 190$ cables. However, A_{20} cannot communicate with any of these 29 computers.

- b) **With exactly 191 cables, all computers must be able to communicate with each other.** By the Pigeonhole Principle, we conclude that at least one brand B computer connects to all 20 brand A computers. Without loss of generality, we assume that B_1 connects to all 20 brand A computers. On the other hand, we conclude that at least 11 brand A computers connect to all 10 brand B computers. Without loss of generality, we assume that $A_1, A_2, \dots, A_{11}$ connect to all 10 brand B computers.

Therefore, any two computers must be able to communicate with each other:

- Between A_i and A_j , where $i, j \in \{1, 2, \dots, 20\}$ and $i \neq j$. The relay is $A_i \leftrightarrow B_1 \leftrightarrow A_j$.
- Between B_i and B_j , where $i, j \in \{1, 2, \dots, 10\}$ and $i \neq j$. The relay is $B_i \leftrightarrow \{A_1, A_2, \dots, A_{11}\} \leftrightarrow B_j$.
- Between A_i and B_j , where $i \in \{1, 2, \dots, 20\}$ and $j \in \{1, 2, \dots, 10\}$. The relay is $A_i \leftrightarrow B_1 \leftrightarrow \{A_1, A_2, \dots, A_{11}\} \leftrightarrow B_j$.

17. (AMC 12 12A Fall 2021) For how many ordered pairs (b, c) of positive integers does neither $x^2 + bx + c = 0$ nor $x^2 + cx + b = 0$ have two distinct real solutions?
 (A) 4 (B) 6 (C) 8 (D) 12 (E) 16

Answer: B

Solution 1 (Casework)

A quadratic equation does not have two distinct real solutions if and only if the **discriminant** is nonpositive. We conclude that:

- a) Since $x^2 + bx + c = 0$ does not have real solutions, we have $b^2 \leq 4c$.
- b) Since $x^2 + cx + b = 0$ does not have real solutions, we have $c^2 \leq 4b$.

Squaring the first inequality, we get $b^4 \leq 16c^2$. Multiplying the second inequality by 16, we get $16c^2 \leq 64b$. Combining these results, we get

$$b^4 \leq 16c^2 \leq 64b.$$

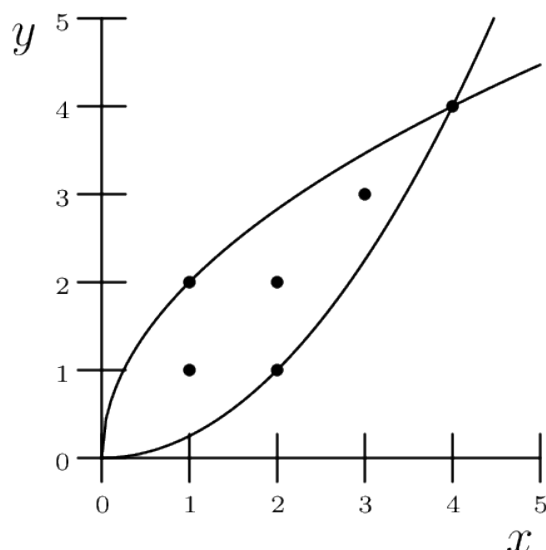
Since $b^4 \leq 64b$, it follows that $b \leq 4$. We apply casework to the value of b :

- If $b = 1$, then $1 \leq 16c^2 \leq 64$, from which $c = 1, 2$.
- If $b = 2$, then $16 \leq 16c^2 \leq 128$, from which $c = 1, 2$.
- If $b = 3$, then $81 \leq 16c^2 \leq 192$, from which $c = 3$.
- If $b = 4$, then $256 \leq 16c^2 \leq 256$, from which $c = 4$.

Together, there are **(B) 6** ordered pairs (b, c) , namely $(1, 1), (1, 2), (2, 1), (2, 2), (3, 3)$, and $(4, 4)$.

Solution 2 (Graphing)

Similar to Solution 1, use the discriminant to get $b^2 \leq 4c$ and $c^2 \leq 4b$. These can be rearranged to $c \geq \frac{1}{4}b^2$ and $b \geq \frac{1}{4}c^2$. Now, we can roughly graph these two inequalities, letting one of them be the x axis and the other be y . The graph of solutions should be above the parabola and under its inverse, meaning we want points on the graph or in the first area enclosed by the two graphs:



We are looking for lattice points (since b and c are positive integers), of which we can count **(B) 6**.

Solution 3 (Graphing)

We need to solve the following system of inequalities:

$$\begin{cases} b^2 - 4c \leq 0 \\ c^2 - 4b \leq 0 \end{cases}.$$

Feasible solutions are in the region formed between two parabolas $b^2 - 4c = 0$ and $c^2 - 4b = 0$.

Define $f(b) = \frac{b^2}{4}$ and $g(b) = 2\sqrt{b}$. Therefore, all feasible solutions are in the region formed between the graphs of these two functions.

For $b = 1$, we have $f(b) = \frac{1}{4}$ and $g(b) = 2$. Hence, the feasible c are 1, 2.

For $b = 2$, we have $f(b) = 1$ and $g(b) = 2\sqrt{2}$. Hence, the feasible c are 1, 2.

For $b = 3$, we have $f(b) = \frac{9}{4}$ and $g(b) = 2\sqrt{3}$. Hence, the feasible c is 3.

For $b = 4$, we have $f(b) = 4$ and $g(b) = 4$. Hence, the feasible c is 4.

For $b > 4$, we have $f(b) > g(b)$. Hence, there is no feasible c .

Putting all cases together, the correct answer is **(B) 6**.

Solution 4 (Oversimplified but Risky)

A quadratic equation $Ax^2 + Bx + C = 0$ has one real solution if and only if $B^2 - 4AC = 0$. Similarly, it has imaginary solutions if and only if $B^2 - 4AC < 0$. We proceed as following:

We want both $x^2 + bx + c$ to be 1 value or imaginary and $x^2 + cx + b$ to be 1 value or imaginary. $x^2 + 4x + 4$ is one such case since $\sqrt{b^2 - 4ac}$ is 0. Also, $x^2 + 3x + 3$, $x^2 + 2x + 2$, $x^2 + x + 1$ are always imaginary for both b and c . We also have $x^2 + x + 2$ along with $x^2 + 2x + 1$ since the latter has one solution, while the first one is imaginary. Therefore, we have **(B) 6** total ordered pairs of integers.

Solution 5 (Quick and Easy)

We see that $b^2 \leq 4c$ and $c^2 \leq 4b$. WLOG, assume that $b \geq c$. Then we have that $b^2 \leq 4c \leq 4b$, so $b^2 \leq 4b$ and therefore $b \leq 4$, also meaning that $c \leq 4$. This means that we only need to try 16 cases. Now we can get rid of the assumption that $b \geq c$, because we want ordered pairs. For $b = 1$ and $b = 2$, $c = 1$ and $c = 2$ work. When $b = 3$, c can only be 3, and when $b = 4$, only $c = 4$ works, for a total of (B) 6 ordered pairs of integers.

Solution 6 (Fastest)

We need both $b^2 \leq 4c$ and $c^2 \leq 4b$.

If $b = c$ then the above become $b^2 \leq 4b \iff b \leq 4$, so we have four solutions (k, k) , where $k = 1, 2, 3, 4$.

If $b < c$ then we only need $c^2 \leq 4b$ since it implies $b^2 < 4c$. Now $c^2 \leq 4b \leq 4(c-1) \implies (c-2)^2 \leq 0 \implies c = 2$, so $b = 1$. We plug $b = 1, c = 2$ back into $c^2 \leq 4b$ and it works. So there is another solution $(1, 2)$.

By symmetry, if $b > c$ then $(b, c) = (2, 1)$.

Therefore the total number of solutions is (B) 6.

Solution 7 (Shortest)

Since $b^2 - 4c \leq 0$ and $c^2 - 4b \leq 0$, adding the two together yields $b^2 + c^2 \leq 4(c + b)$. Obviously, this is not true if either b or c get too large, and they are equal when $b = c = 4$, so the greatest pair is $(4, 4)$ and both numbers must be lesser for further pairs. For there to be two distinct real solutions, we can test all these pairs where (b, c) are less than 4 (except for the already valid solution) on the original quadratics, and we find the working pairs are $(1, 1)$, $(2, 1)$, $(1, 2)$, $(2, 2)$, $(3, 3)$, $(4, 4)$ meaning there are (B) 6 pairs.

- youtube.com/indianmathguy

18. (AMC 12 12A Fall 2021) Each of 20 balls is tossed independently and at random into one of 5 bins. Let p be the probability that some bin ends up with 3 balls, another with 5 balls, and the other three with 4 balls each. Let q be the probability that every bin ends up with 4 balls. What is $\frac{p}{q}$?
- (A) 1 (B) 4 (C) 8 (D) 12 (E) 16

Answer: E

Solution 1 (Multinomial Coefficients)

For simplicity purposes, we assume that the balls and the bins are both distinguishable.

Recall that there are 5^{20} ways to distribute 20 balls into 5 bins. For p , we choose one of the 5 bins to have 3 balls and another one of the 4 bins to have 5 balls. We get

$$p = \frac{5 \cdot 4 \cdot \binom{20}{3,5,4,4,4}}{5^{20}} \text{ and } q = \frac{\binom{20}{4,4,4,4,4}}{5^{20}}.$$

Therefore, the answer is

$$\frac{p}{q} = \frac{5 \cdot 4 \cdot \binom{20}{3,5,4,4,4}}{\binom{20}{4,4,4,4,4}} = \frac{5 \cdot 4 \cdot \frac{20!}{3! \cdot 5! \cdot 4! \cdot 4! \cdot 4!}}{\frac{20!}{4! \cdot 4! \cdot 4! \cdot 4! \cdot 4!}} = \frac{5 \cdot 4 \cdot (4! \cdot 4! \cdot 4! \cdot 4! \cdot 4!)}{3! \cdot 5! \cdot 4! \cdot 4! \cdot 4!} = \frac{5 \cdot 4 \cdot 4}{5} = \span style="border: 1px solid black; padding: 2px;">(E) 16.$$

MRENTHUSIASM Jesshuang mathboy282

Solution 2

For simplicity purposes, we assume that the balls and the bins are both distinguishable.

Let $q = \frac{x}{a}$, where a is the total number of combinations and x is the number of cases where every bin ends up with 4 balls.

We can take 1 ball from one bin and place it in another bin so that some bin ends up with 3 balls, another with 5 balls, and the other three with 4 balls each. Note that one configuration of 4–4–4–4–4 corresponds to $5 \cdot 4 \cdot 4 = 80$ configurations of 3–5–4–4–4. On the other hand, one configuration of 3–5–4–4–4 corresponds to 5 configurations of 4–4–4–4–4.

Therefore, we have

$$p = \frac{80}{5} \cdot \frac{x}{a} = 16 \cdot \frac{x}{a},$$

from which $\frac{p}{q} = \boxed{\text{(E)} 16}$.

Solution 3 (Binomial Coefficients)

Since both of the cases will have 3 bins with 4 balls in them, we can leave those out. There are $2 \cdot \binom{5}{2} = 20$ ways to choose where to place the 3 and the 5. After that, there are $\binom{8}{3} = 56$ ways to put the 3 and 5 balls being put into the bins. For the 4, 4, 4, 4, 4 case, after we canceled the 4, 4, 4 out, we have $\binom{8}{4} = 70$ ways to put the 4 balls inside the bins. Therefore, we have $\frac{56 \cdot 20}{70}$ which is equal to $8 \cdot 2 = \boxed{\text{(E)} 16}$.

Solution 4 (Set Theory / Graph Theory)

Construct the set A consisting of all possible 3–5–4–4–4 bin configurations, and construct set B consisting of all possible 4–4–4–4–4 configurations. If we let N be the total number of configurations possible, it's clear we want to solve for $\frac{p}{q} = \frac{\frac{|A|}{N}}{\frac{|B|}{N}} = \frac{|A|}{|B|}$.

Consider drawing an edge between an element in A and an element in B if it is possible to reach one configuration from the other by moving a single ball (Note this process is reversible.). Let us consider the total number of edges drawn.

For any element in A , we may choose one of the 5 balls in the 5-bin and move it to the 3-bin to get a valid element in B . This implies the number of edges is $5|A|$.

On the other hand, for any element in B , we may choose one of the 20 balls and move it to one of the other 4-bins to get a valid element in A . This implies the number of edges is $80|B|$.

We equate the expressions to get $5|A| = 80|B|$, from which $\frac{|A|}{|B|} = \frac{80}{5} = \boxed{\text{(E)} 16}$.

19. (AMC 12 12A Fall 2021) Let x be the least real number greater than 1 such that $\sin(x) = \sin(x^2)$, where the arguments are in degrees. What is x rounded up to the closest integer?

(A) 10 (B) 13 (C) 14 (D) 19 (E) 20

Answer: B

Solution 1

The smallest x to make $\sin(x) = \sin(x^2)$ would require $x = x^2$, but since x needs to be greater than 1, these solutions are not valid.

The next smallest x would require $x = 180 - x^2$, or $x^2 + x = 180$.

After a bit of guessing and checking, we find that $12^2 + 12 = 156$, and $13^2 + 13 = 182$, so the solution lies between 12 and 13, making our answer **(B) 13**.

Note: One can also solve the quadratic and estimate the radical.

Note: the answer is 13 because 182 is far more close to 180 than 156 and as x is positive then it is much closer, so the answer is 13

Note: solving: $x^2 + x - 180 = 0$ so $x = \frac{-1 \pm \sqrt{721}}{2}$ clearly only $x = \frac{-1 + \sqrt{721}}{2}$ is feasible here. we make a rough estimate of 721 being 729 (as 729 is the closet perfect square), so it is $\frac{-1 + 27}{2} = 13$. in addition, we could try to estimate with 12 (which is the worth case), $26^2 = 4 \cdot 169 = 676$ which yeilds us $x = \frac{-1 + 26}{2} = 12.5$ which closely, it is still 13 (12.5 is close to 13)

therefore the answer is undeinably 13. majorly it is some addition onto the original basis

Solution 2

For choice **(A)**, we have

$$\begin{aligned} \left| \sin x - \sin(x^2) \right| &= \left| \sin 10^\circ - \sin((10^2)^\circ) \right| \\ &= \left| \sin 10^\circ - \sin 100^\circ \right| \\ &= \left| \sin 10^\circ - \sin 80^\circ \right| \\ &= \sin 80^\circ - \sin 10^\circ. \end{aligned}$$

For choice **(B)**, we have

$$\begin{aligned} \left| \sin x - \sin(x^2) \right| &= \left| \sin 13^\circ - \sin((13^2)^\circ) \right| \\ &= \left| \sin 13^\circ - \sin 169^\circ \right| \\ &= \left| \sin 10^\circ - \sin 11^\circ \right| \\ &= \sin 11^\circ - \sin 10^\circ. \end{aligned}$$

For choice **(C)**, we have

$$\begin{aligned} \left| \sin x - \sin(x^2) \right| &= \left| \sin 14^\circ - \sin((14^2)^\circ) \right| \\ &= \left| \sin 14^\circ - \sin 196^\circ \right| \\ &= \left| \sin 14^\circ + \sin 16^\circ \right| \\ &= \sin 14^\circ + \sin 16^\circ. \end{aligned}$$

For choice **(D)**, we have

$$\begin{aligned} \left| \sin x - \sin(x^2) \right| &= \left| \sin 19^\circ - \sin((19^2)^\circ) \right| \\ &= \left| \sin 19^\circ - \sin 361^\circ \right| \\ &= \left| \sin 19^\circ - \sin 1^\circ \right| \\ &= \sin 19^\circ - \sin 1^\circ. \end{aligned}$$

For choice **(E)**, we have

$$\left| \sin x - \sin(x^2) \right| = \left| \sin 20^\circ - \sin((20^2)^\circ) \right|$$

$$\begin{aligned} &= |\sin 20^\circ - \sin 400^\circ| \\ &= |\sin 20^\circ - \sin 40^\circ| \\ &= \sin 40^\circ - \sin 20^\circ. \end{aligned}$$

Therefore, the answer is **(B) 13**, as $\sin 11^\circ - \sin 10^\circ$ is the closest to 0.

Easy Solution with Intermediate Value Theorem We know that $x^2 - x = 180 - 2x + 360k$ or $x^2 - x = 360k$ from looking at the sin values on the unit circle. With these equations, we can just insert the values that are given in the 5 answers.

Let's try A first.

We know that one of the equations should be greater on the left, and another equation we make should be greater on the right for the solution to be in between the values inserted into the equations. The two numbers it is in between is 9 and 10 since it is rounded up.

$$100 - 10 \text{ vs } 180 - 20, 81 - 9 \text{ vs } 180 - 18.$$

This does not satisfy what we would like to find.

Now let's try B.

Between 12 and 13;

$$144 - 12 \text{ vs } 180 - 24, 169 - 13 \text{ vs } 180 - 26.$$

Now this satisfies our hunt for the solution. The answer is **(B)**

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Solution by Emily Q

(edited by A7456321)

20. (AMC 12 12A Fall 2021) For each positive integer n , let $f_1(n)$ be twice the number of positive integer divisors of n , and for $j \geq 2$, let $f_j(n) = f_1(f_{j-1}(n))$. For how many values of $n \leq 50$ is $f_{50}(n) = 12$? **(A) 7** **(B) 8** **(C) 9** **(D) 10** **(E) 11**

Answer: D

Solution 1

First, we can test values that would make $f(x) = 12$ true. For this to happen x must have 6 divisors, which means its prime factorization is in the form pq^2 or p^5 , where p and q are prime numbers. Listing out values less than 50 which have these prime factorizations, we find 12, 18, 20, 28, 44, 45, 50 for pq^2 , and just 32 for p^5 . Here 12 especially catches our eyes, as this means if one of $f_i(n) = 12$, each of $f_{i+1}(n), f_{i+2}(n), \dots$ will all be 12. This is because $f_{i+1}(n) = f(f_i(n))$ (as given in the problem statement), so were $f_i(n) = 12$, plugging this in we get $f_{i+1}(n) = f(12) = 12$, and thus the pattern repeats. Hence, as long as for a i , such that $i \leq 50$ and $f_i(n) = 12$, $f_{50}(n) = 12$ must be true, which also immediately makes all our previously listed numbers, where $f_1(x) = 12$, possible values of n .

We also know that if $f_1(x)$ were to be any of these numbers, x would satisfy $f_{50}(n)$ as well. Looking through each of the possibilities aside from 12, we see that $f_1(x)$ could only possibly be equal to 20 and 18, and still have x less than or equal to 50. This is because if $f_1(x) = 20$ or 18, $f_2(x) = 12$. This would mean x must have 10, or 9 divisors, and testing out, we see that x will then be of the form p^4q , or p^2q^2 . The only two values less than or equal to 50 would be 48 and 36 respectively.

From here there are no more possible values, so tallying our possibilities we count (D) 10 values (Namely 12, 18, 20, 28, 32, 36, 44, 45, 48, 50).

Solution 2 (Rigorous reasoning on why there cannot be any other solutions)

First, take note that the maximum possible value of $f_1(n)$ for $1 \leq n \leq k$ increases as k increases. When we increase k , we either include a new value of n that has more divisors than any previous values of n , increasing the maximum of $f_1(x)$, or none of the new n have more divisors, which would leave the maximum of $f_1(x)$ the same as before. In other words, $f_1(x)$ is increasing. This means that $f_1(x)$ decreases when we decrease k .

Also, let $f_1(n) = 2d(n)$ where $d(n)$ is the number of divisors of n .

Since $n \leq 50$, $f_1(n) \leq 20$. This maximum occurs when $d(n) = 10 \implies n = 2^4 \cdot 3 = 48$. Next, since $f_1(n) \leq 20$, $f_1(f_1(n)) \leq 12 \implies f_2(n) \leq 12$. This maximum occurs when $d(f_1(n)) = 6 \implies n = 2 \cdot 3^2 = 18, n = 2^2 \cdot 3 = 12$. Since $f_2(n) \leq 12$, $f_1(f_2(n)) \leq 12 \implies f_3(n) \leq 12$, once again. This maximum again occurs when $d(f_2(n)) = 6 \implies f_2(n) = 2^2 \cdot 3 = 12$. Notice that, from this point on, for $k \geq 3$, $f_k(n) \leq 12$ by a straightforward induction argument.

On a hunch, we investigate what happens when $f_2(n) < 12$, as this is the first time when $f_k(n)$ is bounded to a maximum of 12. Suppose for the sake of contradiction that $f_2(n) < 12$. Then, $f_3(n) < 12$, as $f_2(n) = 12$ is the only number that would maximize $f_3(n)$ for $f_2(n) \leq 12$. As a result, since $f_1(n)$ is increasing, and because 12 is the only value for which f_1 is maximized, f_1 "steps down" from a maximum of $6 \cdot 2 = 12$ if $n < 12$. This means we must have that $f_1(f_3(n)) < f_1(12) = 12 \implies f_4(n) < 12$. We continue applying f_1 on both sides (which is possible since f_1 is increasing) until we reach $f_{50}(n)$, giving us that $f_{50}(n) < 12$. However, $f_{50}(n) = 12$, which is a contradiction. Thus, $f_2(n) = 12$.

Now, we can finally solve for the solutions with a clean conscience - we proceed similarly to Solution 1. $f_2(n) = 12 \implies f_1(f_1(n)) = 12 \implies d(f_1(n)) = 6$. $d(f_1(n)) = 6$. This gives us two cases. First, we have the case where $f_1(n) = p^2 \cdot q$ where p and q are primes. Second, we have the case where $f_1(n) = p^5$ where p is a prime. For both cases, since $f_1(n) \leq 20$, $f_1(n)$ can only be 12, 18, or 20.

If $f_1(n) = 12$, then $d(n) = 6 \implies n = p^5, p^2 \cdot q \implies n \in \{12, 18, 20, 28, 32, 44, 45, 50\}$, resulting in 8 solutions.

If $f_1(n) = 18$, then $d(n) = 9 \implies n = p^8, p^2 \cdot q^2 \implies n = 36$, giving us one more solution. Finally, $f_1(n) = 20 \implies d(n) = 10 \implies n = p^9, p^4 \cdot q \implies n = 48$.

Thus, in total, we have (D)10 solutions.

Solution 3

Observation 1: $f_1(12) = 12$.

Hence, if n has the property that $f_j(n) = 12$ for some j , then $f_k(n) = 12$ for all $k > j$.

Observation 2: $f_1(8) = 8$.

Hence, if n has the property that $f_j(n) = 8$ for some j , then $f_k(n) = 8$ for all $k > j$.

Case 1: $n = 1$.

We have $f_1(n) = 2$, $f_2(n) = f_1(2) = 4$, $f_3(n) = f_1(4) = 6$, $f_4(n) = f_1(6) = 8$. Hence, Observation 2 implies $f_{50}(n) = 8$.

Case 2: n is prime.

We have $f_1(n) = 4$, $f_2(n) = f_1(4) = 6$, $f_3(n) = f_1(6) = 8$. Hence, Observation 2 implies $f_{50}(n) = 8$.

Case 3: The prime factorization of n takes the form p_1^2 .

We have $f_1(n) = 6$, $f_2(n) = f_1(6) = 8$. Hence, Observation 2 implies $f_{50}(n) = 8$.

Case 4: The prime factorization of n takes the form p_1^3 .

We have $f_1(n) = 8$. Hence, Observation 2 implies $f_{50}(n) = 8$.

Case 5: The prime factorization of n takes the form p_1^4 .

We have $f_1(n) = 10$, $f_2(n) = f_1(10) = 8$. Hence, Observation 2 implies $f_{50}(n) = 8$.

Case 6: The prime factorization of n takes the form p_1^5 .

We have $f_1(n) = 12$. Hence, Observation 1 implies $f_{50}(n) = 12$.

In this case the only n is $2^5 = 32$.

Case 7: The prime factorization of n takes the form p_1p_2 .

We have $f_1(n) = 8$. Hence, Observation 2 implies $f_{50}(n) = 8$.

Case 8: The prime factorization of n takes the form $p_1p_2^2$.

We have $f_1(n) = 12$. Hence, Observation 1 implies $f_{50}(n) = 12$.

In this case, all n are 12, 18, 20, 28, 44, 45, and 50.

Case 9: The prime factorization of n takes the form $p_1p_2^3$.

We have $f_1(n) = 16$, $f_2(n) = f_1(16) = 10$, $f_3(n) = f_1(10) = 8$. Hence, Observation 2 implies $f_{50}(n) = 8$.

Case 10: The prime factorization of n takes the form $p_1p_2^4$.

We have $f_1(n) = 20$, $f_2(n) = f_1(20) = 12$. Hence, Observation 1 implies $f_{50}(n) = 12$.

In this case, the only n is 48.

Case 11: The prime factorization of n takes the form $p_1^2p_2^2$.

We have $f_1(n) = 18$, $f_2(n) = f_1(18) = 12$. Hence, Observation 1 implies $f_{50}(n) = 12$.

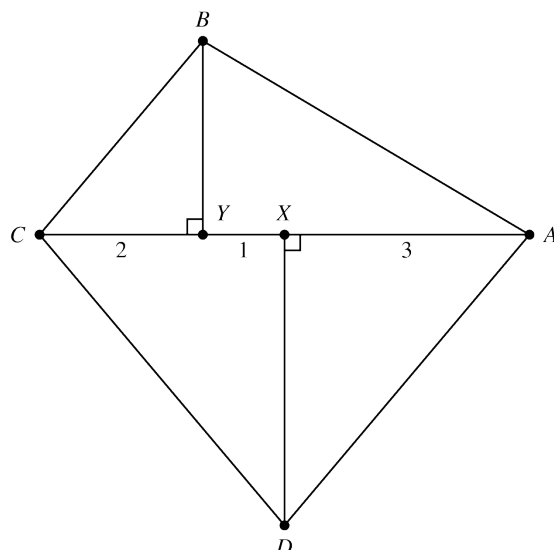
In this case, the only n is 36.

Case 12: The prime factorization of n takes the form $p_1p_2p_3$.

We have $f_1(n) = 16$, $f_2(n) = f_1(16) = 10$, $f_3(n) = f_2(10) = 8$. Hence, Observation 2 implies $f_{50}(n) = 8$.

Putting all cases together, the number of feasible $n \leq 50$ is (D) 10.

21. (AMC 12 12A Fall 2021) Let $ABCD$ be an isosceles trapezoid with $\overline{BC} \parallel \overline{AD}$ and $AB = CD$. Points X and Y lie on diagonal $\overline{AC}$ with X between A and Y , as shown in the figure. Suppose $\angle AXD = \angle BYC = 90^\circ$, $AX = 3$, $XY = 1$, and $YC = 2$. What is the area of $ABCD$?



- (A) 15 (B) $5\sqrt{11}$ (C) $3\sqrt{35}$ (D) 18 (E) $7\sqrt{7}$

Answer: C

Solution 1

First realize that $\triangle BCY \sim \triangle DAX$. Thus, because $CY : XA = 2 : 3$, we can say that $BY = 2s$ and $DX = 3s$. From the Pythagorean Theorem, we have $AB^2 = (2s)^2 + 4^2 = 4s^2 + 16$ and $CD^2 = (3s)^2 + 3^2 = 9s^2 + 9$. Because $AB = CD$, from the problem statement, we have that

$$4s^2 + 16 = 9s^2 + 9.$$

Solving, gives $s = \frac{\sqrt{7}}{\sqrt{5}}$. To find the area of the trapezoid, we can compute the area of $\triangle ABC$ and add it to the area of $\triangle ACD$. Thus, the area of the trapezoid is $\frac{1}{2} \cdot 2 \cdot \frac{\sqrt{7}}{\sqrt{5}} \cdot 6 + \left(\frac{1}{2} \cdot 3 \cdot \frac{\sqrt{7}}{\sqrt{5}} \cdot 6 \right) = \frac{15\sqrt{7}}{\sqrt{5}} = 3\sqrt{35}$. Thus, the answer is (C) $3\sqrt{35}$.

Solution 2

We put the graph to a coordinate plane. We put X to the origin, AC to the x -axis, and DX to the y -axis.

We have the coordinates of the following points: $X = (0, 0)$, $A = (3, 0)$, $Y = (-1, 0)$, $C = (-3, 0)$.

Denote $BY = u$, $DX = v$. Hence, $B = (-1, u)$, $D = (0, -v)$.

Hence, the slope of BC is $m_{BC} = \frac{u}{2}$. The slope of AD is $m_{AD} = \frac{v}{3}$.

Because $BC \parallel AD$, $m_{BC} = m_{AD}$. Hence,

$$\frac{u}{2} = \frac{v}{3}. \quad (1)$$

In $\triangle AYB$, following from the Pythagorean theorem, we have $AB^2 = AY^2 + BY^2 = 4^2 + u^2$.

In $\triangle CXD$, following from the Pythagorean theorem, we have $CD^2 = CX^2 + XD^2 = 3^2 + v^2$.

Because $AB = CD$,

$$4^2 + u^2 = 3^2 + v^2. \quad (2)$$

Solving Equations (1) and (2), we get $u = \frac{2\sqrt{35}}{5}$, $V = \frac{3\sqrt{35}}{5}$.

Therefore,

$$\begin{aligned} \text{Area } ABCD &= \text{Area } \triangle ABC + \text{Area } \triangle ADC \\ &= \frac{1}{2}AC \cdot BY + \frac{1}{2}AC \cdot DX \\ &= \frac{1}{2}AC (BY + DX) \\ &= 3\sqrt{35}. \end{aligned}$$

Therefore, the answer is (C) $3\sqrt{35}$.

Solution 3

Since both $\overline{BY}$ and $\overline{DX}$ are perpendicular to $\overline{AC}$, we have $BD^2 = XY^2 + (BY + DX)^2$.

Since the trapezoid is equilateral, $BD = AC = 6$. Hence, $BY + DX = \sqrt{6^2 - 1^2} = \sqrt{35}$.

Notice that both triangles ABC and ADC share a common base $\overline{AC}$.

Therefore,

$$[ABCD] = [ABC] + [ADC] = \frac{1}{2}AC \cdot (BY + DX) = \frac{1}{2} \cdot 6 \cdot \sqrt{35} = \boxed{\text{(C) } 3\sqrt{35}}.$$

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22. (AMC 12 12A Fall 2021) Azar and Carl play a game of tic-tac-toe. Azar places an X in one of the boxes in a 3-by-3 array of boxes, then Carl places an O in one of the remaining boxes. After that, Azar places an X in one of the remaining boxes, and so on until all 9 boxes are filled or one of the players has 3 of their symbols in a row - horizontal, vertical, or diagonal - whichever comes first, in which case that player wins the game. Suppose the players make their moves at random, rather than trying to follow a rational strategy, and that Carl wins the game when he places his third O . How many ways can the board look after the game is over?

(A) 36 (B) 112 (C) 120 (D) 148 (E) 160

Answer: D

Solution 1

We need to find out the number of configurations with 3 O and 3 X with 3 O in a row, and 3 X not in a row.

Case 1: 3 O are in a horizontal row or a vertical row.

Step 1: We determine the row that 3 O occupy.

The number of ways is 6.

Step 2: We determine the configuration of 3 X .

The number of ways is $\binom{6}{3} - 2 = 18$. (6 open spots to put the X and subtracting cases where they fill row/column)

In this case, following from the rule of product, the number of ways is $6 \cdot 18 = 108$.

Case 2: 3 O are in a diagonal row.

Step 1: We determine the row that 3 O occupy.

The number of ways is 2.

Step 2: We determine the configuration of 3 X .

The number of ways is $\binom{6}{3} = 20$.

In this case, following from the rule of product, the number of ways is $2 \cdot 20 = 40$.

Putting all cases together, the total number of ways is $108 + 40 = 148$.

Therefore, the answer is (D) 148.

Solution 2 (Complementary)

First count all the ways to have 3 O in a row and 3 X . The O can be arranged in 8 possible ways (3 horizontal, 3 vertical, 2 diagonal), and in each case, the X can be arranged in $\binom{6}{3}$ ways, so there are $8 \cdot \binom{6}{3} = 160$ arrangements.

Now exclude the ones in which the X are also in a row column. Either the X and O both form rows, which can be achieved in 6 ways, or they both form columns, which can also be achieved in 6 ways. Excluding these 12 arrangements, the answer is shown to be (D) 148.

23. (AMC 12 12A Fall 2021) A quadratic polynomial with real coefficients and leading coefficient 1 is called *disrespectful* if the equation $p(p(x)) = 0$ is satisfied by exactly three real numbers. Among all the disrespectful quadratic polynomials, there is a unique such polynomial $\tilde{p}(x)$ for which the sum of the roots is maximized. What is $\tilde{p}(1)$?

(A) $\frac{5}{16}$ (B) $\frac{1}{2}$ (C) $\frac{5}{8}$ (D) 1 (E) $\frac{9}{8}$

Answer: A

Solution 1 (Standard Form)

Let r_1 and r_2 be the roots of $\tilde{p}(x)$. Then, $\tilde{p}(x) = (x - r_1)(x - r_2) = x^2 - (r_1 + r_2)x + r_1r_2$. The solutions to $\tilde{p}(\tilde{p}(x)) = 0$ is the union of the solutions to

$$\tilde{p}(x) - r_1 = x^2 - (r_1 + r_2)x + (r_1r_2 - r_1) = 0$$

and

$$\tilde{p}(x) - r_2 = x^2 - (r_1 + r_2)x + (r_1r_2 - r_2) = 0.$$

Note that one of these two quadratics has one solution (a double root) and the other has two as there are exactly three solutions. WLOG, assume that the quadratic with one root is $x^2 - (r_1 + r_2)x + (r_1r_2 - r_1) = 0$. Then, the discriminant is 0, so $(r_1 + r_2)^2 = 4r_1r_2 - 4r_1$. Thus, $r_1 - r_2 = \pm 2\sqrt{-r_1}$, but for $x^2 - (r_1 + r_2)x + (r_1r_2 - r_2) = 0$ to have two solutions, it must be the case that $r_1 - r_2 = -2\sqrt{-r_1}(\star)$. It follows that the sum of the roots of $\tilde{p}(x)$ is $2r_1 + 2\sqrt{-r_1}$, whose maximum value occurs when $r_1 = -\frac{1}{4}(\star)$. Solving for r_2 yields $r_2 = \frac{3}{4}$. Therefore, $\tilde{p}(x) = x^2 - \frac{1}{2}x - \frac{3}{16}$, so

$$\tilde{p}(1) = \boxed{\text{(A)} \frac{5}{16}}.$$

<u>**Remarks**</u>

- For $x^2 - (r_1 + r_2)x + (r_1r_2 - r_2) = 0$ to have two solutions, the discriminant $(r_1 + r_2)^2 - 4r_1r_2 + 4r_2$ must be positive. From here, we get that $(r_1 - r_2)^2 > -4r_2$, so $-4r_1 > -4r_2 \implies r_1 < r_2$. Hence, $r_1 - r_2$ is negative, so $r_1 - r_2 = -2\sqrt{-r_1}$.

- Set $\sqrt{-r_1} = x$. Now $r_1 + \sqrt{-r_1} = -x^2 + x$, for which the maximum occurs when $x = \frac{1}{2} \rightarrow r_1 = -\frac{1}{4}$.

Solution 2 (Vertex Form)

Let $p(x) = (x - h)^2 + k$ for some real constants h and k . Suppose that $p(x)$ has real roots r and s .

Since $p(p(x)) = 0$, we conclude that $p(x) = r$ or $p(x) = s$. Without loss of generality, we assume that $p(x) = r$ has two real solutions and $p(x) = s$ has one real solution. Therefore, we have $k = s$, from which

$$p(x) = (x - h)^2 + s.$$

As $p(s) = 0$, we expand the left side to obtain $(s - h)^2 + s = 0$, or

$$s^2 - (2h - 1)s + h^2 = 0. \quad (\star)$$

Since $(\star)$ has real solutions for s , the discriminant is nonnegative: $(2h - 1)^2 - 4h^2 \geq 0$. We solve this inequality to get $h \leq \frac{1}{4}$.

Either by the axis of symmetry or Vieta's formulas, note that $r + s = 2h$. As we wish to maximize $2h$, we maximize h . Substituting $h = \frac{1}{4}$ into $(\star)$, we obtain $s^2 + \frac{1}{2}s + \frac{1}{16} = 0$. We factor the left side to get $(s + \frac{1}{4})^2 = 0$, or $s = -\frac{1}{4}$.

Finally, the unique such polynomial is

$$\tilde{p}(x) = \left(x - \frac{1}{4}\right)^2 - \frac{1}{4},$$

from which $\tilde{p}(1) = \boxed{\text{(A)} \frac{5}{16}}$.

Solution 2.5 (Vertex Form but Done Differently)

Continuing from the previous solution's reasoning, we know that the y -coordinate of the vertex must also be a root of the polynomial. Thus, $\tilde{p}(x) = (x - \frac{s+r}{2})^2 + s$, where r and s are roots of the polynomial.

We also know that in order for $\tilde{p}(x) = r$ to have two solutions, $y = r$ must intersect $\tilde{p}(x)$ above its minimum/vertex. Thus, $r > s$. Let $r = s + a$. Then, $\tilde{p}(x) = (x - \frac{2s+a}{2})^2 + s$.

Since s is a root, $\tilde{p}(s) = 0$ so $(s - \frac{2s+a}{2})^2 + s = 0 \implies \frac{a^2}{4} + s = 0 \implies s = -\frac{1}{4}a^2$.

We wish to maximize $s + s + a = 2s + a = -\frac{1}{2}a^2 + a$.

This quadratic is maximized at $a = 1$ (when converting the quadratic to vertex form). Then, since $\tilde{p}(s + a) = 0 \implies \tilde{p}(s + 1) = 0$, $(s + 1 - s - \frac{1}{2})^2 + s = 0 \implies \frac{1}{4} + s = 0 \implies s = -\frac{1}{4}$.

Thus, $\tilde{p}(x) = (x - \frac{1}{4})^2 - \frac{1}{4}$. So, $\tilde{p}(1) = \boxed{\text{(A)} \frac{5}{16}}$.

Solution 3 (Symmetry)

Let $\tilde{p}(x) = (x - h)^2 + k$. We seek to maximize the sum of the roots $2h$, so we maximize h .

Let $P(x) = \tilde{p}(\tilde{p}(x)) = ((x - h)^2 + k - h)^2 + k$. Note that $P(x)$ is symmetric about $x = h$.

$P(x) = 0$ has 3 real solutions, Due to the complex conjugate theorem, $P(x)$ must have 4 real roots. Therefore, $P(x)$ must have exactly 1 double root.

This root cannot be to the left or to the right of $x = h$, as the symmetry of the function would mean that there would be another double root reflected across the $x = h$. It follows that the double root could only be situated at $x = h$.

$0 = P(h) = ((h - h)^2 + k - h)^2 + k = (k - h)^2 + k$. Expanding and writing this out in terms of k , $k^2 + (1 - 2h)k + h^2 = 0$.

In order for this to have a solution, the discriminant has to be non-negative. In other words, $(1 - 2h)^2 - 4h^2 \geq 0$.

This simplifies to $1 - 4h \geq 0$, or $h \leq \frac{1}{4}$.

As we seek to maximize h , we set $h = \frac{1}{4}$ and see that $k = -\frac{1}{4}$.

Therefore, $\tilde{p}(x) = \left(x - \frac{1}{4}\right)^2 - \frac{1}{4}$, and $\tilde{p}(1) = \left(1 - \frac{1}{4}\right)^2 - \frac{1}{4} = \frac{9}{16} - \frac{1}{4} = \boxed{\text{(A)} \frac{5}{16}}$.

Solution 4 (Discriminant)

Equation $p(p(x)) = 0$ is equivalent to the following system of equations:

$$\begin{aligned} p(u) &= 0, \\ p(x) - u &= 0. \end{aligned}$$

Denote $p(x) = x^2 + px + q$.

Denote by r_1 and r_2 two roots of $p(x) = 0$.

Because $p(p(x)) = 0$ has three real roots, we must have the properties that r_1 and r_2 are real with $r_1 \neq r_2$. Without loss of generality, we assume $r_1 < r_2$.

We notice that all roots of $p(p(x)) = 0$ are the collection of all roots of $p(x) - r_1 = 0$ and all roots of $p(x) - r_2 = 0$.

Because each of these two equations is quadratic, it has two roots (may be identical). To get a total number of three roots, one equation must have two identical roots.

Because $r_1 < r_2$, equation $p(x) - r_1 = 0$ has two identical roots. Hence, the discriminant of this equation satisfies

$$p^2 - 4(q - r_1) = 0.$$

Because $r_1 < r_2$, $r_1 = \frac{-p - \sqrt{p^2 - 4q}}{2}$.

Hence, the above equation can be written as

$$p^2 - 4\left(q - \frac{-p - \sqrt{p^2 - 4q}}{2}\right) = 0.$$

This can be reorganized as

$$(p^2 - 4q) - 2\sqrt{p^2 - 4q} - 2p = 0. \quad (1)$$

Define $x = \sqrt{p^2 - 4q}$. Hence, the value of p should ensure that equation $x^2 - 2x - 2p = 0$ has at least one real nonnegative root.

This condition can be satisfied if the discriminant of this equation is nonnegative: $(-2)^2 - 4 \cdot 1 \cdot (-2p) \geq 0$. Hence, $p \geq -\frac{1}{2}$.

Now, we are ready to find $\tilde{p}(x)$.

Following from Vieta's formula, $r_1 + r_2 = -p$. Hence, to get $r_1 + r_2$ maximized, we need to find smallest p .

Because $p \geq -\frac{1}{2}$, the smallest p is $-\frac{1}{2}$. Plugging this value into Equation (1), we get $\sqrt{p^2 - 4q} = 1$. Hence, $q = -\frac{3}{16}$.

Thus, $\tilde{p}(x) = x^2 - \frac{1}{2}x - \frac{3}{16}$. Therefore, $\tilde{p}(1) = \boxed{\text{(A)} \frac{5}{16}}$.

Solution 5 (Factored Form)

The disrespectful function $p(x)$ has leading coefficient 1, so it can be written in factored form as $(x - r)(x - s)$. Now the problem states that all $p(x)$ must satisfy $p(p(x)) = 0$. Plugging our form in, we get

$$((x - r)(x - s) - r)((x - r)(x - s) - s) = 0.$$

The roots of this equation are $(x - r)(x - s) = r$ and $(x - r)(x - s) = s$. By the fundamental theorem of algebra, each root must have two roots for a total of four possible values of x yet the problem states that this equation is satisfied by three values of x . Therefore one equation must give a double root. Without loss of generality, let the equation $(x - r)(x - s) = r$ be the equation that produces the double root. Expanding gives $x^2 - (r + s)x + rs - r = 0$. We know that if there is a double root to this equation, the discriminant must be equal to zero, so $(r + s)^2 - 4(rs - r) = 0 \implies r^2 + 2rs + s^2 - 4rs + 4r = 0 \implies r^2 - 2rs + s^2 + 4r = 0$.

From here two solutions can progress.

Solution 5.1 (Quadratic Formula)

We can rewrite $r^2 - 2rs + s^2 + 4r = 0$ as $(r - s)^2 + 4r = 0$. Let's keep our eyes on the ball; we want to find the disrespectful quadratic that maximizes the sum of the roots, which is $r + s$. Let this be equal to a new variable, m , so that our problem is reduced to maximizing this variable. We can rewrite our equation in terms of m as $(2r - m)^2 + 4r = 0 \implies m^2 - 4rm + 4r^2 + 4r = 0$.

This is a quadratic in m , so we can use the quadratic formula:

$$m = \frac{4r \pm \sqrt{16r^2 - 4(4r^2 + 4r)}}{2} = 2r \pm \sqrt{-4r} = 2(r \pm \sqrt{-r}).$$

It will be easier to think without the square root, so let $q = \sqrt{-r}$. We can rewrite the equation as $m = 2(-q^2 \pm q)$. We want to maximize m , so we take the plus value of the right-hand-side of the equation. Then,

$$m = 2(-q^2 + q) \implies m = -2q(q - 1).$$

To maximize m , we find the vertex of the right-hand side of the equation. The vertex of $-2q(q - 1)$ is the average of the roots of the equation which is $\frac{0+1}{2} = \frac{1}{2}$. This means that $r = -q^2 \implies r = -\frac{1}{4}$ and $m = -2q(q - 1) \implies m = \frac{1}{2}$. Therefore, $m - r = s \implies s = \frac{3}{4}$.

Solution 5.2 (Derivation-Rotated Conics)

We see that the equation $r^2 - 2rs + s^2 + 4r = 0$ is in the form of the general equation of a rotated conic

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0.$$

Because $B^2 - 4AC = (-2)^2 - 4(1)(1) = 0$, this rotated conic is a parabola.

The definition of a parabola is the locus of all points that are equidistant from a point (focus) and line (directrix). Let the focus and directrix of this particular parabola be (a, b) and $y = mx + c$. Then we can try to find the general form of a rotated parabola in terms of a, b, m , and c .

The distance between two points (x, y) and (a, b) is $\sqrt{(x-a)^2 + (y-b)^2}$. Therefore this is the distance from any point on the parabola to the focus.

The distance from a point (x, y) to a line $y = mx + c \implies mx - y + c = 0$ is $\frac{|mx - y + c|}{\sqrt{m^2 + 1}}$.

We can set these two equal to each other and we get

$$\sqrt{(x-a)^2 + (y-b)^2} = \frac{|mx - y + c|}{\sqrt{m^2 + 1}}.$$

Squaring both sides of the equation, we get

$$(x-a)^2 + (y-b)^2 = \frac{(mx - y + c)^2}{m^2 + 1}.$$

Expanding both sides of the equation gives

$$x^2 - 2ax + a^2 + y^2 - 2by + b^2 = \frac{m^2x^2 + y^2 + c^2 + 2mxc - 2yc - 2mxy}{m^2 + 1}.$$

Multiplying both sides of the equation by $m^2 + 1$ and rearranging gives

$$x^2 + 2mx + m^2y^2 - 2((m^2 + 1)a + mc)x - 2((m^2 + 1)b - c)y + (m^2 + 1)(a^2 + b^2) - c^2.$$

Now we can compare to our rotated parabola, $r^2 - 2rs + s^2 + 4r = 0$. From this, $-2 = 2m$ or $m = -1$. From here we have a system of three equations:

$$\begin{aligned} -2((m^2 + 1)a + mc) &= 4, \\ -2((m^2 + 1)b - c) &= 0, \\ (m^2 + 1)(a^2 + b^2) - c^2 &= 0. \end{aligned}$$

Plugging in $m = -1$ we get

$$\begin{aligned} -2(2a - c) &= 4, \\ -2(2b - c) &= 0, \\ 2(a^2 + b^2) - c^2 &= 0. \end{aligned}$$

Solving for the first equation, $c = 2 + 2a$.

Subtracting the first two equations, $-4a + 4b = 4 \implies b = a + 1$.

Plugging into the third equation, $2a^2 + 2a^2 + 4a + 2 = c^2 \implies 4a^2 + 4a + 2 = c^2$.

Substituting c in, we get $4a^2 + 4a + 2 = 4a^2 + 8a + 4 \implies 4a + 2 = 0 \implies a = -\frac{1}{2}$.

Now $b = a + 1 = \frac{1}{2}$ and $c = 2 + 2a = 1$.

This means that the focus of the parabola is $\left(-\frac{1}{2}, \frac{1}{2}\right)$ and the directrix is $y = -x + 1$. The maximum value of $r + s$ would lie at the vertex of the parabola, which is the midpoint of the focus and the foot of the focus at the directrix. The line that the vertex and focus lie on is perpendicular to the directrix, so it has slope 1. It can be written as $y = x + d$ and must go through $\left(-\frac{1}{2}, \frac{1}{2}\right)$ so $d = 1$. This perpendicular line intersects the directrix, so to find the point at which this foot occurs, we set the equation of the lines equal to each other:

$$\begin{aligned}y &= x + 1, \\y &= -x + 1.\end{aligned}$$

Adding, we get $2y = 2$ or $y = 1$ and $x = 0$. The vertex of the parabola is now at the midpoint of $\left(-\frac{1}{2}, \frac{1}{2}\right)$ and $(0, 1)$ which is $\left(-\frac{1}{4}, \frac{3}{4}\right)$.

Therefore, we have $r = -\frac{1}{4}$ and $s = \frac{3}{4}$.

Solutions 5.1 and 5.2 Rejoined

Now that we know the roots of $\tilde{p}(1)$, we can plug in our equation:

$$(x - r)(x - s) = \left(1 - \left(-\frac{1}{4}\right)\right) \left(1 - \frac{3}{4}\right) = \frac{5}{4} \cdot \frac{1}{4} = \frac{5}{16} = \boxed{\text{(A)} \frac{5}{16}}.$$

KingRavi

Solution 6 (Factored Form)

Let $p(x) = (x - p)(x - q)$. Then,

$$p(p(x)) = ((x - p)(x - q) - p)((x - p)(x - q) - q) = 0,$$

which means that either $(x - p)(x - q) - p = 0$ or $(x - p)(x - q) - q = 0$. Both of these equations are quadratics, so $p(p(x))$ has four roots, unless there's a double root.

Without loss of generality, let $(x - p)(x - q) - p$ be the expression that produces the double root, so its discriminant is zero. When expanded,

$$(x - p)(x - q) - p = x^2 - (p + q)x + pq - p = 0.$$

The value of x is irrelevant to the discriminant, which is $(-(p + q))^2 - 4(pq - p)$. Setting this equal to zero and simplifying, this equation becomes $p^2 - 2pq + q^2 + 4p = 0$, which is a quadratic in p .

Now, we seek the value of p . The previous quadratic is equivalent to $p^2 - (2q - 4)p + q^2 = 0$. Using the quadratic formula by having $a = 1$, $b = -(2q - 4)$, and $c = q^2$, we have

$$p = \frac{2q - 4 \pm \sqrt{-(2q - 4)^2 - 4(1)(q^2)}}{2(1)} = \frac{2q - 4 \pm 4\sqrt{1 - q}}{2} = q - 2 + 2\sqrt{1 - q}.$$

Our focus is on maximizing $p + q$, so we need the maximum values of p and q respectively (by taking the positive square root). Adding q , we see that

$$p + q = 2 \left(q - 1 + \sqrt{1 - q} \right).$$

Let $k = \sqrt{1-q}$. Then, $-k^2 = q - 1$, so $p + q = 2(-k^2 + k)$.

The graph of $-k^2 + k$ is a parabola that opens up downwards, and has its maximum value at y -value of the vertex. The x coordinate of the vertex is the average of the two roots, which in this case, are 0 and 1, so the average of these two is $\frac{1}{2}$. This means that the maximum value of $-k^2 + k$ occurs at $k = \frac{1}{2}$.

Substituting $k = \sqrt{1-q}$, we see that $q = \frac{3}{4}$. Since $p = q - 2 + 2\sqrt{1-q}$, we can plug $q = \frac{3}{4}$ in, to see that $p = -\frac{1}{4}$.

Because the roots of this quadratic are $\frac{3}{4}$ and $-\frac{1}{4}$, our quadratic is $\left(x - \frac{3}{4}\right)\left(x + \frac{1}{4}\right)$.

$\tilde{p}(1) = \frac{5}{16}$, so the answer is (A) $\frac{5}{16}$.

Solution 7 (Vieta's Formulas)

Since $\tilde{p}(x)$ is a disrespectful polynomial, it must satisfy $\tilde{p}(\tilde{p}(x))$. Let $\tilde{p}(x)$ have roots x_1, x_2 . So, the equation $\tilde{p}(\tilde{p}(x))$ becomes two separate equations:

$$x^2 + bx + c = x_1,$$

$$x^2 + bx + c = x_2.$$

Let r_1, r_2, r_3, r_4 be the roots of $\tilde{p}(\tilde{p}(x))$. Since there are 3 real roots, then either one of them is complex, or two of them are the same. It is known that if complex number w is a root to a certain polynomial, then so is $\bar{w}$. So, if one root to $\tilde{p}(\tilde{p}(x))$ is complex, then another root should also be complex — we can't have that. So, two roots are equal; WLOG, assume it is $r_3 = r_4$. That means the roots to $\tilde{p}(\tilde{p}(x))$ are r_1, r_2, r_3, r_3 .

Now, if the roots of $x^2 + bx + c = x_1$ are r_1 and r_3 and the roots of $x^2 + bx + c = x_2$ are r_2 and r_3 , then we have $-b = r_1 + r_3 = r_2 + r_3$ giving $r_1 = r_2$ by **Vieta's Formula**. That can't happen either. So, $x^2 + bx + c = x_1$ must have roots r_1, r_2 and $x^2 + bx + c = x_2$ must have a double root r_3 . Again by Vieta's, we have

$$r_1 + r_2 = -b,$$

$$2r_3 = -b.$$

That implies $r_2 - r_3 = r_3 - r_1$. So, let $r_1 = r_3 - k$ and $r_2 = r_3 + k$. Also, from $2r_3 = -b$, we have $r_3 = -\frac{b}{2}$. By Vieta's again, we have

$$r_1 \cdot r_2 = c - x_1 = r_1 \cdot r_2 = (r_3 - k)(r_3 + k) = r_3^2 - k^2,$$

$$r_3^2 = c - x_2.$$

Subtracting yields $x_1 - x_2 = k^2$. Adding that with $x_1 + x_2 = -b$ brings us to

$$x_1 = \frac{k^2 - b}{2},$$

$$x_2 = \frac{-k^2 - b}{2}.$$

Thus, $c = x_1 \cdot x_2 = \frac{b^2 - k^4}{4}$. Now, $c - x_2 = r_3^2$ by Vieta's. Plugging in $c, x_2, r_3 = -\frac{b}{2}$ gives

$$\frac{b^2 - k^4}{4} + \frac{k^2 + b}{2} = \frac{b^2 - k^4 + 2k^2 - 2b}{4} = \frac{b^2}{4}.$$

Simplifying yields $2b = k^4 - 2k^2 = (k^2 - 1)^2 - 1 \geq -1$ by **Trivial Inequality**. Thus, $\min(b) = -\frac{1}{2}$. Since $\tilde{p}(x)$ have the maximum value to the sum of the roots, which

is $-b$, we want to minimize b . That means that b for $\tilde{p}(x)$ is $\frac{1}{2}$. That means $k^2 = 1$ and $c = -\frac{3}{16}$, and thus giving us $\tilde{p}(1) = 1 + b + c = 1 - \frac{1}{2} - \frac{3}{16} = \boxed{\text{(A)} \frac{5}{16}}$.

Solution 8 (Calculus)

Let $p(x) = (x - r_1)(x - r_2)$. Then

$$p(p(x)) = ((x - r_1)(x - r_2) - r_1)((x - r_1)(x - r_2) - r_2)$$

As noted in other solutions, in order for $p(p(x))$ to have three (distinct) real roots, one of the roots must be repeated twice.

For a general polynomial $q(x)$ with an n -times repeated root s , write $q(x) = (x - s)^n u(x)$, where $u(x)$ is another polynomial. Taking the derivative,

$$\begin{aligned} q'(x) &= n(x - s)^{n-1}u(x) + (x - s)^n u'(x) \\ &= (x - s)^{n-1} [nu(x) + (x - s)u'(x)] \end{aligned}$$

This shows that a repeated root / linear factor will appear in the derivative as well.

Returning to the problem, consider the derivative of $p(p(x))$, which is

$$[p(p(x))]' = p'(p(x))p'(x) = [2(x - r_1)(x - r_2) - r_1 - r_2] [2x - r_1 - r_2]$$

By the observation above, the repeated root of $p(p(x))$ must be a root of this derivative. Therefore, a search through its roots will eventually give information to resolve r_1 and r_2 . Setting the derivative equal to zero yields the candidate equations

$$2(x - r_1)(x - r_2) - r_1 - r_2 = 0 \quad \text{or} \quad 2x - r_1 - r_2 = 0$$

Case 1: Starting with the first equation, if x_0 is a solution, then rearrange to obtain

$$(x_0 - r_1)(x_0 - r_2) = \frac{r_1 + r_2}{2}$$

Then

$$\begin{aligned} p(p(x_0)) &= ((x_0 - r_1)(x_0 - r_2) - r_1)((x_0 - r_1)(x_0 - r_2) - r_2) \\ &= \left(\frac{r_1 + r_2}{2} - r_1\right) \left(\frac{r_1 + r_2}{2} - r_2\right) \\ &= \frac{1}{4}(r_1 - r_2)^2 = 0 \end{aligned}$$

This shows that $r_1 = r_2$. However, this gives $p(x) = (x - r_1)^2$ and $p(p(x)) = [(x - r_1)^2 - r_1]^2$, which only has at most two distinct real roots. This contradicts the three-root condition given by the problem.

Case 2: The failure of Case 1 shows that the second equation must contain the repeated root. The solution for the second equation is readily $x_0 = (r_1 + r_2)/2$. Then,

$$p(p(x_0)) = \left[\left(\frac{r_1 + r_1}{2} - r_1\right) \left(\frac{r_1 + r_2}{2} - r_2\right) - r_1\right] \left[\left(\frac{r_1 + r_2}{2} - r_1\right) \left(\frac{r_1 + r_2}{2} - r_2\right) - r_2\right] = 0$$

Solving for r_1 yields four candidate equations:

$$r_1 = r_2 \pm 2\sqrt{-r_2} \quad \text{or} \quad r_1 = r_2 \pm 2\sqrt{1 - r_2} - 2$$

Since it is desired to maximise the sum of roots, add r_2 to both sides to consider

$$r_1 + r_2 = 2r_2 \pm 2\sqrt{-r_2} \quad \text{or} \quad r_1 + r_2 = 2r_2 \pm 2\sqrt{1-r_2} - 2$$

For the first family of solutions, the "plus" version is immediately larger, so maximise

$$f(r_2) = 2r_2 + 2\sqrt{-r_2}$$

Taking the derivative and setting equal to zero,

$$f'(r_2) = 2 - \frac{1}{\sqrt{-r_2}} = 0$$

yielding $r_2 = -1/4$ and $f(-1/4) = 1/2$, which can be verified to be a maximum. Since $f(r_2) = r_1 + r_2$, $r_1 = 3/4$. A similar calculation on the second family of solutions yields the complementary version $r_1 = -1/4$ and $r_2 = 3/4$.

In summary, this calculation shows that $\tilde{p}(x) = \left(x + \frac{1}{4}\right) \left(x - \frac{3}{4}\right)$. Finally, $\tilde{p}(1) =$

$$\left(1 + \frac{1}{4}\right) \left(1 - \frac{3}{4}\right) = \boxed{\text{(A)} \frac{5}{16}}.$$

Solution 9 (Lagrange Multipliers)

First note that $p(x)$ must have two distinct roots, else $p(p(x))$ will have fewer than 3 roots. Let these be a, b . Since $p(x)$ has 3 real roots, and complex roots come in pairs, there must be a double root. So there must be a_1, a_2 s.t $p(a_1) = p(a_2) = a$ and b_1, b_2 s.t $p(b_1) = p(b_2) = b$ but two of a_1, a_2, b_1, b_2 must be the same. WLOG either $a_1 = a_2$ or $a_1 = b_1$. In the latter, case, then,

$$\begin{aligned} p(x) &= a + (x - a_1)(x - a_2) \\ p(x) &= b + (x - b_1)(x - b_2) \\ &= b + (x - a_1)(x - b_2) \\ \implies -(a_1 + a_2)x &= -(a_1 + b_2)x \\ \implies a_2 &= b_2 \end{aligned}$$

which implies $a = b$ which is not possible. Hence assume WLOG $a_1 = a_2$. Then note that

$$\begin{aligned} p(x) &= (x - a)(x - b) \\ &= a + (x - a_1)^2 \\ -(a + b)x &= -2a_1x, \end{aligned}$$

so

$$\begin{aligned} p(x) &= a + \left(x - \frac{a+b}{2}\right)^2 \\ x^2 - (a+b)x + ab &= x^2 - (a+b)x + \left(a + \frac{1}{4}(a^2 + 2ab + b^2)\right) \\ 0 &= a^2 - 2ab + b^2 + 4a. \end{aligned}$$

The problem wants us to maximize the sum of the roots, $a + b$. Thus define $f(a, b) = a + b, g(a, b) = a^2 - 2ab + b^2 + 4a$. By the method of Lagrange Multipliers, we need to solve,

$$\vec{\nabla} f = \lambda \vec{\nabla} g$$

$$g = 0.$$

In the first equation, note,

$$\begin{aligned}\vec{\nabla} f &= \lambda \vec{\nabla} g \\ \langle 1, 1 \rangle &= \lambda \langle 2a - 2b + 4, -2a + 2b \rangle \\ 2a - 2b + 4 &= -2a + 2b \\ a - b &= -1.\end{aligned}$$

Substituting into the second equation,

$$\begin{aligned}g(a, b) &= (a - b)^2 + 4a \\ 0 &= 1 + 4a \\ a &= -\frac{1}{4}, b = \frac{3}{4} \\ \tilde{p}(1) &= \left(1 + \frac{1}{4}\right) \left(1 - \frac{3}{4}\right) = \boxed{\text{(A)} \frac{5}{16}}.\end{aligned}$$

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Solution 10 (Solution 8 but Simpler)

Let $p(x) = (x - r)(x - s)$, having roots r, s . WLOG, set $r < s$. The graph of $p(x)$ will have a vertex at $(\frac{r+s}{2}, p(\frac{r+s}{2}))$. Since $p(p(x)) = 0 \iff p(x) = r$ or $p(x) = s$ has exactly 3 distinct solutions, we know that $r = p(\frac{r+s}{2})$ and $s > p(\frac{r+s}{2})$ providing 1 and 2 solutions respectively for a total of 3 (if both $r, s > p(\frac{r+s}{2})$ there will be 4 solutions and if either $r, s < p(\frac{r+s}{2})$ there will be at most 2 solutions). Then,

$$\begin{aligned}r = p\left(\frac{r+s}{2}\right) &= \left(\frac{s-r}{2}\right) \left(\frac{r-s}{2}\right) = -\frac{(s-r)^2}{4} \\ (s-r)^2 &= -4r \\ s+r &= 2r \pm \sqrt{-4r}\end{aligned}$$

The negative case clearly doesn't work (giving $\max s+r = 0$ where $r, s = 0$), leaving us with $s+r = 2r + \sqrt{-4r}$. Finding the derivative we see that $s+r$ is maximized when $r = -\frac{1}{4}$, which gives $s = \frac{3}{4}$. Plugging these back into $p(x)$ we see

$$\text{that } \tilde{p}(x) = (x + \frac{1}{4})(x - \frac{3}{4}). \text{ Then } \tilde{p}(1) = \boxed{\text{(A)} \frac{5}{16}}.$$

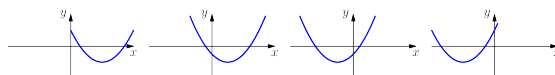
Solution 11 (AM-GM Inequality)

Because $p(x)$ is a quadratic equation with a leading coefficient of 1, let $p(x) = x^2 + ax + b$. Moreover, let r_1, r_2, r_3 be the roots of the equation $p(p(x)) = 0$.

$$\begin{aligned}p(r_1) &= a \\ p(r_2) &= b \\ p(r_3) &= c\end{aligned}$$

$$\begin{aligned}p(a) &= p(b) = p(c) = 0 \\ p(x) &= (x - \alpha)(x - \beta)\end{aligned}$$

The following cases are the possible graphs for $p(x)$.



The second graph is the case that could provide the maximum sum of the roots.

$$\begin{aligned}\therefore \left(x - \frac{\alpha + \beta}{2}\right)^2 + \alpha &= (x - \alpha)(x - \beta) \\ x^2 - (\alpha + \beta)x + \frac{(\alpha + \beta)^2}{4} + \alpha &= x^2 - (\alpha + \beta)x + \alpha\beta \\ \frac{(\alpha + \beta)^2}{4} + \alpha &= \alpha\beta \\ (\alpha + \beta)^2 &= 4\alpha\beta - 4\alpha = 4\alpha(\beta - 1)\end{aligned}$$

$$\sqrt{\alpha(\beta - 1)} \leq \frac{\alpha + (\beta - 1)}{2}$$

leads to

$$\alpha = \beta - 1.$$

$$\begin{aligned}(\alpha + \beta)^2 &= 4\alpha^2 \\ \alpha &= \beta \text{ or } 3\alpha + \beta = 0\end{aligned}$$

$$\therefore \alpha = -\frac{1}{4}, \beta = \frac{3}{4}$$

$$p(1) = (1 - \alpha)(1 - \beta) = \frac{5}{4} \cdot \frac{1}{4} = \boxed{\text{(A)} \frac{5}{16}}$$

Solution 12: 2 Ways

In this solution are really 2 solutions: my first one, that I did really fast, but was basically solution 1 except I showed that $\tilde{p}(\tilde{p}(x))$ can only have real roots and neither quadratic factor can share any roots. This is too simple, however, so I had to come up with something overkill and unnecessary. So in my second solution, I explicitly list root cases, use derivatives when the optimization could be done with algebra, etc.

Solution 12.1: The Sane Way

Let a and b be the roots of $\tilde{p}(x)$. We have that

$$\tilde{p}(x) = (x - a)(x - b) \implies \tilde{p}(\tilde{p}(x)) = ((x - a)(x - b) - a)((x - a)(x - b) - b).$$

For this to have 3 real solutions, one quadratic factor must have a discriminant of 0 and the other discriminant must be positive. From this, it can quickly be shown that a and b must be real: Assume a and b are nonreal such that $a + b$ and ab are real; then neither quadratic factor can have any real roots because $\tilde{p}(x) = x^2 - (a + b)x + ab$ must be real for all $x \in \mathbb{R}$, so for all $x \in \mathbb{R}$ the equations $\tilde{p}(x) = a$ and $\tilde{p}(x) = b$ cannot be satisfied. Then, it quite immediately follows that for the discriminants to differ, $a \neq b$ so $p(x) = a$ and $p(x) = b$ cannot be satisfied

simultaneously, which ensures neither quadratic factor shares a root. Thus, without loss of generality:

$$x^2 - (a+b)x + ab - a = 0 \text{ for 2 distinct } x \in \mathbb{R} \implies (a+b)^2 - 4(ab-a) = a^2 + b^2 - 2ab + 4a = 0 \implies (a-b)^2 = 4a$$

and

$$x^2 - (a+b)x + ab - b = 0 \text{ for 1 } x \in \mathbb{R} \implies (a+b)^2 - 4(ab-b) = a^2 + b^2 - 2ab + 4b > 0 \implies (a-b)^2 = -4a > 0$$

Hence,

$$b - a = 2\sqrt{-a} \implies a + b = 2(a + \sqrt{-a}).$$

We want to maximize $a + \sqrt{-a}$ for $a \leq 0$. Substitute $x = \sqrt{-a}$ so we maximize $-x^2 + x$ for $x \geq 0$. We have $-x^2 + x = -x(x-1)$ which is maximized at $x = \frac{1}{2} \implies a = -\frac{1}{4} \implies \max(a + \sqrt{-a}) = \frac{1}{4}$. Then $\max(a+b) = \frac{1}{2}$, $a = -\frac{1}{4} \implies b = \frac{3}{4}$. Hence

$$\tilde{p}(x) = x^2 - \frac{1}{2}x - \frac{3}{16} \implies \tilde{p}(1) = \boxed{\text{(A)} \frac{5}{16}}$$

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Solution 12.2: Projectile Motion and General Inefficiency

Let $q(x)$ be a disrespectful quadratic. It follows that the graph of $q(x)$ is a convex parabola, and thus the graph of $-q(x)$ is a concave parabola. Suppose $q(x) = (x-a)(x-b)$ where a and b are real numbers. Then:

$$q(q(x)) = (q(x)-a)(q(x)-b).$$

The graph of $q(q(x))$ can be viewed as essentially graphing $q(t)$ where $t \in \text{ran}(q)$. The following cases yield different amounts of real numbers such that $q(q(x)) = 0$:

Case 1: 4 real roots We need the equations

$$q(x) = a \quad \text{and} \quad q(x) = b$$

to have 2 real roots each.

Case 2: 3 real roots We need the equations

$$q(x) = a \quad \text{and} \quad q(x) = b$$

to have 2 real roots and 1 real root respectively, or 1 real root and 2 real roots respectively.

Case 3: 2 real roots We need the equations

$$q(x) = a \quad \text{and} \quad q(x) = b$$

to have 1 real roots each.

Case 4: 1 real root We need the equations

$$q(x) = a \quad \text{and} \quad q(x) = b$$

to have 1 real root and no real roots respectively, or no real roots and 1 real root respectively.

=Case 5: no real roots We need the equations

$$q(x) = a \quad \text{and} \quad q(x) = b$$

to both have no real roots.

Case 2 is what we want. For the $q(x) = a$ and $q(x) = b$ to have different amounts of real solutions, it immediately follows that $a \neq b$. It can also be quickly shown that to have a disrespectful polynomial, a and b must be real: since any disrespectful quadratic has real coefficients, all real inputs yield real outputs and so the equations $q(x) = a$ and $q(x) = b$ can't have any real solutions unless a and b are real. Let

$$\tilde{p}(x) = (x - \alpha)(x - \beta),$$

and fix $\alpha < \beta$ for simplicity. Define the k -shift of $g(x)$ as $g(x - k)$. It follows that $-\tilde{p}(x)$ is concave parabola such that $\max(-\tilde{p}) > 0$, so $-\tilde{p}(x)$ is the α -shift of the path of a projectile that experiences negligible air resistance launched from a horizontal surface with constant downward acceleration. A projectile launched with speed u at an angle θ to the horizontal has initial horizontal velocity $u \cos(\theta)$, on which no relevant horizontal drag, friction, or gravitational force effects and thus the projectile has horizontal position

$$p_h(t) = ut \cos(\theta)$$

after t seconds. Then, the projectile has initial vertical velocity $u \sin(\theta)$ and has a vertical acceleration g downwards and thus a total vertical velocity of $u \sin(\theta) - gt$ and a vertical position of

$$p_v(t) = ut \sin(\theta) - \frac{1}{2}gt^2.$$

We can now see that setting $p'_v(t) = 0$ yields that $t = \frac{u \sin(\theta)}{g}$ so letting $H_{\max} = \max(p_v)$ denote the maximum height,

$$H_{\max} = \frac{u^2 \sin^2(\theta)}{g} - \frac{u^2 \sin^2(\theta)}{2g} = \frac{u^2 \sin^2(\theta)}{2g}.$$

Then, to find the range R (horizontal distance the projectile travels) we need $p_v(t) = 0$ such that $t \neq 0$ so

$$u \sin(\theta) - \frac{gt}{2} = 0 \implies t = \frac{2u \sin(\theta)}{g} \implies R = \frac{2u^2 \sin(\theta) \cos(\theta)}{g} = \frac{u^2 \sin(2\theta)}{g}.$$

Now, we can see that modeling $-\tilde{p}(x)$ as the α -shift of such a trajectory, $\beta - \alpha$ is the range of the projectile and so we are maximizing

$$f(\alpha, \theta, u) = \alpha + (\alpha + R) = 2\alpha + \frac{u^2 \sin(2\theta)}{g}$$

subject to some constraints where g is a positive real constant. Then, the equation $\tilde{p}(\tilde{p}(x)) = 0$ can be rewritten as

$$-\tilde{p}(\tilde{p}(x)) = -(\tilde{p}(x) - \alpha) \left(\tilde{p}(x) - \alpha - \frac{u^2 \sin(2\theta)}{g} \right).$$

We now use the constraints of case 2: we either have that $\tilde{p}(x) - \alpha$ has 2 real roots and $\tilde{p}(x) - \alpha - \frac{u^2 \sin(2\theta)}{g}$ has 1, or vice versa. It can be seen that

$$-\alpha > -\alpha - \frac{u^2 \sin(2\theta)}{g}$$

(which was immediate from $\alpha < \beta$ anyways) so the vertex of $\tilde{p}(x) - \alpha - \frac{u^2 \sin(2\theta)}{g}$ lies below the vertex of $\tilde{p}(x) - \alpha$. Thus $\tilde{p}(x) - \alpha$ has to have its vertex on the x -axis and $\tilde{p}(x) - \alpha - \frac{u^2 \sin(2\theta)}{g}$ has to have its vertex below the x -axis. Since $\tilde{p}(x) - \alpha$ has its vertex on the x -axis it must be expressible in the form $(x - h)^2 = x^2 - 2hx + h^2$ for some real h , and be zero at this h . Thus we'd have

$$0 = x^2 - \left(2\alpha + \frac{u^2 \sin(2\theta)}{g}\right)x + \alpha^2 + \frac{\alpha u^2 \sin(2\theta)}{g} - \alpha = x^2 - 2hx + h^2.$$

Comparing linear coefficients instantly yields

$$h = \alpha + \frac{u^2 \sin(2\theta)}{2g}.$$

Comparing our constant term and using our expression for h yields

$$h^2 = \alpha^2 + \frac{u^2 \sin(2\theta)}{g} + \frac{u^4 \sin^2(2\theta)}{4g^2} = \alpha^2 + \frac{u^2 \sin(2\theta)}{g} - \alpha \implies -4\alpha = \frac{u^2 \sin^2(2\theta)}{g^2} \implies 2\sqrt{-\alpha} = \frac{u \sin(2\theta)}{g}$$

Thus to maximize f we can just maximize

$$f_1(\alpha) = 2\alpha + 2\sqrt{-\alpha} \quad (\alpha \leq 0)$$

We used common derivatives/integrals for the projectile motion but we're not doing that here for no reason

$$\frac{df}{d\alpha} = \lim_{h \rightarrow 0} \frac{2(\alpha + h) + 2\sqrt{-\alpha - h} - 2\alpha - 2\sqrt{-\alpha}}{h} = 2 + 2 \lim_{h \rightarrow 0} \frac{\sqrt{-\alpha - h} - \sqrt{-\alpha}}{h} = 2 + 2 \lim_{h \rightarrow 0} \frac{-h}{h(\sqrt{-\alpha - h} + \sqrt{-\alpha})}$$

So

$$\frac{df}{d\alpha} = 0 \implies \frac{1}{\sqrt{-\alpha}} = 2 \implies \sqrt{-\alpha} = \frac{1}{2} \implies \alpha = -\frac{1}{4}.$$

We then have that $\max(f) = \alpha + \beta = -\frac{1}{2} + 1 = \frac{1}{2}$ so $\beta = \frac{3}{4}$. Thus

$$\tilde{p}(x) = \left(x + \frac{1}{4}\right)\left(x - \frac{3}{4}\right) \implies \tilde{p}(1) = \frac{5}{4} \cdot \frac{1}{4} = \boxed{\text{(A)} \frac{5}{16}}$$

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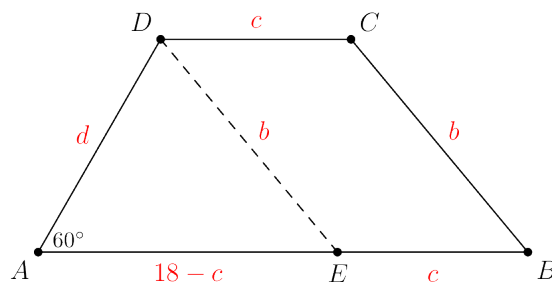
24. (AMC 12 12A Fall 2021) Convex quadrilateral $ABCD$ has $AB = 18$, $\angle A = 60^\circ$, and $\overline{AB} \parallel \overline{CD}$. In some order, the lengths of the four sides form an arithmetic progression, and side $\overline{AB}$ is a side of maximum length. The length of another side is a . What is the sum of all possible values of a ?

(A) 24 (B) 42 (C) 60 (D) 66 (E) 84

Answer: E

Solution 1

Let E be a point on $\overline{AB}$ such that $BCDE$ is a parallelogram. Suppose that $BC = ED = b$, $CD = BE = c$, and $DA = d$, so $AE = 18 - c$, as shown below.



We apply the Law of Cosines to $\triangle ADE$:

$$\begin{aligned} AD^2 + AE^2 - 2 \cdot AD \cdot AE \cdot \cos 60^\circ &= DE^2 \\ d^2 + (18 - c)^2 - d(18 - c) &= b^2 \\ (18 - c)^2 - d(18 - c) &= b^2 - d^2 \\ (18 - c)(18 - c - d) &= (b + d)(b - d). \end{aligned} \quad (\star)$$

Let k be the common difference of the arithmetic progression of the side-lengths. It follows that b, c , and d are $18 - k, 18 - 2k$, and $18 - 3k$, in some order. It is clear that $0 \leq k < 6$.

If $k = 0$, then $ABCD$ is a rhombus with side-length 18, which is valid.

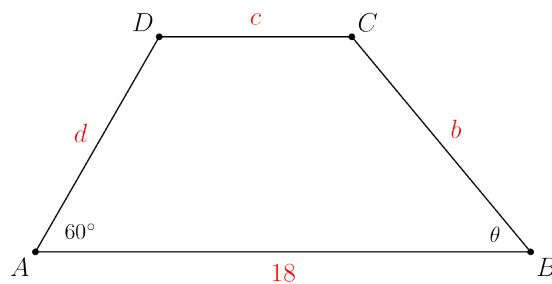
If $k \neq 0$, then we have six cases:

- $(b, c, d) = (18 - k, 18 - 2k, 18 - 3k)$ Note that $(\star)$ becomes $2k(5k - 18) = (36 - 4k)(2k)$, from which $k = 6$. So, this case generates no valid solutions (b, c, d) .
- $(b, c, d) = (18 - k, 18 - 3k, 18 - 2k)$ Note that $(\star)$ becomes $3k(5k - 18) = (36 - 3k)k$, from which $k = 5$. So, this case generates $(b, c, d) = (13, 3, 8)$.
- $(b, c, d) = (18 - 2k, 18 - k, 18 - 3k)$ Note that $(\star)$ becomes $k(4k - 18) = (36 - 5k)k$, from which $k = 6$. So, this case generates no valid solutions (b, c, d) .
- $(b, c, d) = (18 - 2k, 18 - 3k, 18 - k)$ Note that $(\star)$ becomes $3k(4k - 18) = (36 - 3k)(-k)$, from which $k = 2$. So, this case generates $(b, c, d) = (14, 12, 16)$.
- $(b, c, d) = (18 - 3k, 18 - k, 18 - 2k)$ Note that $(\star)$ becomes $k(3k - 18) = (36 - 5k)(-k)$, from which $k = 9$. So, this case generates no valid solutions (b, c, d) .
- $(b, c, d) = (18 - 3k, 18 - 2k, 18 - k)$ Note that $(\star)$ becomes $2k(3k - 18) = (36 - 4k)(-2k)$, from which $k = 18$. So, this case generates no valid solutions (b, c, d) .

Together, the sum of all possible values of a is $18 + (13 + 3 + 8) + (14 + 12 + 16) =$
(E) 84.

Solution 2

Let b, c , and d denote the sides BC, CD , and AD respectively.



Since $AB \parallel CD$, we get

$$\frac{\sqrt{3}}{2}d = b \sin \theta \quad \text{and} \quad \frac{1}{2}d + c + b \cos \theta = 18.$$

Using $b^2 \sin^2 \theta + b^2 \cos^2 \theta = b^2$, we eliminate θ from above to get $(36 - 2c - d)^2 + 3d^2 = 4b^2$, which rearranges to $(36 - 2c - d)^2 - d^2 = 4(b^2 - d^2)$, and, upon factoring, yields

$$(18 - c)(18 - c - d) = (b + d)(b - d). \quad (5)$$

We divide into two cases, depending on whether c is the smallest side.

If c is not the smallest side then $18 - c = \pm(b - d)$. If $c = 18$, we get a rhombus of side 18, so one possible value is $a = 18$. Otherwise, we can cancel the common factor from (1). After rearranging we get

$$18 - c = -b \quad \text{or} \quad 18 - c = b + 2d.$$

The first condition is false because $-b < 0 < 18 - c$; the second condition is false because $b + 2d > |b - d| = 18 - c$.

If c is the smallest side, then $18 - c = \pm 3(b - d)$. Assuming $c < 18$ we can cancel common factors in (1) to get

$$8b = 13d \quad \text{or} \quad 8b = 7d.$$

The first condition yields the solution $(c, d, b) = (3, 8, 13)$ and the second condition yields the solution $(c, b, d) = (12, 14, 16)$.

Together, the sum of all possible values of a is $18 + (3 + 8 + 13) + (12 + 14 + 16) =$
(E) 84.

Solution 3

Denote $x = AD$, $\theta = \angle B$. Hence, $BC = \frac{\sqrt{3}}{2} \cdot \frac{x}{\sin \theta}$, $DC = 18 - \frac{x}{2} - \frac{\sqrt{3}}{2}x \cot \theta$.

Case 1: $DC = AD = BC = AB$.

This is a rhombus. So each side has length 18.

For the following cases, we consider four sides that have distinct lengths. To make their lengths an arithmetic sequence, we must have $\theta \neq 120^\circ$.

Therefore, in the subsequent analysis, we exclude the solution $\theta = 120^\circ$.

Case 2: $DC < AD < BC < AB$.

Because the lengths of these sides form an arithmetic sequence, we have the following system of equations:

$$AB - BC = BC - AD = AD - DC.$$

Hence,

$$18 - \frac{\sqrt{3}}{2} \cdot \frac{x}{\sin \theta} = \frac{\sqrt{3}}{2} \cdot \frac{x}{\sin \theta} - x = x - \left(18 - \frac{x}{2} - \frac{\sqrt{3}}{2}x \cot \theta\right).$$

By solving this system of equations, we get $(\cos \theta, \sin \theta, x) = \left(\frac{11}{13}, \frac{4\sqrt{3}}{13}, 8\right)$.

Thus, in this case, $DC = 3$, $AD = 8$, $BC = 13$.

Case 3: $DC < BC < AD < AB$.

Because the lengths of these sides form an arithmetic sequence, we have the following system of equations:

$$AB - AD = AD - BC = BC - DC.$$

Hence,

$$18 - x = x - \frac{\sqrt{3}}{2} \cdot \frac{x}{\sin \theta} = \frac{\sqrt{3}}{2} \cdot \frac{x}{\sin \theta} - \left(18 - \frac{x}{2} - \frac{\sqrt{3}}{2}x \cot \theta\right).$$

By solving this system of equations, we get $(\cos \theta, \sin \theta, x) = \left(-\frac{1}{7}, \frac{4\sqrt{3}}{7}, 16\right)$.

Thus, in this case, $DC = 12$, $AD = 16$, $BC = 14$.

Case 4: $BC < CD < AD < AB$.

By doing the similar analysis, we can show there is no solution in this case.

Case 5: $BC < AD < CD < AB$.

By doing the similar analysis, we can show there is no solution in this case.

Case 6: $AD < CD < BC < AB$.

By doing the similar analysis, we can show there is no solution in this case.

Case 7: $AD < BC < CD < AB$.

By doing the similar analysis, we can show there is no solution in this case.

Therefore, the sum of all possible values of a is

$$18 + (3 + 8 + 13) + (12 + 14 + 16) = 84.$$

Therefore, the answer is **(E)** 84.

25. (AMC 12 12A Fall 2021) Let $m \geq 5$ be an odd integer, and let $D(m)$ denote the number of quadruples (a_1, a_2, a_3, a_4) of distinct integers with $1 \leq a_i \leq m$ for all i such that m divides $a_1 + a_2 + a_3 + a_4$. There is a polynomial

$$q(x) = c_3x^3 + c_2x^2 + c_1x + c_0$$

such that $D(m) = q(m)$ for all odd integers $m \geq 5$. What is c_1 ? **(A)** -6 **(B)** -1 **(C)** 4 **(D)** 6

Answer: E

Solution 1 (Complete Residue System)

For a fixed value of m , there is a total of $m(m-1)(m-2)(m-3)$ possible ordered quadruples (a_1, a_2, a_3, a_4) .

Let $S = a_1 + a_2 + a_3 + a_4$. We claim that exactly $\frac{1}{m}$ of these $m(m-1)(m-2)(m-3)$ ordered quadruples satisfy that m divides S :

Since $\gcd(m, 4) = 1$, we conclude that

$$\{k + 4(0), k + 4(1), k + 4(2), \dots, k + 4(m-1)\}$$

is the complete residue system modulo m for all integers k .

Given any ordered quadruple (a'_1, a'_2, a'_3, a'_4) in modulo m , it follows that exactly one of these m ordered quadruples has sum 0 modulo m :

Ordered Quadruple	Sum Modulo m
(a'_1, a'_2, a'_3, a'_4)	$S' + 4(0)$
$(a'_1 + 1, a'_2 + 1, a'_3 + 1, a'_4 + 1)$	$S' + 4(1)$
$(a'_1 + 2, a'_2 + 2, a'_3 + 2, a'_4 + 2)$	$S' + 4(2)$
$\dots$	$\dots$
$(a'_1 + m - 1, a'_2 + m - 1, a'_3 + m - 1, a'_4 + m - 1)$	$S' + 4(m - 1)$

We conclude that $D(m) = \frac{1}{m} \cdot [m(m-1)(m-2)(m-3)] = (m-1)(m-2)(m-3)$, so

$$q(x) = (x-1)(x-2)(x-3) = c_3x^3 + c_2x^2 + c_1x + c_0.$$

By Vieta's Formulas, we get $c_1 = 1 \cdot 2 + 1 \cdot 3 + 2 \cdot 3 = \boxed{\text{(E) } 11}$.

Solution 2 (Symmetric Congruent Numbers and Interpolation)

Define

$$b_i = \begin{cases} a_i & \text{if } 1 \leq a_i \leq \frac{m-1}{2} \\ a_i - m & \text{if } \frac{m-1}{2} + 1 \leq a_i \leq m-1 \\ 0 & \text{if } a_i = m \end{cases}.$$

Hence, b_i is a one-to-one and onto function of a_i , and the range of b_i is $\left\{-\frac{m-1}{2}, \dots, \frac{m-1}{2}\right\}$.

Therefore, to solve this problem, it is equivalent for us to count the number of tuples (b_1, b_2, b_3, b_4) that are all distinct and satisfy $m|b_1 + b_2 + b_3 + b_4$.

Denote by $d(m)$ the number of such tuples that are also subject to the constraint $b_1 < b_2 < b_3 < b_4$.

Hence, $D(m) = 4!d(m) = 24d(m)$.

We do the following casework analysis to compute $d(m)$.

Case 1: There is one 0 in (b_1, b_2, b_3, b_4) .

Denote by $d_{1i}(m)$ the number of tuples with $b_i = 0$.

By symmetry, $d_{11}(m) = d_{14}(m)$ and $d_{12}(m) = d_{13}(m)$.

Case 2: There is no 0 in (b_1, b_2, b_3, b_4) .

Denote by $d_{2i}(m)$ the number of tuples with i positive entries.

By symmetry, $d_{20}(m) = d_{24}(m)$ and $d_{21}(m) = d_{23}(m)$.

Therefore,

$$\begin{aligned} D(m) &= 24d(m) \\ &= 24 \left(\sum_{i=1}^4 d_{1i}(m) + \sum_{i=0}^4 d_{2i}(m) \right) \\ &= 24(2d_{11}(m) + 2d_{12}(m) + 2d_{24}(m) + 2d_{23}(m) + d_{22}(m)). \end{aligned}$$

Now, we compute $D(m)$ for $m = 5, 7, 9, 11$.

SCENARIO $m = 5$.

We have $b_i \in \{-2, \dots, 2\}$.

Case 1 Case 1.1: $b_1 = 0$

We cannot have 3 distinct positive integers. So $d_{11}(5) = 0$.

Case 1.2: $b_2 = 0$

Because there are 2 positive integers, we must have $b_3 = 1, b_4 = 2$. Hence, $b_1 = -3$. However, this is out of the range of b_i . Thus, $d_{12}(5) = 0$.

Case 2 Case 2.1: $b_1 > 0$

We cannot have 4 distinct positive integers. So $d_{24}(5) = 0$.

Case 2.2: $b_1 < 0 < b_2$

We cannot have 3 distinct positive integers. So $d_{23}(5) = 0$.

Case 2.3: $b_2 < 0 < b_3$

The only solution is $(b_1, b_2, b_3, b_4) = (-2, -1, 1, 2)$. So $d_{22}(5) = 1$.

Therefore, $D(5) = 24$.

SCENARIO $m = 7$.

We have $b_i \in \{-3, \dots, 3\}$.

Case 1 Case 1.1: $b_1 = 0$

We have no feasible solution. Thus, $d_{11}(7) = 0$.

Case 1.2: $b_2 = 0$

The only solution is $(b_1, b_3, b_4) = (-3, 1, 2)$. Thus, $d_{12}(7) = 1$.

Case 2 Case 2.1: $b_1 > 0$

We cannot have 4 distinct positive integers. So $d_{24}(7) = 0$.

Case 2.2: $b_1 < 0 < b_2$

To get 3 distinct positive integers, we have $(b_2, b_3, b_4) = (1, 2, 3)$. This implies $b_1 = -6$. However, this is out of the range of b_1 . So $d_{23}(7) = 0$.

Case 2.3: $b_2 < 0 < b_3$

We have $d_{22}(7) = \binom{3}{2} = 3$.

Therefore, $D(7) = 24 \cdot 5$.

SCENARIO $m = 9$.

We have $b_i \in \{-4, \dots, 4\}$.

Case 1 Case 1.1: $b_1 = 0$

The only solution is $(b_2, b_3, b_4) = (2, 3, 4)$. Thus, $d_{11}(9) = 1$.

Case 1.2: $b_2 = 0$

The feasible solutions are $(b_1, b_3, b_4) = (-3, 1, 2), (-4, 1, 3)$. Thus, $d_{12}(9) = 2$.

Case 2 Case 2.1: $b_1 > 0$

There is no feasible solution. So $d_{24}(9) = 0$.

Case 2.2: $b_1 < 0 < b_2$

To get 3 distinct positive integers, we have $b_2 + b_3 + b_4 \geq 1 + 2 + 3 = 6$. This implies $b_1 = -6$. However, this is out of the range of b_1 . So $d_{23}(9) = 0$.

Case 2.3: $b_2 < 0 < b_3$

We have $d_{22}(9) = 8$.

Therefore, $D(9) = 24 \cdot 14$.

SCENARIO $m = 11$.

We have $b_i \in \{-5, \dots, 5\}$.

Case 1 Case 1.1: $b_1 = 0$

The only solution is $(b_2, b_3, b_4) = (2, 4, 5)$. Thus, $d_{11}(11) = 1$.

Case 1.2: $b_2 = 0$

The feasible solutions are $(b_1, b_3, b_4) = (-3, 1, 2), (-4, 1, 3), (-5, 1, 4), (-5, 2, 3)$. Thus, $d_{12}(11) = 4$.

Case 2 Case 2.1: $b_1 > 0$

The only feasible solution is $(b_1, b_2, b_3, b_4) = (1, 2, 3, 5)$. So $d_{24}(11) = 1$.

Case 2.2: $b_1 < 0 < b_2$

The only feasible solution is $(b_1, b_2, b_3, b_4) = (-1, 3, 4, 5)$. So $d_{23}(11) = 1$.

Case 2.3: $b_2 < 0 < b_3$

We have $d_{22}(11) = 16$.

Therefore, $D(11) = 24 \cdot 30$.

We know that $q(m) = D(m)$ for odd $m \geq 5$.

Plugging $m = 5, 7, 9, 11$ into this equation, we get

$$c_3 5^3 + c_2 5^2 + c_1 5 + c_0 = 24 \cdot 1 \quad (1.1)$$

$$c_3 7^3 + c_2 7^2 + c_1 7 + c_0 = 24 \cdot 5 \quad (1.2)$$

$$c_3 9^3 + c_2 9^2 + c_1 9 + c_0 = 24 \cdot 14 \quad (1.3)$$

$$c_3 11^3 + c_2 11^2 + c_1 11 + c_0 = 24 \cdot 30 \quad (1.4)$$

Now, we solve this system of equations. Taking $\frac{(1.2)-(1.1)}{2}$, $\frac{(1.3)-(1.2)}{2}$, $\frac{(1.4)-(1.2)}{2}$, we get

$$c_3 109 + c_2 12 + c_1 = 48 \quad (2.1)$$

$$c_3 193 + c_2 16 + c_1 = 108 \quad (2.2)$$

$$c_3 301 + c_2 20 + c_1 = 192 \quad (2.3)$$

Taking $\frac{(2.2)-(2.1)}{4}$, $\frac{(2.3)-(2.2)}{4}$, we get

$$c_3 21 + c_2 = 15 \quad (3.1)$$

$$c_3 27 + c_2 = 21 \quad (3.2)$$

Taking $\frac{(3.2)-(3.1)}{6}$, we get $c_3 = 1$.

Plugging c_3 into Equation (3.1), we get $c_2 = -6$.

Plugging c_2 and c_3 into Equation (2.1), we get $c_1 = 11$.

Therefore, the answer is (E) 11.

Solution 3

As before, note that we have $m(m-1)(m-2)$ numbers we can choose as a, b, c . From here, there is exactly one possible value of $1 \leq d \leq m$ that could make $S = a + b + c + d$ divisible by m . However, there is a $\frac{3}{m}$ chance that this value of d has already been chosen as a, b or c . Thus our polynomial is $m(m-1)(m-2)\left(1 - \frac{3}{m}\right) = m(m-1)(m-2)\left(\frac{m-3}{m}\right) = (m-1)(m-2)(m-3)$. By Vieta's, $c_1 = 2 + 3 + 6 = \span style="border: 1px solid black; padding: 2px;">(E) 11.$

Solution 4

We can solve this by directly counting quadruples and deriving a closed formula for $D(m)$.

For an odd integer $m \geq 5$, we need to count ordered quadruples (a_1, a_2, a_3, a_4) of distinct integers where $1 \leq a_i \leq m$ and m divides their sum.

Consider how many total ordered quadruples of distinct integers exist in the range $[1, m]$:- We choose 4 distinct numbers from m possible values: $\binom{m}{4}$ - We arrange these 4 numbers in order: $4!$

Therefore, total quadruples $= \binom{m}{4} \cdot 4! = \frac{m!}{(m-4)!}$

Now, these quadruples can be partitioned into m groups based on the remainder of their sum when divided by m . By symmetry, these remainders are uniformly distributed modulo m .

Therefore, exactly $\frac{1}{m}$ of all quadruples have a sum divisible by m :

$$D(m) = \frac{1}{m} \cdot \frac{m!}{(m-4)!} = \frac{m(m-1)(m-2)(m-3)}{m} = (m-1)(m-2)(m-3)$$

Expanding this expression:

$$(m-1)(m-2)(m-3) = m^3 - 6m^2 + 11m - 6$$

Comparing with $q(x) = c_3x^3 + c_2x^2 + c_1x + c_0$, we identify: $c_3 = 1$, $c_2 = -6$, $c_1 = 11$, $c_0 = -6$

Therefore, $c_1 = 11$, and the answer is (E)11.

Solution 5 (Principle of Inclusion-Exclusion)

If we do not have to worry about whether or not the numbers are distinct, then there would be m^3 possible quadruples. This is because the first 3 numbers can be anything, and then the last number is uniquely chosen so that the sum is a multiple of m .

Now, we must subtract the quadruples with at least two of the same number. There are $\binom{4}{2}$ ways to choose the positions for the two identical numbers, m ways to choose the number for these two positions, and m ways to choose a number for another position. The last position is uniquely determined as before. Thus, our expression becomes $m^3 - \binom{4}{2} \cdot m \cdot m = m^3 - 6m^2$.

However, we have subtracted the cases in which there are three identical numbers too many times. Each such case was subtracted $\binom{3}{2} = 3$ times, and there are $\binom{4}{3} \cdot m = 4m$ such cases. Thus, we need to add back $8m$.

We also subtracted the cases in which there are two pairs of two identical numbers too many times. There are $\frac{\binom{4}{2}m}{2} = 3m$ such cases. This is because we first choose the two positions for the first two identical numbers, then multiply by m to determine which number goes in these positions. The last two positions can be determined uniquely because m is odd. We divide by 2 because we counted each case twice. In our original counting, we subtracted each such case twice, so we add back $3m$. In total, our expression is now $m^3 - 6m^2 + 8m + 3m = m^3 - 6m^2 + 11m$.

We can stop here, because we know that the number of times $(5, 5, 5, 5)$ was overcounted is just a constant. Thus, the answer is $\boxed{(E)11}$.

§7 AMC 12 12B 2022

1. (AMC 12 12B 2022) Define $x \diamond y$ to be $|x - y|$ for all real numbers x and y . What is the value of

$$(1 \diamond (2 \diamond 3)) - ((1 \diamond 2) \diamond 3)?$$

- (A) -2 (B) -1 (C) 0 (D) 1 (E) 2

Answer: A

Solution 1

We have

$$(1 \diamond (2 \diamond 3)) - ((1 \diamond 2) \diamond 3) = |1 - |2 - 3|| - ||1 - 2| - 3| \quad (6)$$

$$= |1 - 1| - |1 - 3| \quad (7)$$

$$= 0 - 2 \quad (8)$$

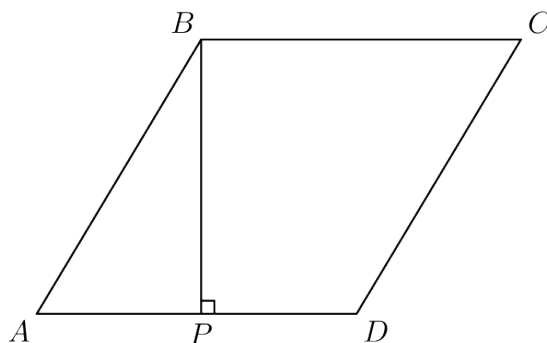
$$= \boxed{(A) -2}. \quad (9)$$

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Solution 2

Observe that the $\diamond$ function is simply the positive difference between two numbers. Thus, we evaluate: the difference between 2 and 3 is 1; the difference between 1 and 1 is 0; the difference between 1 and 2 is 1; the difference between 1 and 3 is 2; and finally, $0 - 2 = \boxed{(A) -2}$.

2. (AMC 12 12B 2022) In rhombus $ABCD$, point P lies on segment $\overline{AD}$ so that $\overline{BP} \perp \overline{AD}$, $AP = 3$, and $PD = 2$. What is the area of $ABCD$? (Note: The figure is not drawn to scale.)



- (A) $3\sqrt{5}$ (B) 10 (C) $6\sqrt{5}$ (D) 20 (E) 25

Answer: D

Solution

Notice that $BC = AD = AP + PD = 3 + 2 = 5$. Now, because the question tells us $ABCD$ is a rhombus, $BC = AB = 5$. Thus, by Pythagorean Theorem on APB , $BP = 4$. Therefore,

$$[ABCD] = AD \times BP = 5 \times 4 = \boxed{\text{D } 20}$$

DUODUOLUO (First time using $\LaTeX$, may have some minor mistakes)

3. (AMC 12 12B 2022) How many of the first ten numbers of the sequence 121, 11211, 1112111, ... are prime numbers?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Answer: A

Solution 1 (Generalization)

The n th term of this sequence is

$$\sum_{k=n}^{2n} 10^k + \sum_{k=0}^n 10^k = 10^n \sum_{k=0}^n 10^k + \sum_{k=0}^n 10^k = (10^n + 1) \sum_{k=0}^n 10^k.$$

It follows that the terms are

$$\begin{aligned} 121 &= 11 \cdot 11, \\ 11211 &= 101 \cdot 111, \\ 1112111 &= 1001 \cdot 1111, \\ &\vdots \end{aligned}$$

Therefore, there are $\boxed{\text{(A) } 0}$ prime numbers in this sequence.

Solution 2 (Detailed Explanation of Solution 1)

Denote this sequence as a_n , then we can find that

$$\begin{aligned} a_1 &= 121 = 10^2 + 2 \cdot 10 + 1 = (10^2 + 10) + (10 + 1), \\ a_2 &= 11211 = (10^4 + 10^3 + 10^2) + (10^2 + 10 + 1), \\ a_3 &= 1112111 = (10^6 + 10^5 + 10^4 + 10^3) + (10^3 + 10^2 + 10 + 1), \\ &\vdots \end{aligned}$$

So, we can induct that the general term is

$$\begin{aligned} a_n &= (10^{2n} + 10^{2n-1} + \dots + 10^{n+1} + 10^n) + (10^n + 10^{n-1} + \dots + 10 + 1) \\ &= 10^n \cdot (10^n + 10^{n-1} + \dots + 10 + 1) + (10^n + 10^{n-1} + \dots + 10 + 1) \\ &= (10^n + 1) \sum_{k=0}^n 10^k. \end{aligned}$$

Therefore, there are $\boxed{\text{(A) } 0}$ prime numbers in this sequence.

Solution 3 (Simple Sums)

Observe how

$$\begin{aligned} 121 &= 110 + 11, \\ 11211 &= 11100 + 111, \\ 1112111 &= 1111000 + 1111, \\ &\vdots \end{aligned}$$

all take the form of

$$\underbrace{111\dots}_{n+1}\underbrace{00\dots}_n + \underbrace{111\dots}_{n+1} = \underbrace{111\dots}_{n+1}(10^n + 1).$$

Factoring each of the sums, we have

$$11(10 + 1), 111(100 + 1), 1111(1000 + 1), \dots$$

respectively. With each number factored, there are **(A) 0** primes in the set.

Solution 4 (Educated Guess)

Note that 121 is divisible by 11 and 11211 is divisible by 3. Because this is Problem 6 of the AMC 10, we assume we do not need to check two-digit prime divisibility or use obscure theorems. Therefore, the answer is **(A) 0**.

4. (AMC 12 12B 2022) For how many values of the constant k will the polynomial $x^2 + kx + 36$ have two distinct integer roots?
(A) 6 **(B) 8** **(C) 9** **(D) 14** **(E) 16**

Answer: B

Solution 1

Let p and q be the roots of $x^2 + kx + 36$. By **Vieta's Formulas**, we have $p + q = -k$ and $pq = 36$.

It follows that p and q must be distinct factors of 36. The possibilities of $\{p, q\}$ are

$$\pm\{1, 36\}, \pm\{2, 18\}, \pm\{3, 12\}, \pm\{4, 9\}.$$

Each unordered pair gives a unique value of k . Therefore, there are **(B) 8** values of k , corresponding to $\mp 37, \mp 20, \mp 15, \mp 13$, respectively.

Solution 2

Note that k must be an integer. Using the **quadratic formula**, $x = \frac{-k \pm \sqrt{k^2 - 144}}{2}$. Since 4 divides 144 evenly, k and $k^2 - 144$ have the same parity, so x is an integer if and only if $k^2 - 144$ is a perfect square.

Let $k^2 - 144 = n^2$. Then, $(k + n)(k - n) = 144$. Since k is an integer and 144 is even, $k + n$ and $k - n$ must both be even. Assuming that k is positive, we get 5 possible values of $k + n$, namely 2, 4, 8, 6, 12, which will give distinct positive values of k , but $k + n = 12$ gives $k + n = k - n$ and $n = 0$, giving 2 identical integer roots. Therefore, there are 4 distinct positive values of k . Multiplying that by 2 to take the negative values into account, we get $4 \cdot 2 =$ **(B) 8** values of k .

Solution 3 (Pythagorean Triples)

Proceed similar to Solution 2 and deduce that the discriminant of $x^2 + kx + 36$ must be a perfect square greater than 0 to satisfy all given conditions. Seeing something like $k^2 - 144$ might remind us of a right triangle, where k is the hypotenuse, and 12 is a leg. There are four ways we could have this: a 9-12-15 triangle, a 12-16-20 triangle, a 5-12-13 triangle, and a 12-35-37 triangle.

Multiply by 2 to account for negative k values (since k is being squared), and our answer is **(B) 8**.

Solution 4

Since $36 = 2^2 \cdot 3^2$, that means there are $(2 + 1)(2 + 1) = 9$ possible factors of 36. Since $6 \cdot 6$ violates the distinct root condition, subtract 1 from 9 to get 8. Each sum is counted twice, and we count of those twice for negatives. This cancels out, so we get **(B) 8**.

Solution 5 (Discriminant and Factor Rainbow)

Since to have two solutions, the discriminant must be greater than 0, $k^2 - 144 > 0$ so k must be greater than 12 or less than -12 (since it's squared). Next, we know that when factoring, the two numbers must multiply to 36. So you can create a factor rainbow (remember from elementary school) which has the pairs of integers that multiply to 36 (1, 2, 3, 4, 6, 6, 9, 12, 18, 36) (you just look at the position and the one on the opposite side, for example 2 is the second value, and 18 is the second last value, and they multiply to 36). See that there are 4 pairs of values which add up to greater than 12 ($6 + 6$ is equal to 12, not greater), and we look at the negative values of them too (since k is squared) we get **(B) 8** total values.

5. (AMC 12 12B 2022) The point $(-1, -2)$ is rotated 270° counterclockwise about the point $(3, 1)$. What are the coordinates of its new position?
- (A) $(-3, -4)$ (B) $(0, 5)$ (C) $(2, -1)$ (D) $(4, 3)$ (E) $(6, -3)$

Answer: B

Solution 1 (Cartesian Plane)

$(-1, -2)$ is 4 units west and 3 units south of $(3, 1)$. Performing a counterclockwise rotation of 270° , which is equivalent to a clockwise rotation of 90° , the answer is 3 units west and 4 units north of $(3, 1)$, or **(B) $(0, 5)$** .

Solution 2 (Complex Numbers)

We write $(-1, -2)$ as $-1 - 2i$. We'd like to rotate about $(3, 1)$, which is $3 + i$ in the complex plane, by an angle of $270^\circ = \frac{3\pi}{2}$ rad counterclockwise.

The formula for rotating the complex number z about the complex number w by an angle of θ counterclockwise is given as $(z - w)e^{\theta i} + w$. Plugging in our values $z = -1 - 2i, w = 3 + i, \theta = \frac{3\pi}{2}$, we evaluate the expression as $((-1 - 2i) - (3 + i))e^{\frac{3\pi i}{2}} + (3 + i) = (-4 - 3i)(-i) + (3 + i) = 4i - 3i^2 + 3 + i = 4i - 3 + 3 + i = 5i$, which corresponds to **(B) $(0, 5)$** on the Cartesian plane.

Solution 3 (Cheese)

A 270° rotation counterclockwise is equivalent to a 90° rotation clockwise, which means the resultant point should be around quadrant 2. Looking at the answer choices the only option that works in **(B) $(0, 5)$** .

6. (AMC 12 12B 2022) Consider the following 100 sets of 10 elements each:

$$\begin{aligned} &\{1, 2, 3, \dots, 10\}, \\ &\{11, 12, 13, \dots, 20\}, \\ &\{21, 22, 23, \dots, 30\}, \\ &\vdots \\ &\{991, 992, 993, \dots, 1000\}. \end{aligned}$$

How many of these sets contain exactly two multiples of 7?

- (A) 40 (B) 42 (C) 43 (D) 49 (E) 50

Answer: B

Solution 1 (Competition)

There are $\left\lfloor \frac{1000}{7} \right\rfloor = 142$ numbers divisible by 7. We split these into 100 sets containing 10 numbers each, giving us 1.42 multiples of 7 per set. After the first set, the numbers come evenly, and we multiply 100 by $1.42 - 1 = \boxed{\text{(B) } 42}$.

(Genuinely genius solution MW)

Solution 2 (Casework)

We apply casework to this problem. The only sets that contain two multiples of seven are those for which:

- The multiples of 7 are 1 (mod 10) and 8 (mod 10). That is, the first and eighth elements of such sets are multiples of 7. The first element is $1 + 10k$ for some integer $0 \leq k \leq 99$. It is a multiple of 7 when $k = 2, 9, 16, \dots, 93$.
- The multiples of 7 are 2 (mod 10) and 9 (mod 10). That is, the second and ninth elements of such sets are multiples of 7. The second element is $2 + 10k$ for some integer $0 \leq k \leq 99$. It is a multiple of 7 when $k = 4, 11, 18, \dots, 95$.
- The multiples of 7 are 3 (mod 10) and 0 (mod 10). That is, the third and tenth elements of such sets are multiples of 7. The third element is $3 + 10k$ for some integer $0 \leq k \leq 99$. It is a multiple of 7 when $k = 6, 13, 20, \dots, 97$.

Each case has $\left\lfloor \frac{100}{7} \right\rfloor = 14$ sets. Therefore, the answer is $14 \cdot 3 = \boxed{\text{(B) } 42}$.

Solution 3 (Find A Pattern)

We find a pattern.

$$\begin{aligned} &\{1, 2, 3, \dots, 10\}, \\ &\{11, 12, 13, \dots, 20\}, \\ &\{21, 22, 23, \dots, 30\}, \\ &\vdots \\ &\{991, 992, 993, \dots, 1000\}. \end{aligned}$$

We can figure out that the first set has 1 multiple of 7. The second set also has 1 multiple of 7. The third set has 2 multiples of 7. The fourth set has 1 multiple of 7. The fifth set has 2 multiples of 7. The sixth set has 1 multiple of 7. The

seventh set has 1 multiple of 7. The eighth set has 2 multiples of 7. Disregarding the first set and then calculating this pattern further, we can see (reasonably) that it repeats for each 1 sets. We see that the pattern for the number of multiples per 7 sets (again, disregarding the first set) goes: 1, 2, 1, 2, 1, 1, 2. So, for every 7 sets after the first, there are three sets with 2 multiples of 7. We calculate $\left\lfloor \frac{100-1}{7} \right\rfloor$ and multiply that by 3. (We also disregard the remainder of 1 since it doesn't add any extra sets with 2 multiples of 7.). We get $14 \cdot 3 = \boxed{\text{(B) } 42}$.

Solution 4 (Fastest)

Each set contains exactly 1 or 2 multiples of 7 (Because it will need at least $2 \cdot 7 = 14$ numbers in a group for 3 or more multiples of 7).

There are $\frac{1000}{10} = 100$ total sets and $\left\lfloor \frac{1000}{7} \right\rfloor = 142$ multiples of 7. Thus, there are $142 - 100 = \boxed{\text{(B) } 42}$ sets with 2 multiples of 7.

Solution 5 (Simple Counting, Similar to Solution 1)

Consecutive multiples of 7 must differ by 7. So, if a set $\{\dots 1, \dots 2, \dots 3, (\dots), \dots 8, \dots 9, \dots 0\}$ contains two multiples of 7, they must end with the digits 1 and 8, 2 and 9, or 3 and 0. This reduces the problem to counting the amount of multiples of 7 less than 1000 that end with 1, 2, and 3.

The first multiple of 7 that ends with 1 is 21. The next multiple that ends with 1 occurs 70 later, since that is the smallest multiple of 7 we can add to 21 without affecting the last digit. The greatest number of 70's we can add to 21 while keeping it less than 1000 is 13, because $21 + 70(13) = 931$. Therefore, the set of multiples of 7 less than 1000 ending with 1 is $\{21 + 70(0), 21 + 70(1), \dots, 21 + 70(13)\}$, meaning there are 14 of these particular multiples. We can use the same reasoning to count the multiples of 7 that end with 2 and 3.

The first multiple of 7 that ends with 2 is 42. The greatest number of 70's we can add to 42 here is also 13, since $42 + 70(13) = 952$. The set of multiples of 7 less than 1000 ending with 2 is $\{42 + 70(0), 42 + 70(1), \dots, 42 + 70(13)\}$, giving 14 multiples.

The first multiple of 7 that ends with 3 is 63. The greatest number of 70's we can add to 63 here is yet again 13, since $63 + 70(13) = 973$. The set of multiples of 7 less than 1000 ending with 3 is $\{63 + 70(0), 63 + 70(1), \dots, 63 + 70(13)\}$, giving another 14 multiples.

In total, there are $14 + 14 + 14 = 42$ of these multiples, and so $\boxed{\text{(B) } 42}$ sets with two multiples of 7.

Solution 6 (Very Fast System of Equations)

Let a be the number of sets with 1 multiple of 7 and b be the number of sets with 2 multiples of 7. Note that it is impossible for a set to have more than two multiples of 7.

Since there are a total of 100 sets, $a + b = 100$. Also, since there are $\left\lfloor \frac{1000}{7} \right\rfloor = 142$ multiples of 7 between 1 and 1000, we must have $a + 2b = 142$.

Solving the system of equations

$$a + b = 100$$

$$a + 2b = 142$$

for b gives us the answer of **(B) 42**.

Solution 7 (reasonable count)

(Similar to Solution 3, but a little more intuitive and less numbers.) Note that this system loops every cycle of length 70, or 7 such sets. From 1 to 70, the multiples of 7 are 7, 14, 21, 28, 35, 42, 49, 56, 63, and 70; note that (21, 28), (42, 49), and (63, 70) are in the same sets. Thus, for every 7 sets, we have 3 sets with exactly two multiples of 7. We have 100 sets, which is $98 + 2 = 7 \cdot 14 + 2$; the first 98 sets contain $\frac{3}{7}(7 \cdot 14) = 42$ desired sets. The last two sets comprise the integers from 981 to 999; the multiples of 7 here are 987 and 994. Neither of the two last sets contains two multiples of 7, so our answer is simply **(D) 42**.

Solution 8 (Pigeonhole Principle, very fast)

Consider any consecutive seven sets beginning on one more than a multiple of 70. There are guaranteed to be 10 multiples of 7 in these seven sets. Note that no set may have three multiples of 7. This means that some three of the ten sets have 2 multiples of seven. There are 14 such groups of seven sets, and then the sets $\{981, 982, \dots, 989\}$ and $\{991, 992, \dots\}$. These last two sets are congruent to $\{1, 2, \dots, 9\}$ and $\{11, 12, \dots, 19\}$ modulo 7, thus making it easy to check that each only has one multiple of 7. Therefore, there are $14 \cdot 3 = \mathbf{(D) 42}$ sets with 2 multiples of 7.

7. (AMC 12 12B 2022) Camila writes down five positive integers. The unique mode of these integers is 2 greater than their median, and the median is 2 greater than their arithmetic mean. What is the least possible value for the mode?
- (A) 5 (B) 7 (C) 9 (D) 11 (E) 13

Answer: D

Solution 1 (Variables)

Let M be the median. It follows that the two largest integers are both $M + 2$.

Let a and b be the two smallest integers such that $a < b$. The sorted list is

$$a, b, M, M + 2, M + 2.$$

Since the median is 2 greater than their arithmetic mean, we have $\frac{a+b+M+(M+2)+(M+2)}{5} + 2 = M$, or

$$a + b + 14 = 2M.$$

Note that $a + b$ must be even. We minimize this sum so that the arithmetic mean, the median, and the unique mode are minimized. Let $a = 1$ and $b = 3$, from which $M = 9$ and $M + 2 = \mathbf{(D) 11}$.

Solution 2 (Elimination)

We can also easily test all the answer choices. (This strategy is generally good to use for multiple-choice questions if you don't have a concrete method to proceed with!)

For answer choice **(A)**, the mode is 5, the median is 3, and the arithmetic mean is 1. However, we can quickly see this doesn't work, as there are five integers, and they can't have an arithmetic mean of 1 while having a mode of 5.

Trying answer choice **(B)**, the mode is 7, the median is 5, and the arithmetic mean is 3. From the arithmetic mean, we know that all the numbers have to sum to 15. We know three of the numbers: $_, _, 5, 7, 7$. This exceeds the sum of 15.

Now we try answer choice **(C)**. The mode is 9, the median is 7, and the arithmetic mean is 5. From the arithmetic mean, we know that the list sums to 25. Three of the numbers are $_, _, 7, 9, 9$, which is exactly 25. However, our list needs positive integers, so this won't work.

Since we were really close on answer choice **(C)**, we can intuitively feel that the answer is probably going to be **(D)**. We can confirm this by creating a list that satisfies the problem and choose **(D)** : 1, 3, 9, 11, 11.

So, our answer is **(D)** 11.

8. (AMC 12 12B 2022) What is the graph of $y^4 + 1 = x^4 + 2y^2$ in the coordinate plane?

(A) two intersecting parabolas **(B)** two nonintersecting parabolas **(C)** two intersecting circles
(D) a circle and a hyperbola **(E)** a circle and two parabolas

Answer: D

Solution 1

Since the equation has even powers of x and y , let $y' = y^2$ and $x' = x^2$. Then $y'^2 + 1 = x'^2 + 2y'$. Rearranging gives $y'^2 - 2y' + 1 = x'^2$, or $(y' - 1)^2 = x'^2$. There are two cases: $y' \leq 1$ or $y' > 1$.

If $y' \leq 1$, taking the square root of both sides gives $1 - y' = x'$, and rearranging gives $x' + y' = 1$. Substituting back in $x' = x^2$ and $y' = y^2$ gives us $x^2 + y^2 = 1$, the equation for a circle.

Similarly, if $y' > 1$, we take the square root of both sides to get $y' - 1 = x'$, or $y' - x' = 1$, which is equivalent to $y^2 - x^2 = 1$, a hyperbola.

Hence, our answer is **(D)** a circle and a hyperbola.

(Solutions 1 and 2 are in essence the same; Solution 1 lets $(x', y') = (x^2, y^2)$ for convenience, but the two solutions are otherwise identical.)

Solution 2 (Factoring)

We can subtract $2y^2$ from both sides and factor the left side to get $(y^2 - 1)^2 = x^4$. If we take the square root of both sides, we are left with two equations: $y^2 - 1 = x^2$ and $y^2 - 1 = -x^2$. In the former case, we get $y^2 - x^2 = 1$, which is the formula for a hyperbola. This means that a hyperbola will appear in the graph of the equation. In the latter case, we get $x^2 + y^2 = 1$, which is the equation of a circle which will also appear in the graph. Looking at our options, this is the only valid answer. Hence, the answer is **(D)** a circle and a hyperbola.

9. (AMC 12 12B 2022) The sequence $a_0, a_1, a_2, \dots$ is a strictly increasing arithmetic sequence of positive integers such that

$$2^{a_7} = 2^{27} \cdot a_7.$$

What is the minimum possible value of a_2 ?

(A) 8 **(B)** 12 **(C)** 16 **(D)** 17 **(E)** 22

Answer: B

Solution 1

We can rewrite the given equation as $2^{a_7-27} = a_7$. Hence, a_7 must be a power of 2 and larger than 27. The first power of 2 that is larger than 27, namely 32, does satisfy the equation: $2^{32-27} = 2^5 = 32$. In fact, this is the only solution; 2^{a_7-27} is exponential whereas a_7 is linear, so their graphs will not intersect again.

Now, let the common difference in the sequence be d . Hence, $a_0 = 32 - 7d$ and $a_2 = 32 - 5d$. To minimize a_2 , we maximize d . Since the sequence contains only positive integers, $32 - 7d > 0$ and hence $d \leq 4$. When $d = 4$, $a_2 = \boxed{\text{(B)} 12}$.

10. (AMC 12 12B 2022) Regular hexagon $ABCDEF$ has side length 2. Let G be the midpoint of $\overline{AB}$, and let H be the midpoint of $\overline{DE}$. What is the perimeter of $GCHF$?

(A) $4\sqrt{3}$ (B) 8 (C) $4\sqrt{5}$ (D) $4\sqrt{7}$ (E) 12

Answer: D

Solution 1

Let the center of the hexagon be O . $\triangle AOB$, $\triangle BOC$, $\triangle COD$, $\triangle DOE$, $\triangle EOF$, and $\triangle FOA$ are all equilateral triangles with side length 2. Thus, $CO = 2$, and $GO = \sqrt{AO^2 - AG^2} = \sqrt{3}$. By symmetry, $\angle COG = 90^\circ$. Thus, by the Pythagorean theorem, $CG = \sqrt{2^2 + \sqrt{3}^2} = \sqrt{7}$. Because $CO = OF$ and $GO = OH$, $CG = HC = FH = GF = \sqrt{7}$. Thus, the solution to our problem is $\sqrt{7} + \sqrt{7} + \sqrt{7} + \sqrt{7} = \boxed{\text{(D)} 4\sqrt{7}}$.

Solution 2

Consider triangle AFG . Note that $AF = 2$, $AG = 1$, and $\angle GAF = 120^\circ$ because it is an interior angle of a regular hexagon. (See note for details.)

By the Law of Cosines, we have:

$$FG^2 = AG^2 + AF^2 - 2 \cdot AG \cdot AF \cdot \cos \angle GAF$$

$$FG^2 = 1^2 + 2^2 - 2 \cdot 1 \cdot 2 \cdot \cos 120^\circ$$

$$FG^2 = 5 + 4 \cdot \left(\frac{1}{2}\right)$$

$$FG^2 = 7$$

$$FG = \sqrt{7}.$$

By SAS Congruence, triangles AFG , BCG , CDH , and EFH are congruent, and by CPCTC, quadrilateral $GCHF$ is a rhombus. Therefore, the perimeter of $GCHF$ is $4 \cdot FG = \boxed{\text{(D)} 4\sqrt{7}}$.

Note: The sum of the interior angles of any polygon with n sides is given by $180^\circ(n - 2)$. Therefore, the sum of the interior angles of a hexagon is 720° , and each interior angle of a regular hexagon measures $\frac{720^\circ}{6} = 120^\circ$.

Solution 3

We use a coordinates approach. Letting the origin be the center of the hexagon, we can let $A = (-1, \sqrt{3})$, $B = (1, \sqrt{3})$, $C = (2, 0)$, $D = (1, -\sqrt{3})$, $E = (-1, -\sqrt{3})$, $F = (-2, 0)$. Then, $G = (0, \sqrt{3})$ and $H = (0, -\sqrt{3})$.

We use the distance formula four times to get CH , HF , FG , and GC .

$$CH^2 = (2 - 0)^2 + (0 - (-\sqrt{3}))^2 = 7 \rightarrow CH = \sqrt{7},$$

$$\begin{aligned} HF^2 &= (0 - (-2))^2 + (-\sqrt{3} - 0)^2 = 7 \rightarrow HF = \sqrt{7}, \\ FG^2 &= (-2 - 0)^2 + (0 - \sqrt{3})^2 = 7 \rightarrow FG = \sqrt{7}, \\ GC^2 &= (0 - 2)^2 + (\sqrt{3} - 0)^2 = 7 \rightarrow GC = \sqrt{7}. \end{aligned}$$

Thus, the perimeter of $GCHF = CH + HF + FG + GC = \sqrt{7} + \sqrt{7} + \sqrt{7} + \sqrt{7} =$
(D) $4\sqrt{7}$.

Note: the last part of this solution could have been simplified by noting that $CH = HF = FG = GC = \sqrt{7}$.

Solution 4

Note that triangles $\triangle GAF$, $\triangle FEH$, $\triangle HDC$, and $\triangle CBG$ are all congruent, since they have side lengths of 1 and 2 and an included angle of 120° .

By the Law of Cosines,

$$FG = \sqrt{1^2 + 2^2 - 2 \cdot 1 \cdot 2 \cdot \frac{1}{2}} = \sqrt{7}.$$

Therefore, $FG + GC + CH + HF = 4 \cdot FG =$ **(D)** $4\sqrt{7}$.

-Benedict T (countmath1)

Solution 5 (Answer Choices + Pythagorean Theorem Extension)

Like the previous solutions, note that $\triangle GAF$, $\triangle FEH$, $\triangle HDC$, and $\triangle CBG$ are all congruent by SAS. It follows that quadrilateral $GCHF$ is a rhombus.

Recall the Pythagorean Theorem, which states $a^2 + b^2 = c^2$ for all right triangles, where c is the hypotenuse of the triangle. However, by drawing a quick diagram of an obtuse triangle, we can see that $a^2 + b^2 < c^2$, in any given obtuse triangle.

Since $ABCDEF$ is a regular hexagon, all of its angles are obtuse. It follows that $\triangle GAF$ is an obtuse triangle. Using the extended Pythagorean Theorem for obtuse triangles, we have:

$$AG^2 + AF^2 < GF^2 \implies 1^2 + 2^2 < GF^2 \implies \sqrt{5} < GF$$

Since $GCHF$ is a rhombus, the perimeter is $4GF$. This eliminates all answer choices but D and E , since in all of those options $GF < \sqrt{5}$. Lastly, E is eliminated due to the triangle inequality, as $1 + 2$ is not greater than $12/4 = 3$.

Hence, the answer is **(D)** $4\sqrt{7}$.

11. (AMC 12 12B 2022) Let $f(n) = \left(\frac{-1+i\sqrt{3}}{2}\right)^n + \left(\frac{-1-i\sqrt{3}}{2}\right)^n$, where $i = \sqrt{-1}$. What is $f(2022)$?

(A) -2 (B) -1 (C) 0 (D) $\sqrt{3}$ (E) 2

Answer: E

Solution 1

Converting both summands to exponential form,

$$\begin{aligned} -1 + i\sqrt{3} &= 2e^{\frac{2\pi i}{3}}, \\ -1 - i\sqrt{3} &= 2e^{-\frac{2\pi i}{3}} = 2e^{\frac{4\pi i}{3}}. \end{aligned}$$

Notice that the two terms in the problem are two of the third roots of unity (that is, both of them equal 1 when raised to the power of 3). When we replace the summands with their exponential form, we get

$$f(n) = \left(e^{\frac{2\pi i}{3}}\right)^n + \left(e^{\frac{4\pi i}{3}}\right)^n.$$

When we substitute $n = 2022$, we get

$$f(2022) = \left(e^{\frac{2\pi i}{3}}\right)^{2022} + \left(e^{\frac{4\pi i}{3}}\right)^{2022}.$$

We can rewrite 2022 as $3 \cdot 674$, how does that help?

$$f(2022) = \left(e^{\frac{2\pi i}{3}}\right)^{3 \cdot 674} + \left(e^{\frac{4\pi i}{3}}\right)^{3 \cdot 674} = \left(\left(e^{\frac{2\pi i}{3}}\right)^3\right)^{674} + \left(\left(e^{\frac{4\pi i}{3}}\right)^3\right)^{674} = 1^{674} + 1^{674} = \boxed{\text{(E)} 2}.$$

Since any third root of unity must cube to 1.

Solution 2 (Eisenstein Units)

The numbers $\frac{-1+i\sqrt{3}}{2}$ and $\frac{-1-i\sqrt{3}}{2}$ are both **Eisenstein Units** (along with 1), denoted as ω and ω^2 , respectively. They have the property that when they are cubed, they equal to 1. Thus, we can immediately solve:

$$\omega^{2022} + \omega^{2 \cdot 2022} = \omega^{3 \cdot 674} + \omega^{3 \cdot 2 \cdot 674} = 1^{674} + 1^{2 \cdot 674} = \boxed{\text{(E)} 2}.$$

mathboy100

Solution 3 (Quick and Easy)

We begin by recognizing this form looks similar to the definition of cosine:

$$\cos(x) = \frac{e^{ix} + e^{-ix}}{2}.$$

We can convert our two terms into exponential form to find

$$f(n) = \left(e^{\frac{2\pi i}{3}}\right)^n + \left(e^{-\frac{2\pi i}{3}}\right)^n = e^{\frac{2\pi in}{3}} + e^{-\frac{2\pi in}{3}}.$$

This simplifies nicely:

$$f(n) = 2 \cos\left(\frac{2\pi n}{3}\right).$$

Thus,

$$f(2022) = 2 \cos\left(\frac{2\pi(2022)}{3}\right) = 2 \cos(1348\pi) = \boxed{\text{(E)} 2}.$$

Solution 4 (Third-order Homogeneous Linear Recurrence Relation)

Notice how this looks like the closed form of the Fibonacci sequence except different roots. This is motivation to turn this closed formula into a recurrence relation. The base of the exponents are the roots of the characteristic equation $r^3 - 1 = 0$. So we have

$$\begin{aligned} a_0 &= 2 \\ a_1 &= -1 \\ a_2 &= -1 \\ a_n &= a_{n-3} \end{aligned}$$

Every time n is multiple of 3 as is true when $n = 2022$, $a_n = \boxed{\text{(E)} 2}$ lopkiloinm

Solution 5 (Polynomial + Recursion)

Let $a = \frac{-1+i\sqrt{3}}{2}$ and $b = \frac{-1-i\sqrt{3}}{2}$. We know that $a + b = -1$ and $a \cdot b = 1$. Therefore, a and b are the roots of $x^2 + x + 1 = 0$. By the factor theorem, $a^2 + a + 1 = 0$ and $b^2 + b + 1 = 0$. Multiply the first equation by a^{n-2} and the second equation by b^{n-2} . This gives us $a^n + a^{n-1} + a^{n-2} = 0$ and $b^n + b^{n-1} + b^{n-2} = 0$. Adding both equations together we get $a^n + b^n + a^{n-1} + b^{n-1} + a^{n-2} + b^{n-2} = 0$. This is the same as $f(n) + f(n-1) + f(n-2) = 0$. Therefore, $f(n) = -f(n-1) - f(n-2)$. Plugging in $n = 1, 2, 3, 4, 5$, and 6 , we get $f(n) = -1, -1, 2, -1, -1, 2$. Therefore we know that if n is a multiple of 3, then $f(n)$ is 2. Since 2022 is a multiple of 3, our answers is $E)2$. vpeddi18

Solution 6 (SO FAST)

Converting the two terms into rectangular form,

$$f(2022) = \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)^{2022} + \left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} \right)^{2022}.$$

By DeMoivre's Theorem,

$$f(2022) = \left(\cos \left(\frac{2\pi}{3} \cdot 2022 \right) + i \sin \left(\frac{2\pi}{3} \cdot 2022 \right) \right) + \left(\cos \left(\frac{4\pi}{3} \cdot 2022 \right) + i \sin \left(\frac{4\pi}{3} \cdot 2022 \right) \right).$$

Note that $\cos \pi \cdot k = 1$ if k is even and -1 if k is odd, and that $\sin \pi \cdot k = 0$ for all integers k .

All arguments are even in the second equation for $f(2022)$, so the two cos terms are equal to 1, and the two sin terms are equal to 0.

Therefore the answer is $1 + 1 = \boxed{\text{(E)} 2}$.

-Benedict T (countmath1)

Solution 7 (No Euler's Identity)

Observe that $\left(\frac{-1+i\sqrt{3}}{2}\right)^2 = \frac{-1-i\sqrt{3}}{2}$ and $\left(\frac{-1-i\sqrt{3}}{2}\right)^2 = \frac{-1+i\sqrt{3}}{2}$.

It follows that $\left(\frac{-1+i\sqrt{3}}{2}\right)^4 = \frac{-1+i\sqrt{3}}{2}$ and $\left(\frac{-1-i\sqrt{3}}{2}\right)^4 = \frac{-1-i\sqrt{3}}{2}$.

The binary expansion of 2022 is

$$2^{10} + 2^9 + 2^8 + 2^7 + 2^6 + 2^5 + 2^2 + 2^1$$

which can be rewritten as

$$(4^5 + 4^4 + 4^3 + 4^1) + 2(4^4 + 4^3 + 4^2 + 4^0)$$

Thus,

$$\left(\frac{-1+i\sqrt{3}}{2}\right)^{2022} = \left(\left(\frac{-1+i\sqrt{3}}{2}\right)^4\right)^4 \cdot \left(\left(\frac{-1+i\sqrt{3}}{2}\right)^4\right)^2$$

or

$$\left(\frac{-1+i\sqrt{3}}{2}\right)\left(\frac{-1-i\sqrt{3}}{2}\right)$$

Similarly,

$$\left(\frac{-1-i\sqrt{3}}{2}\right)^{2022} = \left(\frac{-1-i\sqrt{3}}{2}\right)\left(\frac{-1+i\sqrt{3}}{2}\right)$$

so

$$f(2022) = 2\left(\frac{-1 + i\sqrt{3}}{2}\right)\left(\frac{-1 - i\sqrt{3}}{2}\right) = \boxed{\text{(E)} 2}$$

12. (AMC 12 12B 2022) Kayla rolls four fair 6-sided dice. What is the probability that at least one of the numbers Kayla rolls is greater than 4 and at least two of the numbers she rolls are greater than 2?

(A) $\frac{2}{3}$ (B) $\frac{19}{27}$ (C) $\frac{59}{81}$ (D) $\frac{61}{81}$ (E) $\frac{7}{9}$

Answer: D

Solution 1 (Complementary Counting)

We will subtract from one the probability that the first condition is violated and the probability that *only* the second condition is violated, being careful not to double-count the probability that both conditions are violated.

For the first condition to be violated, all four dice must read 4 or less, which happens with probability $\left(\frac{2}{3}\right)^4 = \frac{16}{81}$.

For the first condition to be met but the second condition to be violated, at least one of the dice must read greater than 4, but less than two of the dice can read greater than 2. Therefore, one of the four die must read 5 or 6, while the remaining three dice must read 2 or less, which happens with probability $\binom{4}{1} \left(\frac{1}{3}\right) \left(\frac{1}{3}\right)^3 = 4 \cdot \frac{1}{3} \cdot \frac{1}{27} = \frac{4}{81}$.

Therefore, the overall probability of meeting both conditions is $1 - \frac{16}{81} - \frac{4}{81} =$

$$\boxed{\text{(D)} \frac{61}{81}}.$$

Solution 2 (Direct and Complementary Counting)

There are either 0, 1, 2, 3, or 4 dice that have values of more than 4. The probability of getting 0 is $\left(\frac{2}{3}\right)^4 = \frac{16}{81}$, the probability of getting 1 is $4 \cdot \left(\frac{1}{3}\right) \left(\frac{2}{3}\right)^3 = \frac{32}{81}$, and the probability of getting 2 or greater is $1 - \frac{16}{81} - \frac{32}{81} = \frac{11}{27}$.

It is obvious that the probability of getting at least two numbers greater than 2 is 1 if we have 2 numbers greater than 4.

Let us calculate the probability of getting at least two numbers greater than 2 if one die is greater than 4 by using complementary counting. We already have one die that is greater than 2, and the probability that a die that is less than 5 is greater than 2 is $\frac{1}{2}$. Thus, our probability is $1 - \left(\frac{1}{2}\right)^3 = \frac{7}{8}$.

$$\text{Finally, our total probability is } \frac{11}{27} + \frac{7}{8} \cdot \frac{32}{81} = \boxed{\text{(D)} \frac{61}{81}}$$

Solution 3 (Complementary Counting and PIE)

We use Solution 1's idea of complementary counting. However, we will find the probability that the first condition is violated and the probability that the second condition is violated, then subtract the overlap (first and second condition both violated). We can do this because of PIE.

As in Solution 1, the probability that the **first condition** is violated is $\left(\frac{2}{3}\right)^4 = \frac{16}{81}$.

The probability that the **second condition** is violated (regardless of the first condition) can be broken into two cases.

Case 1: exactly one number rolled is greater than 2

Then, there are four choices for which die is the one greater than 2, $\frac{2}{3}$ chance for whether it reads 3, 4, 5, or 6, and $\left(\frac{1}{3}\right)^3 = \frac{1}{27}$ for the other three die that must read 1 or 2. The probability is then $4 \left(\frac{2}{3}\right) \left(\frac{1}{27}\right) = \frac{8}{81}$. **Case 2: exactly zero numbers rolled are greater**

Then, all four dice are either 1 or 2, so the probability of this is $\left(\frac{1}{3}\right)^4 = \frac{1}{81}$.

Now, we look at the **overlap**. In other words, we want to find the probability none of the rolls are greater than 4 and less than two of the rolls are greater than 2. We split into the same two cases again.

Case 1: exactly one number rolled is greater than 2 (and none of the rolls are greater t

There are still 4 choices for this die, and it must be 3 or 4, with probability $\frac{1}{3}$. The other must read 1 or 2, with probability $\frac{1}{27}$, so this case yields $4 \left(\frac{1}{3}\right) \left(\frac{1}{27}\right) = \frac{4}{81}$.

Case 2: exactly zero numbers rolled are greater than 2 (and none of the rolls are great

The first statement implies the second statement, so this case yields $\frac{1}{81}$ as from our previous Case 2.

Now we've got everything! Our answer is $1 - \left(\frac{16}{81} + \left(\frac{8}{81} + \frac{1}{81}\right) - \left(\frac{4}{81} + \frac{1}{81}\right)\right) = \boxed{\text{(D)} \frac{61}{81}}$.

Solution 4 (Direct Counting with Subtraction)

Let X be the number of rolls greater than 4 (i.e. 5 or 6), and Y be the number of rolls greater than 2 (i.e. 3, 4, 5, or 6). We want to compute $P(X \geq 1 \text{ and } Y \geq 2)$.

We first compute $P(X \geq 1)$. The complement is that no roll is greater than 4, so every roll is in 1,2,3,4. The probability of this is $(4/6)^4 = 16/81$. Thus,

$$P(X \geq 1) = 1 - 16/81 = 65/81.$$

This already guarantees $Y \geq 1$, but we still need to enforce $Y \geq 2$. Therefore, we subtract the outcomes where $X \geq 1$ but $Y = 1$.

The only way for $X \geq 1$ and $Y = 1$ to occur is if exactly one die is greater than 2, and that die is also greater than 4. In other words, one die is 5 or 6, and the other three dice are 1 or 2.

There are 4 choices for which die is ≥ 5 , and that die has probability $2/6 = 1/3$. The other three dice must be in 1,2, each with probability $1/3$. Therefore, the probability of this "bad" case is

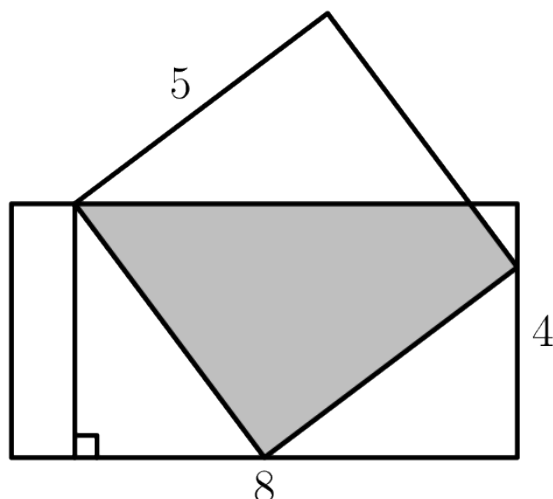
$$4 * (1/3) * (1/3)^3 = 4/81.$$

Finally, subtracting gives

$$P(X \geq 1 \text{ and } Y \geq 2) = 65/81 - 4/81 = 61/81.$$

Thus, the answer is $\boxed{\text{(D)} 61/81}$. FastOne

13. (AMC 12 12B 2022) The diagram below shows a rectangle with side lengths 4 and 8 and a square with side length 5. Three vertices of the square lie on three different sides of the rectangle, as shown. What is the area of the region inside both the square and the rectangle?

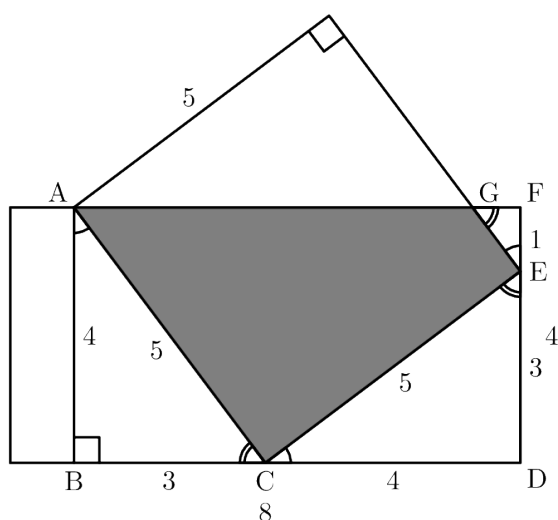


- (A) $15\frac{1}{8}$ (B) $15\frac{3}{8}$ (C) $15\frac{1}{2}$ (D) $15\frac{5}{8}$ (E) $15\frac{7}{8}$

Answer: D

Solution 1

Let us label the points on the diagram.



By doing some angle chasing using the fact that $\angle ACE$ and $\angle CEG$ are right angles, we find that $\angle BAC = \angle DCE = \angle FEG$. Similarly, $\angle ACB = \angle CED = \angle EGF$. Therefore, $\triangle ABC \sim \triangle CDE \sim \triangle EFG$.

As we are given a rectangle and a square, $AB = 4$ and $AC = 5$. Therefore, $\triangle ABC$ is a 3-4-5 right triangle and $BC = 3$.

CE is also 5. So, using the similar triangles, $CD = 4$ and $DE = 3$.

$EF = DF - DE = 4 - 3 = 1$. Using the similar triangles again, EF is $\frac{1}{4}$ of the corresponding AB . So,

$$[\triangle EFG] = \left(\frac{1}{4}\right)^2 \cdot [\triangle ABC]$$

$$\begin{aligned}
 &= \frac{1}{16} \cdot 6 \\
 &= \frac{3}{8}.
 \end{aligned}$$

Finally, we have

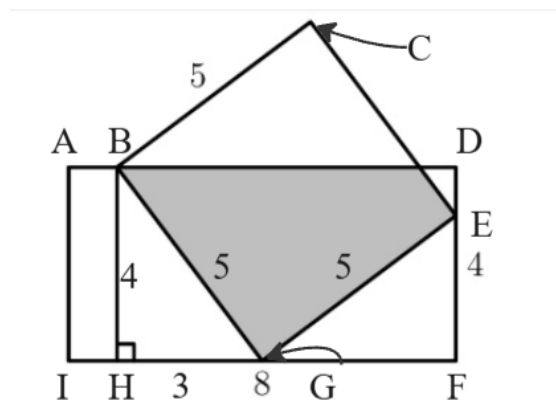
$$\begin{aligned}
 [ACEG] &= [ABDF] - [\triangle ABC] - [\triangle CDE] - [\triangle EFG] \\
 &= 7 \cdot 4 - \frac{1}{2} \cdot 3 \cdot 4 - \frac{1}{2} \cdot 3 \cdot 4 - \frac{3}{8} \\
 &= 28 - 6 - 6 - \frac{3}{8} \\
 &= \boxed{\text{(D)} 15\frac{5}{8}}.
 \end{aligned}$$

Solution 2 (Clever)

(Refer to the diagram above) Proceed the same way as Solution 1 until you get all of the side lengths. Then, it is clear that due to the answer choices, we only need to find the fractional part of the shaded area. The area of the whole rectangle is integral, as is the area of $\triangle ABC$, $\triangle CDE$, and the rectangle to the far left of the diagram. The area of EFG is $\frac{3}{8}$ and thus the fractional part of the answer is $\frac{5}{8}$.

The only answer choice that has $\frac{5}{8}$ in it is **(D)** $15\frac{5}{8}$

Solution 3 (Intuitive Risking)



We can quickly find out through Pythagorean Triples that $GH = 3$. Now, we make an educated guess that G is the midpoint of FI , and therefore $HI = 1$.

Noice! So, now what? Well, FG is then 4, and EF would then be 3.

Given that info, we also see that $DE = 1$.

Our area of the rectangle $ADFI$ is $8 \times 4 = 32$. We then see that we must subtract rectangle $ABHI = 4 \times 1 = 4$, triangle $BHG = \frac{1}{2} \times 3 \times 4 = 6$, and triangle $GFE = \frac{1}{2} \times 4 \times 3 = 6$. This gives $32 - (4 + 6 + 6) = 16$.

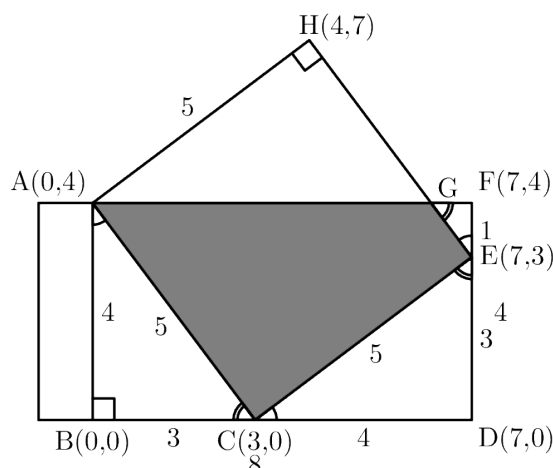
We make a guess that the intersection between B and D , call it J , to E is exactly $5/4$. Then, through the Pythagorean theorem, $JD = 1/2$, and the area is $\frac{1}{2} \times \frac{1}{2} \times 1 = \frac{1}{4}$. So, the area of that mini triangle is $\frac{1}{4}$, in reality we know it is bigger. We subtract

(D) $15\frac{5}{8}$

Take risks everyone! You might do great things like solve an AMC problem :D!

Solution 4 (Coordinate Geometry)

Same diagram as Solution 1, but added point H , which is $(4, 7)$. I also renamed all the points to form coordinates using B as the origin.



In order to find the area, point G 's coordinates must be found. Notice how EH and AG intercept at point G . This means that we need to find the equations for EH and AG and make a system of linear equations.

Using the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$, we get the slope for EH , which means $m = \frac{3-7}{7-4} = -\frac{4}{3}$

Then, by using point slope form. $y - y_1 = m(x - x_1)$. We can say that the equation for EH is $y - 7 = -\frac{4}{3}(x - 4)$ or in this case, $y = -\frac{4}{3}x + 12\frac{1}{3}$.

And it is easy to figure out that the equation for AG is $y = 4$.

The best way to solve the system of linear equations is to substitute the y for the 4 in equation EH . $4 = -\frac{4}{3}x + 12\frac{1}{3}$, so $x = 6\frac{1}{4}$ and $y = 4$ This would mean $G\left(6\frac{1}{4}, 4\right)$.

Since we have our G coordinate, we can continue with Solution 3, with the area of the trapezoid $\left(\frac{EG+AC}{2}\right)(CE)$, where $EG = \frac{5}{4}$ (using distance formula for E to G), $AC = 5$, and $CE = 5$.

By substitution, we get $\left(\frac{5+5}{2}\right)(5) = \boxed{(\mathbf{D})\ 15\frac{5}{8}}$.

Solution 5 (FAST)

Notice the small triangle in the upper right corner is a $3-4-5$ triangle. Then that triangle is similar to the big triangle by AA similarity. From that, do similar triangles and you find that the longer leg of the small triangle is 1. Then you find that the triangle below is $3-4-5$, so the side length of the rectangle (without the outer rectangle) is 7. afterwards you just add the half of the square + the remaining triangle which can be found by multiplying base and height (in which we already know)

I found the same solution when solving this, but essentially, we know all triangles in the picture are similar, and we know the lower left hand triangle is a $3-4-5$ triangle, and it is similar to the top triangle, where 4 corresponds to 5, therefore the unknown side length of that triangle is $15/4$. Continuing on, the area of the white part of the square is $5 * 15/4 * 1/2 = 75/8$, $25 - 75/8 = \boxed{\text{(D)}}$

-kaiser

Solution 6 (Fastest Similar Triangles)

For reference, use the points labelled in the diagram of Solution 3. Let the point one unit to the right of A be A' (so that A' is one of the vertices of the square). The square means $A'C = 5$, so we get a $3-4-5$ triangle $A'BC$.

$$m\angle BA'C = 90^\circ - m\angle CA'G = m\angle GA'H.$$

Therefore $\triangle GA'H$ is proportional to a $3-4-5$ triangle, with HG corresponding to 3 and $A'H$ corresponding to 4. By similar triangles, we find

$$HG = A'H \cdot \frac{3}{4} = \frac{15}{4}.$$

$$\text{then, finally, } [A'CEG] = [A'CEH] - [\triangle GA'H] = 5^2 - \frac{1}{2} \cdot \frac{15}{4} \cdot 5 =$$

$$\boxed{\text{(D) } 15\frac{5}{8}}.$$

14. (AMC 12 12B 2022) The graph of $y = x^2 + 2x - 15$ intersects the x -axis at points A and C and the y -axis at point B . What is $\tan(\angle ABC)$?

- (A) $\frac{1}{7}$ (B) $\frac{1}{4}$ (C) $\frac{3}{7}$ (D) $\frac{1}{2}$ (E) $\frac{4}{7}$

Answer: E

Solution 1 (Dot Product)

First, find $A = (-5, 0)$, $B = (0, -15)$, and $C = (3, 0)$. Create vectors $\overrightarrow{BA}$ and $\overrightarrow{BC}$. These can be reduced to $\langle -1, 3 \rangle$ and $\langle 1, 5 \rangle$, respectively. Then, we can use the dot product to calculate the cosine of the angle (where $\theta = \angle ABC$) between them:

$$\begin{aligned} \langle -1, 3 \rangle \cdot \langle 1, 5 \rangle &= 15 - 1 = \sqrt{10}\sqrt{26}\cos(\theta), \\ \implies \cos(\theta) &= \frac{7}{\sqrt{65}}. \end{aligned}$$

Thus,

$$\tan(\angle ABC) = \sqrt{\frac{65}{49} - 1} = \boxed{\text{(E) } \frac{4}{7}}.$$

Solution 2

Note that $y = x^2 + 2x - 15$ intersects the x -axis at points $(-5, 0)$ and $(3, 0)$. Without loss of generality, let these points be A and C respectively. Also, the graph intersects the y -axis at point $B = (0, -15)$.

Let point $O = (0, 0)$. It follows that $\triangle AOB$ and $\triangle BOC$ are right triangles.

We have

$$\tan(\angle ABC) = \tan(\angle ABO + \angle OBC) = \frac{\tan(\angle ABO) + \tan(\angle OBC)}{1 - \tan(\angle ABO) \cdot \tan(\angle OBC)} = \frac{\frac{1}{5} + \frac{1}{3}}{1 - \frac{1}{15}} = \boxed{\text{(E) } \frac{4}{7}}.$$

Alternatively, we can use the **Pythagorean Theorem** to find that $AB = 5\sqrt{10}$ and $BC = 3\sqrt{26}$ and then use the $A = \frac{1}{2}ab \sin \angle C$ **area formula for a triangle** and the **Law of Cosines** to find $\tan(\angle ABC)$.

Solution 3

Like above, we set A to $(-5, 0)$, B to $(0, -15)$, and C to $(3, 0)$, then finding via the Pythagorean Theorem that $AB = 5\sqrt{10}$ and $BC = 3\sqrt{26}$. Using the Law of Cosines, we see that

$$\cos(\angle ABC) = \frac{AB^2 + BC^2 - AC^2}{2AB \cdot BC} = \frac{250 + 234 - 64}{30\sqrt{260}} = \frac{7}{\sqrt{65}}.$$

Then, we use the identity $\tan^2(x) = \sec^2(x) - 1$ to get

$$\tan(\angle ABC) = \sqrt{\frac{65}{49} - 1} = \boxed{\text{(E)} \frac{4}{7}}.$$

Alternatively, we could observe that $\sin(\angle ABC) = \sqrt{1 - \cos^2(\angle ABC)} = \sqrt{1 - \left(\frac{7}{\sqrt{65}}\right)^2} = \sqrt{\frac{16}{65}} = \frac{4}{\sqrt{65}}$, so $\tan(\angle ABC) = \frac{\sin(\angle ABC)}{\cos(\angle ABC)} = \boxed{\text{(E)} \frac{4}{7}}$.

Solution 4

We can reflect the figure, but still have the same angle. This problem is the same as having points $D(0, 0)$, $E(3, 15)$, and $F(-5, 15)$, where we're solving for angle FED. We can use the formula for $\tan(a - b)$ to solve now where a is the x -axis to angle F and b is the x -axis to angle E . $\tan(a) = \text{slope of line } DF = -3$ and $\tan(b) = \text{slope of line } DE = 5$. Plugging these values into the $\tan(a - b)$ formula, we get $(-3 - 5)/(1 + (-3 \cdot 5))$ which is $\boxed{\text{(E)} \frac{4}{7}}$.

Solution 5

We use the formula $[ABC] = \frac{1}{2}ab \sin C$.

Note that $\triangle ABC$ has side-lengths $AB = 5\sqrt{10}$ and $BC = 3\sqrt{26}$ from Pythagorean theorem, with the area being $\frac{1}{2} \cdot 8 \cdot 15$.

We equate the areas together to get:

$$\frac{1}{2} \cdot 8 \cdot 15 = \frac{1}{2} \cdot 5\sqrt{10} \cdot 3\sqrt{26} \cdot \sin B,$$

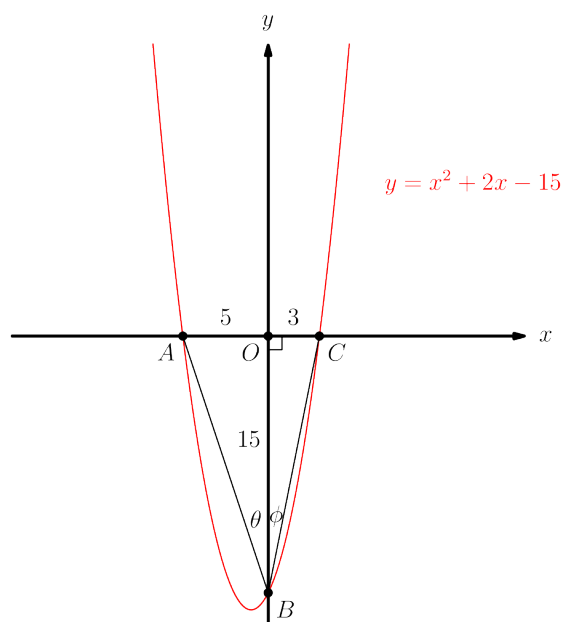
from which $\sin B = \frac{8}{\sqrt{260}}$.

From Pythagorean Identity, $\cos B = \frac{14}{\sqrt{260}}$.

Then we use $\tan B = \frac{\sin B}{\cos B}$, to obtain $\tan B = \frac{8}{14} = \boxed{\text{(E)} \frac{4}{7}}$.

- SAHANWIJETUNGA

Solution 6 (Complex Numbers)



From $x^2 + 2x - 15 = (x - 3)(x + 5)$, we may assume, without loss of generality, that x -intercepts of the given parabola are $A(-5, 0)$ and $C(3, 0)$. And, point B has coordinates $(0, -15)$. Consider complex numbers $z = 3 + i$ and $w = 5 + i$ whose arguments are $\theta := \angle OBA$ and $\phi := \angle OBC$, respectively. Notice that $\angle ABC = \theta + \phi$ is the argument of the product zw which is

$$zw = (3 + i)(5 + i) = 14 + 8i.$$

Hence

$$\tan \angle ABC = \frac{\operatorname{Im}(zw)}{\operatorname{Re}(zw)} = \frac{8}{14} = \boxed{\text{(E)} \frac{4}{7}}.$$

15. (*AMC 12 12B 2022*) One of the following numbers is not divisible by any prime number less than 10. Which is it?

(A) $2^{606} - 1$ (B) $2^{606} + 1$ (C) $2^{607} - 1$ (D) $2^{607} + 1$ (E) $2^{607} + 3^{607}$

Answer: C

Solution 1 (Modular Arithmetic)

For (A) modulo 3,

$$\begin{aligned} 2^{606} - 1 &\equiv (-1)^{606} - 1 \\ &\equiv 1 - 1 \\ &\equiv 0. \end{aligned}$$

Thus, $2^{606} - 1$ is divisible by 3.

For (B) modulo 5,

$$\begin{aligned} 2^{606} + 1 &\equiv 2^{\operatorname{Rem}(606, \phi(5))} + 1 \\ &\equiv 2^{\operatorname{Rem}(606, 4)} + 1 \\ &\equiv 2^2 + 1 \\ &\equiv 0. \end{aligned}$$

Thus, $2^{606} + 1$ is divisible by 5.

For **(D)** modulo 3,

$$\begin{aligned} 2^{607} + 1 &\equiv (-1)^{607} + 1 \\ &\equiv -1 + 1 \\ &\equiv 0. \end{aligned}$$

Thus, $2^{607} + 1$ is divisible by 3.

For **(E)** modulo 5,

$$\begin{aligned} 2^{607} + 3^{607} &\equiv 2^{607} + (-2)^{607} \\ &\equiv 2^{607} - 2^{607} \\ &\equiv 0. \end{aligned}$$

Thus, $2^{607} + 3^{607}$ is divisible by 5.

Therefore, the answer is **(C)** $2^{607} - 1$.

Solution 2 (Factoring)

We have

$$\begin{aligned} 2^{606} - 1 &= 4^{303} - 1 = (4 - 1)(4^{302} + 4^{301} + 4^{300} + \cdots + 4^0), \\ 2^{606} + 1 &= 4^{303} + 1 = (4 + 1)(4^{302} - 4^{301} + 4^{300} - \cdots + 4^0), \\ 2^{607} + 1 &= (2 + 1)(2^{606} - 2^{605} + 2^{604} - \cdots + 2^0), \\ 2^{607} + 3^{607} &= (2 + 3)(2^{606} \cdot 3^0 - 2^{605} \cdot 3^1 + 2^{604} \cdot 3^2 - \cdots + 2^0 \cdot 3^{606}). \end{aligned}$$

We conclude that **(A)** is divisible by 3, **(B)** is divisible by 5, **(D)** is divisible by 3, and **(E)** is divisible by 5.

Since all of the other choices have been eliminated, we are left with **(C)** $2^{607} - 1$.

Solution 3 (Elimination)

Mersenne Primes are primes of the form $2^n - 1$, where n is prime. Using the process of elimination, we can eliminate every option except for **(A)** and **(C)**. Clearly, 606 isn't prime, so the answer must be **(C)** $2^{607} - 1$.

Note: A organization of computer scientist and mathematicians named Great Internet Mersenne Prime Search (GIMPS) search for the worlds biggest prime numbers. More information can be found on <https://www.mersenne.org/primes/> hash-brown2009

Solution 3a (Elimination)

We examine option E first. 2^{607} has a units digit of 8 (Taking the units digit of the first few powers of two gives a pattern of 2, 4, 8, 6, 2, 4, 8, 6, 2, 4, 8, 6, $\dots$) and 3^{607} has a units digit of 7 (Taking the units digit of the first few powers of three gives a pattern of 3, 9, 7, 1, 3, 9, 7, 1, 3, 9, 7, 1, $\dots$). Adding 7 and 8 together, we get 15, which is a multiple of 5, meaning that $2^{607} + 3^{607}$ is divisible by 5.

Next, we examine option D. We take the first few powers of 2 added with 1:

$$\begin{aligned} 2^1 + 1 &= 3 \\ 2^2 + 1 &= 5 \end{aligned}$$

$$2^3 + 1 = 9$$

$$2^4 + 1 = 17$$

$$2^5 + 1 = 33$$

$$2^6 + 1 = 65$$

$$2^7 + 1 = 129$$

We see that the odd powers of 2 added with 1 are multiples of three. If we continue this pattern, $2^{607} + 1$ will be divisible by 3. (The reason why this pattern works: When you multiply $2 \equiv 2 \pmod{3}$ by 2, you obtain $4 \equiv 1 \pmod{3}$. Multiplying by 2 again, we get $1 \cdot 2 \equiv 2 \pmod{3}$. We see that in every cycle of two powers of 2, it goes from $2 \pmod{3}$ to $1 \pmod{3}$ and back to $2 \pmod{3}$.)

Next, we examine option B. We see that 2^{606} has a units of digits of 4 (Taking the units digit of the first few powers of two gives a pattern of 2, 4, 8, 6, 2, 4, 8, 6, 2, 4, 8, 6, $\dots$). Adding 1 to 4, we get 5. Since $2^{606} + 1$ has a units digit of 5, it is divisible by 5.

Lastly, we examine option A. Using the difference of cubes factorization $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$, we have $2^{606} - 1^3 = (2^{202} - 1)(2^{404} + 2^{202} + 1)$. Since $2^{404} + 2^{202} + 1 \equiv 0 \pmod{3}$ (Every term in the sequence is equivalent to $1 \pmod{3}$), $2^{606} - 1$ is divisible by 3.

Since we have eliminated every option except C, $\boxed{(C) \ 2^{607} - 1}$ is not divisible by any prime less than 10.

Solution 3b (Elimination + Number Theory)

We know that the prime numbers less than 10 are 2, 3, 5 and 7. We can start by testing if any of the answer choices are divisible by 2. We see that they are all sums of one even number and one odd number, which is simply odd. So, we cannot exclude any answer choices so far. Now, let's check divisibility by 3. We can use the fact that $2 \equiv -1 \pmod{3}$ to our advantage:

(A) $2^{606} - 1 \equiv (-1)^{606} - 1 \equiv 1 - 1 \equiv 0 \pmod{3}$ (B) $2^{606} + 1 \equiv (-1)^{606} + 1 \equiv 1 + 1 \equiv 2 \pmod{3}$ (C) $2^{607} - 1 \equiv (-1)^{607} - 1 \equiv -1 - 1 \equiv -2 \pmod{3}$ (D) $2^{607} + 1 \equiv (-1)^{607} + 1 \equiv -1 + 1 \equiv 0 \pmod{3}$ (E) $2^{607} + 3^{607} \equiv 2^{607} \equiv (-1)^{607} \equiv -1 \pmod{3}$

So, we eliminate choices A and D from divisibility by 3. Now, we move onto divisibility by 5. We can use cycling of powers to find useful remainders. Let's start with choice B.

(B) $2^{606} + 1$

We see that the remainders of powers of 2 when divided by 5 cycle in a pattern: 2, 4, 3, 1, 2, 4, 3, 1 $\dots$. Since the pattern cycles every 4 terms, we use modulo 4 to simplify 606, getting that $606 \equiv 2 \pmod{4}$. So, we get that 2^{606} is congruent modulo 5 to the second term of our pattern, so $2^{606} \equiv 4 \pmod{5}$. Thus, $2^{606} + 1 \equiv 4 + 1 \equiv 5 \equiv 0 \pmod{5}$. We now eliminate choice B and compare choices C and E.

(E) $2^{607} + 3^{607}$

Looking at choice E, we see that we have to do similar cycling for powers of 3. We get a pattern of 3, 4, 2, 1, 3, 4, 2, 1 $\dots$. Since this pattern also cycles every 4 terms, we use modulo 4 to simplify 607, getting that $607 \equiv 3 \pmod{4}$. Both the power of 3 and the power of 2 have an exponent of 607, so we use the third term (since we just found that $607 \equiv 3 \pmod{4}$) in each corresponding 4 term pattern to get that

$2^{607} + 3^{607} \equiv 3 + 2 \equiv 5 \equiv 0 \pmod{5}$. We eliminate choice E, and we are left with the correct answer: choice (C) $2^{607} - 1$

Solution 3c (Elimination, Number Theory, and Function Analysis)

Let's start by calculating the unit's digit of each answer choice.

2^{606} has a unit's digit of 4, so choices A and B have unit's digits of 3 and 5 respectively.

2^{607} has a unit's digit of 8, so choices C and D have unit's digits of 7 and 9 respectively.

3^{607} has a unit's digit of 7. Combining with the statement above, choice E has a unit's digit of 5.

So, none of the answer choices are divisible by 2. But, we can eliminate choices B and E because they both have unit's digit of 5, and therefore, are divisible by 5.

We are left now with choices A, C, and D. Let's now apply function analysis.

Look at choices A and C again. They share the form $2^n - 1$. Notice that if n is even, it follows that $2^n - 1$ is divisible by 3. Since 606 is even, choice A is divisible by 3. However, our answer cannot be divisible by 3, so eliminate A. This leaves us with choices C and D.

Look at choice D again. It has the form $2^n + 1$. Notice that if n is odd, it follows that $2^n + 1$ is divisible by 3. Since 607 is odd, choice D is divisible by 3. However, since the answer cannot be divisible by 3, we eliminate D.

At this point, you would choose choice C, but let's now test divisibility by 7. Notice that if n is a multiple of 3, it follows that $2^n - 1$ is divisible by 7. However, since 607 is not a multiple of 3, choice C is not divisible by 7.

Since it is not divisible by any prime less than 10, (C) $2^{607} - 1$ is the correct answer.

Solution 4 (modulo 15 elimination)

We proceed by taking the solutions modulo 15, allowing us to eliminate both multiples of 3 and multiples of 5.

First, we must identify cycles of powers of 2 modulo 15. $2^0 \equiv 1$, $2^1 \equiv 2$, $2^2 \equiv 4$, $2^3 \equiv 8$, $2^4 \equiv 1 \pmod{15}$. Thus, the pattern repeats every 4 powers: $2^n \equiv \begin{cases} 1 & \text{if } n \equiv 0 \pmod{4}, \\ 2 & \text{if } n \equiv 1 \pmod{4}, \\ 4 & \text{if } n \equiv 2 \pmod{4}, \\ 8 & \text{if } n \equiv 3 \pmod{4}. \end{cases}$

Since $606 \equiv 2 \pmod{4}$ and $607 \equiv 3 \pmod{4}$, $2^{606} \equiv 4 \pmod{15}$, $2^{607} \equiv 8 \pmod{15}$.

Now let's look at cycles of powers of 3 mod 15. $3^0 \equiv 1$, $3^1 \equiv 3$, $3^2 \equiv 9$, $3^3 \equiv 12$, $3^4 \equiv 6$, $3^5 \equiv 3 \pmod{15}$. The cycle from here repeats every 4 terms for

$$n \geq 1: 3^n \equiv \begin{cases} 6 & \text{if } n \equiv 0 \pmod{4}, \\ 3 & \text{if } n \equiv 1 \pmod{4}, \\ 9 & \text{if } n \equiv 2 \pmod{4}, \\ 12 & \text{if } n \equiv 3 \pmod{4}. \end{cases}$$

Since $607 \equiv 3 \pmod{4}$, $3^{607} \equiv 12 \pmod{15}$.

We then evaluate each option modulo 15.

$$(A) : 2^{606} - 1 \equiv 4 - 1 \equiv 3 \pmod{15},$$

$$(B) : 2^{606} + 1 \equiv 4 + 1 \equiv 5 \pmod{15},$$

$$(C) : 2^{607} - 1 \equiv 8 - 1 \equiv 7 \pmod{15},$$

$$(D) : 2^{607} + 1 \equiv 8 + 1 \equiv 9 \pmod{15},$$

$$(E) : 2^{607} + 3^{607} \equiv 8 + 12 \equiv 20 \equiv 5 \pmod{15}.$$

The residues divisible by 3 or 5 modulo 15 are 0, 3, 5, 6, 9, 10, 12. Among our results, only 7 (from option (C)) is not among these, eliminating all the other answer choices.

(C) $2^{607} - 1$ is the number not divisible by any prime less than 10.

Loquacious_Autodidact

Solution 5 (Contradiction; Less than 1 minute if you're good at number theory)

This is like solution 1, but don't find the remainder, we assume it.

The primes less than 10 are 2, 3, 5, and 7.

Option choice A: Let $2^{606} \equiv 1 \pmod{3}$ as it clearly isn't divisible by 2. Then, because $2^2 \equiv 1 \pmod{3}$, it follows that 2^{606} satisfies the modulo. Option A is not possible.

Option choice B ends in 5, and thus not possible.

Option choice C: It is obviously not divisible by 2. We set $2^{607} \equiv 1 \pmod{3}$, which we see is not possible through Fermat's Little Theorem. We see 2^{607} doesn't end in 5. We set $2^{607} \equiv 1 \pmod{7}$, and through Fermat's Little Theorem $2^6 \equiv 1 \pmod{7}$. It follows that 607 isn't a factor of 6, and thus does not hold the desired claim, a contradiction.

At this point, for speed purposes, one would circle C and move on. However, given the directory that this is a solution, I shall show the last two cases are not possible.

Option choice D: It is obviously not divisible by 2. We set $2^{607} \equiv 2 \pmod{3}$. Through properties of modulo 3 we have that $(-1)^{607} \equiv -1 \pmod{3}$, which is true. Thus, D is not plausible.

Option choice E is divisible by 5, and thus not possible. $\square$

Additionally, for option D. One could notice that $2^{607} \equiv 2 \pmod{3}$ in the following way. We analyze the pattern. $2^1 \equiv 2 \pmod{3}$. $2^3 \equiv 2 \pmod{3}$. $2^5 \equiv 2 \pmod{3}$. $2^{2n+1} \pmod{3} \equiv 4^n \cdot 2^1 \pmod{3} \equiv 1^n \cdot 2 \pmod{3} \equiv 2 \pmod{3}$. 607 is odd, and thus $2^{607} \equiv 2 \pmod{3}$, as desired. $\square$

Solution 6 (Extremely Illegal Cheese)

All of the other solutions are extremely long, but this is how to cheese it using some methods that MIGHT be unreliable: First, turn 606 to 0 and 607 to 1. Next, Calculate each value, giving you $A = 0B = 2C = 1D = 3E = 5$. Now since 2, 3, and 5 are prime, they can be divided by a prime (themselves). 0 doesn't really have multiples, so you can literally ignore it, and since 1 is only divisible by 1, which isn't prime, you get the correct answer, (C).

16. (AMC 12 12B 2022) Suppose x and y are positive real numbers such that

$$x^y = 2^{64} \text{ and } (\log_2 x)^{\log_2 y} = 2^7.$$

What is the greatest possible value of $\log_2 y$?

- (A) 3 (B) 4 (C) $3 + \sqrt{2}$ (D) $4 + \sqrt{3}$ (E) 7

Answer: C

Solution 1

Take the base-two logarithm of both equations to get

$$y \log_2 x = 64 \quad \text{and} \quad (\log_2 y)(\log_2 \log_2 x) = 7.$$

Now taking the base-two logarithm of the first equation again yields

$$\log_2 y + \log_2 \log_2 x = 6.$$

It follows that the real numbers $r := \log_2 y$ and $s := \log_2 \log_2 x$ satisfy $r + s = 6$ and $rs = 7$. Solving this system yields

$$\{\log_2 y, \log_2 \log_2 x\} \in \{3 - \sqrt{2}, 3 + \sqrt{2}\}.$$

Thus the largest possible value of $\log_2 y$ is $3 + \sqrt{2} \implies \boxed{\text{(C)} \ 3 + \sqrt{2}}$.

Solution 2

$$x^y = 2^{64} \Rightarrow y \log_2 x = 64 \Rightarrow \log_2 x = \frac{64}{y}.$$

Substitution into $(\log_2 x)^{\log_2 y} = 2^7$ yields

$$\left(\frac{64}{y}\right)^{\log_2 y} = 2^7 \Rightarrow \log_2 y \log_2 \frac{64}{y} = 7 \Rightarrow \log_2 y(6 - \log_2 y) = 7 \Rightarrow \log_2^2 y - 6 \log_2 y + 7 = 0.$$

Solving for $\log_2 y$ yields $\log_2 y = 3 - \sqrt{2}$ or $3 + \sqrt{2}$, and we take the greater value

$$\boxed{\text{(C)} \ 3 + \sqrt{2}}.$$

Solution 3

Let $x = 2^a$, $y = 2^b$. We have $(2^a)^{2^b} = 2^{64} \Rightarrow 2^{a \cdot 2^b} = 2^{64} \Rightarrow a \cdot 2^b = 64$, and $a^b = 128$.

Then, from eq 1, $a = 64 \cdot 2^{-b}$, and substituting in to eq 2, $(64 \cdot 2^{-b})^b = 64^b \cdot 2^{-b^2} = 2^{6b} \cdot 2^{-b^2} = 2^{6b-b^2} = 2^7$. Thus, $6b - b^2 = 7$.

Solving for b using the quadratic formula gets $b = 3 \pm \sqrt{2}$. Since we are looking for $\log_2 y$ which equals b , we put $\boxed{\text{(C)} \ 3 + \sqrt{2}}$ as our answer.

17. (AMC 12 12B 2022) How many 4×4 arrays whose entries are 0s and 1s are there such that the row sums (the sum of the entries in each row) are 1, 2, 3, and 4, in some order, and the column sums (the sum of the entries in each column) are also 1, 2, 3, and 4, in some order? For example, the array

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

satisfies the condition.

- (A) 144 (B) 240 (C) 336 (D) 576 (E) 624

Answer: D

Solution 1 (One-to-One Correspondence)

Note that the arrays and the sum configurations have one-to-one correspondence. Furthermore, the row sum configuration and the column sum configuration are independent of each other. Therefore, the answer is $(4!)^2 = \boxed{\text{(D) } 576}$.

Remark

For any given sum configuration, we can uniquely reconstruct the array it represents. Conversely, for any array, it is clear that we can determine the unique sum configuration associated with it. Therefore, this establishes a one-to-one correspondence between the arrays and the sum configurations.

Solution 2 (Linear Transformation and Permutation)

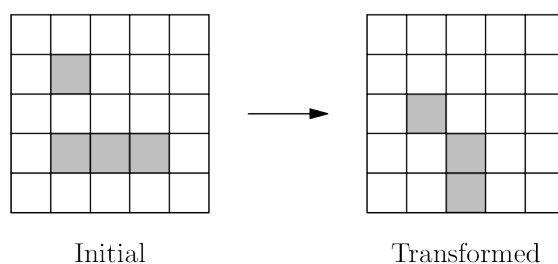
In this problem, we call a matrix that satisfies all constraints given in the problem a feasible matrix.

First, we observe that if a matrix is feasible, and we swap two rows or two columns to get a new matrix, then this new matrix is still feasible.

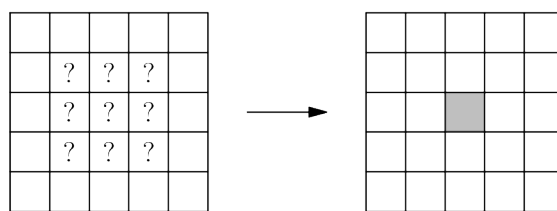
Therefore, any feasible matrix can be obtained through a sequence of such swapping operations from a feasible matrix where for all

18. (AMC 12 12B 2022) Each square in a 5×5 grid is either filled or empty, and has up to eight adjacent neighboring squares, where neighboring squares share either a side or a corner. The grid is transformed by the following rules:
- Any filled square with two or three filled neighbors remains filled.
 - Any empty square with exactly three filled neighbors becomes a filled square.
 - All other squares remain empty or become empty.

A sample transformation is shown in the figure below.



Suppose the 5×5 grid has a border of empty squares surrounding a 3×3 subgrid. How many initial configurations will lead to a transformed grid consisting of a single filled square in the center after a single transformation? (Rotations and reflections of the same configuration are considered different.)



Initial

Transformed

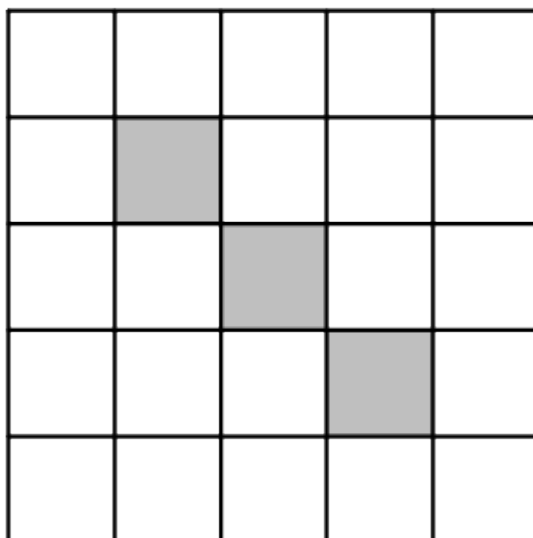
- (A) 14 (B) 18 (C) 22 (D) 26 (E) 30

Answer: C

Solution

There are two cases for the initial configuration:

- a) The center square is filled. Exactly two of the eight adjacent neighboring squares of the center are filled. Clearly, the only possibility is that the squares along one diagonal are filled, as shown below:



2 Configurations

In this case, there are 2 possible initial configurations. All rotations and reflections are considered.

- b) The center square is empty. Exactly three of the eight adjacent neighboring squares of the center are filled. The possibilities are shown below:



4 Configurations



4 Configurations



4 Configurations



8 Configurations

In this case, there are $4 + 4 + 4 + 8 = 20$ possible initial configurations. All rotations and reflections are considered.

Together, the answer is $2 + 20 = \boxed{(C) 22}$.

Solution Alternate

Key Idea:

Only the 3×3 inner grid can initially contain filled squares, since the outer border is empty. We want exactly the center square to be filled after one step.

Case 1: The center is initially filled.

A filled square remains filled precisely when it has 2 or 3 filled neighbors. We examine the possible choices of filled neighbors of the center and require that no other square becomes filled.

This gives 2 valid configurations, counting rotations/reflections.

Case 2: The center is initially empty.

For the center to become filled, it must have exactly 3 filled neighbors. There are several possible shapes for these three neighbors. Counting the configurations of each shape gives

$$2 + 4 + 4 + 4 + 8 = 22.$$

The total number of configurations is

$$\begin{aligned} 2 + (2 + 4 + 4 + 4 + 8) &= 2 + 22 \\ &= \boxed{24}. \end{aligned}$$

Therefore the answer is $\boxed{(C) 22}$.

Lunar paradox (talk) 18:06, 24 August 2026 (EDT)

Remark This problem is based off of Conway's Game of Life, a cellular automaton created by British mathematician John Conway in 1970. The game follows the exact same pattern of rules as this problem, but is played on an infinite grid of cells.

19. (AMC 12 12B 2022) In $\triangle ABC$ medians $\overline{AD}$ and $\overline{BE}$ intersect at G and $\triangle AGE$ is equilateral. Then $\cos(C)$ can be written as $\frac{m\sqrt{p}}{n}$, where m and n are relatively prime positive integers and p is a positive integer not divisible by the square of any prime. What is $m + n + p$? (A) 44 (B) 48 (C) 52 (D) 56 (E) 60

Answer: A

Solution 1 (Law of Cosines)

Let $AG = AE = EG = 2x$. Since E is the midpoint of $\overline{AC}$, we must have $EC = 2x$.

Since the centroid splits the median in a $2 : 1$ ratio, $GD = x$ and $BG = 4x$.

Applying Law of Cosines on $\triangle ADC$ and $\triangle AGB$ yields $AB = \sqrt{28}x$ and $CD = BD = \sqrt{13}x$. Finally, applying Law of Cosines on $\triangle ABC$ yields $\cos(C) = \frac{5}{2\sqrt{13}} = \frac{5\sqrt{13}}{26}$. The requested sum is $5 + 13 + 26 = 44$.

Solution 2 (Law of Cosines: One Fewer Step)

Let $AG = 1$. Since $\frac{BG}{GE} = 2$ (as G is the centroid), $BE = 3$. Also, $EC = 1$ and $\angle BEC = 120^\circ$. By the law of cosines (applied on $\triangle BEC$), $BC = \sqrt{13}$.

Applying the law of cosines again on $\triangle BEC$ gives $\cos \angle C = \frac{1+13-9}{2\sqrt{13}} = \frac{5\sqrt{13}}{26}$, so the answer is $\boxed{\text{(A) } 44}$.

Solution 3 (Law of Cosine)

Let $AG = AE = GE = CE = 1$. Since G is the centroid, $DG = \frac{1}{2}$, $BG = 2$.

$$\angle BGD = \angle AGE = 60^\circ$$

By the Law of Cosine in $\triangle BGD$

$$BD^2 = BG^2 + DG^2 - 2 \cdot BG \cdot DG \cdot \cos \angle BGD$$

$$BD = \sqrt{2^2 + \left(\frac{1}{2}\right)^2 - 2 \cdot 2 \cdot \frac{1}{2} \cdot \cos \angle BGD} = \frac{\sqrt{13}}{2}, \quad CD = \frac{\sqrt{13}}{2}$$

By the Law of Cosine in $\triangle ACD$

$$AD^2 = AC^2 + CD^2 - 2 \cdot AC \cdot CD \cdot \cos \angle C$$

$$\cos \angle C = \frac{AC^2 + CD^2 - AD^2}{2 \cdot AC \cdot CD} = \frac{2^2 + \left(\frac{\sqrt{13}}{2}\right)^2 - \left(\frac{3}{2}\right)^2}{2 \cdot 2 \cdot \frac{\sqrt{13}}{2}} = \frac{5\sqrt{13}}{26}$$

$$5 + 13 + 26 = \boxed{\text{(A) } 44}$$

Solution 4 (Barycentric Coordinates)

Using reference triangle $\triangle AGE$, we can let

$$A = (1, 0, 0), G = (0, 1, 0), E = (0, 0, 1), C = (-1, 0, 2), D = \left(-\frac{1}{2}, \frac{3}{2}, 0\right), B = (0, 3, -2).$$

If we move A, B, C each over by $(1, 0, -2)$, leaving $\angle C$ unchanged, we have

$$A = (2, 0, -2), B = (1, 3, -4), C = (0, 0, 0).$$

The angle θ between vectors $\overrightarrow{CA}$ and $\overrightarrow{CB}$ satisfies

$$\cos \theta = \frac{(2)(1) + (0)(3) + (-2)(-4)}{\sqrt{2^2 + 0^2 + (-2)^2} \sqrt{1^2 + 3^2 + (-4)^2}} = \frac{10}{\sqrt{8} \cdot 26} = \frac{10}{4\sqrt{13}} = \frac{5\sqrt{13}}{26},$$

giving the answer, $5 + 13 + 26 = \boxed{\text{(A) } 44}$.

Solution 5 (Coordinate Bash)

Let A be at $(0, 0)$, with E at $(1, 0)$ and G at $(\frac{1}{2}, \frac{\sqrt{3}}{2})$. Because $\triangle AGE$ is equilateral, the equation of line AD is $y = x\sqrt{3}$, and the equation for BE is $y = -x\sqrt{3} + \sqrt{3}$. Because BE is a median, we know that C is at $(2, 0)$.

Therefore, the equation for line BC is $y = m(x - 2)$, where m is the slope of the line. Since AD is also a median, D is the midpoint of $\overline{BC}$.

To find m , solve for the coordinates of D in terms of m :

$$x\sqrt{3} = mx - 2m$$

$$x(m - \sqrt{3}) = 2m$$

$$x = \frac{2m}{m - \sqrt{3}}$$

The y -value of D is then $y = x\sqrt{3}$:

$$y = \frac{2m\sqrt{3}}{m - \sqrt{3}}$$

Because D is the midpoint of $\overline{BC}$ and $y_C = 0$, the y -value of B is double that of D :

$$y_B = \frac{4m\sqrt{3}}{m - \sqrt{3}}$$

Now use the equation for line BE to express B :

$$mx - 2m = -x\sqrt{3} + \sqrt{3}$$

$$x(m + \sqrt{3}) = 2m + \sqrt{3}$$

$$x_B = \frac{2m + \sqrt{3}}{m + \sqrt{3}}$$

Plug this back into BE to find y_B :

$$y_B = \frac{-2m - \sqrt{3}}{m + \sqrt{3}} + \sqrt{3} = \frac{-m\sqrt{3}}{m + \sqrt{3}}$$

Equating the two expressions for y_B :

$$-\frac{m\sqrt{3}}{m + \sqrt{3}} = \frac{4m\sqrt{3}}{m - \sqrt{3}}$$

$$-\sqrt{3}m^2 + 3m = 4\sqrt{3}m^2 + 12m$$

$$5\sqrt{3}m^2 + 9m = 0$$

$$m(5m\sqrt{3} + 9) = 0$$

Since $m \neq 0$, we have

$$m = -\frac{3\sqrt{3}}{5}.$$

Now find B :

$$-x\sqrt{3} + \sqrt{3} = -\frac{3\sqrt{3}}{5}x + \frac{6\sqrt{3}}{5}$$

$$\frac{-2\sqrt{3}}{5}x = \frac{\sqrt{3}}{5}$$

$$x_B = -\frac{1}{2}$$

Plug this into BE to get

$$y_B = \frac{3\sqrt{3}}{2}.$$

Since we're asked to find $\cos \angle C$, extend $\overline{AC}$ to F where $F = (-\frac{1}{2}, 0)$, forming right triangle $\triangle BFC$. Using the Pythagorean theorem for $\overline{BC}$:

$$\left(\frac{5}{2}\right)^2 + \left(\frac{3\sqrt{3}}{2}\right)^2 = BC^2, \quad BC = \sqrt{13}.$$

Thus

$$\cos \angle C = \frac{\frac{5}{2}}{\sqrt{13}} = \frac{5\sqrt{13}}{26}.$$

The problem asks for $m + n + p$, so $5 + 13 + 26 = \boxed{44}$.

20. (AMC 12 12B 2022) Let $P(x)$ be a polynomial with rational coefficients such that when $P(x)$ is divided by the polynomial $x^2 + x + 1$, the remainder is $x + 2$, and when $P(x)$ is divided by the polynomial $x^2 + 1$, the remainder is $2x + 1$. There is a unique polynomial of least degree with these two properties. What is the sum of the squares of the coefficients of that polynomial?

(A) 10 (B) 13 (C) 19 (D) 20 (E) 23

Answer: E

Solution 1 (Experimentation)

Given that all the answer choices and coefficients are integers, we hope that $P(x)$ has positive integer coefficients.

Throughout this solution, we will express all polynomials in base x . E.g. $x^2 + x + 1 = 111_x$.

We are given:

$$111a + 12 = 101b + 21 = P(x).$$

We add 111 and 101 to each side and balance respectively:

$$111(a - 1) + 123 = 101(b - 1) + 122 = P(x).$$

We make the unit's digits equal:

$$111(a - 1) + 123 = 101(b - 2) + 223 = P(x).$$

We now notice that:

$$111(a - 11) + 1233 = 101(b - 12) + 1233 = P(x).$$

Therefore $a = 11_x = x + 1$, $b = 12_x = x + 2$, and $P(x) = 1233_x = x^3 + 2x^2 + 3x + 3$. 3 is the minimal degree of $P(x)$ since there is no way to influence the x 's digit in $101b + 21$ when b is an integer. The desired sum is $1^2 + 2^2 + 3^2 + 3^2 = \boxed{\text{(E) } 23}$

P.S. The four computational steps can be deduced through quick experimentation.

Solution 2

Let $P(x) = Q(x)(x^2 + x + 1) + x + 2$, then $P(x) = Q(x)(x^2 + 1) + xQ(x) + x + 2$, therefore $xQ(x) + x + 2 \equiv 2x + 1 \pmod{x^2 + 1}$, or $xQ(x) \equiv x - 1 \pmod{x^2 + 1}$. The rule says we can change any x^2 into -1 . This is because $x^2 \equiv -1 \pmod{x^2 + 1}$ (since $x^2 - (-1) = x^2 + 1$, which is a multiple of $x^2 + 1$). To find the least possible degree $Q(x)$, it would be logical to first assume it's a constant. Let's say $Q(x) = c$, which makes $xQ(x) = cx$. Is cx congruent to $x - 1$? The difference is $cx - (x - 1) = (c - 1)x + 1$. This is a polynomial of degree 1, so it's not a multiple of $x^2 + 1$ (a degree 2 polynomial). So, a constant won't work. Could $Q(x)$ be a simple degree 1 polynomial? Let's try $Q(x) = x + 1$. Let's check if $Q(x) = x + 1$ works. Calculate $xQ(x)$: $x(x + 1) = x^2 + x$.

Now, apply our rule: we can replace x^2 with -1 . Therefore, $x^2 + x$ becomes $-1 + x$, which is $x - 1$. This shows that $xQ(x)$ is congruent to $x - 1$. Since we showed that no constant polynomial works, and we found a degree 1 polynomial that does work, the minimum degree must be 1. The simplest polynomial of degree 1 that satisfies the condition is $Q(x) = x + 1$, and expanding gives $P(x) = x^3 + 2x^2 + 3x + 3$. Summing the squares of coefficients gives **(E) 23**

Solution 3

Let $P(x) = (x^2 + x + 1)Q_1(x) + x + 2$, then $P(x) = (x^2 + 1)Q_1(x) + xQ_1(x) + x + 2$. Also, $P(x) = (x^2 + 1)Q_2(x) + 2x + 1$.

We infer that $Q_1(x)$ and $Q_2(x)$ have same degree, we can assume $Q_1(x) = x + a$, and $Q_2(x) = x + b$, since $P(x)$ has least degree. If this cannot work, we will try quadratic, etc.

Then we get: $(x^2 + 1)(Q_1(x) - Q_2(x)) + xQ_1(x) - x + 1 = 0$

The constant term gives us: $(Q_1(x) - Q_2(x)) + 1 = 0$

So $Q_1(x) - Q_2(x) = -1$

Substituting this in gives: $-(x^2 + 1) + xQ_1(x) - x + 1 = 0$

Solving this equation, we get $Q_1(x) = x + 1$

Plugging this into our original equation we get $P(x) = x^3 + 2x^2 + 3x + 3$

Verify this works with $P(x) = (x^2 + 1)Q_2(x) + 2x + 1$

Therefore the answer is $1^2 + 2^2 + 3^2 + 3^2 = \mathbf{(E) 23}$

Solution 4 (Undetermined Coefficients)

Notice that we cannot have the quotients equal to some constants, since the same constant will yield different constant terms for $P(x)$ (which is bad) and different constants will yield different first coefficients (also bad). Thus, we try setting the quotients equal to linear terms (for minimizing degree).

Let $P(x) = (x^2 + x + 1)(ax + b) + (x + 2)$ and $P(x) = (x^2 + 1)(ax + c) + (2x + 1)$. The quotients have the same x coefficient, since $P(x)$ must have the same x^3 coefficient in both cases. Expanding, we get

$$P(x) = ax^3 + (a + b)x^2 + (a + b + 1)x + (b + 2)$$

and

$$P(x) = ax^3 + cx^2 + (a + 2)x + (c + 1).$$

Equating coefficients, we get $b + 2 = c + 1$, $a + b + 1 = a + 2$, and $a + b = c$. From the second equation, we get $b = 1$, then substituting into the first, $c = 2$. Finally, from $a + b = c$, we have $a = 1$. Now, $P(x) = (x^2 + x + 1)(ax + b) + (x + 2) = (x^2 + x + 1)(x + 1) + (x + 2) = x^3 + 2x^2 + 3x + 3$ and our answer is

$$1^2 + 2^2 + 3^2 + 3^2 = \boxed{\text{(E)} 23}.$$

Solution 5 (Quick, but Not Quicker Than Solution 2)

We construct the following equations in terms of $P(x)$ and the information given by the problem:

$$(1) \quad P(x) = (x^2 + x + 1) \cdot Q(x) + x + 2$$

$$(2) \quad P(x) = (x^2 + 1) \cdot R(x) + 2x + 1$$

Upon inspection, $Q(x)$ and $R(x)$ cannot be constant, so the smallest possible degree of $P(x)$ is 3, and both $Q(x)$ and $R(x)$ are linear.

Let $Q(x) = x - q$ and $R(x) = x - r$. We know there will be values for q and r that make the below equation hold, so we can assume that $P(x)$ has a leading coefficient of 1.

Substituting these values in, and setting (1) and (2) equal to each other,

$$(x^2 + x + 1)(x - q) + x + 2 = (x^2 + 1)(x - r) + 2x + 1.$$

We plug in $x = 0$, yielding $r + 1 = q$. Substituting this value into the above equation,

$$(x^2 + x + 1)(x - r - 1) + x + 2 = (x^2 + 1)(x - r) + 2x + 1.$$

Letting $x = 1$, we conclude that $r = -2$, so $R(x) = x + 2$. Therefore,

$$P(x) = (x^2 + 1)(x + 2) + 2x + 1 = x^3 + 2x^2 + 3x + 3.$$

The requested sum is

$$1^2 + 2^2 + 3^2 + 3^2 = \boxed{\text{(E)} 23}$$

-Benedict T (countmath1)

Solution 6 (Similar to Solution 3)

By remainder theorem, the polynomial can be written as follows.

$$P(x) = (x^2 + x + 1)Q_1(x) + x + 2 = (x^2 + 1)Q_2(x) + 2x + 1$$

This is a timed exam, we can use the information given by answer choices. The answer choices tell us this is the polynomial with integer coefficients, and we need to find the polynomial with the least degree so we can assume both $Q_1(x)$ and $Q_2(x)$ are linear (the coefficient of x should be same).

Then we can write $P(x)$ as a cubic polynomial.

$$P(x) = (x^2 + x + 1)(ax + b) + x + 2 = (x^2 + 1)(ax + c) + 2x + 1$$

Substituting $x = 0, 1, -1$ to determine the value of a and b .

We have:

$$P(0) = b + 2 = c + 1$$

$$P(1) = 3a + 3b + 3 = 2a + 2c + 3$$

$$P(-1) = -a + b + 1 = -2a + 2c - 1$$

We can solve the simultaneous equations: $a = 1, b = 1, c = 2$

Hence, $P(x) = (x^2 + x + 1)(x + 1) + x + 2 = x^3 + 2x^2 + 3x + 3$. The answer is $1^2 + 2^2 + 3^2 + 3^2 = \boxed{\text{(E)} 23}$

Solution 7 (Bounding with Answer Choices)

We are given:

$$P(x) \equiv x + 2 \pmod{x^2 + x + 1} \quad P(x) \equiv 2x + 1 \pmod{x^2 + 1}$$

Since the problem asks us about the coefficients of the polynomial, we should evaluate $P(x)$ at 1.

Our congruences now become:

$$P(1) \equiv 1 + 2 \pmod{(1)^2 + (1) + 1} \implies P(1) \equiv 3 \pmod{3} \implies P(1) \equiv 0 \pmod{3}.$$

$$P(1) \equiv 2(1) + 1 \pmod{(1)^2 + 1} \implies P(1) \equiv 3 \pmod{2} \implies P(1) \equiv 1 \pmod{2}.$$

We now know that the coefficients of the polynomial sum to an odd multiple of 3.

We now begin bounding the value of the sum of the coefficients squared. For our answer, the sum of the coefficients squared must be between 10 and 23. Trying a quadratic polynomial, if $P(1) = 3$, then the maximum of the coefficients squared is 9 ($3^2 + 0^2 + 0^2$ is the maximum, any other permutation can be trivially shown to be lower). If $P(1) = 9$, the minimum of the coefficients squared is 27 ($3^2 + 3^2 + 3^2$, once again, any other permutation can be trivially shown to be higher). Any higher value of $P(1)$ obviously will not work, so $P(x)$ cannot be a quadratic.

Since a quadratic did not work, we now move up to a cubic. We start with $P(1) = 3$. Similarly to the quadratic, there is no way to get an answer within the range of the answer choices.

However, moving up to $P(1) = 9$, we get that the minimum of the coefficients squared is $81/4$ (which we get from all 4 coefficients being equal at $9/4$). Since this is the minimum, any other answers will be higher. We get that $\boxed{\text{(E)} 23}$ is our answer, since it is the only answer choice greater than $81/4$.

-SwordOfJustice

Solution 8

Since the divisors $x^2 + x + 1$ and $x^2 + 1$ are both powers of 2, and the remainders $x + 2$ and $2x + 1$ are both linear, we can assume that $P(x)$ has a minimum degree of 3.

We can then set the coefficients of the polynomial as $P(x) = ax^3 + bx^2 + cx + d$.

Then, using long division, we divide $P(x) = ax^3 + bx^2 + cx + d$ by $x^2 + x + 1$ and obtain a remainder of $(c - b)x + (a + d - b)$, which is equal to $x + 2$

We also divide $P(x) = ax^3 + bx^2 + cx + d$ by $x^2 + 1$ and obtain a remainder of $(c - a)x + (d - b)$, which is equal to $2x + 1$.

From here we can set up a systems of equations:

$$c - b = 1 \quad a + d - b = 2 \quad c - a = 2 \quad d - b = 1$$

Thus giving us the solutions $a = 1, b = 2, c = 3, d = 3$ and hence $P(x) = x^3 + 2x^2 + 3x + 3$

Squaring the coefficients, then adding them, $1^2 + 2^2 + 3^2 + 3^2$ we get the final answer (E) 23

Solution 9

We assume $P(x)$ is a monic cubic polynomial. Now, we set $P(x) = x^3 + ax^2 + bx + c$, and set the equations:

$$x^3 + ax^2 + bx + c = Q_1(x)(x^2 + x + 1) + x + 2 \quad x^3 + ax^2 + bx + c = Q_2(x)(x^2 + 1) + 2x + 1$$

Plugging in $x = i$ for the second equation, we get $-i - a + bi + c = 2i + 1$, and matching coefficients gives $c - a = 1$ and $b - 1 = 2 \rightarrow b = 3$. Now, we can guess values for a and c based on the answer choices. We have that the sum of the squares of the current known coefficients is $1^2 + 3^2 = 10$, and guessing $c = 1, a = 0$ clearly doesn't match up with the answer choices, so we guess $c = 3, a = 2$ and find that the answer is (E) 23

Solution 10 (EZ CRT)

If we let $x = 10$, we just get a CRT problem. Note: this only works because all the coefficients are integers between 0 and 9. Our problem now is to find n such that

$$n \equiv 12 \pmod{111}$$

$$n \equiv 21 \pmod{101}$$

After using some careful Euclidean Algorithm (or just eyeing it), $n = 1233$, so our answer is $1^2 + 2^2 + 3^2 + 3^2 = \span style="border: 1px solid black; padding: 2px;">(E) 23.$

Solution 11 (Polynomial CRT)

$x^2 + x + 1$ and $x^2 + 1$ are relatively prime according to Bézout's theorem.

$P(x) \equiv U(x) + V(x) \pmod{(x^2 + x + 1)(x^2 + 1)}$, where

$$U(x) \equiv x + 2 \pmod{x^2 + x + 1}, \quad U(x) \equiv 0 \pmod{x^2 + 1}$$

$$V(x) \equiv 2x + 1 \pmod{x^2 + 1}, \quad V(x) \equiv 0 \pmod{x^2 + x + 1}$$

Notice that

$$x^2 + 1 \equiv -x \pmod{x^2 + x + 1} \tag{10}$$

$$(x^2 + 1)(x + 1) \equiv -x(x + 1) \equiv 1 \pmod{x^2 + x + 1} \tag{11}$$

$$U(x) \equiv -(x^2 + 1) + 2(x^2 + 1)(x + 1) \pmod{(x^2 + x + 1)(x^2 + 1)} \tag{12}$$

$$U(x) \equiv 2x^3 + x^2 + 2x + 1 \pmod{(x^2 + x + 1)(x^2 + 1)} \tag{13}$$

And

$$x^2 + x + 1 \equiv x \pmod{x^2 + 1} \tag{14}$$

$$(x^2 + x + 1)x \equiv x^2 \equiv -1 \pmod{x^2 + 1} \tag{15}$$

$$V(x) \equiv 2(x^2 + x + 1) - (x^2 + x + 1)x \pmod{(x^2 + x + 1)(x^2 + 1)} \tag{16}$$

$$V(x) \equiv -x^3 + x^2 + x + 2 \pmod{(x^2 + x + 1)(x^2 + 1)} \tag{17}$$

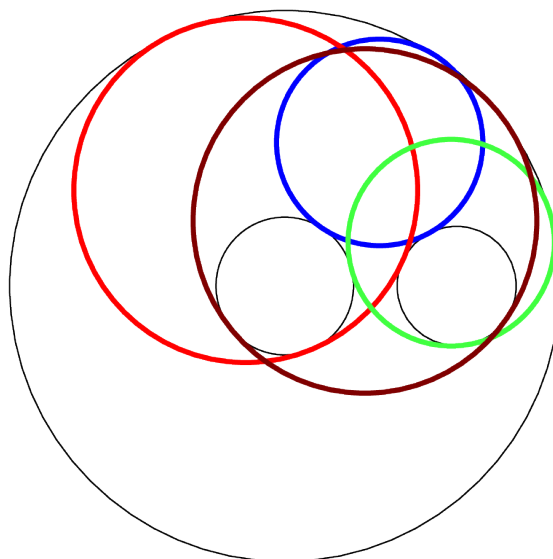
Therefore $P(x) \equiv U(x) + V(x) \equiv x^3 + 2x^2 + 3x + 3 \pmod{(x^2 + x + 1)(x^2 + 1)}$, the answer is $1^2 + 2^2 + 3^2 + 3^2 = \span style="border: 1px solid black; padding: 2px;">(E) 23$

21. (AMC 12 12B 2022) Let S be the set of circles in the coordinate plane that are tangent to each of the three circles with equations $x^2 + y^2 = 4$, $x^2 + y^2 = 64$, and $(x - 5)^2 + y^2 = 3$. What is the sum of the areas of all circles in S ?

(A) 48π (B) 68π (C) 96π (D) 102π (E) 136π

Answer: E

Solution 1



The circles match up as follows: Case 1 is brown, Case 2 is blue, Case 3 is green, and Case 4 is red. Let $x^2 + y^2 = 64$ be circle O , $x^2 + y^2 = 4$ be circle P , and $(x - 5)^2 + y^2 = 3$ be circle Q . All the circles in S are internally tangent to circle O . There are four cases with two circles belonging to each:

- * P and Q are internally tangent to S .
- * P and Q are externally tangent to S .
- * P is externally and Circle Q is internally tangent to S .
- * P is internally and Circle Q is externally tangent to S .

Consider Cases 1 and 4 together. Since circles O and P have the same center, the line connecting the center of S and the center of O will pass through the tangency point of both S and O and the tangency point of S and P . This line will be the diameter of S and have length $r_P + r_O = 10$. Therefore the radius of S in these cases is 5.

Consider Cases 2 and 3 together. Similarly to Cases 1 and 4, the line connecting the center of S to the center of O will pass through the tangency points. This time, however, the diameter of S will have length $r_P - r_O = 6$. Therefore, the radius of S in these cases is 3.

The set of circles S consists of 8 circles - 4 of which have radius 5 and 4 of which have radius 3. The total area of all circles in S is $4(5^2\pi) + 4(3^2\pi) = 136\pi \Rightarrow \boxed{\text{(E)}}$.

-naman12

Solution 2

We denote by C_1 the circle that has the equation $x^2 + y^2 = 4$. We denote by C_2 the circle that has the equation $x^2 + y^2 = 64$. We denote by C_3 the circle that has the equation $(x - 5)^2 + y^2 = 3$.

We denote by C_0 a circle that is tangent to C_1 , C_2 and C_3 . We denote by (u, v) the coordinates of circle C_0 , and r the radius of this circle.

From the graphs of circles C_1 , C_2 , C_3 , we observe that if C_0 is tangent to all of them, then C_0 must be internally tangent to C_2 . We have

$$u^2 + v^2 = (8 - r)^2. \quad (1)$$

We do the following casework analysis in terms of the whether C_0 is externally tangent to C_1 and C_3 .

Case 1: C_0 is externally tangent to C_1 and C_3 .

We have

$$u^2 + v^2 = (r + 2)^2 \quad (2)$$

and

$$(u - 5)^2 + v^2 = (r + \sqrt{3})^2. \quad (3)$$

Taking (2) - (1), we get $r + 2 = 8 - r$. Thus, $r = 3$. We can further compute (omitted here) that there exist feasible (u, v) with this given r .

Case 2: C_1 is internally tangent to C_0 and C_3 is externally tangent to C_0 .

We have

$$u^2 + v^2 = (r - 2)^2 \quad (2)$$

and

$$(u - 5)^2 + v^2 = (r + \sqrt{3})^2. \quad (3)$$

Taking (2) - (1), we get $r - 2 = 8 - r$. Thus, $r = 5$. We can further compute (omitted here) that there exist feasible (u, v) with this given r .

Case 3: C_1 is externally tangent to C_0 and C_3 is internally tangent to C_0 .

We have

$$u^2 + v^2 = (r + 2)^2 \quad (2)$$

and

$$(u - 5)^2 + v^2 = (r - \sqrt{3})^2. \quad (3)$$

Taking (2) - (1), we get $r + 2 = 8 - r$. Thus, $r = 3$. We can further compute (omitted here) that there exist feasible (u, v) with this given r .

Case 4: C_1 is internally tangent to C_0 and C_3 is internally tangent to C_0 .

We have

$$u^2 + v^2 = (r - 2)^2 \quad (2)$$

and

$$(u - 5)^2 + v^2 = (r - \sqrt{3})^2. \quad (3)$$

Taking (2) - (1), we get $r - 2 = 8 - r$. Thus, $r = 5$. We can further compute (omitted here) that there exist feasible (u, v) with this given r .

Because the graph is symmetric with the x -axis, and for each case above, the solution of v is not 0. Hence, in each case, there are two congruent circles whose centers are symmetric through the x -axis.

Therefore, the sum of the areas of all the circles in S is $2(3^2\pi + 5^2\pi + 3^2\pi + 5^2\pi) =$
(E) 136π .

Solution 3

The problem of counting the number of circles tangent to a given number of other circles, lines, or points is called Apollonius's Problem. Here, there are three circles; solutions to Apollonius's problem tell us that there are 8 possible tangent circles. When looking for all eight, we can notice that the diagram is symmetrical about the x -axis, so we only need to find four different circles and then reflect them to the other side.

After drawing in those circles, we can use any of the above methods to find the radii of the circle. The important step is to realize that the segment between tangent circles' centers must pass through the point of tangency. From here, we can prove that the center of the circle with equation $x^2 + y^2 = 4$ and the center of the circle tangent to all three must both lie on the radius of $x^2 + y^2 = 64$, allowing us to quickly find the radius of the tangent circle.

For the tangent circles mostly above the x -axis, there are two of radius 5, and two of radius 3. The total area is $50\pi + 18\pi = 68\pi$. As due to symmetry there are four other circles below the x -axis, the total area is 136π , so we choose **(E)**.

22. (AMC 12 12B 2022) Ant Amelia starts on the number line at 0 and crawls in the following manner. For $n = 1, 2, 3$, Amelia chooses a time duration t_n and an increment x_n independently and uniformly at random from the interval $(0, 1)$. During the n th step of the process, Amelia moves x_n units in the positive direction, using up t_n minutes. If the total elapsed time has exceeded 1 minute during the n th step, she stops at the end of that step; otherwise, she continues with the next step, taking at most 3 steps in all. What is the probability that Amelia's position when she stops will be greater than 1?

(A) $\frac{1}{3}$ (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$ (E) $\frac{5}{6}$

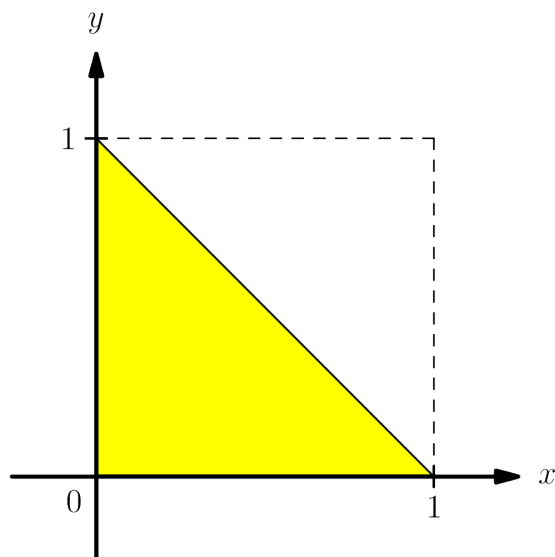
Answer: C

Solution 1 (Geometric Probability)

Let x and y be random variables that are independently and uniformly distributed in the interval $(0, 1)$. Note that

$$P(x + y \leq 1) = \frac{\frac{1}{2} \cdot 1^2}{1^2} = \frac{1}{2},$$

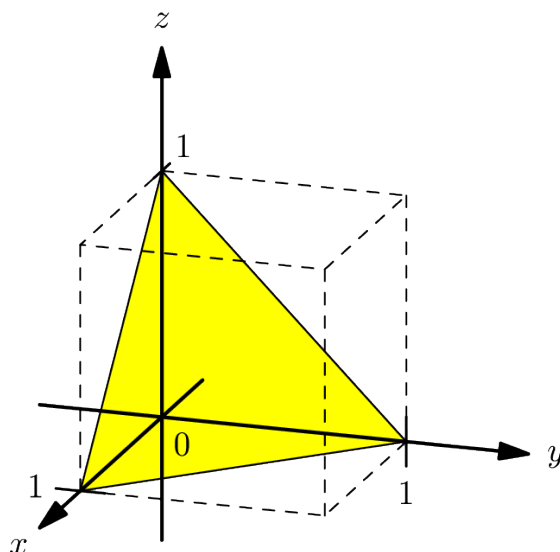
as shown below:



Let x , y , and z be random variables that are independently and uniformly distributed in the interval $(0, 1)$. Note that

$$P(x + y + z \leq 1) = \frac{\frac{1}{3} \cdot \left(\frac{1}{2} \cdot 1^2\right) \cdot 1}{1^3} = \frac{1}{6},$$

as shown below:



We have two cases:

- a) Amelia takes exactly 2 steps. We need $x_1 + x_2 > 1$ and $t_1 + t_2 > 1$. So, the probability is

$$P(x_1 + x_2 > 1) \cdot P(t_1 + t_2 > 1) = \left(1 - \frac{1}{2}\right) \cdot \left(1 - \frac{1}{2}\right) = \frac{1}{4}.$$

- b) Amelia takes exactly 3 steps. We need $x_1 + x_2 + x_3 > 1$ and $t_1 + t_2 \leq 1$. So, the probability is

$$P(x_1 + x_2 + x_3 > 1) \cdot P(t_1 + t_2 \leq 1) = \left(1 - \frac{1}{6}\right) \cdot \frac{1}{2} = \frac{5}{12}.$$

Together, the answer is $\frac{1}{4} + \frac{5}{12} = \boxed{(C) \frac{2}{3}}$.

Solution 2 (Generalization and Induction)

We can in fact find the probability that any number of randomly distributed numbers on the interval $[0, 1]$ sum to more than 1 using geometric probability, as shown in the video below.

If we graph the points that satisfy $x + y < 1$, $0 < x, y < 1$, we get the triangle with points $(0, 0)$, $(1, 0)$, and $(0, 1)$. If we graph the points that satisfy $x + y + z < 1$, $0 < x, y, z < 1$, we get the tetrahedron with points $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$.

Of course, the probability of either of these cases happening is simply the area/volume of the points we graphed divided by the total area of the graph, which is always 1 (this would be much simpler than my calculus proof above).

Thus, we can now solve for the probability that the sum is less than one for n numbers using induction.

Claim: The probability that the sum is less than one is $\frac{1}{n!}$.

Base Case: For just 1 number, the probability is 1.

Induction step: Suppose that the probability for n numbers is $\frac{1}{n!}$. We will prove that the probability for $n + 1$ numbers is $\frac{1}{(n+1)!}$. To prove this, we consider that the area of an $n + 1$ -dimensional tetrahedron is simply the area/volume of the base times the height divided by $n + 1$.

Of course, the area of the base is $\frac{1}{n!}$, and the height is 1, and thus, we obtain $\frac{1}{n! \cdot (n+1)} = \frac{1}{(n+1)!}$ as our volume (this may be hard to visualize for higher dimensions). The induction step is complete.

The probability of the sum being less than 1 is $\frac{1}{n!}$, and the probability of the sum being more than 1 is $\frac{n! - 1}{n!}$. This trivializes the problem. The answer is

$$\frac{1}{2} \cdot \frac{2! - 1}{2!} + \frac{1}{2} \cdot \frac{3! - 1}{3!} = \boxed{(C) \frac{2}{3}}.$$

Solution 3 (Observations)

There are two cases: Amelia takes two steps or three steps.

The former case has a probability of $\frac{1}{2}$, as stated in Solution 1, and thus the latter also has a probability of $\frac{1}{2}$.

The probability that Amelia passes 1 after two steps is also $\frac{1}{2}$, as it is symmetric to the probability above.

Thus, if the probability that Amelia passes 1 after three steps is x , our total probability is $\frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot x$. We know that $0 < x < 1$, and it is relatively obvious that $x > \frac{1}{2}$ (because the probability that $x > \frac{3}{2}$ is $\frac{1}{2}$). This means that our total probability is between $\frac{1}{2}$ and $\frac{3}{4}$, non-inclusive, so the only answer choice that fits is

$$\boxed{(C) \frac{2}{3}}.$$

Remark (Calculus) It is not immediately clear why three random numbers between 0 and 1 have a probability of $\frac{5}{6}$ of summing to more than 1. Here is a proof:

Let us start by finding the probability that two random numbers between 0 and 1 have a sum of more than x , where $0 \leq x \leq 1$.

Suppose that our two numbers are y and z . Then, the probability that $y > x$ (which means that $y + z > x$) is $1 - x$, and the probability that $y < x$ is x .

If $y < x$, the probability that $y + z > x$ is $1 - x + y$. This is because the probability that $y + z < x$ is equal to the probability that $z < x - y$, which is $x - y$, so our total probability is $1 - (x - y) = 1 - x + y$.

Let us now find the average of the probability that $y + z > x$ when $y < x$. Since y is a random number between 0 and x , its average is $\frac{x}{2}$. Thus, our average is $1 - x + \frac{x}{2} = 1 - \frac{x}{2}$.

Hence, our total probability is equal to

$$1(1 - x) + \left(1 - \frac{x}{2}\right)(x) = 1 - \frac{1}{2}x^2.$$

Now, let us find the probability that three numbers uniformly distributed between 0 and 1 sum to more than 1.

Let our three numbers be a , b , and c . Then, the probability that $a + b + c > 1$ is equal to the probability that $b + c$ is greater than $1 - a$, which is equal to $1 - \frac{1}{2}(1 - a)^2$.

To find the total probability, we must average over all values of a . This average is simply equal to the area under the curve $1 - \frac{1}{2}(1 - x)^2$ from 0 to 1, all divided by 1. We can compute this value using integrals:

$$\begin{aligned} \frac{\int_0^1 1 - \frac{1}{2}(1 - x)^2 dx}{1} &= \int_0^1 1 - \frac{1}{2}(1 - x)^2 dx \\ &= 1 - \frac{1}{2} \int_0^1 (1 - x)^2 dx \\ &= 1 - \frac{1}{2} \int_0^1 x^2 dx \\ &= 1 - \frac{1}{2} \left(\frac{1}{3}\right) \\ &= \frac{5}{6}. \end{aligned}$$

For those who don't know calculus, $\int_m^n f(x) dx$ is the area under the curve $f(x)$ from m to n .

Remark (Rigorous Calculus) In the language of probability, we have three random variables X , Y , Z , each independent and uniformly distributed over the interval $(0, 1)$. We are interested in the probability that $X + Y + Z > 1$, or $P(X + Y + Z > 1)$.

It follows from the definition of the joint probability density function $f(x, y, z)$ that (we assume that $0 < x, y, z < 1$):

$$P(X + Y + Z > 1) = \iiint_{x+y+z>1} f(x, y, z) dx dy dz. \quad (18)$$

Since X, Y, Z are independent, we have $f(x, y, z) = f_X(x) * f_Y(y) * f_Z(z)$, where $f_X(x), f_Y(y), f_Z(z)$ represent the probability densities of X, Y, Z respectively. Recall

that X, Y, Z are uniformly distributed over the interval $(0, 1)$. Hence, $f_X(x) = f_Y(y) = f_Z(z) = 1$.

It remains to evaluate the following triple integral

$$\iiint_{x+y+z>1} dx \, dy \, dz = \frac{5}{6}. \quad (19)$$

Therefore, $P(X + Y + Z > 1) = \frac{5}{6}$.

respect to whoever would do this during the comp

-guy_who_likes_frogs

23. (AMC 12 12B 2022) Let $x_0, x_1, x_2, \dots$ be a sequence of numbers, where each x_k is either 0 or 1. For each positive integer n , define

$$S_n = \sum_{k=0}^{n-1} x_k 2^k$$

Suppose $7S_n \equiv 1 \pmod{2^n}$ for all $n \geq 1$. What is the value of the sum

$$x_{2019} + 2x_{2020} + 4x_{2021} + 8x_{2022}?$$

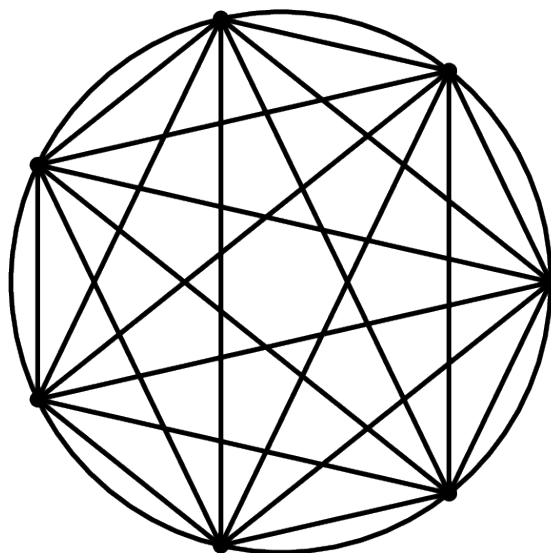
- (A) 6 (B) 7 (C) 12 (D) 14 (E) 15

Answer: A

Solution 1

In binary

24. (AMC 12 12B 2022) The figure below depicts a regular 7-gon inscribed in a unit circle.



What is the sum of the 4th powers of the lengths of all 21 of its edges and diagonals?

- (A) 49 (B) 98 (C) 147 (D) 168 (E) 196

Answer: C

Solution 1 (Complex Numbers)

There are 7 segments whose lengths are $2 \sin \frac{\pi}{7}$, 7 segments whose lengths are $2 \sin \frac{2\pi}{7}$, 7 segments whose lengths are $2 \sin \frac{3\pi}{7}$.

Therefore, the sum of the 4th powers of these lengths is

$$\begin{aligned}
 7 \cdot 2^4 \sin^4 \frac{\pi}{7} + 7 \cdot 2^4 \sin^4 \frac{2\pi}{7} + 7 \cdot 2^4 \sin^4 \frac{3\pi}{7} &= \frac{7 \cdot 2^4}{(2i)^4} \left(e^{i\frac{\pi}{7}} - e^{-i\frac{\pi}{7}} \right)^4 + \frac{7 \cdot 2^4}{(2i)^4} \left(e^{i\frac{2\pi}{7}} - e^{-i\frac{2\pi}{7}} \right)^4 + \frac{7 \cdot 2^4}{(2i)^4} \left(e^{i\frac{3\pi}{7}} - e^{-i\frac{3\pi}{7}} \right)^4 \\
 &= 7 \left(e^{i\frac{4\pi}{7}} - 4e^{i\frac{2\pi}{7}} + 6 - 4e^{-i\frac{2\pi}{7}} + e^{-i\frac{4\pi}{7}} \right) \\
 &\quad + 7 \left(e^{i\frac{8\pi}{7}} - 4e^{i\frac{4\pi}{7}} + 6 - 4e^{-i\frac{4\pi}{7}} + e^{-i\frac{8\pi}{7}} \right) \\
 &\quad + 7 \left(e^{i\frac{12\pi}{7}} - 4e^{i\frac{6\pi}{7}} + 6 - 4e^{-i\frac{6\pi}{7}} + e^{-i\frac{12\pi}{7}} \right) \\
 &= 7 \left(e^{i\frac{4\pi}{7}} + e^{i\frac{8\pi}{7}} + e^{i\frac{12\pi}{7}} + e^{-i\frac{4\pi}{7}} + e^{-i\frac{8\pi}{7}} + e^{-i\frac{12\pi}{7}} \right) \\
 &\quad - 7 \cdot 4 \left(e^{i\frac{2\pi}{7}} + e^{i\frac{4\pi}{7}} + e^{i\frac{6\pi}{7}} + e^{-i\frac{2\pi}{7}} + e^{-i\frac{4\pi}{7}} + e^{-i\frac{6\pi}{7}} \right) \\
 &\quad + 7 \cdot 6 \cdot 3 \\
 &= 7 \left(e^{i\frac{4\pi}{7}} + e^{-i\frac{6\pi}{7}} + e^{-i\frac{2\pi}{7}} + e^{-i\frac{4\pi}{7}} + e^{i\frac{6\pi}{7}} + e^{i\frac{2\pi}{7}} \right) \\
 &\quad - 7 \cdot 4 \left(e^{i\frac{2\pi}{7}} + e^{i\frac{4\pi}{7}} + e^{i\frac{6\pi}{7}} + e^{-i\frac{2\pi}{7}} + e^{-i\frac{4\pi}{7}} + e^{-i\frac{6\pi}{7}} \right) \\
 &\quad + 7 \cdot 6 \cdot 3 \\
 &= -7 + 7 \cdot 4 + 7 \cdot 6 \cdot 3 \\
 &= \boxed{\text{(C) } 147},
 \end{aligned}$$

where the fourth from the last equality follows from the property that

$$\begin{aligned}
 e^{i\frac{2\pi}{7}} + e^{i\frac{4\pi}{7}} + e^{i\frac{6\pi}{7}} + e^{-i\frac{2\pi}{7}} + e^{-i\frac{4\pi}{7}} + e^{-i\frac{6\pi}{7}} &= e^{-i\frac{6\pi}{7}} \sum_{j=0}^6 e^{i\frac{2\pi j}{7}} - 1 \\
 &= 0 - 1 \\
 &= -1.
 \end{aligned}$$

Steven Chen (Professor Chen Education Palace, www.professorchenedu.com)

Solution 2 (Trigonometry)

There are 7 segments whose lengths are $2 \sin \frac{\pi}{7}$, 7 segments whose lengths are $2 \sin \frac{2\pi}{7}$, 7 segments whose lengths are $2 \sin \frac{3\pi}{7}$.

Therefore, the sum of the 4th powers of these lengths is

$$\begin{aligned}
 7 \cdot 2^4 \sin^4 \frac{\pi}{7} + 7 \cdot 2^4 \sin^4 \frac{2\pi}{7} + 7 \cdot 2^4 \sin^4 \frac{3\pi}{7} &= 7 \cdot 2^4 \left(\frac{1 - \cos \frac{2\pi}{7}}{2} \right)^2 + 7 \cdot 2^4 \left(\frac{1 - \cos \frac{4\pi}{7}}{2} \right)^2 + 7 \cdot 2^4 \left(\frac{1 - \cos \frac{6\pi}{7}}{2} \right)^2 \\
 &= 7 \cdot 2^2 \left(1 - 2 \cos \frac{2\pi}{7} + \cos^2 \frac{2\pi}{7} \right) + 7 \cdot 2^2 \left(1 - 2 \cos \frac{4\pi}{7} + \cos^2 \frac{4\pi}{7} \right) + 7 \cdot 2^2 \left(1 - 2 \cos \frac{6\pi}{7} + \cos^2 \frac{6\pi}{7} \right) \\
 &= 7 \cdot 2^2 \cdot 3 - 7 \cdot 2^3 \left(\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} \right) + 7 \cdot 2^2 \left(\cos^2 \frac{2\pi}{7} + \cos^2 \frac{4\pi}{7} + \cos^2 \frac{6\pi}{7} \right) \\
 &= 7 \cdot 2^2 \cdot 3 - 7 \cdot 2^3 \left(\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} \right) + 7 \cdot 2^2 \left(\frac{1 + \cos \frac{4\pi}{7}}{2} + \frac{1 + \cos \frac{8\pi}{7}}{2} + \frac{1 + \cos \frac{12\pi}{7}}{2} \right) \\
 &= 7 \cdot 2^2 \cdot 3 - 7 \cdot 2^3 \left(\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} \right) + 7 \cdot 2 \cdot 3 + 7 \cdot 2 \left(\cos \frac{4\pi}{7} + \cos \frac{8\pi}{7} + \cos \frac{12\pi}{7} \right)
 \end{aligned}$$

$$\begin{aligned}
 &= 7 \cdot 2^2 \cdot 3 - 7 \cdot 2^3 \left(\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} \right) + 7 \cdot 2 \cdot 3 + 7 \cdot 2 \left(\cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} + \cos \frac{2\pi}{7} \right) \\
 &= 7 \cdot 2 \cdot 3 (2 + 1) - 7 \cdot 2 (4 - 1) \left(\cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} + \cos \frac{2\pi}{7} \right) \\
 &= 7 \cdot 2 \cdot 3 (2 + 1) - 7 \cdot 2 (4 - 1) \cdot \left(-\frac{1}{2} \right) \\
 &= \boxed{\text{(C) } 147},
 \end{aligned}$$

where the second from the last equality follows from the property that

$$\cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} + \cos \frac{2\pi}{7} = -\frac{1}{2}.$$

For explanation see [supplement](#).

Solution 3 (Complex Numbers and Trigonometry)

As explained in Solutions 1 and 2, what we are trying to find is $7 \cdot 2^4 \sin^4 \frac{\pi}{7} + 7 \cdot 2^4 \sin^4 \frac{2\pi}{7} + 7 \cdot 2^4 \sin^4 \frac{3\pi}{7}$. Using trig we get

$$\begin{aligned}
 &\sin^4 \frac{\pi}{7} + \sin^4 \frac{2\pi}{7} + \sin^4 \frac{3\pi}{7} \\
 &= \sin^2 \frac{\pi}{7} \left(1 - \cos^2 \frac{\pi}{7} \right) + \sin^2 \frac{2\pi}{7} \left(1 - \cos^2 \frac{2\pi}{7} \right) + \sin^2 \frac{3\pi}{7} \left(1 - \cos^2 \frac{3\pi}{7} \right) \\
 &= \sin^2 \frac{\pi}{7} - \left(\frac{1}{2} \sin \frac{2\pi}{7} \right)^2 + \sin^2 \frac{2\pi}{7} - \left(\frac{1}{2} \sin \frac{4\pi}{7} \right)^2 + \sin^2 \frac{3\pi}{7} - \left(\frac{1}{2} \sin \frac{6\pi}{7} \right)^2 \\
 &= \sin^2 \frac{\pi}{7} - \frac{1}{4} \sin^2 \frac{2\pi}{7} + \sin^2 \frac{2\pi}{7} - \frac{1}{4} \sin^2 \frac{4\pi}{7} + \sin^2 \frac{3\pi}{7} - \frac{1}{4} \sin^2 \frac{6\pi}{7} \\
 &= \frac{3}{4} \left(\sin^2 \frac{\pi}{7} + \sin^2 \frac{2\pi}{7} + \sin^2 \frac{3\pi}{7} \right) \\
 &= \frac{3}{4} \cdot \frac{1}{2} \left(1 - \cos \frac{2\pi}{7} + 1 - \cos \frac{4\pi}{7} + 1 - \cos \frac{6\pi}{7} \right) \\
 &= \frac{3}{4} \cdot \frac{1}{2} \left(3 - \left(-\frac{1}{2} \right) \right) \\
 &= \frac{21}{16}.
 \end{aligned}$$

Like in the second solution, we also use the fact that $\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} = -\frac{1}{2}$, which admittedly might need some explanation.

For explanation see [supplement](#).

Notice that

$$\begin{aligned}
 \cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} &= \frac{1}{2} \left(e^{\frac{2i\pi}{7}} + e^{-\frac{2i\pi}{7}} \right) + \frac{1}{2} \left(e^{\frac{4i\pi}{7}} + e^{-\frac{4i\pi}{7}} \right) + \frac{1}{2} \left(e^{\frac{6i\pi}{7}} + e^{-\frac{6i\pi}{7}} \right) \\
 &= \frac{1}{2} \left(e^{\frac{2i\pi}{7}} + e^{-\frac{2i\pi}{7}} + e^{\frac{4i\pi}{7}} + e^{-\frac{4i\pi}{7}} + e^{\frac{6i\pi}{7}} + e^{-\frac{6i\pi}{7}} + 1 \right) - \frac{1}{2}
 \end{aligned}$$

In the brackets we have the sum of the roots of the polynomial $x^7 - 1 = 0$. These sum to 0 by [Vieta's formulas](#), and the desired identity follows. See [Roots of unity](#) if you have not seen this technique.

Going back to the question:

$$7 \cdot 2^4 \sin^4 \frac{\pi}{7} + 7 \cdot 2^4 \sin^4 \frac{2\pi}{7} + 7 \cdot 2^4 \sin^4 \frac{3\pi}{7} = 7 \cdot 2^4 \left(\sin^4 \frac{\pi}{7} + \sin^4 \frac{2\pi}{7} + \sin^4 \frac{3\pi}{7} \right) = 7 \cdot 2^4 \cdot \frac{21}{16} = \boxed{\text{(C) } 147}.$$

obscene_kangaroo

Solution 4 (Trigonometry)

This solution follows the same steps as the trigonometry solutions (Solutions 2 and 3), except it gives an alternate way to prove the statement below true without complex numbers:

$$\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} = -\frac{1}{2}$$

$$\begin{aligned} S &= \cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7}, \\ S^2 &= \cos^2 \frac{2\pi}{7} + \cos^2 \frac{4\pi}{7} + \cos^2 \frac{6\pi}{7} + 2 \cos \frac{2\pi}{7} \cos \frac{4\pi}{7} + 2 \cos \frac{2\pi}{7} \cos \frac{6\pi}{7} + 2 \cos \frac{4\pi}{7} \cos \frac{6\pi}{7} \\ &= \left(\frac{1 + \cos \frac{4\pi}{7}}{2} \right) + \left(\frac{1 + \cos \frac{8\pi}{7}}{2} \right) + \left(\frac{1 + \cos \frac{12\pi}{7}}{2} \right) + 2 \cos \frac{2\pi}{7} \cos \frac{4\pi}{7} + 2 \cos \frac{2\pi}{7} \cos \frac{6\pi}{7} + 2 \cos \frac{4\pi}{7} \cos \frac{6\pi}{7} \\ &= \frac{1}{2}(3 + S) + \left(\cos \frac{6\pi}{7} + \cos \frac{2\pi}{7} \right) + \left(\cos \frac{8\pi}{7} + \cos \frac{4\pi}{7} \right) + \left(\cos \frac{10\pi}{7} + \cos \frac{2\pi}{7} \right) \\ &= \frac{1}{2}(3 + S) + 2 \cos \frac{2\pi}{7} + 2 \cos \frac{4\pi}{7} + 2 \cos \frac{6\pi}{7} \\ &= \frac{1}{2}(3 + S) + 2S. \end{aligned}$$

We end up with

$$2S^2 - 5S - 3 = 0.$$

Using the quadratic formula, we find the solutions for S to be $-\frac{1}{2}$ and 3. Because 3 is impossible, $S = -\frac{1}{2}$. With this result, following similar to steps to Solutions 2 and 3 will get (C) 147

Solution 5 (Law of Cosines)

Let x , y , and z be the lengths of the chords with arcs $\frac{2\pi}{7}$, $\frac{4\pi}{7}$ and $\frac{6\pi}{7}$ respectively. Then by the law of cosines we get:

$$\begin{aligned} x^2 &= 2 \left(1 - \cos \frac{2\pi}{7} \right), \\ y^2 &= 2 \left(1 - \cos \frac{4\pi}{7} \right), \\ z^2 &= 2 \left(1 - \cos \frac{6\pi}{7} \right). \end{aligned}$$

The answer is then just $7(x^4 + y^4 + z^4)$ (since there's 7 of each diagonal/side), obtained by summing the squares of the above equations and then multiplying by 7.

$$\begin{aligned} &7 \cdot 2^2 \left(\left(1 - \cos \frac{2\pi}{7} \right)^2 + \left(1 - \cos \frac{4\pi}{7} \right)^2 + \left(1 - \cos \frac{6\pi}{7} \right)^2 \right) \\ &= 7 \cdot 4 \left(\left(1 - 2 \cos \frac{2\pi}{7} + \cos^2 \frac{2\pi}{7} \right) + \left(1 - 2 \cos \frac{4\pi}{7} + \cos^2 \frac{4\pi}{7} \right) + \left(1 - 2 \cos \frac{6\pi}{7} + \cos^2 \frac{6\pi}{7} \right) \right) \\ &= 7 \cdot 4 \left(3 - 2 \left(\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} \right) + \left(\cos^2 \frac{2\pi}{7} + \cos^2 \frac{4\pi}{7} + \cos^2 \frac{6\pi}{7} \right) \right) \end{aligned}$$

$$\begin{aligned}
 &= 7 \cdot 4 \left(3 - 2 \left(\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} \right) + \frac{1}{2} \left(1 + \cos \frac{4\pi}{7} + 1 + \cos \frac{8\pi}{7} + 1 + \cos \frac{12\pi}{7} \right) \right) \\
 &= 7 \cdot 4 \left(\frac{9}{2} - 2 \left(\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} \right) + \frac{1}{2} \left(\cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} + \cos \frac{2\pi}{7} \right) \right) \\
 &= 7 \cdot 4 \left(\frac{9}{2} - \frac{3}{2} \left(\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} \right) \right) \\
 &= 7 \cdot 4 \left(\frac{9}{2} - \frac{3}{2} \left(-\frac{1}{2} \right) \right) \\
 &= \boxed{\text{(C) } 147}.
 \end{aligned}$$

(Use the identity that $\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} = -\frac{1}{2}$.)

For explanation of this identity see [supplement](#).

- SAHANWIJETUNGA

Solution 6 (Ruler Measure)

Hope you had a ruler handy! This problem can be done with a ruler and basic estimation.

First, measuring the radius of the circle obtains 2.9 cm (when done on the paper version). Thus, any other measurement we get for the sides/diagonals should be divided by 2.9.

Measuring the sides of the circle gets 2.5 cm. The shorter diagonals are 4.5 cm, and the longest diagonals measure 5.6 cm. Thus, we'd like to estimate

$$7 \left(\frac{2.5}{2.9} \right)^4 + 7 \left(\frac{4.5}{2.9} \right)^4 + 7 \left(\frac{5.6}{2.9} \right)^4.$$

We know $\left(\frac{2.5}{2.9} \right)^4$ is slightly less than 1. Let's approximate it as 1 for now. Thus, $7 \left(\frac{2.5}{2.9} \right)^4 \approx 7$.

Next, $\left(\frac{4.5}{2.9} \right)^4$ is slightly more than $\left(\frac{4.5}{3} \right)^4$. We know $\left(\frac{4.5}{3} \right)^4 = 1.5^4 = \frac{81}{16}$, slightly more than 5, so we can approximate $\left(\frac{4.5}{2.9} \right)^4$ as 5.5. Thus, $7 \left(\frac{4.5}{2.9} \right)^4 \approx 38.5$.

Finally, $\left(\frac{5.6}{2.9} \right)^4$ is slightly less than $\left(\frac{5.6}{2.8} \right)^4 = 2^4 = 16$. We say it's around 15, so then $7 \left(\frac{5.6}{2.9} \right)^4 \approx 105$.

Adding what we have, we get $105 + 38.5 + 1 = 144.5$ as our estimate. We see $\boxed{\text{(C) } 147}$ is very close to our estimate, so we have successfully finished the problem.

Solution 7 (Pythagorean Theorem and Trig)

First, we put the figure in the coordinate plane with the center of the circle at the origin and a vertex on the positive x-axis. Thus, the coordinates of the vertices will be the terminal points of integer multiples of the angle $\frac{2\pi}{7}$, which are

$$\left(\cos \frac{2\pi n}{7}, \sin \frac{2\pi n}{7} \right)$$

for integer n . Then, we notice there are three types of diagonals: the ones with chords of arcs $\frac{2\pi}{7}$, $\frac{4\pi}{7}$, and $\frac{6\pi}{7}$. We notice there are here are 7 of each type of diagonal.

Then, we use the **pythagorean theorem** to find the distance from $(\cos \frac{2\pi n}{7}, \sin \frac{2\pi n}{7})$ to $(1, 0)$:

$$\begin{aligned} \left(\sqrt{\left(\cos \frac{2\pi n}{7} - 1 \right)^2 + \sin^2 \frac{2\pi n}{7}} \right)^4 &= \left(\sqrt{\cos^2 \frac{2\pi n}{7} + \sin^2 \frac{2\pi n}{7} - 2 \cos \frac{2\pi n}{7} + 1} \right)^4 \\ &= \left(\sqrt{2 - 2 \cos \frac{2\pi n}{7}} \right)^4 \\ &= 4 \cos^2 \frac{2\pi n}{7} - 8 \cos \frac{2\pi n}{7} + 4. \end{aligned}$$

By the cosine double angle identity, $\cos 2\theta = 2 \cos^2 \theta - 1$. This means that $2 \cos^2 \theta = \cos 2\theta + 1$. Substituting this in,

$$4 \cos^2 \frac{2\pi n}{7} - 8 \cos \frac{2\pi n}{7} + 4 = 2 \left(\cos \frac{4\pi n}{7} + 1 \right) - 8 \cos \frac{2\pi n}{7} + 4 = 2 \cos \frac{4\pi n}{7} - 8 \cos \frac{2\pi n}{7} + 6.$$

Summing this up for $n = 1, 2, 3$,

$$\begin{aligned} &2 \cos \frac{4\pi}{7} - 8 \cos \frac{2\pi}{7} + 6 + 2 \cos \frac{8\pi}{7} - 8 \cos \frac{4\pi}{7} + 6 + 2 \cos \frac{12\pi}{7} - 8 \cos \frac{6\pi}{7} + 6 \\ &= 2 \left(\cos \frac{4\pi}{7} + \cos \frac{8\pi}{7} + \cos \frac{12\pi}{7} \right) - 8 \left(\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} \right) + 18 \\ &= 2 \left(\cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} + \cos \frac{2\pi}{7} \right) - 8 \left(\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} \right) + 18 \\ &= 2(-0.5) - 8(-0.5) + 18 = 21. \end{aligned}$$

(These equalities are based on $\cos \theta = \cos(2\pi - \theta)$ and $\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} = -\frac{1}{2}$.) (For explanation of this see **supplement**.) Finally, because there are 7 of each type of diagonal, the answer is $7 \cdot 21 = \boxed{147}$.

Solution 8 (Roots of Unity)

Place the figure on the complex plane and let $w = e^{\frac{2\pi}{7}}$ (a 7th root of unity). The vertices of the 7-gon are $w^0, w^1, w^2, \dots, w^6$. We wish to find

$$\sum_{i=0}^5 \sum_{j=i+1}^6 |w^i - w^j|^4 = \frac{1}{2} \sum_{i=0}^6 \sum_{j=0}^6 |w^i - w^j|^4.$$

The second expression is more convenient to work with. The factor of $\frac{1}{2}$ is because it double-counts each edge (and includes when $i = j$, but the length is 0 so it doesn't matter).

Recall the identity $|z|^2 = z\bar{z}$. Since $|w| = 1$, $\bar{w} = w^{-1}$, and

$$\begin{aligned} |w^i - w^j|^4 &= ((w^i - w^j)(w^{-i} - w^{-j}))^2 \\ &= (2 - w^{i-j} - w^{j-i})^2 \\ &= 4 + w^{2i-2j} + w^{2j-2i} + 2(-2w^{i-j} - 2w^{j-i} + 1) \\ &= 6 - 4w^{i-j} - 4w^{j-i} + w^{2i-2j} + w^{2j-2i}. \end{aligned}$$

But notice that, for a fixed value of i , over all values of j from 0 to 6, by properties of modular arithmetic with $w^7 = w^0$ (or expanding), each of the $w^\bullet$ -terms takes on every value in $w^0, w^1, w^2, \dots, w^6$ exactly once. Since $\sum_{j=0}^6 w^j = 0$,

$$\begin{aligned} \frac{1}{2} \sum_{i=0}^6 \sum_{j=0}^6 |w^i - w^j|^4 &= \frac{1}{2} \sum_{i=0}^6 \sum_{j=0}^6 (6 - 4w^{i-j} - 4w^{j-i} + w^{2i-2j} + w^{2j-2i}) \\ &= \frac{1}{2} \sum_{i=0}^6 \left(\sum_{j=0}^6 6 - 4 \sum_{j=0}^6 w^{i-j} - 4 \sum_{j=0}^6 w^{j-i} + \sum_{j=0}^6 w^{2i-2j} + \sum_{j=0}^6 w^{2j-2i} \right) \\ &= \frac{1}{2} \sum_{i=0}^6 \left(\sum_{j=0}^6 6 - 4 \sum_{j=0}^6 w^j - 4 \sum_{j=0}^6 w^j + \sum_{j=0}^6 w^j + \sum_{j=0}^6 w^j \right) \\ &= \frac{1}{2} \sum_{i=0}^6 \sum_{j=0}^6 6 \\ &= \frac{1}{2} \cdot 7 \cdot 7 \cdot 6 \\ &= \boxed{\text{(C) } 147}. \end{aligned}$$

Fun fact: This generalizes to any regular n -gon with $n \geq 3$ to obtain

$$\frac{1}{2} \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} 6 = 3n^2,$$

though for even n , w^{2i-2j} and w^{2j-2i} take on every value in $w^0, w^2, w^4, \dots, w^{n-2}$ exactly twice. These are the $\frac{n}{2}$ th roots of unity, so they still sum to 0.

This also explains why $n < 3$ doesn't follow the pattern. For $n = 2$ (one edge of length 2), since $w^{2i-2j} = (w^2)^{i-j} = 1^{i-j} = 1$ and similarly for w^{2j-2i} , the constant term in the summation becomes 8 instead of 6. For $n = 1$ (no edges), the constant term becomes 0.

-maxamc

Solution 9 (Inductive Reasoning)

This is how I solve this problem:

It's easy to solve for 3-gon, 4-gon, and 6-gon inscribed in a unit circle. (Okay, it's just the weird names for triangle, square, and hexagon)

For 3-gon, the sum is equal to 3 times the 4th power of an edge. Thus,

$$S_3 = 3 \cdot (\sqrt{3})^4 = 27.$$

For 4-gon, the sum is equal to 4 times the 4th power of an edge, and 2 times the 4th power of the diagonal. Thus,

$$S_4 = 4 \cdot (\sqrt{2})^4 + 2 \cdot (2)^4 = 48.$$

For 6-gon, the sum is equal to 6 times the 4th power of an edge, 6 times the 4th power of the short diagonal, and 3 times the 4th power of the long diagonal. Thus,

$$S_6 = 6 \cdot (1)^4 + 6 \cdot (\sqrt{3})^4 + 3 \cdot (2)^4 = 108.$$

Then, I quickly noticed that $27 = 3 \cdot 3^2$, $48 = 3 \cdot 4^2$, and $108 = 3 \cdot 6^2$. So reasonably, it will work out this formula, $S_n = 3n^2$. (This step is purely out of guessing, maybe have a look at Solution 8 for more info...)

By inductive reasoning, we got $S_7 = 3 \cdot 7^2 = \boxed{\text{(C)} 147}$.

- Prof. Joker

Solution 10 (cheese)

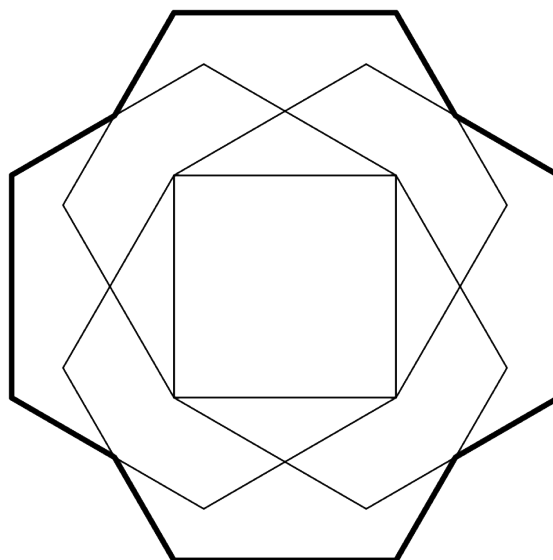
There are 7 segments with length $2 \sin \frac{\pi}{7}$, 7 segments with length $2 \sin \frac{2\pi}{7}$, and 7 segments with length $2 \sin \frac{3\pi}{7}$. The sum is $112(\sin^4 \frac{\pi}{7} + \sin^4 \frac{2\pi}{7} + \sin^4 \frac{3\pi}{7})$. Note that $\frac{3}{7} = 0.425871\dots$, so $\sin \frac{3\pi}{7} \approx \sin \frac{\pi}{2}$, but a little less (The actual value is approximately 0.975), and based on the answer choices, we can conclude that our answer will be a little greater than 1, so we make the educated guess of $\boxed{\text{(C)} 147}$

Supplement (Explanation of why $\cos(2\pi/7) + \cos(4\pi/7) + \cos(6\pi/7) = -1/2$) The sum of all roots of unity is 0, therefore, the sum of the real parts of the roots of unity is 0.

Consider the roots of unity of $x^n - 1 = 0$ where n is odd on the unit circle, the cosine (real parts) of all the roots under the x-axis is the same as the cosine of all the roots above the x-axis because of symmetry. Hence, 2 times the sum of the cosines of all the roots above the x-axis plus 1 is 0.

In the context of this problem, $2(\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7}) + 1 = 0$. Hence, $\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} = -\frac{1}{2}$

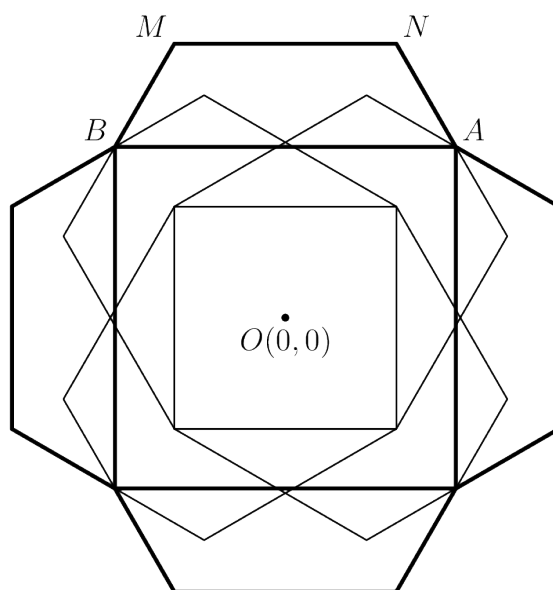
25. (AMC 12 12B 2022) Four regular hexagons surround a square with a side length 1, each one sharing an edge with the square, as shown in the figure below. The area of the resulting 12-sided outer nonconvex polygon can be written as $m\sqrt{n} + p$, where m , n , and p are integers and n is not divisible by the square of any prime. What is $m + n + p$?



- (A) -12 (B) -4 (C) 4 (D) 24 (E) 32

Answer: B

Solution 1 (Coord bash)



Refer to the diagram above.

Let the origin be at the center of the square, A be the intersection of the top and right hexagons, B be the intersection of the top and left hexagons, and M and N be the top points in the diagram.

By symmetry, A lies on the line $y = x$. The equation of line AN is $y = -x\sqrt{3} + \frac{3\sqrt{3}}{2} - \frac{1}{2}$ (due to it being one of the sides of the top hexagon). Thus, we can solve for the coordinates of A by finding the intersection of the two lines:

$$\begin{aligned} x &= -x\sqrt{3} + \frac{3\sqrt{3} - 1}{2} \\ x(\sqrt{3} + 1) &= \frac{3\sqrt{3} - 1}{2} \\ x &= \frac{3\sqrt{3} - 1}{2} \cdot \frac{1}{\sqrt{3} + 1} \\ &= \frac{3\sqrt{3} - 1}{2(\sqrt{3} + 1)} \cdot \frac{\sqrt{3} - 1}{\sqrt{3} - 1} \\ &= \frac{10 - 4\sqrt{3}}{4} \\ &= \frac{5}{2} - \sqrt{3} \\ \therefore A &= \left(\frac{5}{2} - \sqrt{3}, \frac{5}{2} - \sqrt{3} \right). \end{aligned}$$

This means that we can find the length AB , which is equal to $2(\frac{5}{2} - \sqrt{3}) = 5 - 2\sqrt{3}$. We will next find the area of trapezoid $ABMN$. The lengths of the bases are 1 and $5 - 2\sqrt{3}$, and the height is equal to the y -coordinate of M minus the y -coordinate of A . The height of the hexagon is $\sqrt{3}$ and the bottom of the hexagon lies on the line $y = \frac{1}{2}$. Thus, the y -coordinate of M is $\sqrt{3} - \frac{1}{2}$, and the height is $2\sqrt{3} - 3$. We can now find the area of the trapezoid:

$$[ABMN] = (2\sqrt{3} - 3) \left(\frac{1 + 5 - 2\sqrt{3}}{2} \right)$$

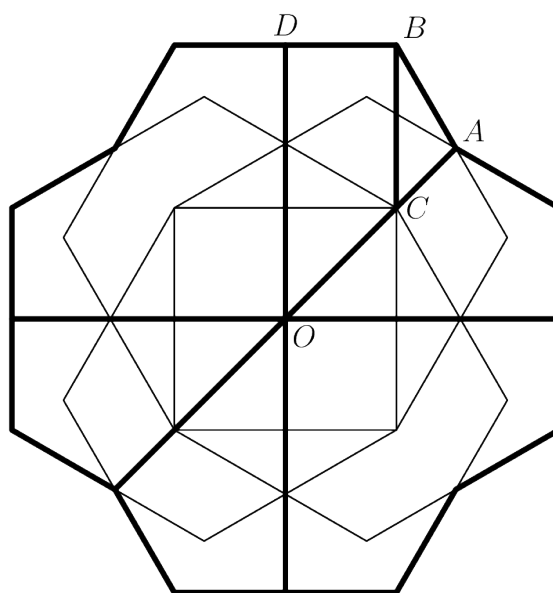
$$\begin{aligned}
 &= (2\sqrt{3} - 3)(3 - \sqrt{3}) \\
 &= 6\sqrt{3} + 3\sqrt{3} - 9 - 6 \\
 &= 9\sqrt{3} - 15.
 \end{aligned}$$

The total area of the figure is the area of a square with side length AB plus four times the area of this trapezoid:

$$\begin{aligned}
 \text{Area} &= (5 - 2\sqrt{3})^2 + 4(9\sqrt{3} - 15) \\
 &= 37 - 20\sqrt{3} + 36\sqrt{3} - 60 \\
 &= 16\sqrt{3} - 23.
 \end{aligned}$$

Our answer is $16 + 3 - 23 = \boxed{\text{(B)} - 4}$.

Solution 2



Begin by dividing the figure as shown above. Clearly, the entire figure has 8-fold symmetry. Therefore, we can calculate the area of $ODBA$ and multiply it by 8. We split $[ODBA]$ into $[ODBC] + [ABC]$.

Knowing the side length of the hexagon is 1, we can use 30-60-90 triangles within the hexagon to find the total distance between opposite edges is $2 \cdot \frac{\sqrt{3}}{2} = \sqrt{3}$. Thus, $OD = \sqrt{3} - \frac{1}{2}$ and $BC = \sqrt{3} - 1$. Recognizing $DB = \frac{1}{2}$ and $ODBC$ is a trapezoid,

$$8 \cdot [ODBC] = 2 \left(\sqrt{3} - \frac{1}{2} + \sqrt{3} - 1 \right) = 4\sqrt{3} - 3.$$

Next, we aim to find $[ABC]$. By angle chasing, we find $\angle A = 105^\circ$, $\angle B = 30^\circ$, and $\angle C = 45^\circ$. We can use the law of sines to find AB :

$$\frac{\sin(105^\circ)}{\sqrt{3} - 1} = \frac{\sin(45^\circ)}{AB} \implies AB = \frac{\sqrt{6} - \sqrt{2}}{2 \sin(105^\circ)}.$$

We may not know what $\sin(105^\circ)$ is by memory, but we can cleverly calculate it using a common trig identity:

$$\begin{aligned}
 \sin(105^\circ) &= \sin(60^\circ + 45^\circ), \\
 &= \sin(60^\circ)\cos(45^\circ) + \cos(60^\circ)\sin(45^\circ), \\
 &= \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2} + \frac{1}{2} \cdot \frac{\sqrt{2}}{2}, \\
 &= \frac{\sqrt{6} + \sqrt{2}}{4}.
 \end{aligned}$$

With some simplification, we'll find $AB = 4 - 2\sqrt{3}$. Now, we can easily calculate $8 \cdot [ABC]$ as

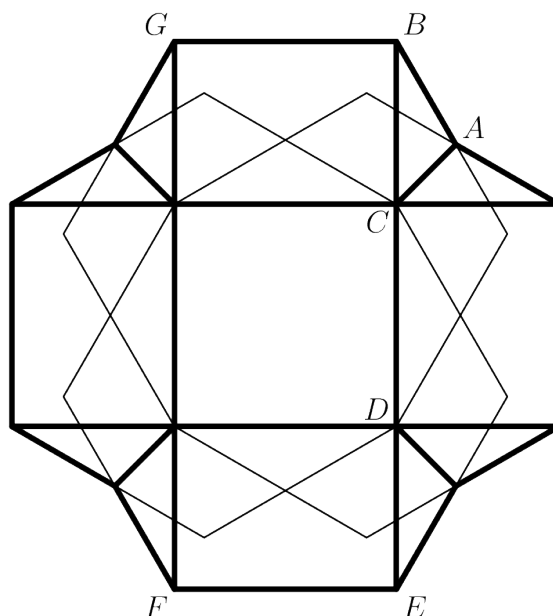
$$8 \cdot \frac{1}{2} \cdot (\sqrt{3} - 1)(4 - 2\sqrt{3}) \sin(30^\circ) = 12\sqrt{3} - 20.$$

Thus, the area of the dodecagon is $8 \cdot [ODBA] = 8 \cdot [ODBC] + 8 \cdot [ABC] = 4\sqrt{3} - 3 + 12\sqrt{3} - 20 = 16\sqrt{3} - 23$.

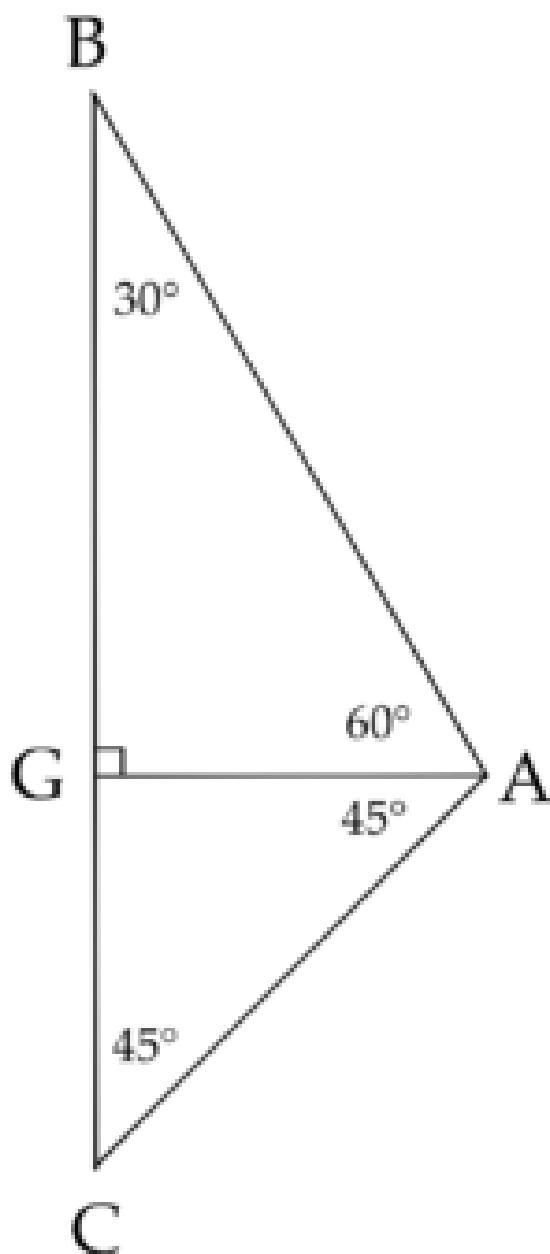
Finally, we find

$$16 + 3 - 23 = \boxed{\text{(B)} - 4}.$$

Solution 3 (Euclidean Geometry)



Divide this polygon into a 'cross' and 8 triangles like $\triangle ABC$. $BD = 2 \cdot \frac{\sqrt{3}}{2} = \sqrt{3}$, $BE = 2BD - CD = 2\sqrt{3} - 1$, $EF = 1$, $[BEFG] = 1 \cdot (2\sqrt{3} - 1) = 2\sqrt{3} - 1$, $[\text{cross}] = 2 \cdot [BEFG] - 1 = 2(2\sqrt{3} - 1) - 1 = 4\sqrt{3} - 3$



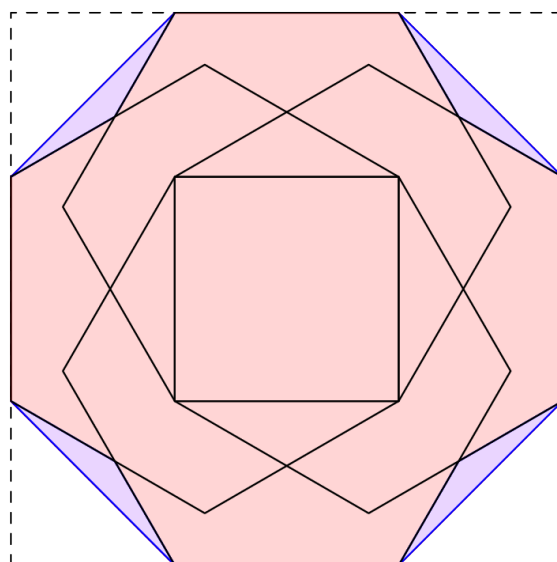
$\angle ABC = 120^\circ - 90^\circ = 30^\circ$, $\angle ACB = \frac{90^\circ}{2} = 45^\circ$, $\angle BAC = 180^\circ - 30^\circ - 45^\circ = 105^\circ$,
 $BC = BD - CD = \sqrt{3} - 1$. Draw a perpendicular line AG to side BC .

$BG + CG = BC$, $AG \cdot \sqrt{3} + AG = \sqrt{3} - 1$, $AG = \frac{\sqrt{3}-1}{\sqrt{3}+1} = \frac{(\sqrt{3}-1)^2}{2} = \frac{4-2\sqrt{3}}{2} = 2 - \sqrt{3}$,
 $[ABC] = \frac{1}{2} \cdot AG \cdot BC = \frac{1}{2}(2 - \sqrt{3})(\sqrt{3} - 1)$ [12-sided outer polygon] = [cross] + $8 \cdot$
 $[ABC] = 4\sqrt{3} - 3 + 8 \cdot \frac{1}{2}(2 - \sqrt{3})(\sqrt{3} - 1) = 4\sqrt{3} - 3 + 4(3\sqrt{3} - 5) = 16\sqrt{3} - 23$

$$16 + 3 - 23 = \boxed{\text{(B)} - 4}$$

Solution 4

We calculate the area as the area of the red octagon minus the four purple congruent triangles:



We first find the important angles in the figure. We note that 2 adjacent hexagons are rotated 90° with respect to the other, so the angles between any sides is 150° . In particular, as the purple triangles are isosceles, they have angles 150° , 15° , and 15° , and the octagon is equiangular (all its angles are 135°). Thus, we can draw a square around the octagon, and we note that the “cut out” triangles are all isosceles right triangles.

Now, we calculate the side length of the square. Note that the hexagon has a height of $\sqrt{3}$, so the length of a side of the square is $2\sqrt{3} - 1$. In particular, the horizontal/vertical sides of the octagon have length 1, so the legs of the isosceles triangles are

$$\frac{2\sqrt{3} - 1 - 1}{2} = \sqrt{3} - 1$$

Thus, the area of the octagon is

$$(2\sqrt{3} - 1)^2 - 4 \cdot \frac{1}{2}(\sqrt{3} - 1)^2 = 5$$

Now, we calculate the area of one of the four isosceles triangles. The base of the triangle is $(\sqrt{3} - 1)\sqrt{2}$, so the area is

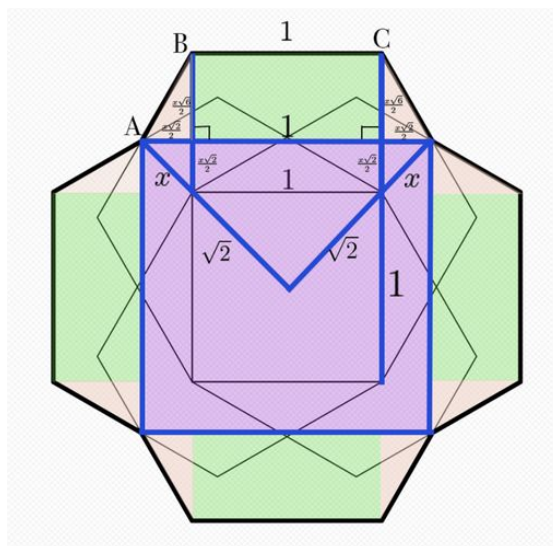
$$\frac{1}{4} \text{baselength}^2 \cdot \tan(\text{baseangle}) = \frac{1}{4}((\sqrt{3} - 1)\sqrt{2})^2 \cdot \tan 15^\circ = \frac{1}{4}(8 - 4\sqrt{3})(2 - \sqrt{3}) = 7 - 4\sqrt{3}$$

Thus, the area of the dodecagon is

$$5 - 4(7 - 4\sqrt{3}) = 16\sqrt{3} - 23$$

Thus the answer is $16 + 3 - 23 = -4$, or **(B)**.

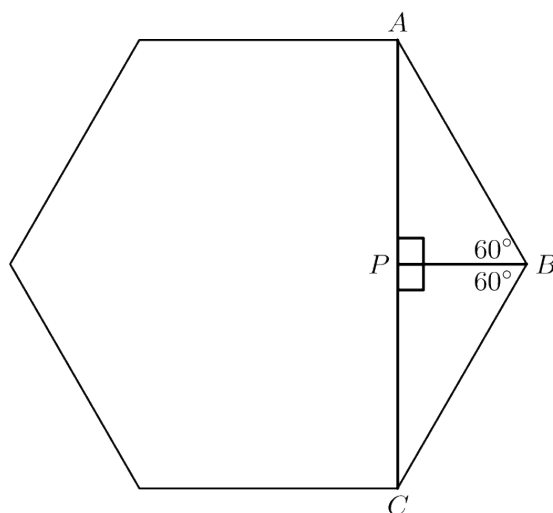
Solution 5 (No Trig, No Coordbash)



Note that each of the green sections is a rectangle, so its interior angles are all 90° . Since $\angle ABC = 120^\circ$, every one of the orange sections is a $30 - 60 - 90$ right triangle.

Define x to be the distance from the corner of the square with side length 1 to the corner of the larger blue square. Due to the sides of the two squares being parallel to each other, the large blue triangle is a $45 - 45 - 90$ right triangle. By *AAA* similarity, the smaller blue triangles are also $45 - 45 - 90$, and have side lengths of $\frac{x\sqrt{2}}{2}$, $\frac{x\sqrt{2}}{2}$, and x . By $30 - 60 - 90$ triangle relations, the largest altitude of the orange triangle is $\frac{x\sqrt{6}}{2}$.

Now, we can find the height of the hexagon to obtain an equation in terms of x . Consider a hexagon with side length 1, where point P is the foot of the perpendicular dropped from B , bisecting $\angle ABC$:



Note that triangles $\triangle APB$ and $\triangle CPB$ are congruent $30 - 60 - 90$ triangles, by *SAS* congruence. Since the side length of this hexagon is 1, the length of AP is $\frac{1}{2} \cdot \sqrt{3} = \frac{\sqrt{3}}{2}$, by $30 - 60 - 90$ triangle relations. The height of the hexagon is twice this value, or $\sqrt{3}$.

The height is also equal to the sum of the values along the long blue line, in the first diagram. Therefore, $1 + \frac{x\sqrt{2}}{2} + \frac{x\sqrt{6}}{2} = \sqrt{3}$. Solving and rationalizing, $x = 2\sqrt{2} - \sqrt{6}$.

The area of the dodecagon is equal to the sum of the areas of the four rectangles, eight orange triangles, and purple square. In terms of x , this is

$$8 \cdot \frac{1}{2} \cdot \frac{x\sqrt{2}}{2} \cdot \frac{x\sqrt{6}}{2} + 4 \cdot 1 \cdot \frac{x\sqrt{6}}{2} + (x\sqrt{2} + 1)^2.$$

Plugging in $x = 2\sqrt{2} - \sqrt{6}$, the area of the dodecagon is $16\sqrt{3} - 23$. Therefore, the answer is $16 + 3 - 23 = \boxed{\text{(B)} - 4}$.

-Benedict T (countmath1)

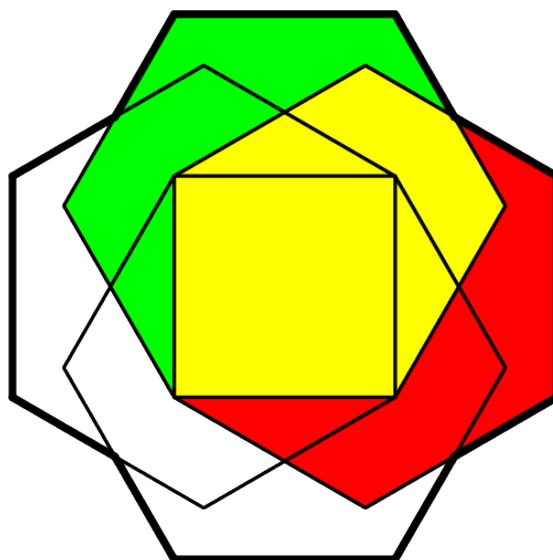
Solution 6 (PIE bash)

Obviously, each of the hexagon have area of $\frac{3\sqrt{3}}{2}$, which gives a total of $6\sqrt{3}$ (without accounting for the overcounted overlapped areas). Now we calculate the area of the intersection of two hexagons. There are two cases :

Case 1: The hexagons are on the opposite sides

In this case, we see that the two hexagons' intersection is the middle square plus two triangles on its side. The square have area 1. Each of the triangles are $120 - 30 - 30$, and hence its short side is $\frac{\sqrt{3}}{3}$ because its longer side is 1. Therefore, each of the small triangles have area of $\frac{\sqrt{3}}{12}$, so each of the intersections have $1 + \frac{\sqrt{3}}{6}$ area. Since there are 2 pairs of hexagons on opposite side of each other, that gives a total area of $2 + \frac{\sqrt{3}}{3}$.

Case 2: The hexagons are adjacent



In this case, their intersection is a convex hexagon. We cut it in half and we're left with a quadrilateral with two sides with length one (because it shares these sides with one of the hexagons, and for this reason the angle between the two sides is 120 degrees) and one short side and one long side. Here, we divide the area even further. We draw a line that divide the quadrilateral into a triangle with side lengths 1, 1, and $\sqrt{3}$ (because it is $30 - 30 - 120$) and another triangle with unknown side lengths. Simple angle chasing yields the triangle with unknown side lengths is a

15 – 75 – 90 triangle. If you remembered your sines and cosines, you know that $\sin 15 = \frac{\sqrt{6}-\sqrt{2}}{4}$ or you can calculate it by using the fact that $\sin 15 = \sin 45 - 30$. Also, $\sin 75 = \frac{\sqrt{6}+\sqrt{2}}{4}$. Now, because the 15 – 75 – 90 triangle have one side with length $\sqrt{3}$ (it shares the side with the other triangle) and it corresponds to the 75 degrees angle, we have

$$\frac{\sqrt{3}}{\frac{\sqrt{6}+\sqrt{2}}{4}} = \frac{x}{\frac{\sqrt{6}-\sqrt{2}}{4}}$$

$$\implies x = \frac{\sqrt{3}(\sqrt{6}-\sqrt{2})}{\sqrt{6}+\sqrt{2}} = 2\sqrt{3}-3$$

where x denote the shorter leg of the 15 – 75 – 90 triangle. Hence, that triangle have area

$$\frac{\sqrt{3}}{2}x = \frac{\sqrt{3}}{2}(2\sqrt{3}-3) = \frac{6-3\sqrt{3}}{2}$$

The isosceles 120 – 30 – 30 triangle obviously have area $\frac{\sqrt{3}}{4}$, and that gives a total of $\frac{12-5\sqrt{3}}{4}$. We multiply that by 2 because $\frac{12-5\sqrt{3}}{4}$ is only half of the intersection of the hexagons. We multiply the result by 4 because there are 4 pairs of adjacent hexagons. That gives $24-10\sqrt{3}$, which is it for Case 2.

Now that we got the total area of the intersection of two hexagons, we find the area of the intersection of three hexagons. Note that the intersection of any three hexagons is just the middle square plus one small triangle on the side. Again, the middle square have area 1 and the triangle have area $\frac{\sqrt{3}}{12}$, so the total is $1 + \frac{\sqrt{3}}{12}$. We multiply that by $\binom{4}{3}$ to get $4 + \frac{\sqrt{3}}{3}$.

Then, we calculate the area of the intersection of all four hexagons, which is just the middle square with area 1.

Finally, we apply PIE. The total area is

$$6\sqrt{3} - [(2 + \frac{\sqrt{3}}{3}) + (24 - 10\sqrt{3})] + (4 + \frac{\sqrt{3}}{3}) - 1 = 16\sqrt{3} - 23$$

so our final answer is $16 + 3 - 23 = \boxed{(B) - 4}$. ■ Ddk001

§8 AMC 12 12A 2022

1. (AMC 12 12A 2022) What is the value of

$$3 + \frac{1}{3 + \frac{1}{3 + \frac{1}{3}}}$$

- (A) $\frac{31}{10}$ (B) $\frac{49}{15}$ (C) $\frac{33}{10}$ (D) $\frac{109}{33}$ (E) $\frac{15}{4}$

Answer: D

Solution 1

We have

$$\begin{aligned}
 3 + \frac{1}{3 + \frac{1}{3 + \frac{1}{3}}} &= 3 + \frac{1}{3 + \frac{1}{\left(\frac{10}{3}\right)}} \\
 &= 3 + \frac{1}{3 + \frac{3}{10}} \\
 &= 3 + \frac{1}{\left(\frac{33}{10}\right)} \\
 &= 3 + \frac{10}{33} \\
 &= \boxed{\text{(D)} \frac{109}{33}}.
 \end{aligned}$$

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Solution 2

Continued fractions with integer parts q_i and numerators all 1 can be calculated as

$$\frac{[q_0, q_1, q_2, \dots, q_n]}{[q_1, q_2, \dots, q_n]}$$

where

$$\begin{aligned}
 [] &= 1 \\
 [q_0, q_1, q_2, \dots, q_n] &= q_0[q_1, q_2, \dots, q_n] + [q_2, \dots, q_n]
 \end{aligned}$$

Solution 3

Finite continued fractions of form $n + \frac{1}{n + \frac{1}{n + \dots}} = \frac{x}{y}$ have linear combinations of x, y that solve Pell's Equation. Specifically, the denominator y and numerator x are solutions to the Diophantine equation $(n^2 + 4)\left(\frac{y}{2}\right)^2 - \left(x - \frac{ny}{2}\right)^2 = \pm 1$. So for this problem in particular, the denominator y and numerator x are solutions to the Diophantine equation $13\left(\frac{y}{2}\right)^2 - \left(x - \frac{3y}{2}\right)^2 = \pm 1$. That leaves two answers. Since the number of 1's in the continued fraction is odd, we further narrow it down to $13\left(\frac{y}{2}\right)^2 - \left(x - \frac{3y}{2}\right)^2 = -1$, which only leaves us with 1 answer and that is

$$(x, y) = (109, 33) \text{ which means } \boxed{\text{(D)} \frac{109}{33}}.$$

(Note: Integer solutions increase exponentially, so our next solution will have a numerator greater than $3^2(109)$. Therefore, when you don't see numerators greater than $3^2(109)$ in the answer choices, this method should be fine.)

2. (AMC 12 12A 2022) The sum of three numbers is 96. The first number is 6 times the third number, and the third number is 40 less than the second number. What is the absolute value of the difference between the first and second numbers?
- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Answer: E

Solution 1

Let x be the third number. It follows that the first number is $6x$, and the second number is $x + 40$.

We have

$$6x + (x + 40) + x = 8x + 40 = 96,$$

from which $x = 7$.

Therefore, the first number is 42, and the second number is 47. Their absolute value of the difference is $|42 - 47| = \boxed{\text{(E)} 5}$.

Solution 2

Solve this using a system of equations. Let x, y , and z be the three numbers, respectively. We get three equations:

$$x + y + z = 96,$$

$$x = 6z,$$

$$z = y - 40.$$

Rewriting the third equation gives us $y = z + 40$, so we can substitute x as $6z$ and y as $z + 40$.

Therefore, we get

$$6z + (z + 40) + z = 96$$

$$8z + 40 = 96$$

$$8z = 56$$

$$z = 7.$$

Substituting 7 in for z gives us $x = 6z = 6(7) = 42$ and $y = z + 40 = 7 + 40 = 47$.

So, the answer is $|x - y| = |42 - 47| = \boxed{\text{(E)} 5}$.

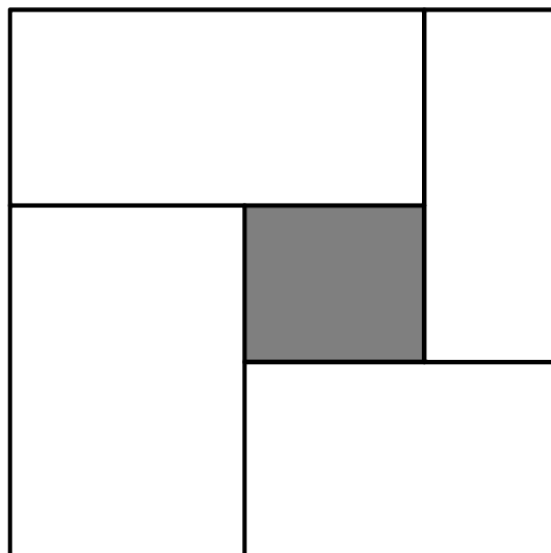
Solution 3

In accordance with Solution 2,

$$y = z + 40, x = 6z \implies |x - y| = |6z - z - 40| = 5|z - 8| \implies \boxed{\text{(E)} 5}.$$

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3. (*AMC 12 12A 2022*) Five rectangles, A , B , C , D , and E , are arranged in a square as shown below. These rectangles have dimensions 1×6 , 2×4 , 5×6 , 2×7 , and 2×3 , respectively. (The figure is not drawn to scale.) Which of the five rectangles is the shaded one in the middle?



- (A) A (B) B (C) C (D) D (E) E

Answer: B

Solution 1 (Area and Perimeter of Square)

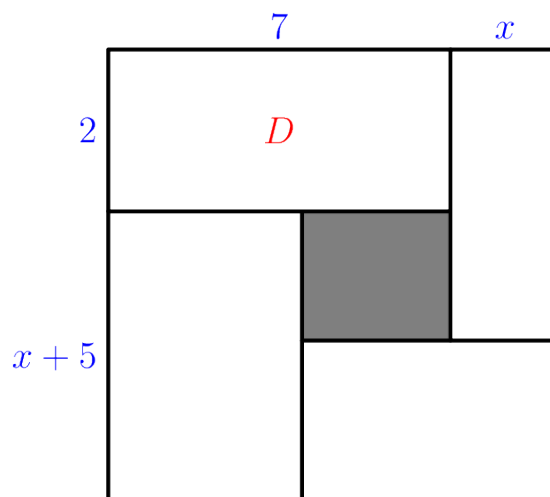
The area of this square is equal to $6+8+30+14+6 = 64$, and thus its side lengths are 8. The sum of the dimensions of the rectangles are $2+7+5+6+2+3+1+6+2+4 = 38$. Thus, because the perimeter of the square is 32, the rectangle on the inside must have a perimeter of $6 \cdot 2 = 12$. The only rectangle that works is **(B) B** .

Solution 2 (Perimeter of Square)

Note that the perimeter of the square is the sum of four pairs of dimensions (eight values are added). Adding up all dimensions gives us $2+7+5+6+2+3+1+6+2+4 = 38$. We know that as the square's side length is an integer, the perimeter must be divisible by 4. Testing out by subtracting all five pairs of dimensions from 38, only 2×4 works since $38 - 2 - 4 = 32 = 8 \cdot 4$, which corresponds with **(B) B** .

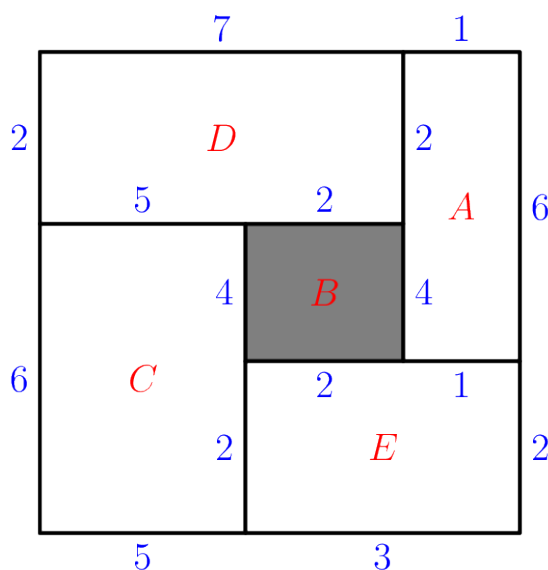
Solution 3 (Observations)

Note that rectangle D must be on the edge. Without loss of generality, let the top-left rectangle be D , as shown below:



It is clear that $x = 1$, so we can determine Rectangle A .

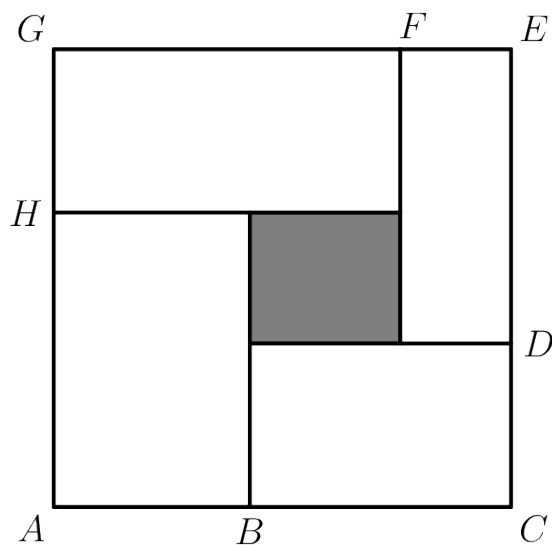
Continuing with a similar process, we can determine Rectangles C , E , and B , in this order. The answer is **(B)** B as shown below.



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Solution 4 (Observations)

Let's label some points:



By finding the dimensions of the middle rectangle, we need to find the dimensions of the other four rectangles. We have a rule:

$$AB + BC = CD + DE = EF + FG = GH + AH.$$

Let's make a list of all dimensions of the rectangles from the diagram. We have to fill in the dimensions from up above, but still apply to the rule:

$$AB \times AH$$

$$CD \times BC$$

$$EF \times DE$$

$$GH \times FG$$

By applying the rule, we get $AB = 2, BC = 6, CD = 5, DE = 3, EF = 2, FG = 6, GH = 1$, and $AH = 7$.

By substitution, we get this list

$$2 \times 7$$

$$5 \times 6$$

$$2 \times 3$$

$$1 \times 6$$

(This also tells us that the diagram is not drawn to scale.)

Notice how the only dimension not used in the list was 2×4 and that corresponds with B so the answer is, **(B) B**.

4. (AMC 12 12A 2022) The least common multiple of a positive integer n and 18 is 180, and the greatest common divisor of n and 45 is 15. What is the sum of the digits of n ?
- (A) 3 (B) 6 (C) 8 (D) 9 (E) 12

Answer: B

Solution 1

Note that

$$18 = 2 \cdot 3^2,$$

$$180 = 2^2 \cdot 3^2 \cdot 5,$$

$$45 = 3^2 \cdot 5$$

$$15 = 3 \cdot 5.$$

Let $n = 2^a \cdot 3^b \cdot 5^c$. It follows that:

- a) From the least common multiple condition, we have

$$\text{lcm}(n, 18) = \text{lcm}(2^a \cdot 3^b \cdot 5^c, 2 \cdot 3^2) = 2^{\max(a,1)} \cdot 3^{\max(b,2)} \cdot 5^{\max(c,0)} = 2^2 \cdot 3^2 \cdot 5,$$

from which $a = 2, b \in \{0, 1, 2\}$, and $c = 1$.

- b) From the greatest common divisor condition, we have

$$\text{gcd}(n, 45) = \text{gcd}(2^a \cdot 3^b \cdot 5^c, 3^2 \cdot 5) = 2^{\min(a,0)} \cdot 3^{\min(b,2)} \cdot 5^{\min(c,1)} = 3 \cdot 5,$$

from which $b = 1$.

Together, we conclude that $n = 2^2 \cdot 3 \cdot 5 = 60$. The sum of its digits is $6+0 = \mathbf{(B) 6}$.

Solution 2

The options for $\text{lcm}(x, 18) = 180$ are 20, 60, and 180. The options for $\text{gcd}(y, 45) = 15$ are 15, 30, 60, 75, etc. We see that 60 appears in both lists; therefore, $6+0 = \mathbf{(B) 6}$.

Remark If you ignore or mess up the LCM, and get $n = 15$, you'll still get the correct answer.

5. (AMC 12 12A 2022) The taxicab distance between points (x_1, y_1) and (x_2, y_2) in the coordinate plane is given by

$$|x_1 - x_2| + |y_1 - y_2|.$$

For how many points P with integer coordinates is the taxicab distance between P and the origin less than or equal to 20?

- (A) 441 (B) 761 (C) 841 (D) 921 (E) 924

Answer: C

Solution 1 (Arithmetic Series)

Let us consider the number of points for a certain x -coordinate. For any x , the viable points are in the range $[-20 + |x|, 20 - |x|]$. This means that our total sum is equal to

$$\begin{aligned} 1 + 3 + 5 + \cdots + 41 + 39 + 37 + \cdots + 1 &= (1 + 3 + 5 + \cdots + 39) + (1 + 3 + 5 + \cdots + 41) \\ &= 20^2 + 21^2 \\ &= 29^2 \\ &= \boxed{\text{(C) } 841}. \end{aligned}$$

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Solution 2 (Arithmetic Series With Symmetry)

Since the second point is the origin, this is equivalent to finding all points (x, y) such that $|x| + |y| \leq 20$. Due to the absolute values, the set of all such points will be symmetric about the origin meaning we can focus on the first quadrant and multiply by 4.

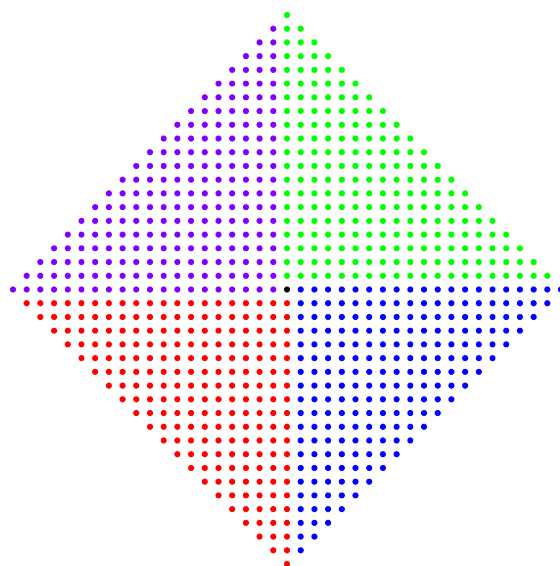
To avoid overcounts, ignore points on the axes. This means $x, y > 0$. If $x = 1$, there are 19 solutions for y ($y = 1, 2, 3, \dots, 19$). If $x = 2$, there are 18 solutions. This pattern repeats until $x = 19$, at which point there is 1 solution for y .

So we get $19 + 18 + 17 + \cdots + 1 = \frac{19(20)}{2} = 190$ points in the first quadrant. Multiplying by 4 gives 760. Now, the x axis has $y = 0$ which gives $|x| \leq 20$, meaning there are 41 solutions. This is the same with the y axis, but we overcounted the origin by 1.

Our final answer is $760 + 41 + 41 - 1 = \boxed{\text{(C) } 841}$.

Solution 3 (Triangular Numbers)

This solution refers to the **Diagram** section.

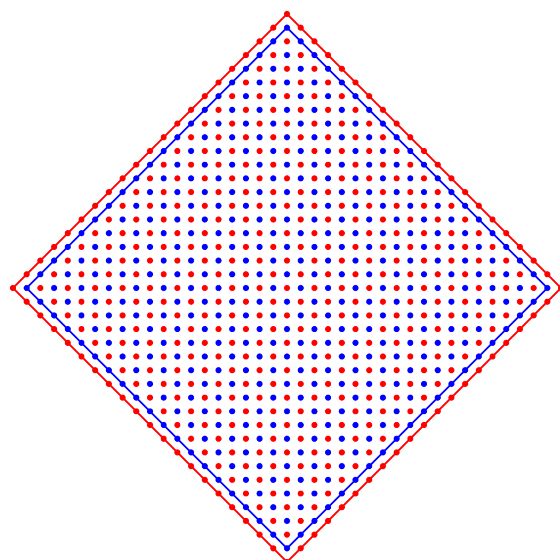


The problem can be visualized as depicted on the right split equally into four "triangular" parts excluding the origin. The "triangular" parts are identical the ones that would be used in a visual proof of the formula for triangular numbers. Becuase of this the number of points in each part is equal to $\frac{n(n+1)}{2}$ where n is the length of a "leg" of the "triangle" which is 20 for this problem. Substituting and computing, we get 210. Multiplying by 4 and adding 1 to account for all parts and the origin, we get $210 \cdot 4 + 1 = \boxed{\text{(C)} 841}$.

Solution 4 (Two Square Arrays)

This solution refers to the **Diagram** section.

As shown below, the taxicab distance between each red point and the origin is even, and the taxicab distance between each blue point and the origin is odd.



Note that the red array consists of $21^2 = 441$ points, and the blue array consists of $20^2 = 400$ points.

Together, the answer is $441 + 400 = \boxed{\text{(C)} 841}$.

Solution 5 (Pick's Theorem)

Let $P = (x, y)$. Since the problem asks for taxicab distances from the origin, we want $|x| + |y| \leq 20$. The graph of all solutions to this equation on the xy -plane is a square with vertices at $(0, \pm 20)$ and $(\pm 20, 0)$ (In order to prove this, one can divide the sections of this graph into casework on the four quadrants, and tie together the resulting branches.) We want the number of lattice points on the border of the square and inside the square. Each side of the square goes through an equal number of lattice points, so if we focus on one side going from $(0, 20)$ to $(20, 0)$, we can see that it goes through 21 points in total. In addition, each of the vertices gets counted twice, so the total number of border points is $21 \cdot 4 - 4 = 80$. Also, the area of the square is 800, so when we plug this information inside Pick's theorem, we get $800 = i + \frac{80}{2} - 1 \implies i = 761$. Then our answer is $761 + 80 = \boxed{\text{(C)} 841}$.

Solution 6 (Stars and Bars)

Instead of considering all points with integer coordinates, first consider points with nonnegative coordinates only. Then, we want $x + y \leq 20$ where x and y are nonnegative integers. We can introduce a third variable, say z , such that $z = 20 - (x + y)$. Note that counting the number of ways to have $x + y + z = 20$ is the same as counting the number of ways to have $x + y \leq 20$. Therefore, by stars and bars, there are $\binom{20 + 3 - 1}{3 - 1} = 231$ solutions with nonnegative integer coordinates.

Then, we can copy our solutions over to the other four quadrants. First, so as not to overcount, we remove all points on the axes. There are $20 + 20 + 1 = 41$ such points with nonnegative integer coordinates. We multiply the 190 remaining points by 4 to get 760 points that are not on the axes. Then, we can add back the 41 nonnegative points on the axes, as well as the 40 other points on the negative axes to get $760 + 41 + 40 = \boxed{\text{(C)} 841}$.

Solution 7 (Fast, Cheese)

The number of lattice points is roughly equal to the area of a rhombus with diagonals of length 41. The area of the quadrilateral is $41^2/2 = 840.5$. The closest answer is $\boxed{\text{(C)} 841}$.

6. (AMC 12 12A 2022) A data set consists of 6 (not distinct) positive integers: 1, 7, 5, 2, 5, and X . The average (arithmetic mean) of the 6 numbers equals a value in the data set. What is the sum of all positive values of X ?

(A) 10 (B) 26 (C) 32 (D) 36 (E) 40

Answer: D

Solution (Casework)

First, note that $1 + 7 + 5 + 2 + 5 = 20$. There are 3 possible cases:

Case 1: the mean is 5.

$$X = 5 \cdot 6 - 20 = 10.$$

Case 2: the mean is 7.

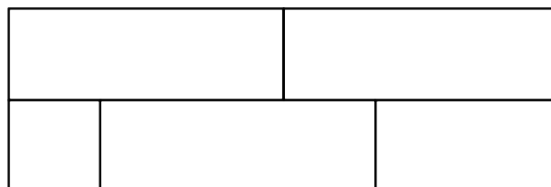
$$X = 7 \cdot 6 - 20 = 22.$$

Case 3: the mean is X .

$$X = \frac{20+X}{6} \Rightarrow X = 4.$$

Therefore, the answer is $10 + 22 + 4 = \boxed{\text{(D)} 36}$.

7. (AMC 12 12A 2022) A rectangle is partitioned into 5 regions as shown. Each region is to be painted a solid color - red, orange, yellow, blue, or green - so that regions that touch are painted different colors, and colors can be used more than once. How many different colorings are possible?



- (A) 120 (B) 270 (C) 360 (D) 540 (E) 720

Answer: D

Solution 1

The top left rectangle can be 5 possible colors. Then the bottom left region can only be 4 possible colors, and the bottom middle can only be 3 colors since it is next to the top left and bottom left. Similarly, we have 3 choices for the top right and 3 choices for the bottom right, which gives us a total of $5 \cdot 4 \cdot 3 \cdot 3 \cdot 3 = \boxed{\text{(D)} 540}$.

Note that the number of choices for a rectangle is $5 - n$ when there are n rectangles adjacent to it. GoldemGem4848

Solution 2 (Casework)

Case 1: All the rectangles are different colors. It would be $5! = 120$ choices.

Case 2: Two rectangles that are the same color. Grouping these two rectangles as one gives us $5 \cdot 4 \cdot 3 \cdot 2 = 120$. But, you need to multiply this number by three because the same-colored rectangles can be chosen at the top left and bottom right, the top right and bottom left, or the bottom right and bottom left, which gives us a grand total of 360.

Case 3: We have two sets of rectangles chosen from these choices (top right bottom left, top left bottom right) that have the same color. However, the choice of the bottom left and bottom right does not work for this case, as the second pair would be chosen from two touching rectangles. Again, grouping the same-colored rectangles gives us $5 \cdot 4 \cdot 3 = 60$.

Therefore, we have $120 + 360 + 60 = \boxed{\text{(D)} 540}$.

Solution 3 (Different Kind Of Casework)

Let us first choose the color of the bottom-middle rectangle, as it is touching all the other rectangles and has to be a different color than all of them. This rectangle has 5 possibilities. To make things easier for us, let's assume we chose the color red. Now, let's choose the color of the top-left rectangle. This has 4 options: orange, yellow, blue, or green (it can't be the same color as the middle rectangle, which we assumed was red). Let's assume this is orange. Now, we know that the bottom-left rectangle only has 3 choices: yellow, blue, or green. Let's assume this is colored yellow. Now, let's do casework on the color of the bottom right-rectangle. It either can be the same color as the top-left (orange), or different. CASE 1: (Its the same color) We then know that the bottom-right rectangle is also colored orange. We

now know that the top-left rectangle only has 3 choices: yellow, blue, or green (it can't be orange or red). The sum of the possibilities of this case is $5 \cdot 4 \cdot 3 \cdot 3$.

CASE 2: (Its a different color) We then know that the bottom-right rectangle has 3 options (yellow, blue, or green, since it can't be red, and it can't be orange since we already covered that in the case 1). Lets assume it is colored yellow. We know have 3 choices for the top-right rectangle (blue or green). The sum of the possibilities of this case is $5 \cdot 4 \cdot 3 \cdot 3 \cdot 2$

We can then sum up the possibilities of our two cases, which is $5 \cdot 4 \cdot 3 \cdot 3 + 5 \cdot 4 \cdot 3 \cdot 3 \cdot 2 = 5 \cdot 4 \cdot 3(3 + 3 \cdot 2) = 60 \cdot 9 = \boxed{\text{(D)} 540}$

8. (AMC 12 12A 2022) The infinite product

$$\sqrt[3]{10} \cdot \sqrt[3]{\sqrt[3]{10}} \cdot \sqrt[3]{\sqrt[3]{\sqrt[3]{10}}} \dots$$

evaluates to a real number. What is that number?

- (A) $\sqrt{10}$ (B) $\sqrt[3]{100}$ (C) $\sqrt[4]{1000}$ (D) 10 (E) $10\sqrt[3]{10}$

Answer: A

Solution 1

We can write $\sqrt[3]{10}$ as $10^{\frac{1}{3}}$. Similarly, $\sqrt[3]{\sqrt[3]{10}} = (10^{\frac{1}{3}})^{\frac{1}{3}} = 10^{\frac{1}{3^2}}$.

By continuing this, we get the form

$$10^{\frac{1}{3}} \cdot 10^{\frac{1}{3^2}} \cdot 10^{\frac{1}{3^3}} \dots,$$

which is

$$10^{\frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \dots}.$$

Using the formula for an infinite geometric series $S = \frac{a}{1-r}$, we get

$$\frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \dots = \frac{\frac{1}{3}}{1 - \frac{1}{3}} = \frac{1}{2}.$$

Thus, our answer is $10^{\frac{1}{2}} = \boxed{\text{(A)} \sqrt{10}}$.

- phuang1024

Solution 2

We can write this infinite product as L (we know from the answer choices that the product must converge):

$$L = \sqrt[3]{10} \cdot \sqrt[3]{\sqrt[3]{10}} \cdot \sqrt[3]{\sqrt[3]{\sqrt[3]{10}}} \dots$$

If we raise everything to the third power, we get:

$$L^3 = 10 \cdot \sqrt[3]{10} \cdot \sqrt[3]{\sqrt[3]{10}} \dots = 10L \implies L^3 - 10L = 0 \implies L \in \{0, \pm\sqrt{10}\}.$$

Since L is positive (as it is an infinite product of positive numbers), it must be that

$$L = \boxed{\text{(A)} \sqrt{10}}.$$

Solution 3

Move the first term inside the second radical. We get

$$\sqrt[3]{10} \cdot \sqrt[3]{\sqrt[3]{10}} \cdot \sqrt[3]{\sqrt[3]{\sqrt[3]{10}}} \cdots = \sqrt[3]{10 \sqrt[3]{10}} \cdot \sqrt[3]{\sqrt[3]{\sqrt[3]{10}}} \cdots$$

Do this for the third radical as well:

$$\sqrt[3]{10 \sqrt[3]{10}} \cdot \sqrt[3]{\sqrt[3]{\sqrt[3]{10}}} \cdots = \sqrt[3]{10 \sqrt[3]{10} \sqrt[3]{\sqrt[3]{10}}} \cdots = \sqrt[3]{10 \sqrt[3]{10 \sqrt[3]{10} \cdots}}.$$

It is clear what the pattern is. Setting the answer as P , we have

$$P = \sqrt[3]{10P},$$

from which $P = \boxed{\sqrt{10}}$.

Solution 4

Set the product equal to P . We get

$$\sqrt[3]{10} \cdot \sqrt[3]{\sqrt[3]{10}} \cdot \sqrt[3]{\sqrt[3]{\sqrt[3]{10}}} \cdots = P$$

Since this is an infinite product, there may exist a clever manipulation where we set two different expressions involving P equal, from which we could solve for a number. Since all terms are raised to the $\frac{1}{3}$ power, we can cube both sides of our equation. We would get

$$10 \cdot \sqrt[3]{10} \cdot \sqrt[3]{\sqrt[3]{10}} \cdot \sqrt[3]{\sqrt[3]{\sqrt[3]{10}}} \cdots = P^3$$

From this, we know that $\frac{P^3}{10} = P$, and to this equation, the only solution is $P = \boxed{\sqrt{10}}$

9. (AMC 12 12A 2022) On Halloween 31 children walked into the principal's office asking for candy. They can be classified into three types: Some always lie; some always tell the truth; and some alternately lie and tell the truth. The alternaters arbitrarily choose their first response, either a lie or the truth, but each subsequent statement has the opposite truth value from its predecessor. The principal asked everyone the same three questions in this order.

"Are you a truth-teller?" The principal gave a piece of candy to each of the 22 children who answered yes.

"Are you an alternater?" The principal gave a piece of candy to each of the 15 children who answered yes.

"Are you a liar?" The principal gave a piece of candy to each of the 9 children who answered yes.

How many pieces of candy in all did the principal give to the children who always tell the truth?

- (A) 7 (B) 12 (C) 21 (D) 27 (E) 31

Answer: A

Solution 1

Note that:

- Truth-tellers would answer yes-no-no to the three questions in this order.
- Liars would answer yes-yes-no to the three questions in this order.
- Alternaters who responded truth-lie-truth would answer no-no-no to the three questions in this order.
- Alternaters who responded lie-truth-lie would answer yes-yes-yes to the three questions in this order.

Suppose that there are T truth-tellers, L liars, and A alternaters who responded lie-truth-lie.

The conditions of the first two questions imply that

$$\begin{aligned} T + L + A &= 22, \\ L + A &= 15. \end{aligned}$$

Subtracting the second equation from the first, we have $T = 22 - 15 = \boxed{(A) 7}$.
Remark

The condition "31 children" and the third question are extraneous. However, they imply $A = 9$ and $L = 6$, from which there are $31 - T - L - A = 9$ alternaters who responded truth-lie-truth.

Solution 2

Consider when the principal asks "Are you a liar?": The truth tellers truthfully say no, and the liars lie and say no. This leaves only alternaters who lie on this question to answer yes. Thus, all 9 children that answered yes are alternaters that falsely answer Questions 1 and 3, and truthfully answer Question 2. The rest of the alternaters, however many there are, have the opposite behavior.

Consider the second question, "Are you an alternater?": The truth tellers again answer no, the liars falsely answer yes, and alternaters that truthfully answer also say yes. From the previous part, we know that 9 alternaters truthfully answer here. Because only liars and 9 alternaters answer yes, we can deduce that there are $15 - 9 = 6$ liars.

Consider the first question, "Are you a truth teller?": Truth tellers say yes, liars also say yes, and alternaters that lie on this question also say yes. From the first part, we know that 9 alternaters lie here. From the previous part, we know that there are 6 liars. Because only the number of truth tellers is unknown here, we can deduce that there are $22 - 9 - 6 = 7$ truth tellers.

The final question is how many pieces of candy did the principal give to truth tellers. Because truth tellers answer yes on only the first question, we know that all 7 of them said yes once, resulting in $\boxed{(A) 7}$ pieces of candy.

Solution 3 (case analysis)

There are T truthtellers, L liars, and A alternaters. We translate from logic to math via **formalization**.

From statement 3 we see that $A \geq 9$. From statement 2 we have that $A + L = 24$, and from statement 1 we have that $T + L = 13$.

We also know that $T + L + A = 31$. Having a system of three, we see that case 3 is redundant and unnecessary. Therefore, solving

$$T + L + A = 31$$

$$T + L = 13$$

$$A + L = 24$$

the solution $(T, A, L) = (\boxed{7}, 18, 6)$. This is exactly solution 1 and 2 except faster and mathematically intuitive rather than logically intuitive.

Solution 4 (logical deduction via answer choices)

Note that liars will always lie and truth-tellers will always tell the truth. Therefore, when he asks, "Are you a liar", only the alternators who answered "Yes" to question 1 will answer question 3 as "Yes". Therefore, we currently know 9 alternators exist.

Now when he asks, "Are you are an alternator", we know the remaining alternators and the liars will both say "Yes". Therefore, we know that the remaining alternators and liars answer yes to that. In addition, when he asks "Are you a truth-teller", all the 9 alternators will say yes, so we have 13 people, some of which are truth-tellers, and some of which are liars. This reduces our answer to either 7 or 12.

Now, say we have 7 truth-tellers. Then, that must mean we have 6 liars. Therefore, we have $9 + 9 = 18$ alternators, for a total of 31, which is satisfactory. On the other hand, if we have 12 truth-tellers, that must mean we have 1 liar, meaning we have $9 + 14 = 23$ alternators. That count is 36, which is a contradiction. The answer is $\boxed{7}$.

Remark (Fake Solve) This problem is broken in an interesting way that helps the test-taker. Since the true answers alternate for alternators, you can still get the correct answer if you misinterpret the problem as "alternators alternate their yes/no answers, but not their truth values". Suppose that A alternators answer yes-no-yes, and A' alternators answer no-yes-no. You still get $A = A' = 9$.

In addition, you can even get the correct answer with the misinterpretation "alternators have two types: alternators of the same type give the same arbitrary answers to the three questions, and alternators of different types give the opposite arbitrary answers to the three questions".

It's also notable that the misinterpretation makes the problem harder, so that the solution actually relies on all the information. This suggests that the question-writer may have been mistaken but got lucky.

10. (AMC 12 12A 2022) How many ways are there to split the integers 1 through 14 into 7 pairs such that in each pair, the greater number is at least 2 times the lesser number?

(A) 108 (B) 120 (C) 126 (D) 132 (E) 144

Answer: E

Solution 1 (Multiplication Principle)

Clearly, the integers from 8 through 14 must be in different pairs, so are the integers from 1 through 7. Note that 7 must pair with 14.

We pair the numbers 1, 2, 3, 4, 5, 6 with the numbers 8, 9, 10, 11, 12, 13 systematically:

- 6 can pair with either 12 or 13.
- 5 can pair with any of the three remaining numbers from 10, 11, 12, 13.
- 1, 2, 3, 4 can pair with the other four remaining numbers from 8, 9, 10, 11, 12, 13 without restrictions.

Together, the answer is $2 \cdot 3 \cdot 4! = \boxed{\text{(E)} 144}$.

Solution 2 (Multiplication Principle)

As said in Solution 1, clearly, the integers from 8 through 14 must be in different pairs.

We know that 8 or 9 can pair with any integer from 1 to 4, 10 or 11 can pair with any integer from 1 to 5, and 12 or 13 can pair with any integer from 1 to 6. Thus, 8 will have 4 choices to pair with, 9 will then have 3 choices to pair with (9 cannot pair with the same number as the one 8 pairs with). 10 cannot pair with the numbers 8 and 9 has paired with but can also now pair with 5, so there are 3 choices. 11 cannot pair with 8's, 9's, or 10's paired numbers, so there will be 2 choices for 11. 12 can pair with an integer from 1 to 5 that hasn't been paired with already, or it can pair with 6. 13 will only have one choice left, and 7 must pair with 14.

So, the answer is $4 \cdot 3 \cdot 3 \cdot 2 \cdot 2 \cdot 1 \cdot 1 = \boxed{\text{(E)} 144}$.

Solution 3 (Generalization)

The integers $x \in \{8, \dots, 14\}$ must each be the larger elements of a distinct pair.

Assign partners in decreasing order for $x \in \{7, \dots, 1\}$:

Note that 7 must pair with 14: **1 choice**.

For $5 \leq x \leq 7$, the choices are $\{2x, \dots, 14\} - \{\text{previous choices}\}$. As x decreases by 1, The minuend increases by 2 elements, and the subtrahend increases by 1 element, so the difference increases by 1, yielding **3! combined choices**.

After assigning a partner to 5, there are no invalid pairings for yet-unpaired numbers, so there are **4! ways** to choose partners for $\{1, 2, 3, 4\}$.

The answer is $3! \cdot 4! = \boxed{\text{(E)} 144}$.

In general, for $1, \dots, 2n$, the same logic yields answer: $\left\lfloor \frac{n}{2} \right\rfloor! \cdot \left\lceil \frac{n}{2} \right\rceil!$

11. (AMC 12 12A 2022) What is the product of all real numbers x such that the distance on the number line between $\log_6 x$ and $\log_6 9$ is twice the distance on the number line between $\log_6 10$ and 1?
- (A) 10 (B) 18 (C) 25 (D) 36 (E) 81

Answer: E

Solution 1

Let $a = 2 \cdot |\log_6 10 - 1| = |\log_6 9 - \log_6 x| = \left| \log_6 \frac{9}{x} \right|$.

$\pm a = \log_6 \frac{9}{x} \implies 6^{\pm a} = b^{\pm 1} = \frac{9}{x} \implies x = 9 \cdot b^{\pm 1} \cdot 9b^{-1} = \boxed{81}$.

Solution 2

First, notice that there must be two such numbers: one greater than $\log_6 9$ and one less than it. Furthermore, they both have to be the same distance away, namely $2(\log_6 10 - 1)$. Let these two numbers be $\log_6 a$ and $\log_6 b$. Because they are equidistant from $\log_6 9$, we have $\frac{\log_6 a + \log_6 b}{2} = \log_6 9$. Using log properties, this simplifies to $\log_6 \sqrt{ab} = \log_6 9$. We then have $\sqrt{ab} = 9$, so $ab = \boxed{\text{(E)} 81}$.

Solution 3 (Logarithmic Rules and Casework)

In effect we must find all x such that $|\log_6 9 - \log_6 x| = 2d$ where $d = \log_6 10 - 1$.

Notice that by log rules

$$d = \log_6 10 - 1 = \log_6 \frac{10}{6}$$

Using log rules again,

$$2d = 2 \log_6 \frac{10}{6} = \log_6 \frac{25}{9}$$

Now we proceed by casework for the distinct values of x .

Case 1

$$\log_6 9 - \log_6 x_1 = 2d$$

Subbing in for $2d$ and using log rules,

$$\log_6 \frac{9}{x_1} = \log_6 \frac{25}{9}$$

From this we may conclude that

$$\frac{9}{x_1} = \frac{25}{9} \implies x_1 = \frac{81}{25}$$

Case 2

$$\log_6 9 - \log_6 x_2 = -2d$$

Subbing in for $-2d$ and using log rules,

$$\log_6 \frac{9}{x_2} = \log_6 \frac{9}{25}$$

From this we conclude that

$$\frac{9}{x_2} = \frac{9}{25} \implies x_2 = 25$$

Finding the product of the distinct values, $x_1 x_2 = \boxed{\text{(E)} 81}$

12. (AMC 12 12A 2022) Let M be the midpoint of $\overline{AB}$ in regular tetrahedron $ABCD$. What is $\cos(\angle CMD)$?

(A) $\frac{1}{4}$ (B) $\frac{1}{3}$ (C) $\frac{2}{5}$ (D) $\frac{1}{2}$ (E) $\frac{\sqrt{3}}{2}$

Answer: B

Solution 1 (Right Triangles)

Without loss of generality, let the edge-length of $ABCD$ be 2. It follows that $MC = MD = \sqrt{3}$.

Let O be the center of $\triangle ABD$, so $\overline{CO} \perp \overline{MOD}$. Note that $MO = \frac{1}{3}MD = \frac{\sqrt{3}}{3}$.

In right $\triangle CMO$, we have

$$\cos(\angle CMD) = \frac{MO}{MC} = \boxed{\text{(B)} \frac{1}{3}}.$$

MRENTHUSIASM

Solution 2 (Law of Cosines)

Without loss of generality, let the edge-length of $ABCD$ be 2. It follows that $CM = DM = \sqrt{3}$.

By the Law of Cosines,

$$\cos(\angle CMD) = \frac{CM^2 + DM^2 - CD^2}{2(CM)(DM)} = \boxed{\text{(B)} \frac{1}{3}}.$$

jamesl123456

Solution 3 (Double Angle Identities)

As done above, let the edge-length equal 2 (usually better than 1 because we can avoid fractions when dropping altitudes). Notice that the triangle stated in the question has two side-lengths that are the altitudes of two equilateral triangles. By dropping the equilateral triangles' altitude and using 30° - 60° - 90° properties, we find that the other two sides are equal to $\sqrt{3}$. Now by dropping the main triangle's altitude, we see it equals $\sqrt{2}$ from the Pythagorean Theorem. we can use the Double Angle Identities for Cosine. Doing this, we obtain

$$\cos(\angle CMD) = \frac{2}{3} - \frac{1}{3} = \boxed{\text{(B)} \frac{1}{3}}.$$

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Solution 4 (Vector Methods)

Without loss of generality, let tetrahedron $ABCD$ lie in three-dimensional space with vertices $A(-1, 1, 1)$, $B(1, 1, -1)$, $C(1, -1, 1)$, and $D(-1, -1, -1)$. Let point M be located at $(0, 1, 0)$.

$$\vec{MD} = \langle -1, -2, -1 \rangle$$

$$\vec{MC} = \langle 1, -2, 1 \rangle$$

We know that the dot product of two vectors equals the product of their magnitudes multiplied by the cosine of the angle between them:

$$\vec{MD} \cdot \vec{MC} = (-1)(1) + (-2)(-2) + (-1)(1) = -1 + 4 - 1 = 2$$

Compute the magnitudes:

$$\|\vec{MD}\| = \sqrt{(-1)^2 + (-2)^2 + (-1)^2} = \sqrt{1 + 4 + 1} = \sqrt{6}$$

$$\|\vec{MC}\| = \sqrt{(1)^2 + (-2)^2 + (1)^2} = \sqrt{1 + 4 + 1} = \sqrt{6}$$

Then:

$$\vec{MD} \cdot \vec{MC} = \|\vec{MD}\| \cdot \|\vec{MC}\| \cdot \cos(\angle CMD)$$

$$2 = \sqrt{6} \cdot \sqrt{6} \cdot \cos(\angle CMD) = 6 \cos(\angle CMD)$$

Finally:

$$\cos(\angle CMD) = \frac{2}{6} = \boxed{\text{(B)} \frac{1}{3}}.$$

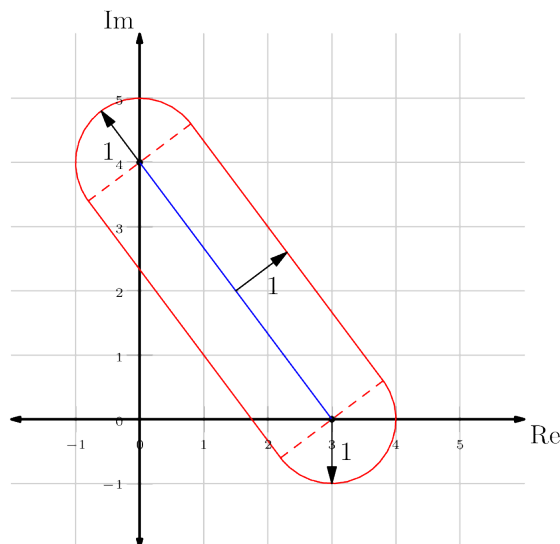
TylerTrikowsky

13. (AMC 12 12A 2022) Let $\mathcal{R}$ be the region in the complex plane consisting of all complex numbers z that can be written as the sum of complex numbers z_1 and z_2 , where z_1 lies on the segment with endpoints 3 and $4i$, and z_2 has magnitude at most 1. What integer is closest to the area of $\mathcal{R}$?

(A) 13 (B) 14 (C) 15 (D) 16 (E) 17

Answer: A

Solution



If z is a complex number and $z = a + bi$, then the magnitude (length) of z is $\sqrt{a^2 + b^2}$. Therefore, z_1 has a magnitude of 5. If z_2 has a magnitude of at most one, that means for each point on the segment given by z_1 , the bounds of the region $\mathcal{R}$ could be at most 1 away. Along the line, excluding the endpoints, a rectangle with a width of 2 and a length of 5, the magnitude, would be formed. At the endpoints, two semicircles will be formed with a radius of 1 for a total area of $\pi \approx 3$. Therefore, the total area is $5(2) + \pi \approx 10 + 3 = \boxed{\text{(A) } 13}$.

14. (AMC 12 12A 2022) What is the value of

$$(\log 5)^3 + (\log 20)^3 + (\log 8)(\log 0.25)$$

where $\log$ denotes the base-ten logarithm?

(A) $\frac{3}{2}$ (B) $\frac{7}{4}$ (C) 2 (D) $\frac{9}{4}$ (E) 3

Answer: C

Solution 1

Let $\log 2 = x$. The expression then becomes

$$(1 + x)^3 + (1 - x)^3 + (3x)(-2x) = \boxed{2}.$$

-bluelinfish

Solution 1.1

(Elaboration motivation behind Sol. 1)

Note that 5, 20, 8, and 0.25 are all products and quotients of exponents of 2 and 5, and the base of the logarithms is $10 = 2 \times 5$; this strongly hints at some sort of major simplification using the addition and subtraction rules of logarithms so we can convert all the different arguments of the logs into 1 common argument for easy algebra.

Note that we can write all of the following expressions in the following ways:

$$\begin{aligned}\log 5 &= \log \frac{10}{2} = \log 10 - \log 2 = 1 - \log 2 \\ \log 20 &= \log(2 \cdot 10) = \log 2 + \log 10 = \log 2 + 1 \\ \log 8 &= \log(2^3) = 3 \log 2 \\ \log 0.25 &= \log(2^{-2}) = -2 \log 2\end{aligned}$$

Thus, let $a = \log 2$, and proceed as in solution 1.

Solution 2

Using sum of cubes

$$\begin{aligned} & (\log 5)^3 + (\log 20)^3 \\ &= (\log 5 + \log 20)((\log 5)^2 - (\log 5)(\log 20) + (\log 20)^2) \\ &= 2((\log 5)^2 - (\log 5)(2 \log 2 + \log 5) + (2 \log 2 + \log 5)^2)\end{aligned}$$

Let $x = \log 5$ and $y = \log 2$, so $x + y = 1$

The entire expression becomes

$$\begin{aligned} & 2(x^2 - x(2y + x) + (2y + x)^2) - 6y^2 \\ &= 2(x^2 + 2xy + 4y^2 - 3y^2) \\ &= 2(x + y)^2 = \boxed{2}\end{aligned}$$

Solution 3 (Estimates)

We can estimate the solution. Using $\log(2) \approx 0.3$, $\log(20) = \log(10) + \log(2) = 1 + 0.3 \approx 1.3$, $\log(8) \approx 0.9$ and $\log(.25) = \log(1) - \log(4) = 0 - 0.6 \approx -0.6$, we have

Solution 4(log bash)

Using log properties, we combine the terms to make our expression equal to $\log((5^{\log^2 5}) \cdot (20^{\log^2 20}) \cdot 8^{\log \frac{1}{4}})$. By exponent properties, we separate the part with base 20 to become $20^{\log^2 5} \cdot 20^{\log^2 20 - \log^2 5}$. Then, we substitute this into the original expression to get $\log((5^{\log^2 5}) \cdot 20^{\log^2 5} \cdot 20^{\log^2 20 - \log^2 5} \cdot 8^{\log \frac{1}{4}}) = \log((100^{\log^2 5}) \cdot 20^{\log^2 20 - \log^2 5} \cdot 8^{\log \frac{1}{4}})$. Because $100^{\log^2 5} = 25^{\log^2 5}$, and $\log^2 20 - \log^2 5 = (\log 20 + \log 5)(\log 20 - \log 5) = \log 100 \cdot \log 4 = 2 \log 4$, this expression is equal to $\log(25^{\log^2 5} \cdot 400^{\log 4} \cdot 8^{\log \frac{1}{4}})$. We perform the step with the base combining on 25 and 400 to get $25^{\log^2 5} \cdot 400^{\log 4} = 25^{\log^2 5 - \log 4} \cdot 10000^{\log 4} = 25^{\log \frac{5}{4}} \cdot 256$. Putting this back into the whole equation gives $\log(25^{\log \frac{5}{4}} \cdot 256 \cdot 8^{\log \frac{1}{4}})$. One last base merge remains - but $25 \cdot 8$ isn't a power of 10. We can rectify this by converting $8^{\log \frac{1}{4}}$ to $(4^{\frac{3}{2}})^{\log \frac{1}{4}} = 4^{\log \frac{1}{8}}$. Finally, we complete this arduous process by performing the base merge on $\log(25^{\log \frac{5}{4}} \cdot 256 \cdot 4^{\log \frac{1}{8}})$. We get $25^{\log \frac{5}{4}} \cdot 4^{\log \frac{1}{8}} = 25^{\log \frac{5}{4} - \log \frac{1}{8}} \cdot 100^{\log \frac{1}{8}} = 25^{\log 10} \cdot \frac{1}{64} = \frac{25}{64}$. Putting this back into that original equation one last time, we get $\log(256 \cdot \frac{25}{64}) = \log 100 = \boxed{2}$. aop2014

15. (AMC 12 12A 2022) The roots of the polynomial $10x^3 - 39x^2 + 29x - 6$ are the height, length, and width of a rectangular box (right rectangular prism). A new rectangular box is formed by lengthening each edge of the original box by 2 units. What is the volume of the new box?

(A) $\frac{24}{5}$ (B) $\frac{42}{5}$ (C) $\frac{81}{5}$ (D) 30 (E) 48

Answer: D

Solution 1 (Vieta's Formulas)

Let a, b, c be the three roots of the polynomial. The lengthened prism's volume is

$$V = (a+2)(b+2)(c+2) = abc + 2ac + 2ab + 2bc + 4a + 4b + 4c + 8 = abc + 2(ab + ac + bc) + 4(a + b + c) + 8.$$

By Vieta's formulas, we know that a cubic polynomial $Ax^3 + Bx^2 + Cx + D$ with roots a, b, c satisfies:

$$\begin{aligned} a + b + c &= -\frac{B}{A} = \frac{39}{10}, \\ ab + ac + bc &= \frac{C}{A} = \frac{29}{10}, \\ abc &= -\frac{D}{A} = \frac{6}{10}. \end{aligned}$$

We can substitute these into the expression, obtaining

$$V = \frac{6}{10} + 2\left(\frac{29}{10}\right) + 4\left(\frac{39}{10}\right) + 8 = \boxed{\text{(D) } 30}.$$

- phuang1024

Solution 2 (Guessing Roots)

From the answer choices, we can assume the roots are rational numbers, and therefore this polynomial should be easily factorable. The coefficients of x must multiply to 10, so these coefficients must be 5, 2, 1 or 10, 1, in some order. We can try one at a time, and therefore write the factored form as follows:

$$(5x - p)(2x - q)(x - r).$$

Note that p, q, r have to multiply to 6, so they must be either 3, 2, 1 or 6, 1, 1 in some order. Again, we can try one at a time in different positions and see if they multiply correctly. We try $(5x - 2)(2x - 1)(x - 3)$ and multiply the x -terms, and sure enough they add up to 29. You can try to add up the x^2 terms and they add up to -39 . Therefore the roots are $\frac{2}{5}, \frac{1}{2}$ and 3. Now if you add 2 to each root, you get the volume is $\frac{12}{5} \cdot \frac{5}{2} \cdot 5 = 6 \cdot 5 = 30 = \boxed{\text{(D) } 30}$.

Solution 3 (Rational Root Theorem Bash)

We can find the roots of the cubic using the Rational Root Theorem, which tells us that the rational roots of the cubic must be in the form $\frac{p}{q}$, where p is a factor of the constant (-6) and q is a factor of the leading coefficient (10). Therefore, p is $\pm(1, 2, 3, 6)$ and q is $\pm(1, 2, 5, 10)$.

Doing Synthetic Division, we find that 3 is a root of the cubic:

$$\begin{array}{r|rrrr} 3 & 10 & -39 & 29 & -6 \\ & & 30 & -27 & 6 \\ \hline & 10 & -9 & 2 & 0 \end{array}.$$

Then, we have a quadratic $10x^2 - 9x + 2$. Using the Quadratic Formula, we can find the other two roots:

$$x = \frac{9 \pm \sqrt{(-9)^2 - 4(10)(2)}}{2 \cdot 10},$$

which simplifies to $x = \frac{1}{2}, \frac{2}{5}$.

To find the new volume, we add 2 to each of the roots we found:

$$(3 + 2) \cdot \left(\frac{1}{2} + 2\right) \cdot \left(\frac{2}{5} + 2\right).$$

Simplifying, we find that the new volume is (D) 30.

-MathWizard09

Solution 4

Let $P(x) = 10x^3 - 39x^2 + 29x - 6$, and let a, b, c be the roots of $P(x)$. The roots of $P(x - 2)$ are then $a + 2, b + 2, c + 2$, so the product of the roots of $P(x - 2)$ is the area of the desired rectangular prism.

$P(x - 2)$ has leading coefficient 10 and constant term $P(0 - 2) = P(-2) = 10(-2)^3 - 39(-2)^2 + 29(-2) - 6 = -300$.

Thus, by Vieta's Formulas, the product of the roots of $P(x - 2)$ is $\frac{-(-300)}{10} =$
(D) 30.

-Orange_Quail_9

Solution 5

Let $P(x) = 10x^3 - 39x^2 + 29x - 6$. This can be factored as $P(x) = 10(x - a)(x - b)(x - c)$, where a, b , and c are the roots of $P(x)$. We want $V = (a + 2)(b + 2)(c + 2)$.

"Luckily" $P(-2) = 10(-2 - a)(-2 - b)(-2 - c) = -10V$. $P(-2) = -300$, giving $V =$
(D) 30.

-Oxymoronic15

(It's not just lucky. If $P(x)$ has roots $x \in \{r_i\}$, $Q(x) = P(x - 2)$ has roots $y \in \{r_i + 2\}$. By Vieta, the product V of the roots is the negation of the constant term divided by the leading coefficient 10, which is $-Q(0)/10$, which is $-P(0 - 2)/10$. -oinava)

Solution 6 (Very Nice Vieta Solve)

If we let the roots of the cubic polynomial be p, q, r , then we can write what we want in terms of p, q, r . So, we have

$$(p+2)(q+2)(r+2) = pqr + 2pq + 2qr + 4q + 2pr + 4p + 4r + 8 = pqr + 2(pq + qr + pr) + 4(q + p + r) + 8.$$

Now, we can just use Vieta's to finish up from here! We have $pqr = -(-\frac{6}{10}) = \frac{3}{5}$.

Now, we can find the sum of the pairwise roots which is just $pq + qr + pr = \frac{29}{10}$

and then multiply by 2 which gives $\frac{29}{5}$. Looking good so far! Now, we need to find

the sum of the roots which is just $p + q + r = \frac{39}{10}$ and then multiply by 4 which

gives $\frac{78}{5}$. Almost there! Now, we just add 8 to that sum, and then sum it all up to

get (D) 30

-jb2015007

Note: This is the same as Solution 1.

16. (AMC 12 12A 2022) A triangular number is a positive integer that can be expressed in the form $t_n = 1 + 2 + 3 + \cdots + n$, for some positive integer n . The three smallest triangular numbers that are also perfect squares are $t_1 = 1 = 1^2$, $t_8 = 36 = 6^2$, and $t_{49} = 1225 = 35^2$. What is the sum of the digits of the fourth smallest triangular number that is also a perfect square?

(A) 6 (B) 9 (C) 12 (D) 18 (E) 27

Answer: D

Solution 1(Pell Equation)

We have $t_n = \frac{n(n+1)}{2}$. If t_n is a perfect square, then it can be written as $\frac{n(n+1)}{2} = k^2$, where k is a positive integer.

Thus, $n(n+1) = 2k^2$. Rearranging, we get $(2n+1)^2 - 2(2k)^2 = 1$, a Pell equation (see https://artofproblemsolving.com/wiki/index.php/Pell_equation). So $\frac{2n+1}{2k}$ must be a truncation of the continued fraction for $\sqrt{2}$:

$$\begin{aligned} 1 + \frac{1}{2} &= \frac{2 \cdot 1 + 1}{2 \cdot 1} \\ 1 + \frac{1}{2 + \frac{1}{2}} &= \frac{2 \cdot 8 + 1}{2 \cdot 6} \\ 1 + \frac{1}{2 + \frac{1}{2 + \frac{1}{2}}} &= \frac{2 \cdot 49 + 1}{2 \cdot 35} \\ 1 + \frac{1}{2 + \frac{1}{2 + \frac{1}{2 + \frac{1}{2}}}} &= \frac{2 \cdot 288 + 1}{2 \cdot 204} \end{aligned}$$

Therefore, $t_{288} = \frac{288 \cdot 289}{2} = 204^2 = 41616$, so the answer is $4+1+6+1+6 = \boxed{\text{(D) } 18}$.

- Steven Chen (Professor Chen Education Palace, www.professorchenedu.com)

Edited by wzs26843545602 Edited by dad

Solution 1a (Alt. Pell)

As in Solution 1, we find that $(2n+1)^2 - 8k^2 = 1$. Let $q = 2n+1$; applying the difference of squares, we see that $(q - 2\sqrt{2}k)(q + 2\sqrt{2}k) = 1$. For this Pell equation, note that $q + 2\sqrt{2}k$ and $q - 2\sqrt{2}k$ are both algebraic integers that are conjugates of each other. WLOG let $z = q - 2\sqrt{2}k$ so that $\bar{z} = q + 2\sqrt{2}k$; we know that $z\bar{z} = 1$. It is clear that $\overline{ab} = \bar{a}\bar{b}$, and it immediately follows that $(z^n)(\bar{z}^n) = 1$. In essence, for a solution z to $z\bar{z} = 1$, then any integer power of z is also a solution. This is particularly useful in this problem: we are given that $n = 1 \implies q = 3$ and $k = 1$ is a solution, so $z = 3 - 2\sqrt{2}$. From our result, we know that $z^2 = (3 - 2\sqrt{2})^2 = 17 - 2\sqrt{2} \cdot 6$ is also a solution; this particular solution is $n = 8$ and $k = 6$, which is the second smallest desired number. Similarly, z^3 gives us the third-smallest desired number, so z^4 must give us our desired answer. We can calculate: $z^4 = (17 - 2\sqrt{2} \cdot 6)^2 = 289 + 288 - 2 \cdot 17 \cdot 6 \cdot 2\sqrt{2} = 577 - 2\sqrt{2} \cdot 204$.

This lets $n = 288$ and $k = 204$; we thus know that $T_{288} = 204^2 = 41616$, which has digit sum (D) 18. QED. $\square$

Solution 2 (Bash)

As mentioned above, $t_n = \frac{n(n+1)}{2}$. If t_n is a perfect square, one of two things must occur when the fraction is split into a product. Either $\frac{n}{2}$ and $n+1$ must both be squares, or n and $\frac{n+1}{2}$ must both be squares, and thus the search for the next perfect square triangular number can be narrowed down by testing values of n that are close to or are perfect squares. After some work, we reach $n = 288$, 1 less than 289, and $t_{288} = \frac{288 \cdot 289}{2} = 144 \cdot 289 = 41616$. This product is a perfect square, and thus the sum of the digits of the fourth smallest perfect square triangular number is therefore $4 + 1 + 6 + 1 + 6 = \span style="border: 1px solid black; padding: 2px;">(D) 18.$

Solution 3

According to the problem, we want to find integer p such $\frac{n(n+1)}{2} = p^2$, after expanding, we have $n^2 + n = 2p^2$, $4n^2 + 4n = 8p^2$, $(2n+1)^2 - 8p^2 = 1$, we call $2n+1 = q$, the equation becomes $q^2 - 8p^2 = 1$, obviously $(q, p) = (3, 1)$ is the elementary solution for this pell equation, thus the forth smallest solution set $q_4 + 2\sqrt{2}p_4 = (3 + 2\sqrt{2})^4 = 577 + 408\sqrt{2}$, which indicates $p = 204$, $p^2 = 41616$ leads to 18

Solution 4

If $\binom{n}{2}$ is a square, then $\binom{(2n-1)^2}{2}$ is also a square. We can prove this quite simply:

$$\begin{aligned} & \binom{(2n-1)^2}{2} \\ &= \frac{(2n-1)^2 \cdot ((2n-1)^2 - 1)}{2} \\ &= \frac{(2n-1)^2 \cdot (2n \cdot (2n-2))}{2} \\ &= (2n-1)^2 \cdot 4 \binom{n}{2}. \end{aligned}$$

Therefore, $\binom{(2n-1)^2}{2}$ is a square. Note that $T_n = \binom{n+1}{2}$. We can easily check all smaller possibilities using a bit of casework, and they don't work. Our solution is thus $\binom{289}{2} = 204^2 = 41616$, and so the answer is 18.

Solution 5

We want to find integer n_i and m_i such that $t_{n_i} = \frac{n_i(n_i+1)}{2} = m_i^2$, $n_0 = 0$.

We use the formula $\sqrt{n_{i+1}} = \sqrt{2n_i} + \sqrt{n_i + 1}$ and get

$$\begin{aligned} n_1 &= (\sqrt{2n_0} + \sqrt{n_0 + 1})^2 = (0 + 1)^2 = 1, \\ n_2 &= (\sqrt{2n_1} + \sqrt{n_1 + 1})^2 = (\sqrt{2} + \sqrt{1 + 1})^2 = 8, \\ n_3 &= (\sqrt{2n_2} + \sqrt{n_2 + 1})^2 = (\sqrt{16} + \sqrt{8 + 1})^2 = 49, \\ n_4 &= (\sqrt{2n_3} + \sqrt{n_3 + 1})^2 = (\sqrt{98} + \sqrt{49 + 1})^2 = ((7 + 5)\sqrt{2})^2 = 288, \\ n_5 &= (\sqrt{2n_4} + \sqrt{n_4 + 1})^2 = (\sqrt{576} + \sqrt{289})^2 = (24 + 17)^2 = 1681, \\ n_6 &= (\sqrt{2n_5} + \sqrt{n_5 + 1})^2 = (41\sqrt{2} + \sqrt{1682})^2 = ((41 + 29)\sqrt{2})^2 = 9800, \dots \end{aligned}$$

Therefore, $t_{n_4} = t_{288} = \frac{288 \cdot 289}{2} = 41616 \implies 4 + 1 + 6 + 1 + 6 = \span style="border: 1px solid black; padding: 2px;">(D) 18$

- Pell's equation (simple solutions)

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Solution 6(Fast and Simple)

This is an easy one to tackle by just noticing a pattern. Write out all the perfect squares before 300, and then notice that 4 and 9 are what are the factors of the 8th term. 25 and 49 are factors of the 49th term. 4 multiplied by 2 is one less than 9. 25 multiplied by 2 is one more than 49. Well, then we keep on writing out the perfect squares and then hunt for any that can be multiplied by two and then has an absolute value of 1 valued difference. 144 and 289 satisfy this. 288th term would give us these factors. By the way, this isn't very rigorous.

No formula needed.

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17. (AMC 12 12A 2022) Suppose a is a real number such that the equation

$$a \cdot (\sin x + \sin(2x)) = \sin(3x)$$

has more than one solution in the interval $(0, \pi)$. The set of all such a that can be written in the form

$$(p, q) \cup (q, r),$$

where p, q , and r are real numbers with $p < q < r$. What is $p + q + r$?

- (A) -4 (B) -1 (C) 0 (D) 1 (E) 4

Answer: A

Solution 1

We are given that $a \cdot (\sin x + \sin(2x)) = \sin(3x)$

Using the sine double angle formula combine with the fact that $\sin(3x) = \sin x \cdot (4\cos^2 x - 1)$, which can be derived using sine angle addition with $\sin(2x + x)$, we have

$$a \cdot (\sin x + 2\sin x \cos x) = \sin x \cdot (4\cos^2 x - 1)$$

Since $\sin x \neq 0$ as it is on the open interval $(0, \pi)$, we can divide out $\sin x$ from both sides, leaving us with

$$a \cdot (1 + 2\cos x) = 4\cos^2 x - 1$$

Now, distributing a and rearranging, we achieve the equation

$$4\cos^2 x - 2a\cos x - (1 + a) = 0$$

which is a quadratic in $\cos x$.

Applying the quadratic formula to solve for $\cos x$, we get

$$\cos x = \frac{2a \pm \sqrt{4a^2 + 4(4)(1 + a)}}{8}$$

and expanding the terms under the radical, we get

$$\cos x = \frac{2a \pm \sqrt{4a^2 + 16a + 16}}{8}$$

Factoring, since $4a^2 + 16a + 16 = (2a + 4)^2$, we can simplify our expression even further to

$$\cos x = \frac{a \pm (a + 2)}{4}$$

Now, solving for our two solutions, $\cos x = -\frac{1}{2}$ and $\cos x = \frac{a+1}{2}$.

Since $\cos x = -\frac{1}{2}$ yields a solution that is valid for all a , that being $x = \frac{2\pi}{3}$, we must now solve for the case where $\frac{a+1}{2}$ yields a valid value.

As $x \in (0, \pi)$, $\cos x \in (-1, 1)$, and therefore $\frac{a+1}{2} \in (-1, 1)$, and $a \in (-3, 1)$.

There is one more case we must consider inside this interval though, the case where $\frac{a+1}{2} = -\frac{1}{2}$, as this would lead to a double root for $\cos x$, yielding only one valid solution for x . Solving for this case, $a \neq -2$.

Therefore, combining this fact with our solution interval, $a \in (-3, -2) \cup (-2, 1)$, so the answer is $-3 - 2 + 1 = \boxed{\text{(A)} -4}$.

- DavidHovey

Solution 2

We can optimize from the step from

$$a \cdot (1 + 2 \cos x) = 4 \cos^2 x - 1$$

in solution 1 by writing

$$a = \frac{4 \cos^2 x - 1}{1 + 2 \cos x} = 2 \cos x - 1$$

and then get

$$\cos x = \frac{a + 1}{2}.$$

Now, solving for our two solutions, $\cos x = -\frac{1}{2}$ and $\cos x = \frac{a+1}{2}$.

Since $\cos x = -\frac{1}{2}$ yields a solution that is valid for all a , that being $x = \frac{2\pi}{3}$, we must now solve for the case where $\frac{a+1}{2}$ yields a valid value.

As $x \in (0, \pi)$, $\cos x \in (-1, 1)$, and therefore $\frac{a+1}{2} \in (-1, 1)$, and $a \in (-3, 1)$.

There is one more case we must consider inside this interval though, the case where $\frac{a+1}{2} = -\frac{1}{2}$, as this would lead to a double root for $\cos x$, yielding only one valid solution for x . Solving for this case, $a \neq -2$.

Therefore, combining this fact with our solution interval, $a \in (-3, -2) \cup (-2, 1)$, so the answer is $-3 - 2 + 1 = \boxed{\text{(A)} -4}$.

- Dan

Remark: If you get to $a \in (-3, 1)$, then you can observe from the answer choices that q must be $-2, 1$, or something greater than 1. Then you can guess that $q = -2$ since q can't be 1.

Solution 3

Use the sum to product formula to obtain $2a \cdot \sin \frac{3x}{2} \cos \frac{x}{2} = \sin 3x$. Use the double angle formula on the RHS to obtain $a \cdot \sin \frac{3x}{2} \cos \frac{x}{2} = \sin \frac{3x}{2} \cos \frac{3x}{2}$. From here, it is obvious that $x = \frac{2\pi}{3}$ is always a solution, and thus we divide by $\sin \frac{3x}{2}$ to get

$$a \cdot \cos \frac{x}{2} = \cos \frac{3x}{2}$$

We wish to find all a such that there is at least one more solution to this equation distinct from $x = \frac{2\pi}{3}$. Letting $y = \cos \frac{x}{2}$, and noting that $\cos \frac{3x}{2} = 4y^3 - 3y$, we can rearrange our equation to $4y^3 = y(3 + a)$. The smallest value x where $y = 0$ is π , which is not in our domain so we divide by y to obtain $4y^2 = a + 3$. By the trivial inequality, $a + 3 \geq 0$. Furthermore, $y \neq 0$, so $a + 3 > 0$. Also, if $a = -2$, then the solution to this equation would be shared with $x = \frac{2\pi}{3}$, so there would only be one distinct solution. Finally, because $y \leq 1$ due to the restrictions of a sine wave, and that $y \neq 1$ due to the restrictions on x , we have $-3 < a < 1$ with $a \neq -2$. Thus, $p = -3, q = -2, r = 1$, so our final answer is $-3 + (-2) + 1 = \boxed{\text{(A)} -4}$.

18. (AMC 12 12A 2022) Let T_k be the transformation of the coordinate plane that first rotates the plane k degrees counterclockwise around the origin and then reflects the plane across the y -axis. What is the least positive integer n such that performing the sequence of transformations $T_1, T_2, T_3, \dots, T_n$ returns the point $(1, 0)$ back to itself?

(A) 359 (B) 360 (C) 719 (D) 720 (E) 721

Answer: A

Solution 1 (Polar Coords)

Let $P = (r, \theta)$ be a point in polar coordinates, where θ is in degrees.

Rotating P by k° counterclockwise around the origin gives the transformation $(r, \theta) \rightarrow (r, \theta + k^\circ)$. Reflecting P across the y -axis gives the transformation $(r, \theta) \rightarrow (r, 180^\circ - \theta)$. Note that

$$\begin{aligned} T_k(P) &= (r, 180^\circ - \theta - k^\circ), \\ T_{k+1}(T_k(P)) &= (r, \theta - 1^\circ). \end{aligned}$$

We start with $(1, 0^\circ)$ in polar coordinates. For the sequence of transformations $T_1, T_2, T_3, \dots, T_k$, it follows that

- After T_1 , we have $(1, 179^\circ)$.
- After T_2 , we have $(1, -1^\circ)$.
- After T_3 , we have $(1, 178^\circ)$.
- After T_4 , we have $(1, -2^\circ)$.
- After T_5 , we have $(1, 177^\circ)$.
- After T_6 , we have $(1, -3^\circ)$.
- ...
- After T_{2k-1} , we have $(1, 180^\circ - k^\circ)$.
- After T_{2k} , we have $(1, -k^\circ)$.

The least such positive integer k is 180. Therefore, the least such positive integer n is $2k - 1 = \boxed{\text{(A)} 359}$.

Solution 2

Note that since we're reflecting across the y -axis, if the point ever makes it to $(-1, 0)$ then it will flip back to the original point. Note that after T_1 the point will be 1 degree clockwise from the negative x -axis. Applying T_2 will rotate it to be 1

degree counterclockwise from the negative x -axis, and then flip it so that it is 1 degree clockwise from the positive x -axis. Therefore, after every 2 transformations, the point rotates 1 degree clockwise. To rotate it so that it will rotate 179 degrees clockwise will require $179 \cdot 2 = 358$ transformations. Then finally on the last transformation, it will rotate on to $(-1, 0)$ and then flip back to its original position. Therefore, the answer is $358 + 1 = 359 = \boxed{\text{(A) } 359}$.

Solution 3

So, you don't know polar coordinates (Solution 1) nor came up with Solution 2's logic, then, here we go!

The easy solution is to notice that, without reflections and taking $k = 1$, there are 360 turns that need to be made.

Now take $k = 2$. We see that there are 180 rotations, so the number of rotations is just $360/k$.

Now include the reflections over the y axis.

Say that each time we reflect a point over the y axis, we draw a line.

Each time we reflect over the y axis, we see more and more lines that generate. For $k = 1$, there are quite obviously 360 of these lines.

However, one can see that this works for every single case, because when we reflect it over the y axis and make a k° rotation, we fill up all the missed gaps until we reach the end of a sequence - I prove this on section 1.1.

But wait! We overcounted by one line! Notice how I said there were 360 lines, but the line $x = 1$ intersects $y = 0$ is already given to you, and that is when $T_k = 0$, which is the point $(1, 0)$! So our final answer is $360 - 1 = \boxed{\text{(A) } 359}$.

Proof A way you can prove such an assumption is by taking ideas like addition. We know that eventually there exists a number N such that $l \times N = m$, where m has a units digit of l . Utilizing the same approach, we can find that this actually works.

Solution 4

In degrees:

Starting with $n = 0$, the sequence goes $0 \rightarrow 179 \rightarrow 359 \rightarrow 178 \rightarrow 358 \rightarrow 177 \rightarrow 357 \rightarrow \dots$

We see that it takes 2 steps to downgrade the point by 1° . Since the 1st point in the sequence is 179, the answer is $1 + 2(179) = \boxed{\text{(A) } 359}$.

Solution 5 (Simple)

We can consider the rotations and reflections separately. For the rotations, each rotation turns it by the next natural number. Thus the total number of degrees turned would be a triangle number. We test the smallest number, 359 first, and we get that it turns $\frac{(1+359)359}{2} = 180n$, where n is an integer. Thus, the point would be rotated to $(-1, 0)$. We may be tempted to dismiss this option but we haven't considered the reflections. Each reflection acts as a 180° rotation, so every two reflections cancel. However, 359 is odd so we have to reflect $(-1, 0)$, taking us to $(1, 0)$, which is what we want. Thus we get $\boxed{\text{(A) } 359}$.

Solution 6 (Complex)

Rotations and reflections in 2D are very nice to describe using complex numbers in polar form. Reflections and rotations also don't affect the length (modulus or absolute value) of a complex number. This motivates us to set $z = \exp(i\theta)$, starting with $z = 1 + 0i = \exp(i0)$. Then, we can describe rotations and reflections about the y -axis using these formulas:

$$\text{Rotation}(\exp(i\theta), k) : \exp(i\theta) \mapsto \exp(i(\theta + k))$$

$$\text{Reflection}(\exp(i\theta)) : \exp(i\theta) \mapsto \exp(i(180^\circ - \theta)) = -\exp(i(-\theta))$$

If we apply two successive iterations, we see a simplification:

$$T_k(\exp(i\theta)) = -\exp(-i(\theta + k))$$

$$\begin{aligned} T_{k+1}(T_k(\exp(i\theta))) &= T_{k+1}(-\exp(i(-\theta - k))) \\ &= -(-\exp(-i(-\theta - k + (k + 1)))) \\ &= \exp(i(\theta - 1)) \end{aligned}$$

We also can calculate $T_1(\exp(i0)) = \exp(179^\circ)$. Thus, the point $(1, 0)$ gets sent back to when all double iterations after 1 cancel 179° . $179 - \left(\frac{n-1}{2}\right) = 0$ so $n = 1 + 2 \cdot 179 = \boxed{\text{(A) } 359}$.

Solution 7 (Matrices)

Similar to the above solution, one can use rotation matrices to compute it.

$$\left(\text{i.e. } \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \right)$$

For the reflection, use the reflection matrix (i.e. $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$)

Then proceed like Solution 5.

Solution 8 (Very simple and quick)

Let's test a few small cases for T_k . When $k = 1$, the point will move $(1 + 180)^\circ$. When $k = 2$, the point will move $(2 + 180)^\circ$. We can see that T_k will result in the point moving $(k + 180)^\circ$ from its previous location. Since we want the point to return to itself, the total number of degrees it has rotated must be a multiple of 360° . This means $T_1 + T_2 + T_3 + \dots + T_k \equiv 0 \pmod{360}$. Using the formula for the sum of the first k integers, this can be expressed as

$$180k + \frac{k(k+1)}{2} \equiv 0 \pmod{360}$$

Testing the answer choices gives $\boxed{\text{(A) } 359}$.

19. (AMC 12 12A 2022) Suppose that 13 cards numbered $1, 2, 3, \dots, 13$ are arranged in a row. The task is to pick them up in numerically increasing order, working repeatedly from left to right. In the example below, cards $1, 2, 3$ are picked up on the first pass, 4 and 5 on the second pass, 6 on the third pass, $7, 8, 9, 10$ on the fourth pass, and $11, 12, 13$ on the fifth pass. For how many of the $13!$ possible orderings of the cards will the 13 cards be picked up in exactly two passes?

7 11 8 6 4 5 9 12 1 13 10 2 3

- (A) 4082 (B) 4095 (C) 4096 (D) 8178 (E) 8191

Answer: D

Solution 1 (Casework)

For $1 \leq k \leq 12$, suppose that cards $1, 2, \dots, k$ are picked up on the first pass. It follows that cards $k + 1, k + 2, \dots, 13$ are picked up on the second pass.

Once we pick the spots for the cards on the first pass, there is only one way to arrange all 13 cards.

For each value of k , there are $\binom{13}{k} - 1$ ways to pick the k spots for the cards on the first pass: We exclude the arrangement

1 2 3 4 5 6 7 8 9 10 11 12 13

in which the cards are arranged such that the first pass consists of all 13 cards.

Therefore, the answer is

$$\sum_{k=1}^{12} \left[\binom{13}{k} - 1 \right] = \left[\sum_{k=1}^{12} \binom{13}{k} \right] - 12 = \left[\sum_{k=0}^{13} \binom{13}{k} \right] - 14 = 2^{13} - 14 = \boxed{\text{(D) } 8178}.$$

Solution 2 (Casework)

Since the 13 cards are picked up in two passes, the first pass must pick up the first n cards and the second pass must pick up the remaining cards m through 13. Also note that if m , which is the card that is numbered one more than n , is placed before n , then m will not be picked up on the first pass since cards are picked up in order. Therefore we desire m to be placed before n to create a second pass, and that after the first pass, the numbers m through 13 are lined up in order from least to greatest.

To construct this, n cannot go in the n th position because all cards 1 to $n - 1$ will have to precede it and there will be no room for m . Therefore n must be in slots $n + 1$ to 13. Let's do casework on which slot n goes into to get a general idea for how the problem works.

Case 1: With n in spot $n + 1$, there are n available slots before n , and there are $n - 1$ cards preceding n . Therefore the number of ways to reserve these slots for the $n - 1$ cards is $\binom{n}{n-1}$. Then there is only 1 way to order these cards (since we want them in increasing order). Then card m goes into whatever slot is remaining, and the $13 - m$ cards are ordered in increasing order after slot $n + 1$, giving only 1 way. Therefore in this case there are $\binom{n}{n-1}$ possibilities.

Case 2: With n in spot $n + 2$, there are $n + 1$ available slots before n , and there are $n - 1$ cards preceding n . Therefore the number of ways to reserve slots for these cards are $\binom{n+1}{n-1}$. Then there is one way to order these cards. Then cards m and $m + 1$ must go in the remaining two slots, and there is only one way to order them since they must be in increasing order. Finally, cards $m + 2$ to 13 will be ordered in increasing order after slot $n + 1$, which yields 1 way. Therefore, this case has $\binom{n+1}{n-1}$ possibilities.

I think we can see a general pattern now. With n in slot x , there are $x - 1$ slots to distribute to the previous $n - 1$ cards, which can be done in $\binom{x-1}{n-1}$ ways. Then the remaining cards fill in in just 1 way. Since the cases of n start in slot $n + 1$ and end in slot 13, this sum amounts to:

$$\binom{n}{n-1} + \binom{n+1}{n-1} + \binom{n+2}{n-1} + \cdots + \binom{12}{n-1}$$

for any n .

Hmmm ... where have we seen this before?

We use wishful thinking to add a term of $\binom{n-1}{n-1}$:

$$\binom{n-1}{n-1} + \binom{n}{n-1} + \binom{n+1}{n-1} + \binom{n+2}{n-1} + \cdots + \binom{12}{n-1}$$

This is just the hockey stick identity! Applying it, this expression is equal to $\binom{13}{n}$. However, we added an extra term, so subtracting it off, the total number of ways to order the 13 cards for any n is

$$\binom{13}{n} - 1$$

Finally, to calculate the total for all n , we sum from $n = 0$ to 13. This yields us:

$$\begin{aligned} \sum_{n=0}^{13} \left(\binom{13}{n} - 1 \right) &\implies \sum_{n=0}^{13} \binom{13}{n} - \sum_{n=0}^{13} 1 \\ &\implies 2^{13} - 14 = 8192 - 14 = 8178 = \boxed{\text{(D)} 8178}. \end{aligned}$$

Solution 3 (Recursion)

To solve this problem, we can use recursion on n . Let A_n be the number of arrangements for n numbers. Now, let's look at how these arrangements are related to A_{n-1} by case work on the first number a_1 .

If $a_1 = 1$, the remaining $n - 1$ numbers from 2 to n are arranged in the same way just like number 1 to $n - 1$ in the case of $n - 1$ numbers. So there are A_{n-1} arrangements.

If $a_1 = 2$, then we need to choose 1 position from position 2 to $n - 1$ to put 1, and all remaining numbers must be arranged in increasing order, so there are $\binom{n-1}{1}$ such arrangements.

If $a_1 = k$, then we need to choose $k - 1$ positions from position 2 to $n - 1$ to put $1, 2, \dots, k - 1$, and all remaining numbers must be arranged in increasing order, so there are $\binom{n-1}{k-1}$ such arrangements.

So we can write

$$A_n = A_{n-1} + \binom{n-1}{1} + \binom{n-1}{2} + \cdots + \binom{n-1}{n-1}$$

which can be simplified to

$$A_n = A_{n-1} + 2^{n-1} - 1$$

Replace each card by its label, A or B , giving a row of x A 's and $13 - x$ B 's. For each such row of labels, there is exactly 1 way to assign the numbers back onto the cards: inherently, the A -positions must read $1, 2, \dots, x$ from left to right and the B -positions must read $x + 1, \dots, 13$ from left to right. So the arrangements we want correspond one-to-one with the label rows, and the number of label rows is

$$\frac{13!}{x!(13-x)!} = \binom{13}{x},$$

choosing which x of the 13 positions get an A . However, there is an arrangement with all the A cards in a row followed by all the B cards. Since order is fixed to be increasing, such a row reconstructs to $1, 2, \dots, 13$, which takes 1 pass instead of 2, meaning we must subtract one case per value of x , or 12.

Since arrangements of A and B between different values of x are disjoint, we can sum without worrying about overcounting:

$$\binom{13}{1} + \binom{13}{2} + \dots + \binom{13}{12} - 12.$$

The sum looks awfully similar to the Pascal's Triangle row-sum identity:

$$\sum_{k=0}^{13} \binom{13}{k} = \binom{13}{0} + \binom{13}{1} + \dots + \binom{13}{13} = 2^{13}$$

With a little bit of wishful thinking and rearranging, we get

$$\begin{aligned} \binom{13}{0} + \binom{13}{1} + \dots + \binom{13}{13} - \binom{13}{0} - \binom{13}{13} - 12 &= 2^{13} - \binom{13}{0} - \binom{13}{13} - 12 \\ \binom{13}{1} + \binom{13}{2} + \dots + \binom{13}{12} - 12 &= 2^{13} - \binom{13}{0} - \binom{13}{13} - 12 \\ &= 8192 - 1 - 1 - 12 \\ &= \boxed{\text{(D) } 8178}. \end{aligned}$$

20. (AMC 12 12A 2022) Isosceles trapezoid $ABCD$ has parallel sides $\overline{AD}$ and $\overline{BC}$, with $BC < AD$ and $AB = CD$. There is a point P in the plane such that $PA = 1, PB = 2, PC = 3$, and $PD = 4$. What is $\frac{BC}{AD}$? (A) $\frac{1}{4}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) $\frac{2}{3}$ (E) $\frac{3}{4}$

Answer: B

Solution 1 (Reflections + Ptolemy's Theorem)

Consider the reflection

21. (AMC 12 12A 2022) Let

$$P(x) = x^{2022} + x^{1011} + 1.$$

Which of the following polynomials is a factor of $P(x)$?

- (A) $x^2 - x + 1$ (B) $x^2 + x + 1$ (C) $x^4 + 1$ (D) $x^6 - x^3 + 1$ (E) $x^6 + x^3 + 1$

Answer: E

Solution 1

$P(x) = x^{2022} + x^{1011} + 1$ is equal to $\frac{x^{3033}-1}{x^{1011}-1}$ by difference of powers.

Therefore, the answer is a polynomial that divides $x^{3033} - 1$ but not $x^{1011} - 1$.

Note that any polynomial $x^m - 1$ divides $x^n - 1$ if and only if m is a factor of n .

The prime factorizations of 1011 and 3033 are $3 * 337$ and $3^2 * 337$, respectively.

Hence, $x^9 - 1$ is a divisor of $x^{3033} - 1$ but not $x^{1011} - 1$.

By difference of powers, $x^9 - 1 = (x^3 - 1)(x^6 + x^3 + 1)$. Therefore, the answer is $\boxed{E}$.

Solution 2

We simply test roots for each, as 2022, 1011 are multiples of three, we need to make sure the roots are in the form of $e^{i\frac{k\pi}{9}}$, so we only have to look at D, E .

If we look at choice E , $x = e^{i\frac{\pm 2\pi}{9}}$ which works perfectly, the answer is just E .

Solution 3

Let $x^{1011} = u$, now we can rewrite our polynomial as $u^2 + u + 1$. Using the quadratic formula to solve for the roots of this polynomial, we have

$$x^{1011} = \frac{-1 \pm i\sqrt{3}}{2}$$

Looking at our answer choices, we want to find a polynomial whose roots satisfy this expression. Since the expression $x^6 + x^3 + 1$ is in a similar form to our original polynomial, except with x^3 in place of x^{1011} , this would be a good place to start. Solving for the roots of x^3 in a similar fashion,

$$x^3 = \frac{-1 \pm i\sqrt{3}}{2}$$

for the solution we are testing. Now notice that we can rewrite the roots of x^3 as

$$x^3 = \text{cis } \frac{2\pi}{3}, \text{cis } \frac{4\pi}{3}$$

Both of which are third roots of unity. We want to now check if this value of x^3 satisfies $x^{1011} = \frac{-1 \pm i\sqrt{3}}{2}$. Notice that $x^{1011} = (x^3)^{112 \cdot 3} \cdot x^3$, and since both values of x^3 are roots of unity, we can simplify the expression we want satisfied to the expression to $x^{1011} = x^3$. Since both values of x^3 are also values of x^{1011} , the roots for our $x^6 + x^3 + 1$ are also roots of $x^{2022} + x^{1011} + 1$, meaning that

$$x^6 + x^3 + 1 | x^{2022} + x^{1011} + 1$$

so Therefore, the answer is $\boxed{E}$.

- DavidHovey

Solution 4 (Describe the Roots)

We know that a monic polynomial q divides a monic polynomial p if and only if all the roots of q are roots of p . Since

$$P(x) = x^{2022} + x^{1011} + 1 = \frac{x^{3033} - 1}{x^{1011} - 1}$$

, the roots of P are the 3033rd roots of unity that aren't 1011th roots of unity.

Now, note that:

- 1: The roots of polynomial A are the primitive 6th roots of unity.
- 2: The roots of polynomial B are the primitive cube roots of unity.
- 3: The roots of polynomial C are the primitive 8th roots of unity.
- 4: The roots of polynomial D are the primitive 18th roots of unity.
- 5: The roots of polynomial E are the primitive 9th roots of unity.

However, since 6, 8, and 18 don't divide 3033, the roots of polynomial A are not all 3033rd roots of unity, and the same is true for polynomials C and D , eliminating choices A , C and D . Also, since 3 divides 1011, the roots of polynomial B are all 1011th roots of unity, eliminating choice B . That leaves choice $\boxed{E}$, and we can confirm that this is correct by noticing that 9 divides 3033 but not 1011. From that, we can see that the roots of polynomial E are 3033rd roots of unity but not 1011th roots of unity, so they are all roots of P . Therefore, E divides P .

Solution 5 (Simple Elimination)

Put the value $x = -1$. This gives $P(-1) = (-1)^{2022} + (-1)^{1011} + 1 = 1$. This automatically eliminates choices A , C and D since they do not form a factor of 1 at $x = -1$.

Solution 5.1: Now, put $x = \omega$ (omega) at choice B . We get that,

$$\omega^2 + \omega + 1 = 0$$

Thus choice B has $(x - \omega)$ as a factor. If choice B were to be a factor of $P(x)$, $(x - \omega)$ would also have to be a factor of $P(x)$, which is clearly not the case, as $P(\omega) \neq 0$. This eliminates choice B , leaving us with answer $\boxed{E}$.

Solution 5.2 (Mods): Plug in 2 for x at choice B . $2^2 + 2 \cdot 2 + 1 = 7$. The cycle of mods for 2 and 7 is 2, 4, 1. Both 2022 and 1011 are divisible by 3, so we have $1 + 1 + 1 = 3$. 3 is not divisible by 7, so we eliminate choice B . This leaves us with $\boxed{(E)x^6 + x^3 + 1}$.

Solution 5.2 Bread 10

Solution 5a (Elimination but slightly different)

Like Solution 5, let $x = -1$ which eliminates the choices of A , C , and D as they do not divide $P(-1) = 1$ as they form 3, 2, 3 respectively by letting $x = -1$.

This leaves us with only 2 choices, B and E . Notice that letting $x = 0$ or $x = 1$ still make these answer choices work and the other values will leave large numbers for us to check which is not feasible in a 75 minutes math competition.

However, we notice answer choice B is quadratic so if answer choice B divides the given polynomial, then the roots of the quadratic must also be roots of the polynomial.

Through quadratic formula, we find the roots of this quadratic as $\frac{-1 + i\sqrt{3}}{2}$ and $\frac{-1 - i\sqrt{3}}{2}$.

We notice that these roots can be written nicely in polar form $\cos(\frac{2\pi}{3}) + i\sin(\frac{2\pi}{3})$ or $\cos(\frac{4\pi}{3}) + i\sin(\frac{4\pi}{3})$.

We plug either one of these (preferably the root on the left as it is smaller) and see that the polynomial doesn't equal 0 suggesting that B is not the correct answer choice.

As we only have one answer choice left, we choose $\boxed{E}$

Solution 5b (elimination but even faster and cheesier)

We first substitute 1 and -1 and eliminate A , C , and D . We are now left with B and E . When $x = 3$, $B = 13$; we then calculate $3^{2022} + 3^{1011} + 1 \pmod{13}$. Note that 2022 and 1011 are both multiples of 3, and $3^3 = 27 \equiv 1 \pmod{13}$. Thus, $3^{2022} \equiv 3^{1011} \equiv 3^3 \equiv 1 \pmod{13}$, so it turns out that the given expression is $1 + 1 + 1 = 3 \pmod{13}$. We need it to be 0, and since $3 \not\equiv 0$, then B is not the correct answer; $\boxed{(E) \ x^6 + x^3 + 1}$ is correct.

22. (AMC 12 12A 2022) Let c be a real number, and let z_1 and z_2 be the two complex numbers satisfying the equation $z^2 - cz + 10 = 0$. Points z_1 , z_2 , $\frac{1}{z_1}$, and $\frac{1}{z_2}$ are the vertices of (convex) quadrilateral $\mathcal{Q}$ in the complex plane. When the area of $\mathcal{Q}$ obtains its maximum possible value, c is closest to which of the following?

(A) 4.5 (B) 5 (C) 5.5 (D) 6 (E) 6.5

Answer: A

Solution 1

Because c is real, $z_2 = \bar{z}_1$. We have

$$\begin{aligned} 10 &= z_1 z_2 \\ &= z_1 \bar{z}_1 \\ &= |z_1|^2, \end{aligned}$$

where the first equality follows from Vieta's formula.

Thus, $|z_1| = \sqrt{10}$.

We have

$$\begin{aligned} c &= z_1 + z_2 \\ &= z_1 + \bar{z}_1 \\ &= 2\operatorname{Re}(z_1), \end{aligned}$$

where the first equality follows from Vieta's formula.

Thus, $\operatorname{Re}(z_1) = \frac{c}{2}$.

We have

$$\begin{aligned} \frac{1}{z_1} &= \frac{1}{10} \cdot \frac{10}{z_1} \\ &= \frac{1}{10} \cdot \frac{z_1 z_2}{z_1} \\ &= \frac{z_2}{10} \\ &= \frac{\bar{z}_1}{10}. \end{aligned}$$

where the second equality follows from Vieta's formula.

We have

$$\begin{aligned}\frac{1}{z_2} &= \frac{1}{10} \cdot \frac{10}{z_2} \\ &= \frac{1}{10} \cdot \frac{z_1 z_2}{z_2} \\ &= \frac{z_1}{10}.\end{aligned}$$

where the second equality follows from Vieta's formula.

Therefore,

$$\begin{aligned}\text{Area } Q &= \frac{1}{2} |\operatorname{Re}(z_1)| \cdot 2 |\operatorname{Im}(z_1)| \cdot \left(1 - \frac{1}{10^2}\right) \\ &= \frac{1}{2} |c| \sqrt{10 - \frac{c^2}{4}} \left(1 - \frac{1}{10^2}\right) \\ &= \frac{1 - \frac{1}{10^2}}{4} \sqrt{c^2 (40 - c^2)} \\ &\leq \frac{1 - \frac{1}{10^2}}{4} \cdot \frac{c^2 + (40 - c^2)}{2} \\ &= \frac{1 - \frac{1}{10^2}}{4} \cdot 20,\end{aligned}$$

where the inequality follows from the AM-GM inequality, and it is augmented to an equality if and only if $c^2 = 40 - c^2$. Thus, $|c| = 2\sqrt{5} \approx \boxed{\text{(A) } 4.5}$.

Solution 2 (Trapezoid)

Since c , which is the sum of roots z_1 and z_2 , is real, $z_1 = \overline{z_2}$.

Let $z_1 = a + bi$. Then $z_2 = a - bi$. Note that the product of the roots is 10 by Vieta's, so $z_1 z_2 = (a + bi)(a - bi) = a^2 + b^2 = 10$.

Thus, $\frac{1}{z_2} = \frac{1}{a-bi} \cdot \frac{a+bi}{a+bi} = \frac{a+bi}{a^2+b^2} = \frac{a+bi}{10}$. With the same process, $\frac{1}{z_1} = \frac{a-bi}{10}$.

So, our four points are $a + bi$, $\frac{a+bi}{10}$, $a - bi$, and $\frac{a-bi}{10}$. WLOG let $a + bi$ be in the first quadrant and graph these four points on the complex plane. Notice how quadrilateral Q is a trapezoid with the real axis as its axis of symmetry. It has a short base with endpoints $\frac{a+bi}{10}$ and $\frac{a-bi}{10}$, so its length is $\left|\frac{a+bi}{10} - \frac{a-bi}{10}\right| = \frac{b}{5}$. Likewise, its long base has endpoints $a + bi$ and $a - bi$, so its length is $|(a + bi) - (a - bi)| = 2b$.

The height, which is the distance between the two lines, is the difference between the real values of the two bases $\implies h = a - \frac{a}{10} = \frac{9a}{10}$.

Plugging these into the area formula for a trapezoid, we are trying to maximize $\frac{1}{2} \cdot \left(2b + \frac{b}{5}\right) \cdot \frac{9a}{10} = \frac{99ab}{100}$. Thus, the only thing we need to maximize is ab .

With the restriction that $a^2 + b^2 = (a - b)^2 + 2ab = 10$, ab is maximized when $a = b = \sqrt{5}$.

Remember, c is the sum of the roots, so $c = z_1 + z_2 = 2a = 2\sqrt{5} = \sqrt{20} \approx \boxed{4.5}$

Solution 3 (Fast)

Like the solutions above we can know that $|z_1| = |z_2| = \sqrt{10}$ and $z_1 = \overline{z_2}$.

Let $z_1 = \sqrt{10}e^{i\theta}$ where $0 < \theta < \pi$, then $z_2 = \sqrt{10}e^{-i\theta}$, $\frac{1}{z_1} = \frac{1}{\sqrt{10}}e^{-i\theta}$, $\frac{1}{z_2} = \frac{1}{\sqrt{10}}e^{i\theta}$.

On the basis of symmetry, the area A of Q is the difference between two isocles triangles, so

$2A = 10 \sin 2\theta - \frac{1}{10} \sin 2\theta \leq 10 - \frac{1}{10}$. The inequality holds when $2\theta = \frac{\pi}{2}$, or $\theta = \frac{\pi}{4}$.

Thus, $c = 2\operatorname{Re} z_1 = 2\sqrt{10} \cos \frac{\pi}{4} = \sqrt{20} \approx \boxed{\text{(A) } 4.5}$.

Solution 4 (Calculus Finish)

Like in Solution 3, we find that $Q = \frac{1}{2} \cdot \left(2b + \frac{b}{5}\right) \cdot \frac{9a}{10} = \frac{99}{100}ab$, thus, Q is maximized when ab is maximized. $ab = a \cdot \sqrt{10 - a^2} = \sqrt{10a^2 - a^4}$, let $f(a) = \sqrt{10a^2 - a^4}$.

By the Chain Rule and the Power Rule, $f'(a) = \frac{1}{2} \cdot (10a^2 - a^4)^{-\frac{1}{2}} \cdot (10(2a) - 4a^3) = \frac{20a - 4a^3}{2\sqrt{10a^2 - a^4}} = \frac{10a - 2a^3}{\sqrt{10a^2 - a^4}}$ $f'(a) = 0$, $10a - 2a^3 = 0$, $a \neq 0$, $a^2 = 5$, $a = \sqrt{5} \therefore f'(a) = 0$ when $a = \sqrt{5}$, $f'(a)$ is positive when $a < \sqrt{5}$, and $f'(a)$ is negative when $a > \sqrt{5} \therefore f(a)$ has a local maximum when $a = \sqrt{5}$.

Notice that $ab = \frac{(a+b)^2 - (a^2 + b^2)}{2} = \frac{c^2 - 10}{2}$, $\frac{c^2 - 10}{2} = 5$, $c = \sqrt{2 \cdot 5 + 10} = \sqrt{20} \approx \boxed{\text{(A) } 4.5}$

Solution 5 (calculus but it's bash)

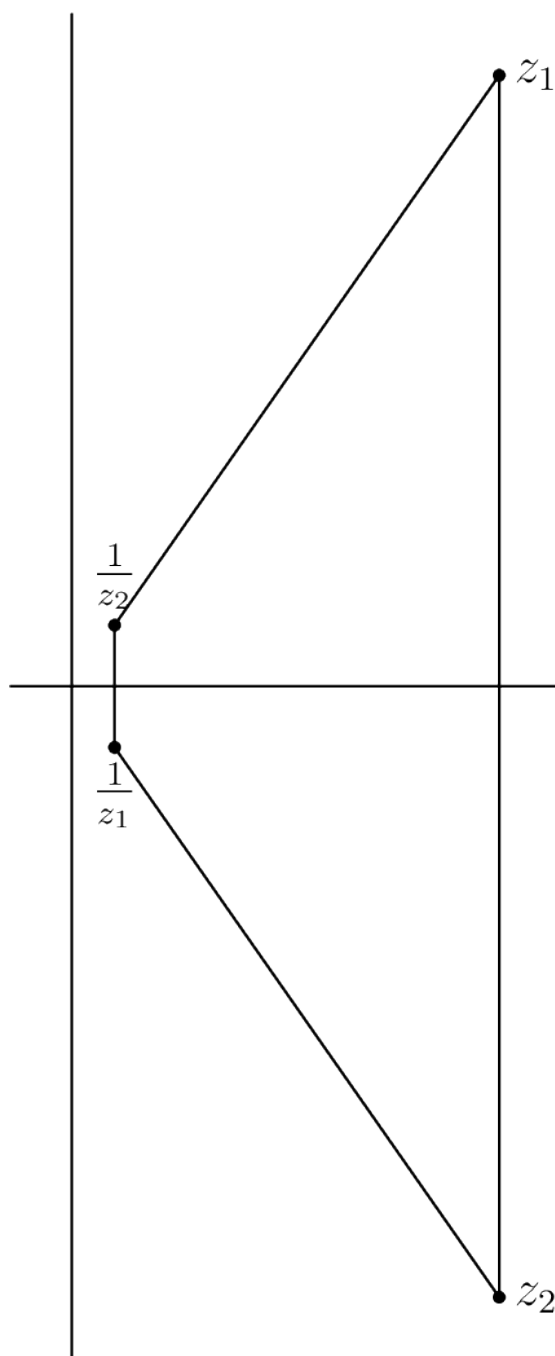
Note that $z = \frac{c}{2} \pm \frac{\sqrt{c^2 - 40}}{2}$, so let $z_1 = \frac{c}{2} + \frac{\sqrt{c^2 - 40}}{2}$ and $z_2 = \frac{c}{2} - \frac{\sqrt{c^2 - 40}}{2}$. Taking a look at the answer choices, they range between $c = 4.5$ to $c = 6.5$, and in that range, c^2 is always less than 40. Thus, $c^2 - 40 < 0$ for our possible answer choices; we can then rewrite z_1 and z_2 as $\frac{c}{2} + \frac{\sqrt{40 - c^2}}{2}i$ and $\frac{c}{2} - \frac{\sqrt{40 - c^2}}{2}i$, respectively, with real coefficients.

Let us compute $\frac{1}{z}$:

$$\frac{1}{z} = \frac{1}{\frac{c}{2} \pm \frac{\sqrt{c^2 - 40}}{2}} = \frac{\frac{c}{2} \mp \frac{\sqrt{c^2 - 40}}{2}}{\left(\frac{c}{2}\right)^2 - \left(\frac{\sqrt{c^2 - 40}}{2}\right)^2} = \frac{\frac{c}{2} \mp \frac{\sqrt{c^2 - 40}}{2}}{\frac{c^2}{4} - \frac{c^2 - 40}{4}} = \frac{2c \mp 2\sqrt{c^2 - 40}}{40} = \frac{c \mp \sqrt{c^2 - 40}}{20}.$$

$$\text{Then, } \frac{1}{z_1} = \frac{c - \sqrt{c^2 - 40}}{20} = \frac{c}{20} - \frac{\sqrt{40 - c^2}}{20}i \text{ while } \frac{1}{z_2} = \frac{c + \sqrt{c^2 - 40}}{20} = \frac{c}{20} + \frac{\sqrt{40 - c^2}}{20}i.$$

In the complex plane, we can draw a rough sketch of $z_1, z_2, \frac{1}{z_1}, \frac{1}{z_2}$:



Note that, here, our quadrilateral is an isosceles trapezoid. The shorter base length is $\left(\frac{\sqrt{40-c^2}}{20} - \left(-\frac{\sqrt{40-c^2}}{20} \right) \right) = \frac{1}{10}\sqrt{40-c^2}$.

The longer base length is $\left(\frac{\sqrt{40-c^2}}{2} - \left(-\frac{\sqrt{40-c^2}}{2} \right) \right) = \sqrt{40-c^2}$.

The average of the two bases is $\frac{11}{20}\sqrt{40-c^2}$.

The height of our trapezoid (which is horizontal parallel to the x -axis in our diagram above) is simply $\frac{c}{2} - \frac{c}{20} = \frac{9}{20}c$.

Since the area of a trapezoid is the product of the average of its bases and its

height, we conclude that our trapezoid's area is $\frac{99}{400}c\sqrt{40-c^2}$, which is a function of c . Thus, let $A(c) = \frac{99}{400}c\sqrt{40-c^2}$.

Taking the derivative (FINALLY we get to the Calculus part haha, this definitely wasn't clickbait :3 trust me) with respect to c , we find that $\frac{dA}{dc} = \frac{99}{400} \left(\sqrt{40-c^2} + c \left(\frac{-2c}{2\sqrt{40-c^2}} \right) \right) = \frac{99}{400} \left(\sqrt{40-c^2} - \frac{c^2}{\sqrt{40-c^2}} \right)$.

To find an extremum, we set the derivative equal to zero:

$$\begin{aligned} \frac{dA}{dc} &= \frac{99}{400} \left(\sqrt{40-c^2} - \frac{c^2}{\sqrt{40-c^2}} \right) \\ 0 &= \frac{99}{400} \left(\sqrt{40-c^2} - \frac{c^2}{\sqrt{40-c^2}} \right) \\ \sqrt{40-c^2} - \frac{c^2}{\sqrt{40-c^2}} &= 0 \\ \sqrt{40-c^2} &= \frac{c^2}{\sqrt{40-c^2}} \\ \left(\sqrt{40-c^2} \right)^2 &= c^2 \\ c^2 &= 40 - c^2 \\ 2c^2 &= 40 \\ c^2 &= 20 \\ c &= \sqrt{20} \\ &\approx 4.47 \end{aligned}$$

Clearly, this is very close to (A) 4.5, so we are done. QED.

Solution 6 (AM-GM Inequality)

First and foremost, because z_n is a complex number, and the value of z could be found in terms of c , the given quadratic equation could be solved. WLOG, $z_1 = \frac{c+\sqrt{c^2-40}}{2}$ and $z_2 = \frac{c-\sqrt{c^2-40}}{2}$. If $c^2 - 40 \geq 0$, no quadrilateral will form. Thereby, it could be inferred that $c^2 - 40 < 0$.

Each complex number could be represented as a point.

$$\begin{aligned} z_1 &\left(\frac{c}{2}, \frac{\sqrt{40-c^2}}{2} \right) \\ z_2 &\left(\frac{c}{2}, \frac{-\sqrt{40-c^2}}{2} \right) \\ \frac{1}{z_1} &\left(\frac{c}{20}, \frac{c-\sqrt{40-c^2}}{20} \right) \\ \frac{1}{z_2} &\left(\frac{c}{20}, \frac{c+\sqrt{40-c^2}}{20} \right) \end{aligned}$$

Notice that vertices form a trapezoid. Therefore, the area of the trapezoid Q could be represented as

$$\frac{\left(\frac{c}{2} - \frac{c}{20}\right) \left\{ \left(\frac{\sqrt{40-c^2}}{2} + \frac{\sqrt{40-c^2}}{2}\right) + \left(\frac{c+\sqrt{40-c^2}}{20} - \frac{c-\sqrt{40-c^2}}{20}\right) \right\}}{2}.$$

Since we are finding the maximum value, any desired constant could be multiplied. Multiplying 80 and other constants to the equation above may provide a simplified form.

$$\begin{aligned} & (9c)(20\sqrt{40-c^2} + 2\sqrt{40-c^2}) \\ \Rightarrow & 9c(22\sqrt{40-c^2}) \\ \Rightarrow & c\sqrt{40-c^2} \end{aligned}$$

We can use AM-GM Inequality! According to the AM-GM Inequality,

$$40 - c^2 + c^2 \geq 2\sqrt{(40 - c^2) \cdot c^2}.$$

$\sqrt{(40 - c^2) \cdot c^2}$ reaches maximum when $40 - c^2 = c^2$. In another words, $c = \pm 2\sqrt{5}$. Using 2.2 for the approximation of $\sqrt{5}$, $\max(c) \approx 4.4$. Thus, the closest value in the answer choice is **(A) 4.5**.

23. (AMC 12 12A 2022) Let h_n and k_n be the unique relatively prime positive integers such that

$$\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} = \frac{h_n}{k_n}.$$

Let L_n denote the least common multiple of the numbers $1, 2, 3, \dots, n$. For how many integers with $1 \leq n \leq 22$ is $k_n < L_n$?

- (A) 0 (B) 3 (C) 7 (D) 8 (E) 10

Answer: D

Solution 1

We are given that

$$\sum_{i=1}^n \frac{1}{i} = \frac{1}{L_n} \sum_{i=1}^n \frac{L_n}{i} = \frac{h_n}{k_n}.$$

Since $k_n < L_n$, we need $\gcd\left(\sum_{i=1}^n \frac{L_n}{i}, L_n\right) > 1$.

For all primes p such that $p \leq n$, let $\nu_p(L_n) = e \geq 1$ be the exponent of the largest power of p that divides L_n .

It is clear that $L_n \equiv 0 \pmod{p}$, so we test whether $\sum_{i=1}^n \frac{L_n}{i} \equiv 0 \pmod{p}$. Note that

$$\sum_{i=1}^n \frac{L_n}{i} \equiv \sum_{i=1}^{\lfloor \frac{n}{p^e} \rfloor} \frac{L_n}{ip^e} \pmod{p^e} \equiv \sum_{i=1}^{\lfloor \frac{n}{p^e} \rfloor} \frac{L_n}{ip^e} \pmod{p}.$$

We construct the following table for $\nu_p(L_n) = e$:

Case of (p, e)	Interval of n	Sum	$\stackrel{?}{\equiv} 0 \pmod{p}$
(2, 1)	[2, 3]	$L_n/2$	
(2, 2)	[4, 7]	$L_n/4$	
(2, 3)	[8, 15]	$L_n/8$	
(2, 4)	[16, 22]	$L_n/16$	
(3, 1)	[3, 5]	$L_n/3$	
	[6, 8]	$L_n/3 + L_n/6$	✓
(3, 2)	[9, 17]	$L_n/9$	
	[18, 22]	$L_n/9 + L_n/18$	✓
(5, 1)	[5, 9]	$L_n/5$	
	[10, 14]	$L_n/5 + L_n/10$	
	[15, 19]	$L_n/5 + L_n/10 + L_n/15$	
	[20, 22]	$L_n/5 + L_n/10 + L_n/15 + L_n/20$	✓
(7, 1)	[7, 13]	$L_n/7$	
	[14, 20]	$L_n/7 + L_n/14$	
	[21, 22]	$L_n/7 + L_n/14 + L_n/21$	
(11, 1)	[11, 21]	$L_n/11$	
	{22}	$L_n/11 + L_n/22$	
(13, 1)	[13, 22]	$L_n/13$	
(17, 1)	[17, 22]	$L_n/17$	
(19, 1)	[19, 22]	$L_n/19$	

Note that:

- If the **Sum** column has only one term, then it is never congruent to 0 modulo p .
- If p and q are positive integers such that $p \geq q$, then L_p is a multiple of L_q . Therefore, for a specific case, if the sum is congruent to 0 modulo p for the smallest element in the interval of n , then it is also congruent to 0 modulo p for all other elements in the interval of n .

Together, there are (D) 8 such integers n , namely

$$6, 7, 8, 18, 19, 20, 21, 22.$$

MRENTHUSIASM

Solution 2

Notice that $\sum_{i=1}^n \frac{1}{i} = \frac{\sum_{i=1}^n \frac{L_n}{L_n i}}$. Thus, in order for L_n to reduce to a smaller value, the numerator and denominator must share a common factor.

We can start by testing some values of n , and quickly observe that $n = 6$ is the first number that satisfies the desired conditions. In particular, $\sum_{i=1}^6 \frac{1}{i} = \frac{\sum_{i=1}^6 \frac{L_6}{L_6 i}}{L_6} = \frac{147}{60} = \frac{49}{20}$. Since a factor of three is shared, we are motivated to observe factors of three in the numerator $\sum_{i=1}^6 \frac{60}{i}$. Notice that $\nu_3(L_6) = 1$, since 3^1 is the greatest power of three that occurs in 1, 2, 3, 4, 5, 6. Consider the sum in the numerator in

mod 3. We have $(\frac{60}{1}, \frac{60}{2}, \frac{60}{3}, \frac{60}{4}, \frac{60}{5}, \frac{60}{6}) \rightarrow (0, 0, 2, 0, 0, 1)$. Adding $0+0+2+0+0+1$ we get 0 mod 3, which is why we are able to simplify $\frac{147}{60}$.

Thus, we should be taking the numerator mod p , where p is a prime.

Quickly, we wish to show that the numerator and denominator can never share a factor of 2. Say we have least common multiple L_n . Let $\nu_2(L_n) = e$. In other words, let 2^e be the largest power of 2 such that 2^e divides L_n . Notice that among all numbers 1 through n , only one of them can be a multiple of 2^e , being 2^e itself. The next multiple of 2^e would be $2 \cdot 2^e = 2^{e+1}$, contradicting the fact that $\nu_2(L_n) = e$. This means that in the sum $\sum_{i=1}^n \frac{L_n}{i}$, there will be 1 value that is 1 mod 2, resulting from $\frac{L_n}{2^e}$, and the rest are all 0s. Clearly, the numerator will then be 1 mod 2, which is not a multiple of 2.

When $p = 3$, $n = 6, 7, 8, 18, 19, 20, 21, 22$ will all work, and this can be easily tested through taking the numerator mod 3. $p = 5$ is only satisfied when the L_n contains $(5, 10, 15, 20)$, so when $n \geq 20$. However, this is just an overlap with the previous values, so there are **(D) 8**.

24. (AMC 12 12A 2022) How many strings of length 5 formed from the digits 0, 1, 2, 3, 4 are there such that for each $j \in \{1, 2, 3, 4\}$, at least j of the digits are less than j ? (For example, 02214 satisfies this condition because it contains at least 1 digit less than 1, at least 2 digits less than 2, at least 3 digits less than 3, and at least 4 digits less than 4. The string 23404 does not satisfy the condition because it does not contain at least 2 digits less than 2.)

(A) 500 (B) 625 (C) 1089 (D) 1199 (E) 1296

Answer: E

Solution 1 (Parking Functions)

For some n , let there be $n + 1$ parking spaces counterclockwise in a circle. Consider a string of n integers $c_1 c_2 \dots c_n$ each between 0 and n , and let n cars come into this circle so that the i th car tries to park at spot c_i , but if it is already taken then it instead keeps going counterclockwise and takes the next available spot. After this process, exactly one spot will remain empty.

Then the strings of n numbers between 0 and $n - 1$ that contain at least k integers $< k$ for $1 \leq k \leq n$ are exactly the set of strings that leave spot n empty. Also note for any string $c_1 c_2 \dots c_n$, we can add 1 to each $c_i \pmod{n + 1}$ to shift the empty spot counterclockwise, meaning for each string there exists exactly one j with $0 \leq j \leq n$ so that $(c_1 + j)(c_2 + j) \dots (c_n + j)$ leaves spot n empty. This gives there are $\frac{(n+1)^n}{n+1} = (n+1)^{n-1}$ such strings.

Plugging in $n = 5$ gives **(E) 1296** such strings.

Solution 2 (OmegaLearn)

Consider some string that satisfies the problem statement, with a 0's, b 1's, c 2's, d 3's, and e 4's, so $a + b + c + d + e = 5$. There are exactly $\frac{5!}{a!b!c!d!e!}$ ways to order the given string. This number will be divisible by 5 unless one of a, b, c, d, e is 5, which implies that the rest are 0. This corresponds to the string having 5 of one digit. But the only possible string that has 1 distinct digit is 00000. Thus, the number of strings must be 1 (mod 5). The only answer choice that satisfies this is

(E) 1296.

Solution 3 (Casework)

Note that a valid string must have at least one 0.

We perform casework on the number of different digits such strings can have. For each string, we list the digits in ascending order, then consider permutations:

- a) The string has 1 different digit. The only possibility is 00000.

There is 1 string in this case.

- b) The string has 2 different digits. We have the following table:

Digits	01	02	03	04	Row's Count
00001	00002	00003	00004		$4 \cdot \frac{5!}{4!1!} = 20$
00011	00022	00033			$3 \cdot \frac{5!}{3!2!} = 30$
00111	00222				$2 \cdot \frac{5!}{2!3!} = 20$
01111					$1 \cdot \frac{5!}{1!4!} = 5$

There are $20 + 30 + 20 + 5 = 75$ strings in this case.

- c) The string has 3 different digits. We have the following table:

Digits	012	013	014	023	024	034	Row's Count
00012	00013	00014	00023	00024	00034		$6 \cdot \frac{5!}{3!1!1!} = 120$
00112	00113	00114	00223	00224			$5 \cdot \frac{5!}{2!2!1!} = 150$
00122	00133		00233				$3 \cdot \frac{5!}{2!1!2!} = 90$
01112	01113	01114					$3 \cdot \frac{5!}{1!3!1!} = 60$
01122	01133						$2 \cdot \frac{5!}{1!2!2!} = 60$
01222							$1 \cdot \frac{5!}{1!1!3!} = 20$

There are $120 + 150 + 90 + 60 + 60 + 20 = 500$ strings in this case.

- d) The string has 4 different digits. We have the following table:

Digits	0123	0124	0134	0234
00123	00124	00134	00234	
01123	01124	01134		
01223	01224			
01233				

There are $10 \cdot \frac{5!}{2!1!1!1!} = 600$ strings in this case.

- e) The string has 5 different digits. **There are $5! = 120$ strings in this case.**

Together, the answer is $1 + 75 + 500 + 600 + 120 = \boxed{\text{(E)} 1296}$.

Solution 4 (Recursive Equations Approach)

Denote by $N(p, q)$ the number of p -digit strings formed by using numbers $0, 1, \dots, q$, where for each $j \in \{1, 2, \dots, q\}$, at least j of the digits are less than j .

We have the following recursive equation:

$$N(p, q) = \sum_{i=0}^{p-q} \binom{p}{i} N(p-i, q-1), \quad \forall p \geq q \text{ and } q \geq 1$$

and the boundary condition $N(p, 0) = 1$ for any $p \geq 0$.

By solving this recursive equation, for $q = 1$ and $p \geq q$, we get

$$\begin{aligned} N(p, 1) &= \sum_{i=0}^{p-1} \binom{p}{i} N(p-i, 0) \\ &= \sum_{i=0}^{p-1} \binom{p}{i} \\ &= \sum_{i=0}^p \binom{p}{i} - \binom{p}{p} \\ &= 2^p - 1. \end{aligned}$$

For $q = 2$ and $p \geq q$, we get

$$\begin{aligned} N(p, 2) &= \sum_{i=0}^{p-2} \binom{p}{i} N(p-i, 1) \\ &= \sum_{i=0}^{p-2} \binom{p}{i} (2^{p-i} - 1) \\ &= \sum_{i=0}^p \binom{p}{i} (2^{p-i} - 1) - \sum_{i=p-1}^p \binom{p}{i} (2^{p-i} - 1) \\ &= \sum_{i=0}^p \left(\binom{p}{i} 1^i 2^{p-i} - \binom{p}{i} 1^i 1^{p-i} \right) - p \\ &= (1+2)^p - (1+1)^p - p \\ &= 3^p - 2^p - p. \end{aligned}$$

For $q = 3$ and $p \geq q$, we get

$$\begin{aligned} N(p, 3) &= \sum_{i=0}^{p-3} \binom{p}{i} N(p-i, 2) \\ &= \sum_{i=0}^{p-3} \binom{p}{i} (3^{p-i} - 2^{p-i} - (p-i)) \\ &= \sum_{i=0}^p \binom{p}{i} (3^{p-i} - 2^{p-i} - (p-i)) - \sum_{i=p-2}^p \binom{p}{i} (3^{p-i} - 2^{p-i} - (p-i)) \\ &= \sum_{i=0}^p \left(\binom{p}{i} 1^i 3^{p-i} - \binom{p}{i} 1^i 2^{p-i} - \binom{p}{i} (p-i) \right) - \frac{3}{2}p(p-1) \end{aligned}$$

$$\begin{aligned}
 &= (1+3)^p - (1+2)^p - \frac{d(1+x)^p}{dx} \Big|_{x=1} - \frac{3}{2}p(p-1) \\
 &= 4^p - 3^p - 2^{p-1}p - \frac{3}{2}p(p-1).
 \end{aligned}$$

For $q = 4$ and $p = 5$, we get

$$\begin{aligned}
 N(5, 4) &= \sum_{i=0}^1 \binom{5}{i} N(5-i, 3) \\
 &= N(5, 3) + 5N(4, 3) \\
 &= \boxed{\text{(E) } 1296}.
 \end{aligned}$$

Solution 6 (Observations)

The number of strings is $(n+1)^{(n-1)}$ as shown by Solution 1 (Parking Function), which is always equivalent to $1 \pmod n$. Thus you can choose $\boxed{\text{(E) } 1296}$.

However, you **CANNOT** prove that fact from this *incorrect* argument:

Let the set of all valid sequences be S . Notice that for any sequence $\{a_1, a_2, a_3, a_4, a_5\}$ in S , the 5 cyclic permutations

$$\begin{aligned}
 &\{a_1, a_2, a_3, a_4, a_5\} \\
 &\{a_2, a_3, a_4, a_5, a_1\} \\
 &\{a_3, a_4, a_5, a_1, a_2\} \\
 &\{a_4, a_5, a_1, a_2, a_3\} \\
 &\{a_5, a_1, a_2, a_3, a_4\}
 \end{aligned}$$

must also belong in S . However, one must consider the edge case all 5 elements are the same (only $\{0, 0, 0, 0, 0\}$), in which case all sequences listed are equivalent.

Alas, one must also consider cases like 10101 that only have 3 distinct cyclic permutations, so we cannot get a useful constraint from this view, unless *all* the cases are categorized and counted.

Thus it is NOT implied that $|S| \equiv 1 \pmod 5$, so it is not justified to choose $\boxed{\text{(E) } 1296}$ by inspection of answer options.

If you solve $S(n)$ for smaller n by bashing, you can find $|S(1)| = 1$, $|S(2)| = 3$, $|S(3)| = 16$, $|S(4)| = 125$, which leads to an Engineer's Induction guess that $|S(n)| \equiv 1 \pmod n$, or the exact formula $(n+1)^{(n-1)}$

Note: I'm pretty sure this solution works. Firstly, 10101 does have 5 distinct cyclic permutations, namely 10101, 11010, 01101, 10110, and 01011. Secondly, any sequence of 5 elements must have either 1 or a multiple of 5 distinct cyclic permutations. If a sequence of 5 elements has only 1 cyclic permutation, that must be the case with $(0, 0, 0, 0, 0)$, as noted (since 0 is necessary in our sequence when $j = 1$). If the sequence is cyclic every k "rotations/shifts," then it is *ONLY* going to be at the starting sequence when you shift it $0, k, 2k, \dots$ times. Since shifting any sequence 5 times takes it back to itself, we must have $k \mid 5 \implies k = 5$ (since we already considered $k = 1$ with $(0, 0, 0, 0, 0)$), so the cycle must be 5. So really, this solution does work, is not by coincidence, and is a very nice (albeit cheap) solution.

TL;DR: The sentence "Alas, one must also consider cases like 10101 that only have 3 distinct cyclic permutations" is incorrect, for these cases do not actually exist. It's trivial to check 10101 has 5 cyclic permutations.

Solution 5 (Mod 5 Trick Redeemed)

Solution 4 tried to observe the answer modulo 5 to easily solve the problem, but apparently had faulty logic. This solution is still completely viable though:

Notice that for any valid set $\{a_1, a_2, a_3, a_4, a_5\}$, if there is at least one element in the set that is unique (i.e. there is at least one digit in the set that is found nowhere else in the set), then the number of distinct permutations of the set is clearly divisible by 5 (alternatively there is no factor of 5 in the denominator of the multinomial coefficient unless an element is repeated 5 times). Therefore to evaluate the answer modulo 5, we only need to look at sets where each element has a multiplicity of at least 2 (i.e. appears twice or more in the set).

These sets are of the form $\{a, a, b, b, b\}$ and $\{a, a, a, a, a\}$. The first set can be permuted in $\binom{5}{2,3} = 10 \equiv 0 \pmod{5}$, and the second set can be permuted one way, and the only set of the form $\{a, a, a, a, a\}$ is $\{0, 0, 0, 0, 0\}$. Therefore the answer is congruent to 1 (mod 5) and you CAN choose (E) 1296.

Solution 7 (Casework Bash)

We list out all possibilities for the string, sorted in increasing order. Note that 1 cannot appear before the 2nd position, 2 cannot appear before the 3rd position, and so on. We also make a list of the number of possible permutations of each sorted string.

00000 - 1
 00001 - 5
 00002 - 5
 00003 - 5
 00004 - 5
 00011 - 10
 00012 - 20
 00013 - 20
 00014 - 20
 00022 - 10
 00023 - 20
 00024 - 20
 00033 - 10
 00034 - 20
 00111 - 10
 00112 - 30
 00113 - 30
 00114 - 30
 00122 - 30
 00123 - 60

00124 - 60
 00133 - 30
 00134 - 60
 00222 - 10
 00223 - 30
 00224 - 30
 00233 - 30
 00234 - 60
 01111 - 5
 01112 - 20
 01113 - 20
 01114 - 20
 01122 - 30
 01123 - 60
 01124 - 60
 01133 - 30
 01134 - 60
 01222 - 20
 01223 - 60
 01224 - 60
 01233 - 60
 01234 - 120

Finally, adding all of these up, we obtain $1 \cdot 1 + 5 \cdot 5 + 5 \cdot 10 + 10 \cdot 20 + 10 \cdot 30 + 10 \cdot 60 + 1 \cdot 120 = \boxed{\text{(E) 1296}}$. (Or alternatively, we could notice as in solution 5 that the only possible string with only 1 permutation is 00000, and thus the answer must be congruent to 1 modulo 5)

The key to organizing all of these cases is to increase the digits one by one, starting from the right and iterating through all the possible cases.

Solution 8 (Complementary Counting)

Let us call the condition where you require at least j digits less than j Rule j . For a sequence to break Rule 1, it must contain no 0s. There are $4^5 = 1024$ ways to do this. For a sequence to break Rule 2 but not Rule 1, it must contain exactly 1 0. However, this one 0 contributes toward completing the rule, meaning all other digits must be greater than 2. After placing the one 0 in any of 5 slots, there are 4 slots left with 3 options each, giving a total of $5 \cdot 3^4 = 405$ ways. For a sequence to break Rule 3 but neither Rule 2 or 1, it must have two digits less than 2 (either (0, 0) or (0, 1)) and all other digits must be 3 or 4. There are a total of 30 ways to arrange the first two digits and 2^3 ways to arrange the latter three, leaving 240 ways to do this. Finally, for a sequence to only break rule 4, it must have three digits less than 3. They could be (0, 1, 2) (60 arr.), (0, 1, 1); (0, 0, 1); (0, 0, 2) (30 arr. each), or (0, 0, 0) (10 arr.) This gives 160 possible arrangements overall.

There are in total $5^5 = 3125$ possible arrangements, so by adding all of the invalid states together and subtracting from 3125, we get our answer, which is (E) 1296

Great idea! Bigbrain_2009

Solution 9 (Recursion)

We will use recursion to solve this problem. Let us break up the problem's condition into criteria, with criterion k being the requirement that at least k of the digits are less than k . Define A_n as the number of strings of length n formed from the digits $0, 1, 2, \dots, n-1$ such that at least j of the digits are less than j for all integers j ; $1 \leq j \leq n-1$.

Note that the final criterion any string counted as apart of A_{n-1} is required to meet is criterion $n-2$. However, all strings that A_{n-1} counts will also meet criterion $n-1$. This is due to the maximum value of such strings' $n-1$ digits being $n-2$. Initially, we have $A_1 = 1$. We check cases and find $A_2 = 3$. To generate recursion, consider the compliment of the problem's condition. Without the condition, we are counting strings of length n with n choices for each digit. There are of course n^n such strings. To count the number of strings that do not meet at least one criterion, we denote a string as k -bad if the first criterion the string violates is criterion k . For example, the string 33040 is 3-bad because it meets criteria 1 and 2, but violates criterion 3. Similarly, the string 13223 is 1-bad because it does contain 1 digit less than 1. We can count the number of strings that are k -bad by first choosing and assigning $k-1$ of the string's n spots to a string counted as apart of A_{k-1} . This can be done in $\binom{n}{k-1} \cdot A_{k-1}$ ways. This $(k-1)$ digit string will meet all criteria up to criterion $k-1$. To ensure criterion k is not met, we fill up the remaining $n-(k-1)$ spots with a digit from k to $n-1$, inclusive. This can be done in $(n-k)^{n+1-k}$ ways. We sum over all integers k , $1 \leq k \leq n-1$ to count all strings that do not meet at least one criterion.

$$\sum_{k=1}^{n-1} \binom{n}{k-1} A_{k-1} (n-k)^{n+1-k}$$

This is the same as

$$\sum_{k=2}^n \binom{n}{k-2} A_{k-2} (n-k+1)^{n+2-k} = \sum_{k=2}^n \binom{n}{n+2-k} A_{k-2} (n-k+1)^{n+2-k}$$

We substitute $n+2-k$ for k as the expression takes on the same set of values. We have

$$\sum_{k=2}^n \binom{n}{n+2-(n+2-k)} A_{(n+2-k)-2} (n-(n+2-k)+1)^{n+2-(n+2-k)} = \sum_{k=2}^n \binom{n}{k} A_{n-k} (k-1)^k$$

Therefore,

$$A_n = n^n - \sum_{k=2}^n \binom{n}{k} A_{n-k} (k-1)^k$$

Setting $A_0 = 1$ and $A_1 = 1$, we use the recursion to derive $A_2 = 3$, $A_3 = 16$, $A_4 = 125$, $A_5 = 1296$. In our problem, we were counting 5 digits strings. Thus, the answer is (E) 1296

Solution 10 (Parity Cheese)

We note that there are 6 different multisets of digits that have an odd number of permutations, namely, 00000, 00001, 00002, 00003, 0004, 01111. We know that there cannot be any other possibilities since, if any digit appears 2 or 3 times, there are $\binom{5}{2} = \binom{5}{3} = 10$ ways to place those two digits, and thus the total number of permutations must be even. Additionally, if there are at least two digits that appear once, there are $5 \cdot 4 = 20$ ways to place them and the total number of permutations will be even. Hence, we can have only at most 1 digit that appears once, and the rest must appear either 4 or 5 times (or 0). From here we may arrive at the above 6 configurations. Since there is an even number of odd cases, we know that the final answer will be even. After a little bit of casework we observe that 500 is far too low, and thus the answer is 1296.

25. (AMC 12 12A 2022) A circle with integer radius r is centered at (r, r) . Distinct line segments of length c_i connect points $(0, a_i)$ to $(b_i, 0)$ for $1 \leq i \leq 14$ and are tangent to the circle, where a_i , b_i , and c_i are all positive integers and $c_1 \leq c_2 \leq \dots \leq c_{14}$. What is the ratio $\frac{c_{14}}{c_1}$ for the least possible value of r ?

(A) $\frac{21}{5}$ (B) $\frac{85}{13}$ (C) 7 (D) $\frac{39}{5}$ (E) 17

Answer: E

Solution 1

Suppose that with a pair (a_i, b_i) the circle is an excircle. Then notice that the hypotenuse must be $(r - x) + (r - y)$, so it must be the case that

$$a_i^2 + b_i^2 = (2r - a_i - b_i)^2.$$

Similarly, if with a pair (a_i, b_i) the circle is an incircle, the hypotenuse must be $(x - r) + (y - r)$, leading to the same equation.

Notice that this equation can be simplified through SFFT to

$$(a_i - 2r)(b_i - 2r) = 2r^2.$$

Thus, we want the smallest r such that this equation has at least 14 distinct pairs (a_i, b_i) for which this holds. The obvious choice to check is $r = 6$. In this case, since $2r^2 = 2^3 \cdot 3^2$ has 12 positive factors, we get 12 pairs, and we get another two if the factors are $-8, -9$ or vice versa. One can check that for smaller values of r , we do not even get close to 14 possible pairs.

When $r = 6$, the smallest possible c -value is clearly when the factors are negative. When this occurs, $a_i = 4, b_i = 3$ (or vice versa), so the minimal c is 5. The largest possible c -value occurs when the largest of a_i and b_i are maximized. This occurs when the factors are 72 and 1, leading to $a_i = 84, b_i = 13$ (or vice-versa), leading to a maximal c of 85.

Hence the answer is $\frac{85}{5} = \boxed{17}$.

Solution 2

Case 1: The tangent and the origin are on the opposite sides of the circle.

In this case, $a, b > 2r$.

We can easily prove that

$$a + b - 2r = c. \quad (1)$$

Recall that $c = \sqrt{a^2 + b^2}$.

Taking square of (1) and reorganizing all terms, (1) is converted as

$$(a - 2r)(b - 2r) = 2r^2.$$

Case 2: The tangent and the origin are on the same sides of the circle.

In this case, $0 < a, b < r$.

We can easily prove that

$$2r - a - b = c. \quad (2)$$

Recall that $c = \sqrt{a^2 + b^2}$.

Taking square of (2) and reorganizing all terms, (2) is converted as

$$(a - 2r)(b - 2r) = 2r^2.$$

Putting both cases together, for given r , we look for solutions of a and b satisfying

$$(a - 2r)(b - 2r) = 2r^2,$$

with either $a, b > 2r$ or $0 < a, b < r$.

Now, we need to find the smallest r , such that the number of feasible solutions of (a, b) is at least 14.

For equation

$$uv = 2r^2,$$

we observe that the R.H.S. is not a perfect square. Thus, the number of positive (u, v) is equal to the number of positive divisors of $2r^2$.

Second, for each feasible positive solution (u, v) , its opposite $(-u, -v)$ is also a solution. However, (u, v) corresponds to a feasible solution if (a, b) with $a = u + 2r$ and $b = v + 2r$, but $(-u, -v)$ may not lead to a feasible solution if (a, b) with $a = 2r - u$ and $b = 2r - v$.

Recall that we are looking for r that leads to at least 14 solutions. Therefore, the above observations imply that we must have r , such that $2r^2$ has least 7 positive divisors.

Following this guidance, we find the smallest r is 6. This leads to the following solutions:

$$(a_1, b_1, c_1) = (3, 4, 5), (a_2, b_2, c_2) = (4, 3, 5).$$

$$(a_3, b_3, c_3) = (20, 21, 29), (a_4, b_4, c_4) = (21, 20, 29).$$

$$(a_5, b_5, c_5) = (18, 24, 30), (a_6, b_6, c_6) = (24, 18, 30).$$

$$(a_7, b_7, c_7) = (16, 30, 34), (a_8, b_8, c_8) = (30, 16, 34).$$

$$(a_9, b_9, c_9) = (15, 36, 39), (a_{10}, b_{10}, c_{10}) = (36, 15, 39).$$

$$(a_{11}, b_{11}, c_{11}) = (14, 48, 50), (a_{12}, b_{12}, c_{12}) = (48, 14, 50).$$

$$(a_{13}, b_{13}, c_{13}) = (13, 84, 85), (a_{14}, b_{14}, c_{14}) = (84, 13, 85).$$

Therefore, $\frac{c_{14}}{c_1} = \boxed{\text{(E) 17}}$.

Solution 3

As c_i is the length of the segment $(0, a_i)$ and $(b_i, 0)$ $a_i^2 + b_i^2 = c_i^2$. The equation for the line that passes through $(0, a_i)$ and $(b_i, 0)$ is $a_i x + b_i y - a_i b_i = 0$.

By the point-line distance formula from point (r, r) to line $a_i x + b_i y - a_i b_i = 0$

$$r = \frac{|a_i r + b_i r - a_i b_i|}{\sqrt{a_i^2 + b_i^2}}, \quad c_i = \frac{|a_i r + b_i r - a_i b_i|}{r} = \left| a_i + b_i - \frac{a_i b_i}{r} \right|$$

$$\because c_i \text{ is an integer, } \therefore r | a_i b_i$$

To find c_1 and c_{14} for the smallest r , we need Pythagorean triples (a_i, b_i, c_i) with relatively prime elements from the smallest. Notice that we only need to find 7 triples with $a_i < b_i$, the 7 other triples will simply be (b_i, a_i, c_i) . Using the formula $2mn, m^2 - n^2, m^2 + n^2$ to generate primitive triplets, where $m^2 + n^2$ is c_i , we have:

c_1 is when $m = 2, n = 1 \implies c_1 = 2^2 + 1^2 = 5$ $c_7 = c_8 = \dots = c_{14}$ is when $m = 7, n = 6 \implies c_{14} = 7^2 + 6^2 = 85$

Therefore, $\frac{c_{14}}{c_1} = \frac{85}{5} = \boxed{\text{(E) } 17}$.

§9 AMC 12 12B 2023

1. (AMC 12 12B 2023) Mrs. Jones is pouring orange juice into four identical glasses for her four sons. She fills the first three glasses completely but runs out of juice when the fourth glass is only $\frac{1}{3}$ full. What fraction of a glass must Mrs. Jones pour from each of the first three glasses into the fourth glass so that all four glasses will have the same amount of juice?

(A) $\frac{1}{12}$ (B) $\frac{1}{4}$ (C) $\frac{1}{6}$ (D) $\frac{1}{8}$ (E) $\frac{2}{9}$

Answer: C

Solution 1

The first three glasses each have a full glass. Let's assume that each glass has "1 unit" of juice. It won't matter exactly how much juice everyone has because we're dealing with ratios, and that wouldn't affect our answer. The fourth glass has a glass that is one third. So the total amount of juice will be $1 + 1 + 1 + \frac{1}{3} = \frac{10}{3}$. If

we divide the total amount of juice by 4, we get $\frac{5}{6}$, which should be the amount of juice in each glass. This means that each of the first three glasses will have to contribute $1 - \frac{5}{6} = \boxed{\text{(C) } \frac{1}{6}}$ to the fourth glass.

Solution 2 (unnecessary numerical values)

Given that the first three glasses are full and the fourth is only $\frac{1}{3}$ full, let's represent their contents with a common denominator, which we'll set as 6. This makes the first three glasses $\frac{6}{6}$ full, and the fourth glass $\frac{2}{6}$ full.

To equalize the amounts, Mrs. Jones needs to pour juice from the first three glasses into the fourth. Pouring $\frac{1}{6}$ from each of the first three glasses will make them all $\frac{5}{6}$ full. Thus, all four glasses will have the same amount of juice. Therefore, the

answer is $\boxed{\text{(C) } \frac{1}{6}}$.

Solution 3 (simple algebra)

We let x denote how much juice we take from each of the first 3 children and give to the 4th child.

We can write the following equation: $1 - x = \frac{1}{3} + 3x$, since each value represents how much juice each child (equally) has in the end. (Each of the first three children now have $1 - x$ juice, and the fourth child has $3x$ more juice on top of their initial $\frac{1}{3}$.)

Solving, we see that $x = \boxed{\text{(C)} \frac{1}{6}}$.

2. (AMC 12 12B 2023) Carlos went to a sports store to buy running shoes. Running shoes were on sale, with prices reduced by 20% on every pair of shoes. Carlos also knew that he had to pay a 7.5% sales tax on the discounted price. He had 43 dollars. What is the original (before discount) price of the most expensive shoes he could afford to buy?

(A) 46 (B) 50 (C) 48 (D) 47 (E) 49

Answer: B

Solution 1

Let the price originally be x . Then, after a 20 percent discount, the price is now $x - \frac{1}{5}x = \frac{4}{5}x$.

From the discounted price $\frac{4}{5}x$, we now take $\frac{7.5}{100}$ of $\frac{4}{5}x$ and add it to $\frac{4}{5}x$, giving us $\frac{4}{5}x + \left(\frac{7.5}{100}\right)\left(\frac{4}{5}x\right) = \frac{4}{5}x + \frac{30}{500}x = \frac{4}{5}x + \frac{3}{50}x = \frac{43}{50}x$.

Now we write the inequality $\frac{43}{50}x \leq \$43$ and multiply by $\frac{50}{43}$ on both sides to get $x \leq \$50$.

We want the greatest original price, which would be 50 or option choice B.

Solution 2 (easy)

We can create the equation:

$$0.8x \cdot 1.075 = 43$$

using the information given. This is because x , the original price, got reduced by 20%, or multiplied by 0.8, and it also got multiplied by 1.075 on the discounted price. Solving that equation, we get

$$\frac{4}{5} \cdot x \cdot \frac{43}{40} = 43$$

$$\frac{4}{5} \cdot x \cdot \frac{1}{40} = 1$$

$$\frac{1}{5} \cdot x \cdot \frac{1}{10} = 1$$

$$x = \boxed{50}$$

Solution 3 (One-Step Equation)

The discounted shoe is 20% off the original price. So that means $1 - 0.2 = 0.8$. There is also a 7.5% sales tax charge, so $0.8 * 1.075 = 0.86$. Now we can set up the equation $0.86x = 43$, and solving that we get $x = \boxed{\text{(B)} 50}$ kabbybear

Solution 4

Let the original price be x dollars. After the discount, the price becomes $80\% \cdot x$ dollars. After tax, the price becomes $80\% \times (1 + 7.5\%) = 86\% \cdot x$ dollars. So, $43 = 86\% \cdot x$, $x = \boxed{\text{(B)} \$50}$.

Solution 5

We can assign a variable c to represent the original cost of the shoes. Next, we set up the equation $80\% \cdot 107.5\% \cdot c = 43$. We can solve this equation for c and get $\boxed{\text{(B)} \$50}$.

Solution 6 (Intuition and Guessing)

We know the discount price will be $5/4$, and 0.075 is equal to $3/40$. So we look at answer choice **(B)**, see that the discount price will be 40 , and with sales tax applied it will be 43 , so the answer choice is $\boxed{\text{(B)} \$50}$.

3. (AMC 12 12B 2023) A $3 - 4 - 5$ right triangle is inscribed in circle A , and a $5 - 12 - 13$ right triangle is inscribed in circle B . What is the ratio of the area of circle A to the area of circle B ?

(A) $\frac{9}{25}$ (B) $\frac{1}{9}$ (C) $\frac{1}{5}$ (D) $\frac{25}{169}$ (E) $\frac{4}{25}$

Answer: D

Solution 1

Because the triangles are right triangles, we know the hypotenuses are diameters of circles A and B . Thus, their radii are 2.5 and 6.5 (respectively). Square the two numbers and multiply π to get 6.25π and 42.25π as the areas of the circles. Multiply 4 on both numbers to get 25π and 169π . Cancel out the π , and lastly, divide, to get your answer = $\boxed{\text{(D)} \frac{25}{169}}$.

Solution 1.5 (Way FASTER! Use in the contest)

With right triangles inscribed in circles, the hypotenuse must be the diameter. Therefore, the ratio of radii is $\frac{5}{13}$ which means the area ratio is just $\frac{5^2}{13^2}$ so, $\boxed{\text{(D)} \frac{25}{169}}$.

-Mismatchedcubing/Andrew__

Note for Solution 1 The reason why the right triangle has its hypotenuse on the diameter is because of its properties with the circle and the central angle theorem. The central angle theorem states that an angle inscribed on a circle is just arc length in degrees. The diameter of a circle splits the circle into two halves with arc of 180° , and therefore the angle inscribed in that arc is just $180/2 = 90$ degrees. The longest side of a triangle is opposite to the longest angle, and therefore the length of the hypotenuse is just the diameter of the circle.

Solution 2

Since the arc angle of the diameter of a circle is 90 degrees, the hypotenuse of each these two triangles is respectively the diameter of circles A and B .

Therefore the ratio of the areas equals the radius of circle A squared : the radius of circle B squared = $0.5 \times$ the diameter of circle A , squared : $0.5 \times$ the diameter of circle B , squared = the diameter of circle A , squared : the diameter of circle B , squared = $\boxed{\text{(D)} \frac{25}{169}}$.

Solution 3

The ratio of areas of circles is the same as the ratios of the diameters squared (since they are similar figures). Since this is a right triangle the hypotenuse of each triangle will be the diameter of the circle. This yields the expression $\frac{5^2}{13^2} = \boxed{\text{(D)} \frac{25}{169}}$.

4. (*AMC 12 12B 2023*) Jackson's paintbrush makes a narrow strip with a width of 6.5 millimeters. Jackson has enough paint to make a strip 25 meters long. How many square centimeters of paper could Jackson cover with paint?
- (A) 162,500 (B) 162.5 (C) 1,625 (D) 1,625,000 (E) 16,250

Answer: C

Solution 1

6.5 millimeters is equal to 0.65 centimeters. 25 meters is 2500 centimeters. The answer is 0.65×2500 , so the answer is $\boxed{\text{(C)} 1,625}$.

Solution 2 (Scientific Form)

6.5 millimeters can be represented as 65×10^{-2} centimeters. 25 meters is 25×10^2 centimeters. Multiplying out these results in $(65 \times 10^{-2}) \times (25 \times 10^2)$, which is 65×25 making the answer $\boxed{\text{(C)} 1,625}$.

Remark A good way to remember metric conversions is with the phrase "**King Henry Died By Drinking Chocolate Milk**" (Kilo, Hecto, Deka, Basic units, Deci, Centi, Milli). As you read from left to right you divide by 10. So 1 meter (which is a basic unit) is equivalent to 1000 millimeters, or 100 centimeters.

5. (*AMC 12 12B 2023*) You are playing a game. A 2×1 rectangle covers two adjacent squares (oriented either horizontally or vertically) of a 3×3 grid of squares, but you are not told which two squares are covered. Your goal is to find at least one square that is covered by the rectangle. A "turn" consists of you guessing a square, after which you are told whether that square is covered by the hidden rectangle. What is the minimum number of turns you need to ensure that at least one of your guessed squares is covered by the rectangle?
- (A) 3 (B) 5 (C) 4 (D) 8 (E) 6

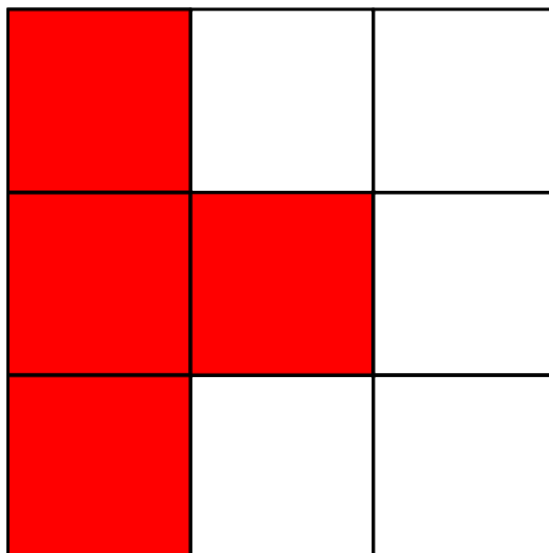
Answer: C

Solution 1

Notice that the 3×3 square grid has a total of 12 possible 2×1 rectangles.

Suppose you choose the middle square for one of your turns. The middle square is covered by 4 rectangles, and each of the remaining 8 squares is covered by a maximum of 2 uncounted rectangles. This means that the number of turns is at least $1 + \frac{12-4}{2} = 1 + 4 = 5$.

Now suppose you don't choose the middle square. The squares on the middle of the sides are covered by at most 3 uncounted rectangles, and the squares on the corners are covered by at most 2 uncounted rectangles. In this case, we see that the least number of turns needed to account for all 12 rectangles is $12 \div 3 = 4$. To prove that choosing only side squares indeed does cover all 12 rectangles, we need to show that the 3 rectangles per square that cover each side square do not overlap. Drawing the rectangles that cover one square, we see they form a *T* shape and they do not cover any other side square. Hence, our answer is 4.



Solution 2

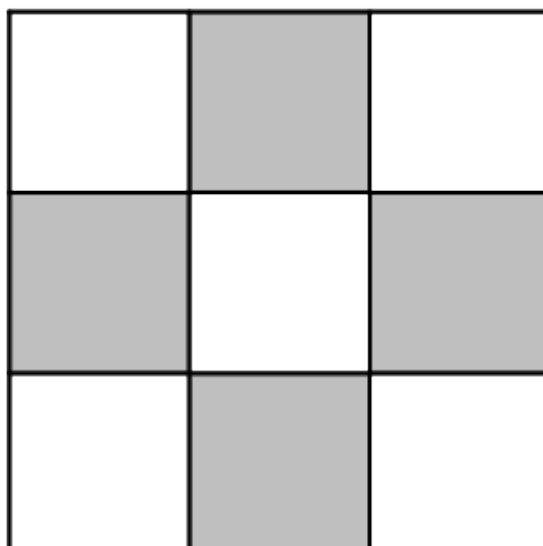
First, note that since the rectangle covers 2 squares, we only need to guess squares that are not adjacent to any of our other guesses. To minimize the amount of guesses, each of our guessed squares should try to touch another guess on one vertex and one vertex only. There are only two ways to do this: one with 5 guesses, and one with 4. Since the problem is asking for the minimum number, the answer is (C) 4.

Solution 3

Since the hidden rectangle can only hide two adjacent squares, we may think that we eliminate 8 squares and we're done, but think again. This is the AMC 10, so there must be a better solution (also note that every other solution choice is below 8 so we're probably not done) So, we think again, we notice that we haven't used the adjacent condition, and then it clicks. If we eliminate the four squares with only one edge on the boundary of the 9x9 square. We are left with 5 diagonal squares, since our rectangle can't be diagonal, we can ensure that we find it in 4 moves. So our answer is : (C) 4

Solution 4 (Checkerboard)

The 3×3 grid can be colored like a checkerboard with alternating black and white squares. Let the top left square be white, and the rest of the squares alternate colors, as shown below:



Each 2×1 rectangle always covers 1 white square and 1 black square. You can ensure that at least one of your guessed squares is covered by the rectangle by choosing either each of the white squares (5 turns) or each of the black squares (4 turns).

Since it is ideal to be the most efficient with our turns, by choosing all the black squares, we guarantee that one of the (C) 4 squares are of the 2×1 rectangle.

(Minor edits by NSAoPS)

Solution 5 (Logic)

We realize that every 2×1 rectangle must contain an edge and no more than one edge. There are a total of four edges so the answer is (C) 4. darrenn.cp

Solution 6 (Very Fast)

Draw the 3×3 grid. We want to minimize our options, so we look at the square shared by the most 2×1 rectangles. This is the center square. So, to guarantee our success, we choose the four edges from the center square, so our answer is C.4

6. (AMC 12 12B 2023) When the roots of the polynomial

$$P(x) = (x - 1)^1(x - 2)^2(x - 3)^3 \cdots (x - 10)^{10}$$

are removed from the number line, what remains is the union of 11 disjoint open intervals. On how many of these intervals is $P(x)$ positive?

- (A) 3 (B) 7 (C) 6 (D) 4 (E) 5

Answer: C

Solution 1

The expressions to the power of even powers are always positive, so we don't need to care about those. We only need to care about $(x - 1)^1(x - 3)^3(x - 5)^5(x - 7)^7(x - 9)^9$. We need 0, 2, or 4 of the expressions to be negative. The 9 through 10 interval and 10 plus interval make all of the expressions positive. The 5 through 6 and 6 through 7 intervals make two of the expressions negative. The 1 through 2 and 2 through 3 intervals make four of the expressions negative. There are (C) 6 intervals.

Solution 2

The roots of the factorized polynomial are intervals from numbers 1 to 10. We take each interval as being defined as the number behind it. To make the function positive, we need to have an even number of negative expressions. Real numbers raised to even powers are always positive, so we only focus on $x - 1$, $x - 3$, $x - 5$, $x - 7$, and $x - 9$. The intervals 1 and 2 leave 4 negative expressions, so they are counted. The same goes for intervals 5, 6, 9, and 10. Intervals 3 and 4 leave 3 negative expressions and intervals 7 and 8 leave 1 negative expression. The solution is the number of intervals which is **(C) 6**.

Solution 3

We can use the turning point behavior at the roots of a polynomial graph to find out the amount of intervals that are positive.

First, we evaluate any value on the interval $(-\infty, 1)$. Since the degree of $P(x)$ is $1 + 2 + \dots + 9 + 10 = \frac{10 \times 11}{2} = 55$, and every term in $P(x)$ is negative, multiplying 55 negatives gives a negative value. So $(-\infty, 0)$ is a negative interval.

We know that the roots of $P(x)$ are at 1, 2, ..., 10. When the degree of the term of each root is odd, the graph of $P(x)$ will pass through the graph and change signs, and vice versa. So at $x = 1$, the graph will change signs; at $x = 2$, the graph will not, and so on.

This tells us that the interval (1, 2) is positive, (2, 3) is also positive, (3, 4) is negative, (4, 5) is also negative, and so on, with the pattern being +, +, -, -, +, +, -, -, ...

The positive intervals are therefore (1, 2), (2, 3), (5, 6), (6, 7), (9, 10), and (10, ∞), for a total of **(C) 6**.

Solution 4

Denote by I_k the interval $(k - 1, k)$ for $k \in \{2, 3, \dots, 10\}$ and I_1 the interval $(-\infty, 1)$.

Therefore, the number of intervals that $P(x)$ is positive is

$$\begin{aligned} 1 + \sum_{i=1}^{10} \mathbb{I} \left\{ \sum_{j=i}^{10} j \text{ is even} \right\} &= 1 + \sum_{i=1}^{10} \mathbb{I} \left\{ \frac{(i+10)(11-i)}{2} \text{ is even} \right\} \\ &= 1 + \sum_{i=1}^{10} \mathbb{I} \left\{ \frac{-i^2 + i + 110}{2} \text{ is even} \right\} \\ &= 1 + \sum_{i=1}^{10} \mathbb{I} \left\{ \frac{i^2 - i}{2} \text{ is odd} \right\} \\ &= \mathbf{(C) 6}. \end{aligned}$$

Solution 5: Graphing

Recall two key facts about the roots of a polynomial:- If a root has an odd multiplicity (e.g. the root appears an odd number of times), then the graph will cross the x-axis.- If a root has an even multiplicity (e.g. the root appears an even number of times), then the graph will bounce off the x-axis.

Sketching the graph and noting the multiplicity of the roots, we see that there are **(C) 6** positive intervals.

Solution 6: Solution 3 with powers

When the range of $P(x)$ is from $-\infty$ to 1, the sign is negative because in $(x-1)^1(x-2)^2(x-3)^3\cdots(x-10)^{10}$, the terms corresponding to $x = 1, 3, 5, 7$, and 9 are all negative, and since they have odd exponents, they contribute a negative sign. The other terms with even exponents (like $(x-2)^2$, $(x-4)^4$, etc.) are positive regardless of the sign inside, so they don't affect the overall sign. However, in the range 1 to 2, the power of 1 flips, causing this range to be positive. For the range 2 - 3, the sign of any even inside the exponent doesn't matter, so this range stays positive. Continuing this pattern, We see the pattern is $+, +, -, -, \dots$ and so on. there are 3 complete cycles, so $3 * 2 = \boxed{\text{(C) 6}}$ BlueKite

7. (AMC 12 12B 2023) For how many integers n does the expression

$$\sqrt{\frac{\log(n^2) - (\log n)^2}{\log n - 3}}$$

represent a real number, where $\log$ denotes the base 10 logarithm?

- (A) 900 (B) 3 (C) 902 (D) 2 (E) 901

Answer: E

Solution

We have

$$\begin{aligned}\sqrt{\frac{\log(n^2) - (\log n)^2}{\log n - 3}} &= \sqrt{\frac{2\log n - (\log n)^2}{\log n - 3}} \\ &= \sqrt{\frac{(\log n)(2 - \log n)}{\log n - 3}}.\end{aligned}$$

Because n is an integer and $\log n$ is well defined, n must be a positive integer.

Case 1: $n = 1$ or 10^2 .

The above expression is 0. So these are valid solutions.

Case 2: $n \neq 1, 10^2$.

Thus, $\log n > 0$ and $2 - \log n \neq 0$. To make the above expression real, we must have $2 < \log n < 3$. Thus, $100 < n < 1000$. Thus, $101 \leq n \leq 999$. Hence, the number of solutions in this case is 899.

Putting all cases together, the total number of solutions is $\boxed{\text{(E) 901}}$.

Solution (Solution 1 Simplified)

Notice $\log(n^2)$ can be written as $2\log(n)$. Setting $a = \log(n)$, the equation becomes $\sqrt{\frac{2a-a^2}{a-3}}$ which can be written as $\sqrt{\frac{a(2-a)}{a-3}}$

Case 1: $a \geq 3$ The expression is undefined when $a = 3$. For $a > 3$, it is trivial to see that the denominator is positive and the numerator is negative, thus resulting in no real solutions.

Case 2: $2 \leq a < 3$ For $a = 2$, the numerator is zero, giving us a valid solution. When $a > 2$, both the denominator and numerator are negative so all real values of a in this interval is a solution to the equation. All integers of n that makes this true are between 10^2 and $10^3 - 1$. There are 900 solutions here.

Case 3: $0 < a < 2$ The numerator will be positive but the denominator is negative, no real solutions exist.

Case 4: $a = 0$ The expression evaluates to zero, 1 valid solution exists.

Case 5: $a < 0$ All values for $a < 0$ requires $0 < n < 1$, no integer solutions exist.

Adding up the cases: $900 + 1 = \boxed{\text{(E) 901}}$

Solution 3 (3 degree polynomial graph)

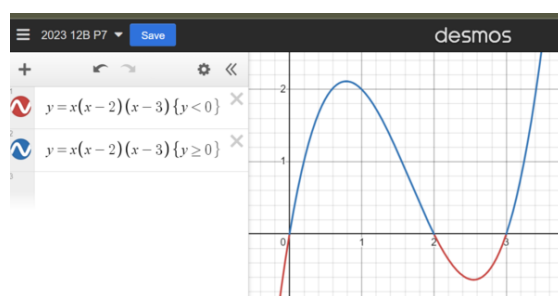
for $\frac{a(2-a)}{a-3} > 0$,

transform it into $a(a-2)(a-3) < 0$,

use the following graph to quickly confirm

1) $a < 0$ or

2) $2 < a < 3$



8. (AMC 12 12B 2023) How many nonempty subsets B of $\{0, 1, 2, 3, \dots, 12\}$ have the property that the number of elements in B is equal to the least element of B ? For example, $B = \{4, 6, 8, 11\}$ satisfies the condition.
- (A) 256 (B) 136 (C) 108 (D) 144 (E) 156

Answer: D

Solution

There is no way to have a set with 0. If a set is to have its lowest element as 1, it must have only 1 element: 1. If a set is to have its lowest element as 2, it must have 2, and the other element will be chosen from the natural numbers between 3 and 12, inclusive. To calculate this, we do $\binom{10}{1}$. If the set is to have its lowest element as 3, the other 2 elements will be chosen from the natural numbers between 4 and 12, inclusive. To calculate this, we do $\binom{9}{2}$. We can see a pattern emerge, where the top is decreasing by 1 and the bottom is increasing by 1. In other words, we have to add $1 + \binom{10}{1} + \binom{9}{2} + \binom{8}{3} + \binom{7}{4} + \binom{6}{5}$. This is $1 + 10 + 36 + 56 + 35 + 6 = \boxed{\text{(D) 144}}$.

Note: In general, i.e. when the number is not 13, the answer is F_{n-1} . -Mr Sharkman

9. (AMC 12 12B 2023) What is the area of the region in the coordinate plane defined by $||x| - 1| + ||y| - 1| \leq 1$? (A) 2 (B) 8 (C) 4 (D) 15 (E) 12

Answer: B

Solution 1

First consider, $|x - 1| + |y - 1| \leq 1$. We can see that it is a square with a side length of $\sqrt{2}$ (diagonal 2). The area of the square is $\sqrt{2}^2 = 2$.

Next, we insert an absolute value sign into the equation and get $|x-1|+||y|-1| \leq 1$. This will double the square reflecting over x-axis.

So now we have 2 squares.

Finally, we add one more absolute value and obtain $||x|-1|+||y|-1| \leq 1$. This will double the squares as we reflect the 2 squares we already have over the y-axis.

Concluding, we have 4 congruent squares. Thus, the total area is $4 \cdot 2 = \boxed{\text{(B)} 8}$

Solution 2 (Graphing)

We first consider the lattice points that satisfy $||x|-1| = 0$ and $||y|-1| = 1$. The lattice points satisfying these equations are $(1, 0), (1, 2), (1, -2), (-1, 0), (-1, 2)$, and $(-1, -2)$. By symmetry, we also have points $(0, 1), (2, 1), (-2, 1), (0, -1), (2, -1)$, and $(-2, -1)$ when $||x|-1| = 1$ and $||y|-1| = 0$. Graphing and connecting these points, we form 5 squares. However, we can see that any point within the square in the middle does not satisfy the given inequality (take $(0, 0)$, for instance). As noted in the above solution, each square has a diagonal 2 for an area of $\frac{2^2}{2} = 2$, so the total area is $4 \cdot 2 = \boxed{\text{(B)} 8}$.

Note This problem is very similar to a past AIME problem (1997 P13)

https://artofproblemsolving.com/wiki/index.php/1997_AIME_Problems/Problem_13

Solution 3 (Logic)

The value of $|x|$ and $|y|$ can be a maximum of 1 when the other is 0. Therefore the value of x and y range from -2 to 2. This forms a diamond shape which has area $4 \times \frac{2^2}{2}$ which is $\boxed{\text{(B)} 8}$.

Solution 4

We start by considering the graph of $|x| + |y| \leq 1$. To get from this graph to $||x|-1| + ||y|-1| \leq 1$ we have to translate it by ± 1 on the x axis and ± 1 on the y axis.

Graphing $|x| + |y| \leq 1$ we get a square with side length of $\sqrt{2}$, so the area of one of these squares is just 2.

We have to multiply by 4 since there are 4 combinations of shifting the x and y axis.

So we have 2×4 which is $\boxed{\text{(B)} 8}$.

Solution 5 (Desperate)

There are 2 sets of 2 absolute value bars. Each of those 2 absolute value bars can take on 2 values, so we have $2 \cdot 2 \cdot 2 = 8$ cases. We guess that the answer is divisible by 8. The only answer choice that is divisible by 8 is $\boxed{\text{(B)} 8}$.

10. (AMC 12 12B 2023) In the xy -plane, a circle of radius 4 with center on the positive x -axis is tangent to the y -axis at the origin, and a circle with radius 10 with center on the positive y -axis is tangent to the x -axis at the origin. What is the slope of the line passing through the two points at which these circles intersect?

(A) $\frac{2}{7}$ (B) $\frac{3}{7}$ (C) $\frac{2}{\sqrt{29}}$ (D) $\frac{1}{\sqrt{29}}$ (E) $\frac{2}{5}$

Answer: E

Solution 1

The center of the first circle is $(4, 0)$. The center of the second circle is $(0, 10)$. Thus, the slope of the line that passes through these two centers is $-\frac{10}{4} = -\frac{5}{2}$.

Because this line is the perpendicular bisector of the line that passes through two intersecting points of two circles, the slope of the latter line is $\frac{-1}{-\frac{5}{2}} = \frac{2}{5}$. **(E)**

Solution 2 (Coordinate Geometry)

The first circle can be written as $(x - 4)^2 + y^2 = 4^2$ we'll call this equation (1) The second can be written as $x^2 + (y - 10)^2 = 10^2$, we'll call this equation (2)

Expanding (1):

$$\begin{aligned} x^2 - 8x + 4^2 + y^2 &= 4^2 \\ x^2 - 8x + y^2 &= 0 \end{aligned}$$

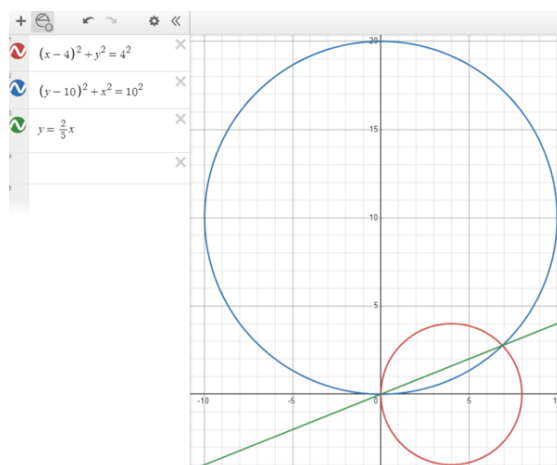
Expanding (2)

$$\begin{aligned} x^2 + y^2 - 20y + 10^2 &= 10^2 \\ x^2 + y^2 - 20y &= 0 \end{aligned}$$

Now we can set the equations equal to each other:

$$\begin{aligned} x^2 - 8x + y^2 &= x^2 + y^2 - 20y \\ \frac{8}{20}x &= y \\ \frac{2}{5}x &= y \end{aligned}$$

This is in slope intercept form therefore the slope is **(E)** $\frac{2}{5}$.



Solution 3 (Polar Coordinates)

In polar coordinates, a circle of radius r tangent to the y -axis with center on the positive x -axis has equation $r = 2a \cos \theta$, and a circle tangent to the x -axis with center on the positive y -axis has equation $r = 2b \sin \theta$.

The first circle (radius 4, center $(4, 0)$) has polar equation $r = 8 \cos \theta$, and the second circle (radius 10, center $(0, 10)$) has polar equation $r = 20 \sin \theta$.

The origin lies on both circles, so it is one intersection point. Setting the equations equal for the other:

$$8 \cos \theta = 20 \sin \theta \implies \tan \theta = \frac{8}{20} = \frac{2}{5}$$

Since $\tan \theta = \frac{y}{x}$, both intersection points (the origin and the point at angle θ) lie on the line through the origin with slope $\tan \theta$. Therefore, the slope of the line through the two intersection points is $\boxed{\frac{2}{5}}$.

11. (AMC 12 12B 2023) What is the maximum area of an isosceles trapezoid that has legs of length 1 and one base twice as long as the other?

(A) $\frac{5}{4}$ (B) $\frac{8}{7}$ (C) $\frac{5\sqrt{2}}{4}$ (D) $\frac{3}{2}$ (E) $\frac{3\sqrt{3}}{4}$

Answer: D

Solution 1

Let the trapezoid be $ABCD$ with $AD = BC = 1$, $AB = x$, $CD = 2x$. Extend AD and BC to meet at point E . Then, notice $\triangle ABE \sim \triangle DCE$ with side length ratio $1 : 2$ and $AE = BE = 1$. Thus, $[DCE] = 4 \cdot [ABE]$ and $[ABCD] = [DCE] - [ABE] = \frac{3}{4} \cdot [DCE]$.

The problem reduces to maximizing the area of $[DCE]$, an isosceles triangle with legs of length 2. Analyzing the sine area formula, this is clearly maximized when $\angle DEC = 90^\circ$, so $[DCE] = 2$ and $[ABCD] = \frac{3}{4} \cdot 2 = \boxed{\frac{3}{2}}$.

-PIDay

Solution 2

Denote by x the length of the shorter base. Thus, the height of the trapezoid is

$$\sqrt{1^2 - \left(\frac{x}{2}\right)^2}.$$

Thus, the area of the trapezoid is

$$\begin{aligned} \frac{1}{2} (x + 2x) \sqrt{1^2 - \left(\frac{x}{2}\right)^2} &= \frac{3}{4} \sqrt{x^2 (4 - x^2)} \\ &\leq \frac{3}{4} \frac{x^2 + (4 - x^2)}{2} \\ &= \boxed{\text{(D)} \frac{3}{2}}, \end{aligned}$$

where the inequality follows from the AM-GM inequality and it is binding if and only if $x^2 = 4 - x^2$.

Solution 3 (Calculus)

Derive the expression for area

$$A = \frac{3}{4} x \sqrt{4 - x^2}$$

as in the solution above. To find the minimum, we can take the derivative with respect to x :

$$\frac{dA}{dx} = \frac{3}{4}\sqrt{4-x^2} + \left(\frac{3x}{4}\right) \frac{-2x}{2\sqrt{4-x^2}} = \frac{6-3x^2}{2\sqrt{4-x^2}}.$$

This expression is equal to zero when $x = \pm\sqrt{2}$, so A has two critical points at $\pm\sqrt{2}$. But given the bounds of the problem, we can conclude $x = \sqrt{2}$ maximizes A (alternatively you can do first derivative test). Plugging that value back in, we get

$$A_{\max} = \boxed{\text{(D)} \frac{3}{2}}.$$

(Slightly Simpler)

Or rewrite the expression for area to be

$$A = \frac{3}{4}\sqrt{4x^2 - x^4}$$

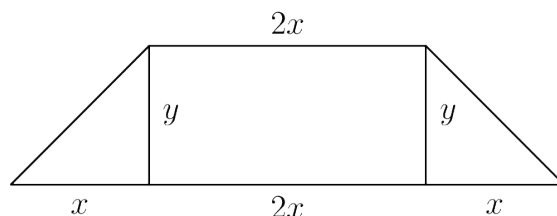
Now to find the minimum, we can just find the minimum of what's inside the square root (since the square root function is increasing). Take the derivative of $4x^2 - x^4$,

$$f'(x) = 8x - 4x^3.$$

This is equal to zero at $x = 0, \pm\sqrt{2}$ but the solution must be positive so $x = \sqrt{2}$.

Solution 4 (Trigonometry)

Let the length of the shorter base of the trapezoid be $2x$ and the height of the trapezoid be y .



Each leg has length 1 if and only if $x^2 + y^2 = 1$, where x and y are positive real numbers. The general solution to this equation is

$$(x, y) = (\cos t, \sin t)$$

for any number $0 < t < \frac{\pi}{2}$ so that x and y are positive. The area to maximize is

$$\frac{1}{2}(2x + 4x)y = 3xy$$

Hence, we maximize $3 \sin t \cos t$ for $0 < t < \frac{\pi}{2}$.

$$\begin{aligned} 3xy &= 3 \sin t \cos t \\ &= \frac{3}{2}(2 \sin t \cos t) \\ &= \frac{3}{2} \sin(2t) \end{aligned}$$

The maximum of $\sin(2t)$ is 1, thus the maximum of $3xy$ is $\boxed{\text{(D)} \frac{3}{2}}$ which occurs at $t = \frac{\pi}{4}$, satisfying the inequality $0 < t < \frac{\pi}{2}$.

Solution 5

Denote x and $2x$ as the two bases. We can straightforwardly find that the height of the trapezoid is $\sqrt{1 - \frac{x^2}{4}}$ by the Pythagorean theorem.

The area of this trapezoid is then given by the expression $\frac{x+2x}{2} \cdot \sqrt{1 - \frac{x^2}{4}}$.

Let m be the maximum value that this expression achieves. Since x is positive and assuming $m \geq 1$, we can perform the following operations to maximize $16m^2$, thus m^2 , and thus m . We have

$$\begin{aligned}\frac{x+2x}{2} \cdot \sqrt{1 - \frac{x^2}{4}} &= m \\ \frac{9x^2}{4} \cdot \frac{4-x^2}{4} &= m^2 \\ -9x^4 + 36x^2 &= 16m^2.\end{aligned}$$

The maximum value of this quartic occurs at $x^2 = \frac{-36}{2(-9)} = 2$, or when $x = \sqrt{2}$.

It follows that the area of the trapezoid is equal to

$$\frac{3\sqrt{2}}{2} \cdot \frac{1}{\sqrt{2}} = \boxed{\text{(D)} \frac{3}{2}}.$$

-Benedict T (countmath1)

12. (AMC 12 12B 2023) For complex numbers $u = a + bi$ and $v = c + di$, define the binary operation $\otimes$ by

$$u \otimes v = ac + bdi.$$

Suppose z is a complex number such that $z \otimes z = z^2 + 40$. What is $|z|$?

- (A) 2 (B) 5 (C) $\sqrt{5}$ (D) $\sqrt{10}$ (E) $5\sqrt{2}$

Answer: E

Solution 1

let $z = a + bi$.

$$z \otimes z = a^2 + b^2i.$$

This is equal to $z^2 + 40 = a^2 - b^2 + 40 + 2abi$

Since the real values have to be equal to each other, $a^2 - b^2 + 40 = a^2$. Simple algebra shows $b^2 = 40$, so $b = \pm 2\sqrt{10}$.

The imaginary components must also equal each other, meaning $b^2 = 2ab$, or $b = 2a$. This means $a = \frac{b}{2} = \pm\sqrt{10}$.

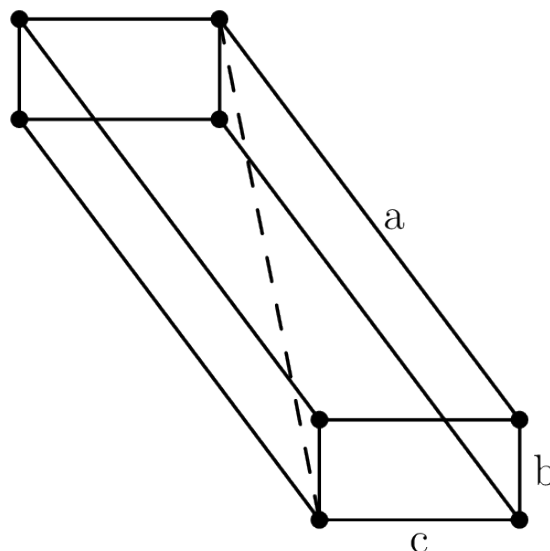
Thus, the magnitude of z is $\sqrt{a^2 + b^2} = \sqrt{10 + 40} = \sqrt{50} = 5\sqrt{2} = \boxed{\text{(E)} 5\sqrt{2}}$

13. (AMC 12 12B 2023) A rectangular box P has distinct edge lengths a , b , and c . The sum of the lengths of all 12 edges of P is 13, the sum of the areas of all 6 faces of P is $\frac{11}{2}$, and the volume of P is $\frac{1}{2}$. What is the length of the longest interior diagonal connecting two vertices of P ?

(A) 2 (B) $\frac{3}{8}$ (C) $\frac{9}{8}$ (D) $\frac{9}{4}$ (E) $\frac{3}{2}$

Answer: D

Solution 1 (algebraic manipulation)



We can create three equations using the given information.

$$4a + 4b + 4c = 13$$

$$2ab + 2ac + 2bc = \frac{11}{2}$$

$$abc = \frac{1}{2}$$

We also know that we want $\sqrt{a^2 + b^2 + c^2}$ because that is the length that can be found from using the Pythagorean Theorem. We cleverly notice that $a^2 + b^2 + c^2 = (a + b + c)^2 - 2(ab + ac + bc)$. We know that $a + b + c = \frac{13}{4}$ and $2(ab + ac + bc) = \frac{11}{2}$,

so $a^2 + b^2 + c^2 = \left(\frac{13}{4}\right)^2 - \frac{11}{2} = \frac{169-88}{16} = \frac{81}{16}$. So our answer is $\sqrt{\frac{81}{16}} = \boxed{\text{(D)} \frac{9}{4}}$.

Solution 2 (Vieta's)

We use the equations from Solution 1 and manipulate it a little:

$$a + b + c = \frac{13}{4}$$

$$ab + ac + bc = \frac{11}{4}$$

$$abc = \frac{1}{2}$$

Notice how these are the equations for the vieta's formulas for a polynomial with roots of a , b , and c . Let's create that polynomial. It would be $x^3 - \frac{13}{4}x^2 + \frac{11}{4}x - \frac{1}{2}$.

Multiplying each term by 4 to get rid of fractions, we get $4x^3 - 13x^2 + 11x - 2$. Notice how the coefficients add up to 0. Whenever this happens, that means that $(x - 1)$ is a factor and that 1 is a root. After using synthetic division to divide $4x^3 - 13x^2 + 11x - 2$ by $x - 1$, we get $4x^2 - 9x + 2$. Factoring that, you get $(x - 2)(4x - 1)$. This means that this polynomial factors to $(x - 1)(x - 2)(4x - 1)$ and that the roots are 1, 2, and $1/4$. Since we're looking for $\sqrt{a^2 + b^2 + c^2}$, this is equal to $\sqrt{1^2 + 2^2 + \frac{1}{4}^2} = \sqrt{\frac{81}{16}} = \boxed{\text{(D)} \frac{9}{4}}$

Solution 3 (Newton's Sums)

Same formal process of Solution 2, once we have $4x^3 - 13x^2 + 11x - 2$. Let $s_2 = a^2 + b^2 + c^2$ and $s_1 = a + b + c = \frac{13}{4}$. Applying **Newton's Sums**, we have

$$(4)s_2 + (-13)s_1 + 2 * 11 = 0$$

. We can solve that $s_2 = \frac{81}{16}$, and the interior diagonal would be

$$\sqrt{s_2} = \frac{9}{4}$$

Solution 4 (Cheese Method)

Incorporating the solution above, we know $a + b + c = 13/4 \Rightarrow a + b + c > 3$. The side lengths are larger than $1 \cdot 1 \cdot 1$ (a unit cube). The side length of the interior of a unit cube is $\sqrt{3}$, and we know that the side lengths are larger than $1 \cdot 1 \cdot 1$, so that means the diagonal has to be larger than $\sqrt{3}$, and the only answer choice larger than $\sqrt{3} \Rightarrow \boxed{\text{(D)} \frac{9}{4}}$

Note that the real number $\sqrt{3}$ is around 1.73. Option A is also greater than $\sqrt{3}$ meaning there are two options greater than $\sqrt{3}$. Option A is an integer so educationally guessing we arrive at answer D $\Rightarrow \boxed{\text{(D)} \frac{9}{4}}$

14. (AMC 12 12B 2023) For how many ordered pairs (a, b) of integers does the polynomial $x^3 + ax^2 + bx + 6$ have 3 distinct integer roots?

(A) 5 (B) 6 (C) 8 (D) 7 (E) 4

Answer: A

Solution

Denote three roots as $r_1 < r_2 < r_3$. Following from Vieta's formula, $r_1 r_2 r_3 = -6$.

Case 1: All roots are negative.

We have the following solution: $(-3, -2, -1)$.

Case 2: One root is negative and two roots are positive.

We have the following solutions: $(-3, 1, 2)$, $(-2, 1, 3)$, $(-1, 2, 3)$, $(-1, 1, 6)$.

Putting all cases together, the total number of solutions is $\boxed{\text{(A)} 5}$.

15. (AMC 12 12B 2023) Suppose a , b , and c are positive integers such that

$$\frac{a}{14} + \frac{b}{15} = \frac{c}{210}.$$

Which of the following statements are necessarily true?

I. If $\gcd(a, 14) = 1$ or $\gcd(b, 15) = 1$ or both, then $\gcd(c, 210) = 1$.

II. If $\gcd(c, 210) = 1$, then $\gcd(a, 14) = 1$ or $\gcd(b, 15) = 1$ or both.

III. $\gcd(c, 210) = 1$ if and only if $\gcd(a, 14) = \gcd(b, 15) = 1$.

(A) I, II, and III (B) I only (C) I and II only (D) III only (E) II and III only

Answer: E

Solution 1 (Guess and check + Contrapositive)

We examine each of the conditions.

The first condition is false. A simple counterexample is $a = 3$ and $b = 5$. The corresponding value of c is 115. Since $\gcd(3, 14) = 1$, condition I would imply that $\gcd(c, 210) = 1$. However, $\gcd(115, 210)$ is clearly not 1 (they share a common factor of 5). Condition I is false so that we can rule out choices A, B, and C.

We now decide between the two answer choices D and E. What differs between them is the validity of condition II, so it suffices to check II simply.

We look at statement II's contrapositive to prove it. The contrapositive states that if $\gcd(a, 14) \neq 1$ and $\gcd(b, 15) \neq 1$, then $\gcd(c, 210) \neq 1$. In other words, if a shares some common factor that is not 1 with 14 and b shares some common factor that is not 1 with 15, then c also shares a common factor that is not 1 with 210. Let's say that $a = a' \cdot n$, where a' is a factor of 14 not equal to 1. (So a' is the common factor.)

We can rewrite the given equation as $15a + 14b = c \implies 15(a'n) + 14b = c$. We can express 14 as $a' \cdot n'$, for some positive integer n' (this n' can be 1). We can factor a' out to get $a'(15n + bn') = c$.

Since all values in this equation are integers, c must be divisible by a' . Since a' is a factor of 14, a' must also be a factor of 210, a multiple of 14. Therefore, we know that c shares a common factor with 210 (which is a'), so $\gcd(c, 210) \neq 1$. This is what II states, so therefore II is true.

Thus, our answer is (E) II and III only. Technodoggo AVM2023 (Minor Edits - Fixes in the first paragraph and grammar edits)

Solution 2

The equation given in the problem can be written as

$$15a + 14b = c. \quad (1)$$

First, we prove that Statement I is not correct.

A counter example is $a = 1$ and $b = 3$. Thus, $\gcd(c, 210) = 3 \neq 1$.

Second, we prove that Statement III is correct.

First, we prove the "if part."

Suppose $\gcd(a, 14) = 1$ and $\gcd(b, 15) = 1$. However, $\gcd(c, 210) \neq 1$.

Thus, c must be divisible by at least one factor of 210. W.L.O.G, we assume c is divisible by 2.

Modulo 2 on Equation (1), we get that $2|a$. This is a contradiction with the condition that $\gcd(a, 14) = 1$. Therefore, the "if part in Statement III is correct."

Second, we prove the "only if part."

Suppose $\gcd(c, 210) \neq 1$. Because $210 = 14 \cdot 15$, there must be one factor of 14 or 15 that divides c . W.L.O.G, we assume there is a factor $q > 1$ of 14 that divides c . Because $\gcd(14, 15) = 1$, we have $\gcd(q, 15) = 1$. Modulo q on Equation (1), we have $q|a$.

Because $q|14$, we have $\gcd(a, 14) \geq q > 1$.

Analogously, we can prove that $\gcd(b, 15) > 1$.

Third, we prove that Statement II is correct.

This is simply a special case of the “only if *part of Statement III*. So we omit the proof.

All analyses above imply **(E) II and III only**.

Solution 3 (Answer Choices)

It can easily be shown that statement I is false (a counterexample would be $a = 1, b = 5, c = 85$), meaning the only viable answer choices are D and E. Since both of these answer choices include statement III, this means III is true. Since III is true, this actually implies that statement II is true, as III is just a stronger version of statement II (or its contrapositive, to be precise). Therefore the answer is **(E) II and III only**.

Solution 4

First, we will show that statement I is false. A simple counterexample is $a = 2$ and $b = 2$. Then, c is even, and it is not relatively prime with 210.

Now, we will focus on statement III. It's pretty easy to realize that if either a isn't relatively prime to 14 or b isn't relatively prime to 15, it will always share factors with 210 because $c = 15a + 14b$ (think $15 \cdot 14 + 14$). Therefore, statement III must be true.

Since statement III is a stricter case of statement II, the last two statements are true, hence choose **(E) II and III only**.

Solution 5 (Rigorous)

The condition in the problem is equivalent to

$$15a + 14b = c.$$

From here we can use the properties of the greatest common divisor to get

$$\gcd(c, 14) = \gcd(15a + 14b, 14) = \gcd(15a, 14) = \gcd(a, 14)$$

and

$$\gcd(c, 15) = \gcd(15a + 14b, 15) = \gcd(14b, 15) = \gcd(b, 15)$$

From here it's not hard to get

$$\gcd(c, 210) = \gcd(c, 15) * \gcd(c, 14) = \gcd(a, 14) * \gcd(b, 15)$$

Hence $\gcd(c, 210) = 1$ is equivalent to $\gcd(a, 14) = 1$ and $\gcd(b, 15) = 1$. This means the answer must be **(E) II and III only**.

Solution 6 (Intuitive Risking)

We know that $\frac{a}{14} + \frac{b}{15} = \frac{c}{210}$. So, therefore, $\frac{a+14b}{15} = \frac{c}{1}$, and $15a + 14b = 210c$.

We could see that some value of a and another value of b gives a different value of c . **WLOG**, we give an intuitive risk, saying that $a + b = c$. Our goal is to just prove statements, not find solutions.

Statement I:

If $\gcd(a, 14) = 1$ or $\gcd(b, 15) = 1$ or both, then $\gcd(c, 210) = 1$.

We have $a + b = c$.

Then if $\gcd(a, 14) = 1$, we say $a = 1$

So then $\gcd(b, 14) \neq 1$, so we say $b = 2$.

$1 + 2 = 3$, and $\gcd(3, 210) = 1$ is false. So statement I is incorrect.

Statement II:

If $\gcd(c, 210) = 1$, then $\gcd(a, 14) = 1$ or $\gcd(b, 15) = 1$ or both.

We say $a + b = c$.

For statement II, we need to prove that it works for every odd number that is a multiple of the first number that makes both a and b positive integers and $\gcd(c, 210) = 1$.

This number is 9. The possible pair values for a and b are 5 and 4 respectively. If we try 27, we get for a and b can be 15 and 12. We clearly see this is a pattern, and to get to an odd multiple of 9 we just multiply the sums by the n th multiple of 9. Therefore statement II is always true.

Statement III:

$\gcd(c, 210) = 1$ if and only if $\gcd(a, 14) = \gcd(b, 15) = 1$.

We say $a + b = c$

We prove this by contradiction.

Assume $\gcd(a, 14) \neq \gcd(b, 15) \neq 1$.

Then, that means $\gcd(c, 210)$ is never equal to 1.

So if we take $a = 14$ and $b = 15$, we get that $\gcd(29, 210) \neq 1$. This is obviously false, because 29 is prime.

Therefore statement III is also correct.

Only II and III are correct, leading us to answer choice (E) II and III only.

Solution 7 (Formal Logic Lattice)

Multiply the original equation by 210 to get $15a + 14b = c$.

Denote the propositions $A := (\gcd(a, 14) = 1)$, $B := (\gcd(b, 15) = 1)$, $C := (\gcd(c, 210) = 1)$.

Work on the lattice with set $\{A, B, C\}$ with $\geq$ the order relation given by $\implies$, $\wedge$ the logical minimum (AND), $\vee$ the logical maximum (OR), $\equiv$ logical equivalence $\iff$.

I. $A \vee B \geq C \iff (A \geq C) \wedge (B \geq C)$

II. $C \geq A \vee B$

III. $C \equiv A \wedge B$

Note $A \wedge B \geq A \vee B$, so **III** $\implies$ **II**. Also, **III** $\wedge$ **I** $\implies A \equiv B \equiv C$ by the squeeze lemma. In our case $A \not\equiv B \implies \neg \text{III} \vee \neg \text{I}$ (at least one of these statements is false). (The squeeze lemma refers to $A \wedge B \geq A \vee B \implies A \equiv B$ from decomposing **I** into $A \geq C \equiv A \wedge B \iff A \geq B$ and symmetrically $B \geq A$.)

But notice $\gcd(14, 15) = 1$. Use multiplicativity $\gcd(x, y) = 1 \implies \gcd(z, xy) = \gcd(z, x) \gcd(z, y)$ and Euclidean algorithm $\gcd(x, y) = \gcd(x, y \pm x)$ to get

$$\gcd(c, 210) = \gcd(15a + 14b, 14) \gcd(15a + 14b, 15) = \gcd(a, 14) \gcd(b, 15).$$

Since $\gcd : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}_{\geq 0} \implies \gcd \geq 1$, so $\gcd(c, 210) = 1 \iff \gcd(a, 14) = \gcd(b, 15) = 1$. By the above, **III** is true, **II** is true, **I** is false, so the answer is (E) II and III only.

Solution 8 (Number Theory)

Rewrite as $15a + 14b = c$. Assume I, particularly the "or both" section. this fails if $a = 1$ and $b = 2$, for which we have the gcd is a factor of 2 times et cetera. Due to failure within the "or both" case, note that we can let $a = 2$ for which the first half fails. I is false.

Consider II. Firstly, we show "or both" is a viable case. Consider $\gcd(c, 210) = 1$. Then, $\gcd(14a + 15b, 210) = 1$. So, $14a + 15b$ is odd not a multiple of 3 nor 7. Modulo 3 gives $2a \equiv 0 \pmod{3}$. That implies $a \equiv 0 \pmod{3}$. Also, take modulo 7. Then, $b \equiv 0 \pmod{7}$. $\gcd(14a + 15b, 210) > 1$ holds if a, b is a multiple of either 7 or 3 respectively (or both). This implies that if $\gcd(14a + 15b, 210) = 1$ holds, then none of a, b is a multiple of 7 or 3, meaning that $\gcd(a, 14) = \gcd(b, 15) = 1$, which satisfies the "or both" case of II, and thus II is true.

Consider III. That is literally the "both" case of II, which we proved is always true. The answer is II and III only.

Solution 9 (Set Theory)

We can use set theory to approach this problem, firstly by defining the universal set U to be all possible positive integer ordered pairs (a, b) .

Next, create a Venn diagram that consists of 2 sets: a set A of pairs (a, b) so that $\gcd(a, 14) > 1$, and a set B of all pairs (a, b) so that $\gcd(b, 15) > 1$. In other words, set A includes all pairs where a is a multiple of 2 or 7 and set B includes all pairs where b is a multiple of 3 or 5.

Multiplying the original equation by 210 yields $15a + 14b = c$. Because $210 = 2 \cdot 3 \cdot 5 \cdot 7$, if $\gcd(c, 210) > 1$, a must be a multiple of 2 or 7 or b must be a multiple of 3 or 5. Thus, $\gcd(c, 210) > 1$ directly corresponds to $A \cup B$.

Next, understand what it means when the gcd function equals 1. $\gcd(a, 14) = 1$, for example, means that a is **not** a multiple of 2 or 7. Similarly, $\gcd(b, 15) = 1$ means that b is **not** a multiple of 3 or 5. Note that this means the set of pairs satisfying $\gcd(a, 14) = 1$ is the complement of set A (A'). Similarly, the set of pairs satisfying $\gcd(b, 15) = 1$ is the complement of set B (B') and the set of pairs satisfying $\gcd(c, 210) = 1$ is the complement of $A \cup B$ ($(A \cup B)'$).

Therefore, we can rewrite statements I, II, and III using the complements of set A , set B , and $A \cup B$.

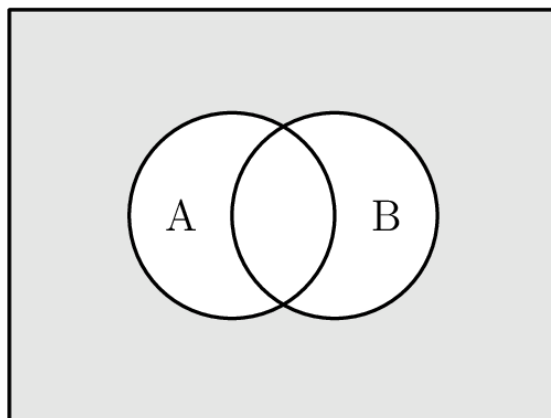
I. $A' \cup B' \subseteq (A \cup B)'$

II. $(A \cup B)' \subseteq A' \cup B'$

III. $(A \cup B)' = A' \cap B'$ $A' \cup B'$, $(A \cup B)'$, and $A' \cap B'$ can be modeled as follows:

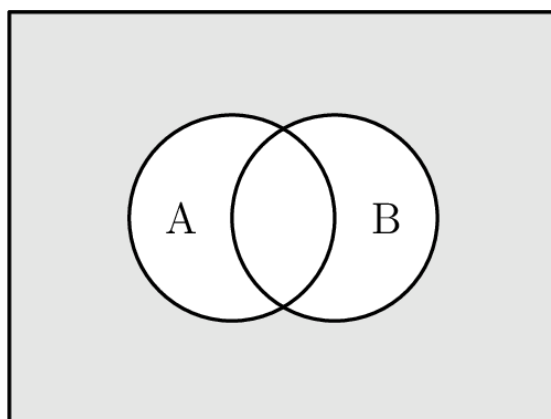
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$$(A \cup B)'$$



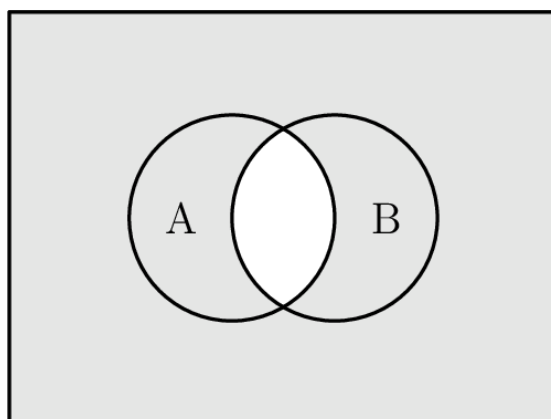
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$$A' \cap B'$$



--	--

$$A' \cup B'$$



If we simply compare these Venn diagrams to the rewritten statements, we can see that I is false and II and III (which turns out to be De Morgan's First Law) are true. Thus, the answer is **(E) II and III only**.

16. (AMC 12 12B 2023) In Coinland, there are three types of coins, each worth 6, 10, and 15. What is the sum of the digits of the maximum amount of money that is impossible to have?
- (A) 8 (B) 10 (C) 7 (D) 11 (E) 9

Answer: D

Solution 1

This problem asks to find largest x that cannot be written as

$$6a + 10b + 15c = x, \quad (1)$$

where $a, b, c \in \mathbb{Z}_+$.

Denote by $r \in \{0, 1\}$ the remainder of x divided by 2. Modulo 2 on Equation (1), we get

By using modulus $m \in \{2, 3, 5\}$ on the equation above, we get

$c \equiv r \pmod{2}$. Following from the **Chicken McNugget Theorem**, we have that any number that is no less than $(3 - 1)(5 - 1) = 8$ can be expressed in the form of $3a + 5b$ with $a, b \in \mathbb{Z}_+$.

Therefore, all even numbers that are at least equal to $2 \cdot 8 + 15 \cdot 0 = 16$ can be written in the form of Equation (1) with $a, b, c \in \mathbb{Z}_+$. All odd numbers that are at least equal to $2 \cdot 8 + 15 \cdot 1 = 31$ can be written in the form of Equation (1) with $a, b, c \in \mathbb{Z}_+$.

The above two cases jointly imply that all numbers that are at least 30 can be written in the form of Equation (1) with $a, b, c \in \mathbb{Z}_+$.

Next, we need to prove that 29 cannot be written in the form of Equation (1) with $a, b, c \in \mathbb{Z}_+$.

Because 29 is odd, we must have $c \equiv 1 \pmod{2}$. Because $a, b, c \in \mathbb{Z}_+$, we must have $c = 1$. Plugging this into Equation (1), we get $3a + 5b = 7$. However, this equation does not have non-negative integer solutions.

All analysis above jointly imply that the largest x that has no non-negative integer solution to Equation (1) is 29. Therefore, the answer is $2 + 9 =$ **(D) 11**.

Solution 2 (Multiples Of 6)

Arrange the positive integers into rows of 6, like this:

01	02	03	04	05	06
07	08	09	10	11	12
13	14	15	16	17	18
19	20	21	22	23	24
25	26	27	28	29	30
31	32	33	34	35	36

⋮

Observe that if any number can be made from a combination of 6s, 10s, and 15s, then every number below it in the same column must also be possible to make, by simply adding 6s.

Thus, we will cross out any numbers that CAN be made as well as all numbers below it.

In column 1, 25 is possible ($10 + 15$) and so is everything below 25.

Column 2 - cross out 20

Column 3 - cross out 15

Column 4 - cross out 10

Column 5 - cross out 35

Column 6 - cross all out.

The maximum number that remains is 29. Answer is $2 + 9 = \boxed{\text{(D) } 11}$.

Solution 3 (Modulo 6)

Let the number of 6 cent coins be a , the number of 10 cent coins be b , and the number of 15 cent coins be c . We get the **Diophantine equation**

$$6a + 10b + 15c = k$$

and we wish to find the largest possible value of k

Construct the following mod 6 table of 6, 10, and 15

number of coins	6	10	15
1	0	4	3
2	0	2	0

There are only 6 possible residues for 6, they are: 0, 1, 2, 3, 4, and 5.

$0 \pmod 6$ We can obtain any value that is $0 \pmod 6$ because we have 6\$ cent coins.

$1 \pmod 6$ We can obtain values that equal $1 \pmod 6$ by using one 10\$ cent coin, or

$2 \pmod 6$ We can obtain values that equal $2 \pmod 6$ by using two 10\$ cent coins, a

\$3 \pmod 6\$ We can obtain values that equal \$3 \pmod 6\$ by using one \$15\$ cent coin, and
 \$4 \pmod 6\$ We can obtain values that equal \$4 \pmod 6\$ by using one \$10\$ cent coin, and
 \$5 \pmod 6\$ We can obtain values that equal \$5 \pmod 6\$ by using two \$10\$ cent coins, and

Hence, the largest value in cents we cannot obtain using 6, 10, and 15 cent coins is
 $29, 2 + 9 = \boxed{\text{(D) } 11}$.

Solution 4

We claim that the largest number that cannot be obtained using 6, 10, and 15 cent coins is 29.

Let's first focus on the combination of 6, 10. As both of them are even numbers, we cannot obtain any odd numbers from these two but requires 15 to sum up to an odd number. Notice that by Chicken McNugget Theorem, the largest even number cannot be obtained by 6, 10 is $2(3 \cdot 5 - 3 - 5) = 14$. Add this with 29, we can easily verify that 29 cannot be obtained by 6, 10, and 15 as it needs at least one odd number, with the remaining part cannot be represented by 6 and 10.

Let's show that any number greater than 29 can be obtained. First, any even numbers greater than 29 can be obtained by 6 and 10 by the Chicken McNugget Theorem. Next, any odd number greater than 29 can be obtained by adding one 15 with some 6s and 10s, which is also shown by the Chicken McNugget Theorem. This completes the proof. So the answer is $2 + 9 = \boxed{\text{(D) } 11}$

Solution 5 (apply Chicken McNugget Theorem twice)

First considering the two terms

$$6a + 10b = 2(3a + 5b)$$

$$3a + 5b = (3 - 1)(5 - 1) + t = 8 + t, t \geq 0$$

Again applying the two variable formula for the terms in brackets we see that

$$6a + 10b + 15c = 2(8 + t) + 15c = 2t + 15c + 16 = (2 - 1)(15 - 1) + s + 16 = 30 + s, s \geq 0$$

so the given expression $6a + 10b + 15c$ produces all numbers from 30 and 29 is the largest number that could not be produced, so the answer is $2 + 9 = \boxed{\text{(D) } 11}$

- Note: Chicken McNugget Theorem: any positive integer $N = (a-1)(b-1) + t$ (a, b co-prime, $t \geq 0$) can always be represented as $N = aP + bQ$ (p, q are non-negative integer)

NOTE (by Mqoi_KOLA)- this method isn't true in general for any three coprime numbers take for example 10, 14, 3 from the method used here one get 49 as the largest possible number which can't be achieved which is false because 11 is unachievable.

17. (AMC 12 12B 2023) Triangle ABC has side lengths in arithmetic progression, and the smallest side has length 6. If the triangle has an angle of 120° , what is the area of ABC ?

- (A) $12\sqrt{3}$ (B) $8\sqrt{6}$ (C) $14\sqrt{2}$ (D) $20\sqrt{2}$ (E) $15\sqrt{3}$

Answer: E

Solution 1

The length of the side opposite to the angle with 120° is longest. We denote its value as x .

Because three side lengths form an arithmetic sequence, the middle-valued side length is $\frac{x+6}{2}$.

Following from the law of cosines, we have

$$6^2 + \left(\frac{x+6}{2}\right)^2 - 2 \cdot 6 \cdot \frac{x+6}{2} \cdot \cos 120^\circ = x^2.$$

By solving this equation, we get $x = 14$. Thus, $\frac{x+6}{2} = 10$.

Therefore, the area of the triangle is

$$\frac{1}{2} \cdot 6 \cdot 10 \cdot \sin 120^\circ = \boxed{\text{(E)} 15\sqrt{3}}.$$

Solution 2

Let the side lengths of $\triangle ABC$ be 6, x , and $2x - 6$, where $6 \leq x \leq 2x - 6$. As $2x - 6$ is the longest side, the angle opposite to it will be 120° .

By the law of Cosine

$$(2x - 6)^2 = 6^2 + x^2 - 2 \cdot 6 \cdot x \cdot \cos 120^\circ$$

$$4x^2 - 24x + 36 = 36 + x^2 + 6x$$

$$3x^2 - 30x = 0$$

$$x^2 - 10x = 0$$

As $x \neq 0$, $x = 10$.

$$\text{Therefore, } [ABC] = \frac{6 \cdot 10 \cdot \sin 120^\circ}{2} = \boxed{\text{(E)} 15\sqrt{3}}$$

Solution 3

Let the side lengths of $\triangle ABC$ be 6, $6 + d$, and $6 + 2d$, where $6 \leq 6 + d \leq 6 + 2d$. As $6 + 2d$ is the longest side, the angle opposite to it will be 120° .

By the law of Cosine

$$(6 + 2d)^2 = 6^2 + (6 + d)^2 - 2 \cdot 6 \cdot (6 + d) \cdot \cos 120^\circ$$

$$4d^2 + 24d + 36 = 36 + 36 + 12d + d^2 + 36 + 6d$$

$$3d^2 + 6d - 72 = 0$$

$$d^2 + 2d - 24 = 0$$

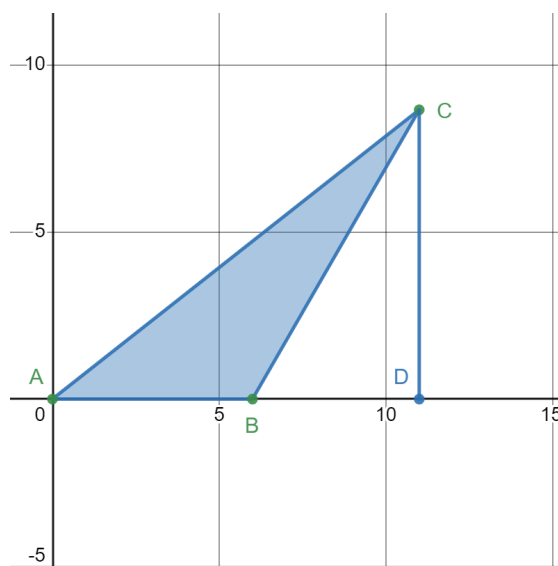
$$(d + 6)(d - 4) = 0$$

As $d \geq 0$, $d = 4$, $6 + d = 10$

$$\text{Therefore, } [ABC] = \frac{6 \cdot 10 \cdot \sin 120^\circ}{2} = \boxed{\text{(E)} 15\sqrt{3}}$$

Solution 4 (Analytic Geometry)

Since the triangle's longest side must correspond to the 120° angle, the triangle is unique. By analytic geometry, we construct the following plot.



We know the coordinates of point A being the origin and B being $(6,0)$. Constructing the line which point C can lay on, here since $\angle B = 120^\circ$, C is on the line

$$y = \sqrt{3}(x - 6).$$

I denote D as the perpendicular line from C to AB , and assume $CD = k$. Here we know $\triangle BCD$ is a $30^\circ - 60^\circ - 90^\circ$ triangle. Hence $DC = \sqrt{3}k$ and $BC = 2k$.

Furthermore, due to the arithmetic progression, we know $AC = 4k - 6$. Hence, in $\triangle ACD$,

$$(4k - 6)^2 = (6 + k)^2 + 3k^2,$$

$$k = 5.$$

Thus, the area is equal to $\frac{1}{2} \cdot 6 \cdot \sqrt{3}k = \boxed{\text{(E)} 15\sqrt{3}}$.

18. (AMC 12 12B 2023) Last academic year Yolanda and Zelda took different courses that did not necessarily administer the same number of quizzes during each of the two semesters. Yolanda's average on all the quizzes she took during the first semester was 3 points higher than Zelda's average on all the quizzes she took during the first semester. Yolanda's average on all the quizzes she took during the second semester was 18 points higher than her average for the first semester and was again 3 points higher than Zelda's average on all the quizzes Zelda took during her second semester. Which one of the following statements cannot possibly be true?
- (A) Yolanda's quiz average for the academic year was 22 points higher than Zelda's.
- (B) Zelda's quiz average for the academic year was higher than Yolanda's.
- (C) Yolanda's quiz average for the academic year was 3 points higher than Zelda's.
- (D) Zelda's quiz average for the academic year equaled Yolanda's.
- (E) If Zelda had scored 3 points higher on each quiz she took, then she would have had the same average for the academic year as Yolanda.

Answer: A

Solution 1

Denote by A_i the average of person with initial A in semester $i \in \{1, 2\}$. Thus, $Y_1 = Z_1 + 3$, $Y_2 = Y_1 + 18$, $Y_2 = Z_2 + 3$.

Denote by A_{12} the average of person with initial A in the full year. Thus, Y_{12} can be any number in (Y_1, Y_2) and Z_{12} can be any number in (Z_1, Z_2) .

Therefore, the impossible solution is (A) Yolanda's quiz average for the academic year was 22 points

Solution 2

We can use process of elimination by finding possible solutions to answer choices. Let y_1 and y_2 be the number of quizzes Yolanda took in the first and second semesters, respectively. Define z_1 and z_2 similarly for Zelda.

Answer choice B is satisfied by $(y_1, y_2, z_1, z_2) = (289, 1, 1, 289)$.

Answer choice C and E are both satisfied by $(y_1, y_2, z_1, z_2) = (17, 17, 17, 17)$.

Answer choice D is satisfied by $(y_1, y_2, z_1, z_2) = (7, 5, 5, 7)$.

Therefore the answer is (A) Yolanda's quiz average for the academic year was 22 points higher than

Solution 3 (Algebra)

Let Yolanda's average for semester 1 be y_1 , the number of quizzes Yolanda took in semester 1 be n_1 , Zelda's average for semester 1 be z_1 , the number of quizzes Zelda took in semester 1 be k_1 , Yolanda's average for semester 2 be y_2 , the number of quizzes Yolanda took in semester 2 be n_2 , Zelda's average for semester 2 be z_2 , the number of quizzes Zelda took in semester 2 be k_2 , Yolanda's average for the entire year be y , Zelda's average for the entire year be z .

From the problem we know that

$$y_1 = z_1 + 3, \quad y_2 = z_2 + 3$$

$$y_2 = y_1 + 18, \quad z_2 = z_1 + 18$$

$$y = \frac{y_1 n_1 + y_2 n_2}{n_1 + n_2} = \frac{y_1 n_1 + (y_1 + 18) n_2}{n_1 + n_2} = y_1 + \frac{18 n_2}{n_1 + n_2}$$

$$z = \frac{z_1 k_1 + z_2 k_2}{k_1 + k_2} = \frac{z_1 k_1 + (z_1 + 18) k_2}{k_1 + k_2} = z_1 + \frac{18 k_2}{k_1 + k_2}$$

$$y - z = y_1 + \frac{18 n_2}{n_1 + n_2} - z_1 - \frac{18 k_2}{k_1 + k_2} = y_1 - z_1 + 18 \left(\frac{n_2}{n_1 + n_2} - \frac{k_2}{k_1 + k_2} \right) = 3 + 18 \left(\frac{n_2}{n_1 + n_2} - \frac{k_2}{k_1 + k_2} \right)$$

$\frac{n_2}{n_1 + n_2}$ at most is 1, $\frac{k_2}{k_1 + k_2}$ is at least 0, meaning that $\frac{n_2}{n_1 + n_2} - \frac{k_2}{k_1 + k_2}$ is at most 1.

Therefore,

$$y - z \leq 3 + 18 = 21$$

Hence, (A) Yolanda's quiz average for the academic year was 22 points higher than Zelda's is not possible.

Solution 4

Denote y_1 and y_2 as the quiz averages of Yolanda in the 1st and 2nd semesters, respectively. Similarly, denote z_1 and z_2 as the quiz averages of Zelda in the 1st and 2nd semesters.

We have $y_1 = z_1 + 3$, so $y_1 - 3 = z_1$. We also know that $y_2 = 18 + y_1 = 3 + z_2$, implying $z_2 = 15 + y_1$.

The average quiz scores for both students must lie between the averages of each semester, i.e $\text{Avg}_y \in [y_1, 18 + y_1]$, and $\text{Avg}_z \in [y_1 - 3, y_1 + 15]$. Since $z_2 > z_1$, $y_2 > y_1$, and $y_1 > z_1$, we have $\min\{\text{Avg}\} = \min\{\text{Avg}_z\} = z_1$ and $\max\{\text{Avg}\} = \max\{\text{Avg}_y\} = y_2$. Therefore the maximum difference between the two yearly averages is

$$|y_2 - z_1| = |(y_1 + 18) - (y_1 - 3)| = 21 < 22.$$

Therefore, (A) Yolanda's quiz average for the academic year was 22 points higher than Zelda's is not possible.

-Benedict T (countmath1)

Solution 5

Let

$$\text{Yolanda's 1st Semester Average} = \frac{a_1 + \cdots + a_n}{n}$$

$$\text{Yolanda's 2nd Semester Average} = \frac{a'_1 + \cdots + a'_n}{n'}$$

$$\text{Zelda's 1st Semester Average} = \frac{b_1 + \cdots + b_m}{m}$$

$$\text{Zelda's 2nd Semester Average} = \frac{b'_1 + \cdots + b'_m}{m'}.$$

According to the problem, $\frac{a_1 + \cdots + a_n}{n} + 18 = \frac{a'_1 + \cdots + a'_n}{n'}$.

Moreover, $\frac{a_1 + \cdots + a_n}{n} = \frac{b_1 + \cdots + b_m}{m} + 3$ and $\frac{a'_1 + \cdots + a'_n}{n'} = \frac{b'_1 + \cdots + b'_m}{m'} + 3$ are true.

In another words,

$$\frac{a_1 + \cdots + a_n}{n} = \frac{a'_1 + \cdots + a'_n - 18n'}{n'} = \frac{b_1 + \cdots + b_m + 3m}{m} = \frac{b'_1 + \cdots + b'_m - 15m'}{m'}.$$

The relationship leads to

$$\frac{a_1 + \cdots + a_n + a'_1 + \cdots + a'_n - 18n'}{n + n'} = \frac{b_1 + \cdots + b_m + b'_1 + \cdots + b'_m + 3m - 15m'}{m + m'}, \text{ or}$$

$$\frac{a_1 + \cdots + a_n + a'_1 + \cdots + a'_n}{n + n'} - \frac{18n'}{n' + n} - \frac{3m}{m + m'} + \frac{15m'}{m + m'} = \frac{b_1 + \cdots + b_m + b'_1 + \cdots + b'_m}{m + m'}.$$

For choice (A) to be true, $-\frac{18n'}{n' + n} - \frac{3m}{m + m'} + \frac{15m'}{m + m'}$ must be -22 .

Let $n' = kn$ and $m' = k'm$.

$$-\frac{18n'}{n' + n} - \frac{3m}{m + m'} + \frac{15m'}{m + m'} = -\frac{18kn}{n(k + 1)} - \frac{3m}{m(k' + 1)} + \frac{15k'm}{m(k' + 1)} = -\frac{18kn}{n(k + 1)} + \frac{3m(5k' - 1)}{m(k' + 1)}$$

n, n', m, m', k and k' must be non-zero integer to satisfy the condition in the question. While the equation above is a multiple of three, 22 is not. Therefore, choice

(A) cannot be true.

19. (AMC 12 12B 2023) Each of 2023 balls is placed in one of 3 bins. Which of the following is closest to the probability that each of the bins will contain an odd number of balls?

(A) $\frac{2}{3}$ (B) $\frac{3}{10}$ (C) $\frac{1}{2}$ (D) $\frac{1}{3}$ (E) $\frac{1}{4}$

Answer: E

Solution 1 (Systematic Algebraic Way)

Suppose the numbers are a_1 , a_2 , and a_3 . First, we try to calculate the amount of ways for all three balls to be placed in a bin so the number of balls in each bin is odd. $a_1 + a_2 + a_3 = 2023$ and each bin has at least one ball because they are positive odd numbers. Changing the equation, we see that $\frac{[(a_1)+1]}{2} + \frac{[(a_2)+1]}{2} + \frac{[(a_3)+1]}{2} = \frac{(2023+3)}{2}$. Let $a_1 + 1 = 2b_1$, $a_2 + 1 = 2b_2$, and $a_3 + 1 = 2b_3$. Thus $b_1 + b_2 + b_3 = 1013$. We can also see that b_1 , b_2 , and b_3 are all positive. Using the positive version of stars and bars, we get $\binom{1013-1}{3-1} = \binom{1012}{2}$ choices.

Now, we want to find the total amount of cases. Using the non-negative version of stars and bars, we find that the total is $\binom{2023+3-1}{3-1} = \binom{2025}{2}$.

Now we need to calculate $\binom{1012}{2} / \binom{2025}{2}$, which is just $\frac{1012 \cdot 1011 \cdot 2}{2 \cdot 2025 \cdot 2024}$. Cancelling the twos, we get $\frac{1012 \cdot 1011}{2025 \cdot 2024}$. This is roughly equal to $\frac{1}{4}$. The answer is (E) $\frac{1}{4}$.

Solution 2 (Pairing)

There are $\left\lceil \frac{2023}{2} \right\rceil = 1012$ odd numbers between 1 and 2023.

Out of these, there are $\left\lfloor \frac{1012}{3} \right\rfloor = 337$ ways to evenly distribute each ball into the first bin, then, WLOG, we say there is $337 - 2 = 335$ for the second and $337 + 2 = 339$ for the third bin.

Out of these, there are $\left\lfloor \frac{337}{2} \right\rfloor = 168$ odd sums of pairs for the first bin, then $\left\lfloor \frac{335}{2} \right\rfloor = 167$ for the second, and $\left\lfloor \frac{339}{2} \right\rfloor = 169$ for the third. The sum of all of these is $168 + 167 + 169 = 504$ total distributions.

From here, we need to find the probability over 2023 total distributed balls, which is simply $504/2023 \approx 500/2000 = 1/4 \Rightarrow$ (E) $\frac{1}{4}$.

This is not an exact probability. It is simply a solution for general competition math. For the exact way to calculate the probability (highly not recommended), please refer to Solution 8.

Solution 3

We first examine the possible arrangements for parity of number of balls in each box for 2022 balls.

If a E denotes an even number and a O denotes an odd number, then the distribution of balls for 2022 balls could be EEE , EOO , OEO , or OOE . With the insanely overpowered magic of cheese, we assume that each case is about equally likely.

From EEE , it is not possible to get to all odd by adding one ball; we could either get OEE , EOE , or EEO . For the other 3 cases, though, if we add a ball to the exact right place, then it'll work.

For each of the working cases, we have 1 possible slot the ball can go into (for OEO , for example, the new ball must go in the center slot to make OOO) out of the 3

slots, so there's a $\frac{1}{3}$ chance. We have a $\frac{3}{4}$ chance of getting one of these working cases, so our answer is $\frac{3}{4} \cdot \frac{1}{3} = \boxed{\text{(E)} \frac{1}{4}}.$

Solution 4

2023 is an arbitrary large number. So, we proceed assuming that an arbitrarily large number of balls have been placed.

For an odd-numbered amount of balls case, the 3 bins can only be one of these 2 combinations:

OEE (OEE,EOE,EEO)

OOO (OOO)

Let the probability of achieving the *OOO* case to be $P(OOO) = p$ and any of the *OEE* permutations to be $P(OEE) = 1 - p$.

Because the amount of balls is arbitrarily large, $P(OOO) = p$ even after another two balls are placed.

There are two cases for which placing another two balls results in *OOO*:

OOO: The two balls are placed in the same bin (*OOO* → *OOE* → *OOO*)

OEE: The two balls are placed in the two even bins (*OEE* → *OOE* → *OOO*)

So,

$$P(OOO) = P(OOO) * \frac{1}{3} + P(OEE) * \frac{2}{3} * \frac{1}{3} \quad p = p * \frac{1}{3} + (1 - p) * \frac{2}{3} * \frac{1}{3} \quad \frac{8}{9}p = \frac{2}{9}$$

$$p = \boxed{\text{(E)} \frac{1}{4}}$$

Solution 5

We use the generating functions approach to solve this problem. Define $\Delta = \{(a, b, c) \in \mathbb{Z}_+ : a + b + c = 2023\}$.

We have

$$(x + y + z)^{2023} = \sum_{(a,b,c) \in \Delta} \binom{2023}{a, b, c} x^a y^b z^c.$$

First, we set $x \leftarrow 1, y \leftarrow 1, z \leftarrow 1$. We get

$$3^{2023} = \sum_{(a,b,c) \in \Delta} \binom{2023}{a, b, c} 1. \quad (1)$$

Second, we set $x \leftarrow 1, y \leftarrow -1, z \leftarrow 1$. We get

$$1 = \sum_{(a,b,c) \in \Delta} \binom{2023}{a, b, c} (-1)^b. \quad (2)$$

Third, we set $x \leftarrow 1, y \leftarrow 1, z \leftarrow -1$. We get

$$1 = \sum_{(a,b,c) \in \Delta} \binom{2023}{a, b, c} (-1)^c. \quad (3)$$

Fourth, we set $x \leftarrow 1$, $y \leftarrow -1$, $z \leftarrow -1$. We get

$$-1 = \sum_{(a,b,c) \in \Delta} \binom{2023}{a,b,c} (-1)^{b+c}. \quad (4)$$

Taking $\frac{(1)-(2)-(3)+(4)}{4}$, we get

$$\begin{aligned} \frac{3^{2023} - 1 - 1 + (-1)}{4} &= \frac{1}{4} \sum_{(a,b,c) \in \Delta} \binom{2023}{a,b,c} (1 - (-1)^b - (-1)^c + (-1)^{b+c}) \\ &= \frac{1}{4} \sum_{(a,b,c) \in \Delta} \binom{2023}{a,b,c} (1 - (-1)^b) (1 - (-1)^c) \\ &= \sum_{\substack{(a,b,c) \in \Delta \\ a,b,c \text{ are odds}}} \binom{2023}{a,b,c}. \end{aligned}$$

The last expression above is the number of ways to get all three bins with odd numbers of balls. Therefore, this happens with probability

$$\frac{\frac{3^{2023} - 1 - 1 + (-1)}{4}}{3^{2023}} \approx \boxed{\text{(E)} \frac{1}{4}}.$$

Solution 6

Four even-odd splittings divides 2023 in to three, namely (O, O, O) , (E, E, O) , (E, O, E) , and (O, E, E) . Here if we define a "move" as relocated one ball, then we will notice in each case, that a random "move" will be evenly likely to be one of the other three splittings. Hence by Group theory (or by intuition), we will find the structure of this splitting is V_4 group, and it's symmetric for all four elements in this Group.

Thus, no matter what is the initial starting point, four cases will be evenly likely to appear when repeated many times. The answer is $\boxed{\text{(E)} \frac{1}{4}}$.

Solution 7 (Exact Probability)

Let $p(n)$ be the probability of all 3 bins having odd number of balls when we randomly put $2n + 1$ balls into these bins. We need find out $p(1011)$. Clearly $p(0) = 0$. For $n > 1$, there are only two scenarios for the first $2n - 1$ balls: **Case 1** all 3 bins are odd; **Case 2** only 1 bin is odd and the other 2 bins are even.

For case 1 which has a probability $p(n - 1)$, in order to get all 3 bins to be odd, we need to put the last 2 balls in the same bin. This has a probability of $\frac{1}{3}$.

For case 2 which has a probability of $1 - p(n - 1)$, in order to get all 3 bins to be odd, we need to put the last 2 balls into the 2 even bins. The first one has a probability of $\frac{2}{3}$ to be put in an even bin, and the second has a probability of $\frac{1}{3}$ to be put the in the last even bin, so the probability is $\frac{2}{9}$ to get all 3 bins to be odd in this case.

So we have $p(n) = \frac{1}{3}p(n - 1) + \frac{2}{9}(1 - p(n - 1)) = \frac{1}{9}p(n - 1) + \frac{2}{9}$.

Solving for the fixed point: $x = \frac{1}{9}x + \frac{2}{9}$ yields $x = \frac{1}{4}$, and this helps us the rewrite the above as $p(n) - \frac{1}{4} = \frac{1}{9}(p(n - 1) - \frac{1}{4})$.

Therefore $p(n) - \frac{1}{4} = \frac{1}{9^n}(p(0) - \frac{1}{4})$. Using $p(0) = 0$, we have $p(n) = \frac{1}{4}(1 - \frac{1}{9^n})$.

The probability for 2023 balls is $\frac{1}{4}(1 - \frac{1}{9^{1011}})$, and this is closest to $\boxed{\text{(E)} \frac{1}{4}}$.

Solution 8 (Recursion, rigorous)

We can use recursion to figure out an expression that determines the probability of all the bins being odd.

Define P_n to be the probability that with $2n - 1$ balls, randomly inserting them makes all 3 bins have an odd count. We would first want to find $P_{n+1} = f(P_n)$ for some function f .

Now, we can say that if we have an O-O-O combination, where O denotes odd and E denotes even, then to add 2 more balls to keep it O-O-O, we need to put them in the same bin. This would result in a $\frac{1}{3}$ probability of staying O-O-O.

If we have an E-E-O combination, then we need to add 2 more balls, one in each of the even bins. This would result in a $\frac{2}{9}$ probability of turning to O-O-O. Now, we can write a function of P_{n+1} in terms of P_n .

$$P_{n+1} = \frac{1}{3}P_n + \frac{2}{9} - \frac{2}{9}P_n = \frac{1}{9}P_n + \frac{2}{9}$$

Now, we want to express $P_{n+1} = g(n)$ for some function g . Here, the $\frac{2}{9}$ is a problem because now we can't just put it as a geometric sequence. However, we can rewrite it as followed:

$$P_{n+1} = \frac{1}{9}P_n + \frac{2}{9} \quad P_{n+1} - \frac{1}{4} = \frac{1}{9}(P_n - \frac{1}{4})$$

Now, we can let $Q_n = P_n - \frac{1}{4}$. We can rewrite that as $Q_{n+1} = \frac{1}{9}Q_n$.

We can note that $Q_1 = P_1 - \frac{1}{4}$. Since $P_1 = 0$ (you only have one ball, so two of the bins have to be even), $Q_1 = -\frac{1}{4}$. Therefore, $Q_n = Q_1(\frac{1}{9})^{n-1} = -\frac{1}{4} \cdot (\frac{1}{9})^{n-1}$.

Substituting, we can find that $P_n = -\frac{1}{4} \cdot (\frac{1}{9})^{n-1} + \frac{1}{4}$.

2023 is the 1012th odd number, so we can substitute $n = 1012$ into the equation.

$$-\frac{1}{4} \cdot (\frac{1}{9})^{1011} + \frac{1}{4} \approx 0 + \frac{1}{4} = \frac{1}{4}.$$

Therefore, the answer is $\boxed{\text{(E)} \frac{1}{4}}$

Solution 9 (Generating Functions instead of stars and bars)

Define

$$\begin{aligned} [x^{2023}]G(x) &= (x^1 + x^3 + \dots + x^{2023} + \dots)^3 \\ [x^{2023}]G(x) &= x^3(1 + x^2 + x^4 + \dots + x^{2022} + \dots)^3 \\ [x^{2020}]G(x) &= \left(\frac{1}{1 - x^2}\right)^3 \\ [x^{2020}]G(x) &= (1 - x^2)^{-3} \\ [x^{2020}]G(x) &= \left(1 + \frac{3 \cdot 2}{2 \cdot 1}(x^2)^1 + \dots + \frac{1012 \cdot 1011}{2 \cdot 1}(x^2)^{1010}\right) \\ [x^{2020}] &= \frac{1012 \cdot 1011}{2 \cdot 1} \end{aligned}$$

Similarly we can compute the total number of cases which is: $\frac{2025 \cdot 2022}{2 \cdot 1}$

So, the probability is:

$$= \frac{\frac{1012 \cdot 1011}{2 \cdot 1}}{\frac{2025 \cdot 2022}{2 \cdot 1}}$$

$$= \frac{1012 \cdot 1011}{2025 \cdot 2024}$$

$$\approx \frac{1}{4}$$

Therefore, the answer is $\boxed{\text{(E)} \frac{1}{4}}$

Solution 10 (Short On Time)

One must not forget that the balls are placed randomly. With intuition, we can assume that the probability of getting OOO , OOE , OEE , and EEE , where O denotes odd numbers and E denotes even numbers, would nearly be the same. Since there are 4 possible outcomes and 1 of them is desired, we can conclude that the likelihood of getting OOO is $\frac{1}{4}$ and that the correct answer choice is $\boxed{\text{(E)} \frac{1}{4}}$.

Note: Use this type of reasoning only when a logical solution doesn't work out for you.

Another Note (not by Aqf243): This may not be the best solution because 2023 is an odd number. Hence, realistically, the probability of getting OOE is 0, allowing us to remove it from our list. This is because adding 2 odd numbers (represented as 2 numbers $x \in \{x \mid x \bmod 2 = 1\}$) always results in an even number.

$\therefore$

$$x_1 = 1 \bmod 2 \quad x_2 = 1 \bmod 2 \quad \therefore$$

$$x_1 + x_2 = 1 + 1 \bmod 2 = 2 \bmod 2 = 0 \bmod 2$$

for any number $x \in \{x \mid x \bmod 2 = 0\}$, x is an even number.

Similarly, adding two even numbers creates another even number

$$x_1 = 0 \bmod 2$$

$$x_2 = 0 \bmod 2 \quad x_1 + x_2 = 0 + 0 \bmod 2 = 0 \bmod 2$$

So adding 2 odd numbers and an even number ALWAYS results in an even number.

Basically, Aqf243 got a little lucky in this situation...

Solution 11 (Notice a pattern)

Let the bins be called a , b , and c . We want the number of odd positive integer solutions that satisfy $a + b + c = 2023$. We know from stars and bars that the total number of integer solutions is $\binom{2023+3-1}{3-1} = \binom{2025}{2}$. If $a = 1$, b can be any odd number from 1 to 2021. There are $(2021+1)/2 = 1011$ choices for b . If $a = 3$, there would be 1010 choices for b , and so on. c is always constrained by a and b so our answer is just the number of choices b has for each a , which is the sum of all integers from 1 to 1011 = $\frac{1011 \cdot 1012}{2}$. So we get $\frac{1011 \cdot 1012/2}{\binom{2025}{2}}$ (around 0.249), which is

closest to 0.25 or $\boxed{\text{(E)} \frac{1}{4}}$.

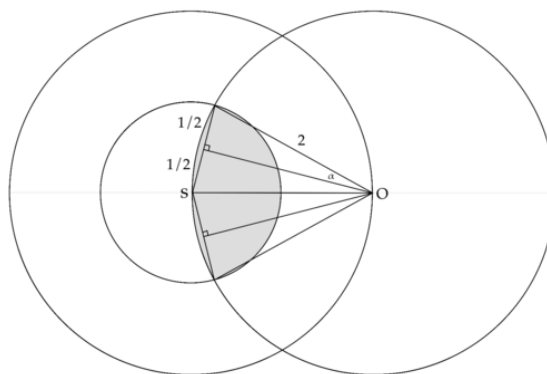
NOTE: The probability we came to in this solution and in other solutions that use Stars and Bars is not exact. Stars and Bars treats all outcomes to be equally likely when in reality some distributions are more likely than others (see clarification above), but the reason we still use it here is because it gives us a good average and the answer choices are pretty far apart so it's not very significant to our approximation.

20. (AMC 12 12B 2023) Cyrus the frog jumps 2 units in a direction, then 2 more in another direction. What is the probability that he lands less than 1 unit away from his starting position?

(A) $\frac{1}{6}$ (B) $\frac{1}{5}$ (C) $\frac{\sqrt{3}}{8}$ (D) $\frac{\arctan \frac{1}{2}}{\pi}$ (E) $\frac{2 \arcsin \frac{1}{4}}{\pi}$

Answer: E

Solution 1



Let Cyrus's starting position be S . WLOG, let the place Cyrus lands at for his first jump be O . From O , Cyrus can reach all the points on $\odot O$. The probability that Cyrus will land less than 1 unit away from S is $\frac{4\alpha}{2\pi}$.

$$\sin \alpha = \frac{\frac{1}{2}}{2} = \frac{1}{4}, \quad \alpha = \arcsin \frac{1}{4}$$

Therefore, the answer is

$$\frac{4 \arcsin \frac{1}{4}}{2\pi} = \boxed{\text{(E)} \frac{2 \arcsin \frac{1}{4}}{\pi}}$$

Solution 2

Denote by A_i the position after the i th jump. Thus, to fall into the region centered at A_0 and with radius 1, $\angle A_2 A_1 A_0 < 2 \arcsin \frac{1/2}{2} = 2 \arcsin \frac{1}{4}$.

Therefore, the probability is

$$\frac{2 \cdot 2 \arcsin \frac{1}{4}}{2\pi} = \boxed{\text{(E)} \frac{2 \arcsin \frac{1}{4}}{\pi}}.$$

Solution 3(coord bash)

Let the origin be the starting point of frog. Then, WLOG assume that after the first jump, it is at the point $(2,0)$. Then, the range of all possible places the frog can jump to at its second jump is the circle with equation $(x-2)^2 + y^2 = 2^2$. If it landed 1 unit within its starting point (the origin), then it is inside the circle $x^2 + y^2 = 1$. We clearly want the intersection point. So we're trying to solve the system of equations $x^2 + y^2 = 1$ and $(x-2)^2 + y^2 = 2^2$. We have $x = \frac{1}{4}$, so $y = \pm \frac{\sqrt{15}}{4}$. Therefore, our desired answer would be $\frac{\arcsin \frac{\sqrt{15}}{8}}{\pi}$ (the angle we want divided by 2π). Since

$$\arcsin \frac{\sqrt{15}}{8} = \arccos \frac{7}{8} = \arccos \left(1 - 2 \cdot \left(\frac{1}{4}\right)^2\right) = 2 \arcsin \frac{1}{4}$$

where the last step holds by the double angle formula, our answer is $(E) \frac{2 \arcsin \frac{1}{4}}{\pi}$. Ddk001

Solution 4 - Law of Cosines and Double Angle Formula

Let A be Cyrus's starting point, B be the first point he jumps to ($AB = 2$), and C be the second point he jumps to ($BC = 2$). Let angle ABC be k , such that $AC = 1$. The probability of $AC < 1$ would therefore be $\frac{2k}{360}$ (since C could be on either side of AB so there are two possible areas of having $AC < 1$) which simplifies to $\frac{k}{180}$. Converting to radians gives us $\frac{k}{\pi}$. To find k , we use the law of cosines.

$$AC^2 = AB^2 + BC^2 - 2 \cdot AB \cdot BC \cdot \cos k$$

$$1^2 = 2^2 + 2^2 - 2 \cdot 2 \cdot 2 \cdot \cos k$$

$$1 = 4 + 4 - 8 \cos k$$

$$8 \cos k = 7$$

$$\cos k = \frac{7}{8}$$

$$k = \arccos \left(\frac{7}{8}\right) = \arcsin \left(\sqrt{1 - \left(\frac{7}{8}\right)^2}\right) = \arcsin \left(\sqrt{\frac{15}{64}}\right) = \arcsin \left(\frac{1}{4} \sqrt{\frac{15}{4}}\right) = \arcsin \left(\frac{1}{4} \sqrt{1 - \left(\frac{1}{4}\right)^2}\right)$$

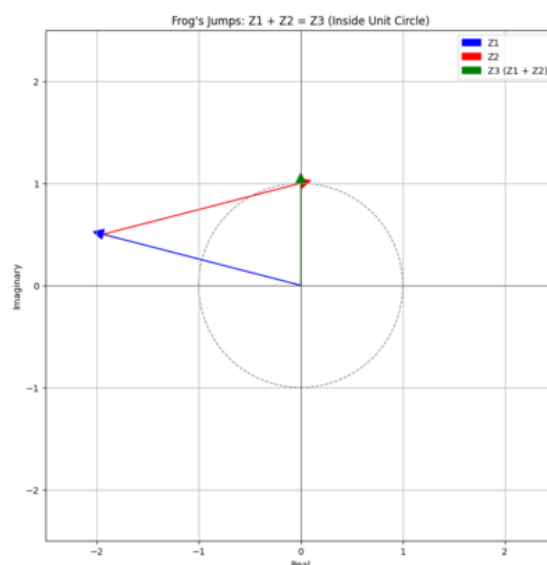
$$= \arcsin \left(\sin \left(\arcsin \left(\frac{1}{4}\right)\right) \cos \left(\arcsin \left(\frac{1}{4}\right)\right)\right) = \arcsin \left(\sin \left(2 \arcsin \left(\frac{1}{4}\right)\right)\right) = 2 \arcsin \left(\frac{1}{4}\right)$$

The probability is

$$\frac{k}{\pi} = \frac{2 \arcsin \left(\frac{1}{4}\right)}{\pi}$$

which is E .

Solution 5 - Complex Number and Double Angle Formula



The frog jumps 2 units in one direction, which we can represent as $z_1 = 2e^{ia}$, where a is the angle in radians. Then, the frog jumps another 2 units in a different direction, represented as $z_2 = 2e^{ib}$, where b is the angle in radians.

The total displacement from the starting position is $z_3 = z_1 + z_2 = 2e^{ia} + 2e^{ib}$.

We need to find the condition under which $|z_3| \leq 1$. This translates to:

$$|2e^{ia} + 2e^{ib}| \leq 1$$

$$|e^{ia} + e^{ib}| \leq \frac{1}{2}$$

$$\begin{aligned} e^{ia} + e^{ib} &= \cos(a) + i\sin(a) + \cos(b) + i\sin(b) \\ &= (\cos(a) + \cos(b)) + i(\sin(a) + \sin(b)) \end{aligned}$$

The magnitude squared of this sum is:

$$\begin{aligned} |e^{ia} + e^{ib}|^2 &= (\cos(a) + \cos(b))^2 + (\sin(a) + \sin(b))^2 \\ &= \cos^2(a) + 2\cos(a)\cos(b) + \cos^2(b) + \sin^2(a) + 2\sin(a)\sin(b) + \sin^2(b) \\ &= 2 + 2(\cos(a)\cos(b) + \sin(a)\sin(b)) \\ &= 2 + 2\cos(a - b) \end{aligned}$$

Thus:

$$\begin{aligned} 2 + 2\cos(a - b) &\leq \frac{1}{4} \\ 2\cos(a - b) &\leq -\frac{7}{4} \\ \cos(a - b) &\leq -\frac{7}{8} \end{aligned}$$

Therefore, our desired answer would be $\frac{\arccos \frac{7}{8}}{\pi}$ (the angle we want divided by 2π). Since

$$\arccos \frac{7}{8} = \arccos \left(1 - 2 \cdot \left(\frac{1}{4} \right)^2 \right) = 2 \arcsin \frac{1}{4}$$

where the last step holds by the double angle formula, our answer is $\boxed{(E) \frac{2 \arcsin \frac{1}{4}}{\pi}}$.

21. (AMC 12 12B 2023) A lampshade is made in the form of the lateral surface of the frustum of a right circular cone. The height of the frustum is $3\sqrt{3}$ inches, its top diameter is 6 inches, and its bottom diameter is 12 inches. A bug is at the bottom of the lampshade and there is a glob of honey on the top edge of the lampshade at the spot farthest from the bug. The bug wants to crawl to the honey, but it must stay on the surface of the lampshade. What is the length in inches of its shortest path to the honey?

(A) $6 + 3\pi$ (B) $6 + 6\pi$ (C) $6\sqrt{3}$ (D) $6\sqrt{5}$ (E) $6\sqrt{3} + \pi$

Answer: E

Solution

We augment the frustum to a circular cone. Denote by O the apex of the cone. Denote by A the bug and B the honey.

By using the numbers given in this problem, the height of the cone is $6\sqrt{3}$. Thus, $OA = 12$ and $OB = 6$.

We unfold the lateral face. So we get a circular sector. The radius is 12 and the length of the arc is 12π . Thus, the central angle of this circular sector is 180° .

Because A and B are opposite in the original frustum, in the unfolded circular cone, $\angle AOB = \frac{180^\circ}{2} = 90^\circ$.

Notice that a feasible path between A and B can only fall into the region with the range of radii between $OB = 6$ and $OA = 12$. Therefore, we cannot directly connect A and B and must make a detour. Denote by AC a tangent to the circular sector with radius 6 that meets it at point C . Therefore, the shortest path between A and B consists of a segment AC and an arc from C to B .

Because $OA = 12$, $OC = 6$ and $\angle OCA = 90^\circ$, we have $AC = \sqrt{OA^2 - OC^2} = 6\sqrt{3}$ and $\angle AOC = 60^\circ$. This implies $\angle COB = \angle AOB - \angle AOC = 30^\circ$. Therefore, the length of the arc between C and B is $OB \cdot \pi \cdot \frac{\angle COB}{180^\circ} = \pi$. Therefore, the shortest distance between A and B is **(E)** $6\sqrt{3} + \pi$.

22. (AMC 12 12B 2023) A real-valued function f has the property that for all real numbers a and b ,

$$f(a+b) + f(a-b) = 2f(a)f(b).$$

Which one of the following cannot be the value of $f(1)$? **(A)** 0 **(B)** 1 **(C)** -1
(D) 2 **(E)** -2

Answer: E

Solution 1

Substituting $a = b$ we get

$$f(2a) + f(0) = 2f(a)^2$$

Substituting $a = 0$ we find

$$2f(0) = 2f(0)^2 \implies f(0) \in \{0, 1\}.$$

This gives

$$f(2a) = 2f(a)^2 - f(0) \geq 0 - 1$$

Plugging in $a = \frac{1}{2}$ implies $f(1) \geq -1$, so answer choice **(E) -2** is impossible.

Solution 2

First, we set $a \leftarrow 0$ and $b \leftarrow 0$. Thus, the equation given in the problem becomes $[f(0) + f(0) = 2f(0) \times f(0).]$

Thus, $f(0) = 0$ or 1 .

Case 1: $f(0) = 0$.

We set $b \leftarrow 0$. Thus, the equation given in the problem becomes $[2f(a) = 0.]$

Thus, $f(a) = 0$ for all a .

Case 2: $f(0) = 1$.

We set $b \leftarrow a$. Thus, the equation given in the problem becomes

$$[f(2a) + 1 = 2(f(a))^2.]$$

Thus, for any a ,

$$\begin{aligned} f(2a) &= -1 + 2(f(a))^2 \\ &\geq -1. \end{aligned}$$

Therefore, an infeasible value of $f(1)$ is (E) -2.

Solution 3

The relationship looks suspiciously like a product-to-sum identity. In fact,

$$\cos(\alpha)\cos(\beta) = \frac{1}{2}(\cos(\alpha - \beta) + \cos(\alpha + \beta))$$

which is basically the relation. So we know that $f(x) = \cos(x)$ is a valid solution to the function. However, if we define $x = ay$, where a is arbitrary, the above relation should still hold for $f(x) = \cos(ay) = \cos(a(1))$ so any value in $[-1, 1]$ can be reached, so choices A , B , and C are incorrect.

In addition, using the similar formula for hyperbolic cosine, we know

$$\cosh(\alpha)\cosh(\beta) = \frac{1}{2}(\cosh(\alpha - \beta) + \cosh(\alpha + \beta))$$

The range of $\cosh(ay)$ is $[1, \infty)$ so choice D is incorrect.

Therefore, the remaining answer is choice (E) -2.

Solution 4 (Recognition)

Similar to solution 3 we know that $\cos(a - b)\cos(a + b) = \cos(a)\cos(b)$. Doing testing gives us that any function in the form $\cos(bx)$ works. We now see that only 2 and -2 are the candidates to be impossible. Solving the functional equation the regular way, we see that $f(0) = 1$, $f(-x) = x$. We eventually get a squared term (from even more testing), and see that -2(E) is the answer because negative square roots are impossible.

23. (AMC 12 12B 2023) When n standard six-sided dice are rolled, the product of the numbers rolled can be any of 936 possible values. What is n ?

(A) 11 (B) 6 (C) 8 (D) 10 (E) 9

Answer: A

Solution 1

We start by trying to prove a function of n , and then we can apply the function and equate it to 936 to find the value of n .

It is helpful to think of this problem in the format $(1 + 2 + 3 + 4 + 5 + 6) \cdot (1 + 2 + 3 + 4 + 5 + 6) \dots$. Note that if we represent the scenario in this manner, we can think of picking a 1 for one factor and then a 5 for another factor to form their product - this is similar thinking to when we have the factorized form of a polynomial. Unfortunately this is not quite accurate to the problem because we can reach the same product in many ways: for example for $n = 2$, 4 can be reached

by picking 1 and 4 or 2 and 2. However, this form gives us insights that will be useful later in the problem.

Note that there are only 3 primes in the set $\{1, 2, 3, 4, 5, 6\}$: 2, 3, and 5. Thus if we're forming the product of possible values of a dice roll, the product has to be written in the form $2^h \cdot 3^i \cdot 5^j$ (the choice of variables will become clear later), for integer nonnegative values h, i, j . So now the problem boils down to how many distinct triplets (h, i, j) can be formed by taking the product of n dice values.

We start our work on representing j : the powers of 5, because it is the simplest in this scenario because there is only one factor of 5 in the set. Because of this, having j fives in our prime factorization of the product is equivalent to picking j factors from the polynomial $(1 + \dots + 6) \dots$ and choosing each factor to be a 5. Now that we've selected j factors, there are $n - j$ factors remaining to choose our powers of 3 and 2.

Suppose our prime factorization of this product contains i powers of 3. These powers of 3 can either come from a 3 factor or a 6 factor, but since both 3 and 6 contain only one power of 3, this means that a product with i powers of 3 corresponds directly to picking i factors from the polynomial, each of which is either 3 or 6 (but this distinction doesn't matter when we consider only the powers of 3).

Now we can reframe the problem again. Our method will be as follows: Suppose we choose an arbitrary pair (i, j) that match the requirements, corresponding to the number of 3's and the number of 5's our product will have. Then how many different h values for the powers of 2 are possible?

In the $i + j$ factors we have already chosen, we obviously can't have any factors of 2 in the j factors with 5. However, we can have a factor of 2 pairing with factors of 3, if we choose a 6. The maximal possible power of 2 in these i factors is thus 2^i , which occurs when we pick every factor to be 6.

We now have $n - i - j$ factors remaining, and we want to allocate these to solely powers of 2. For each of these factors, we can choose either a 1, 2, or 4. Therefore the maximal power of 2 achieved in these factors is when we pick 4 for all of them, which is equivalent to $2^{2(n-i-j)}$.

Now if we multiply this across the total n factors (or n dice) we have a total of $2^{2n-2i-2j} \cdot 2^i = 2^{2n-i-2j}$, which is the maximal power of 2 attainable in the product for a pair (i, j) . Now note that every power of 2 below this power is attainable: we can simply just take away a power of 2 from an existing factor by dividing by 2. Therefore the powers of 2, and thus the h value ranges from $h = 0$ to $h = 2n - i - 2j$, so there are a total of $2n + 1 - i - 2j$ distinct values for h for a given pair (i, j) .

Now to find the total number of distinct triplets, we must sum this across all possible i s and j s. Lets take note of our restrictions on i, j : the only restriction is that $i + j \leq n$, since we're picking factors from n dice.

$$\sum_{i+j \leq n} 2n + 1 - i - 2j = \sum_{i+j \leq n} 2n + 1 - \sum_{i+j \leq n} i + 2j$$

We start by calculating the first term. $2n + 1$ is constant, so we just need to find out how many pairs there are such that $i + j \leq n$. Set i to 0: j can range from 0 to n , then set i to 1: j can range from 0 to $n - 1$, etc. The total number of pairs is thus $n + 1 + n + n - 1 + \dots + 1 = \frac{(n+1)(n+2)}{2}$. Therefore the left summation

evaluates to

$$\frac{(2n+1)(n+1)(n+2)}{2}$$

Now we calculate $\sum_{i+j \leq n} i + 2j$. This simplifies to $\sum_{i+j \leq n} i + 2 \cdot \sum_{i+j \leq n} j$. Note that because $i + j = n$ is symmetric with respect to i, j , the sum of i in all of the pairs will be equal to the sum of j in all of the pairs. Thus this is equal to calculating $3 \cdot \sum_{i+j \leq n} i$.

In the pairs, $i = 1$ appears for j ranging between 0 and $n - 1$ so the sum here is $1 \cdot (n)$. Similarly $i = 2$ appears for j ranging from 0 to $n - 2$, so the sum is $2 \cdot (n - 1)$. If we continue the pattern, the sum overall is $(n) + 2 \cdot (n - 1) + 3 \cdot (n - 2) + \cdots + (n) \cdot 1$. We can rearrange this as $((n) + (n - 1) + \cdots + 1) + ((n - 1) + (n - 2) + \cdots + 1) + ((n - 2) + (n - 3) + \cdots + 1) + \cdots + 1$

$$= \frac{(n)(n+1)}{2} + \frac{(n-1)(n)}{2} + \cdots + 1$$

We can write this in easier terms as $\sum_{k=0}^n \frac{(k)(k+1)}{2} = \frac{1}{2} \cdot \sum_{k=0}^n k^2 + k$

$$\begin{aligned} &= \frac{1}{2} \cdot \left(\sum_{k=0}^n k^2 + \sum_{k=0}^n k \right) \\ &= \frac{1}{2} \cdot \left(\frac{(n)(n+1)(2n+1)}{6} + \frac{(n)(n+1)}{2} \right) \\ &= \frac{1}{2} \cdot \left(\frac{(n)(n+1)(2n+1)}{6} + \frac{3n(n+1)}{6} \right) = \frac{1}{2} \cdot \frac{n(n+1)(2n+4)}{6} \\ &= \frac{n(n+1)(n+2)}{6} \end{aligned}$$

We multiply this by 3 to obtain that

$$\sum_{i+j \leq n} i + 2j = \frac{n(n+1)(n+2)}{2}$$

Thus our final answer for the number of distinct triplets (h, i, j) is:

$$\begin{aligned} \sum_{i+j \leq n} 2n+1-i-2j &= \frac{(2n+1)(n+1)(n+2)}{2} - \frac{n(n+1)(n+2)}{2} \\ &= \frac{(n+1)(n+2)}{2} \cdot (2n+1-n) = \frac{(n+1)(n+2)}{2} \cdot (n+1) \\ &= \frac{(n+1)^2(n+2)}{2} \end{aligned}$$

Now most of the work is done. We set this equal to 936 and prime factorize. $936 = 12 \cdot 78 = 2^3 \cdot 3^2 \cdot 13$, so $(n+1)^2(n+2) = 936 \cdot 2 = 2^4 \cdot 3^2 \cdot 13$. Clearly 13 cannot be anything squared and $2^4 \cdot 3^2$ is a perfect square, so $n+2 = 13$ and $n = 11 = \boxed{A}$

Solution 2

The product can be written as

$$2^a 3^b 4^c 5^d 6^e = 2^{a+2c+e} 3^{b+e} 5^d.$$

Therefore, we need to find the number of ordered tuples $(a + 2c + e, b + e, d)$ where a, b, c, d, e are non-negative integers satisfying $a + b + c + d + e \leq n$. We denote this number as $f(n)$.

Denote by $g(k)$ the number of ordered tuples $(a + 2c + e, b + e)$ where $(a, b, c, e) \in \Delta_k$ with $\Delta_k \triangleq \{(a, b, c, e) \in \mathbb{Z}_+^4 : a + b + c + e \leq k\}$.

Thus,

$$\begin{aligned} f(n) &= \sum_{d=0}^n g(n-d) \\ &= \sum_{k=0}^n g(k). \end{aligned}$$

Next, we compute $g(k)$.

Denote $i = b + e$. Thus, for each given i , the range of $a + 2c + e$ is from 0 to $2k - i$. Thus, the number of $(a + 2c + e, b + e)$ is

$$\begin{aligned} g(k) &= \sum_{i=0}^k (2k - i + 1) \\ &= \frac{1}{2} (k + 1) (3k + 2). \end{aligned}$$

Therefore,

$$\begin{aligned} f(n) &= \sum_{k=0}^n g(k) \\ &= \sum_{k=0}^n \frac{1}{2} (k + 1) (3k + 2) \\ &= \frac{3}{2} \sum_{k=0}^n (k + 1)^2 - \frac{1}{2} \sum_{k=0}^n (k + 1) \\ &= \frac{3}{2} \cdot \frac{1}{6} (n + 1) (n + 2) (2n + 3) - \frac{1}{2} \cdot \frac{1}{2} (n + 1) (n + 2) \\ &= \frac{1}{2} (n + 1)^2 (n + 2). \end{aligned}$$

By solving $f(n) = 936$, we get $n = \boxed{\text{(A) 11}}$.

Solution 3 (Cheese)

The product can be written as

$$2^x 3^y 5^z$$

Letting $n = 1$, we get $(x, y, z) = (0, 0, 0), (0, 0, 1), (0, 1, 0), (1, 0, 0), (1, 1, 0), (2, 0, 0)$, 6 possible values. But if the only restriction of the product is that $2x \leq n, y \leq n, z \leq n$, we can get $(2 + 1)(1 + 1)(1 + 1) = 12$ possible values. We calculate the ratio

$$r = \frac{\text{possible values of real situation}}{\text{possible values of ideal situation}} = \frac{6}{12} = 0.5.$$

Letting $n = 2$, we get $(x, y, z) = (0, 0, 0), (0, 0, 1), (0, 0, 2), (0, 1, 0), (0, 1, 1), (1, 0, 0), (1, 0, 1), (1, 1, 0), (1, 1, 1), (2, 0, 0), (2, 0, 1), (2, 1, 0), (2, 2, 0), (3, 0, 0), (3, 1, 0), (4, 0, 0)$, 17 possible values. The

number of possibilities in the ideal situation is $5 * 3 * 3 = 45$, making $r = 17/45 \approx 0.378$.

Now we can predict the trend of r : as n increases, r decreases. Letting $n = 3$, you get possible values of ideal situation $= 7 * 4 * 4 = 112$. $n = 4$, the number $= 9 * 5 * 5 = 225$. $n = 5$, the number $= 11 * 6 * 6 = 396$. $n = 6$, the number $= 13 * 7 * 7 = 637$, $637 < 936$ so 6 is not the answer. $n = 7$, the number $= 15 * 8 * 8 = 960$. $n = 8$, the number $= 17 * 9 * 9 = 1377$, but $1377 * 0.378 \approx 521$ still much smaller than 936. $n = 9$, the number $= 19 * 10 * 10 = 1900$, but $1900 * 0.378 \approx 718$ still much smaller than 936. $n = 10$, the number $= 21 * 11 * 11 = 2541$, but $2541 * 0.378 \approx 962$ still much smaller than 936. $n = 11$, the number $= 23 * 12 * 12 = 3312$, $3312 * 0.378 \approx 1252$ is much bigger than 936.

Check calculation: $n = 11$, the number $= 23 * 12 * 12 = 3312$, $3312 * 0.378 \approx 1252$ is much bigger than 936.

Solution 4 (Easy computation)

The key observation is if $P = 2^a \cdot 3^b \cdot 5^c$, then given b and c , a can take any value from 0 to $b + 2d$ where $d = n - b - c$ is the number of rolls which is neither divisible by 3 nor 5. We are left to calculate

$$\sum_{b+c \leq n} (b + 2d + 1) = \sum_{b+c+a=n} (b + 2d + 1).$$

By symmetry, $\sum_{b+c+d=n} d = \sum_{b+c+d=n} c$. Therefore,

$$\sum_{b+c \leq n} (b + 2d + 1) = \sum_{b+c \leq n} (b + c + d + 1) = \sum_{b+c \leq n} (n + 1) = (n + 1) \binom{n+2}{2}.$$

The rest is the same as above.

Solution 5 ("normalize" the product) by FireSirius

The difficulty is that a product can be formed by different combinations of 1, 2, ..., n. So the key thought of this solution is to set a "normalized" combination for each product.

$$P = 1^{k_1} \cdot 2^{k_2} \cdot 3^{k_3} \cdot 4^{k_4} \cdot 5^{k_5} \cdot 6^{k_6}, \text{ with } k_1 + k_2 + \dots + k_6 = n.$$

For a fixed number of dice, it is easy to see that there are three basic "transformation": $1*6 = 2*3$, $1*4 = 2*2$, and $2*6 = 3*4$, which does not change the number of dice. Other transformation like $1*6*6 = 3*3*4$ can be viewed as a $1*6 = 2*3$ combined with a $2*6 = 3*4$.

So we can transform any given combination into a "normalised" combination in this way: step 1, if there are 1 and 6, then transform them into $2*3$; step 2, if there are both 1 and 4, then transform them into a $2*2$; step 3: if there are both 2 and 6, then transform them into $3*4$. Until no more basic transformation can be made.

Now begin casework with k_1 :

case 1 ($k_1 \leq k_6$): all "1"s cancel with "6"s, so the product contains only 2, 3, 4, 5, 6. And now:

If "2"s are no more than remaining "6"s: all "2"s cancel with "6"s, so the "normalized" product consists of only "3456", with the total number of dice unchanged. No transformation can be made now and it is the same with n stars and 3 bars, so there are C_{n+3}^3 ways.

If "2"s are no less than remaining "6"s: all "6" cancels with "2", so the product consist of only "2345". Again, there are C_{n+3}^3 ways.

case 2 ($k_1 \geq k_6$): all "6" cancels with "1", the remaining numbers are: 1, 2, 3, 4, 5; now if "1"s are no less than "4"s, then it can be transformed into product with only "1235"; there are, again, C_{n+3}^3 ways. if "1"s are no more than "4", then it is product with only 2, 3, 4, 5, which is already counted before.

So there are three major cases: "2345", "3456" and "1235", each with C_{n+3}^3 ways. There are overlaps: $3^{k_3}4^{k_4}5^{k_5}$, which has C_{n+2}^2 ways, are counted in both "2345" and "3456", and $2^{k_2}3^{k_3}5^{k_5}$ are counted in both "2345" and "1235".

So, the final number is:

$$3C_{n+3}^3 - 2C_{n+2}^2 = \frac{(n+1)^2(n+2)}{2},$$

and now any possible product is counted exactly once. And then it is easily to find out $n = \boxed{\text{(A) 11}}$.

Solution 6 (defining sequences)

We define:

a_n is the number of possible products of the results of n dice being rolled.

Now consider a modified five-sided die that excludes 5, having faces 1, 2, 3, 4, and 6. Let the number of possible products of the results of n of these modified dice be equal to b_n .

If in the n rolls of the dice, we roll 5 zero times, then the number of distinct products is equal to b_n . If we roll 5 once, then the number of distinct products is equal to b_{n-1} . In general, if we roll 5 i times, then the number of distinct products is equal to b_{n-i} . Between the different cases of how many times we roll 5, the products are guaranteed to be distinct. This is because in their prime factorizations, the powers of 5 are different, and by the Fundamental Theorem of Arithmetic, the products must be different as well.

In other words, we have found:

$$a_n = \sum_{i=0}^n b_i$$

Now let us focus on b_n , and from now on, until otherwise specified, the die will refer to the modified, 5-sided die. In these n rolls of the dice, if we roll 3 zero times and 6 zero times, then the product must be a power of 2. It can range from 2^0 to 4^n , which is equal to 2^{2n} . Therefore, there are $2n+1$ possible products. If we either roll 3 once or 6 once (not both), then the product must be 3 times a power of 2. This power of 2 can range from 2^0 to $2^{2(n-1)+1} = 2^{2n-1}$, when we roll one 6 and all other rolls are 4. Therefore, there are $2n$ products. In general, if we roll 3 and 6 a combined i times, then the product must be in the form $2^k * 3^i$, and k ranges from 0 to $2(n-i) + i = 2n - i$, resulting in $2n - i + 1$ products.

In other words, we have found:

$$b_n = \sum_{i=0}^n 2n - i + 1$$

We can now calculate b_n . Writing out the full sum, we get:

$$b_n = (2n+1) + (2n) + \cdots + (n+1)$$

This simplifies to:

$$b_n = \frac{(n+1)(3n+2)}{2} = \frac{3n^2 + 5n + 2}{2}$$

Now, finally, we can calculate a_n :

$$a_n = \sum_{i=0}^n b_i$$

$$\begin{aligned}
 &= \sum_{i=0}^n \frac{3i^2 + 5i + 2}{2} \\
 &= \sum_{i=0}^n \frac{3i^2}{2} + \sum_{i=0}^n \frac{5i}{2} + \sum_{i=0}^n 1 \\
 &= \frac{3}{2} * \frac{n(n+1)(2n+1)}{6} + \frac{5}{2} * \frac{n(n+1)}{2} + (n+1) \\
 &= (n+1) \left(\frac{n(2n+1)}{4} + \frac{5n}{4} + 1 \right) \\
 &= (n+1) \left(\frac{2n^2 + 6n + 4}{4} \right) \\
 &= \frac{(n+1)^2(n+2)}{2}
 \end{aligned}$$

Finally, plugging in $a_n = 936$, we get that $n = \boxed{\text{(A) 11}}$.

24. (AMC 12 12B 2023) Suppose that a, b, c and d are positive integers satisfying all of the following relations.

$$abcd = 2^6 \cdot 3^9 \cdot 5^7$$

$$\text{lcm}(a, b) = 2^3 \cdot 3^2 \cdot 5^3$$

$$\text{lcm}(a, c) = 2^3 \cdot 3^3 \cdot 5^3$$

$$\text{lcm}(a, d) = 2^3 \cdot 3^3 \cdot 5^3$$

$$\text{lcm}(b, c) = 2^1 \cdot 3^3 \cdot 5^2$$

$$\text{lcm}(b, d) = 2^2 \cdot 3^3 \cdot 5^2$$

$$\text{lcm}(c, d) = 2^2 \cdot 3^3 \cdot 5^2$$

What is $\text{gcd}(a, b, c, d)$?

- (A) 30 (B) 45 (C) 3 (D) 15 (E) 6

Answer: C

Solution 1

Denote by $\nu_p(x)$ the number of prime factor p in number x .

We index Equations given in this problem from (1) to (7).

First, we compute $\nu_2(x)$ for $x \in \{a, b, c, d\}$.

Equation (5) implies $\max\{\nu_2(b), \nu_2(c)\} = 1$. Equation (2) implies $\max\{\nu_2(a), \nu_2(b)\} = 3$. Equation (6) implies $\max\{\nu_2(b), \nu_2(d)\} = 2$. Equation (1) implies $\nu_2(a) + \nu_2(b) + \nu_2(c) + \nu_2(d) = 6$.

Therefore, all above jointly imply $\nu_2(a) = 3$, $\nu_2(d) = 2$, and $(\nu_2(b), \nu_2(c)) = (0, 1)$ or $(1, 0)$.

Second, we compute $\nu_3(x)$ for $x \in \{a, b, c, d\}$.

Equation (2) implies $\max\{\nu_3(a), \nu_3(b)\} = 2$. Equation (3) implies $\max\{\nu_3(a), \nu_3(c)\} = 3$. Equation (4) implies $\max\{\nu_3(a), \nu_3(d)\} = 3$. Equation (1) implies $\nu_3(a) + \nu_3(b) + \nu_3(c) + \nu_3(d) = 9$.

Therefore, all above jointly imply $\nu_3(c) = 3$, $\nu_3(d) = 3$, and $(\nu_3(a), \nu_3(b)) = (1, 2)$ or $(2, 1)$.

Third, we compute $\nu_5(x)$ for $x \in \{a, b, c, d\}$.

Equation (5) implies $\max\{\nu_5(b), \nu_5(c)\} = 2$. Equation (2) implies $\max\{\nu_5(a), \nu_5(b)\} = 3$. Thus, $\nu_5(a) = 3$.

From Equations (5)-(7), we have either $\nu_5(b) \leq 1$ and $\nu_5(c) = \nu_5(d) = 2$, or $\nu_5(b) = 2$ and $\max\{\nu_5(c), \nu_5(d)\} = 2$.

Equation (1) implies $\nu_5(a) + \nu_5(b) + \nu_5(c) + \nu_5(d) = 7$. Thus, for $\nu_5(b)$, $\nu_5(c)$, $\nu_5(d)$, there must be two 2s and one 0.

Therefore,

$$\begin{aligned} \gcd(a, b, c, d) &= \prod_{p \in \{2, 3, 5\}} p^{\min\{\nu_p(a), \nu_p(b), \nu_p(c), \nu_p(d)\}} \\ &= 2^0 \cdot 3^1 \cdot 5^0 \\ &= \boxed{(C) \ 3}. \end{aligned}$$

Solution 2 (GCD/LCM Comparison)

We are given that $abcd = 2^6 \cdot 3^9 \cdot 5^7$, and several LCM relations involving pairs of these numbers. Notice that

$$\text{lcm}(a, b) \cdot \text{lcm}(c, d) = (2^3 \cdot 3^2 \cdot 5^3) \cdot (2^2 \cdot 3^3 \cdot 5^2) = 2^5 \cdot 3^5 \cdot 5^5.$$

Comparing this product with $abcd$, we see that

$$abcd = 2^6 \cdot 3^9 \cdot 5^7 = \text{lcm}(a, b) \cdot \text{lcm}(c, d) \cdot 2^1 \cdot 3^4 \cdot 5^2.$$

This additional factor $2^1 \cdot 3^4 \cdot 5^2$ must be accounted for by the overlaps among a, b, c, d , which are their common divisors.

The key observation is that the only prime with a leftover exponent greater than 3 is 3 (specifically, 3^4), which suggests that all four numbers share at least one factor of 3.

For primes 2 and 5, the leftover exponents are too small (1 and 2, respectively) to be shared by all four numbers, since the GCD exponent must be the minimum exponent among a, b, c, d .

Thus, the only possible nontrivial common factor among a, b, c, d is 3.

Therefore,

$$\gcd(a, b, c, d) = \boxed{3}.$$

- (A) $4 - \sqrt{5}$ (B) $\sqrt{5} - 1$ (C) $8 - 3\sqrt{5}$ (D) $\frac{\sqrt{5}+1}{2}$ (E) $\frac{2+\sqrt{5}}{3}$

Solution 1

Because of their similarity, the ratio of the inner pentagon's area to that of the outer pentagon can be represented by

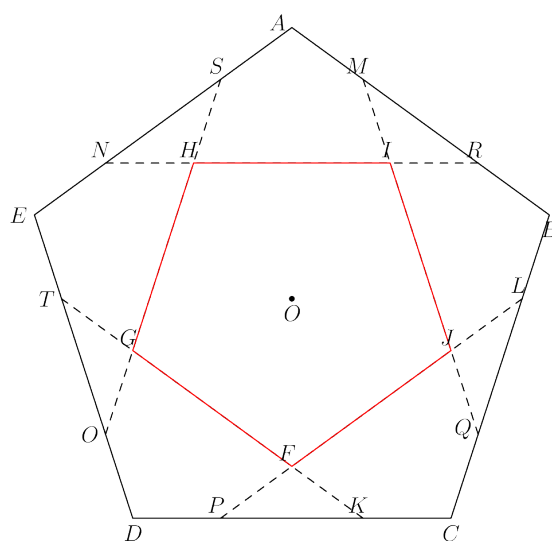
Option 1: Knowledge Remember that $\sin 54 = \frac{1+\sqrt{5}}{4}$.

Option 3: Golden Ratio/Triangle It is well known that a 36-36-108 triangle has side lengths with ratio $1 : 1 : \varphi$

$$\sin^2 54 = \frac{3+\sqrt{5}}{8}$$
$$[Z] = \frac{1}{4 \sin^2 54} [ABCDE] = \frac{2(1+\sqrt{5})}{3+\sqrt{5}} = \sqrt{5} - 1$$

-Dissmo Thegoat

Solution 2

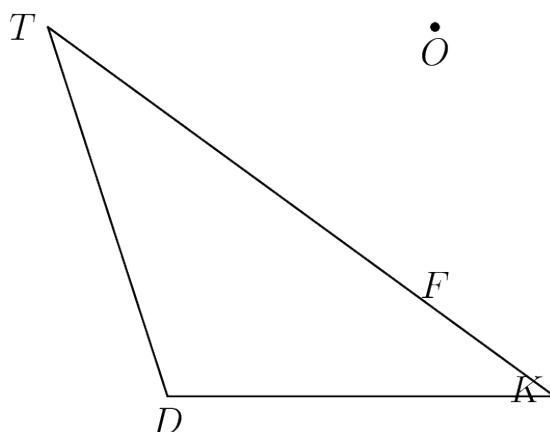


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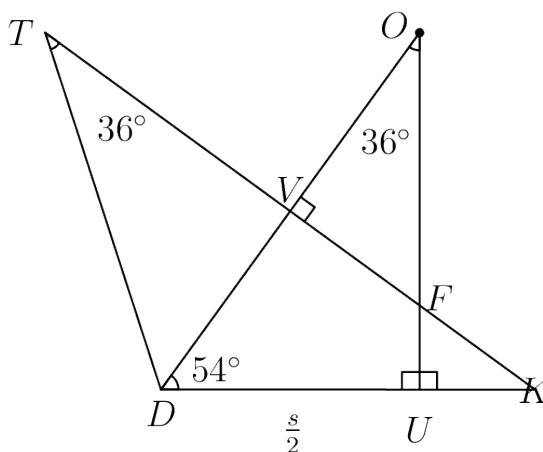
The area outside the red pentagon is the sum of the larger isosceles triangles, but this double counts the overlapping regions of the small isosceles triangles, so we have to subtract those out.

We have $[FGHIJ] = [ABCDE] - (5 \cdot [DKT] - 5 \cdot [PFK])$

Lets focus on finding the area of each individual triangle:



Notice that we have no information about the side length, so instead we let the side length be s . Now we can drop an altitude from O to the base of the triangle, and we know this altitude must split the base of the pentagon in half, so we can create a right triangle. Furthermore, draw a line from O to D . This must bisect angle D which is 108 degrees, so we create 36 – 54 – 90 triangles. Specifically, we know $\angle ODK = 54^\circ$, $\angle DOU = 36^\circ$, and $\angle DTK = 36^\circ$ because $\triangle DTK$ is isosceles and we know the vertex angle is 108° . We encode this information in the diagram below:



Since $\triangle DKT$ is isosceles, the area of $\triangle DVT$ is half the area of $\triangle DKT$. Similarly, the area of $\triangle UFK$ is half that of $\triangle PFK$. Thus:

$$[FGHIJ] = [ABCDE] - (5 \cdot [DKT] - 5 \cdot [PFK]) \implies [FGHIJ] = [ABCDE] - (10 \cdot [DVT] - 10 \cdot [UFK])$$

We also know that since we dropped an altitude from O to U , the area of $\triangle ODU$ must be half of a fifth of the total area of the pentagon. Therefore we can rewrite the above equation as

$$[FGHIJ] = 10 \cdot ([ODU] - [DVT] + [UFK])$$

Now notice that $\triangle ODU \sim \triangle TDV \sim \triangle KFU$ by AA similarity. Therefore, if we can write the areas of the latter two triangles as a ratio of the first triangle, we can express the whole equation in terms of $[ODU]$, and by extension $[ABCDE]$, which we know. To find these ratios, we can find the side length ratios and square them because the triangles are similar.

We already know $DU = \frac{s}{2}$, so let's try to find it's analogous side for $\triangle TDV$ and $\triangle KFU$. These sides are DV and FU , respectively.

First, $\frac{s}{2} = OD \cdot \cos(54^\circ)$, so $OD = \frac{s}{2} \cdot \sec(54^\circ)$. Then notice that $DV = \frac{OD}{2}$ because we have to fold D to hit O , so the folding crease has to be exactly halfway between O and D . Therefore,

$$DV = \frac{s}{4} \cdot \sec(54^\circ) \implies \frac{DU}{DV} = \frac{\frac{s}{2}}{\frac{s}{4} \cdot \sec(54^\circ)} = 2 \cdot \cos(54^\circ)$$

Now the ratio between the area of two similar triangles is the square of the ratio of their analogous side lengths. Thus

$$\begin{aligned} \frac{[ODU]}{[TDV]} &= 4 \cdot \cos^2(54^\circ) \implies \frac{10 \cdot [ODU]}{10 \cdot [TDV]} = 4 \cdot \cos^2(54^\circ) \\ \implies \frac{[ABCDE]}{10 \cdot [TDV]} &= 4 \cdot \cos^2(54^\circ) \implies \frac{\sqrt{5} + 1}{10 \cdot [TDV]} = 4 \cdot \cos^2(54^\circ) \\ \implies 10 \cdot [TDV] &= \frac{\sqrt{5} + 1}{4 \cdot \cos^2(54^\circ)} \implies 10 \cdot [TDV] = \frac{\sqrt{5} + 1}{4} \cdot \sec^2(54^\circ) \end{aligned}$$

Now let's move on and calculate the ratio of the other side length. Calculating FU is slightly trickier. First, we find TD : $TD \cdot \cos(54^\circ) = DV = \frac{s}{4} \sec(54^\circ) \implies TD = \frac{s}{4} \cdot \sec^2(54^\circ)$. Now since $\triangle DTK$ is isosceles, $TD = DK$ and $UK = DK - DU = TD - DU = \frac{s}{4} \cdot \sec^2(54^\circ) - \frac{s}{2} = \frac{s}{4} \cdot (\sec^2(54^\circ) - 2) = \frac{s}{4} \cdot (\tan^2(54^\circ) - 1)$.

Now $FU = UK \cdot \tan(36^\circ) \implies FU = \frac{s}{4} \cdot (\tan^2(54^\circ) - 1) \cdot \tan(36^\circ)$. Now note that $\tan(x) \cdot \tan(90 - x) = 1$ because opposite over adjacent cancel each other out in a right triangle. Thus, $FU = \frac{s}{4} \cdot (\tan(54^\circ) - \tan(36^\circ))$

Now,

$$\begin{aligned} \frac{DU}{FU} &= \frac{\frac{s}{2}}{\frac{s}{4} \cdot (\tan(54^\circ) - \tan(36^\circ))} = \frac{2}{\tan(54^\circ) - \tan(36^\circ)} \\ \implies \frac{[DU]}{[FU]} &= \frac{4}{(\tan(54^\circ) - \tan(36^\circ))^2} \implies 10 \cdot [FU] = \frac{1 + \sqrt{5}}{4} \cdot (\tan(54^\circ) - \tan(36^\circ))^2 \end{aligned}$$

Now we go back to our first equation and plug in our values:

$$\begin{aligned} [FGHIJ] &= [ABCDE] - (10 \cdot [DVT] - 10 \cdot [UFK]) \implies [FGHIJ] = 1 + \sqrt{5} - \frac{\sqrt{5} + 1}{4} \cdot \sec^2(54^\circ) + \frac{1 + \sqrt{5}}{4} \cdot (\tan(54^\circ) - \tan(36^\circ))^2 \\ \implies [FGHIJ] &= \frac{1 + \sqrt{5}}{4} \cdot (4 - \sec^2(54^\circ) + (\tan(54^\circ) - \tan(36^\circ))^2) \end{aligned}$$

Note $(\tan(x) - \tan(90 - x))^2 = \tan^2(x) - 2 \cdot \tan(x) \cdot \tan(90 - x) + \tan^2(90 - x) = \tan^2(x) - 2 + \tan^2(90 - x)$.

Also note that $\tan^2(x) + 1 = \sec^2(x)$. Thus

$$[FGHIJ] = \frac{1 + \sqrt{5}}{4} \cdot (2 - \sec^2(54^\circ) + \tan^2(54^\circ) + \tan^2(36^\circ))$$

$$\implies [FGHIJ] = \frac{1 + \sqrt{5}}{4} \cdot (2 + (-1) + \tan^2(36^\circ)) \implies [FGHIJ] = \frac{1 + \sqrt{5}}{4} \cdot (1 + \tan^2(36^\circ))$$

Now all that remains is to find $\tan^2(36^\circ)$. We can use the tan addition formula to find the general form of $\tan(5x)$ or remember question 25 from this year's AMC 12A. We have that

$$\tan(5x) = \frac{5\tan(x) - 10\tan^3(x) + \tan^5(x)}{1 - 10\tan^2(x) + 5\tan^4(x)}$$

Plug in $x = 36$. Then we have

$$\tan(180^\circ) = \frac{5\tan(36^\circ) - 10\tan^3(36^\circ) + \tan^5(36^\circ)}{1 - 10\tan^2(36^\circ) + 5\tan^4(36^\circ)}$$

Now let $y = \tan(36^\circ)$. We have the equation

$$\frac{5y - 10y^3 + y^5}{1 - 10y^2 + 5y^4} = 0 \implies 5y - 10y^3 + y^5 = 0$$

$$\implies 5 - 10y^2 + y^4 = 0 \implies 5 - 10z + z^2 = 0$$

Where we let $z = y^2$. Using the quadratic formula, we have

$$z = \frac{10 \pm \sqrt{80}}{2} = 5 \pm 2\sqrt{5}$$

Now since $y = \tan(36^\circ)$, $z = \tan^2(36^\circ)$, which is what we were looking for. Notice that $\tan(0^\circ) = 0$ and $\tan(45^\circ) = 1$, so $\tan(36^\circ)$ is between 0 and 1, and so is its square. Thus $z = 5 - 2\sqrt{5}$, not the other root.

Finally:

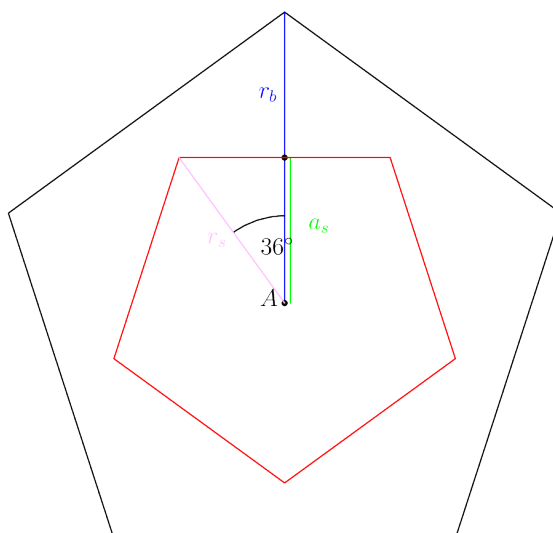
$$[FGHIJ] = \frac{1 + \sqrt{5}}{4} \cdot (1 + \tan^2(36^\circ)) = \frac{1 + \sqrt{5}}{4} \cdot (1 + 5 - 2\sqrt{5})$$

$$\implies [FGHIJ] = \frac{1 + \sqrt{5}}{4} \cdot (6 - 2\sqrt{5}) = \frac{4\sqrt{5} - 4}{4} = \sqrt{5} - 1$$

Therefore,

$$[FGHIJ] = \sqrt{5} - 1 = \boxed{B}$$

Solution 3



Let r_b and r_s be the circumradius of the big and small pentagon, respectively. Let a_s be the apothem of the smaller pentagon and A_s and A_b be the areas of the smaller and larger pentagon, respectively.

From the diagram:

$$\begin{aligned}
 \cos 36^\circ &= \frac{a_s}{r_s} = \frac{\phi}{2} = \frac{\sqrt{5}+1}{4} \\
 a_s &= \frac{r_b}{2} \\
 A_s &= \left(\frac{r_s}{r_b}\right)^2 A_b \\
 &= \left(\frac{a_s}{\cos 36^\circ r_b}\right)^2 (1+\sqrt{5}) \\
 &= \left(\frac{a_s}{\frac{\phi}{2} r_b}\right)^2 (1+\sqrt{5}) \\
 &= \left(\frac{1}{2\frac{\phi}{2}}\right)^2 (1+\sqrt{5}) \\
 &= \left(\frac{2}{\sqrt{5}+1}\right)^2 (1+\sqrt{5}) \\
 &= \frac{4}{\sqrt{5}+1} \\
 &= \frac{4(\sqrt{5}-1)}{(\sqrt{5}+1)(\sqrt{5}-1)} \\
 &= \sqrt{5}-1
 \end{aligned}$$

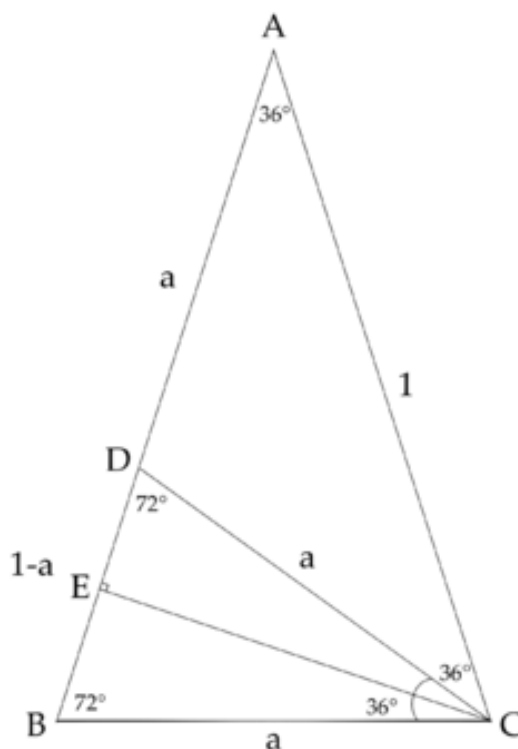
$$\boxed{\text{(B)} \sqrt{5}-1}$$

Solution 4

Interestingly, we find that the pentagon we need is the one that is represented by the intersection of perpendicular bisectors of the connection from the center of the pentagon to one vertex. Through similar triangles and the golden ratio, we find that the side length ratio of the two pentagons is $\frac{\sqrt{5}-1}{2}$. Thus, the answer is $(\sqrt{5} + 1) \cdot \left(\frac{\sqrt{5}-1}{2}\right)^2 = \sqrt{5} - 1$.

B andliu766

Supplement (Calculating $\sin 54^\circ / \cos 36^\circ$ from Scratch) Method 1:



Construct golden ratio triangle $\triangle ABC$ with $\angle A = 36^\circ$, $\angle B = \angle C = 72^\circ$ and $\triangle BCD$ with $\angle C = 36^\circ$, $\angle DBC = \angle BDC = 72^\circ$. WLOG, let $AB = AC = 1$, $BC = CD = AD = a$, $BD = 1 - a$. $\triangle BAC \sim \triangle BCD$

$$\frac{AC}{BC} = \frac{BC}{BD}, \quad \frac{1}{a} = \frac{a}{1-a}, \quad 1-a = a^2, \quad a^2 + a - 1 = 0$$

$$a = \frac{-1 + \sqrt{1^2 - 4(-1)}}{2} = \frac{\sqrt{5} - 1}{2}$$

$$\cos 36^\circ = \cos \angle A = \frac{AE}{AC} = \frac{1-a}{2} + a = \frac{a+1}{2} = \frac{\frac{\sqrt{5}-1}{2} + 1}{2} = \frac{\sqrt{5} + 1}{4}$$

$$\sin 54^\circ = \cos 36^\circ = \frac{\sqrt{5} + 1}{4}$$

Method 2:

As explained [here](#), $\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$

$$\cos \frac{2\pi}{5} - \cos \frac{\pi}{5} = -\frac{1}{2}$$

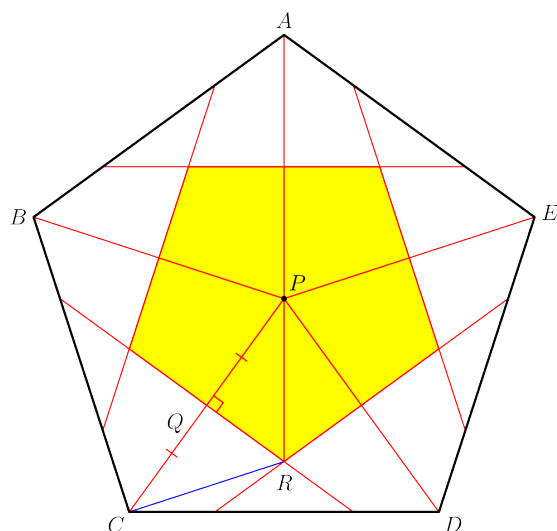
$$2\left(\cos \frac{\pi}{5}\right)^2 - 1 - \cos \frac{\pi}{5} = -1/2$$

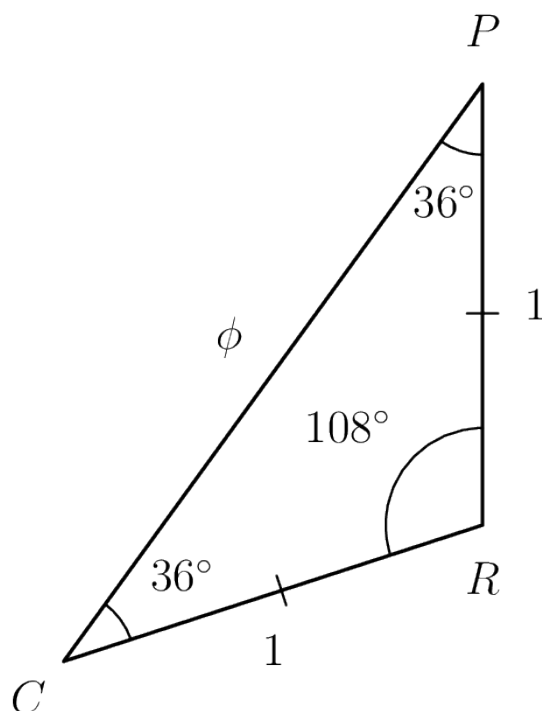
$$4\left(\cos \frac{\pi}{5}\right)^2 - 2\cos \frac{\pi}{5} - 1 = 0$$

$$\cos 36^\circ = \cos \frac{\pi}{5} = \frac{2 + \sqrt{2^2 + 4 \cdot 4}}{8} = \frac{1 + \sqrt{5}}{4}$$

Solution 5 (Much easier if you notice the one line construction + use properties of the golden ratio)

We will start with a similar diagram to all of the other submitted solutions with ONE difference: We will connect line segment CR to form a 36° - 36° - 108° golden triangle $\triangle CPR$. We know this is a 36° - 36° - 108° golden triangle because QR is the perpendicular bisector of CP (in order to fold vertex C onto the center P), so $\triangle CPR$ must be isosceles, with angle $\angle CPR = \frac{360^\circ}{10} = 36^\circ$ due to the regular pentagon.





Rather than trigonometry, it is well known that $\frac{CP}{PR} = \frac{\phi}{1}$ due to the golden triangle. Since both pentagons are regular, they are similar. Therefore the ratio of the areas is equal to the ratio of the radii squared (rather than using the ratio of the sides of the pentagon). Let x be the area of the smaller pentagon. Note that we are given the larger pentagon has area 2ϕ .

We will also use a well known identity of the golden ratio: $\frac{1}{\phi} = \phi - 1$ (and for completeness sake for any student reading this in the future looking for other golden ratio identities: $\phi^2 = \phi + 1$) to simplify expressions involving the golden ratio rather than to do the algebra on the square roots. Then:

$$\begin{aligned}\frac{x}{2\phi} &= \left(\frac{PR}{CP}\right)^2 \\ \frac{x}{2\phi} &= \left(\frac{1}{\phi}\right)^2 \\ \frac{x}{2} &= \frac{1}{\phi} \\ \frac{x}{2} &= \phi - 1 \\ x &= 2\phi - 2 \\ x &= 2\left(\frac{1 + \sqrt{5}}{2}\right) - 2 \\ x &= 1 + \sqrt{5} - 2 \\ x &= \sqrt{5} - 1\end{aligned}$$

B

§10 AMC 12 12A 2023

1. (*AMC 12 12A 2023*) Cities A and B are 45 miles apart. Alicia lives in A and Beth lives in B . Alicia bikes towards B at 18 miles per hour. Leaving at the same time, Beth bikes toward A at 12 miles per hour. How many miles from City A will they be when they meet?

(A) 20 (B) 24 (C) 25 (D) 26 (E) 27

Answer: E

Solution 1

This is a $d = st$ problem, so let x be the time it takes to meet. We can write the following equation:

$$12x + 18x = 45$$

Solving gives us $x = 1.5$. The $18x$ is Alicia so $18 \times 1.5 = \boxed{\text{(E) 27}}$

Solution 2

The relative speed of the two is $18 + 12 = 30$, so $\frac{3}{2}$ hours would be required to travel 45 miles. $d = st$, so $x = 18 \cdot \frac{3}{2} = \boxed{\text{(E) 27}}$

Solution 3

Since 18 mph is $\frac{3}{2}$ times 12 mph, Alicia will travel $\frac{3}{2}$ times as far as Beth. If x is the distance Beth travels,

$$\frac{3}{2}x + x = 45$$

$$\frac{5}{2}x = 45$$

$$x = 18$$

Since this is the amount Beth traveled, the amount that Alicia traveled was

$$45 - 18 = \boxed{\text{(E) 27}}$$

Solution 4

Alice and Barbara close in on each other at 30mph. Since they are 45 miles apart, they will meet in $t = d/s = 45\text{miles} / 30\text{mph} = 3/2$ hours. We can either calculate the distance Alice travels at 18mph or the distance Barbara travels at 12mph; since we want the distance from Alice, we go with the former. Alice (and Barbara) will meet in $1 \frac{1}{2}$ hours at $18\text{mph} \times 3/2 \text{ hours} = 27$ miles from A. $\boxed{\text{(E) 27}}$

Solution 5 (Under 20 seconds)

We know that Alice approaches Beth at 18 mph and Beth approaches Alice at 12 mph. If we consider that if Alice moves 18 miles at the same time Beth moves 12 miles $\rightarrow$ 15 miles left. Alice then moves 9 more miles at the same time as Beth moves 6 more miles. Alice has moved 27 miles from point A at the same time that Beth has moved 18 miles from point B, meaning that Alice and Beth meet 27 miles from point A.

Solution 6 (simple linear equations)

We know that Beth starts 45 miles away from City A, let's create two equations: Alice-
 $\rightarrow 18t = d$ Beth- $\rightarrow -12t + 45 = d$ [-12 is the slope; 45 is the y-intercept]

Solve the system:

$$18t = -12t + 45 \quad 30t = 45 \quad t = 1.5$$

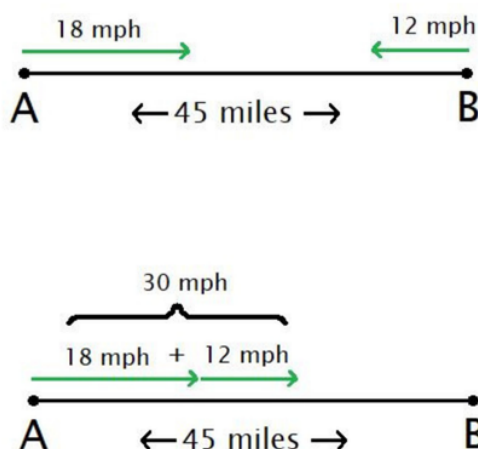
So, $18(1.5) = \boxed{\text{(E) } 27}$

Solution 7

Since Alicia and Beth's speeds are constant, they are directly proportional to their distances covered, so the ratio of their speeds is equal to the ratio of their covered distances. Since Alicia travels $\frac{18}{30} = \frac{3}{5}$ of their combined speed, she travels $\frac{3}{5} \cdot 45 = \boxed{\text{(E) } 27}$ of the total distance.

-Benedict T (countmath1)

Solution 8



Note that Alicia and Beth must have travelled 45 miles together in order for them to meet together. While travelling the 45 miles, Alicia came closer towards Beth and Beth came closer towards Alicia. Since Alicia is travelling faster than Beth, we know that they will meet slightly closer towards city B from the middle. Also, the distance remaining between Alicia and Beth as they bike toward each other is proportional to their combined velocity. The combined velocity is $18t + 12t = 30t$. The time it takes to travel 45 miles going at 30 miles per hour can be calculated using simple algebra.

Let t be the time it takes for Alicia and Beth to meet together.

$$18t + 12t = 45$$

$$30t = 45$$

$$t = \frac{3}{2}$$

It takes $\frac{3}{2}$ hours of cycling by both Alicia and Beth at their respective biking speeds in order to meet together. Also, note that the question asks for the distance from City A they will be when they meet. This is the same as the distance travelled by Alicia from City A. Alicia travels at 18 mph, which turns into a distance = speed $\times$ time question.

$$d = st$$

$$d = 18 \times \frac{3}{2}$$

$$d = 27$$

The distance from City A when Alicia and Beth meet is **(E) 27**.

Note - Even though it might seem like there is a lot of work for this problem, it would typically take a very short time to think this entire process out.

2. (AMC 12 12A 2023) The weight of $\frac{1}{3}$ of a large pizza together with $3\frac{1}{2}$ cups of orange slices is the same weight of $\frac{3}{4}$ of a large pizza together with $\frac{1}{2}$ cups of orange slices. A cup of orange slices weigh $\frac{1}{4}$ of a pound. What is the weight, in pounds, of a large pizza?

(A) $1\frac{4}{5}$ (B) 2 (C) $2\frac{2}{5}$ (D) 3 (E) $3\frac{3}{5}$

Answer: A

Solution 1 (Substitution)

Use a system of equations. Let x be the weight of a pizza and y be the weight of a cup of orange slices. We have

$$\frac{1}{3}x + \frac{7}{2}y = \frac{3}{4}x + \frac{1}{2}y.$$

Rearranging, we get

$$\begin{aligned}\frac{5}{12}x &= 3y, \\ x &= \frac{36}{5}y.\end{aligned}$$

Plugging in $\frac{1}{4}$ pounds for y by the given gives $\frac{9}{5} = \mathbf{(A) 1\frac{4}{5}}$.

Solution 2

Let: p be the weight of a pizza. o be the weight of a cup of orange.

From the problem, we know that $o = \frac{1}{4}$.

Write the equation below:

$$\frac{1}{3}p + \frac{7}{2} \cdot \frac{1}{4} = \frac{3}{4}p + \frac{1}{2} \cdot \frac{1}{4}$$

Solving for p : $\frac{5}{12}p = \frac{3}{4}p - \frac{9}{8}$ $p = \frac{9}{5} = \mathbf{(A) 1\frac{4}{5}}$.

Solution 3

$\frac{P}{3} + \frac{7}{2}R = \frac{3}{4}P + \frac{R}{2}$ where P is the pizza weight and R is the weight of cup of oranges. Since oranges weigh $\frac{1}{4}$ pound per cup, the oranges on the LHS weigh $\frac{7}{2}$ cups $\times \frac{1}{4}$ pounds/cup = $\frac{7}{8}$ pound, and those on the RHS weigh $\frac{1}{2}$ cup $\times \frac{1}{4}$ pounds/cup = $\frac{1}{8}$ pound.

So $\frac{P}{3} + \frac{7}{8}$ pound = $\frac{3}{4}P + \frac{1}{8}$ pound; $\frac{P}{3} + \frac{3}{4}$ pound = $\frac{3}{4}P$.

Multiplying both sides by $\text{lcm}(3, 4) = 12$, we have $4P + 9 = 9P$; $5P = 9$; $P = \text{weight of a large pizza} = \frac{9}{5}$ pounds = $\mathbf{(A) 1\frac{4}{5}}$ pounds.

3. (AMC 12 12A 2023) How many positive perfect squares less than 2023 are divisible by 5?

(A) 8 (B) 9 (C) 10 (D) 11 (E) 12

Answer: A

Solution 1 (slightly refined)

Since $\lfloor \sqrt{2023} \rfloor = 44$, there are $\lfloor \frac{44}{5} \rfloor = \boxed{\text{(A) } 8}$ perfect squares less than 2023 that are divisible by 5.

Solution 2

Since 5 is square-free, each solution must be divisible by $5^2 = 25$. We take $\lfloor \frac{2023}{25} \rfloor = 80$ and see that there are $\boxed{\text{(A) } 8}$ positive perfect squares no greater than 80.

Solution 3

Since the perfect squares have to be divisible by 5, then we know it has to be 5 times some number squared ($5 \cdot x^2$). With this information, you can figure out every single product of 5 and another number squared to count how many perfect squares are divisible by 5 that are less than 2023. (EX: $5^2 = 25$, $10^2 = 100$, ..., $40^2 = 1600$) With this you get a max of 40^2 , or $\lfloor \frac{44}{5} \rfloor = \boxed{\text{(A) } 8}$ solutions.

Solution 4

The way of BlueShadow refined:

All it takes is to recall that 45 squared is 2025, and 45 is 5×9 . So all the squares of 5×1 , 5×2 , 5×3 so on are divisible by 5. So the answer is 8. It can be done even if one does not remember that 45 squared is 2025, all it takes is intuition. One can easily see mentally that 5×8 that is 40 squared is 1600, and then one has to do just one more computation and see that 5×9 that is 45 squared exceeds 2023, so the answer is 8.

Note that you can find the square of any number that ends in 5 by taking the number 5 more than it and the number 5 less than it, multiplying those together, and adding 25. For example, to calculate the square of 45, you do $40 \times 50 = 2000$, and $2000 + 25 = 2025$.

And then you know that it goes up from 1 to 44, and the multiples of five are 5, 10..., 35, 40, dividing each of them by 5, you get 1, 2..., 7, 8, therefore, the answer is $\boxed{\text{(A) } 8}$

4. (AMC 12 12A 2023) How many digits are in the base-ten representation of $8^5 \cdot 5^{10} \cdot 15^5$?

(A) 14 (B) 15 (C) 16 (D) 17 (E) 18

Answer: E

Solution 1

Prime factorizing this gives us $2^{15} \cdot 3^5 \cdot 5^{15} = 10^{15} \cdot 3^5 = 243 \cdot 10^{15}$.

10^{15} has 16 digits and $243 = 2.43 \cdot 10^2$ gives us 3 more digits. $16 + 2 = \boxed{\text{(E) } 18}$
 $2.43 \cdot 10^{17}$ has 18 digits

Solution 2 (Only if you don't know how to do the rest of the problems and have about 20 minutes left, IT IS NOT recommended)

Multiplying it out, we get that $8^5 \cdot 5^{10} \cdot 15^5 = 24300000000000000$. Counting, we have the answer is **(E) 18** andliu766

Solution 3 (Similar to Solution 1)

All the exponents have a common factor of 5 which we can factor out. This leaves us with $(8 \cdot 5^2 \cdot 15)^5 = (3000)^5 = (3 \cdot 1000)^5$. We can then distribute the power leaving us with $3^5 \cdot 10^{3 \cdot 5} = 243 \cdot 10^{15}$. This would be 243 followed by 15 zeros resulting in our answer being $15 + 3 = \mathbf{(E)18}$ leon_oiler

Solution 4 (log's)

We have 8^5 , 5^{10} , and 15^5 , if we do a logarithm base of 10 we can figure out how many digits are in the expression when multiplied. So we have $((\log(5^{10} \cdot 8^5 \cdot 15^5)))$ which using some rules we have $((10\log(5)) + (5\log(8)) + (5\log(15)))$, using the fact that $\log(5) = 0.7, \log(8) = 0.9, \log(15) = 1.2$, multiplying it out we get $10 \cdot 0.7 + 5 \cdot 0.9 + 1.2 \cdot 5$, and since $\log(1)=0$, as this is false as 1 is a digit and not 0, we add one and then take the floor to get us **(E)18** Randomuser456All values stated are approximated

5. (AMC 12 12A 2023) Janet rolls a standard 6-sided die 4 times and keeps a running total of the numbers she rolls. What is the probability that at some point, her running total will equal 3? (A) $\frac{2}{9}$ (B) $\frac{49}{216}$ (C) $\frac{25}{108}$ (D) $\frac{17}{72}$ (E) $\frac{13}{54}$

Answer: B

Solution 1 (Casework)

There are 3 cases where the running total will equal 3: one roll; two rolls; or three rolls:

Case 1:The chance of rolling a running total of 3, namely (3) in exactly one roll is $\frac{1}{6}$.

Case 2:The chance of rolling a running total of 3 in exactly two rolls, namely (1, 2) and (2, 1) is $\frac{1}{6} \cdot \frac{1}{6} \cdot 2 = \frac{1}{18}$.

Case 3:The chance of rolling a running total of 3 in exactly three rolls, namely (1, 1, 1) is $\frac{1}{6} \cdot \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{216}$.

Using the rule of sum we have $\frac{1}{6} + \frac{1}{18} + \frac{1}{216} = \mathbf{(B) \frac{49}{216}}$.

Solution 2 (Brute Force)

Because there is only a maximum of 3 rolls we must count (running total = 3 means there can't be a fourth roll counted), we can simply list out all of the probabilities.

If we roll a 1 on the first, the rolls that follow must be 2 or 1,1, with the following results not mattering. This leaves a probability of $\frac{1}{6} \times \frac{1}{6} + \frac{1}{6} \times \frac{1}{6} \times \frac{1}{6} = \frac{7}{216}$.

If we roll a 2 on the first, the roll that follows must be 1, resulting in a probability of $\frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$.

If we roll a 3 on the first, the following rolls do not matter, resulting in a probability of $\frac{1}{6}$. Any roll greater than three will result in a running total greater than 3 no matter what, so those cases can be ignored. Summing the answers, we have

$$\frac{7}{216} + \frac{1}{36} + \frac{1}{6} = \frac{7+6+36}{216} = \mathbf{(B) \frac{49}{216}}$$

Solution 3

Consider sequences of 4 integers with each integer between 1 and 6, the number of permutations of 6 numbers is $6^4 = 1296$.

The following 4 types of sequences that might generate a running total of the numbers to be equal to 3 (x, y, or z denotes any integer between 1 and 6).

Sequence #1, (1, 1, 1, x): there are 6 possible sequences.

Sequence #2, (1, 2, x, y): there are $6^2 = 36$ possible sequences.

Sequence #3, (2, 1, x, y): there are $6^2 = 36$ possible sequences.

Sequence #4, (3, x, y, z): there are $6^3 = 216$ possible sequences.

Out of 1296 possible sequences, there are a total of $6 + 36 + 36 + 216 = 294$ sequences

that qualify. Hence, the probability is $294/1296 = \boxed{\text{(B)} \frac{49}{216}}$

Solution 4 (cheese-ish)

You have to add $1/216$ for the case 1-1-1, so it should have denominator 216. Therefore, the answer is B: $49/216$

-Supernova77

6. (AMC 12 12A 2023) Points A and B lie on the graph of $y = \log_2 x$. The midpoint of $\overline{AB}$ is $(6, 2)$. What is the positive difference between the x -coordinates of A and B ?

(A) $2\sqrt{11}$ (B) $4\sqrt{3}$ (C) 8 (D) $4\sqrt{5}$ (E) 9

Answer: D

Solution 1

Let $A(6 + m, 2 + n)$ and $B(6 - m, 2 - n)$, since $(6, 2)$ is their midpoint. Thus, we must find $2m$. We find two equations due to A, B both lying on the function $y = \log_2 x$. The two equations are then $\log_2(6 + m) = 2 + n$ and $\log_2(6 - m) = 2 - n$. Now add these two equations to obtain $\log_2(6 + m) + \log_2(6 - m) = 4$. By logarithm rules, we get $\log_2((6 + m)(6 - m)) = 4$. By raising 2 to the power of both sides, we obtain $(6 + m)(6 - m) = 16$. We then get

$$36 - m^2 = 16 \rightarrow m^2 = 20 \rightarrow m = 2\sqrt{5}$$

. Since we're looking for $2m$, we obtain $(2)(2\sqrt{5}) = \boxed{\text{(D)} 4\sqrt{5}}$

Solution 2

We have $\frac{x_A + x_B}{2} = 6$ and $\frac{\log_2 x_A + \log_2 x_B}{2} = 2$. The first equation becomes $x_A + x_B = 12$, and the second becomes $\log_2(x_A x_B) = 4$, so $x_A x_B = 16$. Then

$$\begin{aligned} |x_A - x_B| &= \sqrt{(x_A + x_B)^2 - 4x_A x_B} \\ &= \boxed{\text{(D)} 4\sqrt{5}}. \end{aligned}$$

Solution 3

Basically, we can use the midpoint formula

assume that the points are (x_1, y_1) and (x_2, y_2)

assume that the points are $(x_1, \log_2(x_1))$ and $(x_2, \log_2(x_2))$

midpoint formula is $(\frac{x_1 + x_2}{2}, \frac{\log_2(x_1) + \log_2(x_2)}{2})$

thus $x_1 + x_2 = 12$ and $x_1 x_2 = 16$. Also, $\log_2(x_1) + \log_2(x_2) = 4\log_2(x_1) + \log_2(12 - x_1) = \log_2(16) \log_2((12x_1 - x_1^2)/16) = 0$

since $2^0 = 1$ so,

$12x_1 - x_1^2 = 16$ for simplicity let's say $x_1 = x$ $12x - x^2 = 16$. We rearrange to get $x^2 - 12x + 16 = 0$.

put this into quadratic formula and you should get

$$x_1 = 6 + 2\sqrt{5} \text{ Therefore, } x_1 = 6 + 2\sqrt{5} - (6 - 2\sqrt{5})$$

$$\text{which equals } 6 - 6 + 4\sqrt{5} = \boxed{\text{(D)} 4\sqrt{5}}$$

Solution 4

Similar to above, but solve for $x = 2^y$ in terms of y :

$$(2^y + 2^{2+(2-y)})/2 = 6 \quad 2^y + 2^{4-y} = 12 \quad (2^y)^2 + 2^4 = 12(2^y) \quad x^2 - 12x + 16 = 0$$

Distance between roots of the quadratic is the discriminant: $\sqrt{12^2 - 4(1)(16)} = \sqrt{80} = \boxed{\text{(D)} 4\sqrt{5}}$

Solution 5 (Similar to Solution 3)

Let the two points on the curve $y = \log_2 x$ be $A(x_1, \log_2 x_1)$ and $B(x_2, \log_2 x_2)$. By the midpoint formula, the coordinates of the midpoint are given by:

$$\left(\frac{x_1 + x_2}{2}, \frac{\log_2 x_1 + \log_2 x_2}{2} \right)$$

Given that the midpoint is $(6, 2)$, we obtain the following system of equations:

$$x_1 + x_2 = 12 \tag{20}$$

$$\log_2(x_1 x_2) = 4 \implies x_1 x_2 = 16 \tag{21}$$

By Vieta's formulas, x_1 and x_2 are the roots of the quadratic equation:

$$x^2 - 12x + 16 = 0$$

Solving via the quadratic formula yields the roots:

$$x = \frac{12 \pm \sqrt{144 - 64}}{2} = 6 \pm 2\sqrt{5}$$

Without loss of generality, let $x_1 = 6 + 2\sqrt{5}$ and $x_2 = 6 - 2\sqrt{5}$. The horizontal distance between the two points is:

$$|x_1 - x_2| = (6 + 2\sqrt{5}) - (6 - 2\sqrt{5}) = \boxed{\text{(D)} 4\sqrt{5}} \quad \blacksquare$$

7. (AMC 12 12A 2023) A digital display shows the current date as an 8-digit integer consisting of a 4-digit year, followed by a 2-digit month, followed by a 2-digit date within the month. For example, Arbor Day this year is displayed as 20230428. For how many dates in 2023 will each digit appear an even number of times in the 8-digital display for that date?

(A) 5 (B) 6 (C) 7 (D) 8 (E) 9

Answer: E

Solution 1 (Casework)

Do careful casework by each month. Make sure to start with 2023. In the month and the date, we need a 0, a 3, and two digits repeated (which has to be 1 and 2 after consideration). After the casework, we get **(E) 9**. For curious readers, the numbers (in chronological order) are:

202301132023013120230223202303112023032220231013202310312023110320231130

Solution 2

There is one 3, so we need one more (three more means that either the month or units digit of the day is 3). For the same reason, we need one more 0. By inspection, we can see that the other two digits must be a repeated digit that is not 3 or 0.

If 3 is the units digit of the month, then the 0 must be the units digit of the month, since there is no month ending with 3 that is more than March. In this case, the last two digits must match (11, 22). This gives 2023 – 03 – 11 and 2023 – 03 – 22, so 2 successes.

If 3 is the tens digit of the day, then 0 can be in any of the remaining three places. If it is in the ones digit of the the day, then the month must be 11, which gives 2023 – 11 – 30. If it is in the ones digit of the month, then the month must be 10, in which case the day must be 31, which gives 2023 – 10 – 31. If 0 is in the tens digit of the month, then the month must be 01, which gives 2023 – 01 – 31. Together, 2023 – 11 – 30, 2023 – 10 – 31, and 2023 – 01 – 31 make 3 successes.

If 3 is the units digit of the day, then 0 could go in any of the 3 remaining slots again. If it's the tens digit of the day, then the other digits must be 1. If 0 is the units digit of the month, then the other two slots must both be 1. If 0 is the tens digit of the month, then the other two slots can be either both 1 or both 2. In total, there are 4 successes here.

Summing through all cases, there are $2 + 3 + 4 = \mathbf{(E) 9}$ dates.

-Benedict T (countmath1)

-Edits by Cat_Of_Logic

Solution 3

We start with 2023 – – – – we need an extra 0 and an extra 3. So we have at least one of those extras in the days, except we can have the month 03. We now have 6 possible months 01, 02, 03, 10, 11, 12. For month 1 we have two cases, we now have to add in another 1, and the possible days are 13, 31. For month 2 we need an extra 2 so we can have the day 23 note that we can't use 32 because it is too large. Now for month 3 we can have any number and multiply it by 11 so we have the solution 11, 22. For October we need a 1 and a 3 so we have 13, 31 as our choices. For November we have two choices which are 03, 30. Now for December we have 0 options. Summing $2 + 1 + 2 + 2 + 2$ we get **(E) 9** solutions.

8. (AMC 12 12A 2023) Maureen is keeping track of the mean of her quiz scores this semester. If Maureen scores an 11 on the next quiz, her mean will increase by 1. If she scores an 11 on each of the next three quizzes, her mean will increase by 2. What is the mean of her quiz scores currently?

(A) 4 (B) 5 (C) 6 (D) 7 (E) 8

Answer: D

Solution 1

Let a represent the amount of tests taken previously and x the mean of the scores taken previously.

We can write the following equations:

$$\frac{ax + 11}{a + 1} = x + 1 \quad (1)$$

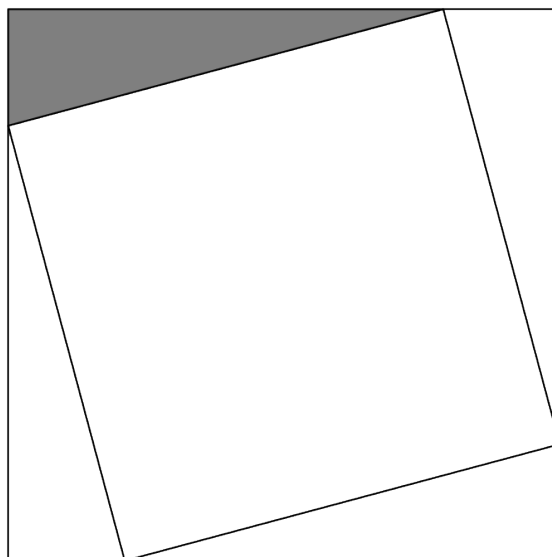
$$\frac{ax + 33}{a + 3} = x + 2 \quad (2)$$

Multiplying equation (1) by $(a + 1)$ and solving, we get:

$$ax + 11 = ax + a + x + 1$$

$$11 = a + x + 1$$

9. (AMC 12 12A 2023) A square of area 2 is inscribed in a square of area 3, creating four congruent triangles, as shown below. What is the ratio of the shorter leg to the longer leg in the shaded right triangle?



- (A) $\frac{1}{5}$ (B) $\frac{1}{4}$ (C) $2 - \sqrt{3}$ (D) $\sqrt{3} - \sqrt{2}$ (E) $\sqrt{2} - 1$

Answer: C

Solution 1

The side lengths of the inner square and outer square are $\sqrt{2}$ and $\sqrt{3}$ respectively. Let the shorter side of our triangle be x , thus the longer leg is $\sqrt{3} - x$. Hence, by the Pythagorean Theorem, we have

$$\begin{aligned} (\sqrt{3} - x)^2 + x^2 &= (\sqrt{2})^2 \\ 3 - 2\sqrt{3}x + x^2 + x^2 &= 2 \\ 2x^2 - 2\sqrt{3}x + 1 &= 0. \end{aligned}$$

By the quadratic formula, we find that $x = \frac{\sqrt{3} \pm 1}{2}$, so the answer is $\frac{\sqrt{3} - 1}{\sqrt{3} + 1} =$

(C) $2 - \sqrt{3}$.

Solution 2 (Area)

Looking at the diagram, we know that the square inscribed in the square with area 3 has area 2. Thus, the area outside of the small square is $3 - 2 = 1$. This area is composed of 4 congruent triangles, so we know that each triangle has an area of $\frac{1}{4}$.

From solution 1, the base (short side of the triangle) has a length x and the height $\sqrt{3} - x$, which means that $\frac{x(\sqrt{3}-x)}{2} = \frac{1}{4}$.

We can turn this into a quadratic equation: $x^2 - x\sqrt{3} + \frac{1}{2} = 0$.

By using the quadratic formula, we get $x = \frac{\sqrt{3} \pm 1}{2}$. Therefore, the answer is $\frac{\sqrt{3}-1}{\sqrt{3}+1} =$

$$\boxed{\text{(C)} \ 2 - \sqrt{3}}.$$

(Clarity formatting edits by Technodoggo)

Solution 3

Let x be the ratio of the shorter leg to the longer leg, and y be the length of longer leg. The length of the shorter leg will be xy .

Because the sum of two legs is the side length of the outside square, we have $xy + y = \sqrt{3}$, which means $(xy)^2 + y^2 + 2xy^2 = 3$. Using the Pythagorean Theorem for the shaded right triangle, we also have $(xy)^2 + y^2 = 2$. Solving both equations, we get $2xy^2 = 1$. Using $y^2 = \frac{1}{2x}$ to substitute y in the second equation, we get $x^2 \cdot \frac{1}{2x} + \frac{1}{2x} = 2$. Hence, $x^2 - 4x + 1 = 0$. By using the quadratic formula, we get $x = 2 \pm \sqrt{3}$. Because x be the ratio of the shorter leg to the longer leg, it is always less than 1. Therefore, the answer is $\boxed{\text{(C)} \ 2 - \sqrt{3}}$.

Solution 4

The side length of the bigger square is equal to $\sqrt{3}$, while the side length of the smaller square is $\sqrt{2}$. Let x be the shorter leg and y be the longer one. Clearly, $x + y = \sqrt{3}$, and $xy = \frac{1}{2}$. Using Vieta's to build a quadratic, we get

$$x^2 - \sqrt{3}x + \frac{1}{2} = 0.$$

Solving, we get $x = \frac{\sqrt{3}-1}{2}$ and $y = \frac{\sqrt{3}+1}{2}$. Thus, we find $\frac{x}{y} = \frac{\sqrt{3}-1}{\sqrt{3}+1} = \frac{(\sqrt{3}-1)^2}{(\sqrt{3}+1)(\sqrt{3}-1)} = \frac{4-2\sqrt{3}}{2} = \boxed{\text{(C)} \ 2 - \sqrt{3}}.$

Solution 5

Let θ be the angle opposite the smaller leg. We want to find $\tan \theta$.

The area of the triangle is $\frac{1}{2} (\sqrt{2} \sin \theta) (\sqrt{2} \cos \theta) = \frac{1}{2} \sin 2\theta = \frac{1}{4}$, which implies $\sin 2\theta = \frac{1}{2}$, or $\theta = 15^\circ$. Therefore $\tan \theta = \boxed{\text{(C)} \ 2 - \sqrt{3}}$

Solution 6

Allow a and b to be the sides of a triangle. WLOG, suppose $a > b$. We want to find $\frac{b}{a}$. Notice that the area of a triangle is $\frac{3-2}{4}$, which results in $\frac{1}{4}$. Thus, $ab = \frac{1}{2}$. However, the square of the hypotenuse of this triangle is $a^2 + b^2$, but also 2. We can write b as $\frac{1}{2a}$, and then plug it in. We get $a^2 + \frac{1}{4a^2} = 2$, so $4a^4 - 8a^2 + 1 = 0$. Applying the quadratic formula, $a^2 = \frac{8 \pm 4\sqrt{3}}{2}$, or $4 \pm 2\sqrt{3}$. However, since a and b must both be solutions of the quadratic, since both equations were cyclic. Since $a > b$, then $a^2 = 4 + 2\sqrt{3}$, and $b^2 = 4 - 2\sqrt{3}$. To find $\frac{b}{a}$, we can simply find the

square root of $\frac{b^2}{a^2}$. This is $\sqrt{\frac{4-2\sqrt{3}}{4+2\sqrt{3}}} = \sqrt{\frac{2-\sqrt{3}}{2+\sqrt{3}}} = \sqrt{(2-\sqrt{3})(2-\sqrt{3})} = \boxed{2-\sqrt{3}}$, so the answer is **(C)**. - Sepehr2010

Solution 7 (Manipulation)

Let a be the length of the shorter leg and b be the longer leg. By the Pythagorean theorem, we can derive that $a^2 + b^2 = 2$. Using area we can also derive that $2ab = 1$. $(a+b)^2 = 3$ as given in the diagram, we can find that $(a-b)^2 = 1$ because $4ab = 2$. This means that $b+a = \sqrt{3}$ and $b-a = 1$. Adding the equations gives $b = \frac{\sqrt{3}+1}{2}$ and when b is plugged in $a = \frac{\sqrt{3}-1}{2}$. Rationalizing the denominators gives us **(C) $2 - \sqrt{3}$** .

Solution 8

Let s be the length of the shorter leg, l the longer leg, and t the desired ratio. We have:

$$s^2 + l^2 = 2 \text{ and } \frac{1}{2}sl = \frac{1}{4}$$

The former dividing the latter gives:

$$t + \frac{1}{t} = 4$$

Solve this for t . Multiply by t :

$$t^2 + 1 = 4t$$

Subtract $-4t$ to get a quadratic:

$$t^2 - 4t + 1 = 0$$

Use the quadratic formula to get:

$$2 \pm \sqrt{3}$$

And since the smaller one is the first one, the solution has to be less than 1, so the answer is the one less than 1, the answer is **(C) $2 - \sqrt{3}$**

10. (AMC 12 12A 2023) Positive real numbers x and y satisfy $y^3 = x^2$ and $(y-x)^2 = 4y^2$. What is $x+y$?

(A) 12 (B) 18 (C) 24 (D) 36 (E) 42

Answer: D

Solution 1

Because $y^3 = x^2$, set $x = a^3$, $y = a^2$ ($a \neq 0$). Put them in $(y-x)^2 = 4y^2$ we get $(a^2(a-1))^2 = 4a^4$ which implies $a^2 - 2a + 1 = 4$. Solve the equation to get $a = 3$ or -1 . Since x and y are positive, $a = 3$ and $x+y = 3^3 + 3^2 = \boxed{\text{(D)}36}$.

Solution 2

Let's take the second equation and square root both sides. This will obtain $y-x = \pm 2y$. Solving the case where $y-x = +2y$, we'd find that $x = -y$. This is known to be false because both x and y have to be positive, and $x = -y$ implies that at least one of the variables is not positive. So we instead solve the case where $y-x = -2y$. This means that $x = 3y$. Inputting this value into the first equation, we find $y^3 = (3y)^2$, from which $y = 9$.

This means that $x = 3y = 3(9) = 27$. Therefore, $x + y = 9 + 27 = \boxed{36}$

Solution 3: Quadratic formula

First, expand:

$$(y - x)^2 = 4y^2 \quad y^2 - 2xy + x^2 = 4y^2 \quad y^2 - 2yx + x^2 = 4y^2 \quad x^2 - 2xy - 3y^2 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

consider $a=1, b=-2y, c=-3y^2$

$$x = \frac{2y \pm \sqrt{(-2y)^2 - 4(1)(-3y^2)}}{2a} \quad x = \frac{2y \pm \sqrt{16y^2}}{2} \quad \frac{2y+4y}{2} \text{ or } \frac{2y-4y}{2} \quad x = y + 2y \text{ and } x = y - 2y$$

$$x = 3y \text{ and } x = -y$$

we can see both x and y will be positive in $x = 3y$

now do same as solution 2

$$y^3 = (3y)^2$$

$$y^3 = 9y^2$$

$$y = 9$$

This means that $x = 3y = 3(9) = 27$. Therefore, $x + y = 9 + 27 = \boxed{36}$

alternate solving similar to solution 5(soapedya): you can also factor at $x^2 - 2xy - 3y^2 = 0$ into $(x - 3y)(x + y) = 0$

Since $x + y = 0$ would result in one of them being negative (or just look at the answer choices; it is not 0), $x - 3y = 0$ so $x = 3y$. plugging back into $y^3 = x^2$ gives $y = 9$ and then $x = 27$, so $x + y$ is $\boxed{36}$.

Solution 4: Substitution

Since $a^2 = |a|^2$, we can rewrite the second equation as $(x - y)^2 = 4y^2$

Let $u = x + y$. The second equation becomes

$$(u - 2y)^2 = 4y^2$$

$$u^2 - 4uy = 0$$

$$u = 4y$$

$$x + y = 4y$$

$$x = 3y.$$

Substituting this into the first equation, we have

$$y^3 = (3y)^2,$$

so $x = 9$.

Hence $x = 27$ and $x + y = \boxed{(D)36}$.

-Benedict T (countmath1)

Solution 5: Difference of Squares

We will use the difference of squares in the second equation.

$$(y - x)^2 = 4y^2$$

$$(y - x)^2 - (2y)^2 = 0$$

$$(y - x - 2y)(y - x + 2y) = 0$$

$$-(x + y)(3y - x) = 0$$

Since x and y are positive, $x+y$ is non-zero. Thus,

$$3y = x$$

Substituting into the first equation:

$$y^3 = x^2$$

$$y^3 = 9y^2$$

$$y = 9, x = 27 \rightarrow x + y = \boxed{\text{(D)}36}$$

11. (AMC 12 12A 2023) What is the degree measure of the acute angle formed by lines with slopes 2 and $\frac{1}{3}$?

(A) 30 (B) 37.5 (C) 45 (D) 52.5 (E) 60

Answer: C

Solution 1

Remind that slope $= \frac{\Delta y}{\Delta x} = \tan \theta$ where θ is the angle between the slope and x -axis.
 $k_1 = 2 = \tan \alpha$, $k_2 = \frac{1}{3} = \tan \beta$. The angle formed by the two lines is $\alpha - \beta$.
 $\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta} = \frac{2 - 1/3}{1 + 2 \cdot 1/3} = 1$. Therefore, $\alpha - \beta = \boxed{\text{(C)}45^\circ}$.

Solution 2

We can take any two lines of this form, since the angle between them will always be the same. Let's take $y = 2x$ for the line with slope of 2 and $y = \frac{1}{3}x$ for the line with slope of $1/3$. Let's take 3 lattice points and create a triangle. Let's use $(0, 0)$, $(1, 2)$, and $(3, 1)$. The distance between the origin and $(1, 2)$ is $\sqrt{5}$. The distance between the origin and $(3, 1)$ is $\sqrt{10}$. The distance between $(1, 2)$ and $(3, 1)$ is $\sqrt{5}$. We notice that we have a triangle with 3 side lengths: $\sqrt{5}$, $\sqrt{5}$, and $\sqrt{10}$. This forms a 45-45-90 triangle, meaning that the angle is $\boxed{45^\circ}$.

Solution 2.b

WLOG, allow the line with slope 2 to pass through point $O = (0, 0)$ and point $A = (1, 2)$, and the line with slope $\frac{1}{3}$ to pass through point O and point $B = (3, 1)$. Notice that the slope of line $\overline{AB}$ is $\frac{1-2}{3-1} = -\frac{1}{2}$ and is hence perpendicular to line $\overline{OA}$. Drawing $\triangle OAB$, we have $\overline{OA} = \overline{AB} = \sqrt{(1)^2 + (2)^2} = \sqrt{5}$, so $\triangle OAB$ is an isosceles right triangle. Therefore, the desired angle is $\angle AOB = \boxed{\text{(C)} 45^\circ}$.

Solution 3 (Law of Cosines)

Follow Solution 2 up until the lattice points section. Let's use $(0, 0)$, $(2, 4)$, and $(9, 3)$. The distance between the origin and $(2, 4)$ is $\sqrt{20}$. The distance between the origin and $(9, 3)$ is $\sqrt{90}$. The distance between $(2, 4)$ and $(9, 3)$ is $\sqrt{50}$. Using the Law of Cosines, we see the $50 = 90 + 20 - 2 \times \sqrt{20} \times \sqrt{90} \times \cos(\theta)$, where θ is the angle we are looking for.

Simplifying, we get $-60 = -2 \times (\sqrt{20}) \times (\sqrt{90}) \times \cos(\theta)$.

$$30 = \sqrt{1800} \times \cos(\theta).$$

$$30 = 30\sqrt{2} \times \cos(\theta).$$

$$\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2} = \cos(\theta).$$

$$\text{Thus, } \theta = \boxed{(C)45^\circ}$$

Solution 4 (Vector Bash)

We can set up vectors $\vec{a} = \langle 1, 2 \rangle$ and $\vec{b} = \langle 3, 1 \rangle$ to represent the two lines. We know that $\frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|} = \cos \theta$. Plugging the vectors in gives us $\cos \theta = \frac{5}{5\sqrt{2}} = \frac{1}{\sqrt{2}}$. From this we get that $\theta = \boxed{(C)45^\circ}$.

Solution 5 (Complex Numbers)

Let $Z_1 = 3 + i$ and $Z_2 = 1 + 2i$

$$Z_2 = Z_1 \cdot re^{i\theta}$$

$$1 + 2i = (3 + i) \cdot re^{i\theta}$$

$$1 + 2i = (3 + i) \cdot r(\cos \theta + i \sin \theta)$$

$$1 + 2i = 3r \cos \theta - r \sin \theta + 3ri \sin \theta + ri \cos \theta$$

From this we have:

$$1 = 3r \cos \theta - r \sin \theta \tag{22}$$

$$2 = r \cos \theta + 3r \sin \theta \tag{23}$$

To solve this we must compute r

$$\begin{aligned} r &= \frac{|Z_2|}{|Z_1|} \\ &= \frac{\sqrt{5}}{\sqrt{10}} \\ &= \frac{\sqrt{2}}{2} \end{aligned}$$

Using elimination we have: $3 \cdot (2) - (1)$

$$\begin{aligned} 5 &= 10r \sin \theta \\ \frac{1}{2r} &= \sin \theta \\ \frac{1}{2 \cdot \frac{\sqrt{2}}{2}} &= \sin \theta \\ \frac{\sqrt{2}}{2} &= \sin \theta \\ \theta &= \boxed{(C)45^\circ} \end{aligned}$$

Solution 5.b (Complex Numbers - Simpler)

Let $Z_1 = 3 + i$ and $Z_2 = 1 + 2i$

$$\begin{aligned} Z_2 &= Z_1 \cdot re^{i\theta} \\ 1 + 2i &= (3 + i) \cdot re^{i\theta} \\ \frac{1 + 2i}{3 + i} &= re^{i\theta} \\ \frac{(1 + 2i)(3 - i)}{(3 + i)(3 - i)} &= re^{i\theta} \\ \frac{3 + 2 + 6i - i}{(3 + i)(3 - i)} &= re^{i\theta} \\ \tan(\theta) &= 5/5 \\ \theta &= \boxed{(C)45^\circ} \end{aligned}$$

Solution 6

The lines $y = 2x$, $y = \frac{1}{3}x$, and $x = 3$ form a large right triangle and a small right triangle. Call the angle that is formed by the x-axis and the line $y = 2x$ α , and call the angle that is formed by the x-axis and the line $y = \frac{1}{3}x$ β . We try to find $\sin(\alpha - \beta)$ first, and then try to see if any of the answer choices match up.

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \sin \beta \cos \alpha.$$

Using soh-cah-toa, we find that $\sin \alpha = \frac{2}{\sqrt{5}}$, $\sin \beta = \frac{1}{\sqrt{10}}$, $\cos \alpha = \frac{1}{\sqrt{5}}$, and $\cos \beta = \frac{3}{\sqrt{10}}$.

Plugging it all in, we find that $\sin(\alpha - \beta) = \frac{5}{\sqrt{50}}$, which is equivalent to $\frac{\sqrt{2}}{2}$. Since $\sin(\alpha - \beta) = \frac{\sqrt{2}}{2}$, we get that $\alpha - \beta = 45^\circ$. Therefore, the answer is $\boxed{(C)45^\circ}$.

Solution 7 (Cheese)

AMC 12 does not allow graph paper or protractor. <https://amc-reg.maa.org/manual/AMC1012B.pdf>

Using a makeshift ruler, draw an accurate to-scale diagram. You can do this by simply drawing the two lines such that they intersect at the origin. Then, measure the angle by eye or by folding paper to observe that they form a 45 degree angle. The answer is $\boxed{45^\circ}$.

12. (AMC 12 12A 2023) What is the value of

$$2^3 - 1^3 + 4^3 - 3^3 + 6^3 - 5^3 + \dots + 18^3 - 17^3?$$

- (A) 2023 (B) 2679 (C) 2941 (D) 3159 (E) 3235

Answer: D

Solution 1

To solve this problem, we will be using difference of cube, sum of squares and sum of arithmetic sequence formulas.

$$\begin{aligned} &2^3 - 1^3 + 4^3 - 3^3 + 6^3 - 5^3 + \dots + 18^3 - 17^3 \\ &= (2-1)(2^2+1\cdot2+1^2) + (4-3)(4^2+4\cdot3+3^2) + (6-5)(6^2+6\cdot5+5^2) + \dots + (18-17)(18^2+18\cdot17+17^2) \\ &= (2^2+1\cdot2+1^2) + (4^2+4\cdot3+3^2) + (6^2+6\cdot5+5^2) + \dots + (18^2+18\cdot17+17^2) \end{aligned}$$

$$= 1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2 \dots + 17^2 + 18^2 + 1 \cdot 2 + 3 \cdot 4 + 5 \cdot 6 + \dots + 17 \cdot 18$$

$$= \frac{18(18+1)(36+1)}{6} + 1 \cdot 2 + 3 \cdot 4 + 5 \cdot 6 + \dots + 17 \cdot 18$$

we could rewrite the second part as $\sum_{n=1}^9 (2n-1)(2n) \quad (2n-1)(2n) = 4n^2 - 2n$

$$\sum_{n=1}^9 4n^2 = 4\left(\frac{9(9+1)(18+1)}{6}\right) \quad \sum_{n=1}^9 -2n = -2\left(\frac{9(9+1)}{2}\right)$$

Hence,

$$1 \cdot 2 + 3 \cdot 4 + 5 \cdot 6 + \dots + 17 \cdot 18 = 4\left(\frac{9(9+1)(18+1)}{6}\right) - 2\left(\frac{9(9+1)}{2}\right)$$

Adding everything up:

$$2^3 - 1^3 + 4^3 - 3^3 + 6^3 - 5^3 + \dots + 18^3 - 17^3 = \frac{18(18+1)(36+1)}{6} + 4\left(\frac{9(9+1)(18+1)}{6}\right) - 2\left(\frac{9(9+1)}{2}\right)$$

$$= 3(19)(37) + 6(10)(19) - 9(10) = 2109 + 1140 - 90 = \boxed{\text{(D) } 3159}$$

Solution 2

Think about $2^3 + 4^3 + 6^3 + \dots + 18^3$. Once we factor out $2^3 = 8$, we get $1^3 + 2^3 + \dots + 9^3$, which can be found using the sum of cubes formula, $\left(\frac{n(n+1)}{2}\right)^2$. Now think about $1^3 + 3^3 + \dots + 17^3$. This is just the previous sum subtracted from the total sum of 18 cubes. So now we have the two things we need to add. The sum of all the even cubes is $8 \cdot \left(\frac{90}{2}\right)^2 \rightarrow 8 \cdot 2025 = 16200$. The sum of all cubes from 1^3 to 18^3 is $\left(\frac{18 \cdot 19}{2}\right)^2 = 29241$. The sum of the odd cubes is then $29241 - 16200 = 13041$. Thus we get $16200 - 13041 = \boxed{\text{(D) } 3159}$ amcrunner

Solution 2 (a bit faster)

Using the same sum of cubes formula, we can rewrite as $2(2^3 + 4^3 + \dots + 18^3) - (1^3 + 2^3 + \dots + 18^3) = 2(2^3)(1^3 + \dots + 9^3) - (1^3 + \dots + 18^3)$

Solution 3

For any real numbers x and y , $x^3 - y^3 = (x - y)(x^2 + xy + y^2) = (x - y)((x - y)^2 + 3xy) = (x - y)^3 + 3xy(x - y)$.

When $x = y + 1$, with the above formula, we will get $x^3 - y^3 = 1^3 + 3xy = 1 + 3xy$.

Therefore,

$$2^3 - 1^3 + 4^3 - 3^3 + \dots + 18^3 - 17^3 = (1 + 3 \cdot 1 \cdot 2) + (1 + 3 \cdot 3 \cdot 4) + \dots + (1 + 3 \cdot 17 \cdot 18)$$

$$= 9 + 3 \cdot (1 \cdot 2 + 3 \cdot 4 + \dots + 17 \cdot 18) = 9 + 3 \cdot (2 + 12 + 30 + 56 + 90 + 132 + 182 + 240 + 306)$$

$$= 9 + 3 \cdot 1050 = \boxed{\text{(D) } 3159}$$

Alternatively, to avoid the long sum,

$$(1 \cdot 2 + 3 \cdot 4 + \dots + 17 \cdot 18)$$

$$= (2^2)(9(10)(19)/6) - (2)(9(10)/2)$$

$$= (2)(9)(10)(2(19/6) - 1/2)$$

$$= 180(35/6) = 35(30) = 1050$$

Solution 4

We rewrite the sum as

$$\sum_{k=1}^9 (2k)^3 - (2k-1)^3$$

$$= \sum_{k=1}^9 12k^2 - 6k + 1$$

$$= 12 \sum_{k=1}^9 k^2 - 6 \sum_{k=1}^9 k + 9$$

$$\begin{aligned}
 &= 12 \cdot \frac{9 \cdot 10 \cdot 19}{6} - 6 \cdot \frac{9 \cdot 10}{2} + 9 \\
 &= 3420 - 270 + 9 = \boxed{\text{(D) } 3159}
 \end{aligned}$$

-Benedict T (countmath1)

Solution 4 (answer choices)

We see $= 12 \cdot \frac{9 \cdot 10 \cdot 19}{6} - 6 \cdot \frac{9 \cdot 10}{2} + 9 = 9 \cdot (12 \cdot \frac{10 \cdot 19}{6} - 6 \cdot \frac{10}{2} + 1)$ which is clearly a multiple of 9. The only answer choice which is a multiple of 9 is (D) 3159 Ilaggo2432

Solution 5 (Bash)

$$2^3 - 1^3 + 4^3 - 3^3 + 6^3 - 5^3 + \dots + 18^3 - 17^3 = 8 - 1 + 64 - 27 + 216 - 125 + 512 - 343 + 1000 - 729 + 1728 - 1331 + 2744 - 2197 + 4096 - 3375 + 5832 - 4913 = \boxed{\text{(D) } 3159}$$

Solution 6

Reduce all terms mod 9. This yields:

$$2^3 - 1^3 + 4^3 - 3^3 + 6^3 - 5^3 + \dots + 18^3 - 17^3 \equiv -1 - 1 + 1 - 0 + 0 - -1 + -1 - 1 + 1 - 0 + 0 - -1 + -1 - 1 + 1 - 0 + 0 - -1 \pmod{9} \equiv 0 \pmod{9}$$

The only answer choice which is also $\equiv 0 \pmod{9}$ is (D) 3159

Solution 7

$$x^3 - y^3 = (x - y)(x^2 + xy + y^2)$$

Notice xy is an inverse variation upon x, y so all values $1(2), 3(4), 5(6), \dots, 17(18)$ form a quadratic where $x = 1, 2, 3, \dots$ gives outputs $y = 2, 12, 30, \dots$. Through principle of finite differences for the quadratic of the form $ax^2 + bx + c$, we have $a = 2F_2$ where F_2 is the second finite difference. It follows $F_2 = 8$ so $a = 4$. Solving, we have

$$4 + b + c = 2$$

$$16 + 2b + c = 12$$

$$b = -2, c = 0$$

.

So the quadratic $4x^2 - 2x$ represents the requested values of the inverse variation.

Otherwise, we wish to find

$$\begin{aligned}
 &\sum_{k=0}^{18} k^2 + \sum_{k=0}^9 4k^2 - 2k \\
 &= \frac{18 \cdot 19 \cdot 37}{6} + 4 \sum_{k=0}^9 k^2 + (-2 \sum_{k=0}^9 k) \\
 &= 2109 + 4 \cdot 285 - 90 \\
 &= \boxed{3159}
 \end{aligned}$$

.

13. (AMC 12 12A 2023) In a table tennis tournament every participant played every other participant exactly once. Although there were twice as many right-handed players as left-handed players, the number of games won by left-handed players was 40% more than the number of games won by right-handed players. (There were no ties and no ambidextrous players.) What is the total number of games played?

(A) 15 (B) 36 (C) 45 (D) 48 (E) 66

Answer: B

Solution 1

We know that the total amount of games must be the sum of games won by left and right handed players. Then, we can write $g = l + r$, and since $l = 1.4r$, $g = 2.4r$. Given that r and g are both integers, $g/2.4$ also must be an integer. From here we can see that g must be divisible by 12, leaving only answers B and D. Now we know the formula for how many games are played in this tournament is $n(n-1)/2$, the sum of the first $n-1$ numbers. Now, setting 36 and 48 equal to the equation will show that two consecutive numbers must have a product of 72 or 96. Clearly, $72 = 8 * 9$, so the answer is **(B) 36**.

Solution 2

First, we know that every player played every other player, so there's a total of $\binom{n}{2}$ games since each pair of players forms a bijection to a game. Therefore, that rules out D. Also, if the right-handed players won a total of x games, the left-handed players must have won a total of $\frac{7}{5}x$ games, meaning that the total number of games played was $\frac{12}{5}x$. Thus, the total number of games must be divisible by 12. Therefore leaving only answer choices B and D. Since answer choice D doesn't satisfy the first condition, the only answer that satisfies both conditions is **(B) 36**.

A simpler way is to know that the result must be a number in the form $\binom{n}{2}$ games. This eliminates D. Then write an expression that equals the answer, $1.4x + x = n$ games total. Only 48 and 36 satisfy so 36 is the answer which is **(B) 36**.

Solution 3

Let r be the amount of games the right-handed won. Since the left-handed won $1.4r$ games, the total number of games played can be expressed as $(2.4)r$, or $12/5r$, meaning that the answer is divisible by 12. This brings us down to two answer choices, B and D. We note that the answer is some number x choose 2. This means the answer is in the form $x(x-1)/2$. Since answer choice D gives $48 = x(x-1)/2$, and $96 = x(x-1)$ has no integer solutions, we know that **(B) 36** is the only possible choice.

Solution 4 (Rigid solving, no cheese)

Here is the rigid way to prove that 36 is the only answer. Let the number of left-handed players be n , so the number of right-handed players is $2n$. The number of games won by the left-handed players comes in two ways:

- The games played by two left-left pairs, which is $\frac{n(n-1)}{2}$, and
- The games played by left-right pairs, which we'll call x .

Note that $x \leq 2n^2$, which is the total number of games played by left-right pairs. Using

the same logic for right-right pairs and right-left pairs, we have that

$$\frac{\frac{n(n-1)}{2} + x}{\frac{2n(2n-1)}{2} + 2n^2 - x} = 1.4,$$

which gives

$$x = \frac{17n^2}{8} - \frac{3n}{8}.$$

We know that $x \leq 2n^2$, applying that becomes

$$\begin{aligned}\frac{17n^2}{8} - \frac{3n}{8} &\leq 2n^2 \\ 17n^2 - 3n &\leq 16n^2 \\ n^2 - 3n &\leq 0 \\ n &\leq 3.\end{aligned}$$

(We can safely divide by n because it must be positive). So the total number of players p can only be 3, 6, and 9.

Since the total number of games $p(p-1)/2$ is $1.4 + 1 = \frac{12}{5}$ times a non-negative integer number of games won by righties, $p(p-1)$ must be a multiple of $12(2) = 24$. Among $\{3, 6, 9\}$, only $p = 9$ satisfies this condition, so the total number of games is $\frac{(9)(8)}{2} = \boxed{\text{(B) } 36}$.

Solution 5 (Cheese)

If there are n players, the total number of games played must be $\binom{n}{2}$, so it has to be a triangular number. The ratio of games won by left-handed to right-handed players is $7 : 5$, so the number of games played must also be divisible by 12. Finally, we notice that only $\boxed{\text{(B) } 36}$ satisfies both of these conditions.

Solution 5.1 (Slightly Slower)

We know that the number of games played must be a triangular number, which eliminates **(D)**48, then we see that there are twice as many right-handed players as left-handed ones. Which means that the total number of players must be divisible by 3, and since each player can't play themselves, we can add 1 to the largest number in the sequence, since $1 + 2 + 3 + \dots + 8$ is the only one which the amount of players is divisible by 3 (9 players), $\boxed{\text{(B) } 36}$ is the answer

Solution 6 (Cheese 2)

Call the number of games x , the number of games won by right handed player R , and by left handed players L . $L+R=x$ and $L=1.4 R$. Therefore, $x/2.4$ must be an integer, which leaves only 36 and 48. Then, note x must be half the product of two consecutive numbers (the number of possible combinations of two people) and we can eliminate 48.

How is this cheese... it is literally the first solution but scaled down

-Shreyansh

EXACTLY... the first solution is also cheese.

-AndrewZhong2012

cheddar

-PerseverePlayer

Solution 7

Let the number of people be n . This means that there will be $\frac{n(n-1)}{2}$ games played. We can now eliminate (D) because it's not a triangular number. The problem states that there are twice as many right handed people as left handed people with no ambidextrous people. Based on this, we can say that the number of people (or n) has to be divisible by 3. Because the number of games won by left-handed people is 40% more than the number of games won by right-handed people, we can set the ratio between the number of games won by left-handed to right-handed as $\frac{7}{5}$. The sum of the numerator and the denominator is 12. This means that the number of games must be divisible by 12. Thus, the only answer choice that fits the conditions above is answer **(B) 36**.

Solution 8(Pairing) - easiest noncheese

Case 1: in-group pairing

In every pair where both players use the same hand, the win rate is always 50%. This is because the total number of such pairs is even — any factorial other than the trivial case 1! is even, and 1 is insufficient to form a pair. Thus, no player is left out, and in each pair exactly one player wins and one loses, giving a win-loss ratio of 1 : 1. In other words, this case completely cancels out due to its balanced win-loss ratio.

Case 2: cross-group pairing

For convenience, let left-handed players be denoted by L and right-handed players by R . The problem gives the ratio $R : L = 1 : 2$. This means each L plays $2L$ times as many games as an R . From this, the total number of cross-group games is $2L \times L = 2L^2$.

The problem also states that the ratio of L wins to R wins is 7 : 5, so the total number of wins in this case must be divisible by $5 + 7 = 12$. We therefore require $2L^2$ to be divisible by 12. Simplifying gives $L^2 \mid 6$. The smallest perfect square divisible by 6 is **(B) 36**.

Solution 9(Subgroups)

Let L and R be the number of left-handed and right-handed players, respectively. The total number of games played in the tournament can be broken down into three distinct parts based on the matchups: games between two left-handed players, games between two right-handed players, and games between a left-handed and a right-handed player. This gives the total game count formula:

$$\text{Total Games} = LR + \frac{L(L-1)}{2} + \frac{R(R-1)}{2}$$

Every game played results in exactly one win. In the matches where left-handed players play each other, a left-handed player must win. This guarantees $\frac{L(L-1)}{2}$ wins for the left-handed group. Similarly, matches where right-handed players play each other guarantee $\frac{R(R-1)}{2}$ wins for the right-handed group.

For the LR games played between a left-handed player and a right-handed player, let W be the number of games won by the left-handed players. The remaining $(LR - W)$ games in this section must be won by the right-handed players. We can now write expressions for the total wins of each group:

$$\begin{aligned} \text{Total Left Wins } (W_L) &= \frac{L(L-1)}{2} + W \\ \text{Total Right Wins } (W_R) &= \frac{R(R-1)}{2} + LR - W \end{aligned}$$

The problem states that there are twice as many right-handed players as left-handed players, so we substitute $R = 2L$ into our equations:

$$W_L = \frac{L(L-1)}{2} + W$$

$$W_R = \frac{2L(2L-1)}{2} + L(2L) - W = L(2L-1) + 2L^2 - W = 4L^2 - L - W$$

We are given that left-handed players won 40% more games than right-handed players, which means $W_L = 1.4 \times W_R$. Setting up the equation:

$$\frac{L(L-1)}{2} + W = 1.4(4L^2 - L - W)$$

To clear the fraction and decimal, we multiply the entire equation by 10:

$$5L(L-1) + 10W = 14(4L^2 - L - W)$$

$$5L^2 - 5L + 10W = 56L^2 - 14L - 14W$$

Grouping the W terms on the left side and the L terms on the right side yields:

$$24W = 51L^2 - 9L$$

Dividing the entire equation by 3 simplifies it to:

$$8W = 17L^2 - 3L \implies W = \frac{17L^2 - 3L}{8}$$

Since W represents a physical number of wins, it must be a non-negative integer. Additionally, W cannot be greater than the total number of inter-group games, meaning $W \leq 2L^2$. Testing small integer values for L :

- If $L=1$: $W = \frac{14}{8}$, which is not an integer.
- If $L=2$: $W = \frac{62}{8}$, which is not an integer.
- If $L=3$: $W = \frac{153-9}{8} = \frac{144}{8} = 18$. This is a valid integer, and it satisfies $18 \leq 2(3)^2 = 18$.

With $L=3$, the number of right-handed players is $R = 6$, giving a total of 9 players. Substituting these into the original total games formula:

$$\text{Total Games} = (3)(6) + \frac{3(2)}{2} + \frac{6(5)}{2} = 18 + 3 + 15 = \boxed{\text{(B)} 36}$$

14. (AMC 12 12A 2023) How many complex numbers satisfy the equation $z^5 = \bar{z}$, where $\bar{z}$ is the conjugate of the complex number z ?
- (A) 2 (B) 3 (C) 5 (D) 6 (E) 7

Answer: E

Solution 1

When $z^5 = \bar{z}$, there are two conditions: either $z = 0$ or $z \neq 0$. When $z \neq 0$, since $|z^5| = |\bar{z}|$, $|z| = 1$. $z^5 \cdot z = z^6 = \bar{z} \cdot z = |z|^2 = 1$. Consider the $r(\cos \theta + i \sin \theta)$ form, when

$z^6 = 1$, there are 6 different solutions for z . Therefore, the number of complex numbers satisfying $z^5 = \bar{z}$ is (E)7.

Solution 2

Let $z = re^{i\theta}$. We now have $\bar{z} = re^{-i\theta}$, and want to solve

$$r^5 e^{5i\theta} = re^{-i\theta}.$$

From this, we have $r = 0$ as a solution, which gives $z = 0$. If $r \neq 0$, then we divide by it, yielding

$$r^4 e^{5i\theta} = e^{-i\theta}.$$

Dividing both sides by $e^{-i\theta}$ yields $r^4 e^{6i\theta} = 1$. Taking the magnitude of both sides tells us that $r^4 = 1$, so $r^2 = \pm 1$. However, if $r^2 = -1$, then $r = \pm i$, but r must be real. Therefore, $r^2 = 1$.

Multiplying both sides by r^2 ,

$$r^6 \cdot e^{6i\theta} = z^6 = 1.$$

Each of the 6th roots of unity is a solution to this, so there are $6 + 1 =$ (E) 7 solutions.

-Benedict T (countmath 1)

Solution 3 (Rectangular Form, similar to Solution 1)

Let $z = a + bi$.

Then, our equation becomes: $(a + bi)^5 = a - bi$

Note that since every single term in the expansion contains either an a or b , simply setting $a = 0$ and $b = 0$ yields a solution.

Now, considering the other case that either a or b does not equal 0:

Multiplying both sides by $a + bi$ (or z), we get: $(a + bi)^6 = a^2 + b^2$ (since $i^2 = -1$).

Substituting z back into the left hand side, we get: $z^6 = a^2 + b^2$

Note that this will have 6 distinct, non-zero solutions since in this case, we consider that either a or b is not 0, and these are simply the sixth roots of a positive real number.

Adding up the solutions, we get $1 + 6 =$ (E)7

-SwordOfJustice

Solution 4

Using the fact that $z\bar{z} = |z|^2$, we rewrite our equation as $z^6 = |z|^2$. Now, let r represent $|z|$. We know that $r^6 = r^2$; hence, we have $r^6 - r^2 = r^2(r^4 - 1) = 0$.

From here, we have two cases: $r = 0$, or $r = 1$. In the case that $r = 0$, we have $z^6 = 0$ hence $z = 0$. This gives one solution. Alternatively, if $r = 1$, then we have $z^6 = 1$, giving 6 solutions for each root of unity.

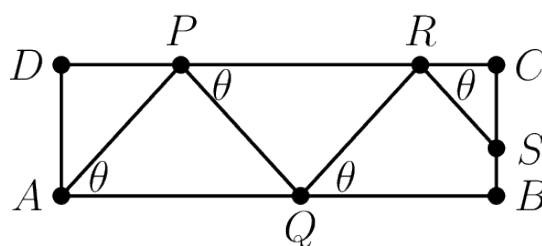
Therefore, the answer is $6 + 1 =$ (E)7.

- xHypotenuse

Solution 5 (with motivation)

Immediately, we know that if z is a root of unity, then $\bar{z} = \frac{1}{z}$. This only holds if z is a root of unity of has $|z| = 1$. In addition, if z is a root of unity, then z^n is also a root of unity. So, if $z^5 = \frac{1}{z}$, we require $z^6 = 1$, and that implies all sixth roots of unity. However, 0 is also a solution, hence we have $\boxed{7}$ values. We can prove no other z can satisfy this requirement as a conjugational rotation on the complex plane is a 180 degree rotation of a point, and no roots of unity besides the sixth roots can map, after a 180 degree rotation, modular 5 to the same point as the conjugate.

15. (AMC 12 12A 2023) Usain is walking for exercise by zigzagging across a 100-meter by 30-meter rectangular field, beginning at point A and ending on the segment $\overline{BC}$. He wants to increase the distance walked by zigzagging as shown in the figure below ($APQRS$). What angle θ $\angle PAB = \angle QPC = \angle RQB = \dots$ will produce a length that is 120 meters? (This figure is not drawn to scale. Do not assume that the zigzag path has exactly four segments as shown; there could be more or fewer.)

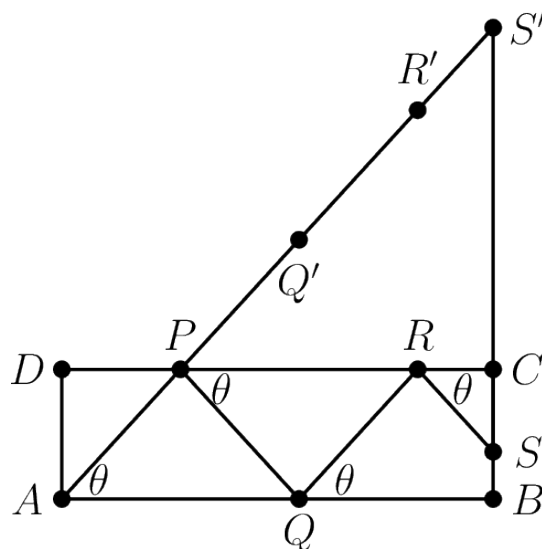


- (A) $\arccos \frac{5}{6}$ (B) $\arccos \frac{4}{5}$ (C) $\arccos \frac{3}{10}$ (D) $\arcsin \frac{4}{5}$ (E) $\arcsin \frac{5}{6}$

Answer: A

Solution 1

By "unfolding" $APQRS$ into a straight line, we get a right angled triangle ABS' .



It follows that

$$\cos(\theta) = \frac{100}{120}$$

$$\theta = \boxed{\text{(A) } \arccos\left(\frac{5}{6}\right)}.$$

lptoggled

Solution 2

Drop an altitude from P to AB and let its base be x . Note that if we repeat this for Q and R , all four right triangles (including $\triangle RSC$) will have the same trig ratios. By proportion, the hypotenuse AP is $\frac{x}{100}(120) = \frac{6}{5}x$, so $\cos \theta = \frac{x}{(\frac{6}{5}x)} = \frac{5}{6} \Rightarrow \theta = \boxed{\text{(A)} \arccos \frac{5}{6}}$.

Solution 3 (Trig Bash)

We can let x be the length of one of the full segments of the zigzag. We can then notice that $\sin \theta = \frac{30}{x}$. By Pythagorean Theorem, we see that $DP = \sqrt{x^2 - 900}$. This implies that:

$$RC = 100 - 3\sqrt{x^2 - 900}.$$

We also realize that $RS = 120 - 3x$, so this means that:

$$\cos \theta = \frac{100 - 3\sqrt{x^2 - 900}}{120 - 3x}.$$

We can then substitute $x = \frac{30}{\sin \theta}$, so this gives:

$$\begin{aligned} \cos \theta &= \frac{100 - 3\sqrt{x^2 - 900}}{120 - 3x} \\ &= \frac{100 - 3\sqrt{\frac{900}{\sin^2 \theta} - 900}}{120 - \frac{90}{\sin \theta}} \\ &= \frac{100 - 90\sqrt{\csc^2 \theta - 1}}{120 - \frac{90}{\sin \theta}} \\ &= \frac{100 - \frac{90}{\tan \theta}}{120 - \frac{90}{\sin \theta}} \\ &= \frac{100 \sin \theta - 90 \cos \theta}{120 \sin \theta - 90} \\ &= \frac{10 \sin \theta - 9 \cos \theta}{12 \sin \theta - 9} \end{aligned}$$

Now we have:

$$\cos \theta = \frac{10 \sin \theta - 9 \cos \theta}{12 \sin \theta - 9},$$

meaning that:

$$12 \sin \theta \cos \theta - 9 \cos \theta = 10 \sin \theta - 9 \cos \theta \implies \cos \theta = \frac{10}{12} = \frac{5}{6}.$$

This means that $\theta = \arccos\left(\frac{5}{6}\right)$, giving us $\boxed{\text{A}}$

why would you EVER do this to yourself

- PerseverePlayer

Solution 4 (No Trig)

Let x be the length of DP . Apply the Pythagoras theorem on $\triangle ADP$ to get $AP = \sqrt{900 + x^2}$, which is also the length of every zigzag segment.

There are $\frac{100}{x}$ such segments. Thus the total length formed by the zigzags is

$$\frac{100}{x} \times \sqrt{900 + x^2} = 120$$

$$\sqrt{900 + x^2} = \frac{6}{5}x$$

$$900 + x^2 = \frac{36}{25}x^2$$

$$x = \frac{150}{\sqrt{11}} = DP$$

$$AP = \sqrt{900 + x^2} = \frac{180}{\sqrt{11}}$$

$$\cos \theta = \frac{DP}{AP} = \frac{5}{6}$$

$$\theta = \boxed{\text{(A) } \arccos\left(\frac{5}{6}\right)}$$

(note that $\frac{100}{x}$ is not an integer, but it doesn't matter because of similar triangles. The length of the incomplete segment is always proportionate to the length of the incomplete base)

Solution 5 (Intuitive and Quick)

Imagine that Usain walks at a constant speed. The horizontal component of Usain's velocity does not change. (Imagine a beam of light reflecting off of mirrors. A mirror only changes the velocity of light in the direction perpendicular to the mirror.) The horizontal component of Usain's velocity divided by his total velocity must be $\frac{100}{120}$. Therefore

$$\theta = \boxed{\text{(A) } \arccos\left(\frac{5}{6}\right)}$$

. numerophile

Solution 6

Although the diagram is not fully accurate, we can use it to some extent.

It's given that the length of the rectangle is 100 and the total length of the path Usain is taking is 120, so Usain will walk $\frac{120}{100} = \frac{6}{5}$ longer than he would if he were just to walk along the rectangle. Therefore for each point along his path, he will travel $\frac{6}{5}$ farther than he would if he were just to walk along the rectangle.

Drop a perpendicular from point P to point Q on $\overline{AB}$. Let $AQ = x$: it follows that $QP = 30$ and $PA = \frac{6}{5}x$. By the Pythagorean Theorem,

$$x^2 + 900 = \frac{36}{25}x^2,$$

$$\text{so } x = \frac{150}{\sqrt{11}} \text{ and } \frac{6x}{5} = \frac{180}{\sqrt{11}}.$$

$$\text{Now, } \cos \theta = \frac{\frac{150}{\sqrt{11}}}{\frac{180}{\sqrt{11}}} = \frac{5}{6} \text{ and } \theta = \boxed{\text{(A) } \arccos\left(\frac{5}{6}\right)}.$$

-Benedict T (countmath1)

Solution 7 (Very Bashy)

First, we test that we need 3 segments. If there were only two segments, then the length of each one would be $\sqrt{50^2 + 30^2} = 10\sqrt{34}$. Then the length of two segments would be $20\sqrt{34}$. Since $\sqrt{34}$ is a bit less than 6, it is safe to say we need 3 segments. The length of the first two segments would be $\frac{30 \cdot 2}{\sin \theta} = \frac{60}{\sin \theta}$. The width remaining for the third segment would be $100 - \frac{30 \cdot 2}{\tan \theta}$ and thus the length of the third segment would be $\frac{100 - \frac{30 \cdot 2}{\tan \theta}}{\cos \theta}$. Now, we have a very bashy equation that we have to solve.

$$\frac{60}{\sin \theta} + \frac{100 - \frac{60}{\tan \theta}}{\cos \theta} = 120 \quad \frac{3}{\sin \theta} + \frac{5 - \frac{3}{\tan \theta}}{\cos \theta} = 6 \quad \frac{3}{\sin \theta} + \frac{5 - \frac{3}{\frac{\sin \theta}{\cos \theta}}}{\cos \theta} = 6 \quad \frac{3}{\sin \theta} + \frac{5 - \frac{3 \cdot \cos \theta}{\sin \theta}}{\cos \theta} = 6 \quad \frac{3}{\sin \theta} + \frac{5}{\cos \theta} - \frac{3}{\sin \theta} = 6 \quad \frac{5}{\cos \theta} = 6 \quad \cos \theta = \frac{5}{6} \quad \theta = \arccos \frac{5}{6}$$

Thus, the answer is (A) $\arccos \left(\frac{5}{6} \right)$

16. (AMC 12 12A 2023) Consider the set of complex numbers z satisfying $|1 + z + z^2| = 4$. The maximum value of the imaginary part of z can be written in the form $\frac{\sqrt{m}}{n}$, where m and n are relatively prime positive integers. What is $m + n$?

(A) 20 (B) 21 (C) 22 (D) 23 (E) 24

Answer: B

Solution 1

Note that the third roots of unity exclusion 1 are symmetric about $\Re$ axis. Hence, transpose the locus of z onto the real number line. Then, their magnitude is just their distance on the plane. Distance in this case refers to a negative or positive direction based on the vector values of z .

We identify two cases, $1 + z + z^2 = 4$ or $1 + z + z^2 = -4$. From here, evaluate each case separately.

Firstly $z \neq 1$ and even if $z = 1$ then $1 + z + z^2 \neq 4, -4$. So no constraints exist on what z is besides maximality upon imaginary roots.

In case 1, the quadratic $z^2 + z - 3$ has discriminant 13 which is positive, and therefore no imaginary roots exist. The imaginary part $\Im(z) = 0$ here.

In case 2, the quadratic $z^2 + z + 5$ has discriminant -19 which implies two strict conjugational imaginary roots exist. Through the quadratic formula, the roots are $z = \frac{-1 \pm i\sqrt{19}}{2}$ so $\Im(\max(z)) = \frac{\sqrt{19}}{2} \implies \boxed{21}$.

Note Further justification of the transposition occurs when one evaluates the Cassini Oval.

Alternate If one doesn't believe in the transposition, one can solve the problem as follows:

Rewrite the polynomial as $|(z - \omega)(z - \omega^2)| = 4$ where $\omega = e^{i\frac{2}{3}}$. Then, $|z - \omega||z - \omega^2| = 4$. The figure is an oval with foci ω and ω^2 . That implies $\Re(z) = -\frac{1}{2}$. Then, rewrite $z = -\frac{1}{2} + yi$. We have two vertical distances $|z - \omega| = |y - \frac{\sqrt{3}}{2}|$ and $|z - \omega^2| = |y + \frac{\sqrt{3}}{2}|$. So, $(y - \frac{\sqrt{3}}{2})(y + \frac{\sqrt{3}}{2}) = 4 \implies y = \frac{\sqrt{19}}{2} \implies \boxed{21}$.

Solution 2

First, substitute in $z = a + bi$.

$$\begin{aligned} |1 + (a + bi) + (a + bi)^2| &= 4 \\ |(1 + a + a^2 - b^2) + (b + 2ab)i| &= 4 \\ (1 + a + a^2 - b^2)^2 + (b + 2ab)^2 &= 16 \\ (1 + a + a^2 - b^2)^2 + b^2(1 + 4a + 4a^2) &= 16 \end{aligned}$$

Let $p = b^2$ and $q = 1 + a + a^2$

$$\begin{aligned} (q - p)^2 + p(4q - 3) &= 16 \\ p^2 - 2pq + q^2 + 4pq - 3p &= 16 \end{aligned}$$

We are trying to maximize $b = \sqrt{p}$, so we'll turn the equation into a quadratic to solve for p in terms of q .

$$\begin{aligned} p^2 + (2q - 3)p + (q^2 - 16) &= 0 \\ p &= \frac{(-2q + 3) \pm \sqrt{-12q + 73}}{2} \end{aligned}$$

We want to maximize p . Since q is always negatively contributing to p 's value, we want to minimize q .

Due to the trivial inequality: $q = 1 + a + a^2 = (a + \frac{1}{2})^2 + \frac{3}{4} \geq \frac{3}{4}$

If we plug q 's minimum value in, we get that p 's maximum value is

$$p = \frac{(-2(\frac{3}{4}) + 3) + \sqrt{-12(\frac{3}{4}) + 73}}{2} = \frac{\frac{3}{2} + 8}{2} = \frac{19}{4}$$

Then

$$b = \frac{\sqrt{19}}{2}$$

and

$$m + n = \boxed{\text{(B)} 21}$$

- CherryBerry

Solution 3

We are given that $1 + z + z^2 = c$ where c is some complex number with magnitude 4. Rearranging the quadratic to standard form and applying the quadratic formula, we have

$$z = \frac{-1 \pm \sqrt{1^2 - 4(1)(1 - c)}}{2} = \frac{-1 \pm \sqrt{4c - 3}}{2}.$$

The imaginary part of z is maximized when $c = -4$. (Why? See note below.)

Thus $z = -\frac{1}{2} \pm i\frac{\sqrt{19}}{2}$, and so the answer is $\boxed{\text{(B)} 21}$.

Note: You can cheese/fakesolve c by assuming c is real. But there is a good reason for it, within the parameters of this problem:

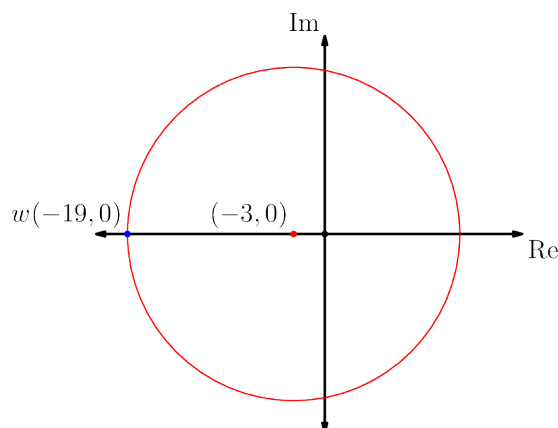
c lies on a circle of radius 4, centered at the origin. Thus, $4c - 3$ is also a circle, centered at -3 . Luckily, the center of this adjusted circle is a negative real number, so the most negative point on that circle ($w = -19$) has the largest magnitude (and so its square

root also has the largest magnitude of any square root), and its positive square root is the most purely imaginary square root ("vertical"). [We ignore the alternative negative square root, since the problem asks for the most positive value.]

Thus, $\text{Im}(\sqrt{w}) = r \sin(\theta)$ has both the largest r and largest $\sin(\theta)$ of any square root of $4c - 3$, and so the largest product $r \sin(\theta)$. (See note2 for more details)

[This argument and overall solution would not work if -3 were instead a positive or non-real number, which would happen if the original problem had $|C + Bz + Az^2| = 4$ with $B^2 - 4AC > 0$.] oinava

Note 2:

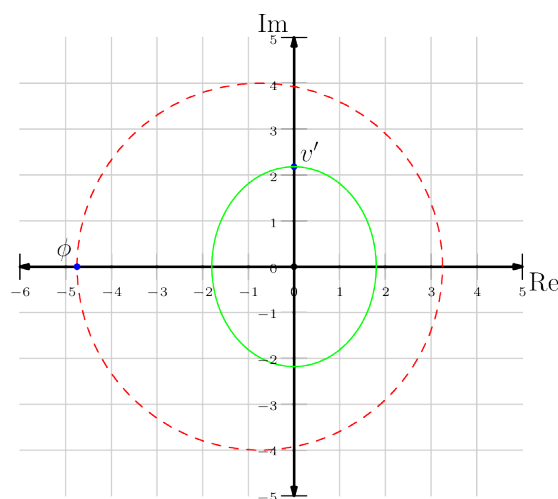


The circle in the graph is all the possible values of $4c - 3$. Let $4c - 3 = re^{i\theta}$.

$4c - 3 = re^{i\theta} \implies \sqrt{4c - 3} = \sqrt{re^{i\theta}} = \sqrt{r}e^{i\frac{\theta}{2}}$. Therefore, $\text{Im}(\sqrt{4c - 3}) = \sqrt{r} \sin(\frac{\theta}{2})$.

At $w(-19, 0)$ as in the graph, $\theta = \pi$ and $r = 19$. So $\sqrt{r}$ and $\sin(\frac{\theta}{2})$ are all at their maximum, which means that $\text{Im}(\sqrt{4c - 3})$ is at its maximum.

Solution 4 (Geometry + Logic)



We can write the given condition as

$$\left| \left(z + \frac{1}{2} \right)^2 + \frac{3}{4} \right| = 4.$$

Letting $u = \left(z + \frac{1}{2}\right)^2$, the equation $\left|u + \frac{3}{4}\right| = 4$ equates to the circle centered at $-\frac{3}{4}$ with radius 4 in the complex plane, call it ω . Thus the locus of $\left(z + \frac{1}{2}\right)^2$ is ω . Let $v = z + \frac{1}{2}$, and since the $+\frac{1}{2}$ does not change z 's imaginary part, we now need to find v with the largest imaginary part such that v^2 lies on ω .

Note that the point on ω with largest magnitude is $19/4$ and has argument π , call it ϕ (The leftmost point on ω). The value v' with positive imaginary part such that $(v')^2 = \phi$ has an argument of $\frac{\pi}{2}$ and a magnitude of $\frac{\sqrt{19}}{2}$.

Since across all values of v the imaginary part is given by $r \sin \theta$ and v' has the largest possible r and the largest possible value of $\sin \theta$, it must have the largest imaginary part. This can non-rigorously be seen by sketching the oval which is the locus of v .

This gives $19 + 2 \Rightarrow \boxed{\text{(B) } 21}$

Solution 5

To start, we factor $1 + z + z^2$ to get:

$$\left| \left(z - \frac{-1 + \sqrt{3}i}{2} \right) \cdot \left(z - \frac{-1 - \sqrt{3}i}{2} \right) \right| = 4.$$

Note that since the magnitude of a product of complex numbers is equal to the product of the magnitudes:

$$\left| \left(z + \frac{1 - \sqrt{3}i}{2} \right) \right| \cdot \left| \left(z + \frac{1 + \sqrt{3}i}{2} \right) \right| = 4.$$

Now, we substitute $z = a + bi$ (Note that we are trying to maximize b):

$$\left| a + \frac{1}{2} + \left(b + \frac{\sqrt{3}}{2} \right) i \right| \cdot \left| a + \frac{1}{2} + \left(b - \frac{\sqrt{3}}{2} \right) i \right| = 4.$$

Since we are trying to maximize b , we want the real parts of the components to be as small as possible, which we can do by setting $a = -\frac{1}{2}$. This leaves us with:

$$\begin{aligned} \left| \left(b + \frac{\sqrt{3}}{2} \right) i \right| \cdot \left| \left(b - \frac{\sqrt{3}}{2} \right) i \right| &= 4 \\ \left(b + \frac{\sqrt{3}}{2} \right) \left(b - \frac{\sqrt{3}}{2} \right) &= 4 \\ b^2 - \frac{3}{4} &= 4 \\ b^2 &= \frac{19}{4} \\ b &= \frac{\sqrt{19}}{2}. \end{aligned}$$

This gives us $19 + 2 = \boxed{\text{(B) } 21}$.

Solution 6

$z^2 + z + 1$ reminds me of $(z - 1)(z^2 + z + 1) = z^3 - 1$. Complex numbers z that satisfies $|1 + z + z^2| = 4$ either comply to $1 + z + z^2 = 4$ or $1 + z + z^2 = -4$. In another words, complex numbers z that satisfy $(z - 1)(z^2 + z + 1) = 4(z - 1)$ or $(z - 1)(z^2 + z + 1) = -4(z - 1)$ corresponds to complex numbers z that satisfies $|1 + z + z^2| = 4$.

Case I: $(z - 1)(z^2 + z + 1) = 4(z - 1)(z - 1)(z^2 + z + 1) = 4(z - 1)$ could be rewritten as $z^3 - 1 = 4(z - 1)$.

$$z^3 - 4z + 3 = 0$$

By substituting 1 for z and using synthetic division, it is evident that $z^3 - 4z + 3 = (z - 1)(z^2 + z - 3) = 0$.

$$z = 1 \text{ or } z = \frac{-1 \pm \sqrt{13}}{2}$$

Case II: $(z - 1)(z^2 + z + 1) = -4(z - 1)(z - 1)(z^2 + z + 1) = -4(z - 1)$ could be rewritten as $z^3 - 1 = -4(z - 1)$.

$$z^3 + 4z - 5 = 0$$

By substituting 1 for z and using synthetic division, it is evident that $z^3 + 4z - 5 = (z - 1)(z^2 + z + 5) = 0$.

$$z = 1 \text{ or } z = \frac{-1 \pm \sqrt{-19}}{2}$$

Determining the Maximum Value of the Imaginary Part of z It is important to note the 1 cannot be a value for z because while multiplying $(z - 1)$ to both sides, 1 became a solution for z . Therefore, four values for possible z are $\frac{-1 \pm \sqrt{13}}{2}$ and $\frac{-1 \pm \sqrt{-19}}{2}$. Because m and n are positive integers, the only possible value for $m + n$ is $19 + 2$, or $\boxed{(B) 21}$. MaPhyCom

17. (AMC 12 12A 2023) Flora the frog starts at 0 on the number line and makes a sequence of jumps to the right. In any one jump, independent of previous jumps, Flora leaps a positive integer distance m with probability $\frac{1}{2^m}$. What is the probability that Flora will eventually land at 10?

(A) $\frac{5}{512}$ (B) $\frac{45}{1024}$ (C) $\frac{127}{1024}$ (D) $\frac{511}{1024}$ (E) $\frac{1}{2}$

Answer: E

Solution 1

Initially, the probability of landing at 10 and landing past 10 (summing the infinite series) are exactly the same. Landing before 10 repeats this initial condition, with a different irrelevant scaling factor. Therefore, the probability must be $\boxed{(E) \frac{1}{2}}$.

Solution 2

Let's denote $f(n)$ as the probability of reaching n from 0. We immediately see that $f(0) = 1$, and $f(1) = \frac{1}{2}$, since there's only one way to get to 1 from 0. Just jump!

Now, let's write an expression for $f(10)$. Suppose we know $f(9), f(8), \dots, f(2)$.

The probability of reaching 10 from some integer $k < 10$ will be $\frac{1}{2^{10-k}}$ (use the formula given in the problem!) The probability of reaching that integer k from 0 is going to be $f(k)$. Then, the probability of going from $0 \rightarrow \underbrace{k \rightarrow 10}_{\text{one jump}}$ will be

$$\mathbb{P}(\text{Reaching 10 from 0 while passing through } k) = f(k) \cdot \frac{1}{2^{10-k}}$$

We want the probability of reaching 10 from anywhere though, so what we can do is sum over all passing points k , i.e.

$$f(10) = \sum_{k=0}^9 \mathbb{P}(\text{Reaching 10 from 0 while passing through } k) = \sum_{k=0}^9 \frac{1}{2^{10-k}} \cdot f(k)$$

Now that we have expressed our problem formally, we can actually start solving it!

Let's calculate $f(9)$ (our expression is actually a general fact, not just limited to 10).

$$f(9) = \sum_{k=0}^8 \frac{1}{2^{9-k}} \cdot f(k)$$

$$\frac{f(9)}{2} = \sum_{k=0}^8 \frac{1}{2^{10-k}} \cdot f(k)$$

Hmm, we see that the first 8 terms of $\frac{f(9)}{2}$ are exactly the first 8 terms of $f(10)$. Let's substitute it in.

$$f(10) = \frac{1}{2} \cdot f(9) + \sum_{k=0}^8 \frac{1}{2^{10-k}} \cdot f(k)$$

$$f(10) = \frac{1}{2} \cdot f(9) + \frac{1}{2} \cdot f(9)$$

$$f(10) = f(9)$$

Isn't that interesting. Turns out, this reasoning can be extended all the way to $f(10) = f(9) = \dots = f(2) = f(1)$.

It breaks at $f(1) \neq f(0)$, since $f(1) = \frac{1}{2}f(0)$. Anyway, with this, we see that the answer is just

Solution 3

In order to find the probability of landing on 10, we must multiply the amount of successful combinations by the probability of those combinations. Notice for any successful combination of steps, the probability must always be $\frac{1}{2^{10}}$. Now, we only need to find the amount of possibilities for steps since we know the probability of each combination occurring is the same. This can be done using sticks and stones $C_0^9 + C_1^9 + C_2^9 + \dots + C_9^9 = 2^9$.

Hence the final answer is $\frac{2^9}{2^{10}}$ or $\boxed{\text{(E)} \frac{1}{2}}$

Note that an alternative way of thinking the number of possible steps is that for each number from 1 to 9, there are 2 possibilities: Flora lands on this number or Flora doesn't land on this number. Thus there are 2^9 total possible combination of steps

Solution 3.1

Like said above, the probability of each successful combination is always $\frac{1}{2^{10}}$. A simple way of thinking about counting each combination is visualizing a series. Let's make a series with 10 terms, where 1 means that Flora touches the lilypad and 0 means that she does not: $a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8, a_9, a_{10}$. In this set, a_i is either 0 or 1. Note that the last lilypad, a_{10} must always be 1 for a successful combination. Now we just need to find the number of different ways to make a_1 through a_9 0 or 1. This is obviously $2^9 = 512$.

$$\frac{512}{1024} = \boxed{\text{(E)} \frac{1}{2}}$$

Solution 4 (engineer's induction)

The probability frog lands on 1 is trivially $\frac{1}{2}$.

The probability frog lands on 2 is $\frac{1}{4} + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}$, from the two cases 0-2 and 0-1-2.

The probability frog lands on 3 is $\frac{1}{8} + 2 \cdot \frac{1}{2} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}$ from the cases 0-3, 0-1-3 and 0-2-3, 0-1-2-3.

The probability frog lands on 4 is $\frac{1}{16} + 2 \cdot \frac{1}{2} \cdot \frac{1}{8} + \frac{1}{4} \cdot \frac{1}{4} + 3 \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}$ from the cases 0-4, 0-1-4 and 0-3-4, 0-2-4, 0-1-2-4 and 0-1-3-4 and 0-2-3-4, 0-1-2-3-4.

It looks like the probability is $\frac{1}{2}$ regardless of the ending number. Therefore, we choose

$$\boxed{(E) \frac{1}{2}}$$

Solution 5

No matter what the probability of getting to the end has a probability of $\frac{1}{1024}$

So the thing is how many ways to jump from the first spot to the last spot.

Given it requires n steps to reach the end, there are $\binom{10-1}{n-1} = \binom{9}{n-1}$ ways to get the end.

$$\sum_{n=1}^{10} \binom{9}{n-1} = 2^9 = 512 \implies \frac{1}{2}$$

Solution 7 (Mathematician's Induction)

A more rigorous proof to solution 4.

Let P_n represents the probability of Flora landing on number n

We hypothesize that $P_n = \frac{1}{2}, \forall n \geq 1$

Base case: $P_1 = \frac{1}{2}$ as shown in solution 4

Induction steps:

$$P_n = \frac{P_{n-1}}{2} + \frac{P_{n-2}}{4} + \dots + \frac{P_{n-m}}{2^m} \dots + \frac{P_0}{2^n}$$

We assume that $P_{n-1} = P_{n-2} = \dots = P_1 = \frac{1}{2}$ and $P_0 = 1$ is true

$$P_n = \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{2^{n-1}} + \frac{1}{2^n} + \frac{1}{2^n}$$

(it's obvious here that this will add up to $\frac{1}{2}$ but to be 100% rigorous we do sum of geometric sequences)

$$P_n = \frac{\frac{1}{4}(1 - (\frac{1}{2})^{n-1})}{1 - \frac{1}{2}} + \frac{1}{2^n}$$

$$P_n = \frac{2^{n-1} - 1}{2^n} + \frac{1}{2^n}$$

$$P_n = \frac{2^{n-1}}{2^n} = \frac{1}{2}$$

Thus, since our base case is true, by induction every $P_n = \frac{1}{2}$ for $n \geq 1$ and $P_{10} = \boxed{(E) \frac{1}{2}}$

Solution 8 (Genfuncs OP)

Let $f(z) = \sum_{k=1}^{\infty} (\frac{z}{2})^k = \frac{\frac{z}{2}}{1 - \frac{z}{2}} = \frac{z}{2-z}$. The coefficient of z^n in $f(z)$ is the probability

of reaching n in exactly one step. Now, write $g(z) = \sum_{k=1}^{\infty} f(z)^k = \frac{f(z)}{1-f(z)}$. Then, the coefficient of z^n in $g(z)$ is the probability of reaching n at some point, since we either do it in one step, two steps, etc. Now, we simplify

$$g(z) = \frac{\frac{z}{2-z}}{1 - \frac{z}{2-z}} = \frac{z}{2-2z} = \frac{1}{2} \cdot \frac{z}{1-z} = \sum_{n=1}^{\infty} \frac{1}{2} z^n.$$

Therefore, the probability of eventually reaching n is $\frac{1}{2}$ for all $n \geq 1$, so our answer is

$$\boxed{(\mathbf{E})\frac{1}{2}}.$$

Solution 9 (Stars and Bars)

We can start off by doing casework for the amount of jumps we take to reach 10:

$$C_1 : \text{Onejump} C_1 :$$

and then after doing a few, we realize that the probability is just equal to:

$$\frac{\text{Number of ways to split 10 into positive integers}}{2^{10}} \quad (25)$$

Now, we can just use stars and bars for each case which we find to be $\binom{9}{0} + \binom{9}{1} + \dots + \binom{9}{8} + \binom{9}{9} = 2^9$, so our answer is just $\frac{2^9}{2^{10}} = \boxed{(\mathbf{E})\frac{1}{2}}$ -jb2015007

Solution 10 (Casework on the Number of 1-Jumps)

Landing on 10 requires a sequence of jumps $(m_1, m_2, \dots, m_k)$ of positive integers with $m_1 + m_2 + \dots + m_k = 10$ so it is, an **ordered composition** of 10. The probability of any particular such sequence occurring is

$$\frac{1}{2^{m_1}} \cdot \frac{1}{2^{m_2}} \cdots \frac{1}{2^{m_k}} = \frac{1}{2^{m_1 + \dots + m_k}} = \frac{1}{2^{10}},$$

so *every* composition of 10 contributes exactly $\frac{1}{1024}$ to the answer. Hence

$$P(\text{land on 10}) = \frac{\#\{\text{compositions of 10}\}}{1024}.$$

We count compositions of 10 by casework on j , the number of jumps equal to 1.

Suppose exactly j jumps equal 1, and the rest (all ≥ 2) form a partition of $N = 10 - j$. For each partition of N into parts ≥ 2 with multiplicities $v_1, v_2, \dots$ (and b total parts), the number of ways to arrange the full jump sequence (the j ones together with the b big jumps, accounting for repeated part sizes) is

$$\frac{(b+j)!}{j! v_1! v_2! \cdots}.$$

Summing over all partitions of N into parts ≥ 2 gives the total count for that value of j .

Case $j = 0$ ($N = 10$): partitions 5+3+2, 3+3+2+2, 4+2+2+2, 6+2+2, 4+4+2, 4+3+3, 8+2, 7+3, 6+4, give

$$6 + 6 + 4 + 3 + 3 + 3 + 2 + 2 + 2 + 1 + 1 + 1 = 34.$$

Case $j = 1$ ($N = 9$): partitions 4+3+2, 3+2+2+2, 5+2+2, 7+2, 6+3, 5+4, 3+3+3, 9 give

$$24 + 20 + 12 + 6 + 6 + 6 + 4 + 2 = 80.$$

Case $j = 2$ ($N = 8$): partitions $4+2+2$, $3+3+2$, $2+2+2+2$, $6+2$, $5+3$, $4+4$, 8 give

$$30 + 30 + 15 + 12 + 12 + 6 + 3 = 108.$$

Case $j = 3$ ($N = 7$): partitions $3+2+2$, $5+2$, $4+3$, 7 give

$$60 + 20 + 20 + 4 = 104.$$

Case $j = 4$ ($N = 6$): partitions $2+2+2$, $4+2$, $3+3$, 6 give

$$35 + 30 + 15 + 5 = 85.$$

Case $j = 5$ ($N = 5$): partitions $3+2$, 5 give

$$42 + 6 = 48.$$

Case $j = 6$ ($N = 4$): partitions $2+2$, 4 give

$$28 + 7 = 35.$$

Case $j = 7$ ($N = 3$): partition 3 gives

$$8.$$

Case $j = 8$ ($N = 2$): partition 2 gives

$$9.$$

Case $j = 9$ ($N = 1$): no partition of 1 into parts ≥ 2 exists, so this case contributes 0 .

Case $j = 10$ ($N = 0$): the empty partition (all ten jumps are 1 's) gives

$$1.$$

Adding every case:

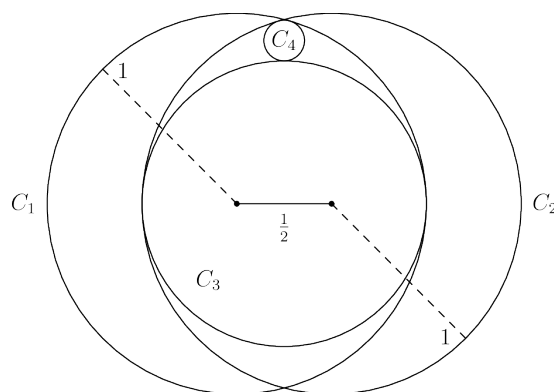
$$34 + 80 + 108 + 104 + 85 + 48 + 35 + 8 + 9 + 0 + 1 = 512.$$

Therefore,

$$P(\text{land on } 10) = \frac{512}{1024} = \frac{1}{2}.$$

The answer is **(E)** $\frac{1}{2}$.

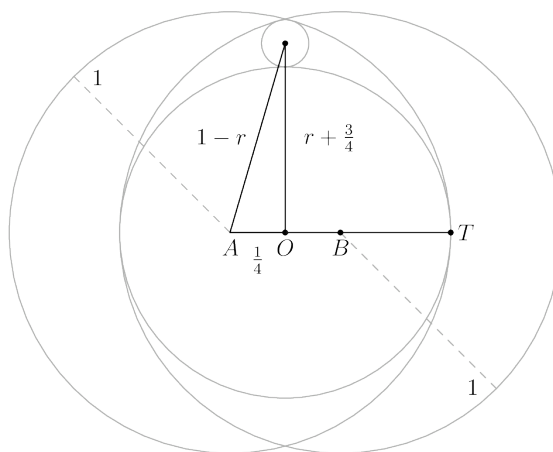
18. (*AMC 12 12A 2023*) Circle C_1 and C_2 each have radius 1 , and the distance between their centers is $\frac{1}{2}$. Circle C_3 is the largest circle internally tangent to both C_1 and C_2 . Circle C_4 is internally tangent to both C_1 and C_2 and externally tangent to C_3 . What is the radius of C_4 ?



- (A) $\frac{1}{14}$ (B) $\frac{1}{12}$ (C) $\frac{1}{10}$ (D) $\frac{3}{28}$ (E) $\frac{1}{9}$

Answer: D

Solution



Let O be the center of the midpoint of the line segment connecting both the centers, say A and B .

Let the point of tangency with the inscribed circle and the right larger circles be T .

Then $OT = BO + BT = BO + AT - \frac{1}{2} = \frac{1}{4} + 1 - \frac{1}{2} = \frac{3}{4}$.

Since C_4 is internally tangent to C_1 , center of C_4 , C_1 and their tangent point must be on the same line.

Now, if we connect centers of C_4 , C_3 and C_1/C_2 , we get a right angled triangle.

Let the radius of C_4 equal r . With the pythagorean theorem on our triangle, we have

$$\left(r + \frac{3}{4}\right)^2 + \left(\frac{1}{4}\right)^2 = (1 - r)^2$$

Solving this equation gives us

$$r = \boxed{\text{(D)} \frac{3}{28}}$$

Solution 2: Estimation

Draw 2 lines from the middle segment's endpoints to an intersection of C_1 and C_2 . Draw the perpendicular bisector of the middle segment until it reaches one of the intersections. We see this becomes a right triangle with a hypotenuse of 1 and one leg length of $\frac{1}{4}$.

Using the Pythagorean theorem, the other leg length is $\frac{\sqrt{15}}{4}$. Also drawing C_1 's radius along the segment of $\frac{1}{2}$ allows us to find the that radius of the middle circle is $\frac{3}{4}$.

Finally notice that $r_{C_3} + 2r_{C_4}$ is really close to $\frac{\sqrt{15}}{4}$. $2r_{C_4} = \frac{\sqrt{15} - 3}{4}$, so $r_{C_4} = \frac{\sqrt{15} - 3}{8}$.

This is really close to $\frac{1}{8}$ and the closest answer to that is D) $\frac{3}{28}$

19. (AMC 12 12A 2023) What is the product of all the solutions to the equation

$$\log_{7x} 2023 \cdot \log_{289x} 2023 = \log_{2023x} 2023?$$

(A) $(\log_{2023} 7 \cdot \log_{2023} 289)^2$ (B) $\log_{2023} 7 \cdot \log_{2023} 289$ (C) 1

(D) $\log_7 2023 \cdot \log_{289} 2023$ (E) $(\log_7 2023 \cdot \log_{289} 2023)^2$

Answer: C

Solution 1

For $\log_{7x} 2023 \cdot \log_{289x} 2023 = \log_{2023x} 2023$, transform it into $\frac{\ln 289 + \ln 7}{\ln 7 + \ln x} \cdot \frac{\ln 289 + \ln 7}{\ln 289 + \ln x} = \frac{\ln 289 + \ln 7}{\ln 289 + \ln 7 + \ln x}$. Replace $\ln x$ with y . Because we want to find the product of all solutions of x , it is equivalent to finding the exponential of the sum of all solutions of y . Change the equation to standard quadratic equation form, the term with 1 power of y is canceled. By using Vieta, we see that since there does not exist a by term, $\sum y = 0$ and $\prod x = e^0 = \boxed{\text{(C)}1}$.

Solution 2 (Same idea as Solution 1 with easily understood steps)

$$\log_{7x} 2023 \cdot \log_{289x} 2023 = \log_{2023x} 2023$$

Rearranging it give us:

$$\log_{2023} 7x \cdot \log_{2023} 289x = \log_{2023} 2023x$$

$$(\log_{2023} 7 + \log_{2023} x)(\log_{2023} 289 + \log_{2023} x) = (\log_{2023} 2023 + \log_{2023} x)$$

let $\log_{2023} x$ be a , we get

$$(\log_{2023} 7 + a)(\log_{2023} 289 + a) = 1 + a$$

$$a^2 + (\log_{2023} 7 + \log_{2023} 289)a + \log_{2023} 7 \cdot \log_{2023} 289 = 1 + a$$

$$a^2 + \log_{2023} 7 \cdot \log_{2023} 289 - 1 = 0$$

by Vieta's Formulas,

$$a_1 + a_2 = 0$$

$$\log_{2023} x_1 + \log_{2023} x_2 = 0$$

$$\log_{2023} x_1 x_2 = 0$$

$$x_1 x_2 = \boxed{(C)1}$$

Solution 3

Similar to solution 1, change the bases first

$$\frac{\ln 289 + \ln 7}{\ln 7 + \ln x} \cdot \frac{\ln 289 + \ln 7}{2 \ln 17 + \ln x} = \frac{\ln 289 + \ln 7}{\ln 7 + 2 \ln 17 + \ln x}$$

Cancel and cross multiply to get

$$(\ln 7 + 2 \ln 17)(\ln 7 + 2 \ln 17 + \ln x) = (\ln 7 + \ln x)(2 \ln 17 + \ln x)$$

Simplify to get

$$\begin{aligned} (\ln x)^2 &= 4(\ln 17)^2 + 2 \ln 17 \ln 7 + (\ln 7)^2 \\ \ln x &= \pm \sqrt{4(\ln 17)^2 + 2 \ln 17 \ln 7 + (\ln 7)^2} \end{aligned}$$

The sum of all possible $\ln x$ is 0, thus the product of all solutions of x is $\boxed{(C)1}$

Solution 4(Fakesolve)

We take the reciprocal of both sides:

$$\frac{1}{\log_{7x} 2023} \cdot \frac{1}{\log_{289x} 2023} = \frac{1}{\log_{2023x} 2023}.$$

Using logarithm properties, we have

$$\log_{2023} 7x \cdot \log_{2023} 289x = \log_{2023} 2023x.$$

Simplify to obtain

$$2023x^2 = 2023x,$$

from which we have $x = \boxed{(C)1}$

This solution works for this problem by chance but do note that the simplification step to get

$$2023x^2 = 2023x$$

is not how log properties work and that the actual solutions for x are

$$x = e^{\pm \sqrt{\ln^2(7) + \ln(17)\ln(2092529)}}$$

(as shown in solution 3) which multiply to 1

Solution 5(Similar to solution 4, Fakesolve)

First, we take the reciprocal of both sides. We get

$$\frac{1}{\log_{7x} 2023} \cdot \frac{1}{\log_{289x} 2023} = \frac{1}{\log_{2023x} 2023}.$$

Flip the logarithms to get

$$\log_{2023} 7x \cdot \log_{2023} 289x = \log_{2023} 2023x.$$

Now we can use $\log_a bc = \log_a b + \log_a c$. We get

$$\log_{2023} 7 + \log_{2023} x + \log_{2023} 289 + \log_{2023} x = \log_{2023} 2023 + \log_{2023} x.$$

The $\log_{2023} 7 + \log_{2023} 289$ and $\log_{2023} 2023$ terms cancel, giving

$$2 \cdot \log_{2023} x = \log_{2023} x$$

so now we are sure that $\log_{2023} x = 0$, so the only solution is $x = \boxed{\text{(C)}1}$.

Solution 6 (Bogus Solution)

We can begin by changing the logarithms' base to x and simplifying:

$$\frac{(\log_x 2023)^2}{\log_x 7 + 1} \cdot \frac{1}{\log_x 289 + 1} = 1$$

After that, we can multiply the denominator on the LHS on both sides, resulting in:

$$(\log_x 2023)^2 = (\log_x 7 + 1)(\log_x 289 + 1)$$

$$(\log_x 2023)^2 = \log_x 2023 + \log_x 7 + \log_x 289 + 1$$

$$(\log_x 2023)^2 = 2 \log_x 2023 + 1$$

Now, we can substitute some variable a as $\log_x 2023$, giving the quadratic $a^2 - 2a - 1 = 0$.

By Vieta's, the product of the roots is 1, thus the solution is $\boxed{\text{(C)}1}$.

20. (AMC 12 12A 2023) Rows 1, 2, 3, 4, and 5 of a triangular array of integers are shown below:

$$\begin{array}{ccccccc} & & & & 1 & & & & \\ & & & & & & 1 & & \\ & & & 1 & & & & & 1 \\ & & 1 & & 3 & & & & 1 \\ & 1 & & 5 & & 5 & & & 1 \\ 1 & & 7 & & 11 & & 7 & & 1 \end{array}$$

Each row after the first row is formed by placing a 1 at each end of the row, and each interior entry is 1 greater than the sum of the two numbers diagonally above it in the previous row. What is the units digit of the sum of the 2023 numbers in the 2023rd row?

- (A) 1 (B) 3 (C) 5 (D) 7 (E) 9

Answer: C

Solution 1

First, let $R(n)$ be the sum of the n th row. Now, with some observation and math instinct, we can guess that $R(n) = 2^n - n$.

Now we try to prove it by induction,

$$R(1) = 2^n - n = 2^1 - 1 = 1 \text{ (works for base case)}$$

$$R(k) = 2^k - k \quad R(k+1) = 2^{k+1} - (k+1) = 2(2^k) - k - 1$$

By definition from the question, the next row is always:

Double the sum of last row (Imagine each number from last row branches off toward left and right to the next row), plus # of new row, minus 2 (minus leftmost and rightmost's 1)

$$\text{Which gives us : } 2(2^k - k) + (k+1) - 2 = 2(2^k) - k - 1$$

Hence, proven

$$\text{Last, simply substitute } n = 2023, \text{ we get } R(2023) = 2^{2023} - 2023$$

$$\text{Last digit of } 2^{2023} \text{ is } 8, 8 - 3 = \boxed{\text{(C) } 5}$$

Solution 2

In a normal Pascal's Triangle with a row's elements being $a_1, a_2, a_3, \dots, a_n$, the following row's elements will sum to:

$$a_1 + (a_1 + a_2) + (a_2 + a_3) \dots (a_{n-1} + a_n) + a_n = 2(a_1 + a_2 + a_3 + \dots + a_n).$$

So the sum of the elements in the next row are double that of the previous row in a Pascal's Triangle.

Now, in this specific problem, for every consecutive pair of numbers in the previous row, an extra 1 is added. There are $n - 1$ pairs in the previous row to add ones to, but we have to subtract 1 because of the two 1s on the sides which don't change. Let S_n represent the sum of the elements on the n th row. Then we have:

$$S_n = 2(S_{n-1}) + n - 1 - 1 = 2(S_{n-1}) + n - 2.$$

So

$$S_{2023} = 2S_{2022} + 2021 = 2(2(S_{2021}) + 2020) + 2021 = 2(2(2S_{2020} + 2019) + 2020) + 2021 = \dots$$

Notice the pattern. At S_{2021} we had:

$$S_{2023} = 2^2 \cdot S_{2021} + 2^1 \cdot 2020 + 2^0 \cdot 2021$$

And at S_{2020} we had:

$$S_{2023} = 2^3 \cdot S_{2020} + 2^2 \cdot 2019 + 2^1 \cdot 2020 + 2^0 \cdot 2021$$

Aside from the first term, the exponents and constant always sum to the same number. We can generalize S_{2023} based on where we stop for S_n :

$$S_{2023} = 2^k S_{2023-k} + \sum_{j=0}^{k-1} 2^j (2021 - j)$$

Let $k = 2022$:

$$S_{2023} = 2^{2022} S_1 + \sum_{j=0}^{2021} 2^j (2021 - j) = 2^{2023} + \sum_{j=0}^{2021} 2^j \cdot 2021 - \sum_{j=0}^{2021} 2^j \cdot j$$

We apply the sum of geometric sequences formula to the first part of the summation:

$$\sum_{j=0}^{2021} 2^j \cdot 2021 = 2021(2^{2022} - 1)$$

Now we expand the second part of the summation:

$$\sum_{j=0}^{2021} 2^j \cdot j = 2^0 \cdot 0 + 2^1 \cdot 1 + \dots + 2021 \cdot 2^{2021}$$

Then multiply this by 2 to get another expression:

$$2 \sum_{j=0}^{2021} 2^j \cdot j = 2^1 \cdot 0 + 2^2 \cdot 1 + \dots + 2021 \cdot 2^{2022}$$

Subtract the two expressions:

$$\begin{aligned} - \sum_{j=0}^{2021} 2^j \cdot j &= 0 \cdot 2^0 + (1 - 0) \cdot 2^1 + (2 - 1) \cdot 2^2 + \dots + (2021 - 2020) \cdot 2^{2021} - 2021 \cdot 2^{2022} = \\ &= 2^0 + 2^1 + 2^2 + \dots + 2^{2021} - 2021 \cdot 2^{2022} \end{aligned}$$

This can again be simplified using the geometric sequence sum formula to finally get

$$\sum_{j=0}^{2021} 2^j \cdot j = 2020 \cdot 2^{2022} + 2$$

Plugging this back into the overall expression finally gives us:

$$S_{2023} = 2^{2022} S_1 + 2021(2^{2022} - 1) - 2020 \cdot 2^{2022} - 2$$

Since S_1 is just 1 we have:

$$S_{2023} = 2^{2022} + 2^{2022} - 2023 = 2^{2023} - 2023.$$

Take this mod 10 to get the last digit: $2^{2023} - 2023 \equiv 8 - 3 \equiv \boxed{5} \pmod{10}$.

Solution 3

Let the sum of the numbers in row 2022 be S_{2022} . Let each number in row 2022 be x_i where $1 \leq i \leq 2022$.

Then

$$\begin{aligned} S_{2023} &= 1 + (x_1 + x_2 + 1) + (x_2 + x_3 + 1) + \dots + (x_{2021} + x_{2022} + 1) + 1 \\ S_{2023} &= x_1 + 2(S_{2022} - x_1 - x_{2022}) + 2023 + x_{2022} \\ S_{2023} &= 2S_{2022} + 2021 \end{aligned}$$

From this we can establish:

$$\begin{aligned} S_n &= 2S_{n-1} + n - 2 \\ S_{n-1} &= 2S_{n-2} + n - 3 \end{aligned}$$

$$S_n - S_{n-1} = 2S_{n-1} - 2S_{n-2} + 1$$

Let $B_n = S_n - S_{n-1}$

From this we have:

$$\begin{aligned} B_n + 1 &= 2(B_{n-1} + 1) \\ B_n &= 2^{n-1} - 1 \\ S_n - S_{n-1} &= 2^{n-1} - 1 \\ S_n &= 2^{n-1} + 2^{n-2} + \dots + 2^1 - (n-1) + S_1 \\ S_n &= 2^n - n \\ S_{2023} &= 2^{2023} - 2023 \end{aligned}$$

The problem requires us to find the last digit of S_{2023} . We can use modular arithmetic.

$$\begin{aligned} S_{2023} &= 2^{2023} - 2023 \\ 2^{2023} - 2023 &\equiv 8 - 3 \pmod{10} \\ &\equiv \boxed{\text{(C)} 5} \pmod{10} \end{aligned}$$

Solution 3.b simpler

$$\begin{aligned} S_n &= 2S_{n-1} + n - 2 \\ S_n + An + B &= 2(S_{n-1} + A(n-1) + B) \\ A &= 1, B = 0 \\ S_n + n &= 2(S_{n-1} + n - 1) = 2^{n-1}(S_1 + 1) \end{aligned}$$

continue with the rest from solution 2

Solution 4 (Recursion)

Let the sum of the n^{th} row be S_n .

For each of the $n - 2$ non-1 entries in the n^{th} row, they are equal to the sum of the 2 numbers diagonally above it in the $n - 1^{th}$ row plus 1. Therefore all $n - 3$ non-1 entries in the $n - 1^{th}$ row appear twice in the sum of the $n - 2$ non-1 entries in the n^{th} row, the two 1s on each end of the $n - 1^{th}$ row only appear once in the sum of the $n - 2$ non-1 entries in the n^{th} row. Additionally, additional 1s are placed at each end of the n^{th} row. Hence, $S_n = 2(S_{n-1} - 1) + n - 2 + 1 + 1 = 2S_{n-1} + n - 2$

$$\begin{array}{ccccc} & & 1 & & \\ & 1 & & 1 & \\ 1 & & 1 & & 1 \\ & 1 & 3 & 1 & \\ & 1 & 5 & 5 & 1 \\ & 1 & 5 & 1 & 5 & 1 \end{array} \quad \begin{array}{ccccc} & & 1 & & \\ & 1 & & 1 & \\ 1 & & 1 & & 1 \\ & 1 & 3 & 1 & \\ & 1 & 5 & 5 & 1 \\ & 1 & 5 & 5 & 1 & 1 \end{array} \quad \begin{array}{ccccc} & & 1 & & \\ & 1 & & 1 & \\ 1 & & 1 & & 1 \\ & 1 & 3 & 1 & \\ & 1 & 5 & 5 & 1 \\ & 1 & 5 & 5 & 1 & 1 \end{array}$$

$S_4 = 1 + 5 + 5 + 1 = 12$, $S_5 = 1 + 7 + 11 + 7 + 1 = 27$. By using the recursive formula, $S_5 = 2 \cdot S_4 + 5 - 2 = 27$

$$S_n = 2S_{n-1} + n - 2$$

$$S_n + n = 2S_{n-1} + 2n - 2$$

$$S_n + n = 2(S_{n-1} + n - 1)$$

$S_n + n$ is a geometric sequence by a ratio of 2 $\therefore S_1 = 1 \therefore S_n + n = 2^n, S_n = 2^n - n$
 $S_{2023} = 2^{2023} - 2023$

The unit digit of powers of 2 is periodic by a cycle of 4 digits: 2, 4, 8, 6. $2023 = 3 \pmod{4}$, the unit digit of 2^{2023} is 8.

Therefore, the unit digit of S_{2023} is $8 - 3 = \boxed{\text{(C)} 5}$

Solution 5

Consider Pascal's triangle as the starting point. In the Pascal's triangle depicted below, the sum of the numbers in the n th row is $2^{(n-1)}$. For the 2023rd row in the Pascal's triangle, the sum of numbers is 2^{2022} .

$$\begin{array}{c}
 1 \\
 1 \quad 1 \\
 1 \quad 2 \quad 1 \\
 1 \quad 3 \quad 3 \quad 1 \\
 1 \quad 4 \quad 6 \quad 4 \quad 1
 \end{array}$$

For the triangular array of integers in the problem, 1 is added to each interior entry, which propagates to two numbers diagonally below it in the next row, making the sum of numbers in the next row to increase by 2. And the addition of 1 continues to propagate to the next row, which makes the sum of numbers in the row below the next to increase by 4.

Examine the following triangular array of 1's, which indicates the 1's being added to each position of corresponding numbers in the Pascal's triangle.

$$\begin{array}{c}
 0 \\
 0 \quad 0 \\
 0 \quad 1 \quad 0 \\
 0 \quad 1 \quad 1 \quad 0 \\
 0 \quad 1 \quad 1 \quad 1 \quad 0
 \end{array}$$

For the 3rd row, 1 is added to the original number in the same position in the Pascal's triangle. And the addition of 1 in the 3rd row makes the sum of numbers in the 4th row to increase by 2, and makes the sum of numbers in the 5th row to increase by 4, and so

forth. Therefore, the addition of a 1 in the 3rd row makes the sum of numbers in the 2023rd row to increase by $2^{2023-3} = 2^{2020}$. And similarly, the addition of each 1 in the 4th row makes the number of numbers in the 2023rd row to increase by $2^{2023-4} = 2^{2019}$. The 1's being added between the 3rd and 2022nd rows impact on the sum of numbers in the 2023rd row to increase by $2^{2020} + 2 \cdot 2^{2019} + 3 \cdot 2^{2018} + \dots + 2020 \cdot 2^1 = \sum_{n=1}^{2020} (n \cdot 2^{(2021-n)})$.

Therefore, the sum of numbers in the 2023rd row is the aggregation of the sum of numbers in the 2023rd row in the Pascal's triangle, the impact of addition of 1's between the 3rd row and the 2022nd row, and the addition of 1's on 2021 interior entries in the 2023rd row, which is $2^{2022} + \sum_{n=1}^{2020} (n \cdot 2^{(2021-n)}) + 2021$.

Because $2^{2022} = 2^{2020+2} = 16^{505} \cdot 2^2 = 16^{505} \cdot 4$ and 16^k will always ends with 6 as the unit digit, the unit digit of 2^{2022} is 4.

For $\sum_{n=1}^{2020} (n \cdot 2^{(2021-n)})$, we can solve the sum of geometric sequence to be $2^{2022} - 4 - 2020 \cdot 2$, which has 0 for the unit digit.

Therefore, for the sum of numbers in the 2023rd row, which is $2^{2022} + \sum_{n=1}^{2020} (n \cdot 2^{(2021-n)}) + 2021$, its unit digit is $4 + 0 + 1 = \boxed{(C) 5}$.

Solution 6 (mod bash)

Observe that: $S_n = 2S_{n-1} + (n - 2)$
 $= 2S_{n-1} + D_n$
 $= 2S_{n-1} + D_{n-1} + 1$ where $D_1 = -1$, and $D_n = D_{n-1} + 1$ for $n > 1$.

Make a table of values, $(S_n, D_n) \pmod 2$ and $(S_n, D_n) \pmod 5$, a until the cycle completes for each. ($\pmod 2$ has period 1; $\pmod 5$ has period 20). (In both cases, there is no "head" that is not part of the cycle.)

Mod 2: $S_{2023} \equiv S_{2023 \pmod 2} \equiv S_1 = (1 \cdot 2 + 0) \cdot 0 + 1 \equiv 1 \pmod 2$

Mod 5: $S_{2023} \equiv S_{2023 \pmod{20}} \equiv S_3 = (1 \cdot 2 + 0) \cdot 2 + 1 \equiv 5 \pmod 5$

Thus $S_{2023} \pmod{10} = \boxed{(C) 5}$.

Solution 7: Newton's Little Formula

The triangle provided by the question may be rewritten with Pascal's Triangle being the starting point.

$$\begin{array}{ccccccc}
 & & & & 1 & & \\
 & & & & & & 1 \\
 & & 1 & & & 1 & \\
 & & & 1 & & 2 + \textcolor{red}{1} & & 1 \\
 & & & & 1 & & 3 + \textcolor{red}{2} & & 3 + \textcolor{red}{2} & & 1 \\
 & & 1 & & & 3 + \textcolor{red}{2} & & 3 + \textcolor{red}{2} & & & 1 \\
 & & & 1 & & 4 + \textcolor{red}{3} & & 4 + \textcolor{red}{5} & & 4 + \textcolor{red}{3} & & 1 \\
 & & & & & & & & & & & \\
 & & & & & & & & & & & \vdots
 \end{array}$$

The numbers in Pascal's Triangle could be written with combinations. Moreover, the following equation is true.

$$\sum_{k=0}^n {}^nC_k = 2^n$$

Therefore, the sum of every number in n^{th} row of a Pascal's Triangle is 2^n .

Another triangle with the red colored number could be written to evaluate the sum of the red numbers in each row.

$$\begin{array}{ccccc}
 & & 1 & & \\
 & 2 & & 2 & \\
 & 3 & & 5 & & 3 \\
 & & & \vdots & & \\
 \end{array}$$

The sum of numbers in each line could be investigated.

$$\begin{array}{ccccccc}
 1 & & 4 & & 11 & & 26 & & 53 & \dots \\
 & 3 & & 7 & & 15 & & 27 & \dots \\
 & & 4 & & 8 & & 12 & \dots \\
 & & & 4 & & 4 & \dots
 \end{array}$$

Using Newton's Little Formula, it is evident that the following equation is true.

$$a_n = 1 \cdot {}_{n-1}C_0 + 3 \cdot {}_{n-1}C_1 + 4 \cdot {}_{n-1}C_2 + 4 \cdot {}_{n-1}C_3$$

With all the tools, the last digit, or the remainder when divided by 10, of the sum of the numbers in the 2023rd row may be computed. Let x be the last digit. Moreover, a_{2020} is equivalent to the sum of red numbers in 2023rd row of the original triangle.

$$\begin{aligned}
 2^{2020} + 1 \cdot {}_{2020}C_0 + 3 \cdot {}_{2020}C_1 + 4 \cdot {}_{2020}C_2 + 4 \cdot {}_{2020}C_3 &\equiv x \pmod{10} \\
 &\equiv 2^{2022} + 1 + 3 \cdot 2020 + 2 \cdot 2020 \cdot 2019 \\
 &\quad + 2 \cdot 2020 \cdot 673 \cdot 2018 \pmod{10} \\
 &= 2^{2022} + 1 \pmod{10} \\
 &= 4 + 1 \pmod{10} \\
 &= 5 \pmod{10}
 \end{aligned}$$

Answer: $\boxed{(C)5}$

21. (AMC 12 12A 2023) If A and B are vertices of a polyhedron, define the distance $d(A, B)$ to be the minimum number of edges of the polyhedron one must traverse in order to connect A and B . For example, if $\overline{AB}$ is an edge of the polyhedron, then $d(A, B) = 1$, but if $\overline{AC}$ and $\overline{CB}$ are edges and $\overline{AB}$ is not an edge, then $d(A, B) = 2$. Let Q , R , and S be randomly chosen distinct vertices of a regular icosahedron (regular polyhedron made up of 20 equilateral triangles). What is the probability that $d(Q, R) > d(R, S)$?

(A) $\frac{7}{22}$ (B) $\frac{1}{3}$ (C) $\frac{3}{8}$ (D) $\frac{5}{12}$ (E) $\frac{1}{2}$

Answer: A

Solution 1

To find the total amount of vertices we first find the amount of edges, and that is $\frac{20 \times 3}{2}$. Next, to find the amount of vertices we can use Euler's characteristic, $V - E + F = 2$, and therefore the amount of vertices is 12

So there are $\binom{12}{3} = 220$ ways to choose 3 distinct points.

Now, the furthest distance we can get from one point to another point in an icosahedron is 3. Which gives us a range of $1 \leq d(Q, R), d(R, S) \leq 3$

With some case work, we get two cases:

Case 1: $d(Q, R) = 3; d(R, S) = 1, 2$

Since we have only one way to choose Q, that is, the opposite point from R, we have one option for Q and any of the other points could work for S.

Then, we get $12 \times 1 \times 10 = 120$ (ways to choose R $\times$ ways to choose Q $\times$ ways to choose S)

Case 2: $d(Q, R) = 2; d(R, S) = 1$

We can visualize the icosahedron as 4 rows, first row with 1 vertex, second row with 5 vertices, third row with 5 vertices and fourth row with 1 vertex. We set R as the one vertex on the first row, and we have 12 options for R. Then, Q can be any of the 5 points on the third row and finally S can be one of the 5 points on the second row.

Therefore, we have $12 \times 5 \times 5 = 300$ (ways to choose R $\times$ ways to choose Q $\times$ ways to choose S)

In total, we have 420 ways, but we must divide by 3! to account for permutations, giving us $70/220 = \boxed{\text{(A)} \frac{7}{22}}$.

Solution 2 (Cheese + Actual way)

In total, there are $\binom{12}{3} = 220$ ways to select the points. However, if we look at the denominators of B, C, D , they are 3, 8, 12 which are not divisors of 220. Also $\frac{1}{2}$ is impossible as cases like $d(Q, R) = d(R, S)$ exist. The only answer choice left is $\boxed{\text{(A)} \frac{7}{22}}$

Note: this cheese is actually wrong because the total number of ways to select the points is actually $12 \times 11 \times 10 = 1320$ as order matters, so all denominators are possible. Rather, you can arrive at the same conclusion by fixing R WLOG, leading to $11 \times 10 = 110$ ways in total, which works for the original cheese.. awesomeguy856

(Actual way)

Fix an arbitrary point, to select the rest 2 points, there are $11 \cdot 10 = 110$ ways. To make $d(Q, R) = d(R, S), d = 1/2$. Which means there are in total $2 \cdot 5 \cdot 4 = 40$ ways to make the distance the same. $\frac{1}{2}(1 - \frac{40}{110}) = \boxed{\text{(A)} \frac{7}{22}}$ bluesoul

Solution 3

We can imagine the icosahedron as having 4 layers. 1 vertex at the top, 5 vertices below connected to the top vertex, 5 vertices below that which are 2 edges away from the top vertex, and one vertex at the bottom that is 3 edges away. WLOG because the icosahedron is symmetric around all vertices, we can say that R is the vertex at the top. So now, we just need to find the probability that S is on a layer closer to the top than Q. We can do casework on the layer S is on to get

$$\frac{5}{11} \cdot \frac{6}{10} + \frac{5}{11} \cdot \frac{1}{10} = \frac{35}{110} = \frac{7}{22}$$

So the answer is (A) $\frac{7}{22}$.

-awesomeparrot

Solution 4

We can actually see that the probability that $d(Q, R) > d(R, S)$ is the exact same as $d(Q, R) < d(R, S)$ because $d(Q, R)$ and $d(R, S)$ have no difference. (In other words, we can just swap Q and S, meaning that can be called the same probability-wise.) Therefore, we want to find the probability that $d(Q, R) = d(R, S)$.

WLOG, we can rotate the icosahedron so that R is the top of the icosahedron. Then we can divide this into 2 cases:

1. They are on the second layer

There are 5 ways to put one point, and 4 ways to put the other point such that $d(Q, R) = d(R, S) = 1$. So, there are $5 \cdot 4 = 20$ ways to put them on the second layer.

2. They are on the third layer

There are 5 ways to put one point, and 4 ways to put the other point such that $d(Q, R) = d(R, S) = 2$. So, there are $5 \cdot 4 = 20$ ways to put them on the third layer.

The total number of ways to choose P and S are $11 \cdot 10 = 110$ (because there are 12 vertices), so the probability that $d(Q, R) = d(R, S)$ is $\frac{20+20}{110} = \frac{4}{11}$.

Therefore, the probability that $d(Q, R) > d(R, S)$ is $\frac{1-\frac{4}{11}}{2} = \span style="border: 1px solid black; padding: 2px;">(A) $\frac{7}{22}$$

Solution 5

We know that there are 20 faces. Each of those faces has 3 borders (since each is a triangle), and each edge is used as a border twice (for each face on either side). Thus, there are $\frac{20 \cdot 3}{2} = 30$ edges.

Euler's formula states that $V - E + F = 2$ for all convex polyhedra, so we know that $2 - F + E = 12$ vertices.

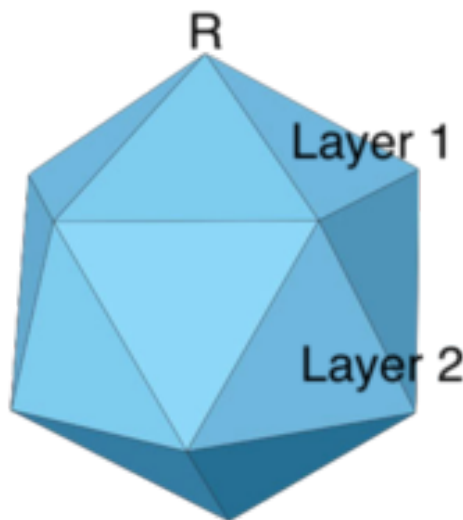
The answer can be counted by first counting the number of possible paths that will yield $d(Q, R) > d(R, S)$ and dividing it by $12 \cdot 11 \cdot 10$ (or $\binom{12}{3}$, depending on the approach). In either case, one will divide by 11 somewhere in the denominator. We can then hope that there will be no factor of 11 in the numerator (which would cancel the 11 in the denominator out), and answer the only option that has an 11 in the denominator:

(A) $\frac{7}{22}$.

Additional note by "Fruitz": One can eliminate 1/2 by symmetry if you swap the new sign.

Another note by "andliu766": A shorter way to find the number of vertices and edges is to use the fact that the MAA logo is an icosahedron. :)

Solution 6 (Case Work(similar to 3))



WLOG, let R be at the top-most vertex of the icosahedron. There are 2 cases where $d(Q, R) > d(R, S)$.

Case 1: Q is at the bottom-most vertex

If Q is at the bottom-most vertex, no matter where S is, $d(Q, R) > d(R, S)$. The probability that Q is at the bottom-most vertex is $\frac{1}{11}$

Case 2: Q is at the second layer

If Q is at the second layer, S must be at the first layer, for $d(Q, R) > d(R, S)$ to be true. The probability that Q is at the second layer, and S is at the first layer is $\frac{5}{11} \cdot \frac{5}{10} = \frac{5}{22}$

$$\frac{1}{11} + \frac{5}{22} = \boxed{\text{(A)} \frac{7}{22}}$$

Supplement First, we have 20 faces of equilateral triangles. But each edge is shared by two faces hence there are thus $\frac{20 \cdot 3}{2} = 30$ edges.

Now by Euler's formula $V - E + F = 2 \Rightarrow V - 30 + 20 = 2 \Rightarrow V = 12$.

Furthermore, the number of triangles (n) that meet at one vertex can be found as follows. Note that there are 12 total vertices. Without accounting for overcounting, total number of vertices = number of faces $\cdot$ vertices per face = $20 \cdot 3 = 60$. But since we now know that there are only 12 vertices and that each vertex is shared by n triangles, you find that there are 5 triangles that meet at one vertex.

Once we pick R as the topmost vertex, there are 11 possible vertices remaining for Q . Continue as follows in this solution.

Solution 7 (efficient)

Since the icosahedron is symmetric polyhedron, we can rotate it so that R is on the topmost vertex. Since Q and S basically the same, we can first count the probability that $d(Q, R) = d(R, S)$.

Case 1 : $d(Q, R) = d(R, S) = 1$

There are 5 points P such that $d(Q, P) = 1$. There is $5 \times 4 = \boxed{20}$ ways to choose Q and S in this case.

Case 2 : $d(Q, R) = d(R, S) = 2$

There are 5 points P such that $d(Q, P) = 2$. There is $5 \times 4 = \boxed{20}$ ways to choose Q and S in this case.

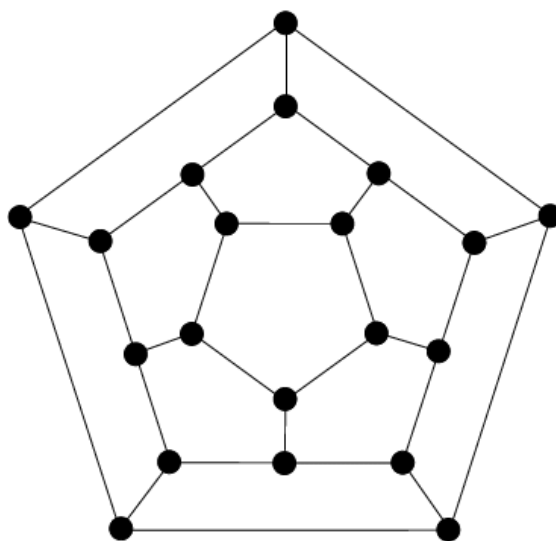
Case 3 : $d(Q, R) = d(R, S) = 3$

There is 1 point P such that $d(Q, P) = 3$. There is $1 \times 0 = \boxed{0}$ ways to choose Q and S in this case.

Final solution

There are 11 points P that are distinct from R . There is $11 \times 10 = \boxed{110}$ ways to choose Q and S . There is $20 + 20 + 0 = \boxed{40}$ ways to choose Q and S such that $d(Q, R) = d(R, S)$. There is $\frac{110-40}{2} = 35$ ways to choose Q and S such that $d(Q, R) > d(R, S)$. The probability that $d(Q, R) > d(R, S)$ is therefore $\frac{35}{110} = \frac{7}{22}$ which corresponds to answer choice **A**.

Solution 8 (Dodecahedron Schlegel (Planar Graph) Diagram)



Instead of thinking this problem as vertices of a icosahedron, think of it as faces of a dodecahedron. Our middle face is R and we first choose Q and S to be the faces with distance 1 from the middle face. There are $\binom{5}{2}$ ways of doing that. Then we choose Q and S to be the faces with distance 2 from the middle face. There are $\binom{5}{2}$ ways of doing that. Our answer is therefore $\frac{1}{2} \left(1 - \frac{2\binom{5}{2}}{\binom{11}{2}} \right) = \boxed{\text{(A)} \frac{7}{22}}$

Note: You can do the same analysis on the vertices of the icosahedron using its diagram, which is less to draw because it has fewer (12) vertices and the same number (30) of edges. But this diagram is either less symmetrical or non-planar. (There's no good place to put the last vertex (diametrically opposite the first/central vertex), except in the same place as the first vertex.)

Solution 9 (Symmetry)

Note that the vertices of an icosahedron are in a $1 - 5 - 5 - 1$ configuration.

We will denote the probability of event A happening as $P(A)$

We should also note that $P(d(Q, R) > d(R, S)) + P(d(Q, R) < d(R, S)) + P(d(Q, R) = d(R, S)) = 1$

Also, $P(d(Q, R) > d(R, S)) = P(d(Q, R) < d(R, S))$ since Q and S are basically the same.

Therefore, what we are trying to find is just $\frac{1 - P(d(Q, R) = d(R, S))}{2}$ which can easily be computed when we fix R at the top vertex in the $1 - 5 - 5 - 1$ structure.

Since all the points are distinct (stated in problem), the conditions are only satisfied where Q and S are at 5s of the $1 - 5 - 5 - 1$ structure.

$P(d(Q, R) = d(R, S)) = 2 \cdot \frac{5}{11} \cdot \frac{4}{10} = \frac{4}{11}$ since we have to choose one of the 5 out of 11 possible vertices and another one of 4 out of 10 two times.

And the answer is $\frac{1 - \frac{4}{11}}{2} = \boxed{\text{(A)} \frac{7}{22}}$

22. (AMC 12 12A 2023) Let f be the unique function defined on the positive integers such that

$$\sum_{d|n} d \cdot f\left(\frac{n}{d}\right) = 1$$

for all positive integers n , where the sum is taken over all positive divisors of n . What is $f(2023)$?

- (A) -1536 (B) 96 (C) 108 (D) 116 (E) 144

Answer: B

Solution 1 (Very Thorough)

First, we note that $f(1) = 1$, since the only divisor of 1 is itself.

Then, let's look at $f(p)$ for p a prime. We see that

$$\sum_{d|p} d \cdot f\left(\frac{p}{d}\right) = 1$$

$$1 \cdot f(p) + p \cdot f(1) = 1$$

$$f(p) = 1 - p \cdot f(1)$$

$$f(p) = 1 - p$$

Nice.

Now consider $f(p^k)$, for $k \in \mathbb{N}$.

23. (AMC 12 12A 2023) How many ordered pairs of positive real numbers (a, b) satisfy the equation

$$(1 + 2a)(2 + 2b)(2a + b) = 32ab?$$

- (A) 0 (B) 1 (C) 2 (D) 3 (E) an infinite number

Answer: B

Solution 1: AM-GM Inequality

Using the AM-GM inequality on the two terms in each factor on the left-hand side, we get

$$(1 + 2a)(2 + 2b)(2a + b) \geq 8\sqrt{2a \cdot 4b \cdot 2ab} = 32ab,$$

This means the equality condition must be satisfied. Therefore, we must have $1 = 2a = b$, so the only solution is **(B) 1**.

Solution 2: Sum Of Squares

Equation $(1 + 2a)(2 + 2b)(2a + b) = 32ab$ is equivalent to

$$b(2a - 1)^2 + 2a(b - 1)^2 + (2a - b)^2 = 0,$$

where $a, b > 0$. Therefore $2a - 1 = b - 1 = 2a - b = 0$, so $(a, b) = (\frac{1}{2}, 1)$. Hence the answer is **(B) 1**.

Solution 3:

$$(1 + 2a)(1 + b)(2a + b) = 16ab,$$

let $x = 2a, y = b$, then it becomes $(1+x)(1+y)(x+y) = 8xy$, or $(1+x+y+xy)(x+y) = 8xy$.

Let $\alpha = x + y, \beta = xy$, it becomes $(1 + \alpha + \beta)\alpha = 8\beta$,

notice we have $\alpha^2 - 4\beta = (x - y)^2 \geq 0$, now $\beta = \frac{\alpha(1+\alpha)}{(8-\alpha)}$ $\alpha^2 \geq 4\beta = 4\alpha(1 + \alpha)/(8 - \alpha)$ (notice we must have $8 - \alpha > 0$), $(8 - \alpha)\alpha \geq 4(1 + \alpha)$, $(\alpha - 2)^2 \leq 0$, $\alpha = 2$, and $\beta = 1$, so $x = y = 1$ and $a = \frac{1}{2}, b = 1$ is the only solution.

answer is **(B)**

Solution 4 Algebraic Method

Let $m = 2a + b$ and $z = ab$. The given equation is $(1 + 2a)(2 + 2b)(2a + b) = 32ab$. Expanding the first two terms as $(2 + 2b + 4a + 4ab)$ and substituting m, z :

$$(2m + 4z + 2)m = 32z \implies m^2 + (2z + 1)m - 16z = 0 \quad (1)$$

Since $2a$ and b are positive reals, they are the roots of $t^2 - mt + 2z = 0$. For real roots, the discriminant $D = m^2 - 8z \geq 0$, so $m^2 \geq 8z$.

From (1), we have $16z = m^2 + (2z + 1)m$. Substituting this into $16z \leq 2m^2$:

$$m^2 + (2z + 1)m \leq 2m^2 \implies (2z + 1)m \leq m^2$$

Since $m > 0$, we have $m \geq 2z + 1$. Substituting this boundary back into (1):

$$(2z + 1)^2 + (2z + 1)(2z + 1) - 16z \leq 0$$

$$2(4z^2 + 4z + 1) - 16z \leq 0 \implies 8z^2 - 8z + 2 \leq 0 \implies 2(2z - 1)^2 \leq 0$$

This forces $(2z - 1)^2 = 0$, so $z = \frac{1}{2}$. Plugging $z = \frac{1}{2}$ into (1) gives $m^2 + 2m - 8 = 0$, so $m = 2$ (as $m > 0$). Solving $2a + b = 2$ and $ab = 1/2$ yields $(a, b) = (1/2, 1)$. This is the only solution, so the answer is **(B)**.

24. (AMC 12 12A 2023) Let K be the number of sequences $A_1, A_2, \dots, A_n$ such that n is a positive integer less than or equal to 10, each A_i is a subset of $\{1, 2, 3, \dots, 10\}$, and A_{i-1} is a subset of A_i for each i between 2 and n , inclusive. For example, $\{\}, \{5, 7\}, \{2, 5, 7\}, \{2, 5, 7\}, \{2, 5, 6, 7, 9\}$ is one such sequence, with $n = 5$. What is the remainder when K is divided by 10?

(A) 1 (B) 3 (C) 5 (D) 7 (E) 9

Answer: C**Solution 1**

Consider any sequence with n terms. Every number has such choices: never appear, appear the first time in the first spot, appear the first time in the second spot ..., and appear the first time in the n th spot, which means every number has $(n + 1)$ choices to show up in the sequence. Consequently, for each sequence with length n , there are $(n + 1)^{10}$ possible ways.

Thus, the desired value is $\sum_{n=1}^{10} (n + 1)^{10} \equiv \boxed{\text{(C) } 5} \pmod{10}$.

Solution 2

Let $f(x, \ell)$ be the number of sequences $A_1, A_2, \dots, A_\ell$ such that each A_i is a subset of $\{1, 2, \dots, x\}$, and A_i is a subset of A_{i+1} for $i = 1, 2, \dots, \ell - 1$. Then $f(x, 1) = 2^x$ and $f(0, \ell) = 1$.

If $\ell \geq 2$ and $x \geq 1$, we need to get a recursive formula for $f(x, \ell)$: If $|A_1| = i$, then A_1 has C_x^i possibilities, and the subsequence $\{A_i\}_{2 \leq i \leq \ell}$ has $f(x - i, \ell - 1)$ possibilities. Hence

$$f(x, \ell) = \sum_{i=0}^x C_x^i f(x - i, \ell - 1).$$

By applying this formula and only considering modulo 10, we get $f(1, 2) = 3, f(1, 3) = 4, f(1, 4) = 5, f(1, 5) = 6, f(1, 6) = 7, f(1, 7) = 8, f(1, 8) = 9, f(1, 9) = 0, f(1, 10) = 1, f(2, 2) = 9, f(2, 3) = 6, f(2, 4) = 5, f(2, 5) = 6, f(2, 6) = 9, f(2, 7) = 4, f(2, 8) = 1, f(2, 9) = 0, f(2, 10) = 1, f(3, 2) = 7, f(3, 3) = 4, f(3, 4) = 5, f(3, 5) = 6, f(3, 6) = 3, f(3, 7) = 2, f(3, 8) = 9, f(3, 9) = 0, f(3, 10) = 1, f(4, 2) = 1, f(4, 3) = 6, f(4, 4) = 5, f(4, 5) = 6, f(4, 6) = 1, f(4, 7) = 6, f(4, 8) = 1, f(4, 9) = 0, f(4, 10) = 1, f(5, 2) = 3, f(5, 3) = 4, f(5, 4) = 5, f(5, 5) = 6, f(5, 6) = 7, f(5, 7) = 8, f(5, 8) = 9, f(5, 9) = 0, f(5, 10) = 1, f(6, 2) = 9, f(6, 3) = 6, f(6, 4) = 5, f(6, 5) = 6, f(6, 6) = 9, f(6, 7) = 4, f(6, 8) = 1, f(6, 9) = 0, f(6, 10) = 1, f(7, 2) = 7, f(7, 3) = 4, f(7, 4) = 5, f(7, 5) = 6, f(7, 6) = 3, f(7, 7) = 2, f(7, 8) = 9, f(7, 9) = 0, f(7, 10) = 1, f(8, 2) = 1, f(8, 3) = 6, f(8, 4) = 5, f(8, 5) = 6, f(8, 6) = 1, f(8, 7) = 6, f(8, 8) = 1, f(8, 9) = 0, f(8, 10) = 1, f(9, 2) = 3, f(9, 3) = 4, f(9, 4) = 5, f(9, 5) = 6, f(9, 6) = 7, f(9, 7) = 8, f(9, 8) = 9, f(9, 9) = 0, f(9, 10) = 1, f(10, 2) = 9, f(10, 3) = 6, f(10, 4) = 5, f(10, 5) = 6, f(10, 6) = 9, f(10, 7) = 4, f(10, 8) = 1, f(10, 9) = 0, f(10, 10) = 1.$

Lastly, we get $K \equiv \sum_{i=1}^{10} f(10, i) \equiv \boxed{\text{(C) } 5} \pmod{10}$. Mahthir Marshad

Solution 3 (Cheese, but this time it actually works)

Seeing that all the answers are different modulus 5, and that 10 is divisible by 5, we cheese this problem.

Let $A_1, A_2, \dots, A_n$ be one sequence satisfying the constraints of the problem. Let $b_1, b_2, \dots, b_n, b_{n+1}$ be the sequence of nonnegative integers such that A_k has b_k elements for all $1 \leq k \leq n$, and $b_{n+1} = 10$. Note that we can generate the number of valid sequences of A by first generating all sequences of b such that $b_i \leq b_{i+1}$ for all $1 \leq i < n$, then choosing the elements from A_{k+1} that we keep in A_k , given the sequence of b as the restraint for the number of elements. For each sequence $b_1, b_2, \dots, b_{n+1}$, there are $\prod_{i=2}^{n+1} \binom{b_i}{b_{i-1}}$ corresponding sequences for A . Now, consider two cases - either all terms in b are either 0, 5, or 10, or there is at least one term in b that is neither 0, 5 nor 10. In the second case, consider the last term in b that is not 10 or 5, say b_m . However,

that implies $b_{m+1} = 10$, and so the number of corresponding sequences of A is $\binom{10}{b_m} \cdot$ something or $\binom{5}{b_m} \cdot$ something, which is always a multiple of 5. Therefore, we only need to consider sequences of b where each term is 0, 5, or 10. If all terms in b are 0 or 10, then for each $1 \leq n \leq 10$ there are $n + 1$ sequences of b (since there are $n + 1$ places to turn from 0 to 10), for a total of $2 + 3 + \cdots + 11 = 65 \equiv 0 \pmod{5}$. If there exists at least one term $b_k = 5$, then we use stars and bars to count the number of sequences of b , and each sequence of b corresponds to $\binom{10}{5}$ sequences of A . For each n , we must have at least one term of 5. After that, there are $n - 1$ stars and 2 bars (separating 0 to 5 and 5 to 10), so that is $\binom{n+1}{2}$ sequences of b . So the sum is $\binom{2}{2} + \binom{3}{2} + \cdots + \binom{11}{2} = \binom{12}{3} \equiv 0 \pmod{5}$. Therefore, the answer is $0 \pmod{5}$, and it must be **(C) 5**

Solution 4

We observe that in each sequence, if element $e \in A_i$, then $e \in A_j$ for all $j \geq i$. Therefore, to determine a sequence with a fixed length n , we only need to determine the first set A_i that each element in $\{1, 2, \dots, 10\}$ is inserted into, or an element is never inserted into any subset.

We have

$$\begin{aligned} K &= \sum_{n=1}^{10} (n+1)^{10} \\ &= \sum_{m=2}^{11} m^{10}. \end{aligned}$$

Recalling or noticing that $x^n \equiv x^{n \bmod 4} \pmod{10}$, then, Modulo 10, we have

$$\begin{aligned} K &\equiv \sum_{m=2}^{11} m^2 \\ &\equiv \sum_{m=1}^{11} m^2 - 1^2 \\ &\equiv \frac{11 \cdot (11+1)(2 \cdot 11+1)}{6} - 1 \\ &\equiv 505 \\ &\equiv \boxed{\text{(C) } 5}. \end{aligned}$$

Slightly easier, observe that $11^{10} \equiv 1^{10} \pmod{10}$, so, working $\pmod{10}$, we have

$$\begin{aligned} K &\equiv \sum_{m=2}^{11} m^2 \\ &\equiv \sum_{m=1}^{10} m^2 \\ &\equiv \frac{10 \cdot (10+1)(2 \cdot 10+1)}{6} \\ &\equiv \frac{10 \cdot 11 \cdot 21}{6} \\ &\equiv 5 \cdot 11 \cdot 7 \end{aligned}$$

$$\equiv \boxed{5}$$

Solution 5 (Constructive Counting)

(See 2021 AIME II P6 for a better understanding; it covers similar material, albeit with a bit more manipulation)

We try this for $n = 1$. Clearly, this is just the number of subsets, giving 2^{10} sequences.

For $n = 2$, we have 3 choices for each element in the list:

1. The element is in both A_1 and A_2 .
2. The element is in only A_2 and NOT in A_1 .
3. The element is in neither A_1 nor A_2 .

Hence, we have 3^{10} sequences.

For $n = 3$, we follow a similar logic; we have 4 choices for each element in the list:

1. The element is in all of A_3 , A_2 , and A_1 .
2. The element is in only A_3 and A_2 .
3. The element is in only A_3 .
4. The element is in none of the above.

Hence, we have 4^{10} sequences.

Continuing so, the total number of sequences sums to $2^{10} + 3^{10} + 4^{10} + \dots + 11^{10}$. By using mod patterns to find the units digit, we easily arrive at $\boxed{\text{(C)} 5}$ as our answer.

25. (AMC 12 12A 2023) There is a unique sequence of integers $a_1, a_2, \dots, a_{2023}$ such that

$$\tan 2023x = \frac{a_1 \tan x + a_3 \tan^3 x + a_5 \tan^5 x + \dots + a_{2023} \tan^{2023} x}{1 + a_2 \tan^2 x + a_4 \tan^4 x + \dots + a_{2022} \tan^{2022} x}$$

whenever $\tan 2023x$ is defined. What is a_{2023} ? (A) -2023 (B) -2022 (C) -1 (D) 1 (E) 2023

Answer: C

Solution 1

$$\begin{aligned} \cos 2023x + i \sin 2023x &= (\cos x + i \sin x)^{2023} \\ &= \cos^{2023} x + \binom{2023}{1} \cos^{2022} x (i \sin x) + \binom{2023}{2} \cos^{2021} x (i \sin x)^2 + \binom{2023}{3} \cos^{2020} x (i \sin x)^3 \\ &\quad + \dots + \binom{2023}{2022} \cos x (i \sin x)^{2022} + (i \sin x)^{2023} \\ &= \cos^{2023} x + i \binom{2023}{1} \cos^{2022} x \sin x - \binom{2023}{2} \cos^{2021} x \sin^2 x - i \binom{2023}{3} \cos^{2020} x \sin^3 x \\ &\quad + \binom{2023}{4} \cos^{2019} x \sin^4 x + \dots + \binom{2023}{2022} \cos x \sin^{2022} x - i \sin^{2023} x \end{aligned}$$

By equating real and imaginary parts:

$$\cos 2023x = \cos^{2023} x - \binom{2023}{2} \cos^{2021} x \sin^2 x + \cdots - \binom{2023}{2022} \cos x \sin^{2022} x$$

$$\sin 2023x = \binom{2023}{1} \cos^{2022} x \sin x - \binom{2023}{3} \cos^{2020} x \sin^3 x + \cdots - \sin^{2023} x$$

$$\begin{aligned} \tan 2023x &= \frac{\sin 2023x}{\cos 2023x} = \frac{\binom{2023}{1} \cos^{2022} x \sin x - \binom{2023}{3} \cos^{2020} x \sin^3 x + \cdots - \sin^{2023} x}{\cos^{2023} x - \binom{2023}{2} \cos^{2021} x \sin^2 x + \cdots - \binom{2023}{2022} \cos x \sin^{2022} x} \\ &= \frac{\binom{2023}{1} \frac{\cos^{2022} x \sin x}{\cos^{2023} x} - \binom{2023}{3} \frac{\cos^{2020} x \sin^3 x}{\cos^{2023} x} + \cdots - \frac{\sin^{2023} x}{\cos^{2023} x}}{\frac{\cos^{2023} x}{\cos^{2023} x} - \binom{2023}{2} \frac{\cos^{2021} x \sin^2 x}{\cos^{2023} x} + \cdots - \binom{2023}{2022} \frac{\cos x \sin^{2022} x}{\cos^{2023} x}} \\ &= \frac{\binom{2023}{1} \tan x - \binom{2023}{3} \tan^3 x + \cdots - \tan^{2023} x}{1 - \binom{2023}{2} \tan^2 x + \cdots - \binom{2023}{2022} \tan^{2022} x} \end{aligned}$$

$$a_{2023} = \boxed{(C) - 1}$$

This problem is the same as problem 7.64 in the Art of Problem Solving textbook Precalculus chapter 7 that asks to prove $\tan nx = \frac{\binom{n}{1} \tan x - \binom{n}{3} \tan^3 x + \binom{n}{5} \tan^5 x - \binom{n}{7} \tan^7 x + \cdots}{1 - \binom{n}{2} \tan^2 x + \binom{n}{4} \tan^4 x - \binom{n}{6} \tan^6 x + \cdots}$

Solution 2 (Formula of tanx)

Note that $\tan kx = \frac{\binom{k}{1} \tan x - \binom{k}{3} \tan^3 x + \cdots \pm \binom{k}{k} \tan^k x}{\binom{k}{0} \tan^0 x - \binom{k}{2} \tan^2 x + \cdots \pm \binom{k}{k-1} \tan^{k-1} x}$, where k is odd and the sign of each term alternates between positive and negative. To realize this during the test, you should know the formulas of $\tan 2x$, $\tan 3x$, and $\tan 4x$, and can notice the pattern from that. The expression given essentially matches the formula of $\tan kx$ exactly. a_{2023} is evidently equivalent to $\pm \binom{2023}{2023}$, or 1. However, it could be positive or negative. Notice that in the numerator, whenever the exponent of the tangent term is congruent to 1 mod 4, the term is positive. Whenever the exponent of the tangent term is 3 mod 4, the term is negative. 2023, which is assigned to k , is congruent to 3 mod 4. This means that the term of $\binom{k}{k} \tan^k x$ is $\boxed{(C) - 1}$.

Notice: If you have time and don't know $\tan 3x$ and $\tan 4x$, you'd have to keep deriving $\tan kx$ until you see the pattern.

Solution 3

For odd n , we have

$$\begin{aligned} \tan nx &= \frac{\sin nx}{\cos nx} \\ &= \frac{\frac{1}{2i} (e^{inx} - e^{-inx})}{\frac{1}{2} (e^{inx} + e^{-inx})} \\ &= -i \frac{e^{inx} - e^{-inx}}{e^{inx} + e^{-inx}} \\ &= -i \frac{(\cos x + i \sin x)^n - (\cos x - i \sin x)^n}{(\cos x + i \sin x)^n + (\cos x - i \sin x)^n} \\ &= \frac{-2i \sum_{m=0}^{(n-1)/2} \binom{n}{2m+1} (\cos x)^{n-2m-1} (i \sin x)^{2m+1}}{2 \sum_{m=0}^{(n-1)/2} \binom{n}{2m} (\cos x)^{n-2m} (i \sin x)^{2m}} \end{aligned}$$

$$\begin{aligned}
&= \frac{\frac{1}{(\cos x)^n} \sum_{m=0}^{(n-1)/2} \binom{n}{2m+1} (\cos x)^{n-2m-1} (i \sin x)^{2m+1}}{i \frac{1}{(\cos x)^n} \sum_{m=0}^{(n-1)/2} \binom{n}{2m} (\cos x)^{n-2m} (i \sin x)^{2m}} \\
&= \frac{\sum_{m=0}^{(n-1)/2} \binom{n}{2m+1} (i \tan x)^{2m+1}}{i \sum_{m=0}^{(n-1)/2} \binom{n}{2m} (i \tan x)^{2m}} \\
&= \frac{\sum_{m=0}^{(n-1)/2} \binom{n}{2m+1} (\tan x)^{2m+1} i^{2m+1}}{\sum_{m=0}^{(n-1)/2} \binom{n}{2m} (\tan x)^{2m} i^{2m+1}} \\
&= \frac{\sum_{m=0}^{(n-1)/2} \binom{n}{2m+1} (\tan x)^{2m+1} (-1)^m}{\sum_{m=0}^{(n-1)/2} \binom{n}{2m} (\tan x)^{2m} (-1)^m}.
\end{aligned}$$

Thus, for $n = 2023$, we have

$$\begin{aligned}
a_{2023} &= \binom{2023}{2023} (-1)^{(2023-1)/2} \\
&= (-1)^{1011} \\
&= \boxed{\text{(C) } -1}.
\end{aligned}$$

Solution 4

We can use recursion to find a pattern for this problem. Notice that,

$$\begin{aligned}
\tan x &= \tan x \\
\tan 2x &= \frac{\tan x + \tan x}{1 - \tan^2 x} = \frac{2 \tan x}{1 - \tan^2 x} \\
\tan 3x &= \frac{\tan 2x + \tan x}{1 - \tan 2x \tan x} = \frac{3 \tan x - \tan^3 x}{1 - 3 \tan^2 x}
\end{aligned}$$

The coefficient of the highest degree term seems to be always ± 1 . Now, we prove this by an imcomplete mathematical induction.

- Firstly, we suppose n is odd, and $\tan nx = \frac{a_1 \tan x + a_3 \tan^3 x + \cdots + a_n \tan^n x}{a_0 + a_2 \tan^2 x + \cdots + a_{n-1} \tan^{n-1} x}$, then

$$\begin{aligned}
\tan(n+1)x &= \frac{\tan nx + \tan x}{1 - \tan nx \tan x} \\
&= \frac{\frac{a_1 \tan x + a_3 \tan^3 x + \cdots + a_n \tan^n x}{a_0 + a_2 \tan^2 x + \cdots + a_{n-1} \tan^{n-1} x} + \tan x}{1 - \frac{a_1 \tan x + a_3 \tan^3 x + \cdots + a_n \tan^n x}{a_0 + a_2 \tan^2 x + \cdots + a_{n-1} \tan^{n-1} x} \tan x} \\
&= \frac{a_1 \tan x + a_3 \tan^3 x + \cdots + a_n \tan^n x + a_0 \tan x + a_2 \tan^3 x + \cdots + a_{n-1} \tan^n x}{a_0 + a_2 \tan^2 x + \cdots + a_{n-1} \tan^{n-1} x - a_1 \tan^2 x - a_3 \tan^4 x - \cdots - \boxed{-a_n \tan^{n+1} x}}
\end{aligned}$$

The coefficient of the highest degree term becomes $-a_n$.

- Secondly, we suppose n is even, and $\tan nx = \frac{a_1 \tan x + a_3 \tan^3 x + \cdots + a_{n-1} \tan^{n-1} x}{a_0 + a_2 \tan^2 x + \cdots + a_n \tan^n x}$, then

$$\begin{aligned}
 \tan(n+1)x &= \frac{\tan nx + \tan x}{1 - \tan nx \tan x} \\
 &= \frac{\frac{a_1 \tan x + a_3 \tan^3 x + \cdots + a_{n-1} \tan^{n-1} x}{a_0 + a_2 \tan^2 x + \cdots + a_n \tan^n x} + \tan x}{1 - \frac{a_1 \tan x + a_3 \tan^3 x + \cdots + a_{n-1} \tan^{n-1} x}{a_0 + a_2 \tan^2 x + \cdots + a_n \tan^n x} \tan x} \\
 &= \frac{a_1 \tan x + a_3 \tan^3 x + \cdots + a_{n-1} \tan^{n-1} x + a_0 \tan x + a_2 \tan^3 x + \cdots + a_n \tan^{n+1} x}{a_0 + a_2 \tan^2 x + \cdots + a_n \tan^n x - a_1 \tan^2 x - a_3 \tan^4 x - \cdots - a_{n-1} \tan^n x}
 \end{aligned}$$

The coefficient of the highest degree term remains a_n .

When $n = 1$, $a_n = 1$. During the process of n increasing to 2023, a_n changed its sign a total of $2022 \div 2 = 1011$ times.

Hence, $a_{2023} = 1 \times (-1)^{1011} = \boxed{(\text{C}) - 1}$

Solution 5 (2 minutes, easy)

Notice that $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$, and $\tan 3x = \frac{3 \tan x - \tan^3 x}{1 - 3 \tan^2 x}$. Since this is the AMC, conjecture that some pattern is to be found within the last term of the expansion of $\tan nx$. Guess that the sign of the last term alternates between $+$ and $-$. Indeed, for odd coefficients the last term has a negative sign, so choose $\boxed{(\text{C}) - 1}$

§11 AMC 12 12B 2024

- (AMC 12 12B 2024) In a long line of people arranged left to right, the 1013th person from the left is also the 1010th person from the right. How many people are in the line?

(A) 2021 (B) 2022 (C) 2023 (D) 2024 (E) 2025

Answer: B

Solution for Certain China Testpapers

If the person is the 1015th from the left, that means there is 1014 people to their left. If the person is the 1010th from the right, that means there is 1009 people to their right. Therefore, there are $1014 + 1 + 1009 = \boxed{(\text{D}) 2024}$ people in line.

Solution 1

If the person is the 1013th from the left, that means there is 1012 people to their left. If the person is the 1010th from the right, that means there is 1009 people to their right. Therefore, there are $1012 + 1 + 1009 = \boxed{(\text{B}) 2022}$ people in line.

Solution 2

The person is 1013rd person from the left is also the 1010th person from the right, so the same person is counted twice.

Therefore, there are $1013 + 1010 - 1 = \boxed{(\text{B}) 2022}$ people in line.

$\boxed{(\text{B}) 2022}$

Solution 3

We can look at a smaller case where it is the 4th person from the left and the 2nd person from the right. Listing out the people as numbers gives us a list of

1, 2, 3, 4, 5. We see that the total number of people is the position from the left + the position from the right - 1, which in the case above is $4 + 2 - 1 = 5$. Plugging in 1013 and 1010 gives us $1013 + 1010 - 1 = \boxed{\text{(B)} 2022}$

Solution 4

Since the person is the 1013th person from the left and the 1010th person from the right, there are 1009 people after the 1013th person.

Therefore, there are $1013 + 1009 = \boxed{\text{(B)} 2022}$ people in line.

(Note: Similarly, you can perceive this information as 1012 people to the left of 1010 people and proceed.)

Solution 5 (Think of a Pattern)

If you are second going from the left and third going from the right, there are the third person out of four people p_1, p_2, p_3, p_4 .

We can simply add $1013 + 1010$ and subtract 1 since you are counting yourself twice, giving $1013 + 1009 = \boxed{\text{(B)} 2022}$

2. (AMC 12 12B 2024) What is $10! - 7! \cdot 6!$?

(A) -120 (B) 0 (C) 120 (D) 600 (E) 720

Answer: B

Solution 1

$$10! = 10 \cdot 9 \cdot 8 \cdot 7! = 720 \cdot 7! \quad 6! \cdot 7! = 720 \cdot 7!$$

Therefore, the equation is equal to $720 \cdot 7! - 720 \cdot 7! = \boxed{\text{(B)} 0}$

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$$0 - 5! = \boxed{\text{(A)} -120}$$

Solution 2

Factoring out $7!$ gives

$$7!(10 \cdot 9 \cdot 8 - 1 \cdot 6!).$$

Since $10 \cdot 9 \cdot 8 = 6! = 720$, the answer is $\boxed{\text{(B)} 0}$

Factoring $6!$ also works, it just makes the expression in the parenthesis a little harder to compute.

Solution 3

Note that $10! - 7! \cdot 6!$ must be divisible by 7, since

$$10! - 7! \cdot 6! = 7!(10 \cdot 9 \cdot 8 \cdot 6!) = 7 \cdot (6! \cdot 10 \cdot 9 \cdot 8 \cdot 6!)$$

and $\boxed{\text{(B)} 0}$ is the only option divisible by 7.

Solution 4

$10! - 7! \cdot 6!$ can be split into two parts, $10!$ and $7! \cdot 6!$. We can break the $6!$ into $(2 \cdot 4)(3 \cdot 5 \cdot 6)$. The $2 \cdot 4$ part makes 8, and the $3 \cdot 5 \cdot 6$ part makes 90, which is $9 \cdot 10$. We still have the $7!$, and we can multiply it by $8 \cdot 9 \cdot 10$. This is clearly equivalent to $10!$, so our solution is $10! - 10! = \boxed{\text{(B)} 0}$.

Solution 5

$10! = 3,628,800$, $7! = 5,040$, and $6! = 720$. $5,040 \cdot 720 = 3,628,800$. Therefore, $3,628,800 - 3,628,800 = \boxed{\text{(B) } 0}$.

-pepper2831

-minor edits by blakezhang

3. (*AMC 12 12B 2024*) For how many integer values of x is $|2x| \leq 7\pi$? (A) 16 (B) 17 (C) 19

Answer: E

Solution for Certain China Testpapers

Use the fact that $\pi \approx 3.14$, and thus you can get $6\pi \approx 18.84$. We could easily see that the answer is $\{-9, -8, \dots, 8, 9\} \Rightarrow \boxed{\text{(C) } 19}$

Solution 1

$\pi = 3.14159\dots$ is slightly less than $\frac{22}{7} = 3.\overline{142857}$. So $7\pi \approx 21.9$. The inequality expands to be $-21.9 \leq 2x \leq 21.9$. We find that x can take the integer values between -10 and 10 inclusive. There are $\boxed{\text{E. } 21}$ such values.

Note that if you did not know whether π was greater than or less than $\frac{22}{7}$, then you might perform casework. In the case that $\pi > \frac{22}{7}$, the valid solutions are between -11 and 11 inclusive: 23 solutions. Since, 23 is not an answer choice, we can be confident that $\pi < \frac{22}{7}$, and that $\boxed{\text{E. } 21}$ is the correct answer.

Test advice: If you are in the test and do not know if $\frac{22}{7}$ is bigger or smaller than π , you can use the extremely sophisticated method of just dividing $\frac{22}{7}$ via long division. Once you get to 3.142 you realise that you don't need to divide further since $\pi = 3.1416$ when rounded to 4 decimal places. Therefore, you do not include 11 and -11 and the answer is 21.

Solution 2

7π is incredibly close to 22, but doesn't reach it. This can be both computed by using $\pi \approx 3.142 \Rightarrow 7 \cdot 3.142 = 21.994 < 22$ or assumed. Therefore, including both positive and negative values, the answer is $\{-10, -9, \dots, 9, 10\} \Rightarrow \boxed{\text{(E) } 21}$.
Tacos_are_yummy_1

4. (*AMC 12 12B 2024*) Balls numbered $1, 2, 3, \dots$ are deposited in 5 bins, labeled A, B, C, D , and E , using the following procedure. Ball 1 is deposited in bin A , and balls 2 and 3 are deposited in B . The next three balls are deposited in bin C , the next 4 in bin D , and so on, cycling back to bin A after balls are deposited in bin E . (For example, 22, 23, $\dots$, 28 are deposited in bin B at step 7 of this process.) In which bin is ball 2024 deposited?

(A) A (B) B (C) C (D) D (E) E

Answer: D

Solution 1

Consider the triangular array of numbers:

1

2, 3

4, 5, 6

7, 8, 9, 10

11, 12, 13, 14, 15

$\vdots$

The numbers in a row congruent to 1 mod 5 will be in bucket A. Similarly, the numbers in a row congruent to 2, 3, 4, 0 mod 5 will be in buckets B, C, D, and E respectively. Note that the n^{th} row ends with the n^{th} triangle number, $\frac{n(n+1)}{2}$.

We must find values of n that make $\frac{n(n+1)}{2}$ close to 2024.

$$\frac{n(n+1)}{2} \approx 2024$$

$$n(n+1) \approx 4048$$

$$n^2 \approx 4048$$

$$n \approx 63$$

Trying $n = 63$ we find that $\frac{n(n+1)}{2} = 2016$. Since 2016 will be the last ball in row 63, ball 2024 will be in row 64. Since $64 \equiv 4 \pmod{5}$, ball 2024 will be placed in bucket D. D.

Solution 2

Here balls are going in bins in arithmetic progression.

Sum of no. of balls before 2024th ball shall be less than 2024.

Therefore,

$$2024 > \frac{n}{2} \times [2 \times (1) + (n-1) \times 1]$$

$$4048 > n(n+1)$$

Here greatest integral value of n can be 63.

$$63 \equiv 3 \pmod{5}$$

On 63rd go 63 balls will be dropped in bin c, ball 2016 will be dropped in bin C.

So on 64th go 2024th ball will be dropped in bin D.

Hence, Option (D) D

5. (AMC 12 12B 2024) In the following expression, Melanie changed some of the plus signs to minus signs:

$$1 + 3 + 5 + 7 + \cdots + 97 + 99$$

When the new expression was evaluated, it was negative. What is the least number of plus signs that Melanie could have changed to minus signs?

- (A) 14 (B) 15 (C) 16 (D) 17 (E) 18

Answer: B

Solution 1

Recall that the sum of the first n odd numbers is n^2 . Thus

$$1 + 3 + 5 + 7 + \cdots + 97 + 99 = 50^2 = 2500.$$

If we want to minimize the number of sign flips to make the number negative, we must flip the signs corresponding to the values with largest absolute value. This will result in the inequality

$$1 + 3 + 5 + \cdots + (2n - 3) + (2n - 1) - (2n + 1) - (2n + 3) - \cdots - 97 - 99 < 0.$$

The positive section of the sum will contribute n^2 , and the negative section will contribute $-(2500 - n^2) = (n^2 - 2500)$. The inequality simplifies to

$$n^2 + (n^2 - 2500) < 0$$

$$2n^2 < 2500$$

$$n^2 < 1250$$

The greatest positive value of n satisfying the inequality is $n = 35$, corresponding to 35 positive numbers, and B. 15 negatives.

ALEX

Solution 2

The formula for the sum of all odd positive integers from 1 to $2n - 1$ is

$$1 + 3 + 5 + \cdots + 2n - 3 + 2n - 1 = n^2.$$

Therefore, the given sum evaluates to

$$99 = 2(50) - 1 \implies 50^2 = 2500.$$

Since we're looking for the minimum possible sign changes, we focus on the largest numbers in the set to change to negative.

If we change the sign of a number n to negative, then the sum decreases by $2n$. Therefore, we're looking for a subset of numbers that add to greater than $2500/2 = 1250$.

Now we look at the answer choices.

(A) 14 means that we're changing the signs of the numbers $\{73, 75, \dots, 99\}$, and $73 + 75 + \cdots + 99 = \frac{(73 + 99) \cdot (\frac{99 - 73}{2} + 1)}{2} = 1204$.

Now this so slightly happens to be less than 1250, and (B) 15 means we're adding 71 to the set, too. Since $71 > 1250 - 1204 = 46$, then the answer is (B) 15

Solution 3

Note that as mentioned above, $1 + 3 + 5 \dots + 2n - 1 = n^2$. So, the sum of the first 50 odd numbers is equal to 2500, and we need to find the smallest integer such that $2k^2 < 50^2$. Taking the square root, $k = 25\sqrt{2}$, keeping in mind we need to take the floor $2k^2$ needs to be slightly less than 2500. Using an approximation of $\sqrt{2} = 1.4$, $25 * \frac{7}{5}$ is 35, so our answer is (B) 15 aleyang

Solution 4 (Simple Algebra)

As mentioned above, $1 + 3 + 5 \dots + 2n - 1 = n^2$. This will be especially important in this solution. The sum of the original expression is 2500, so the combined (positive) sum of the negatives must be greater than half of this, i.e. 1250. We are looking to minimize the number of negative numbers, so we only convert the last numbers to negatives. Let $99 = o_1 = 100 - 1$, $97 = o_2 = 100 - 3$, and for any n , $o_n = 100 - (2n - 1)$. Then the sum of all numbers up till o_n is $100n - n^2$. This sum needs to be greater than 1250, i.e.:

$$100n - n^2 > 1250$$

Testing out n from the solution set yields the minimum value of $n = 15$. The answer is therefore (B) 15.

6. (*AMC 12 12B 2024*) The national debt of the United States is on track to reach $5 \cdot 10^{13}$ dollars by 2033. How many digits does this number of dollars have when written as a numeral in base 5? (The approximation of $\log_{10} 5$ as 0.7 is sufficient for this problem.)
- (A) 18 (B) 20 (C) 22 (D) 24 (E) 26

Answer: B

Solution 1

Generally, the number of digits of number n in base b is

$$\lfloor \log_b n \rfloor + 1.$$

In this question, it is $\lfloor \log_5(5 \times 10^{13}) \rfloor + 1$. Note that

$$\begin{aligned} \log_5(5 \times 10^{13}) &= 1 + \frac{13}{\log_{10} 5} \\ &\approx 1 + \frac{13}{0.7} \\ &\approx 19.5 \end{aligned}$$

Hence, our answer is (B) 20

Solution 2

We see that $5 \times 10^{13} = 2^{13} \cdot 5^{14}$ and $2^{13} = 8192$. Converting this to base 5 gives us 230232 (trust me it doesn't take that long). So the final number in base 5 is 230232 with 14 zeroes at the end, which gives us $6 + 14 = 20$ digits. So the answer is (B) 20.

Note - Base Conversion Step

To convert the number 8192 from base 10 to base 5, we follow these steps:

1. Divide the number by 5 repeatedly, noting the quotient and remainder each time.
2. Stop when the quotient becomes 0, then read the remainders from bottom to top.

$$8192 \div 5 = 1638 \text{ remainder } 2$$

$$1638 \div 5 = 327 \text{ remainder } 3$$

$$327 \div 5 = 65 \text{ remainder } 2$$

$$65 \div 5 = 13 \text{ remainder } 0$$

$$13 \div 5 = 2 \text{ remainder } 3$$

$$2 \div 5 = 0 \text{ remainder } 2$$

Now, reading the remainders from bottom to top: 2, 3, 0, 2, 3, 2.

Thus, 8192 in base 5 is:

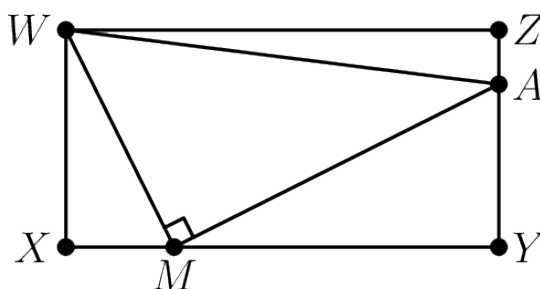
Note - an approximation you can use here to convert 8192 to base 5: since $5^5 = 3125$ and $5^6 = 15625$, we automatically know that you need 6 digits to represent 2^{13} in base 5.

Solution 3

$5 \times 10^{13} = 5 \times (5^{13} \times 2^{13}) = 2^{13} \times 5^{14} = 8192 \times 5^{14}$. $5^5 = 3125$ and $5^6 = 15625$ (or just notice that it must be > 8192) $\implies 5^5 < 8192 < 5^6 \implies 5^{19} < 5 \times 10^{13} < 5^{20}$.

Since an integer x has n base- a digits when it satisfies $a^{n-1} \leq x < a^n$, it follows that 5×10^{13} requires **(B) 20** base-5 digits.

7. (AMC 12 12B 2024) In the figure below $WXYZ$ is a rectangle with $WX = 4$ and $WZ = 8$. Point M lies $\overline{XY}$, point A lies on $\overline{YZ}$, and $\angle WMA$ is a right angle. The areas of $\triangle WXM$ and $\triangle WAZ$ are equal. What is the area of $\triangle WMA$?



- (A) 13 (B) 14 (C) 15 (D) 16 (E) 17

Answer: C

Solution 1

We know that $WX = 4$, $WZ = 8$, so $YZ = 4$ and $YX = 8$. Since $\angle WMA = 90^\circ$, triangles WXM and MYA are similar. Therefore, $\frac{WX}{MY} = \frac{XM}{YA}$, which gives $\frac{4}{8-XM} = \frac{XM}{4-ZA}$. We also know that the areas of triangles WXM and WAZ are equal, so $WX \cdot XM = WZ \cdot ZA$, which implies $4 \cdot XM = 8 \cdot ZA$. Substituting this

into the previous equation, we get $\frac{4}{8-2ZA} = \frac{2ZA}{4-ZA}$, yielding $ZA = 1$ and $XM = 2$. Thus,

$$\triangle WMA = 4 \cdot 8 - \frac{4 \cdot 2}{2} - \frac{8 \cdot 1}{2} - \frac{6 \cdot 3}{2} = \boxed{\text{(C) } 15}$$

Solution 2

Let $XM = b$, $ZA = a$, $4 \cdot b = 8 \cdot a$, $b = 2a$,

$$WM^2 + AM^2 = AW^2$$

$$(b^2 + 4^2) + (4 - a)^2 + (8 - b)^2 = (a^2 + 8^2)$$

$$a = 1, b = 2,$$

$$\triangle WMA = \text{area}(WXYZ) - \triangle WZA - \triangle WXM - \triangle MYA = 32 - 4 - 4 - 9 = \boxed{\text{(C) } 15}$$

Solution 3 (Pythagorean Theorem)

Assign ZA as x , then AY as $4 - x$. Assign XM as y and MY as $8 - y$. Since triangles WXM and WZA are equal in area, we can say $8x = 4y$, so $y = 2x$. Then, therefore, XM is $2x$ and MY has length $8 - 2x$. We can use the Pythagorean theorem to find WM , which is actually $\sqrt{(2x)^2 + 4^2} = \sqrt{4x^2 + 16}$. We don't factor it yet - we are going to find x again using the Pythagorean Theorem. Similarly, finding MA is just the square root of the squares of AY and MY individually, or $\sqrt{(8 - 2x)^2 + (4 - x)^2} = \sqrt{64 - 32x + 4x^2 + 16 - 8x + x^2} = \sqrt{5x^2 - 40x + 80}$. Then simply, WA is really $\sqrt{x^2 + 64}$.

Now we have the three sides of the right triangle: $\sqrt{4x^2 + 16}$, $\sqrt{5x^2 - 40x + 80}$, and $\sqrt{x^2 + 64}$. Per the Pythagorean theorem again, we can see $(4x^2 + 16) + (5x^2 - 40x + 80) = (x^2 + 64)$. Combining like terms gives us $8x^2 - 40x + 32 = 0$, then dividing by 8 gives $x^2 - 5x + 4 = 0$. As this elementary and well-known quadratic gives us the roots of 1 and 4, we can see it is a bit weird to have $x = 4$, as then point Z is point A . So we'll assume $x = 1$. We have two legs of the triangle by plugging in the sides with x in them, given that $x = 1$: $\sqrt{20}$ and $\sqrt{45}$. We should know that $20 \cdot 45 = 900$, and $\sqrt{900} = 30$. Dividing by 2 reveals us our answer:

$$\boxed{\text{(C) } 15}$$

note: to further prove that 1 is the correct root, note that setting A at Z makes WZA have area 0. Therefore, WXM also has to have an area of 0. This makes it impossible for angle WMA to be a right angle, as WMA would simply become a right triangle with half the area of $WXYZ$, making MWA the right angle.

Solution 4 (Similar Triangles)

We are given $WX = 4$, $WZ = 8$. $\triangle WXM$ and $\triangle MYA$ have equal area, so let $XM = 2x$ and $ZA = x$. $MY = 8 - 2x$ and $AY = 4 - x$. From this, we can conclude that $\frac{MY}{AY} = \frac{8-2x}{4-x} = \frac{2}{1}$

Since WM intersects parallel lines WZ and XY , $\angle ZWM = \angle WMZ$. $\angle ZWM + \angle MWX = 90^\circ$, so $180^\circ - 90^\circ = \angle WMZ + \angle AMY$. Thus, $\angle MWX = \angle AMY$ and $\triangle WXM \sim \triangle MYA$ due to AA Similarity.

Corresponding sides of similar triangles are proportional, so $\frac{WX}{XM} = \frac{MY}{AY}$ or $\frac{4}{2x} = \frac{2}{1}$. It is clear that $2x = 2$, and $x = 1$. Now, all we have to do is subtract the area of the rectangle by each of the three triangles.

$$\Delta WMA = 8 \cdot 4 - \left(\frac{1}{2} \cdot 4 \cdot 2\right) - \left(\frac{1}{2} \cdot 8 \cdot 1\right) - \left(\frac{1}{2} \cdot 6 \cdot 3\right) \quad \Delta WMA = 32 - 4 - 4 - 9$$

$$\Delta WMA = \boxed{\text{(C) } 15}$$

Solution 5 (Desperate)

Taking a look at the diagram, points M and A seem to divide $\overline{XY}$ and $\overline{YZ}$ into $1 : 3$ ratios. With this assumption in mind, $XM = 2$, $MY = 6$, $YA = 3$, and $AZ = 1$. This means that the areas of $\triangle WXM$ and $\triangle WZA$ are $\frac{4(2)}{2} = 4$ and $\frac{8(1)}{2} = 4$ respectively, and the area of $\triangle MAY$ is $\frac{6(3)}{2} = 9$. The area of rectangle $WXYZ = 4(8) = 32$, so subtracting the areas of triangles WXM , WZA , and MAY , we get that the area of $\triangle WMA = 32 - 4 - 4 - 9 = \boxed{\text{(C) } 15}$

China Test Solution (Finding $\triangle MAY$) From solution 3, instead of finding the area of $\triangle WMA$, we instead find the area of $\triangle MAY$. Then $x = 1$ then we have $MA = 8 - 2x = 6$. Again, since $AY = 4 - x$, then $AY = 4 - 1 = 3$. The area of a triangle with legs 3 and 6 is $\frac{3 \cdot 6}{2} = \boxed{9}$.

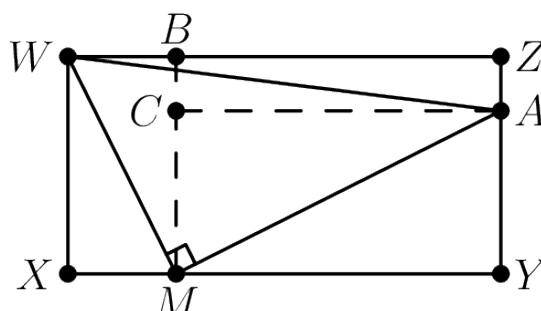
-minor edits by fireball9746

can someone pls explain why $\triangle MAY$ has the same area as $\triangle WMA$

^ To the person above: Under the problem, there's a note that says some tests in China asked for the area of $\triangle MAY$. - trevian1

Solution 6 (cheese)

Of course I don't endorse this method, but...



By looking at partitioned rectangles in the diagram, it's clear that the area of triangle WMA is very close to half the area of the overall rectangle $WXYZ$. So $\boxed{\text{(C) } 15}$ is our best bet.

Obvious note:

The answer is less than half of the overall area of rectangle $WXYZ$, as triangle AWZ clearly sticks out past rectangle $ACBZ$ into rectangle $XMBW$ by the smallest margin, so therefore, the next smallest answer past (C) 16, is $\boxed{\text{(C) } 15}$.

8. (AMC 12 12B 2024) What value of x satisfies

$$\frac{\log_2 x \cdot \log_3 x}{\log_2 x + \log_3 x} = 2?$$

- (A) 25 (B) 32 (C) 36 (D) 42 (E) 48

Answer: C

Solution 1

We have

$$\begin{aligned}\log_2 x \cdot \log_3 x &= 2(\log_2 x + \log_3 x) \\ 1 &= \frac{2(\log_2 x + \log_3 x)}{\log_2 x \cdot \log_3 x} \\ 1 &= 2\left(\frac{1}{\log_3 x} + \frac{1}{\log_2 x}\right) \\ 1 &= 2(\log_x 3 + \log_x 2) \\ \log_x 6 &= \frac{1}{2} \\ x^{\frac{1}{2}} &= 6 \\ x &= 36\end{aligned}$$

so (C) 36

Solution 2 (Change of Base)

$$\begin{aligned}\frac{\log_2 x \cdot \log_3 x}{\log_2 x + \log_3 x} &= 2 \\ \log_2 x \cdot \log_3 x &= 2(\log_2 x + \log_3 x) \\ \log_2 x \cdot \log_3 x &= 2 \log_2 x + 2 \log_3 x \\ \frac{\log x}{\log 2} \cdot \frac{\log x}{\log 3} &= 2 \frac{\log x}{\log 2} + 2 \frac{\log x}{\log 3} \\ \frac{(\log x)^2}{\log 2 \cdot \log 3} &= \frac{2 \log x \cdot \log 3 + 2 \log x \cdot \log 2}{\log 2 \cdot \log 3} \\ (\log x)^2 &= 2 \log x \cdot \log 3 + 2 \log x \cdot \log 2 \\ (\log x)^2 &= 2 \log x (\log 2 + \log 3) \\ \log x &= 2(\log 2 + \log 3) \\ x &= 10^{2(\log 2 + \log 3)} \\ x &= (10^{\log 2} \cdot 10^{\log 3})^2 \\ x &= (2 \cdot 3)^2 = 6^2 = \boxed{\text{(C) } 36}\end{aligned}$$

Solution 3 (Using Variables)

Let $a = \log_2 x$ and $b = \log_3 x$. This gives us the equation

$$\frac{ab}{a+b} = 2.$$

Then, from our definitions of a and b , $2^a = x$ and $3^b = x$, so $2^a = 3^b$. Taking the logarithm base 3 of both sides of this equation gives us $\log_3 2^a = b$, hence $a \log_3 2 = b$. Now, we substitute $a \log_3 2$ for b in the equation, which gives

$$\frac{a \cdot a \log_3 2}{a + a \log_3 2} = 2.$$

Notice that we can factor out an a in the numerator and denominator, if $a \neq 0$, and doing so yields

$$\frac{a \log_3 2}{1 + \log_3 2} = 2.$$

We know that $1 = \log_3 3$, so putting that in gives us

$$\frac{a \log_3 2}{\log_3 3 + \log_3 2} = 2 \implies \frac{a \log_3 2}{\log_3 6} = 2.$$

So, $a = 2 \cdot \frac{\log_3 6}{\log_3 2}$, which, using the change of base formula, is equivalent to $2 \cdot \log_2 6$, thus,

$$a = 2 \cdot \log_2 6 = \log_2 6^2 = \log_2 36.$$

Finally, using our original definition of a , we have

$$a = \log_2 x = \log_2 36,$$

so $x = \boxed{\text{(C) } 36}$.

Solution 4 (Educated Guess)

Prime factorize all the options to get

- A) 5^2
- B) 2^5
- C) $2^2 3^2$
- D) $2 \cdot 3 \cdot 7$
- E) $2^4 3$

Since the LHS of the equation

$$\frac{\log_2 x \cdot \log_3 x}{\log_2 x + \log_3 x} = 2$$

is symmetric in 2 and 3 we can guess that the answer has equal powers of 2 and 3 to get $\boxed{\text{(C) } 36}$. (Note: you can verify this quickly by changing all the logs base 3 to logs base 2 or vice versa) GoldemGem4848

9. (AMC 12 12B 2024) A dartboard is the region B in the coordinate plane consisting of points (x, y) such that $|x| + |y| \leq 8$. A target T is the region where $(x^2 + y^2 - 25)^2 \leq 49$. A dart is thrown and lands at a random point in B . The probability that the dart lands in T can be expressed as $\frac{m}{n} \cdot \pi$, where m and n are relatively prime positive integers. What is $m + n$? (A) 39 (B) 71 (C) 73 (D) 75 (E) 135

Answer: B

Solution 1

Inequalities of the form $|x| + |y| \leq 8$ are well-known and correspond to a square in space with centre at origin and vertices at $(8, 0)$, $(-8, 0)$, $(0, 8)$, $(0, -8)$. The diagonal length of this square is clearly 16, so it has an area of

$$\frac{1}{2} \cdot 16 \cdot 16 = 128$$

Now,

$$(x^2 + y^2 - 25)^2 \leq 49$$

Converting to polar form,

$$r^2 - 25 \leq 7 \implies r \leq \sqrt{32},$$

and

$$r^2 - 25 \geq -7 \implies r \geq \sqrt{18}.$$

The intersection of these inequalities is the circular region T for which every circle in T has a radius between $\sqrt{18}$ and $\sqrt{32}$, inclusive. The area of such a region is thus $\pi(32 - 18) = 14\pi$. The requested probability is therefore $\frac{14\pi}{128} = \frac{7\pi}{64}$, yielding $(m, n) = (7, 64)$. We have $m + n = 7 + 64 = \boxed{\text{(B) } 71}$.

-AbhiSood1234, countmath1

Solution 2 (Calculus)

Expressing the Area of Region B

Region B consists of points where $|x| + |y| \leq 8$

In each quadrant, this can be expressed by the following functions:

First quadrant: $y = 8 - x$ Second quadrant: $y = 8 + x$ Third quadrant: $y = -8 - x$ Fourth quadrant: $y = -8 + x$

In the first quadrant, x ranges from 0 to 8, and y ranges from 0 to $8 - x$. Thus, the area in the first quadrant is:

$$\begin{aligned} \text{Area of first quadrant} &= \int_0^8 \int_0^{8-x} dy \, dx \\ &= \int_0^8 [y]_{y=0}^{y=8-x} dx = \int_0^8 (8 - x) dx \\ &= \left[8x - \frac{x^2}{2} \right]_0^8 = 64 - 32 = 32 \end{aligned}$$

The total area of region B is:

$$\text{Area of } B = 4 \times 32 = 128$$

Expressing the Area of Region T Region T is defined by the inequality $(x^2 + y^2 - 25)^2 \leq 49$, which can be rewritten as:

$$18 \leq x^2 + y^2 \leq 32$$

To find the area, we switch to polar coordinates with $x = r \cos \theta$ and $y = r \sin \theta$, where $x^2 + y^2 = r^2$. Here, r ranges from $\sqrt{18}$ to $\sqrt{32}$, and θ ranges from 0 to 2π .

The area of T can then be found by:

$$\begin{aligned} \text{Area of } T &= \int_0^{2\pi} \int_{\sqrt{18}}^{\sqrt{32}} r \, dr \, d\theta \\ &= \int_0^{2\pi} \left[\frac{r^2}{2} \right]_{r=\sqrt{18}}^{r=\sqrt{32}} d\theta = \int_0^{2\pi} \left(\frac{32}{2} - \frac{18}{2} \right) d\theta \\ &= \int_0^{2\pi} 7 \, d\theta = 14\pi \end{aligned}$$

The probability P that a dart lands in region T is the area of T divided by the area of B :

$$P = \frac{\text{Area of } T}{\text{Area of } B} = \frac{14\pi}{128} = \frac{7\pi}{64}$$

So the probability is of the form $\frac{m}{n}\pi$, where $m = 7$ and $n = 64$, so $m + n = 7 + 64 = 71$.

(B) 71

By the way never pull out calculus on the amc10. It is time consuming and easy to make mistakes on [minor edits by hao11]

Solution 3

Area under this graph

$$|x| + |y| \leq 8 \Rightarrow (8 \times \sqrt{2})^2 = 128$$

$$(x^2 + y^2 - 25)^3 - 7 \leq 0 \Rightarrow (x^2 + y^2 - 25 - 7) \times (x^2 + y^2 - 25 + 7) \leq 0$$

$$\therefore x^2 + y^2 \in [18, 32]$$

Area under such curves is;

$$\therefore A = \pi \times (32 - 18) = 14 \times \pi$$

$$P = \frac{14 \times \pi}{128} = \frac{7 \times \pi}{64} = \frac{m}{n}$$

$$m + n = 7 + 64 = 71$$

i.e, option (B)

10. (AMC 12 12B 2024) A list of 9 real numbers consists of 1, 2.2, 3.2, 5.2, 6.2, and 7, as well as x, y, z with $x \leq y \leq z$. The range of the list is 7, and the mean and median are both positive integers. How many ordered triples (x, y, z) are possible?
- (A) 1 (B) 2 (C) 3 (D) 4 (E) infinitely many

Answer: C

Solution 1 (Main)

First Case

We start off by knowing that there must exist an ordered pair $(0, y, z)$ and a big term 7 such that the range is satisfied and the median could also be satisfied. Because $x \leq y \leq z$, we say $x = 0$. Then, we get the sum 24.8. We need $\frac{24.8 + y + z}{9} \in \mathbb{Z}$. This means that we need the nearest multiple of 9, which is 36, and we get $y + z = 11.2$. We need one of these to be the median, WLOG say y . Then, $y \in \{4, 5\}$. So if $y = 4$,

we get $z = 7.2$, and if $y = 5$, we get 6.2 . We see that if $z = 6.2$, the range will still remain as 7 , and therefore the one ordered triple $(0, 5, 6.2)$ satisfies this.

We know that the median can be y , so we have $y \in \{4, 5\}$. We do casework on 4 and 5 .

Case 1

Say $y = 4$. Then, because we already know there exists an ordered triple where z is not the largest, there must exist at least one ordered triple where z is, so we say $z - x = 7$. We also know that the mean must be divisible by 9 . Quickly summing each number up we get 24.8 . The next number divisible by 9 is 36 . We add y to get 28.8 . We then know $x + z = 36 - 28.8 = 7.2$. The smallest values give $x = 0.1$ and $z = 7.1$. This satisfies our constraints.

We don't want any errors, so we check the next sum, which is 45 . We get then $7.2 = x + z$ and $z - x = 7$. This gives $x = 4.6$ and $z = 11.6$. This is a contradiction, as $x \leq y \leq z$, and the values incrementally become too large, and therefore we only have one ordered triple for this case, which is $(0.1, 4, 7.1)$

Case 2

We now check $y = 5$. This makes our sum 29.8 . The nearest multiple of 9 is 36 , again. This gives us $6.2 = x + z$ and $z - x = 7$. Solving gives us -0.4 and 6.6 . This however doesn't work, as 6.6 is not the greatest value.

We decide to check the next multiple of 9 , which is 45 . This gives us $15.2 = x + z$ and $z - x = 7$. This gives us $x = 4.1$ and $z = 11.1$. This also doesn't work, as x becomes incrementally larger, and therefore there are no ordered triples that satisfy this.

Last Case

We have one more case, and that is x is not the first number, but z is the last, meaning $z - 1 = 7$, and $z = 8$. This means that $\frac{24.8+x+y+z}{9} \in \mathbb{Z}$. So, if $z = 8$, we get 32.8 . We again see that we need the nearest multiple of 9 which is 36 , and we get $3.2 = x + y$. We see that x and y still need to be a integer median and 3.2 is too small, so 36 cannot work. We try 45 , and see $12.2 = x + y$. After some trial and error, we see that $x = 6$ and $y = 6.2$, giving us the ordered triple $(6, 6.2, 8)$. Continuing as before shows us that x again incrementally increases, and therefore we have only one ordered triple.

We have (C) 3 ordered triples. These are $(0.1, 4, 7.1)$, $(0, 5, 6.2)$, and $(6, 6.2, 8)$.

Solution 2 (Circular Arrangements)

This is a more advanced solution, but it gives the answer really fast. Plus, it is always better to expand your knowledge.

This solution is so overpowered that you don't even need to sum the decimals up. :-)

Consider the arrangements like a circle. There are 3 terms; x , y , and z , so we arrange them into a circle. We use the circular gap theorem to solve this, which is

$$C(nA) = \frac{n}{k} \binom{n-k-1}{k-1}$$

We see that there are 3 terms. We need either 1 adjacent from the other, two adjacent from one, or three adjacent.

Also keep in note that if the combinatorial part of $C(nA)$ is invalid, it is just 0.

We then have $k \in \{1, 2, 3\}$.

We then have 3 cases,

Case 1: $k \in \{1\}$

We have $\frac{3}{1} \cdot \binom{3-1-1}{1-1} = \frac{3}{1} \cdot 1 = 3$ ways to do this.

Case 2: $k \in \{2\}$

We have $\frac{3}{2} \cdot \binom{3-2-1}{2-1}$. The combinatorial part of this is $\binom{0}{1}$, which is invalid and therefore 0. Therefore there are $\frac{3}{2} \times 0 = 0$ cases.

Case 3: $k \in \{3\}$

We have $\frac{3}{3} \cdot \binom{3-3-1}{3-1}$. The combinatorial part of this is $\binom{-1}{2}$, and again, this is invalid, and therefore 0. There is then $1 \times 0 = 0$ cases.

Our answer is $3 + 0 + 0 = \boxed{\text{(C)} 3}$

Disclaimer Solution 2 This theorem requires practice to use and when to use it. For example, if they said that this list is non circular or is strictly linear, then we may not solve it like this. However, these specifications have not been said in this problem.

For disclaimer, don't use this formula everywhere. A clear example is that in a problem involving linear arrangements, this method will never work.

Below are some practice problems that I highly recommend you try so you can understand this theorem and its applications better.

Check them below.

2014 AMC 10B Q24

2017 AMC 10B Q18

1983 AIME Q7

Also, it is helpful to note $C_n(A) = \frac{n}{k} \binom{n-k-1}{k-1} = \binom{n}{2k}$.

Solution 3 (AMC 10)

The sum of the six existing numbers is 24.8. For the mean to be an integer, the sum of the remaining three numbers x, y, z and the remaining six must be divisible by 9; thus $x + y + z$ must equal either 2.2, 11.2, 20.2. However, one of x, y, z must be the median since the middle four numbers in the original set are all non-integral. As a result, we can eliminate 2.2 since the sum would be too small to allow for one of x, y, z to be the median (notice that negative numbers are not permitted since the range would be greater than 7). This leaves two cases.

For the $x + y + z = 20.2$ case, notice that all three must be less than or equal to 8 due to the range restriction. However, also notice that y and z must be on the higher end of the range, meaning that the new median would have to fall between the new middle numbers 5.2 and 6.2. Thus $x = 6$, and we can manipulate the numbers to make the range 7 by making $z = 8$ and thus $y = 6.2$, providing one case in $(6, 6.2, 8)$.

For the $x + y + z = 11.2$ case, we note that the average of the three would fall near the median, signaling that one of the numbers would be the median, one would fall lesser than the median, and one would reach greater than the median. Thus we let the median be y . Since adding a number to each side of the set would not change

the median, we know that the median must fall between the two middle numbers 3.2, 5.2. Then we find two triples (0.1, 4, 7.1) and (0, 5, 6.2), resulting in **(C) 3**.

Solution 4 (AMC 12)

We can do casework on the range. We have four possible cases:

$$x < 1, z < 7 \quad x > 1, z < 7 \quad x < 1, z > 7 \quad x > 1, z > 7$$

These encapsulate all possible values of x and z we can choose, so we're not leaving anything out. As we'll see, the problem will become simpler this way. Since we want integer mean, first note the sum of the values given to be 24.8: when we add x, y, z , it must be a multiple of 9 to yield an integer mean. Also remember that the median of an increasing list of 9 numbers is the fifth number.

Case 1:

Since $x < 1, z < 7$, x must be the minimum of this list of numbers and 7 is the max: so the range is $7 - x$. This must be equal to 7, so $x = 0$. Now we try to make the median an integer. With $x = 0$, the fifth smallest number currently is 5.2, which is clearly not an integer. But if we stick y or z in between 3.2 and 5.2, then the median will be an integer. So let one of y, z be 4: then the sum of all the numbers we have so far is 28.8. To make this a multiple of 9, the last number has to be 7.2 to add up to 36. However, we said that $z < 7$, so this violates the bounds of this case. So then we set $y = 5$, so our sum is 29.9: then $z = 36 - 29.9 = 6.2$. Here everything checks out, so for this case there is one solution: (0, 5, 6.2).

Case 2:

Since $x > 1$ and $z < 7$, the minimum is 1 and the maximum is 7. Thus the range is always 6, but we need the range to be 7, so this case has no solutions.

Case 3:

We have $x < 1, z > 7$. Our minimum is x and our maximum is z , so the range is $z - x = 7$. Since $x < 1$, the median still falls on 5.2, so y can be either 4 or 5 to make the median an integer. For now, suppose $y = 4$. Our sum is now $28.8 + x + z$, which we set to the nearest multiple of 9: 36. Thus $x + z = 7.2$. Now we have a system of equations: we add them to get $2z = 14.2 \implies z = 7.1 \implies x = 0.1$. Since the values of z and x satisfy our case, this is a possible solution.

Now suppose $y = 5$. Our sum is $29.8 + x + z$, which must be equal to 36, so $x + z = 6.2$. Adding the equations, we have $2z = 13.2 \implies z = 6.6$. But this violates our assumption that $z > 7$, so this has no solutions.

Therefore this case has one solution: (0.1, 4, 7.1).

Case 4:

Since $x > 1, z > 7$, the max is z and the min is 1. Thus $z - 1 = 7 \implies z = 8$. Now the median is between 5.2 and 6.2, so x or y must take the value 6. Our sum is now $24.8 + 6 + 8 = 38.8$, and adding the last number has to equal 45. Thus the last number is 6.2, and this yields one solution: (6, 6.2, 8).

Therefore we've found three solutions. These cases encompass all possible real values of x and z , so we know we've left nothing out. The answer is **(C) 3**.

Solution 5

We can start doing casework on the numbers. Notice that the median can either be four, five or six because there has to be one or two numbers on one side. We can

start bashing everything out, starting from letting $x = a, y = 4, z = 7 + a$. Now plugging into the sequence, we get that $a, 1, 2.2, 3.2, 4, 5.2, 6.2, 7, 7 + a$. Notice that the sum of the numbers not including x, y, z is 24.8. To make this mean an integer, $x + y + z$ must have a decimal 0.2 when added up. Now, plugging in our values into the mean gets us $35.8 + 2a = 9k$ for some integer k . Notice that $0 \leq a \leq 1$, so the only value for a is 0.1, so we found our first ordered triplet, $(0.1, 4, 7.1)$. Let's make the median 5 now. Then we have to put x and one side and z on the other to "balance" the numbers, or make it so that there are 4 numbers on each side of 5. Now, we can see that x is 0, and solving for z , we get that $z = 6.2$. We get our second triple is $(0, 5, 6.2)$. Now coming to our third case scenario, we use 6 as the median. Clearly it means that x has to be the median, so that y and z can be on the same side as x .

$$1, 2.2, 3.2, 5.2, 6, 6.2, y, 7, z$$

We will use this diagram. Notice that z has to be 8, and solving for y yields $y = 6.2$, so we found our third and last triplet, $(6, 6.2, 8)$, meaning that $\boxed{C}$ is the right answer.

Solution 6 (Cheese)

Answer choice E is too big and answer choices A,B are both too small, so get rid of A,B, and E. Then, after checking the answer choices respectively, you get 3 as the final answer. Therefore, the answer is answer choice $\boxed{(C) 3}$.

Note: The answer to AMC problems is NEVER, EVER infinitely many. Also, the top part wasn't mine, it was somebody else.

Solution 7 (A (still interesting but slightly better) cheese)

Ignore the fact that $x \leq y \leq z$. Then we know that there should be some ordered pair $(0, y, z)$, $(x, 0, z)$, and $(x, y, 0)$, where we can permute x, y , or z such that they do satisfy $x \leq y \leq z$. Our answer is then $\boxed{(C) 3}$.

11. (AMC 12 12B 2024) Let $x_n = \sin^2(n^\circ)$. What is the mean of $x_1, x_2, x_3, \dots, x_{90}$?
- (A) $\frac{11}{45}$ (B) $\frac{22}{45}$ (C) $\frac{89}{180}$ (D) $\frac{1}{2}$ (E) $\frac{91}{180}$

Answer: E

Solution 1

Add up x_1 with x_{89} , x_2 with x_{88} , and x_i with x_{90-i} . Notice

$$x_i + x_{90-i} = \sin^2(i^\circ) + \sin^2((90-i)^\circ) = \sin^2(i^\circ) + \cos^2(i^\circ) = 1$$

by the Pythagorean identity. Since we can pair up 1 with 89 and keep going until 44 with 46, we get

$$x_1 + x_2 + \dots + x_{90} = 44 + x_{45} + x_{90} = 44 + \left(\frac{\sqrt{2}}{2}\right)^2 + 1^2 = \frac{91}{2}$$

Hence the mean is $\boxed{(E) \frac{91}{180}}$

Solution 2

We can add a term x_0 into the list, and the total sum of the terms won't be affected since $x_0 = 0$. Once x_0 is added into the list, the average of the 91 terms is clearly

$\frac{1}{2}$. Hence the total sum of the terms is $\frac{91}{2}$. To get the average of the original 90, we merely divide by 90 to get $\frac{91}{180}$. Hence the mean is $\boxed{\text{(E)} \frac{91}{180}}$

This method is called constructing a variable (although most of you already know).

Solution 3 (Inductive Reasoning)

If we use radians to rewrite the question, we have: $x_n = \sin^2\left(\frac{n\pi}{2 \times 90}\right)$. Notice that 90 have no specialty beyond any other integers, so we can use some inductive processes.

If we change 90 to 2:

$$\frac{\sin^2\left(\frac{\pi}{4}\right) + \sin^2\left(\frac{2\pi}{4}\right)}{2} = \frac{\left(\frac{1}{\sqrt{2}}\right)^2 + (1)^2}{2} = \frac{\frac{1}{2} + 1}{2} = \frac{3}{4}.$$

If we change 90 to 3:

$$\frac{\sin^2\left(\frac{\pi}{6}\right) + \sin^2\left(\frac{2\pi}{6}\right) + \sin^2\left(\frac{3\pi}{6}\right)}{3} = \frac{\left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2 + (1)^2}{3} = \frac{\frac{1}{4} + \frac{3}{4} + 1}{3} = \frac{2}{3} = \frac{4}{6}.$$

By intuition, although not rigorous at all, we can guess out the solution if we change 90 into k , we get $\frac{k+1}{2k}$. Thus, if we plug in $k = 90$, we get $\frac{90+1}{2 \times 90} = \boxed{\text{(E)} \frac{91}{180}}$

Solution 4

Using identity

$$\sin(90 - \theta) = \cos \theta$$

$$\frac{x_1 + x_2 + x_3 + \dots + x_{90}}{90} = \frac{\sin^2 1 + \sin^2 2 + \sin^2 3 + \dots + \sin^2 90}{90} = A$$

$$A = \frac{\sin^2 1 + \cos^2 1 + \sin^2 2 + \cos^2 2 + \dots + \sin^2 44 + \cos^2 44 + \sin^2 45 + \sin^2 90}{90}$$

$$A = \frac{44 + \frac{1}{2} + 1}{90} = \frac{91}{180}$$

Solution 4

Note that $\sin^2(x) = \frac{1 - \cos(2x)}{2}$. We want to determine $\frac{1}{90} \sum_{n=1}^{90} \sin^2(n^\circ)$.

$$\begin{aligned}
&= \frac{1}{90} \sum_{n=1}^{90} \frac{1 - \cos(2n)}{2} \\
&= \frac{1}{2} - \frac{1}{180} \sum_{n=1}^{90} \cos(2n)
\end{aligned}$$

Graphing $\cos(x)$, we can pair $\cos(2^\circ) + \cos(178^\circ) = 0$ and so on. We are left with $\cos(90^\circ) + \cos(180^\circ) = -1$.

Our answer is $\frac{1}{2} + \frac{1}{180} = \boxed{\text{(E)} \frac{91}{180}}$

Solution 5 (Integral Approximation)

Let $f(n) = \sin^2(n^\circ)$

The function $f(n) = \sin^2(n)$ is continuous for all real values of n . We want to find the average value of $f(n)$ over the interval: $[\frac{\pi}{180}, \frac{\pi}{2}]$

While there are 90 discrete terms meaning $f(n)$ isn't actually continuous, the interval is very small compared to 90, so calculating the average over EVERY term in the interval will be very close to calculating the average over 90 terms, so we can instead calculate the average value from $\frac{\pi}{180}$ to $\frac{\pi}{2}$

The average value of $f(n)$ is $\frac{1}{\frac{\pi}{2} - \frac{\pi}{180}} \int_{\frac{\pi}{180}}^{\frac{\pi}{2}} \sin^2(x) dx = \frac{180}{89\pi} \int_{\pi/180}^{\pi/2} \sin^2(x) dx$

We can solve this integral using the power-reduction identity: $\sin^2(x) = \frac{1 - \cos(2x)}{2}$

$$\begin{aligned}
\frac{180}{89\pi} \int_{\pi/180}^{\pi/2} \frac{1 - \cos(2x)}{2} dx &= \frac{180}{89\pi} \left(\frac{x}{2} - \frac{\sin(2x)}{4} \right) \Big|_{\pi/180}^{\pi/2} = \frac{180}{89\pi} \left[\left(\frac{\pi}{4} - 0 \right) - \left(\frac{\pi}{360} - \frac{\sin(\frac{\pi}{90})}{4} \right) \right] \\
&= \frac{180}{89\pi} \left(\frac{89\pi}{360} + \frac{\sin(\frac{\pi}{90})}{4} \right)
\end{aligned}$$

The Taylor series of $\sin(\theta) = \theta - \frac{\theta^3}{6} + \frac{\theta^5}{120} - \dots$

Because $\sin(\frac{\pi}{90})$ is a small angle, we can use a first-order Taylor approximation and approximate $\sin(\frac{\pi}{90}) \approx \frac{\pi}{90}$

Hence, our result is $\frac{180}{89\pi} \left(\frac{89\pi}{360} + \frac{\pi}{360} \right) = \frac{180}{89\pi} \cdot \frac{90\pi}{360} = \frac{180}{89\pi} \cdot \frac{\pi}{4} = \frac{45}{89}$

While $\frac{45}{89}$ isn't an option, it is greater than $\frac{1}{2}$, and out of the given options, only $\frac{91}{180}$ is greater than $\frac{1}{2}$ (We approximated $\sin(\frac{\pi}{90})$ with a first-order Taylor approximation and calculated the average value over the interval instead of over 90 discrete values, but it's close enough to $\frac{91}{180}$).

So the answer is $\boxed{\text{(E)} \frac{91}{180}}$

-sourdeepdeb

12. (AMC 12 12B 2024) Suppose z is a complex number with positive imaginary part, with real part greater than 1, and with $|z| = 2$. In the complex plane, the four points $0, z, z^2$, and z^3 are the vertices of a quadrilateral with area 15. What is the imaginary part of z ?

(A) $\frac{3}{4}$ (B) 1 (C) $\frac{4}{3}$ (D) $\frac{3}{2}$ (E) $\frac{5}{3}$

Answer: D

Solution 1

By making a rough estimate of where z , z^2 , and z^3 are on the complex plane, we can draw a pretty accurate diagram (like above.)

Here, points Z_1 , Z_2 , and Z_3 lie at the coordinates of z , z^2 , and z^3 respectively, and O is the origin.

We're given $|z| = 2$, so $|z^2| = |z|^2 = 4$ and $|z^3| = |z|^3 = 8$. This gives us $OZ_1 = 2$, $OZ_2 = 4$, and $OZ_3 = 8$.

Additionally, we know that $\angle Z_1OZ_2 \cong \angle Z_2OZ_3$ (since every power of z rotates around the origin by the same angle.) We set these angles equal to θ .

We have that

$$\begin{aligned} [OZ_1Z_2Z_3] &= [OZ_1Z_2] + [OZ_2Z_3] \\ &= \frac{1}{2} \cdot 2 \cdot 4 \sin \theta + \frac{1}{2} \cdot 4 \cdot 8 \sin \theta \\ &= 4 \sin \theta + 16 \sin \theta \\ &= 20 \sin \theta \end{aligned}$$

Since this is equal to 15, we have $20 \sin \theta = 15$, so $\sin \theta = \frac{3}{4}$.

$$\text{Thus, } \text{Im}(z) = |z| \sin(\theta) = 2\left(\frac{3}{4}\right) = \boxed{\text{(D)} \frac{3}{2}}.$$

Solution 2 (Shoelace Theorem)

We have the vertices:

0 at $(0, 0)$, z at $(2 \cos \theta, 2 \sin \theta)$, z^2 at $(4 \cos 2\theta, 4 \sin 2\theta)$, z^3 at $(8 \cos 3\theta, 8 \sin 3\theta)$

The Shoelace formula for the area is:

$$\begin{aligned} &= \frac{1}{2} |x_1y_2 + x_2y_3 + x_3y_4 + x_4y_1 - (y_1x_2 + y_2x_3 + y_3x_4 + y_4x_1)| \\ &= \frac{1}{2} |0 + 2 \cos \theta \cdot 4 \sin 2\theta + 4 \cos 2\theta \cdot 8 \sin 3\theta - (2 \sin \theta \cdot 4 \cos 2\theta + 4 \sin 2\theta \cdot 8 \cos 3\theta)| \\ &= \frac{1}{2} |8 \cos \theta \sin 2\theta + 32 \cos 2\theta \sin 3\theta - 8 \sin \theta \cos 2\theta - 32 \sin 2\theta \cos 3\theta| \\ &= \frac{1}{2} |(8 \cos \theta \sin 2\theta - 8 \sin \theta \cos 2\theta) + (32 \cos 2\theta \sin 3\theta - 32 \sin 2\theta \cos 3\theta)| \\ &= \frac{1}{2} |8 \sin(2\theta - \theta) + 32 \sin(3\theta - 2\theta)| \\ &= \frac{1}{2} |8 \sin \theta + 32 \sin \theta| \\ &= \frac{1}{2} |40 \sin \theta| \end{aligned}$$

Given that the area is 15:

$$\frac{1}{2} |40 \sin \theta| = 15.$$

$$20 |\sin \theta| = 15 \implies |\sin \theta| = \frac{3}{4}.$$

Since θ corresponds to a complex number z with a positive imaginary part, we have:

$$\sin \theta = \frac{3}{4}.$$

$$\text{Imaginary part} = 2 \sin \theta = 2 \times \frac{3}{4} = \boxed{\text{(D)} \frac{3}{2}}.$$

Solution 3 (No Trig)

Let $z = a + bi$, so $z^2 = a^2 + 2abi - b^2$ and $z^3 = a^3 + 3a^2bi - 3ab^2 - b^3i$. Therefore, converting $0, z, z^2, z^3$ from complex coordinates to Cartesian coordinates gives us the following.

$$(0, 0)$$

$$(a, b)$$

$$(a^2 - b^2, 2ab)$$

$$(a^3 - 3ab^2, 3a^2b - b^3)$$

The Shoelace Theorem tells us that the area is

$$\begin{aligned} & \frac{1}{2} \left| \left[(0)(b) + (a)(2ab) + (a^2 - b^2)(3a^2b - b^3) + (a^3 - 3ab^2)(0) \right] - \left[(0)(a) + (b)(a^2 - b^2) + (2ab)(a^3 - 3ab^2) + (3a^2b - b^3)(0) \right] \right| \\ &= \frac{1}{2} \left| \left[(0) + (2a^2b) + (3a^4b - a^2b^3 - 3a^2b^3 + b^5) + (0) \right] - \left[(0) + (a^2b - b^3) + (2a^4b - 6a^2b^3) + (0) \right] \right| \\ &= \frac{1}{2} \left| [3a^4b - 4a^2b^3 + b^5 + 2a^2b] - [2a^4b - 6a^2b^3 + a^2b - b^3] \right| \\ &= \frac{1}{2} |a^4b + 2a^2b^3 + b^5 + a^2b + b^3|. \end{aligned}$$

We know that $|z| = |a + bi| = \sqrt{a^2 + b^2} = 2$, so $a^2 = 4 - b^2$. Substituting this gives us this:

$$\begin{aligned} & \frac{1}{2} \left| (4 - b^2)^2b + 2(4 - b^2)b^3 + b^5 + (4 - b^2)b + b^3 \right| \\ &= \frac{1}{2} \left| (16b - 8b^3 + b^5) + (8b^3 - 2b^5) + b^5 + (4b - b^3) + b^3 \right| \\ &= \frac{1}{2} |0b^5 + 0b^3 + 20b| \\ &= 15. \end{aligned}$$

In other words,

$$|10b| = 15$$

$$b = (\mathbf{D}) \frac{3}{2}.$$

13. (AMC 12 12B 2024) There are real numbers x, y, h and k that satisfy the system of equations

$$x^2 + y^2 - 6x - 8y = h$$

$$x^2 + y^2 - 10x + 4y = k$$

What is the minimum possible value of $h + k$?

- (A) -54 (B) -46 (C) -34 (D) -16 (E) 16

Answer: C

Solution 1 (Easy and Fast)

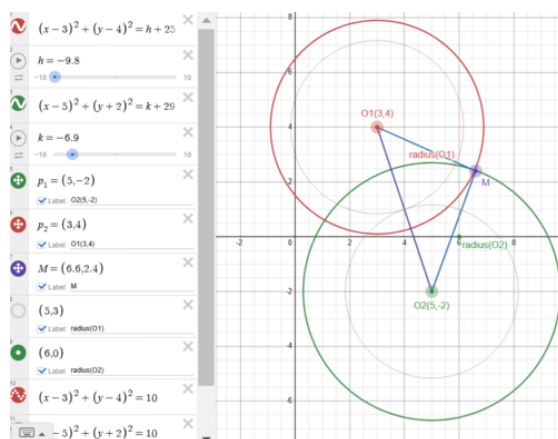
Adding up the first and second equation, we get:

$$\begin{aligned} h + k &= 2x^2 + 2y^2 - 16x - 4y \\ &= 2x^2 - 16x + 2y^2 - 4y \\ &= 2(x^2 - 8x) + 2(y^2 - 2y) \\ &= 2(x^2 - 8x + 16) - (2)(16) + 2(y^2 - 2y + 1) - (2)(1) \\ &= 2(x - 4)^2 + 2(y - 1)^2 - 34 \end{aligned}$$

All squared values must be greater than or equal to 0. As we are aiming for the minimum value, we set the two squared terms to be 0.

This leads to $\min(h + k) = 0 + 0 - 34 = \boxed{(\mathbf{C}) - 34}$

Solution 2 (Coordinate Geometry and AM-GM Inequality)



$$(x - 3)^2 + (y - 4)^2 = h + 25$$

$$(x - 5)^2 + (y + 2)^2 = k + 29$$

The distance between 2 circle centers is

$$(O_1O_2)^2 = (5 - 3)^2 + (4 - (-2))^2 = 40$$

The 2 circles must intersect given there exists one or more pairs of (x,y) , connecting O_1O_2 and any pair of the 2 circle intersection points gives us a triangle with 3 sides, then

$$\text{radius}(O_1) + \text{radius}(O_2) \geq O_1O_2$$

$$\sqrt{h+25} + \sqrt{k+29} \geq 2 \cdot \sqrt{10}$$

Note that they will be equal if and only if the circles are tangent,

Applying the AM-GM inequality $(2(a^2 + b^2) \geq (a + b)^2)$ in the steps below, we get

$$h+k+54 = (h+25)+(k+29) = \sqrt{(h+25)^2} + \sqrt{(k+29)^2} \geq \frac{(\sqrt{h+25} + \sqrt{k+29})^2}{2} \geq \frac{(2\sqrt{10})^2}{2} = 20.$$

Therefore, $h + k \geq 20 - 54 = \boxed{C - 34}$.

Solution 3

$$h = (x-3)^2 + (y-4)^2 - 25$$

$$k = (x-5)^2 + (y+2)^2 - 29$$

$$\begin{aligned} h+k &= (x-3)^2 + (x-5)^2 + (y-4)^2 + (y+2)^2 - 54 \\ h+k &= (2x^2 - 16x + 34) + (2y^2 - 4y + 20) - 54 \end{aligned}$$

$$(2x^2 - 16x + 34)_{\min} \Rightarrow \frac{-D}{4a} = \frac{-(256 - 272)}{8} = 2$$

$$(2y^2 - 4y + 20)_{\min} \Rightarrow \frac{-D}{4a} = \frac{-(16 - 160)}{8} = 18$$

$$(h+k)_{\min} = 2 + 18 - 54 = -34$$

$$(h+k)_{\min} = 2 + 18 - 54 = -34$$

Hence option (C) -34

Solution 4 (Very Fast)

Begin by completing the square for each equation:

$$(x-3)^2 + (y-4)^2 = h+25$$

$$(x-5)^2 + (y+2)^2 = k+29$$

We notice that $x = 4$ minimizes the sum of $(x-3)^2$ and $(x-5)^2$, and $y = 1$ minimizes the sum of $(y-4)^2$ and $(y+2)^2$. Plug those values into both equations to

get $1 + 9 = h + 25$ and $1 + 9 = k + 29$. Add the two equations to get $20 = h + k + 54$. Therefore, the minimum value of $h + k$ is $\boxed{(\text{C}) - 34}$.

Solution 5 (Gradient)

Let $f(x)$ be $h + k$. $f(x) = 2x^2 + 2y^2 - 16x - 4y$. To minimize, set the gradient to 0. The gradient is $[4x - 16, 4y - 4]$, so $x = 4$ and $y = 1$. Plugging this into $f(x)$, the minimum value is $\boxed{(\text{C}) - 34}$.

14. (AMC 12 12B 2024) How many different remainders can result when the 100th power of an integer is divided by 125?

(A) 1 (B) 2 (C) 5 (D) 25 (E) 125

Answer: B

Solution 1

First note that the Euler's totient function of 125 is 100. We can set up two cases, which depend on whether a number is relatively prime to 125.

If n is relatively prime to 125, then $n^{100} \equiv 1 \pmod{125}$ because of Euler's Totient Theorem.

If n is not relatively prime to 125, it must have a factor of 5. Express n as $5m$, where m is some integer. Then $n^{100} \equiv (5m)^{100} \equiv 5^{100} \cdot m^{100} \equiv 125 \cdot 5^{97} \cdot m^{100} \equiv 0 \pmod{125}$.

Therefore,

15. (AMC 12 12B 2024) A triangle in the coordinate plane has vertices $A(\log_2 1, \log_2 2)$, $B(\log_2 3, \log_2 4)$, and $C(\log_2 7, \log_2 8)$. What is the area of $\triangle ABC$?

(A) $\log_2 \frac{\sqrt{3}}{7}$ (B) $\log_2 \frac{3}{\sqrt{7}}$ (C) $\log_2 \frac{7}{\sqrt{3}}$ (D) $\log_2 \frac{11}{\sqrt{7}}$ (E) $\log_2 \frac{11}{\sqrt{3}}$

Answer: B

Solution 1 (Shoelace Theorem)

We rewrite: $A(0, 1)B(\log_2 3, 2)C(\log_2 7, 3)$.

From here we setup Shoelace Theorem and obtain: $\frac{1}{2}(2(\log_2 3) - \log_2 7)$.

Following log properties and simplifying gives $\boxed{(\text{B}) \log_2 \frac{3}{\sqrt{7}}}$.

Solution 2 (Determinant)

To calculate the area of a triangle formed by three points $A(x_1, y_1)$, $B(x_2, y_2)$, and $C(x_3, y_3)$ on a Cartesian coordinate plane, you can use the following formula:

$$\text{Area} = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

The coordinates are: $A(0, 1)$, $B(\log_2 3, 2)$, $C(\log_2 7, 3)$

Taking a numerical value into account:

$$\text{Area} = \frac{1}{2} |0 \cdot (2 - 3) + \log_2 3 \cdot (3 - 1) + \log_2 7 \cdot (1 - 2)|$$

Simplify:

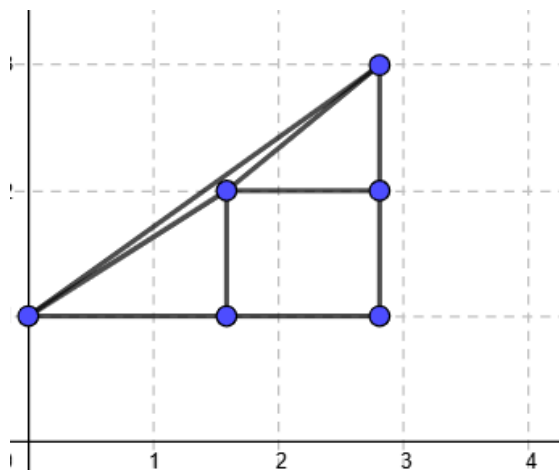
$$= \frac{1}{2} |0 + \log_2 3 \cdot 2 + \log_2 7 \cdot (-1)|$$

$$= \frac{1}{2} \left| \log_2(3^2) - \log_2 7 \right|$$

$$= \frac{1}{2} \left| \log_2 \frac{9}{7} \right|$$

Thus, the area is: $\text{Area} = \frac{1}{2} \left| \log_2 \frac{9}{7} \right| = \boxed{\text{(B)} \log_2 \frac{3}{\sqrt{7}}}$

Solution 3 (Geometry)



In the graph above, the biggest triangle has legs $\log_2 7$ and 2. Its area is then $\log_2 7$. There are 3 smaller shapes as well:

1. Triangle 1. Legs = $\log_2 3$ and 1. Area = $\frac{1}{2} \log_2 3 = \log_2 \sqrt{3}$;
2. Triangle 2. Legs = $\log_2 \frac{7}{3}$ and 1. Area = $\frac{1}{2} \log_2 \frac{7}{3} = \log_2 \frac{\sqrt{7}}{\sqrt{3}}$;
3. Rectangle. Legs = $\log_2 \frac{7}{3}$ and 1. Area = $\log_2 \frac{7}{3}$;

The area of the triangle is therefore $\log_2 7 - \log_2 \sqrt{3} - \log_2 \frac{\sqrt{7}}{\sqrt{3}} - \log_2 \frac{7}{3}$. This is equivalent to $\log_2 7 * \frac{1}{\sqrt{3}} * \frac{\sqrt{3}}{\sqrt{7}} * \frac{3}{7}$ which simplifies to $\boxed{\text{(B)} \log_2 \frac{3}{\sqrt{7}}}$.

Solution 4 (Cross Product)

The area is half of the cross product between two of the vectors. These will be $[\log_2 3, 1]$ and $[\log_2 7, 2]$. The cross product is $2\log_2 3 - \log_2 7$, and half of this is

$$\boxed{\text{(B)} \log_2 \frac{3}{\sqrt{7}}}.$$

16. (AMC 12 12B 2024) A group of 16 people will be partitioned into 4 indistinguishable 4-person committees. Each committee will have one chairperson and one secretary. The number of different ways to make these assignments can be written as $3^r M$, where r and M are positive integers and M is not divisible by 3. What is r ?
- (A) 5 (B) 6 (C) 7 (D) 8 (E) 9

Answer: A

Solution 1

There are $\binom{16}{4}$ ways to choose the first committee, $\binom{12}{4}$ ways to choose the second, $\binom{8}{4}$ for the third, and $\binom{4}{4} = 1$ for the fourth. Since the committees are indistinguishable, we need to divide the product by $4!$. Thus the 16 people can be grouped in

$$\frac{1}{4!} \binom{16}{4} \binom{12}{4} \binom{8}{4} \binom{4}{4} = \frac{16!}{(4!)^5}$$

ways.

In each committee, there are $4 \cdot 3 = 12$ ways to choose the chairperson and secretary, so 12^4 ways for all 4 committees. Therefore, there are

$$\frac{16!}{(4!)^5} 12^4$$

total possibilities.

Since $16!$ contains 6 factors of 3, $(4!)^5$ contains 5, and 12^4 contains 4, $r = 6 - 5 + 4 =$ **(A) 5**.

Solution 2 (Multinomial Coefficients)

There are $\binom{16}{4,4,4,4}$ ways to choose the 4 committees. You have to divide by another $4!$ since the order of the committees does not matter. Furthermore, in each committee, you can have $4 \cdot 3$ ways to choose chairperson and secretary. Hence a total of $\lfloor \frac{16}{3} \rfloor + \lfloor \frac{16}{9} \rfloor + 4 - 5 = 5$.

Solution 3

There will be 16 ways to pick the chairperson of the first committee, then 15 to pick the secretary, and lastly $\binom{14}{2}$ ways to pick the other two members of the first committee. Similarly, we can complete the rest of the terms as follows:

$$\frac{(16)(15)\binom{14}{2}(12)(11)\binom{10}{2}(8)(7)\binom{6}{2}(4)(3)\binom{2}{2}}{4!}$$

We notice the numerator has at most 3^6 , and the denominator has just 3. Thus, the value of r in question is **(A) 5**.

Solution 4

We can convert the question into arranging people in a line of length 16.

First, arrange all 16 people in a line. There are $16!$ possible ways to do this. Then, divide the line into 4 groups of 4, which represent the committees. Since the committees are indistinguishable, we divide by the number of ways to permute the 4 groups, $4!$.

Within each group of 4, the first two positions in the line are assigned as the chairperson and the secretary, while the remaining two are ordinary members. Since the roles of chairperson and secretary are distinct, but the order of the remaining two members does not matter, we divide by 2 for each group. Thus, we divide by 2^4 to account for all 4 committees.

The total number of ways to assign people into committees with roles is given by:

$$\frac{16!}{4! \cdot 2^4}.$$

(This is equivalent to Solution 1's $\frac{16!}{(4!)^5} \cdot 12^4$.)

To find r , we use Legendre's Formula to count the total power of 3 in $16!$:

$$\left\lfloor \frac{16}{3} \right\rfloor + \left\lfloor \frac{16}{9} \right\rfloor = 5 + 1 = 6.$$

Each $4!$ contributes one factor of 3, and since we divide by $4!$ a total of 1 times, we subtract 1 from 6:

$$6 - 1 = 5.$$

Thus, the value of r is $\boxed{5}$. ryk

17. (AMC 12 12B 2024) Integers a and b are randomly chosen without replacement from the set of integers with absolute value not exceeding 10. What is the probability that the polynomial $x^3 + ax^2 + bx + 6$ has 3 distinct integer roots?

(A) $\frac{1}{240}$ (B) $\frac{1}{221}$ (C) $\frac{1}{105}$ (D) $\frac{1}{84}$ (E) $\frac{1}{63}$

Answer: C

Solution 1

Since $-10 \leq a, b \leq 10$, there are 21 integers to choose from, and $P(21, 2) = 21 \times 20 = 420$ equally likely ordered pairs (a, b) .

Applying Vieta's formulas,

$$x_1 \cdot x_2 \cdot x_3 = -6 \quad x_1 + x_2 + x_3 = -a \quad x_1 \cdot x_2 + x_1 \cdot x_3 + x_3 \cdot x_2 = b$$

Cases:

- (1) $(x_1, x_2, x_3) = (-1, 1, 6), b = -1, a = -6$ valid
- (2) $(x_1, x_2, x_3) = (1, 2, -3), b = -7, a = 0$ valid
- (3) $(x_1, x_2, x_3) = (1, -2, 3), b = -5, a = -2$ valid
- (4) $(x_1, x_2, x_3) = (-1, 2, 3), b = 1, a = -4$ valid
- (5) $(x_1, x_2, x_3) = (-1, -2, -3), b = 11$ invalid

the total event space is $21 \cdot (21 - 1)$ (choice of select a times choice of selecting b given no-replacement)

hence, our answer is $\frac{4}{21 \cdot 20} = \boxed{\text{(C)} \frac{1}{105}}$

Solution 1.1 (desperation)

As obtained in Solution 1, we get that there are $P(21, 2) = 420$ equally likely ordered pairs (a, b) , which means that the denominator will likely be a factor of 420, which leaves answers C and D, and if you are lucky enough, you can guess that the answer is $\boxed{\text{(C)} \frac{1}{105}}$. If you're not lucky enough, however, that's a math instinct skill issue, and I recommend you get instinct first before you do math.

18. (AMC 12 12B 2024) The Fibonacci numbers are defined by $F_1 = 1$, $F_2 = 1$, and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 3$. What is

$$\frac{F_2}{F_1} + \frac{F_4}{F_2} + \frac{F_6}{F_3} + \cdots + \frac{F_{20}}{F_{10}}?$$

(A) 318 (B) 319 (C) 320 (D) 321 (E) 322

Answer: B

Solution 1 (Brute Force)

The first 20 terms are $F_n = 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, 377, 610, 987, 1597, 2584, 4181, 6765$

So the answer is $1 + 3 + 4 + 7 + 11 + 18 + 29 + 47 + 76 + 123 = \boxed{(B)319}$.

- Do not do this unless you have no other option or no time to find a smarter way

Solution 1.1 (Sidenote)

Note that $1, 3, 4, 7, 11, 18, 29, 47, 76, 123, \dots$ is a variation of Fibonacci sequence, so you if you found the pattern, you wouldn't have had to divided them all out.

Solution 1.a (Modified Brute Force)

Write out the first few Fibonacci numbers: $1, 1, 2, 3, 5, 8, 13, 21, 34, 55$. (Doesn't matter how many, just need few to show a pattern).

The first few terms of the new series are:

$$\frac{1}{1} + \frac{3}{1} + \frac{8}{2} + \frac{21}{3} + \dots$$

$$\Rightarrow 1 + 3 + 4 + 7 + \dots$$

Assuming that the pattern repeats, this is a generalized Fibonacci series; hence the rest of the series is as follows:

$$1 + 3 + 4 + 7 + 11 + 18 + 29 + 47 + 76 + 123 = \boxed{B. 319}.$$

Please do NOT do this unless you have no choice; brute force should never be a solving method unless you a) cheese or b) have no time/brainpower left.

Solution 2 (Pattern)

Plug in a few numbers to see if there is a pattern. List out a few Fibonacci numbers, and then try them on the equation. You'll find that $\frac{F_2}{F_1} = \frac{1}{1} = 1$, $\frac{F_4}{F_2} = \frac{3}{1} = 3$, $\frac{F_6}{F_3} = \frac{8}{2} = 4$, and $\frac{F_8}{F_4} = \frac{21}{3} = 7$. The pattern is that then ten fractions are in their own leapfrog sequence with the starting two terms being 1 and 3, which can be written as $G_1 = 1, G_2 = 3, G_n = G_{n-1} + G_{n-2}$ for $n \geq 3$. The problem is asking for the sum of the ten terms $G_1 + G_2 + G_3 + \dots + G_{10}$, and after adding up the first ten terms $1 + 3 + 4 + 7 + 11 + 18 + 29 + 47 + 76 + 123$, you arrive at the solution $\boxed{(B) 319}$.

Note The terms $1, 3, 4, 7, \dots, 76, 123$ are actually part of a sequence called the Lucas Numbers.

Solution 3 (Characteristic Equation and Recurrence Relationship)

Using characteristic equation $x^2 = x + 1$ and starting terms $F_1 = 1, F_2 = 1$, we get $F_n = \frac{1}{\sqrt{5}}(A^n - B^n)$, $A = \frac{1+\sqrt{5}}{2}$, $B = \frac{1-\sqrt{5}}{2}$

Define a new sequence (this is actually a Lucas sequence)

$$L_n = \frac{F_{2n}}{F_n} = \frac{A^{2n} - B^{2n}}{A^n - B^n} = A^n + B^n$$

Given that L_n has the same 2 roots (A,B) as the characteristic equation $x^2 = x + 1$ it implies that $L_n = L_{n-1} + L_{n-2}$. It can also be verified below. Noticing $A^2 = A + 1$ and $B^2 = B + 1$, Therefore,

$$L_n = A^n + B^n = A^2 \cdot A^{n-2} + B^2 \cdot B^{n-2} = (A+1) \cdot A^{n-2} + (B+1) \cdot B^{n-2} = (A^{n-1} + B^{n-1}) + (A^{n-2} + B^{n-2}) =$$

Calculating the first 10 terms using the recurrence relation $L_n = L_{n-1} + L_{n-2}$ and the initial values $L_1 = 1$ and $L_2 = 3$, we get:

$$L_1 = 1, \quad L_2 = 3, \quad L_3 = 4, \quad L_4 = 7, \quad L_5 = 11$$

$$L_6 = 18, \quad L_7 = 29, \quad L_8 = 47, \quad L_9 = 76, \quad L_{10} = 123$$

So the answer is $1 + 3 + 4 + 7 + 11 + 18 + 29 + 47 + 76 + 123 = \boxed{\text{(B)} 319}$.

Solution 4 (Generic Lucas Sequence Sum Formula)

$$S_k = \sum_{n=1}^k L_n = \sum_{n=1}^k (A^n + B^n) = \sum_{n=1}^k A^n + \sum_{n=1}^k B^n = \frac{A(1 - A^k)}{1 - A} + \frac{B(1 - B^k)}{1 - B}$$

Recall the sum formula for a geometric series:

$$\sum_{n=1}^k r^n = r + r^2 + \cdots + r^k = \frac{r(1 - r^k)}{1 - r}, \quad \text{for } r \neq 1$$

To simplify further, notice that: $A = \frac{1+\sqrt{5}}{2}$ and $B = \frac{1-\sqrt{5}}{2}$ are roots of $x^2 = x + 1$, so:

$$1 - A = B \quad \text{and} \quad 1 - B = A.$$

$$S_k = \frac{A(1 - A^k)}{B} + \frac{B(1 - B^k)}{A} = \frac{A^{k+2} - A^2 + B^{k+2} - B^2}{A \cdot B} = \frac{A^{k+2} + B^{k+2} - (A^2 + B^2)}{A \cdot B} = A^{k+2} + B^{k+2} - 3$$

This formula $S_k = A^{k+2} + B^{k+2} - 3$ gives us a direct way to calculate the sum of L_n for any k .

$$L_{2n} = (A^n + B^n)^2 - 2 \cdot (AB)^n = L_n^2 - 2 \cdot (-1)^n$$

$$L_{2n+1} = (A^n + B^n)(A^{n+1} + B^{n+1}) - (AB)^n(A + B) = L_n L_{n+1} - (-1)^n$$

$$L_{12} = (L_6)^2 - 2 = (L_3^2 + 2)^2 - 2 = (4^2 + 2)^2 - 2 = 324 - 2 = 322$$

Hence,

$$\sum_{n=1}^{10} L_n = L_{12} - 3 = 322 - 3 = \boxed{\text{(B)} 319}$$

- Please do not do this on an actually exam in AMC 10 and only do it when you have no time in AMC 12

Solution 5 (Fibonacci Identities; Fastest)

If you are well-versed in identities relating Fibonacci numbers (or spot this pattern by induction) then you may see that

$$F_{2n} = F_n(F_{n-1} + F_{n+1})$$

Then we can say that the sum required is just $(F_0 + F_1 + \dots + F_9) + (F_2 + F_3 + \dots + F_{11})$ (and we take $F_0 = 0$). There is not much we can do to aid our calculation, so we directly calculate the answer using the sequence 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89. You should get (B)319.

(If you are interested, this identity can be proved using the inductive process as well as using the fact $F_n F_{n+1} - F_{n-2} F_{n-1} = F_{2n-1}$.)

-liszstepos

Solution 6

This solution is motivated by solution 3.

The explicit formula for Fibonacci numbers is

$$F_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^n$$

Therefore,

$$F_{2n} = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^{2n} - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^{2n}$$

Using the explicit formulas, the equation for $\frac{F_{2n}}{F_n}$ could be computed.

$$\begin{aligned} \frac{F_{2n}}{F_n} &= \frac{\frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^n}{\frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^{2n} - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^{2n}} \\ &= \left(\frac{1+\sqrt{5}}{2} \right)^n + \left(\frac{1-\sqrt{5}}{2} \right)^n \end{aligned}$$

The square roots in the numbers $\frac{1+\sqrt{5}}{2}$ and $\frac{1-\sqrt{5}}{2}$ could be removed by adding or multiplying the numbers. Therefore, let $\alpha = \frac{1+\sqrt{5}}{2}$ and $\beta = \frac{1-\sqrt{5}}{2}$.

Notice that $\alpha + \beta = 1$ and $\alpha\beta = -1$. In other words, α and β are the roots of the equation $x^2 - x - 1 = 0$.

$$\begin{aligned} \alpha^2 &= \alpha + 1 \\ \beta^2 &= \beta + 1 \\ \alpha^2 \alpha^n &= \alpha^{n+1} + \alpha^n \\ \beta^2 \beta^n &= \beta^{n+1} + \beta^n \end{aligned}$$

Let $a_i = \alpha^i$ and $b_i = \beta^i$ for convenience. Notice that $a_{n+2} + b_{n+2} = a_{n+1} + b_{n+1} + a_n + b_n$. Let $f_n = a_n + b_n$. Observe that $f_{n+2} = f_{n+1} + f_n$.

$f_1 = 1$ and $f_2 = 3$ are known. Moreover, f_n share the property of Fibonacci numbers. Note that every sequence F_n that forms a new term by adding two previous terms have the following property.

$$\sum_{k=1}^n F_k = F_{n+2} - F_2 \quad (n \geq 1)$$

Therefore,

$$\sum_{k=1}^{10} F_k = F_{12} - 3.$$

Moreover, $\{f_n\}_{n=1}^{\infty} := \{1, 3, 4, 7, 11, 18, 29, 47, 76, 123, 199, 322, \dots\}$. Therefore, $322 - 3 = \boxed{\text{(B) } 319}$.

Solution 7 (Lots of Identities)

Of course, olympiad algebra mains recall that for any $n \geq 0$,

$$\frac{F_{2n}}{F_n} = L_n$$

where L_n denotes the Lucas numbers. A well known Fibonacci identity is that

$$\sum_{k=1}^n F_k = F_{n+2} - F_2 \quad (n \geq 1)$$

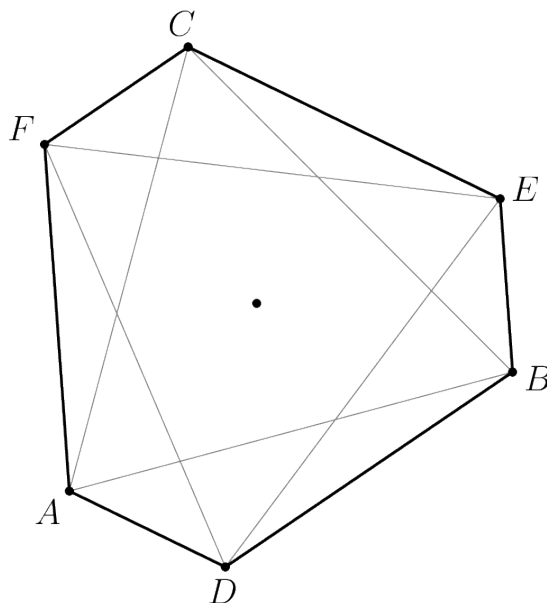
Now we also know

$$L_n = F_{n-1} + F_{n+1}$$

which further nukes the problem if you do not feel like calculating Lucas numbers. Either way, we should get an answer of $\boxed{\text{(B) } 319}$.

Remark: All proofs of these identities are on Wikipedia.

19. (AMC 12 12B 2024) Equilateral $\triangle ABC$ with side length 14 is rotated about its center by angle θ , where $0 < \theta < 60^\circ$, to form $\triangle DEF$. See the figure. The area of hexagon $ADBECF$ is $91\sqrt{3}$. What is $\tan \theta$?



- (A) $\frac{3}{4}$ (B) $\frac{5\sqrt{3}}{11}$ (C) $\frac{4}{5}$ (D) $\frac{11}{13}$ (E) $\frac{7\sqrt{3}}{13}$

Answer: B

Solution 1

Let O be circumcenter of the equilateral triangle

Easily get $OF = \frac{14\sqrt{3}}{3} \cdot 2 \cdot (\triangle OFC + \triangle OCE) = OF^2 \cdot \sin(\theta) + OF^2 \cdot \sin(120 - \theta)$

$$= \frac{14^2 \cdot 3}{9} (\sin(\theta) + \sin(120 - \theta))$$

$$= \frac{196}{3}(\sin(\theta) + \sin(120 - \theta))$$

$$= 2 \cdot \frac{1}{3} \cdot (ADBE CF) = 2 \cdot \frac{91\sqrt{3}}{3}$$

$$\sin(\theta) + \sin(120 - \theta) = \frac{13\sqrt{3}}{14}$$

$$\sin(\theta) + \frac{\sqrt{3}}{2} \cos(\theta) + \frac{\sqrt{1}}{2} \sin(\theta) = \frac{13\sqrt{3}}{14}$$

$$\sqrt{3} \sin(\theta) + \cos(\theta) = \frac{13}{7}$$

$$\cos(\theta) = \frac{13}{7} - \sqrt{3} \sin(\theta)$$

$$\frac{169}{49} - \frac{26\sqrt{3}}{7} \sin(\theta) + 4 \sin^2(\theta) = 1$$

$$\sin(\theta) = \frac{5\sqrt{3}}{14} \text{ or } \frac{4\sqrt{3}}{7}$$

$\frac{4\sqrt{3}}{7}$ is invalid given $\theta \leq 60^\circ$, $\sin(\theta) < \sin(60^\circ) = \frac{\sqrt{3}}{2} = \frac{\sqrt{3} \cdot 3.5}{7}$

$$\cos(\theta) = \frac{11}{14}$$

$$\tan(\theta) = \frac{5\sqrt{3}}{11} \boxed{B}$$

Solution 2

From $\triangle ABC$'s side lengths of 14, we get

$$OF = OC = OE = \frac{14\sqrt{3}}{3}.$$

We let $\angle FOC = \theta$ And $\angle EOC = 120 - \theta$

The answer would be $3([\triangle FOC] + [\triangle COE])$

Which area $\triangle FOC = \frac{1}{2} \left(\frac{14\sqrt{3}}{3} \right)^2 \sin(\theta)$

And area $\triangle COE = \frac{1}{2} \left(\frac{14\sqrt{3}}{3} \right)^2 \sin(120 - \theta)$

So we have that

$$3 \cdot \frac{1}{2} \cdot \left(\frac{14\sqrt{3}}{3} \right)^2 (\sin(\theta) + \sin(120 - \theta)) = 91\sqrt{3}$$

Which means

$$\sin(\theta) + \sin(120 - \theta) = \frac{91\sqrt{3}}{98}$$

$$\frac{1}{2} \cos(\theta) + \frac{\sqrt{3}}{2} \sin(\theta) = \frac{91}{98}$$

$$\sin(\theta + 30) = \frac{91}{98}$$

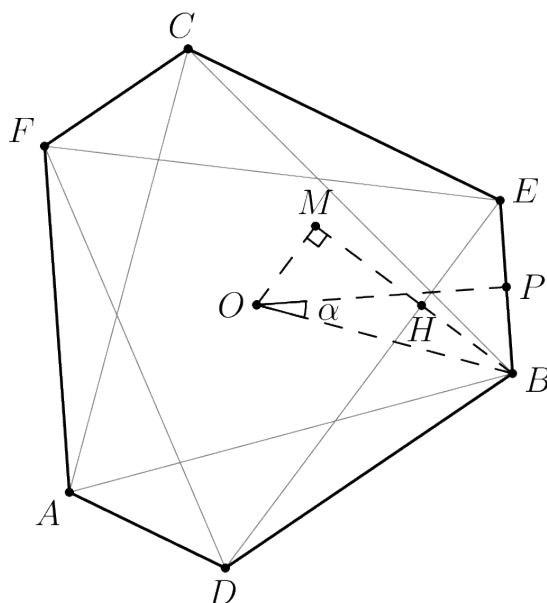
$$\cos(\theta + 30) = \frac{21\sqrt{3}}{98}$$

$$\tan(\theta + 30) = \frac{91}{21\sqrt{3}} = \frac{91\sqrt{3}}{63}$$

Now, $\tan(\theta)$ can be calculated using the addition identity, which gives the answer of

$$(B) \frac{5\sqrt{3}}{11}.$$

Solution 3 (No Trig Manipulations)



Let the circumcenter of the circle inscribing this polygon be O . The area of the equilateral triangle is $\frac{\sqrt{3}}{4} * 196 = 49\sqrt{3}$. The area of one of the three smaller triangles, say $\triangle DBE$ is $14\sqrt{3}$. Let BH be the altitude of $\triangle DBE$, so if we extend BH to point M where $MO \perp BM$, we get right triangle $\triangle OMB$. Note that the height $BH = 2\sqrt{3}$, computed given the area and side length 14, so $MB = MH + HB = \frac{7\sqrt{3}}{3} + 2\sqrt{3} = \frac{13\sqrt{3}}{3}$. $OB = \frac{14\sqrt{3}}{3}$ so Pythag gives $OM = \sqrt{OB^2 - MB^2} = 3$. This means that $HE = 7 - OM = 4$, so Pythag gives $BE = 2\sqrt{7}$. Let $\frac{\theta}{2} = \alpha$ and the midpoint of BE be P so that $BP = PE = \sqrt{7}$, so that Pythag on $\triangle OPE$ gives $OP = \sqrt{OE^2 - PE^2} = \sqrt{\frac{175}{3}}$. Then $\tan \alpha = \frac{\sqrt{7}}{\sqrt{\frac{175}{3}}} = \frac{\sqrt{3}}{5}$. Then

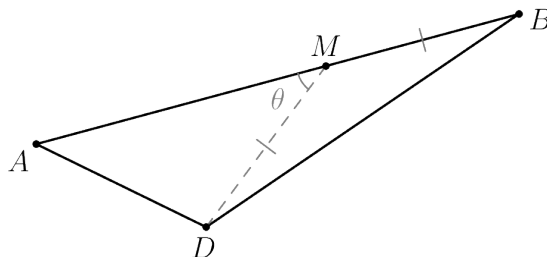
$$\tan 2\alpha = \tan \theta = \frac{\frac{2\sqrt{3}}{5}}{1 - \frac{3}{25}} = \frac{5\sqrt{3}}{11}.$$

-Magnetoininja

Solution 4

$$[\triangle ABD] = \frac{[ADBE CF] - [\triangle ABC]}{3} = \frac{91\sqrt{3} - 49\sqrt{3}}{3} = 14\sqrt{3}$$

Let M be the intersection of AB and DE . Since DMB is isosceles and $\angle AMD = \theta$, we have $\angle ABD = \theta/2$. Also, all of the hexagon's internal angles are equal, so $\angle ADB = 120^\circ$.



Using the side-angle-side area formula:

$$14\sqrt{3} = \frac{1}{2} \cdot 14 \cdot BD \cdot \sin\left(\frac{\theta}{2}\right) \Rightarrow BD = \frac{2\sqrt{3}}{\sin(\theta/2)}.$$

Apply Law of Sines on $\triangle ABD$ with $\angle DAB = 60^\circ - \theta/2$:

$$\frac{14}{\sin 120^\circ} = \frac{BD}{\sin(60^\circ - \theta/2)}$$

$$\frac{28}{\sqrt{3}} = \frac{2\sqrt{3}}{\sin(\theta/2) \sin(60^\circ - \theta/2)}.$$

$$\sin(\theta/2) \sin(60^\circ - \theta/2) = \frac{3}{14}.$$

Using trig identities:

$$\sqrt{\frac{1 - \cos(\theta)}{2}} \cdot (\sin(60^\circ) \cos(\theta/2) - \sin(\theta/2) \cos(60^\circ)) = \frac{3}{14}$$

$$\sqrt{\frac{1 - \cos(\theta)}{2}} \cdot \left(\frac{\sqrt{3}}{2} \cos(\theta/2) - \frac{1}{2} \sin(\theta/2) \right) = \frac{3}{14}$$

$$\sqrt{\frac{1 - \cos(\theta)}{2}} \cdot \left(\frac{\sqrt{3}}{2} \sqrt{\frac{1 + \cos(\theta)}{2}} - \frac{1}{2} \sqrt{\frac{1 - \cos(\theta)}{2}} \right) = \frac{3}{14}$$

$$\left(\frac{\sqrt{3}}{2} \sqrt{\frac{1 - \cos^2(\theta)}{4}} - \frac{1}{2} \left(\frac{1 - \cos(\theta)}{2} \right) \right) = \frac{3}{14}$$

$$\frac{\sqrt{3} \sin \theta}{4} - \frac{(1 - \cos \theta)}{4} = \frac{3}{14}$$

$$\sqrt{3} \sin \theta + \cos \theta = \frac{13}{7}$$

$$\cos \theta = \frac{13}{7} - \sqrt{3} \sin \theta.$$

Substitute into $\sin^2 \theta + \cos^2 \theta = 1$:

$$\left(\frac{13}{7} - \sqrt{3} \sin \theta \right)^2 + \sin^2 \theta = 1.$$

$$4 \sin^2 \theta - \frac{26\sqrt{3}}{7} \sin \theta + \frac{120}{49} = 0.$$

Solving:

$$\sin \theta = \frac{5\sqrt{3}}{14} \quad (\text{valid root}), \quad \cos \theta = \frac{11}{14} \Rightarrow \tan \theta = \frac{5\sqrt{3}}{11}.$$

20. (AMC 12 12B 2024) Suppose A , B , and C are points in the plane with $AB = 40$ and $AC = 42$, and let x be the length of the line segment from A to the midpoint of $\overline{BC}$. Define a function f by letting $f(x)$ be the area of $\triangle ABC$. Then the domain of f is an open interval (p, q) , and the maximum value r of $f(x)$ occurs at $x = s$. What is $p + q + r + s$?

(A) 909 (B) 910 (C) 911 (D) 912 (E) 913

Answer: C

Solution 1

Let the midpoint of BC be M , and let the length $BM = CM = a$. We know there are limits to the value of x , and these limits can probably be found through the triangle inequality. But the triangle inequality relates the third side length BC to AC and AB , and doesn't contain any information about the median. Therefore we're going to have to write the side BC in terms of x and then use the triangle inequality to find bounds on x .

We use Stewart's theorem to relate BC to the median AM : $man + dad = bmb + cnc$. In this case $m = \frac{a}{2}$, $n = \frac{a}{2}$, $a = m + n$, $d = x$, $b = 42$, $c = 40$.

Therefore we get the equation $2a^3 + 2ax^2 = a \cdot 42^2 + a \cdot 40^2$ $2a^2 + 2x^2 = 42^2 + 40^2$.

Notice that since $20 - 21 - 29$ is a pythagorean triple, this means $2a^2 + 2x^2 = 58^2$.

$$\begin{aligned} \Rightarrow a^2 &= \frac{58^2}{2} - x^2 \\ \Rightarrow a &= \sqrt{\frac{58^2}{2} - x^2} \end{aligned}$$

By triangle inequality, $2a + 40 > 42 \Rightarrow a > 1$ and $40 + 42 > 2a \Rightarrow a < 41$

Let's tackle the first inequality:

$$\begin{aligned} \sqrt{\frac{58^2}{2} - x^2} > 1 &\Rightarrow x^2 < \frac{58^2}{2} - 1 \\ \Rightarrow x^2 < \frac{40^2 + 42^2}{2} - 1 &\Rightarrow x^2 < 41^2 \end{aligned}$$

Here we use the property that $\frac{x^2 + (x+2)^2}{2} - 1 = (x+1)^2$.

Therefore in this case, $x < 41$.

For the second inequality,

$$\sqrt{\frac{58^2}{2} - x^2} < 41 \Rightarrow x^2 > \frac{58^2}{2} - 41^2$$

$$\implies x^2 > \frac{58^2}{2} - 1 + 1 - 41^2$$

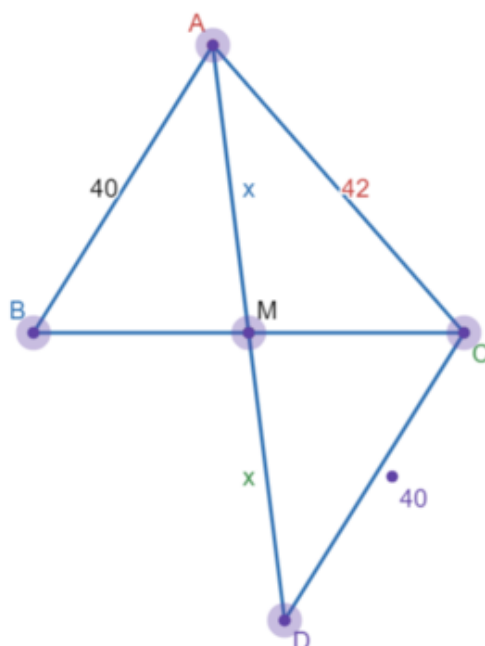
$$\implies x^2 > 41^2 + 1 - 41^2 \implies x^2 > 1 \implies x > 1$$

Therefore we have $1 < x < 41$, so the domain of $f(x)$ is $(1, 41)$.

The area of this triangle is $\frac{1}{2}42 \cdot 40 \cdot \sin(\theta)$. The maximum value of the area occurs when the triangle is right, i.e. $\theta = 90^\circ$. Then the area is $\frac{1}{2} \cdot 40 \cdot 42 = 840$. The length of the median of a right triangle is half the length of it's hypotenuse, which squared is $40^2 + 42^2 = 58^2$. Thus the length of x is 29.

Our final answer is $1 + 41 + 840 + 29 = \boxed{911} \text{ (C)}$

Solution 2 (Geometry)



Let midpoint of BC as M , extends AM to D and $MD = x$, triangle ACD has 3 sides $(40, 42, 2x)$, based on triangle inequality,

$$42 - 40 < 2x < 42 + 40$$

$$1 < x < 41$$

so

$$p = 1, q = 41$$

$$2 \cdot f(x) = 40 \cdot 42 \cdot \sin(A) \leq 2 \cdot 840$$

so $r = 840$ which is achieved when $A = 90^\circ$, then $\angle ACD = 90^\circ$

$$(2x)^2 = 40^2 + 42^2$$

$$x = 29$$

$$s = 29$$

$$p + q + s + r = 1 + 41 + 29 + 840 = \boxed{(C) \ 911}$$

Solution 3 (Trigonometry)

Let $A = (0, 0)$, $B = (b, 0)$, $C = (c \cos \theta, c \sin \theta)$

$$M = \left(\frac{b + c \cos \theta}{2}, \frac{c \sin \theta}{2} \right).$$

$$AM = x = \sqrt{\left(\frac{b + c \cos \theta}{2} \right)^2 + \left(\frac{c \sin \theta}{2} \right)^2} = \frac{\sqrt{c^2 + 2bc \cos \theta + b^2}}{2}.$$

When $\cos \theta = 1$:

$$x = \frac{\sqrt{(c+b)^2}}{2} = \frac{c+b}{2} = \frac{42+40}{2} = 41.$$

When $\cos \theta = -1$:

$$x = \frac{\sqrt{(c-b)^2}}{2} = \frac{c-b}{2} = \frac{42-40}{2} = 1.$$

The domain of $f(x)$ is the open interval:

$$\boxed{(1, 41)}.$$

The rest follows Solution 2

Solution 4 (Apollonius)

Here's a faster way to solve this problem using Apollonius's Theorem (which is a special case of Stewart's Theorem for medians). In this case, $x^2 = \frac{1}{4} * (2AB^2 + 2AC^2 - BC^2)$. So, $x^2 = 1682 - \frac{1}{4} * BC^2$.

We know that, by the Triangle Inequality, $2 < BC < 82$. Applying these to Apollonius, we have that the minimum value of x is $x = 1$ and the maximum value is $x = 41$ (both cannot be reached, however).

The rest of the solution follows Solution 1.

Solution 5 (Median length formula)

Let the midpoint of BC be D . Then, by the Median Length Formula: $2 * AD^2 = AB^2 - BD^2 + AC^2 - CD^2$. If we let $BC = 2n$ and $AD = x$, then we get the relationship that: $x = \sqrt{1682 - n^2}$. By the Triangle Inequality $2 < BC < 82$, so $1 < n < 41$. This means that the domain of x is $(\sqrt{1}, \sqrt{1681}) = (1, 41)$.

The rest follows Solution 1.

Solution 6 (AM-GM Inequality)

By letting BC equal $2a$, we can use Heron's formula to calculate the area. Notice the semi-perimeter is just $\frac{40+42+2a}{2}$ which is just $a + 41$. Next, by Heron's formula, the area of ABC is: $\sqrt{(a+41)(a+1)(a-1)(41-a)}$ which simplifies to the $\sqrt{(a^2-1)(41^2-a^2)}$. We now know that the domain of $f(x)$ is just the domain of $\sqrt{(a^2-1)(41^2-a^2)}$. This domain is very easy to calculate. We see that $a^2 > 1$ and $a^2 < 41^2$. Because a is always positive, we see that a is in the open interval $(1, 41)$. Now, we find the maximum of $f(x)$. By the AM-GM inequality, we

have: $\frac{((a^2-1)+(41^2-a^2))}{2} \geq \sqrt{(a^2-1)(41^2-a^2)}$. Simplifying and letting $\sqrt{(a^2-1)(41^2-a^2)} = f(x)$, we get that $f(x) \leq \frac{41^2-1}{2} = 840$. We know by AM-GM that $f(x) = 840$ if and only if $a^2-1 = 41^2-a^2$. Solving, $a = 29$. Therefore, we have found the domain of f is the open interval $(1, 41)$ and the maximum of f is 840 which occurs at $x = 29$ (Apply Stewart's to triangle ABC when knowing that $BC = 58$.) Adding these up, we get $1 + 41 + 840 + 29 = 911$ or $\boxed{C}$.

Solution 7 (Cheese (Barely Any Risk and So Much Easier))

We can easily see that $\text{Dom}(f) = (1, 41)$ by considering extreme cases of degenerate triangles (B is on AC or A is on BC). Then, the maximum area is extremely likely to be when the triangle is right, which happens when $BC = 58$ ($20 - 21 - 29$ triangle, but scaled up). The area of this triangle is 840. Since $40 \approx 42$, the median length is approximately equal to the altitude length, which is $\frac{1680}{58} = \frac{840}{29} = \frac{870-30}{29} = 30 - \frac{30}{29} \approx 29$. Thus, the answer is $1 + 41 + 840 + 29 = \boxed{(C) 911}$.

21. (AMC 12 12B 2024) The measures of the smallest angles of three different right triangles sum to 90° . All three triangles have side lengths that are primitive Pythagorean triples. Two of them are $3 - 4 - 5$ and $5 - 12 - 13$. What is the perimeter of the third triangle?

(A) 40 (B) 126 (C) 154 (D) 176 (E) 208

Answer: C

Solution 1

Let α and β be the smallest angles of the $3 - 4 - 5$ and $5 - 12 - 13$ triangles respectively. We have

$$\tan(\alpha) = \frac{3}{4} \text{ and } \tan(\beta) = \frac{5}{12}$$

Then

$$\tan(\alpha + \beta) = \frac{\frac{3}{4} + \frac{5}{12}}{1 - \frac{3}{4} \cdot \frac{5}{12}} = \frac{56}{33}$$

Let θ be the smallest angle of the third triangle. Consider

$$\tan 90^\circ = \tan((\alpha + \beta) + \theta) = \frac{\frac{56}{33} + \tan \theta}{1 - \frac{56}{33} \cdot \tan \theta}$$

In order for this to be undefined, we need

$$1 - \frac{56}{33} \cdot \tan \theta = 0$$

so

$$\tan \theta = \frac{33}{56}$$

Hence the base side lengths of the third triangle are 33 and 56. By the Pythagorean Theorem, the hypotenuse of the third triangle is 65, so the perimeter is $33 + 56 + 65 = \boxed{(C) 154}$.

Solution 2 (Complex Number)

The smallest angle of $3 - 4 - 5$ triangle can be viewed as the argument of $4 + 3i$, and the smallest angle of $5 - 12 - 13$ triangle can be viewed as the argument of $12 + 5i$.

Hence, if we assume the ratio of the two shortest length of the last triangle is $1 : k$ (k being some rational number), then we can derive the following formula of the sum of their argument. Since their argument adds up to $\frac{\pi}{2}$, it's the argument of i . Hence,

$$(4 + 3i)(12 + 5i)(k + i) = ni,$$

where n is some real number.

Solving the equation, we get

$$56k - 33 = 0, \quad 33k + 56 = n.$$

Hence $k = \frac{33}{56}$

Since the sidelength of the theird triangle are co-prime integers, two of its sides are 33 and 56. And the last side is $\sqrt{33^2 + 56^2} = 65$, hence, the parameter of the third triangle if $33 + 56 + 65 = \boxed{\text{(C)} 154}$.

Solution 3 (Another Trig)

Denote the smallest angle of the $3 - 4 - 5$ triangle as α , the smallest angle of the $5 - 12 - 13$ triangle as β , and the smallest angle of the triangle we are trying to solve for as θ . We then have

$$\alpha + \beta + \theta = 90$$

$$\alpha + \beta = 90 - \theta$$

$$\sin(\alpha + \beta) = \sin(90 - \theta) = \cos \theta$$

$$\cos \theta = \sin \alpha \cos \beta + \cos \alpha \sin \beta = \left(\frac{3}{5}\right)\left(\frac{12}{13}\right) + \left(\frac{4}{5}\right)\left(\frac{5}{13}\right) = \frac{56}{65}$$

Taking the hypotenuse to be 65 and one of the legs to be 56, we compute the last leg to be $\sqrt{65^2 - 56^2} = \sqrt{(65 - 56)(65 + 56)} = \sqrt{9 \cdot 121} = 3 \cdot 11 = 33$

Giving us a final answer of $65 + 56 + 33 = \boxed{\text{(C)} 154}$.

Solution 3.1 (Different Flavor of the same thing)

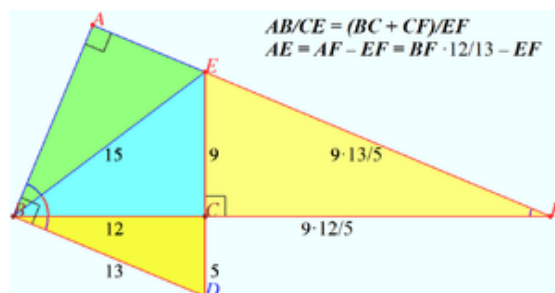
Consider $\sin(\theta) = \sin(90 - (\alpha + \beta)) = \cos(\alpha + \beta) = \frac{33}{65}$ using the cosine addition identity. Instead of using the Pythagorean theorem, we can use Euclid's formula since we're dealing with primitive triples.

$$65 = a^2 + b^2$$

$$33 = a^2 - b^2$$

Combining that, we get $a = 7$ and $b = 4$. Using this, we can get that the other leg must be $2ab = 56$. We add the lengths and get that the perimeter is $56 + 65 + 33 = \boxed{154}$.

Solution 4 (Similarity)



Let's arrange the triangles $BCD(5 - 12 - 13)$, $BCE(9 - 12 - 15)$ and ABE as shown in the diagram.

$$F = AE \cap BC.$$

$$AE \perp AB, DB \perp AB \implies \triangle BCD \sim \triangle FCE \sim \triangle FAB \implies$$

$$EF = \frac{9 \cdot 13}{5}, CF = \frac{9 \cdot 12}{5}, BF = BC + CF = \frac{12 \cdot 14}{5},$$

$$\frac{AB}{CE} = \frac{BF}{EF} \implies AB = \frac{12 \cdot 14}{13},$$

$$AE = AF - EF = BF \cdot \frac{12}{13} - EF = \frac{99}{13} \implies$$

$$AE : AB : BE = 99 : 12 \cdot 14 : 15 \cdot 13 = 33 : 56 : 65 \implies 65 + 56 + 33 = \boxed{\text{(C) } 154}$$

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Solution 5 (Complex)

Suppose the triangle has legs a, b . We want

$$\arctan \frac{3}{4} + \arctan \frac{5}{12} + \arctan \frac{a}{b} = \frac{\pi}{2}.$$

This is equivalent to

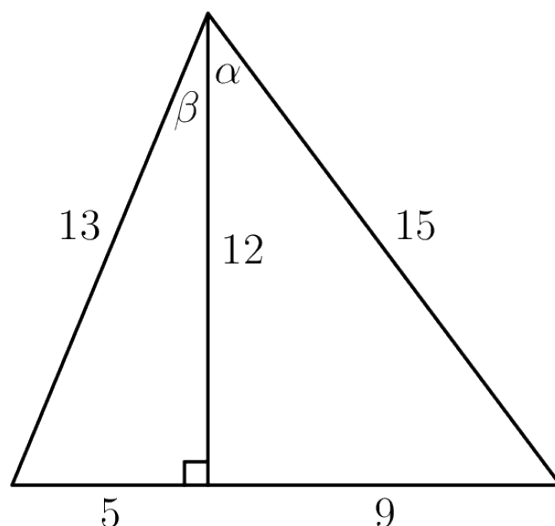
$$\begin{aligned} \arg e^{i \arctan \frac{3}{4}} \cdot e^{i \arctan \frac{5}{12}} \cdot e^{i \arctan \frac{a}{b}} &= \frac{\pi}{2} \\ \arg(4 + 3i)(12 + 5i)(b + ai) &= \frac{\pi}{2} \\ \arg((33b - 56a) + (33a + 56b)i) &= \frac{\pi}{2} \end{aligned}$$

Since the argument of this complex number is $\frac{\pi}{2}$, its real part must be 0. Matching real and imaginary parts yields $33b = 56a$, or $b = \frac{56}{33}a$. The smallest pair (a, b) that works is $(33, 56)$, which yields a hypotenuse of 65. The perimeter of this triangle is $33 + 56 + 65 = \boxed{\text{(C) } 154}$.

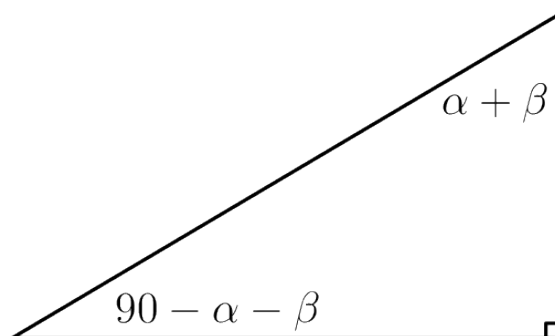
-Benedict T (countmath1)

Solution 6 (Law of Cosines)

We can start by scaling the $3 - 4 - 5$ right triangle up by a factor of 3 and "gluing" it to the $5 - 12 - 13$ triangle's longer leg. Let α , β , and θ be the smallest angles in the $3 - 4 - 5$, $5 - 12 - 13$ and unknown triangle respectively. We can construct the following diagram of the two known triangles.



We can also construct a diagram for the third, unknown triangle like so. We know that $\alpha + \beta + \theta = 90$, and therefore that $\theta = 90 - \alpha - \beta$. We also know that the other acute angle in this third triangle will have a measure of $\alpha + \beta$ by the triangle angle sum theorem.



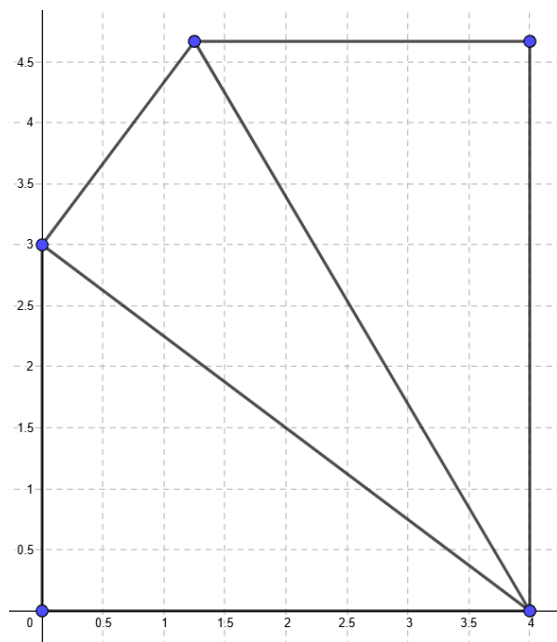
We can use the law of cosines on the triangle in the first diagram to get the equation $14^2 = 15^2 + 13^2 - 2(13)(15)\cos(\alpha + \beta)$. Isolating $\cos(\alpha + \beta)$, we get $-198 = -390\cos(\alpha + \beta)$, which further simplifies to $\cos(\alpha + \beta) = \frac{198}{390}$. Since the third triangle has to be a primitive Pythagorean triple, we must take this trig ratio into its most simple form, namely $\cos(\alpha + \beta) = \frac{33}{65}$. Using this information in our second diagram, we know that the smaller, adjacent leg must have length 33, and the hypotenuse must have length 65. We can then use the Pythagorean theorem to find the other, unknown leg, which has length 56. Adding these three lengths together, we find that the perimeter of this right triangle is $33 + 56 + 65 = \boxed{\text{(C) } 154}$.

Solution 6.1 (Faster Ending)

Instead of computing $\sqrt{65^2 - 33^2}$ to find the second leg in the unknown triangle by hand, we can use process of elimination. **(B)** 126 and **(C)** 154 are the only answers within the realm of possibility, because **(A)** 40 would entitle a triangle with a negative side length, and **(D)** 176 and **(E)** 208 would require legs greater than the length of the hypotenuse. The answer choice **(B)** 126 would force the second leg to have a length of 28, which is smaller than the smallest leg in the triangle. (We know that the leg with length 33 is the smallest leg in the triangle by the side-angle relationship theorem, because it is opposite the smallest angle in the

triangle.) Therefore, the only valid answer choice remaining is **(C) 154**.

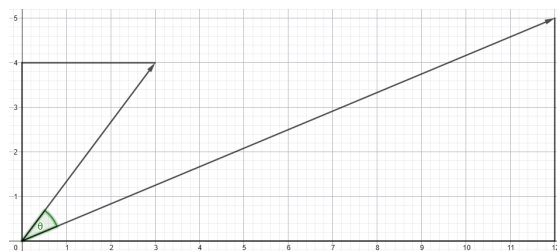
Solution 7 (No Trig, Coord Bash)



Set up a coordinate system. Let $(0, 0)$, $(4, 0)$, and $(0, 3)$ be the vertices of the base $3-4-5$ right triangle. In this case, the three smallest angles will all be at $(4, 0)$, and one of the coordinates of the unknown triangle has to lie on the line $x = 4$. Now, scale down the $5-12-13$ triangle by $\frac{5}{12}$ so that the new sides are $\frac{25}{12}-5-\frac{65}{12}$, and place the side with length 5 at the coordinates $(0, 3)$ and $(4, 0)$. The line passing through these two points can be written as $y = -\frac{3}{4}x + 3$, so the perpendicular of this line at $(0, 3)$ can be written as $y = \frac{4}{3}x + 3$. Since the length of the other side is $\frac{25}{12}$, after drawing smaller $3-4-5$ right triangles, we find that the third coordinate of the $\frac{25}{12}-5-\frac{65}{12}$ is at $(\frac{5}{4}, \frac{14}{3})$. This coordinate will be one of the coordinates for our unknown triangle. We can place the other coordinate of the unknown triangle at $(4, \frac{14}{3})$ and the third is, by definition, at $(4, 0)$. The distance from $(\frac{5}{4}, \frac{14}{3})$ to $(4, \frac{14}{3})$ is $\frac{11}{4}$, and the distance from $(4, \frac{14}{3})$ to $(4, 0)$ is $\frac{14}{3}$, and from before, the distance from $(\frac{5}{4}, \frac{14}{3})$ to $(4, 0)$ is $\frac{65}{12}$. Scaling up the sides so that they are integers, we see that the side lengths make a $33-56-65$ right triangle. The perimeter is then $33 + 56 + 65 = \mathbf{(C) 154}$.

Solution 8 (Linear Algebra)

Because the three smallest angles add up to 90° , you can imagine putting the $3-4-5$ and $5-12-13$ triangles on the y -axis and x -axis, respectively, in the first quadrant. So that all 3 triangles will have their smallest angle meet at the origin and exactly fill up the first quadrant.



Now the $3-4-5$ and $5-12-13$ triangles describe the vectors $\langle 3, 4 \rangle$ and $\langle 12, 5 \rangle$ with the smallest angle of the third triangle, θ , between them. We can find $\cos(\theta)$ by taking the dot product and dividing by the magnitudes of the vectors (which are the hypotenuses that we know to be 5 and 13).

$$\cos \theta = \frac{\langle 3, 4 \rangle \cdot \langle 12, 5 \rangle}{\|\langle 3, 4 \rangle\| \|\langle 12, 5 \rangle\|} = \frac{3 \cdot 12 + 4 \cdot 5}{5 \cdot 13} = \frac{56}{65}$$

This ratio $\cos \theta = \frac{56}{65}$ describes the third triangle to have a hypotenuse with length 65, one leg with length 56, and another leg with length $\sqrt{65^2 - 56^2} = \sqrt{(65-56)(65+56)} = \sqrt{9 \cdot 121} = 33$ by Pythagorean and difference of squares.

The perimeter is then $65 + 56 + 33 = \boxed{\text{(C) } 154}$

22. (AMC 12 12B 2024) Let $\triangle ABC$ be a triangle with integer side lengths and the property that $\angle B = 2\angle A$. What is the least possible perimeter of such a triangle?
 (A) 13 (B) 14 (C) 15 (D) 16 (E) 17

Answer: C

Solution 1

Let $AB = c$, $BC = a$, $AC = b$. According to the law of sines,

$$\frac{b}{a} = \frac{\sin \angle B}{\sin \angle A} = 2 \cos \angle A$$

According to the law of cosines,

$$\cos \angle A = \frac{b^2 + c^2 - a^2}{2bc}$$

Hence,

$$\frac{b}{a} = \frac{b^2 + c^2 - a^2}{bc}$$

This simplifies to $b^2 = a(a+c)$. We want to find the positive integer solution (a, b, c) to this equation such that a, b, c forms a triangle, and $a + b + c$ is minimized. We proceed by casework on the value

23. (AMC 12 12B 2024) A right pyramid has regular octagon $ABCDEFGH$ with side length 1 as its base and apex V . Segments $\overline{AV}$ and $\overline{DV}$ are perpendicular. What is the square of the height of the pyramid?
 (A) 1 (B) $\frac{1+\sqrt{2}}{2}$ (C) $\sqrt{2}$ (D) $\frac{3}{2}$ (E) $\frac{2+\sqrt{2}}{3}$

Answer: B

Solution 1

To find the height of the pyramid, we need the length from the center of the octagon (denote as I) to its vertices and the length of AV .

From symmetry, we know that $\overline{AV} = \overline{DV}$, therefore $\triangle AVD$ is an isosceles right triangle. Denote $\overline{AV}$ as x so that $\overline{AD} = x\sqrt{2}$. Dropping the altitudes from B and C to AD in isosceles trapezoid $ABCD$ (we know this from the fact that it is a

regular octagon) reveals that $\overline{AD} = 1 + 2(\sqrt{2}/2) = 1 + \sqrt{2}$ and $\overline{AV} = (\overline{AD})/\sqrt{2} = (\sqrt{2} + 2)/2$.

To find the length IA , we cut the octagon into 8 triangles, each with a smallest angle of 45 degrees. Using the law of cosines on $\triangle AIB$ reveals that $IA^2 = (2 + \sqrt{2})/2$.

Finally, using the Pythagorean theorem, we can find that $\overline{IV}^2 = \overline{AV}^2 - \overline{IA}^2 = ((2 + \sqrt{2})/2)^2 - (2 + \sqrt{2})/2 = \boxed{(1 + \sqrt{2})/2}$ which is answer choice $\boxed{B}$.

username2333 hashbrown2009

Solution 2 (Less computation)

Let O be the center of the regular octagon. Connect AD , and let I be the midpoint of line segment AD . It is easy to see that $VI = \frac{1}{2}AD = \frac{1+\sqrt{2}}{2}$ and $OI = \frac{1}{2}AH = \frac{1}{4}$. Hence,

$$\begin{aligned} VO^2 &= VI^2 - OI^2 \\ &= \left(\frac{1 + \sqrt{2}}{2}\right)^2 - \frac{1}{4} \\ &= \frac{1 + \sqrt{2}}{2} \end{aligned}$$

Hence, the answer is $\boxed{B}$.

Solution 3

ABCD is isosceles trapezium so,

$$AD = 1 + \sqrt{2}$$

let h be height of pyramid. let O be point where height touches base.

Perpendicular distance from O to AD is 0.5 .

So, by pythagorus theorem in triangle ABD

$$(1 + \sqrt{2})^2 = (VA)^2 + (VD)^2 = (2VA)^2 \Rightarrow (VA)^2 = \frac{3 + 2\sqrt{2}}{2}$$

Height of triangle ABD

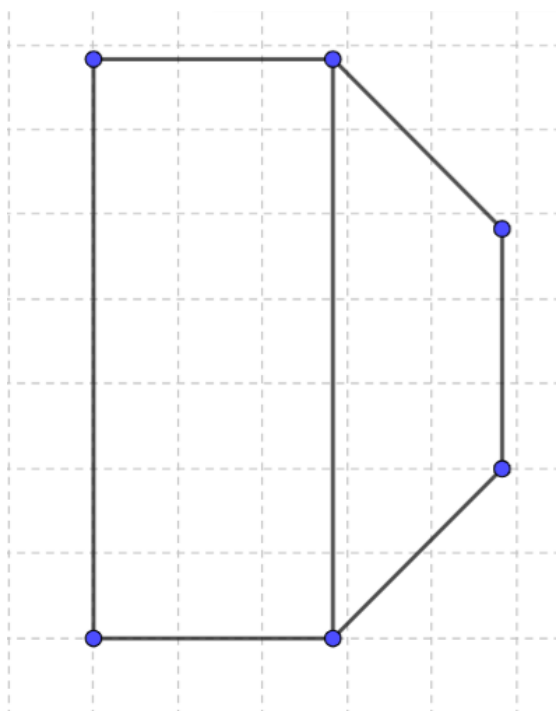
$$\left(\frac{1 + \sqrt{2}}{2}\right)^2 + y^2 = \frac{3 + 2\sqrt{2}}{2} \Rightarrow y^2 = \frac{3 + 2\sqrt{2}}{4}$$

Height of pyramid.

$$h^2 = y^2 \cdot (0.5)^2 \Rightarrow h^2 = \frac{3 + 2\sqrt{2}}{4} \Rightarrow h = \frac{1 + \sqrt{2}}{2}$$

Hence, option (B)

Solution 4 (Easy Geometry)



Firstly, scale each of the sides up by $2\sqrt{2}$, and split the octagon into half. Next, divide the octagon into a rectangle and the isosceles trapezoid, and let A and D be the points where the shapes coincide. Also, let I be the center of the rectangle (which is also the center of the octagon.) By drawing $45-45-90$ triangles in the trapezoid, we see that $AD = 4 + 2\sqrt{2}$. Now, consider the length of AI . It is half of the diagonal of the rectangle, and since the diagonal is just $\sqrt{(2\sqrt{2})^2 + (4 + 2\sqrt{2})^2} = \sqrt{8 + 24 + 16\sqrt{2}} = \sqrt{32 + 16\sqrt{2}}$, the length of AI is just $\sqrt{8 + 4\sqrt{2}}$. Now, consider triangle ADV . The apex of a regular pyramid is defined to be equidistant from every vertex, and since segments AV and DV are perpendicular, then triangle ADV is a $45-45-90$ right triangle with the right angle at V . This means that both AV and DV can be written as $\frac{AD}{\sqrt{2}} = \frac{4+2\sqrt{2}}{\sqrt{2}} = 2\sqrt{2} + 2$. Finally, since IV is perpendicular to base $ABCDEFGH$, then AIV is a right triangle. Applying Pythagorean Theorem yields $IV^2 = (2\sqrt{2} + 2)^2 - (8 + 4\sqrt{2}) = 4 + 4\sqrt{2}$. However, initially we had scaled each of the sides by $2\sqrt{2}$, so we have to divide the answer by the square of that, so the real answer is then $\frac{1+\sqrt{2}}{2}$. Therefore the answer is then

$$\boxed{\text{(B)} \frac{1 + \sqrt{2}}{2}}$$

Solution 5 (Vectors)

Consider the vectors $\vec{AV}$ and $\vec{DV}$. If we use a coordinate plane where one of the axes is parallel to one of the sides of the octagon, we can calculate each of the vectors to be

$$\vec{AV} = \left\langle \frac{1}{2}, \frac{1 + \sqrt{2}}{2}, h \right\rangle$$

$$\vec{DV} = \left\langle \frac{1}{2}, \frac{-1 - \sqrt{2}}{2}, h \right\rangle$$

Now, we must have $\vec{AV} \cdot \vec{DV} = 0$ if the vectors are perpendicular to each other, so

$$\left\langle \frac{1}{2}, \frac{-1 - \sqrt{2}}{2}, h \right\rangle \cdot \left\langle \frac{1}{2}, \frac{1 + \sqrt{2}}{2}, h \right\rangle = \frac{1}{4} - \frac{3 + 2\sqrt{2}}{4} + h^2 = 0$$

$$h^2 = \frac{2 + 2\sqrt{2}}{4} = \frac{1 + \sqrt{2}}{2}$$

Yielding answer choice $\boxed{B}$.

==Only B and D looks normal , guess one using

24. (AMC 12 12B 2024) What is the number of ordered triples (a, b, c) of positive integers, with $a \leq b \leq c \leq 9$, such that there exists a (non-degenerate) triangle $\triangle ABC$ with an integer inradius for which a , b , and c are the lengths of the altitudes from A to $\overline{BC}$, B to $\overline{AC}$, and C to $\overline{AB}$, respectively? (Recall that the inradius of a triangle is the radius of the largest possible circle that can be inscribed in the triangle.)

(A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Answer: B

Solution 1

First we derive the relationship between the inradius of a triangle r , and its three altitudes a, b, c . Using an area argument, we can get the following well known result

$$\left(\frac{AB + BC + AC}{2} \right) r = A$$

where AB, BC, AC are the side lengths of $\triangle ABC$, and A is the triangle's area. Substituting $A = \frac{1}{2} \cdot AB \cdot c$ into the above we get

$$\frac{r}{c} = \frac{AB}{AB + BC + AC}$$

Similarly, we can get

$$\frac{r}{b} = \frac{AC}{AB + BC + AC}$$

$$\frac{r}{a} = \frac{BC}{AB + BC + AC}$$

Hence,

$$\frac{1}{r} = \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \tag{26}$$

Note that there exists a unique, non-degenerate triangle with altitudes a, b, c if and only if $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are the side lengths of a non-degenerate triangle, i.e., $\frac{1}{b} + \frac{1}{c} > \frac{1}{a}$.

With this in mind, it remains to find all positive integer solutions (r, a, b, c) to the above such that $\frac{1}{b} + \frac{1}{c} > \frac{1}{a}$, and $a \leq b \leq c \leq 9$. We do this by doing casework on the value of r .

Since r is a positive integer, $r \geq 1$. Since $a \leq b \leq c \leq 9$, $\frac{1}{r} \geq \frac{1}{3}$, so $r \leq 3$. The only possible values for r are 1, 2, and 3.

Case 1: $r = 1$

For this case, we can't have $a \geq 4$, since $\frac{1}{a} + \frac{1}{b} + \frac{1}{c}$ would be too small. When $a = 3$, we must have $b = c = 3$. When $a \leq 2$, we would have $\frac{1}{b} + \frac{1}{c} \leq \frac{1}{a}$, which doesn't work. Hence this case only yields one valid solution $(1, 3, 3, 3)$

Case 2: $r = 2$

For this case, we can't have $a \geq 7$, for the same reason as in Case 1. When $a = 6$, we must have $b = c = 6$. When $a = 5$, we must have $b = 5, c = 10$ or $b = 10, c = 5$. Regardless, 10 appears, so it is not a valid solution. When $a \leq 4$, $\frac{1}{b} + \frac{1}{c} \leq \frac{1}{a}$. Hence, this case also only yields one valid solution $(2, 6, 6, 6)$

Case 3: $r = 3$

The only possible solution is $(3, 9, 9, 9)$, and clearly it is a valid solution.

Hence the only valid solutions are $(1, 3, 3, 3)$, $(2, 6, 6, 6)$, $(3, 9, 9, 9)$, and our answer is **(B) 3**

Solution 2 (cheese+does not actually work)

Note that some equilateral triangles with irrational side lengths will have rational altitudes and as a result, a rational inradius. Therefore, the ordered pairs $(a, b, c) = (3, 3, 3)$, $(a, b, c) = (6, 6, 6)$, and $(a, b, c) = (9, 9, 9)$ yield integer inradii, and we can make an educated guess that the answer is **(B) 3**

25. (AMC 12 12B 2024) Pablo will decorate each of 6 identical white balls with either a striped or a dotted pattern, using either red or blue paint. He will decide on the color and pattern for each ball by flipping a fair coin for each of the 12 decisions he must make. After the paint dries, he will place the 6 balls in an urn. Frida will randomly select one ball from the urn and note its color and pattern. The events "the ball Frida selects is red" and "the ball Frida selects is striped" may or may not be independent, depending on the outcome of Pablo's coin flips. The probability that these two events are independent can be written as $\frac{m}{n}$, where m and n are relatively prime positive integers. What is m ? (Recall that two events A and B are independent if $P(A \text{ and } B) = P(A) \cdot P(B)$.)

(A) 243 (B) 245 (C) 247 (D) 249 (E) 251

Answer: A

Solution 1

Let a be the number of balls that are both striped and red, x be the number of balls that are striped but blue, and y be the number of balls that are red but dotted. Then there must be $6 - a - x - y$ balls that are dotted and blue.

Let A be the event, "the ball Frida selects is red", and B be the event, "the ball Frida selects is striped". A and B are independent if and only if $\frac{a}{6} = \frac{a+x}{6} \cdot \frac{a+y}{6}$, i.e.,

$$6a = (a+x)(a+y)$$

Before we continue, I'd like to clarify that the sample space S , under which we are computing probability that A and B are independent, consists of 4^6 total outcomes, each equally likely. Notice that once given a, x, y , there are $\binom{6}{a} \binom{6-a}{x} \binom{6-a-x}{y} = \frac{6!}{a!x!y!(6-a-x-y)!}$ corresponding outcomes.

Now, it remains to find all nonnegative integer solutions (a, x, y) to the above equation such that $a + x + y < 6$, add up all the corresponding outcomes, and

then divide by 4^6 . We can find the solutions relatively easily by doing casework on $a = 0, 1, 2, 3, 4, 5, 6$. Long story short, one finds the solutions to be

$$\begin{aligned} &(0, 0, 0), (0, 0, 1) \cdots (0, 0, 6) \\ &(0, 1, 0), (0, 2, 0) \cdots (0, 6, 0) \\ &(1, 5, 0), (1, 0, 5), (1, 2, 1), (1, 1, 2) \\ &(2, 4, 0), (2, 0, 4), (2, 2, 1), (2, 1, 2) \\ &(3, 3, 0), (3, 0, 3) \\ &(4, 2, 0), (4, 0, 2) \\ &(5, 1, 0), (5, 0, 1) \\ &(6, 0, 0) \end{aligned}$$

There are little tricks we can use to make the process of adding up the corresponding outcomes faster. First, we notice that the first, second, and last row of solutions contributes $2^7 = 128$ outcomes. Then we can take advantage of the symmetry between x, y to add up the rest of the outcomes. In the end we get

$$128 + 2(6 + 180 + 15 + 180 + 20 + 15 + 6) = 972$$

Hence, the probability is $\frac{972}{4^6} = \frac{243}{2^{10}}$. Therefore, the answer is (A) 243

Solution 2 Symmetry + Observations = 5 min solve

Let S be the number of balls with stripes, R the number of red balls, and N the number of balls that are red and have stripes. Now, $P(A \text{ and } B) = P(A) \cdot P(B) \Rightarrow \frac{N}{6} = \frac{S \cdot R}{36}$, or $6N = SR$.

Make the trivial observation that whenever S or R equals 0, or both equal 6, there is one unique N satisfying the equality. Notice for $S = 0$ and R going from 0 to 6, we have a total probability of $\frac{2^6}{2^{12}}$ due to binomial theorem. By symmetry, we must multiply by 2 to account for switching R and S , and another 2 for the case where $S = R = 6$. Finally, we subtract the four overlapping cases $(0, 0), (0, 6), (6, 0)$, and $(6, 6)$, where each pair has format (S, R) . Thus, the total probability is $\frac{4 \cdot 2^6 - 4}{2^{12}}$.

Now, there are 4 more pairs (S, R) that satisfy the condition, namely $(2, 3), (3, 4), (3, 2)$, and $(4, 3)$.

Again, make the trivial observation that each of these have the same probability due to R/S invariance and similar distribution.

For a single pair, take $(2, 3)$ as an example. We know $N = 1$, and we fix S . The probability is therefore $\frac{\binom{6}{3}\binom{3}{1}\binom{3}{1}}{2^{12}} = \frac{180}{2^{12}}$.

Adding all of them up, we get $\frac{4 \cdot 2^6 - 4 + 180 \cdot 4}{2^{12}} = \frac{2^6 + 179}{2^{10}} = \frac{243}{1024}$. Thus, the answer is A.

§12 AMC 12 12A 2024

1. (*AMC 12 12A 2024*) What is the value of $9901 \cdot 101 - 99 \cdot 10101$? (A) 2 (B) 20 (C) 200 (D) 2000

Answer: A

Solution 1 (Direct Computation)

The likely fastest method will be direct computation. $9901 \cdot 101$ evaluates to 1000001 and $99 \cdot 10101$ evaluates to 999999. The difference is (A) 2.

Solution by [juwushu](#).

Solution 2 (Distributive Property)

We have

$$\begin{aligned} 9901 \cdot 101 - 99 \cdot 10101 &= (10000 - 99) \cdot 101 - 99 \cdot (10000 + 101) \\ &= 10000 \cdot 101 - 99 \cdot 101 - 99 \cdot 10000 - 99 \cdot 101 \\ &= (10000 \cdot 101 - 99 \cdot 10000) - 2 \cdot (99 \cdot 101) \\ &= 2 \cdot 10000 - 2 \cdot 9999 \\ &= \span style="border: 1px solid black; padding: 0 2px;">(A) 2\end{aligned}$$

MRENTHUSIASM

Solution 3 (Solution 1 but Distributive)

Note that $9901 \cdot 101 = 9901 \cdot 100 + 9901 = 990100 + 9901 = 1000001$ and $99 \cdot 10101 = 100 \cdot 10101 - 10101 = 1010100 - 10101 = 999999$, therefore the answer is (A) 2.

Solution 4 (Modular Arithmetic)

Evaluating the given expression $(\text{mod } 10)$ yields $1 - 9 \equiv 2 \pmod{10}$, so the answer is either (A) or (D). Evaluating $(\text{mod } 101)$ yields $0 - 99 \equiv 2 \pmod{101}$. Because answer (D) is $202 = 2 \cdot 101$, that cannot be the answer, so we choose choice (A) 2.

Solution 5 (Process of Elimination)

We simply look at the units digit of the problem we have (or take mod 10)

$$9901 \cdot 101 - 99 \cdot 10101 \equiv 1 \cdot 1 - 9 \cdot 1 = 2 \pmod{10}.$$

Since the only answers with 2 in the units digit are (A) and (D), we can then continue if you are desperate to use guess and check or an actually valid method to find the answer is (A) 2.

Solution 6 (Faster Distribution)

Observe that $9901 = 9900 + 1 = 99 \cdot 100 + 1$ and $10101 = 10100 + 1 = 101 \cdot 100 + 1$

$$\begin{aligned} \Rightarrow 9901 \cdot 101 - 99 \cdot 10101 &= ((9900 \cdot 101) + (1 \cdot 101)) - ((99 \cdot 10100) + (99 \cdot 1)) \\ &= (99 \cdot 100 \cdot 101) + 101 - (99 \cdot 100 \cdot 101) - 99 \\ &= 101 - 99 \\ &= \span style="border: 1px solid black; padding: 0 2px;">(A) 2\end{aligned}$$

Solution 7 (Cubes)

Let $x = 100$. Then, we have

$$\begin{aligned} 101 \cdot 9901 &= (x + 1) \cdot (x^2 - x + 1) = x^3 + 1, \\ 99 \cdot 10101 &= (x - 1) \cdot (x^2 + x + 1) = x^3 - 1. \end{aligned}$$

Then, the answer can be rewritten as $(x^3 + 1) - (x^3 - 1) = \boxed{\text{(A)} 2}$.

Solution 8 (Super Fast)

It's not hard to observe and express 9901 into $99 \cdot 100 + 1$, and 10101 into $101 \cdot 100 + 1$. We then simplify the original expression into $(99 \cdot 100 + 1) \cdot 101 - 99 \cdot (101 \cdot 100 + 1)$, which could then be simplified into $99 \cdot 100 \cdot 101 + 101 - 99 \cdot 100 \cdot 101 - 99$, which we can get the answer of $101 - 99 = \boxed{\text{(A)} 2}$.

Solution 9 (Estimation) *VERY FAST*

Notice that the answer choices are significantly different in value. This allows us to estimate the answer. 9901 is about 10000, and 101 is about 100. 99 is about 100, and 10101 is about 10000. Computing, we get $10000 \cdot 100 - 100 \cdot 10000 = 0$. The closest answer to this estimation is $\boxed{\text{(A)} 2}$.

Solution 10

We can see that the units digit of the expression is $1 - 9 \Rightarrow 11 - 9 \Rightarrow 2$, elimination options B, C, and E. Next, notice that $(9901)(101)$ is divisible by 101 while $(99)(10101)$ is not divisible by 101 (to see this, notice that 101 is prime, and $10101 = 10100 + 1$, so is not divisible by 101). This means that the final answer is not divisible by 101, eliminating (D), so the answer is $\boxed{\text{(A)} 2}$.

2. (AMC 12 12A 2024) A model used to estimate the time it will take to hike to the top of the mountain on a trail is of the form $T = aL + bG$, where a and b are constants, T is the time in minutes, L is the length of the trail in miles, and G is the altitude gain in feet. The model estimates that it will take 69 minutes to hike to the top if a trail is 1.5 miles long and ascends 800 feet, as well as if a trail is 1.2 miles long and ascends 1100 feet. How many minutes does the model estimates it will take to hike to the top if the trail is 4.2 miles long and ascends 4000 feet?
- (A) 240 (B) 246 (C) 252 (D) 258 (E) 264

Answer: B

Solution 1

Plug in the values into the equation to give you the following two equations:

$$\begin{aligned} 69 &= 1.5a + 800b, \\ 69 &= 1.2a + 1100b. \end{aligned}$$

Solving for the values a and b gives you that $a = 30$ and $b = \frac{3}{100}$. These values can be plugged back in showing that these values are correct. Now, using the given length of the trail, 4.2, and the given vertical increase, 4000, we get a final answer of $\boxed{\text{(B)} 246}$.

Solution by [juwushu](#).

Minor edits by ParticlePhysics and TigerSenju

Solution 2

Alternatively, observe that using $a = 10x$ and $b = \frac{y}{100}$ makes the numbers much closer to each other in terms of magnitude.

Plugging in the new variables:

$$\begin{aligned} 69 &= 15x + 8y, \\ 69 &= 12x + 11y. \end{aligned}$$

The solution becomes more obvious in this way, with $15 + 8 = 12 + 11 = 23$, and since $23 \cdot 3 = 69$, we determine that $x = y = 3$.

The question asks us for $4.2a + 4000b = 42x + 40y$. Since $x = y$, we have $(40 + 42) \cdot 3 = \boxed{\text{(B) } 246}$.

3. (AMC 12 12A 2024) The number 2024 is written as the sum of not necessarily distinct two-digit numbers. What is the least number of two-digit numbers needed to write this sum?

(A) 20 (B) 21 (C) 22 (D) 23 (E) 24

Answer: B

Solution 1

Since we want the least number of two-digit numbers, we maximize the two-digit numbers by choosing as many 99s as possible. Since $2024 = 99 \cdot 20 + 44 \cdot 1$, we choose twenty 99s and one 44, for a total of $\boxed{\text{(B) } 21}$ two-digit numbers.

Solution 2

We claim the answer is 21. This can be achieved by adding twenty 99's and a 44. To prove that the answer cannot be less than or equal to 20, we note that the maximum value of the sum of 20 or less two digit numbers is $20 \cdot 99 = 1980$, which is smaller than 2024, so we are done. Thus, the answer is $\boxed{\text{(B) } 21}$.

Solution 3 (Same as Solution 1 but Using $100=99+1$)

$2024 = 100 \cdot 20 + 24$. Since $100 = 99 + 1$, $2024 = (99 + 1) \cdot 20 + 24 = 99 \cdot 20 + 1 \cdot 20 + 24 = 99 \cdot 20 + 44$. Therefore a total of $\boxed{\text{(B) } 21}$ two-digit numbers are needed.

Solution 4

The maximum 2-digit number is 99, but try 100. $2024 \div 100$ is a little more than 20, and the remainder is less than 100, by intuition, so there's 20+ the remainder 1 = $\boxed{\text{(B) } 21}$.

4. (AMC 12 12A 2024) What is the least value of n such that $n!$ is a multiple of 2024?

(A) 11 (B) 21 (C) 22 (D) 23 (E) 253

Answer: D

Solution

Note that $2024 = 2^3 \cdot 11 \cdot 23$ in the prime factorization. Since $23!$ is a multiple of 2^3 , 11, and 23, we conclude that $23!$ is a multiple of 2024. Therefore, we have $n = \boxed{\text{(D) } 23}$. Remark

Memorizing the prime factorization of the current year is useful for the AMC 8/10/12 Exams.

Remark

Note that 2024 is $2025 - 1$, which is $45^2 - 1 = (45 + 1)(45 - 1) = 46 \cdot 44$.

5. (AMC 12 12A 2024) A data set containing 20 numbers, some of which are 6, has mean 45. When all the 6s are removed, the data set has mean 66. How many 6s were in the original data set?

(A) 4 (B) 5 (C) 6 (D) 7 (E) 8

Answer: D

Solution

Because the set has 20 numbers and mean 45, the sum of the terms in the set is $45 \cdot 20 = 900$.

Let there be s sixes in the set.

Then, the mean of the set without the sixes is $\frac{900-6s}{20-s}$. Equating this expression to 66 and solving yields $s = 7$, so we choose answer choice **(D) 7**.

Solution 2

Let S be the sum of the data set without the sixes and x be the number of sixes. We are given that $\frac{S+6x}{20} = 45$ and $\frac{S}{20-x} = 66$; the former equation becomes $S+6x = 900$ and the latter $S = 1320 - 66x$. Since we want x , we equate the two equations and see that $900 - 6x = 1320 - 66x \implies 60x = 420 \implies x = \mathbf{(D) 7}$.

Solution 3

Suppose there are x sixes. Then the sum of all the numbers can be written as $(20-x) \cdot 45 + 6x$

Then, the mean of this set is $\frac{(20-x) \cdot 45 + 6x}{20} = 45$. Solving this, we get $x = \mathbf{(D) 7}$

6. (AMC 12 12A 2024) The product of three integers is 60. What is the least possible positive sum of the three integers?

(A) 2 (B) 3 (C) 5 (D) 6 (E) 13

Answer: B

Solution 1

We notice that the optimal solution involves two negative numbers and a positive number. Thus we may split 60 into three factors and choose negativity. We notice that $10 \cdot 6 \cdot 1 = 10 \cdot (-6) \cdot (-1) = 60$, and trying other combinations does not yield lesser results so the answer is $10 - 6 - 1 = \mathbf{(B) 3}$.

Solution 2

We have $abc = 60$. Let a be positive, and let b and c be negative. Then we need $a > |b+c|$. If $a = 6$, then $|b+c|$ is at least 7, so this doesn't work. If $a = 10$, then $(b, c) = (-6, -1)$ works, giving $10 - 7 = \mathbf{(B) 3}$

Solution 3

We can see that the most optimal solution would be 1 positive integer and 2 negative ones (as seen in solution 1). Let the three integers be x , y , and z , and let x be positive and y and z be negative. If we want the optimal solution, we want the negative numbers to be as large as possible. so the answer should be $-60 \cdot -1 \cdot 1$, where $-60 - 1 + 1 = -60...$ right?

No! Our sum must be positive, so that would be invalid! We see that -30, -20, -15, and -10 are too less to allow the sum to be positive. For example, $-15 = z \cdot -15xy = 60$, so $xy = -4$. For xy to be the most positive, we will have 4 and -1. Yet,

$-15 + 4 - 1$ is still negative. After -10 , the next factor of 60 would be -6 . if $z = -6$, $xy = -10$. This might be positive! Now, if we have $z = -6$, $y = -1$, and $x = 10$, $x + y + z = 3$. It cannot be smaller because $x = 5$ and $y = -2$ would result in $x + y + z$ being negative. Therefore, our answer would be (B) 3.

Tiny minor edits made by SuperVince1

7. (AMC 12 12A 2024) In $\triangle ABC$, $\angle ABC = 90^\circ$ and $BA = BC = \sqrt{2}$. Points $P_1, P_2, \dots, P_{2024}$ lie on hypotenuse $\overline{AC}$ so that $AP_1 = P_1P_2 = P_2P_3 = \dots = P_{2023}P_{2024} = P_{2024}C$. What is the length of the vector sum

$$\overrightarrow{BP_1} + \overrightarrow{BP_2} + \overrightarrow{BP_3} + \dots + \overrightarrow{BP_{2024}}?$$

(A) 1011 (B) 1012 (C) 2023 (D) 2024 (E) 2025

Answer: D

Solution 1 (technical vector bash)

Let us find an expression for the x - and y -components of $\overrightarrow{BP_i}$. Note that $AP_1 + P_1P_2 + \dots + P_{2023}P_{2024} + P_{2024}C = AC = 2$, so $AP_1 = P_1P_2 = \dots = P_{2023}P_{2024} = P_{2024}C = \frac{2}{2025}$. All of the vectors $\overrightarrow{AP_1}$, $\overrightarrow{P_1P_2}$, and so on up to $\overrightarrow{P_{2024}C}$ are equal;

moreover, they equal $\mathbf{v} = \left\langle \frac{1}{\sqrt{2}} \cdot \frac{2}{2025}, -\frac{1}{\sqrt{2}} \cdot \frac{2}{2025} \right\rangle = \left\langle \frac{\sqrt{2}}{2025}, -\frac{\sqrt{2}}{2025} \right\rangle$.

We now note that $\overrightarrow{AP_i} = i\mathbf{v} = \left\langle \frac{i\sqrt{2}}{2025}, -\frac{i\sqrt{2}}{2025} \right\rangle$ (i copies of $\mathbf{v}$ added together). Fur-

thermore, note that $\overrightarrow{BP_i} = \overrightarrow{BA} + \overrightarrow{AP_i} = \langle 0, \sqrt{2} \rangle + \left\langle \frac{i\sqrt{2}}{2025}, -\frac{i\sqrt{2}}{2025} \right\rangle = \left\langle \frac{i\sqrt{2}}{2025}, \sqrt{2} - \frac{i\sqrt{2}}{2025} \right\rangle$.

We want $\sum_{i=1}^{2024} \overrightarrow{BP_i}$'s length, which can be determined from the x - and y -components. Note that the two values should actually be the same - in this problem, everything is symmetric with respect to the line $x = y$, so the magnitudes of the x - and y -components should be identical. The x -component is easier to calculate.

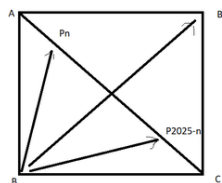
$$\sum_{i=1}^{2024} (\overrightarrow{BP_i})_x = \sum_{i=1}^{2024} \frac{i\sqrt{2}}{2025} = \frac{\sqrt{2}}{2025} \sum_{i=1}^{2024} i = \frac{\sqrt{2}}{2025} \cdot \frac{2024 \cdot 2025}{2} = 1012\sqrt{2}.$$

One can similarly evaluate the y -component and obtain an identical answer; thus, our desired length is $\sqrt{(1012\sqrt{2})^2 + (1012\sqrt{2})^2} = \sqrt{4 \cdot 1012^2} = \boxed{\text{(D) } 2024}$.

Solution 2

Notice that the average vector sum is 1. Multiplying the 2024 by 1, our answer is D

Solution 3 (Pair Sum)



Let point B reflect over $\overleftrightarrow{AC} \rightarrow B'$

We can see that for all n ,

$$|\overrightarrow{BP_n} + \overrightarrow{BP_{2025-n}}| = |\overrightarrow{BB'}| = 2.$$

As a result,

$$|\overrightarrow{BP_1} + \overrightarrow{BP_2} + \dots + \overrightarrow{BP_{2024}}| = 2 \cdot 1012 = \boxed{\text{(D) } 2024}$$

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edited by [luckuso](#)

Solution 4

Using the Pythagorean theorem, we can see that the length of the hypotenuse is 2. There are 2024 equally-spaced points on AC , so there are 2025 line segments along that hypotenuse. $\frac{2}{2025}$ is the length of each line segment. We get

$$\frac{2}{2025} + \frac{4}{2025} + \dots + \frac{4048}{2025} = \frac{2}{2025} \cdot \frac{2024 \cdot 2025}{2} = \boxed{\text{(D) } 2024}.$$

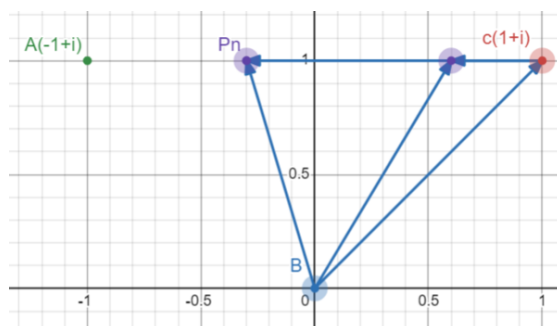
NOTE: This solution seems to have misunderstood the problem and counted line segments from A , arriving at the correct answer by luck.

Solution 5 (Physics-Inspired)

Let B be the origin, and set the x and y axes so that the x axis bisects $\angle ABC$, and the y axis is parallel to $\overleftrightarrow{AC}$. Notice that the endpoints of each vector all lie on $i = 1$, so each vector is of the form $1i + xj$. Furthermore, observe that for each $v_k = 1i + xj$, we have $v_{2024-k} = 1i - xj$, by properties of reflections about the x -axis: therefore $v_k + v_{2024-k} = 2i$. Since there are 1012 pairs, the resultant vector is $1012 \cdot 2i$, the magnitude of which is $\boxed{\text{(D) } 2024}$.

–Benedict T (countmath1)

Solution 6 (Complex Number)



Let B be the origin, place C at $C = 1 + i$ $\overrightarrow{CP_1} = re^{i\theta}$ $\overrightarrow{CP_n} = nre^{i\theta}$

Now we'll find $re^{i\theta} \frac{|\overrightarrow{AC}|}{\text{Number of Equal Segments}}$

$$= \frac{2}{2025} e^{i\pi}$$

$= -\frac{2}{2025}$ P_1 to P_{2024} can be written as such:

$$P_1 = C + \overrightarrow{CP_1} \quad P_2 = C + \overrightarrow{CP_2}$$

...

$$P_{2024} = C + \overrightarrow{CP_{2024}}$$

We want to find the sum of the complex numbers:

$$P_1 + P_2 + \dots + P_{2024} = 2024c + re^{i\theta}(1 + 2 + \dots + 2024) = 2024c + \frac{2024 \cdot 2025}{2} \cdot re^{i\theta}$$

Now we can plug in our value for C and $re^{i\theta}$ $2024c + \frac{2024 \cdot 2025}{2} \cdot re^{i\theta} = 2024(1+i) - 2024 = 2024i$

So the length is (D) 2024

Solution 7 (Extreme Case)

Notice that we can put points P_i with odd numbers i on C and those with even numbers i on A . So the sum of vectors $\overrightarrow{BP_1} + \overrightarrow{BP_2} + \overrightarrow{BP_3} + \dots + \overrightarrow{BP_{2024}}$ is just $1012 \times (\overrightarrow{BC} + \overrightarrow{BA})$ and the length of vector sum is (D) 2024

8. (AMC 12 12A 2024) How many angles θ with $0 \leq \theta \leq 2\pi$ satisfy $\log(\sin(3\theta)) + \log(\cos(2\theta)) = 0$?

(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Answer: A

Solution 1

Note that this is equivalent to $\sin(3\theta)\cos(2\theta) = 1$, which is clearly only possible when $\sin(3\theta) = \cos(2\theta) = \pm 1$. (If either one is between 1 and -1 , the other one must be greater than 1 or less than -1 to offset the product, which is impossible for sine and cosine.) They cannot be both -1 since we cannot take logarithms of negative numbers, so they are both $+1$. Then 3θ is $\frac{\pi}{2}$ more than a multiple of 2π and 2θ is a multiple of 2π , so θ is $\frac{\pi}{6}$ more than a multiple of $\frac{2}{3}\pi$ and also a multiple of π . However, a multiple of $\frac{2}{3}\pi$ will always have a denominator of 1 or 3, and never 6; it can thus never add with $\frac{\pi}{6}$ to form an integral multiple of π . Thus, there are (A) 0 solutions.

Solution 1.1 (less words)

$$\log(\sin(3\theta)) + \log(\cos(2\theta)) = 0$$

$$\log(\sin(3\theta)\cos(2\theta)) = 0$$

$$\sin(3\theta)\cos(2\theta) = 1$$

$$\text{Since } -1 \leq \sin(x), \cos(x) \leq 1 \Rightarrow \sin(3\theta) = \cos(2\theta) = \pm 1$$

BUT note that $\log(-1)$ is not real

$$\Rightarrow \sin(3\theta) = \cos(2\theta) = 1$$

$$3\theta = \frac{\pi}{2} + 2\pi n; 2\theta = 2\pi m (m, n \in \mathbb{Z})$$

$$\theta = \frac{\pi}{6} + \frac{2\pi n}{3}; \theta = \pi m$$

$\Rightarrow \theta$ has no solution

Giving us (A) 0.

Solution 2

Let $f(\theta) = \log(\sin(3\theta))$ and let $g(\theta) = \log(\cos(2\theta))$.

Note that $-1 \leq \sin(3\theta), \cos(2\theta) \leq 1$. Because the logarithm of a nonpositive number is not real, the functions f and g only exist when $\sin(3\theta)$ and $\cos(2\theta)$ are positive, respectively. Furthermore, because the logarithm of any positive real number less than 1 is negative, the only case where the function $f + g$ could equal 0 is if $f(\theta) = g(\theta) = 0$, which only occurs when their respective sine and cosine expressions are both equal to 1.

Thus, we have these two equations, where m and n are any integers:

$$\begin{aligned} \sin(3\theta) = 1 &\Rightarrow 3\theta = \frac{\pi}{2} + 2\pi m \Rightarrow \theta = \frac{\pi}{6} + \frac{2\pi}{3}m = \frac{4m+1}{6}\pi \\ \cos(2\theta) = 1 &\Rightarrow 2\theta = 2\pi n \Rightarrow \theta = \pi n \end{aligned}$$

Because m is an integer, $4m+1$ is odd, and so $\frac{4m+1}{6}$ is never an integer. Therefore, by the first equation, θ can never be an integer multiple of π . Thus, because the second equation requires that θ be an integer multiple of π , these two equations cannot both be satisfied, and so there are no solutions to the given equation.

Thus, we choose answer choice (A) 0.

Solution 3 (solution 1.1 with graphing)

$$\log(\sin(3\theta)) + \log(\cos(2\theta)) = 0$$

$$\log(\sin(3\theta)) = -\log(\cos(2\theta)) = \log(\sec(2\theta))$$

$$\therefore \text{let } y = \sin(3\theta) = \sec(2\theta), y > 0$$

Sketching the graphs $y = \sin(3x)$ and $y = \sec(2x)$ for $y > 0$, we see there are no intersections.

Thus there are (A) 0 angles that work.

9. (AMC 12 12A 2024) Let M be the greatest integer such that both $M + 1213$ and $M + 3773$ are perfect squares. What is the units digit of M ?

(A) 1 (B) 2 (C) 3 (D) 6 (E) 8

Answer: E

Solution 1

Let $M + 1213 = P^2$ and $M + 3773 = Q^2$ for some positive integers P and Q . We subtract the first equation from the second, then apply the difference of squares:

$$(Q + P)(Q - P) = 2560.$$

Note that $Q + P$ and $Q - P$ have the same parity, and

10. (AMC 12 12A 2024) Let α be the radian measure of the smallest angle in a 3–4–5 right triangle. Let β be the radian measure of the smallest angle in a 7–24–25 right triangle. In terms of α , what is β ?

(A) $\frac{\alpha}{3}$ (B) $\alpha - \frac{\pi}{8}$ (C) $\frac{\pi}{2} - 2\alpha$ (D) $\frac{\alpha}{2}$ (E) $\pi - 4\alpha$

Answer: C

Solution 1

We are given that

$$\tan \alpha = \frac{3}{4} \text{ and } \tan \beta = \frac{7}{24}.$$

We have

$$\begin{aligned} \tan(\alpha + \beta) &= \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} \\ &= \frac{\frac{3}{4} + \frac{7}{24}}{1 - \frac{3}{4} \cdot \frac{7}{24}} \\ &= \frac{4}{3}. \end{aligned}$$

It follows that

$$\begin{aligned} \alpha + \beta &= \tan^{-1} \left(\frac{4}{3} \right) \\ &= \frac{\pi}{2} - \tan^{-1} \left(\frac{3}{4} \right) \\ &= \frac{\pi}{2} - \alpha. \end{aligned}$$

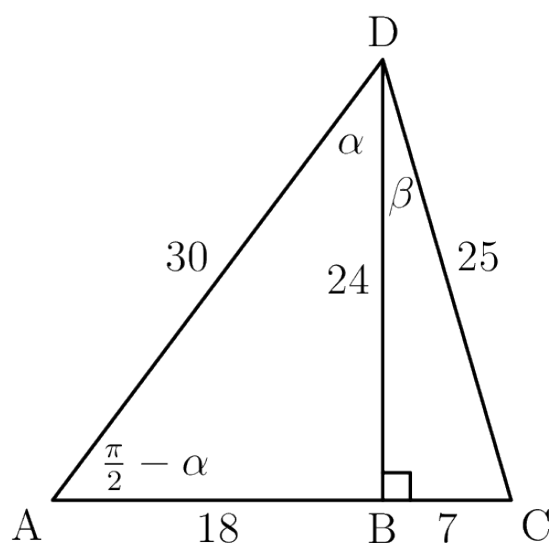
Therefore, the answer is

$$\beta = \boxed{\text{(C)} \frac{\pi}{2} - 2\alpha}.$$

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Solution 2: Scaling and combining triangles

We can scale the 3-4-5 triangle up by a factor of 6 to make its side lengths 18, 24, and 30, then glue its side of length 24 to the corresponding side in the 7-24-25 triangle:



Angles $\angle DAB$ and $\angle BDA$ are complementary in $\triangle ABD$, so $\angle DAB = \frac{\pi}{2} - \alpha$. We also have $AC = 18 + 7 = 25 = CD$, so $\triangle ACD$ is isosceles. That means that its base angles $\angle CDA$ and $\angle CAD$ are congruent, so $\alpha + \beta = \frac{\pi}{2} - \alpha$, and hence

$$\beta = \boxed{(C) \frac{\pi}{2} - 2\alpha}.$$

Solution 3: Trial and Error

Another approach to solving this problem is trial and error, comparing the sine of the answer choices with $\sin \beta = \frac{7}{25}$. Starting with the easiest sine to compute from the answer choices (option choice D). We get:

$$\begin{aligned} \sin\left(\frac{\alpha}{2}\right) &= \sqrt{\frac{1 - \cos \alpha}{2}} \\ &= \sqrt{\frac{1 - \frac{4}{5}}{2}} \\ &= \sqrt{\frac{1}{10}} \\ &\neq \frac{7}{25} \end{aligned}$$

The next easiest sine to compute is option choice C.

$$\begin{aligned} \sin\left(\frac{\pi}{2} - 2\alpha\right) &= \sin\left(\frac{\pi}{2}\right) \cos(2\alpha) \\ &= \cos 2\alpha \\ &= \cos^2 \alpha - \sin^2 \alpha \\ &= \frac{16}{25} - \frac{9}{25} \\ &= \frac{7}{25} \end{aligned}$$

Since $\sin\left(\frac{\pi}{2} - 2\alpha\right)$ is equal to $\sin \beta$, option choice C is the correct answer. amshah

Solution 4:

$$\sin(B) = \frac{24}{25} = 2 \cdot \frac{12}{25} = 2 \cdot \frac{3}{5} \cdot \frac{4}{5} = 2 \cdot \sin(A) \cdot \cos(A) = \sin(2A) = \cos(90^\circ - 2A)$$

Therefore

$$\beta = \boxed{(C) \frac{\pi}{2} - 2\alpha}$$

Added Note: It should be $\cos(B) = \frac{24}{25} = 2 \cdot \frac{12}{25} = 2 \cdot \frac{3}{5} \cdot \frac{4}{5} = 2 \cdot \sin(A) \cdot \cos(A) = \sin(2A) = \cos(90^\circ - 2A)$. $\sin(B)$ is actually $\frac{7}{25}$!

Hello 2d world (talk) 08:45, 22 August 2025 (EDT)

Solution 5: Ptolemy (no trig)

Let AB have length 15, BC have length 20, AC length 25, AD length 7 and CD length 24. Let x be the measure of segment BD. Thus the measure of angle ACB is α and the measure of angle ACD is β . ABCD is a cyclic quadrilateral because angle ABC and angle ADC are right angles. Using Ptolemy's theorem on this quadrilateral yields $25x = 15 \cdot 24 + 7 \cdot 20 = 500$, or $x = 20$. This means triangle CBD is isosceles. The perpendicular bisector of CD passes through the center (O)

of the circle on which ABCD lies and also passes through B. Let the intersection of the perpendicular bisector of CD and CD be point P. The measure of angle OBC is the same as the measure of the angle OCB which is α , so the measure of angle BOC is $\pi - 2\alpha$, so the measure of angle COP is 2α . Triangle COP is a right triangle with angle OCP being the same as angle ACD (β), angle COP being 2α , and angle CPO being $\frac{\pi}{2}$. So:

$$\beta + 2\alpha + \frac{\pi}{2} = \pi$$

$$\beta = \boxed{(C) \frac{\pi}{2} - 2\alpha}$$

Ilaggo2432

Solution 6(rough value)

given in the question, $\sin(a)$ is $3/5$ and $\sin(b)$ is $7/25$, estimate the angle by using the arcsin function. $\arcsin 3/5$ is around 0.643 rad and $\arcsin 7/25$ is around 0.283. Note that $\pi/2$ is closest to 1.57 rad, try one by one and option c suggests that

$$1.57 = 2(0.643) + 0.283.$$

My first amc 12 editmine too

Solution 7 (Angle Addition)

$\sin(\alpha) \cos(\beta) + \sin(\beta) \cos(\alpha) = \sin(\alpha + \beta) = \frac{4}{5}$. Noticing $\frac{4}{5} = \cos(\alpha) = \sin(\frac{\pi}{2} - \alpha)$

gives us $\alpha + \beta = \frac{\pi}{2} - \alpha$ so $\boxed{(C) \frac{\pi}{2} - 2\alpha}$ KEVIN_LIU

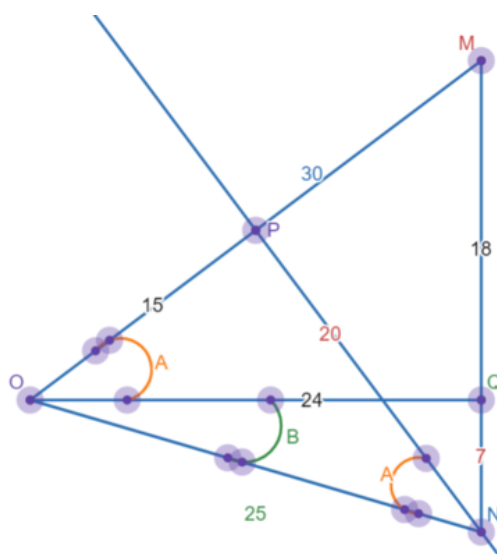
Solution 8 (Geometry Solution Without Words)

$$\angle A + \angle B + \angle A = \frac{\pi}{2}$$

Therefore

$$\beta = \boxed{(C) \frac{\pi}{2} - 2\alpha}$$

luckuso



Solution 9 (Complex Number)

Define $Z_A = 4 + 3i = 5e^{iA}$

and $Z_B = 24 + 7i = 25e^{iB}$

Looking at the answer choices, we start by testing out an easier choice, such as C.

$$(4 + 3i)^2 = 16 + 9 \cdot i^2 + 2 \cdot 12 \cdot i = 16 - 9 + 24i = 7 + 24i$$

$$\text{Finally: } 25^2 \cdot e^{i(2A+B)} = 25e^{i2A} \cdot 25e^{iB} = (5e^{iA})^2 \cdot 25e^{iB} = (7 + 24i)(24 + 7i) = 25^2 i = 25^2 e^{i\frac{\pi}{2}}$$

This proves that $\angle 2A + \angle B = \frac{\pi}{2}$. (Given that $\angle A < \frac{\pi}{2}$ and $\angle B < \frac{\pi}{2}$)

Therefore

$$\beta = \boxed{(C) \frac{\pi}{2} - 2\alpha}$$

Solution 10 (Double Angle + Co-function Identity)

$$\cos(\alpha) = \frac{4}{5} \quad \text{and} \quad \sin(\alpha) = \frac{3}{5}$$

Use a double angle identity to calculate $\cos(2\alpha)$:

$$\cos(2\alpha) = \cos^2(\alpha) - \sin^2(\alpha) = \left(\frac{4}{5}\right)^2 - \left(\frac{3}{5}\right)^2 = \frac{16}{25} - \frac{9}{25} = \frac{7}{25}$$

We know that:

$$\sin(\beta) = \frac{7}{25}$$

Equate both to each other:

$$\cos(2\alpha) = \sin(\beta)$$

Apply the co-function identity, $\cos(2\alpha) = \sin\left(\frac{\pi}{2} - 2\alpha\right)$:

$$\sin\left(\frac{\pi}{2} - 2\alpha\right) = \sin(\beta)$$

Hence,

$$\beta = \boxed{(C) \frac{\pi}{2} - 2\alpha}$$

Solution 11 (don't do this)

$$\frac{3}{5} \approx \alpha - \frac{\alpha^3}{6}$$

$$-5\alpha^3 + 30\alpha - 18 \approx 0$$

$$\alpha \approx 0.64464$$

$$\frac{7}{25} \approx \beta - \frac{\beta^3}{6}$$

$$-25\beta^3 + 150\beta - 42 \approx 0$$

$$\beta \approx 0.28381$$

$$\frac{\pi}{2} - 2 \cdot 0.64464 \approx 0.28151632679 \approx \beta$$

$$\beta = \boxed{(\mathbf{C}) \frac{\pi}{2} - 2\alpha}$$

Solution 12

$$\frac{4}{5} \approx 1 - \frac{\alpha^2}{2}$$

$$5\alpha^2 \approx 2$$

$$\alpha \approx \sqrt{\frac{2}{5}} \approx 0.63246$$

$$\frac{24}{25} \approx 1 - \frac{\beta^2}{2}$$

$$25\beta^2 \approx 2$$

$$\beta \approx \sqrt{\frac{2}{25}} \approx 0.28284$$

$$\frac{\pi}{2} - 2 \cdot 0.63246 \approx 0.30587632679 \approx \beta$$

$$\beta = \boxed{(\mathbf{C}) \frac{\pi}{2} - 2\alpha}$$

11. (AMC 12 12A 2024) There are exactly K positive integers b with $5 \leq b \leq 2024$ such that the base- b integer 2024_b is divisible by 16 (where 16 is in base ten). What is the sum of the digits of K ?

(A) 16 (B) 17 (C) 18 (D) 20 (E) 21

Answer: D

Solution 1

$2024_b = 2b^3 + 2b + 4 \equiv 0 \pmod{16} \implies b^3 + b + 2 \equiv 0 \pmod{8}$. If b is even, then $b + 2 \equiv 0 \pmod{8} \implies b \equiv 6 \pmod{8}$. If b is odd, then $b^2 \equiv 1 \pmod{8}^* \implies b^3 + b + 2 \equiv 2b + 2 \pmod{8}$ so $2b + 2 \equiv 0 \pmod{8} \implies b + 1 \equiv 0 \pmod{4} \implies b \equiv 3, 7 \pmod{8}$. Now $8 \mid 2024$ so $\frac{3}{8} \cdot 2024 = 759$ but one of the answers we got from that, 3, is too small, so $759 - 1 = 758 \implies \boxed{(\mathbf{D}) 20}$.

- The reason why $b^2 \equiv 1 \pmod{8}$ (if b is odd) is that we can write b as $2k + 1$, where k is any number (such that the problem conditions are met). Plugging this into $b^2 \equiv 1 \pmod{8}$, we get $(2k + 1)^2 = 4k^2 + 1 + 4k = 4k(k + 1) + 1 \equiv 1 \pmod{8}$. Since $4k$ is, of course, always a multiple of 4 and since $k + 1$ is even, $4k(k + 1)$ is divisible by 8, and adding 1 creates a remainder of 1.

Solution 2

$$2024_b \equiv 0 \pmod{16}$$

$$2b^3 + 2b + 4 \equiv 0 \pmod{16}$$

$$b^3 + b + 2 \equiv 0 \pmod{8}$$

Clearly, b is either even or odd. If b is even, let $b = 2a$.

$$\begin{aligned}(2a)^3 + 2a + 2 &\equiv 0 \pmod{8} \\ 8a^3 + 2a + 2 &\equiv 0 \pmod{8} \\ 0 + 2a + 2 &\equiv 0 \pmod{8} \\ a + 1 &\equiv 0 \pmod{4} \\ a &\equiv 3 \pmod{4}\end{aligned}$$

Thus, one solution is $b = 2(4x + 3) = 8x + 6$ for some integer x , or $b \equiv 6 \pmod{8}$.

What if b is odd? Then let $b = 2a + 1$:

$$\begin{aligned}(2a + 1)^3 + 2a + 1 + 2 &\equiv 0 \pmod{8} \\ 8a^3 + 12a^2 + 6a + 1 + 2a + 1 + 2 &\equiv 0 \pmod{8} \\ 8a^3 + 12a^2 + 8a + 4 &\equiv 0 \pmod{8} \\ 4a^2 + 4 &\equiv 0 \pmod{8} \\ a^2 &\equiv 1 \pmod{2}\end{aligned}$$

This simply states that a is odd. Thus, the other solution is $b = 2(2x + 1) + 1 = 4x + 3$ for some integer x , or $b \equiv 3 \pmod{4}$.

We now simply must count the number of integers between 5 and 2024, inclusive, that are $6 \pmod{8}$ or $3 \pmod{4}$. Note that the former case comprises even numbers only while the latter is only odd; thus, there is no overlap and we can safely count the number of each and add them.

In the former case, we have the numbers $6, 14, 22, 30, \dots, 2022$; this list is equivalent to $8, 16, 24, 32, \dots, 2024 \cong 1, 2, 3, 4, \dots, 253$, which comprises 253 numbers. In the latter case, we have the numbers $7, 11, 15, 19, \dots, 2023 \cong 4, 8, 12, 16, \dots, 2020 \cong 1, 2, 3, 4, \dots, 505$, which comprises 505 numbers. There are 758 numbers in total, so our answer is $7 + 5 + 8 = \boxed{\text{(D)} 20}$.

Solution 3

Note that $2024b = 2b^3 + 2b + 4$ is to be divisible by 16, which means that $b^3 + b + 2$ is divisible by 8.

If $b = 0$, then $b^3 + b + 2 \equiv (0)^3 + (0) + 2 \equiv 2$ is not divisible by 8.

If $b = 1$, then $b^3 + b + 2 \equiv (1)^3 + (1) + 2 \equiv 4$ is not divisible by 8.

If $b = 2$, then $b^3 + b + 2 \equiv (2)^3 + (2) + 2 \equiv 4$ is not divisible by 8.

If $b = 3$, then $b^3 + b + 2 \equiv (3)^3 + (3) + 2 \equiv (8 + 1) \cdot 3 + (3) + 2 \equiv 0$ is divisible by 8.

If $b = 4$, then $b^3 + b + 2 \equiv (4)^3 + (4) + 2 \equiv 0 + 4 + 2 \equiv 6$ is not divisible by 8.

If $b = 5$, then $b^3 + b + 2 \equiv (-3)^3 + (-3) + 2 \equiv (8 + 1) \cdot 3 + (-3) + 2 \equiv 2$ is not divisible by 8.

If $b = 6$, then $b^3 + b + 2 \equiv (-2)^3 + (-2) + 2 \equiv -8 + (-2) + 2 \equiv 0$ is divisible by 8.
 If $b = 7$, then $b^3 + b + 2 \equiv (-1)^3 + (-1) + 2 \equiv -1 + (-1) + 2 \equiv 0$ is divisible by 8.
 Therefore, for every 8 values of b , 3 of them will make $b^3 + b + 2$ divisible by 8.
 Therefore, since 2024 is divisible by 8, $\frac{3}{8} \cdot 2024 = 759$ values of b , but this includes $b = 3$, which does not satisfy the given inequality. Therefore, the answer is

$$759 - 1 = 758 \rightarrow 7 + 5 + 8 = \boxed{(D) 20}$$

Tacos_are_yummy_1

More detail by [luckuso](#)

Solution 4

$$2024_b = 2 * b^3 + 2 * b + 4$$

$$2024_{(b+8)} = 2 * (b + 8)^3 + 2 * (b + 8) + 4$$

$$2024_{(b+8)} - 2024_b = 2 * (8) * (b^2 + 8b + 64) + 2 * 8 = 16 * (b^2 + 8b + 64) + 16$$

$$2024_{(b+8)} - 2024_b \equiv 0 \pmod{16}$$

$$2024_{(b+8)} \equiv 2024_b \pmod{16}$$

$$2024_0 \equiv 4 \pmod{16}$$

$$2024_1 \equiv 8 \pmod{16}$$

$$2024_2 \equiv 6 \pmod{16}$$

$$2024_3 \equiv 0 \pmod{16}$$

$$2024_4 \equiv 12 \pmod{16}$$

$$2024_5 \equiv 8 \pmod{16}$$

$$2024_6 \equiv 0 \pmod{16}$$

$$2024_7 \equiv 0 \pmod{16}$$

We need $b \equiv 3 \pmod{8}$ or $b \equiv 6 \pmod{8}$ or $b \equiv 7 \pmod{8}$

$\lfloor (2024 - 3)/8 \rfloor + \lfloor (2024 - 6)/8 \rfloor + \lfloor (2024 - 7)/8 \rfloor + 3 = 759$ take away one because $b = 3$ is out of range, so $758 \Rightarrow 7 + 8 + 5 = \boxed{(D) 20}$

Solution 4.1: Alternate Ending

One can arrange 2024 into different "blocks" consisting of 8 numbers each, particularly $\{0, 1, 2, 3, 4, 5, 6, 7\}$, $\{8, 9, 10, 11, 12, 13, 14, 15\}$...

These just represent different mod (8) class residues. So, you can just divide 2024 by 8 blocks giving us 253 total blocks with 0 extra semi-blocks.

We see that $2024_3 \pmod{8}$, $2024_6 \pmod{8}$, and $2024_7 \pmod{8}$ all appear exactly 253 times each, so we have $253 \times 3 = 759$ different values of K .

However, we must not include 2024_3 , as the bases b range from $5 \leq b \leq 2024$, therefore we have $759 - 1 = 758$ cases.

Our answer is $7 + 5 + 8 = \boxed{(D) 20}$.

Solution 5 (Factoring)

Start similar to other solutions.

$$b^3 + b + 2 \equiv 0 \pmod{8}$$

Except now we notice the following:

$$\begin{aligned} b^3 + 8 + b + 2 &\equiv 8 \pmod{8} \equiv 0 \pmod{8} \\ (b+2)(b^2 - 2b + 4) + b + 2 &\equiv 0 \pmod{8} \\ (b+2)(b^2 - 2b + 5) &\equiv 0 \pmod{8} \end{aligned}$$

Notice that $b+2$ and $b^2 - 2b + 5$ have opposite parity, and when multiplied, is divisible by 8. Thus, either $b+2$ or $b^2 - 2b + 5$ is divisible by 8.

$b+2$ is obvious, where $b+2 \mid 8$ when $b \equiv 6 \pmod{8}$

When testing for $b^2 - 2b + 5 \mid 8$, we find $b \equiv 3, 7 \pmod{8}$

Proceed similar to the other solutions, $\lfloor \frac{3}{8} \cdot 2024 \rfloor - 1 = 759 - 1 = 758$, where

$$7 + 5 + 8 = \boxed{\text{(D)} 20}$$

Also noting this is an example of a misplaced problem this would be far to easy for even P 18 of amc 10, deserved to be Q5-6

Solution 6

$$2024_b = 2b^3 + 2b + 4 \equiv 0 \pmod{16} \rightarrow 2b^3 + 2b \equiv 12 \pmod{16}.$$

Dividing everything by 2 gives:

$$b^3 + b \equiv 6 \pmod{\frac{16}{\gcd(16, 2)}} \rightarrow b(b^2 + 1) \equiv 6 \pmod{8}$$

.

Then proceed in the same manner as the solutions above to get a final answer of $\boxed{\text{(D)} 20}$.

12. (AMC 12 12A 2024) The first three terms of a geometric sequence are the integers a , 720, and b , where $a < 720 < b$. What is the sum of the digits of the least possible value of b ? (A) 9 (B) 12 (C) 16 (D) 18 (E) 21

Answer: E

Solution 1

For a geometric sequence, we have $ab = 720^2 = 2^8 3^4 5^2$, and we can test values for b . We find that $b = 768$ and $a = 675$ works, and we can test multiples of 5 in between the two values. Finding that none of the multiples of 5 divide 720^2 besides 720 itself, we know that the answer is $7 + 6 + 8 = \boxed{\text{(E)} 21}$.

(Note: To find the value of b without bashing, we can observe that $2^8 = 256$, and that multiplying it by 3 gives us 768, which is really close to 720. YTH)

Note: The reason why $ab = 720^2$ is because $b/720 = 720/a$. Rearranging this gives $ab = 720^2$

Note: Another reason that $ab = 720^2$ is because the $\sqrt{ab} = 720$ (as the middle term in a geometric series is always the geometric mean [the geometric mean is the

square root of the product of the first and last terms of the series]) and squaring on both sides results in $ab = 720^2$.

Solution 2

$720 = 2^4 \cdot 3^2 \cdot 5$. We want to find factors x and y where $y > x$ such that $\frac{y}{x}$, the ratio, is minimized. To do this, x should be as large as possible, and y should be just slightly larger than x . Additionally, to ensure a and b are both integers, x and y should both be a factor of 720.

After experimenting, we see this is achieved when $y = 16$ and $x = 15$, which means our value of b is $720 \cdot \frac{16}{15} = 768$. Therefore, the sum is $7 + 6 + 8 = \boxed{\text{(E) } 21}$.

Solution 3 (Similar to previous solutions)

To minimize the value of b , where it has to be an integer, and it has to be greater than 720, we can express the common ratio as $\frac{n+1}{n}$, where the value has to be greater than 1, and n , and $n+1$ have to be factors of 720. Since the bigger the denominator gets, the smaller the value of the fraction, we essentially have to find the biggest value for n , where itself and $n+1$ are factors of 720. From here, we can check whether $n(n+1) = 720$ yields an integer root, which it doesn't. So, then we check the next biggest factor of 720, which is 360. $n(n+1) = 360$, this doesn't have an integer root either. So, then we check the next biggest factor which is 240, $n(n+1) = 240$, which we get 15 as a root. This means the common ratio is $\frac{16}{15}$.

We then multiply $\frac{16}{15}$ times 720 and add up the digits getting $\boxed{\text{(E) } 21}$.

Solution 4

Let the common ratio of the geometric sequence be $\frac{x}{y}$, with $x > y$. This means that $720(\frac{x}{y})$ and $720(\frac{y}{x})$ must both be integers, therefore x and y are both factors of 720. We would achieve the smallest ratio $\frac{x}{y}$ if x and y are consecutive, so by listing out the factors of 720, we find that 1, 2, 3, 4, 5, 6, 8, 9, 10, 15, 16 are the only consecutive factors (any factors larger than these would result in the ratio simplifying, making it larger than just using consecutive integers). 15 and 16 are the largest, so we find the common ratio to be $\frac{16}{15}$, making $b = 720(\frac{16}{15})$ giving us 768. The sum of its digits is $\boxed{\text{(E) } 21}$.

Solution 5 (similar to other solutions)

We look for the smallest possible increasing ratio in the geometric sequence for a , 720, and b . First, we find the prime factorization of 720, which is really easy $720 = 2^4 \cdot 3^2 \cdot 5$. Then, we look for the closest possible fraction-ratio out of the primes, and after a quick look, it's easy to see it is 16 and 15. Now that we found our ratio, we multiply 720 by $\frac{16}{15}$ to get 768, so the sum of the digits of the least possible value of b is 21.

NOTE: It's not $9/8$ because we want it to be as close as possible to 1 gabrielle.esther

13. (AMC 12 12A 2024) The graph of $y = e^{x+1} + e^{-x} - 2$ has an axis of symmetry. What is the reflection of the point $(-1, \frac{1}{2})$ over this axis?
- (A) $(-1, -\frac{3}{2})$ (B) $(-1, 0)$ (C) $(-1, \frac{1}{2})$ (D) $(0, \frac{1}{2})$ (E) $(3, \frac{1}{2})$

Answer: D

Solution 1

The line of symmetry is probably of the form $x = a$ for some constant a . A vertical line of symmetry at $x = a$ for a function f exists if and only if $f(a - b) = f(a + b)$; we substitute $a - b$ and $a + b$ into our given function and see that we must have

$$e^{a-b+1} + e^{-(a-b)} - 2 = e^{a+b+1} + e^{-(a+b)} - 2$$

for all real b . Simplifying:

$$\begin{aligned} e^{a-b+1} + e^{-(a-b)} - 2 &= e^{a+b+1} + e^{-(a+b)} - 2 \\ e^{a-b+1} + e^{b-a} &= e^{a+b+1} + e^{-a-b} \\ e^{a-b+1} - e^{-a-b} &= e^{a+b+1} - e^{b-a} \\ e^{-b} (e^{a+1} - e^{-a}) &= e^b (e^{a+1} - e^{-a}). \end{aligned}$$

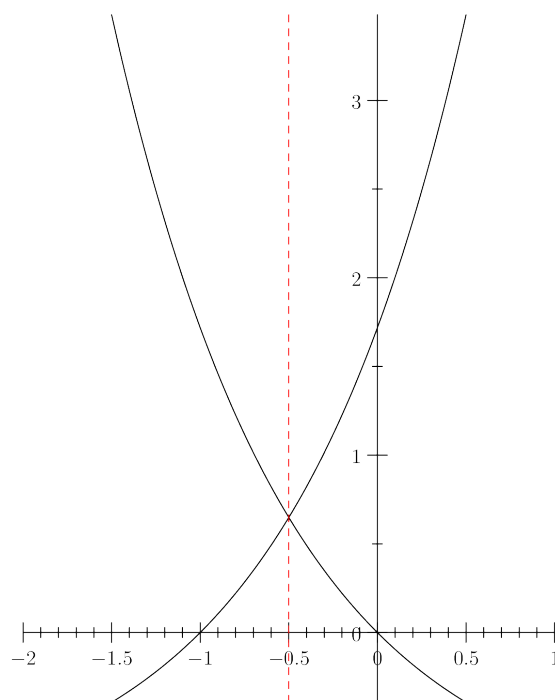
If $e^{a+1} - e^{-a} \neq 0$, then $e^{-b} = e^b$ for all real b ; this is clearly impossible, so let $e^{a+1} - e^{-a} = 0 \implies a + 1 = -a \implies a = -\frac{1}{2}$. Thus, our line of symmetry is $x = -\frac{1}{2}$, and reflecting $\left(-1, \frac{1}{2}\right)$ over this line gives **(D)** $\left(0, \frac{1}{2}\right)$.

Solution 2 (Graphing)

Consider the graphs of $y = e^{x+1} - 1$ and $y = e^{-x} - 1$. A rough sketch will show that they intercept somewhere between -1 and 0 and the axis of symmetry is vertical.

Thus, **(D)** $\left(0, \frac{1}{2}\right)$ is the only possible answer.

Note: You can more rigorously think about the solution by noting that since the derivative of the power that e is raised to in one equation is equal to the derivative of the power that e is raised to multiplied by -1 and both equations are subtracted by 1, then the sum of both equations will be the same from one side of the interception to the other. Setting both equations equal to each other, it is trivial to see $x = -1/2$, giving us the axis of symmetry.



(graph by Technodoggo)

Solution 3 (Derivatives)

Since y is a proper function of x , the axis of symmetry is vertical (and so answer choices A and B are impossible). Where a function crosses a vertical axis of symmetry, it must have either a vertical asymptote (clearly impossible for this given function, since linear functions and exponential function are total functions) or a local maximum or minimum (derivative 0).

So the derivative is equal to 0: $\frac{d[e^{x+1} + e^{-x} - 2]}{dx} = 0 = e^{x+1} - e^{-x}$

We have $x + 1 = -x$ hence $x = -\frac{1}{2}$.

Reflect $(-1, \frac{1}{2})$ over $x = -\frac{1}{2}$, giving answer choice **(D)** $(0, \frac{1}{2})$.

Solution 4

$$f(0) = e - 1$$

$$f(1) = e^2 + e^{-1} - 2$$

$$f(-1) = 1 + e - 2 = e - 1$$

$$f(-2) = e^{-1} + e^2 - 2$$

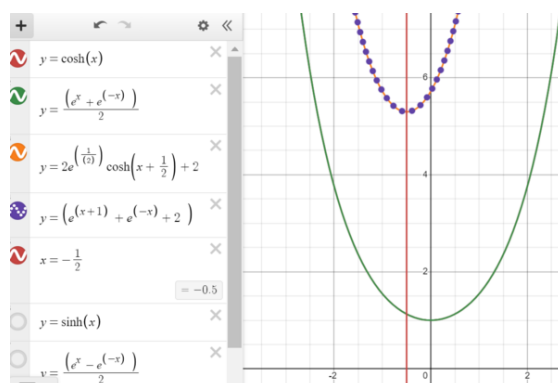
so $f(0) = f(-1)$, $f(1) = f(-2)$ then $f(x)$ is symmetric about $x = -1/2$ point $(-1, 1/2)$ reflects over axis $x = -1/2$ is point $(0, 1/2)$ answer choice D luckuso

Solution 5

Notice that this is simply a transformation of $\cosh(x)$, specifically $2e^{\frac{1}{2}} \cosh(x + \frac{1}{2}) + 2$. We know that $\cosh(x)$ has a parabolic shape with an axis of symmetry $x = 0$. Since this is simply a transformation of $\frac{1}{2}$ to the left of the origin, we have the axis of

symmetry is $x = -\frac{1}{2}$ giving us **(D)** $(0, \frac{1}{2})$

Solution 5 (Graph)



Solution 6

$$f(-x-1) = e^{-x-1+1} + e^{-(-x-1)} - 2 = e^{-x} + e^{x+1} - 2 = f(x)$$

So the axis is $\frac{x+(-x-1)}{2} = -1/2$

Point $(-1, 1/2)$ reflects over the axis $x = -1/2$ to the point **(D)** $\left(0, \frac{1}{2}\right)$

Solution 7

Setting the axis of symmetry as $x = a$ implies $f(2a - x) = f(x)$

$$\begin{aligned} f(2a - x) &= f(x) \\ e^{2a-x+1} + e^{-(2a-x)} - 2 &= e^{x+1} + e^{-x} - 2 \end{aligned}$$

Analyzing the terms e^{2a-x+1} and e^{-x} tells us that $2a + 1 = 0$

This reveals that $a = -1/2$

Finally, point $(-1, 1/2)$ reflected over the axis $x = -1/2$ is the point **(D)** $\left(0, \frac{1}{2}\right)$

Solution 8

First identify that the graphs are both forms of the parent function e^x and will not have a slant asymptote, so we can eliminate answer choices with a different y-value and the same x-value as this function can not have different y value for a same x value.

The answer choices that we can still consider are: **(D)** $\left(0, \frac{1}{2}\right)$ **(E)** $\left(3, \frac{1}{2}\right)$

Remember that the constant term does not matter for an axis of symmetry, as it simply shifts the function(s) up or down.

Hence, we can use the answer choices and plug in valid values of x for $e^{x+1} = e^{-x}$.

The answer for x is then $-1/2$, and the answer is **(D)** $\left(0, \frac{1}{2}\right)$

14. (AMC 12 12A 2024) The numbers, in order, of each row and the numbers, in order, of each column of a 5×5 array of integers form an arithmetic progression of length 5. The numbers in positions $(5, 5)$, $(2, 4)$, $(4, 3)$, and $(3, 1)$ are 0, 48, 16, and 12, respectively. What number is in position $(1, 2)$?

$$\begin{bmatrix} . & ? & . & . & . \\ . & . & . & 48 & . \\ 12 & . & . & . & . \\ . & . & 16 & . & . \\ . & . & . & . & 0 \end{bmatrix}$$

- (A) 19 (B) 24 (C) 29 (D) 34 (E) 39

Answer: C

Solution 1 (Arithmetic Sequences)

Start from the 0 square. Going up, let the common difference be a , and going left, let the common difference be b . Therefore, we have

$$\begin{bmatrix} . & ? & . & . & 4a \\ . & . & . & 48 & 3a \\ 12 & . & . & . & 2a \\ . & . & 16 & . & a \\ 4b & 3b & 2b & b & 0 \end{bmatrix}$$

Looking at the third column, we can see that the common difference going up is $16 - 2b$. We fill this in:

$$\begin{bmatrix} . & ? & 64 - 6b & . & 4a \\ . & . & 48 - 4b & 48 & 3a \\ 12 & . & 32 - 2b & . & 2a \\ . & . & 16 & . & a \\ 4b & 3b & 2b & b & 0 \end{bmatrix}$$

Looking at the second row, 48 has two values beside it, so we can write

$$48 = \frac{48 - 4b + 3a}{2} \rightarrow 96 = 48 - 4b + 3a \rightarrow 48 = -4b + 3a,$$

and we can do the same with the third row, which gives

$$32 - 2b = \frac{12 + 2a}{2} \rightarrow 32 - 2b = 6 + a \rightarrow 26 = a + 2b.$$

Now we have the system of equations

$$48 = -4b + 3a$$

$$26 = a + 2b,$$

and solving it gives $a = 20, b = 3$, therefore we can now fill in the grid with actual numbers. But before doing that, note that we're only looking for a value in the first row, and because we already have two known values in that row, we can find the common difference for that row and not focus on anything else.

Focusing only on the first row yields

$$\begin{bmatrix} . & ? & 46 & . & 80 \end{bmatrix}$$

This means that the common difference from right to left is $\frac{80 - 46}{2} = 17$. Therefore, the desired value is $46 - 17 = \boxed{(C) 29}$ Tacos_are_yummy_1

Solution 2 (Some Basic Algebra and Answer Choices)

Assume the number in position $(3, 3)$ is x .

$$\begin{bmatrix} . & ? & . & . & . \\ . & . & . & 48 & . \\ 12 & . & x & . & . \\ . & . & 16 & . & . \\ . & . & . & . & 0 \end{bmatrix}$$

Then, the integer in position $(2, 3)$ will be $2x - 16$, since $2x - 16$ and 16 average out to x . (They form an arithmetic sequence).

$$\begin{bmatrix} . & ? & . & . & . \\ . & . & 2x - 16 & 48 & . \\ 12 & . & x & . & . \\ . & . & 16 & . & . \\ . & . & . & . & 0 \end{bmatrix}$$

Similarly, the integer in position $(3, 2)$ is $0.5x + 6$, and the integer in position $(2, 2)$ is $4x - 80$.

$$\begin{bmatrix} . & ? & . & . & . \\ . & 4x - 80 & 2x - 16 & 48 & . \\ 12 & 0.5x + 6 & x & . & . \\ . & . & 16 & . & . \\ . & . & . & . & 0 \end{bmatrix}$$

This means the number $7.5x - 166$ is in position $(1, 2)$.

$$\begin{bmatrix} . & 7.5x - 166 & . & . & . \\ . & 4x - 80 & 2x - 16 & 48 & . \\ 12 & 0.5x + 6 & x & . & . \\ . & . & 16 & . & . \\ . & . & . & . & 0 \end{bmatrix}$$

The only answer choice that makes x an integer is $\boxed{(C) 29}$

Solution 3 (Bashy)

Let's look at the bottom-left corner. There can be 3 possible cases for the bottom row. Either the numbers are negative, all 0, or positive.

Let's assume all the numbers are negative. We know that the numbers in between the ? will be large, so we want to minimize it, because the largest answer choice is only 39.

$$\begin{bmatrix} . & ? & . & . & . \\ . & . & . & 48 & . \\ 12 & . & . & . & . \\ . & . & 16 & . & . \\ -4 & -3 & -2 & -1 & 0 \end{bmatrix}$$

To fill in the 1st and 3rd column, we simply find the common differences. We get:

$$\begin{bmatrix} 28 & ? & 70 & . & . \\ 20 & . & 52 & 48 & . \\ 12 & . & 34 & . & . \\ 4 & . & 16 & . & . \\ -4 & -3 & -2 & -1 & 0 \end{bmatrix}$$

Finding the $?$, we notice that it is 49, which is bigger than any of the other answer choices, so the bottom row can't be negative. (As the common difference for the bottom row gets larger, so does the common difference for the 1st and 3rd column, making the $?$ bigger, and because the smallest answer is 39 while the minimum value for negative numbers is 49, it can't be negative.)

Now let's assume that all the numbers are 0. By just filling in the 1st and 3rd column, we find that it is:

$$\begin{bmatrix} 24 & ? & 64 & . & . \\ 18 & . & 48 & 48 & . \\ 12 & . & 32 & . & . \\ 6 & . & 16 & . & . \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

which makes the $? = 44$. 44 is not an answer choice, so the bottom row cannot be all 0's.

Now, we know that the bottom row must be positive. Here's where the bashing begins. Lucky for us, the numbers must all be integers, cutting down on the number of cases we need to test. Starting with the bottom row having a common difference of 1, we get

$$\begin{bmatrix} . & ? & . & . & . \\ 16 & 30 & 44 & 48 & . \\ 12 & 21 & 30 & 39 & 48 \\ 8 & 12 & 16 & 20 & 24 \\ 4 & 3 & 2 & 1 & 0 \end{bmatrix}$$

At this point we notice that the second row does not form an arithmetic progression, so we test out the next case, with the common difference being 2. We get

$$\begin{bmatrix} . & ? & . & . & . \\ 14 & 27 & 40 & 48 & . \\ 12 & 20 & 28 & 36 & 44 \\ 10 & 13 & 16 & 19 & 22 \\ 8 & 6 & 4 & 2 & 0 \end{bmatrix}$$

At this point we notice that the second row does not form an arithmetic progression, so we test out the next case, with the common difference being 3. We get

$$\begin{bmatrix} 12 & 29 & 46 & 63 & 80 \\ 12 & 24 & 36 & 48 & 60 \\ 12 & 19 & 26 & 33 & 40 \\ 12 & 14 & 16 & 18 & 20 \\ 12 & 9 & 6 & 3 & 0 \end{bmatrix}$$

, giving us the answer (C) 29

Solution 4 (Arithmetic Sequences)

Let the bottom row of the grid have arithmetic difference d . Then the matrix becomes

$$\begin{bmatrix} . & ? & . & . & 4d \\ . & . & . & 48 & 3d \\ 12 & . & . & . & 2d \\ . & . & 16 & . & d \\ . & . & . & . & 0 \end{bmatrix}$$

Using the middle column, we have the middle item equal to $32 - 2d$.

$$\begin{bmatrix} . & ? & . & . & 4d \\ . & . & . & 48 & 3d \\ 12 & . & 32 - 2d & . & 2d \\ . & . & 16 & . & d \\ . & . & . & . & 0 \end{bmatrix}$$

Filling in the middle row (which has a common difference of $10 - d$)

$$\begin{bmatrix} . & ? & . & . & . \\ . & . & . & 48 & . \\ 12 & 22 - d & 32 - 2d & 42 - 3d & 52 - 4d \\ . & . & 16 & . & . \\ 4d & 3d & 2d & d & 0 \end{bmatrix}$$

The arithmetic sequence in the 4th column (2nd from the right) then has a common difference of $-3d - 6$, so we

$$\begin{bmatrix} . & ? & . & . & . \\ . & . & . & 48 & . \\ 12 & 22 - d & 32 - 2d & 42 - 3d & 52 - 4d \\ . & . & 16 & 36 - 6d & . \\ 4d & 3d & 2d & 30 - 9d & 0 \end{bmatrix} = \begin{bmatrix} . & ? & . & . & . \\ . & . & . & 48 & . \\ 12 & 22 - d & 32 - 2d & 42 - 3d & 52 - 4d \\ . & . & 16 & 36 - 6d & . \\ 4d & 3d & 2d & d & 0 \end{bmatrix}$$

From the cell in the 4th column, 5th row we have $30 - 9d = d$, or $d = 3$. Finally, using the 2nd column, $? + 3d = 2(22 - d)$, or $? + 9 = 38 \Rightarrow ? = \span style="border: 1px solid black; padding: 0 5px;">(C) 29$

Solution 5

We subtract 12 from each element of the array and denote $a_{i,j}$ the number in position (i, j) , $a_{3,1} = 0, a_{4,3} = 4, a_{2,4} = 36, a_{5,5} = -12$. Denoting $a_{3,2} = x \implies$ that in third row we get $(0, x, 2x, 3x, 4x)$. We use the property of an arithmetic progression and get

$$a_{4,4} = 2a_{3,4} - a_{2,4} = 6x - 36,$$

$$a_{4,5} = 2a_{4,4} - a_{3,4} = 12x - 76,$$

$$a_{5,5} = 2a_{4,5} - a_{3,5} \implies -12 = 2(12x - 76) - 4x \implies x = 7 = a_{3,2}, a_{3,3} = 14.$$

$$\begin{bmatrix} . & . & . & . & . \\ . & . & . & 36 & . \\ 0 & x & 2x & 3x & 4x \\ . & . & 4 & 6x - 36 & 12x - 76 \\ . & . & . & . & -12 \end{bmatrix}$$

$$a_{2,3} = 2a_{3,3} - a_{4,4} = 2 \cdot 14 - 4 = 24,$$

$$a_{2,2} = 2a_{2,3} - a_{2,4} = 2 \cdot 24 - 36 = 12,$$

$$a_{1,2} = 2a_{2,2} - a_{3,2} = 2 \cdot 12 - 7 = 17.$$

$$\begin{bmatrix} . & 17 & . & . & . \\ . & 12 & 24 & 36 & . \\ 0 & 7 & 14 & . & . \\ . & . & . & . & . \\ . & . & . & . & . \end{bmatrix}$$

We add 12 and get the number in position (1, 2) as $17 + 12 = 29$.

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Solution 6 (Bilinear Form with Coordinate Translation)

A fundamental structural property of grids with arithmetic progressions in both rows and columns is that every entry can be expressed in the bilinear form:

$$A_{m,n} = amn + bm + cn + d$$

where m and n are the row and column indices. This representation captures the essential two-dimensional linear structure of the problem.

Solving the Problem To minimize computational complexity, we employ coordinate translation. Set the known value 12 at position (3, 1) as our origin by defining $x = m - 3$ and $y = n - 1$. The given data points become:

- $A_{0,0} = 12$ (from position (3, 1))
- $A_{-1,3} = 48$ (from position (2, 4))
- $A_{1,2} = 16$ (from position (4, 3))
- $A_{2,4} = 0$ (from position (5, 5))

We seek the value at $(-2, 1)$, corresponding to the original position (1, 2).

Substituting into $A_{x,y} = axy + bx + cy + d$:

- a) $A_{0,0} = d = 12$
- b) $A_{-1,3} = -3a - b + 3c + 12 = 48 \implies -3a - b + 3c = 36$
- c) $A_{1,2} = 2a + b + 2c + 12 = 16 \implies 2a + b + 2c = 4$
- d) $A_{2,4} = 8a + 2b + 4c + 12 = 0 \implies 4a + b + 2c = -6$

Subtracting equation (3) from equation (4) eliminates both b and c :

$$(4a + b + 2c) - (2a + b + 2c) = -6 - 4 \implies 2a = -10 \implies a = -5$$

With $a = -5$ in equation (3): $-10 + b + 2c = 4 \implies b + 2c = 14$

With $a = -5$ in equation (2): $15 - b + 3c = 36 \implies -b + 3c = 21$

Adding these equations: $5c = 35 \implies c = 7$, hence $b = 0$.

Our formula is therefore $A_{x,y} = -5xy + 7y + 12$. Evaluating at $(-2, 1)$:

$$A_{-2,1} = -5(-2)(1) + 7(1) + 12 = 10 + 7 + 12 = 29$$

The answer is (C) 29.

Verification of the Formula To confirm this bilinear form universally satisfies the arithmetic progression property, let $A_{m,n} = amn + bm + cn + d$.

For fixed row m , the column difference is:

$$A_{m,n+1} - A_{m,n} = am(n+1) + bm + c(n+1) + d - (amn + bm + cn + d) = am + c$$

This is constant with respect to n , confirming each row forms an arithmetic progression.

For fixed column n , the row difference is:

$$A_{m+1,n} - A_{m,n} = a(m+1)n + b(m+1) + cn + d - (amn + bm + cn + d) = an + b$$

This is constant with respect to m , confirming each column forms an arithmetic progression.

Two Distinct Derivations of the Formula The bilinear form can be derived through two fundamentally different mathematical perspectives, each revealing unique structural insights.

Derivation 1: Algebraic and Functional Perspective

This approach exploits the nested linear structure without invoking difference operators.

Since every row is an arithmetic progression, for fixed n , the function $A_{m,n}$ must be linear in m :

$$A_{m,n} = u_n \cdot m + v_n$$

where coefficients u_n and v_n depend only on n .

To determine the functional forms of u_n and v_n , evaluate at specific rows:

Setting $m = 0$: $A_{0,n} = v_n$. Since column 0 is an arithmetic progression, v_n must be linear in n . Write $v_n = cn + d$.

Setting $m = 1$: $A_{1,n} = u_n + v_n$. Since column 1 is also an arithmetic progression, $u_n + v_n$ must be linear in n . Given that v_n is already linear, u_n must itself be linear in n . Write $u_n = an + b$.

Substituting back:

$$A_{m,n} = (an + b)m + (cn + d) = amn + bm + cn + d$$

This derivation reveals that the bilinear form emerges naturally from the compositional structure of linear dependencies in both dimensions.

Derivation 2: Discrete Calculus Perspective

This approach uses difference operators, illuminating the discrete analog of Clairaut's theorem on equality of mixed partial derivatives.

Define forward difference operators:

- $\Delta_m A_{m,n} = A_{m+1,n} - A_{m,n}$ (row difference)
- $\Delta_n A_{m,n} = A_{m,n+1} - A_{m,n}$ (column difference)

The arithmetic progression conditions state:

- $\Delta_n A_{m,n} = r_m$ depends only on m
- $\Delta_m A_{m,n} = c_n$ depends only on n

Consider the mixed second difference $M_{m,n} = \Delta_m \Delta_n A_{m,n}$:

- Applying Δ_m to r_m : $M_{m,n} = r_{m+1} - r_m$ depends only on m
- Applying Δ_n to c_n : $M_{m,n} = c_{n+1} - c_n$ depends only on n

Since $M_{m,n}$ depends only on m and only on n simultaneously, it must be constant. Denote $M_{m,n} = a$.

Therefore:

- $c_{n+1} - c_n = a \implies c_n = an + b$ (column differences form an AP)
- $r_{m+1} - r_m = a \implies r_m = am + c$ (row differences form an AP)

Reconstructing $A_{m,n}$ from $A_{0,0} = d$ via discrete path integration:

$$A_{m,n} = A_{0,0} + \sum_{j=0}^{n-1} \Delta_n A_{0,j} + \sum_{i=0}^{m-1} \Delta_m A_{i,n}$$

$$A_{m,n} = d + \sum_{j=0}^{n-1} r_0 + \sum_{i=0}^{m-1} c_n = d + nc + m(an + b) = amn + bm + cn + d$$

This derivation establishes a rigorous framework connecting the commutativity of discrete differences to the bilinear structure, providing a powerful generalization technique for higher-dimensional problems.

Complex systems of equations From the given information, we can ensure that we find the correct answer by creating a 4-variable, 4-equation system to solve for $(1, 2)$.

Let the common difference moving side to side from row 3 be d , the common difference moving vertically from 48 be y , the common difference moving vertically from 16 be z , and the common difference moving side to side from 0 be j .

We can then write 4 equations and solve for each difference.

$$12 + 3x = 48 + y \quad 16 - z = 12 + 2x \quad 16 + z = 0 - 2j \quad 48 + 3y = 0 - j$$

We can solve each variable in terms of y procedurally until we can get x in terms of y , and solve each from there. We get that $j = -3y - 48$. We then plug this into equation 3.

Doing this, we find that $z = 6y + 80$.

We then plug this into equation 2 and find that $x = -3y - 38$.

We then substitute our value for x into equation 1. We find that $y = -15$.

We then plug this value into our equations and see that $x = 7$, $j = -3$, and $z = -10$.

Now we can see that the number directly to the left of 48 must be 36, as we add 20, or two times z . This means the common difference of this row is 12. We move another digit to the left and see that the number directly above $(1, 2)$ will be 24.

The digit directly below this will be 19, as we add x to 12 or 7. This means that to get to $(1, 2)$ we must add 5 to 24.

$$24 + 5 = \boxed{29}.$$

15. (AMC 12 12A 2024) The roots of $x^3 + 2x^2 - x + 3$ are p, q , and r . What is the value of

$$(p^2 + 4)(q^2 + 4)(r^2 + 4)?$$

- (A) 64 (B) 75 (C) 100 (D) 125 (E) 144

Answer: D

Solution 1

You can factor $(p^2 + 4)(q^2 + 4)(r^2 + 4)$ as $(p + 2i)(p - 2i)(q + 2i)(q - 2i)(r + 2i)(r - 2i)$.

For any polynomial $f(x)$, you can create a new polynomial $f(x + a)$, which will have roots that instead have the value a subtracted.

Substituting $x - 2i$ and $x + 2i$ into x for the first polynomial, gives you $10i - 5$ and $-10i - 5$ as c for both equations. Multiplying $10i - 5$ and $-10i - 5$ together gives you **(D) 125**.

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Solution 2

Let $f(x) = x^3 + 2x^2 - x + 3 = (x - p)(x - q)(x - r) = -(p - x)(q - x)(r - x)$. Then

$$(p^2 + 4)(q^2 + 4)(r^2 + 4) = (p + 2i)(p - 2i)(q + 2i)(q - 2i)(r + 2i)(r - 2i) = f(2i)f(-2i).$$

We find that $f(2i) = -8i - 8 - 2i + 3 = -10i - 5$ and $f(-2i) = 8i - 8 + 2i + 3 = 10i - 5$, so

$$f(2i)f(-2i) = (-5 - 10i)(-5 + 10i) = (-5)^2 - (10i)^2 = 25 + 100 = \boxed{\text{(D) 125}}.$$

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Solution 3

First, denote that

$$p + q + r = -2, pq + pr + qr = -1, pqr = -3$$

Then we expand the expression

$$\begin{aligned} & (p^2 + 4)(q^2 + 4)(r^2 + 4) \\ &= (pqr)^2 + 4((pq)^2 + (pr)^2 + (qr)^2) + 4^2(p^2 + q^2 + r^2) + 4^3 \\ &= (-3)^2 + 4((pq + pr + qr)^2 - 2pqr(p + q + r)) + 4^2((p + q + r)^2 - 2(pq + pr + qr)) + 4^3 \\ &= (-3)^2 + 4((-1)^2 - 2(-3)(-2)) + 4^2((-2)^2 - 2(-1)) + 4^3 \\ &= \boxed{\text{(D) 125}} \end{aligned}$$

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Solution 4 (Newton's Sums and Vieta's)

Expand the expression:

$$(p^2 + 4)(q^2 + 4)(r^2 + 4) = (p^2q^2r^2) + 4(q^2r^2 + p^2r^2 + p^2q^2) + 16(p^2 + q^2 + r^2) + 64.$$

By Vieta's we have $pqr = -3$. Therefore, $p^2q^2r^2 = (-3)^2 = 9$. We use Newton's Sums to quickly compute $p^2 + q^2 + r^2$:

$$S_2 + 2S_1 + 2(-1) = 0$$

By Vieta's again, $S_1 = p + q + r = -2$ which means $S_2 = p^2 + q^2 + r^2 = 6$.

Now all we need is $q^2r^2 + p^2r^2 + p^2q^2$. Vieta's also tells us that $pq + qr + pr = -1$, so we can take this equation and square both sides:

$$(pq + qr + pr)^2 = q^2r^2 + p^2r^2 + p^2q^2 + 2(qpr^2 + q^2rp + p^2rq) = 1.$$

We know $pqr = -3$ so we can substitute this in:

$$(pq + qr + pr)^2 = q^2r^2 + p^2r^2 + p^2q^2 - 6(r + p + q) = 1.$$

We also know $p + q + r = -2$, which means $q^2r^2 + p^2r^2 + p^2q^2 = -11$.

So the answer is just $9 + 4(-11) + 16(6) + 64 = \boxed{125}$.

Solution 5 (Reduction of power)

The motivation for this solution is the observation that $(p + c)(q + c)(r + c)$ is easy to compute for any constant c , since $(p + c)(q + c)(r + c) = -f(-c)$ (*), where f is the polynomial given in the problem. The idea is to transform the expression involving p^2, q^2, r^2 into one involving p, q, r .

Since p is a root of f ,

$$p^3 + 2p^2 - p + 3 = 0 \implies p^3 + 2p^2 = p^2(p + 2) = p - 3,$$

which gives us that $p^2 = \frac{p-3}{p+2}$. Then

$$p^2 + 4 = \frac{p-3}{p+2} + 4 = \frac{5p+5}{p+2} = 5 \cdot \frac{p+1}{p+2}.$$

Since q and r are also roots of f , the same analysis holds, so

$$\begin{aligned} (p^2 + 4)(q^2 + 4)(r^2 + 4) &= \left(5 \cdot \frac{p+1}{p+2}\right) \left(5 \cdot \frac{q+1}{q+2}\right) \left(5 \cdot \frac{r+1}{r+2}\right) \\ &= 125 \frac{(p+1)(q+1)(r+1)}{(p+2)(q+2)(r+2)} \\ &= 125 \frac{-f(-1)}{-f(-2)} \\ &= 125 \cdot 1 \\ &= \boxed{\text{(D)} 125}. \end{aligned}$$

(*) This is because

$$(p + c)(r + c)(q + c) = (-1)^3(-c - p)(-c - r)(-c - q) = -f(-c),$$

since

$$f(x) = (x - p)(x - q)(x - r)$$

for all x .

Solution 6 (Cheesing it out)

Expanding the expression

$$(p^2 + 4)(q^2 + 4)(r^2 + 4)$$

gives us

$$(pqr)^2 + 4p^2q^2 + 4p^2r^2 + 4q^2r^2 + 16p^2 + 16q^2 + 16r^2 + 64$$

Notice that everything other than $(pqr)^2$ is a multiple of 4. Solving for $(pqr)^2$ using Vieta's formulas, we get 9. Since 9 is 1 (mod 4), the answer should be as well. The only answer that is 1 (mod 4) is **(D) 125**.

Solution 7

Suppose $y = x^2 + 4$

then $x = \pm\sqrt{y-4}$. Substitute $x = \sqrt{y-4}$ into $x^3 + 2x^2 - x + 3 = 0$ (It is same for $x = -\sqrt{y-4}$ because the squares in $(p^2 + 4)(q^2 + 4)(r^2 + 4)$)

$(\sqrt{y-4})^3 + 2(\sqrt{y-4})^2 - \sqrt{y-4} + 3 = 0 \implies (y-5)^2(y-4) = (-2y+5)^2$ whose constant is 125

according to Vieta's theorem, $y_1y_2y_3 = 125 \implies (x_1^2+4)(x_2^2+4)(x_3^2+4) = 125 \implies (p^2+4)(q^2+4)(r^2+4) = \mathbf{(D) 125}$.

16. (AMC 12 12A 2024) A set of 12 tokens — 3 red, 2 white, 1 blue, and 6 black — is to be distributed at random to 3 game players, 4 tokens per player. The probability that some player gets all the red tokens, another gets all the white tokens, and the remaining player gets the blue token can be written as $\frac{m}{n}$, where m and n are relatively prime positive integers. What is $m + n$?

(A) 387 (B) 388 (C) 389 (D) 390 (E) 391

Answer: C

Solution 1A (Trivial/Easy solve)

We have $\binom{12}{4,4,4}$ ways to handle the red/white/blue tokens distribution on the denominator. Now we simply $\binom{6}{1} \binom{5}{2} (3!)$ for the numerator in order to handle the black tokens and distinguishable persons. The solution is therefore $\frac{6 \cdot 6 \cdot 10}{70 \cdot 45 \cdot 11} = \frac{4}{385}$ or $4 + 385 = \mathbf{(C) 389}$.

Remarks - Notice we let tokens and persons be distinguishable to increase ease of calculations

Solution 1B (12fact bash)

We have $12!$ total possible arrangements of 12 distinct tokens. If we imagine the first 4 tokens of our arrangement go to the first player, the next 4 go to the second, and the final 4 go to the third, then we can view this problem as counting the number of valid arrangements.

Firstly, the tokens are not all distinct, so we multiply by $3!$, $2!$, $1!$, and $6!$ to account for the fact that the red, white, blue, and black tokens, respectively can switch around from where they are.

Letting R denote red, W denote white, B denote blue, and L denote black, then our arrangement must be something like $RRRLWWLLBLLL$. The three players are arbitrary, so we multiply by $3!$; then, the player who gets the reds has $\binom{4}{1} = 4$

possible arrangements, the player who gets the whites has $\binom{4}{2} = 6$ possibilities, and the player who gets the blacks has $\binom{4}{3} = 4$ possibilities. Our total on top is thus $3! \cdot 2! \cdot 1! \cdot 6! \cdot 3! \cdot 4 \cdot 6 \cdot 4$, and the denominator is $12!$. Firstly, we have the $6!$ in the numerator cancel out part of the denominator; we thus have the following:

$$\frac{3 \cdot 2 \cdot 2 \cdot 3 \cdot 2 \cdot 4 \cdot 6 \cdot 4}{12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 \cdot 7} = \frac{2^8 3^3}{2^6 3^3 5 \cdot 7 \cdot 11} = \frac{4}{385}.$$

Our answer is $4 + 385 = \boxed{(C) 389}$.

Solution 2

Assume all of them are distinct even though some have the same color,

Total possibility = $C_4^{12} C_4^8 C_4^4$ (choosing 4 random token for each person)

Next, assume that all the token are already in 3 different groups (Note: $3!$ Ways to do so since 3 people)

We then distribute the 6 distinct black token into these 3 different groups (So 1,2,3 token for each group)

There are a total of $3! \cdot C_3^6 C_2^3 C_1^1$ ways in doing so

Thus the answer is $3! \cdot \frac{6!}{1!2!3!} / \frac{12!}{4!4!4!} = \frac{4}{385}$

So the answer is $\boxed{(C) 389}$

Solution 3

We first assume there are designated red, white, and blue token players that will receive all of their respective one.

Consider each non-black token: The probability of the red player getting the first red token is $\frac{4}{12}$, because each player has 4 empty token "slots" for a total of 12. It follows that the probability of the player receiving all 3 red tokens is $\left(\frac{4}{12}\right) \left(\frac{3}{11}\right) \left(\frac{2}{10}\right)$, the white token player is $\left(\frac{4}{9}\right) \left(\frac{3}{8}\right)$, and the blue token player is $\frac{4}{7}$.

The combined probability is $\left(\frac{4}{12}\right) \left(\frac{3}{11}\right) \left(\frac{2}{10}\right) \left(\frac{4}{9}\right) \left(\frac{3}{8}\right) \left(\frac{4}{7}\right) = \frac{2}{1155}$.

Finally, we multiply the probability by $3! = 6$ to remove our initial assumption to get $\frac{4}{385}$.

The requested sum is $4 + 385 = \boxed{(C) 389}$.

Solution 4

Process start, **first** player get tokens. The probability that he gets 3 red tokens and 1 black token is $\frac{3 \cdot 2 \cdot 1 \cdot 6}{12 \cdot 11 \cdot 10 \cdot 9} = \frac{1}{330}$.

There is $\binom{4}{1} = 4$ possible arrangements (RRRB,RRBR,RBRR,BRRR) and 3 possibilities who is the first, so the probability that **some** player gets 3 red tokens and 1 black token is $\frac{4 \cdot 3}{330} = \frac{2}{55}$.

After that **second** player get tokens. The probability that he gets 2 white tokens and 2 black tokens is $\frac{2 \cdot 1 \cdot 5 \cdot 4}{8 \cdot 7 \cdot 6 \cdot 5} = \frac{1}{42}$.

There is $\binom{4}{2} = 6$ possible arrangements (WWBB, WBWB, WBBW, BBWW, BWBW, BWBW) and 2 possibilities who is the second, so the probability that **some** player gets 2 white tokens and 2 black tokens is $\frac{6 \cdot 2}{42} = \frac{2}{7}$.

The third player gets last tokens - 1 blue and 3 black tokens.

The desired probability is $\frac{2 \cdot 2}{55 \cdot 7} = \frac{4}{385} \implies 4 + 385 = \boxed{\text{(C) } 389}$.

To check the result suppose that first and some) player gets 1 blue and 3 black tokens. The probability is $\frac{1 \cdot 6 \cdot 5 \cdot 4}{12 \cdot 11 \cdot 10 \cdot 9} \cdot 4 \cdot 3 = \frac{4}{33}$.

The probability that second (and some) player gets 3 red tokens and 1 black token is $\frac{3 \cdot 2 \cdot 1 \cdot 3}{8 \cdot 7 \cdot 6 \cdot 5} \cdot 4 \cdot 2 = \frac{3}{35}$.

The desired probability is $\frac{4 \cdot 3}{33 \cdot 35} = \frac{4}{385}$.

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Solution 5

Arranging these 12 tokens in a row, there are total of $\frac{12!}{3!2!1!6!}$ arrangements.

Player one takes the first to the fourth tokens, player two takes the fifth to the eighth, and player three takes the rest. There are 6 cases of desired arrangements:

Case 1. Player one has three red tokens (and one black), player 2 has two white tokens and player 3 has 1 blue token. There are total of $\binom{4}{3}\binom{4}{2}\binom{4}{1} = 4 \cdot 6 \cdot 4$ arrangements in this case.

Other cases (such as player 1 has two white tokens, player 2 has three red, and player has one blue token) are all similar.

The number of desired arrangements for all cases is $6 \cdot (4 \cdot 6 \cdot 4)$.

The target probability is $\frac{6 \cdot (4 \cdot 6 \cdot 4)}{12! / (3! 2! 1! 6!)} = \frac{4}{385}$.

-J.Z.

Solution 6 (Alternate Solution 3)

We make no assumptions about which player receives which tokens.

Let us first distribute the red tokens. It does not matter who the first token goes to, but the second one must go to the same player, in 1 of their 3 open spots, out of the total 11 remaining spots. Similarly, the third red token must go in one of the 2 open spots out of 10 remaining.

Distributing the white tokens next, the first token can go in any of the 9 remaining spots except for the 1 spot of the red token player. The second white token must then go in the 3 open spots of this white token player out of 8 spots.

Lastly, the blue token must go in 1 of the 4 open spots of the last player, out of a total of 7 spots. The remaining 6 black tokens can be distributed however.

Thus, our probability is $\frac{3}{11} \cdot \frac{2}{10} \cdot \frac{8}{9} \cdot \frac{3}{8} \cdot \frac{4}{7} = \frac{4}{385}$, and so our answer is $4 + 385 = \boxed{\text{(C) } 389}$.

17. (AMC 12 12A 2024) Integers a , b , and c satisfy $ab + c = 100$, $bc + a = 87$, and $ca + b = 60$. What is $ab + bc + ca$?

(A) 212 (B) 247 (C) 258 (D) 276 (E) 284

Answer: D

Solution 1

Subtracting the first two equations yields $(a - c)(b - 1) = 13$. Notice that both factors are integers, so $b - 1$ could equal one of 13, 1, -1, -13 and $b = 14, 2, 0, -12$. We consider each case separately:

For $b = 0$, from the second equation, we see that $a = 87$. Then $87c = 60$, which is not possible as c is an integer, so this case is invalid.

For $b = 2$, we have $2c + a = 87$ and $ca = 58$, which by experimentation on the factors of 58 has no solution, so this is also invalid.

For $b = 14$, we have $14c + a = 87$ and $ca = 46$, which by experimentation on the factors of 46 has no solution, so this is also invalid.

Thus, we must have $b = -12$, so $a = 12c + 87$ and $ca = 72$. Thus $c(12c + 87) = 72$, so $c(4c + 29) = 24$. We can simply trial and error this to find that $c = -8$ so then $a = -9$. The answer is then $(-9)(-12) + (-12)(-8) + (-8)(-9) = 108 + 96 + 72 =$
(D) 276.

By Antony Huang Very Minor Edit by lucassf12

Solution 2

Adding up first two equations:

$$(a + c)(b + 1) = 187$$

$$b + 1 = \pm 11, \pm 17$$

$$b = -12, 10, -18, 16$$

Subtracting equation 1 from equation 2:

$$(a - c)(b - 1) = 13$$

$$b - 1 = \pm 1, \pm 13$$

$$b = 0, 2, -12, 14$$

$$\Rightarrow b = -12$$

Which implies that $a + c = -17$ from $(a + c)(b + 1) = 187$

Giving us that $a + b + c = -29$

Therefore, $ab + bc + ac = 100 + 87 + 60 - (a + b + c) =$ **(D) 276**

Solution 3 (Guess and check)

The idea is that you could guess values for c , since then a and b are factors of $100 - c$. The important thing to realize is that a , b , and c are all negative. Then, this can be solved in a few minutes, giving the solution $(-9, -12, -8)$, which gives the answer **(D)276** andliu766

Solution 4

$$ab + c = 100 \tag{27}$$

$$bc + a = 87 \tag{28}$$

$$ca + b = 60 \tag{29}$$

$$(1) + (2) \implies ab + c + bc + a = (a + c)(b + 1) = 187 \implies b + 1 = \pm 11, \pm 17$$

$$(1) - (2) \implies ab + c - bc - a = (a - c)(b - 1) = 13 \implies b - 1 = \pm 1, \pm 13$$

Note that $(b + 1) - (b - 1) = 2$, and the only possible pair of results that yields this is $b - 1 = -13$ and $b + 1 = -11$, so $a + c = -17$.

Therefore,

Solution 5

$$ab + c = 100 \tag{30}$$

$$bc + a = 87 \tag{31}$$

$$ca + b = 60 \tag{32}$$

$$(1) - (2) \implies ab + c - bc - a = (a - c)(b - 1) = 13$$

$$(2) - (3) \implies bc + a - ca - b = (b - a)(c - 1) = 27$$

$$(3) - (1) \implies ca + b - ab - c = (c - b)(a - 1) = -40$$

There are 3 ordered triplets of (a, b, c) : $(5, 14, 4)$, $(-3, -12, -3)$, $(-9, -12, -8)$.

However, only the last ordered triplet meets all three equations.

Therefore, $ab + ba + ac = -9 \cdot -12 + -12 \cdot -8 + -8 \cdot -9 = \boxed{\text{(D)} 276}$.

Solution 6 (Elimination)

Before we start, keep in mind that the problem is asking for the sum $ab + bc + ac$. This is nothing but $100 + 87 + 60 - a - b - c$, or $247 - (a + b + c)$.

To solve the problem, we systematically test the options using elimination:

Let's first check options A and B, since they only happen when a, b , and c sum to 35 or 0. We begin by testing three positive values, but none satisfy the equation when there is a plus sign. For example, $(12, 8, 4)$ satisfies $ab + c = 100$, but does not satisfy $bc + a = 87$, or $ac + b = 60$. If $a + b + c = 0$, then not all of the numbers can be positive or negative, so this would not work. From this observation, we conclude that the answer cannot be **A** or **B**.

Now let's test the next option, option C. Option **C** states $ab + bc + ca = 258$. If true, then:

$$a + b + c = -11$$

This sum is too large. Furthermore, if all three numbers are negative, the solution still fails. For example, testing $(-4, -5, -2)$ confirms the equation is not satisfied, as we get results that are too small. Thus, we eliminate option **C**.

Finally, let's test the last two options: D and E. For option **E**, the sum $a + b + c$ would be:

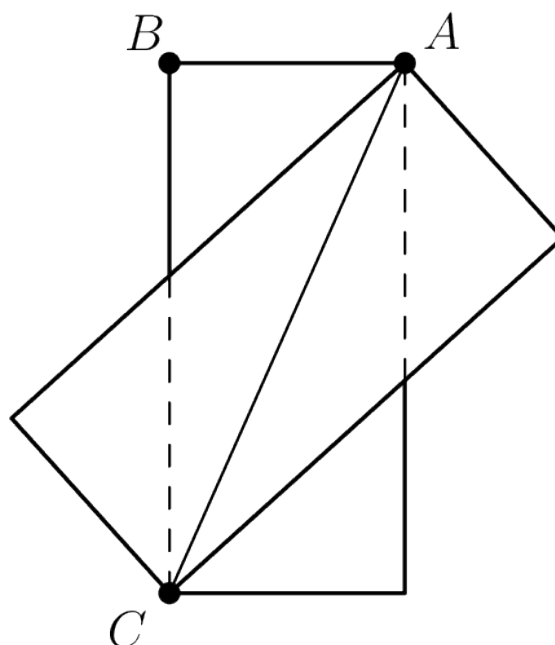
$$247 - 284 = -37$$

Testing values such as $(-11, -12, -14)$, the resulting sums $ab + c$, $bc + a$, and $ac + b$ are far too large to satisfy the equation. Therefore, **E** is also eliminated.

Once we have this answer, we still need to verify it by testing out numbers: Finally, we test option **D**. Using $ab + bc + ca = 276$, we get that $a + b + c = -29$. Also note that a , b , and c all have to be different, because the sums from the three equations are all different. We want to get the three closest values of a , b , and c such that they are all different, and the sum $a + b + c = -29$. The values $(-9, -12, -8)$ are the closest three numbers. When we try them, they satisfy all three equations. So, the correct answer is: $ab + ba + ac = -9 \cdot -12 + -12 \cdot -8 + -8 \cdot -9 = \boxed{\text{(D) } 276}$.

I NEED EASIER SOLUTIONS HERE:

18. (AMC 12 12A 2024) On top of a rectangular card with sides of length 1 and $2 + \sqrt{3}$, an identical card is placed so that two of their diagonals line up, as shown ($\overline{AC}$, in this case).



Continue the process, adding a third card to the second, and so on, lining up successive diagonals after rotating clockwise. In total, how many cards must be used until a vertex of a new card lands exactly on the vertex labeled B in the figure?

- (A) 6 (B) 8 (C) 10 (D) 12 (E) No new vertex will land on B .

Answer: A

Solution 1

Let the midpoint of AC be P .

We see that no matter how many moves we do, P stays where it is.

Now we can find the angle of rotation ($\angle APB$) per move with the following steps:

$$AP^2 = \left(\frac{1}{2}\right)^2 + \left(1 + \frac{\sqrt{3}}{2}\right)^2 = 2 + \sqrt{3}$$

$$1^2 = AP^2 + AP^2 - 2(AP)(AP) \cos \angle APB$$

$$1 = 2(2 + \sqrt{3})(1 - \cos \angle APB)$$

$$\cos \angle APB = \frac{3 + 2\sqrt{3}}{4 + 2\sqrt{3}}$$

$$\cos \angle APB = \frac{3 + 2\sqrt{3}}{4 + 2\sqrt{3}} \cdot \frac{4 - 2\sqrt{3}}{4 - 2\sqrt{3}}$$

$$\cos \angle APB = \frac{2\sqrt{3}}{4} = \frac{\sqrt{3}}{2}$$

$$\angle APB = 30^\circ$$

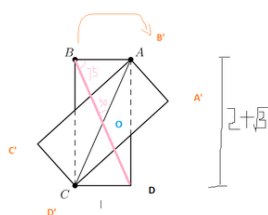
Since Vertex C is the closest one and

$$\angle BPC = 360 - 180 - 30 = 150$$

Vertex C will land on Vertex B when $\frac{150}{30} + 1 = \boxed{(A) 6}$ cards are placed.

(someone insert diagram maybe)

Solution 2



Let AC intersect BD at O,

We want to find $\angle AOB$

Since $\tan(75^\circ) = 2 + \sqrt{3} = \frac{AD}{AB}$, $\angle CBD = \angle BCA = 15^\circ$

$$\angle AOB = \angle CBD + \angle BCA = 30^\circ$$

So each time we rotate BD to AC for 30° , and we need to rotate $180^\circ/30^\circ = 6$ times to overlap a point with B

Therefore, the answer is **(A)** 6

Note: If you don't remember $\tan(75^\circ)$ $\tan(75^\circ) = \frac{\tan(45^\circ)+\tan(30^\circ)}{1-\tan(45^\circ)\cdot\tan(30^\circ)} = \frac{1+\frac{1}{\sqrt{3}}}{1-1\cdot\frac{1}{\sqrt{3}}}$
 $= \frac{(\sqrt{3}+1)^2}{(\sqrt{3})^2-1} = 2 + \sqrt{3}$

Solution 3 (Inscribing in Circle)

By the Pythagorean Theorem, $AC = \sqrt{(2 + \sqrt{3})^2 + 1} = \sqrt{8 + 4\sqrt{3}}$. So we have

$$\arcsin \angle BCA = \frac{1}{\sqrt{8+4\sqrt{3}}} = \frac{1}{\sqrt{(\sqrt{6}+\sqrt{2})^2}} = \frac{\sqrt{6}-\sqrt{2}}{4}.$$

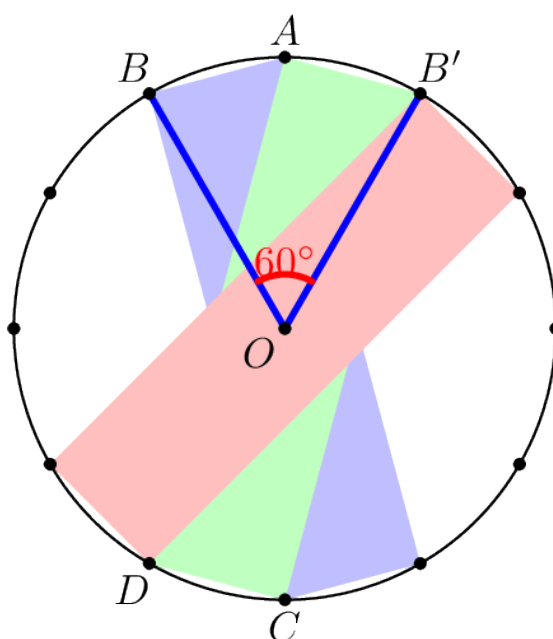
We note that

$$\sin 15^\circ = \cos 75^\circ = \cos(30^\circ + 45^\circ) = \cos 30^\circ \cos 45^\circ - \sin 45^\circ \sin 30^\circ = \frac{\sqrt{6} - \sqrt{2}}{4}.$$

Therefore, $\angle BCA = 15^\circ$.

Call the rotated rectangle given in the problem's diagram $AB'CD$. Since $\angle ACB' = \angle BCA = 15^\circ$, we have $\angle BCB' = 30^\circ$. So the angle formed between B and B' is $30^\circ \cdot 2 = 60^\circ$.

In general, when the rectangle is rotated, the angle formed between the left vertex of the original rectangle and the right vertex of the new rectangle is 60° . So we can inscribe the rectangles in a circle with $2 \cdot 360/60 = 12$ equally spaced points, and the rotated rectangles will have vertices on these points.



We get the equation $P = 2(R + 1)$, where R is the number of rectangles used and P is the number of points used on the circle for those rectangles (for example, 1 rectangle uses 4 points, 2 rectangles use 6, etc). The vertex will land on B after we have used all 12 points on the circle plus the additional 2 points we get from creating a new rotated rectangle, so $P = 12 + 2 = 14$. Then $R = \boxed{\text{(A) } 6}$.

Solution 4(In case you have no time and that's what I did)

$\tan 15 = \frac{\sin 15}{\cos 15} = \frac{1}{2 + \sqrt{3}}$ and it eliminates all options except 6 and 12. After one rotation it has turned 30° , so to satisfy the problem, divide $\frac{180}{30}$ and get $\boxed{\text{A. } 6}$.

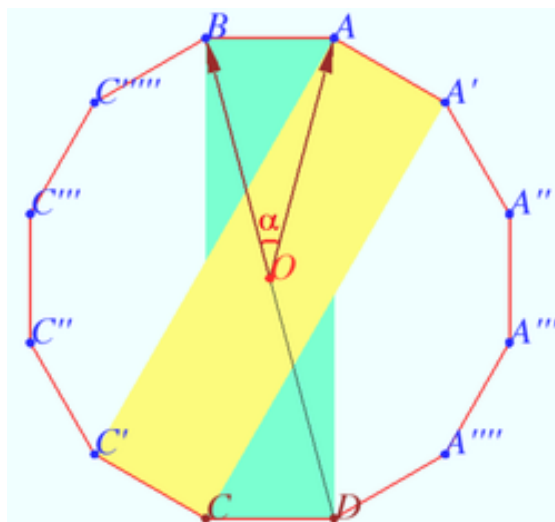
Solution 4.5

Memorize that in a 15-75-90 right triangle, the ratios of the side lengths are 1, $2 + \sqrt{3}$, and $\sqrt{2} + \sqrt{6}$. So, we have that the diagonal of the rectangle forms a 15 degree angle. Drawing it out we see the answer is $\boxed{\text{(A) } 6}$, and this makes sense because 15 times 6 is 90, thus rotating a vertex back to B.

Solution 5 (the simplest solution ever)

Look at the picture and draw the next one and continue draw down the line and then when you first hit B, count how many rectangles you've drawn (excluding the first which hasn't been rotated), and you should get **(A) or 6** as the answer.

Solution 6



Process is the rotation around the center of the card point O at the angle $\alpha = \angle AOB$.

$$AO = BO = R, BD^2 = 4R^2 = AB^2 + AD^2 = 4 \cdot (2 + \sqrt{3}).$$

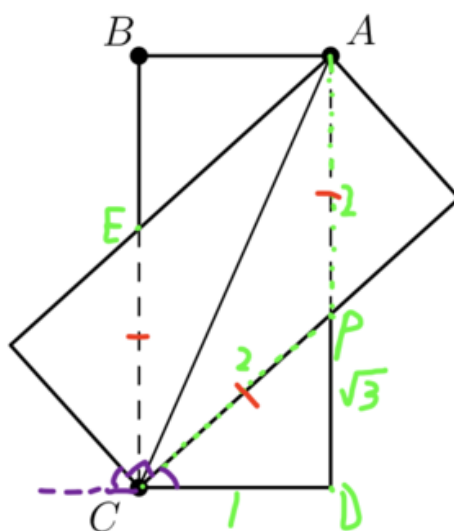
By applying the Law of Cosines, we get

$$2R^2(1 - \cos \alpha) = AB^2 \implies \cos \alpha = \frac{\sqrt{3}}{2} \implies$$

$$\alpha = 30^\circ \implies \boxed{\text{(A) or 6}}.$$

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Solution 7 (No Trig needed)



Let E and P be the intersection points, by symmetry axis of AC (since two rectangles are identical), we get:

$$CP = CE = PA$$

Now assume there exists a point F on AD , where $FD = \sqrt{3}$

Thus:

$$AF = (2 + \sqrt{3}) - \sqrt{3} = 2$$

Connecting CF , we see that:

$$CF = 2$$

So F must collide with P , and triangle PCD is a 30° - 60° - 90° triangle, so:

$\angle PCD = 60^\circ$ and external angle would be 30°

$$\implies \boxed{\text{(A) or 6}}.$$

Solution 8 (Similar to Solution 1)

Let the vertex of the second rectangular card closest to C be E and the unnamed vertex of the first card be F .

First, notice how the second rectangular card is a reflection of the first rectangular card about BC rotated about C . This means that the diagonal AC from the second rectangular card would have been rotated an angle $2\angle BCA$ clockwise. This also means that side EC would have also been rotated an angle $2\angle BCA$ clockwise, so the angle external to $\angle FCE$ would equal $2\angle BCA$. When this process is repeated, a regular polygon is formed with half the number of sides of the polygon being the answer.

To find $2\angle BCA$, we can reflect triangle ABC about BC to form an isosceles triangle. Then

$$CA^2 = 1 + (2 + \sqrt{3})^2 = 8 + 4\sqrt{3}$$

By the law of cosine:

$$(2(1))^2 = CA + CA - 2(CA)(CA)(\cos(2\angle BCA))$$

$$4 = (2CA^2)(1 - \cos(2\angle BCA))$$

$$4 = (2(8 + 4\sqrt{3}))(1 - \cos(2\angle BCA))$$

$$\frac{1}{4 + 2\sqrt{3}} = 1 - \cos(2\angle BCA)$$

$$\frac{3 + 2\sqrt{3}}{4 + 2\sqrt{3}} = \cos(2\angle BCA)$$

Simplifying:

$$\frac{3 + 2\sqrt{3}}{4 + 2\sqrt{3}} * \frac{4 - 2\sqrt{3}}{4 - 2\sqrt{3}} = \cos(2\angle BCA)$$

$$\frac{2\sqrt{3}}{4} = \cos(2\angle BCA)$$

$$\cos(2\angle BCA) = \frac{\sqrt{3}}{2}$$

$$2\angle BCA = 30^\circ$$

The resulting polygon from repeating the process would have $\frac{360}{30} = 12$ sides and $\frac{12}{2} = \boxed{\text{(A) } 6}$ cards would have to have been placed.

Solution 9

The answer choices imply an even number of successive diagonals, which motivates us to try the double angle formula $\tan$. Let $\alpha = \angle BCA$. Then $\tan \alpha = \frac{1}{2+\sqrt{3}}$ and

$$\tan 2\alpha = \frac{2 \cdot \tan \alpha}{1 - \tan^2 \alpha} = \frac{2 \cdot \frac{1}{2+\sqrt{3}}}{1 - \left(\frac{1}{2+\sqrt{3}}\right)^2} = -\sqrt{3}$$

We have $2\alpha = 120^\circ \Rightarrow \alpha = 60^\circ$, so our answer is $\frac{360^\circ}{60^\circ} = \boxed{\text{(A) } 6}$.

19. (AMC 12 12A 2024) Cyclic quadrilateral $ABCD$ has lengths $BC = CD = 3$ and $DA = 5$ with $\angle CDA = 120^\circ$. What is the length of the shorter diagonal of $ABCD$?

(A) $\frac{31}{7}$ (B) $\frac{33}{7}$ (C) 5 (D) $\frac{39}{7}$ (E) $\frac{41}{7}$

Answer: D

Solution 1

First, $\angle CBA = 60^\circ$ by properties of cyclic quadrilaterals.

Let $AC = u$. Apply the **Law of Cosines** on $\triangle ACD$:

$$u^2 = 3^2 + 5^2 - 2(3)(5) \cos 120^\circ$$

$$u = 7$$

Let $AB = v$. Apply the Law of Cosines on $\triangle ABC$:

$$7^2 = 3^2 + v^2 - 2(3)(v) \cos 60^\circ$$

$$v = \frac{3 \pm 13}{2}$$

$$v = 8$$

By **Ptolemy's Theorem**,

$$AB \cdot CD + AD \cdot BC = AC \cdot BD$$

$$8 \cdot 3 + 5 \cdot 3 = 7BD$$

$$BD = \frac{39}{7}$$

Since $\frac{39}{7} < 7$, The answer is $\boxed{\text{(D) } \frac{39}{7}}$.

Solution 2 (Law of Cosines + Law of Sines)

Draw diagonals AC and BD . By Law of Cosines,

$$AC^2 = 3^2 + 5^2 - 2(3)(5) \cos \left(\frac{2\pi}{3} \right)$$

$$\begin{aligned} &= 9 + 25 + 15 \\ &= 49. \end{aligned}$$

Since AC is positive, taking the square root gives $AC = 7$. Let $\angle BDC = \angle CBD = x$. Since $\triangle BCD$ is isosceles, we have $\angle BCD = 180 - 2x$. Notice we can eventually solve BD using the Extended Law of Sines:

$$\frac{BD}{\sin(180 - 2x)} = 2r,$$

where r is the radius of the circumcircle $ABCD$. Since $\sin(180 - 2x) = \sin(2x) = 2\sin(x)\cos(x)$, we simplify our equation:

$$\frac{BD}{2\sin(x)\cos(x)} = 2r.$$

Now we just have to find $\sin(x)$, $\cos(x)$, and $2r$. Since $ABCD$ is cyclic, we have $\angle CBD = \angle CAD = x$. By Law of Cosines on $\triangle ADC$, we have

$$3^2 = 7^2 + 5^2 - 70\cos(x).$$

Thus, $\cos(x) = \frac{13}{14}$. Similarly, by Law of Sines on $\triangle ACD$, we have

$$\frac{7}{\sin\left(\frac{2\pi}{3}\right)} = 2r.$$

Hence, $2r = \frac{14\sqrt{3}}{3}$. Now, using Law of Sines on $\triangle BCD$, we have $\frac{3}{\sin(x)} = 2r = \frac{14\sqrt{3}}{3}$, so $\sin(x) = \frac{3\sqrt{3}}{14}$. Therefore,

$$\frac{BD}{2\left(\frac{3\sqrt{3}}{14}\right)\left(\frac{13}{14}\right)} = \frac{14\sqrt{3}}{3}.$$

Solving, $BD = \frac{39}{7}$, so the answer is (D) $\frac{39}{7}$.

Solution 3 (Law of Cosines + Cyclic Quadrilateral Property)

Draw diagonals AC and BD . First, use Law of Cosines to get that

$$\begin{aligned} AC^2 &= 3^2 + 5^2 - 2(3)(5)\cos(120^\circ) \\ &= 9 + 25 + 15 \\ &= 49. \end{aligned}$$

Thus, $AC = 7$. Since $ABCD$ is cyclic, $\angle CAD = \angle CBD$, so Law of Cosines once again with respect to $\angle CAD$ on triangle ACD leads to

$$\begin{aligned} 9 &= 5^2 + 7^2 - 2(7)(5)\cos\theta \\ &= 74 - 70\cos\theta. \end{aligned}$$

Solving yields $\cos\theta = \frac{13}{14}$. Finally, in $\triangle CBD$, we have $BD = 6\cos\theta \implies$ (D) $\frac{39}{7}$.

Solution 4 (Law of Cosines+Law of Sines+Trig Identities)

Let $\angle BCA = x, \angle DCA = y$. If we know $\cos(x + y)$ we can compute BD . Notice that

$$\cos(x + y) = \cos(x) \cos(y) - \sin(x) \sin(y)$$

. Now it remains to find all 4 terms in this equation. Applying Law of Cosines on triangle ABC to find $\cos(x)$, we find that $\cos(x) = -\frac{6}{42} = -\frac{1}{7}$. Similarly we find that $\cos(y) = \frac{11}{14}$. Now we compute $\sin(x)$ and $\sin(y)$. Applying Law of Sines on triangle ABC we see that $\frac{\sin(x)}{8} = \frac{\sin(\frac{\pi}{3})}{7}$, or $\sin(x) = \frac{4\sqrt{3}}{7}$. Similarly $\sin(y) = \frac{5\sqrt{3}}{14}$. Now $\cos(x + y) = -\frac{71}{98}$. Let $BD = k$, we see that $k^2 = 3^2 + 3^2 + 2 * 3 * 3(\frac{71}{98})$. Solving for k yields $k = \frac{39}{7}$.

20. (AMC 12 12A 2024) Points P and Q are chosen uniformly and independently at random on sides $\overline{AB}$ and $\overline{AC}$, respectively, of equilateral triangle $\triangle ABC$. Which of the following intervals contains the probability that the area of $\triangle APQ$ is less than half the area of $\triangle ABC$? (A) $[\frac{3}{8}, \frac{1}{2}]$ (B) $(\frac{1}{2}, \frac{2}{3}]$ (C) $(\frac{2}{3}, \frac{3}{4}]$ (D) $(\frac{3}{4}, \frac{7}{8}]$ (E) $(\frac{7}{8}, 1]$

Answer: D

Solution 1

Let $\overline{AP} = x$ and $\overline{AQ} = y$. Applying the sine formula for a triangle's area, we see that

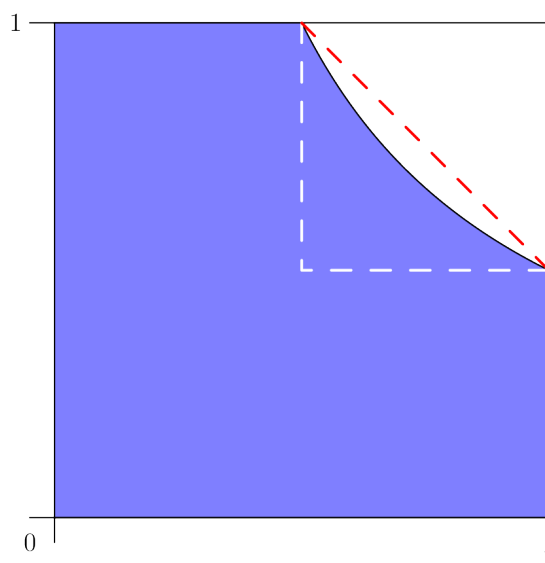
$$[\triangle APQ] = \frac{1}{2} \cdot x \cdot y \sin(\angle PAQ) = \frac{xy}{2} \sin(60^\circ) = \frac{\sqrt{3}}{4} xy.$$

Without loss of generality, we let $AB = BC = CA = 1$, and thus $[\triangle ABC] = \frac{\sqrt{3}}{4}$;

we therefore require $\frac{\sqrt{3}}{4} xy \leq \frac{1}{2} \cdot \frac{\sqrt{3}}{4} \implies xy \leq \frac{1}{2}$ for $0 \leq x, y \leq 1$. (Note: You can skip most of this by using triangle area ratios A_MatheMagician.)

A quick rough sketch of $y = \frac{1}{2x}$ on the square given by $x, y \in [0, 1]$ reveals that the curve intersects the boundaries at $(0.5, 1)$ and $(1, 0.5)$, and it is actually quite (very) obvious that the area bounded by the inequality $xy \leq 0.5$ and the aforementioned unit square is more than $\frac{3}{4}$ but less than $\frac{7}{8}$ (cf. the diagram below). Thus, our

answer is **(D)** $(\frac{3}{4}, \frac{7}{8}]$.



Solution 2

WLOG let $AB = AC = 1$

$$\frac{AP \cdot AQ \cdot \sin 60}{2} < \frac{1 \cdot 1 \cdot \sin 60}{4}$$

$$AP \cdot AQ < \frac{1}{2}$$

Which we can express as $xy < \frac{1}{2}$.

We follow by graphing the equation $xy < \frac{1}{2}$ on $x, y \in [0, 1]$, and we follow the same way solution 1 does.

We see that the probability is slightly less than $\frac{7}{8}$ but definitely greater than $\frac{3}{4}$

Thus answer choice $\boxed{(D) \left(\frac{3}{4}, \frac{7}{8}\right]}$

Solution 3

The actual probability can be found by using calculus, and taking the integral.

$$P = \int_{0.5}^1 \frac{1}{2x} dx + 0.5 = \frac{\ln 2 + 1}{2} \approx 0.84657.$$

Thus, since $\frac{3}{4} < 0.84655 < \frac{7}{8}$, we get $\boxed{(D) \left(\frac{3}{4}, \frac{7}{8}\right]}$ lptoggled

Note that if you do not know an approximation for $\ln 2$, you can use the following power series formula:

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

Using this formula to four terms gives the approximation $\ln 2 \approx \frac{19}{24}$, which is good enough to solve the problem.

Another note If you're feeling up to it, you can assume that the curve is actually a 90° sector of a circle with radius $\frac{1}{2}$. Doing so yields

$$P = 1 - \pi \frac{\left(\frac{1}{2}\right)^2}{4} \approx 0.8037,$$

which is clearly within the interval D . This is also an under-estimate, since the curve of $\frac{1}{2x}$ is not as steep on the interval $x \in (0.5, 1)$.

21. (AMC 12 12A 2024) Suppose that $a_1 = 2$ and the sequence (a_n) satisfies the recurrence relation

$$\frac{a_n - 1}{n - 1} = \frac{a_{n-1} + 1}{(n - 1) + 1}$$

for all $n \geq 2$. What is the greatest integer less than or equal to

$$\sum_{n=1}^{100} a_n^2?$$

- (A) 338,550 (B) 338,551 (C) 338,552 (D) 338,553 (E) 338,554

Answer: B

Solution 1

Multiply both sides of the recurrence to find that $n(a_n - 1) = (n - 1)(a_{n-1} + 1) = (n - 1)(a_{n-1} - 1) + (n - 1)(2)$.

Let $b_n = n(a_n - 1)$. Then the previous relation becomes

$$b_n = b_{n-1} + 2(n - 1)$$

We can rewrite this relation for values of n until 1 and use telescoping to derive an explicit formula:

$$b_n = b_{n-1} + 2(n - 1)$$

$$b_{n-1} = b_{n-2} + 2(n - 2)$$

$$b_{n-2} = b_{n-3} + 2(n - 3)$$

.

.

.

$$b_2 = b_1 + 2(1)$$

Summing the equations yields:

$$b_n + b_{n-1} + \cdots + b_2 = b_{n-1} + b_{n-2} + \cdots + b_1 + 2((n - 1) + (n - 2) + \cdots + 1)$$

$$b_n - b_1 = 2 \cdot \frac{n(n - 1)}{2}$$

$$b_n - 1 = n(n - 1)$$

$$b_n = n(n - 1) + 1$$

Now we can substitute a_n back into our equation:

$$n(a_n - 1) = n(n - 1) + 1$$

$$a_n - 1 = n - 1 + \frac{1}{n}$$

$$a_n = n + \frac{1}{n}$$

$$a_n^2 = n^2 + \frac{1}{n^2} + 2$$

Thus the sum becomes

$$\sum_{n=1}^{100} a_n^2 = \sum_{n=1}^{100} n^2 + \sum_{n=1}^{100} \frac{1}{n^2} + \sum_{n=1}^{100} 2$$

We know that $\sum_{n=1}^{100} n^2 = \frac{100 \cdot 101 \cdot 201}{6} = 338350$, and we also know that $\sum_{n=1}^{100} 2 = 200$, so the requested sum is equivalent to $\sum_{n=1}^{100} \frac{1}{n^2} + 338550$. All that remains is to calculate $\sum_{n=1}^{100} \frac{1}{n^2}$, and we know that this value lies between 1 and 2 (see the note below for a proof). Thus,

$$\sum_{n=1}^{100} a_n^2 = \sum_{n=1}^{100} \frac{1}{n^2} + 338550 < 2 + 338550 = 338552$$

$$\sum_{n=1}^{100} a_n^2 = \sum_{n=1}^{100} \frac{1}{n^2} + 338550 > 1 + 338550 = 338551$$

so

$$\sum_{n=1}^{100} a_n^2 \in (338551, 338552)$$

and thus the answer is (B) 338551.

Note: $\sum_{n=1}^{100} \frac{1}{n^2} < \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} < 2$. It is obvious that the sum is greater than 1 (since it contains $\frac{1}{1^2}$ as one of its terms).

If you forget this and have to derive this on the exam, here is how:

$$\begin{aligned} \sum_{n=1}^{100} \frac{1}{n^2} &< \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{1}{1^2} + \left(\frac{1}{2^2} + \frac{1}{3^2} \right) + \left(\frac{1}{4^2} + \frac{1}{5^2} + \frac{1}{6^2} + \frac{1}{7^2} \right) + \cdots \\ &< 1 + \left(\frac{1}{2^2} + \frac{1}{2^2} \right) + \left(\frac{1}{4^2} + \frac{1}{4^2} + \frac{1}{4^2} + \frac{1}{4^2} \right) + \left(\frac{1}{8^2} + \frac{1}{8^2} + \frac{1}{8^2} + \frac{1}{8^2} + \frac{1}{8^2} + \frac{1}{8^2} + \frac{1}{8^2} + \frac{1}{8^2} \right) + \cdots \\ &= 1 + \frac{2}{2^2} + \frac{4}{4^2} + \frac{8}{8^2} + \cdots \\ &= 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = 2 \end{aligned}$$

and it is clear that $\sum_{n=1}^{100} \frac{1}{n^2} < 2$. eevee9406

Solution 2

According to the equation given,

$$\frac{a_n - 1}{n - 1} = \frac{a_{n-1} + 1}{n}$$

$$na_n - n = (n - 1)a_{n-1} + n - 1$$

$$na_n - (n - 1)a_{n-1} = 2n - 1$$

Suppose $\{b_n\}$, $b_n = na_n$, then $b_n - b_{n-1} = 2n - 1$, where $b_1 = 1 \cdot a_1 = 2$, then we obtain that

$$b_n = b_1 + (b_2 - b_1) + (b_3 - b_2) + \cdots + (b_n - b_{n-1})$$

$$b_n = 2 + 3 + 5 + \cdots + (2n - 1) = 1 + n^2$$

$$a_n = \frac{1+n^2}{n} = n + \frac{1}{n}$$

Hence,

$$\begin{aligned} \sum_{n=1}^{100} a_n^2 &= \sum_{n=1}^{100} \left(n + \frac{1}{n}\right)^2 \\ &= \sum_{n=1}^{100} \left(n^2 + 2 + \frac{1}{n^2}\right) \\ &= \sum_{n=1}^{100} n^2 + \sum_{n=1}^{100} 2 + \sum_{n=1}^{100} \frac{1}{n^2} \\ &= \frac{1}{6} \cdot 100 \cdot (100+1) \cdot (100 \cdot 2 + 1) + 100 \cdot 2 + \sum_{n=1}^{100} \frac{1}{n^2} \\ &= 338350 + 200 + \sum_{n=1}^{100} \frac{1}{n^2} \\ &= 338550 + \sum_{n=1}^{100} \frac{1}{n^2} \end{aligned}$$

Notice that,

$$\begin{aligned} \sum_{n=1}^{100} \frac{1}{n^2} &= 1 + \sum_{n=2}^{100} \frac{1}{n^2} > 1 \\ \sum_{n=1}^{100} \frac{1}{n^2} &< 1 + \sum_{n=2}^{100} \frac{1}{n(n-1)} = 1 + 1 - \frac{1}{100} < 2 \quad (\text{telescope}) \end{aligned}$$

so

$$\sum_{n=1}^{100} a_n^2 \in (338551, 338552)$$

and thus the answer is (B) 338551.

Solution 3 (lazy + quick)

We'll first try to isolate a_n in terms of a_{n-1} .

$$\begin{aligned} (a_n - 1)(n - 1 + 1) &= (n - 1)(a_{n-1} + 1) \quad a_n - 1 = \frac{(n-1)(a_{n-1}+1)}{n} \quad a_n = \frac{n-1}{n}(a_{n-1} + 1) + 1 \\ a_n &= \frac{n-1}{n}(a_{n-1}) + \frac{n-1}{n} + 1 \quad a_n = \frac{n-1}{n}(a_{n-1}) + \frac{2n-1}{n} \end{aligned}$$

Now, as with many, many of these large summation problems, if we just evaluate the first few values in the series, a pattern should emerge quickly. Here it works out well since our product on the LHS cancels out.

$$a_1 = 2 \quad a_2 = \frac{1}{2}(2) + \frac{3}{2} = \frac{5}{2} \quad a_3 = \frac{2}{3}\left(\frac{5}{2}\right) + \frac{5}{3} = \frac{10}{3} \quad a_4 = \frac{3}{4}\left(\frac{10}{3}\right) + \frac{7}{4} = \frac{17}{4}$$

Here it becomes glaringly obvious that $a_n = \frac{n^2+1}{n} = n + \frac{1}{n}$.

So, $a_n^2 = n^2 + \frac{1}{n^2} + 2$.

We proceed with the same summation strategy as Solution 1 and get our answer of

(B) 338551.

- Note: You only have find the answer's units digit from the answer choices; that's $0 + 1 + 0$ for each sum, giving choice B.

Solution 4 (transform)

$$\begin{aligned} na_n - n &= (n-1)a_{n-1} + n - 1 \\ n \cdot a_n - n^2 &= (n-1)a_{n-1} - n^2 + 2n - 1 \\ n \cdot a_n - n^2 &= (n-1)a_{n-1} - (n-1)^2 \end{aligned}$$

Set

$$\begin{aligned} b_n &= na_n - n^2 \\ na_n - n^2 &= b_n = b_{n-1} = \dots = b_1 = 1 \\ a_n &= \frac{1}{n} + n \\ a_n^2 &= \frac{1}{n^2} + n^2 + 2 \end{aligned}$$

$$Sum = 1^2 + 2^2 + \dots + 100^2 + 2 \cdot 100 + 1^2 + 1/2^2 + \dots + 1/100^2 = 100 \cdot 101 \cdot 201 / 6 + 200 + 1 + \dots = 338551.xxx \quad (\text{E})$$

Solution 5 (transform)

According to the equation given,

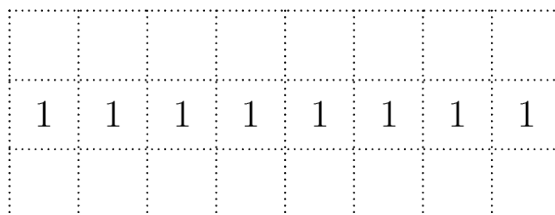
$$\frac{a_n - 1}{n - 1} = \frac{a_{n-1} + 1}{n}$$

$$na_n = (n-1)a_{n-1} + 2n - 1 = (n-2)a_{n-2} + (2n-3) + (2n-1) = 1 \cdot a_1 + 1 + 3 + \dots + 2n - 1 = 1 + \frac{1 + (2n-1) \cdot n}{2}$$

$$a_n = n + \frac{1}{n}$$

The rest continues similar to Solution 1 or 2

22. (AMC 12 12A 2024) The figure below shows a dotted grid 8 cells wide and 3 cells tall consisting of $1'' \times 1''$ squares. Carl places 1-inch toothpicks along some of the sides of the squares to create a closed loop that does not intersect itself. The numbers in the cells indicate the number of sides of that square that are to be covered by toothpicks, and any number of toothpicks are allowed if no number is written. In how many ways can Carl place the toothpicks?



- (A) 130 (B) 144 (C) 146 (D) 162 (E) 196

Answer: C

Solution 1 (Best Motivated)

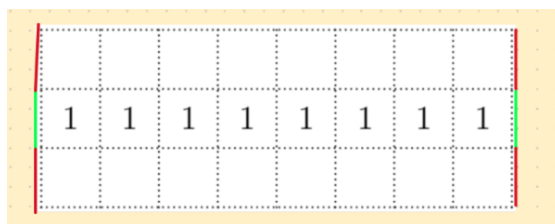
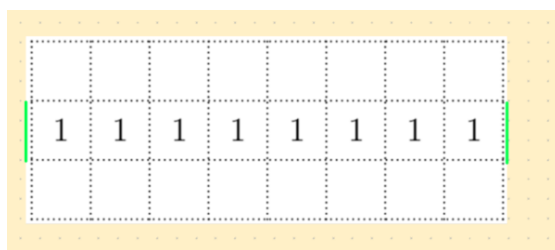
Observations:

1. You can't have a vertical line in any place other than the first two columns and the last two columns. If we did, we would then have at least one of the middle cells with toothpicks along more than one side, which would violate the conditions of the problem.

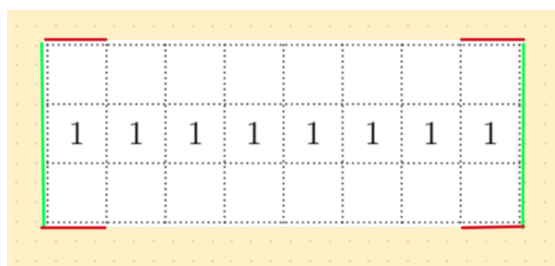
2. There are two cases that look completely different. We can have a long horizontal box that spans all eight cells either on top of or below the middle cells, or we have to have a shape that looks like a rectangle, except with a few places "pushed" in.

Thus, using casework, we can split the task of finding those rectangles with squiggly edges into 3 cases.

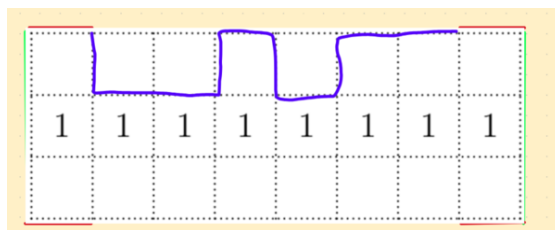
For case 1, we just assume that the green lines shown below are given (always have toothpicks on them). In effect, we will use all eight columns.



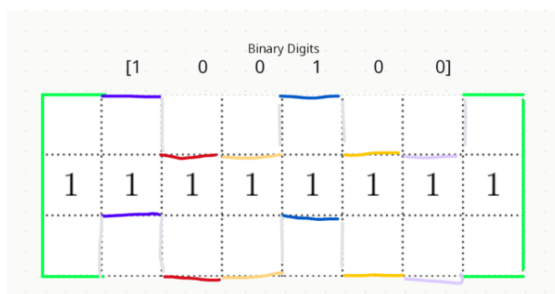
The only toothpicks we can place that will connect to the red lines are to go horizontally inward:



Now, concentrate on the first row of squares. A toothpick can be placed on either the bottom or top and connected to a continuous squiggle by adding vertical toothpicks:

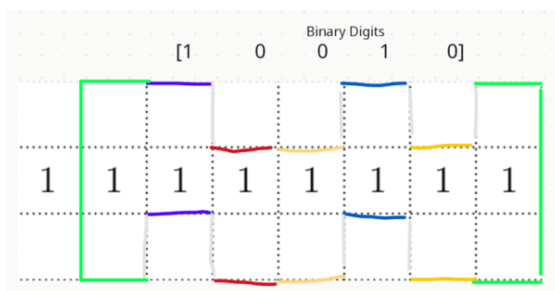
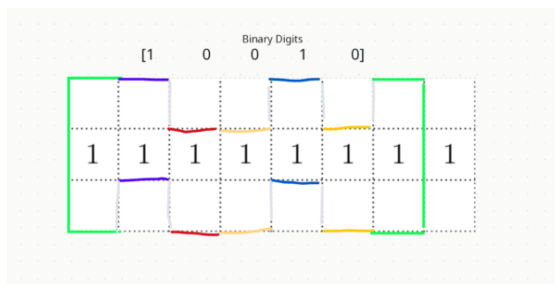


How many squiggles are possible?



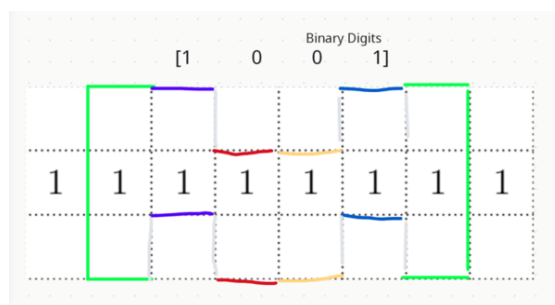
We can summarize this by giving a high squiggle position a 1 and a low position a 0, thus we have a 6-digit binary sequence. Thus, we can have $2^6 = 64$ ways to make this squiggle. (The binary is not absolutely necessary, but it works.)

Case 2: We can also pull in one of the sides, thus we can have a squiggle with 5 binary digits, which only uses the first or last 7 columns:



Here, we only have 5 binary digits to work with, so there are $2^5 = 32$ ways to make this squiggle for each individual subcase. There are two subcases, one with the first 7 columns, and the other with the last 7, so we have a total of $32 \cdot 2 = 64$ arrangements in this case.

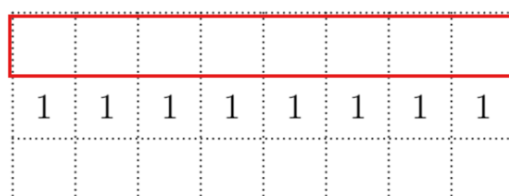
Case 3: We can use an even smaller section. Using only the middle 6 columns gives us a 4-wide squiggle:



Thus, there are $2^4 = 16$ ways to make this squiggle.

These three cases together cover all loops of this form. If we try to bring the square bracket like shapes on each side any closer, there will be some middle cells that do not touch any toothpicks. Adding up all our cases for these types of shapes: $64 + 32 + 32 + 16 = 144$.

However, there are two more ways to draw a qualifying shape:



We can draw a rectangle like that in the first row or third row. Thus, we have a total of $144 + 2 = \boxed{\text{(C) } 146}$ ways.

A note to (potential) editors: This answer was not made to be concise or especially professional. It was made to explicitly explain this problem in a way so that it is easy to understand and follow.

Notes: Remember these are the ONLY possible cases. It is impossible to cross through the rows of boxes of ones (in a snake like pattern) to connect the loop around the bottom since then the loop would intersect itself.

We are not undercounting by only counting the binary digits on the top. Consider the top squiggle. Each middle square must and can only come in contact with one toothpick, so if the top squiggle doesn't touch a specific middle cell, the bottom squiggle must ensure that all squares touch one toothpick. juwushu

Solution 2 (Cheese)

Notice that for any case where the closed loop does not connect from the top side of the ones and bottom side of the ones, there are two of these cases. A cheese solution can be found from this; noting that B and C are the only two options close to each other, and, being two apart, with people likely to forget this case, $\boxed{\text{(C) } 146}$ is likely to be the correct answer. Cheese solution done by juwushu.

Solution 3 (Observation)

We have 2 cases where the loop does not go through the middle.

If the loop goes through the middle, we must have a full column on $(0, 8), (0, 7), (1, 8), (1, 7)$. Then we have 6, 5, 5, 4 empty middle squares. For each one we can have one on top or one on bottom, so $2^6 + 2^5 + 2^5 + 2^4 = 144$. Notice that for each case

of fixed toothpicks, there is only one way to form the loop. Then we just add $2 + 144 = \boxed{\text{(C)} 146}$.

23. (AMC 12 12A 2024) What is the value of

$$\tan^2 \frac{\pi}{16} \cdot \tan^2 \frac{3\pi}{16} + \tan^2 \frac{\pi}{16} \cdot \tan^2 \frac{5\pi}{16} + \tan^2 \frac{3\pi}{16} \cdot \tan^2 \frac{7\pi}{16} + \tan^2 \frac{5\pi}{16} \cdot \tan^2 \frac{7\pi}{16}?$$

(A) 28 (B) 68 (C) 70 (D) 72 (E) 84

Answer: B

Solution 1 (Trigonometric Identities)

First, notice that

$$\begin{aligned} \tan^2 \frac{\pi}{16} \cdot \tan^2 \frac{3\pi}{16} + \tan^2 \frac{\pi}{16} \cdot \tan^2 \frac{5\pi}{16} + \tan^2 \frac{3\pi}{16} \cdot \tan^2 \frac{7\pi}{16} + \tan^2 \frac{5\pi}{16} \cdot \tan^2 \frac{7\pi}{16} \\ = (\tan^2 \frac{\pi}{16} + \tan^2 \frac{7\pi}{16})(\tan^2 \frac{3\pi}{16} + \tan^2 \frac{5\pi}{16}) \end{aligned}$$

Here, we make use of the fact that

$$\begin{aligned} \tan^2 x + \tan^2 \left(\frac{\pi}{2} - x \right) \\ = (\tan x + \tan \left(\frac{\pi}{2} - x \right))^2 - 2 \\ = \left(\frac{\sin x}{\cos x} + \frac{\sin \left(\frac{\pi}{2} - x \right)}{\cos \left(\frac{\pi}{2} - x \right)} \right)^2 - 2 \\ = \left(\frac{\sin x \cos \left(\frac{\pi}{2} - x \right) + \sin \left(\frac{\pi}{2} - x \right) \cos x}{\cos x \cos \left(\frac{\pi}{2} - x \right)} \right)^2 - 2 \\ = \left(\frac{\sin \frac{\pi}{2}}{\cos x \cos \left(\frac{\pi}{2} - x \right)} \right)^2 - 2 \\ = \left(\frac{1}{\cos x \sin x} \right)^2 - 2 \\ = \left(\frac{2}{\sin 2x} \right)^2 - 2 \\ = \frac{4}{\sin^2 2x} - 2 \end{aligned}$$

Hence,

$$\begin{aligned} (\tan^2 \frac{\pi}{16} + \tan^2 \frac{7\pi}{16})(\tan^2 \frac{3\pi}{16} + \tan^2 \frac{5\pi}{16}) \\ = \left(\frac{4}{\sin^2 \frac{\pi}{8}} - 2 \right) \left(\frac{4}{\sin^2 \frac{3\pi}{8}} - 2 \right) \end{aligned}$$

Note that

$$\sin^2 \frac{\pi}{8} = \frac{1 - \cos \frac{\pi}{4}}{2} = \frac{2 - \sqrt{2}}{4}$$

$$\sin^2 \frac{3\pi}{8} = \frac{1 - \cos \frac{3\pi}{4}}{2} = \frac{2 + \sqrt{2}}{4}$$

Hence,

$$\begin{aligned} & \left(\frac{4}{\sin^2 \frac{\pi}{8}} - 2 \right) \left(\frac{4}{\sin^2 \frac{3\pi}{8}} - 2 \right) \\ &= \left(\frac{16}{2 - \sqrt{2}} - 2 \right) \left(\frac{16}{2 + \sqrt{2}} - 2 \right) \\ &= (14 + 8\sqrt{2})(14 - 8\sqrt{2}) \\ &= 68 \end{aligned}$$

Therefore, the answer is (B) 68.

Solution 2 (Another Identity)

First, notice that

$$\begin{aligned} & \tan^2 \frac{\pi}{16} \cdot \tan^2 \frac{3\pi}{16} + \tan^2 \frac{\pi}{16} \cdot \tan^2 \frac{5\pi}{16} + \tan^2 \frac{3\pi}{16} \cdot \tan^2 \frac{7\pi}{16} + \tan^2 \frac{5\pi}{16} \cdot \tan^2 \frac{7\pi}{16} \\ &= (\tan^2 \frac{\pi}{16} + \tan^2 \frac{7\pi}{16})(\tan^2 \frac{3\pi}{16} + \tan^2 \frac{5\pi}{16}) \end{aligned}$$

Here, we make use of the fact that

$$\begin{aligned} \tan^2 x + \tan^2 \left(\frac{\pi}{2} - x \right) &= (\tan x - \tan \left(\frac{\pi}{2} - x \right))^2 + 2 \\ &= \left(\tan \left(\frac{\pi}{2} - 2x \right) \cdot (1 + \tan x \tan \left(\frac{\pi}{2} - x \right)) \right)^2 + 2 \quad (\text{difference of two tan}) \\ &= \left(\tan \left(\frac{\pi}{2} - 2x \right) \cdot (1 + 1) \right)^2 + 2 \\ &= 4 \tan^2 \left(\frac{\pi}{2} - 2x \right) + 2 \end{aligned}$$

Hence,

$$\begin{aligned} (\tan^2 \frac{\pi}{16} + \tan^2 \frac{7\pi}{16})(\tan^2 \frac{3\pi}{16} + \tan^2 \frac{5\pi}{16}) &= (4 \tan^2 \left(\frac{\pi}{2} - \frac{\pi}{16} \cdot 2 \right) + 2)(4 \tan^2 \left(\frac{\pi}{2} - \frac{3\pi}{16} \cdot 2 \right) + 2) \\ &= (4 \tan^2 \frac{3\pi}{8} + 2)(4 \tan^2 \frac{\pi}{8} + 2) \\ &= 16 \tan^2 \frac{3\pi}{8} \cdot \tan^2 \frac{\pi}{8} + 8(\tan^2 \frac{3\pi}{8} + \tan^2 \frac{\pi}{8}) + 4 \end{aligned}$$

$$\begin{aligned}
 &= 16 + 8(4 \tan^2(\frac{\pi}{2} - \frac{\pi}{8} \cdot 2) + 2) + 4 \\
 &= 16 + 8(4 \tan^2 \frac{\pi}{4} + 2) + 4 \\
 &= 16 + 8(4 + 2) + 4 \\
 &= 68
 \end{aligned}$$

Therefore, the answer is (B) 68.

Solution 3 (Complex Numbers)

Let $\theta = \frac{\pi}{16}$. Then,

$$y = e^{8i\theta} = e^{\frac{\pi}{2}i} = (\cos \theta + i \sin \theta)^8 = 0 + i.$$

Expanding by using a binomial expansion,

$$\Re(y) = \cos^8 \theta - 28 \cos^6 \theta \sin^2 \theta + 70 \cos^4 \theta \sin^4 \theta - 28 \cos^2 \theta \sin^6 \theta + \sin^8 \theta = 0.$$

Divide by $\cos^8 \theta$ and notice we can set $\frac{\sin \theta}{\cos \theta} = x$ where $x = \tan(\theta)$. Then, define $f(x)$ so that

$$f(x) = 1 - 28x^2 + 70x^4 - 28x^6 + x^8.$$

Notice that we can have $(\cos \theta_k + i \sin \theta_k)^8 = 0 \pm i$ because we are only considering the real parts. We only have this when $k \equiv 1, 3 \pmod{4}$, meaning $k \equiv 1 \pmod{2}$. This means that we have $k = 1, 3, 5, 7, 9, 11, 13, 15$ as unique roots (we get them from $k\theta \in [0, \pi]$) and by using the fact that $\tan(\pi - \theta) = -\tan \theta$, we get

$$x \in \{\tan \theta, -\tan \theta, \tan(3\theta), -\tan(3\theta), \tan(5\theta), -\tan(5\theta), \tan(7\theta), -\tan(7\theta)\}$$

Since we have a monic polynomial, by the Fundamental Theorem of Algebra,

$$f(x) = (x - \tan \theta)(x + \tan \theta)(x - \tan(3\theta))(x + \tan(3\theta))(x - \tan(5\theta))(x + \tan(5\theta))(x - \tan(7\theta))(x + \tan(7\theta))$$

$$f(x) = (x^2 - \tan^2 \theta)(x^2 - \tan^2(3\theta))(x^2 - \tan^2(5\theta))(x^2 - \tan^2(7\theta))$$

Looking at the x^4 term in the expansion for $f(x)$ and using vietas gives us

$$\begin{aligned}
 &\tan^2 \theta \tan^2(3\theta) + \tan^2 \theta \tan^2(5\theta) + \tan^2 \theta \tan^2(7\theta) + \tan^2(3\theta) \tan^2(5\theta) \\
 &\quad + \tan^2(3\theta) \tan^2(7\theta) + \tan^2(5\theta) \tan^2(7\theta) = \frac{70}{1} = 70.
 \end{aligned}$$

Since $\tan(\frac{\pi}{2} - \theta) = \cot \theta$ and $\tan \theta \cot \theta = 1$

$$\tan^2 \theta \tan^2(7\theta) = \tan^2(3\theta) \tan^2(5\theta) = 1.$$

Therefore

$$\tan^2 \theta \tan^2(3\theta) + \tan^2 \theta \tan^2(5\theta) + \tan^2(3\theta) \tan^2(7\theta) + \tan^2(5\theta) \tan^2(7\theta) + 2 = 70.$$

$$\tan^2 \theta \tan^2(3\theta) + \tan^2 \theta \tan^2(5\theta) + \tan^2(3\theta) \tan^2(7\theta) + \tan^2(5\theta) \tan^2(7\theta) = \boxed{\text{(B) } 68}$$

Solution 5 (Transformation)

Set $x = \pi/16$, $7x = \pi/2 - x$, set $C7 = \cos^2(7x)$, $C5 = \cos^2(5x)$, $C3 = \cos^2(3x)$, $C = \cos^2(x)$, $S2 = \sin^2(2x)$, $S6 = \sin^2(6x)$, etc.

First, notice that

$$\begin{aligned}
 & \tan^2 x \cdot \tan^2 3x + \tan^2 3x \cdot \tan^2 5x + \tan^2 3x \cdot \tan^2 7x + \tan^2 5x \cdot \tan^2 7x \\
 &= (\tan^2 x + \tan^2 7x)(\tan^2 3x + \tan^2 5x) \\
 &= \left(\frac{1}{C} - 1 + \frac{1}{C7} - 1\right)\left(\frac{1}{C3} - 1 + \frac{1}{C5} - 1\right) \\
 &= \left(\frac{C + C7}{C \cdot C7} - 2\right)\left(\frac{C3 + C5}{C3 \cdot C5} - 2\right) \\
 &= \left(\frac{1}{C \cdot S} - 2\right)\left(\frac{1}{C3 \cdot S3} - 2\right) \\
 &= \left(\frac{4}{S2} - 2\right)\left(\frac{4}{S6} - 2\right) \\
 &= 4\left(\frac{2 - S2}{S2}\right)\left(\frac{2 - S6}{S6}\right) \\
 &= 4\left(\frac{4 - 2 \cdot S2 - S \cdot S6}{S2 \cdot S6} + 1\right) \\
 &= 4 + \frac{8}{S2 \cdot S6} \\
 &= 4 + \frac{32}{S4} \\
 &= 4 + 64 \\
 &= 68
 \end{aligned}$$

Solution 6 (Half angle formula twice)

So from the question we have:

$$\begin{aligned}
 & \tan^2 \frac{\pi}{16} \cdot \tan^2 \frac{3\pi}{16} + \tan^2 \frac{\pi}{16} \cdot \tan^2 \frac{5\pi}{16} + \tan^2 \frac{3\pi}{16} \cdot \tan^2 \frac{7\pi}{16} + \tan^2 \frac{5\pi}{16} \cdot \tan^2 \frac{7\pi}{16} \\
 &= (\tan^2 \frac{\pi}{16} + \tan^2 \frac{7\pi}{16})(\tan^2 \frac{3\pi}{16} + \tan^2 \frac{5\pi}{16})
 \end{aligned}$$

Using $\tan^2 \frac{\theta}{2} = \frac{1 - \cos \theta}{1 + \cos \theta}$

$$= \left(\frac{1 + \cos \frac{\pi}{8}}{1 - \cos \frac{\pi}{8}} + \frac{1 + \cos \frac{7\pi}{8}}{1 - \cos \frac{7\pi}{8}}\right)\left(\frac{1 + \cos \frac{3\pi}{8}}{1 - \cos \frac{3\pi}{8}} + \frac{1 + \cos \frac{5\pi}{8}}{1 - \cos \frac{5\pi}{8}}\right)$$

Using $\cos \theta = -\cos(\pi - \theta)$

$$\begin{aligned}
 &= \left(\frac{1 + \cos \frac{\pi}{8}}{1 - \cos \frac{\pi}{8}} + \frac{1 - \cos \frac{\pi}{8}}{1 + \cos \frac{\pi}{8}}\right)\left(\frac{1 + \cos \frac{3\pi}{8}}{1 - \cos \frac{3\pi}{8}} + \frac{1 - \cos \frac{3\pi}{8}}{1 + \cos \frac{3\pi}{8}}\right) \\
 &= \left(\frac{(1 + \cos \frac{\pi}{8})^2 + (1 - \cos \frac{\pi}{8})^2}{1 - \cos^2 \frac{\pi}{8}}\right)\left(\frac{(1 + \cos \frac{3\pi}{8})^2 + (1 - \cos \frac{3\pi}{8})^2}{1 - \cos^2 \frac{3\pi}{8}}\right) \\
 &= \left(\frac{2 + 2 \cos^2 \frac{\pi}{8}}{1 - \cos^2 \frac{\pi}{8}}\right)\left(\frac{2 + 2 \cos^2 \frac{3\pi}{8}}{1 - \cos^2 \frac{3\pi}{8}}\right)
 \end{aligned}$$

Using $\cos^2 \frac{\theta}{2} = \frac{1+\cos \theta}{2}$

$$\begin{aligned}
 &= \left(\frac{2+1+\cos \frac{\pi}{4}}{1-\frac{1+\cos \frac{\pi}{4}}{2}} \right) \left(\frac{2+1+\cos \frac{3\pi}{4}}{1-\frac{1+\cos \frac{3\pi}{4}}{2}} \right) \\
 &= \left(\frac{12+2\sqrt{2}}{2-\sqrt{2}} \right) \left(\frac{12-2\sqrt{2}}{2+\sqrt{2}} \right) \\
 &= \frac{136}{2} = \boxed{\text{B) } 68}
 \end{aligned}$$

Solution 7 (Find each individual tan)

The half angle formula for $\tan^2$ is $\tan^2 \frac{\theta}{2} = \frac{1-\cos \theta}{1+\cos \theta}$ and the half angle formula for cosine is $\cos \frac{\theta}{2} = \sqrt{\frac{1+\cos \theta}{2}}$. We can use this to find each tan:

$$\cos(\pi/8) = \sqrt{\frac{1+\cos(\pi/4)}{2}} = \frac{\sqrt{2+\sqrt{2}}}{2}$$

$$\tan^2(\pi/16) = \frac{1-\cos(\pi/8)}{1+\cos(\pi/8)} = \frac{2-\sqrt{2+\sqrt{2}}}{2+\sqrt{2+\sqrt{2}}}$$

$$\cos(3\pi/8) = \sqrt{\frac{1+\cos(3\pi/4)}{2}} = \frac{\sqrt{2-\sqrt{2}}}{2}$$

$$\tan^2(3\pi/16) = \frac{1-\cos(3\pi/8)}{1+\cos(3\pi/8)} = \frac{2-\sqrt{2-\sqrt{2}}}{2+\sqrt{2-\sqrt{2}}}$$

$$\cos(5\pi/8) = -\sqrt{\frac{1+\cos(5\pi/4)}{2}} = -\frac{\sqrt{2-\sqrt{2}}}{2}$$

$$\tan^2(5\pi/16) = \frac{1-\cos(5\pi/8)}{1+\cos(5\pi/8)} = \frac{2+\sqrt{2-\sqrt{2}}}{2-\sqrt{2-\sqrt{2}}}$$

$$\cos(7\pi/8) = -\sqrt{\frac{1+\cos(7\pi/4)}{2}} = -\frac{\sqrt{2+\sqrt{2}}}{2}$$

$$\tan^2(7\pi/16) = \frac{1-\cos(7\pi/8)}{1+\cos(7\pi/8)} = \frac{2+\sqrt{2+\sqrt{2}}}{2-\sqrt{2+\sqrt{2}}}$$

The problem's expression can be factored as

$$\left(\tan^2 \frac{\pi}{16} + \tan^2 \frac{7\pi}{16} \right) \left(\tan^2 \frac{3\pi}{16} + \tan^2 \frac{5\pi}{16} \right).$$

So the answer is

$$\begin{aligned}
 & \left(\frac{2 - \sqrt{2 + \sqrt{2}}}{2 + \sqrt{2 + \sqrt{2}}} + \frac{2 + \sqrt{2 + \sqrt{2}}}{2 - \sqrt{2 + \sqrt{2}}} \right) \cdot \left(\frac{2 - \sqrt{2 - \sqrt{2}}}{2 + \sqrt{2 - \sqrt{2}}} + \frac{2 + \sqrt{2 - \sqrt{2}}}{2 - \sqrt{2 - \sqrt{2}}} \right) = \\
 & \left(\frac{12 + 2\sqrt{2}}{2 - \sqrt{2}} \right) \cdot \left(\frac{12 - 2\sqrt{2}}{2 + \sqrt{2}} \right) = \\
 & (14 + 8\sqrt{2}) \cdot (14 - 8\sqrt{2}) = \\
 & 196 - 128 = \boxed{68}.
 \end{aligned}$$

Solution 8 (single formula)

$$\cot \alpha - \tan \alpha = 2 \cot 2\alpha \implies \cot^2 \alpha + \tan^2 \alpha = 4 \cot^2 2\alpha + 2.$$

We use $\alpha = \frac{\pi}{16}$ for $(\tan^2 \frac{\pi}{16} + \tan^2 \frac{7\pi}{16})(\tan^2 \frac{3\pi}{16} + \tan^2 \frac{5\pi}{16})$.

$$\begin{aligned}
 & (\tan^2 \alpha + \cot^2 \alpha)(\tan^2(\frac{\pi}{4} - \alpha) + \cot^2(\frac{\pi}{4} - \alpha)) = (4 \cot^2 2\alpha + 2)(4 \cot^2(\frac{\pi}{2} - 2\alpha) + 2) = \\
 & = 4 \cdot (4 + 2 \tan^2 2\alpha + 2 \cot^2 2\alpha + 1) = 20 + 8 \cdot (4 \cot^2 4\alpha + 2) = 68. \blacksquare
 \end{aligned}$$

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Solution 9 (Vietas)

As the above solutions noted, we can factor the expression into $(\tan^2 \frac{\pi}{16} + \tan^2 \frac{7\pi}{16})(\tan^2 \frac{3\pi}{16} + \tan^2 \frac{5\pi}{16})$.

Before we directly solve this problem, let's analyze the roots of $\tan(4 \tan^{-1} x) = 1$, or equivalently using tangent expansion formula, $\frac{1-6x^2+x^4}{4x-4x^3} = 1$, which implies $x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$. Now note that the roots of this equation are precisely $\tan \frac{\pi}{16}, \tan \frac{5\pi}{16}, \tan \frac{9\pi}{16}, \tan \frac{13\pi}{16}$, so the second symmetric sum of these four numbers is 6 by Vieta's. Thus, we have

$$\tan \frac{\pi}{16} \tan \frac{5\pi}{16} + \tan \frac{\pi}{16} \tan \frac{9\pi}{16} + \tan \frac{\pi}{16} \tan \frac{13\pi}{16} + \tan \frac{5\pi}{16} \tan \frac{9\pi}{16} + \tan \frac{5\pi}{16} \tan \frac{13\pi}{16} + \tan \frac{9\pi}{16} \tan \frac{13\pi}{16} = 6$$

Upon further inspection, $\tan \frac{\pi}{16} \tan \frac{9\pi}{16} + \tan \frac{5\pi}{16} \tan \frac{13\pi}{16} = -2$ using the fact that $\tan(x) * \tan(x + \pi/2) = -1$. Hence, we have

$$\begin{aligned}
 & \tan \frac{\pi}{16} \tan \frac{5\pi}{16} - 1 + \tan \frac{\pi}{16} \tan \frac{13\pi}{16} + \tan \frac{5\pi}{16} \tan \frac{9\pi}{16} - 1 + \tan \frac{9\pi}{16} \tan \frac{13\pi}{16} = 6 \\
 & \tan \frac{\pi}{16} \tan \frac{5\pi}{16} + \tan \frac{\pi}{16} \tan \frac{13\pi}{16} + \tan \frac{5\pi}{16} \tan \frac{9\pi}{16} + \tan \frac{9\pi}{16} \tan \frac{13\pi}{16} = 8 \\
 & (\tan \frac{\pi}{16} + \tan \frac{9\pi}{16})(\tan \frac{5\pi}{16} + \tan \frac{13\pi}{16}) = 8
 \end{aligned}$$

Now, we return to the problem statement, where we see a similar squared sum. We use this motivation to square our equation above to obtain

$$(\tan^2 \frac{\pi}{16} + \tan^2 \frac{9\pi}{16} - 2)(\tan^2 \frac{5\pi}{16} + \tan^2 \frac{13\pi}{16} - 2) = 64$$

$$(\tan^2 \frac{\pi}{16} + \tan^2 \frac{9\pi}{16})(\tan^2 \frac{5\pi}{16} + \tan^2 \frac{13\pi}{16}) - 2(\tan^2 \frac{\pi}{16} + \tan^2 \frac{5\pi}{16} + \tan^2 \frac{9\pi}{16} + \tan^2 \frac{13\pi}{16}) + 4 = 64$$

$$(\tan^2 \frac{\pi}{16} + \tan^2 \frac{9\pi}{16})(\tan^2 \frac{5\pi}{16} + \tan^2 \frac{13\pi}{16}) - 2(\tan^2 \frac{\pi}{16} + \tan^2 \frac{5\pi}{16} + \tan^2 \frac{9\pi}{16} + \tan^2 \frac{13\pi}{16}) = 60$$

Then, use the fact that $\tan^2 x = \tan^2 \pi/2 - x$ to get

$$(\tan^2 \frac{\pi}{16} + \tan^2 \frac{7\pi}{16})(\tan^2 \frac{3\pi}{16} + \tan^2 \frac{5\pi}{16}) - 2(\tan^2 \frac{\pi}{16} + \tan^2 \frac{5\pi}{16} + \tan^2 \frac{9\pi}{16} + \tan^2 \frac{13\pi}{16}) = 60$$

Hold on; the first term is exactly what we are solving for! It thus suffices to find $\tan^2 \frac{\pi}{16} + \tan^2 \frac{5\pi}{16} + \tan^2 \frac{9\pi}{16} + \tan^2 \frac{13\pi}{16}$. Fortunately, this is just $S_1^2 - 2S_2$ (Where S_n is the nth symmetric sum), with relation to roots of $x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$. By Vieta's, this is just $(-4)^2 - 2(6) = 4$.

Finally, we plug this value into our equation to obtain

$$(\tan^2 \frac{\pi}{16} + \tan^2 \frac{7\pi}{16})(\tan^2 \frac{3\pi}{16} + \tan^2 \frac{5\pi}{16}) - 2(4) = 60$$

$$(\tan^2 \frac{\pi}{16} + \tan^2 \frac{7\pi}{16})(\tan^2 \frac{3\pi}{16} + \tan^2 \frac{5\pi}{16}) = \boxed{68}$$

Alternate proof of the two tangent squares formula We want to simplify $\tan^2(x) + \tan^2(\frac{\pi}{2} - x)$. We make use of the fact that $\tan(\frac{\pi}{2} - x) = \cot(x)$. Then, the expression becomes $\tan^2(x) + \cot^2(x)$. Notice we can write: $(\tan x + \cot x)^2$ as $\tan^2(x) + \cot^2(x) + 2$ as tangent and cotangent are reciprocals of each other. Then, the sum of the tangent and cotangent can be simplified to $\frac{\sec^2 x}{\tan x}$. Using the fact that secant is the reciprocal of cosine and tangent is the ratio of sine and cosine, we can simplify that expression to $\frac{1}{\sin x \cos x}$. So, we have that: $\tan^2(x) + \cot^2(x) = \frac{1}{(\sin x \cos x)^2} - 2$ which can be simplified to: $2\left(\frac{2}{\sin^2(2x)} - 1\right)$ or $\frac{4}{\sin^2(2x)} - 2$ as stated in earlier solutions.

Solution 10 (Options)

For this question, there are five options: 28, 68, 70, 72, and 84. Since 28 is too small and 84 is too large, these two options can be eliminated. At this point, only three options remain. While one could make a guess here, it is better to analyze these remaining options further. When we divide each of these three options by 2, we get 34, 35, and 36 respectively. Option C (70) should be eliminated because, after division by 2, it is the only odd number among the three results. Last but not least, Option D (72) appears excessively frequently in AMC answer choices. Therefore, it is reasonable to conjecture that the Mathematical Association of America (MAA) will avoid using this option again for AMC, leading to the elimination of D. As a result, the only remaining option is B (68).

Curator note: Upon extensive investigation, we have been forced to conclude that the above solution is pure undistilled yap

24. (AMC 12 12A 2024) A *disphenoid* is a tetrahedron whose triangular faces are congruent to one another. What is the least total surface area of a disphenoid whose faces are scalene triangles with integer side lengths?
- (A) $\sqrt{3}$ (B) $3\sqrt{15}$ (C) 15 (D) $15\sqrt{7}$ (E) $24\sqrt{6}$

Answer: D

Solution 1 (Definition of disphenoid)

Notice that any scalene *acute* triangle can be the faces of a *disphenoid*. (See proof in Solution 2.)

As a result, we simply have to find the smallest area a scalene acute triangle with integer side lengths can take on. This occurs with a 4, 5, 6 triangle (notice that if you decrease the value of any of the sides the resulting triangle will either be isosceles, degenerate, or non-acute). For this triangle, the semiperimeter is $\frac{15}{2}$, so by Heron's Formula:

$$\begin{aligned} A &= \sqrt{\frac{15}{2} \cdot \frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2}} \\ &= \sqrt{\frac{15^2 \cdot 7}{16}} \\ &= \frac{15}{4}\sqrt{7} \end{aligned}$$

The surface area is simply four times the area of one of the triangles, or (D) $15\sqrt{7}$.

Solution 2 (Disphenoid in Box)

Let the side lengths of one face of the disphenoid be a, b, c . By the definition of a disphenoid with scalene faces, opposite sides must be the same length. Then the disphenoid can be constructed in a rectangular box with dimensions p, q, r such that a, b, c are the 3 different diagonal lengths of the faces of the box and no two sides are parallel (someone feel free to insert a diagram). Then we have the system

$$p^2 + q^2 = a^2$$

$$p^2 + r^2 = b^2$$

$$q^2 + r^2 = c^2$$

for positive integers a, b, c and positive p, q, r .

Solving for p, q, r , we have

$$p^2 = \frac{a^2 + b^2 - c^2}{2}$$

$$q^2 = \frac{a^2 - b^2 + c^2}{2}$$

$$r^2 = \frac{-a^2 + b^2 + c^2}{2}$$

(Notice that, by law of cosines (or by Pythagorean Inequality), these (parallelepiped box side lengths squared) p^2, q^2, r^2 each have the same sign as the cosine of the angle at a vertex of a triangular face of the disphenoid. Since they are squares and so non-negative, every angle is non-obtuse. Further, since they are squares of positive side lengths, every angle is acute. So a, b, c are the side lengths of an *acute* triangle.)

WLOG, let $a < b < c$. For $a < 4$, a^2 is less than or equal to the gap between squares greater than a^2 , so all such triangles are non-acute and fail.

The next smallest case works: $4^2 + 5^2 > 6^2$ so $a, b, c = 4, 5, 6$.

Using Heron's Formula, the minimum total surface area of the disphenoid is

$$\boxed{\text{(D)} \ 15\sqrt{7}}.$$

Bonus: by Heron's Formula, the area of an $x-1, x, x+1$ triangle is $x\sqrt{3(x-2)(x+2)}/4 = x\sqrt{3(x^2-4)}/4$.

(Acute triangle observations and bonus formula by oinava)

Solution 3 (Volume formula)

Recall the formula for a disphenoid is $72V^2 = (-a^2+b^2+c^2)(a^2-b^2+c^2)(a^2+b^2-c^2)$, where V is the volume and a, b, c are the sides of each face. Because $72V^2$ is a positive number, $(-a^2+b^2+c^2)(a^2-b^2+c^2)(a^2+b^2-c^2)$ must also be positive which is enough to rule out 2, 3, 4 and etc, leaving us with a 4-5-6 triangle for a surface area of $\boxed{\text{(D)} \ 15\sqrt{7}}$

-PaixiaoLover

Solution 4 (Finding the Smallest Scalene Triangle with Integer Sides)

Claim 1: The four triangles are acute

Proof of Claim 1:

Option A: Definition of disphenoid $\square$

Option B: Suppose coordinates $A(x, y, z), B(x, -y, -z), C(-x, y, -z)$ and $D(-x, -y, z)$, with $x, y, z > 0$. It follows the figure $ABCD$ is a disphenoid. Every face is congruent, so it suffices to show ABC is acute. Note $AB^2 = 4(y^2 + z^2), AC^2 = 4(x^2 + z^2), BC^2 = 4(x^2 + y^2)$. Suppose AB is the longest side WLOG, then $y^2 + z^2 \geq x^2 + z^2, y^2 + z^2 \geq x^2 + y^2$ implying $z \geq x$ and $y \geq x$. Now, $AB^2 = 4(y^2 + z^2) < 4(x^2 + y^2 + x^2 + z^2) = AC^2 + BC^2$ so $AB^2 < AC^2 + BC^2$. This is the converse of Pythagorean Inequality which directly implies ABC is acute.

Claim 2 : The smallest possible acute scalene triangle has side lengths 4,5,6.

Proof of Claim 2:

Suppose $a < b < c$ are the sides of a triangle. Then, since the triangle is acute, $c^2 < a^2 + b^2$, and the triangle inequality gives $c < a + b$. We prove no triangle with perimeter less than 15 works.

If $a = 1$ then $1 + b > c$ forces $c \leq b$, a contradiction. If $a = 2$ then $b \geq 3$ and the only such triangle is of lengths 2,3,4. But this triangle is obtuse, a contradiction. So $a \geq 3$, but the perimeter is at most 14. So $b + c \leq 11$, and because $b > 3, b \geq 4$ and the two triples are 3,4,5 and 3,4,6, one of which is right and the other is obtuse. Therefore, the minimum perimeter is 15, in which the smallest triangle has lengths 4,5,6.

2024 AMC 12A Q24:

All that is left is to find the surface area. The semi perimeter $s = \frac{15}{2}$, so the area through Heron's is $\frac{15\sqrt{7}}{4}$ and times the four sides the total area is $\boxed{15\sqrt{7}}$.

25. (AMC 12 12A 2024) A graph is *symmetric* about a line if the graph remains unchanged after reflection in that line. For how many quadruples of integers (a, b, c, d) , where $|a|, |b|, |c|, |d| \leq 5$ and c and d are not both 0, is the graph of

$$y = \frac{ax + b}{cx + d}$$

symmetric about the line $y = x$?

- (A) 1282 (B) 1292 (C) 1310 (D) 1320 (E) 1330

Answer: B

Solution 1 (Inverse Function)

Symmetric about the line $y = x$ implies that the inverse function $y^{-1} = y$. Then we split the question into several cases to find the final answer.

Case 1: $c = 0$

Then $y = \frac{a}{d}x + \frac{b}{d}$ and $y^{-1} = \frac{d}{a}x - \frac{b}{a}$. Giving us $\frac{a}{d} = \frac{d}{a}$ and $\frac{b}{d} = -\frac{b}{a}$

Therefore, we obtain 2 subcases: $b \neq 0, a + d = 0$ and $b = 0, a^2 = d^2$

Case 2: $c \neq 0$

Then $y^{-1} = \frac{b-dx}{cx-a} = \frac{(cx-a)(-\frac{d}{c})+b-\frac{ad}{c}}{cx-a} = -\frac{d}{c} + \frac{b-\frac{ad}{c}}{cx-a}$

And $y = \frac{(cx+d)(\frac{a}{c})+b-\frac{ad}{c}}{cx+d} = \frac{a}{c} + \frac{b-\frac{ad}{c}}{cx+d}$

So $\frac{a}{c} = -\frac{d}{c}$, or $a = -d$ ($c \neq 0$), and substitute that into $\frac{b-\frac{ad}{c}}{cx-a} = \frac{b-\frac{ad}{c}}{cx+d}$ gives us:

$bc - ad \neq 0$ (Otherwise $y = \frac{a}{c}$, $y^{-1} = -\frac{d}{c} = \frac{a}{c}$, and is not symmetric about $y = x$)

Therefore we get three cases:

Case 1.1: $c = 0, b \neq 0, d \neq 0, a + d = 0$

We have 10 choice of b , 10 choice of d and each choice of d has one corresponding choice of a . In total $10 \times 10 = 100$ ways.

Case 1.2: $c = 0, b = 0, d \neq 0, a^2 = d^2$

We have 10 choice for d ($d \neq 0$), each choice of d has 2 corresponding choice of a , thus $10 \times 2 = 20$ ways.

Case 2: $c \neq 0, bc - ad \neq 0, a = -d$ $a = 0$: $10 \times 10 = 100$ ways.

$a = \pm 1$: $(11 \times 10 - 2) \times 2 = 216$ ways.

$a = \pm 2$: $(11 \times 10 - 6) \times 2 = 208$ ways.

$a = \pm 3$: $(11 \times 10 - 2) \times 2 = 216$ ways.

$a = \pm 4$: $(11 \times 10 - 2) \times 2 = 216$ ways.

$a = \pm 5$: $(11 \times 10 - 2) \times 2 = 216$ ways.

In total $100 + 20 + 216 \times 4 = 1172$ ways.

So the answer is $100 + 20 + 1172 = \boxed{\text{(B)} 1292}$

Solution 2 (Rotation + Edge Cases)

First, observe that the only linear functions that are symmetric about $y = x$ are $y = x$ and $y = -x + k$, where k is some constant.

We perform a 45° counterclockwise rotation of the Cartesian plane. Let (x, y) be sent to (u, v) . Then u and v are the real and imaginary parts of $(x + yi)(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i)$ respectively, which gives

$$u = \frac{x - y}{\sqrt{2}}$$

$$v = \frac{x + y}{\sqrt{2}}$$

so

$$x = \frac{v+u}{\sqrt{2}}$$

$$y = \frac{v-u}{\sqrt{2}}$$

The rotated function is symmetric about the y -axis, so the equation holds after replacing all instances of u with $-u$ (this is just switching the values of x and y which is a reflection over $y = x$, but working in terms of (u, v) allows more cancellations in the following calculations).

Writing x and y in terms of u and v , we have

$$\frac{v-u}{\sqrt{2}} = \frac{a(v+u) + b\sqrt{2}}{c(v+u) + d\sqrt{2}}$$

$$\frac{v+u}{\sqrt{2}} = \frac{a(v-u) + b\sqrt{2}}{c(v-u) + d\sqrt{2}}$$

Multiplying both equations by $\sqrt{2}$ and subtracting the second equation from the first equation gives $d = -a$. Since a, b, c, d are integers between -5 and 5 , this gives $11^3 = 1331$ combinations. We need to subtract the edge cases that don't work, namely all undefined functions and linear functions except $y = x$ and $y = -x + k$. Consider the following cases:

Case 1: a, b, c, d are all nonzero. Then the function is linear when $ax + b$ is a multiple of $cx + d$, or $\frac{a}{b} = \frac{c}{-a}$.

If $a = \pm 1$, $(b, c) = (1, -1)$ or $(-1, 1)$; there are $2 * 2 = 4$ ways.

If $a = \pm 2$, there are 12 ways.

If $a = \pm 3$, there are 4 ways.

If $a = \pm 4$, there are 4 ways.

If $a = \pm 5$, there are 4 ways.

In total, this case has 28 combinations.

Case 2: $a = b = d = 0$ or $a = c = d = 0$

If $a = b = d = 0$ then c can take on 11 values, and if $a = c = d = 0$, then b can take on 11 values, but $a = b = c = d = 0$ is counted twice so this case has $11 + 11 - 1 = 21$ combinations.

Finally, we need to add the case where $y = x$, which occurs when $a = d$ and $b = c = 0$. a can be any integer from -5 to 5 except 0 , so this case has 10 combinations. Since $y = -x + k$ occurs when $a = -d$ and $c = 0$, this case has already been counted.

Thus, the answer is $1331 - 28 - 21 + 10 = \boxed{\text{(B) } 1292}$.

Solution 3 (Asymptotes)

There are two cases: when $c = 0$ and when $c \neq 0$.

Case 1: $c = 0$. If $c = 0$, then $y = \frac{ax+b}{d} = \frac{a}{d}x + \frac{b}{d}$. This is the equation of a line, and the only lines symmetric about $y = x$ are those perpendicular to $y = x$ (i.e.

those with slope -1) and $y = x$ itself. To have a slope of -1 , we need $a = -d \neq 0$, and b can be any of its 11 possibilities from -5 to 5 . There are $11 \cdot 10 = 110$ possibilities here. For the function to be $y = x$, we need $a = d \neq 0$ and $b = 0$. There are 10 possibilities here. Thus, our total for Case 1 is $110 + 10 = 120$ possibilities.

Case 2: $c \neq 0$. When $c \neq 0$, we will first consider the case in which the graph is a hyperbola. Clearly, for this hyperbola to be symmetric about $y = x$, the intersection of its horizontal and vertical asymptotes must be on $y = x$. The location of the horizontal asymptote is $y = \frac{a}{c}$, and the vertical asymptote occurs at $x = -\frac{d}{c}$. These asymptotes intersect on $y = x$ when $\frac{a}{c} = -\frac{d}{c}$, or, more simply, when $a = -d$.

If the asymptotes intersect on $y = x$, then the hyperbola must be symmetric about $y = x$. This is true because for any hyperbola with perpendicular asymptotes, we can rotate and translate the coordinate plane in a certain way such that that hyperbola has an equation of the form $x^2 - y^2 = a^2$. Then, the hyperbola's asymptotes would intersect at the origin, and it would be symmetric about the coordinate axes (because it makes a distinction neither between x and $-x$ nor y and $-y$). The coordinate axes are the bisectors of the angles formed by the asymptotes, and the hyperbola is symmetric about them. Thus, because the angles formed by our hyperbola's asymptotes are bisected by $y = x$, our hyperbola must be symmetric about $y = x$.

Thus, with the conditions that $a = -d$ and $c \neq 0$, there are $11 \cdot 11 \cdot 10 \cdot 1 = 1210$ possibilities for (a, b, c, d) . However, not all of these ordered quadruples produce hyperbolas. If $b = a = d = 0$ or $\frac{a}{c} = \frac{b}{d}$, then the quadruples produce horizontal lines with a hole when the denominator equals 0. As seen in Case 1, these lines, with slope 0, cannot be symmetric about $y = x$.

For the subcase where $b = a = d = 0$, there are 10 possibilities for $c \neq 0$, which gives us 10 wrongly counted quadruples.

For the subcase where $\frac{a}{c} = \frac{b}{d}$, we wrongly counted cases where $d = -a$. Here, $-a^2 = bc$ by cross-multiplication. The casework on the possible values of a below counts the number of triples (a, b, c) with $|a|, |b|, |c| \leq 5$ which satisfy this condition.

If $a = \pm 1$, $bc = -1$, which yields $2 \cdot 2 \cdot 1 = 4$ possibilities.

If $a = \pm 2$, $bc = -4$, which yields $2 \cdot 2 \cdot 3 = 12$ possibilities.

If $a = \pm 3$, $bc = -9$, which yields $2 \cdot 2 \cdot 1 = 4$ possibilities. (recall that $|b|, |c| \leq 5$)

If $a = \pm 4$, $bc = -16$, which yields $2 \cdot 2 \cdot 1 = 4$ possibilities.

If $a = \pm 5$, $bc = -25$, which yields $2 \cdot 2 \cdot 1 = 4$ possibilities.

Adding the above values together for this subcase yields $4 + 12 + 4 + 4 + 4 = 28$ wrongly counted quadruples.

Subtracting the wrongly counted quadruples from our count for Case 2 yields $1210 - 10 - 28 = 1172$.

Adding the possibilities for Case 1 and Case 2 yields our final answer of $120 + 1172 =$
(B) 1292 possible quadruples.

Solution 4

Note that the condition is equivalent to having $f(f(x)) = x$.

So we have: $\frac{a(\frac{ax+b}{cx+d})+b}{c(\frac{ax+b}{cx+d})+d} = x \Rightarrow x^2(ac+cd) + x(d^2-a^2) - (ab+bd) = 0$

Thus we require:

$ac + cd = 0 \rightarrow c = 0$ or $a = -d$ $d^2 - a^2 = 0 \rightarrow a = d$ or $a = -d$ $ab + bd = 0 \rightarrow b = 0$ or $a = -d$

Note that if $a = -d$ then all 3 cases work and give $11^3 = 1331$ solutions.

If instead $c = 0$ and $a \neq -d$ then we require $a = d$ and $b = 0$ which then give 10 solutions.

Now, we must remove all extraneous cases. This is when $x(ac + cd) + bc + d^2 = 0$ (note this includes the case where $c = d = 0$).

So this is equivalent to having both $ac + cd = 0$ and $bc + d^2 = 0$.

If $c = d = 0$ we have 11 solutions.

If $c = 0$ and $d \neq 0$ then we have 0 solutions.

If $c \neq 0$ and $d = 0$ then we require $a = 0$ and $b = 0$ so we have 10 solutions.

And if $c \neq 0$ and $d \neq 0$ we have $a = -d$ and $d^2 = -bc$, see that if $|d| = 1, 3, 4, 5$ we have 2 solutions and if $|d| = 2$ we have 6 solutions, so a total of $2(2+6+2+2+2) = 28$ solutions.

Thus, the final answer is $1331 + 10 - 11 - 10 - 28 = \boxed{\text{(B)} 1292}$

Solution 5

Proceed the same way as the other solutions keeping in mind that $y = y^{-1}$. Then, I got: $\frac{(b-dx)}{(cx-a)} = \frac{(ax+b)}{(cx+d)}$. Cross multiplying and matching coefficients, we get: $-cd(x^2) = ac(x^2)$; $bd = -ab$; and $-x(d^2) = -x(a^2)$. Then, I broke it up into 4 cases:

Case 1: $b \neq 0, c = 0, a = -d$.

Case 2: $b = 0, c = 0, a^2 = d^2$.

Case 3: $b \neq 0, c \neq 0, a = -d$.

Case 4: $b = 0, c \neq 0, a = -d$.

I first accounted the total number of cases for everything to work WITHOUT any restrictions. So for Case 1, there would be 10 possibilities for b and 11 for the part where $a = -d$. So a total of 110 cases. Similarly, in the same fashion, Case 2 would yield 21 cases. Case 3 would yield 1100 cases and Case 4 would bring a total of 110 cases. So, our total possibilities right now is $110 + 21 + 1100 + 110 = 1341$. But wait! Notice a lot of these cases would bring the denominator of y to be 0 namely $cx + d$ to be 0. We don't want this! So we have to subtract out all the cases that bring $cx + d$ to be 0. Notice for Case 1, $c = 0$ so if $d = 0$, we have a bad case. We shall keep track of all the "bad" cases. For Case 1, any quadruple that includes $d = 0$ is "bad". So no restrictions upon b since a depends upon d . Therefore, we have 10 bad cases so far remember, $b \neq 0$ for Case 1.

We now proceed to Case 2. Notice we only have 1 "bad" case namely $a = 0$. This is because a and d depend on each other again.

Now Case 3. This will have quite a bit of "bad" cases as $c \neq 0$. Firstly, clearly $a = 0$ would again have some overcounting going on so we would have to subtract out 10 again as the value of b doesn't matter. Then, we start off with $c = -5$. Obviously, then $d = 5$ would yield a bad case IF $x = 1$. Now we see what happens is $x = 2$. Then, the denominator would be $2c + d$. For this to be 0, $2c = -d$. So all even values of d would be considered "bad". Thus, we would have $d = -2, -4, 2, 4$ to be "bad" cases which yields a subtotal of 4 cases. Now, the same thing would happen if $x = 3$ except all multiples of 3 would be considered "bad". So $d = -3, 3$ would

be "bad" or a subtotal of 2 cases. Then, we see the pattern. Multiples of 4 could also be eliminated for $x = 4$ which would yield $d = -4, 4$ but these overcount the multiples of 2 so no need to worry about them. Lastly, multiples of 5 wouldn't work if $x = 5$ so $d = -5, 5$ is "bad" or 2 cases. Then, $d = -1, 1$ could also be eliminated. Then, at last $d = 0$ would also not work. Finally, a total of $10 + 4 + 2 + 2 + 1$ cases would be "bad" which yields a subtotal of 19 cases. As you would see, Case 4 has the exact same restrictions on c but not b . So we can predict it would yield the same number of "bad" cases as Case 3. Finally, we could add up all of the "bad" cases we have: $10 + 1 + 19 + 19 = 49$ "bad" cases. What we did here is also known as counting up all the "asymptotes".

Our answer should be $1341 - 49 = 1292$ or **(B) 1292**.

Cases Case 1: $d = -a \wedge c \neq 0$, $10 \times 11 \times 11 = 1210$ possibilities

Excluded from Case 1: $d = -a \wedge c \neq 0 \wedge d^2 = -bc$, $10 + 4 + 12 + 4 + 4 + 4 = 38$ possibilities

Case 2: $d = -a \wedge c = 0 \wedge d \neq 0$, $10 \times 11 = 110$ possibilities

Case 3: $d = a \wedge c = 0 \wedge d \neq 0 \wedge b = 0$, $10 = 10$ possibilities

Answer: $1210 - 38 + 110 + 10 = 1292$

§13 AMC 12 12B 2025

- (AMC 12 12B 2025) The instructions on a 350-gram bag of coffee beans say that proper brewing of a large mug of pour-over coffee requires 20 grams of coffee beans. What is the greatest number of properly brewed large mugs of coffee that can be made from the coffee beans in that bag?

(A) 16 (B) 17 (C) 18 (D) 19 (E) 20

Answer: B

Solution 1

$\frac{350}{20}$ is 17.5. We round down to find the maximum he can make (he cannot brew half of a cup), so the answer is **(B) 17**.

Solution 2

We can use 340-grams if we use 17 mugs. This is because $20 \cdot 17 = 340$. We cannot use 18 because that would mean using 360-grams. The answer is **(B) 17**.

Note that you can also find this by finding $\lfloor \frac{350}{20} \rfloor = 17$.

Solution 3

Dividing 350 by 20, we get 17 with a remainder of 10. However, 10 grams cannot make a properly brewed large mug of coffee, as stated in the problem. We can only count fully brewed mugs of coffee, hence the answer is **(B) 17**.

Solution 4 - Cheese (Sort of)

We multiply 20 by each of the answer choices to see if we can find which one is closest and under 350. First, we try solution A, which is 16. $16 \cdot 20 = 320$, which is 30 off from 350, so A cannot be the answer, because we can add another 20. Moving on to answer choice B, $17 \cdot 20 = 340$. We notice that $350 - 340 = 10$, and so we cannot add another 20 to get closer but less than 350. (The next number, 18, would result in $20 \cdot 18 = 360$, which is more than 350.) So, the answer is **(B) 17**.

(note - the actual cheese solution would likely include every single answer choice, but I use a bit of intellect in this solution)

2. (*AMC 12 12B 2025*) Jerry wrote down the ones digit of each of the first 2025 positive squares: 1, 4, 9, 6, 5, 6, $\dots$. What is the sum of all the numbers Jerry wrote down?

(A) 9025 (B) 9070 (C) 9090 (D) 9115 (E) 9160

Answer: D

Solution 1

By a modulo 10 argument, the ones digits repeat with period 10 in the following order:

$$1, 4, 9, 6, 5, 6, 9, 4, 1, 0$$

The sum of the numbers can be verified to be 45.

There are 202 periods that occur from 1 to 2025, and there are five extra numbers, those being 1, 4, 9, 6, 5, corresponding to 2021, 2022, 2023, 2024, 2025. The sum of these numbers is 25.

Hence, the total is

$$202 \cdot 45 + 25 = 9090 + 25 = \boxed{\text{(D)} 9115}$$

Solution 2 (Slightly more rigorous than Solution 1, but same idea)

Let $x \equiv n \pmod{10}$. We have $x^2 \equiv n^2 \pmod{10}$. So, the numbers must repeat in patterns of 10. Listing them out gives:

$$1, 4, 9, 6, 5, 6, 9, 4, 1, 0.$$

These numbers sum to 45, so the sum of the first 2020 numbers is $45 \cdot 202 = 9090$. Since we need 5 more numbers, we add $1 + 4 + 9 + 6 + 5 = 25$ to our answer to get

(D) 9115.

Solution 3 (Similar to the other solutions but without the use of mods)

The ones digit of a square number is determined by the ones digit of the base. We can find the pattern by looking at the ones digits of the first ten positive squares:

$$1^2 = 1$$

$$2^2 = 4$$

$$3^2 = 9$$

$$4^2 = 16 \quad (\text{ones digit } 6)$$

$$5^2 = 25 \quad (\text{ones digit } 5)$$

$$6^2 = 36 \quad (\text{ones digit } 6)$$

$$7^2 = 49 \quad (\text{ones digit } 9)$$

$$8^2 = 64 \quad (\text{ones digit } 4)$$

$$9^2 = 81 \quad (\text{ones digit } 1)$$

$$10^2 = 100 \quad (\text{ones digit } 0)$$

The sequence of ones digits is:

$$1, 4, 9, 6, 5, 6, 9, 4, 1, 0$$

This pattern of ten digits repeats for every subsequent group of ten squares (e.g., 11^2 has the same ones digit as 1^2 , 12^2 as 2^2 , etc.).

$$1 + 4 + 9 + 6 + 5 + 6 + 9 + 4 + 1 + 0 = 45$$

We now divide by 10 as the pattern repeats every 10 terms: $2025/10$ results in 202 full cycles with a remainder of 5. The sum of the 202 full cycles is:

$$202 * 45 = 9090$$

However, we must add back 5 digits, as we had a remainder of 5 when we divided 2025 by 10. Therefore, the sum of the remaining 5 digits is:

$$1 + 4 + 9 + 6 + 5 = 25$$

Thus, the total sum is:

$$9090 + 25 = 9115$$

Therefore, the answer is (D) 9115.

Solution 4: Modular Arithmetic; Ultra-Rigorous, Not Recommended

We want the sum of the ones digits of $1^2, 2^2, 3^2, \dots, 2025^2$. The normal way is to notice the pattern repeats every 10, but instead we will approach this with more modular arithmetic than any sane person would tolerate.

Let $f(n) = n^2 \pmod{10}$, which extracts the ones digit of n^2 . We want

$$S = \sum_{n=1}^{2025} f(n).$$

First observe the basic periodicity result: Since $n^2 \pmod{10}$ depends only on $n \pmod{10}$, we have

$$f(n+10) \equiv f(n) \pmod{10}.$$

We will pretend this is deep.

Now compute all residues $n \pmod{10}$ and their corresponding $n^2 \pmod{10}$:

$$\begin{aligned} 0^2 &\equiv 0 \\ 1^2 &\equiv 1 \\ 2^2 &\equiv 4 \\ 3^2 &\equiv 9 \\ 4^2 &\equiv 6 \\ 5^2 &\equiv 5 \\ 6^2 &\equiv 6 \\ 7^2 &\equiv 9 \\ 8^2 &\equiv 4 \\ 9^2 &\equiv 1. \end{aligned}$$

Instead of immediately summing these, we treat the 10-term sequence as a modular function over the ring $\mathbb{Z}/10\mathbb{Z}$. Define the sequence $a_k = (k^2 \pmod{10})$ for $k \in \mathbb{Z}/10\mathbb{Z}$. We want the sum

$$T = \sum_{k=0}^9 a_k = 0 + 1 + 4 + 9 + 6 + 5 + 6 + 9 + 4 + 1 = 45.$$

Next, determine how many full cycles of length 10 lie in the first 2025 integers. Compute

$$2025 \mod 10 \equiv 5.$$

Then

$$2025 = 10 \cdot 202 + 5.$$

So we have 202 complete cycles plus 5 extra terms.

Sum contributed by the full cycles:

$$202 \cdot 45.$$

We now compute this entirely modulo 10 even though we need the actual sum. First,

$$202 \equiv 2 \pmod{10}, \quad 45 \equiv 5 \pmod{10},$$

so

$$202 \cdot 45 \equiv 2 \cdot 5 = 10 \equiv 0 \pmod{10}.$$

This tells us nothing useful, so abandon that and compute normally:

$$202 \cdot 45 = (200 \cdot 45) + (2 \cdot 45) = 9000 + 90 = 9090.$$

Now handle the leftover 5 terms: $n = 1, 2, 3, 4, 5$. Compute $n^2 \mod 10$ again even though we already have it:

$$1^2 \equiv 1, \quad 2^2 \equiv 4, \quad 3^2 \equiv 9, \quad 4^2 \equiv 6, \quad 5^2 \equiv 5.$$

Sum:

$$1 + 4 + 9 + 6 + 5 = 25.$$

The full sum is

$$S = 9090 + 25 = 9115.$$

To unnecessarily verify using modular decomposition, write

$$S \equiv \sum_{n=1}^{2025} n^2 \pmod{10}.$$

We can compute the entire sum of squares formula modulo 10 and compare. Using

$$\sum_{n=1}^N n^2 = \frac{N(N+1)(2N+1)}{6},$$

take $N = 2025$. Reduce everything modulo 10 first:

$$2025 \equiv 5, \quad 2026 \equiv 6, \quad 4051 \equiv 1 \pmod{10}.$$

Now compute

$$5 \cdot 6 \cdot 1 = 30.$$

Thus

$$\sum_{n=1}^{2025} n^2 \equiv 30 \cdot 6^{-1} \pmod{10}.$$

Compute the inverse of 6 modulo 10. Solve $6x \equiv 1 \pmod{10}$. No such x exists because $\gcd(6, 10) = 2 \neq 1$. Therefore this modular-verification method is unusable, so we immediately stop and simply trust the earlier cycle-counting method.

Thus the sum of the ones digits of the first 2025 squares is

$$9115.$$

3. (AMC 12 12B 2025) What is the value of $i(i-1)(i-2)(i-3)$, where $i = \sqrt{-1}$?
 (A) $6 - 5i$ (B) $-10i$ (C) $10i$ (D) -10 (E) 10

Answer: D

Solution

We find that $i(i-3) = -1 - 3i$ and $(i-1)(i-2) = -1 - 3i + 2 = 1 - 3i$, so

$$i(i-1)(i-2)(i-3) = i(i-3) \cdot (i-1)(i-2) \tag{33}$$

$$= (-1 - 3i)(1 - 3i) \tag{34}$$

$$= -(1 + 3i)(1 - 3i) \tag{35}$$

$$= -(1^2 - (3i)^2) \tag{36}$$

$$= -(1 + 9) \tag{37}$$

$$= \boxed{\text{(D)} - 10} \tag{38}$$

4. (AMC 12 12B 2025) The value of the two-digit number $\underline{a} \underline{b}$ in base seven equals the value of the two-digit number $\underline{b} \underline{a}$ in base nine. What is $a + b$?
 (A) 7 (B) 9 (C) 10 (D) 11 (E) 14

Answer: A

Solution 1

By definition of bases, $\underline{a} \underline{b}$ base seven is equal to $7a + b$, and $\underline{b} \underline{a}$ base nine is equal to $9b + a$. Therefore, we must have $7a + b = 9b + a$, or $6a = 8b$, or $3a = 4b$. This means e.g. $3a$ is a multiple of 4 (as a is an integer), and since 3 and 4 are coprime, it follows simply that a is a multiple of 4, i.e. $a = 4k$ for some integer k . Hence $4b = 3a = 3 \cdot 4k \iff b = 3k$, i.e. the integer solutions are parametrised by

$(a, b) = (4k, 3k)$, but of course in base 7 we must have $a, b < 7$, yielding $a = 4$ and $b = 3$ as the only solution. Thus their sum is $a + b = 4 + 3 = \boxed{\text{(A)} 7}$.

Remark The first equation comes from the following idea. In base 10, a two-digit number can be represented as 10 times the tens digit plus the units digit, or $10^1 \cdot a + 10^0 \cdot b$. If we insert the base numbers into this expression for $\underline{a} \underline{b}$ and $\underline{b} \underline{a}$, we get $7^1 \cdot a + b = 9^1 \cdot b + a$. The rest of the solution is above.

Note that this method applies to any number of digits as well as decimals. Olympiad-Master

Solution 2 (Answer Choices)

Since a and b are both digits base 7, they must be from 0 to 6. Furthermore, $\underline{a}$ must be strictly greater than b because two-digit representations in base 7 exceed the values of those in base 9. Starting from testing (A), we see that the combination $(a, b) = (4, 3)$ satisfies the condition. The answer is $\boxed{\text{(A)} 7}$.

Solution 3 (Not too bad- Simple Modular Arithmetic Bash)

We interpret the two-digit numbers literally:

In base 7:

$$\overline{a} \overline{b} = 7a + b.$$

In base 9:

$$\overline{b} \overline{a} = 9b + a.$$

The condition is

$$7a + b = 9b + a.$$

We want $a + b$, but we will milk modular arithmetic as much as possible.

Start with the equation:

$$7a + b = 9b + a.$$

Rearrange:

$$7a - a = 9b - b,$$

$$6a = 8b,$$

$$3a = 4b.$$

This already says $\frac{a}{b} = \frac{4}{3}$. But instead of finishing, we now do highly unnecessary modular analysis.

First, reduce the equation modulo 3:

$$3a - 4b \equiv 0 \pmod{3}.$$

Compute residues:

$$3a \equiv 0 \pmod{3} \text{ always.}$$

$$4b \equiv b \pmod{3}.$$

So:

$$0 - b \equiv 0 \pmod{3} \Rightarrow b \equiv 0 \pmod{3}.$$

Thus $b \in \{0, 3, 6\}$ (since digits in base 7 must be $0 \leq b \leq 6$).

Now reduce the same equation modulo 4:

Original condition $3a = 4b$. Modulo 4:

$$3a \equiv 0 \pmod{4}.$$

Since $\gcd(3, 4) = 1$, the multiplicative inverse of 3 modulo 4 is 3, because

$$3 \cdot 3 = 9 \equiv 1 \pmod{4}.$$

Multiply both sides of

$$3a \equiv 0 \pmod{4}$$

by 3:

$$a \equiv 0 \pmod{4}.$$

Thus $a \in \{0, 4\}$, because digits in base 7 satisfy $0 \leq a \leq 6$.

We now have:

$$a \in \{0, 4\}, b \in \{0, 3, 6\}.$$

Next impose the original equation $3a = 4b$ exactly, not just modularly.

Test combinations:

$a = 0$:

$$3a = 0, 4b = 4b.$$

For equality we need $b = 0$. That gives $(a, b) = (0, 0)$, but this represents the “two-digit” numbers 00_7 and 00_9 . Technically valid but trivial; we check the problem expectation: since it asks “What is $a + b$?”, zero is possible but let’s check the other option.

$a = 4$: Solve

$$3 \cdot 4 = 4b,$$

$$12 = 4b,$$

$$b = 3.$$

Check that digits are valid:

$a = 4$ is allowed in base 7.

$b = 3$ is allowed in base 7 and base 9.

Now confirm the original equation explicitly using base values:

Left side (base 7 number $\overline{ab}$):

$$7a + b = 7 \cdot 4 + 3 = 28 + 3 = 31.$$

Right side (base 9 number $\overline{ba}$):

$$9b + a = 9 \cdot 3 + 4 = 27 + 4 = 31.$$

Equality holds.

Thus the nontrivial valid solution is

$$(a, b) = (4, 3).$$

Compute:

$$a + b = 4 + 3 = 7.$$

Final answer:7.

Solution 4 (Fast)

Write both integers as linear expressions and equate them

$$7a + b = 9b + a$$

To find solutions to this, we can put everything on one side so it equals 0

$$6a - 8b = 0$$

Now, assuming a coordinate form (a, b) you can quite easily see that $(8, 6)$ is a solution. However, this violates the fact that a and b are less than 7. You can see that just taking half of them gets you $(4, 3)$, and that no other solution would work without exceeding, so, the answer is

(A) 7.

5. (AMC 12 12B 2025) Positive integers x and y satisfy the equation $57x + 22y = 400$. What is the least possible value of $x + y$?

(A) 10 (B) 11 (C) 13 (D) 14 (E) 15

Answer: E

Solution 1

Let $x + y = a$. Then we have the equation

$$35x + 22a = 400.$$

Because the other two terms are divisible by 5, $22a$ must be divisible by 5 too. Specifically, a is divisible by 5. Let $a = 5b$ and substitute:

$$35x + 110b = 400$$

$$7x + 22b = 80.$$

After some analysis, we find that if $b = 3$, $x = 2$. In fact, this is the only solution with positive integer solutions. Therefore, $a = x + y = 5 \cdot 3 =$ **(E) 15**.

Solution 1.5 (EXTREMELY EASY)

$22y$ will always be even. $57x$ must be even for the sum to be an even 400, so x is even.

Moreover, $x \leq 7$ because $\left\lfloor \frac{400}{57} \right\rfloor = 7$.

So the possible values of x are 2, 4, and 6.

$$400 - 57 \cdot 2 = 286 = 22 \cdot 13 \text{ works.}$$

$$400 - 57 \cdot 4 = 172 \text{ is not a multiple of } 22.$$

$$400 - 57 \cdot 6 = 58 \text{ is not a multiple of } 22.$$

Thus, $x = 2$ and $y = 13$. $x + y =$ **(E) 15**

- unhappyfarmer

Solution 2 (Extended Euclid Algorithm)

We first solve for integers x and y such that $57x + 22y = 400$. Use the Extended Euclid Algorithm to find that

$$x_1 = -5, y_1 = 13.$$

We multiply both x_1 and y_1 by 400. Since we require both x and y to be positive integers, let

$$x_k = -5 \cdot 400 + 22k$$

$$y_k = 13 \cdot 400 - 57k.$$

We can check this method of writing x_k and y_k is valid by substitution into the original equation. Finally, we add x_k and y_k , to find

$$-5 \cdot 400 + 13 \cdot 400 - 35k.$$

We want to minimize this value, and after evaluation we find $3200 - 35k$. Minimizing this value over the positive integers, we find $k = 91$, which yields **(E) 15**.

Solution 3 (Mod 11)

Taking mod 11, we see that $2x \equiv 4 \pmod{11} \iff x \equiv 2 \pmod{11}$. This forces $x = 2$ as bigger numbers make the sum pass 400, so that $y = \frac{400 - 2 \cdot 57}{22} = \frac{200 - 57}{11} = \frac{143}{11} = 13$. So the answer is $x + y = 2 + 13 = \mathbf{(E) 15}$.

Solution 4 (Brute Force)

List all the factors of 57 and 22 that are less than 400

Iterate through and find a solution to find $x = 2$ and $y = 13$

6. (AMC 12 12B 2025) Emmy says to Max, "I ordered 36 math club sweatshirts today." Max asks, "How much did each shirt cost?" Emmy responds, "I'll give you a hint. The total cost was \$A B B .B A, where A and B are digits and $A \neq 0$." After a pause, Max says, "That was a good price." What is $A + B$?

(A) 7 (B) 8 (C) 11 (D) 14 (E) 15

Answer: C

Solution 1

The problem is essentially telling us that A B B .B A is divisible by 36, so it is divisible by 9. Then, by the divisibility by 9 condition,

$$A + B + B + B + A = 2A + 3B \equiv 0 \pmod{9}$$

This means that A must be divisible by 3, or otherwise $2A + 3B$ would not be divisible by 3 and thus 9. Since $A \neq 0$, we must have one of $A = 3, 6, 9$.

But A must be even or else the entire number would not be even and thus would not be divisible by 4. Hence $A = 6$. Then, $2 \cdot 6 + 3B \equiv 0 \pmod{9}$, so $4 + B \equiv 0 \pmod{3}$, and thus $B \equiv 2 \pmod{3}$. This yields $B = 2, 5, 8$, of which $B = 5$ is the only number that allows A B B .B A to be divisible by 4. Thus the answer is $6 + 5 = \mathbf{(C) 11}$.

Solution 2 (Trial and Error)

A number is divisible by 36 if it is divisible by 4, and 9. The divisibility rule for 4 requires the last 2 digits to also be a multiple of 4. The divisibility rule for 9 requires the sum of all digits to be divisible by 9. Using the first condition $A + B$

is a multiple of 4. We can list out all possibilities for this, since both A and B are digits. The possibilities are $(1, 2), (1, 6), \dots, (9, 6)$ for A , and B respectively. There are a total of 2 digits that are A 's and 3 that are B 's. So if the digits sum to 9, then $2A + 3B$ is a multiple of 9. We can try plugging in numbers from our list earlier into the equation $2A + 3B$ to find what values set the expression equal to a multiple of 9. By trying numbers $A = 6$, and $B = 5$. So then $A + B$ yields

$$6 + 5 = \boxed{\text{(C)} 11}.$$

7. (AMC 12 12B 2025) What is the value of

$$\sum_{n=2}^{255} \frac{\log_2(1 + \frac{1}{n})}{(\log_2 n)(\log_2(n+1))}?$$

- (A) $\frac{3}{4}$ (B) $1 - \frac{1}{\log_2 255}$ (C) $\frac{7}{8}$ (D) $\frac{15}{16}$ (E) 1

Answer: C

Solution

By logarithm rules, write

$$\sum_{n=2}^{255} \frac{\log_2\left(1 + \frac{1}{n}\right)}{(\log_2 n)(\log_2(n+1))} = \sum_{n=2}^{255} \frac{\log_2\left(\frac{n+1}{n}\right)}{(\log_2 n)(\log_2(n+1))} \quad (39)$$

$$= \sum_{n=2}^{255} \frac{\log_2(n+1) - \log_2 n}{(\log_2 n)(\log_2(n+1))} \quad (40)$$

$$= \sum_{n=2}^{255} \left(\frac{1}{\log_2 n} - \frac{1}{\log_2(n+1)} \right) \quad (41)$$

$$(42)$$

But this telescopes to $\frac{1}{\log_2 2} - \frac{1}{\log_2 256} = \frac{1}{1} - \frac{1}{8} = \boxed{\text{(C)} \frac{7}{8}}.$

8. (AMC 12 12B 2025) There are integers a and b such that the polynomial $x^3 - 5x^2 + ax + b$ has $4 + \sqrt{5}$ as a root. What is $a + b$?

- (A) 13 (B) 17 (C) 20 (D) 30 (E) 68

Answer: C

Solution 1

Because every coefficient of our polynomial $f(x)$ is an integer, the conjugate of $4 + \sqrt{5}$, $4 - \sqrt{5}$ is also a root.

Let the third and final root be r_3 . Thanks to Vieta's Formulas, we know that the sum of the roots is

$$(r_3) + (4 + \sqrt{5}) + (4 - \sqrt{5}) = 5$$

$$r_3 = -3.$$

We now calculate $f(1) = (1 + 3)(1 - 4 - \sqrt{5})(1 - 4 + \sqrt{5})$.

$$(1 - 4 - \sqrt{5})(1 - 4 + \sqrt{5}) = (-3 - \sqrt{5})(-3 + \sqrt{5}) = 9 - 5 = 4.$$

So $f(1) = 4 \cdot 4 = 16$.

Hence $f(1) = 16 = 1 - 5 + a + b$. Therefore, $a + b = \boxed{20}$.

Solution 2

As in Solution 1, we have that $f(x)$ contains roots $4 + \sqrt{5}$ and its conjugate $4 - \sqrt{5}$, and that $r_3 = -3$, where r_3 is the last root. Hence, we can create a factorization for $f(x)$ and compare coefficients as follows:

$(x + 3)(x - (4 + \sqrt{5}))(x - (4 - \sqrt{5})) = (x + 3)(x^2 - 8x + 11) = x^3 - 8x^2 + 11x + 3x^2 - 24x + 33 = x^3 - 5x^2 - 13x + 33$, and hence by comparing the coefficients we have that $a + b = -13 + 33 = \boxed{20 \text{ (C)}}$.

Solution 3 (Bash)

Since $4 + \sqrt{5}$ is a root of polynomial, by definition

$$(4 + \sqrt{5})^3 - 5(4 + \sqrt{5})^2 + a(4 + \sqrt{5}) + b = 0 = 0 + 0\sqrt{5}$$

Since $a, b \in \mathbb{Z}$, we can expand and compare coefficients,

$$\begin{cases} 64 + 3 \cdot 4 \cdot 5 - 5(4^2 + 5) + 4a + b = 0 \\ 3 \cdot 16 + 5 - 5 \cdot 2 \cdot 4 + a = 0 \end{cases}$$

Solving, $a = -13$, $b = -4a - 19 = 33$, so $a + b = \boxed{\text{(C) } 20}$.

Solution 4

Just find the terms that have a $\sqrt{5}$ in them to find a , then only look at the integer terms to find b .

(I know this way works but can smn else write the actual math to get the answer)

Solution 5

As explained in earlier solutions the given function $x^3 - 5x^2 + ax + b$ has $4 + \sqrt{5}$ has roots $4 + \sqrt{5}$, and $4 - \sqrt{5}$. Therefore it should be divisible by $(4 + \sqrt{5})(4 - \sqrt{5})$ with no remainder.

$$\begin{array}{r} (4 + \sqrt{5})(4 - \sqrt{5}) = x^2 - 8x + 11 \\ x^2 - 8x + 11 \overline{) x^3 - 5x^2 + ax + b} \\ \underline{-(x^3 - 8x^2 + 11x)} \\ 3x^2 + (a - 11)x + b \\ \underline{-(3x^2 - 24x + 33)} \\ (a + 13)x + (b - 33) \end{array}$$

$$(a + 13)x + (b - 33) = 0$$

So in order for $(a + 13)x + (b - 33)$ to be 0 for all values of x $(a + 13)(b - 33)$ must each equal 0. $(a = -13), (b = 33)$. $a + b = \boxed{\text{(C) } 20}$.

9. (AMC 12 12B 2025) What is the tens digit of 6^{6^6} ?

- (A) 1 (B) 3 (C) 5 (D) 7 (E) 9

Answer: C

Solution 1

We wish to find $6^{6^6} \pmod{100}$ and find the tens digit. By the Chinese Remainder Theorem, it suffices to find $6^{6^6} \pmod{4}$ and $6^{6^6} \pmod{25}$ and find the solution to the system. We know that $6^{6^6} \equiv 0 \pmod{4}$, so we just need to find the remainder modulo 25.

By Euler's Totient Theorem, $6^{6^6} \equiv 6^r \pmod{25}$, where $r \equiv 6^6 \pmod{\varphi(25) = 20}$. Hence we need to find $6^6 \pmod{20}$.

Now we again use the Chinese Remainder Theorem. We know that $6^6 \equiv 0 \pmod{4}$ and $6^6 \equiv 1^6 \equiv 1 \pmod{5}$. Testing cases yields $6^6 \equiv 16 \pmod{20}$.

Therefore, we know that $6^{6^6} \equiv 6^{16} \pmod{25}$. Now we can simplify:

$$6^{6^6} \equiv 6^{16} \equiv (6^4)^4 \equiv 1296^4 \equiv (-4)^4 \equiv 256 \equiv 6 \pmod{25}$$

Solving the system $6^{6^6} \equiv 6 \pmod{25}$ and $6^{6^6} \equiv 0 \pmod{4}$ yields $6^{6^6} \equiv 56 \pmod{100}$. Hence the answer is $\boxed{(C) 5}$.

Explanation for Step 2 of Solution 1 For coprime integers a and n , there is a specific identity that's a direct consequence of Euler's Totient Theorem, which is probably what the author used: $a^b \equiv a^{b \bmod \varphi(n)} \pmod{n}$

To show how the identity works in this case, the theorem directly tells us that $6^{20} \equiv 1 \pmod{25}$ because $\varphi(25) = 25 \cdot (1 - \frac{1}{5}) = 20$.

Logically, in order to apply this relationship, it's convenient to define $6^6 = 20k + r$, where k and r are integers. By definition, $r \equiv 6^6 \pmod{20}$ since r is the remainder when divided by 20. So:

$$6^{6^6} \equiv 6^{20k} \cdot 6^r \equiv (6^{20})^k \cdot 6^r \equiv 1^k \cdot 6^r \equiv 1 \cdot 6^r \equiv 6^r \pmod{25}$$

Solution 2

$6^{6^6} = 6^{46656}$ Now, listing out the first few values of the rightmost two digits for power of 6, we get

06, 36, 16, 96, 76, 56, 36, 16, 96, 76, 56, 36...

Now, note that the pattern starting from at 6^2 will continue. Treating the first as a 56 since it won't matter for the result, the pattern turns to 56 every n such that $n = 1 + 5k$, where k is a non-negative integer. So, since $46656 = 1 + 5(9331)$, it will have 56 as its rightmost 2 numbers, giving the answer $\boxed{(C) 5}$.

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Solution 3 (Fast)

Trying the first few powers of 6 gives a pattern with period 5:

$$06, 36, 16, 96, 76, 56, 36, \dots \Rightarrow 0, 3, 1, 9, 7, 5, 3, \dots$$

Notice any power of 6 has a units digit 6. Then $6^6 \equiv 1 \pmod{5}$ and the answer is $\boxed{(C) 5}$ (6^1 's tens digit is 0 and for all 6^{1+5n} for any $n \geq 1$ its tens digit is 5 according to our pattern).

Solution 4

$$6^{6^6} \pmod{100}$$

is the desired result. Note that $6^6 \pmod{100} = 64 \cdot 729 \pmod{25} \equiv 14 \cdot 4 \pmod{25} \equiv 56 \pmod{25} \equiv 6 \pmod{25}$ and $0 \pmod{25}$ so two k satisfy $k \equiv 6 \pmod{25}$ and $k \equiv 0 \pmod{4}$. We see that $N = 100$ so $N_1 = 4$ and $N_2 = 25$ and in addition $4x_1 \equiv 1 \pmod{25}$ is satisfied for $x_1 = 19$ and $x_2 = 1$, so the smallest solution is $6 \cdot 4 \cdot 19 \pmod{100} = 56$.

So we need $6^{56} \pmod{100}$ which is finding $6^{56} \pmod{25}$ and $6^{56} \pmod{4}$. The latter is 0, so the former is $6^{16} \pmod{25}$ or $2^{16} \cdot 3^{16} \pmod{25}$. Note that $2^{10} \equiv -1 \pmod{25}$ so $2^{16} \equiv -64 \pmod{25} \equiv 11 \pmod{25}$ and $3^3 \equiv 2 \pmod{25}$ so $3^{16} \equiv 2^5 \cdot 3 \pmod{25} \equiv 21 \pmod{25}$ so we have $6^{16} \equiv 231 \pmod{25} \equiv 6 \pmod{25}$. Once again, through Chinese Remainder Theorem our $d_i = 4, 25$ respectively and $4x_1 \equiv 1 \pmod{25} \implies x_1 = 19$ and $x_2 \equiv 1 \pmod{4}$ so the smallest solution is $6 \cdot 19 \cdot 4 \equiv 56 \pmod{100}$ so the hundreds digit is $\boxed{5}$.

10. (AMC 12 12B 2025) The altitude to the hypotenuse of a 30° – 60° – 90° right triangle is divided into two segments of lengths $x < y$ by the median to the shortest side of the triangle. What is the ratio $\frac{x}{x+y}$?

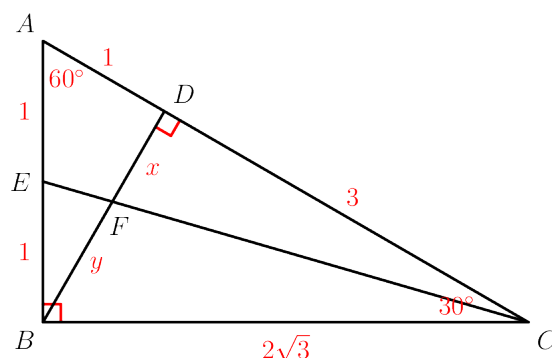
- (A) $\frac{3}{7}$ (B) $\frac{\sqrt{3}}{4}$ (C) $\frac{4}{9}$ (D) $\frac{5}{11}$ (E) $\frac{4\sqrt{3}}{15}$

Answer: A

Solution 1 (Mass Points)

Without loss of generality, let $\triangle ABC$ have side-lengths $AB = 2$, $BC = 2\sqrt{3}$, and $AC = 4$. Let D be the foot of the perpendicular from B to $\overline{AC}$, E be the midpoint of $\overline{AB}$, and F be the intersection of $\overline{CE}$ and $\overline{BD}$. Note that $\triangle ADB$ and $\triangle BDC$ are both 30° – 60° – 90° triangles. From the side-length ratio, we get $AD = 1$ and $DC = 3$.

We obtain the following diagram:



From here, we will proceed with **mass points**. Throughout this solution, we will use W_P to denote the weight of point P .

Let $W_C = 1$. Since $3AD = DC$, it follows that $W_A = 3$ and $W_D = W_C + W_A = 4$. Since $AE = EB$ and $W_A = 3$, it follows that $W_B = 3$.

Now we focus on $\overline{BD}$: Since $W_B = 3$ and $W_D = 4$, we have $\frac{DF}{FB} = \frac{x}{y} = \frac{3}{4}$. Therefore, the answer is

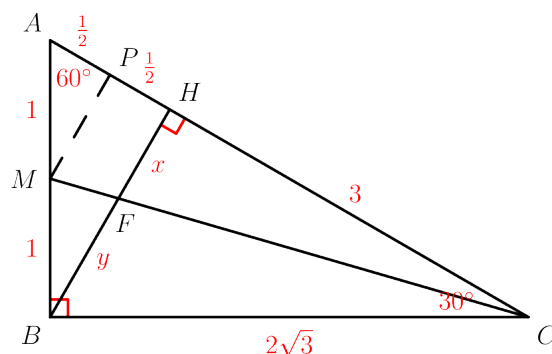
$$\frac{x}{x+y} = \frac{3}{3+4} = \boxed{\text{(A)} \frac{3}{7}}.$$

Solution 2 (Similar Triangles)

This solution refers to the diagram in Solution 1, with the definitions slightly modified.

Let $AB = 2$. Label the foot of the median M and the foot of the altitude H . In $\triangle ABH$, we construct the midsegment $\overline{MP}$, which is parallel to $\overline{BH}$ and half as long as $\overline{BH}$. In addition, let F be the intersection of $\overline{CM}$ and $\overline{BH}$.

We obtain the following diagram:



We can calculate that $PH = \frac{1}{2}$ and $HC = 3$. By similar triangles $\triangle HFC \sim \triangle PMC$, we have $\frac{x}{PM} = \frac{HC}{PC} = \frac{3}{3 + \frac{1}{2}} = \frac{6}{7}$. Also by similar triangles $\triangle PMA \sim \triangle HBA$, we

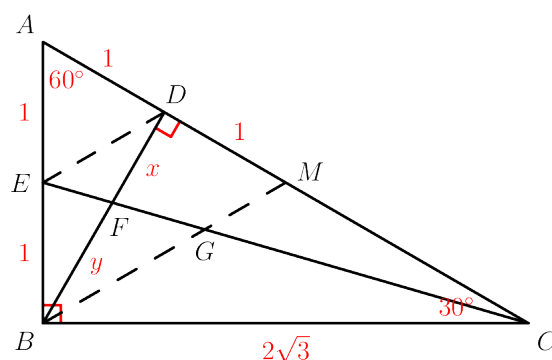
have $\frac{PM}{x+y} = \frac{1}{2}$. Combining these, we get $\frac{x}{x+y} = \boxed{\text{(A)} \frac{3}{7}}$.

Note: Thanks MRENTHUSIASM for original diagram.

Solution 3 (Similar Triangles)

This solution refers to the definitions and the diagram in Solution 1.

Let M be the midpoint of $\overline{AC}$, so $DM = 1$. Let G be the intersection of $\overline{BM}$ and $\overline{CE}$, so G is the centroid of $\triangle ABC$. In addition, we construct $\overline{DE}$, as shown below:



In any right triangle, since the median to the hypotenuse is half as long as the hypotenuse, we get $BM = \frac{1}{2}AC = 2$. Moreover, in any triangle, since the centroid divides each median into parts in the ratio $2 : 1$, with the centroid being twice as close to the midpoint of a side as it is to the opposite vertex, we get $BG = \frac{4}{3}$ and $GM = \frac{2}{3}$.

Note that $\triangle ABM$ is an equilateral triangle with side-length 2. By the SAS similarity, we have $\triangle ABM \sim \triangle AED$. Therefore, we conclude that $ED = 1$ and $\overline{ED} \parallel \overline{BM}$.

By the AA Similarity, we have $\triangle DEF \sim \triangle BGF$, from which $\frac{x}{y} = \frac{1}{4/3} = \frac{3}{4}$. It follows that

$$\frac{x}{x+y} = \frac{3}{3+4} = \boxed{\text{(A)} \frac{3}{7}}.$$

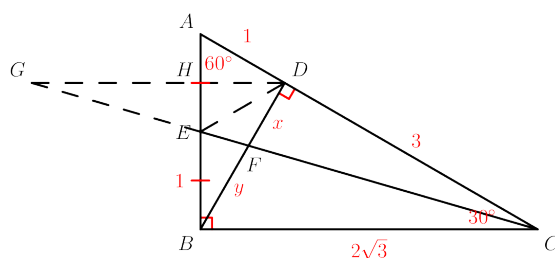
MRENTHUSIASM

Solution 4 (Similar Triangles)

This solution refers to the definitions and the diagram in Solution 1.

Since E is the midpoint of $\overline{AB}$, we have $AE = 1$, and because $\angle A = 60^\circ$ and $AD = 1$, $\triangle AED$ is equilateral. Its perpendicular bisector meets $\overline{AE}$ at H , giving $AH = HE = \frac{1}{2}$ and $DH = \frac{\sqrt{3}}{2}$.

Extend $\overline{CE}$ past E to G so that $\overline{GD}$ is parallel to $\overline{BC}$ and is the perpendicular bisector of $\overline{AE}$, so H lies on both $\overline{GD}$ and $\overline{AE}$:



By AA similarity, $\triangle BCE \sim \triangle HGE$ because $\angle BEC = \angle HEG$ and $\angle BCE = \angle HGE$ ($GD \parallel BC$). Thus $\frac{GH}{BC} = \frac{HE}{BE}$, and with $HE = \frac{1}{2}$, $BE = 1$, and $BC = 2\sqrt{3}$, we get $\frac{GH}{2\sqrt{3}} = \frac{1}{2}$ so $GH = \sqrt{3}$.

Also by AA similarity, $\triangle BCF \sim \triangle DGF$ because $\angle BFC = \angle DFG$ and $\angle BCF = \angle DGF$ ($GD \parallel BC$). Therefore $\frac{DF}{BF} = \frac{DG}{BC}$. Since $DG = DH + HG = \frac{\sqrt{3}}{2} + \sqrt{3} = \frac{3\sqrt{3}}{2}$ and $BC = 2\sqrt{3}$, we obtain $\frac{DF}{BF} = \frac{x}{y} = \frac{\frac{3\sqrt{3}}{2}}{2\sqrt{3}} = \frac{3}{4}$, so $y = \frac{4}{3}x$.

$$\text{Finally, } \frac{x}{x+y} = \frac{x}{x+\frac{4}{3}x} = \frac{x}{\frac{7}{3}x} = \boxed{\text{(A)} \frac{3}{7}}.$$

Solution 5 (Similar Triangles)

This solution refers to the definitions in Solution 1.

Draw a line perpendicular to BC from D , intersecting BC at G , and intersecting CE at H . Let $a = AE$. We know that $\triangle DHF \sim \triangle BEF$ by AA. We also know that $AE = EB = DA$. We find that $BG = \frac{a\sqrt{3}}{2}$, and $GC = BC - BG = 2a\sqrt{3} - \frac{a\sqrt{3}}{2} = \frac{3a\sqrt{3}}{2}$. Because $\triangle CHG \sim \triangle CEB$, $HG = \frac{a}{2a\sqrt{3}} \cdot \frac{3a\sqrt{3}}{2} = \frac{3}{4}a$, and $DG = \frac{3}{2}a$ due to 30-60-90 triangle properties, $DH = \frac{3}{4}a$. Because $\triangle DHF \sim \triangle BEF$, $\frac{DH}{EB} = \frac{DF}{FB}$, $\frac{\frac{3a}{4}}{a} = \frac{3}{4} = \frac{DF}{FB}$, and $\frac{DF}{DB} = \boxed{\text{(A)} \frac{3}{7}}$.

(You can tell DF is shorter than FB visually)

Solution 6 (Coordinates)

WLOG, let the triangle have hypotenuse 12. Situate it in the coordinate plane such the A is at $(0, 0)$, B is at $(6, 0)$, and C is at $(6, 6\sqrt{3})$.

Let h be the altitude from A to BC . From the area formula we get $1/2 \cdot 6 \cdot 6\sqrt{3} = 1/2 \cdot 12 \cdot h$ $h = 3\sqrt{3}$

The shortest side is AB with side length 6, so the median from C to AB will be at $(3, 0)$. Finding the equation of the line through that point to C , find that the slope is $\frac{6\sqrt{3}-0}{6-3} = 2\sqrt{3}$. Plugging the point $(3, 0)$ into $y = 2\sqrt{3}x + b$, we get that $0 = 6\sqrt{3} + b$, which simplifies to $b = -6\sqrt{3}$. So, the equation of the line of the median from C is $y = 2\sqrt{3}x - 6\sqrt{3}$.

The line of the altitude from B to the hypotenuse is perpendicular to the line of the hypotenuse which has slope $6\sqrt{3}/6 = \sqrt{3}$.

So, the altitude line has slope $\frac{-1}{\sqrt{3}} = -\frac{\sqrt{3}}{3}$, and the line passes through the point $B = (6, 0)$. Solving for the equation using similar methods above, we get $y = -\frac{\sqrt{3}}{3}x + 2\sqrt{3}$.

Using the two equations and solving for their intersection, we get $y = 2\sqrt{3}x - 6\sqrt{3} = -\frac{\sqrt{3}}{3}x + 2\sqrt{3}$, which simplifies to $8\sqrt{3} = \frac{7\sqrt{3}}{3}x$, so $x = 24/7$. Using this x value, we get $y = 2\sqrt{3} \cdot 24/7 - 6\sqrt{3} = 6\sqrt{3}/7$.

Finding the distance from this intersection to B , $d = \sqrt{(6 - 24/7)^2 + (0 - 6\sqrt{3}/7)^2} = \sqrt{324/49 + 108/49} = \frac{12\sqrt{3}}{7}$.

This is greater than half of the calculated altitude value from above ($3\sqrt{3}$), so the shorter side length is $3\sqrt{3} - \frac{12\sqrt{3}}{7} = \frac{9\sqrt{3}}{7}$.

Lastly, taking the ratio of the lengths, $\frac{\frac{9\sqrt{3}}{7}}{3\sqrt{3}} = \frac{9}{21} = \boxed{\text{(A)} \frac{3}{7}}$.

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Solution 7 (Symmedian and Bary)

Notice that the smaller right triangle with the original 30° angle can be obtained with a dilation and a reflection over the original triangle's angle bisector. Hence it is not too hard to see that the original median is a symmedian to this smaller triangle. Recall that the barycentric equation for the symmedian is in the form $\frac{x}{a^2} = \frac{y}{b^2}$, and we have $a, b = \sqrt{3}, 2$ in a $30^\circ - 60^\circ - 90^\circ$ triangle. Hence $x : y = 4 : 3$ and we get the ratio $\boxed{\text{(A)} \frac{3}{7}}$, or A relatively quickly.

Solution 8 (Trigonometry Bash)

Let the triangle be called $\triangle ABC$, with $\angle C = 90^\circ$ and $\angle A = 30^\circ$. Then let the altitude from C intersect with AB at a point D , and the median from A intersect BC at the point E . WLOG, let $AB = 2$, $AC = \sqrt{3}$, and $BC = 1$. Then $BE = EC = \frac{1}{2}$, and by similar triangles, $\triangle DBC$ is also $30^\circ - 60^\circ - 90^\circ$, and so $DB = \frac{1}{2}$ and hence $AD = \frac{3}{2}$. Therefore, $DC = \sqrt{3} \cdot \frac{1}{2} = \frac{\sqrt{3}}{2}$. Also, we have that $\tan(\angle EAC) = \frac{1}{2\sqrt{3}} \implies \angle EAC = \arctan\left(\frac{1}{2\sqrt{3}}\right)$. Hence, $\angle BAE = 30^\circ - \arctan\left(\frac{1}{2\sqrt{3}}\right)$. Now we can set up to solve for x ; we have that $\tan(\angle BAE) = \frac{x}{\frac{3}{2}}$. $\tan(\angle BAE) = \frac{\frac{1}{\sqrt{3}} - \frac{1}{2\sqrt{3}}}{1 + \frac{1}{\sqrt{3}} \cdot \frac{1}{2\sqrt{3}}} = \frac{\frac{1}{2\sqrt{3}}}{\frac{4}{6}} = \frac{1}{2\sqrt{3}} \cdot \frac{6}{7} = \frac{\sqrt{3}}{7}$. Hence $x = \frac{3}{2} \cdot \frac{\sqrt{3}}{7} = \frac{3\sqrt{3}}{14}$, and therefore since we

found earlier that $x + y = DC = \frac{\sqrt{3}}{2}$, we have that $\frac{x}{x+y} = \frac{\frac{3\sqrt{3}}{14}}{\frac{\sqrt{3}}{2}} = \boxed{\text{(A)} \frac{3}{7}}$.

Solution 9 (Easier to Understand, Using Similar 30-60-90 Triangles)

Let the triangle be $\triangle ABC$, where $\angle A = 30^\circ$, $\angle B = 60^\circ$, and $\angle C = 90^\circ$. Let the point where the altitude of AB and AB intersect be G . Let the intersection between the median of CB and the altitude of AB be E . $x = EG$ and $y = CE$.

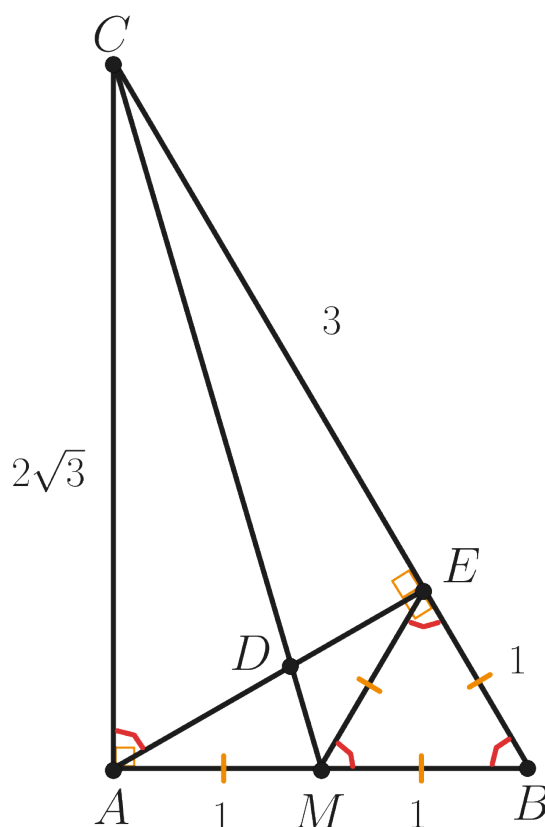
Draw a segment, DF , so that it is perpendicular to AC and passes through E . This creates two small $30^\circ-60^\circ-90^\circ$ triangles, $\triangle ECD$ and $\triangle EFG$. Using the Side-Splitter Thm., we know that $DE = EF$. WLOG, let $DE = EF = 2$. Using the rules of $30^\circ-60^\circ-90^\circ$ triangles, we can find that $x = EG = \frac{2}{2} \cdot \sqrt{3} = \sqrt{3}$ and that $y = CE = \frac{2}{\sqrt{3}} \cdot 2 = \frac{4}{\sqrt{3}} = \frac{4\sqrt{3}}{3}$. Therefore,

$$\frac{x}{x+y} = \frac{\sqrt{3}}{\sqrt{3} + \frac{4\sqrt{3}}{3}} = \frac{\sqrt{3}}{\frac{3\sqrt{3}}{3} + \frac{4\sqrt{3}}{3}} = \frac{\sqrt{3}}{\frac{7\sqrt{3}}{3}} = \sqrt{3} \cdot \frac{3}{7\sqrt{3}} = \frac{3\sqrt{3}}{7\sqrt{3}} = \boxed{\text{(A)} \frac{3}{7}}.$$

Solution 10 (Ruler)

Since rulers are allowed on the AMC 10, we can draw out the figure accurately with a ruler. Draw a $30-60-90$ triangle at any scale that you want (I used 1 inch, $\sqrt{3}$ inches, and 2 inches for my sides), then draw the altitude to the hypotenuse and the median to the shortest side on the triangle. Find the ratio of the short part of the altitude divided by the median and the total length of the altitude. Assuming that you drew accurately, the answer is then $\boxed{\text{(A)} \frac{3}{7}}$.

Solution 11 (Law of Cosines Bash)



WLOG $\overline{AB} = 2$ as the problem statement concerns only lengths ratios. Draw $\overline{EM}$ connecting the median of $\triangle ABE$ to right angle $\angle E$, we know that by inscribing right $\triangle ABE$ in a circle and by Thales theorem, $\overline{ME} = \overline{MA} = \overline{MB}$ the circumradius. $\overline{MB} = \overline{ME} \iff \triangle BEM$ is isosceles, and because $\angle B = 60^\circ$, then $\angle E = 60^\circ$, making $\triangle BEM$ equilateral.

Let $\overline{AD} = x$ and $\overline{DE} = y$. By $30^\circ - 60^\circ - 90^\circ$ special right triangle

$$\sqrt{3} = \overline{AE} = x + y$$

$$\overline{AM} = \sqrt{(2\sqrt{3})^2 + 1^2} = \sqrt{13}$$

$$\overline{CD} = \sqrt{y^2 + 3^2} = \sqrt{y^2 + 9} = \sqrt{x^2 - 2\sqrt{3}x + 12}$$

$$\overline{DM} = \sqrt{x^2 + 1^2 - 2 \cdot 1 \cdot x \cdot \cos 30^\circ} = \sqrt{x^2 - \sqrt{3}x + 1}$$

$$\overline{CD} + \overline{DM} = \overline{AM}$$

$$\sqrt{x^2 - 2\sqrt{3}x + 12} + \sqrt{x^2 - \sqrt{3}x + 1} = \sqrt{13}$$

Algebra:

$$\sqrt{x^2 - 2\sqrt{3}x + 12} = \sqrt{13} - \sqrt{x^2 - \sqrt{3}x + 1}$$

$$x^2 - 2\sqrt{3}x + 12 = x^2 - \sqrt{3}x + 1 + 13 - 2\sqrt{13(x^2 - \sqrt{3}x + 1)}$$

$$-\sqrt{3}x - 2 = -2\sqrt{13(x^2 - \sqrt{3}x + 1)}$$

$$3x^2 + 4\sqrt{3}x + 4 = 4 \cdot 13(x^2 - \sqrt{3}x + 1)$$

$$49x^2 - 56\sqrt{3}x + 48 = 0$$

$$(7x - 4\sqrt{3})^2 = 0$$

$x = \frac{4\sqrt{3}}{7}y = \frac{3\sqrt{3}}{7}x$ is defined to be minimum of those lengths, so $x = \frac{3\sqrt{3}}{7}$. Hence

$$\frac{x}{x+y} = \boxed{\text{(A)} \frac{3}{7}}.$$

Solution 12 (Vectors and Projections)

Let the triangle have vertices A (opposite hypotenuse), B , C . Let the shortest side be AB , so the median is from C to midpoint M of AB . Let h be the altitude from C to hypotenuse BC . Represent everything vectorially:

$\vec{A}, \vec{B}, \vec{C} \in \mathbb{R}^2$. Let the hypotenuse vector be $\vec{u} = \vec{B} - \vec{C}$. The altitude from C is along $\vec{v}$ perpendicular to $\vec{u}$. The median vector is $\vec{w} = \vec{M} - \vec{C}$.

We want the intersection point P along the altitude such that $\vec{P} = \vec{C} + t\vec{v}$ lies on the median: $\vec{P} = \vec{C} + s\vec{w}$. But vectors $\vec{v}$ and $\vec{w}$ are not parallel. Solve for scalar t using projection formula:

$$t = \frac{\vec{w} \cdot \vec{v}}{|\vec{v}|^2}.$$

Take $x = |\vec{CP}| = t|\vec{v}|$ and $x + y = |\vec{v}|$. So

$$\frac{x}{x+y} = \frac{t|\vec{v}|}{|\vec{v}|} = t = \frac{\vec{w} \cdot \vec{v}}{|\vec{v}|^2}.$$

Now we just need vectors $\vec{v}$ and $\vec{w}$. Let the triangle be standard 30-60-90: shortest side along $\vec{AB} = \hat{i}$, height along $\hat{j}$. Then $\vec{v}$ is perpendicular to hypotenuse $\vec{u} = \vec{B} - \vec{C}$, say $\vec{u} = (1, -\sqrt{3})$, so $\vec{v} = (\sqrt{3}, 1)$ (perpendicular). Median: $\vec{w} = \vec{M} - \vec{C} = (1/2, 0) - (0, \sqrt{3}) = (1/2, -\sqrt{3})$.

Compute the projection:

$$t = \frac{\vec{w} \cdot \vec{v}}{|\vec{v}|^2} = \frac{(1/2)(\sqrt{3}) + (-\sqrt{3})(1)}{(\sqrt{3})^2 + 1^2} = \frac{\frac{\sqrt{3}}{2} - \sqrt{3}}{4} = \frac{-\frac{\sqrt{3}}{2}}{4} = -\frac{\sqrt{3}}{8}.$$

Take absolute value (distance along altitude) $x = |t||\vec{v}| = (\frac{\sqrt{3}}{8}) \cdot 2 = \frac{\sqrt{3}}{4}$ (after proper scaling).

Thus the ratio

$$\frac{x}{x+y} = \frac{3}{7}.$$

Answer:

$$\boxed{(A) \frac{3}{7}}.$$

Solution 13 (Straightforward Coordinate Geometry)

Let the triangle have vertices A , B , and C on the 60, 90, and 30 angles, respectively. Let E be the intersection of the median from C and AB , let F be the intersection of the altitude from B to AC , and let P be the intersection of CE and BF . WLOG let A be $(0, 2)$, B be $(0, 0)$, and C be $(2\sqrt{3}, 0)$. In order to find the ratio of PF to BF , we need to find the coordinates of F and P . First, slope of AC is $\frac{-1}{\sqrt{3}}$, so the slope of BF is $\sqrt{3}$. So, the equation for AC is $y = \frac{-1}{\sqrt{3}}x + 2$ and for BF is $y = \sqrt{3}x$. Finally, E is the midpoint of AB , E is at $(0, 1)$, the slope of CE is $\frac{-1}{2\sqrt{3}}$, and the equation for CE is $y = \frac{-1}{2\sqrt{3}}x + 1$. Setting the equations for AC and BF equal, we find the intersection is F at $(\frac{\sqrt{3}}{2}, \frac{3}{2})$. Setting the equations for CE and BF equal, we find the intersection is P at $(\frac{2\sqrt{3}}{7}, \frac{6}{7})$. Because BF goes through the origin, we can find $\frac{BP}{BF}$ by dividing the y -coordinates of P and F , getting $\frac{4}{7}$. Thus,

$$\frac{PF}{BF} = 1 - \frac{BP}{BF} = 1 - \frac{4}{7} = \boxed{(A) \frac{3}{7}}.$$

Solution 14 (Menelaus)

Let $AC = 2$, $AB = 1$, hence $BC = \sqrt{3}$. If we use the diagram in Solution 1, first note that $\triangle ACB \sim \triangle ABD$ by AA with similar ratio 2, hence $AD = \frac{1}{2}$. Now, we use Menelaus' Theorem in $\triangle BAD$ with transversal line EFC :

$$\begin{aligned} \frac{BE}{EA} \cdot \frac{AC}{CD} \cdot \frac{DF}{FB} &= 1 \\ \implies \frac{1}{1} \cdot \frac{2}{\frac{3}{2}} \cdot \frac{x}{y} &= 1 \\ \implies \frac{x}{y} &= \frac{3}{4} \end{aligned}$$

$$\text{So } \frac{x}{x+y} = \boxed{\frac{3}{7}(A)}.$$

11. (AMC 12 12B 2025) Nine athletes, no two of whom are the same height, try out for the basketball team. One at a time, they draw a wristband at random, without replacement, from a bag containing 3 blue bands, 3 red bands, and 3 green bands. They are divided into a blue group, a red group, and a green group. The tallest member of each group is named the group captain. What is the probability that the group captains are the three tallest athletes?

(A) $\frac{2}{9}$ (B) $\frac{2}{7}$ (C) $\frac{9}{28}$ (D) $\frac{1}{3}$ (E) $\frac{3}{8}$

Answer: C

Solution 1

We will arrange the nine athletes in a line, with the first three being in the blue group, the next three being in the red group, and the last three being in the green group. Suppose the three tallest athletes are named A, B, C , in some order.

We have $3!$ ways to choose which group each of these athletes can be in. We then have $3 \cdot 3 \cdot 3$ ways to determine which of the three positions in that group they can take. From here, we must distribute the remaining six athletes in the remaining six spaces, which can be done in $6!$ ways.

There are therefore $6 \cdot 27 \cdot 6!$ favorable outcomes. There are also $9!$ total ways to arrange the athletes with no restrictions. Our answer is $\frac{6 \cdot 27 \cdot 6!}{9!} = \frac{6 \cdot 27}{9 \cdot 8 \cdot 7} = \boxed{\text{(C)} \frac{9}{28}}$.

Solution 2

Let the three tallest athletes be A_1 (tallest), A_2 (second-tallest), and A_3 (third-tallest).

The condition "the captains are A_1, A_2 , and A_3 " is equivalent to the condition " A_1, A_2 , and A_3 must all be in different-colored groups." We can find this probability by considering them drawing bands sequentially. Athlete A_1 (Tallest): A_1 draws a band. It can be any of the 9 bands. A_1 will always be a captain, so the probability is 1. We now have 8 bands left. Athlete A_2 (2nd Tallest): A_2 must draw a band of a different color than A_1 . There are 8 bands left in the bag. A_1 took one band, so 2 bands of that color remain. The other 6 bands (3 of one color, 3 of another) are "good" bands for A_2 . The probability of A_2 drawing a different color is $P(A_2) = \frac{6}{8}$. Athlete A_3 (3rd Tallest): A_3 must draw a band of the third, remaining color. We assume A_1 and A_2 were successful, so there are 7 bands left. A_1 took a band of color 1. A_2 took a band of color 2. There are 3 bands of color 3 (the "good" color) still in the bag. The probability of A_3 drawing the last color is $P(A_3) = \frac{3}{7}$. The total probability is the product of these independent probabilities:

$$P(\text{Total}) = P(A_1) \times P(A_2) \times P(A_3)$$

$$P(\text{Total}) = 1 \times \frac{6}{8} \times \frac{3}{7} = \frac{18}{56}$$

Simplifying the fraction:

$$\frac{18}{56} = \frac{9}{28}$$

The correct option is (C). Jonathanmo

Solution 3 (Counting)

There are $\binom{9}{3}\binom{6}{3}$ ways to assign rubber bands to the athletes. Now we have to find the number of ways for the team captains to be the 3 tallest people. Each of the 3

tallest people must have a different rubber band color, and there are $3! = 6$ ways to permute their individual rubber band colors.

For the other 6 people, there are $\binom{6}{2}\binom{4}{2}$ ways to assign rubber bands to them.

Thus, our probability is $\frac{3! \cdot \binom{6}{2} \binom{4}{2}}{\binom{9}{3} \binom{6}{3}} = \boxed{(C) \frac{9}{28}}$.

Solution 4 (similar to Solution 3)

We have $\binom{9}{3} \cdot \binom{6}{3} \cdot \binom{3}{3} = 1680$ ways to select the people in each team, and we must divide by $3!$ ways to order the teams. This gives us $\frac{1680}{3!} = 280$ total ways to create the teams.

To have the three team captains be the tallest, we need them to be in separate teams. Once placed in their respective teams we have $3! = 6$ ways to orient them. We then have 6 people remaining. There are $\binom{6}{2} \cdot \binom{4}{2} \cdot \binom{2}{2} = 90$ ways to place two more people on each team. However, this overcounts by $3! = 6$, as the orientation of the teams themselves doesn't matter, hence in reality only 15 cases work. We multiply by the orientation of each athlete, which yields $15 \cdot 6 = 90$ cases.

Our probability is $\frac{90}{280} = \boxed{(C) \frac{9}{28}}$.

Solution 5

Assume the tallest person is in the red group. Then the second person has 6 spots to go out of 8 remaining (1 was taken away). Then the third tallest person has 3 spots to go out of 7 remaining. $\frac{6}{8} \cdot \frac{3}{7}$ is the same as $\boxed{(C) \frac{9}{28}}$.

Solution 6 (cheese)

Upon realizing that the denominator is a multiple of 7, and that there are a bunch of 3's in the numerator (just do some CP with the right logic that could even be a bit flawed) we can make an educated guess that the answer must be $\boxed{(C) \frac{9}{28}}$.

Solution 7 (Fast, like Solution 5)

WLOG, assume that the three tallest people are the first ones to draw. Irrespective of where the first person goes, the probability that the second and third person go on different teams is $\frac{6}{8}$ and $\frac{3}{7}$, respectively, yielding $\boxed{(C) \frac{9}{28}}$.

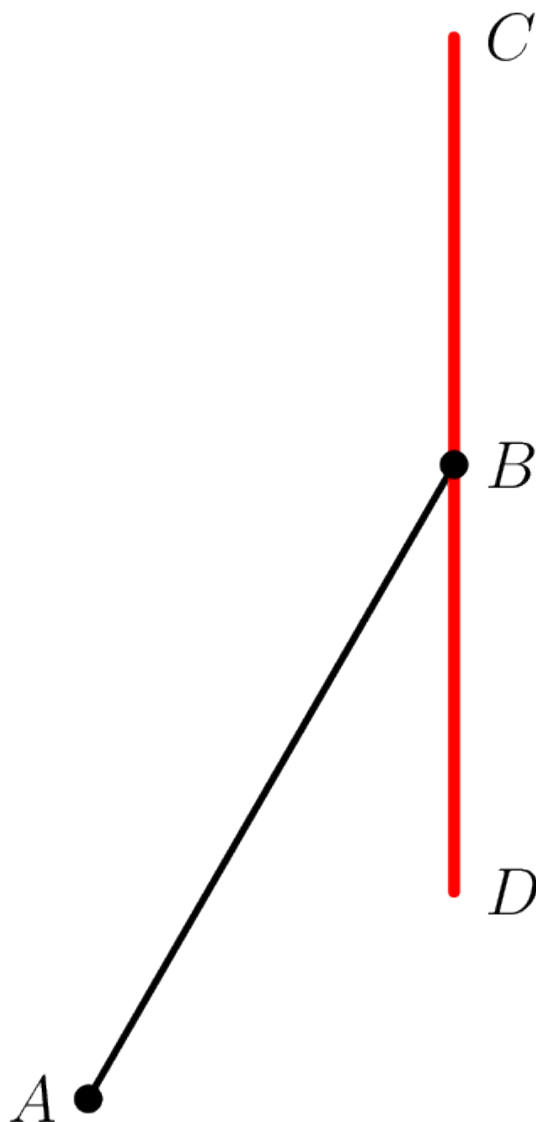
Solution 8 We can visualize the nine athletes as a 3×3 grid, where each row represents a color group and each row contains 3 positions. We want the three tallest athletes to become captains. This happens exactly when each of the three tallest athletes occupies a different row.

Total ways to choose positions for the three tallest athletes: $\binom{9}{3} = 84$

Favorable ways (exactly one of the three tallest athletes in each row): $3^3 = 27$

Thus, our probability is $\frac{27}{84} = \boxed{(C) \frac{9}{28}}$.

12. (AMC 12 12B 2025) The windshield wiper on the driver's side of a large bus is depicted below.



Arm $\overline{AB}$ pivots back and forth around point A , sweeping out an arc of 60° , symmetric about the vertical line through A . The wiper blade $\overline{CD}$ is attached to B at its midpoint and stays vertical as the arm moves. The arm is 3 feet long, and the wiper blade is 3.5 feet tall. What is the area of the windshield cleaned by the wiper, in square feet, to the nearest hundredth? (Assume that the windshield is a flat vertical surface.)

- (A) 9.68 (B) 10.14 (C) 10.50 (D) 11.32 (E) 12.00

Answer: C

Solution 1 (Cavalieri's Principle)

The area cleaned by the wiper follows a thickened curve with vertical ends, where the curve is a 60° arc with radius 3. Since the sides are vertical, by Cavalieri's Principle, the area is equivalent to a rectangle with side lengths 3.5 and the distance between the two vertical ends. Let B' be the result of point B at its leftmost point after it has swept 60° . Then ABB' is equilateral so $BB' = AB = 3$. So the area is $3.5 \cdot 3 = \boxed{\text{(C) } 10.50}$.

Note: if you aren't aware of Cavalieri's Principle, you can still recognize this by

slicing the top curved part of the shape and moving it to the bottom, creating a rectangle with the same area as the original shape.

Solution 2 (Calculus)

As solution 1 kind of explains, the top and bottom of the area swept out follow the exact same path shape, so we can think of the top one as $f(x)$ and the bottom one as $f(x) - 3.5$. As solution 1 points out, the width of the figure is 3. Then, the area between them is $\int_0^3 [f(x) - (f(x) - 3.5)] dx = \boxed{\text{(C)} 10.50}$.

Solution 3

Using the diagram (credit - MRENTHUSIASM), we see that the line covers a certain area that we need to find. However, finding funky circle areas usually requires calculus. Using some intuition, we can see that the lune at the top of the area we need to find is the exact required area to remove and place into the bottom crescent. This transformation shifts the area into a perfect rectangle with one side length equal to the height of the blade:

$$\text{Height} = 3.5$$

To find the other side (the horizontal width), we continue the vertical line $\overline{CD}$ downward at its rightmost position. Then, we draw a perpendicular line from the pivot point A to this extended line $\overline{CD}$, which touches our new location F . Let B' be the leftmost position of the arm after the 60° sweep. If we draw a perpendicular line from B' touching line $\overline{AF}$ at point G , we realize that:

$$\triangle AFC \sim \triangle AB'G \text{ by SAS}$$

Because of the symmetric angles of the sweep, we know that:

$$\angle CAF = \angle B'AG = 60^\circ$$

This makes it a standard 30° - 60° - 90° triangle. By the ratios of a 30° - 60° - 90° triangle, the short leg is half the hypotenuse:

$$FA = AG = 1.5$$

Since the width spans symmetrically from G to F :

$$FG = 3$$

Now we have our rectangle dimensions, so we just multiply them to find the total area:

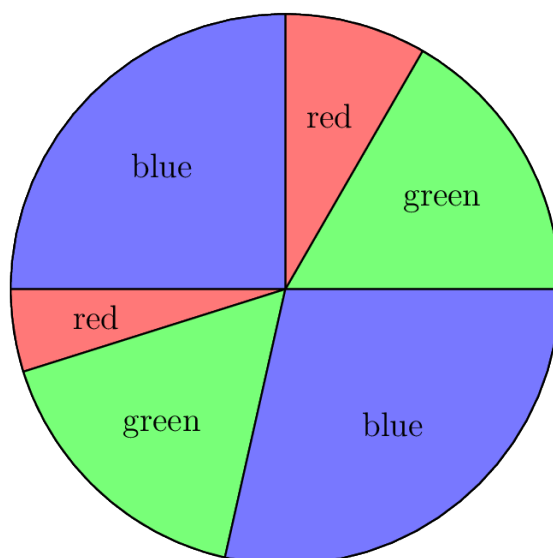
$$\text{Area} = 3 \times 3.5 = 10.5$$

The answer is **(C)**.

Solution 4

We just realize that the sector inside and the sector outside have equal areas so we can just imagine it as a 3.5×3 rectangle, which gives us $\boxed{10.5}$ **(C)**

13. (AMC 12 12B 2025) A circle has been divided into 6 sectors of different sizes. Then 2 of the sectors are painted red, 2 painted green, and 2 painted blue so that no two neighboring sectors are painted the same color. One such coloring is shown below.



How many different colorings are possible?

- (A) 12 (B) 16 (C) 18 (D) 24 (E) 28

Answer: D

Solution 1

Create an arbitrary six slice circular diagram. The first slice has 3 options, and the second has 2, third has 2, fourth has 2, fifth has 2, and the sixth has only 1 color to be chosen.

Understand that if $\mathbb{X}$ is the set containing all possible cases of the circle, then $\mathbb{X}$ has a correlation to another set, call it $\mathbb{K}$, which contains all the colors. This can be quite easily sought as a rotation of one element in $\mathbb{X}$ that accurately maps to another element in $\mathbb{X}$ is considered the same.

In simpler terms, to account for overlapping cases in $\mathbb{X}$, one must divide by 2 (as two colorings that are rotated are considered different, and we don't want that as it defeats the purpose of "unique").

Then, multiplying out we have $3 \times 2^4 = 48$. Dividing by 2 to account for overlap in $\mathbb{X}$, we get $48/2$ or **(D) 24**

Solution 2

Since the sectors are of different sizes, this is equivalent to ordering the colors in a line of six spaces such that the colors at either end must be different and adjacent colors must be different. WLOG let the first color be red and the second be green.

If the third color is also red, then two of the remaining spaces must be blue, so blue must occupy the fourth and sixth spaces. Our coloration is thus *RGRBGB*.

If the third color is blue, then we again consider cases. The fourth color being green forces red and then blue, so we must have *RGBGRB*.

Otherwise, the fourth color would be red, and either of the final two space colorations work, yielding *RGBRBRG* and *RGBRGRB*.

Since the first color can be chosen in 3 ways and the second in 2 ways, and there are a total of 4 colorations after choosing these two colors, the total number of cases is $3 \cdot 2 \cdot 4 = \mathbf{(D) 24}$.

Solution 3

There are two cases on how the colors can be distributed:

Case 1: Every color is diametrically opposite itself. The 6 sectors form 3 diametrically opposite pairs: (1, 4), (2, 5), and (3, 6). In this case, each of the 3 colors must cover one of these pairs. There are $3!$ ways to assign the 3 colors to the 3 pairs.

$$3! = 6$$

Case 2: Only 1 color is diametrically opposite itself. First, there are 3 choices for which color is opposite itself. Second, there are 3 choices for which diametrical pair (e.g., (1, 4), (2, 5), or (3, 6)) this color will cover. Without loss of generality, suppose one color covers sectors 1 and 4. The two remaining colors must then cover the remaining sectors 2, 3, 5, 6. These must be arranged in "crooked" pairs, for example, 2, 6 and 3, 5, so that neither of the remaining colors is also opposite itself. There are 2 ways to assign the two remaining colors to these two crooked pairs. This gives $3 \cdot 3 \cdot 2 = 18$ options in this case. It is impossible to have exactly 0 or 2 diametrically opposite colors. Therefore, the total number of ways is the sum of the two cases: $18 + 6 = \boxed{\text{(D) } 24}$.

Solution 4

(casework based on red sectors)

Case 1) Gaps between red sectors is 2-2: There are $2 \cdot 2 = 4$ ways to order the green and blue sectors between the reds (there is one sector of each color in each gap and two ways to order the two colors per gap). Multiply by 3 to account for ways to rotate the ordering.

Case 2) Gaps between red sectors is 1-3: There are 2 ways to assign Green and Blue for sectors between Reds (two ways to pick color for gap of size 1, which determines color of center sector for gap of size 3 such that all other sector colors are decided). Multiply by 6 to account for rotation.

Answer is $4 \cdot 3 + 2 \cdot 6 = \boxed{\text{(D) } 24}$.

Solution 5 (Burnside's Lemma)

We apply Burnside's lemma to this problem. Use the cyclic group $G = C_6$, it has 6 rotations so $|C_6| = 6$. Burnside's lemma says:

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |X^g|$$

Our goal is to find $n := |X| = |X^e|$, where e denotes the identity. Up to rotation, we can count the number of orbits. Place $2A, 2B, 2C$ on a circle with no 2 adjacent. Start with placing A 's, they can either have gaps of $2-2$ ($A_A_$) or $1-3$ ($A_A_$) for the remaining places. Up to rotation, for $2-2$, we count $ABCABC$ and $2 \times ABCACB$, and for $1-3$ we get $ABACBC$ and $ACABCB$, so there are $|X/G| = 3 + 2 = 5$ orbits up to rotation.

Let r be a single cycle of the elements. By symmetry, these cases are equivalent: $r \equiv r^{-1} \equiv r^5$, $r^2 \equiv r^4$. The cycle index polynomial of C_6 is therefore $Z\{C_6\} = x_1^6 + 2x_6 + 2x_3^2 + x_2^3$.

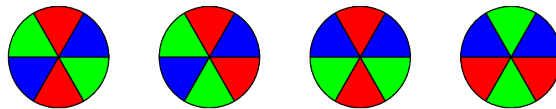
Counting the fixed points, we get

$$5 = \frac{1}{6}(n + 2 \cdot 0 + 2 \cdot 0 + 3!)$$

Solving, $n = 5 \cdot 6 - 3! = \boxed{\text{(D)} 24}$.

Solution 6

You can tackle the problem by forgetting about the different sizes at first. This gets you 4 combinations:



B-R-G-B-R-G-B-R-G-B-G-RB-R-B-G-R-GB-G-B-R-G-R

Now we can multiply by 6 since we have 6 different pieces. $6 \cdot 4 = \boxed{\text{(D)} 24}$.

14. (AMC 12 12B 2025) Consider a decreasing sequence of n positive integers $x_1 > x_2 > \cdots > x_n$ that satisfies the following two conditions:

- The average (arithmetic mean) of the first 3 terms in the sequence is 2025.
- For all $4 \leq k \leq n$, the average of the first k terms in the sequence is 1 less than the average of the first $k - 1$ terms in the sequence.

What is the greatest possible value of n ? (A) 1013 (B) 1014 (C) 1016 (D) 2016 (E) 2017

Answer: B

Solution 1

Since the mean of the first three terms is 2025, their sum is $2025 \cdot 3 = 6075$. Then, incorporating the fourth term makes the mean $2025 - 1 = 2024$, so their sum must be

$2024 \cdot 4 = 8096$, implying that the 4th term is $8096 - 6075 = 2021$. Doing the same for the 5th term,

$2023 \cdot 5 = 10115$, 5th term is

$10115 - 8096 = 2019$

Algebraically, we can model this pattern for the k th term = x as

$$(k-1)(2029-k) + x = (k)(2028-k) \quad 2029k - 2029 - k^2 + k + x = 2028k - k^2 \\ x = 2029 - 2k$$

So the maximum $k \leq n$ for which x is positive is 1014, giving our answer $n =$

(B) 1014

Solution 2

The average of the first 3 terms in the sequence is 2025, which means

$$\frac{x_1 + x_2 + x_3}{3} = 2025,$$

so $x_1 + x_2 + x_3 = 6075$.

For $k = 4$, we have

$$\frac{x_1 + x_2 + x_3 + x_4}{4} = 2025 - 1 = 2024,$$

so $x_1 + x_2 + x_3 + x_4 = 8096$. We know $x_1 + x_2 + x_3 = 6075$, so

$$x_4 = 8096 - 6075 = 2021.$$

For $k = 5$, we have

$$\frac{x_1 + x_2 + x_3 + x_4 + x_5}{5} = 2024 - 1 = 2023.$$

That means $x_1 + x_2 + x_3 + x_4 + x_5 = 10115$. We know $x_1 + x_2 + x_3 + x_4 = 8096$, so

$$x_5 = 10115 - 8096 = 2019.$$

Continuing on for $k = 6, 7, 8, \dots$, we can see that each term decreases by 2. So,

$$x_6 = 2017, x_7 = 2015, x_8 = 2013, \text{ and likewise, } x_3 = 2023, x_2 = 2025, x_1 = 2027.$$

To find the greatest possible value of n , we need to find when the numbers in this sequence become negative. The general term is

$$x_n = 2027 - 2(n - 1).$$

We require $x_n > 0$, so

$$2027 - 2(n - 1) > 0.$$

This simplifies to

$$2027 > 2n - 2 \Rightarrow 2029 > 2n \Rightarrow n < 1014.5.$$

Thus the greatest possible integer value of n is 1014, so the answer is (B) 1014.
anzhuPro

Solution 3

Assume that the first three terms are 2026, 2025, and 2024, respectively (it doesn't matter what the first 3 terms are, as long as their average is 2025). The average of the first four terms is $2025 - 1 = 2024$, which means the fourth term is 2021. Continuing this process, we realize that the next term is 2019 and each subsequent term is 2 fewer. This means the n^{th} term of this sequence where $n > 5$ is the $n - 1^{\text{th}}$ term minus 2. Realizing this is an arithmetic sequence and all the terms must be positive, we get the answer, **(B) 1014**

Solution 4 (Proof of Sol. 3)

If the first k terms average to $2028 - k$, note that they sum to $2028k - k^2$. Then, $k + 1$ terms average to $2027 - k$, so they sum to $-k^2 + 2026k + 2027$, so the $k + 1^{\text{th}}$ term is $(-k^2 + 2026k + 2027) - (2028k - k^2)$, where $k \geq 3$, so $k + 1^{\text{th}}$ term is $2027 - 2k$. Note that for this to be positive, $k \leq 1013$, so $k + 1^{\text{th}}$ term is the 1014th term, answer is 1014.

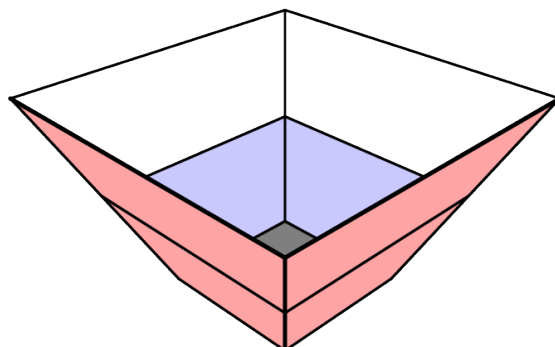
Solution 5 (basic calculus)

From the previous solution, we derive the sum of the first k terms to be equal to $2028k - k^2$. Taking a derivative of this, we get $2028 - 2k$. Since the only local extreme is at $k = 1014$, answer must be **(B) 1014**

Solution 6

Let's pretend that k does not have to be at least 4. Then the for $k = 1$ the average would be 2027. To ensure that the next average is 2026 the second number must be 2025, and the pattern must continue since it's like median. Therefore you can work backwards from the last number being 1 since it always drops by 2 each time so you can say like $2n - 1 = 2027$ so $n =$ **(B) 1014**

15. (AMC 12 12B 2025) A container has a 1×1 square bottom, a 3×3 open square top, and four congruent trapezoidal sides, as shown. Starting when the container is empty, a hose that runs water at a constant rate takes 35 minutes to fill the container up to the midline of the trapezoids.



How many more minutes will it take to fill the remainder of the container?

- (A) 70 (B) 85 (C) 90 (D) 95 (E) 105

Answer: D

Solution 1

Extend the edges pointing downw

16. (AMC 12 12B 2025) An analog clock starts at midnight and runs for 2025 minutes before stopping. What is the tangent of the acute angle between the hour hand and the minute hand when the clock stops?

- (A) 0 (B) $\sqrt{2} - 1$ (C) $2 - \sqrt{2}$ (D) $\frac{\sqrt{2}}{2}$ (E) $3 - \sqrt{2}$

Answer: B

Solution 1

When divided by 60, 2025 yields 33, with a remainder of 45. Since $33 \equiv 9 \pmod{24}$, the clock is now at 9 : 45.

At this time, the hour is 75% over, so the hour hand is also 75% of the way between 9 and 10. There is $\frac{360^\circ}{12} = 30^\circ$ angle between 9 and 10, and since the hour hand is 75% of the way, it is $30^\circ \cdot \frac{3}{4} = \frac{45^\circ}{2}$ from the minute hand, which is pointing directly at 9.

We now just need to find $\tan\left(\frac{45^\circ}{2}\right)$, so let this be x . Using the tangent addition formula, we find that

$$\begin{aligned}\tan(45^\circ) &= \frac{x + x}{1 - x \cdot x} \\ \iff 1 &= \frac{2x}{1 - x^2} \\ \iff x^2 + 2x - 1 &= 0.\end{aligned}$$

Using the quadratic formula, and knowing that x must be positive since $0^\circ < \frac{45^\circ}{2} < 90^\circ$, it follows that $x = \tan\left(\frac{45^\circ}{2}\right) = \boxed{\text{(B)} \sqrt{2} - 1}$.

Note: One could solve this quickly using the *tangent half-angle identity*, which is $\tan\left(\frac{\theta}{2}\right) = \frac{\sin(\theta)}{1 + \cos(\theta)} = \frac{1 - \cos(\theta)}{\sin(\theta)}$. math660

Note: Another interesting way to find the angle between the hands at 9 : 45 is to use the formula $\left(30h - \frac{11}{2}m\right)^\circ$, where h is the hour and m is the minute. In this case, we deduce that the angle is $\left(30 \cdot 9 - \frac{11}{2} \cdot 45\right)^\circ = \left(\frac{540-495}{2}\right)^\circ = \frac{45^\circ}{2}$, as desired.

Solution 2 (Sin-Cos Angle Addition)

The closest multiple of 60 to 2025 is $60 \cdot 33 = 1980$. 33, when divided by 12, leaves remainder 9, and thus the clock depicts hour 9. $2025 - 1980 = 45$ minutes, so the time is 9 : 45. As an hour has 60 minutes, this represents $\frac{45}{60} = 0.75$ of an hour past the 9th hour.

The angle between two successive hours on the clock face is $\frac{360^\circ}{12} = 30^\circ$, so from above, 45 minutes corresponds to $0.75 \cdot 30^\circ = 22.5^\circ$. Since double 22.5° is 45° , we can use sine and cosine double angle (or addition) identities:

$$\sin(2\theta) = \sin(\theta) \cdot \cos(\theta) + \sin(\theta) \cdot \cos(\theta) = 2 \sin(\theta) \cos(\theta)$$

and

$$\cos(2\theta) = \cos(\theta) \cdot \cos(\theta) - \sin(\theta) \cdot \sin(\theta) = \cos^2(\theta) - \sin^2(\theta).$$

Notice that by dividing $\cos(2\theta)$ by $\sin(2\theta)$, the denominator and numerator cancel a trig function, so we obtain

$$\frac{\cos(2\theta)}{\sin(2\theta)} = \frac{\cos^2(\theta) - \sin^2(\theta)}{2 \sin(\theta) \cos(\theta)},$$

which reduces to

$$\frac{2}{\tan(2\theta)} = \frac{\cos(\theta)}{\sin(\theta)} - \frac{\sin(\theta)}{\cos(\theta)}.$$

Now setting $\theta = 22.5^\circ$, and recalling that $\tan(45^\circ) = 1$, we deduce that

$$2 = \frac{1}{\tan(22.5^\circ)} - \tan(22.5^\circ).$$

Letting $x = \tan(22.5^\circ)$ and using the quadratic formula, this becomes

$$2x = 1 - x^2 \iff x^2 + 2x - 1 = 0 \iff x = \frac{-2 \pm \sqrt{4+4}}{2} = \pm\sqrt{2} - 1.$$

But we must have $x > 0$, since $0^\circ < 22.5^\circ < 90^\circ$, and thus we have $x =$

(B) $\sqrt{2} - 1$.

Solution 3 (Root Approximation, No Trig Identities)

Following from Solutions 1 and 2, we get the time to be 9 : 45 and the angle between the hour and minute hands to be 22.5° .

Moreover, we know that this angle satisfies $22.5^\circ < 30^\circ$ and thus $\tan(22.5^\circ) < \tan(30^\circ)$, where $\tan(30^\circ) = \frac{\sin(30^\circ)}{\cos(30^\circ)} = \frac{\left(\frac{1}{2}\right)}{\left(\frac{\sqrt{3}}{2}\right)} = \frac{1}{\sqrt{3}}$.

Thus we need to find an answer choice that is less than $\frac{1}{\sqrt{3}}$, and we also know that **(A)** is wrong since certainly $\tan(22.5^\circ) > 0$. The other options involve $\sqrt{2}$, which we can approximate using the 'Babylonian method':

$$x_{n+1} = \frac{x_n + \frac{S}{x_n}}{2},$$

where in this recurrence relation, x_n is the current guess for $\sqrt{S}$ and x_{n+1} would be the next approximated guess. As such, $\sqrt{2} \approx \frac{1+\frac{1}{2}}{2} = 1.5$, or if you memorized the value then you can take $\sqrt{2} \approx 1.414$. For $\sqrt{3}$ the approximation would be $\sqrt{3} \approx \frac{2+\frac{3}{2}}{2} = 1.75$. Hence, $\tan(30^\circ) = \frac{1}{\sqrt{3}} \approx \frac{1}{1.75} = \frac{4}{7} \approx 0.571$. Using these values we can check the approximate value of every answer choice (having already eliminated choice **(A)** as explained above):

$$\text{(B)} \quad \sqrt{2} - 1 \approx 1.414 - 1 = 0.414,$$

$$\text{(C)} \quad 2 - \sqrt{2} \approx 2 - 1.414 = 0.586,$$

$$\text{(D)} \quad \frac{\sqrt{2}}{2} \approx \frac{1.414}{2} = 0.707, \text{ and}$$

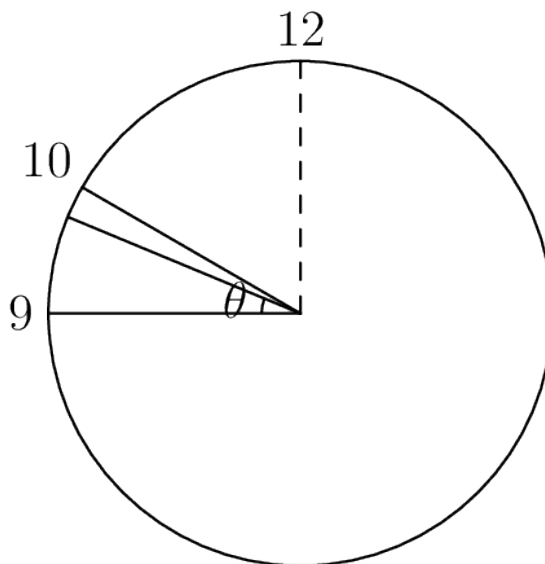
$$\text{(E)} \quad 3 - \sqrt{2} \approx 3 - 1.414 = 1.586.$$

As seen above, only choice **(B)** satisfies the criterion that the answer is less than approximately 0.571. (Even though choice **(C)** may seem to work with e.g. a $\sqrt{2} \approx 1.5$ approximation, it is clear geometrically that $\tan(22.5^\circ)$ is noticeably different from $\tan(30^\circ)$, so such a close value can't be correct.)

Accordingly, the answer is indeed **(B)** $\sqrt{2} - 1$.

Solution 4 (Very Fast)

To start, we need to find out what time of day it is. We know there are $24 \cdot 60 = 1440$ minutes in a day, so we only have $2025 - 1440 = 585$ minutes that we care about. We see that this is 60 minutes 9 times with an extra 45 minutes tacked on, so the time is 9 : 45. Thus, we have the arms of the clock face in this configuration:



Then, since the angle between any two consecutive hours is $\frac{2\pi}{12} = \frac{\pi}{6}$, we know that $\theta = \frac{45}{60} \cdot \frac{\pi}{6} = \frac{3}{4} \cdot \frac{\pi}{6} = \frac{\pi}{8}$ (which is indeed the *acute* angle between the hour hand and the minute hand, as required by the problem). Therefore, using a standard

half-angle identity, the answer is precisely

$$\tan\left(\frac{\pi}{8}\right) = \frac{\sin\left(\frac{\pi}{4}\right)}{\cos\left(\frac{\pi}{4}\right) + 1} = \frac{\left(\frac{1}{\sqrt{2}}\right)}{\left(\frac{1}{\sqrt{2}} + 1\right)} = \frac{1}{1 + \sqrt{2}} = \frac{1 \cdot (\sqrt{2} - 1)}{(1 + \sqrt{2}) \cdot (\sqrt{2} - 1)} = \frac{\sqrt{2} - 1}{2 - 1} = \boxed{\text{(B)} \sqrt{2} - 1}.$$

17. (AMC 12 12B 2025) Each of the 9 squares in a 3×3 grid is to be colored red, blue, or yellow in such a way that each red square shares an edge with at least one blue square, each blue square shares an edge with at least one yellow square, and each yellow square shares an edge with at least one red square. Colorings that can be obtained from one another by rotations and/or reflections are to be considered the same. How many different colorings are possible?

(A) 3 (B) 9 (C) 12 (D) 18 (E) 27

Answer: C

Solution 1 (Casework)

Denote 1 = red, 2 = blue, 3 = yellow. We need $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$.

WLOG place 1 in the center $(0, 0)$, 2 on the left edge $(-1, 0)$, 3 on the top-left corner $(-1, 1)$.

$$\begin{bmatrix} 3 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

3 must see 1, so the top edge $(0, 1)$ must also have 1. Then, 1 must see 2, so the top-right corner $(1, 1)$ becomes 2, which must see 3, so the right edge $(1, 0)$ must have 3.

$$\begin{bmatrix} 3 & 1 & 2 \\ 2 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

Now this bottom-right corner can vary either as 1, 2 or 3. Cases on $(1, -1)$: If 1:

$$\begin{bmatrix} 3 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 2 & 1 \end{bmatrix}$$

but the 3 needs a 1 and does not have it, so there are 0 cases. If 2:

$$\begin{bmatrix} 3 & 1 & 2 \\ 2 & 1 & 3 \\ b & a & 2 \end{bmatrix}$$

if $a = 1$, b can be 1 or 3. If $a = 2$, $b = 3$, but the 3 needs a 1 and can't get it. If $a = 3$, $b = 1, 2$, so there are 4 cases in total. If 3:

$$\begin{bmatrix} 3 & 1 & 2 \\ 2 & 1 & 3 \\ 2 & 1 & 3 \end{bmatrix}$$

Since the 2 in the bottom left corner does not have a 3 nearby, there are 0 cases.

WLOG the center fixes a factor of 3, so the answer is $4 \cdot 3 = \boxed{\text{(C)} 12}$.

Solution 2 (Complex Circular Counting)

This solution is complex, but gives the answer in less than a couple minutes and saves the bearing of casework, so proceed as a last resort.

Also yes this solution is inspired from 3Blue1Brown's Olympiad Counting video.

We will find the arrangement that contains the least possible occurrences of a color a and the most occurrences of a color b , in which we will then use the circular gap theorem $C_n A$ on all integers x such that $x \in [a, b]$ and $(a, b, x) \in \mathbb{Z}^+$.

The average number of colors M is simply $9/3 = 3$ (9 slots divided by 3 colors). So we see that the interval contains $a < 3 < b$. We also have that the minimal number of colors is clearly 1 (it doesn't take much to show a case that contains only one color). The challenge is to find the maximal number, which is b .

We see that there must be at least one of each color, so b can be a maximum of 6. However, we can see that 6 clearly cannot be b , as if we have a configuration of 6 colors, that would mean that through the constraints in the problem, the other three colors are already uniquely determined. However, now there is no slots in the grid, and we cannot satisfy all the requirements, so $b \leq 5$.

However, we also find that $b \neq 5$, as the four colors, say blue already determine 2 other colors, say yellow, which determine another 2 colors, say red. However, now there is one color left! This color must be blue (as there are 5), but if there is one blue and no yellow to satisfy the condition, then the case is simply impossible. Therefore, $b \leq 4$.

We can try to do the same thing for 4, but we will see that $b = 4$ satisfies the prerequisites.

Therefore, $a = 1$ and $b = 4$, in which we then use the circular gap theorem

$$C_n A = \frac{n}{k} \binom{n-k-1}{k-1}$$

For $n = x : x \in [a, b]$, $(a, b, x) \in \mathbb{Z}^+$ and $0 < k \leq \lfloor \frac{n}{2} \rfloor : k \in \mathbb{Z}^+$

And so, we do so!

For $n = 1$, $C_n A = 1$ (note that every $\binom{n}{0} = 1$)

For $n = 2$, $C_n A = 2$

For $n = 3$, $C_n A = 3$

For $n = 4$, $C_n A = 6$

Our answer is just the sum of all four of these, which is $6 + 3 + 2 + 1$ or (C) 12

btw the actual theorem name for $C_n A$ isn't actually the Circular Gap Theorem. I just call it that because I honestly don't know the name of it. It is a pretty neat theorem however, and I first learned it from this AIME problem here: [1983 AIME Q7 Sol. 9](#).

It is apparently a closed form for the number of ways to write n as the sum of k positive circular cycles for $n \geq 2$, and for $0 < n \leq 1$, the value is just 1.

Solution 3 (Alternative Casework)

We begin by WLOG-ing the center box as red, denoted R.

	R	

We will then consider the upper edge box, for now we'll assume it to be blue (it will be clear later)

	B	
	R	

Now we consider the upper right box, which is forced to be yellow as it would be redundant due to reflection through the middle column.

	B	Y
	R	

Now, working in a clockwise direction, the right edge box is forced to be red, the bottom corner therefore is forced to be blue, and the bottom edge is forced to be yellow.

	B	Y
	R	R
	Y	B

Now we must consider the bottom left corner. If we set the bottom left corner to red, the left edge therefore must be blue, but then nothing can be filled in for the top left box (red causes left edge to violate the blue $\rightarrow$ yellow rule, blue fails for a multitude of reasons, and yellow violates the yellow $\rightarrow$ red rule).

?	B	Y
B	R	R
R	Y	B

Yellow also doesn't work, since we get:

B	B	Y
R	R	R
Y	Y	B

but then the blue top-left corner has no adjacent yellow box. On the other hand, blue does give 4 valid configurations:

Either (1):

R	B	Y
R	R	R
B	Y	B

or (2):

Y	B	Y
R	R	R
B	Y	B

or (3):

R	B	Y
Y	R	R
B	Y	B

or (4):

B	B	Y
Y	R	R
B	Y	B

We can now explain why it's valid to assume that the upper edge box is blue: if it were yellow instead, the grid would simply be the reflection of one of the above configurations through the middle row, precisely because the bottom edge box is yellow (as this would then become the upper edge box under the reflection). If the upper edge box were red, either the upper-left or the upper-right box would have to be blue, and so by reflection through the middle column (just like above), we can assume that the upper-right box specifically is blue. Again similarly to above, we can now fill in the rest of the entire grid, giving one of the following 5 configurations:

Either (A):

Y	R	B
B	R	Y
R	R	B

or (B):

R	R	B
B	R	Y
Y	R	B

or (C):

Y	R	B
B	R	Y
Y	R	B

or (D):

Y	R	B
B	R	Y
R	Y	B

or (E):

Y	R	B
B	R	Y
B	Y	B

But (B) is simply the reflection of (A) in the bottom row, so we can eliminate it. After that, we observe that (A) is a rotation of (1) above, (C) is a rotation of (2), (D) is a rotation of (3), and (E) is a rotation of (4), so we can in fact eliminate all of these too.

Accordingly, having taken the central box as red, there are 4 distinct possible configurations (precisely (1) to (4) above), and by symmetry, that number of possible configurations will clearly be the same if the central box is blue or yellow instead. Thus there are 3 possible choices for the central box, followed by 4 ways to complete the grid in each of these cases, so the total number of possible grids is $3 \cdot 4 = \boxed{\text{(C)} \ 12}$.

18. (AMC 12 12B 2025) Awnik repeatedly plays a game that has a probability of winning of $\frac{1}{3}$. The outcomes of the games are independent. What is the expected value of the number of games he will play until he has both won and lost at least once?

(A) $\frac{5}{2}$ (B) 3 (C) $\frac{16}{5}$ (D) $\frac{7}{2}$ (E) $\frac{15}{4}$

Answer: D

Solution 1

Let the probability of a win be p and the probability of a loss be q . If the first game is a win, then we must find the expected number of further games to get a loss, which will be $\frac{1}{q}$. In addition to the first game played, there will be $1 + \frac{1}{q}$ games played. Therefore, the expected number of games needed to get a win on game 1 and then a loss is $(p) \left(1 + \frac{1}{q}\right)$.

Similarly, if the first game is a loss, we need to find the expected number of further games to get a win, which will be $\frac{1}{p}$. There will be a total of $1 + \frac{1}{p}$ games played. Therefore, the expected number of games needed to get a loss on game 1 and then a win is $(q) \left(1 + \frac{1}{p}\right)$.

The answer is

$$(p) \left(1 + \frac{1}{q}\right) + (q) \left(1 + \frac{1}{p}\right)$$

$$\left(\frac{1}{3}\right) \left(1 + \frac{3}{2}\right) + \left(\frac{2}{3}\right) (1 + 3) = \frac{5}{6} + \frac{8}{3} = \boxed{\frac{7}{2}}.$$

Solution 2 (First-step analysis)

Note that if we stop after getting at least one win and one loss, then the stopping condition is the first state change. If we start with a win, then we count the expected number of rounds until we lose and vice versa.

We use first-step analysis:

$$\mathbb{E}[T] = 1 + \sum_{s \in \Omega} \mathbb{P}(s) \mathbb{E}[T - 1 | s]$$

Here, we have $\mathbb{P}(W) = \frac{1}{3}, \mathbb{P}(L) = \frac{2}{3}$.

Using first-step analysis on each condition expectation, by the Markov property

$$\mathbb{E}[T|W] = 1 + \mathbb{P}(W)\mathbb{E}[T|W] + \mathbb{P}(L)(0) = 1 + \mathbb{P}(W)\mathbb{E}[T|W]$$

Solving, $\mathbb{E}[T|W] = \frac{1}{1-\mathbb{P}(W)} = \frac{1}{\mathbb{P}(L)}$. Similarly, $\mathbb{E}[T|L] = \frac{1}{\mathbb{P}(W)}$. Write $p = \mathbb{P}(W)$. $q = \mathbb{P}(L) = 1 - p$.

$$\mathbb{E}[T] = 1 + \mathbb{P}(W)\mathbb{E}[T|W] + \mathbb{P}(L)\mathbb{E}[T|L] = 1 + p \left(\frac{1}{q}\right) + q \left(\frac{1}{p}\right) = 1 + \frac{p}{q} + \frac{q}{p} = 1 + \frac{p^2 + q^2}{pq}$$

We can simplify using $(p + q)^2 = 1 = p^2 + q^2 + 2pq$, giving $p^2 + q^2 = 1 - 2pq$. Therefore,

$$\mathbb{E}[T] = 1 + \frac{1 - 2pq}{pq} = 1 + \frac{1}{pq} - 2 = \frac{1}{pq} - 1$$

Substituting $p = \frac{1}{3}, q = \frac{2}{3}$ yields $3 \cdot \frac{3}{2} - 1 = \boxed{\text{(D)} \frac{7}{2}}$.

Solution 3 (Geometric Series and Engineer's Induction)

We first conduct a little casework regarding the total number of games played.

2 games played: $2 * (1/3 * 2/3 + 2/3 * 1/3) = 8/9$

3 games played: $3 * ((2/3)^2 * 1/3 + (1/3)^2 * 2/3) = 18/27$

4 games played: $4 * ((2/3)^3 * 1/3 + (1/3)^3 * 2/3) = 40/81$

5 games played: $5 \cdot ((2/3)^4 \cdot 1/3 + (1/3)^4 \cdot 2/3) = 90/243$

etc...

We observe that the pairs $(8/9, 40/81)$ and $(18/27, 90/243)$ all have a common ratio of $5/9$. We speculate that the sum can be comprised of two infinite geometric series, all with a common ratio of $5/9$, one with a first term of $8/9$ and one with a first term of $18/27$. Computing the sum of the two yields 2 and $3/2$, respectively. Thus, the answer is $2 + 3/2 = 7/2$.

Note: I just realized that this method is flawed, the later cases don't match this geometric pattern, yet it still achieves the right answer

Solution 4 (Easiest with States)

Let E_0, E_{1W}, E_{1L} denote the expected number of remaining games needed from the states of no wins or losses, ≥ 1 win + no losses, ≥ 1 loss + no wins. Then we may write

$$\begin{aligned} E_0 &= 1 + \frac{1}{3}E_{1W} + \frac{2}{3}E_{1L} \\ E_{1W} &= 1 + \frac{2}{3} \cdot 0 + \frac{1}{3}E_{1W} \rightarrow E_{1W} = \frac{3}{2} \\ E_{1L} &= 1 + \frac{1}{3} \cdot 0 + \frac{2}{3}E_{1L} \rightarrow E_{1L} = 3 \end{aligned}$$

Solving gives

$$E_0 = 1 + \frac{1}{3} \cdot \frac{3}{2} + \frac{2}{3} \cdot 3 = \boxed{\frac{7}{2}}$$

Solution 5

If the game is played for n turns, we can easily observe the following two cases.

Either he wins the first $n - 1$ turns and loses on the n th turn Or he loses the first $n - 1$ turns and wins on the n th turn

This is motivated from looking at smaller cases and noting that if the goal has to be achieved in exactly n moves, he can't immediately start with winning the first turn and losing the second turn since that finishes the game in 2 turns and not necessarily n turns.

The expected value is just the probability multiplied by n .

So we have

$$n((\frac{1}{3})^{n-1} \cdot \frac{2}{3} + (\frac{2}{3})^{n-1} \cdot \frac{1}{3}) = \frac{n(2+2^{n-1})}{3^n}.$$

This expression needs to be summed from $n = 2$ to infinity.

We rewrite the expression as

$$2 \cdot n(\frac{1}{3})^n + \frac{1}{2} \cdot n(\frac{2}{3})^n$$

We claim that the standard arithmetico-geometric series nx^n has a sum of $\frac{x}{(1-x)^2}$ for $|x| < 1$.

Proof: We start with $1 + x + x^2 + \dots = \frac{1}{1-x}$. Taking the derivative with respect to x on both sides gives

$1 + 2x + 3x^2 + \dots$ which is just the sum of nx^{n-1} and this equals $\frac{1}{(1-x)^2}$. To match nx^n we multiply both sides by x and get that the sum of nx^n from $n = 1$ to infinity

is $\frac{x}{(1-x)^2}$. However, we're summing from $n = 2$ not $n = 1$ so we subtract $1 \cdot x^1 = x$ from both sides to get the desired sum of

$$\frac{x^2(2-x)}{(1-x)^2}$$

Now we plug in $x = \frac{1}{3}$ and $\frac{2}{3}$ to get

$$2 \cdot \frac{(\frac{1}{3})^2 \cdot (2 - \frac{1}{3})}{(\frac{2}{3})^2} = \frac{5}{6} \text{ for the first part of the sum.}$$

Now for the second part we have

$$\frac{1}{2} \cdot \left(\frac{\frac{2}{3}}{(1 - \frac{2}{3})^2} - \frac{2}{3} \right) = \frac{8}{3} \text{ for this second part.}$$

Finally we have

$$\frac{5}{6} + \frac{8}{3} = \boxed{\frac{7}{2}}$$

Solution 6 (Infinite Series, Sol 1 more in depth)

Denote p the probability of a win and q the probability of a loss.

Assume he begins with a win. That is a p chance. After the first game, he can lose with expectation $1q$, win then lose with expectation $2pq$, win then win then lose with expectation $3p^2q$, so on so forth. That is a

$$\sum_{k=1}^{\infty} np^{k-1}q = q \cdot \sum_{k=1}^{\infty} np^{n-1} = q \cdot \frac{1}{q^2} = \frac{1}{q}$$

expectation. In addition to the win, the expected number of games is $p(1 + \frac{1}{q})$.

Assume a loss. That is a q chance. After the first game, he can win with expectation $1p$, lose-win $2qp$, so on which gives $\frac{1}{p}$. The expected result here is $q(1 + \frac{1}{p})$.

The requested result is

$$\left(\frac{1}{3}\right)\left(1 + \frac{3}{2}\right) + \left(\frac{2}{3}\right)(1 + 3) = \frac{5}{6} + \frac{8}{3} = \boxed{\frac{7}{2}}.$$

19. (AMC 12 12B 2025) A rectangular grid of squares has 141 rows and 91 columns. Each square has room for two numbers. Horace and Vera each fill in the grid by putting the numbers from 1 through $141 \times 91 = 12,831$ into the squares. Horace fills the grid horizontally: he puts 1 through 91 in order from left to right into row 1, puts 92 through 182 into row 2 in order from left to right, and continues similarly through row 141. Vera fills the grid vertically: she puts 1 through 141 in order from top to bottom into column 1, then 142 through 282 into column 2 in order from top to bottom, and continues similarly through column 91. How many squares get two copies of the same number?

(A) 7 (B) 10 (C) 11 (D) 12 (E) 19

Answer: C

Solution 1

Let's say n is one of the numbers that got written twice in the same box. Suppose it is at column x and row y . We will write an expression for n in terms of x and y in two ways: from Horace's perspective and Vera's perspective.

From Horace's perspective, $n = (y - 1)(141) + x$. This is because there are $y - 1$ full rows before n , and we then need x more numbers to reach n . Similarly, Vera will say $n = (x - 1)(91) + y$.

We now have the Diophantine equation

$$(y - 1)(141) + x = (x - 1)(91) + y$$

$$141y - 141 + x = 91x - 91 + y$$

$$140y = 90x + 50$$

$$14y = 9x + 5$$

One such solution is, of course, $x = y = 1$. We find further solutions by adding 14 to x and 9 to y . For example, the second solution is $(15, 10)$. This continues until $(141, 91)$ is reached. There are $\boxed{11}$ ordered pairs in this list.

Solution 1 (Easier way to count)

The Diophantine equation $14y - 9x = 5$ is in the form $ax + by = c$ has primary solution $(x_0, y_0) = (1, 1)$. Also, $h = \gcd(a, b) = \gcd(14, -5) = \gcd(14, 5) = 1$, and thus the solutions are of the form

$$y = 1 + 9k$$

$$x = 1 + 14k$$

for some k .

Then, notice that because $k \geq 0$, $1 \leq x \leq 141$, and $1 \leq y \leq 91$, the largest value possible is $(x, y) = (141, 91)$, and thus

$$14k \leq 140$$

$$9k \leq 90$$

.

In which $k \leq 10$ in both cases. Thus there are 10 positive integers k that satisfy the following. Including 0, there are $10 + 1 = \boxed{11}$.

Solution 2 (0-indexing)

For these types of problems, we essentially analyze residues, so it is more convenient to work with 0-indexing: $c = 0, 1, 2, \dots, 90$, $r = 0, 1, 2, \dots, 140$.

Then $n = 91r + c + 1 = 141c + r + 1 \iff 9r = 14c$. We must have $(r, c) = (0, 0), (14, 9), (28, 18), \dots, (140, 90)$ so $\boxed{\text{(C) } 11}$.

Solution 2 (faster ending)

After indexing to 0, we imagine a line that passes from $(0, 0)$ to $(140, 90)$. We know we are looking for all lattice points that lie on this line (this intuitively makes sense, because any points above the line would have column numbers far ahead of the rows, and vice versa).

Therefore, we know that the number of points must divide both 140 and 90. The GCD of the two is 10. We also account for point $(0, 0)$, and get that $10 + 1 = \boxed{\text{(C) } 11}$.

Solution 3

Observe that Horaces number can be expressed as the remainder of the row number plus the column number so $H(\text{Horaces Number}) = 91(c - 1)(\text{column addition}) + r(\text{remainder of division})$ and do the same for Vera $V = 141(r - 1) + c$. The condition is $H = V$ so set the equations equal to each other. $91(c - 1) + r = 141(r - 1) + c$ so $91c - 91 + r = 141r - 141 + c$, $140r = 90c + 50$. This is a linear Diophantine equation and the 1st solution is $(1, 1)$ where the solution repeats every $\text{lcm}(90, 140) = 1260$ there are 11 solutions so **(C)11** is correct.

20. (AMC 12 12B 2025) A frog hops along the number line according to the following rules.

- It starts at 0.
- If it is at 0, then it moves to 1 with probability $\frac{1}{2}$ and it disappears with probability $\frac{1}{2}$.
- For $n = 1, 2$, or 3 , if it is at n , then it moves to $n + 1$ with probability $\frac{1}{4}$, it moves to $n - 1$ with probability $\frac{1}{4}$, and it disappears with probability $\frac{1}{2}$.

What is the probability that the frog reaches 4? **(A)** $\frac{1}{101}$ **(B)** $\frac{1}{100}$ **(C)** $\frac{1}{99}$ **(D)** $\frac{1}{98}$ **(E)** $\frac{1}{97}$

Answer: E

Solution 1 (conditional probabilities of reaching 4)

We will solve this using the **law of total probability** (otherwise apparently known in Olympiad folklore as the method of "states", presumably reflecting the fact that these are kids who won't yet have studied formal/axiomatic probability theory). Firstly, we note that the probabilities of the frog taking each of its possible moves always depend only on its current position, and not on how it got there - in other words, the frog, just like coins or dice, 'has no memory'. Therefore, we can unambiguously define p_n as the probability of reaching 4 given that the frog is currently at position n . We want to find p_0 (since the frog starts at 0), and of course $p_4 = 1$.

For each value of n , we now consider the conditional probabilities of reaching 4 given that the frog takes each of the possible moves from its current position. For instance, if the frog is at 3, it can go to $3 - 1 = 2$ with probability $\frac{1}{4}$, and if it does, the conditional probability of it now reaching 4 from its new position at 2 is p_2 by definition; or it can go to $3 + 1 = 4$, also with probability $\frac{1}{4}$, and if it does, the conditional probability is of course $p_4 = 1$; or it can disappear with probability $\frac{1}{2}$, and if it does, it can never reach 4, i.e. the conditional probability is 0.

By analogous arguments at each of the other possible values of n , and then by applying the law of total probability, we deduce that

$$\begin{aligned} p_3 &= \frac{1}{4}p_4 + \frac{1}{4}p_2 \\ p_2 &= \frac{1}{4}p_3 + \frac{1}{4}p_1 \\ p_1 &= \frac{1}{4}p_2 + \frac{1}{4}p_0, \text{ and} \\ p_0 &= \frac{1}{2}p_1. \end{aligned}$$

Finally, by solving this system of equations, it follows that $p_0 = \boxed{\text{(E)} \frac{1}{97}}$ (and incidentally, the probabilities for the frog's other possible starting positions are $p_1 = \frac{2}{97}, p_2 = \frac{7}{97}$, and $p_3 = \frac{26}{97}$).

Solution 2 (conditional probabilities of taking single steps)

Let p_n be the probability that the frog eventually reaches position n , given that it is at position $n - 1$. (This is well-defined because the frog 'has no memory', just as explained in Solution 1.)

From position 0, the frog moves to 1 with probability $\frac{1}{2}$, or disappears with probability $\frac{1}{2}$. Hence $p_1 = \frac{1}{2}$.

From position 1, the frog moves to 2 with probability $\frac{1}{4}$, back to 0 with probability $\frac{1}{4}$, or disappears with probability $\frac{1}{2}$. Thus, by the law of total probability (again similarly to in Solution 1) and using $p_1 = \frac{1}{2}$ from above, we obtain

$$p_2 = \frac{1}{4} + \frac{1}{4} \cdot \frac{1}{2} p_2 = \frac{1}{4} + \frac{1}{8} p_2 \iff \frac{7}{8} p_2 = \frac{1}{4} \iff p_2 = \frac{2}{7}.$$

We also now observe that if the frog is currently at 1 and wants to eventually reach 3, then it will necessarily have to pass through 2 first, since all of its possible moves only increase or decrease its position by 1. Accordingly, if the frog is at 2 and chooses to move back to 1, then the conditional probability that it eventually reaches 3 from its new position at 1 is equivalent to the probability that it eventually reaches 2 starting at 1 *and then* eventually reaches 3 starting at 2. That in turn reduces to the product of these two conditional probabilities: that of reaching 2 starting at 1 is $p_2 = \frac{2}{7}$ from above, while that of reaching 3 starting at 2 is p_3 .

Using the law of total probability (analogously to above) at position 2, it follows that

$$p_3 = \frac{1}{4} + \frac{1}{4} \cdot \frac{2}{7} p_3 = \frac{1}{4} + \frac{1}{14} p_3 \iff \frac{13}{14} p_3 = \frac{1}{4} \iff p_3 = \frac{7}{26}.$$

Now from position 3, using $p_3 = \frac{7}{26}$ as just found above, we deduce that

$$p_4 = \frac{1}{4} + \frac{1}{4} \cdot \frac{7}{26} p_4 \iff \frac{97}{104} p_4 = \frac{1}{4} \iff p_4 = \frac{26}{97}.$$

Lastly, again because all of the frog's possible moves only increase or decrease its position by 1 (as explained above), the required probability that it eventually reaches 4 starting at 0 is equivalent to the probability that it eventually reaches 1 starting at 0, *then* reaches 2 starting at 1, *then* 3 starting at 2, *and then* 4 starting at 3. Similarly to above, this reduces to the product of the conditional probabilities

$$p_1, p_2, p_3, \text{ and } p_4, \text{ i.e. } \frac{1}{2} \cdot \frac{2}{7} \cdot \frac{7}{26} \cdot \frac{26}{97} = \boxed{\text{(E)} \frac{1}{97}}.$$

Solution 3 (recurrence relation)

As in Solution 1, let p_n be the probability of reaching 4 given that the frog is currently at position n , and observe that for $n = 1, 2, 3$, the frog can either go to position $n + 1$ with probability $\frac{1}{4}$, in which case the conditional probability of it reaching 4 is precisely p_{n+1} by definition; or position $n - 1$, also with probability $\frac{1}{4}$, giving a conditional probability of reaching 4 of p_{n-1} , again by definition; or else it disappears with probability $\frac{1}{2}$, in which case the conditional probability of reaching 4 is obviously 0. Therefore, by the law of total probability, we have the recurrence relation

$$p_n = \frac{1}{4} p_{n+1} + \frac{1}{4} p_{n-1} + \frac{1}{2} \cdot 0 \iff 4p_n = p_{n+1} + p_{n-1} \iff p_{n+1} - 4p_n + p_{n-1} = 0, \quad \text{for } n = 1, 2, 3.$$

Moreover, one boundary condition is obviously $p_4 = 1$, while the other concerns p_0 : since the frog, starting at position 0, moves to 1 with probability $\frac{1}{2}$, in which case the conditional probability of it reaching 4 will be precisely p_1 , or else disappears, also with probability $\frac{1}{2}$, the law of total probability again gives

$$p_0 = \frac{1}{2} \cdot p_1 + \frac{1}{2} \cdot 0 \iff 2p_0 = p_1.$$

We can now simply use standard methods to solve the above recurrence relation, subject to these two boundary conditions.

Accordingly, the characteristic equation for this recurrence relation is $\lambda^2 - 4\lambda + 1 = 0 \iff (\lambda - 2)^2 = 2^2 - 1 \iff \lambda = 2 \pm \sqrt{3}$, so its general solution is given by

$$p_n = A(2 + \sqrt{3})^n + B(2 - \sqrt{3})^n$$

for some constants A and B . Imposing the boundary conditions now yields

$$2p_0 = p_1 \iff 2(A \cdot 1 + B \cdot 1) = A(2 + \sqrt{3}) + B(2 - \sqrt{3}) \iff A\sqrt{3} - B\sqrt{3} = 0 \iff A = B,$$

and hence

$$p_4 = 1 \iff A(2 + \sqrt{3})^4 + A(2 - \sqrt{3})^4 = 1 \iff A(97 + 56\sqrt{3} + 97 - 56\sqrt{3}) = 1 \iff A = B = \frac{1}{194}.$$

This means the particular solution is precisely

$$p_n = \frac{1}{194}(2 + \sqrt{3})^n + \frac{1}{194}(2 - \sqrt{3})^n = \frac{1}{194}((2 + \sqrt{3})^n + (2 - \sqrt{3})^n),$$

and so, again as in Solution 1, the required probability is $p_0 = \frac{1}{194} \cdot (1 + 1) =$

$$\boxed{\text{(E)} \quad \frac{1}{97}}.$$

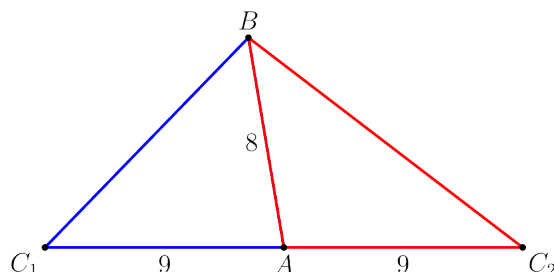
Ebai

21. (AMC 12 12B 2025) Two non-congruent triangles have the same area. Each triangle has sides of length 8 and 9, and the third side of each triangle has integer length. What is the sum of the lengths of the third sides?

(A) 20 (B) 22 (C) 24 (D) 26 (E) 28

Answer: C

Solution 1



Let the angles between the 8 and 9 sides be θ_1 and θ_2 . Relating the two triangles' areas, we get

$$\frac{1}{2} \cdot 8 \cdot 9 \cdot \sin \theta_1 = \frac{1}{2} \cdot 8 \cdot 9 \cdot \sin \theta_2$$

$$\sin \theta_1 = \sin \theta_2$$

For this to be true, $\theta_1 + \theta_2 = 180$. We will now apply Law of Cosines to get an expression for s_1 and s_2 in terms of θ_1 and θ_2 .

$$s_1^2 = 8^2 + 9^2 - (2)(8)(9) \cos(\theta_1)$$

$$s_2^2 = 8^2 + 9^2 - (2)(8)(9) \cos(180 - \theta_1).$$

Noting that $\cos \theta_1 = -\cos(180 - \theta_1)$, we can add the two equations to get that

$$s_1^2 + s_2^2 = 290.$$

We now need two perfect squares that add to 290. After some analysis, we find $121 + 169 = 290$. Therefore, $s_1 = 11$ and $s_2 = 13$, so our answer is $11 + 13 = \boxed{24}$. Note that the $\cos \theta_1 = 1/6$. Also, note that $289 + 1 = 290$, but 17 and 1 are not valid side lengths due to the Triangle Inequality.

Solution 2 (Heron)

Let the third sides be a and b . Say the third sides is x , where x is a dummy variable. Then, by Heron's Formula, the area of the triangle is

$$\sqrt{\frac{x+17}{2} \cdot \frac{x+1}{2} \cdot \frac{x-1}{2} \cdot \frac{17-x}{2}} = \frac{1}{4} \sqrt{(17^2 - x^2)(x^2 - 1^2)}$$

Since the areas are the same, this value when $x = a$ is the same as this value when $x = b$. In other words,

$$\frac{1}{4} \sqrt{(17^2 - a^2)(a^2 - 1^2)} = \frac{1}{4} \sqrt{(17^2 - b^2)(b^2 - 1^2)}$$

equivalent to

$$(289 - a^2)(a^2 - 1) = (289 - b^2)(b^2 - 1) = c$$

for some constant c .

But if we consider the quadratic

$$(289 - y)(y - 1) = c$$

we know that the two roots must be $y = a^2$ and $y = b^2$. Expanding the quadratic, we get

$$y^2 - 290y + 289 + c = 0$$

By Vieta's Formulas, we must have $a^2 + b^2 = 290$. Casework yields $a = 11, b = 13$ (or vice versa) as the only solutions, so the sum is $a + b = \boxed{(C) 24}$.

Solution 3 (Slight difference to Solution 1)

Denote the angle that is between side lengths 8 and 9 as angle A , the other triangle will be $180^\circ - A$, because by the area formula of $\frac{1}{2}ab \sin C$, the $\sin C$ needs to remain constant while a and b stay the same to keep the area the same. Denote the shorter third side as x and longer third side as y .

By applying the Law of Cosines, $y^2 = 8^2 + 9^2 - 2 \cdot 8 \cdot 9 \cdot \cos(180 - A)$, which gives $y^2 = 145 + 144 \cos A$. Similarly, $x^2 = 145 - 144 \cos A$. Since the question says the third lengths are both integers, try testing values of $\cos A$. Notice when $\cos A = \frac{1}{6}$, $y^2 = 169$ and $x^2 = 121$, giving the lengths as 13 and 11 respectively. The required answer is then $13 + 11 = \boxed{(C) 24}$.

Solution 4 (Sine/Cosine Law + Quadratic Residues)

As in other solutions, use sine law to check $\beta = 180 - \alpha$, $\cos \alpha = -\cos \beta$. Apply cosine law $x^2, y^2 = 8^2 + 9^2 \mp 2(8)(9) \cos \alpha = 145 \mp 144 \cos \alpha$ and verify integer solutions to $x^2 + y^2 = 2(8^2 + 9^2) = 290$.

Since $\alpha > 0$, we need $\cos \alpha < 1$ so $(1, 289) = (1^2, 17^2)$ is not valid. WLOG $2 \leq x < y \leq 16$. To help with the bash, note the quadratic residues mod 3 and 4 are 0, 1, so it must be $1 + 1 \equiv 2$, so we only need to check odd cases not divisible by 3, so $x = 5, 7, 11, 13$.

$290 - 25 = 265$ bad

$290 - 49 = 241$ bad

$290 - 121 = 169$ good, stop by symmetry.

So $(x, y) = (11, 13)$, giving $x + y = 11 + 13 = \boxed{\text{(C) } 24}$.

Solution 5 (Simple Tabling)

Consider arbitrary such triangle with third side x . By Heron's formula, the area of this triangle is

$$\sqrt{\left(\frac{17+x}{2}\right)\left(\frac{17-x}{2}\right)\left(\frac{1+x}{2}\right)\left(\frac{-1+x}{2}\right)}$$

By the triangle inequality, we have $2 \leq x \leq 16$, which means there are few enough possibilities of x that tabling becomes a viable strategy. Consider setting up a table such as:

Row	$17 - x$	$x + 17$	$x + 1$	$x - 1$
A_1	15	19	3	1
A_2	14	20	4	2
A_3	13	21	5	3
A_4	12	22	6	4
A_5	11	23	7	5
A_6	10	24	8	6
A_7	9	25	9	7
A_8	8	26	10	8
A_9	7	27	11	9
A_{10}	6	28	12	10
A_{11}	5	29	13	11
B_1	4	30	14	12
B_2	3	31	15	13
B_3	2	32	16	14
B_4	1	33	17	15

As x goes from $2 \rightarrow 16$, we know the area will increase, peak, and then decrease. Thus, the two possible side lengths with the same area must occur on opposite sides of the side length s that maximizes the area. The area is maximized when 8 and 9 are legs of a right triangle (Why?*), giving a side length of $\sqrt{8^2 + 9^2} = \sqrt{145}$. Thus, $12 < s < 13$, and we can divide the table in two parts, split at $x = s$. Denote the top part A and the bottom part B .

The final step is to find some rows A_i and B_j of the table that match in product. This is easier than it seems at first if we proceed by eliminating rows from B :

- B_2 contains a factor of 31, which is not present in A
- B_3 contains 11 factors of 2, which does not occur for any row in A

- B_4 contains a factor of 17, which is not present in A

Thus, B_1 is the only valid row remaining, occurring at $x = 13$. We quickly** find the matching row in A to be A_{10} , occurring at $x = 11$. Thus, we arrive at an answer of $13 + 11 = \boxed{\text{(C) } 24}$.

(*) Think about how you would maximize an area using $\frac{1}{2}ab\sin(C)$.

(**) As B_1 is so close to the maximum, it is likely that the corresponding row in A will be as well, so if we start at A_{11} and work our way up, we find A_{10} on our second try.

Solution 6 (Easiest Accessible to Everyone but Tedious with only pythagorean therom and little trigonometry involved)

Let's set up a diagram with the side of length 9 being the base. The side of length 8 is placed at an angle α ($0 < \alpha < 90^\circ$) to the base. If we drop a perpendicular from the top vertex to the base of 9, we form a right triangle where the height is $8\sin\alpha$ and the remaining piece of the base for the third side is $9 - 8\cos\alpha$.

By the Pythagorean theorem, the third side of this first triangle is the hypotenuse, which evaluates to:

$$\sqrt{(9 - 8\cos\alpha)^2 + (8\sin\alpha)^2} = \sqrt{81 - 144\cos\alpha + 64(\cos^2\alpha + \sin^2\alpha)} = \sqrt{145 - 144\cos\alpha}$$

To construct another non-congruent triangle with the exact same area, we can reflect the side of length 8 across that perpendicular height (Why?*). Using a similar method with the Pythagorean theorem, the base becomes $9 + 8\cos\alpha$, and the third side of this second triangle is:

$$\sqrt{(9 + 8\cos\alpha)^2 + (8\sin\alpha)^2} = \sqrt{145 + 144\cos\alpha}$$

Because the third side is strictly an integer length, $145 - 144\cos\alpha$ must be one of the 12 perfect squares from 1 to 144 (since $\alpha > 0$ and $\alpha < 90^\circ$). We can use brute-force trial and error to find a value of $144\cos\alpha$ that makes both expressions perfect squares. Testing our options, two possibilities for $144\cos\alpha$ emerge that satisfy both:

- $144\cos\alpha = 144$ yields third side lengths of $\sqrt{145 - 144} = 1$ and $\sqrt{145 + 144} = 17$

- $144\cos\alpha = 24$ yields third side lengths of $\sqrt{145 - 24} = 11$ and $\sqrt{145 + 24} = 13$

However, 1 and 17 do not work because they violate the triangle inequality, forming degenerate triangles ($8 + 1 = 9$ and $8 + 9 = 17$). Thus, 13 and 11 are the only possible two valid side lengths. We arrive at an answer of $11 + 13 = \boxed{\text{(C) } 24}$.

(*) This preserves the base of 9 and the height of $8\sin\alpha$, ensuring the area remains identical! Man, what can I say? Mamba out. When I see you again.

22. (AMC 12 12B 2025) What is the greatest possible area of the triangle in the complex plane with vertices $2z$, $(1+i)z$, and $(1-i)z$, where z is a complex number satisfying $|4z - 2| = 1$?

- (A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) $\frac{9}{16}$ (D) $\frac{3}{4}$ (E) 1

Answer: C

Solution 1

Because we can factor z from each vertex, the area of the triangle is equal to $|z|^2$ multiplied by the area of the triangle with vertices $2, 1+i, 1-i$.

The vertical side of that triangle has length 2 and the altitude to that side has length 1, so the area is 1. We now just need the maximum possible value of $|z|^2$.

The equation $|4z - 2| = 1$ can be rewritten as $|z - 1/2| = 1/4$. Therefore, it exists as a circle with radius $1/4$ and center at $(1/2, 0)$ on the complex plane. The maximum possible magnitude of a complex number on this circle occurs with $\frac{3}{4}$, with magnitude $\frac{3}{4}$. We can justify this because the line connecting the origin and the complex number $\frac{3}{4}$ goes through the center of the circle.

Our answer is then $\left(\frac{3}{4}\right)^2 = \boxed{\frac{9}{16}}$.

Solution 2

Rewrite the second and the third complex number, we get: $\sqrt{2}e^{i\frac{\pi}{4}}z, \sqrt{2}e^{-i\frac{\pi}{4}}z$. This means that the other two complex numbers both form a 45° angle with complex numbers both form $2z$. Also notice that the magnitude of the first complex number is $2|z|$, and the second and third complex numbers both have magnitude $\sqrt{2}|z|$, this means that these three points form an isosceles right triangle. Therefore the area for this triangle is simply just $\frac{1}{2} \cdot \sqrt{2}|z| \cdot \sqrt{2}|z| = |z|^2$. Now since we have $|4z - 2| = 1$, we can set

$$4z - 2 = e^{i\theta}, \theta \in [0, 2\pi)$$

Therefore

$$z = \frac{1}{2} + \frac{e^{i\theta}}{4}$$

,

$$|z| = \frac{1}{4}|2 + e^{i\theta}| = \frac{1}{4}\sqrt{5 + 4\cos\theta} \leq \frac{3}{4}$$

$\Rightarrow$

$$Area \leq \boxed{(C) \frac{9}{16}}$$

23. (AMC 12 12B 2025) Let S be the set of all integers $z > 1$ such that for all pairs of nonnegative integers (x, y) with $x < y < z$, the remainder when $2025x$ is divided by z is less than the remainder when $2025y$ is divided by z . What is the sum of the elements of S ?

(A) 3041 (B) 3542 (C) 3750 (D) 4044 (E) 4319

Answer: E

Solution 1

Notice that there are z distinct remainders modulo z . However, if we let $R(x)$ denote the remainder when $2025x$ is divided by z , we know that by the problem condition,

$$R(0) < R(1) < R(2) < \cdots < R(z-1)$$

But there are only z distinct remainders, and each of the z terms above must correspond to a distinct remainder, so we must have $R(k) = k$ for all $k = 0, 1, \dots, z-1$. Then $2025k \equiv k \pmod{z}$, so $2024k \equiv 0 \pmod{z}$. Since k can vary, this is equivalent to $2024 \equiv 0 \pmod{z}$, or $z \mid 2024$.

Therefore, the values z that satisfy the property are the factors of 2024, so we simply need to find the sum of the factors of 2024 and subtract 1 at the end since $z \neq 1$. But $2024 = 2^3 \cdot 11 \cdot 23$, so the sum of the factors minus 1 is

$$(1 + 2 + 4 + 8)(1 + 11)(1 + 23) - 1 = 15 \cdot 12 \cdot 24 - 1 = 4320 - 1 = \boxed{\text{(E) } 4319}.$$

Solution 2 (Fast)

Notice that $2025x \equiv x \pmod{z} \forall z \mid 2024$. Since $x < y < z$, all pairs (x, y) satisfy the remainder condition when $z \mid 2024$ except for $z = 1$. Recall that $2024 = 2^3 \cdot 11 \cdot 23$, so $\sum z = (15)(12)(24) - 1 = \boxed{\text{(E) } 4319}$, which is the largest answer choice.

Solution 3 (Rigorous)

We will show that the z 's satisfying the statement are actually all $z > 1$ and $2025 \equiv 1 \pmod{z}$.

We first provide the construction. Let the remainder of 2025 divided by z be p . We have $p < z$. Assume that $p \geq 1$ in the following steps.

$$2025x \equiv px \pmod{z}.$$

$$2025y \equiv py \pmod{z}.$$

The aim of the construction is to make the remainder of px upon division by z as large as possible, and to make the remainder of py upon division by z as small as possible.

The Construction

Set x to be the largest positive integer satisfying $px < z$.

Set y equal to $x + 1$.

Proof of why the construction works

$$py = p(x + 1) \geq z.$$

Now calculate the remainder of $2025y$ divided by z . We have $p(x + 1) = px + p \leq 2px < 2z$, so the remainder of $p(x + 1)$ upon division by z is simply $p(x + 1) - z = px + p - z < px$.

The remainder of $2025x$ divided by z is px , while the remainder of $2025y$ divided by z is less than px . This is a success in finding a counterexample, but we have to ensure that this pair of (x, y) satisfies the condition $x < y < z$.

Since $px < z$, $x < \frac{z}{p}$, $y < \frac{z}{p} + 1$. If $p \geq 2$, we have $y < \frac{z}{2} + 1$, $y < \frac{z}{2} + 1 < z$ for all $z \geq 3$. For $z = 2$, a simple manual check of $x = 0$ and $y = 1$ works as this is the only possible pair.

This explains why the construction fails when $p = 1$. If $p = 1$ we have $x = z - 1$ and $y = z$, but the condition requires $y < z$.

In fact, for $p = 1$, $2025x \equiv x \pmod{z}$, $2025y \equiv y \pmod{z}$. The remainders are simply x and y themselves, the problem statement is automatically true for any pair of (x, y) with $x < y$.

For $p = 0$, $2025x \equiv 0 \pmod{z}$, $2025y \equiv 0 \pmod{z}$. The statement is always false because both remainders are always 0.

In the end, we identify and add all the possible values of z together.

$$2025 = kz + 1 \text{ for some positive integer } k \implies kz = 2024 \implies kz = 2^3 \times 11 \times 23$$

In the end, do not forget that $z = 1$ is not included. The answer is $4320 - 1 = 4319$.

24. (AMC 12 12B 2025) How many real numbers satisfy the equation $\sin(20\pi x) = \log_{20}(x)$?
- (A) 199 (B) 200 (C) 398 (D) 399 (E) 400

Answer: D

Solution 1

Let $f(x) = \sin(20\pi x)$ and $g(x) = \log_{20}(x)$. Note that g passes through $(\frac{1}{20}, -1)$ and $(1, 0)$ and $(20, 1)$; these are the extrema and midpoint of f . We want to find the number of intersections of f and g .

Let an occurrence of sine passing under the x -axis a *down-dip*, and similarly define an *up-dip*. We find that the period of f is $\frac{1}{10}$, so between $x = \frac{1}{20}$ and $x = 1$ the number of periods is 9.5. The first period indeed counts, so effectively we have 10 down-dips in this interval. Each down-dip contributes 2 to the total, so we have 20 total intersections.

From $x = 1$ to $x = 20$, there are 190 periods, each of which also contributes 2 to the total due to the up-dips. Therefore, this case contributes 380 points to the total.

But $(1, 0)$ is counted in both cases, so the total is $20 + 380 - 1 = \boxed{\text{(D)} 399}$.

Solution 2 (AoPS Precalculus)

Note that the number of solutions to $\sin(20\pi x) = \log_{20}(x)$ is the same as asking the number of intersection points of the graphs $\sin(20\pi x)$ and $\log_{20}(x)$.

We begin by understanding the range of $\sin(20\pi x)$. The range is just $-1 \leq \sin(20\pi x) \leq 1$.

For the graph $\log_{20}(x)$ to intersect $\sin(20\pi x)$, we need $\log_{20}(x)$ to also be in the range $-1 \leq \log_{20}(x) \leq 1$, which is simplified into $\frac{1}{20} \leq x \leq 20$.

The period of $\sin(20\pi x)$ is $\frac{2\pi}{20\pi} = \frac{1}{10}$. Thus in every 1 unit, there are 10 periods.

Also, the x intersections of both graphs are the same. $\sin(20\pi x) = 0$ indicates $x = \frac{1}{2} + k$. Also, $\log_{20}(x) = 0$ if $x = 1$, so we have $k = 0$ and we must consider two cases, either $\frac{1}{20} \leq x \leq 1$ and $1 < x \leq 20$.

Case 1: $\frac{1}{20} \leq x \leq 1$.

For this case, we first must understand one thing. Does $\log_{20}(x)$ intersect the first time $\sin(20\pi x)$ dips below the x -axis for $x \geq 0$ 0 times, 1 time, or 2 times?

Well, notice that if we plug in $x = \frac{1}{20}$, $\sin(20\pi x) = 0$ and $\log_{20}(x) = -1$, so because $\log_{20}(x)$ is increasing, it will intersect the graph of $\sin(20\pi x)$ twice.

Taking this into consideration, in each period, half the period is positive whereas the other half is negative. Then, because there are 10 periods between $\frac{1}{20} \leq x \leq 1$, there will be 10 negative periods.

Then, the graph $\log_{20}(x)$ intersects these 10 periods twice, giving us a total of 20 intersections in this case.

Case 2: $1 < x \leq 20$.

For this case, we see that there is a total of 10 periods per unit, and there are $20 - 1 = 19$ units. Then, because once again there are 10 positive intervals, $\log_{20}(x)$ will intersect each interval twice, giving us a total of 20 intersections per one unit. Multiplying by 19, we get a total of $19 \times 20 = 380$ intersections. However, we have excluded 1, and the first period lasting $1 \leq x \leq \frac{11}{10}$, we see that two intersections

occur, one at the point (1,0) and another at some random point. We already considered the point (1,0) last case, so we subtract 1 from the total to obtain $380 - 1 = 379$.

Thus the total number of solutions to the equation is $20 + 379 = \boxed{\text{(D) } 399}$.

Solution 3 (quick but risky)

Each period of the sin function is $\frac{1}{10}$, so 200 periods, 2 intersections each (just draw it out) - 400. But there is 1 overlap in two periods at (1, 0), so subtract 1 -> 399 (can check end behavior if time) ScoutViolet

Solution 4 (Unit-period parametrization)

Note that $-1 \leq \sin 20\pi x = \log_{20} x \leq 1$, forcing $\frac{1}{20} \leq x \leq 20$. For simplicity, let $2\pi t = 20\pi x$ or $t = 10x \in [\frac{1}{2}, 200]$. Since $2\pi t$ is the unit-period parametrization, there are 200 full periods on $[0, 200]$, and we check the closed intervals $[\frac{1}{2}, 1], [1, 2], \dots, [199, 200]$. By sketching the graph, we see that when sine and log share the same sign, the sine will weave through the log twice for the non-degenerate case. The endpoints are also non-degenerate after verifying $\log(\frac{1}{20}) = -1, \log_{20} 20 = 1$ whereas $\sin 20\pi(\frac{1}{20}) = \sin 20\pi(200) = 0$.

Because $\log_{20}(x = 1) = 0 = \log_{20}(t = 10)$ lands perfectly on an integer, by the intermediate value theorem, log will only have at most two signs among $-, 0, +$ in each full period closed interval. We count $200 \cdot 2 = 400$ intersections. However, $t = 10$ is shared among $[9, 10]$ and $[10, 11]$ so we must subtract the overlap to get $400 - 1 = \boxed{\text{(D) } 399}$.

Solution 5 (Risky)

We know that the period of $\sin(20\pi x)$ is $\frac{2\pi}{20\pi} = \frac{1}{10}$, and that the amplitude of the graph is 1. Now, we need to find the values of x for which the graph of $y = \log_{20} x$ intersects with the upper and lower extrema of $\sin(20\pi x)$, or with $y = 1$ and $y = -1$. At $y = -1$, $x = 20^{-1} = \frac{1}{20}$, and at $y = 1$, $x = 20^1 = 20$. We know that for each period of the sin graph, there will be 2 intersections, and there are $\frac{20}{\frac{1}{10}} = 200$ periods of the sin graph from 0 to 20. Because there are 3 answer choices close to 400, we can test with smaller bounds using a line, and count the points. The line from $(\frac{1}{20}, -1)$ to $(\frac{1}{2}, 1)$ intersected the graph $y = \sin(20\pi x)$ 9 times, and there were 5 periods between $\frac{1}{2}$ and 0, thus we can say the answer is $\boxed{\text{(D) } 399}$

25. (AMC 12 12B 2025) Three concentric circles have radii 1, 2, 3. An equilateral triangle with side length s has one vertex on each circle. What is s^2 ?
 (A) 6 (B) $\frac{25}{4}$ (C) $\frac{13}{2}$ (D) $\frac{27}{4}$ (E) 7

Answer: E

Solution 1

Let $\triangle ABC$ and center O be such that $OA = 1, OB = 2, OC = 3$. Suppose the length of the side of ABC is s . Noticing that $OA + OB = OC$, we suspect that $OACB$ is a cyclic quadrilateral. A quadrilateral is cyclic if and only if Ptolemy's Theorem holds true:

$$OA \cdot CB + OB \cdot AC \stackrel{?}{=} OC \cdot AB \tag{43}$$

$$1 \cdot s + 2 \cdot s = 3 \cdot s. \tag{44}$$

Because Ptolemy's is true, $OACB$ is cyclic. By properties of cyclic quadrilaterals, $\angle AOB = 180^\circ - \angle BCA = 120^\circ$. Applying Law of Cosines on AOB , we get

$$s^2 = 1^2 + 2^2 - 2 \cdot 1 \cdot 2 \cdot \cos 120^\circ \quad (45)$$

$$= 1 + 4 + 2 = \boxed{7}. \quad (46)$$

Solution 2

Call the point that lies on the circle $r = 1$ as A , $r = 2$ as B and $r = 3$ as C .

We could set $A = (1, 0)$, $B = (2 \cos B, 2 \sin B)$ and $C = (3 \cos C, 3 \sin C)$ $s = BA = CB = CA$.

$$BA = (2 \cos B - 1)^2 + (2 \sin B)^2 = 5 - 4 \cos B$$

$$CA = (3 \cos C - 1)^2 + (3 \sin C)^2 = 10 - 6 \cos C$$

$$CB = (3 \cos C - 2 \cos B)^2 + (3 \sin C - 2 \sin B)^2 = 13 - 12 \cos(B - C)$$

Let $\cos B = x$ and $\cos C = y$ for easier representation

Because $5 - 4 \cos B = 10 - 6 \cos C$, hence $x = \frac{6y-5}{4}$

Also, $5 - 4 \cos B = 13 - 12 \cos(B - C)$, which $12 \cos(B - C) = 8 + 4 \cos B = 8 + 4 \cdot \frac{6y-5}{4} = 3 + 6y$, leading to $\cos(B - C) = \frac{1+2y}{4}$

Noting that

$$\cos(B - C) = xy + \sqrt{1-x^2}\sqrt{1-y^2} = \frac{1+2y}{4}$$

$$1 - x^2 = 1 - \left(\frac{6y-5}{4}\right)^2 = \frac{-36y^2 + 60y - 9}{16}$$

let $K = \sqrt{1-x^2}\sqrt{1-y^2}$, which the equation $xy + \sqrt{1-x^2}\sqrt{1-y^2} = \frac{1+2y}{4}$ can be simplified to $\frac{y(6y-5)}{4} + K = \frac{1+2y}{4}$

By rearranging, the equation $\frac{y(6y-5)}{4} + K = \frac{1+2y}{4}$ turns into $K = \frac{-6y^2+7y+1}{4}$

Squaring both sides, which

$$K^2 = (1 - x^2)(1 - y^2) = \frac{(-6y^2 + 7y + 1)^2}{16}$$

Substituting in $1 - x^2 = \frac{-36y^2+60y-9}{16}$, we get

$$(-6y^2 + 7y + 1)^2 = (-36y^2 + 60y - 9)(1 - y^2)$$

By expanding, $LHS = 36y^4 - 84y^3 + 37y^2 + 14y + 1$ and $RHS = 36y^4 - 60y^3 - 27y^2 + 60y - 9$

Rearrange it, we get

$$12y^3 - 32y^2 + 23y - 5 = 0$$

By inspection (which y cant be greater than 1 because it is a cosine value), let's try $y = \frac{1}{2}$: it is a root.

By polynomial division, this becomes

$$(2y - 1)(6y^2 - 13y + 5) = 0$$

The other 2 solutions from the quadratic are $\frac{1}{2}$ and $\frac{5}{3}$ (greater than 1, rejected), meaning that the only y possible value is $\frac{1}{2}$.

Because length of $CA = (3 \cos C - 1)^2 + (3 \sin C)^2 = 10 - 6 \cos C$, hence the answer is $10 - 6(1/2) = 10 - 3 = \boxed{\text{(E)} 7}$.

Solution 3 (rotation)

Let A be a point on the circle with radius 1, B be a point on the circle with radius 2, C be a point on the circle with radius 3. Call the shared center for these three circles O . First let's check if O is in $\triangle ABC$. Rotate OA clockwise by 60° to OA' , then it is equivalent to check if point A' is outside $\triangle ABC$. Assume that A' is outside $\triangle ABC$, easy to prove that $\triangle AOC \cong \triangle AA'B$. Therefore $OC = A'B = 3$, in $\triangle OA'B$,

$$OA' + OB = 1 + 2 = 3 = A'B$$

This violates the triangle inequality. Thus A' must be inside $\triangle ABC$ in order to let the shape make sense. Easy to prove $\triangle AOB \cong \triangle AA'C$, therefore $A'C = 2$. And by rotation $OA' = 1$. Also notice that $OC = 3$, which means $OA' + A'C = OC$, this means O, A', C must be collinear. Since $\angle AOA' = \angle AA'O = 60^\circ$, $\angle AA'C = 120^\circ$, then use Law of Cosines in $\triangle AA'C$:

$$s^2 = AC^2 = 1^2 + 2^2 - 2 \cdot 1 \cdot 2 \cdot \cos 120^\circ = \boxed{\text{(E)} 7}$$

Augustusgao

Solution 4

Let A be a point on the circle with radius 1, B be a point on the circle with radius 2, C be a point on the circle with radius 3. Call the shared center for these three circles O . First let's check if O is in $\triangle ABC$. Rotate OA clockwise by 60° to OA' , then it is equivalent to check if point A' is outside $\triangle ABC$. Assume that A' is outside $\triangle ABC$, easy to prove that $\triangle AOC \cong \triangle AA'B$. Therefore $OC = A'B = 3$, in $\triangle OA'B$,

$$OA' + OB = 1 + 2 = 3 = A'B$$

This violates the triangle inequality. Thus A' must be inside $\triangle ABC$ in order to let the shape make sense. Easy to prove $\triangle AOB \cong \triangle AA'C$, therefore $A'C = 2$. And by rotation $OA' = 1$. Also notice that $OC = 3$, which means $OA' + A'C = OC$, this means O, A', C must be collinear. Since $\angle AOA' = \angle AA'O = 60^\circ$, $\angle AA'C = 120^\circ$, then use Law of Cosines in $\triangle AA'C$:

$$s^2 = AC^2 = 1^2 + 2^2 - 2 \cdot 1 \cdot 2 \cdot \cos 120^\circ = \boxed{\text{(E)} 7}$$

Solution 5

Let O be the common center and let the vertices of the triangle A, B, C satisfy $OA = 1, OB = 2, OC = 3$, with side length $AB = BC = CA = s$. Let $x = s^2$. Let $\alpha = \angle AOB, \beta = \angle BOC, \gamma = \angle COA$, so $\alpha + \beta + \gamma = 360^\circ$. By the Law of Cosines,

$$x = 1^2 + 2^2 - 2 \cdot 1 \cdot 2 \cos \alpha = 5 - 4 \cos \alpha \Rightarrow \cos \alpha = \frac{5 - x}{4}$$

$$x = 2^2 + 3^2 - 2 \cdot 2 \cdot 3 \cos \beta = 13 - 12 \cos \beta \Rightarrow \cos \beta = \frac{13 - x}{12}$$

$$x = 3^2 + 1^2 - 2 \cdot 3 \cdot 1 \cos \gamma = 10 - 6 \cos \gamma \Rightarrow \cos \gamma = \frac{10 - x}{6}$$

For angles summing to 360° , we have

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma - 2 \cos \alpha \cos \beta \cos \gamma = 1$$

Substitute $\cos \alpha, \cos \beta, \cos \gamma$ in terms of x and plug in answer choices (don't bother trying to solve it), we get $x = \boxed{7}$.

This solution is flawed. $\alpha + \beta + \gamma = 360^\circ$ isn't true.

Solution 6

Let ω_1 be a circle of radius 1 centered at $O = (0, 0)$. Let ω_2 be a circle of radius 2 centered at $(0, 0)$. Let A be the point $(0, 3)$. Let B be some point on ω_1 . Let C be some point in the plane such that ABC is equilateral. Our goal is to find some C that lies on ω_2 .

Because $\angle BAC = 60^\circ$ and $AB = AC$, we see that C must be the rotation of B around A , either 60° clockwise or counterclockwise. Note that because the locus of all points B is ω_1 , the locus of all points C is the two circles formed by rotating ω_1 60° around A , either direction.

We want a C that lies on ω_2 , so we naturally intersect ω_2 with one of these two circles (I chose the one that's rotated counterclockwise because the numbers are nicer). We could easily calculate the intersections using coord geo, but notice that the problem asks for a unique s ; this means that the circle is tangent to ω_2 . The tangent point, which is the only C that works, lies on the line connecting the centers. The center (denote it O') of (half of) C 's locus is the center of ω_2 rotated 60° counterclockwise. Because $AO = AO'$ and $\angle OAO' = 60^\circ$, we see $\triangle OAO'$ is equilateral, meaning $\angle AOO' = \angle AOC = 60^\circ$, which means by extension that C is 30° counterclockwise from the horizontal. $OC = 2$ as C lies on ω_2 , so the coordinates of C are at $(2 \cos 30^\circ, 2 \sin 30^\circ) = (\sqrt{3}, 1)$. A is at $(0, 3)$, and a quick distance formula gives us $s^2 = (C.x - A.x)^2 + (C.y - A.y)^2 = 3 + 4 = \boxed{7}$. JoeyG

Solution 7 (complex)

WLOG assume that the triangle is $\triangle ABC$, and that $A = -1, B = a + bi$. Hence $C = (B - A)(\frac{1}{2} + \frac{\sqrt{3}}{2}i) + A = \frac{1}{2}(a + 1 + bi)(1 + \sqrt{3}i) - 1$. Since $|B| = 2, a^2 + b^2 = 4$ and since $|C| = 3, |2C| = 6$, so $|(a + 1 + bi)(1 + \sqrt{3}i) - 2| = 6$. Expanding yields $|(a - \sqrt{3}b - 1) + i(\sqrt{3}a + b + \sqrt{3})| = 6$, hence $(a - \sqrt{3}b - 1)^2 + (\sqrt{3}a + b + \sqrt{3})^2 = 36$, or $4a^2 + 4b^2 + 4a + 4\sqrt{3}b + 4 = 36$, so $a^2 + b^2 + a + \sqrt{3}b + 1 = 9$. But $a^2 + b^2 = 4$, hence $a + \sqrt{3}b = 4, a^2 + b^2 = 4$. We can pretty quickly guess the solution of $a = 1, b = \sqrt{3}$ (drawing it out shows pretty quickly that it's tangent). Hence $|B - A| = (\sqrt{3})^2 + (2)^2 = s^2 = 7$. benjamintontungtungungsahur

§14 AMC 12 12A 2025

1. (AMC 12 12A 2025) Andy and Betsy both live in Mathville. Andy leaves Mathville on his bicycle at 1:30, traveling due north at a steady 8 miles per hour. Betsy leaves on her bicycle from the same point at 2:30, traveling due east at a steady 12 miles per hour. At what time will they be exactly the same distance from their common starting point?

(A) 3:30 (B) 3:45 (C) 4:00 (D) 4:15 (E) 4:30

Answer: E**Solution 1**

At 2:30, Andy is 8 miles ahead. For every hour that they both travel, Betsy gains 4 miles on Andy. Therefore, it will take her 2 more hours to be the same distance from the starting point as Andy, which occurs at **(E) 4:30**.

$$8h = 12(h - 1) \Rightarrow 8h = 12h - 12 \Rightarrow 4h = 12 \Rightarrow h = 3$$

Since Andy started at 1:30, the catch-up time is 4:30. Answer: **(E)**.

Alternatively, from Betsy's perspective: $8(h + 1) = 12h \Rightarrow 8h + 8 = 12h \Rightarrow h = 2$
Same result: **(E) 4:30**.

Solution 2 (Simplest Version)

Andy goes 8 mph, Betsy goes 12 mph.

We know that Betsy starts 1 hour after Andy, so she will be 8 miles behind.

$$12x - 8 = 8x \quad 4x = 8 \rightarrow x = 2$$

2 hours after Betsy's start time is **4:30**

-XaberTooth

Solution 3

Try the answer choices:

At 3:30: Andy = 16, Betsy = 12 At 3:45: Andy = 18, Betsy = 15 At 4:00: Andy = 20, Betsy = 18 At 4:15: Andy = 22, Betsy = 21 At 4:30: Andy = 24, Betsy = 24 You can visualize this using a table:

Time	Andy - 8 mph	Betsy - 12 mph
3:30	16	12
3:45	18	15
4:00	20	18
4:15	22	21
4:30	24	24

Thus the correct answer is **(E) 4:30**.

Solution 4

Since $\text{lcm}(8, 12) = 24$, the time for Andy to reach 24 miles is $24/8 = 3$ hours.

$$1:30 + 3:00 = \textbf{(E) 4:30}.$$

Note: Alternatively we can find that the time for Betsy to reach 24 miles is $\frac{24}{12} = 2$ hours. So we have

$$2:30 + 2:00 = \textbf{(E) 4:30}.$$

Solution 5

Andy goes $8x$ miles, Betsy goes $12(x - 1)$ miles. Set equal:

$$8x = 12(x - 1) \Rightarrow x = 3$$

Thus the answer is **(E) 4:30**.

Solution 6

Using constant rates:

Time	Andy – 8 mph	Betsy – 12 mph
1:30 – Starting	0 miles	0 miles
2:30	8 miles	0 miles
3:30	16 miles	12 miles
4:30	24 miles	24 miles

Thus the answer is (E) 4:30.

Solution 7 (Calculus Overkill)

Let $D_A(t) = 8t$, $D_B(t) = 12(t - 1)$ for $t \geq 1$.

Difference: $F(t) = 12(t - 1) - 8t = 4t - 12$.

Set $F(t) = 0$: $4t - 12 = 0$ $t = 3$ hours.

$1:30 + 3 = 4:30$.

-apex304

Solution 8 (Easy to understand, fast)(Detailed)

Let t denote time in hours measured from 1:30 PM:

$t :=$ hours after 1 : 30 PM.

Thus $t = 0$ at 1:30 PM, $t = 1$ at 2:30 PM, and the physically relevant domain for both riders having left is $t \geq 1$.

Velocity/position (vectorized). Choose orthonormal basis $\{i, j\}$ with i east and j north. Define piecewise position vectors $r_A(t), r_B(t) \in \mathbb{R}^2$ (for all $t \geq 0$):

$r_A(t) = 8t j = (0, 8t)$,

$$r_B(t) = \begin{cases} (0, 0), & 0 \leq t < 1, \\ 12(t - 1) i, & t \geq 1, \end{cases} = (12 \max\{t - 1, 0\}, 0).$$

Define scalar distance functions (Euclidean norm / 2-norm):

$$D_A(t) := \|r_A(t)\|_2 = \sqrt{0^2 + (8t)^2} = 8t, D_B(t) := \|r_B(t)\|_2 = 12 \max\{t - 1, 0\}.$$

We seek t with $D_A(t) = D_B(t)$ and $t \geq 1$. For $t \geq 1$ the equality reduces to the linear equation

$$8t = 12(t - 1).$$

Bring like terms together:

$$8t = 12t - 12 \implies 12 = 12t - 8t.$$

Compute the difference $12t - 8t$ digit-by-digit: $12t - 8t = (12 - 8)t = 4t$. So

$$12 = 4t.$$

Divide 12 by 4 carefully: $12 \div 4 = 3$. Hence

$$t = 3.$$

- (AMC 12 12A 2025) A box contains 10 pounds of a nut mix that is 50 percent peanuts, 20 percent cashews, and 30 percent almonds. A second nut mix containing 20 percent peanuts, 40 percent cashews, and 40 percent almonds is added to the

box resulting in a new nut mix that is 40 percent peanuts. How many pounds of cashews are now in the box?

- (A) 3.5 (B) 4 (C) 4.5 (D) 5 (E) 6

Answer: B

Solution 1

We are given that $0.2(10) = 2$ pounds of cashews in the first box.

Denote the pounds of nuts in the second nut mix as x .

$$5 + 0.2x = 0.4(10 + x)$$

$$5 + 0.2x = 4 + 0.4x$$

$$5 - 4 = 0.4x - 0.2x$$

$$0.2x = 1$$

$$x = 5$$

Thus, we have 5 pounds of the second mix.

$$0.4(5) + 2 = 2 + 2 = \boxed{\text{(B)} 4}$$

Solution 2

Let the number of pounds of nuts in the second nut mixture be x . Therefore, we get the equation $0.5 \cdot 10 + 0.2 \cdot x = 0.4(x + 10)$. Solving it, we then get $x = 5$. Therefore, the amount of cashews in the two bags is $(0.2 \cdot 10) + (0.4 \cdot 5) = 4$, so our answer choice is $\boxed{\text{(B)} 4}$.

Solution 3

The percent of peanuts in the first mix is 10% away from the total percentage of peanuts, and the percent of peanuts in the second mix is 20% away from the total percentage. This means the first mix has twice as many nuts as the second mix, so the second mix has 5 pounds. $0.20 \cdot 10 + 0.40 \cdot 5 = 4$ pounds of cashews. So our answer is, $\boxed{\text{(B)} 4}$

The weight of the second box of nut mix can be calculated by: $(0.2x + 5)/(10 + x) = 0.4$, where the numerator is equal to the weight of the peanuts, the denominator is equal to the total weight of the boxes, and x is equal to the weight of the second box.

Multiplying both sides by $10 + x$, you get $0.2x + 5 = 4 + 0.4x$

Isolating the variable: $0.2x = 1 \Rightarrow x = 5$

Calculating the weight of the cashews: $5(0.4) + 10(0.2) \Rightarrow 2 + 2 \Rightarrow 4$

The answer is $\boxed{\text{(B)} 4}$.

Solution 4

Note that we can set the information given in the problem into a table shown below:

Peanuts	Cashews	Almonds
5	2	3
$\frac{2}{10}x$	$\frac{4}{10}x$	$\frac{4}{10}x$

We are given that the new nut mix will contain 40% peanuts. Hence, $5 + \frac{2}{10}x$ is 40% of the total mix which is $10 + x$. Solving the equation $5 + \frac{2}{10}x = \frac{2}{5} \cdot (10 + x)$ yields $x = 5$. Therefore, the number of cashews in the new mix is equal to $2 + \frac{4}{5} \cdot 5 = \boxed{\text{(B)}4}$.

Solution 5

Write composition vectors as column vectors (mass fractions multiplied by mass):

Initial mass vector (peanuts, cashews, almonds) for 10 lb:

$$\mathbf{v}_0 = 10 \begin{pmatrix} 0.50 \\ 0.20 \\ 0.30 \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \\ 3 \end{pmatrix}.$$

Added mix per unit mass:

$$\mathbf{u} = \begin{pmatrix} 0.20 \\ 0.40 \\ 0.40 \end{pmatrix}.$$

After adding x lb, total mass vector is

$$\mathbf{v}(x) = \mathbf{v}_0 + x\mathbf{u} = \begin{pmatrix} 5 \\ 2 \\ 3 \end{pmatrix} + x \begin{pmatrix} 0.2 \\ 0.4 \\ 0.4 \end{pmatrix}.$$

Peanut fraction requirement:

$$\frac{\mathbf{e}_1^\top \mathbf{v}(x)}{\mathbf{1}^\top \mathbf{v}(x)} = 0.4,$$

where $\mathbf{e}_1 = (1, 0, 0)^\top$ and $\mathbf{1} = (1, 1, 1)^\top$. This is exactly the same scalar equation,

$$\frac{5 + 0.2x}{10 + x} = 0.4,$$

leading to $x = 5$ and hence

$$\mathbf{v}(5) = \begin{pmatrix} 5 \\ 2 \\ 3 \end{pmatrix} + 5 \begin{pmatrix} 0.2 \\ 0.4 \\ 0.4 \end{pmatrix} = \begin{pmatrix} 6 \\ 4 \\ 5 \end{pmatrix}.$$

Thus cashews = 4 lb.

Solution 6: Very easy algebra and straightforward!

Set up the ratios for the initial and the additional mixtures. The ratio of Peanuts:Cashews:Almonds (denoted as P:C:A) would initially be $5x : 2x : 3x$ for some factor x . Then for the second mixture it would be $2x : 4x : 4x$. If we add these we get $7x:6x:7x$. Now we know that the percentage of peanuts is 40%. But we also know that the number of pounds of peanuts and almonds is the same. $40\% + 40\% = 80\%$ and so the number of cashews makes up 20% of the mixture. If you divide the factor x by the whole ratio you get $7 : 6 : 7$. These values sum to 20 and 20% of 20 is 4.

Video Solutions

3. (*AMC 12 12A 2025*) A team of students is going to compete against a team of teachers in a trivia contest. The total number of students and teachers is 15. Ash, a cousin of one of the students, wants to join the contest. If Ash plays with the students, the average age on that team will increase from 12 to 14. If Ash plays with the teachers, the average age on that team will decrease from 55 to 52. How old is Ash?

(A) 28 (B) 29 (C) 30 (D) 32 (E) 33

Answer: A

Solution 1

When Ash joins a team, the team's average is pulled towards his age. Let A be Ash's age and N be the number of people on the student team. This means that there are $15 - N$ people in the teacher team. Let us write an expression for the change in the average for each team.

The students originally had an average of 12, which became 14 when Ash joined, so there was an increase of 2. The term $A - 12$ represents how much older Ash is compared to the average of the students'. If we divide this by $N + 1$, which is the number of people on the student team when Ash joins, we get the average change per team member once Ash is added. Therefore,

$$\frac{A - 12}{N + 1} = 2.$$

Similarly, for teachers, the average was originally 55, which decreased by 3 to become 52 when Ash joined. Intuitively, $55 - A$ represents how much younger Ash is than the average age of the teachers. Dividing this by the expression $(15 - N) + 1$, which is the new total number of people on the teacher team, represents the average change per team member once Ash joins. We can write the equation

$$\frac{55 - A}{16 - N} = 3.$$

To solve the system, multiply equation (1) by $N + 1$, and similarly multiply equation (2) by $16 - N$. Then add the equations together, canceling A , leaving equation $43 = 50 - N$. From this we get $N = 7$ and $A = \boxed{28}$.

Solution 2

As shown above, we get the equation

$$\frac{55 - A}{16 - N} = 3.$$

Rearranging we get

$$\frac{55 - A}{3} = 16 - N.$$

Therefore, since N must be an integer, $55 - A$ must be divisible by 3. Plugging in values gets us $A = \boxed{28}$.

Solution 3

Another way is to say that there are S students and T teachers and Ash's age is A . We can create the equations

$$(1)S + T = 15 \tag{47}$$

$$(2) \frac{12S + A}{S + 1} = 14 \quad (48)$$

$$(3) \frac{55T + A}{T + 1} = 52 \quad (49)$$

Equation (2) simplifies to $A = 2S + 14$ and equation (3) simplifies to $A = -3T + 52$. Multiply equation (2) by 3 to get equation

$$(4) 3A = 6S + 42 \quad (50)$$

and equation (3) by -2 to get

$$(5) -2A = 6T - 104 \quad (51)$$

Add (4) and (5) to get $A = 6(S + T) - 62$. After substituting equation (1) and simplifying, you get $A = \boxed{28}$ or answer choice $\boxed{A}$.

4. (AMC 12 12A 2025) Agnes writes the following four statements on a blank piece of paper.

- At least one of these statements is true.
- At least two of these statements are true.
- At least two of these statements are false.
- At least one of these statements is false.

Each statement is either true or false. How many false statements did Agnes write on the paper?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Answer: B

Solution 1

We first number all the statements:

- 1) At least one of these statements is true. 2) At least two of these statements are true. 3) At least two of these statements are false. 4) At least one of these statements is false.

We can immediately see that statement 4 must be true, as it would contradict itself if it were false. Similarly, statement 1 must be true, as all the other statements must be false if it were false, which is contradictory because statement 4 is true. Since both 1 and 4 are true, statement 2 has to be true. Therefore, statement 3 is the only false statement, making the answer $\boxed{(B) 1}$. -Rainjs

Solution 2

Statements *I*, *II*, and *IV* are true, while statement *III* is false. Hence, there are 3 true statements and $\boxed{(B) 1}$ false statement. This result can be checked by examining the statements individually again.

Statements *I* and *II* will be true because there are $3 \geq 2$ true statements. Statement *IV* is also true because there is $1 \geq 1$ false statement. Finally, statement *III* is false because there are $1 \not\geq 2$ false statements.

Note: This solution does not give a path to the solution, but rather only how to check the answer after the answer has been identified.

Solution 3

Let's say there are T true statements. We know that T can be any integer from 0 to 4. Numbering the 4 given statements in order as $S1$ to $S4$, we can represent $S1$ as $T \geq 1$, $S2$ as $T \geq 2$, and since there are $(4 - T)$ false statements, $S3$ as $4 - T \geq 2 \iff T \leq 2$, and $S4$ as $4 - T \geq 1 \iff T \leq 3$.

Accordingly, we now consider the possible cases for the value of T . If $T = 0$, then the true statements are $S3$ and $S4$, i.e. there are 2 true statements in total. But that means $T = 2 \neq 0$, giving a contradiction.

Similarly, if $T = 1$, then the true statements are $S1$, $S3$, and $S4$, i.e. there are 3 true statements in total. This means $T = 3 \neq 1$, again giving a contradiction.

If $T = 2$, then all 4 statements are true, yielding the contradiction $T = 4 \neq 2$.

Finally, however, if $T = 3$, then the true statements are $S1$, $S2$, and $S4$. Hence there are again 3 true statements in total, but now as $T = 3$ in the first place, this is actually consistent and not a contradiction.

For completeness, we also observe that if $T = 4$, then the true statements are only $S1$ and $S2$, i.e. $T = 2 \neq 4$, and so this case again leads to a contradiction.

Thus we conclude that the only possible value is $T = 3$, which gives a consistent solution when $S1$, $S2$, and $S4$ are true and $S3$ is false (as noted above). It follows that the number of false statements is $\boxed{\text{(B) } 1}$, namely precisely $S3$ (and indeed, $T = 3$ gives $4 - T = 4 - 3 = 1$, as expected).

Solution 4

Suppose Statement I is false, then none of the Statements are true, which contradicts the fact that a false Statement III or IV is telling the truth. Therefore, Statement I is true and assume Statement II is false.

Statement II thus implies that only Statement I was the truth, and the rest, false. But then, there are 3 false statements but then Statement III and Statement IV are telling the truth. So Statement II is also true.

Now, if Statement III is true, then both Statement III and Statement IV is false, contradicting the fact that it is true. Statement III is hence false and Statement IV tells the truth since Statement III lied so indeed, there are at least one lie. There are a total of 3 truths and 1 lie, making the answer $\boxed{\text{(B) } 1}$.

Solution 5 (Quick)

I tried setting that the statements regarding the true ones (I and II) were true and regarding the false ones (III and IV) were false, and it clearly doesn't work. It doesn't work because statement III was true, not false as I set it to be originally.

What if we make that statement (Statement III) true too? Well, then we have 3 T and 1 F statements, so statement IV is false, statement III is true, and statements I and II are true too.

So there is $\boxed{\text{(B) } 1}$ false statement.

Solution 6 (Plug in Answer Choices)

For this question, what I did was to simply test each of the answer choices.

Let's start with (A) 0 false statements:

We can check the answer of 0 false statements against each of the statements.

Statement I: "At least one of these statements is true."

This statement is false, which already invalidates the answer of (A) 0 false statements, as we already have a false statement.

Let's move onto (B) 1 false statement:

We can check once again this answer against each of the statements.

Statement I: "At least one of these statements is true."

This statement is true, as when we have 1 false statement, we have 3 true statements, and therefore this is true.

Statement II: "At least two of these statements are true."

Once again, this statement is true, as when we have 1 false statement, we have 3 true statements, and thus this is also true.

Statement III: "At least two of these statements are false."

This statement is false, as we have one false statement, not 2 and above.

Statement IV: "At least one of these statements is false."

This statement is true, as we have one false statement.

So, after checking the answer choice of (B) 1 false statement across the 4 statements, we get three true statements and one false statement, making this statement the correct answer. I will not dive into the other 3 answer choices, as we have already gotten the answer for this question, and if you were doing this way of solving on the test, you would most likely not test the other 3 answers either for the sake of saving time.

Solution 7 (Step by Step, Simple)

Start by going statement by statement and checking if they can be false. If statement 1 were false statement 4 would be true leading to a contradiction so statement 1 is true. If statement 2 were false 4 would be true and one would be true making statement 2 true. If statement 3 was false and we know statement 1 and statement 2 were true and 4 can be true and it works. Therefore, the correct answer is (B) 1.

–TFEA(would also appreciate if somebody could please latex my proof).–Iceboy1221(i wrote it using latex)

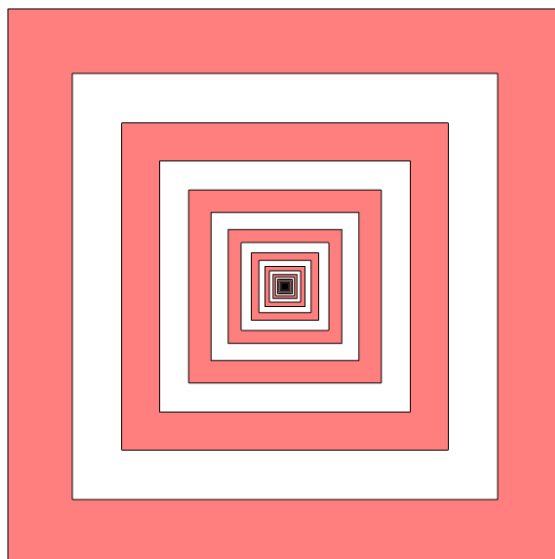
Solution 8 (for Chinese Committee Paper)

Due to reasons to prevent cheating, there was a variant for test takers in China, which involves slight iterations in the questions and answer choices. For this particular problem, the question asks for the number of **true** statements that Agnes wrote.

Therefore, utilizing any of the methods in the solutions above, we find that there is one false statement, therefore there are three true statements.

Therefore, the correct answer choice is **3**.

5. (AMC 12 12A 2025) In the figure below, the outside square contains infinitely many squares, each of them with the same center and sides parallel to the outside square. The ratio of the side length of a square to the side length of the next inner square is k , where $0 < k < 1$. The spaces between squares are alternately shaded as shown in the figure (which is not necessarily drawn to scale).



The area of the shaded portion of the figure is 64% of the area of the original square. What is k ?

- (A) $\frac{3}{5}$ (B) $\frac{16}{25}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$ (E) $\frac{4}{5}$

Answer: D

Solution 1

Let the side length of the largest square be a , so it has area a^2 . Hence, the second-largest square has area a^2k^2 , the third-largest has a^2k^4 , and so on.

It follows that the total shaded area is

$$a^2 - a^2k^2 + a^2k^4 - a^2k^6 + \dots = a^2(1 - k^2 + k^4 - k^6 + \dots) = a^2 \frac{1}{1 + k^2}.$$

The ratio of the area of the shaded region to that of the original square is then

$$\frac{a^2 \frac{1}{1+k^2}}{a^2} = \frac{1}{1+k^2} = \frac{64}{100}$$

$$\implies 64 + 64k^2 = 100 \implies k^2 = \frac{36}{64} \implies k = \boxed{\text{(D)} \frac{3}{4}}.$$

Remark We can just let $a = 1$ because the question deals with ratios, meaning that there wouldn't be a loss of generality if we let the side length equal some value, getting the same answer $\boxed{D}$. This was done in solution 3.

Solution 2

Let the side length of the first square be a and the second square be b . The area of the original square is a^2 . The area of the outermost shaded region is $a^2 - b^2$. Let $\frac{a}{b} = r$. We have $b = \frac{a}{r}$. So the outermost shaded region area becomes:

$$a^2 - b^2 = a^2 - \frac{a^2}{r^2}$$

Now let the next square's side lengths be c and d . Similarly, $c = \frac{b}{r} = \frac{a}{r^2}$ and $d = \frac{c}{r} = \frac{a}{r^3}$ and the area of the next shaded region becomes:

$$c^2 - d^2 = \frac{a^2}{r^4} - \frac{a^2}{r^6}$$

Notice the pattern of adding 4 to the exponent. If this sequence continues infinitely, we ultimately get:

$$(a^2 + \frac{a^2}{r^4} + \frac{a^2}{r^8} + \dots) - (\frac{a^2}{r^2} + \frac{a^2}{r^6} + \dots)$$

Which can be simplified using the infinite geometric sequence formula. The problem also tells us that all of this equals $64\%a^2$, or $\frac{16}{25}a^2$:

$$\frac{a^2}{1 - \frac{1}{r^4}} - \frac{\frac{a^2}{r^2}}{1 - \frac{1}{r^4}} = \frac{16}{25}a^2$$

$$\frac{1}{1 - \frac{1}{r^4}} - \frac{\frac{1}{r^2}}{1 - \frac{1}{r^4}} = \frac{16}{25}$$

$$r^2 = \frac{16}{9} \implies r = \frac{4}{3}$$

Since the problem tells us the ratio must be less than one, we take the reciprocal to finally get

$$k = \frac{1}{r} = \boxed{(D) \frac{3}{4}}.$$

Solution 3 (Faster)

Let the outside square be of side length 1, and let $p = k^2$. Then the area of the square is $1 - p + p^2 - p^3 + p^4 - p^5 \dots$ or $(1 - p)(1 + p^2 + p^4 + p^6 + \dots)$. Then, using geometric series, we get $(1 - p)\left(\frac{1}{1 - p^2}\right) = \frac{1}{1 + p}$. This is equal to $\frac{64}{100}$. Therefore,

$$1 + p = \frac{25}{16}, \text{ so } p = 9/16 \text{ and } k = \frac{3}{4}. \text{ Therefore, the answer is } \boxed{(D) \frac{3}{4}}.$$

Solution 4 (The fastest, no sequences/series)

Since this is an infinite fixed pattern, we can actually take a huge shortcut. Suppose we take off the outer "layer" and then swap the colors. We have essentially created a smaller copy of the original diagram, specifically with a $k : 1$ ratio of side lengths. Assume the original diagram had 100 units of area. Then, based on the givens, the original diagram had 64 units of shaded area and 36 units of unshaded area, meaning our new smaller diagram has 36 units of shaded area. Since the ratio of corresponding side lengths between our new diagram and the original diagram is k ,

$$\text{the ratio of corresponding areas is } k^2. \text{ Thus, we get } k^2 = \frac{36}{64}, \text{ so } k = \boxed{(D) \frac{3}{4}}.$$

Note: the explanation seems complicated to explain, but the thought process while doing it is extremely fast, much faster than any of the methods involving a series.

Solution 5 (based very closely on solution 4)

Similarly to solution 4, we can imagine "peeling" off the outer layer to get an inner figure that is similar to the outer figure. Letting the large square have side 1, we know the outer shaded layer has area $1 - k^2$. The shaded part of the inner figure (with the outer shaded part removed now must make up $\frac{36}{100} = \frac{9}{25}$ of the inner area of k^2 . Together, the inner shaded part and the outer shaded part must add up to $\frac{64}{100}$, as they make up 64% of the square. Thus, we get the equation $1 - k^2 + \frac{9k}{25} = 16/25$. Solving this gives $k = 3/4$ Neber100

Solution 6 (The longest way)

Notes that: The sum S can be written as:

$$\begin{aligned} S &= 1^2 - k^2 + (k^2)^2 - (k^3)^2 + (k^4)^2 - (k^5)^2 + (k^6)^2 - \dots \\ &= k^0 - k^2 + k^4 - k^6 + k^8 - k^{10} + \dots \\ &= (k^0 + k^4 + k^8 + k^{12} + \dots) - (k^2 + k^6 + k^{10} + k^{14} + \dots) \end{aligned}$$

So we have:

$$\begin{aligned} S &= \lim_{n \rightarrow \infty} \left[1 \times \frac{1 - (k^4)^n}{1 - k^4} - k^2 \times \frac{1 - (k^4)^n}{1 - k^4} \right] \\ &= \frac{1}{1 - k^4} - \frac{k^2}{1 - k^4} \\ &= \frac{1 - k^2}{1 - k^4} \\ &= \frac{1 - k^2}{(1 + k^2)(1 - k^2)} \\ &= \frac{1}{1 + k^2} \end{aligned}$$

We are given the condition that:

$$\therefore \frac{1}{1 + k^2} = 64\% = \frac{64}{100} = \frac{16}{25}$$

We can now solve for k :

$$\begin{aligned} \therefore \frac{1}{1 + k^2} &= \frac{16}{25} \\ 1 + k^2 &= \frac{25}{16} \\ k^2 &= \frac{25}{16} - 1 \\ k^2 &= \frac{25}{16} - \frac{16}{16} \\ k^2 &= \frac{9}{16} \\ k &= \pm \sqrt{\frac{9}{16}} \\ k &= \pm \frac{3}{4} \end{aligned}$$

Therefore, ignore the negative value (Because $0 < k < 1$), so the final answer is $\boxed{\frac{3}{4}}$.

We are done :)

Solution 7

Assume that the side length for the largest square in the diagram is 1. Since the ratio of the next inner square to the largest square is k , we get that the side length for the second largest square is k , k^2 for the third largest square, and k^3 for the fourth largest square. So, we can write that

$$(1 - k^2) + (k^4 - k^6) + \dots = \frac{64}{100}$$

by using the diagram since the area of a square with side length s is s^2 . This is a geometric sequence with first term $1 - k^2$ and ratio k^4 . So, we can sum the geometric series for

$$\frac{1 - k^2}{1 - k^4} = \frac{64}{100}.$$

Cross multiplying, we have that

$$100 - 100k^2 = 64 - 64k^4.$$

Moving the terms to one side, we have $64k^4 - 100k^2 + 36 = 0$, and we can factor as

$$(64k^2 - 36)(k^2 - 1) = 0.$$

Since $k < 1$, $k^2 - 1 = 0$ is invalid. So, we have

$$64k^2 - 36 = 0,$$

and simplifying we get $k^2 = \frac{36}{64}$. Since $k > 0$,

$$k = \frac{6}{8} = \boxed{\frac{3}{4}}.$$

Solution 8 (Cheese)

Measure the side length of the first square to be 2 inches. Measure the side length of the second square to be 1.5 inches.

$$\frac{1.5}{2} = \frac{3}{4} = \boxed{(D) \frac{3}{4}}$$

Solution 9 (Fastest and shortest, similar to solution 4, no series)

Since the figure is self-similar, the nonshaded area is $100\% - 64\% = 36\%$ of the largest square and 64% of the square with side length k , which has area k^2 . Thus $36\% = 64\%k^2$, $k^2 = \frac{36}{64}$, so $k = \frac{6}{8} = \boxed{(D) \frac{3}{4}}$.

Solution 10 (Series Subtraction)

Assume initial length 1 since the problem just uses ratios. We see that each side length shrinks by factor k , then they are squared to get the areas, which really just multiplies their exponents by 2. We add our shaded areas and subtract our un-shaded areas, yielding the alternating series

$$1 - k^2 + k^4 - k^6 \dots$$

We express this as two infinite series, one being subtracted from the other

$$\sum_{n=0}^{\infty} k^{4n} - \sum_{n=0}^{\infty} k^{4n+2}$$

Rewrite using the infinite geometric series formula and simplify as far as possible

$$\frac{1}{1 - k^4} - \frac{k^2}{1 - k^4}$$

$$\frac{1 - k^2}{1 - k^4} = \frac{1 - k^2}{(1 + k^2)(1 - k^2)} = \frac{1}{1 + k^2}$$

We know this is equal to $\frac{64}{100} = \frac{16}{25}$

$$\frac{1}{1 + k^2} = \frac{16}{25}$$

$$16k^2 + 16 = 25$$

$$16k^2 = 9$$

$$k^2 = \frac{9}{16}$$

We must only consider the positive root

$$k = \boxed{\text{(D)} \frac{3}{4}}.$$

shockfront99

6. (*AMC 12 12A 2025*) Six chairs are arranged around a round table. Two students and two teachers randomly select four of the chairs to sit in. What is the probability that the two students will sit in two adjacent chairs and the two teachers will also sit in two adjacent chairs?

(A) $\frac{1}{6}$ (B) $\frac{1}{5}$ (C) $\frac{2}{9}$ (D) $\frac{3}{13}$ (E) $\frac{1}{4}$

Answer: B

Solution 1 (20 seconds)

Pair two students together and put them adjacent on any two seats. There are 6 ways to do this. Considering one of these cases (they are all the same), there are 4 seats left, in which we wish to arrange the teachers together. So pair the teachers together and put them adjacent on any two seats not already occupied by two of the students. There are 3 ways to do this. For all 6 cases, there are $6 \times 3 = 18$ favorable outcomes.

The number of ways to arrange the 2 students and 2 teachers is $\binom{6}{2} \times \binom{4}{2} = 90$.

Our probability is $\frac{18}{90} = \boxed{\text{(B)} \frac{1}{5}}$

Solution 2

We first count the number of ways to place 4 distinct people into 6 distinct chairs: $6 \cdot 5 \cdot 4 \cdot 3 = 360$.

We now count how many favorable assignments there are. There are 6 ways to choose an adjacent pair of chairs for the two students. We can arrange the two students in those two chairs in 2 ways.

After those two adjacent chairs are taken, there are 4 chairs left, with 3 adjacent pairs among them. We choose one of these pairs, arranging the teachers for $3 \cdot 2 = 6$ ways.

There are $6 \cdot 2 \cdot 6 = 72$ favorable arrangements.

The probability is therefore $\frac{72}{360} = \boxed{\frac{1}{5}}$

Solution 3

We split the problem into cases of where the first teacher and first student sits. The first student sits in a seat with 1 probability

Case 1: The first teacher sits next to the first student. There's are 2 ways to do this so this will happen with $\frac{2}{5}$ chance. Now, there is one valid seat for the second student to sit in with probability $\frac{1}{4}$ and and one valid valid seat for the second teacher to sit in with probability $\frac{1}{3}$. Total probability of this case is $\frac{2}{5} \cdot \frac{1}{4} \cdot \frac{1}{3} = \frac{1}{30}$

Case 2: The first teacher sits on the opposite end of the circle with $1/5$ chance. This means the second student and the second teacher each have 2 valid spots to sit in, with probability $\frac{2}{4}$ and $\frac{2}{3}$. Total probability is $\frac{1}{5} \cdot \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{15}$.

Case 3: The first teacher sits one chair away from the first student with $\frac{2}{5}$ probability because there are 2 seats that are one chair away.

Case 3.1:

The second student sits in between the first teacher and first student with $\frac{1}{4}$ chance. The second teacher only has one valid seat, with $\frac{1}{3}$ chance to sit in it. Total probability is $\frac{1}{4} \cdot \frac{1}{3} = \frac{1}{12}$.

Case 3.2:

The second student doesn't sit in between the teacher and the student with a $\frac{1}{4}$ chance. The second teacher can sit in 2 valid seats with a $\frac{2}{3}$ chance. Total probability is $\frac{1}{4} \cdot \frac{2}{3} = \frac{1}{6}$.

Total probability for case 3 is $\frac{2}{5} \cdot (\frac{1}{12} + \frac{1}{6}) = \frac{1}{10}$.

Adding up all the cases, you get $\frac{1}{30} + \frac{1}{15} + \frac{1}{10} = \boxed{\frac{1}{5}}$.

Solution 4

We consider how many ways there are to place the students and teachers at random at the table. This is a word arrangement in a circle, and so there are $\frac{(6-1)!}{2! \cdot 2! \cdot 2!} = 15$ different ways of doing so, assuming that teachers, students and empty spaces are indistinguishable between themselves.

Now, we wish to find how many favorable arrangements there are, so suppose we fix two adjacent teachers. Then, there are three "spots" (blocks of two seats) two adjacent students could occupy: two on the side of either teacher, and one such that no student is sitting adjacent to a teacher.

Thus, the probability is $\frac{3}{15} = \boxed{\frac{1}{5}}$.

Solution 5 (Step-by-step, needs verification)

Because this is in a circle, rotating around doesn't increase or decrease the probability (so pin one seat and the others will not rotate).

Hence, label the spots 1-6 in clockwise. Let student 1 sit in spot 1. Student 2 can therefore sit in either spot 2 or 6, which the probability is $2/5$. For explanation purposes, put student 2 in position 2.

Here then requires a case breakdown.

Case 1: Teacher 1 sits in spot 3, hence teacher 2 needs to sit in spot 4 to satisfy. This case totals up with $\frac{2}{5} \cdot \frac{1}{4} \cdot \frac{1}{3} = \frac{1}{30}$.

Case 2: Teacher 1 sits in spot 4, hence teacher 2 needs to sit in spot 3 or 5 to satisfy. This case totals up with $\frac{2}{5} \cdot \frac{1}{4} \cdot \frac{2}{3} = \frac{1}{15}$.

Case 3: Teacher 1 sits in spot 5, hence teacher 2 needs to sit in spot 4 or 6 to satisfy. This case totals up with $\frac{2}{5} \cdot \frac{1}{4} \cdot \frac{2}{3} = \frac{1}{15}$.

Case 4: Teacher 1 sits in spot 6, hence teacher 2 needs to sit in spot 5 to satisfy. This case totals up with $\frac{2}{5} \cdot \frac{1}{4} \cdot \frac{1}{3} = \frac{1}{30}$.

By adding the case probabilities, we have $(\frac{1}{30}) \cdot 2 + (\frac{1}{15}) \cdot 2 = \boxed{\frac{1}{5}}$.

Note: Feel free to delete if there is a logic error. Mitsuihisashi14

Solution 6 (Quickest)

For the two teachers' seatings, there are 6 ways that they are adjacent out of a combination of $n = \binom{6}{2}$ ways. This probability is $\frac{6}{15} = \frac{2}{5}$. For the two students' seatings, there are 3 ways to choose the two seats out of $n = \binom{4}{2}$ ways. This probability is $\frac{3}{6} = \frac{1}{2}$. Answer is $\frac{2}{5} \cdot \frac{1}{2} = \boxed{\frac{1}{5}}$.

Solution 7 (Probability = Desired Arrangements/Total arrangements)

Let the seats be numbered 1, 2, 3, 4, 5, 6 in clockwise order. Then assume placing a group of 2 (either the two students of the two teachers) in seats 1 and 2, this group of two can be permuted (or in this case just swpped) in $2! = 2$ ways. Now the second group may sit in the seats 3, 4 or 4, 5 or 5, 6 giving a factor of 3. This group can also be permuted in $2! = 2$ ways. This gives 12 ways for the 4 people to be seated under the condition. Without the condition there are $\frac{(6-1)!}{2} = 60$ ways for the 4 people to be seated (the division by $2!$ comes from the two empty seats being considered the same). Finally, the desired probability is:

$$\frac{12}{60} = \boxed{\frac{1}{5}}$$

Solution 8

For the two students, you can seat the first one in any seat, followed by the second one with a $\frac{2}{5}$ possibility of satisfying the condition. After this, we have two cases.

Case 1: The first teacher sits next to the two students. This can happen $\frac{1}{2}$ of the time and the other teacher has a $\frac{1}{3}$ chance of sitting next to the first one. This gives us a $\frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$ chance for this case.

Case 2: The first teacher sits away from the two students. This can also happen $\frac{1}{2}$ of the time and the other teacher has a $\frac{2}{3}$ chance of sitting next to the first one. This gives us a $\frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}$ chance for the second case.

Now, we can find the desired probability as: $(\frac{1}{3} + \frac{1}{6}) \cdot \frac{2}{5} = \frac{1}{2} \cdot \frac{2}{5} = \boxed{\frac{1}{5}}$

7. (AMC 12 12A 2025) In a certain alien world, the maximum running speed v of an organism is dependent on its number of toes n and number of eyes m . The relationship can be expressed as $v = kn^am^b$ centimeters per hour, where k , a , and b are integer constants. In a population where all organisms have 5

toes, $\log v = 4 + 2 \log m$; and in a population where all organisms have 25 eyes, $\log v = 4 + 4 \log n$, where the logarithms are base 10. What is $k + a + b$?

- (A) 20 (B) 21 (C) 22 (D) 23 (E) 24

Answer: C

Solution 1 (logs and system of equations)

Substituting v in the equation where $n = 5$, we have:

$$\log(k \cdot 5^a m^b) = 4 + 2 \log m.$$

Using logarithmic properties, we can write this as:

$$\log(k \cdot 5^a m^b) = \log(10^4 \cdot m^2)$$

We can do this with the other equation where $m=25$:

$$\log(k \cdot n^a 25^b) = \log(10^4 \cdot n^4)$$

Now we can get rid of the logs on both sides and are left with the following system of equations:

$$k \cdot 5^a m^b = 10^4 m^2 \quad k \cdot n^a 25^b = 10^4 n^4$$

Notice that in the first equation, we can change m arbitrarily, so we know that the exponent of m must be the same - hence $b = 2$. Similarly, from the second equation, we get $a = 4$. 10^4 can be written as $2^4 \cdot 5^4$, which means that $k = 2^4 = 16$. Thus the answer is $2 + 4 + 16 = \boxed{(C)22}$.

-Cyrus825 ScoutViolet (mostly minor fixes) knight10 (minor fixes)

Solution 2

We first try to simplify both log equations, and then we bring in the equation for velocity.

For the equation representing organisms with 5 toes:

$$\begin{aligned} \log v &= 4 + 2 \log m & \log v &= \log 10^4 + \log m^2 & \log v &= \log(10^4 \cdot m^2) & v &= 10^4 \cdot m^2 \\ k \cdot 5^a m^b &= 10^4 \cdot m^2 \end{aligned}$$

We do the same with the logarithm equation representing organisms with 25 eyes to get:

$$\log v = 4 + 4 \log n \quad v = 10^4 \cdot n^4 \quad k \cdot n^a (25^b) = 10^4 \cdot n^4$$

Now we need to figure out what a , b , and k are. Looking at the first equation (or the second), we notice that we can easily match factors. There are two m s on the right side so we set $b = 2$ so there are two m s on the left side. Now we need $10^4 = k \cdot 5^a = 5^4 \cdot 2^4$. There are four 5s on one side, so we need $a = 4$ to get four 5s on the other. Now, $k = 2^4 = 16$. This solution works for the first equation.

Checking if these values work with the second equation, it does, so our answer is $2 + 4 + 16 = \boxed{22}$

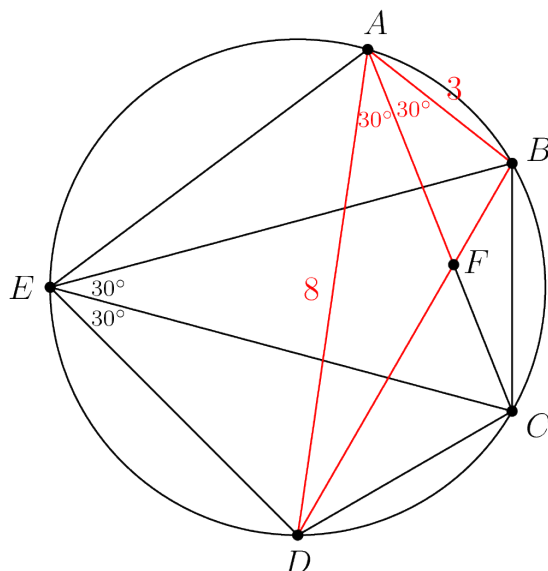
8. (AMC 12 12A 2025) Pentagon $ABCDE$ is inscribed in a circle, and $\angle BEC = \angle CED = 30^\circ$. Let line AC and line BD intersect at point F , and suppose that $AB = 9$ and $AD = 24$. What is BF ?

- (A) $\frac{57}{11}$ (B) $\frac{59}{11}$ (C) $\frac{60}{11}$ (D) $\frac{61}{11}$ (E) $\frac{63}{11}$

Answer: E

Solution 1

We will scale down the diagram by a factor of 3 so that $AB = 3$ and $AD = 8$. Since $\angle BEC = 30^\circ$, it follows that $\angle BAC = \angle BDC = 30^\circ$ as they all subtend the same arc. Similarly, since $\angle CED = 30^\circ$, it follows that $\angle CAD = \angle CBD = 30^\circ$ as well. We obtain the following diagram:



Note that $\triangle ABD$ has $\angle BAD = 60^\circ$. Applying Law of Cosines, we get

$$\begin{aligned} BD^2 &= AB^2 + AD^2 - 2AB \cdot AD \cdot \cos 60^\circ \\ &= 9 + 64 - 2 \cdot 3 \cdot 8 \cdot \frac{1}{2} \\ &= 49, \end{aligned}$$

from which $BD = 7$.

From here, we wish to find BF . As AF is the angle bisector of $\angle BAD$, we apply the Angle Bisector Theorem:

$$\frac{AB}{BF} = \frac{AD}{DF}$$

$$\frac{3}{BF} = \frac{8}{7 - BF}.$$

Solving for BF , we get $BF = \frac{21}{11}$. Remember to scale the figure back up by a factor of 3, so our answer is $\frac{21}{11} \cdot 3 = \boxed{\text{(E)} \frac{63}{11}}$.

Note Instead of taking Angle Bisector one could also take area. The area of DAB is $6\sqrt{3}$. Let $AF = s$ and so through area formulae $2s + \frac{3s}{4} = \frac{11s}{4} = 6\sqrt{3}$ so $s = \frac{24\sqrt{3}}{11}$.

Through Law of Cosines, $BF = \frac{21}{11}$ so scaled we have $\boxed{\frac{63}{11}}$

Solution 2 Law of (Co)Sine

From cyclic quadrilateral $CDAE$, we have $\angle CAD = \angle CED = 30^\circ$. Since $ABDE$ is also cyclic, we have $\angle BAD = \angle BED = 60^\circ$, so,

$$\angle BAC = \angle BAD - \angle CAD = 60^\circ - 30^\circ = 30^\circ.$$

Using Law of Cosines on $\triangle ABD$, we get

$$BD^2 = 9^2 + 24^2 - 2(9)(24)\cos(60^\circ).$$

Solving, we get $BD = 21$. Next, let $\overline{BF} = x$, and $\angle AFB = \theta$, which means $\overline{FD} = 21 - x$ and $\angle AFD = 180 - \theta$. Using Law of Sines on $\triangle AFB$, we have

$$\frac{9}{\sin \theta} = \frac{x}{\sin 30^\circ}.$$

Solving for $\sin \theta$, we get $\sin \theta = \frac{9}{2x}$. Now we apply the Law of Sines to $\triangle AFD$. We have

$$\frac{24}{\sin(180 - \theta)} = \frac{21 - x}{\sin 30^\circ}.$$

Since $\sin(180 - \theta) = \sin(\theta)$, and $\sin \theta = \frac{9}{2x}$, we have

$$\frac{16x}{3} = 42 - 2x.$$

Solving for x gives $x = \frac{63}{11}$ or **(E)** $\frac{63}{11}$.

Solution 3 (Ptolemy's + Similarity)

We have $ABCDE$ cyclic, so $\angle BAC = \angle CAD = \angle BEC = 30^\circ$. Hence cyclic quadrilateral $ABCD$ has $\angle BAD = 60^\circ$. Law of Cosines on triangle BAD gives $\overline{BD}^2 = 9^2 + 24^2 - 2 \cdot 9 \cdot 24 \cos 60^\circ$. Hence $\overline{BD} = 21$. Since triangle BCD is a 120-30-30 triangle, we can use law of sines or just memorize ratios to get $\overline{BC} = \overline{CD} = 7\sqrt{3}$. Now Ptolemy's on $ABCD$ yields $7\sqrt{3}(9 + 24) = 21\overline{AC}$. Hence $\overline{AC} = 11\sqrt{3}$. Now notice that $\angle BCF = \angle ACB$, and $\angle CBF = \angle CAB = 30^\circ$. Hence triangles CBF and CAB are similar, and $\frac{\overline{BF}}{\overline{BC}} = \frac{\overline{AB}}{\overline{AC}}$, so $\frac{\overline{BF}}{7\sqrt{3}} = \frac{9}{11\sqrt{3}}$ and $\overline{BF} = \frac{63}{11}$, or **(E)** $\frac{63}{11}$.

Solution 4 (Area Angle Bisector Length)

As shown in previous solutions, the angles subtending the same arcs give us $\angle BAC = \angle BEC = 30^\circ$ and $\angle CAD = \angle CED = 30^\circ$. This tells us that

$$\angle BAD = \angle BAC + \angle CAD = 60^\circ,$$

and line AF acts as the interior angle bisector of $\angle BAD$.

We can find the length of the angle bisector AF by splitting the area of $\triangle ABD$ into the sum of the areas of $\triangle ABF$ and $\triangle AFD$:

$$\text{Area}(\triangle ABD) = \text{Area}(\triangle ABF) + \text{Area}(\triangle AFD).$$

Using the sine formula for area (Area = $\frac{1}{2}ab \sin C$), we have:

$$\frac{1}{2}(AB \cdot AD) \sin 60^\circ = \frac{1}{2}(AB \cdot AF) \sin 30^\circ + \frac{1}{2}(AD \cdot AF) \sin 30^\circ.$$

Substituting the known values $AB = 9$ and $AD = 24$:

$$\frac{1}{2}(9 \cdot 24) \frac{\sqrt{3}}{2} = \frac{1}{2}(9 \cdot AF) \frac{1}{2} + \frac{1}{2}(24 \cdot AF) \frac{1}{2}.$$

Multiplying the entire equation by 4 to clear the denominators yields:

$$\begin{aligned} 216\sqrt{3} &= 9(AF) + 24(AF) \\ 216\sqrt{3} &= 33(AF) \\ AF &= \frac{216\sqrt{3}}{33} = \frac{72\sqrt{3}}{11}. \end{aligned}$$

Now, we apply the Law of Cosines on $\triangle ABF$ to solve directly for BF :

$$BF^2 = AB^2 + AF^2 - 2(AB \cdot AF) \cos 30^\circ.$$

Substituting $AB = 9$, $AF = \frac{72\sqrt{3}}{11}$, and $\cos 30^\circ = \frac{\sqrt{3}}{2}$:

$$\begin{aligned} BF^2 &= 9^2 + \left(\frac{72\sqrt{3}}{11}\right)^2 - 2 \cdot 9 \cdot \frac{72\sqrt{3}}{11} \cdot \frac{\sqrt{3}}{2} \\ BF^2 &= 81 + \frac{15552}{121} - \frac{1944}{11}. \end{aligned}$$

Finding a common denominator of 121:

$$\begin{aligned} BF^2 &= \frac{81 \cdot 121 + 15552 - 1944 \cdot 11}{121} \\ BF^2 &= \frac{9801 + 15552 - 21384}{121} \\ BF^2 &= \frac{3969}{121}. \end{aligned}$$

Taking the square root of both sides ($\sqrt{3969} = 63$ and $\sqrt{121} = 11$), we find:

$$BF = \frac{63}{11}.$$

Thus, the answer is (E) $\frac{63}{11}$.

9. (AMC 12 12A 2025) Let w be the complex number $2 + i$, where $i = \sqrt{-1}$. What real number r has the property that r , w , and w^2 are three collinear points in the complex plane?

(A) $\frac{3}{4}$ (B) 1 (C) $\frac{7}{5}$ (D) $\frac{3}{2}$ (E) $\frac{5}{3}$

Answer: E

Solution 1 (Rectangular Form)

We begin by calculating w^2 :

$$w^2 = (2 + i)^2 = 4 + 4i - 1 = 3 + 4i.$$

Values on the complex plane can easily be represented as points on the Cartesian plane, so we go ahead and do that so we are in a more familiar place. Translating onto the Cartesian plane, we have the points $(2, 1)$ and $(3, 4)$. The slope of the line passing through these points is $\frac{4-1}{3-2} = 3$, so the equation of this line is

$$y = 3(x - 2) + 1$$

$$y = 3x - 5.$$

We want the real number that passes through this line, which is equivalent to the x -intercept. This occurs when $y = 0$, so the x -intercept of this line is $x = \boxed{\text{(E)} \frac{5}{3}}$.

Solution 2 (Polar Form)

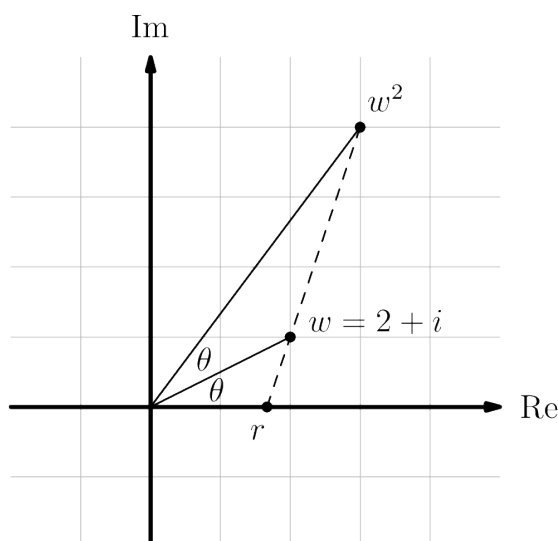
Recall that the slope of a line is $m = \tan \phi$, where ϕ is the angle formed by the line and the positive x -axis.

Note that $|w| = \sqrt{2^2 + 1^2} = \sqrt{5}$. In polar coordinates, let $w = \sqrt{5} \operatorname{cis} \theta$. It follows that $\tan \theta = \frac{1}{2}$.

By De Moivre's Theorem, we have $w^2 = 5 \operatorname{cis}(2\theta)$, from which

$$\tan(2\theta) = \frac{2 \tan \theta}{1 - \tan^2 \theta} = \frac{4}{3}.$$

We obtain the following diagram:



Since $|w^2| = 5$, we have $w^2 = 3 + 4i$ by a 3-4-5 triangle. The complex numbers w and w^2 correspond to the points $(2, 1)$ and $(3, 4)$, respectively, and r corresponds to the x -intercept of this line. In slope-intercept form, the line containing these two points is $y = 3x - 5$. Therefore, the x -intercept is $r = \boxed{\text{(E)} \frac{5}{3}}$.

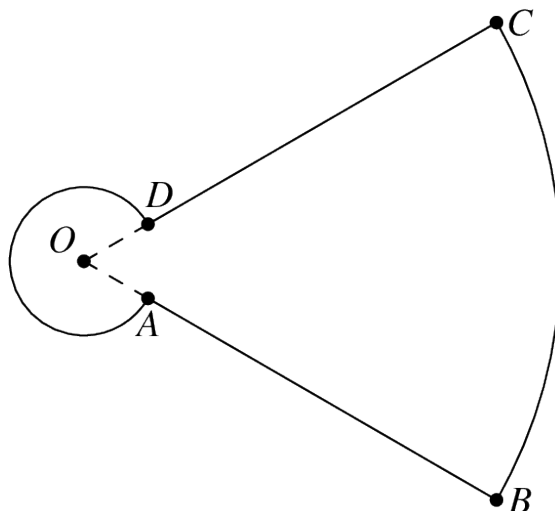
Solution 3 (Coordinate Conversion, Similar to Solution 1)

Determine $2 + i = (2, 1)$ on the real cartesian plane and notice $(2 + i)^2 = 3 + 4i$ takes the point $(3, 4)$. The slope of the line through the two points is $3x - 5 = y$ so $y = 0$ gives $x = \boxed{\frac{5}{3}}$. Because the x axis and the $\Re(x)$ axis are identical, we are finished.

The solution is here to show a different point of logic for the x axis.

10. (AMC 12 12A 2025) In the figure shown below, major arc $\widehat{AD}$ and minor arc $\widehat{BC}$ have the same center, O . Also, A lies between O and B , and D lies between O and

C . Major arc $\widehat{AD}$, minor arc $\widehat{BC}$, and each of the two segments $\overline{AB}$ and $\overline{CD}$ have length 2π . What is the distance from O to A ?



- (A) 1 (B) $1 - \pi + \sqrt{\pi^2 + 1}$ (C) $\frac{\pi}{2}$ (D) $\frac{\sqrt{\pi^2 + 1}}{2}$ (E) 2

Answer: B

Solution 1 (Simple)

The ratio between the radius and the arc length is constant. For the inner circle, the radius, which we will denote as r , has a corresponding arc length of $2\pi r - 2\pi$ (the circumference minus the major arc length). For the outer circle, the radius,

11. (AMC 12 12A 2025) The orthocenter of a triangle is the concurrent intersection of the three (possibly extended) altitudes. What is the sum of the coordinates of the orthocenter of the triangle whose vertices are $A(2, 31)$, $B(8, 27)$, and $C(18, 27)$?
- (A) 5 (B) 17 (C) $10 + 4\sqrt{17} + 2\sqrt{13}$ (D) $\frac{113}{3}$ (E) 54

Answer: A

Solution 1 (coordinate geometry)

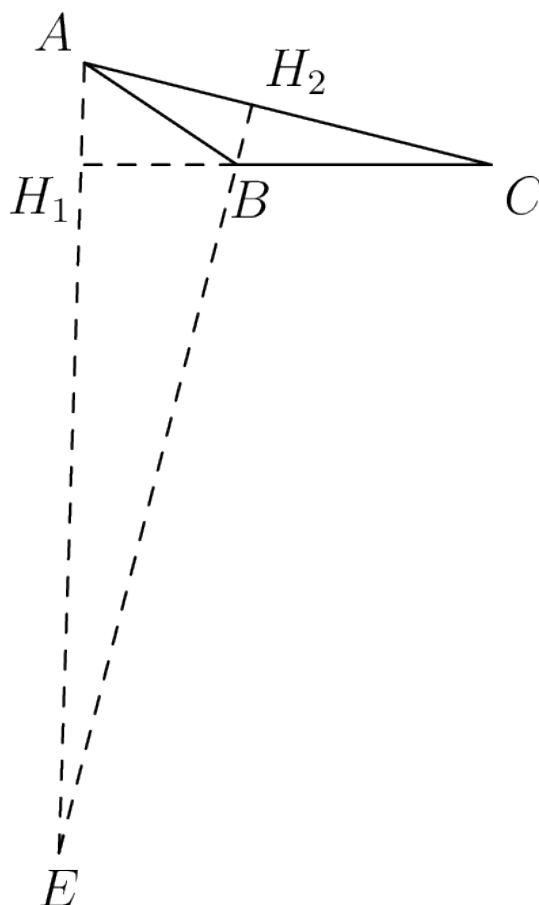
As we are given the coordinates of all 3 vertices, we can compute the slope (or gradient) of each side, and thus the slope of each altitude since it must be perpendicular to the corresponding side. We can therefore determine the Cartesian equation of each altitude using this slope and a point on it (namely, the vertex opposite to the corresponding side), after which we only need to find the point of intersection of at least 2 of these lines.

Accordingly, since BC is horizontal (as B and C both have the same y -coordinate, namely 27), we observe that the altitude through A perpendicular to BC will be vertical. As A has x -coordinate 2, the equation of this altitude is therefore precisely $x = 2$. Now, as explained above, we need to determine the equation of 1 more altitude, so we arbitrarily choose the one through C perpendicular to AB . The slope of AB is $\frac{27-31}{8-2} = -\frac{2}{3}$, so because the slopes of perpendicular lines (other than a horizontal and a vertical line) have product -1 , the slope of this altitude is $\frac{-1}{(-\frac{2}{3})} = \frac{3}{2}$. Hence, as this altitude passes through C , its equation in point-slope form is $y - 27 = \frac{3}{2}(x - 18)$, which becomes $y - 27 = \frac{3}{2}x - 27$, and thus reduces

to $y = \frac{3}{2}x$. Consequently, solving this equation simultaneously with $x = 2$ (from above) gives the coordinates of the orthocenter as simply $(2, \frac{3}{2} \cdot 2) = (2, 3)$, and so, finally, the answer is $2 + 3 = \boxed{\text{(A)} 5}$.

Alternative: using the altitude through B While we arbitrarily chose to use the altitudes through A and C in the solution above, we could of course use any 2 of the 3 altitudes to find the coordinates of the orthocenter as their point of intersection. It's also a good idea to check our answer from above by verifying that $(2, 3)$ indeed lies on the third altitude, namely that through B perpendicular to AC . Therefore, we compute the slope of AC as $\frac{27-31}{18-2} = -\frac{1}{4}$, so that of the altitude is $\frac{-1}{(-\frac{1}{4})} = 4$ (again using the 'slopes have product -1 ' result referred to above). We deduce that the equation of this altitude in point-slope form is $y - 27 = 4(x - 8) \iff y - 27 = 4x - 32 \iff y = 4x - 5$, and solving this equation simultaneously with either $x = 2$ or $y = \frac{3}{2}x$ from above indeed gives the solution $x = 2, y = 4 \cdot 2 - 5 = \frac{3}{2} \cdot 2 = 3$. In other words, the coordinates of the orthocenter are precisely $(2, 3)$, confirming our answer from above.

Solution 2 (trigonometry)



Let H_1 be the foot of the altitude from A to BC , H_2 similarly be the foot of the altitude from B to AC , and E be the point of intersection of the lines AH_1 and BH_2 . Then we have $\angle EH_1B = \angle EH_2A = 90^\circ$, so since angles in a triangle sum to 180° , we obtain

$$\angle EBH_1 = 180^\circ - \angle EH_1B - \angle BEH_1$$

$$\begin{aligned}
 &= 180^\circ - 90^\circ - \angle BEH_1 && \text{(from above)} \\
 &= 180^\circ - \angle EH_2A - \angle BEH_1 && \text{(again from above)} \\
 &= 180^\circ - \angle EH_2A - \angle AEH_2 && \text{(as } H_1 \text{ lies on } AE \text{ and } B \text{ lies on } H_2E, \text{ so } \angle BEH_1 = \angle AEH_2) \\
 &= \angle EAH_2.
 \end{aligned}$$

Moreover, $\angle EAH_2$ is also the same angle as $\angle H_1AC$, since (once again) H_1 lies on AE and H_2 lies on AC . As H_1 has the same x -coordinate as A and the same y -coordinate as C , its coordinates are precisely $(2, 27)$, so we now deduce

$$\begin{aligned}
 \tan \angle EBH_1 &= \tan \angle EAH_2 = \tan \angle H_1AC \\
 &= \frac{CH_1}{AH_1} && \text{(as } \angle AH_1C = 90^\circ, \text{ i.e. triangle } \triangle AH_1C \text{ is right-angled at } H_1) \\
 &= \frac{18-2}{31-27} && \text{(using the coordinates of } H_1 \text{ from above)} \\
 &= 4.
 \end{aligned}$$

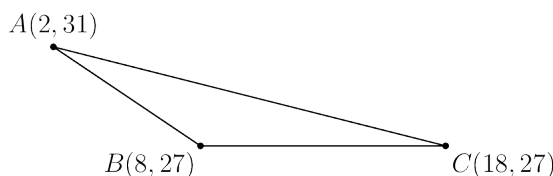
But again as $\angle EH_1B = 90^\circ$, triangle $\triangle EH_1B$ is also right-angled at H_1 , so, similarly to above,

$$\begin{aligned}
 \tan \angle EBH_1 &= \frac{EH_1}{BH_1} \\
 &= \frac{EH_1}{8-2} && \text{(again using the coordinates of } H_1 \text{ from above)} \\
 \iff \frac{EH_1}{6} &= 4 \iff EH_1 = 24.
 \end{aligned}$$

Hence, as E lies vertically *below* H_1 , it finally follows that its coordinates are $(2, 27 - EH_1) = (2, 27 - 24) = (2, 3)$, and thus the answer is $2 + 3 = \boxed{\text{(A) } 5}$.

Solution 3 (Graph analysis)

WLOG assume BC is base as BC is collinear on the same line $y = 27$. Our triangle looks as such:



One altitude has slope undefined and equation $x = 2$. Another such altitude defined h_B is perpendicular to AC who has slope $\frac{-1}{4}$, so h_B has slope 4. It passes through $(8, 27)$, so $y = 4x + b$ and $27 = 32 + b$ so $b = -5$ and therefore $y = 4x - 5$ and when $x = 2$, $y = 8 - 5 = 3$. The sum is $\boxed{5}$.

OR

Translate point B to $(0, 0)$ and so $A(-6, 4), C(10, 0)$. The line $x = -6$ and $y = 4x$ exists, whose intersection is at $(-6, -24)$. Adding back $(x, y) \rightarrow (x + 8, y + 27)$ we obtain $H(2, 3)$ whose sum is $\boxed{5}$.

Solution 4 (Finding Slope among other Things)

The orthocenter of a triangle is located at the intersection of its altitudes. Altitudes are lines that cross from a vertex to the opposite side, such that they form a 90° angle where they cross the base. We can use this fact and our knowledge of slope to find where two of these lines intersect.

Start by looking at line BC , a horizontal line with a slope of 0. A line perpendicular to this that crosses point A must have the equation $x = 2$.

We now do the same for line AB . The line passing through AB has a slope of

$$m = \frac{27 - 31}{8 - 2} \Rightarrow -\frac{2}{3}.$$

The line perpendicular to this has a slope of $\frac{3}{2}$. This altitude will also cross point C . Using point-slope form, and substituting in $x = 2$ from our first altitude, we find the y -coordinate of the intersection:

$$y = \frac{3}{2}(x - 18) + 27 \Rightarrow \frac{3}{2}(2 - 18) + 27 = -24 + 27 = 3.$$

Thus, the orthocenter is located at $(2, 3)$. The sum of the coordinates is

$$2 + 3 = \boxed{5}$$

$$5 = \boxed{A}$$

Solution 5 (Vectors)

The orthocenter of a triangle is the intersection of any 2 of the altitudes. This is helpful because we can choose any two altitudes (some may be more convenient than others).

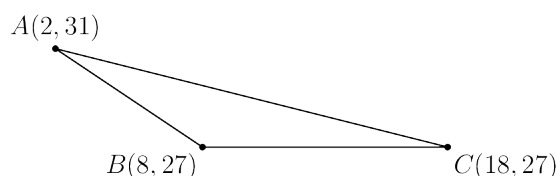


Diagram by pinotation

Notice it is convenient to use the altitude drawn from vertex A , because the line perpendicular to side BC is simply parallel to the y -axis.

The line parallel to the y -axis that crosses vertex A is simply $x = 2$. For the next altitude, we continue with vectors

$$\begin{aligned} \text{Let } (x, y) \text{ be point } D \text{ such that } \overrightarrow{AB} \perp \overrightarrow{CD} \\ \Rightarrow (8 - 2, 27 - 31) \cdot (x - 18, y - 27) = 0 \\ \Rightarrow 6(x - 18) + (-4)(y - 27) = 0 \\ \Rightarrow 4y = 6x \\ \Rightarrow y = \frac{3x}{2} \end{aligned}$$

By naming connecting a point D to vertex C and defining it as perpendicular to line $\overrightarrow{AB}$, we know line $\overrightarrow{CD}$ lies on the line containing the altitude from C . But

because they are perpendicular, the dot product between the two vectors is 0, from which we get the line point D lies on (and therefore the altitude).

Now finding the intersection of the 2 lines,

$$x = 2 \quad y = \frac{3}{2}x \implies \text{Orthocenter is } (2, 3)$$

The answer is $2 + 3 = 5 \implies \boxed{A}$

12. (AMC 12 12A 2025) The *harmonic mean* of a collection of numbers is the reciprocal of the arithmetic mean of the reciprocals of the numbers in the collection. For example, the harmonic mean of 4, 4, and 5 is

$$\frac{1}{\frac{1}{3}(\frac{1}{4} + \frac{1}{4} + \frac{1}{5})} = \frac{30}{7}.$$

What is the harmonic mean of all the real roots of the 4050th degree polynomial

$$\prod_{k=1}^{2025} (kx^2 - 4x - 3) = (x^2 - 4x - 3)(2x^2 - 4x - 3)(3x^2 - 4x - 3)\dots(2025x^2 - 4x - 3)?$$

$$(A) -\frac{5}{3} \quad (B) -\frac{3}{2} \quad (C) -\frac{6}{5} \quad (D) -\frac{5}{6} \quad (E) -\frac{2}{3}$$

Answer: B

Solution 1

We will need to determine the sum of the reciprocals of the roots. To find the sum of the reciprocals of the roots p, q of the quadratic $ax^2 + bx + c$, we use Vieta's formulas. Recall that $p + q = -b/a$ and $pq = c/a$. Therefore,

$$\frac{1}{p} + \frac{1}{q} = \frac{p+q}{pq} = \frac{\frac{-b}{a}}{\frac{c}{a}} = \frac{-b}{a} \cdot \frac{a}{c} = \frac{-b}{c},$$

which doesn't depend on a .

The sum of the reciprocals of the roots of the quadratic $x^2 - 4x - 3$ is $\frac{-(-4)}{-3} = -4/3$. The same is true for every quadratic in the form $ax^2 - 4x - 3$. The sum of all the reciprocals of the roots of $ax^2 + bx + c$ is $2025 \cdot \left(-\frac{4}{3}\right)$.

Because we have 2025 quadratics, there are $2 \cdot 2025 = 4050$ total roots. Our answer

$$\text{is } \frac{1}{\frac{1}{4050} \cdot \frac{-4 \cdot 2025}{3}} = \boxed{(B) -\frac{3}{2}}.$$

Solution 2 (similar to solution 1 but with quadratic formula)

We first find the general roots for quadratics in the form $kx^2 - 4x - 3$. Using the quadratic formula we have

$$\begin{aligned} x &= \frac{4 \pm \sqrt{16 + 12k}}{2k} \\ &= \frac{4 \pm 2\sqrt{4 + 3k}}{2k} \\ &= \frac{2 + \sqrt{4 + 3k}}{k}, \frac{2 - \sqrt{4 + 3k}}{k} \end{aligned}$$

Since we are asked to add the reciprocals of all 4050 roots in the harmonic mean, we will first add the general roots in terms of k . We have,

$$\begin{aligned}\frac{k}{2 + \sqrt{4 + 3k}} + \frac{k}{2 - \sqrt{4 + 3k}} &= \frac{k(2 - \sqrt{4 + 3k}) + k(2 + \sqrt{4 + 3k})}{2^2 - (\sqrt{4 + 3k})^2} \\ &= -\frac{4k}{3k} = -\frac{4}{3}.\end{aligned}$$

Thus, each pair of roots add up to $-\frac{4}{3}$, and since there are 2025 pairs of roots, the harmonic mean of the desired expression is

$$\begin{aligned}\frac{1}{\frac{1}{4050} \left(2025 \left(-\frac{4}{3} \right) \right)} &= \frac{1}{\frac{1}{2} \left(-\frac{4}{3} \right)} \\ &= \boxed{-\frac{3}{2}}, \boxed{B}.\end{aligned}$$

Solution 3 (Vieta's only)

We are asked to find $\frac{4050}{\sum_{i=1}^{4050} \frac{1}{r_i}}$. By Vieta's, note that $r_1 \cdot r_2 \dots r_{4050} = -\frac{3^{2025}}{2025!} = k$ (k is a constant). Then, note that we are asked to find $\frac{4050}{\sum_{i=1}^n \frac{r_i}{k}}$, and by Vieta's we get that $\sum_{i=1}^n \frac{k}{r_i} = \frac{2025 \cdot 4 \cdot 3^{2024}}{2025!}$, so substituting this in, we get

$$-\frac{4050}{\frac{2025 \cdot 4 \cdot 3^{2024}}{\frac{2025!}{3^{2025}}}} = -\frac{4050 \cdot 3^{2025}}{4050 \cdot 3^{2024} \cdot 2} = \boxed{-\frac{3}{2}}.$$

Solution 4 (Vieta's Formulas)

We have:

$$\prod_{k=1}^{2025} (kx^2 - 4x - 3) = (x^2 - 4x - 3)(2x^2 - 4x - 3)(3x^2 - 4x - 3) \dots (2025x^2 - 4x - 3)$$

So we can analyze each part of this "snake":

According to Vieta's Formulas:

For the first one: $x^2 - 4x - 3$: We have:

$$\begin{cases} x_1 + x_2 = \frac{4}{1} \\ x_1 x_2 = -\frac{3}{1} \end{cases}$$

So we have:

$$\frac{1}{x_1} + \frac{1}{x_2} = \frac{x_1 + x_2}{x_1 x_2} = -\frac{4}{3}$$

For the second one: $2x^2 - 4x - 3$: We have:

$$\begin{cases} x_3 + x_4 = \frac{4}{2} \\ x_3 x_4 = -\frac{3}{2} \end{cases}$$

So we have:

$$\frac{1}{x_3} + \frac{1}{x_4} = \frac{x_3 + x_4}{x_3 x_4} = -\frac{4}{3}$$

For the third one: $3x^2 - 4x - 3$: We have:

$$\begin{cases} x_5 + x_6 = \frac{4}{3} \\ x_5 x_6 = -\frac{3}{3} \end{cases}$$

So we have:

$$\frac{1}{x_5} + \frac{1}{x_6} = \frac{x_5 + x_6}{x_5 x_6} = -\frac{4}{3}$$

.....

Then we should check the last one. For the 2025th. one: $2025x^2 - 4x - 3$: We have:

$$\begin{cases} x_{4049} + x_{4050} = \frac{4}{2025} \\ x_{4049} x_{4050} = -\frac{3}{2025} \end{cases}$$

So we have:

$$\frac{1}{x_{4049}} + \frac{1}{x_{4050}} = \frac{x_{4049} + x_{4050}}{x_{4049} x_{4050}} = -\frac{4}{3}$$

$$\begin{aligned} \therefore H.M. &= \frac{1}{\frac{1}{4050} \sum_{i=1}^{4050} \frac{1}{x_i}} \\ &= \frac{4050}{2025 \times \left(-\frac{4}{3}\right)} \\ &= -\frac{3}{2} \end{aligned}$$

Therefore, the final answer is $\boxed{-\frac{3}{2}}$.

Solution 5 (Vieta's Formulas)

For a general quadratic polynomial $ax^2 + bx + c = 0$ with roots p and q , the sum of the reciprocals is:

$$\frac{1}{p} + \frac{1}{q} = \frac{p+q}{pq}$$

Using Vieta's formulas ($p+q = -\frac{b}{a}$ and $pq = \frac{c}{a}$):

$$\frac{1}{p} + \frac{1}{q} = \frac{-\frac{b}{a}}{\frac{c}{a}} = -\frac{b}{c}$$

This result demonstrates that the sum of the reciprocals depends only on the coefficients b and c , and is independent of the leading coefficient a . Since the overall polynomial is a product of quadratics, and all these quadratics share the same b and c coefficients, the sum of the reciprocals for the roots of every individual quadratic is the same. We use the coefficients from one of the quadratics, which is $ax^2 - 4x - 3 = 0$. Thus, $b = -4$ and $c = -3$.

$$\sum(\text{reciprocals}) = -\frac{b}{c} = -\frac{(-4)}{(-3)} = -\frac{4}{3}$$

We substitute the calculated sum of the reciprocals into the formula for the harmonic mean of 2 numbers:

$$\text{HM} = \frac{2}{-\frac{4}{3}} = 2 \times \left(-\frac{3}{4}\right) = -\frac{6}{4} = -\frac{3}{2}$$

The harmonic mean of the roots of any single quadratic is $-\frac{3}{2}$. Since the sum of the reciprocals is the same for all quadratics, the harmonic mean of all the roots of the overall polynomial is also $\boxed{\text{(B)} \quad -\frac{3}{2}}$.

Note It is important to note that the question asks for the sum of all *real* roots. We must therefore be careful to make sure that the 4050 roots of the polynomial are all real, and also that they are distinct, as otherwise e.g. double roots would be counted twice, when the problem requires us to only count each root once (i.e. regardless of multiplicity). Accordingly, we compute that each quadratic factor $(kx^2 - 4x - 3)$ (for $1 \leq k \leq 2025$) has discriminant $(-4)^2 - 4 \cdot k \cdot (-3) = 16 + 12k$, which is clearly positive as $k \geq 1$. This means the 2 roots of each quadratic factor are indeed real and distinct (i.e. none of these quadratics has a double root), so it only remains to show that these roots are also distinct from those of all the *other* quadratic factors. Hence, if we assume that a is a common root of 2 different quadratic factors simultaneously, say $(px^2 - 4x - 3)$ and $(qx^2 - 4x - 3)$ with $1 \leq p, q \leq 2025$ and $p \neq q$, then we must have $pa^2 - 4a - 3 = qa^2 - 4a - 3 = 0$, which implies $pa^2 = qa^2$. This equation factorises as $(p - q)a^2 = 0$, and because $p \neq q$, i.e. $p - q \neq 0$, it further reduces to simply $a^2 = 0$, and hence $a = 0$. But then it follows that $pa^2 - 4a - 3 = p \cdot 0^2 - 4 \cdot 0 - 3 = -3 \neq 0$, yielding a contradiction, so a cannot in fact have been a common root of 2 different quadratic factors, and thus we finally deduce that all 4050 roots are distinct, as required. Shadowleafy

Solution 6

Let the polynomial be $f(x)$, and denote the 4050 roots to be $x_1, x_2, \dots, x_{4050}$. Hence,

$$HM = \frac{4050}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_{4050}}}.$$

We can multiply the numerator and denominator of this fraction by $x_1 x_2 \dots x_{4050}$ to create symmetric sums, which yields

$$HM = \frac{4050(x_1 x_2 \dots x_{4050})}{x_2 x_3 \dots x_{4050} + x_1 x_3 \dots x_{4050} + \dots + x_1 x_2 \dots x_{4049}}.$$

By Vieta's Formulas, since $f(x)$ is of even degree, the product of its roots, $x_1 x_2 \dots x_{4050}$, is just the constant term of $f(x)$, call it c_0 . Likewise, the denominator of our harmonic mean, $x_2 x_3 \dots x_{4050} + x_1 x_3 \dots x_{4050} + \dots + x_1 x_2 \dots x_{4049}$, is the negated coefficient of x in the standard form of $f(x)$. Let the coefficient of x in the standard form of $f(x)$ be c_1 . Note that we do not have to worry about dividing by the coefficient of x^{4050} when using Vieta's Formulas, as they will eventually cancel out in our harmonic mean calculations.

So,

$$HM = \frac{4050c_0}{-c_1}.$$

The constant term in $f(x)$ is just $c_0 = (-3)^{2025}$. For the coefficient of the x term in $f(x)$, there are $\binom{2025}{2024} = 2025$ ways to choose 2024 of the trinomials to include a -3 , and the one trinomial not chosen will include a $-4x$. Hence, $c_1 = 2025 \cdot (-3)^{2024} \cdot (-4)$.

Finally,

$$HM = \frac{4050 \cdot (-3)^{2025}}{-2025 \cdot (-3)^{2024} \cdot (-4)} = \frac{2 \cdot (-3)}{-(-4)} = \boxed{\text{(B)} - \frac{3}{2}}.$$

Solution 7 (Cheese)

We can try replacing the 2025 in the problem with smaller numbers like 1 and 2. For 1, we have the quadratic $x^2 - 4x - 3$. Using the quadratic formula, we have the roots are $2 + \sqrt{7}$ and $2 - \sqrt{7}$. The harmonic mean of these two is

$$\frac{2}{\frac{1}{2+\sqrt{7}} + \frac{1}{2-\sqrt{7}}} = \frac{2}{\frac{\sqrt{7}-2}{3} + \frac{2+\sqrt{7}}{-3}} = \frac{2}{\frac{-4}{3}} = -\frac{3}{2}.$$

We notice that this is one of the answer choices. Also, given that the random choice of 2025, and that the rest of the answer choices are also simple fractions, we can

safely guess that the answer is $\boxed{\text{(B)} - \frac{3}{2}}$.

Solution 8 (Symmetry of Unweighted Means)

Note that all the roots are real since for each quadratic,

$$\Delta = b^2 - 4ac = (-4)^2 - 4k(-3) = 16 + 12k > 0$$

for $k \geq 1$. We will use the fundamental theorem of symmetric polynomials. By Vieta's formulas, we have for each quadratic $ax^2 + bx + c = kx^2 - 4x - 3$:

$$\frac{1}{r} + \frac{1}{s} = \frac{r+s}{rs} = \frac{-b/a}{c/a} = -\frac{b}{c} = -\frac{4}{3}$$

Thus, $HM(r, s) = \frac{2}{\frac{1}{r} + \frac{1}{s}} = \frac{2}{-\frac{4}{3}} = -\frac{3}{2}$. Since the unweighted harmonic mean is symmetric in its arguments, we can group terms to get

$$HM(r_1, r_2, r_3, \dots, r_{4050}) = HM\left(HM(r_1, r_2), HM(r_3, r_4), \dots, HM(r_{4049}, r_{4050})\right) = \boxed{\text{(B)} - \frac{3}{2}}$$

Note: For an unweighted symmetric mean M , we have for $d|n$, $M\{x_k\}_{k=1}^n = M\{M\{x_{ds+k}\}_{k=1}^d\}_{s=0}^{n/d-1}$ and $M\{\bar{x}\}_k = \bar{x}$ for constant mean $\bar{x}$. You can also check this individually for HM:

$$HM(x_1, \dots, x_n) = n(x_1^{-1} + \dots + x_n^{-1})^{-1} = \frac{n}{d} \left[\underbrace{d(x_1^{-1} + \dots + x_d^{-1}) + \dots + d(x_{(n-1)d+1}^{-1} + \dots + x_n^{-1})}_{\frac{n}{d} \text{ terms}} \right]$$

or by checking arithmetic mean and using $\text{HM}\{x_k\}_k = \frac{1}{\text{AM}\left\{\frac{1}{x_k}\right\}_k}$.

Solution 9 (Easy-ish to understand)

Note: I will try my best to make this solution look less intimidating to read as the other ones. This is my third time writing a solution, so I'm not very used to the $\text{\LaTeX}$.

The harmonic mean that the question is asking is

$$\frac{1}{\frac{r_1 + r_2 + \dots + r_{4049} + r_{4050}}{4050}}$$

where r_i is between 1 and 4050 and is the reciprocal of all 4050 roots of the polynomial. All we need to find is the sum of the reciprocals of the roots and we can substitute that into this formula.

First of all, notice that there are 2025 quadratic polynomials. They each hold 2 out of the 4050 roots, and the sum of the reciprocals of their roots are $\frac{1}{a} + \frac{1}{b} = \frac{a+b}{ab}$.

Using Vieta's we get $\frac{a+b}{ab} = -\frac{4}{3}$ on the every quadratic because only the coefficient for x^2 changes. Also, $a + b$ and ab have the denominator of a , so when they are divided, the quotient won't change. Therefore, the sums of the reciprocals of the quadratics sum to $2025 \cdot -\frac{4}{3}$.

Now we can substitute:

$$\frac{1}{\frac{2025 \cdot -\frac{4}{3}}{4050}}.$$

This gets us

$$\frac{1}{\frac{2025 \cdot -\frac{4}{3}}{4050}} = \frac{1}{\frac{2025 \cdot (-4)}{3 \cdot 4050}} \tag{52}$$

$$= \frac{1}{\frac{-8100}{12150}} \tag{53}$$

$$= \frac{1}{\frac{-2}{3}} \tag{54}$$

$$= 1 \cdot \frac{3}{-2} \tag{55}$$

$$= -\frac{3}{2} \tag{56}$$

Or...you could just solve the last part yourself easier than my solution!

Solution 10 (reciprocal roots)

We know that if for polynomial $ax^2 + bx + c$ is r_1 and r_2 , then the polynomial with roots $\frac{1}{r_1}$ and $\frac{1}{r_2}$ (reciprocal roots) will be $cx^2 + bx + a$ where the coefficients are flipped (works for any degree polynomial, not just quadratic). Thus, using vieta's formula, the sum of the reciprocal roots for polynomial $kx^2 - 4x - 3$ will be the sum of the roots for polynomial $-3x^2 - 4x + k$ which is $-\frac{-4}{-3} = -\frac{4}{3}$.

There are 2025 polynomials like this, so the sum of roots would be $2025 \cdot (-\frac{4}{3})$. Finding the harmonic mean, we get:

$$\frac{1}{\frac{1}{4050} \cdot \frac{-4 \cdot 2025}{3}} = \frac{1}{\frac{-2}{3}}$$

Or **(B)** $-\frac{3}{2}$

13. (AMC 12 12A 2025) Let $C = \{1, 2, 3, \dots, 13\}$. Let N be the greatest integer such that there exists a subset of C with N elements that does not contain five consecutive integers. Suppose N integers are chosen at random from C without replacement. What is the probability that the chosen elements do not include five consecutive integers?

(A) $\frac{3}{130}$ (B) $\frac{3}{143}$ (C) $\frac{5}{143}$ (D) $\frac{1}{26}$ (E) $\frac{5}{78}$

Answer: D

Solution 1

We first find what N is by figuring out how many numbers we need to take out of the set so that the set does not contain 5 consecutive integers. Since N must be maximized, we must minimize what numbers are removed, and we quickly find that taking two numbers out works. Consider taking out 5 and 10. You are left with $\{1, 2, 3, 4, 6, 7, 8, 9, 11, 12, 13\}$, which does not have a string of 5 consecutive integers.

There are only 3 ways to take out two integers such that the resulting set meets our condition (5 and 10, 5 and 9, or 4 and 9), and $\binom{13}{2} = 78$ total ways to choose such integers. Therefore, the probability is $\frac{1}{26}$.

Minor edits aashrithm29

Very minor LaTeX edit PerseverePlayer

Note: An alternative way to come to the conclusion that there are only three possible subsets of length twelve is as follows:

Let the two numbers removed from C be a and b such that $a < b$. Then, because we cannot have any sequence of five consecutive numbers, $a \leq 5$, $b - a \leq 5$, and $13 - b + 1 \leq 5$. Then, graphing this system in the ab plane, we easily see that there are only three solutions for (a, b) : (5, 10), (4, 9), and (5, 9). Then, simply proceed as described above.

Solution 2

Trying to find a subset that satisfies the condition, we get $\{1, 2, 3, 4, 6, 7, 8, 9, 11, 12, 13\}$, which has $N = 11$ elements. The subsets $\{1, 2, 3, 5, 6, 7, 8, 10, 11, 12, 13\}$ and $\{1, 2, 3, 4, 6, 7, 8, 10, 11, 12, 13\}$ also work. In total, we have 3 subsets and $\binom{13}{11}$ ways to choose 11 elements from C , so the probability is $\frac{3}{\binom{13}{11}} = \frac{1}{26}$. Thus, the

answer is **(D)** anzhuPro

Solution 3

First, let's find the number N : Consider if:

$N = 13$:

It is impossible - It doesn't remove any integer, violating the condition "not contain five consecutive integers".

$N = 12$:

It is also impossible, and still does not satisfy the condition. An example could be $\{1, 2, 3, 4, 6, 7, 8, 9, 10, 11, 12, 13\}$. 1, 2, 3, and 4 doesn't violate the condition, but 6 - 13 are all consecutive, hence N can't be 12.

$N = 11$:

This is possible, because $\frac{11}{3} = 3$ remainder 2. This means that we can have one string of consecutive length 3 and two strings of consecutive length 4. Hence, it could be either:

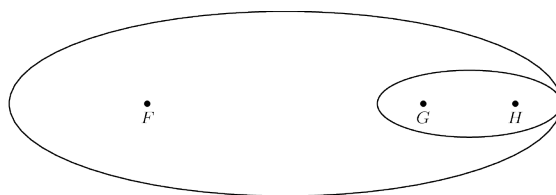
- $\{1, 2, 3, 5, 6, 7, 8, 10, 11, 12, 13\}$ with consecutive integer lengths of (3, 4, 4).
- $\{1, 2, 3, 4, 6, 7, 8, 10, 11, 12, 13\}$ with consecutive integer lengths of (4, 3, 4).
- $\{1, 2, 3, 4, 6, 7, 8, 9, 11, 12, 13\}$ with consecutive integer lengths of (4, 4, 3).

Thus giving 3 cases. The total ways of picking out 2 integers from a set of 13 is $\binom{13}{2} = 78$. Hence, the probability is $\frac{3}{78} = \boxed{\frac{1}{26}}$.

Note: You can also consider the combinations of 1 string of consecutive length 3 and 2 strings of consecutive length 4. The length of 3 has 3 places to go, filling the rest 2 with 4 places each, giving $3/78$ as well, resulting in the same answer.

Formatting and minor edits Logibyte

14. (AMC 12 12A 2025) Points F , G , and H are collinear with G between F and H . The ellipse with foci at G and H is internally tangent to the ellipse with foci at F and G , as shown below.



The two ellipses have the same eccentricity e , and the ratio of their areas is 2025. (Recall that the eccentricity of an ellipse is $e = \frac{c}{a}$, where c is the distance from the center to a focus, and $2a$ is the length of the major axis.) What is e ?

- (A) $\frac{3}{5}$ (B) $\frac{16}{25}$ (C) $\frac{4}{5}$ (D) $\frac{22}{23}$ (E) $\frac{44}{45}$

Answer: D

Solution 1 (Simple)

Let the outer ellipse be ellipse 1, and the inner ellipse be ellipse 2. Let a_1 , b_1 , and c_1 , correspond to the semimajor axis, semiminor axis, and focal distance of ellipse 1, respectively. Similarly, let a_2 , b_2 , and c_2 correspond to the semimajor axis, semiminor axis, and focal distance of ellipse 2, respectively.

Ellipses with the same eccentricity are similar, so $\frac{a_1}{a_2} = \sqrt{2025} = 45$ because the ratio of semimajor axes between similar ellipses is equal to the square root of the ratio between their areas. Notice how

$$a_1 = c_1 + c_2 + a_2.$$

Substituting from the eccentricity equation,

$$a_1 = ea_1 + ea_2 + a_2.$$

Rearranging gives us

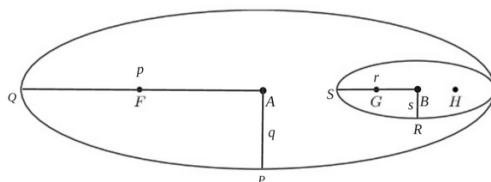
$$a_1 - ea_1 = ea_2 + a_2$$

$$a_1(1 - e) = a_2(1 + e)$$

$$\frac{a_1}{a_2} = \frac{1 + e}{1 - e} = 45.$$

Solving for e then yields $(D) \frac{22}{23}$.

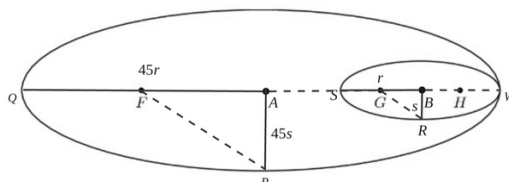
Conic Sections Solution



Let A and B be the centers of the ellipses with foci F, G and foci G, H respectively. Then, say $AQ = p$, $AP = q$, $BS = r$, and $BR = s$. Through area formula of ellipse ($ab\pi$, where a is half the major axis and b is half the minor axis), the area of the ellipse with center at A is πpq and with center at B is πrs . Then, we have $\frac{\pi pq}{\pi rs} = 2025$, in which $\frac{pq}{rs} = 2025$.

Additionally, when two ellipses have the same eccentricity, that essentially means the ellipses are stretched the same amount. Therefore, $\frac{p}{q} = \frac{r}{s}$, in which $ps = qr$, and $p = \frac{qr}{s}$.

From the previous equation $\frac{pq}{rs} = 2025$, we have $pq = 2025rs$, in which $\frac{q^2r}{s} = 2025rs$, in which $q^2r = 2025rs^2$, in which $q^2 = 2025s^2$, and $q = 45s$ (-45 as length cannot be negative). Similarly, $p = 45r$.



Connect F to P and G to R . We form two similar triangles PAF and RBG (through ASA). We just need to determine the length BG , and the problem is solved. Notice that $\frac{RB}{PA} = \frac{BG}{AF}$. because $RB : PA = 45 : 1$, $BG : AF = 1 : 45$, and if $AF = 45x$, $BG = x$. To solve for x , we connect B to the tangency point of both ellipses (W). Then, $AW = 45r$ and $BW = r$, so $AB = 44r$. We found that $AF = 45x$, and through definition of foci, $AG = 45x$. $GB = x$, so $AB = 46x$. However, $AB = 44r$, so $46x = 44r$, $x = \frac{22r}{23}$.

The length of the major axis is $2r$, and because eccentricity is $\frac{c}{a}$, where in our case,

$c = x$ and $a = r$, we have $\frac{22r}{23r}$, or $\frac{22}{23}$.

Solution 2

We label the outer ellipse as Ellipse 1 and the inner ellipse as Ellipse 2. Let the focal distance of Ellipse 1 be $2c_1$ and the major axis be $2a_1$. Similarly, Ellipse 2 has a focal distance of $2c_2$ and a major axis of $2a_2$.

We label the rightmost vertex of both ellipses as I . Because I lies on Ellipse 1, the sum of the lengths of FI and GI equals $2a_1$. The length of FI can be written as $c_2 + a_2 + 2c_1$, and the length of GI can be written as $c_2 + a_2$. Therefore,

$$(c_2 + a_2 + 2c_1) + (c_2 + a_2) = 2a_1$$

$$2c_2 + 2a_2 + 2c_1 = 2a_1$$

$$c_2 + a_2 + c_1 = a_1.$$

Substituting $a_1 = c_1/e$ and $a_2 = c_2/e$, we get

$$c_2 + \frac{c_2}{e} + c_1 = \frac{c_1}{e}$$

$$c_2e + c_2 + c_1e = c_1$$

$$c_2(e + 1) = c_1(1 - e)$$

$$\frac{c_1}{c_2} = \frac{1 + e}{1 - e}$$

We now take into account the information regarding the ratio of the ellipses' areas. Because the area of an ellipse with semi-major axis $2a$ and eccentricity e can be written as $\pi a^2 \sqrt{1 - e^2}$, we get that:

$$\frac{\text{Area}_1}{\text{Area}_2} = \frac{a_1^2 \sqrt{1 - e^2}}{a_2^2 \sqrt{1 - e^2}} = \frac{a_1^2}{a_2^2} = 2025.$$

Therefore, $\frac{a_1}{a_2} = 45$. Because $a_1 = c_1/e$ and $a_2 = c_2/e$, then $\frac{c_1/e}{c_2/e} = \frac{c_1}{c_2} = 45$ as well. Substituting this into the equation found earlier, we get that

$$\frac{c_1}{c_2} = \frac{1 + e}{1 - e} = 45$$

$$1 + e = 45 - 45e$$

$$46e = 44$$

$$e = \boxed{\frac{22}{23}}.$$

Solution 3 (No area formulas!)

In the larger ellipse, let a be half the major axis (also called the semi-major axis) and b be half the minor axis (also called the semi-minor axis), and c be the distance between its center and a focus. Since we are given that $2025 = 45^2$ is the ratio of the areas of the two ellipses, it follows that the linear scale factor of the two is $45 : 1$. Therefore, the semi-major axis of the smaller ellipse is $\frac{1}{45}a$ and its semi-minor axis is $\frac{1}{45}b$. Furthermore, the distance between a focus of the smaller ellipse and its center is $\frac{1}{45}c$. There are now two ways to express the length of the major axis of the large ellipse: $2a$ and $a + c + \frac{1}{45}c + \frac{1}{45}a$. Setting these equal, we now have an expression of degree 1 entirely in terms of a and c . Some algebra gives us:

$$a - \frac{1}{45}a = c + \frac{1}{45}c$$

$$\frac{44}{45}a = \frac{46}{45}c$$

$$44a = 46c$$

$$e = \frac{c}{a} = \boxed{\text{(D)} \frac{22}{23}}$$

Note that this method did not require you to know the area formula of an ellipse (which, incidentally, is πab).

Solution 4 (Similarity)

Parametrize the bigger ellipse with a, c as defined in the problem, and the smaller ellipse with a', c' . Note that the ellipses have the same eccentricity, so they are similar with ratio $k^2 = 2025$ or $k = 45$. We wish to find $e = \frac{c}{a} = \frac{c'}{a'}$.

Looking at the figure, we can decompose the semi-major axis a of the bigger ellipse as

$$a = c + c' + a'$$

Using $c = kc' = 45c'$ and $a = 45a'$ and rearranging gives us:

$$45a' - a' = 44a' = 45c' + c' = 46c'$$

$$\text{Thus } e = \frac{c'}{a'} = \frac{44}{46} = \boxed{\text{(D)} \frac{22}{23}}.$$

15. (AMC 12 12A 2025) A set of numbers is called *sum-free* if whenever x and y are (not necessarily distinct) elements of the set, $x + y$ is not an element of the set. For example, $\{1, 4, 6\}$ and the empty set are sum-free, but $\{1, 4, 5\}$ is not. What is the greatest possible number of elements in a sum-free subset of $\{1, 2, 3, \dots, 20\}$?
- (A) 8 (B) 9 (C) 10 (D) 11 (E) 12

Answer: C

Solution 1

Let our subset be $\{11, 12, 13, \dots, 20\}$. If we add any element from the set $\{1, 2, 3, \dots, 10\}$ to our current subset, we will have to remove at least one element from our subset. We cannot add 10 because if we do, the sum of 10 and itself is 20. Hence, the maximum size of our subset is $\boxed{\text{(C)} 10}$.

Solution 2 (Quicksolve)

Notice that an odd number plus an odd number always results in an even number. Consider the subset of all odd numbers $S = \{1, 3, 5, \dots, 19\}$. The addition of any even number will result in a violation of the rules, so the maximum number of elements in this subset is $\boxed{\text{(C)} 10}$.

Note (Those who know) This problem is essentially the same as [2022 AMC 10B Problem 14](#).

Solution 3

Let $S = \{1, 2, 3, \dots, 20\}$. We are looking for the largest possible sum-free subset $A \subseteq S$.

First, we can show that a sum-free set of size 10 is possible. Consider the set $A_1 = \{11, 12, \dots, 20\}$. The smallest possible sum of two elements (not necessarily

distinct) from this set is $11 + 11 = 22$. Since $22 > 20$, no sum of two elements from A_1 can be in A_1 . The size of A_1 is $20 - 11 + 1 = 10$. Thus, a sum-free set of size 10 does exist.

Next, we prove that no sum-free set $A \subseteq S$ can have 11 or more elements. Let m be the largest element in A . Since A is sum-free and $m \in A$, for any $x \in A$, we must have $m - x \notin A$ (unless $x = m - x$, in which case $x \notin A$). This is because if both x and $m - x$ were in A , their sum $x + (m - x) = m$ would be in A , which is a contradiction.

We can partition the set $\{1, 2, \dots, m\}$ into pairs that sum to m , plus any leftover elements.

Case 1: m is even. Let $m = 2k$. The numbers in $\{1, \dots, m\}$ can be grouped as: $\{(1, 2k - 1), (2, 2k - 2), \dots, (k - 1, k + 1)\}$ along with the numbers k and $m = 2k$.

- A contains m .
- A cannot contain k , because $k + k = 2k = m$, which is in A .
- A can contain at most one number from each of the $k - 1$ pairs.

The maximum number of elements A can contain from $\{1, \dots, m\}$ is $1 + 0 + (k - 1) = k = m/2$. Since $m \leq 20$, the maximum size of A is $m/2 \leq 20/2 = 10$.

Case 2: m is odd. Let $m = 2k - 1$. The numbers in $\{1, \dots, m\}$ can be grouped as: $\{(1, 2k - 2), (2, 2k - 3), \dots, (k - 1, k)\}$ along with $m = 2k - 1$.

- A contains m .
- A can contain at most one number from each of the $k - 1$ pairs.

The maximum number of elements A can contain from $\{1, \dots, m\}$ is $1 + (k - 1) = k$. Since $m = 2k - 1$, we have $k = (m + 1)/2$. The largest possible odd m is 19. If $m = 19$, $k = (19 + 1)/2 = 10$. So the maximum size of A is $(m + 1)/2 \leq (19 + 1)/2 = 10$.

In all cases, a sum-free subset of S can have at most 10 elements. Since we found a set with 10 elements, the greatest possible number of elements is 10.

Solution 4 (Another quicksolve)

Note that selecting integers from $\{11, 12, \dots, 20\}$ also satisfy the constraints of the problem. The minimum sum of any two members of the set is $11 + 11 = 22$ which is greater than any of the elements.

Solution 5 (with quick proof)

As all of the above solutions point out, the set $\{11, \dots, 20\}$ satisfies the required condition, and so the greatest size of a sum-free set is no less than 10. How do we know that 11 isn't possible? Induction is our friend here.

We want to prove that the set $\{1, \dots, 2n\}$ has a sum-free set of size no more than n . It is obvious for $n = 1$. Suppose it is true for $n - 1$. Now let S be a sum-free subset of $\{1, \dots, 2n\}$ and that it has at least $n + 1$ elements. The intersection of S and $\{1, \dots, 2n - 2\}$ is sum-free, so it has no more than $n - 1$ elements. So we must include both $2n - 1$ and $2n$ into S . If so, then 1 and n cannot belong to S .

We are now searching for $n - 1$ more elements. But we can make $n - 2$ 'pigeonholes': $\{2, 2n - 2\}, \{3, 2n - 3\}, \dots, \{n - 1, n + 1\}$, seeing that they can provide no more than $n - 2$ elements because each pigeonhole will have a sum of $2n$.

16. (AMC 12 12A 2025) Triangle $\triangle ABC$ has side lengths $AB = 80$, $BC = 45$, and $AC = 75$. The bisector of $\angle B$ and the altitude to side $\overline{AB}$ intersect at point P . What is BP ?

(A) 18 (B) 19 (C) 20 (D) 21 (E) 22

Answer: D

Solution 1

Let $CD \perp AB$ with foot D . Right triangles ACD and BCD give $AC^2 = AD^2 + CD^2$, $BC^2 = BD^2 + CD^2$, $AC^2 - BC^2 = AD^2 - BD^2 = (AD - BD)(AD + BD)$. Since $AD + BD = AB = 80$ and $AC^2 - BC^2 = 75^2 - 45^2 = 3600$, we get the equation $3600 = 80(AD - BD)$. This equation simplifies to $45 = AD - BD$. We can solve the system of equations $AD + BD = 80$ and $AD - BD = 45$ easily via elimination, and we get $AD = \frac{125}{2}$, $BD = \frac{35}{2}$. $CD^2 = AC^2 - AD^2 = 75^2 - \left(\frac{125}{2}\right)^2 = \frac{6875}{4}$, $CD = \frac{25\sqrt{11}}{2}$.

By Angle Bisector Theorem, $\frac{DP}{PC} = \frac{DB}{BC} = \frac{\frac{35}{2}}{45} = \frac{7}{18}$, $PC = CD - DP$ thus, $18DP = 7(CD - DP)$, $25DP = 7CD$, $DP = \left(\frac{7}{25}\right)CD = \left(\frac{7}{25}\right)\left(\frac{25\sqrt{11}}{2}\right) = \frac{7\sqrt{11}}{2}$. $BP^2 = BD^2 + DP^2 = \left(\frac{35}{2}\right)^2 + \left(\frac{7\sqrt{11}}{2}\right)^2 = \frac{1225}{4} + \frac{49(11)}{4} = \frac{1764}{4} = 441$, thus $BP = \boxed{(D) 21}$.

Solution 2 (Law of Cosines)

Scale this down to a $9 - 15 - 16$ triangle (we will multiply the result by 5 in the end). Note that $9^2 + 16^2 - 18 \cdot 16 \cdot \cos(\angle B) = 15^2$, so $112 = 18 \cdot 16 \cdot \cos(\angle B)$, which simplifies to $\frac{7}{18} = \cos(\angle B)$. Then $\cos(\frac{\angle B}{2}) = \sqrt{\frac{1 + \frac{7}{18}}{2}} = \sqrt{\frac{25}{36}} = \frac{5}{6}$ (positive root since the angle is acute). Therefore, we have $\frac{BD}{BP} = \frac{5}{6}$, assuming that D is the foot of the altitude. There are many ways to proceed from here to find BD . Note that by Heron's formula, the area of the scaled-down triangle is $\sqrt{(20)(4)(5)(11)} = 20\sqrt{11}$. Therefore, $CD = \frac{5}{2}\sqrt{11}$. Using Pythagorean Theorem, we get $BD = \sqrt{81 - \frac{25 \cdot 11}{4}} = \sqrt{\frac{324 - 275}{4}} = \frac{7}{2}$. Therefore, we get $\frac{7}{BP} = \frac{5}{6}$, so $BP = \frac{21}{5}$, and we scale up by 5 to get $\boxed{(D) 21}$. ScoutViolet

Solution 3 (Stewarts)

Let the foot of the altitude coming from C on segment AB be D . Using the fact that CD is a common leg in right triangles $\triangle CDB$ and $\triangle CDA$, we have

$$45^2 - BD^2 = 75^2 - (80 - BD)^2.$$

Expanding gives

$$45^2 = 75^2 - 80^2 + 160BD,$$

so $BD = \frac{35}{2}$. Let the foot of the angle bisector from B to AC be point E . Since BE is the angle bisector of $\angle CBA$, we can use the angle bisector theorem. This gives

$$\frac{45}{CE} = \frac{80}{75 - CE},$$

so $CE = 27$ and $AE = 75 - 27 = 48$. Now we can use Stewart's Theorem to find $\overline{BE}$. We have

$$(48)(45)^2 + (27)(80)^2 = (27)(48)(75) + 75BE^2.$$

To simplify this expression, just divide by the greatest common divisor and solve from there. In the end, we get $BE = 48$. Let $BP = x$, so $PE = 48 - x$. Draw the altitude from E down to AB . Let the foot of this altitude be F . Since $EF \parallel CD$, we have $\triangle AFE \sim \triangle ADC$. Hence, we can write the equation

$$\frac{48}{75} = \frac{AF}{\frac{125}{2}}.$$

Solving gives $AF = 40$, so $FD = \frac{125}{2} - 40 = \frac{45}{2}$. Since $PD \parallel EF$, we also have $\triangle BDP \sim \triangle BFE$, so we have

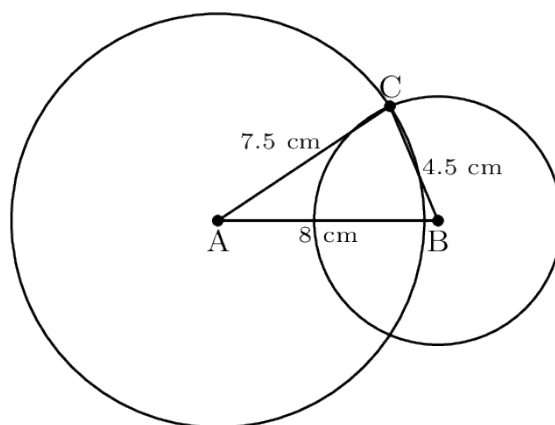
$$\frac{x}{48} = \frac{\frac{35}{2}}{40}.$$

Solving for x gives $x = 21$ or D .

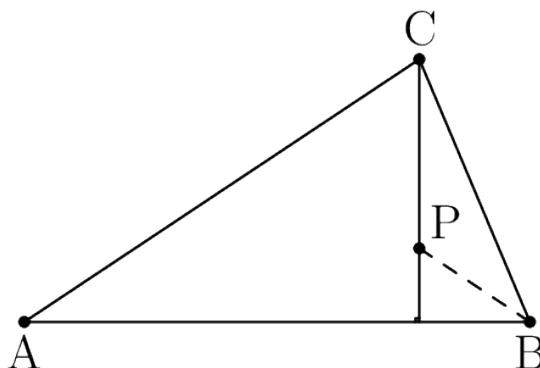
Solution 4 (Rulerbash)

Preface: When we have a problem as such, involving a simple diagram with minimal instructions, I use a method I named "rulerbash". Rulerbash should only be used in specific cases and as a last resort, mainly in the event of a time crunch or a difficult problem.

Start with the longest side, drawing a line with a length of 8 cm (AB). Then, using a compass, draw 2 circles centered around points A and B , 7.5 cm and 4.5 cm radii respectfully. At the point of intersection of these 2 circles, we have point C , completing a perfectly scaled drawing of $\triangle ABC$. (Note the circles are not necessary with a bit of trial and error with the side lengths, they simply offer a way to get it done first try).



After this, dropping the altitude to AB is simple with a ruler and careful placement, and angle bisector can be estimated quite accurately. (Note: Alternately, a compass can be used to accurately place the altitude and the bisector, which are both trivial constructions ZingiberiMaracandae)



After all of this, we can reuse our ruler and measure $BP = 2.1$ cm, and using our scale of $80 = 8$ cm, our final answer is (D) 21.

Other note: you kind of have to draw the angle bisector really accurately for this to work (I tried this) Sakura_kitty

Solution 5 (No Trig)

Let D be the intersection of the altitude from C to AB . To simplify calculations, divide all side lengths by 5, and multiply by 5 again at the end. First, we use Heron's Formula, $\sqrt{s(s-a)(s-b)(s-c)}$, to find the area. Let $[ABC]$ denote the area of $\triangle ABC$. By Heron's Formula,

$$[ABC] = \sqrt{20 \cdot 5 \cdot 4 \cdot 11} = 20\sqrt{11}.$$

Next, we find the altitude CD using the formula for the area of a triangle, $\frac{1}{2}bh = \text{area}$:

$$\frac{1}{2}(16)(CD) = 20\sqrt{11} \Rightarrow CD = \frac{5\sqrt{11}}{2}.$$

We can use the Pythagorean Theorem in $\triangle CDB$ to find DB :

$$DB^2 + \frac{25 \cdot 11}{4} = 81 \Rightarrow 4DB^2 + 275 = 324 \Rightarrow DB^2 = \frac{49}{4},$$

so $DB = \frac{7}{2}$. Next, we use the Angle Bisector Theorem to find PD . Let $x = PC$ and $y = PD$. Since $x + y = \frac{5\sqrt{11}}{2}$, we have $x = \frac{5\sqrt{11}}{2} - y$. From the given ratio,

$$\frac{9}{x} = \frac{7}{2y} \Rightarrow 18y = 7x.$$

Substituting $x = \frac{5\sqrt{11}}{2} - y$,

$$18y = 7 \left(\frac{5\sqrt{11}}{2} - y \right) \Rightarrow 25y = \frac{35\sqrt{11}}{2},$$

so $y = \frac{7\sqrt{11}}{10}$.

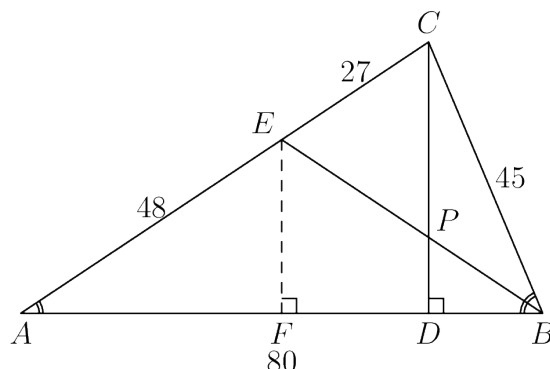
Now, using the Pythagorean Theorem again to find BP :

$$\frac{49 \cdot 11}{100} + \frac{49}{4} = BP^2 \Rightarrow 100BP^2 = 49(11 + 25) = 49 \cdot 36,$$

so $BP = \frac{42}{10} = \frac{21}{5}$.

Finally, multiplying the side lengths by 5 again gives $BP = 21.$, or D.

Solution 6 (Similar Triangle)



Due to Angle Bisector Theorem

$$\frac{CE}{AE} = \frac{BC}{BA} = \frac{45}{80} = \frac{9}{16}$$

$$CE = AC \times \frac{9}{25} = 27$$

$$AE = AC \times \frac{16}{25} = 48$$

Notice that $\frac{CE}{CB} = \frac{CB}{CA} = \frac{3}{5}$, $\triangle CBE \sim \triangle CAB$, $\angle A = \angle CBE = \angle EBF$, so $\triangle ABE$ is isosceles triangle, hence

$$AF = BF$$

Notice that $AF : FD = AE : EC = 16 : 9$, $DB = FB - FD$, then

$$AF : DB = 16 : 7$$

Notice that $\triangle PBD \sim \triangle EAF$, $PB = AE \times \frac{BD}{AF} = 48 \times \frac{7}{16} = \boxed{\text{(D) } 21}$.

Solution 7 (Pythagoras+angle bisector theorem)

Let CD be the altitude to side AB , and let E be the point of intersection of the angle bisector of $\angle B$. Let P be the point of intersection of the angle bisector and the altitude.

By Pythagoras, $CD^2 = BC^2 - BD^2 = AC^2 - AD^2$

We let $BD = x$, so $AD = 80 - x$

Now we have

$$45^2 - x^2 = 75^2 - (80 - x)^2$$

(This looks complicated, but we can see that we are going to have two $-x^2$ terms on both sides, so it cancels out)

Simplifying and rearranging gives us $80 - 2x = 45$, so $x = \frac{35}{2}$

Therefore, $BD = \frac{35}{2}$, and $AD = \frac{125}{2}$. This gives us $CD = \frac{25}{2}\sqrt{11}$

Now, note that in triangle BDC , BP bisects $\angle DBC$

By the angle bisection theorem, we have $\frac{BD}{DP} = \frac{BC}{CP}$

which gives us $\frac{DP}{CP} = \frac{BD}{BC} = \frac{7}{18}$

We have $DP = \frac{7}{25} * CD = \frac{7}{2}\sqrt{11}$ and $CP = \frac{18}{25} * CD = \frac{18}{2}\sqrt{11}$

So

$$BP^2 = BD * BC - DP * CP = \frac{35}{2} * 45 - \frac{7}{2} * \frac{18}{2} * 11 = 441$$

Therefore, $BP = 21$

(This is my first solution, pls point out any mistakes in math or LATEX). backtosq-1

Alternative Ending to Solution 7 As an alternative ending equation to the solution above, I used the Pythagorean Theorem again, giving us: $BP^2 = BD^2 + PD^2 = (\frac{35}{2})^2 + (\frac{7\sqrt{11}}{2})^2 = (\frac{1225}{4}) + (\frac{539}{4}) = \frac{1764}{4} = 441$

Thus, $BP = \boxed{\text{(D)} 21}$

Solution 8 (Pythagora's + Double Angle Identity)

Since all the lengths share a factor of 5, divide by 5 and multiply it back at the end for smaller numbers. Drop the altitude from vertex C to point D on AB . Then comparing the altitude by Pythagora's on both sides gives

$$9^2 - x^2 = 15^2 - (16 - x)^2 = 15^2 - 16^2 + 32x - x^2 = -31 + 32x - x^2$$

Cancelling the $-x^2$ then solving gives us $x = \frac{112}{32} = \frac{7}{2}$. Since $\triangle BDC$ is a right triangle with $\angle B = 2\theta$, we find $\cos(2\theta) = \frac{7}{9} = \frac{7}{18}$.

Let $\theta = \angle PBA = \angle PBC$. Because $\angle B$ is opposite to the leg $9 < 15$ and not the hypotenuse, we must have that 2θ is acute, likewise for θ . Hence, $\cos \theta > 0$.

Thus, using cosine definition on $\triangle BDP$ and the double angle identity $\cos 2\theta = 2\cos^2 \theta - 1$, we get

$$\cos \theta = \frac{7/2}{BP} = \sqrt{\frac{1 + \cos 2\theta}{2}} = \sqrt{\frac{1 + \frac{7}{18}}{2}} = \sqrt{\frac{25}{36}} = \frac{5}{6}$$

Finally, solving for $BP = \frac{7}{2\cos \theta} = \frac{7}{2} \cdot \frac{6}{5} = \frac{21}{5}$. Scaling back to the original diagram by 5 gives $\boxed{\text{(D)} 21}$.

Solution 9 (Law of Cosines + Angle Bisector Length Theorem)

Use Law of Cosines in $\triangle ABC$:

$$\cos B = \frac{80^2 + 45^2 - 75^2}{2 \cdot 80 \cdot 45} = \frac{7}{18}$$

Call E the intersection of the angle bisector of B and line segment AC , and let $CD \perp AB$, $D = CD \cap AB$, therefore $P = CD \cap BE$. In right triangle $\triangle BCD$, use the definition of cosine:

$$\cos B = \frac{BD}{BC} = \frac{7}{18}, \sin B = \sqrt{1 - \cos^2 B} = \frac{5\sqrt{11}}{18}$$

Therefore

$$BD = \frac{35}{2}, CD = \frac{25\sqrt{11}}{2}$$

Use Angle Bisector Theorem in $\triangle BCD$:

$$\frac{BD}{BC} = \frac{PD}{PC} = \frac{7}{18}$$

Therefore,

$$PD = \frac{7}{25}CF = \frac{7\sqrt{11}}{2}, PC = \frac{18}{25}CF = 9\sqrt{11}$$

Use Angle Bisector length formula:

$$BP = \sqrt{BD \cdot BC - PD \cdot PC} = \boxed{\text{(D)} 21}$$

Solution 10

Let the point where the altitude to AB meets AB be D . We can compute D easily. Let $BD = a \implies AD = 80 - a$. So

$$45^2 - a^2 = 75^2 - (80 - a)^2 \implies 45^2 - a^2 = 75^2 - 80^2 + 160a - a^2 \implies 45^2 - 75^2 + 80^2 = 160a \implies -30 \cdot 120 + 80 \cdot 80 = 10 \cdot 16a \implies -360 + 640 = 16a \implies a = \frac{280}{16} = \frac{35}{2}.$$

Now let $\angle DBP = \theta$. Note this is an angle bisector so we can find $\cos(\angle ABC)$ where $\theta = \frac{1}{2}\angle ABC$. So we use Law of Cosines to get

$$80^2 + 45^2 - 160 \cdot 45 \cos(2\theta) = 75^2 \implies 16^2 + 9^2 - 32 \cdot 9 \cdot \cos(2\theta) = 15^2 \implies \cos(2\theta) = \frac{112}{32 \cdot 9} = \frac{7}{18}. \text{ Now note}$$

$$\cos(2\theta) = 2\cos^2(\theta) - 1 = \frac{7}{18} \implies \cos^2(\theta) = \frac{25}{36} \implies \cos(\theta) = \frac{5}{6}.$$

Finally, by $SOH-CAH-TOA$ note that $\cos(\theta) = \frac{BD}{BP} = \frac{5}{6}$ and we know $BD = \frac{35}{2}$ so

$$BP = \frac{6}{5} \cdot \frac{35}{2} = \boxed{21}.$$

17. (AMC 12 12A 2025) The polynomial $(z+i)(z+2i)(z+3i) + 10$ has three roots in the complex plane, where $i = \sqrt{-1}$. What is the area of the triangle formed by these three roots?
- (A) 6 (B) 8 (C) 10 (D) 12 (E) 14

Answer: A

Solution 1 (substitution)

We initially note the symmetry of the product $(z+i)(z+2i)(z+3i)$, in that the factors $(z+i)$ and $(z+3i)$ can be expressed as $(z+2i) \mp i$ respectively. Accordingly, making the substitution $w = z + 2i$ enables us to apply the 'difference of two squares' identity, yielding

$$\begin{aligned} (z+i)(z+2i)(z+3i) + 10 &= (w-i)(w)(w+i) + 10 \\ &= w(w^2 - i^2) + 10 \\ &= w(w^2 + 1) + 10 = w^3 + w + 10. \end{aligned}$$

Having thus eliminated the quadratic term of the given cubic polynomial, we easily observe that -2 is a root of this new cubic (in particular, by the **rational root theorem**, the only possible rational roots are integers that are factors of 10).

It follows that $w^3 + w + 10 = (w+2)(w^2 - 2w + 5)$, and hence, using either the quadratic formula or completing the square, the other two roots are $1+2i$ and $1-2i$. Moreover, since our initial substitution simply shifts all points in the complex plane

by $2i$ (i.e. by 2 units vertically upward) and area is preserved under translation, the required area of the triangle in the z -plane is the same as that of the triangle formed by these three roots in the w -plane.

Now, as $1 + 2i$ and $1 - 2i$ of course have the same real part, the side of the triangle between these two vertices is simply vertical, and has length $2 - (-2) = 4$. The length of the altitude to that side is the perpendicular distance from -2 to the vertical line $\text{physics}\Re(z) = 1$, i.e. the horizontal distance from -2 to 1 , which is $1 - (-2) = 3$. Thus the area of the triangle is $\frac{1}{2} \cdot 4 \cdot 3 = \boxed{\text{(A)} 6}$.

Solution 1.1 (alternative using shoelace formula for last step)

While the area of the triangle (once we have determined its vertices) can be straightforwardly computed using $\text{area} = \frac{1}{2} \cdot \text{base} \cdot \text{perpendicular height}$ as explained above, it is also possible to use the **shoelace formula**, a general result giving the area of any polygon in the plane in terms of the coordinates of its vertices.

Specifically, the vertices -2 , $1 + 2i$, and $1 - 2i$ in the complex plane of course become the points $(-2, 0)$, $(1, 2)$, and $(1, -2)$ in the x - y plane, so the shoelace formula gives the area of the triangle as

$$\frac{1}{2} \cdot |(-2) \cdot 2 + 1 \cdot (-2) + 1 \cdot 0 - 0 \cdot 1 - 2 \cdot 1 - (-2) \cdot (-2)| = \frac{1}{2} \cdot |-12| = \boxed{\text{(A)} 6}.$$

Solution 1.2 (alternative using triangle in w -plane for last two steps)

As an alternative to arguing that the triangle formed by the three roots for w has the same area as the one originally formed by the roots for z , we can simply convert back to the z -plane using $z = w - 2i$. Therefore, as the roots for w are -2 , $1 + 2i$, and $1 - 2i$, those for z are $-2 - 2i$, $(1 + 2i) - 2i = 1$, and $(1 - 2i) - 2i = 1 - 4i$, with the vertical side of the triangle thus being that between 1 and $1 - 4i$.

We can now use either of the two approaches above: for the former, the vertical side again has length $0 - (-4) = 4$ and the corresponding altitude has length $1 - (-2) = 3$, once again giving the area as $\frac{1}{2} \cdot 4 \cdot 3 = \boxed{\text{(A)} 6}$. For the latter, the shoelace formula also gives the area as

$$\frac{1}{2} \cdot |(-2) \cdot 0 + 1 \cdot (-4) + 1 \cdot (-2) - (-2) \cdot 1 - 0 \cdot 1 - (-4) \cdot (-2)| = \frac{1}{2} \cdot |-12| = \boxed{\text{(A)} 6}.$$

Solution 2 (directly finding the roots)

Expanding the given polynomial yields

$$\begin{aligned} (z + i)(z + 2i)(z + 3i) + 10 &= (z^2 + 2iz + iz + 2 \cdot (-1))(z + 3i) + 10 \\ &= (z^2 + 3iz - 2)(z + 3i) + 10 \\ &= z^3 + 3iz^2 + 3iz^2 + 9 \cdot (-1)z - 2z - 6i + 10 \\ &= (z^3 - 11z + 10) + (6z^2 - 6)i, \end{aligned}$$

and so any complex number z will be a root if and only if it is a root of both the real part, $(z^3 - 11z + 10)$, and the imaginary part, $(6z^2 - 6)$.

We immediately observe that 1 is indeed a root of both of them, so by factoring out $(z - 1)$, we obtain $(z + i)(z + 2i)(z + 3i) + 10 = (z - 1)(z^2 + (1 + 6i)z + (-10 + 6i))$,

and now using the quadratic formula, we deduce that the other two roots are given by

$$\begin{aligned} z &= \frac{-(1+6i) \pm \sqrt{(1+6i)^2 - 4 \cdot 1 \cdot (-10+6i)}}{2 \cdot 1} \\ &= \frac{-1-6i \pm \sqrt{1+12i-36+40-24i}}{2} \\ &= \frac{-1-6i \pm \sqrt{5-12i}}{2}. \end{aligned}$$

Noting that $5-12i = (3-2i)^2$ (or else by writing $5-12i = (a+bi)^2$ for some $a, b \in \mathbb{R}$, and then solving for a and b by comparing real and imaginary parts of both sides), the above expression reduces to $z = \frac{-1-6i+3-2i}{2} = \frac{2-8i}{2} = 1-4i$ or $z = \frac{-1-6i-3+2i}{2} = \frac{-4-4i}{2} = -2-2i$. In other words, the three roots are precisely 1, $1-4i$, and $-2-2i$, and thus using either of the approaches in Solution 1.2 above, the area of the triangle is (A) 6.

18. (AMC 12 12A 2025) How many ordered triples (x, y, z) of different positive integers less than or equal to 8 satisfy $xy > z$, $xz > y$, and $yz > x$?

(A) 36 (B) 84 (C) 186 (D) 336 (E) 486

Answer: C

Solution 1

Let $0 < x < y < z \leq 8$; x cannot be 1 because it makes $xy > z \rightarrow y > z$;

$x = 2$, $y = 3$, z can be 4, 5 but not others; $x = 2$, $y = 4$, z can be 5, 6, 7; $x = 2$, $y = 5$, z can be 6, 7, 8; $x = 2$, $y = 6$, z can be 7, 8; $x = 2$, $y = 7$, z can be 8; for $x = 2$, total 11 cases;

Similarly, for $x = 3$, $y = 4, 5, 6, 7$, total 10 cases; for $x = 4$, $y = 5, 6, 7$, total 6 cases; $x = 5$, $y = 6, 7$, 3 cases; $x = 6$, $y = 7$, $z = 8$, 1 cases;

Total = $11 + 10 + 6 + 3 + 1 = 31$. Permutate x, y, z for ordered triple, it is $31 \cdot 6 = 186$, C.

Solution 2

For now, assume $x \leq y \leq z$.

First note that no number can be 0, as it would imply $0y > z$. Similarly, no number can be 1, as it would imply $1y > z$. So we only need to consider numbers between 2 and 8, inclusive.

We may use complementary counting:

Consider when $xy \leq z$. This implies $xy \leq 8$. Some quick calculations gives us the products $2 \cdot 2, 2 \cdot 3, 2 \cdot 4$. We may now calculate the number of times each happens (we are no longer assuming $x \leq y \leq z$):

- $2 \cdot 2$: This case is invalid as it asks for distinct integers.
- $2 \cdot 3$: $2, 3, (n \geq 6)$. Then we have $6 \cdot 3 = 18$ cases.
- $2 \cdot 4$: $2, 4, (n \geq 8)$. Then we have $6 \cdot 1 = 6$ cases.

In total, there are $7 \cdot 6 \cdot 5 = 210$ total cases, so our final answer is $210 - (18 + 6) =$ 186.

Solution 3

Note that if $x, y, z \geq 3$, all pairs work - hence, we have $\binom{6}{3} \cdot 6 = 120$ pairs. Now, note that if $x = 1$, we get $y > z$ and $z > y$, contradiction. Therefore, assume $x = 2$ (since we said $x < 3, x \neq 1$). Note that we need $2y > z > \frac{y}{2}$. WLOG assume $y > z$, we get there are 11 pairs that work (we just need $2z > y$): $(y, z) = (8, 5), (8, 6), (8, 7), (7, 4), (7, 5), (7, 6), (6, 4), (6, 5), (5, 3), (5, 4), (4, 3)$. With $x = 2$, we can re-arrange these in $3! = 6$ ways each, hence the answer is just $120 + 6 \cdot 11 = \boxed{186}$.

Solution 4 (Bounding)

Note that if $x, y, z > 2$, then this must hold, as if it didn't, we would need, WLOG, $x > 3y$, which would imply $x > 8$. Therefore, there are at least $6 \times 5 \times 4 = 120$ solutions. However, we must have $x, y, z \geq 2$, as if any variable is 0, this clearly doesn't work, and $x = 1$ gives $y > z$ and $z > y$, which is impossible. Therefore, there are at most $7 \times 6 \times 5 = 210$ solutions. The only choice that works from here is $\boxed{\text{(C) } 186}$ krithikrokcs

Solution 5 (Compl. Counting)

The total number of cases is $7 \times 6 \times 5 = 210$. Note that no number can be 1, since that would result in cases like $z > y$ and $z < y$ which is impossible.

We can try complementary counting. Since in this problem x, y, z cases are symmetrical, we first consider cases where $xy < z$. The cases are: $(x, y, z) : (2, 3, 6)(2, 3, 7)(2, 3, 8)(2, 4, 8)$. By symmetry, the cases that don't work amounts to $4 \times 6 = 24$ cases. Therefore the total number of cases that work is $210 - 24 = 186$, which is $\boxed{\text{(C) } 186}$. backtosq-1

Solution 6 (Symmetry + Complementary Counting)

Since $\{x, y, z\} \subset \{1, 2, 3, 4, 5, 6, 7, 8\}$, WLOG let $1 \leq x < y < z \leq 8$ ($\times 3! = 6$). Note that the lower bounds make the conditions $y < xz \leq 1z$ and $x < zy < z$ trivial. In addition, $x = 1$ always fails, since it would force $z < xy = y$, a contradiction. Thus, it remains exactly to check $2 \leq x < y < z \leq 8 < xy$.

The only failure cases with $xy \leq 8$ can occur when $(x, y) = (2, 3), (2, 4)$, for which we have corresponding $z \in \{6, 7, 8\}, \{8\}$. The total number of cases is $\binom{7}{3} = 35$. Thus, the number of good cases is $35 - 4 = 31$, and after multiplying by order we get $3! \cdot 31 = \boxed{\text{(C) } 186}$.

19. (AMC 12 12A 2025) Let a, b , and c be the roots of the polynomial $x^3 + kx + 1$. What is the sum

$$a^3b^2 + a^2b^3 + b^3c^2 + b^2c^3 + c^3a^2 + c^2a^3?$$

- (A) $-k$ (B) $-k + 1$ (C) 1 (D) $k - 1$ (E) k

Answer: E

Solution 1

We begin by factoring:

$$\begin{aligned} & a^3b^2 + a^2b^3 + b^3c^2 + b^2c^3 + c^3a^2 + c^2a^3 \\ &= a^2b^2(a + b) + b^2c^2(b + c) + c^2a^2(c + a) \\ &= (a^2b^2 + b^2c^2 + c^2a^2)(a + b + c) - a^2b^2c - b^2c^2a - c^2a^2b \\ &= (a^2b^2 + b^2c^2 + c^2a^2)(a + b + c) - (abc)(ab + bc + ca). \end{aligned} \tag{57}$$

From Vieta's Formulas, we know that $a + b + c = 0$, $ab + bc + ca = k$, and $abc = -1$. Therefore, the answer equals $(a^2b^2 + b^2c^2 + c^2a^2)(0) - (-1)(k) = \boxed{k}$.

Solution 2

Notice that this is $\sum_{cyc} a^2b^2(a + b)$. By Vieta's (just like in solution 1), we have $a + b + c = 0$, so $a + b = -c$, $abc = -1$, and $ab + bc + ac = k$, so this is equivalent to $\sum_{cyc} -a^2b^2c = \sum_{cyc} -(abc)(ab) = \sum_{cyc} ab = k$ by Vieta's. ScoutViolet

Solution 3 (BS)

Let's just pick a nice and convenient a, b, c . Observe that $2 \cdot 2^{-\frac{1}{3}} - 2^{\frac{2}{3}} = 0$, so we can just have $(a, b, c) = (2^{-\frac{1}{3}}, 2^{-\frac{1}{3}}, -2^{\frac{2}{3}})$. So expanding $(x - a)(x - b)(x - c)$ we get $x^3 - 3 \cdot 2^{-\frac{2}{3}} + 1$. Evaluating $a^3b^2 + a^2b^3 + b^3c^2 + b^2c^3 + c^3a^2 + c^2a^3$, or $2a^5 + 2a^2c^3 + 2a^3c^2$, we get $-3 \cdot 2^{-\frac{2}{3}} = k$, so the answer is $\boxed{E}$. (i did this on the actual test)

who is going to think of this on the test bro and change ur username dawg

this solution is acc so fire leave him alone

yo lowk it's acc so fire that the entire tung tung tung sahor family agreed (they told me) and EVERYONE is going to think of this on the test (because, y'know, I totally didn't factor then use Vieta in a few secs *cough *cough) and his username is tuff idk what the first guy was on abt

tbh this sol was my first thought

Solution 4 (Vieta's)

First, note that $a + b + c = 0$, $\sum_{cyc} ab = k$, $abc = -1$. We can factor the original expression into:

$$\sum_{cyc} a^2b^2(a + b) = \sum_{cyc} \left(-\frac{1}{c}\right)^2 (-c) = -\frac{\sum_{cyc} ab}{abc} = \boxed{(E) \ k}.$$

Solution 5 (Newton's Sums) (If you suck at factoring like me)

Note that $a^3b^2 + a^2b^3 + b^3c^2 + b^2c^3 + c^3a^2 + c^2a^3 = (a^3 + b^3 + c^3)(a^2 + b^2 + c^2) - (a^5 + b^5 + c^5)$

By Newton's sums, we have:

$$(a^3 + b^3 + c^3) + k(a + b + c) + 3 = 0$$

$$(a^5 + b^5 + c^5) + k(a^3 + b^3 + c^3) + 1(a^2 + b^2 + c^2) = 0$$

Note that $a^2 + b^2 + c^2 = (a + b + c)^2 - 2(ab + bc + ca)$

From Vieta's formulas, we have $a + b + c = 0$ and $ab + bc + ca = k$

Solving, we get:

$$(a^3 + b^3 + c^3) = -3$$

$$(a^2 + b^2 + c^2) = -2k$$

$$(a^5 + b^5 + c^5) - 3k - 2k = 0 \rightarrow (a^5 + b^5 + c^5) = 5k$$

Therefore, $a^3b^2 + a^2b^3 + b^3c^2 + b^2c^3 + c^3a^2 + c^2a^3 = (-3)(-2k) - 5k = \boxed{(E) \ k}$

Solution 6

Like in solution 1, we begin by factoring:

$$\begin{aligned} & a^3b^2 + a^2b^3 + b^3c^2 + b^2c^3 + c^3a^2 + c^2a^3 \\ &= a^2b^2(a+b) + b^2c^2(b+c) + c^2a^2(c+a) \end{aligned} \quad (58)$$

From Vieta's Formulas, we know that $a+b+c=0$, $ab+bc+ca=k$, and $abc=-1$. square $a+b+c=0$ to get $a^2+b^2+c^2=0$, then isolating one variable each time yields:

$$= -\frac{1}{c^2}(a+b) - \frac{1}{a^2}(b+c) - \frac{1}{b^2}(c+a) \quad (59)$$

Isolating $a+b+c=0$, you get

$$\begin{aligned} &= -\frac{1}{c^2}(-c) - \frac{1}{a^2}(-a) - \frac{1}{b^2}(-b) \\ &= -\left(\frac{ab+bc+ac}{abc}\right) \end{aligned} \quad (60)$$

This simplifies to:

$$\begin{aligned} &= -\left(\frac{k}{-1}\right) \\ &= k \end{aligned} \quad (61)$$

Therefore, the answer equals $\boxed{k}$.

Solution 7 (Vieta + Factorization)

We can easily factor the expression into

$$a^2b^2(a+b) + b^2c^2(b+c) + a^2c^2(a+c)$$

By Vieta, we get $a+b+c=0$, so we can rewrite the expression into $a^2b^2(-c) + b^2c^2(-a) + a^2c^2(-b)$ which is just $-abc(ab+bc+ac)$. By Vieta we also get $abc=-1$ and $ab+bc+ac=k$. Therefore the expression is just $-(-1)k = \boxed{k}$.

Solution 8 (no Vieta + not quite factoring)

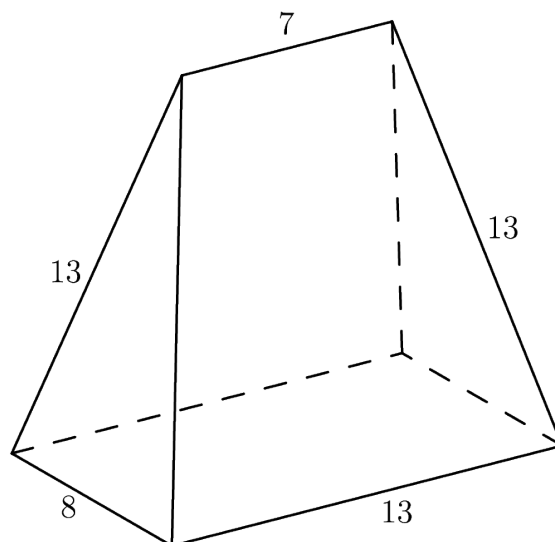
Let the expression we are asked to evaluate equal y .

We have that $x^3 + kx + 1 = (x-a)(x-b)(x-c)$. Expanding the right hand side yields $x^3 - x^2(a+b+c) + x(ac+bc+ab) - abc$. Equating coefficients, $a+b+c=0$, $ac+bc+ab=k$, and $abc=-1$.

We may find the first equation helpful for cancelling things out, so let us fiddle around with it, trying to piece out as much of y as possible: we discover that $(a+b+c)(ab)^2 = a^3b^2 + a^2b^3 + c(ab)^2 = 0$, which shares terms with y . Likewise, $(a+b+c)(bc)^2 = b^3c^2 + b^2c^3 + a(bc)^2 = 0$ and $(a+b+c)(ac)^2 = a^3c^2 + a^2c^3 + b(ac)^2 = 0$.

Adding the three equations gives $y + a(bc)^2 + b(ac)^2 + c(ab)^2 = 0$. Observe that $a(bc)^2 + b(ac)^2 + c(ab)^2 = (abc)(ac+bc+ab) = -1(k)$. Thus $y - k = 0$; $y = \boxed{k}$.

20. (AMC 12 12A 2025) The base of the pentahedron shown below is a 13×8 rectangle, and its lateral faces are two isosceles triangles with base of length 8 and congruent sides of length 13, and two isosceles trapezoids with bases of length 7 and 13 and nonparallel sides of length 13.



What is the volume of the pentahedron?

- (A) 416 (B) 520 (C) 528 (D) 676 (E) 832

Answer: C

Solution 1 (Split Into Three Parts)

Notice that the triangular faces have a slant height of $\sqrt{13^2 - 4^2} = \sqrt{153}$ and that the height is therefore $\sqrt{153 - (\frac{13-7}{2})^2} = 12$. Then we can split the pentahedron into a triangular prism and two pyramids. The pyramids each have a volume of $\frac{1}{3}(3)(8)(12) = 96$ and the prism has a volume of $\frac{1}{2}(8)(12)(7) = 336$. Thus the answer is $336 + 96 \cdot 2 = \boxed{\text{(C) } 528}$

Solution 2

Note that the height is 12 from the previous method. Note that as you go up, the length and width both decrease linearly and reach 0 at the end.

So the answer is $\int_0^{12} (8 - \frac{2}{3}x)(13 - \frac{1}{2}x)dx = \boxed{\text{(C) } 528}$

-KapilTheAngel

Solution (3D Pythagoras)

Let H be the height of the solid. By 3D Pythagoras (recursion of 2D Pythagoras),

$$\left(\frac{8}{2}\right)^2 + \left(\frac{13-7}{2}\right)^2 + H^2 = 13^2 = 4^2 + 3^2 + H^2 = 5^2 + H^2$$

This gives the 5 – 12 – 13 triple, so $H = 12$. Continue as in other solutions.

Note It just reminds about one of old aime problem <https://artofproblemsolving.com/wiki/index.php/ti>
1983 AIME Problem 11. DRA777

21. (AMC 12 12A 2025) There is a unique ordered triple (a, k, m) of nonnegative integers such that

$$\frac{4^a + 4^{a+k} + 4^{a+2k} + \dots + 4^{a+mk}}{2^a + 2^{a+k} + 2^{a+2k} + \dots + 2^{a+mk}} = 964.$$

What is $a + k + m$?

- (A) 8 (B) 9 (C) 10 (D) 11 (E) 12

Answer: A

Solution 1

The numerator can be written as $2^{2a} + 2^{2a+2k} + \dots + 2^{2a+2mk}$, which is actually a sum of geometric series. This can be expressed as $2^{2a} \cdot \frac{1-2^{2k(m+1)}}{1-2^{2k}}$. The denominator in the same way can be expressed as $2^a \cdot \frac{1-2^{k(m+1)}}{1-2^k}$.

Doing some algebra on the top and bottom we get:

$$2^a \cdot \frac{1 + 2^{k(m+1)}}{1 + 2^k} = 964 \quad (62)$$

The prime factorization of 964 is $241 \cdot 2^2$.

Equating $2^a = 2^2$, we get $a = 2$.

Next since 241 is a prime number we equate the second term of the product to 241.

$$\frac{1 + 2^{k(m+1)}}{1 + 2^k} = 241 \quad (63)$$

Doing some algebra we get, that $2^k(2^{km} - 241) = 240$

The closest power of 2 to 241 is 256 which is 2^8 . So setting $km = 8$, we get $2^8 - 241 = 15$ $2^k(15) = 240$, $2^k = 16$, $k = 4$.

This we know that if $k = 4$, then $4m = 8$, so $m = 2$.

Finally, we conclude that $a = 2$, $m = 2$, and $k = 4$.

$4 + 2 + 2 = 8$, so the answer is A

Actually when it comes to $2^k(2^{km} - 241) = 240$; we can directly observe the answer by discovering 2^k is even and $2^{km} - 241$ must be odd. So, the only available divisors of 240 is 16 and 15. And we can have $m = 2$, and $k = 4$.

Solution 2(alternative method)

We start from this equation derived in Solution 1:

$$\frac{1 + 2^{k(m+1)}}{1 + 2^k} = 241 \quad (64)$$

Note that $241 = 256 - 16 + 1 = 16^2 - 16 + 1$. Also note that this looks similar to the second term in the Sum of Cubes factorization $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$. Specifically, when $a = 16$ and $b = 1$, the right term is $16^2 - 16 + 1$. Thus, $\frac{16^3 + 1^3}{16 + 1} = 241$. We rewrite: $\frac{1 + 2^{12}}{1 + 2^4} = 241$

We can then get that $m = 2$ and $k = 4$, and proceed as above.

22. (AMC 12 12A 2025) Three real numbers are chosen independently and uniformly at random between 0 and 1. What is the probability that the greatest of these three numbers is greater than 2 times each of the other two numbers? (In other words, if the chosen numbers are $a \geq b \geq c$, then $a > 2b$.)

(A) $\frac{1}{12}$ (B) $\frac{1}{9}$ (C) $\frac{1}{8}$ (D) $\frac{1}{6}$ (E) $\frac{1}{4}$

Answer: E

Solution 1 (3D geometry)

Let the three numbers be X , Y , and Z , and consider each possible triple (X, Y, Z) as the (Cartesian) coordinates of a point in 3-dimensional space. Then, as each number is chosen independently and uniformly from the interval $[0, 1]$, the total space of possible points (X, Y, Z) is the unit cube $0 \leq X, Y, Z \leq 1$. Using the idea of **geometric probability**, it follows that the required probability equals the volume of the region consisting of all points (X, Y, Z) that satisfy the required condition, divided by the volume of the total space, i.e. the unit cube (which is simply $1^3 = 1$). Accordingly, it only remains to determine the region of possible points (X, Y, Z) and then compute its volume.

For simplicity, we may assume that e.g. X is the maximum - clearly, by symmetry, the corresponding regions where Y is the maximum and where Z is the maximum have the same volume, and these regions overlap only where at least two of X, Y, Z are both the maximum. In such cases, those two variables must be equal (e.g. X and Y will both be the maximum precisely if $X = Y \geq Z$), and so each of these overlaps is either a region within a plane (as e.g. the equation $X = Y$ defines a plane), or a straight line (in the special case where $X = Y = Z$). It follows that the overlaps between the 3 regions all have zero volume, so since each region has the same volume (as explained above), the answer will simply be 3 times the volume of one of these regions, say the one where X is the maximum and the required condition is satisfied.

Within that region, we therefore have $0 \leq X, Y, Z \leq 1$, $X \geq Y$ and $X \geq Z$ for X to be the maximum, and $X > 2Y$ and $X > 2Z$ for the required condition, but this combination of inequalities reduces to simply $0 \leq 2Y, 2Z < X \leq 1$, since we automatically have $2Y \geq Y$ and $2Z \geq Z$ as $Y, Z \geq 0$. This means the boundaries of the region are the lines $X = 2Y$, $X = 2Z$, $X = 0$, $Y = 0$, and $X = 1$, so now by drawing the unit cube as well as those boundaries, we deduce that this region is a square-based pyramid: the base has vertices $(1, 0, 0)$, $(1, 0, \frac{1}{2})$, $(1, \frac{1}{2}, \frac{1}{2})$, and $(1, \frac{1}{2}, 0)$, while the apex of the pyramid is at $(0, 0, 0)$. The square base thus clearly has side length $\frac{1}{2}$, and hence area $(\frac{1}{2})^2 = \frac{1}{4}$; moreover, as the base lies in the plane $X = 1$, the perpendicular height of this pyramid is the same as the perpendicular distance from the apex $(0, 0, 0)$ to that plane, which is obviously $1 - 0 = 1$.

It follows that the volume of this pyramid, i.e. of the region where X is the maximum and the required condition is satisfied, is $\frac{1}{3} \cdot \frac{1}{4} \cdot 1 = \frac{1}{12}$ (using the standard formula for the volume of a pyramid). Finally, as explained above, this means the required probability is $3 \cdot \frac{1}{12} = \boxed{\text{(E)} \frac{1}{4}}$.

Solution 2 (calculus)

Consider three independent uniform random variables X, Y, Z on $[0, 1]$. The goal is to find the probability that the maximum value is greater than twice each of the

other two values. Equivalently, if we order them as $a \geq b \geq c$, this is the probability that $a > 2b$ (which implies $a > 2c$ since $b \geq c$). Therefore, if we assume that X is the maximum (but make no assumption about the order of Y and Z), the required condition reduces (almost surely) to $X > 2Y$ and $X > 2Z$. Conversely, if $X > 2Y$ and $X > 2Z$, then we will necessarily have $X > 2Y \geq Y$ and $X > 2Z \geq Z$ since $Y, Z \geq 0$, and so X will indeed be the maximum.

Accordingly, the probability that X is the maximum and the required condition is satisfied is simply $P(X > 2Y \cap X > 2Z)$, and then the answer will be 3 times this probability, to account for the 3 total cases (the maximum can be X , Y , or Z , and clearly all of these cases are symmetric). Since X, Y, Z are chosen uniformly at random from the interval $[0, 1]$, the marginal distribution of each one is simply the uniform distribution on this interval. As they are also chosen independently, we deduce that their joint distribution is uniform over $[0, 1] \times [0, 1] \times [0, 1]$, i.e. the unit cube, and hence the joint probability density is $\frac{1}{\text{volume of the unit cube}} = \frac{1}{1^3} = 1$.

For the bounds of integration, we can first choose $X \in [0, 1]$, and then we require $Y, Z \in [0, 1]$, $X > 2Y$, and $X > 2Z$; as $X \leq 1$, the latter two conditions will also automatically give $Y, Z < \frac{X}{2} \leq \frac{1}{2} < 1$, so it is both necessary and sufficient that $0 \leq Y, Z < \frac{X}{2}$. Thus we can now compute

$$\begin{aligned} P(X > 2Y \cap X > 2Z) &= \int_0^1 \int_0^{\frac{x}{2}} \int_0^{\frac{x}{2}} 1 \, dz \, dy \, dx \\ &= \int_0^1 \left(\frac{x}{2}\right)^2 dx \\ &= \int_0^1 \frac{x^2}{4} dx \\ &= \frac{1}{4} \left[\frac{1}{3}x^3\right]_0^1 \\ &= \frac{1}{4} \cdot \frac{1}{3} (1^3 - 0^3) \\ &= \frac{1}{12}, \end{aligned}$$

and so, as explained above, the total probability is $3 \cdot \frac{1}{12} = \boxed{\text{(E)} \frac{1}{4}}$.

Solution 2.1 (alternative, fixing the entire order)

Alternatively, instead of just fixing the maximum as X , we can fix the entire order of the 3 variables as e.g. $X > Y > Z$. It then follows that the required condition reduces (almost surely) to $X > 2Y > 2Z$, while conversely, if $X > 2Y > 2Z$, it will indeed necessarily follow that the order is $X > Y > Z$, as $Y \geq 0$ again gives $X > 2Y \geq Y$ and $2Y > 2Z$ gives $Y > Z$.

Hence, similarly to above, the probability that the required condition is satisfied is $P(X > 2Y > 2Z)$, and then as $X > Y > Z$ is just 1 of the total $3! = 3 \cdot 2 \cdot 1 = 6$ possible orders (with all of them clearly being symmetric), the answer will be 6 times this probability. The joint probability density is still 1 as shown above, but now for the bounds of integration, we first choose Y , where we must have $1 \geq X > 2Y$, and thus $0 < Y < \frac{1}{2}$. We can now choose X , again satisfying $X > 2Y$ and of course $0 \leq X \leq 1$, i.e. $2Y < X \leq 1$ (since we then automatically have $X > 2Y \geq 2 \cdot 0 = 0$). Lastly, we need to choose Z such that $Y > Z$ and $0 \leq Z \leq 1$

(as then $X > 2Y$ will automatically also give $X > 2Z$), i.e. $0 \leq Z < Y$ (since $Y \leq 1$ will then automatically give $Z \leq 1$).

Accordingly, we obtain

$$\begin{aligned} P(X > 2Y > 2Z) &= \int_0^{\frac{1}{2}} \int_{2y}^1 \int_0^y 1 \, dz \, dx \, dy \\ &= \int_0^{\frac{1}{2}} y(1 - 2y) \, dy \\ &= \int_0^{\frac{1}{2}} (y - 2y^2) \, dy \\ &= \left[\frac{1}{2}y^2 - \frac{2}{3}y^3 \right]_0^{\frac{1}{2}} \\ &= \left(\frac{1}{2} \cdot \left(\frac{1}{2} \right)^2 - \frac{2}{3} \cdot \left(\frac{1}{2} \right)^3 \right) - (0 - 0) \\ &= \frac{1}{24}, \end{aligned}$$

and so, as explained above (and just as obtained in Solution 2), the required probability is $6 \cdot \frac{1}{24} = \boxed{\text{(E)} \frac{1}{4}}$.

23. (AMC 12 12A 2025) Call a positive integer *fair* if no digit is used more than once, it has no 0s, and no digit is adjacent to two greater digits. For example, 196, 23, and 12463 are fair, but 1546, 320 and 34321 are not. How many fair positive integers are there?

(A) 511 (B) 2584 (C) 9841 (D) 17711 (E) 19682

Answer: C

Solution 1

To satisfy the conditions, a fair integer must have no digit be a local minimum. Let's say we have n distinct digits, with each digit being a number from 1 to 9. To create a fair integer, we begin by placing the largest digit. For the second-largest digit, we can either place this digit to the right or to the left of the string already created. We have these 2 options for the third-largest digit, and so on. Therefore, there are 2^{n-1} valid permutations to create a fair integer.

We must also choose which digits will be in the permutation. If you are creating an n -digit long fair integer, there are $\binom{9}{n}$ ways to pick which digits will be in the number.

Therefore, for each $n \in \{1, 2, \dots, 9\}$, the number of fair integers of length n is:

$$\binom{9}{n} \cdot 2^{n-1}.$$

Summing over all n :

$$\sum_{n=1}^9 \binom{9}{n} \cdot 2^{n-1} = \frac{1}{2} \left(\sum_{n=0}^9 \binom{9}{n} 2^n - 1 \right) = \frac{1}{2} ((1+2)^9 - 1) = \frac{1}{2}(19682) = \boxed{9841}.$$

Note that the Binomial Theorem was used to equate

$$\sum_{n=0}^9 \binom{9}{n} 2^n = (1+2)^9.$$

lprado

Solution 2

Note every fair number will have an increasing string of digits, a maximum digit, then a decreasing string of digits. This is because if it decreases then increases, then the digit in the middle will be less than its adjacent digits.

Let n be the maximum digit. For each number $i < n$, we may either place i before n , after n , or choose not to include it. Note this process will result in a unique number for every case, as the numbers before n must be in increasing order, and the numbers after n must be in decreasing order. Therefore, for each number n , we have 3^{n-1} cases.

Since $n \in \{1, 2, \dots, 9\}$, we have:

$$\sum_{n=1}^9 3^{n-1} = \frac{3^9 - 1}{3 - 1} = \boxed{9841}.$$

Solution 2.5 (Cheese, Goldilocks Zone)

If you consider options D and E too big, and A and B too small, then C is your answer.

(C) 9841 Also E is twice C so C is your answer

- Ameen Patel IrvingtonHighSchoolMathGeeks-Edited by TFEA

Solution 3 (Recursion)

Let $f(n)$ be the number of fair integers given an arbitrary set of n digits. Let a be the smallest of these digits. Notice that a is either the first or last digit, as if it were any other digit, the two digits surrounding it would both be greater. Then, notice that if the remaining $n - 1$ slots form a fair number, so does the number when a is appended. Therefore, $f(n) = 2f(n - 1)$. From here, we may proceed with the calculation in Solution 1 to get the answer of (C) 9841

Solution 4

For any fair integer n , we write some 9-digit base-3 number(s) $f(n) = (\overline{d_1 d_2 \dots d_9})_3$ from left to right as follows:

- $d_k = 0$ if k is not in n ;
- $d_k = 1$ if k is the to the left of all digits greater than k in n ;
- $d_k = 2$ if k is the to the right of all digits greater than k in n .

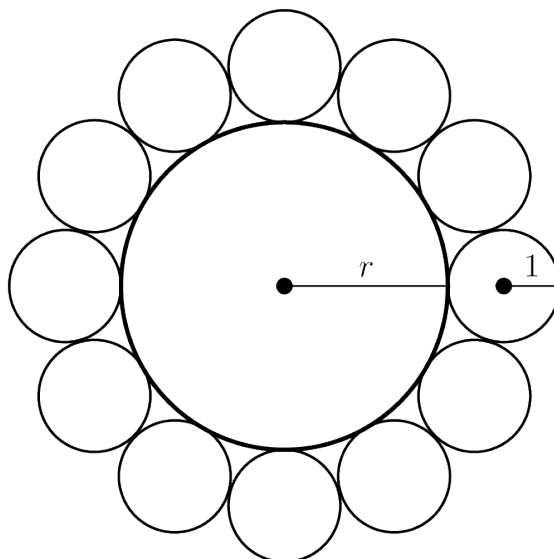
Note that the fairness ensured one and only one of the above three cases happens.

Now if k is the largest digit in n then d_k can be either 1 or 2.

An example: If $n = 136852$, then $f(n) = \overline{121021010}_3$ or $\overline{121021020}_3$.

It's thus easy to see that there is a 1-to-2 mapping from all fair integers to all 9-digit base-3 nonzero numbers. Therefore the answer is $\frac{3^9-1}{2}$.

24. (AMC 12 12A 2025) A circle of radius r is surrounded by 12 circles of radius 1, externally tangent to the central circle and sequentially tangent to each other, as shown. Then r can be written as $\sqrt{a} + \sqrt{b} + c$, where a, b, c are integers. What is $a + b + c$?



- (A) 3 (B) 5 (C) 7 (D) 9 (E) 11

Answer: C

Solution 1 (Sin 15)

Let the center of the large circle be O and the centers of the 12 circles be $A_1, A_2, A_3, \dots, A_{12}$. Triangle OA_1A_2 has side lengths $r + 1, r + 1, 2$, with the angle opposite 2 being $360/12 = 30$. Note that the line connecting A_1 and A_2 goes through their common point of tangency, by definition, which causes A_1A_2 to have a length of 2.

Drawing the angle bisector of the 30 degree angle, we split OA_1A_2 into two congruent right triangles, each with hypotenuse $r + 1$ and side opposite the 15 degree angle 1.

From here, note that $\sin 15 = \frac{\sqrt{6}-\sqrt{2}}{4}$, which be derived using the trigonometric identity $\sin(A - B) = \sin A \cos B - \sin B \cos A$, with $A = 45$ and $B = 30$.

In our right triangle, $\sin 15 = \frac{1}{r+1} = \frac{\sqrt{6}-\sqrt{2}}{4}$. Let $x = r + 1$. Solving for x , we get $x = \frac{4}{\sqrt{6}-\sqrt{2}}$. Rationalizing, we get that $x = \sqrt{6} + \sqrt{2}$.

Remember $x = r + 1 = \sqrt{6} + \sqrt{2}$, so $r = \sqrt{6} + \sqrt{2} - 1$. Therefore, our answer is $6 + 2 - 1 = \boxed{7}$.

Solution 2 (Law of Cosines Bash)

Let the center of the large circle be O and the centers of any two circles be A and B . Triangle OAB has side lengths $r + 1, r + 1, 2$, with the angle opposite 2 being $360/12 = 30$.

Using Law of Cosines, $2^2 = AO^2 + BO^2 - 2AOBO \cos 30$. Plugging in the radius yields $2^2 = (r + 1)^2 + (r + 1)^2 - 2(r + 1)^2 \cdot \frac{\sqrt{3}}{2}$. After solving for r and simplifying, we get $r = \sqrt{6} + \sqrt{2} - 1$. Therefore our answer is 7.

Worked Out Solution $4 = (2 - \sqrt{3})(r + 1)^2$. Divide both sides to get $\frac{4}{2-\sqrt{3}} = (r + 1)^2$. Rationalize and expand to get $8 + 4\sqrt{3} = r^2 + 2r + 1$. Then $r^2 + 2r + (-7 - 4\sqrt{3}) = 0$. Use the quadratic formula to get $r = \frac{-2 \pm \sqrt{4 + 28 + 16\sqrt{3}}}{2}$, or $r = -1 + \sqrt{8 + 4\sqrt{3}}$ (Note: We only take the plus instead of plus-minus because r must be positive). Clearly $c = -1$, and then we need to find some a and b such

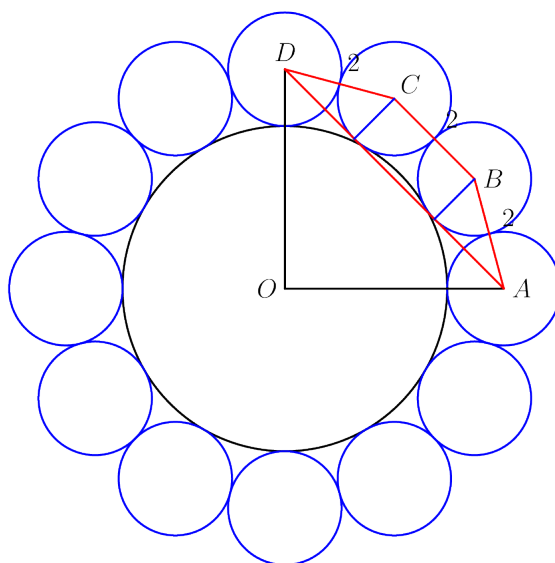
that $\sqrt{a} + \sqrt{b} = \sqrt{8 + 4\sqrt{3}}$. If we square both sides, $a + b + 2\sqrt{ab} = 8 + 4\sqrt{3}$, so $a + b = 8$. Plug in our previously found values to find $a + b + c = 8 + (-1)$. Therefore our answer is 7.

Remark

We could've also noticed that $\sqrt{2 - \sqrt{3}} = \sqrt{\frac{1}{2} \cdot (4 - 2\sqrt{3})} = \frac{1}{\sqrt{2}} * \sqrt{3 - 2\sqrt{3} + 1} = \frac{1}{\sqrt{2}} * (\sqrt{3} - 1)$, and therefore directly gotten $r + 1 = \sqrt{\frac{4}{2 - \sqrt{3}}} = \frac{2}{\frac{1}{\sqrt{2}} * (\sqrt{3} - 1)} = \frac{2\sqrt{2}}{\sqrt{3} - 1}$ and solved from there.

Solution 3 (Easy Geo Method)

We connect the centers of the smaller circles at 0° and 90° as follows:



Note that connecting the centers of the 12 smaller circles forms a regular 12-gon. A regular 12-gon has angles

$$\frac{10 \cdot 180}{12} = 150^\circ,$$

so $\angle OAB = \angle ODC = 75^\circ$. Furthermore, since $OA = OD$ and $\angle AOD = 90^\circ$, we have $\angle OAD = 45^\circ$, so $\angle DAB = 30^\circ$. Next, we drop the perpendiculars from B and C to AD . This creates a 1×2 rectangle and two $30 - 60 - 90$ triangles. Thus,

$$AD = (r + 1)\sqrt{2} = 2 + 2\sqrt{3},$$

and manipulating yields

$$r = -1 + \sqrt{2} + \sqrt{6},$$

so the desired answer is $-1 + 2 + 6 = \boxed{(C) 7}$.

P.S. I posted the asymptote code here. Anyone that wishes to use it can steal :)

25. (AMC 12 12A 2025) Polynomials $P(x)$ and $Q(x)$ each have degree 3 and leading coefficient 1, and their roots are all elements of $\{1, 2, 3, 4, 5\}$. The function $f(x) = \frac{P(x)}{Q(x)}$ has the property that there exist real numbers $a < b < c < d$ such that the

set of all real numbers x such that $f(x) \leq 0$ consists of the closed interval $[a, b]$ together with the open interval (c, d) . How many ordered pairs of polynomials (P, Q) are possible?

- (A) 13 (B) 12 (C) 11 (D) 9 (E) 8

Note: This question in its original form had no answer. It has been modified to have a solution.

Answer: None of the official answer choices were correct.

Solution 1 (Got Wrong)

None of the answer choices on the official test (which asked for the number of possible functions $f(x)$) were correct, but choice E would be correct if this problem asked for the number of pairs of functions $(P(x), Q(x))$.

Let $R(x) = \frac{P(x)}{Q(x)}$. Since $R(x) \leq 0$ on $[a, b]$ but not for values slightly less than a or slightly more than b , $P(x) = 0$ at $x = a$ and $x = b$. Therefore, $P(x) = (x - a)(x - b)(x - r)$ for some $r \in \{1, 2, 3, 4, 5\}$.

Since $R(x) \leq 0$ on (c, d) but not at $x = c$ or $x = d$, $R(x)$ is not continuous at $x = c$ or $x = d$. Therefore, $R(x)$ must be undefined at $x = c$ and $x = d$, that is, $Q(x) = 0$ at $x = c$ and $x = d$. So $Q(x) = (x - c)(x - d)(x - s)$ for some $s \in \{1, 2, 3, 4, 5\}$.

Therefore, $R(x) = \frac{(x-a)(x-b)(x-r)}{(x-c)(x-d)(x-s)}$. Notice that $\frac{(x-a)(x-b)}{(x-c)(x-d)} \leq 0$ only on $[a, b]$ and (c, d) . Therefore, $\frac{x-r}{x-s}$ must be nonnegative for all $x \notin \{a, b, c, d, r, s\}$. This only happens if $r = s$.

Thus $R(x) = \frac{(x-a)(x-b)(x-r)}{(x-c)(x-d)(x-r)}$, which is the same as $\frac{(x-a)(x-b)}{(x-c)(x-d)}$ except that it is undefined at $x = r$. Thus $R(x)$ satisfies the desired property as long as $r \notin [a, b] \cup (c, d)$.

Note that each quintuple (a, b, c, d, r) defines a unique pair of functions $(P(x), Q(x))$.

If $(a, b, c, d) = (1, 2, 3, 4)$, r can be 3, 4, or 5.

If $(a, b, c, d) = (1, 2, 3, 5)$, r can be 3 or 5.

If $(a, b, c, d) = (1, 2, 4, 5)$, r can be 3, 4, or 5.

If $(a, b, c, d) = (1, 3, 4, 5)$, r can be 4 or 5.

If $(a, b, c, d) = (2, 3, 4, 5)$, r can be 1, 4, or 5.

Therefore, there are (E) 13 possible pairs of functions $(P(x), Q(x))$.

-j314andrews

Solution 2 (credit to Sohil Rathi's video solution, pi_is_{3.14}/OmegaLearn)

This is the solution to the original problem, which asked for the number of possible **functions**, not pairs of polynomials.

First, consider the problem if it were talking about two second degree polynomials. We can see that the function

$$f(x) = \frac{(x-a)(x-b)}{(x-c)(x-d)}$$

by itself satisfies the condition of dropping below the x axis over $[a, b] \cup (c, d)$. Now, we need to add one extra $(x - n)$ term each to the numerator and denominator.

Case 1: $f(x) = \frac{(x-a)(x-b)(x-f)}{(x-c)(x-d)(x-f)}$ for some other value f not equal to a, b, c , or d .

Note that f cannot be between a and b or between c and d because this would create a hole in the interval. So the only possibilities are:

- $a = 1, b = 2, c = 3, d = 4, f = 5$
- $a = 1, b = 2, c = 4, d = 5, f = 3$
- $a = 2, b = 3, c = 4, d = 5, f = 1$

This gives us 3 valid functions so far.

Case 2: $f(x) = \frac{(x-a)(x-b)(x-f)}{(x-c)(x-d)(x-f)}$ where f is c or d .

If f equals c or d the resulting function is essentially equivalent to $f(x) = \frac{(x-a)(x-b)}{(x-c)(x-d)}$ because a hole would already exist, so we can safely cancel the $(x-f)$ terms. It doesn't matter whether we choose f to equal c or d because the function will still be the same. There are $\binom{5}{4} = 5$ ways to select the values for a, b, c , and d and 1 way to choose any variable for f to equal, giving $5 \cdot 1 = 5$ valid functions for this case.

The correct answer is $5 + 3 = \boxed{\text{(F)} 8}$ functions.

i tried my best to show what I think was Sohil's thought process, but please feel free to edit this if you can explain it better :)

Solution 3

Solution 1 by user j314andrews is incomplete; **Solution 2** by user grogg007 commits a serious logical error because it only provides functions that satisfy the requirements, without showing that no other functions do. Other places like [1] gives a good solution, but unfortunately it requires calculus. Here I am trying to provide a good solution that does not require calculus.

Let D and E be the set of the real roots of $P(x)$ and $Q(x)$, respectively. Then, outside of the intervals $[a, b]$ and (c, d) , $f(x) > 0$ except when $x \in E$, where $f(x)$ is undefined.

For any function $g(x)$, define a real number t as a **toggling point** for $g(x)$ if there exists a sequence $t_1 < t_2 < \dots < t_N$, with the understanding that $t_0 = -\infty$ and $t_{N+1} = \infty$, such that

- $t = t_k$ for some $1 \leq k \leq N$,
- $g(x)$ is of the same sign (either always positive or always negative) on each interval (t_j, t_{j+1}) , for $j = 0, \dots, N$, and
- $g(x)$ has opposite signs on (t_{k-1}, t_k) and (t_k, t_{k+1}) .

We call these open intervals (t_j, t_{j+1}) ($j = 0, \dots, N$) the **bands** of $g(x)$.

The polynomial $P(x)$ has no toggling point outside of D , and $Q(x)$ has no toggling point outside of E . Now in our problem, $f(x)$ has exactly 4 toggling points: a, b, c, d . Our task is to see how these four points are related to the sets D and E .

Lemma. At each toggling point t for $f(x)$,

- a) either $P(t) = 0$ or $Q(t) = 0$ (or both);
- b) if the multiplicities of the root t of $P(x)$ and $Q(x)$ are m and n , respectively, then $m - n$ is odd.

Proof. (1) if both $P(t)$ and $Q(t)$ are nonzero, then t belongs to a band of $P(x)$ and a band of $Q(x)$. Taking the intersection between these two bands, we get an open interval on which $f(x)$ has the same sign as $f(t)$. Thus t cannot be a toggling point of $f(x)$.

(2) If m is odd, there are real numbers r, s , with $r < t < s$, such that $P(x)$ has the same sign on (r, t) and has the opposite sign on (t, s) . If m is even, there are real numbers r, s , with $r < t < s$, such that $P(x)$ has the same sign on (r, t) and on (t, s) . Similar statements can be made of $Q(x)$, having intervals (u, t) and (t, v) so that $Q(t)$ is of the same sign on (u, t) , and of either a different or the same sign on (t, v) , depending on the parity of n . If both m and n are odd or both m and n are even, then $P(t)/Q(t)$ is of the same sign on the interval $(\max\{r, u\}, t)$ and $(t, \min\{s, v\})$, so t cannot be a toggling point for $f(x)$.

Solution to the problem. Since a is a toggling point for $f(x)$, either $P(a) = 0$ or $Q(a) = 0$, or both. But since $f(a)$ is defined, we have $Q(a) \neq 0$, so $P(a) = 0$. The same can be said of b , so $P(b) = 0$. The multiplicities of a and b must be 1. Otherwise, point (2) of the Lemma would require the multiplicity of one of them to be at least 3, but $P(x)$ is only cubic.

c is a toggling point for $f(x)$, so either $P(c) = 0$ or $Q(c) = 0$, or both. But if $Q(c) \neq 0$, we have $f(c) = 0$, violating the condition. So $Q(c) = 0$. Similarly, $Q(d) = 0$. Concerning their multiplicities, point (2) of the Lemma gives us three cases, given that $Q(x)$ is only cubic:

- (a) c is a single root of $Q(x)$, $P(c) \neq 0$, d is a double root of $Q(x)$ and a single root of $P(x)$;
- (b) switching the roles of c and d ;
- (c) both c and d are single roots of $Q(x)$, and $Q(x)$ has a single root e other than c and d .

Case (a) provides functions in the form of $\frac{(x-a)(x-b)(x-d)}{(x-c)(x-d)^2}$. There are 5 of them because there are 5 ways to choose a, b, c, d from the set $\{1, 2, 3, 4, 5\}$.

Case (b) provides functions in the form of $\frac{(x-a)(x-b)(x-c)}{(x-c)^2(x-d)}$, but they are the same as those provided in case (a). To ensure this, notice that the two expressions have the same domain and they are equal for every x in their domains.

Case (c) requires that $P(e) = 0$, because if $P(e) \neq 0$ then e becomes an extra toggling point for $f(x)$. So this case provides functions in the form of $\frac{(x-a)(x-b)(x-e)}{(x-c)(x-d)(x-e)}$. But since $f(e)$ is undefined, we need to make sure that e does not fall into $[a, b]$ or (c, d) . So e can be 1, 3, 5. In all these cases, we also have $f(e) = \frac{(e-a)(e-b)}{(e-c)(e-d)} > 0$, satisfying the conditions. That's 3 more functions.

So the total number of functions is $5 + 3 = \boxed{8}$.

Lightest (talk) 22:51, 10 November 2025 (EST)

Solution 4 (Multiplicity Analysis)

For the condition $f(x) \leq 0 \iff x \in [a, b] \cup (c, d)$, f must change sign at a, b, c , and d . The key trick is that a rational function can only change sign through zeroes or discontinuities. Since both $P(x)$ and $Q(x)$ are monic cubic polynomials, $f(x) \rightarrow 1$ as $x \rightarrow \pm\infty$. This means $f(x)$ is $+$ for large $|x|$. The required sign pattern must therefore be:

$$f(x) : \quad + \left[\begin{array}{c} - \\ a \end{array} \right] + \left(\begin{array}{c} - \\ b \end{array} \right) + \left(\begin{array}{c} - \\ c \end{array} \right) + \left(\begin{array}{c} - \\ d \end{array} \right) +$$

- For the interval to be closed, $[a, b]$, we must have $f(a) = 0$ and $f(b) = 0$. This implies $P(a) = P(b) = 0$ and, critically, $Q(a) \neq 0$ and $Q(b) \neq 0$ to avoid discontinuity.

- For the interval to be open, (c, d) , the function must be undefined at the endpoints: if $f(c)$ were defined, by the intermediate value theorem $f(c) = 0$, violating the condition. This implies $Q(c) = 0$ and $Q(d) = 0$. Note that $P(c), P(d)$ could be 0, as $\frac{0}{0}$ remains undefined.

From this, we conclude that $a, b, c, d \in \{1, 2, 3, 4, 5\}$. Let the remaining root in be t , so $\{1, 2, 3, 4, 5\} = \{a, b, c, d, t\}$.

Let $p_i, q_i \in \mathbb{Z}_{\geq 0}$ be the multiplicities of a root $i = 1, 2, 3, 4, 5$ in $P(x), Q(x)$, respectively, e.g. $P(x) = (x-1)^{p_1}(x-2)^{p_2}(x-3)^{p_3}(x-4)^{p_4}(x-5)^{p_5}$. Let the multiplicity in $f(x)$ at i be $m_i = p_i - q_i$. A sign change occurs at i if and only if m_i is odd.

- The sign changes at a, b, c, d . Thus, m_a, m_b, m_c, m_d must all be odd.
- The sign does not change at t . Thus, m_t must be even.

From the root analysis, $q_a = 0 \implies m_a = p_a - q_a = p_a$ is odd. Similarly, p_b is odd. Since $\deg(P) = \sum p_i = 3$, and $p_a, p_b \geq 1$ are odd, $m_a = m_b = p_a = p_b = 1$. This forces the roots of $P(x)$ to be distinct: $\{a, b, k\}$ for some $k \in \{1, 2, 3, 4, 5\} \setminus \{a, b\}$ with $p_k = 1$.

We already distributed 2 roots for P and Q each, so there is 1 left to give. Our constraints give $p_c + p_d + p_t = 1$, $p_c, p_d, p_t \in \{0, 1\}$ and $q_t \in \{0, 1\}$, $q_c, q_d \in \{1, 2\}$. We can make some inferences:

- $p_c \in \{0, 1\}$, $q_c \in \{1, 2\}$, $m_c = p_c - q_c \in \{-2, -1, 0\}$. Since m_c is odd, $m_c = -1$. Similarly, $m_d = -1$.
- $p_t \in \{0, 1\}$, $q_t \in \{0, 1\}$, $m_t = p_t - q_t \in \{-1, 0, 1\}$. Since m_t is even, $m_t = 0$.

At this point, we have $(m_a, m_b, m_c, m_d, m_t) = (1, 1, -1, -1, 0)$. But this is exactly the distribution of multiplicities of $\frac{(x-a)(x-b)}{(x-c)(x-d)}$, which is forced from the beginning. Thus, the remaining multiplicity of each of P and Q must cancel into a single bin to create net multiplicity $0 = 1 - 1$, yielding

$$f(x) = \frac{(x-a)(x-b)}{(x-c)(x-d)} \cdot \frac{x-k}{x-k}$$

where $k = c, d, t$, and this creates a hole if there isn't one already. However, the problem is that if there is already a hole (i.e. $k = c, d$), then this factor $\frac{x-k}{x-k}$ changes nothing as f was already undefined at $x = k$, so while it gives a different pair of polynomials, it does not provide a new function f .

Cases $k = c, d$ are symmetric:

WLOG, let $k = c$, and choosing $t = 1, 2, 3, 4, 5$ determines the remaining $1 \leq a < b < c < d \leq 5$. Each t creates a unique pair of polynomials, so there are $2 \cdot \binom{5}{1} = 10$ pairs of polynomials. However, as established, puncturing a hole again on $k = c, d$ does nothing:

$$f(x) = \frac{(x-a)(x-b)}{(x-c)(x-d)} \cdot \frac{x-c}{x-c} = \frac{(x-a)(x-b)}{(x-c)(x-d)} \cdot \frac{x-d}{x-d}$$

So each t creates exactly one new function regardless of whether $k = c$ or $k = d$, i.e. there are only 5 new functions.

Case $k = t$:

Sketching a quick diagram for (sign pattern of f), t cannot puncture $[a, b]$ nor (c, d) , meaning it must lie outside them. All other cases do work, since you can puncture $+$ intervals without affecting $-$ intervals.

$$\begin{cases} t, [a, b], (c, d) \\ [a, b], t, (c, d) \\ [a, b], (c, d), t \end{cases}$$

Hence, there are 3 valid choices of t .

In total, there are $10 + 3 = \boxed{\text{(E)} 13}$ ordered pairs of polynomials (P, Q) but only $\boxed{8}$ different functions created by $f = \frac{P}{Q}$.